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THEORY OF BRAIDS

BY E. ARTIN

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A theory of braids leading to a classification was given in my paper "Theorie der Zöpfe" in vol. 4 of the Hamburger Abhandlungen (quoted as Z). Most of the proofs are entirely intuitive. That of the main theorem in §7 is not even convincing. It is possible to correct the proofs. The difficulties that one encounters if one tries to do so come from the fact that projection of the braid, which is an excellent tool for intuitive investigations, is a very clumsy one for rigorous proofs. This has lead me to abandon projections altogether. We shall use the more powerful tool of braid coordinates and obtain thereby farther reaching results of greater generality.

A few words about the initial definitions. The fact that we assume of a braid string that it ends in a straight line is of course unimportant. It could be replaced by limit assumptions or introduction of infinite points. The present definition was selected because it makes some of the discussions easier and may be replaced any time by another one. I also wish to stress the fact that the definition of s-isotopy is of a provisional character only and is replaced later (Definition 3) by a general notion of isotopy.

More than half of the paper is of a geometric nature. In this part we develop some results that may escape an intuitive investigation (Theorem 7 to 10).

We do not prove (as has been done in Z) that the relations (18) (19) are defining relations for the braid group. We refer the reader to a paper by F. Bohnenblust[1] where a proof of this fact and of many of our results is given by purely group theoretical methods.

Later the proofs become more algebraic. With the developed tools we are able to give a unique normal form for every braid[2] (Theorem 17, fig. 4 and remark following Theorem 18). In Theorem 19 we determine the center of the braid group and finally we give a characterisation of braids of braids.

I would like to mention in this introduction a few of the more important of the unsolved problems:

1) Assume that two braids can be deformed into each other by a deformation of the most general nature including self intersection of each string but avoiding intersection of two different strings. Are they isotopic? One would be inclined to doubt it. Theorem 8 solves, however, a special case of this problem.

2) In Definition 3, we introduce a notion of isotopy that is already very general. What conditions must be put on a many to many mapping so that the result of Theorem 9 still holds?

[1] F. Bohnenblust, The algebraical braid group, Ann. Math., vol. 48, (1947), pp. 127–136.

[2] The freedom of the group of k-pure braids has been proved with other methods in: W. FRÖHLICH, *Über ein spezielles Transformationsproblem bei einer besonderen Klasse von Zöpfen*, Monatshefte für Math. und Physik, vol. 44 (1936), p. 225.

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3) Determine all automorphisms of the braid group.

4) With what braids is a given braid commutative?

5) Decide for any two given braids whether they can be transformed into each other by an inner automorphism of the group. Concerning applications of braid theory this is by far the most important problem.

The last three of our questions seem to require an extensive study of the automorphisms of free groups.

We shall have to consider numerous functions of several variables. All of them are meant to be continuous in all the variables involved so that the statement of continuity is always omitted. Only more stringent conditions shall be mentioned.

Let x, y, z be the Cartesian coordinates of a 3-space. By a braid string we mean a curve that has precisely one point of intersection with each plane $z = a$ so that z may be used as parameter. Denoting by X the two dimensional vector (x, y) we may therefore describe the string by a vector function $X = X(z)$. In addition to that we assume the existence of two constants a, b such that $X(z)$ assumes a constant value X^- for all $z \leqq a$ and a constant value X^+ for all $z \geqq b$. X^- and X^+ are called the ends of the string.

By an n-braid we mean a set of n strings $X_i(z)$ $(i = 1, 2, \cdots n)$ without intersections (hence $X_i(z) \neq X_k(z)$ for $i \neq k$) where the numbering of the strings is considered unessential.

Two n-braids $X_i(z)$ and $Y_i(z)$ are called strongly isotopic (s-isotopic) if n vector functions $X_i(z, t)$ can be found with the following properties: They are defined for all z and for all t of a certain interval $c \leqq t \leqq d$, give an n-braid for each special t and are $X_i(z)$ for $t = c$, $Y_i(z)$ for $t = d$. They are constant in z and t if z is large enough and also constant if $-z$ is large enough. We remark at once that the ends remain fixed.

THEOREM 1. *s-isotopy is reflexive, symmetric and transitive.*

This allows us to unite s-isotopic braids into one class. We also obtain without difficulty:

THEOREM 2. *Let $g(z, t)$ be a numerical function defined for all z and all t of a certain t-interval. Assume that it tends with z to ∞ uniformly in t (the sign is unimportant). If $X_i(z)$ is a braid then the new braids $X_i(g(z, t))$ for different values of t are all s-isotopic. (They need not be s-isotopic to $X_i(z)$ itself.)*

COROLLARY 1. *If the numerical function $g(z)$ satisfies $\lim_{z=+\infty} g(z) = +\infty$ and $\lim_{z=-\infty} g(z) = -\infty$, then the braids $X_i(z)$ and $X_i(g(z))$ are s-isotopic.*

PROOF: Put $g(z, t) = (1 - t)\, z + t\, g(z)$ for $0 \leqq t \leqq 1$.

COROLLARY 2. *$X_i(z)$ is s-isotopic to any z-translation $X_i(z + t)$.*

COROLLARY 3. *The braids $X_i(\,|\,z\,|\,+\,t)$ are s-isotopic among themselves for different values of t. For large positive values of t all braid strings are constant $= X_i^+$. For large negative values of t the part of the braid above the xy-plane looks like a z-translation of $X_i(z)$, the part below this plane like its reflection on the xy-plane.*

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We mention a few theorems of which little use shall be made in this paper, but that are useful if braids are to be studied by means of their projection on the yz-plane from the positive x-direction. By lexicographical arrangement of vectors we mean, as usual, their arrangement according to the size of their y-coordinate and, in case the y-coordinates are equal, according to their x-coordinate.

Two braids are said to have the same z-pattern of projection if the lexicographical arrangement of the vectors $X_i(z)$ is for each value of z the same as that of the vectors $Y_i(z)$.

LEMMA. *Two braids with the same z-pattern of projection are s-isotopic.*

PROOF: Put $X_i(z, t) = (1 - t)X_i(z) + t\,Y_i(z)$ for $0 \leqq t \leqq 1$. Assume we could find values z, t and $i \neq k$ such that $X_i(z, t) = X_k(z, t)$. This means that $(1 - t)(X_i(z) - X_k(z)) + t(Y_i(z) - Y_k(z)) = 0$. Let $X_i(z) - X_k(z) = (a, b)$ and $Y_i(z) - Y_k(z) = (a', b')$. The y-coordinate of our equation shows that b and b' can not have the same sign hence $b = b' = 0$. Now the x-coordinate of the equation leads to $a = a' = 0$. But then $X_i(z)$ would not be a braid.

Two braids $X_i(z)$ and $Y_i(z)$ are said to have the same pattern of projection if a monotonically increasing function $g(z)$ with infinite limits exists such that $X_i(z)$ and $Y_i(g(z))$ have the same z-pattern of projection.

THEOREM 3. *Two braids with the same pattern of projection are s-isotopic.*

PROOF: Use the lemma and Corollary 1 of Theorem 2.

THEOREM 4. *Let d be less than half the minimal distance between two of the strings of $X_i(z)$. Let $Y_i(z)$ be a braid with the same ends as $X_i(z)$ and assume that equally numbered strings of the two braids have at each z level a distance less than d. Then the braids are s-isotopic.*

The proof is done with the same device as in the lemma and is trivial.

DEFINITION: Two braids $X_i(z)$ and $Y_i(z)$ are called composable if:

1) They have the same number of strings.

2) After a suitable change in the numbering of the strings we have $Y_i^+ = X_i^-$. So the upper ends of $Y_i(z)$ must fit the lower ends of $X_i(z)$.

If (after a suitable translation) we join these ends we obtain a new braid which is said to be composed of $X_i(z)$ and $Y_i(z)$ (in this order). The formal definition would be:

Select an a and a b such that $X_i(z + a)$ is constant for negative z and $Y_i(z + b)$ is constant for positive z. Put $Z_i(z) = X_i(z + a)$ for positive z and $= Y_i(z + b)$ for negative z. This braid and any translation of it is called the composed braid. We still have a great freedom in the selection of a, b and the translation. All these braids are however s-isotopic.

If we replace both braids by others that are s-isotopic to them we obtain braids that are s-isotopic to $Z_i(z)$. The proof follows from the fact that after suitable selection of a and b the necessary deformation can be carried out independently in both sections of the composed braid. This leads to:

DEFINITION 1. Two classes of braids A and B are called *composable* if the braids in these classes are composable. The resulting braids form a class denoted by AB. It is to be remarked that if A is composable with B then B may

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not be composable with A. And even if the classes are composable from both sides then the commutative law may not hold. But the associative law does. If A and B as well as AB and C are composable then B and C as well as A and BC are composable and we have $A(BC) = (AB)C$.

THEOREM 5. *The classes of n-braids form a groupoid under our composition.*

PROOF: The postulates that we must verify are:

1) The kind of associative law we just have mentioned.

2) The existence of two classes U and U' for each given class A (dependent on A) such that $AU = U'A = A$. U is obviously the class containing a constant braid with the same lower ends as A and U' the similar class connected with the upper ends of A.

3) The existence of a class A^{-1} such that $A^{-1}A = U$ and $AA^{-1} = U'$. Corollary 3 shows that the reflection of a braid on the x, y plane gives such a class.

4) If A and B are given classes there exists a class C such that AC as well as CB can be formed. This just means to construct an example of a braid with given ends.

If U is one of the unit classes call G_U the set of all A that have U as left as well as right unit. They form a group. If V is another unit and C a class such that $UC = CV = C$ then $G_V = C^{-1}G_UC$ and the transformation thus indicated is an isomorphism. The knowledge of one of these groups reveals the structure of all and as a matter of fact the structure of the whole groupoid. The braids in such a group are those whose upper ends are only a permutation of the lower ends.

Next we prove that s-isotopy can be extended to the whole space:

THEOREM 6. *Let $X_i(z, t)$ be the n functions describing an s-isotopy. Then we can find a function $F(X, z, t)$ defined for all X and z and all necessary t, whose value is a vector, and that has the following properties:*

1) *For any fixed z it is a deformation of the plane. That means that it is a one to one correspondence of the plane if t also is fixed and it is identity for $t = a$ if that is the beginning of the t-interval.*

2) *Should for any special value of z the original functions $X_i(z, t)$ be independent of t, then $F(X, z, t) = X$ for that z and all X and t.*

3) *If a point (X, z) of the 3-space has a sufficiently large distance from the origin then $F(X, z, t) = X$ for all t.*

4) *$F(X_i(z, a), z, t) = X_i(z, t)$. So the deformation of the space moves the braid-strings precisely as the s-isotopy does.*

PROOF: 1) Select an $r > 0$ such that $| X_i(z, t) - X_k(z, t) | < 3r$ for all $i \neq k$ and all z and t. We first construct an auxiliary function $G(X, P_\nu, Q_\nu)$ of X and $2n$ points P_ν, Q_ν ($\nu = 1, 2, \cdots n$) of the plane. The points P_ν are restricted by the condition that their mutual distance shall always be greater than $3r$, the points Q_ν by the condition that Q_i lies in the interior of a circle C_i of radius r around P_i. The value of G shall be X if X is outside of all the circles or on the periphery of one. For $X = P_i$ the value shall be Q_i. If X is in the interior of C_i but different from P_i draw a radius through X and call R its intersection with C_i. Define the functionvalue as that point on the straight line segment

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$R\ Q_i$ that bisects it in the same ratio as X bisects $R\ P_i$. Our function is continuous in all the variables, is a one to one correspondence of the plane for fixed P_ν and Q_ν and reduces to identity if $P_\nu = Q_\nu$ for all ν.

2) Divide the t interval into a finite number of parts $t_i \leqq t \leqq t_{i+1}$ such that the variation of every $X_i(z, t)$ in that interval is less than r for fixed z.

In $a = t_0 \leqq t \leqq t_1$ we define $F(X, z, t) = G(X, X_\nu(z, a), X_\nu(z, t))$. It has all the necessary properties. Assume that we succeeded to define $F(X, z, t)$ for all t of $t_0 \leqq t \leqq t_m$ and to check on the required properties. For $t_m \leqq t \leqq t_{m+1}$ we define:

$$F(X, z, t) = G(F(X, z, t_m), X_\nu(z, t_m), X_\nu(z, t)).$$

For $t = t_m$, we get the old value so it is a continuous continuation. The properties 1, 2, 3 follow immediately. For $X = X_i(z, a)$ we get $F(X, z, t_m) = X_i(z, t_m)$ hence $F(X, z, t) = X_i(z, t)$.

The extension of an s-isotopy to the whole space is not the only use of Theorem 6. We also use it to introduce new coordinates for the points of the space called braid coordinates. They are much more flexible in dealing with braids and the principal tool in the proofs of most of the following theorems.

Let $X_i(z)$ be a given braid, constant for $z \leqq a$ and for $z \geqq b$. Consider the braids $X_i(z + t)$ for $0 \leqq t \leqq b - a$. They are all isotopic and let $F(X, z, t)$ be the extension of this isotopy to the whole space. Then we have:

$$F(X_i(z), z, t) = X_i(z + t), \qquad 0 \leqq t \leqq b - a. \tag{1}$$

With each point (x, z) of the space we associate now a 2 dimensional vector $Y = Y(X, z)$ in the following way:

For $z \leqq a$ let $Y = X$.

For $a \leqq z \leqq b$ let Y be the unique solution of $F(Y, a, z - a) = X$. For $z = a$ we have $F(X, a, 0) = X$, therefore $X = Y$ again.

For $z \geqq b$ put $Y(X, z) = Y(X, b)$.

If z is fixed then the mapping $Y = Y(X, z)$ is a one to one correspondence of the plane that is certainly identity, outside a large circle whose radius does not depend on z. It is identity for all X if $z \leqq a$. For $z \geqq b$ it is in general not identity but at least is the same mapping for all $z \geqq b$.

$F(X_i(a), a, z - a) = X_i(a + (z - a)) = X_i(z)$ if $a \leqq z \leqq b$ because of (1). This shows that the Y for the point $(X_i(z), z)$ of the ith string is $X_i(a) = X_i^-$. The same is true for $z \leqq a$ and for $z \geqq b$ for trivial reasons.

We associate now with the point (X, z) the corresponding combination $\{Y, z\}$ and call it the braid coordinates of that point. They equal the ordinary coordinates for all $z \leqq a$ and also for all large $|X|$. All points on the ith string have the simple braid coordinates $\{X_i^-, z\}$.

Another way to look upon the braid coordinates is this: Interpret them as ordinary coordinates of a point of a 3-space. Then our 3-space is mapped by a one to one correspondence onto this new one and the braid strings are mapped onto vertical lines with the same lower ends.

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Let now u be any real number. The mapping $\{Y, z\} \rightarrow \{Y, z + u\}$ in our old space is a one to one correspondence that has all the essential features of a translation and shall therefore be called translation by u along the braid. Each single string remains fixed as a whole. For large $|X|$ and also if $|z|$ is large in comparison to $|u|$ it is an ordinary translation. For the other points the z-coordinate does at least behave in the ordinary way.

We can of course also find the inverse function $X = H(Y, z)$ that describes the passage from Y back to X. Let now $X(z)$ be any other braidstring (it may intersect the strings of our braid). Apply to it a translation by u along our braid. The braid coordinates of $(X(z), z)$ are $\{Y, z\}$ where $Y = Y(X(z), z)$. The translation moves it into $\{Y, z + u\}$. The ordinary coordinates are $(X', z + u)$ where $X' = H(Y(X(z), z), z + u)$. Looking at the new string as a whole, we may replace z by $z - u$ and obtain the new braidstring

$$X(z, u) = H(Y(X(z - u), z - u), z)$$

such that $(X(z, u), z)$ are the points of the new braidstring. Letting now the u change, we see that we have before us an s-isotopic change where the points move only in horizontal planes. Should the original string not intersect the braidstrings, then its translation does not either and the braid formed out of the old braid by adding our new string undergoes an s-isotopy under translation.

We use this in the following way: Let $X_i(z)$ be an n-braid and replace its nth string by any other string $X'_n(z)$ with the same ends. The new string may intersect $X_n(z)$ but shall not intersect the other strings of our braid. This gives a new n-braid and we apply now to our old braid a translation by a large u along our new braid. What happens? The $n - 1$ first strings remain fixed. If u is sufficiently large then a very low portion of the nth string will now be in the main part of the braid. In that very low portion the string $X_n(z) = X'_n(z)$. The string $X'_n(z)$ does not change under our translation. This shows that in the main portion of our braid $X_n(z)$ has moved into the position $X'_n(z)$. This is of course compensated by the fact that the nth string is now entangled in the other strings above the main portion of the braid. But above the main section of the braid, the first $n - 1$ strings are very simple, namely parallel lines. Remember finally that we have shown in the preceding paragraph that the translation can also be considered as a horizontal motion, as an s-isotopy.

THEOREM 7. *It is possible to apply to a braid an s-isotopy moving one string only whereby this string may be brought into any other position provided this is compensated by a motion of the string above the main section of the braid where the* $n - 1$ *other strings are parallel.*

This suggests the definition:

DEFINITION 2. A braid with the same upper and lower ends is called *i-pure*, if all but the ith string are constant. A class is called i-pure if it contains an i-pure representative.

We see: if B and B' are braids or classes of braids having $n - 1$ strings in common then $B = AB'$ where A is i-pure.

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Another useful notion connected with braid coordinates is that of projection along the braid. Consider the plane $z = z_0$. The mapping $\{Y, z\} \to \{Y, z_0\}$ is this projection. It carries the braidstrings into their intersection with the plane and a point not on the braid into a point different from these intersections.

Let now R_B be the complementary set to B in the 3-space. We introduce the usual notion of homotopy of paths in R_B. Two paths $a = \{Y(t), z(t)\}$, $b = \{Y'(t), z'(t)\}$, $0 \leqq t \leqq 1$ are called homotopic in R_B : $a \sim_B b$ if a function of two variables t and s (interval 0, 1) $\{Y(t, s), z(t, s)\}$ can be found, that is constant for $t = 0$ and $t = 1$, gives the first path for $s = 0$ and the second for $s = 1$. All points of the deformation have to belong to R_B which means simply that the function Y avoids the values X_i^-.

The composition of homotopy classes is introduced in the usual way and leads to a groupoid. If we select a point P in R_B and consider the homotopy classes of those paths that have beginning as well as endpoint at P, we obtain the Poincaré group of R_B. Let z_0 be the z coordinate of P and $a = \{Y(t), z(t)\}$ any element of this group. The projection $a' = \{Y(t), z_0\}$ is homotopic to a as the function $\{Y(t), z_0 + s(z(t) - z_0)\}$ shows. If two paths in this plane are B-homotropic, say by the function $\{Y(t, s), z(t, s)\}$, then the function $\{Y(t, s), z_0\}$ shows that the paths are already homotopic in the plane. The Poincaré group is therefore the same as that of a plane punctured in the n points $X_i(z_0)$. So it is a free group with n generators.

We must now carefully describe the generators we want to use. The plane will be either in the region of z where the braidstrings assume the constant values X_i^- and shall then be called a lower plane or in the region of the X_i^+ when we call it an upper plane. Take an upper plane and draw in it a ray that does not meet any of the upper or lower ends. We intend to take the point P on that ray sufficiently far away. Each of the points X_i^+ shall be connected with the beginning point Q of the ray by a broken line without self intersection such that two of the lines and also the ray have only the endpoint Q in common. By l_i we denote the connection thus established between the beginning point X_i^+ and the point P on the ray. By $l_i(\epsilon)$ we mean the same path but starting with the parameter value ϵ. An orientation of the plane is selected. By $c_i(\epsilon)$ we mean a curve with the winding number 1 around X_i^+ starting and ending at the beginning point of $l_i(\epsilon)$ that stays within a small neighborhood of X_i^+. It is well known that the paths (for small ϵ)

$$t_i = l_i(\epsilon)^{-1} c_i(\epsilon) l_i(\epsilon) \tag{2}$$

are free generators of the Poincaré group of the punctured plane. This pattern of paths is then transferred to all other upper planes by projection (in the ordinary sense) including the point P. We use the same names for the paths in all the upper planes.

In a lower plane we first transfer the ray, the orientation and the point P to it by ordinary projection. Since we now have to take care of the lower ends,

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new paths are selected denoted by l_i', $l_i'(\epsilon)$, $c_i'(\epsilon)$. The corresponding generators shall be called t_i'.

What we intend to do with this setup is roughly this: If two braids are s-isotopic they must by necessity have the same ends. It therefore suffices to consider braids with given ends. For all these braids we use the same pattern of paths in the lower and upper planes. If B is a given braid then we project by braid projection the generators t_i' into the upper plane. We obtain paths $\bar{t}_i$ that are now generators in the upper plane. Therefore they can be expressed in terms of the t_i. It turns out that these expressions are a complete set of invariants for the isotopy classes.

As a first indication that the study of the homotopy classes must give a solution of our problem let us consider the following special case:

Let C be an $(n-1)$-braid. Form two n-braids by inserting in C an nth string in two ways but both time with the same ends. Select a z_0 such that the two strings are equal for $z \leqq z_0$ and call P_0 the point on the nth strings at that z_0-level. In a similar fashion select z_1 and P_1 for the upper end. Consider now the two pieces of the nth strings between P_0 and P_1. If they are homotopic relative to C we may call the two n^{th} strings C-homotopic. Then the following rather surprising theorem holds:

THEOREM 8. *Let B and B' be obtained from C by insertion of an nth string with given ends. If the two strings are C-homotopic then B and B' are s-isotopic. Every homotopy class can be realized by a braid. The converse is also true but will be proved later.*

PROOF: 1) We use braid coordinates of C and express the two inserted strings in terms of these coordinates: $Y_n(z)$, $Y_n'(z)$. The fact that they do not intersect the strings of C means just that the functions avoid the values X_i^-. By assumption there exists a function $\{Y(t, s), z(t, s)\}$ defined in $z_0 \leqq t \leqq z_1$, $0 \leqq s \leqq 1$ describing the homotopy of the strings. So $Y(t, s)$ will avoid the values X_i^-, have fixed beginning and end points for all s and will be $Y_n(t)$ for $s = 0$ and $Y_n'(t)$ for $s = 1$. The method consists now in forgetting about the function $z(t, s)$ altogether and to define a function $Y_n(z, s)$ as equal to $Y(z, s)$ for $z_0 \leqq z \leqq z_1$ and equal to X_i^- for all other z. The function avoids X_i^- and reduces to the given strings for $s = 0$ and $s = 1$. So it gives the required s-isotopy.

2) If $\{Y(t), z(t)\}$ is any curve defined in $z_0 \leqq z \leqq z_1$ that avoids the strings of C and joins P_0 and P_1, put as before $Y_n(z) = Y(z)$ in that interval and $= X_n^-$ for all other z. The function $\{Y(t), s \cdot z(t) + (1-s)t\}$ shows that this n^{th} string is homotopic to the given curve.

REMARK. The s-isotopy of Theorem 8 moves the nth string only.

Let us now return to our upper plane. Join one of the points X_i^+ to P by a curve h that avoids the braid with exception of its beginning point. Define $h(\epsilon)$ as before and let $d(\epsilon)$ be a curve analogous to $c_i(\epsilon)$. Consider the element $t = h(\epsilon)^{-1} d(\epsilon)\, h(\epsilon)$. If we join the beginning point of $l_i(\epsilon)$ to that of $h(\epsilon)$ by a path e that stays in a small neighborhood of X_i^+ then $e^{-1} c_i(\epsilon) e$ is homotopic to

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$d(\epsilon)$. The path $S' = l_i(\epsilon)^{-1} e\, h(\epsilon)$ is a certain element of the group and may be expressed in terms of the t_ν. The element

$$S'^{-1} t_i S' = h(\epsilon)^{-1} e^{-1} l_i(\epsilon) l_i(\epsilon)^{-1} c_i(\epsilon) l_i(\epsilon) l_i(\epsilon)^{-1} e\, h\ (\epsilon)$$

is homotopic to t. Assume now that a similar expression $t = S^{-1} t_i S$ is known from some other source and may even not be in a reduced form. We first perform the possible cancellations in S only; a further simplification of the expression $S^{-1} t_i S$ is then possible only if S begins with a power of t_i. Then S^{-1} ends with the reciprocal of that power and we see that this term may indeed be dropped. This shows that S is uniquely determined but for a power of t_i and we have therefore $S' = t_i^r S$ or

$$l_i(\epsilon)^{-1} e\, h(\epsilon) = t_i^r S. \tag{3}$$

Let us now reinsert in our plane the one point X_i^+ and consider homotopies in that new plane, punctured in $n - 1$ points only. This homotopy shall be denoted by $\sim_i$. It amounts to put $t_i = 1$ in all previous expressions. The resulting element shall still be denoted by S. We obtain:

$$l_i(\epsilon)^{-1} e\, h(\epsilon) \sim_i S.$$

The path $l_i l_i(\epsilon)^{-1} e\, h(\epsilon) h^{-1}$ is i-homotopic to a closed curve starting at X_i^+ and remaining in a small neighborhood of that point. It is therefore i-homotopic to this point X_i^+. This proves the formula

$$\bar{l}_i^{-1} h \sim_i S. \tag{4}$$

Let now B be a braid with the given ends. If we apply braid projection to a generator t_i' of a lower plane onto an upper plane and assume that the point P has been selected sufficiently far out, then the image $\bar{t}_i$ will be an element of the Poincaré group for P. If l_i is the projection of l_i' we obtain equations of the form:

$$\bar{l}_i = S_i^{-1} t_i S_i \tag{5}$$

$$l_i(\epsilon)^{-1} e_i \overline{l_i(\epsilon)} \sim t_i^r S_i \tag{6}$$

$$l_i^{-1} l_i \sim_i S_i. \tag{7}$$

It is to be remarked that the properties of the braid coordinates show that the form of the equation (5) does not depend on the precise location of the upper and the lower plane. It is also clear that it does not depend on ϵ provided it is only small enough. The position of P plays also no role in it provided that it is far enough out. The equations (6) and (7) change of course their meaning and the exponent r may depend on ϵ and e_i.

We may look upon this process in yet another way. Call g the straight line segment that connects P with its projection in the lower plane and put $\tau_i = g t_i' g^{-1}$. They are elements of the Poincaré group for P and as a matter of fact a set of generators. If we subject them to braid projection they will go over

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again into the $\bar{t}_i$. But being elements with the same beginning and end point τ_i is, as we have seen before, homotopic to $\bar{t}_i$. So equation (5) becomes

$$\tau_i \sim S_1^{-1} t_i S_i . \tag{8}$$

We are now ready for a generalized notion of isotopy:

DEFINITION 3. A braid is called *isotopic to another braid* if the space can be mapped into itself in such a way that points on the first braid but no other points of the space are mapped onto points of the second braid. In addition to this we assume that the mapping is identity outside of a certain sphere. Inside that sphere the mapping must of course be continuous but need not be one to one.

Consider now two isotopic braids B and C. Locate the lower and the upper plane outside the sphere and select P so that g is also outside this sphere. The two elements τ_i and its B-projection $\bar{t}_i$ are B-homotopic. The surface connecting the two paths does not meet B. Its image under our mapping will therefore avoid C. This shows that the images of our paths are C-homotopic. But our paths remain fixed. So τ_i and $\bar{t}_i$ are C-homotopic. τ_i on the other hand is C-homotopic to its C-projection. This C-projection is therefore C-homotopic to $\bar{t}_i$ and this proves:

THEOREM 9. *If B is isotopic to C then the exexpressions in formula* (5) *are the same for B and C.*

The i-homotopy of formula (7) may be interpreted as homotopy with respect to the braid resulting from a cancellation of the ith string. Denote by Σ_i the piece of the ith string that starts at the upper plane and ends at the lower plane. The path $\bar{l}_1^{-1}\, \Sigma_i l_i' g^{-1}$ is a closed path starting at P and as such i-homotopic to its projection onto the upper plane. But this projection is obviously the left side of (7). Computing Σ_i out of the resulting homotopy we get:

$$\Sigma_i \sim_i l_i S_i g l_i'^{-1} \tag{9}$$

and this shows that S_i determines the homotopy class of the ith string.

(9) Interprets the ith string but it would not completely explain S_i since it is only an i-homotopy. Let Σ_i' be any path connecting the beginning points of $l_i(\epsilon)$ and $l_i'(\epsilon)$ that stays in the immediate vicinity of the nth string without intersecting it. Its projection is then a curve that may be used as e_i. Now the projection of $l_i(\epsilon)^{-1}\Sigma_i' l_i'(\epsilon) g$ is the left side of (6):

$$t_i' S_i \sim l_i(\epsilon)^{-1}\Sigma_i' l_i'(\epsilon) g \tag{10}$$

which provides the full geometric meaning of S_i. The converse of Theorem 8 is:

THEOREM 10. *Let B and C be two braids with the same ends and with the same first $n-1$ strings. Assume either that B and C are s-isotopic or that they are isotopic or that the expressions* (5) *are the same for both braids. Then the nth strings are n-homotopic and there exists an s-isotopy moving the nth string only.*

PROOF: If they are s-isotopic then they are isotopic because of Theorem 6. If they are isotopic then the expressions (5) are the same because of Theorem 9.

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If (5) is the same even for $i = n$ only then the nth strings are n-homotopic because of formula (9). The remark to Theorem 8 shows the rest of the contention.

THEOREM 11. *If two braids have the same ends and if the expressions* (5) *are the same for both braids then they are s-isotopic.*

PROOF: For $n = 1$ our theorem is trivial since the two strings are 1-homotopic. Let it be proved for braids with $n - 1$ strings. If B and C are two n-braids for which the assumption holds, let B' and C' be the braids resulting from cancellation of the nth string. The expressions (5) for B' and C' are obtained by putting $t_n = 1$. So (5) is also the same for B' and C'. Therefore B' and C' are s-isotopic. Extend the s-isotopy to the whole space and apply this mapping to the braid B. It carries B into an s-isotopic braid D having again the same expressions (5). D and C have now the first $n - 1$ strings in common so that Theorem 10 shows that they are s-isotopic. This completes the proof.

THEOREM 12. *Isotopy and s-isotopy imply each other.*

The proof follows from Theorems 6, 9 and 11.

THEOREM 13. *The expressions* (5) *do not depend on the special braid-coordinates used. They depend even only on the class and give together with the ends a full system of invariants of the class that determines the class completely.*

The proof is now obvious. It is to be remarked, however, that the expressions depend on the selection of generators t_i and t_i'. We must now develop methods that allow the actual computations of these invariants and reveal the structure of our groupoid.

To do so we have first to change our notation slightly. Select in a plane n points X_i, a ray and paths l_i. Up to this point the numbering was considered unessential. Now we get a natural arrangement of our points by starting with our ray and going around Q in the positive sense of rotation (in a neighborhood of Q). The first path that we meet shall be called l_1, the next l_2 and so on. The points X_i are now numbered precisely as the paths leading to them. The very same pattern is now used for the upper planes as well as for the lower ones. The points X_i are now used as lower and upper ends of braids. We restrict ourselves to the investigation of braids whose lower ends are a subset of the X_i and whose upper ends are another subset. These subsets may or may not be the whole set, no restriction being put upon them. If B is such a braid and X_i, X_j the lower respectively upper end of one of its strings we write:

$$j = B(i).$$

Thus B maps a certain subset of the numbers $1, 2, \cdots n$ onto another subset. The numbering of the generators t_i and t_i' so far was connected with the numbering of the string. Now we change that and attach to them the subscript of the point around which they run. We also drop the accent on the t_i' and write uniformly t_i for all the generators in the different planes. This leads to the following situation:

We have a group F before us with the n free generators t_i. For the Poincaré group of a braid with the reference point in an upper or a lower plane, not all

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the generators are used; the Poincaré group of such a plane is therefore considered a subgroup of F generated by a subset of the t_i. Braid projection of a lower unto an upper plane will provide us with an isomorphic mapping of the group in the lower plane onto the group in the upper one. If T is an element of the group in the lower plane then its image under braid projection shall be denoted by $B(T)$. In this new notation (5) takes on the form:

$$B(t_i) = S_j^{-1} t_j S_j \quad \text{where} \quad j = B(i), \tag{11}$$

and where the numbering of all generators has been changed according to our new convention. (11) alone already gives us the isomorphic mapping and in this form contains also the information about the upper and lower ends of the strings of the braid. In case all the points are used for the lower ends, it will be an automorphism of F. Otherwise it maps a subgroup of F onto another subgroup.

Let A and B be two composable braids and form the composed braid in such a way that in AB the part B corresponds to negative, the part A to positive z. Returning for a moment to the interpretation of our projection in terms of the generators τ_i which allow to express projection in terms of homotopies we see:

$$AB(T) = A(B(T)).$$

We project namely a lower plane of AB first onto the plane $z = 0$ and the result onto an upper plane. Making use of the fact that (11) completely determines the class we see:

THEOREM 14. *The groupoid of braid classes whose lower and upper ends are subsets of the X_i is isomorphic to the groupoid of mappings in F indicated by* (11).

Consequently we express the braid class B in form of a substitution

$$B = \begin{pmatrix} t_i \\ S_j^{-1} t_j S_i \end{pmatrix} \tag{12}$$

where t_i runs of course only through certain of the generators. It is convenient to consider also more general substitutions

$$B = \begin{pmatrix} t_i \\ T_i \end{pmatrix}$$

in the free group F where certain t_i are mapped onto power products regardless of whether the substitution is derived from a braid or not. If the substitution is derived from a braid class then we say briefly that it is a braid. Also in this general case we denote by $B(T)$ the result of applying the substitution B onto the power product T.

The braid substitutions have one special property that we must derive. Draw a huge circle in a lower plane starting at the reference point of the Poincaré group and running around the braid. It is well known from the theory of the homotopies in a punctured plane that this element of the Poincaré group is homotopic to the product of the generators t_i (of course only those that we need for our

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braid) taken in the natural arrangement of the subscripts according to their size. Braid projection onto an upper plane carries the circle into a similar circle starting at P. This proves:

THEOREM 15. *If B is a braid then*

$$B\left(\prod t_i\right) = \prod t_j \tag{13}$$

both products taken in the natural arrangement of their subscripts.

Select a subscript $i < n$ and put $X_\nu(z) = X_\nu = \text{const}$ for $\nu \neq i, i+1$. We connect now X_i and X_{i+1} by a broken line starting at X_i and running then parallel to l_i until it comes near the ray; then it runs parallel to l_{i+1} until it comes close to X_{i+1} with which it is then connected. If $X(t)$ $(0 \leqq t \leqq 1)$ is the parametric representation of this line we put

$$X_i(z) = \begin{cases} X_i & \text{for} \quad z \leqq 0 \\ X(z) & \text{for} \quad 0 \leqq z \leqq 1 \\ X_{i+1} & \text{for} \quad z \geqq 1. \end{cases}$$

Then we draw a similar parallel curve between X_{i+1} and X_i running farther out and not intersecting the previous one but at the ends. The string $X_{i+1}(z)$ is explained in a similar fashion than $X_i(z)$ but it has X_{i+1} as lower and X_i as upper end. That this braid carries t_ν into itself if $\nu \neq i, i+1$ is seen by using ordinary projection which shows that τ_ν is homotopic to t_ν. t_i is mapped into a transform of t_{i+1}. To find the transformer we go back to (10). As parallel curve we use one that will under ordinary projection become the parts of $l_i(\epsilon)$ and $l_{i+1}(\epsilon)$ up to the ray. Consider now the right side of (10). Instead of l_i we have to write l_{i+1} in our new notation. The path projects by ordinary projection still into a homotopic path. But this homotopic path is now obviously 1. So t_i is carried into t_{i+1}. The image of t_{i+1} can now be found by a simpler method, namely, by Theorem 15. Since the product of the t_i must remain fixed we find by a simple computation that t_{i+1} is mapped into $t_{i+1}^{-1} t_i t_{i+1}$. The class of this particular braid shall be called σ_i and the corresponding substitution is:

$$\sigma_i = \begin{pmatrix} t_i, & t_{i+1} \\ t_{i+1}, & t_{i+1}^{-1} t_i\, t_{i+1} \end{pmatrix} \tag{14}$$

with the understanding that the generators that are not mentioned are left unchanged. For σ_i^{-1} we have to compute the inverse substitution and an easy computation gives:

$$\sigma_i^{-1} = \begin{pmatrix} t_i, & t_{i+1} \\ t_i\, t_{i+1} t_i^{-1}, & t_i \end{pmatrix}. \tag{15}$$

To check whether a given braid is σ_i it suffices, however, to check the following properties: dropping the ith and $i+1^{\text{st}}$ string we must obtain a unit. In it the $i+1^{\text{st}}$ string must correspond to the unit homotopy. After reinserting it the ith string must have unit homotopy (always using the simpler formula (9)

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rather then (10)). According to our theory these checks already determine the class.

From now on the nature of our proofs will be mostly algebraic.

Consider a rather general substitution B that maps each generator t_i onto a transform $Q_i^{-1}t_kQ_i = T_i$ (in reduced form) of some generator t_k. When is B a braid? The answer is given by the condition of Theorem 15, namely that

$$T_1T_2 \cdots T_n = t_1t_2 \cdots t_n . \tag{16}$$

The necessity is obvious. To prove the sufficiency we assume each Q_i written as a product of terms t_ν^ϵ, $\epsilon = \pm 1$. The number of terms shall be called the length of Q_i and the sum of all these lengths the length of B. If the length of B is 0 then (16) can hold only if each $T_i = t_i$ or $B = 1$ which is the unit braid. So we may assume our contention proved for all braids with smaller length than B. (16) can hold only if some cancellations take place on the left side. Since each T_i is already reduced these cancellations must take place between adjacent factors of the left side. Two cases are conceivable:

1) In a cancellation between two neighbors the middle terms are never affected. Carrying them out in (16) there will be a residue R_i left from each T_i and this residue must contain the middle term of T_i. We obtain:

$$R_1R_2 \cdots R_n = t_1t_2 \cdots t_n$$

and no further cancellation is possible. This proves $R_i = t_i$ = middle term. The terms on the left side of the middle term of T_1 never could be cancelled at all since no factor is on their left. So $Q_1 = 1$. Now there is no further chance for Q_2^{-1} to be cancelled, so it must be 1 too. This shows $T_i = t_i$ so that this case is settled.

2) Or else there are two neighbors T_i and $T_{i+1} = Q_{i+1}^{-1}t_sQ_{i+1}$ such that one or both of the middle terms are reached in a cancellation. They cannot be affected at the same time since their positive exponent prevents it. Now two alternatives are forced upon us:

a) t_k is affected first. Consider the product $T_iT_{i+1}T_i^{-1}$. Because of the special form of the T_r a cancellation is now possible on both sides of T_{i+1}. Carry it out, term by term, on both sides until the middle term of T_i and T_i^{-1} is reached and stop the cancellation at that point even if it is possible to go on. More than half of T_i and T_i^{-1} will have been absorbed, the middle term of T_{i+1} will not yet be reached and T_{i+1} will have lost as many factors as the other two. A few remnants from these factors will remain but they will be shorter than the loss. It is of course very easy to write this down formally. What is important is, that the length of this product is shorter than that of T_{i+1}. Consider now the substitution $B\sigma_i^{-1}$. We find:

$$B\sigma_i^{-1} = \begin{pmatrix} t_\nu, & t_i, & t_{i+1} \\ T_\nu, & T_iT_{i+1}T_i^{-1}, & T_i \end{pmatrix} \qquad \nu \neq i, i+1.$$

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The product of the second line is $T_1T_2 \cdots T_n = t_1t_2 \cdots t_n$. It has still the general form of B but is shorter. So this substitution A is a braid. Since $B = A\,\sigma_i$, we see that B is a braid.

b) t_s is affected first. Then we find that $T_{i+1}^{-1}T_iT_{i+1}$ is shorter than T_i so that

$$B\sigma_i = \begin{pmatrix} t_\nu, & t_i, & t_{i+1} \\ T_\nu, & T_{i+1}, & T_{i+1}^{-1}T_iT_{i+1} \end{pmatrix} \qquad \nu \neq i, i+1$$

is a braid. This proves that $B = A\sigma_i^{-1}$ is a braid.

Our proof also shows that B can be expressed as a powerproduct of the σ_i.

The proof would also have worked if the subscripts i ran through a subset of all indices only. No condition need be put on the t_k and Q_i. (15) has to be replaced by the condition that B leaves $\prod t_i$ (in the natural order) invariant. The braids σ_i have of course to be replaced by the corresponding braids for this subset of ends.

The most general case would be finally this. A subset of the t_i and mappings of the previous kind are given. B carries $\prod t_i$ into $\prod t_j$ where the t_j form another subset also in the natural order. To reduce this case to the previous one let B_1 be a braid having the X_i as lower ends and the X_j as upper ones. The substitution $B\,B_1^{-1}$ maps $\prod t_j$ onto itself and is therefore a braid. So B is a braid. Let (i) and (j) be subsets of the indices both equal in number. Our result shows that

$$B_{(i)(j)} = \begin{pmatrix} t_i \\ t_j \end{pmatrix} \text{ is a braid.} \quad (i, j \text{ natural arrangement}). \tag{17}$$

Our results may be expressed in the theorem:

THEOREM 16. *A substitution is a braid if, and only if, it has the general form of a transformation and if it satisfies the condition of Theorem* 15. *The full group of n-braids has the σ_i as generators. A general braid can be expressed as a product of a braid of the form* (17) *followed by generators like the σ_i but concerning the lower ends of the braid only.*

A simple computation of substitutions shows that the following relations hold between the σ_i:

$$\sigma_i\sigma_k = \sigma_k\sigma_i \qquad \text{if } |\, i - k \,| \geqq 2 \tag{18}$$

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}\,. \tag{19}$$

In Z. I have shown that these relations form a full set of defining relations for the group. The method is geometric and can easily be made rigorous by means of the tools developed in this paper. However a more interesting proof shall be given in a paper by F. Bohnenblust which is essentially algebraic and leads deeper into the theory of the group. I shall therefore omit a proof here especially since no use shall be made of this fact in this paper. All we shall use is that these relations hold.

A simple operation is that of removing a string. What does it mean for the substitution? Let A be the braid and remove the string with X_r as lower and

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X_m as upper end. That means that we have to cancel the column referring to the image of t_r. In addition to that we have to substitute everywhere in the second row $t_m = 1$ since that describes the shrinkage in the homotopy generators.

The inverse problem in a somewhat simplified form would be:

Given a braid in the form (12), assume r does not occur among the i and m not among the j. Form a new braid A by inserting one string with X_r and X_m as respective lower and upper ends. The simplification will consist in the special position which the new string is in, we shall namely get $A(t_r) = t_m$. To do it, enlarge the meaning of B by prescribing of a new substitution C that it shall have the same effect on the t_i as B and map t_r onto t_m. This substitution will in general not be a braid since the condition concerning the product will be violated. Define two new substitutions α and β by their effect on the t_i respectively t_j and t_r and t_r resp t_m :

$$\alpha(t_i) = \begin{cases} t_i & \text{if} \quad i < r \\ t_r^{-1} t_i t_r & \text{if} \quad i > r, \end{cases} \qquad \alpha(t_r) = t_r ;$$

(20)

$$\beta(t_j) = \begin{cases} t_j & \text{if} \quad j < m \\ t_m t_j t_m^{-1} & \text{if} \quad j > m \end{cases}, \qquad \beta(t_m) = t_m .$$

Then $A = \beta C \alpha$ is the desired braid. We first prove that it is a braid by showing that A carries the product

$$\prod_{i<r} t_i \cdot t_r \cdot \prod_{i>r} t_i \quad \text{into} \quad \prod_{j<m} t_j \cdot t_m \cdot \prod_{j>m} t_j .$$

Indeed α carries it into

$$\prod_{i<r} t_i \cdot t_r \cdot t_r^{-1} \prod_{i>r} t_i \cdot t_r = \prod t_i \cdot t_r$$

C transforms it into

$$\prod t_j \cdot t_m = \prod_{j<m} t_j \cdot \prod_{j>m} t_j \cdot t_m$$

and β into

$$\prod_{j<m} t_j \cdot t_m \prod_{j>m} t_j \cdot t_m^{-1} \cdot t_m = \prod_{j<m} t_j \cdot t_m \cdot \prod_{j>m} t_j .$$

It is seen immediately that cancellation of the rth string leads back to B. So A is the desired braid. Obviously $A(t_r) = t_m$.

We may now combine both operations. If A is a braid, we may first cancel the rth string and then reinsert it with the same ends so that it maps t_r onto t_m. The new braid shall be denoted by $A^{(r)}$. Theorem 7 shows that A may be obtained from $A^{(r)}$ by multiplying it from the left by a uniquely determined m-pure braid (m and not r-pure because of our change of notation). That $A^{(r)}$ is uniquely determined by A follows from Theorem 8 since $A^{(r)}(t_r) = t_m$ describes the homotopy class of the rth string. We may write:

$$A = U_m A^{(r)}, \quad U_m \ m\text{-pure}.$$

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We know how to compute $A^{(r)}$ from A and shall see a little later how U_m can be computed.

The elements S_j in their dependency on the braid shall be denoted by $S_j(A)$. In order to have well defined elements before us, we must still make an agreement about the arbitrary power of t_j that still may be added as a left factor. We choose it in such a way that the sum of the exponents of t_j in S_j is 0.

Let now A and B be two composable braids. B maps t_i into $S_j(B)^{-1}t_jS_j(B)$. A maps this into $A(S_j(B))^{-1}\cdot S_k(A)^{-1}\cdot t_k\cdot S_k(A)\cdot A(S_j(B))$, where $k = A(j)$. Since the transformer is determined to within a power of t_k we get to within such a power

$$S_k(AB) = S_k(A)A(S_j(B)), \qquad k = A(j). \tag{21}$$

On the left side and in the first factor on the right the sum is 0; in $S_j(B)$ the sum of the exponents of t_j is 0. A maps it into a power product where the sum of the exponents of t_k is 0. So (21) is correct as it stands.

A rather elementary invariant can be derived from (21). Calling $H_j(A)$ the sum of all exponents in $S_j(A)$ we obtain:

$$H_k(AB) = H_k(A) + H_j(B) \qquad k = A(j). \tag{22}$$

Defining now the "twining number" $T(A)$ as the sum of all $H_k(A)$ we get:

$$T(AB) = T(A) + T(B). \tag{23}$$

Since $T(\sigma_i^\epsilon) = \epsilon$ for $\epsilon = \pm 1$ this invariant can in case of the full group of n-braids also be explained as the sum of all exponents of the σ_i in any expression of A by the σ_i. This allows us to determine the factor commutator group without making use of the fact that the system of relations (18), (19) is complete. Making all σ_i commutative (19) gives the equality of all the σ_i. In the factor commutator group a braid A shrinks therefore to $\sigma_1^{T(A)}$ So this group is infinite cyclic and $T(A)$ gives the position of A in it.

The homology class of a string is obtained from $S_j(A)$ by the substitution of $t_j = 1$. The result may be denoted by $\bar{S}_j(A)$. In order to obtain a formula similar to (21) we must see what effect that substitution has on the second term on the right of (21). A maps t_j onto a transform of t_k. After substituting $t_k = 1$ all terms coming from t_j will disappear. Hence we may substitute $t_j = 1$ in $S_j(B)$. In addition to that, we must also substitute $t_k = 1$ in the result wherever it appears from the rest of the substitution. The same effect is achieved if the braid A is replaced by one where the jth string has been dropped. Let us denote this braid by A_{-j}. Then we have:

$$\bar{S}_k(AB) = \bar{S}_k(A)\cdot A_{-j}(\bar{S}_j(B)), \qquad k = A(j). \tag{24}$$

Consider the special case that A is k-pure. Then $k = j$, A_{-j} a unit. Hence:

$$\bar{S}_k(AB) = \bar{S}_k(A)\cdot\bar{S}_k(B); \text{ if } A \text{ is } k\text{-pure.} \tag{25}$$

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A still more special case is obtained when both A and B are k-pure. Theorem 8 tells that the homotopy class of the kth string together with the ends determines any k-pure braid completely. It also shows that every homotopy class is possible. (25) means therefore that $\bar{S}_k(A)$ gives an isomorphic mapping of the group of k-pure braids with given ends onto the free group of the generators t_i with $i \neq k$. If we denote by A_{ik} the k-pure braid that is mapped onto t_i then the A_{ik} are the generators of our group and the mapping $\bar{S}_k(A)$ means just a replacement of each A_{ik} by t_i. These braids satisfy $A_{ik} = A_{ki}$. We prove this by giving at the same time the full substitution of A_{ik}^{ϵ} for $i < k$ and ϵ any integer.

$$(26)\qquad A_{ik}^{\epsilon} = A_{ki}^{\epsilon} = \begin{pmatrix} & t_r, & & t_i, & t_r, & t_k \\ & t_r, & & t_i C_{ik}, & C_{ik}^{-1} t_r C_{ik}, & C_{ik}^{-1} t_k \\ r < i \quad \text{or} \quad > k, & & & i \leqq r \leqq k, & & \end{pmatrix}$$

$$C_{ik} = (t_i^{-1} t_k^{-1})^{\epsilon} \cdot (t_i t_k)^{\epsilon}.$$

Writing out the critical terms we see that all are transformations of generators. The product property holds so they are braids. $\bar{S}_i(A_{ik}^{\epsilon}) = t_k^{\epsilon}$, $\bar{S}_k(A_{ik}^{\epsilon}) = t_i^{\epsilon}$ and the braid reduces to a unit if we drop either the ith or the kth string. This proves all our contentions.

It is convenient to introduce also the inverse mapping F_k to $\bar{S}_k$. It maps t_i onto A_{ik}. Let it also have a meaning for t_k whose image shall be 1.

Let A be a braid and assume $k = A(j)$. We can write $A = U \cdot A^{(j)}$ where U is k-pure. Making use of (25) we obtain $\bar{S}_k(A) = \bar{S}_k(U)\bar{S}_k(A^{(j)}) = \bar{S}_k(U)$ since $A^{(j)}$ maps by definition t_j onto t_k, whence $\bar{S}_k(A^{(j)}) = 1$. Now applying F_k gives $U = F_k(\bar{S}_k(A))$ or:

$$(27)\qquad A = F_k(\bar{S}_k(A)) \cdot A^{(j)}, \qquad A(j) = k.$$

This is the algebraic form of Theorem 7.

(27) also solves the general question: given a braid B; insert a new string with the given homotopy class $\bar{S}_k(A)$. We have learned to form a braid with the homotopy class 1, it is the braid $A^{(j)}$. If we substitute this and the given $\bar{S}_k(A)$ in (27) we obtain A expressed by the A_{ik} and $A^{(j)}$. Use now (26).

Another application of (24) is this: let A be j-pure and assume of B that $B(t_j) = t_k$ hence $B^{-1}(t_k) = t_j$. Then $\bar{S}_k(B) = \bar{S}_j(B^{-1}) = 1$. This leads to $\bar{S}_k(BAB^{-1}) = B_{-j}(\bar{S}_j(A))$. BAB^{-1} is k-pure so we can apply F_k. This leads to:

$$BAB^{-1} = F_k(B_{-j}(\bar{S}_j(A))).$$

To get a still more general formula let B be now any braid. We first replace in the previous formula B by $B^{(j)}$. On the right side $B_{-j}^{(j)}$ appears. It is followed by the mapping F_k which anyhow maps t_k onto 1. So this braid may be replaced by B itself. Use now (27) on B. We obtain

$$(28)\qquad BAB^{-1} = F_k(\bar{S}_k(B)) \cdot F_k(B(\bar{S}_j(A))) \cdot (F_k(\bar{S}_k(B)))^{-1}, \quad B(j) = k.$$

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B is here completely general so that we have before us the general transformation formula for a j-pure braid A. If A is given as a power product of the A_{ij} then $\bar{S}_j$ is a very trivial mapping and so is F_k. The right side is directly expressed as a power product of the A_{ik}. It is a very powerful formula that allows us to write down transformation formulas whose direct computation would be very painful.

As an application let $A = A_{ij}$ and $B = \sigma_r$. We give only the result of the computation which is now very easy:

$$(29)\qquad \sigma_r A_{ij} \sigma_r^{-1}$$

$$= \begin{cases} A_{ij} & \text{if } r \neq i-1, i, j-1, j \\ A_{i+1,j} & \text{if } r = i \\ A_{j,j-1} A_{i,j-1} A_{j,j-1}^{-1} & \text{if } r = j-1 \end{cases} \text{but } i \neq j-1; \qquad A_{j,j-1} \text{ for } i = j-1$$

$$\begin{cases} A_{ij}^{-1} A_{i-1,j} A_{ij} & \text{if } r = i-1 \\ A_{i,j+1} & \text{if } r = j \end{cases} \text{but } i \neq j+1; \qquad A_{j,j+1} \text{ for } i = j+1.$$

It is to be remarked that the symmetry $A_{ij} = A_{ji}$ gives other expressions for the same transforms. We note the very special cases $r = i, j$. Since a very simple computation gives $A_{i,i+1} = \sigma_i^2$ we obtain for $i < j$ the following explicit expressions of the A_{ij} in terms of the generators σ_r:

$$(30)\qquad A_{ij} = \sigma_i^{-1} \sigma_{i+1}^{-1} \cdots \sigma_{j-2}^{-1} \sigma_{j-1}^2 \sigma_{j-2} \cdots \sigma_i$$

$$= \sigma_{j-1} \sigma_{j-2} \cdots \sigma_{i+1} \sigma_i^2 \sigma_{i+1}^{-1} \cdots \sigma_{j-2}^{-1} \sigma_{j-1}^{-1}.$$

As a second application we study the structure of the group I of n-braids with identity permutation. We fist prove:

LEMMA: *If A is an element of I that maps t_i onto itself for $i \leqq j$ then the S_k for $k > j$ do not depend on these t_i.*

PROOF: Dropping in the substitution A the first j columns will give a substitution that still satisfies the product condition, so it is an $(n-j)$-braid. This proves the lemma.

Now we use (27) for $j = k = 1$. On $A^{(1)}$ we use it for $k = 2$ and so on. Making use each time of the lemma we obtain a unique expression of A:

THEOREM 17. *The A_{ik} are generators of the group I. Every element can be expressed uniquely in the form:*

$$(31)\qquad A = U_1 U_2 \cdots U_{n-1}$$

where each U_j is a uniquely determined power product of the A_{ij} using only those with $i > j$. (Of course $A_{ij} = A_{ji}$).

The simple geometric meaning of this normal form shall be given later when we interpret our results in terms of projection.

What are the defining relations between these generators? Obviously those that permit the change of an arbitrary power product of them into the normal

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form. We must therefore find rules for interchanging factors of U_j with factors of U_i. For this purpose we derive all transformation rules for the expression $A_{rs}^{\epsilon}A_{ik}A_{rs}^{-\epsilon}$ ($\epsilon = \pm 1$). We use (28) with i instead of k for $B = A_{rs}^{\epsilon}$, $A = A_{ik}$. A simple computation yields:

THEOREM 18. *The braid $A_{rs}^{\epsilon}A_{ik}A_{rs}^{-\epsilon}$ ($\epsilon = \pm 1$) is i-pure. The following rules give its expression as i-pure braid:*

1) *If $i = r$ or s then it has already the desired form. Since the i-pure braids form a free group no other expression can be expected.*

2) *If all indices are different and if the pairs r, s and i, k do not separate each other we simply get A_{ik}.*

In all other cases we change, if necessary, first the names r, s in such a way that the arrangement i, r, s as compared with the natural arrangement of these three numbers is a permutation with the same sign as ϵ.

3) *If $k = r$ we get: $A_{is}^{-\epsilon}A_{ir}A_{is}^{\epsilon}$.*

4) *If $k = s$ we obtain:*

$$A_{is}^{-\epsilon}A_{ir}^{-\epsilon}\cdot A_{is}\cdot A_{ir}^{\epsilon}A_{is}^{\epsilon}.$$

5) *If finally the subscripts are all different and if the pairs r, s and i, k separate each other the result is:*

$$A_{is}^{-\epsilon}A_{ir}^{-\epsilon}A_{is}^{\epsilon}A_{ir}^{\epsilon}\cdot A_{ik}\cdot A_{ir}^{-\epsilon}A_{is}^{-\epsilon}A_{ir}^{\epsilon}A_{is}^{\epsilon}.$$

As defining relations the ones where i is the smallest index of all are sufficient. It also suffices to have only $\epsilon = +1$ since $\epsilon = -1$ is just only the inverse automorphism.

For braids whose permutation is not identity, a normal form is also easily obtained. Select to each of the $n!$ permutations π a braid B_π with this permutation. Any braid can then be written as a product like that in Theorem 17 followed by a B_π. This form is again unique.

The operation $A^{(j)}$ obviously satisfies:

$$(AB)^{(j)} = A^{(k)}B^{(j)}, \qquad k = B(j). \tag{32}$$

For the group I, it is therefore a homomorphic mapping and it suffices to know the result for the generators A_{ik}. $A_{ik}^{(r)} = 1$ for $r = i$ or k since A_{ik} is i- and k-pure. It is A_{ik} if $r < i$ or $> k$ as the substitution shows. If r is between i and k we apply (27) and Theorem 18. The result is:

$$A_{ik}^{(r)} = \begin{cases} 1 & \text{for} \quad r = i \text{ or } r = k \\ A_{ik} & \text{if} \quad r \text{ not between } i \text{ and } k \\ A_{ir}A_{ik}A_{ir}^{-1} = A_{kr}^{-1}A_{ki}A_{kr} & \text{if} \quad i < r < k. \end{cases} \tag{33}$$

We return now to the general group of n-braids. Let A be a braid that leaves t_j fixed if either $j < i$ or $j > k$. For the same reason as that in the previous lemma, we find that A maps t_i, $t_{i+1}\cdots t_k$ onto expressions depending on these variables alone. A can therefore be expressed in terms of the generators σ_i,

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$\sigma_{i+1}, \cdots \sigma_{k-1}$. These braids form a subgroup denoted by G_{ik}. They behave as if they were braids of $k - i + 1$ strings (from i to k) only.

Put $c_{ik} = t_i t_{i+1} \cdots t_k$ and consider the following substitution C_{ik}:

$$C_{ik} = \begin{pmatrix} t_r & t_s \\ t_r & c_{ik}^{-1} t_s c_{ik} \\ r < i \text{ or } > k & i \leqq s \leqq k \end{pmatrix}. \tag{34}$$

It is a braid of G_{ik} since the product property holds. It is obviously commutative with any element of G_{ik} hence in the center of this group. If we drop the kth string in C_{ik} and $C_{i,k-1}$ or the ith in C_{ik} and $C_{i+1,k}$ we always obtain identical braids. $C_{i,k-1}$ leaves t_k and $C_{i+1,k}$ leaves t_i fixed. Hence:

$$C_{ik}^{(k)} = C_{i,k-1}, \qquad C_{ik}^{(i)} = C_{i+1,k} \qquad (C_{ii} \text{ means } 1). \tag{35}$$

Formula (27) gives:

$$C_{ik} = A_{ik} A_{i+1,k} \cdots A_{k-1,k} C_{i,k-1} = A_{i,i+1} A_{i,i+2} \cdots A_{ik} C_{i+1,k}. \tag{36}$$

This gives us the explicit expressions by the A_{ik}.

C_{ik} is closely related to a braid D_{ik} of G_{ik} which maps $t_i, t_{i+1} \cdots t_{k-1}$ onto t_{i+1}, $t_{i+2}, \cdots t_k$ but t_k onto $c_{ik}^{-1} t_1 c_{ik}$. We see that

$$C_{ik} = (D_{ik})^{k-i+1}. \tag{37}$$

The product $\sigma_i \sigma_{i+1} \cdots \sigma_{k-1}$ has the same effect on $t_i, t_{i+1}, \cdots t_{k-1}$ as D_{ik}. Because of the product property this suffices to establish equality. Hence:

$$C_{ik} = (\sigma_i \sigma_{i+1} \cdots \sigma_{k-1})^{k-i+1}. \tag{38}$$

THEOREM 19. *If $n = 2$ all braids are commutative. If $n \geqq 3$ let k be one of the numbers $\leqq n$. If B is commutative with every k-pure braid, then B is a power of $C_{1,n}$. This also determines of course the centers of the whole group, of I, G_{ik} and $I_{ik} = I \cap G_{ik}$.*

PROOF: If A is r-pure and $B(r) = s$, then BAB^{-1} is s-pure. Put $A = A_{ik}$. It is pure only for i and k. Since $BA_{ik}B^{-1} = A_{ik}$ we see that B can at most interchange i and k. If $n \geqq 3$ then i may be replaced by another index which shows that k remains fixed. Consequently i remains fixed too so B has identity as permutation.

We now make use of (28) where $j = k$. The braid B in the middle term on the right side came originally from B_{-k} which we introduce again. For A we take any k-pure braid so that the left side is A again. If we then apply the operation $\bar{S}_k$ to both sides we get:

$$\bar{S}_k(A) = \bar{S}_k(B) \cdot B_{-k}(\bar{S}_k(A)) \cdot (\bar{S}_k(B))^{-1}.$$

In this formula $\bar{S}_k(A)$ may be any power product T of the t_i with $i \neq k$. This shows:

$$B_{-k}(T) = d^{-1} T d, \text{ where } d = \bar{S}_k(B).$$

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For $T = T_0 = \prod_{i \neq k} t_i$ we have on the other hand $B_{-k}(T_0) = T_0$. So T_0 is commutative with d. Since T_0 occurs in a generator system of the free group d is a power of T_0 say T_0^r. B_{-k} transforms all T with T_0^r. The same transformation is produced by $(C_{1,n})_{-k}^r$; B_{-k} is therefore this braid. Put now $C = B\,C_{1,n}^{-r}$. We find $C_{-k} = 1$ so C is k-pure. But C is still commutative with all k-pure braids. They form a free group with at least 2 generators whence $C = 1$ or $B = C_{1,n}^r$. This proves the contention.

For our next question we need a certain result about automorphisms of free groups. Let F be a free group with the generators t_i. Divide the subscripts into two classes p and q and in some other way into the classes g and h. We assume that there are at least two t_q and two t_h. Let x be a power product of the t_h that appears among a generator system of F and y a similar power product of the t_q. Since we assumed that there are at least two t_h, x will not be commutative with every t_h. Define now the automorphisms C and D by:

$$(39)\quad C(t_g) = t_g,\quad C(t_h) = x^{-a}t_h x^a;\quad D(t_p) = t_p,\quad D(t_q) = y^{-b}t_q y^b,$$

where a and b are positive integers. We ask for all automorphisms A that satisfy:

$$(40)\qquad D\,A = A\,C.$$

C leaves all t_g as well as x invariant. Their image T under A will satisfy (because of (40)):

$$(41)\qquad D(T) = T.$$

The power products $T = A(t_h)$ give $D(T) = AC(t_h) = A(x^{-a}t_h x^a)$ hence:

$$(42)\qquad D(T) = z^{-a}T\,z^a \text{ where } z = A(x).$$

The equations (42) (41) exhaust (40) and only the condition that A is an automorphism will have to be taken care of.

Denote by the letter P any power product of the t_p alone, by Q one of the t_q and by R one of the t_p and y. Any T can be written in the form:

$$(43)\qquad T = P_1Q_1P_2Q_2\cdots$$

where P_1 may be absent. Then:

$$(44)\qquad D(T) = P_1y^{-b}Q_1y^bP_2\cdots$$

Assume T satisfies (41). Each Q_i must be commutative with y^b. But y occurs in a generator system of F so Q_i is a power of y. Hence T is an R. We get

$$(45)\qquad A(t_g) = R_g,\qquad A(x) = z = R_0.$$

This takes care of (41). Assume now that T satisfies (42). We get:

$$(46)\qquad P_1y^{-b}Q_1y^bP_2y^{-b}Q_2\cdots = z^{-a}P_1Q_1P_2Q_2\cdots z^a.$$

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If every Q is a power of y then T is an R and $D(T) = T$ must be commutative with z^a. But z is the image of a generator so is itself a generator. So T is a power of z. Since one t_h at least will not be commutative with x, its image T will not be commutative with z. So this case does not always happen.

Assume now that Q_i is the first of the Q in (46) that is not a power of y. Then the whole segments on the left of this factor in (46) must be equal since z on the right side is also an R. We obtain: (the earlier Q are powers of y)

$$P_1Q_1P_2\cdots P_iy^{-b} = z^{-a}P_1Q_1\cdots P_i \qquad \text{or}$$

$$z^a = R^{-1}y^bR.$$

Since y and z are generators this is only possible if $a = b$. $R^{-1}yR$ is also a generator and we get:

$$z = R^{-1}yR. \tag{47}$$

With this R put now $T = R^{-1}T_0R$. Because of $D(R) = R$ (42) gives:

$$D(T_0) = y^{-b}T_0y^b. \tag{48}$$

Writing now T_0 in the form (43) we get:

$$P_1y^{-b}Q_1y^bP_2\cdots = y^{-b}P_1Q_1P_2\cdots y^b.$$

The right side shows that P_1 must be absent. But also the presence of P_2 leads to a contradiction. T_0 is therefore a Q. Our results so far are:

$$A(t_g) = R_g\,,\ a = b,\quad A(x) = R^{-1}yR,\quad A(t_h) = R^{-1}Q_hR. \tag{49}$$

A maps the group generated by the t_g and x into the group of the t_p and y. A^{-1} satisfies $A^{-1}D^{-1} = C^{-1}A^{-1}$ where the roles of g and p are interchanged. It maps therefore the group of the t_p and y into the group of the t_g and x. The mappings are therefore one to one and this shows that the number of subscripts g is the same as that of the subscripts p.

We go now back to the braids and ask what automorphisms satisfy:

$$C^a_{ik}A = AC^a_{rs}\,, \qquad a > 0. \tag{50}$$

Our conditions are satisfied. The p are $< i$ or $> k$, the q satisfy $i \leqq q \leqq k$, the g are $< r$ or $> s$, the h satisfying $r \leqq h \leqq s$. $x = c_{rs}$ and $y = c_{ik}$ are generators. We must have $k - i = s - r$.

Assume a little more about the automorphism A namely that it maps each t_j onto a transform of another. Then:

$$A(t_g) = R_p^{-1}t_pR_p\,,\quad A(t_h) = R^{-1}Q_q^{-1}t_qQ_qR,\quad A(c_{rs}) = R^{-1}c_{ik}R \tag{51}$$

where each Q_q is a power product of the t_q, R_p and R power products of the t_p and of c_{ik}.

Split A into two substitutions:

$$B(t_g) = R_p^{-1}t_pR_p\,, \qquad B(t_{r+j}) = R^{-1}t_{i+j}R, \qquad 0 \leqq j \leqq k - i = s - r. \tag{52}$$

$$E(t_p) = t_p\,,\quad E(t_{i+j}) = Q_q^{-1}t_qQ_q \text{ where } r + j \text{ corresponds to } q. \tag{53}$$

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A maps c_{rs} on one hand onto $R^{-1}c_{ik}R$; computed directly onto $\prod R^{-1}Q_q^{-1}t_qQ_qR$ whence $\prod Q_q^{-1}t_qQ_q = c_{ik}$. Now we see that E leaves c_{ik} fixed and therefore

$$A = E\,B. \tag{54}$$

We see at the same time that E is always a braid. The condition that A be an automorphism is therefore that B is one and that E is a braid. Should A be also a braid then B is one and conversely.

In case A is a braid the geometric significance of (52), (53) and (54) is obvious. B behaves as if the ith up to k^{th} string were just only one strand (only the product of the t_h and of the t_q plays a role). A is obtained by weaving the pattern E into the ith up to the kth string, then, considering this partial braid as one string only, interweaving it with the other strings according to pattern B.

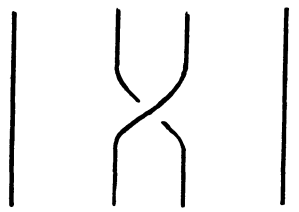


FIGURE 1

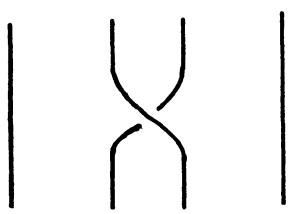


FIGURE 2

The question whether a given braid A can be considered as braid of braids amounts to checking relations of the type (50). It suffices of course to consider $a = 1$. Since we can decide whether or not they hold, this question is decided too.

THEOREM 20. *The number n is a group invariant of the group of braids with n strings.*

PROOF: Theorem 19 shows that $C_{1,n}$ is a generator of the center and therefore together with its reciprocal characterized by an inner property of the group. The number $T(C_{1,n}) = n(n-1)$ is therefore also an invariant since it gives the position in the factor commutator group.

The structure of the group does not depend on the position of the ends. We may therefore put the ends in the special points with the coordinates $x = 0$, $y = i$, $(i = 1, 2, \cdots n)$. As ray for the Poincaré group we may select $x = 1$,

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$y \leqq 0$ and as paths l_i the straight line segments form the beginning point of the ray to the ends. It is advisable to use as orientation of the plane the nega-

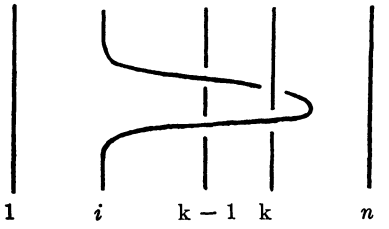


FIGURE 3

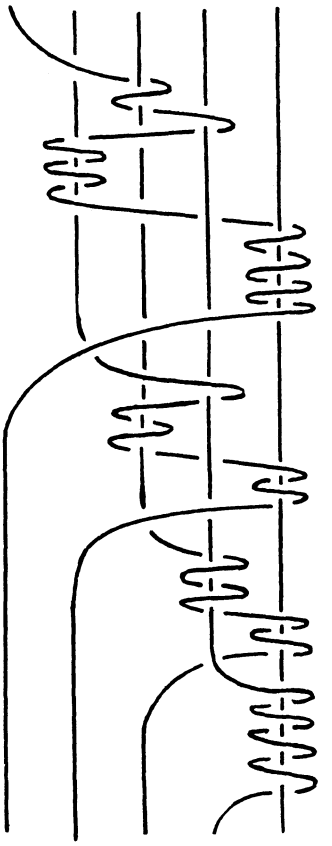


FIGURE 4

tive one, so the sense of rotation from the positive y-axis to the positive x-axis. This fixes all the necessary data and we are now in a position to interpret our results in the projection from the positive x-direction onto the yz-plane.

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Theorem 4 shows us that any braid is isotopic to another one whose strings are broken lines. It leaves us so much freedom that we can assume in addition that the projection is free from any triple points and that no two double points occur at the same z-level.

Figures 1 and 2 show the generators σ_i and σ_i^{-1} in their projection. Corollary 3 to Theorem 2 shows indeed that they are reciprocal. The braid in Fig. 1 maps t_ν obviously onto itself if $\nu \neq i, i+1$. It maps t_i onto t_{i+1} and, because of the product property, that is sufficient to establish its identity. Theorem 3 teaches how to read off from the projection of a braid its expression in terms of the generators σ_i.

Formula (38) shows that C_{ik} is simply the full twist of all the strings from the ith to the kth and that gives the geometric meaning of Theorem 19.

Formula (30) gives now the projection of the generator A_{ij}. It is indicated in Fig. 3.

The geometric meaning of the normal form of Theorem 17 and that mentioned after Theorem 18 is now also clear. Every braid is isotopic to another one whose pattern of projection is especially simple and is indicated in Fig. 4 for a special case. This pattern is unique. The braid in Fig. 4 has the expression:

$$A = A_{13}^{2}A_{14}^{-1}A_{12}^{-3}A_{14}A_{15}^{4}\cdot A_{24}^{-1}A_{23}^{2}A_{24}A_{25}^{-2}\cdot A_{34}^{3}A_{35}^{-2}\cdot A_{34}^{-1}\cdot A_{45}^{-4}.$$

Although it has been proved that every braid can be deformed into a similar normal form the writer is convinced that any attempt to carry this out on a living person would only lead to violent protests and discrimination against mathematics. He would therefore discourage such an experiment.

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ON

THE QUANTUM THEORY OF LINE-SPECTRA

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DEDICATED TO THE MEMORY
OF MY VENERATED TEACHER

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UDGIVET PAA CARLSBERGFONDETS BEKOSTNING

Introduction.

In an attempt to develop certain outlines of a theory of line-spectra based on a suitable application of the fundamental ideas introduced by Planck in his theory of temperature-radiation to the theory of the nucleus atom of Sir Ernest Rutherford, the writer has shown that it is possible in this way to obtain a simple interpretation of some of the main laws governing the line-spectra of the elements, and especially to obtain a deduction of the well known Balmer formula for the hydrogen spectrum[1]) The theory in the form given allowed of a detailed discussion only in the case of periodic systems, and obviously was not able to account in detail for the characteristic difference between the hydrogen spectrum and the spectra of other elements, or for the characteristic effects on the hydrogen spectrum of external electric and magnetic fields. Recently, however, a way out of this difficulty has been opened by Sommerfeld[2]) who, by introducing a suitable generalisation of the theory to a simple type of non-periodic motions and by taking the small variation of the mass of the electron with its velocity into account, obtained an explanation of the fine-structure of the hydrogen lines which was found to be in brilliant conformity with the measurements. Already in his first paper on

[1]) N. Bohr, Phil. Mag., XXVI, pp. 1, 476, 857 (1913), XXVII, p. 506 (1914), XXIX. p. 332 (1915), XXX. p. 394 (1915).

[2]) A. Sommerfeld, Ber. Akad. München, 1915, pp. 425, 459, 1916, p. 131. 1917. p. 83. Ann. de Phys., LI. p. 1 (1916).

this subject, SOMMERFELD pointed out that his theory evidently offered a clue to the interpretation of the more intricate structure of the spectra of other elements. Briefly afterwards EPSTEIN[1]) and SCHWARZSCHILD,[2]) independent of each other, by adapting SOMMERFELD's ideas to the treatment of a more extended class of non-periodic systems obtained a detailed explanation of the characteristic effect of an electric field on the hydrogen spectrum discovered by STARK. Subsequently SOMMERFELD[3]) himself and DEBYE[4]) have on the same lines indicated an interpretation of the effect of a magnetic field on the hydrogen spectrum which, although no complete explanation of the observations was obtained, undoubtedly represents an important step towards a detailed understanding of this phenomenon.

In spite of the great progress involved in these investigations many difficulties of fundamental nature remained unsolved, not only as regards the limited applicability of the methods used in calculating the frequencies of the spectrum of a given system, but especially as regards the question of the polarisation and intensity of the emitted spectral lines. These difficulties are intimately connected with the radical departure from the ordinary ideas of mechanics and electro-

[1]) P. EPSTEIN, Phys. Zeitschr. XVII, p. 148 (1916), Ann. d. Phys. L, p. 489. LI. p. 168 (1916).

[2]) K. SCHWARZSCHILD, Ber. Akad. Berlin, 1916, p. 548.

[3]) A. SOMMERFELD, Phys. Zeitschr. XVII, p. 491 (1916).

[4]) P. DEBYE, Nachr. K. Ges. d. Wiss. Göttingen, 1916, Phys. Zeitschr. XVII, p. 507 (1916).

dynamics involved in the main principles of the quantum theory, and with the fact that it has not been possible hitherto to replace these ideas by others forming an equally consistent and developed structure. Also in this respect, however, great progress has recently been obtained by the work of EINSTEIN[1]) and EHRENFEST.[2]) On this state of the theory it might therefore be of interest to make an attempt to discuss the different applications from a uniform point of view, and especially to consider the underlying assumptions in their relations to ordinary mechanics and electrodynamics. Such an attempt has been made in the present paper, and it will be shown that it seems possible to throw some light on the outstanding difficulties by trying to trace the analogy between the quantum theory and the ordinary theory of radiation as closely as possible.

The paper is divided into four parts.

Part I contains a brief discussion of the general principles of the theory and deals with the application of the general theory to periodic systems of one degree of freedom and to the class of non-periodic systems referred to above.

Part II contains a detailed discussion of the theory of the hydrogen spectrum in order to illustrate the general

[1]) A. EINSTEIN, Verh. d. D. phys. Ges. XVIII, p. 318 (1916), Phys. Zeitschr. XVIII, p. 121 (1917).

[2]) P. EHRENFEST, Proc. Acad. Amsterdam, XVI. p. 591 (1914), Phys. Zeitschr. XV. p. 657 (1914), Ann. d. Phys. LI. p. 327 (1916), Phil. Mag. XXXIII. p. 500 (1917).

considerations.

Part III contains a discussion of the questions arising in connection with the explanation of the spectra of other elements.

Part IV contains a general discussion of the theory of the constitution of atoms and molecules based on the application of the quantum theory to the nucleus atom.

Copenhagen, November 1917.

Part I.

On the general theory.

§ 1. General principles.

The quantum theory of line-spectra rests upon the following fundamental assumptions:

I. *That an atomic system can, and can only, exist permanently in a certain series of states corresponding to a discontinuous series of values for its energy, and that consequently any change of the energy of the system, including emission and absorption of electromagnetic radiation, must take place by a complete transition between two such states. These states will be denoted as the "stationary states" of the system.*

II. *That the radiation absorbed or emitted during a transition between two stationary states is "unifrequentic" and possesses a frequency ν, given by the relation*

$$E' - E'' = h\nu, \tag{1}$$

where h is Planck*'s constant and where E' and E'' are the values of the energy in the two states under consideration.*

As pointed out by the writer in the papers referred to in the introduction, these assumptions offer an immediate interpretation of the fundamental *principle of combination of spectral lines* deduced from the measurements of the frequencies of the series spectra of the elements. According to

the laws discovered by BALMER, RYDBERG and RITZ, the frequencies of the lines of the series spectrum of an element can be expressed by a formula of the type:

$$\nu = f_{\tau''}(n'') - f_{\tau'}(n'), \tag{2}$$

where n' and n'' are whole numbers and $f_\tau(n)$ is one among a set of functions of n, characteristic for the element under consideration. On the above assumptions this formula may obviously be interpreted by assuming that the stationary states of an atom of an element form a set of series, and that the energy in the nth state of the τth series, omitting an arbitrary constant, is given by

$$E_\tau(n) = -hf_\tau(n). \tag{3}$$

We thus see that the values for the energy in the stationary states of an atom may be obtained directly from the measurements of the spectrum by means of relation (1). In order, however, to obtain a theoretical connection between these values and the experimental evidence about the constitution of the atom obtained from other sources, it is necessary to introduce further assumptions about the laws which govern the stationary states of a given atomic system and the transitions between these states.

Now on the basis of a vast amount of experimental evidence, we are forced to assume that an atom or molecule consists of a number of electrified particles in motion, and, since the above fundamental assumptions imply that no emission

of radiation takes place in the stationary states, we must consequently assume that *the ordinary laws of electrodynamics cannot be applied* to these states without radical alterations. In many cases, however, the effect of that part of the electrodynamical forces which is connected with the emission of radiation will at any moment be very small in comparison with the effect of the simple electrostatic attractions or repulsions of the charged particles corresponding to COULOMB's law. Even if the theory of radiation must be completely altered, it is therefore a natural assumption that it is possible in such cases to obtain a close approximation in the description of the motion in the stationary states, by retaining only the latter forces. In the following we shall therefore, as in all the papers mentioned in the introduction, for the present *calculate the motions of the particles in the stationary states as the motions of mass-points according to ordinary mechanics* including the modifications claimed by the theory of relativity, and we shall later in the discussion of the special applications come back to the question of the degree of approximation which may be obtained in this way.

If next we consider a transition between two stationary states, it is obvious at once from the essential discontinuity, involved in the assumptions I and II, that in general it is impossible even approximately to describe this phenomenon by means of ordinary mechanics or to calculate the frequency of the radiation absorbed or emitted by such a process by means of ordinary electrodynamics. On the other hand, from the fact that it has been possible by means of ordinary mechan-

ics and electrodynamics to account for the phenomenon of temperature-radiation in the limiting region of slow vibrations, we may expect that any theory capable of describing this phenomenon in accordance with observations will form some sort of natural generalisation of the ordinary theory of radiation. Now the theory of temperature-radiation in the form originally given by PLANCK confessedly lacked internal consistency, since, in the deduction of his radiation formula, assumptions of similar character as I and II were used in connection with assumptions which were in obvious contrast to them. Quite recently, however, EINSTEIN[1]) has succeeded, on the basis of the assumptions I and II, to give a consistent and instructive deduction of PLANCK's formula by introducing certain supplementary assumptions about the *probability of transition of a system between two stationary states* and about the manner in which this probability depends on the density of radiation of the corresponding frequency in the surrounding space, suggested from analogy with the ordinary theory of radiation. EINSTEIN compares the emission or absorption of radiation of frequency ν corresponding to a transition between two stationary states with the emission or absorption to be expected on ordinary electrodynamics for a system consisting of a particle executing harmonic vibrations of this frequency. In analogy with the fact that on the latter theory such a system will without external excitation emit a radiation of frequency ν, EINSTEIN assumes in

[1]) A. EINSTEIN, loc. cit.

the first place that on the quantum theory there will be a certain probability $A^{n'}_{n''}\,dt$ that the system in the stationary state of greater energy, characterised by the letter n', in the time interval dt will start *spontaneously* to pass to the stationary state of smaller energy, characterised by the letter n''. Moreover, on ordinary electrodynamics the harmonic vibrator will, in addition to the above mentioned independent emission, in the presence of a radiation of frequency ν in the surrounding space, and dependent on the accidental phase-difference between this radiation and the vibrator, emit or absorb radiation-energy. In analogy with this, EINSTEIN assumes secondly that in the presence of a radiation in the surrounding space, the system will on the quantum theory, in addition to the above mentioned probability of spontaneous transition from the state n' to the state n'', possess a certain probability, depending on this radiation, of passing in the time dt from the state n' to the state n'', as well as from the state n'' to the state n'. These latter probabilities are assumed to be proportional to the intensity of the surrounding radiation and are denoted by $\rho_\nu B^{n'}_{n''}\,dt$ and $\rho_\nu B^{n''}_{n'}\,dt$ respectively, where $\rho_\nu\,d\nu$ denotes the amount of radiation in unit volume of the surrounding space distributed on frequencies between ν and $\nu + d\nu$, while $B^{n'}_{n''}$ and $B^{n''}_{n'}$ are constants which, like $A^{n'}_{n''}$, depend only on the stationary states under consideration. EINSTEIN does not introduce any detailed assumption as to the values of these constants, no more than to the conditions by which the different stationary states of a given system are determined or to the "a-priori probabil-

ity" of these states on which their relative occurrence in a distribution of statistical equilibrium depends. He shows, however, how it is possible from the above general assumptions, by means of BOLTZMANN's principle on the relation between entropy and probability and WIEN's well known displacement-law, to deduce a formula for the temperature radiation which apart from an undetermined constant factor coincides with PLANCK's, if we only assume that the frequency corresponding to the transition between the two states is determined by (1). It will therefore be seen that by reversing the line of argument, EINSTEIN's theory may be considered as a very direct support of the latter relation.

In the following discussion of the application of the quantum theory to determine the line-spectrum of a given system, it will, just as in the theory of temperature-radiation, not be necessary to introduce detailed assumptions as to the mechanism of transition between two stationary states. We shall show, however, that the conditions which will be used to determine the values of the energy in the stationary states are of such a type that the frequencies calculated by (1), in the limit where the motions in successive stationary states comparatively differ very little from each other, will tend to coincide with the frequencies to be expected on the ordinary theory of radiation from the motion of the system in the stationary states. In order to obtain the necessary relation to the ordinary theory of radiation in the limit of slow vibrations, we are therefore led directly to certain conclusions about the probability of transition between two stationary

states in this limit. This leads again to certain general considerations about the connection between the probability of a transition between any two stationary states and the motion of the system in these states, which will be shown to throw light on the question of the polarisation and intensity of the different lines of the spectrum of a given system.

In the above considerations we have by an atomic system tacitly understood a number of electrified particles which move in a field of force which, with the approximation mentioned, possesses a potential depending only on the position of the particles. This may more accurately be denoted as a system under constant external conditions, and the question next arises about the variation in the stationary states which may be expected to take place during a variation of the external conditions, e. g. when exposing the atomic system to some variable external field of force. Now, in general, we must obviously assume that this variation cannot be calculated by ordinary mechanics, no more than the transition between two different stationary states corresponding to constant external conditions. If, however, the variation of the external conditions is very slow, we may from the necessary stability of the stationary states expect that the motion of the system at any given moment during the variation will differ only very little from the motion in a stationary state corresponding to the instantaneous external conditions. If now, moreover, the variation is performed at a constant or very slowly changing rate, the forces to which the particles of the system will be exposed will not differ at any moment

from those to which they would be exposed if we imagine that the external forces arise from a number of slowly moving additional particles which together with the original system form a system in a stationary state. From this point of view it seems therefore natural to assume that, with the approximation mentioned, the motion of an atomic system in the stationary states can be calculated by direct application of ordinary mechanics, not only under constant external conditions, but in general also during a slow and uniform variation of these conditions. This assumption, which may be denoted as the principle of the "*mechanical transformability*" of the stationary states, has been introduced in the quantum theory by EHRENFEST[1]) and is, as it will be seen in the following sections, of great importance in the discussion of the conditions to be used to fix the stationary states of an atomic system among the continuous multitude of mechanically possible motions. In this connection it may be pointed out that the principle of the mechanical transformability of the stationary states allows us to overcome a fundamental difficulty which at first sight would seem to be involved in the definition of the energy difference between two stationary states

[1]) P. EHRENFEST, loc. cit. In these papers the principle in question is called the "adiabatic hypothesis" in accordance with the line of argumentation followed by EHRENFEST in which considerations of thermodynamical problems play an important part. From the point of view taken in the present paper, however, the above notation might in a more direct way indicate the content of the principle and the limits of its applicability.

which enters in relation (1). In fact we have assumed that the direct transition between two such states cannot be described by ordinary mechanics, while on the other hand we possess no means of defining an energy difference between two states if there exists no possibility for a continuous mechanical connection between them. It is clear, however, that such a connection is just afforded by EHRENFEST's principle which allows us to transform mechanically the stationary states of a given system into those of another, because for the latter system we may take one in which the forces which act on the particles are very small and where we may assume that the values of the energy in all the stationary states will tend to coincide.

As regards the problem of the statistical distribution of the different stationary states between a great number of atomic systems of the same kind in temperature equilibrium, the number of systems present in the different states may be deduced in the well known way from BOLTZMANN's fundamental relation between entropy and probability, if we know the values of the energy in these states and the *a-priori probability* to be ascribed to each state in the calculation of the probability of the whole distribution. In contrast to considerations of ordinary statistical mechanics we possess on the quantum theory no direct means of determining these a-priori probabilities, because we have no detailed information about the mechanism of transition between the different stationary states. If the a-priori probabilities are known for the states of a given atomic system, however, they may be de-

duced for any other system which can be formed from this by a continuous transformation without passing through one of the singular systems referred to below. In fact, in examining the necessary conditions for the explanation of the second law of thermodynamics EHRENFEST[1]) has deduced a certain general condition as regards the variation of the a-priori probability corresponding to a small change of the external conditions from which it follows, that the a-priori probability of a given stationary state of an atomic system must remain unaltered during a continuous transformation, except in special cases in which the values of the energy in some of the stationary states will tend to coincide during the transformation. In this result we possess, as we shall see, a rational basis for the determination of the a-priori probability of the different stationary states of a given atomic system.

§ 2. Systems of one degree of freedom.

As the simplest illustration of the principles discussed in the former section we shall begin by considering systems of a single degree of freedom, in which case it has been possible to establish a general theory of stationary states. This is due to the fact that *the motion will be simply periodic*, provided the distance between the parts of the system will

[1]) P. EHRENFEST, Phys. Zeitschr. XV p. 660 (1914). The above interpretation of this relation is not stated explicitly by EHRENFEST, but it presents itself directly if the quantum theory is taken in the form corresponding to the fundamental assumption I.

not increase infinitely with the time, a case which for obvious reasons cannot represent a stationary state in the sense defined above. On account of this, the discussion of the mechanical transformability of the stationary states can, as pointed out by EHRENFEST,[1]) for systems of one degree of freedom be based on a mechanical theorem about periodic systems due to BOLTZMANN and originally applied by this author in a discussion of the bearing of mechanics on the explanation of the laws of thermodynamics. For the sake of the considerations in the following sections it will be convenient here to give the proof in a form which differs slightly from that given by EHRENFEST, and which takes also regard to the modifications in the ordinary laws of mechanics claimed by the theory of relativity.

Consider for the sake of generality a conservative mechanical system of s degrees of freedom, the motion of which is governed by HAMILTON's equations:

$$\frac{dp_k}{dt} = -\frac{\partial E}{\partial q_k}, \quad \frac{dq_k}{dt} = \frac{\partial E}{\partial p_k}, \qquad (k = 1, \ldots, s) \qquad (4)$$

where E is the total energy considered as a function of the generalised positional coordinates $q_1, \ldots, q_s$ and the corresponding canonically conjugated momenta $p_1, \ldots, p_s$. If the velocities are so small that the variation in the mass of the particles due to their velocities can be neglected, the p's are

[1]) P. EHRENFEST, loc. cit. Proc. Acad. Amsterdam, XVI, p. 591 (1914).

defined in the usual way by

$$p_k = \frac{\partial T}{\partial q_k}, \qquad (k = 1, \ldots, s)$$

where T is the kinetic energy of the system considered as a function of the generalised velocities $\dot{q}_1, \ldots, \dot{q}_s$ $\left(\dot{q}_k = \frac{dq_k}{dt}\right)$ and of $q_1, \ldots, q_s$. If the relativity modifications are taken into account the p's are defined by a similar set of expressions in which the kinetic energy is replaced by

$$T' = \sum m_0 c^2 \big(1 - \sqrt{1 - v^2/c^2}\big),$$

where the summation is to be extended over all the particles of the system, and v is the velocity of one of the particles and m_0 its mass for zero velocity, while c is the velocity of light.

Let us now assume that the system performs a periodic motion with the period σ, and let us form the expression

$$I = \int_0^\sigma \sum_1^s p_k \dot{q}_k \, dt, \tag{5}$$

which is easily seen to be independent of the special choice of coordinates $q_1, \ldots, q_s$ used to describe the motion of the system. In fact, if the variation of the mass with the velocity is neglected we get

$$I = 2 \int_0^\sigma T \, dt,$$

and if the relativity modifications are included, we get a quite analogous expression in which the kinetic energy is replaced by $T'' = \sum \frac{1}{2} m_0 v^2 \sqrt{1 - v^2/c^2}$.

Consider next some new periodic motion of the system formed by a small variation of the first motion, but which may need the presence of external forces in order to be a mechanically possible motion. For the variation in I we get then

$$\delta I = \int_0^\sigma \sum_1^s (\dot{q}_k \, \delta p_k + p_k \, \delta \dot{q}_k) \, dt + \sum_1^s p_k \, \dot{q}_k \, \delta t \Big|_0^\sigma,$$

where the last term refers to the variation of the limit of the integral due to the variation in the period σ. By partial integration of the second term in the bracket under the integral we get next

$$\delta I = \int_0^\sigma \sum_1^s (\dot{q}_k \, \delta p_k - \dot{p}_k \, \delta q_k) \, dt + \sum_1^s p_k (\dot{q}_k \, \delta t + \delta q_k) \Big|_0^\sigma,$$

where the last term is seen to be zero, because the term in the bracket as well as p_k will be the same in both limits, since the varied motion as well as the original motion is assumed to be periodic. By means of equations (4) we get therefore

$$\delta I = \int_0^\sigma \sum_1^s \left(\frac{\partial E}{\partial p_k} \, \delta p_k + \frac{\partial E}{\partial q_k} \, \delta q_k \right) dt = \int_0^\sigma \delta E \, dt. \qquad (6)$$

Let us now assume that the small variation of the motion is produced by a small external field established at a uniform

rate during a time interval ϑ, long compared with σ, so that the comparative increase during a period is very small. In this case δE is at any moment equal to the total work done by the external forces on the particles of the system since the beginning of the establishment of the field. Let this moment be $t = -\vartheta$ and let the potential of the external field at $t \geq 0$ be given by Ω, expressed as a function of the q's. At any given moment $t > 0$ we have then

$$\delta E = -\int_{-\vartheta}^{0} \frac{\vartheta + t}{\vartheta} \sum_{1}^{s} \frac{\partial \Omega}{\partial q_k} \dot{q}_k \, dt - \int_{0}^{t} \sum_{1}^{s} \frac{\partial \Omega}{\partial q_k} \dot{q}_k \, dt,$$

which gives by partial integration

$$\delta E = \frac{1}{\vartheta} \int_{-\vartheta}^{0} \Omega \, dt - \Omega_t,$$

where the values for the q's to be introduced in Ω in the first term are those corresponding to the motion under the influence of the increasing external field, and the values to be introduced in the second term are those corresponding to the configuration at the time t. Neglecting small quantities of the same order as the square of the external force, however, we may in this expression for δE instead of the values for the q's corresponding to the perturbed motion take those corresponding to the original motion of the system. With this approximation the first term is equal to the mean value of the second taken over a period σ, and we have consequently

$$\int_{0}^{\sigma} \delta E \, dt = 0. \tag{7}$$

From (6) and (7) it follows that I will remain constant during the slow establishment of the small external field, if the motion corresponding to a constant value of the field is periodic. If next the external field corresponding to Ω is considered as an inherent part of the system, it will be seen in the same way that I will remain unaltered during the establishment of a new small external field, and so on. Consequently *I will be invariant for any finite transformation of the system which is sufficiently slowly performed*, provided the motion at any moment during the process is periodic and the effect of the variation is calculated on ordinary mechanics.

Before we proceed to the applications of this result we shall mention a simple consequence of (6) for systems for which every orbit is periodic independent of the initial conditions. In that case we may for the varied motion take an undisturbed motion of the system corresponding to slightly different initial conditions. This gives δE constant, and from (6) we get therefore

$$\delta E = \omega\, \delta I, \tag{8}$$

where $\omega = \dfrac{1}{\sigma}$ is the frequency of the motion. This equation forms a simple relation between the variations in E and I for periodic systems, which will be often used in the following.

Returning now to systems of one degree of freedom, we shall take our starting point from PLANCK's original theory of a *linear harmonic vibrator*. According to this theory the stationary states of a system, consisting of a particle execut-

ing linear harmonic vibrations with a constant frequency ω_0 independent of the energy, are given by the well known relation

$$E = nh\omega_0, \tag{9}$$

where n is a positive entire number, h PLANCK's constant, and E the total energy which is supposed to be zero if the particle is at rest.

From (8) it follows at once that (9) is equivalent to

$$I = \int_0^\sigma p\dot{q}\, dt = \int p\, dq = nh, \tag{10}$$

where the latter integral is to be taken over a complete oscillation of q between its limits. On the principle of the mechanical transformability of the stationary states we shall therefore assume, following EHRENFEST, that (10) holds not only for a PLANCK's vibrator but for *any periodic system of one degree of freedom* which can be formed in a continuous manner from a linear harmonic vibrator by a gradual variation of the field of force in which the particle moves. This condition is immediately seen to be fulfilled by all such systems in which the motion is of oscillating type i. e. where the moving particle during a period passes twice through any point of its orbit once in each direction. If, however, we confine ourselves to systems of one degree of freedom, it will be seen that systems in which the motion is of rotating type, i. e. where the particle during a period passes only once through every point of its orbit, cannot be formed in a

continuous manner from a linear harmonic vibrator without passing through singular states in which the period becomes infinite long and the result becomes ambiguous. We shall not here enter more closely on this difficulty which has been pointed out by EHRENFEST, because it disappears when we consider systems of several degrees of freedom, where we shall see that a simple generalisation of (10) holds for any system for which every motion is periodic.

As regards the application of (9) to statistical problems it was assumed in PLANCK's theory that the different states of the vibrator corresponding to different values of n are *a-priori equally probable*, and this assumption was strongly supported by the agreement obtained on this basis with the measurements of the specific heat of solids at low temperatures. Now it follows from the considerations of EHRENFEST, mentioned in the former section, that the a-priori probability of a given stationary state is not changed by a continuous transformation, and we shall therefore expect that for any system of one degree of freedom the different states corresponding to different entire values of n in (10) are a-priori equally probable.

As pointed out by PLANCK in connection with the application of (9), it is simply seen that statistical considerations, based on the assumption of equal probability for the different states given by (10), will show the necessary relation to considerations of ordinary statistical mechanics in the limit where the latter theory has been found to give results in agreement with experiments. Let the configuration

and motion of a mechanical system be characterised by s independent variables $q_1, \ldots, q_s$ and corresponding momenta $p_1, \ldots, p_s$ and let the state of the system be represented in a $2s$-dimensional phase-space by a point with coordinates $q_1, \ldots, q_s, p_1, \ldots, p_s$. Then, according to ordinary statistical mechanics, the probability for this point to lie within a small element in the phase-space is independent of the position and shape of this element and simply proportional to its volume, defined in the usual way by

$$\delta W = \int dq_1 \ldots dq_s \, dp_1 \ldots dp_s. \tag{11}$$

In the quantum theory, however, these considerations cannot be directly applied, since the point representing the state of a system cannot be displaced continuously in the $2s$-dimensional phase-space, but can lie only on certain surfaces of lower dimensions in this space. For systems of one degree of freedom the phase-space is a two-dimensional surface, and the points representing the states of some system given by (10) will be situated on closed curves on this surface. Now, in general, the motion will differ considerably for any two states corresponding to successive entire values of n in (10), and a simple general connection between the quantum theory and ordinary statistical mechanics is therefore out of question. In the limit, however, where n is large, the motions in successive states will only differ very little from each other, and it would therefore make little difference whether the points representing the systems are distributed

continuously on the phase-surface or situated only on the curves corresponding to (10), provided the number of systems which in the first case are situated between two such curves is equal to the number which in the second case lies on one of these curves. But it will be seen that this condition is just fulfilled in consequence of the above hypothesis of equal a-priori probability of the different stationary states, because the element of phase-surface limited by two successive curves corresponding to (10) is equal to

$$\delta W = \int dp\, dq = \left[\int p\, dq\right]_n - \left[\int p\, dq\right]_{n-1} = I_n - I_{n-1} = h, \tag{12}$$

so that on ordinary statistical mechanics the probabilities for the point to lie within any two such elements is the same. We see consequently that the hypothesis of equal probability of the different states given by (10) gives the same result as ordinary statistical mechanics in all such applications in which the states of the great majority of the systems correspond to large values of n. Considerations of this kind have led DEBYE[1]) to point out that condition (10) might have a general validity for systems of one degree of freedom, already before EHRENFEST, on the basis of his theory of the mechanical transformability of the stationary states, had shown that this condition forms the only rational generalisation of PLANCK's condition (9).

[1]) P. DEBYE, Wolfskehl-Vortrag. Göttingen 1913.

We shall now discuss the relation between the theory of *spectra of atomic systems of one degree of freedom*, based on (1) and (10), and the ordinary theory of radiation, and we shall see that this relation in several respects shows a close analogy to the relation, just considered, between the statistical applications of (10) and considerations based on ordinary statistical mechanics. Since the values for the frequency ω in two states corresponding to different values of n in (10) in general are different, we see at once that we cannot expect a simple connection between the frequency calculated by (1) of the radiation corresponding to a transition between two stationary states and the motions of the system in these states, except in the limit where n is very large, and where the ratio between the frequencies of the motion in successive stationary states differs very little from unity. Consider now a transition between the state corresponding to $n = n'$ and the state corresponding to $n = n''$, and let us assume that n' and n'' are large numbers and that $n' - n''$ is small compared with n' and n''. In that case we may in (8) for δE put $E' - E''$ and for δI put $I' - I''$, and we get therefore from (1) and (10) for the frequency of the radiation emitted or absorbed during the transition between the two states

$$\nu = \frac{1}{h}(E' - E'') = \frac{\omega}{h}(I' - I'') = (n' - n'')\omega. \qquad (13)$$

Now in a stationary state of a periodic system the displacement of the particles in any given direction may always be expressed by means of a FOURIER-series as a sum of har-

monic vibrations:

$$\xi = \sum C_\tau \cos 2\pi(\tau\omega t + c_\tau), \tag{14}$$

where the C's and c's are constants and the summation is to be extended over all positive entire values of τ. On the ordinary theory of radiation we should therefore expect the system to emit a spectrum consisting of a series of lines of frequencies equal to $\tau\omega$, but, as it is seen, this is just equal to the series of frequencies which we obtain from (13) by introducing different values for $n' - n''$. As far as the frequencies are concerned we see therefore that in the limit where n is large there exists a close relation between the ordinary theory of radiation and the theory of spectra based on (1) and (10). It may be noticed, however, that, while on the first theory radiations of the different frequencies $\tau\omega$ corresponding to different values of τ are emitted or absorbed at the same time, these frequencies will on the present theory, based on the fundamental assumption I and II, be connected with entirely different processes of emission or absorption, corresponding to the transition of the system from a given state to different neighbouring stationary states.

In order to obtain the necessary connection, mentioned in the former section, to the ordinary theory of radiation in the limit of slow vibrations, we must further claim that a relation, as that just proved for the frequencies, will, in the limit of large n, hold also for the intensities of the different lines in the spectrum. Since now on ordinary electrodynamics the intensities of the radiations corresponding to different

values of τ are directly determined from the coefficients $C_{|tau}$ in (14), we must therefore expect that for large values of n these coefficients will on the quantum theory determine the *probability of spontaneous transition* from a given stationary state for which $n = n'$ to a neighbouring state for which $n = n'' = n' - \tau$. Now this connection between the amplitudes of the different harmonic vibrations into which the motion can be resolved, characterised by different values of τ, and the probabilities of transition from a given stationary state to the different neighbouring stationary states, characterised by different values of $n' - n''$, may clearly be expected to be of a general nature. Although, of course, we cannot without a detailed theory of the mechanism of transition obtain an exact calculation of the latter probabilities, unless n is large, we may expect that also for small values of n the amplitude of the harmonic vibrations corresponding to a given value of τ will in some way give a measure for the probability of a transition between two states for which $n' - n''$ is equal to τ. Thus in general there will be a certain probability of an atomic system in a stationary state to pass spontaneously to any other state of smaller energy, but if for all motions of a given system the coefficients C in (14) are zero for certain values of τ, we are led to expect that no transition will be possible, for which $n' - n''$ is equal to one of these values.

A simple illustration of these considerations is offered by the linear harmonic vibrator mentioned above in connection with PLANCK's theory. Since in this case C_τ is equal to zero

for any τ different from 1, we shall expect that for this system only such transitions are possible in which n alters by one unit. From (1) and (9) we obtain therefore the simple result that the frequency of any radiation emitted or absorbed by a linear harmonic vibrator is equal to the constant frequency ω_0. This result seems to be supported by observations on the absorption-spectra of diatomic gases, showing that certain strong absorption-lines, which according to general evidence may be ascribed to vibrations of the two atoms in the molecule relative to each other, are not accompanied by lines of the same order of intensity and corresponding to entire multipla of the frequency, such as it should be expected from (1) if the system had any considerable tendency to pass between non-successive states. In this connection it may be noted that the fact, that in the absorption spectra of some diatomic gases faint lines occur corresponding to the double frequency of the main lines,[1]) obtains a natural explanation by assuming that for finite amplitudes the vibrations are not exactly harmonic and that therefore the molecules possess a small probability of passing also between non-successive states.

§ 3. Conditionally periodic systems.

If we consider systems of several degrees of freedom the motion will be periodic only in singular cases and the general conditions which determine the stationary states cannot

[1]) See E. C. Kemble, Phys. Rev., VIII, p. 701, 1916.

therefore be derived by means of the same simple kind of considerations as in the former section. As mentioned in the introduction, however, SOMMERFELD and others have recently succeeded, by means of a suitable generalisation of (10), to obtain conditions for an important class of systems of several degrees of freedom, which, in connection with (1), have been found to give results in convincing agreement with experimental results about line-spectra. Subsequently these conditions have been proved by EHRENFEST and especially by BURGERS[1]) to be invariant for slow mechanical transformations.

To the generalisation under consideration we are naturally led, if we first consider such systems for which the motions corresponding to the different degrees of freedom are dynamically independent of each other. This occurs if the expression for the total energy E in HAMILTON's equations (4) for a system of s degrees of freedom can be written as a sum $E_1 + \cdots + E_s$, where E_k contains q_k and p_k only. An illustration of a system of this kind is presented by a particle moving in a field of force in which the force-components normal to three mutually perpendicular fixed planes are functions of the distances from these planes respectively. Since in such a case the motion corresponding to each degree of freedom in general will be periodic, just as for a system of one degree of

[1]) J. M. BURGERS, Versl. Akad. Amsterdam, XXV, pp. 849, 918, 1055. (1917), Ann. d. Phys. LII. p. 195 (1917), Phil. Mag. XXXIII, p. 514 (1917).

freedom, we may obviously expect that the condition (10) is here replaced by a set of s conditions:

$$I_k = \int p_k \, dq_k = n_k h, \qquad (k = 1, \ldots, \ s) \tag{15}$$

where the integrals are taken over a complete period of the different q's respectively, and where $n_1, \ \ldots, \ n_s$ are entire numbers. It will be seen at once that these conditions are invariant for any slow transformation of the system for which the independency of the motions corresponding to the different coordinates is maintained.

A more general class of systems for which a similar analogy with systems of a single degree of freedom exists and where conditions of the same type as (15) present themselves is obtained in the case where, although the motions corresponding to the different degrees of freedom are not independent of each other, it is possible nevertheless by a suitable choice of coordinates to express each of the momenta p_k as a function of q_k only. A simple system of this kind consists of a particle moving in a plane orbit in a central field of force. Taking the length of the radius-vector from the centre of the field to the particle as q_1, and the angular distance of this radius-vector from a fixed line in the plane of the orbit as q_2, we get at once from (4), since E does not contain q_2, the well known result that during the motion the angular momentum p_2 is constant and that the radial motion, given by the variations of p_1 and q_1 with the time, will be exactly the same as for a system of one degree of freedom. In his funda-

mental application of the quantum theory to the spectrum of a *non-periodic system* SOMMERFELD assumed therefore that the stationary states of the above system are given by two conditions of the form:

$$I_1 = \int p_1\, dq_1 = n_1 h, \quad I_2 = \int p_2\, dq_2 = n_2 h. \tag{16}$$

While the first integral obviously must be taken over a period of the radial motion, there might at first sight seem to be a difficulty in fixing the limits of integration of q_2. This disappears, however, if we notice that an integral of the type under consideration will not be altered by a change of coordinates in which q is replaced by some function of this variable. In fact, if instead of the angular distance of the radius-vector we take for q_2 some continuous periodic function of this angle with period 2π, every point in the plane of the orbit will correspond to one set of coordinates only and the relation between p and q will be exactly of the same type as for a periodic system of one degree of freedom for which the motion is of oscillating type. It follows therefore that the integration in the second of the conditions (16) has to be taken over a complete revolution of the radius-vector, and that consequently this condition is equivalent with the simple condition that the angular momentum of the particle round the centre of the field is equal to an entire multiple of $\frac{h}{2\pi}$. As pointed out by EHRENFEST, the conditions (16) are invariant for such special transformations of the system for which the central symmetry is maintained. This follows

immediately from the fact that the angular momentum in transformations of this type remains invariant, and that the equations of motion for the radial coordinate as long as p_2 remains constant are the same as for a system of one degree of freedom. On the basis of (16), SOMMERFELD has, as mentioned in the introduction, obtained a brilliant explanation of the fine structure of the lines in the hydrogen spectrum, due to the change of the mass of the electron with its velocity.[1]) To this theory we shall come back in Part II.

As pointed out by EPSTEIN[2]) and SCHWARZSCHILD[3]) the

[1]) In this connection it may be remarked that conditions of the same type as (16) were proposed independently by W. WILSON (Phil. Mag. XXIX p. 795 (1915) and XXXI p. 156 (1916)), but by him applied only to the simple Keplerian motion described by the electron in the hydrogen atom if the relativity modifications are neglected. Due to the singular position of periodic systems in the quantum theory of systems of several degrees of freedom this application, however, involves, as it will appear from the following discussion, an ambiguity which deprives the result of an immediate physical interpretation. Conditions analogous to (16) have also been established by PLANCK in his interesting theory of the "physical structure of the phase space" of systems of several degrees of freedom (Verh. d. D. Phys. Ges. XVII p. 407 and p. 438 (1915), Ann. d. Phys. L p. 385, (1916)). This theory, which has no direct relation to the problem of line-spectra discussed in the present paper, rests upon a profound analysis of the geometrical problem of dividing the multiple-dimensional phase space corresponding to a system of several degrees of freedom into "cells" in a way analogous to the division of the phase surface of a system of one degree of freedom by the curves given by (10).

[2]) P. EPSTEIN, loc. cit.

[3]) K. SCHWARZSCHILD, loc. cit.

central systems considered by SOMMERFELD form a special case of a more general class of systems for which conditions of the same type as (15) may be applied. These are the so called *conditionally periodic systems*, to which we are led if the equations of motion are discussed by means of the HAMILTON-JACOBI partial differential equation.[1]) In the expression for the total energy E as a function of the q's and the p's, let the latter quantities be replaced by the partial differential coefficients of some function S with respect to the corresponding q's respectively, and consider the partial differential equation:

$$E\left(q_1, \ldots, \; q_s, \frac{\partial S}{\partial q_1}, \ldots, \; \frac{\partial S}{\partial q_s}\right) = \alpha_1, \tag{17}$$

obtained by putting this expression equal to an arbitrary constant α_1. If then

$$S = F(q_1, \ldots, \; q_s, \alpha_1, \ldots, \; \alpha_s) + C,$$

where $\alpha_2, \ldots, \alpha_s$, and C are arbitrary constants like α_1, is a total integral of (17), we get, as shown by HAMILTON and JACOBI, the general solution of the equations of motion (4) by putting

$$\frac{\partial S}{\partial \alpha_1} = t + \beta_1, \quad \frac{\partial S}{\partial \alpha_k} = \beta_k, \qquad (k = 2, \ldots, \; s) \tag{18}$$

[1]) See f. inst. C. V. L. CHARLIER, Die Mechanik des Himmels, Bd. I, Abt. 2.

and

$$\frac{\partial S}{\partial q_k} = p_k, \qquad (k = 1, \ldots, s) \tag{19}$$

where t is the time and $\beta_1, \ldots, \beta_s$ a new set of arbitrary constants. By means of (18) the q's are given as functions of the time t and the $2s$ constants $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_s$ which may be determined for instance from the values of the q's and $\ddot{q}$'s at a given moment.

Now the class of systems, referred to, is that for which, for a suitable choice of orthogonal coordinates, it is possible to find a total integral of (17) of the form

$$S = \sum_1^s S_k(q_k, \alpha_1, \ldots, \alpha_s), \tag{20}$$

where S_k is a function of the s constants $\alpha_1, \ldots, \alpha_s$ and of q_k only. In this case, in which the equation (17) allows of what is called "separation of variables", we get from (19) that every p is a function of the α's and of the corresponding q only. If during the motion the coordinates do not become infinite in the course of time or converge to fixed limits, every q will, just as for systems of one degree of freedom, oscillate between two fixed values, different for the different q's and depending on the α's. Like in the case of a system of one degree of freedom, p_k will become zero and change its sign whenever q_k passes through one of these limits. Apart from special cases, the system will during the motion never pass twice through a configuration corresponding to the same set

of values for the q's and p's, but it will in the course of time pass within any given, however small, distance from any configuration corresponding to a given set of values $q_1, \ldots, q_s$, representing a point within a certain closed s-dimensional extension limited by s pairs of $(s-1)$-dimensional surfaces corresponding to constant values of the q's equal to the above mentioned limits of oscillation. A motion of this kind is called "conditionally periodic". It will be seen that the character of the motion will depend only on the α's and not on the β's, which latter constants serve only to fix the exact configuration of the system at a given moment, when the α's are known. For special systems it may occur that the orbit will not cover the above mentioned s-dimensional extension everywhere dense, but will, for all values of the α's, be confined to an extension of less dimensions. Such a case we will refer to in the following as a case of "degeneration".

Since for a conditionally periodic system which allows of separation in the variables $q_1, \ldots, q_s$ the p's are functions of the corresponding q's only, we may, just as in the case of independent degrees of freedom or in the case of quasi-periodic motion in a central field, form a set of expressions of the type

$$I_k = \int p_k(q_k, \alpha_1, \ldots, \alpha_s)\, dq_k, \qquad (k = 1, \ldots, s) \qquad (21)$$

where the integration is taken over a complete oscillation of q_k. As, in general, the orbit will cover everywhere dense an s-dimensional extension limited in the characteristic way

mentioned above, it follows that, except in cases of degeneration, a separation of variables will not be possible for two different sets of coordinates $q_1, \ldots, q_s$ and $q'_1, \ldots, q'_s$, unless $q_1 = f_1(q'_1), \ldots, q_s = f_s(q'_s)$, and since a change of coordinates of this type will not affect the values of the expressions (21), it will be seen that the values of the I's are completely determined for a given motion of the system. By putting

$$I_k = n_k h, \qquad (k = l, \ldots, s) \tag{22}$$

where $n_1, \ldots, n_s$ are positive entire numbers, we obtain therefore a *set of conditions which form a natural generalisation of condition* (10) holding for a system of one degree of freedom.

Since the I's, as given by (21), depend on the constants $\alpha_1, \ldots, \alpha_s$ only and not on the β's, the α's may, in general, inversely be determined from the values of the I's. The character of the motion will therefore, in general, be completely determined by the conditions (22), and especially the value for the total energy, which according to (17) is equal to α_1, will be fixed by them. In the cases of degeneration referred to above, however, the conditions (22) involve an ambiguity, since in general for such systems there will exist an infinite number of different sets of coordinates which allow of a separation of variables, and which will lead to different motions in the stationary states, when these conditions are applied. As we shall see below, this ambiguity will not influence the fixation of the total energy in the stationary states, which

is the essential factor in the theory of spectra based on (1) and in the applications of the quantum theory to statistical problems.

A well known characteristic example of a conditionally periodic system is afforded by a particle moving under the influence of the attractions from two fixed centres varying as the inverse squares of the distances apart, if the relativity modifications are neglected. As shown by JACOBI this problem can be solved by a separation of variables if so called elliptical coordinates are used, i. e. if for q_l and q_2 we take two parameters characterising respectively an ellipsoid and a hyperboloid of revolution with the centres as foci and passing through the instantaneous position of the moving particle, and for q_3 we take the angle between the plane through the particle and the centres and a fixed plane through the latter points, or, in closer conformity with the above general description, some continuous periodic function of this angle with period 2π. A limiting case of this problem is afforded by an electron rotating round a positive nucleus and subject to the effect of an additional homogeneous electric field, because this field may be considered as arising from a second nucleus at infinite distance apart from the first. The motion in this case will therefore be conditionally periodic and allow a separation of variables in parabolic coordinates, if the nucleus is taken as focus for both sets of paraboloids of revolution, and their axes are taken parallel to the direction of the electric force. By applying the conditions (22) to this motion EPSTEIN and SCHWARZSCHILD have, as mentioned

in the introduction, independent of each other, obtained an explanation of the effect of an external electric field on the lines of the hydrogen spectrum, which was found to be in convincing agreement with STARK's measurements. To the results of these calculations we shall return in Part II.

In the above way of representing the general theory we have followed the same procedure as used by EPSTEIN. By introducing the so called "angle-variables" well known from the astronomical theory of perturbations, SCHWARZSCHILD has given the theory a very elegant form in which the analogy with systems of one degree of freedom presents itself in a somewhat different manner. The connection between this treatment and that given above has been discussed in detail by EPSTEIN.[1])

As mentioned above the conditions (22), first established from analogy with systems of one degree of freedom, have subsequently been proved generally to be *mechanically invariant for any slow transformation for which the system remains conditionally periodic.* The proof of this in variance has been given quite recently by BURGERS[2]) by means of an interesting application of the theory of contact-transformations based on SCHWARZSCHILD's introduction of angle variables. We shall not enter here on these calculations but shall only consider some points in connection with the

[1]) P. EPSTEIN, Ann. d. Phys. LI, p. 168 (1916). See also Note on page 53 of the present paper.

[2]) J. M. BURGERS, loc. cit. Versl. Akad. Amsterdam, XXV, p. 1055 (1917).

problem of the mechanical transformability of the stationary states which are of importance for the logical consistency of the general theory and for the later applications. In § 2 we saw that in the proof of the mechanical invariance of relation (10) for a periodic system of one degree of freedom, it was essential that the comparative variation of the external conditions during the time of one period could be made small. This may be regarded as an immediate consequence of the nature of the fixation of the stationary states in the quantum theory. In fact the answer to the question, whether a given state of a system is stationary, will not depend only on the motion of the particles at a given moment or on the field of force in the immediate neighbourhood of their instantaneous positions, but cannot be given before the particles have passed through a complete cycle of states, and so to speak have got to know the entire field of force of influence on the motion. If thus, in the case of a periodic system of one degree of freedom, the field of force is varied by a given amount, and if its comparative variation within the time of a single period was not small, the particle would obviously have no means to get to know the nature of the variation of the field and to adjust its stationary motion to it, before the new field was already established. For exactly the same reasons it is a necessary condition for the mechanical invariance of the stationary states of a conditionally periodic system, that the alteration of the external conditions during an interval in which the system has passed approximately through all possible configurations within the above men-

tioned s-dimensional extension in the coordinate-space can be made as small as we like. This condition forms therefore also an essential point in BURGERS' proof of the invariance of the conditions (22) for mechanical transformations. Due to this we meet with a characteristic difficulty when during the transformation of the system we pass one of the cases of degeneration mentioned above, where, for every set of values for the α's, the orbit will not cover the s-dimensional extension everywhere dense, but will be confined to an extension of less dimensions. It is clear that, when by a slow transformation of a conditionally periodic system we approach a degenerate system of this kind, the time-interval which the orbit takes to pass close to any possible configuration will tend to be very long and will become infinite when the degenerate system is reached. As a consequence of this *the conditions (22) will generally not remain mechanically invariant when we pass a degenerate system*, what has intimate connection with the above mentioned ambiguity in the determination of the stationary states of such systems by means of (22).

A typical case of a degenerate system, which may serve as an illustration of this point, is formed by a system of several degrees of freedom for which every motion is *simply periodic*, independent of the initial conditions. In this case, which is of great importance in the physical applications, we have from (5) and (21), for any set of coordinates in which a separation

of variables is possible,

$$I = \int_0^\sigma (p_1\dot{q}_1 + \cdots + p_s\dot{q}_s)\, dt = \kappa_1 I_1 + \cdots + \kappa_s I_s, \qquad (23)$$

where the integration is extended over one period of the motion, and where $\kappa_1, \ldots, \kappa_s$ are a set of positive entire numbers without a common divisor. Now we shall expect that every motion, for which it is possible to find a set of coordinates in which it satisfies (22), will be stationary. For any such motion we get from (23)

$$I = (\kappa_1 n_1 + \cdots + \kappa_s n_s)h = nh, \qquad (24)$$

where n is a whole number which may take all positive values if, as in the applications mentioned below, at least one of the κ's is equal to one. Inversely, if the system under consideration allows of separation of variables in an infinite continuous multitude of sets of coordinates, we must conclude that generally every motion which satisfies (24) will be stationary, because in general it will be possible for any such motion to find a set of coordinates in which it satisfies also (22). It will thus be seen that, for a periodic system of several degrees of freedom, condition (24) forms a simple generalisation of condition (10). From relation (8), which holds for two neighbouring motions of any periodic system, it follows further that the energy of the system will be completely determined by the value of I, just as for systems of one degree of freedom.

Consider now a periodic system in some stationary state satisfying (24), and let us assume that an external field is slowly established at a continuous rate and that the motion at any moment during this process allows of a separation of variables in a certain set of coordinates. If we would assume that the effect of the field on the motion of the system at any moment could be calculated directly by means of ordinary mechanics, we would find that the values of the I's with respect to the latter coordinates would remain constant during the process, but this would involve that the values of the n's in (22) would in general not be entire numbers, but would depend entirely on the accidental motion, satisfying (24), originally possessed by the system. That mechanics, however, cannot generally be applied directly to determine the motion of a periodic system under influence of an increasing external field, is just what we should expect according to the singular position of degenerate systems as regards mechanical transformations. In fact, in the presence of a small external field, the motion of a periodic system will undergo slow variations as regards the shape and position of the orbit, and if the perturbed motion is conditionally periodic these variations will be of a periodic nature. Formally, we may therefore compare a periodic system exposed to an external field with a simple mechanical system of one degree of freedom in which the particle performs a slow oscillating motion. Now the frequency of a slow variation of the orbit will be seen to be proportional to the intensity of the external field, and it is therefore obviously impossible to establish the external

field at a rate so slow that the comparative change of its intensity during a period of this variation is small. The process which takes place during the increase of the field will thus be analogous to that which takes place if an oscillating particle is subject to the effect of external forces which change considerably during a period. Just as the latter process generally will give rise to emission or absorption of radiation and cannot be described by means of ordinary mechanics, we must expect that the motion of a periodic system of several degrees of freedom under the establishment of the external field cannot be determined by ordinary mechanics, but that the field will give rise to effects of the same kind as those which occur during a transition between two stationary states accompanied by emission or absorption of radiation. Consequently we shall expect that, during the establishment of the field, *the system will in general adjust itself in some unmechanical way* until a stationary state is reached in which the frequency (or frequencies) of the above mentioned slow variation of the orbit has a relation to the additional energy of the system due to the presence of the external field, which is of the same kind as the relation, expressed by (8) and (10), between the energy and frequency of a periodic system of one degree of freedom. As it will be shown in Part II in connection with the physical applications, this condition is just secured if the stationary states in the presence of the field are determined by the conditions (22), and it will be seen that these considerations offer a means of fixing the stationary states of a perturbed periodic system also in cases where no separation

of variables can be obtained.

In consequence of the singular position of the degenerate systems in the general theory of stationary states of conditionally periodic systems, we obtain a means of *connecting mechanically two different stationary states of a given system* through a continuous series of stationary states without passing through systems in which the forces are very small and the energies in all the stationary states tend to coincide (comp. page 14). In fact, if we consider a given conditionally periodic system which can be transformed in a continuous way into a system for which every orbit is periodic and for which every state satisfying (24) will also satisfy (22) for a suitable choice of coordinates, it is clear in the first place that it is possible to pass in a mechanical way through a continuous series of stationary states from a state corresponding to a given set of values of the n's in (22) to any other such state for which $\kappa_1 n_1 + \cdots + \kappa_s n_s$ possesses the same value. If, moreover, there exists a second periodic system of the same character to which the first periodic system can be transformed continuously, but for which the set of κ's is different, it will be possible in general by a suitable cyclic transformation to pass in a mechanical way between any two stationary states of the given conditionally periodic system satisfying (22).

To obtain an example of such a cyclic transformation let us take the system consisting of an electron which moves round a fixed positive nucleus exerting an attraction varying as the inverse square of the distance. If we neglect the small relativity corrections, every orbit will be periodic independent of the ini-

tial conditions and the system will allow of separation of variables in polar coordinates as well as in any set of elliptical coordinates, of the kind mentioned on page 36, if the nucleus is taken as one of the foci. It is simply seen that any orbit which satisfies (24) for a value of $n > 1$, will also satisfy (22) for a suitable choice of elliptical coordinates. By imagining another nucleus of infinite small charge placed at the other focus, the orbit may further be transformed into another which satisfies (24) for the same value of n, but which may have any given value for the eccentricity. Consider now a state of the system satisfying (21), and let us assume that by the above means the orbit is originally so adjusted that in plane polar coordinates it will correspond to $n_1 = m$ and $n_2 = n - m$ in (16). Let then the system undergo a slow continuous transformation during which the field of force acting on the electron remains central, but by which the law of attraction is slowly varied until the force is directly proportional to the distance apart. In the final state, as well as in the original state, the orbit of the electron will be closed, but during the transformation the orbit will not be closed, and the ratio between the mean period of revolution and the period of the radial motion, which in the original motion was equal to one, will during the transformation increase continuously until in the final state it is equal to two. This means that, using polar coordinates, the values of κ_1 and κ_2 in (22) which for the first state are equal to $\kappa_1 = \kappa_2 = 1$, will be for the second state $\kappa_1 = 2$ and $\kappa_2 = 1$. Since during the transformation n_1 and n_2 will keep their values, we get therefore in the final state $I = h\big(2m + (n - m)\big) = h(n + m)$. Now in the latter state, the system allows a separation of variables not only in polar coordinates but also in any system of rectangular

Cartesian coordinates, and by suitable choice of the direction of the axes, we can obtain that any orbit, satisfying (24) for a value of $n > l$, will also satisfy (22). By an infinite small change of the force components in the directions of the axes, in such a way that the motions in these directions remain independent of each other but possess slightly different periods, it will further be possible to transform the elliptical orbit mechanically into one corresponding to any given ratio between the axes. Let us now assume that in this way the orbit of the electron is transformed into a circular one, so that, returning to plane polar coordinates, we have $n_1 = 0$ and $n_2 = n + m$, and let then by a slow transformation the law of attraction be varied until again it is that of the inverse square. It will be seen that when this state is reached the motion will again satisfy (24), but this time we will have $I = h(n + m)$ instead of $I = nh$ as in the original state. By repeating a cyclic process of this kind we may pass from any stationary state of the system in question which satisfies (24) for a value of $n > 1$ to any other such state without leaving at any moment the region of stationary states.

The theory of the mechanical transformability of the stationary states gives us a means to discuss the question of the *a-priori probability* of the different states of a conditionally periodic system, characterised by different sets of values for the n's in (22). In fact from the considerations, mentioned in § 1, it follows that, if the a-priori probability of the stationary states of a given system is known, it is possible at once to deduce the probabilities for the stationary states of any other system to which the first system can be transformed continuously without passing through a system of degener-

ation. Now from the analogy with systems of one degree of freedom it seems necessary to assume that, for a system of several degrees of freedom for which the motions corresponding to the different coordinates are dynamically independent of each other, the a-priori probability is the same for all the states corresponding to different sets of n's in (15). According to the above we shall therefore assume that the a-priori probability is the same for all states, given by (22), of any system which can be formed in a continuous way from a system of this kind without passing through systems of degeneration. It will be observed that on this assumption we obtain exactly the same relation to the ordinary theory of statistical mechanics in the limit of large n's as obtained in the case of systems of one degree of freedom. Thus, for a conditionally periodic system, the volume given by (11) of the element of phase-space, including all points $q_1, \ldots, q_s$, $p_1, \ldots, p_s$ which represent states for which the value of I_k given by (21) lies between I_k and $I_k + \delta I_k$, is seen at once to be equal to[1])

$$\delta W = \delta I_1 \, \delta I_2 \ldots \delta I_s, \tag{25}$$

if the coordinates are so chosen that the motion corresponding to every degree of freedom is of oscillating type. The volume of the phase-space limited by s pairs of surfaces, corresponding to successive values for the n's in the conditions (22), will therefore be equal to h^s and consequently be the same for every combination of the n's. In the limit

[1]) Comp. A. Sommerfeld, Ber. Akad. München, 1917, p. 83.

where the n's are large numbers and the stationary states corresponding to successive values for the n's differ only very little from each other, we thus obtain the same result on the assumption of equal a-priori probability of all the stationary states, corresponding to different sets of values of $n_1, \ldots, n_s$ in (22), as would be obtained by application of ordinary statistical mechanics.

The fact that the last considerations hold for every non-degenerate conditionally periodic system suggests the assumption that in general *the a-priori probability will be the same for all the states determined by* (22), even if it should not be possible to transform the given system into a system of independent degrees of freedom without passing through degenerate systems. This assumption will be shown to be supported by the consideration of the intensities of the different components of the STARK-effect of the hydrogen lines, mentioned in the next Part. When we consider a degenerate system, however, we cannot assume that the different stationary states are a-priori equally probable. In such a case the stationary states will be characterised by a number of conditions less than the number of degrees of freedom, and the probability of a given state must be determined from the number of different stationary states of some non-degenerate system which will coincide in the given state, if the latter system is continuously transformed into the degenerate system under consideration.

In order to illustrate this, let us take the simple case of a degenerate system formed by an electrified particle mov-

ing in a plane orbit in a central field, the stationary states of which are given by the two conditions (16). In this case the plane of the orbit is undetermined, and it follows already from a comparison with ordinary statistical mechanics, that the a-priori probability of the states characterised by different combinations of n_1 and n_2 in (16) cannot be the same. Thus the volume of the phase-space, corresponding to states for which I_1 lies between I_1 and $I_1 + \delta I_1$ and for which I_2 lies between I_2 and $I_2 + \delta I_2$, is found by a simple calculation[1]) to be equal to $\delta W = 2I_1\,\delta I_1\,\delta I_2$, if the motion is described by ordinary polar coordinates. For large values of n_1 and n_2, we must therefore expect that the a-priori probability of a stationary state corresponding to a given combination (n_1, n_2) is proportional to n_2. The question of the a-priori probability of states corresponding to small values of the n's has been discussed by SOMMERFELD in connection with the problem of the intensities of the different components in the fine structure of the hydrogen lines (see Part II). From considerations about the volume of the extensions in the phase-space, which might be considered as associated with the states characterised by different combinations (n_1, n_2), SOMMERFELD proposes several different expressions for the a-priori probability of such states. Due to the necessary arbitrariness involved in the choice of these extensions, however, we cannot in this way obtain a rational determination of the a-priori probability of states corre-

[1]) See A. SOMMERFELD, loc. cit.

sponding to small values of n_1 and n_2. On the other hand, this probability may be deduced by regarding the motion of the system under consideration as the degeneration of a motion characterised by three numbers n_1, n_2 and n_3, as in the general applications of the conditions (22) to a system of three degrees of freedom. Such a motion may be obtained for instance by imagining the system placed in a small homogeneous magnetic field. In certain respects this case falls outside the general theory of conditionally periodic systems discussed in this section, but, as we shall see in Part II, it can be simply shown that the presence of the magnetic field imposes the further condition on the motion in the stationary states that the angular momentum round the axis of the field is equal to $n' \frac{h}{2\pi}$, where n' is a positive entire number equal to or less than n_2, and which for the system considered in the spectral problems must be assumed to be different from zero. When regard is taken to the two opposite directions in which the particle may rotate round the axis of the field, we see therefore that for this system a state corresponding to a given combination of n_1 and n_2 in the presence of the field can be established in $2n_2$ different ways. The a-priori probability of the different states of the system may consequently for all combinations of n_1 and n_2 be assumed to be proportional to n_2.

The assumption just mentioned that the angular momentum round the axis of the field cannot be equal to zero is deduced from considerations of systems for which the mo-

tion corresponding to special combinations of the n's in (22) would become physically impossible due to some singularity in its character. In such cases we must assume that no stationary states exist corresponding to the combinations $(n_1, n_2, \ldots, n_s)$ under consideration, and on the above principle of the invariance of the a-priori probability for continuous transformations we shall accordingly expect that the a-priori probability of any other state, which can be transformed continuously into one of these states without passing through cases of degeneration, will also be equal to zero.

Let us now proceed to consider *the spectrum of a conditionally periodic system*, calculated from the values of the energy in the stationary states by means of relation (1). If $E(n_1, \ldots, n_s)$ is the total energy of a stationary state determined by (22) and if ν is the frequency of the line corresponding to the transition between two stationary states characterised by $n_k = n'_k$ and $n_k = n''_k$ respectively, we have

$$\nu = \frac{1}{h}\big[E(n'_1, \ldots, n'_s) - E(n''_1, \ldots, n''_s)\big]. \qquad (26)$$

In general, this spectrum will be entirely different from the spectrum to be expected on the ordinary theory of electrodynamics from the motion of the system. Just as for a system of one degree of freedom we shall see, however, that in the limit where the motions in neighbouring stationary states differ very little from each other, there exists a close relation between the spectrum calculated on the quantum theory and that to be expected on ordinary electrodynamics. As in § 2

we shall further see, that this connection leads to certain general considerations about the probability of transition between any two stationary states and about the nature of the accompanying radiation, which are found to be supported by observations. In order to discuss this question we shall first deduce a general expression for the energy difference between two neighbouring states of a conditionally periodic system, which can be simply obtained by a calculation analogous to that used in § 2 in the deduction of the relation (8).

Consider some motion of a conditionally periodic system which allows of separation of variables in a certain set of coordinates $q_1, \ldots, q_s$, and let us assume that at the time $t = \vartheta$ the configuration of the system will to a close approximation be the same as at the time $t = 0$. By taking ϑ large enough we can make this approximation as close as we like. If next we consider some conditionally periodic motion, obtained by a small variation of the first motion, and which allows of separation of variables in a set of coordinates $q'_1, \ldots, q'_s$ which may differ slightly from the set $q_1, \ldots, q_s$, we get by means of HAMILTON's equations (4), using the coordinates $q'_1, \ldots, q'_s$,

$$\int_0^\vartheta \delta E \, dt = \int_0^\vartheta \sum_1^s \left(\frac{\partial E}{\partial p'_k} \delta p'_k + \frac{\partial E}{\partial q'_k} \delta q'_k \right) dt$$

$$= \int_0^\vartheta \sum_1^s (\dot{q}'_k \, \delta p'_k - \dot{p}'_k \, \delta q'_k) \, dt.$$

By partial integration of the second term in the bracket this

gives:

$$\int_0^{\vartheta} \delta E\, dt = \int_0^{\vartheta} \sum_1^s \delta(p'_k \dot{q}'_k)\, dt - \left| \sum_1^s p'_k\, \delta q'_k \right|_{t=0}^{t=\vartheta}. \qquad (27)$$

Now we have for the unvaried motion

$$\int_0^{\vartheta} \sum_1^s p'_k \dot{q}'_k\, dt = \int_0^{\vartheta} \sum_1^s p_k \dot{q}_k\, dt = \sum_1^s N_k I_k,$$

where I_k is defined by (21) and where N_k is the number of oscillations performed by q_k in the time interval ϑ. For the varied motion we have on the other hand:

$$\int_0^{\vartheta} \sum_1^s p'_k \dot{q}'_k\, dt = \int_{t=0}^{t=\vartheta} \sum_1^s p'_k dq'_k = \sum_1^s N_k I'_k + \left| \sum_1^s p'_k\, \delta q'_k \right|_{t=0}^{t=\vartheta},$$

where the I's correspond to the conditionally periodic motion in the coordinates $q'_1, \ldots, q'_s$, and the δq's which enter in the last term are the same as those in (27). Writing $I'_k - I_k = \delta I_k$, we get therefore from the latter equation

$$\int_0^{\vartheta} \delta E\, dt = \sum_1^s N_k\, \delta I_k. \qquad (28)$$

In the special case where the varied motion is an undisturbed motion belonging to the same system as the unvaried motion we get, since δE will be constant,

$$\delta E = \sum_1^s \omega_k\, \delta I_k, \qquad (29)$$

where $\omega_k = \dfrac{N_k}{\vartheta}$ is the mean frequency of oscillation of q_k between its limits, taken over a long time interval of the same order of magnitude as ϑ. This equation forms a simple generalisation of (8), and in the general case in which a separation of variables will be possible only for one system of coordinates leading to a complete definition of the I's it might have been deduced directly from the analytical theory of the periodicity properties of the motion of a conditionally periodic system, based on the introduction of angle-variables.[1]) From (29) it follows moreover that, if the system allows of a separation of variables in an infinite continuous multitude

[1]) See CHARLIER, Die Mechanik des Himmels, Bd. I Abt. 2, and especially P. EPSTEIN, Ann. d. Phys. LI p. 178 (1916). By means of the well known theorem of JACOBI about the change of variables in the canonical equations of HAMILTON, the connection between the notion of angle-variables and the quantities I, discussed by EPSTEIN in the latter paper, may be briefly exposed in the following elegant manner which has been kindly pointed out to me by Mr. H. A. KRAMERS. Consider the function $S(q_1, \ldots, q_s, I_1, \ldots, I_s)$ obtained from (20) by introducing for the α's their expressions in terms of the I's given by the equations (21). This function will be a many valued function of the q's which increases by I_k if q_k describes one oscillation between its limits and comes back to its original value while the other q's remain constant. If we therefore introduce a new set of variables $w_1, \ldots, w_s$ defined by

$$w_k = \frac{\partial S}{\partial I_k}, \qquad (k = 1, \ldots, s) \tag{1*}$$

it will be seen that w_k increases by one unit while the other w's will come back to their original values if q_k describes one oscillation between its limits and the other q's remain constant. Inversely it will therefore

be seen that the q's, and also the p's which were given by

$$p_k = \frac{\partial S}{\partial q_k}, \qquad (k = i, \ldots, s) \tag{2*}$$

when considered as functions of the I's and w's will be periodic functions of every of the w's with period 1. According to FOURIER's theorem any of the q's may therefore be represented by an s-double trigonometric series of the form

$$q = \sum A_{\tau_1, \ldots, \tau_s} \cos 2\pi(\tau_1 w_1 + \ldots + \tau_s w_s + \alpha_{\tau_1, \ldots, \tau_s}), \tag{3*}$$

where the A's and α's are constants depending on the I's and where the summation is to be extended over all entire values of $\tau_1, \ldots, \tau_s$. On account of this property of the w's, the quantities $2\pi w_1, \ldots, 2\pi w_s$ are denoted as "angle variables". Now from (1*) and (2*) it follows according to the above mentioned theorem of JACOBI (see for instance JACOBI, Vorlesungen über Dynamik § 37) that the variations with the time of the I's and w's will be given by

$$\frac{dI_k}{dt} = -\frac{\partial E}{\partial w_k}, \quad \frac{dw_k}{dt} = \frac{\partial E}{\partial I_k}, \qquad (k = 1, \ldots, s) \tag{4*}$$

where the energy E is considered as a function of the I's and w's. Since E, however, is determined by the I's only we get from (4*), besides the evident result that the I's are constant during the motion, that the w's will vary linearly with the time and can be represented by

$$w_k = \omega_k t + \delta_k, \quad \omega_k = \frac{\partial E}{\partial I_k}, \qquad (k = 1, \ldots, s) \tag{5*}$$

where δ_k is a constant, and where ω_k is easily seen to be equal to the mean frequency of oscillation of q_k. From (5*) equation (28) follows at once, and it will further be seen that by introducing (5*) in (3*) we get the result that every of the q's, and consequently also any one-valued function of the q's, can be represented by an expression of the type (31).

of sets of coordinates, the total energy will be the same for all motions corresponding to the same values of the I's, independent of the special set of coordinates used to calculate these quantities. As mentioned above and as we have al-

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In this connection it may be mentioned that the method of SCHWARZSCHILD of fixing the stationary states of a conditionally periodic system, mentioned on page 36, consists in seeking for a given system a set of canonically conjugated variables $Q_1, \ldots, Q_s, P_1, \ldots, P_s$ in such a way that the positional coordinates of the system $q_1, \ldots, q_s$, and their conjugated momenta $p_1, \ldots, p_s$, when considered as functions of the Q's and P's, are periodic in every of the Q's with period 2π, while the energy of the system depends only on the P's. In analogy with the condition which fixes the angular momentum in SOMMERFELD's theory of central systems SCHWARZSCHILD next puts every of the P's equal to an entire multiple of $\frac{h}{2\pi}$. In contrast to the theory of stationary states of conditionally periodic systems based on the possibility of separation of variables and the fixation of the I's by (22), this method does not lead to an absolute fixation of the stationary states, because, as pointed out by SCHWARZSCHILD himself, the above definition of the P's leaves an arbitrary constant undetermined in every of these quantities. In many cases, however, these constants may be simply determined from considerations of mechanical transformability of the stationary states, and as pointed out by BURGERS (loc. cit. Versl. Akad. Amsterdam XXV p. 1055 (1917)). SCHWARZSCHILD's method possesses on the other hand the essential advantage of being applicable to certain classes of systems in which the displacements of the particles may be represented by trigonometric series of the type (31), but for which the equations of motion cannot be solved by separation of variables in any fixed set of coordinates. An interesting application of this to the spectrum of rotating molecules, given by BURGERS, will be mentioned in Part IV.

ready shown in the case of purely periodic systems by means of (8), the total energy is therefore also in cases of degeneration completely determined by the conditions (22).

Consider now a transition between two stationary states determined by (22) by putting $n_k = n'_k$ and $n_k = n''_k$ respectively, and let us assume that $n'_1, \ldots, n'_s, n''_1, \ldots, n''_s$ are large numbers, and that the differences $n'_k - n''_k$ are small compared with these numbers. Since the motions of the system in these states will differ relatively very little from each other we may calculate the difference of the energy by means of (29), and we get therefore, by means of (1), for the frequency of the radiation corresponding to the transition between the two states

$$\nu = \frac{1}{h}(E' - E'') = \frac{1}{h}\sum_1^s \omega_k (I'_k - I''_k) = \sum_1^s \omega_k (n'_k - n''_k), \quad (30)$$

which is seen to be a direct generalisation of the expression (13) in § 2.

Now, in complete analogy to what is the case for periodic systems of one degree of freedom, it is proved in the analytical theory of the motion of conditionally periodic systems mentioned above that for the latter systems the coordinates $q_1, \ldots, q_s$, and consequently also the displacements of the particles in any given direction, may be expressed as a function of the time by an s-double infinite FOURIER series of the form:

$$\xi = \sum C_{\tau_1, \ldots, \tau_s} \cos 2\pi \left\{ (\tau_1 \omega_1 + \ldots + \tau_s \omega_s) t + c_{\tau_1, \ldots, \tau_s} \right\}, \quad (31)$$

where the summation is to be extended over all positive and negative entire values of the τ's, and where the ω's are the above mentioned mean frequencies of oscillation for the different q's. The constants $C_{\tau_1,\ldots,\ \tau_s}$ depend only on the α's in the equations (18) or, what is the same, on the I's, while the constants $c_{\tau_1,\ldots,\ \tau_s}$ depend on the α's as well as on the β's. In general the quantities $\tau_1\omega_1 + \ldots + \tau_s\omega_s$ will be different for any two different sets of values for the τ's, and in the course of time the orbit will cover everywhere dense a certain s-dimensional extension. In a case of degeneration, however, where the orbit will be confined to an extension of less dimensions, there will exist for all values of the α's one or more relations of the type $m_1\omega_1+\ldots+m_s\omega_s = 0$ where the m's are entire numbers and by the introduction of which the expression (31) can be reduced to a FOURIER series which is less than s-double infinite. Thus in the special case of a system of which every orbit is periodic we have $\frac{\omega_1}{\kappa_1} = \cdots = \frac{\omega_s}{\kappa_s} = \omega$, where the κ's are the numbers which enter in equation (23), and the FOURIER series for the displacements in the different directions will in this case consist only of terms of the simple form $C_\tau \cos 2\pi\{\tau\omega t + c_\tau\}$, just as for a system of one degree of freedom.

On the ordinary theory of radiation, we should expect from (31) that the spectrum emitted by the system in a given state would consist of an s-double infinite series of lines of frequencies equal to $\tau_1\omega_1 + \cdots + \tau_s\omega_s$. In general, this spectrum would be completely different from that given

by (26). This follows already from the fact that the ω's will depend on the values for the constants $\alpha_1, \ldots, \alpha_s$ and will vary in a continuous way for the continuous multitude of mechanically possible states corresponding to different sets of values for these constants. Thus in general the ω's will be quite different for two different stationary states corresponding to different sets of n's in (22), and we cannot expect any close relation between the spectrum calculated on the quantum theory and that to be expected on the ordinary theory of mechanics and electrodynamics. In the limit, however, where the n's in (22) are large numbers, the ratio between the ω's for two stationary states, corresponding to $n_k = n'_k$ and $n_k = n''_k$ respectively, will tend to unity if the differences $n'_k - n''_k$ are small compared with the n's, and as seen from (30) the spectrum calculated by (1) and (22) will in this limit just tend to coincide with that to be expected on the ordinary theory of radiation from the motion of the system.

As far as the frequencies are concerned, we thus see that for conditionally periodic systems there exists a connection between the quantum theory and the ordinary theory of radiation of exactly the same character as that shown in § 2 to exist in the simple case of periodic systems of one degree of freedom. Now on ordinary electrodynamics the coefficients $C_{\tau_1,\ldots,\tau_s}$ in the expression (31) for the displacements of the particles in the different directions would in the well known way determine the intensity and polarisation of the emitted radiation of the corresponding frequency $\tau_1\omega_1 + \ldots + \tau_s\omega_s$. As for systems of one degree of freedom we must therefore

conclude that, in the limit of large values for the n's, the probability of spontaneous transition between two stationary states of a conditionally periodic system, as well as the polarisation of the accompanying radiation, can be determined directly from the values of the coefficient $C_{\tau_1,\ldots,\tau_s}$ in (31) corresponding to a set of τ's given by $\tau_k = n'_k - n''_k$, if $n'_1, \ldots, n'_s$ and $n''_1, \ldots, n''_s$ are the numbers which characterise the two stationary states.

Without a detailed theory of the mechanism of transition between the stationary states we cannot, of course, in general obtain an exact determination of the *probability of spontaneous transition* between two such states, unless the n's are large numbers. Just as in the case of systems of one degree of freedom, however, we are naturally led from the above considerations to assume that, also for values of the n's which are not large, there must exist an intimate connection between the probability of a given transition and the values of the corresponding FOURIER coefficient in the expressions for the displacements of the particles in the two stationary states. This allows us at once to draw certain important conclusions. Thus, from the fact that in general negative as well as positive values for the τ's appear in (31), it follows that we must expect that in general not only such transitions will be possible in which all the n's decrease, but that also transitions will be possible for which some of the n's increase while others decrease. This conclusion, which is supported by observations on the fine structure of the hydrogen lines as well as on the STARK effect, is contrary to the suggestion,

put forward by SOMMERFELD with reference to the essential positive character of the I's, that every of the n's must remain constant or decrease under a transition. Another direct consequence of the above considerations is obtained if we consider a system for which, for all values of the constants $\alpha_1, \ldots, \alpha_s$, the coefficient $C_{\tau_1,\ldots,\tau_s}$ corresponding to a certain set $\tau_1^0, \ldots, \tau_s^0$ of values for the τ's is equal to zero in the expressions for the displacements of the particles in every direction. In this case we shall naturally expect that no transition will be possible for which the relation $n'_k - n''_k = \tau_k^0$ is satisfied for every k. In the case where $C_{\tau_1^0,\ldots,\tau_s^0}$ is equal to zero in the expressions for the displacement in a certain direction only, we shall expect that all transitions, for which $n'_k - n''_k = \tau_k^0$ for every k, will be accompanied by a radiation which is polarised in a plane perpendicular to this direction.

A simple illustration of the last considerations is afforded by the system mentioned in the beginning of this section, and which consists of a particle executing motions in three perpendicular directions which are independent of each other. In that case all the FOURIER coefficients in the expressions for the displacements in any direction will disappear if more than one of the τ's are different from zero. Consequently we must assume that only such transitions are possible for which only one of the n's varies at the same time, and that the radiation corresponding to such a transition will be linearly polarised in the direction of the displacement of the corresponding coordinate. In the special case where the motions in the three directions are simply harmonic, we shall

moreover conclude that none of the n's can vary by more than a single unit, in analogy with the considerations in the former section about a linear harmonic vibrator.

Another example which has more direct physical importance, since it includes all the special applications of the quantum theory to spectral problems mentioned in the introduction, is formed by a conditionally periodic system possessing an axis of symmetry. In all these applications a separation of variables is obtained in a set of three coordinates q_1, q_2 and q_3, of which the first two serve to fix the position of the particle in a plane through the axis of the system, while the last is equal to the angular distance between this plane and a fixed plane through the same axis. Due to the symmetry, the expression for the total energy in HAMILTON's equations will not contain the angular distance q_3 but only the angular momentum p_3 round the axis. The latter quantity will consequently remain constant during the motion, and the variations of q_1 and q_2 will be exactly the same as in a conditionally periodic system of two degrees of freedom only. If the position of the particle is described in a set of cylindrical coordinates z, ρ, ϑ, where z is the displacement in the direction of the axis, ρ the distance of the particle from this axis and ϑ is equal to the angular distance q_3, we have therefore

$$\begin{aligned} z &= \sum C_{\tau_1,\tau_2} \cos 2\pi\big\{(\tau_1\omega_1 + \tau_2\omega_2)t + c_{\tau_1,\tau_2}\big\} \\ \text{and} \quad \rho &= \sum C'_{\tau_1,\tau_2} \cos 2\pi\big\{(\tau_1\omega_1 + \tau_2\omega_2)t + c'_{\tau_1,\tau_2}\big\}, \end{aligned} \tag{32}$$

where the summation is to be extended over all positive and negative entire values of τ_1 and τ_2, and where ω_1 and ω_2 are the mean frequencies of oscillation of the coordinates q_1 and q_2. For the rate of variation of ϑ with the time we have further

$$\begin{aligned}\frac{d\vartheta}{dt} = \dot{q}_3 = \frac{\partial E}{\partial p_3} &= f(q_1, q_2, p_1, p_2, p_3) \\ &= \pm \sum C''_{\tau_1,\tau_2} \cos 2\pi \big\{(\tau_1\omega_1 + \tau_2\omega_2)t + c''_{\tau_1,\tau_2}\big\},\end{aligned}$$

where the two signs correspond to a rotation of the particle in the direction of increasing and decreasing q_3 respectively, and are introduced to separate the two types of symmetrical motions corresponding to these directions. This gives

$$\pm\vartheta = 2\pi\omega_3 t + \sum C'''_{\tau_1,\tau_2} \cos 2\pi \big\{(\tau_1\omega_1 + \tau_2\omega_2)t + c'''_{\tau_1,\tau_2}\big\}, \quad (33)$$

where the positive constant $\omega_3 = \dfrac{1}{2\pi} C''_{0,0}$ is the mean frequency of rotation round the axis of symmetry of the system. Considering now the displacement of the particle in rectangular coordinates x, y and z, and taking as above the axis of symmetry as z-axis, we get from (32) and (33) after a simple contraction of terms

$$\begin{aligned} x &= \rho\cos\vartheta \\ &= \sum D_{\tau_1,\tau_2} \cos 2\pi \big\{(\tau_1\omega_1 + \tau_2\omega_2 + \omega_3)t + d_{\tau_1,\tau_2}\big\} \\ \text{and } y &= \rho\sin\vartheta \\ &= \pm \sum D_{\tau_1,\tau_2} \sin 2\pi \big\{(\tau_1\omega_1 + \tau_2\omega_2 + \omega_3)t + d_{\tau_1,\tau_2}\big\}, \end{aligned} \quad (34)$$

where the D's and d's are new constants, and the summation is again to be extended over all positive and negative values of τ_1 and τ_2.

From (32) and (34) we see that the motion in the present case may be considered as composed of a number of linear harmonic vibrations parallel to the axis of symmetry and of frequencies equal to the absolute values of $(\tau_1\omega_1 + \tau_2\omega_2)$, together with a number of circular harmonic motions round this axis of frequencies equal to the absolute values of $(\tau_1\omega_1 + \tau_2\omega_2 + \omega_3)$, and possessing the same direction of rotation as that of the moving particle or the opposite if the latter expression is positive or negative respectively. According to ordinary electrodynamics the radiation from the system would therefore consist of a number of components of frequency $\tau_1\omega_1 + \tau_2\omega_2$ polarised parallel to the axis of symmetry, and a number of components of frequencies $\tau_1\omega_1 + \tau_2\omega_2 + \omega_3$ and of circular polarisation round this axis (when viewed in the direction of the axis). On the present theory we shall consequently expect that in this case only two kinds of transitions between the stationary states given by (22) will be possible. In both of these n_1 and n_2 may vary by an arbitrary number of units, but in the first kind of transition, which will give rise to a radiation polarised parallel to the axis of the system, n_3 will remain unchanged, while in the second kind of transition n_3 will decrease or increase by one unit and the emitted radiation will be circularly polarised round the axis in the same direction as or the opposite of that of the rotation of the particle respectively.

In the next Part we shall see that these conclusions are supported in an instructive manner by the experiments on the effects of electric and magnetic fields on the hydrogen spectrum. In connection with the discussion of the general theory, however, it may be of interest to show that the formal analogy between the ordinary theory of radiation and the theory based on (1) and (22), in case of systems possessing an axis of symmetry, can be traced not only with respect to frequency relations but also by considerations of *conservation of angular momentum*. For a conditionally periodic system possessing an axis of symmetry the angular momentum round this axis is, with the above choice of coordinates, according to (22) equal to $\frac{I_3}{2\pi} = n_3 \frac{h}{2\pi}$. If therefore, as assumed above for a transition corresponding to an emission of linearly polarised light, n_3 is unaltered, it means that the angular momentum of the system remains unchanged, while if n_3 alters by one unit, as assumed for a transition corresponding to an emission of circularly polarised light, the angular momentum will be altered by $\frac{h}{2\pi}$. Now it is easily seen that the ratio between this amount of angular momentum and the amount of energy $h\nu$ emitted during the transition is just equal to the ratio between the amount of angular momentum and energy possessed by the radiation which according to ordinary electrodynamics would be emitted by an electron rotating in a circular orbit in a central field of force. In fact, if a is the radius of the orbit, ν the frequency of revolu-

tion and F the force of reaction due to the electromagnetic field of the radiation, the amount of energy and of angular momentum round an axis through the centre of the field perpendicular to the plane of the orbit, lost by the electron in unit of time as a consequence of the radiation, would be equal to $2\pi\nu aF$ and aF respectively. Due to the principles of conservation of energy and of angular momentum holding in ordinary electrodynamics, we should therefore expect that the ratio between the energy and the angular momentum of the emitted radiation would be $2\pi\nu$,[1]) but this is seen to be equal to the ratio between the energy $h\nu$ and the angular momentum $\frac{h}{2\pi}$ lost by the system considered above during a transition for which we have assumed that the radiation is circularly polarised. This agreement would seem not only to support the validity of the above considerations but also to offer a direct support, independent of the equations (22), of the assumption that, *for an atomic system possessing an axis of symmetry, the total angular momentum round this axis is equal to an entire multiple of* $\frac{h}{2\pi}$.

A further illustration of the above considerations of the relation between the quantum theory and the ordinary theory of radiation is obtained if we consider a conditionally periodic system subject to the *influence of a small perturbing field of force*. Let us assume that the original system allows of separation of variables in a certain set of coordi-

[1]) Comp. K. Schaposchnikow, Phys. Zeitschr. XV, p. 454 (1914).

nates $q_1, \ldots, q_s$, so that the stationary states are determined by (22). From the necessary stability of the stationary states we must conclude that the perturbed system will possess a set of stationary states which only differ slightly from those of the original system. In general, however, it will not be possible for the perturbed system to obtain a separation of variables in any set of coordinates, but if the perturbing force is sufficiently small the perturbed motion will again be of conditionally periodic type and may be regarded as a superposition of a number of harmonic vibrations just as the original motion. The displacements of the particles in the stationary states of the perturbed system will therefore be given by an expression of the same type as (31) where the fundamental frequencies ω_k and the amplitudes $C_{\tau_1,\ldots,\tau_s}$ may differ from those corresponding to the stationary states of the original system by small quantities proportional to the intensity of the perturbing forces. If now for the original motion the coefficients $C_{\tau_1,\ldots,\tau_s}$ corresponding to certain combinations of the τ's are equal to zero for all values of the constants $\alpha_1, \ldots, \alpha_s$, these coefficients will therefore for the perturbed motion, in general, possess small values proportional to the perturbing forces. From the above considerations we shall therefore expect that, in addition to the main probabilities of such transitions between stationary states which are possible for the original system, there will for the perturbed system exist small probabilities of new transitions corresponding to the above mentioned combinations of the τ's. Consequently we shall expect that the effect of the perturbing field on the

spectrum of the system will consist partly in a small displacement of the original lines, partly in the appearance of new lines of small intensity.

A simple example of this is afforded by a system consisting of a particle moving in a plane and executing harmonic vibrations in two perpendicular directions with frequencies ω_1 and ω_2. If the system is undisturbed all coefficients C_{τ_1,τ_2} will be zero, except $C_{1,0}$ and $C_{0,1}$. When, however, the system is perturbed, for instance by an arbitrary small central force, there will in the FOURIER expressions for the displacements of the particle, in addition to the main terms corresponding to the fundamental frequencies ω_1 and ω_2, appear a number of small terms corresponding to frequencies given by $\tau_1\omega_1 + \tau_2\omega_2$ where τ_1 and τ_2 are entire numbers which may be positive as well as negative. On the present theory we shall therefore expect that in the presence of the perturbing force there will appear small probabilities for new transitions which will give rise to radiations analogous to the so called harmonics and combination tones in acoustics, just as it should be expected on the ordinary theory of radiation where a direct connection between the emitted radiation and the motion of the system is assumed. Another example of more direct physical application is afforded by the effect of an external homogeneous electric field in producing new spectral lines. In this case the potential of the perturbing force is a linear function of the coordinates of the particles and, whatever is the nature of the original system, it follows directly from the general theory of perturbations that

the frequency of any additional term in the expression for the perturbed motion, which is of the same order of magnitude as the external force, must correspond to the sum or difference of two frequencies of the harmonic vibrations into which the original motion can be resolved. With applications of these considerations we will meet in Part II in connection with the discussion of SOMMERFELD's theory of the fine structure of the hydrogen lines and in Part III in connection with the problem of the appearance of new series in the spectra of other elements under the influence of intense external electric fields.

As mentioned we cannot without a more detailed theory of the mechanism of transition between stationary states obtain quantitative information as regards the general question of the intensities of the different lines of the spectrum of a conditionally periodic system given by (26), except in the limit where the n's are large numbers, or in such special cases where for all values of the constants $\alpha_1, \ldots, \alpha_s$ certain coefficients $C_{\tau_1,\ldots,\tau_s}$ in (31) are equal to zero. From considerations of analogy, however, we must expect that it will be possible also in the general case to obtain an *estimate of the intensities* of the different lines in the spectrum by comparing the intensity of a given line, corresponding to a transition between two stationary states characterised by the numbers $n'_1, \ldots, n'_s$ and $n''_1, \ldots, n''_s$ respectively, with the intensities of the radiations of frequencies $\omega_1(n'_1-n''_1)+\cdots+\omega_s(n'_s-n''_s)$ to be expected on ordinary electrodynamics from the motions in these states; although of course this estimate becomes

more uncertain the smaller the values for the n's are. As it will be seen from the applications mentioned in the following Parts this is supported in a general way by comparison with the observations.

Færdig fra Trykkeriet d. 27, April 1918.

Part II.

On the hydrogen spectrum.

§ 1. The simple theory of the series spectrum of hydrogen.

As well known, the frequencies of the lines of the series spectrum of hydrogen may, if we look apart from the fine structure of the single lines revealed by instruments of high dispersive power, be represented by the formula

$$\nu = K\left(\frac{1}{n''^2} - \frac{1}{n'^2}\right), \tag{35}$$

where K is a constant, and n' and n'' a set of two entire numbers, different for the different lines of the spectrum. According to the general principles of the quantum theory of line spectra discussed in the first section of Part I, we shall therefore expect that this spectrum is emitted by a system which possesses a series of stationary states in which the numerical value of the energy in the nth state, omitting an arbitrary constant, with a high degree of approximation is given by

$$E_n = \frac{Kh}{n^2}, \tag{36}$$

where h is Planck's constant which enters in the fundamental relation (1).

Now according to Rutherford's theory of atomic structure, a neutral hydrogen atom must be expected to consist

of an electron and a positive nucleus of a mass very large compared with that of the electron, which move under the influence of a mutual attraction inversely proportional to the square of the distance apart. Assuming that the motion in the stationary states may be determined by ordinary mechanics, and neglecting for the moment the small modifications claimed by the theory of relativity, we find that each of the particles will describe an elliptical orbit with their common centre of gravity at one of the foci, and from the well known laws for a Keplerian motion we have that the frequency of revolution ω and the major axis 2α of the relative orbit of the particles, quite independent of the degree of eccentricity of this orbit, are given by

$$\omega = \sqrt{\frac{2W^3(M+m)}{\pi^2 N^2 e^4 M m}}, \quad 2\alpha = \frac{Ne^2}{W}, \tag{37}$$

where W is the work necessary to remove the electron to infinite distance from the nucleus, while Ne and M are the charge and the mass of the nucleus, and $-e$ and m the charge and the mass of the electron.

As explained in Part I, there will in general be no simple connection between the motion of a system in the stationary states and the spectrum emitted during transitions between these states; such a connection, however, must be expected to exist in the limit where the motions in successive stationary states differ comparatively little from each other. In the present case this connection claims in the first place that the frequency of revolution tends to zero for increasing n.

According to (36) and (37) we may therefore put the value of W in the nth stationary state equal to

$$W_n = \frac{Kh}{n^2}. \tag{38}$$

Moreover, since (35) can be written in the form

$$\nu = (n' - n'')K\,\frac{n' + n''}{n'^2 n''^2},$$

it is seen to be a necessary condition that the frequency of revolution for large values of n is asymptotically given by

$$\omega_n \sim \frac{2K}{n^3}, \tag{39}$$

if we wish that the frequency of the radiation emitted during a transition between two stationary states, for which the numbers n' and n'' are large compared with their difference $n' - n''$, shall tend to coincide with one of the frequencies of the spectrum which on ordinary electrodynamics would be emitted from the system in these states. But from (37) and (38) it will be seen that (39) claims the fulfilment of the relation

$$K = \frac{2\pi^2 N^2 e^4 Mm}{h^3(M+m)} = \frac{2\pi^2 N^2 e^4 m}{h^3(1+m/M)}. \tag{40}$$

As shown in previous papers, this relation is actually found to be fulfilled within the limit of experimental errors

if we put $N = 1$ and for e, m, and h introduce the values deduced from measurements on other phenomena; a result which may be considered as affording a strong support for the validity of the general principles discussed in Part I, as well as for the reality of the atomic model under consideration. Further it was found that, if in formula (35) for the hydrogen spectrum the constant K is replaced by a constant which is four times larger, this formula represents to a high degree of approximation the frequencies of the lines of a spectrum emitted by helium, when this gas is subject to a condensed discharge. This was to be expected on RUTHERFORD's theory, according to which a neutral helium atom contains two electrons and a nucleus of a charge twice that of the nucleus of the hydrogen atom. A helium atom from which one electron is removed will thus form a dynamical system perfectly similar to a neutral hydrogen atom, and may therefore be expected to emit a spectrum represented by (35) if in (40) we put $N = 2$. Moreover a closer comparison of the helium spectrum under consideration with the hydrogen spectrum has shown that the value of the constant K in the former spectrum was not exactly four times as large as that in the latter, but that the ratio between these constants within the limit of experimental errors agreed with the value to be expected from (40), when regard is taken to the different masses of the nuclei of the atoms of hydrogen and helium corresponding to the different atomic weights of these elements.[1])

[1]) For the literature on this subject the reader is referred to the

Introducing the expression for K given by (40) in the formulæ (37) and (38), we find for the values of W, ω and 2α in the stationary states

$$W_n = \frac{1}{n^2}\,\frac{2\pi^2 N^2 e^4 M m}{h^2(M+m)},$$
$$\omega_n = \frac{1}{n^3}\,\frac{4\pi^2 N^2 e^4 M m}{h^3(M+m)}, \quad 2\alpha_n = n^2\,\frac{h^2(M+m)}{2\pi^2 N e^2 M m}. \tag{41}$$

Now for a mechanical system as that under consideration, for which every motion is periodic independent of the initial conditions, we have that the value of the total energy will be completely determined by the value of the quantity I, defined by equation (5) in Part I. As mentioned this follows directly from relation (8), which shows at the same time that for a system for which every motion is periodic the frequency will be completely determined by I or by the energy only. For the value of I in the stationary states of the hydrogen atom we get by means of (8) from (37) and (41), since in this case I will obviously become zero when W becomes infinite,

$$I = \int_{W_n}^{\infty} \frac{dW}{\omega} = \sqrt{\frac{\pi^2 N^2 e^4 M m}{2(M+m)}} \int_{W_n}^{\infty} W^{-3/2}\, dW$$
$$= \sqrt{\frac{2\pi^2 N^2 e^4 M m}{W_n(M+m)}} = nh.$$

papers cited in the introduction.

This result will be seen to be consistent with condition (24) which, as mentioned in Part I, presents itself as a direct generalisation to periodic systems of several degrees of freedom of condition (10) which determines the stationary states of a system of one degree of freedom, and which again on EHRENFEST's principle of the mechanical transformability of the stationary states forms a rational generalisation of PLANCK's fundamental formula (9) for the possible values of the energy of a linear harmonic vibrator.

In this connection it will be observed, that the relation discussed above between the hydrogen spectrum and the motion of the atom in the limit of small frequencies is completely analogous to the general relation, discussed in § 2 in Part I, between the spectrum which on the quantum theory would be emitted by a system of one degree of freedom, the stationary states of which are determined by (10), and the motion of the system in these states. It will at the same time be noted that, in case of hydrogen, this relation implies that the motion of the particles in the stationary states of the atom will not in general be simply harmonic, or in other words that the orbit of the electron will not in general be circular. In fact if the motion of the particles were simply harmonic, as the motion of a PLANCK's vibrator, we should expect on the considerations in Part I that no transition between two stationary states of the atom would be possible for which n' and n'' differ by more than one unit; but this would obviously be inconsistent with the observations, since for instance the lines of the ordinary Balmer series, accord-

ing to the theory, correspond to transitions for which $n'' = 2$ while n' takes the values 3, 4, 5, ... In connection with this consideration it may be remarked that, adopting a terminology well known from acoustics, we may from the point of view of the quantum theory regard the higher members of the Balmer series ($n' = 4, 5, \ldots$) as the "harmonics" of the first member ($n' = 3$), although of course the frequencies of the former lines are by no means entire multipla of the frequency of the latter line.

While in the above way it was possible to obtain a simple interpretation of certain main features of the hydrogen spectrum, it was not found possible in this way to account in detail for such phenomena in which the deviation of the motion of the particles from a simple Keplerian motion plays an essential part. This is the case in the problem of the fine structure of the hydrogen lines, which is due to the effect of the small variation of the mass of the electron with its velocity, as well as in the problems of the characteristic effects of external electric and magnetic fields on the hydrogen lines. As mentioned in the introduction, a progress of fundamental importance in the treatment of such problems was made by SOMMERFELD, who obtained a convincing explanation of the fine structure of the hydrogen lines by means of his theory of the stationary states of central systems, in which the single condition $I = nh$ was replaced by the two conditions (16); and the theory was further developed by EPSTEIN and SCHWARZSCHILD, who on this line established the general theory, based on the conditions (22),

of the stationary states of a conditionally periodic system for which the equations of motion may be solved by means of separation of variables in the HAMILTON-JACOBI partial differential equation. If the hydrogen atom is exposed to a homogeneous electric or to a homogeneous magnetic field, the atom forms a system of this class, and, as shown by EPSTEIN and SCHWARZSCHILD as regards the STARK effect and by SOMMERFELD and DEBYE as regards the ZEEMAN effect, the theory under consideration leads to values for the total energy of the atom in the stationary states, which together with relation (1) lead again to values for the frequencies of the radiations emitted during the transitions between these states, which are in agreement with the measured frequencies of the components into which the hydrogen lines are split up in the presence of the fields. As pointed out in Part I, it is possible moreover to throw light on the question of the intensities and polarisations of these components on the basis of the necessary formal relation between the quantum theory of line spectra and the ordinary theory of radiation in the limit where the motions in successive stationary states differ very little from each other. In the following sections the mentioned problems will be discussed in detail. As regards the fixation of the stationary states we shall not, however, follow the same procedure as used by the authors just mentioned, which rests upon the immediate application of the conditions (22), but it will be shown how the conditions which fix the stationary states of the perturbed atom may be obtained by a direct examination of the small deviations of the motion

of the electron from a simple Keplerian motion. In this way it seems possible to obtain a more direct illustration of the principles discussed in Part I; and we shall see moreover that the treatment in question may be used also in cases where the method of separation of variables cannot be applied.

In Part III the problem of the series spectra of other elements will be treated from a similar point of view. As pointed out by the writer in an earlier paper, a simple explanation of the pronounced analogy between these spectra and the hydrogen spectrum is offered by the fact, that the atomic systems, involved in the emission of the spectra under consideration, in a certain sense may be regarded as a perturbed hydrogen atom. On the other hand, a clue to the interpretation of the characteristic difference between the hydrogen spectrum and the spectra of other elements was first obtained by SOMMERFELD's theory of the stationary states of central systems referred to above. As shown by SOMMERFELD, it is possible on this theory to account in general outlines for the well known laws governing the frequencies of the series spectra of the elements; and, as it will be shown in Part III, it is also possible, on the basis of the formal relation between the quantum theory and the ordinary theory of radiation, in this way to obtain a simple interpretation of the laws governing the remarkable differences in the intensities with which the various series of lines appear, which on the combination principle would constitute the complete spectra under consideration. As regards the detailed discussion of these spectra, however, it is necessary to bear in mind that

the part played by the inner electrons in the atoms of the elements in question forms a far more intricate problem than the perturbing effect of a fixed external field on the hydrogen atom. For the treatment of this problem the theory of conditionally periodic systems based on the conditions (22) does not seem to suffice, while, as it will be shown in Part III, it appears that the method of perturbations exposed in the following lends itself naturally also to this case.

§ 2. The stationary states of a perturbed periodic system.

In Part I it was shown that the problem of the fixation of the stationary states of a periodic system of several degrees of freedom, which is subject to the perturbing influence of a small external field, cannot be treated directly on the basis of the general principle of the mechanical transformability of the stationary states by considering the influence, which on ordinary mechanics a slow establishment of the external field would exert on the motion of some arbitrarily chosen stationary state of the undisturbed system (see Part I, page 41). This is an immediate consequence of the fact, mentioned in the former section, that the stationary states of the perturbed system are characterised by a greater number of extra-mechanical conditions than the stationary states of the undisturbed system. On the other hand, we were led to assume from the general formal relation between the quantum theory of line spectra and the ordinary theory of radiation,

that it is possible to obtain information about the stationary states of the perturbed system from a direct consideration of the slow variations which the periodic orbit undergoes as a consequence of the mechanical effect of the external field on the motion. Thus, if these variations are of periodic or conditionally periodic type, we may expect that, in the presence of the external field, the values for the additional energy of the system in the stationary states are related to the small frequency or frequencies of the perturbations, in a manner analogous to the relation between energy and frequency in the stationary states of an ordinary periodic or conditionally periodic system.

If the equations of motion for the perturbed system can be solved by means of separation of variables, it is easily seen that the relation in question is fulfilled if the stationary states are determined by the conditions (22). Consider thus a system for which every orbit is periodic, and let us assume that in the presence of a given small external field a separation of variables is possible in a certain set of coordinates $q_1, \ldots, q_s$. For the undisturbed system we have then, according to equation (23), that the quantity I, defined by (5), is equal to $\kappa_1 I_1 + \cdots + \kappa_s I_s$, where $I_1, \ldots, I_s$ are defined by (21) and calculated with respect to the set of coordinates just mentioned, and where the κ's are a set of entire positive numbers without a common divisor. For simplicity let us assume that at least one of the κ's, say κ_s, is equal to one, and that consequently, as mentioned on page 40, the number n in (24), which characterises the stationary states of

the undisturbed system, may take all positive values. This condition will be fulfilled in case of all the applications to spectral problems discussed below; it will be seen, however, that the extension to problems where this condition is not fulfilled will only necessitate small modifications in the following considerations. By use of (29) we get now for the difference in the total energy of two slightly different states of the perturbed system

$$\delta E = \sum_1^s \omega_k \, \delta I_k = \omega_s \sum_1^s \kappa_k \, \delta I_k + \sum_1^{s-1} (\omega_k - \kappa_k \omega_s) \, \delta I_k. \quad (42)$$

Since for the undisturbed system $\omega_k = \kappa_k \omega_s$, the differences $\omega_k - \kappa_k \omega_s$ appearing in the last term will, for the perturbed system, be small quantities which will just represent the frequencies of the slow variations which the orbit undergoes in the presence of the external field. These quantities will in the following be denoted by $\mathfrak{v}_k$. Consider now the multitude of states of the perturbed system for which $\sum_1^s \kappa_k I_k$ is equal to nh, where n is a given entire positive number. This multitude will be seen to include all possible stationary states of the perturbed system, which satisfy (22), and the motion of which differs at any moment only slightly from some stationary motion of the undisturbed system, satisfying (24) for the given value of n. Denoting the value of the energy of the undisturbed system in such a state by E_n, and the value of the energy of the perturbed system in a state belonging to the multitude under consideration by $E_n + \mathfrak{E}$,

we get from (42)

$$\delta\mathfrak{E} = \sum_{1}^{s-1} \mathfrak{v}_k \, \delta I_k \tag{43}$$

for the energy difference between two neighbouring states of this multitude. Since this relation has the same form as (29), we see consequently that by putting $I_1, \ldots, I_{s-1}$ equal to entire multipla of h, as claimed by the conditions (22), we obtain exactly the same relation between the additional energy $\mathfrak{E}$ and the small frequencies $\mathfrak{v}_k$, impressed on the system by the external field, as that which holds between the total energy and the fundamental frequencies in the stationary states of a conditionally periodic system of $s-1$ degrees of freedom.

As a simple illustration of these calculations let us consider the system consisting of a particle moving in a plane and subject to an attraction from a fixed point, which varies proportional to the distance apart. If undisturbed, the motion of this system will be periodic independent of the initial conditions, and the particle will describe an elliptical orbit with its centre at the fixed point. Moreover the equations of motion of the undisturbed system may be solved by means of separation of variables in polar coordinates, as well as in any set of rectangular coordinates. In the first case we have, taking for q_1 the length of the radius vector from the fixed point to the particle and for q_2 the angular distance of this radius vector from a fixed direction, $\kappa_1 = 2$ and $\kappa_2 = 1$, while in the second case we have $\kappa_1 = \kappa_2 = 1$. In the presence of an external field the orbit will in general not remain

periodic, but will in the course of time cover a continuous extension of the plane. If the external field is sufficiently small, however, the orbit will at any moment only differ little from a closed elliptical orbit, but in the course of time the lengths and directions of the principal axes of this ellipse will undergo slow variations. In general the perturbed system will not allow of separation of variables, but two cases obviously present themselves in which such a separation is still possible; in the first case the external field is central with the fixed point as centre, and a separation is possible in polar coordinates; in the second case the external field of force is perpendicular to a given line and varies as some function of the distance from this line, and separation is possible in a set of rectangular coordinates with the axes parallel and perpendicular to the given line. In the first case the perturbations will not affect the lengths of the principal axes of the elliptical orbit and will only produce a slow uniform rotation of the directions of these axes, while in the second case the lengths of the principal axes as well as their directions will perform slow oscillations. It will consequently be seen that, by fixing the stationary states of the perturbed system by means of the conditions (22), the cycles of shapes and positions which the orbit of the particle will pass through in the stationary states will be entirely different in the two cases. In both cases, however, it will be seen that the frequency $\mathfrak{v} = \omega_1 - \kappa_1\omega_2$ will be equal to the frequency with which the orbit at regular intervals re-assumes its shape and position. By fixing the stationary states by (22) we obtain therefore, as seen from (43), in both cases that the relation between this frequency and the additional energy of the system due to the presence of the field will be the same as the

relation between energy and frequency in the stationary states of a system of one degree of freedom; and it will be seen that the above considerations offer a dynamical interpretation of the characteristic discontinuity involved in the application of the method of separation of variables to the fixation of the stationary states of perturbed periodic systems.[1])

In general it will not be possible to solve the equations of motion of the perturbed system by means of separation of variables in a fixed set of positional coordinates, but we shall see that the problem of the fixation of the stationary states of the perturbed system may be attacked by a direct examination of the additional energy of the system and its relation

[1]) In this connection it may be of interest to note that the possibility of a rational interpretation of the discontinuity in question would seem to be essentially connected with the form of the principles of the quantum theory adopted in this paper. If for instance the quantum theory is taken in the form proposed by PLANCK in his second theory of temperature radiation, the consequent development to periodic systems of several degrees of freedom would seem to involve a serious difficulty as regards the question of the necessary stability of the temperature equilibrium among a great number of systems for small variations of the external conditions. In fact, in connection with the development of his theory of the "physical structure of the phase space", mentioned in Part I on page 31, in which conditions of the same type as (22) are established, PLANCK has deduced expressions for the total energy of a great number of systems in temperature equilibrium, which, if applied to systems of the same kind as those considered in the above example, show a dependency of this energy on the temperature which is different, according to whether polar coordinates or rectangular coordinates are used as basis for the structure of the phase space.

to the slow variations of the orbit, on the basis of the usual theory of perturbations well known from celestial mechanics. Consider a system for which every orbit, if undisturbed, is periodic independent of the initial conditions, and let us assume that the equations of motion for some set of coordinates $q_1, q_2, \ldots, q_s$ are solved by means of the HAMILTON-JACOBI partial differential equation, given by formula (17) in Part I. The motion of the system is then determined by the equations (18), and the orbit is characterised by means of the constants $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_s$. If now the system is subject to some small external field of force, the motion will no more be periodic, but, defining in the usual way the osculating orbit at a given moment as the periodic orbit which would result if the external forces vanished suddenly at this moment, we find that the constants $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_s$, characterising the osculating orbit, will vary slowly with the time. Assuming for the present that the external forces possess a constant potential Ω given as a function of the q's, we have according to the theory of perturbations that the rates of variation of the orbital constants of the osculating orbit will be given by[1])

$$\frac{d\alpha_k}{dt} = -\frac{d\Omega}{d\beta_k}, \quad \frac{d\beta_k}{dt} = \frac{d\Omega}{d\alpha_k}, \qquad (k = 1, \ldots, s) \qquad (44)$$

where Ω is considered as a function of $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_s$ and t, obtained by introducing for the q's their expressions as

[1]) See f. inst. C. V. L. CHARLIER, Die Mechanik des Himmels, Bd. I, Abt. 1, § 10.

functions of these quantities obtained by solving (18). The equations (44) allow to follow completely the perturbing effect of the external field on the motion of the system. For the problem under consideration, however, a detailed examination of the perturbations is not necessary. In fact, we shall not be concerned with the small deformation of the orbit characterised by the small oscillations of the orbital constants within a time interval of the same order of magnitude as the period of the osculating orbit, but only with the so called "secular perturbations" of the orbit, characterised by the total variation of these constants taken over a time interval long compared with the period of the osculating orbit. As we shall see below, these variations may, with an approximation sufficient for our purpose, be obtained directly by taking mean values on both sides of the equations (44). Before entering on these calculations, however, it may be observed that the part played by the constants α_1 and β_1 differs essentially from that played by the other orbital constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$. Thus from the formulæ (17) and (18) on page 32, it follows that α_1 is the total energy corresponding to the osculating orbit, while β_1 will represent the moment in which the system would pass some distinguished point in this orbit. If for instance we consider the perturbations of a Keplerian motion, we may for β_1 take the so called time of perihelium passage. When discussing the secular perturbations of the shape and position of the orbit, we see therefore in the first place that the variations of β_1 may be left out of consideration. Further, it follows from the

principle of conservation of energy, that $\alpha_1 + \Omega$ will remain constant during the motion, and that consequently during the perturbations α_1 will change only by small quantities of the same order as $\lambda\alpha_1$, where λ denotes a small constant of the same order of magnitude as the ratio between the external forces and the internal forces of the system. Moreover, since the period σ of the undisturbed motion depends on α_1 only, it follows that the period of the osculating orbit will remain constant during the perturbations, with neglect of small quantities of the same order as $\lambda\sigma$. On the other hand it follows from (44) that, in a time interval of the same order as σ/λ, the constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$ will in general undergo variations of the same order of magnitude as the values of these constants themselves.

As mentioned above, the total variations of the constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$, which characterise the *secular perturbations of the shape and position of the orbit*, may be obtained by taking mean values on both sides of the equations (44). Introducing a function ψ of the α's and β's, equal to the mean value of the potential Ω taken over a period σ of the motion of the undisturbed system and defined by the formula

$$\psi = \frac{1}{\sigma}\int_t^{t+\sigma} \Omega\, dt, \tag{45}$$

it is easily seen, since σ depends only on α_1, that the mean values of the partial differential coefficients of Ω with respect to $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$, taken over an approximate period of the perturbed motion, may, if we look apart from

small quantities proportional to λ^2, be replaced by the values of the corresponding partial differential coefficients of ψ at some moment within this period. With the approximation mentioned we get therefore

$$\frac{D\alpha_k}{Dt} = -\frac{\partial\psi}{\partial\beta_k}, \quad \frac{D\beta_k}{Dt} = \frac{\partial\psi}{\partial\alpha_k}, \qquad (k = 2, \ldots, s) \qquad (46)$$

where the differential symbols on the left sides are written to indicate mean values of the rates of variation of the orbital constants during an approximate period of the perturbed motion. From the definition of ψ it follows that this quantity in general will depend on α_1 as well as on $\alpha_2, \ldots, \alpha_s$, $\beta_2, \ldots, \beta_s$, but that it will not depend upon β_1. From the above considerations it follows further that, with the approximation in question, α_1 may be considered as a constant in the expressions on the right sides of (46), while for $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$ we may take a set of values corresponding to some moment within the period to which the mean values on the left sides refer.

It will be seen that the equations (46) allow to follow the secular perturbations during a time interval sufficiently long for the external forces to produce a considerable change in the shape and position of the original orbit, if in the total variations of the orbital constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$ we look apart from small quantities of the same order as the small oscillations of these constants within a single period. As a consequence of the secular variations, the orbit will pass through a cycle of shapes and positions, which will depend

on its original shape and position and on the character of the perturbing field, but not on the intensity of this field. In fact, as seen from (46), the variations in the shape and position of the orbit will remain the same if ψ is multiplied by a constant factor, which will only influence the rate at which these variations are performed. It will further be observed that the problem of determining the secular perturbations by means of (46) consists in solving a set of equations of the same type as the Hamiltonian equations of motion for a system of $s-1$ degrees of freedom. In these equations the quantity ψ plays formally the same part as the total energy in the usual mechanical problem, and in analogy with the principle of conservation of energy it follows directly from (46) that, with neglect of small quantities proportional to λ^2, *the value of ψ will remain constant during the perturbations*, even if the external forces act through a time interval of the same order as σ/λ. In fact, with neglect of small quantities proportional to λ^2, we have

$$\frac{D\psi}{Dt} = \sum_2^s \left(\frac{\partial\psi}{\partial\alpha_k}\frac{D\alpha_k}{Dt} + \frac{\partial\psi}{\partial\beta_k}\frac{D\beta_k}{Dt} \right)$$
$$= \sum_2^s \left(-\frac{\partial\psi}{\partial\alpha_k}\frac{\partial\psi}{\partial\beta_k} + \frac{\partial\psi}{\partial\beta_k}\frac{\partial\psi}{\partial\alpha_k} \right) = 0.$$

Since at any moment ψ will differ only by small quantities proportional to λ^2 from the mean value of the potential of the external forces taken over an approximate period of the perturbed motion, it follows from the above that, with

neglect of small quantities of this order, also the mean value of the inner energy α_1 of the perturbed system, taken over an approximate period, will remain constant during the perturbations, even if the perturbing forces act through a time interval long enough to produce a considerable change in the shape and position of the orbit. In the special case, where the perturbed system allows of separation of variables, this last result may be shown to follow directly from formula (28) in Part I. Taking for the time interval ϑ in this formula the period σ of the undisturbed motion, we get $N_k = \kappa_k$, where $\kappa_1, \ldots, \kappa_s$ are the numbers entering in formula (23). Comparing a given perturbed motion of the system with some undisturbed motion of which it may be regarded as a small variation, we get therefore from (28), with neglect of small quantities proportional to the square of the intensity of the external forces,

$$\int_0^\sigma \delta E \, dt = \sum_1^s \kappa_k \, \delta I_k, \tag{47}$$

where the I's are calculated with respect to a set of coordinates in which a separation can be obtained for the perturbed motion, and where δE is the difference between the total energy of the undisturbed motion and the energy which the system would possess in its perturbed state, if the external forces vanished suddenly at the moment under consideration, and which in the above calculations was denoted by α_1. Now the energy E of the undisturbed motion is determined

completely by the value of $I = \sum \kappa_k I_k$. If therefore the perturbed motion is all the time compared with a neighbouring undisturbed motion of given constant energy, it follows directly from (47), that, with neglect of small quantities of the same order as the square of the external forces, the integral on the left side, taken over an approximate period of the perturbed motion, will remain unaltered during the perturbations through any time interval, however long.

Before proceeding with the applications of the equations (46) which apply to the case of a constant perturbing field, it will be necessary to consider *the effect of a slow and uniform establishment of the external field.* Let us assume that, within the interval $0 < t < \vartheta$ where ϑ denotes a quantity of the same order as σ/λ, the intensity of the external field increases uniformly from zero to the value corresponding to the potential Ω. Since the variation in the perturbing field during a single period will only be a small quantity of the same order as λ^2, we see in the first place that the secular variations of the constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$, with the same approximation as for a constant field, will be given by a set of equations of the same form as (46), with the only difference that ψ is replaced by $\dfrac{t}{\vartheta}\psi$. Moreover it may be shown that in these equations the quantity α_1 may be considered as constant, just as in the equations which hold for a constant perturbing field. In fact the total variation in α_1 at any moment t will be equal to the total work performed by the external forces since the beginning of the

establishment of the perturbing field, and will therefore be given by

$$\Delta_t \alpha_1 = -\int_0^t \frac{t}{\vartheta} \sum_1^s \frac{\partial \Omega}{\partial q_k} \dot{q}_k \, dt = \frac{1}{\vartheta} \int_0^t \Omega \, dt - \frac{t}{\vartheta} \Omega_t, \qquad (48)$$

where the expression on the right side is obtained by partial integration; but, since both terms in this expression are of the same order of magnitude as $\lambda \alpha_1$, we see that the total variation in α_1 within the interval in question will, just as in case of a constant perturbing field, be only a small quantity of this order. We get therefore the result, that, for the same shape and position of the original orbit, the cycle of shapes and positions passed through by the orbit during the increase of the external field will be the same as that which would appear for a constant perturbing field, and that, with neglect of small quantities proportional to λ^2, the value of the function ψ will consequently remain constant during the establishment of the field. With this approximation we get therefore from (48), putting $t = \vartheta$,

$$\Delta_\vartheta \alpha_1 + \Omega_\vartheta = \frac{1}{\vartheta} \int_0^\vartheta \Omega \, dt = \psi,$$

which shows that the change in the total energy of the system, due to the slow and uniform establishment of the external field, is just equal to the value of the function ψ, and consequently equal to the mean value of the potential of the

external forces taken over an approximate period of the perturbed motion. This result may also be expressed by stating, that, with neglect of small quantities proportional to the square of the external forces, the mean value of the inner energy taken over an approximate period of the perturbed motion will be equal to the energy possessed by the system before the establishment of the perturbing field.

Returning now to the problem of the fixation of the stationary states of a periodic system subject to the influence of a small external field of constant potential, we shall base our considerations on the fundamental assumption that these states are distinguished between the continuous multitude of mechanically possible states by a relation between the additional energy of the system due to the presence of the external field and the frequencies of the slow variations of the orbit produced by this field, which is analogous to the relation discussed on page 80 in the special case in which the perturbed system allows of separation of variables in a fixed set of coordinates. On this assumption we shall expect in the first place that, apart from small quantities proportional to λ, the cycles of shapes and positions of the orbit belonging to the stationary states of the perturbed system will depend only on the character of the external field, but not on its intensity. Since now, as shown above, such a cycle will remain unaltered during a slow and uniform increase of the intensity of the external field if the effect of the external forces is calculated by means of ordinary mechanics, we are therefore, with reference to the principle of the mechanical transforma-

bility of the stationary states, led to the conclusion that it is possible by direct application of ordinary mechanics, not only to follow the secular perturbations of the orbit in the stationary states corresponding to a constant external field, but also to calculate the variation in the energy of the system in the stationary states which results from a slow and uniform change in the intensity of this field. If we denote the energy in the stationary states of the perturbed system by $E_n + \mathfrak{E}$, where E_n is the value of the energy in the stationary state of the undisturbed system characterised by a given entire value of n in the condition $I = nh$, we may therefore conclude from the above that *the additional energy* $\mathfrak{E}$ *in the stationary states of the perturbed system will be equal to the value in these states of the function* ψ *defined by* (45)*, if we look apart from small quantities proportional to the square of the intensity of the external forces.* It will be seen that this result is equivalent to the statement, that the mean value of the inner energy taken over an approximate period of the perturbed motion will be equal to the value E_n of the energy in the corresponding stationary state of the undisturbed system. In case of the perturbed system allowing of separation of variables in a fixed set of coordinates, this result may be simply shown to be a direct consequence of the fixation of the stationary states by means of the conditions (22). In fact, if we assume that the undisturbed motion, considered in (47), corresponds to some stationary state, satisfying (24) for a given value of n, and that the perturbed motion is also stationary and satisfies (22), we see that the right side of (47)

will be zero, and we get the result that the mean value of the inner energy in the stationary states of the system, with the approximation mentioned, will not be altered in the presence of the external field.

Due to the above result that the additional energy $\mathfrak{E}$ in the stationary states of the perturbed system, with neglect of small quantities proportional to λ^2, may be taken equal to the value in these states of the function ψ entering in the equations (46) which determine the secular perturbations of the orbits, we are now able to draw further conclusions from the fact, mentioned above, that these equations are of the same type as the Hamiltonian equations of motion for a mechanical system of $s-1$ degrees of freedom. In fact, we see that *the fixation of the stationary states of the perturbed system is reduced to a problem which is formally analogous to the fixation of these states for a mechanical system of less degrees of freedom.* As it will appear from the following applications this problem may, quite independent of the possibility of separation of variables for the perturbed system, be treated directly on the basis of the fundamental relation between energy and frequency in the stationary states of periodic or conditionally periodic systems, discussed in Part I, if only the solution of the equations (46) is of a periodic or conditionally periodic character. In this connection it may once more be emphasised that these equations, according to the manner in which they were deduced, allow to follow the secular perturbations only through a time interval of the same order of magnitude as that sufficient for the external

forces to produce a finite alteration in the shape and position of the orbit. With reference to the necessary stability of the stationary states of an atomic system, it seems justified, however, to conclude that any possible small discrepancy between the motion to be expected from a rigorous application of ordinary mechanics and that determined by a calculation of the secular perturbations, based on the equations (46), cannot cause a material change in the character of the stationary states as fixed by a consideration of the periodicity properties of these perturbations. On the other hand, from the point of view of the general formal relation between the quantum theory and the ordinary theory of radiation, we must be prepared to find that the motion and the energy in the stationary states of a perturbed periodic system, for which we only know that the secular perturbations as determined by (46) are of conditionally periodic type, will not be as sharply defined as the motion and the energy in the stationary states of a conditionally periodic system for which the equations of motion allow of a rigorous solution by means of the method of separation of variables. Thus, if we consider a large number of similar atomic systems of the type in question, we may be prepared to find that the values of the additional energy in a given stationary state will for the different systems deviate from each other by small quantities; but it must be expected that the values of the additional energy for the large majority of systems will differ from the value of ψ, as determined by the method indicated above, only by small quantities proportional to λ^2, and that only

for a small fraction (at most of the same order as λ^2) of the systems the values of the additional energy will show deviations from this value of ψ, which are of the same order as λ.

As to the application of the preceding considerations to special problems, it will be seen in the first place that in case of a *perturbed periodic system possessing two degrees of freedom*, as for instance that considered in the example on page 82, the problem of the fixation of the stationary states of the perturbed system in the presence of a small external field allows of a general solution on the basis of the method developed above, because in this case the secular perturbations will in general be *simply periodic*. In fact, in this case the shape and position of the orbit are characterised by two constants α_2 and β_2, and from the equations (46), which will be analogous to the equations of motion of a system of one degree of freedom, it follows directly that during the perturbations α_2 will be a function of β_2 and that in general these quantities will be periodic functions of the time with a period $\mathfrak{s}$ which, besides on α_1, will depend on the value of ψ only. Considering two slightly different states of the perturbed system for which the corresponding states of the undisturbed system (i. e. the states which would appear if the external forces vanished at a slow and uniform rate) possess the same energy and consequently the same value for the quantity I defined by (5), we get therefore by a calculation completely analogous to that leading to relation (8)

in Part I, which was deduced directly from the Hamiltonian equations, for the difference in the values of the function ψ for these two states

$$\delta\psi = \mathfrak{v}\,\delta\mathfrak{I}, \tag{49}$$

where $\mathfrak{v} = \dfrac{1}{\mathfrak{s}}$ is the frequency of the secular perturbations, and where the quantity $\mathfrak{I}$ is defined by

$$\mathfrak{I} = \int_0^{\mathfrak{s}} \alpha_2 \frac{D\beta_2}{Dt}\, dt = \int \alpha_2\, D\beta_2, \tag{50}$$

where the latter integral is taken over a complete oscillation of β_2. In order to fix the stationary states, it will now be seen in the first place that, among the multitude of states of the perturbed system for which the value of I in the corresponding states of the undisturbed system is equal to nh where n is a given positive integer, the state for which $\mathfrak{I} = 0$ must beforehand be expected to be a stationary state. In fact, for this value of $\mathfrak{I}$, the shape and position of the orbit will not undergo secular perturbations but will remain unaltered for a constant external field as well as during a slow and uniform establishment of this field. In contrast to what in general will take place during a slow establishment of the external field, we may therefore expect that, for this special shape and position of the orbit, a direct application of ordinary mechanics will be legitimate in calculating the effect of the establishment of the field, since there will in this case obviously be nothing to cause the coming into play of some non-mechanical process, connected with the mechanism of

a transition between two stationary states accompanied by the emission or absorption of a radiation of small frequency. With reference to relation (49) we see therefore that, by fixing the stationary states of the perturbed system by means of the condition

$$\mathfrak{J} = \mathfrak{n}h, \tag{51}$$

where $\mathfrak{n}$ is an entire number, we obtain a relation between the additional energy $\mathfrak{E} = \psi$ of the system in the presence of the field and the frequency $\mathfrak{v}$ of the secular perturbations, which is exactly of the same type as that which holds between the energy and frequency in the stationary states of a system of one degree of freedom, and which is expressed by (8) and (10). By means of (51) it is possible, with neglect of small quantities proportional to the square of the perturbing forces, directly to determine the value of the additional energy in the stationary states of a periodic system of two degrees of freedom subject to an arbitrarily given small external field of force, and consequently with this approximation, by use of the fundamental relation (1), to determine the effect of this field on the frequencies of the spectrum of the undisturbed periodic system. In general this effect will consist in a splitting up of each of the spectral lines into a number of components which are displaced from the original position of the line by small quantities proportional to the intensity of the external forces.

When we pass to *perturbed periodic systems of more than two degrees of freedom*, the general problem is more com-

plex. For a given external field, however, it may be possible to choose a set of orbital constants $\alpha_2, \ldots, \alpha_s, \beta_2, \ldots, \beta_s$ in such a way, that during the motion every of the β's will depend on the corresponding β only, while every of the β's will oscillate between two fixed limits. From analogy with the theory of ordinary conditionally periodic systems which allow of separation of variables, the perturbations may in such a case be said to be *conditionally periodic*, and, from a calculation quite analogous to that leading to equation (29) in Part I which is based entirely on the use of the Hamiltonian equations, we get for the difference in ψ for two slightly different states of the perturbed system, for which the value of I in the corresponding states of the undisturbed system is the same,

$$\delta\psi = \sum_{1}^{s-1} \mathfrak{v}_k \, \delta\mathfrak{I}_k, \tag{52}$$

where $\mathfrak{v}_k$ is the mean frequency of oscillation of β_{k+1} between its limits, and where the quantities $\mathfrak{I}_k$ are defined by

$$\mathfrak{I}_k = \int \alpha_{k+1} \, D\beta_{k+1}, \qquad (k = 1, \ldots, \; s-1) \tag{53}$$

where the integral is taken over a complete oscillation of β_{k+1}. In analogy with the expression (31) for the displacements of the particles of an ordinary conditionally periodic system which allows of separation of variables, we get further in the present case that every of the α's and β's may be expressed as a function of the time by a sum of harmonic vibrations of

small frequencies

$$\left.\begin{matrix}\alpha\\ \beta\end{matrix}\right\} = \sum \mathfrak{C}_{\mathfrak{t}_1,\ldots,\ \mathfrak{t}_{s-1}} \cos 2\pi\big\{(\mathfrak{t}_1\mathfrak{v}_1 + \ldots + \mathfrak{t}_{s-1}\mathfrak{v}_{s-1})t + \mathfrak{c}_{\mathfrak{t}_1,\ldots,\ \mathfrak{t}_{s-1}}\big\}, \tag{54}$$

where the $\mathfrak{C}$'s and $\mathfrak{c}$'s are constants, the former of which, besides on I, depend on the $\mathfrak{I}$'s only, and where the summation is to be extended over all positive and negative entire values of the $\mathfrak{t}$'s. If therefore the secular perturbations are conditionally periodic, we may conclude that the stationary states of the perturbed system, corresponding to a given stationary state of the undisturbed system, will be characterised by the $s-1$ conditions

$$\mathfrak{I}_k = \mathfrak{n}_k h, \qquad (k = 1, \ldots,\ s-l) \tag{55}$$

where $\mathfrak{n}_1,\ \ldots,\ \mathfrak{n}_{s-1}$ form a set of entire numbers. In fact, as seen from (52), we obtain in this way a relation between the additional energy and the frequencies of the secular perturbations of exactly the same type as that holding for the energy and frequencies of ordinary conditionally periodic systems and expressed by (22) and (29); moreover we may conclude beforehand that the state in which every of the quantities $\mathfrak{I}_k$, defined by (53), is equal to zero must belong to the stationary states of the perturbed system, because in this case the orbit will not undergo secular perturbations for a constant external field, nor during a slow and uniform establishment of this field. Since the conditions (55), with neglect

of small quantities proportional to the square of the intensities of the external forces, allow to determine the additional energy of the system due to the presence of the external field, we see therefore that the effect of this field on the spectrum of the undisturbed system, if the secular perturbations are conditionally periodic, will consist in a splitting up of each spectral line in a number of components, in analogy with the effect of a perturbing field on the spectrum of a periodic system of two degrees of freedom. In general, however, the perturbations, which a periodic system of more than two degrees of freedom undergoes in the presence of a given external field, cannot be expected to be conditionally periodic and to exhibit periodicity properties of the type expressed by formula (54). In such cases it seems impossible to define stationary states in a way which leads to a complete fixation of the total energy in these states, and we are therefore led to the conclusion, that the effect of the external field on the spectrum will not consist in the splitting up of the spectral lines of the original system into a number of sharp components, but in a diffusion of these lines over spectral intervals of a width proportional to the intensity of the external forces.

In special cases in which the secular perturbations of a perturbed periodic system of more than two degrees of freedom are of conditionally periodic type, it may occur that these perturbations are characterised by a number of fundamental frequencies, which is less than $s-1$. In such cases, in which the perturbed periodic system from analogy with the terminology used in Part I may be said to be *degener-*

ate, the necessary relation between the additional energy and the frequencies of the secular perturbations is secured by a number of conditions less than that given by (55), and the stationary states are consequently characterised by a number of conditions less than s. With a typical example of such systems we meet if, for a perturbed periodic system of more than two degrees of freedom, the secular perturbations are *simply periodic* independent of the initial shape and position of the orbit. In direct analogy to what holds for perturbed periodic systems of two degrees of freedom, the difference between the values of ψ in two slightly different states of the perturbed system, corresponding to the same value of I, will in the present case be given by

$$\delta\psi = \mathfrak{v}\,\delta\mathfrak{J}, \tag{56}$$

where $\mathfrak{v}$ is the frequency of the secular perturbations, and where $\mathfrak{J}$ is defined by

$$\mathfrak{J} = \int_0^{\mathfrak{v}} \sum_2^s \alpha_k \frac{D\beta_k}{Dt}\, dt, \tag{57}$$

where $\mathfrak{s} = 1/\mathfrak{v}$ is the period of the perturbations. We may therefore conclude that the stationary states of the perturbed system, corresponding to a given stationary state of the undisturbed system, will be characterised by the single condition

$$\mathfrak{J} = \mathfrak{n}h, \tag{58}$$

in which $\mathfrak{n}$ is an entire number, and which will be seen to be completely analogous to the condition which fixes the stationary states of ordinary periodic systems of several degrees of freedom.

In the following sections we shall apply the preceding considerations to the problem of the fixation of *the stationary states of the hydrogen atom*, when the relativity modifications are taken into account, and when the atom is exposed to small external fields. In this discussion we shall for the sake of simplicity consider the mass of the nucleus as infinite in the calculations of the perturbations of the orbit of the electron. This involves, in the expression for the additional energy of the system, the neglect of small terms of the same order as the product of the intensity of the external forces with the ratio between the mass of the electron and the mass of the nucleus, but due to the smallness of the latter ratio the error introduced by this simplification will be of no importance in the comparison of the results with the measurements. Since in the case under consideration the system possesses three degrees of freedom, the equations which determine the secular perturbations of the orbit of the electron will correspond to the equations of motion of a system of two degrees of freedom, and it will therefore not be possible to give a general treatment of the problem of the stationary states. Thus, for any given external field, we meet with the question whether the perturbations are conditionally periodic and, if so, in what set of orbital constants this

periodicity may be conveniently expressed. Now, in many spectral problems, the external field possesses *axial symmetry round an axis through the nucleus*, and in this case it is easily shown that the problem of the fixation of the stationary states allows of a general solution. A choice of orbital constants which is suitable for the discussion of this problem, and which is well known from the astronomical theory of planetary perturbations, is obtained by choosing for α_2 the total angular momentum of the electron round the nucleus and for α_3 the component of this angular momentum round the axis of the field. For the set of β's, corresponding to this set of α's, we may take β_2 equal to the angle, which the major axis makes with the line in which the plane of the orbit cuts the plane through the nucleus perpendicular to the axis of the field, and β_3 equal to the angle between this line and a fixed direction in the latter plane. For the problem under consideration it will be seen that, with this choice of constants, the mean value ψ of the potential of the perturbing field will, besides on α_1, generally depend on α_2 and β_2 as well as on α_3, but due to the symmetry round the axis it will obviously not depend on β_3. In consequence of this, the equations (46), which determine the secular perturbations, will possess the same form as the Hamiltonian equations of motion for a particle moving in a plane and subject to a central field of force. Thus corresponding to the conservation of angular momentum for central systems, we get in the first place from (46) that α_3 will remain unaltered during the perturbations. Next corresponding to the simple

periodicity of the radial motion in central systems, we see from (46), if α_3 as well as α_1 is considered as a constant, that during the perturbations α_2 will be a function of β_2 and vary in a simple periodic way with the time. The perturbations of the orbit of the electron produced by an external field which possesses axial symmetry will therefore always be of *conditionally periodic* type, quite independent of the possibility of separation of variables for the perturbed system. As regards the form of the conditions which fix the stationary states, it may be noted, however, that with the choice of orbital constants under consideration the β's will not, as it was assumed for the sake of simplicity in the general discussion on page 100, oscillate between fixed limits, but it will be seen that β_2 during the perturbations may either oscillate between two such limits or increase (or decrease) continuously, while β_3 will always vary in the latter manner. This constitutes, however, only a formal difficulty of the same kind as that mentioned in Part I in connection with the discussion of the conditions (16), which fix the stationary states of a system consisting of a particle moving in a central field of force. Thus from a simple consideration it will be seen that, in complete analogy to the relations (52) and (53), we get in the present case for the difference between the energy of two slightly different states of the perturbed system, which correspond to the same value of I,

$$\delta\psi = \mathfrak{v}_1\,\delta\mathfrak{I}_1 + \mathfrak{v}_2\,\delta I_2, \tag{59}$$

where $\mathfrak{v}_1$ is the frequency with which the shape of the or-

bit and its position relative to the axis of the field repeats itself at regular intervals and which is characterised by the variation of α_2 and β_2, while $\mathfrak{v}_2$ is the mean frequency of rotation of the plane of the orbit round this axis characterised by the variation of β_3, and where $\mathfrak{J}_1$ and $\mathfrak{J}_2$ are defined by the equations

$$\mathfrak{J}_1 = \int \alpha_2 \, D\beta_2, \quad \mathfrak{J}_2 = \int_0^{2\pi} \alpha_3 \, D\beta_3 = 2\pi\alpha_3. \tag{60}$$

In case β_2 varies in an oscillating manner with the time, the first integral must be taken over a complete oscillation of this orbital constant, while, if β_2 during the perturbations increases or decreases continuously, the integral in the expression for $\mathfrak{J}_1$ must be taken over an interval of 2π, just as the integral in the expression for $\mathfrak{J}_2$. By fixing the stationary states of the perturbed system by means of the two conditions[1])

$$\mathfrak{J}_1 = \mathfrak{n}_1 h, \quad \mathfrak{J}_2 = \mathfrak{n}_2 h, \tag{61}$$

[1]) Quite apart from the problem of perturbed periodic systems, the second of these conditions would also follow directly from certain interesting considerations of EPSTEIN (Ber. d. D. Phys. Ges. XIX, p. 116 (1917)) about the stationary states of systems which allow of what may be called "partial separation of variables". In this case it is possible to choose a set of positional coordinates $q_1, \ldots, q_s$ in such a way that, for some of the coordinates, the conjugated momenta may be considered as functions of the corresponding q's only, so that, for these coordinates, quantities I may be defined by (21) in the same way as for systems for which a complete separation of variables can be obtained. From analogy with the theory of the stationary states of the latter systems,

where $\mathfrak{n}_1$ and $\mathfrak{n}_2$ are entire numbers, it will therefore be seen that we obtain the right relation between the additional energy $\mathfrak{E} = \psi$ of the perturbed atom and the frequencies of the secular perturbations of the orbit of the electron. It will moreover be seen that a state in which the electron moves in a circular orbit perpendicular to the axis of the field, and which beforehand must be expected to belong to the stationary states of the perturbed atom since this orbit will not undergo secular perturbations during a uniform establishment of the external field, will be included among the states determined by (61). In fact, if n is the number which characterises the corresponding stationary state of the undisturbed system, this state of the perturbed system will correspond to $\mathfrak{n}_1 = 0$, $\mathfrak{n}_2 = n$ or to $\mathfrak{n}_1 = n$, $\mathfrak{n}_2 = n$, according to whether β_2 during the perturbations oscillates between fixed limits, or increases (or decreases) continuously. As regards the application of the conditions (61) it is of importance to point out that, from considerations of the invariance of the a-priori probability of the stationary states of an atomic sys-

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EPSTEIN proposes therefore the assumption, that some of the conditions to be fulfilled in the stationary states of the systems in question may be obtained by putting the I's thus defined equal to entire multipla of h. It will be seen that, in case of systems possessing an axis of symmetry, this leads to the second of the conditions (61), which expresses the condition that in the stationary states the total angular momentum round the axis must be equal to an entire multiple of $h/2\pi$. As pointed out in Part I on page 64, this condition would also seem to obtain an independent support from considerations of conservation of angular momentum during a transition between two stationary states.

tem during continuous transformations of the external conditions (see Part I, page 14 and page 49), it seems necessary to conclude that no stationary state exists corresponding to $\mathfrak{n}_2 = 0$. For this value of $\mathfrak{n}_2$ the motion of the electron would take place in a plane through the axis, but for certain external fields such motions cannot be regarded as physically realisable stationary states of the atom, since in the course of the perturbations the electron would collide with the nucleus (compare page 134).

A special case of an external field possessing axial symmetry, in which the secular perturbations are very simple, presents itself if the external forces form *a central field with the nucleus at the centre*. In this case the solution of the problem of the fixation of the stationary states is given by SOMMERFELD's general theory of central systems, discussed in Part I, which rests upon the fact that these systems allow of separation of variables in polar coordinates. In connection with the above considerations it may be of interest, however, to consider the problem in question directly from the point of view of perturbed periodic systems, because it presents a characteristic example of a *degenerate* perturbed system. In the present case ψ will, besides on α_1, depend on α_2 only, and from the equations (46) we get therefore the well known result, that the angular momentum of the electron and the plane of its orbit will not vary during the perturbations, and that the only secular effect of the perturbing field will consist in a slow uniform rotation of the direction of the major axis.

For the frequency of this rotation we get from (46)

$$\mathfrak{v} = \frac{1}{2\pi}\frac{D\beta_2}{Dt} = \frac{1}{2\pi}\frac{\delta\psi}{\delta\alpha_2}, \tag{62}$$

from which we get directly for the difference between the values of ψ for two neighbouring states of the perturbed system, for which the corresponding value of I is the same,

$$\delta\psi = 2\pi\mathfrak{v}\,\delta\alpha_2. \tag{63}$$

This relation, which corresponds to (56), is seen to coincide with (59), since in the present case $\mathfrak{v}_2 = 0$ and $\mathfrak{J}_1 = 2\pi\alpha_2$. From (63) it follows that the necessary relation between the additional energy of the atom and the frequency of the perturbations is secured if the stationary states in the presence of a small external central field are characterised by the condition

$$\mathfrak{J} = 2\pi\alpha_2 = \mathfrak{n}h, \tag{64}$$

where $\mathfrak{n}$ is an entire number. This condition, which is equivalent with the second of SOMMERFELD's conditions (16), corresponds to (58) and is seen to coincide with the first of the conditions (61), while the second of the latter conditions in the special case under consideration loses its validity corresponding to the fact that the orientation of the plane of the orbit in space is obviously arbitrary. Since, for a Keplerian motion, the major axis of the orbit depends on the total energy only while the minor axis is proportional to the angular momentum, it will be seen from (64) that the presence of

a small external field imposes the restriction on the motion of the atom in the stationary states, that the minor axis of the orbit of the electron must be equal to an entire multiple of the nth part of the major axis, which was given by $2\alpha_n$ in (41). This result has been pointed out by SOMMERFELD as a consequence of the application of the conditions (16).

In the preceding it has been shown how it is possible to attack the problem of the stationary states of a perturbed periodic system by an examination of the secular perturbations of the shape and position of the orbit, and to fix these states if the perturbations are of periodic or conditionally periodic type. While these considerations allow to determine the possible values for the total energy of the perturbed system and thereby the *frequencies* of the components into which the lines of the spectrum of the undisturbed system are split up in the presence of the external field, it is necessary, however, for the discussion of the *intensities* and *polarisations* of these components to consider more closely the motion of the particles in the perturbed system and the relation of the total energy of this system to the fundamental frequencies which characterise the motion. In the first place it will be seen that, if the secular perturbations as determined by the equations (46) are of conditionally periodic type, the displacements of the particles of the system in any given direction may, with neglect of small quantities proportional to the intensity of the external forces, be represented, within a time interval sufficiently large for these forces to produce a

considerable change in the shape and position of the orbit, as a sum of harmonic vibrations by expressions of the type:

$$\xi = \sum C_{\tau,\mathfrak{t}_1,\ldots,\,\mathfrak{t}_{s-1}} \cos 2\pi\big\{(\tau\omega_P + \mathfrak{t}_1\mathfrak{v}_1 + \cdots + \mathfrak{t}_{s-1}\mathfrak{v}_{s-1})t + c_{\tau,\mathfrak{t}_1,\ldots,\,\mathfrak{t}_{s-1}}\big\}, \tag{65}$$

where the summation is to be extended over all positive and negative entire values of τ, $\mathfrak{t}_1$, …, $\mathfrak{t}_{s-1}$, and where the C's and c's are two sets of constants, the former of which depend only on the values of the quantities $\mathfrak{I}_1$, …, $\mathfrak{I}_{s-1}$ defined by (53) and on the value of the quantity I, which characterises the corresponding state of the undisturbed system which would appear if the external field vanished at a slow and uniform rate. While the quantities $\mathfrak{v}_1$, …, $\mathfrak{v}_{s-1}$ are the same as those which appear in the formula (54), and represent the small frequencies of the secular perturbations of the shape and position of the orbit, the quantity ω_P may be considered as representing the mean frequency of revolution of the particles in their approximately periodic orbit. As regards the total energy of the perturbed system, it may next be proved that, looking apart from small quantities proportional to the square of the intensity of the external forces, the difference in the total energy in two slightly different states of the perturbed system, for which the values of I, $\mathfrak{I}_1$, …, I_{s-1} differ by δI, $\delta\mathfrak{I}_1$, …, $\delta\mathfrak{I}_{s-1}$ respectively, is given by the relation[1])

$$\delta E = \omega_P\,\delta I + \sum_1^{s-1} \mathfrak{v}_k\,\delta\mathfrak{I}_k, \tag{66}$$

which coincides with (52) if $\delta\mathfrak{I} = 0$, and which will be seen to

[1]) From a comparison with formula (8), holding for the energy difference between two neighbouring states of the undisturbed system, and with formula (52), it will be seen that (66) implies the condition $\omega_P = \omega + \partial\psi/\partial I$, where ω is the frequency of revolution in the corresponding state of the undisturbed system characterised by the given value of I, and where, in the partial differential coefficient, ψ is considered as a function of I and $\mathfrak{I}_1, \ldots, \mathfrak{I}_{s-1}$. This relation can be verified by means of a consideration based on the perturbation equations (44), which takes into account the simple relation between α_1 and I for the undisturbed system, as well as the relation between the mean rate of variation of β_1 with the time and the difference between ω_P and ω. We shall not enter, however, on the details of the rather intricate calculations involved in such a consideration, since the problems in question allow of a more elegant treatment by means of another analytical method. Thus it will be shown by Mr. H. A. KRAMERS, in the paper mentioned in the end of § 4, that, quite independent of the possibility of separation of variables for the perturbed system in a fixed set of positional coordinates, the theory of secular perturbations exposed in this section offers—if these perturbations as determined by (46) are of conditionally periodic type—a means of disclosing a set of *angle variables*, which may be used to describe the motion of the perturbed system with the same degree of approximation as that involved in the preceding calculations. According to the definition of angle variables, mentioned in the Note on page 53 in Part I, this means that it is possible, in stead of the positional coordinates $q_1, \ldots, q_s$ of the perturbed system and their conjugated momenta $p_1, \ldots, p_s$ to introduce a new set of s variables in such a way, that the q's and p's are periodic in every of the new variables with period 1, when they are considered as functions of these variables and of their canonically conjugated momenta. These momenta will just coincide with the quantities denoted above by $I, \mathfrak{I}_1, \ldots, \mathfrak{I}_{s-1}$, and the corresponding angle variables may

be completely analogous with formula (29) in Part I, holding for an ordinary conditionally periodic system which allows of separation of variables in a fixed set of positional coordinates; just as (65) is analogous to formula (31) representing the displacements of the particles for such a system. Since moreover, in complete analogy to the conditions (22), the stationary states of the perturbed system are characterised by

$$I = nh, \quad \mathfrak{J}_k = \mathfrak{n}_k h, \qquad (k = 1, \ldots, \ s-1) \qquad (67)$$

we see consequently that, for sufficiently small intensity of the external forces, we obtain in the region of large values of n and of the $\mathfrak{n}$'s a connection between the frequencies of the components of the spectral lines, determined on the quantum

conveniently be denoted by w, $\mathfrak{w}_1$, ..., $\mathfrak{w}_{s-1}$ respectively. Introducing the new variables, the total energy of the perturbed system will be a function of I, $\mathfrak{J}_1$, ..., $\mathfrak{J}_{s-1}$ only, if we look apart from small quantities proportional to λ^2. With this approximation we get consequently by a calculation, analogous to that given in the Note referred to, that the angle variables w, $\mathfrak{w}_1$, ..., $\mathfrak{w}_{s-1}$ may be represented as linear functions of the time within an interval of the same order as σ/λ. Denoting the rates of variation of w, $\mathfrak{w}_1$, ..., $\mathfrak{w}_{s-1}$ by ω, $\mathfrak{v}_1$, ..., $\mathfrak{v}_{s-1}$ respectively, the formulæ (65) and (66) are therefore directly obtained, just as the corresponding formulæ (31) and (29) in Part I. In this connection it will be observed that, due to the possibility of introduction of angle variables, the conditions (67) appear in the same form as that in which the conditions, which fix the stationary states of ordinary conditionally periodic systems which allow of separation of variables, have been formulated by SCHWARZSCHILD, and which, as mentioned in the Note in Part I, has already been applied by BURGERS to certain systems for which such a separation cannot be obtained.

theory by means of relation (1), and those to be expected on ordinary electrodynamics, which is of exactly the same type as the analogous connection, discussed in Part I, in case of ordinary conditionally periodic systems which allow of separation of variables. In perfect analogy with the general considerations in Part I, we are therefore led directly to certain simple conclusions as regards the intensities and polarisations of the components into which the lines of the spectrum of the undisturbed periodic system are split up in the presence of the external field. Thus we shall expect that there will exist an intimate connection between the probability of spontaneous transition between two stationary states of the perturbed system, for which $n = n'$, $\mathfrak{n}_k = \mathfrak{n}'_k$ and $n = n''$, $\mathfrak{n}_k = \mathfrak{n}''_k$ respectively, and the values in these states of the coefficient $C_{\tau,\mathfrak{t}_1,\ldots,\ \mathfrak{t}_{s-1}}$ in the expressions for the displacements of the particles, for which $\tau = n' - n''$ and $\mathfrak{t}_k = \mathfrak{n}'_k - \mathfrak{n}''_k$. If for instance, for a certain set of values of τ and $\mathfrak{t}_1, \ldots, \mathfrak{t}_{s-1}$, the coefficient $C_{\tau,\mathfrak{t}_1,\ldots,\ \mathfrak{t}_{s-1}}$ in the expressions for the displacements in every direction will be equal to zero for all motions of the perturbed system, we shall expect that the corresponding transitions between two stationary states will be impossible in the presence of the given external field; and if this coefficient is zero for the displacements of the particles in a certain direction only, we shall expect that the corresponding transitions will give rise to the emission of a radiation which is polarised in a plane perpendicular to this direction.

With a characteristic example of these considerations we

meet in the case of the spectrum of a hydrogen atom exposed to an external field of force which possesses axial symmetry round an axis through the nucleus. In analogy with the resolution of the motion of an ordinary conditionally periodic system which possesses an axis of symmetry in its constituent harmonic vibrations, discussed in Part I on page 61, it follows from the discussion of the general character of the secular perturbations on page 104 that the motion of the electron in the perturbed atom in this case can be resolved in a number of linear harmonic vibrations parallel to the axis with frequencies $\tau\omega_P + \mathfrak{t}_1\mathfrak{v}_1$ and in a number of circular harmonic rotations perpendicular to the axis with frequencies $\tau\omega_P + \mathfrak{t}_1\mathfrak{v}_1 + \mathfrak{v}_2$. In complete analogy with the considerations in Part I, we are therefore led to conclude that in the present case only two types of transitions between the stationary states of the perturbed atom are possible. In the transitions of the first type $\mathfrak{n}_2$ will remain unaltered and the emitted radiation will give rise to components of the hydrogen lines which will show linear polarisation parallel to the axis. In the transitions of the second type $\mathfrak{n}_2$ will change by one unit and the emitted radiation will show circular polarisation when viewed in the direction of the axis. Remembering that, according to the conditions (61), the angular momentum of the system round the axis in the stationary states is equal to $\mathfrak{n}_2 \frac{h}{2\pi}$, it will be seen moreover that, also in the present case, these conclusions obtain an independent support from a consideration of conservation of angular momentum during

the transitions (Compare Part I page 64).[1]) In the following we will meet with applications of these considerations when discussing the effect of electric and magnetic fields on the hydrogen lines. In the latter case, however, the preceding considerations need some modifications due to the fact, that the external forces acting on the electron cannot be derived from a potential expressed as a function of its positional coordinates; to this point we shall come back in § 5.

Before leaving the general theory of perturbed periodic systems we shall still consider the problem of the effect on the spectrum of a periodic system, undergoing secular perturbations of conditionally periodic type under the influence of a given small external field, if this system is further subject to *the influence of a second external field which is small compared with the first field*, but the perturbing effect of

[1]) *Note added during the proof.* In an interesting paper by A. RUBINOWICZ (Phys. Zeitschr. XIX, p. 441 and p. 465 (1918)) which has just been published, a similar consideration of conservation of angular momentum has been used to draw conclusions, as regards the possibility of transitions between the stationary states of a conditionally periodic system possessing an axis of symmetry, and as regards the character of the polarisation of the radiation accompanying these transitions. In this way RUBINOWICZ has arrived at several of the results discussed in the present paper; in this connection, however, it may be remarked that, from a consideration of conservation of angular momentum, it is not possible, even for systems possessing axial symmetry, to obtain as complete information, as regards the number and polarisation of the possible components, as from a consideration based on the resolution of the motion of the electron in harmonic vibrations.

which is yet large compared with the small effects on the motion, proportional to the square of the intensity of the first perturbing field, which were neglected in the preceding calculations. This problem is closely analogous to the problem, briefly discussed in Part I, of the effect of a small perturbing field on the spectrum of an ordinary conditionally periodic system which allows of separation of variables. As mentioned on page 64, we have in this case, quite independent of the possibility of separation of variables for the perturbed system, that in general the motion under the influence of the external field may still be represented as a sum of harmonic vibrations by a formula of the type (31), if we look apart from small terms proportional to the square of the perturbing forces. Corresponding to this we have in the case under consideration that, independent of the nature of the second external field, the resultant secular perturbations may in general be expressed as a sum of harmonic vibrations of small frequencies of the type (54), if we look apart from small terms of the same order as the product of the secular perturbations produced by the first external field with the square of the ratio between the intensities of the forces due to the first and those due to the second external field. Let us denote this ratio by μ and let, as above, λ represent a small constant of the same order as the ratio between the external forces due to the first field and the internal forces of the system. On the basis of the general relation between energy and frequency in the stationary states, we may then expect that it is possible to fix the motion in these states

for the perturbed periodic system in the presence of both external fields with neglect of small terms of the same order as the largest of the quantities μ^2 and λ, and to fix the corresponding values for the energy with neglect of small terms of the same order as the largest of the quantities $\lambda\mu^2$ and λ^2.[1]) In general, however, the effect on the spectrum of the perturbed system, produced by the second external field, may be calculated without considering the perturbing effect of this field in detail. In fact, it is in general possible, by means of the principle of the mechanical transformability of the stationary states, with the approximation mentioned to determine the alteration of the energy of the system, due to the presence of the second external field, directly from the character of the secular perturbations produced by the first external field only. Thus let us assume that the second field is slowly established at a uniform rate within a time interval of the same order of magnitude as that in which the system will pass approximately through any state belonging to the cycle of shapes and positions, which the orbit passes through in the stationary states in the presence of the first external field only. Denoting a time interval of this order by ϑ and

[1]) In analogy with the considerations on page 97 it may be expected, however, that these limits for the definition of the energy in the stationary states will hold only for the great majority among a large number of atomic systems. Thus in the present case we must be prepared to find that for a small fraction of the systems of the same order as μ^2 (if $\mu^2 > \lambda$) the energy will differ from that fixed by the method under consideration by small quantities of the same order as $\mu\lambda$.

the potential of the first perturbing field by Ω and that of the second by $\Delta\Omega$, we get then, by a calculation quite analogous to that given in Part I on page 17 for the alteration in the mean value of the energy of a periodic system during a slow establishment of a small external field, that the alteration in the mean value of $\alpha_1 + \Omega$ taken over a time interval of the same order as ϑ, due to the establishment of the second external field, will be a small quantity of the same order of magnitude as $\vartheta(\Delta\Omega)^2$; but with the notation used above this means, in general, a small quantity of the same order as $\lambda\mu^2$. It follows consequently that, with this approximation, the alteration in the energy in a given stationary state, due to the presence of the second perturbing field, is equal to the mean value of the potential of this field taken over the cycle of shapes and positions, which the orbit would pass through in the corresponding stationary state of the perturbed system under the influence of the first external field only. In general, the effect on the spectrum will therefore consist in a small displacement of the original components proportional to the intensity of the forces due to the second perturbing field; and as regards the degree of approximation with which these displacements are defined, it will be seen from the above that, if μ is smaller than $\sqrt{\lambda}$, the fixation of the energy in the stationary states in the presence of the second external field, and therefore also the determination of the frequencies of the spectral lines by means of (1), allow of the same degree of approximation as the fixation of the energy in the stationary states of the original perturbed periodic system. If μ is larger

than $\sqrt{\lambda}$, however, the stationary states will in general not be as well defined as for the original system, and from relation (1) we may therefore expect that the components will be diffuse, although, as long as μ remains small compared with unity, the width of the components will remain small compared with the displacements from their positions in the presence of the first external field alone. Only when μ becomes of the same order as unity, the simultaneous effect of both perturbing fields may be expected to consist in a diffusion of the lines of the undisturbed periodic system; unless of course the secular perturbations due to the simultaneous presence of both fields are still of conditionally periodic type, as it may happen in special problems. In certain cases the second external field will not only give rise to small displacements of the original components but also to the appearance of new components of small intensities proportional to μ^2. This occurs if for the original perturbed periodic system, due to some peculiarity of the motion, some of the coefficients $C_{\tau,\mathfrak{t}_1,\ldots,\,\mathfrak{t}_{s-1}}$ in the expressions (65) for the displacements of the particles as a sum of harmonic vibrations, corresponding to certain combinations of the numbers τ, $\mathfrak{t}_1$, $\ldots$, $\mathfrak{t}_{s-1}$, are equal to zero, while in the presence of the second external field these coefficients are small quantities proportional to μ (compare Part I, page 64).[1]) In the preceding con-

[1]) As regards the degree of definition with which the positions of the new components will be determined, we must be prepared to find that the frequencies of these components are only defined with neglect of small quantities proportional to $\lambda\mu$. Compare the detailed discussion

siderations it has been assumed that the perturbed system in the presence of the first external field is non-degenerate. In case, however, this system is *degenerate*, it is obviously impossible, by a direct application of the principle of the mechanical transformability of the stationary states, to determine the alteration in the energy in the stationary states of the system, which will be due to the presence of a second external field small compared with the first field; because, as mentioned, the stationary states of the system, in the presence of this field only, will be determined by a number of conditions which is less than the number s of degrees of freedom, and that consequently the cycles of shapes and positions, which the orbit will pass through in these states, will not be completely determined. For the calculation of the energy in the stationary states it will therefore be necessary to consider the secular perturbing effect of the second external field on these cycles. In the special case where the secular perturbations due to the first field are simply periodic, it wall in this way be seen that the problem of the fixation of the stationary states in the presence of the second external field, by means of the method exposed in this section, may be reduced to the problem of the fixation of the stationary states of a system of $s - 2$ degrees of freedom. If, as in the applications considered below, s is equal to 3, this problem allows of a general solution, and we must therefore expect that in this case the effect on the spectrum of the perturbed

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of the example in § 5 on page 192.

system produced by an arbitrary second external field, which is small compared with the first, will consist in the splitting up of every component into a number of separate components, just as the effect of an arbitrary small external field on the lines of the spectrum of a simple periodic system of two degrees of freedom. We will meet with applications of the above considerations when considering the effect on the hydrogen spectrum of the combined action of different external fields and when considering the effect of an external field on the spectra of other elements, which latter problem will be discussed in Part III.

§ 3. The fine structure of the hydrogen lines.

An instructive application of the calculations in the last section may be made in connection with the fine structure of the hydrogen lines, which, according to SOMMERFELD's theory mentioned in Part I on page 31, may be explained by taking into account the small variation of the mass of the electron with its velocity, claimed by the theory of relativity. In this connection it must first of all be remarked that all the general considerations in the preceding sections, as regards relations between energy and frequency and as regards the mechanical transformability of the stationary states, hold unaltered if the relativity modifications are taken into account. This follows from the fact that the Hamiltonian equations (4), which are taken as a basis for all the previous calculations, may be used to describe the motion also in this case.

If, when the relativity modifications are taken into account, the motion of the system is simply periodic independent of the initial conditions, we shall consequently expect that the stationary states are characterised by the condition $I = nh$ only, and that the energy and frequency are the same for all states corresponding to a given value of n in this equation. Further the stationary states will also in the relativity case be fixed by (22), if the system is conditionally periodic and allows of separation of variables; while the stationary states of a perturbed periodic system, also in the relativity case, will be characterised by the conditions (67), if the secular perturbations are of conditionally periodic type.

Now, when the relativity modifications are taken into account, the motion of the particles in the hydrogen atom will not, as assumed in § 1, be exactly periodic, but the orbit of the electron will be of the same type as that, which would appear on ordinary Newtonian mechanics, if the law of attraction between the particles differed slightly from that of the inverse square. If, for the moment, we consider the mass of the nucleus as infinite, the system will allow of a separation of variables in polar coordinates, and the stationary states may consequently be fixed by the conditions (16). In this way SOMMERFELD obtained an expression for the total energy in the stationary states, which, with neglect of small quantities of higher order than the square of the ratio of the velocity of the electron and the velocity of light c, is given

by[1])

$$E = -\frac{2\pi^2 N^2 e^4 m}{h^2(n_1+n_2)^2}\left[1+\frac{\pi^2 N^2 e^4}{c^2 h^2(n_1+n_2)^2}\left(1+4\frac{n_1}{n_2}\right)\right], \quad (68)$$

where, as in the calculations in § 1, the charge and the mass of the electron are denoted by $-e$ and m, and for sake of generality the charge of the nucleus by Ne. Further n_1 and n_2 are the integers appearing on the right side of the conditions (16) as factors to PLANCK's constant. While n_1 may take the values 0, 1, 2, ..., it will be seen that n_2 can only take the values 1, 2, ..., because in the present case there will obviously not correspond any stationary state to $n_2 = 0$, since in such a state the electron would collide with the nucleus. Introducing the experimental values for e, h and c, it is found that e^2/hc is a small quantity of the same order as 10^{-3}; and, unless N is large number, the second term within the bracket on the right side of (68) will consequently be very small compared with unity. Putting $n_1 + n_2 = n$, it will further be seen that the factor outside the bracket will coincide with the expression for W_n given by (41) in § 1, if we look apart from the small correction due to the finite mass of the nucleus. Due to the presence of the second term within the bracket, we thus see that, for any value of n, formula (68)

[1]) A. SOMMERFELD, Ann. d. Phys. LI, p. 53 (1916). Compare also P. DEBYE, Phys. Zeitschr., XVII, p. 512 (1916). In the special case of circular orbits ($n_1 = 0$), this expression coincides with an expression previously deduced by the writer (Phil. Mag. XXIX p. 332 (1915)), by a direct application of the condition $I = nh$ to these periodic motions.

gives a set of values for E which differ slightly from each other and from $-W_n$. SOMMERFELD's theory leads therefore to a direct explanation of the fact, that the hydrogen lines, when observed by instruments of high dispersive power, are split up in a number of components situated closely to each other; and, by means of formula (68) in connection with relation (1), it was actually found possible, within the limits of experimental errors, to account for the frequencies of the components of this so called *fine structure* of the hydrogen lines. Moreover the theory was supported in the most striking way by PASCHEN's[1]) recent investigation of the fine structure of the lines of the analogous helium spectrum, the frequencies of which are represented approximately by formula (35), if in the expression for K, given by (40), we put $N = 2$. As it should be expected from (68), the components of these lines were found to show frequency differences several times larger than those of the hydrogen lines, and from his measurements PASCHEN concluded, that it was possible on SOMMERFELD's theory to account completely for the frequencies of all the components observed.

We shall not enter here on the details of the calculation leading to (68), but shall only show how this formula may be simply interpreted from the point of view of perturbed periodic systems. Thus, by a simple application of relativistic mechanics, it is found that, if the equation of a Keplerian el-

[1]) F. PASCHEN, Ann. d. Phys. L, p. 901 (1916). See also E. J. EVANS and C. CROXSON, Nature, XCVII, p. 56 (1916).

lipse in polar coordinates is given by $r = f(\vartheta)$, the equation of the orbit of the electron in the case under consideration will be given by $r = f(\gamma\vartheta)$ where γ is a constant given by $\gamma^2 = 1 - \left(\frac{Ne^2}{pc}\right)^2$, in which expression p denotes the angular momentum of the electron round the nucleus.[1]) Now in the stationary states the quantity in the bracket, which is of the same order of magnitude as the ratio between the velocity of the electron and the velocity of light, will be very small, unless N is a large number, and it will therefore be seen that the orbit of the electron can be described as a periodic orbit on which a slow uniform rotation is superposed. Denoting the frequency of revolution in the periodic orbit by ω and the frequency of the superposed rotation by $\mathfrak{v}_R$, we have, with neglect of small quantities of higher order than the square of the ratio between the velocity of the electron and the velocity of light,

$$\mathfrak{v}_R = \omega(1 - \gamma) = \frac{1}{2}\omega\left(\frac{Ne^2}{pc}\right)^2. \tag{69}$$

Comparing this formula with equation (62) and remembering that, with the approximation in question, p may be replaced by the quantity denoted in § 2 by α_2, we see that the frequency of the secular rotation of the orbit will be the same as that which would appear, if the variation of the mass of the electron was neglected, but if the atom was subject to a small external central force the mean value of the potential

[1]) See f. inst. A. SOMMERFELD, loc. cit. p. 47.

of which, taken over a revolution of the electron, was equal to

$$\psi = -\omega \frac{\pi N^2 e^4}{c^2 \alpha_2}. \tag{70}$$

This is simply shown, however, to be equal to the expression for ψ corresponding to a small attractive force varying as the inverse cube of the distance. In fact, let the potential of such a force be given by $\Omega = C/r^2$, where C is a constant and r the length of the radius vector from the nucleus to the electron. By means of the relation $\alpha_2 = mr^2\dot{\vartheta}$, where ϑ is the angular distance of the radius vector from a fixed line in the plane of the orbit, we get then

$$\psi = \frac{1}{\sigma}\int_0^\sigma \frac{C}{r^2}\,dt = \frac{\omega m C}{\alpha_2}\int_0^{2\pi} d\vartheta = \frac{2\pi\omega m C}{\alpha_2},$$

which expression is seen to coincide with (70), if $C = -\dfrac{N^2 e^4}{2c^2 m}$.

If the relativity modifications are taken into account, and if for a moment we would imagine that the nucleus, in addition to its usual attraction, exerted a small repulsion on the electron, proportional to the inverse cube of the distance and equal and opposite to the attraction just mentioned, we would therefore obtain a system for which, with neglect of small quantities of higher order than the square of the ratio between the velocity of the electron and the velocity of light, every orbit would be periodic independent of the initial conditions, and for which consequently the stationary states would be fixed by the single condition $I = nh$. Now

the actual hydrogen atom may obviously be considered as a *perturbed system*, formed by this periodic system, when it is exposed to a small central field for which the value of ψ is given by (70). With the approximation mentioned, we get therefore for the total energy in the stationary states of the atom

$$E = E'_n - \frac{8\pi^4 N^4 e^8 m}{h^4 c^2} \frac{1}{n^3 \mathfrak{n}}, \tag{71}$$

where E'_n is the energy in the stationary states of the periodic system just mentioned, and where the last term is obtained by introducing in (70) the value of α_2 given by (64) and the value of ω_n given by (41), neglecting the small correction due to the finite mass of the nucleus. Remembering that in our notation $n_1 + n_2 = n$ and $n_2 = \mathfrak{n}$, it will be seen that, as regards the small differences in the energy of the different stationary states corresponding to the same value of n, formula (71) gives the same result as SOMMERFELD's formula (68). In fact, comparing (68) and (71), we get

$$E'_n = -\frac{2\pi^2 N^2 e^4 m}{h^2 n^2} \left(1 - \frac{3\pi^2 N^2 e^4}{c^2 h^2 n^2}\right), \tag{72}$$

which is seen to be a function of n only. This expression might also have been deduced directly from the condition $I = nh$ by considering, for instance, a circular orbit, in which case the calculation can be very simply performed.

In connection with the above calculations, it may be remembered that the fixation of the stationary states, leading to the formulæ (68) or (71), is based on the assumption,

that the motion of the electron can be determined as that of a mass point which moves in a conservative field of force, according to the laws of ordinary relativistic mechanics, and that we have looked apart from all such forces which, according to the ordinary theory of electrodynamics, would act on an accellerated charged particle, and which constitute the reaction from the radiation which on this theory would accompany the motion of the electron. Some procedure of this kind, which means a radical departure from the ordinary theory of electrodynamics, is obviously necessary in the quantum theory in order to avoid dissipation of energy in the stationary states. Since we are entirely ignorant as regards the mechanism of radiation, we must be prepared, however, to find that the above treatment will allow to determine the motion in the stationary states, only with an approximation which looks apart from small quantities of the same order as the ratio between the radiation forces in ordinary electrodynamics and the main forces on the electron due to the attraction from the nucleus.[1]) Now it is

[1]) Compare Part I, page 6. It may in this connection be noted that the degree of approximation, involved in the determination of the frequencies of an atomic system by means of relation (1) if in the fixation of the stationary states we look apart from small forces of the same order of magnitude as the radiation forces in ordinary electrodynamics, would appear to be intimately connected with *the limit of sharpness of the spectral lines*, which depends on the total number of waves contained in the radiation emitted during the transition between two stationary states. In fact, from a consideration based on the general connection between the quantum theory and the ordinary theory of radiation, it

easily shown that this ratio will be a small quantity of the same order of magnitude as $N^2 \left(\frac{e^2}{pc}\right)^3$, and it would therefore beforehand seem justified in the expression for the total energy in the stationary states to retain small terms of the same order as the second term in (71), while at the same time it might appear highly questionable, whether, in the complete expression for the total energy in the stationary states deduced by SOMMERFELD and DEBYE on the basis of the conditions (16), it has a physical meaning to retain terms of higher order than those retained in formula (68); unless N is a large number, as in the theory of the Röntgenspectra to be discussed in Part III.

While the preceding considerations, which deal with the determination of the energy in the stationary states of the hydrogen atom, allow to determine *the frequency of the radiation* which would be emitted during a transition between two such states, they leave quite untouched the problem of the actual occurrence of these transitions in the luminous gas, and therefore give no direct information about the *num-*

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seems natural to assume that the rate, at which radiation is emitted during a transition between two stationary states, is of the same order of magnitude as the rate, at which radiation would be emitted from the system in these states according to ordinary electrodynamics. But this will be seen to imply that the total number of waves in question will just be of the same order as the ratio between the main forces acting on the particles of the system and the reaction from the radiation in ordinary electrodynamics.

ber and relative intensities of the components into which the hydrogen lines may be expected to split up as a consequence of the relativity modifications. This problem has recently been discussed by SOMMERFELD,[1]) who in this connection emphasises the importance of the different a-priori probabilities of the stationary states, characterised by different sets of values of the n's in the conditions (16). Thus SOMMERFELD attempts to obtain a measure for the relative intensities of the components of the fine structure of a given line, by comparing the intensities observed with the products of the values of the a-priori probabilities of the two states, involved in the emission of the components under consideration; and he tries in this connection to test different expressions for these a-priori probabilities (See Part I, page 47). In this way, however, it was not found possible to account in a satisfactory manner for the observations; and the difficulty in obtaining an explanation of the intensities on this basis was also strikingly brought out by the fact, that the number and relative intensities of the components observed varied in a remarkable way with the experimental conditions under which the lines were excited. Thus PASCHEN found a greater number of components in the fine structure of the helium lines, mentioned above, when the gas was subject to a condensed interrupted discharge, than when a continuous voltage was applied. It would seem, however, that all the facts observed obtain a simple interpretation on the basis of the general con-

[1]) A. SOMMERFELD, Ber. Akad. München, 1917, p. 83.

siderations about the relation between the quantum theory of line spectra and the ordinary theory of radiation discussed in Part I. According to this relation, we shall assume that the probability, for a transition between two given stationary states to take place, will depend not only on the a-priori probability of these states, which is determining for their occurrence in a distribution of statistical equilibrium, but will also depend essentially on the motion of the particles in these states, characterised by the harmonic vibrations in which this motion can be resolved. Now, in the absence of external forces, the motion of the electron in the hydrogen atom forms a special simple case of the motion of a conditionally periodic system possessing an axis of symmetry, and may therefore be represented by trigonometric series of the type deduced for such motions in Part I. Taking a line through the nucleus perpendicular to the plane of the orbit as z-axis, we get from the calculations on page 59

$$z = \text{const.}$$

and

$$\begin{aligned} x &= \sum C_\tau \cos 2\pi\big\{(\tau\omega_1 + \omega_2)t + c_\tau\big\}, \\ \pm y &= \sum C_\tau \sin 2\pi\big\{(\tau\omega_1 + \omega_2)t + c_\tau\big\}, \end{aligned} \tag{73}$$

where ω_1 is the frequency of the radial motion and ω_2 is the mean frequency of revolution, and where the summation is to be extended over all positive and negative entire values of τ. It will thus be seen that the motion may be considered as a

superposition of a number of circular harmonic vibrations, for which the direction of rotation is the same as, or the opposite of, that of the revolution of the electron round the nucleus, according as the expression $\tau\omega_1 + \omega_2$ is positive or negative respectively. From the relation just mentioned between the quantum theory of line spectra and the ordinary theory of radiation, we shall therefore in the present case expect that, if the atom is not disturbed by external forces, only such transitions between stationary states will be possible, in which the plane of the orbit remains unaltered, and in which the number n_2 in the conditions (16) decreases or increases by one unit; i. e. where the angular momentum of the electron round the nucleus decreases or increases by $h/2\pi$. From the relation under consideration, we shall further expect that there will be an intimate connection between the probability of a spontaneous transition of this type between two stationary states, for which n_1 is equal to n_1' and n_1'' respectively, and the intensity of the radiation of frequency $(n_1' - n_1'')\omega_1 \pm \omega_2$, which on ordinary electrodynamics would be emitted by the atom in these states, and which would depend on the value C_τ of the amplitude of the harmonic rotation, corresponding to $\tau = \pm(n_1' - n_1'')$, which appears in the motion of the electron. Without entering upon a closer examination of the numerical values of these amplitudes, it will directly be seen that the amplitudes of the harmonic rotations, which have the same direction as the revolution of the electron, in general, are considerably larger than the amplitudes of the rotations in the opposite direction, and

we shall accordingly expect that the probability of spontaneous transition will in general be much larger for transitions, in which the angular momentum decreases, than for transitions in which it increases. This expectation is verified by PASCHEN's observations of the fine structure of the helium lines, which show that, for a given line, the components corresponding to the transitions of the former kind are by far the strongest. On PASCHEN's photographs, however, especially in the case of the application of a condensed discharge to the vacuum tube containing the gas, there appear, in addition to the main components corresponding to transitions for which the angular momentum changes by $h/2\pi$, a number of weaker components, corresponding to transitions for which the angular momentum remains unchanged or changes by higher multipla of $h/2\pi$. This fact obtains a simple interpretation on the considerations in Part I on page 64 about the influence of small external forces on the spectrum of a conditionally periodic system. Thus, in the presence of small perturbing forces, the motion will generally not remain in a plane, and in the trigonometric series representing the displacement of the electron in space, there will occur small terms corresponding to frequencies $(\tau_1\omega_1+\tau_2\omega_2)$, where τ_2 may be different from one. In the presence of such forces, we shall therefore expect that, in addition to the regular probabilities of the above mentioned main transitions, there will appear small probabilities for other transitions.[1])

[1]) *Note added during the proof.* As remarked in Part I, this con-

A detailed discussion of these problems will be given in a later paper by Mr. H. A. KRAMERS, who on my proposal has kindly undertaken to examine the resolution of the motion of the electron in its constituent harmonic vibrations more closely, and who has deduced explicit expressions for the amplitudes of these vibrations, not only for the motion of the electron in the undisturbed atom, but also for the perturbed motion in the presence of a small external homogeneous electric field. As it will be shown by KRAMERS, these calculations allow to account in particulars for the observations of the relative intensities of the components of the fine structure of the hydrogen lines and the analogous helium lines, as well as for the characteristic way in which this phenomenon is influenced by the variation of the experimental conditions.

§ 4. The effect of an external electric field on the hydrogen lines.

As mentioned in the introduction, a detailed theory of the characteristic effect of an external homogeneous electric field

sideration obtains a striking confirmation by the observation of the appearance of new series of lines in the ordinary series spectra of helium and other elements, when the atoms are exposed to an intense external electric field. As it will be discussed more closely in Part III, it is possible in this way to account in detail for the manifold results, regarding the appearance of such series in the helium spectrum, which have been published quite recently by J. STARK (Ann. d. Phys. LVI, p. 577 (1918)) and by G. LIEBERT (ibid. LVI, p. 589 and p. 610 (1918)).

on the hydrogen spectrum, discovered by STARK, has been given by EPSTEIN and SCHWARZSCHILD on the basis of the general theory of conditionally periodic systems which allow of separation of variables. Before we enter on the discussion of the results of the calculations of these authors, we shall first, however, show how the problem may be treated in a simple way by means of the considerations about perturbed periodic systems, developed in § 2.

Consider an electron of mass m and charge $-e$, rotating round a positive nucleus of infinite mass and of charge Ne, and subject to a homogeneous electric field of intensity F, and let us for the present neglect the small effect of the relativity modifications. Using rectangular coordinates, and taking the nucleus as origin and the z-axis parallel to the external field, we get for the potential of the system relative to the external field, omitting an arbitrary constant,

$$\Omega = eFz.$$

Calculating now the mean value of Ω over a period σ of the undisturbed motion, we see at once, from considerations of symmetry, that this mean value ψ will depend only on the component of the external electric force in the direction of the major axis of the orbit. We have therefore

$$\psi = eF\cos\varphi\,\frac{1}{\sigma}\int_0^\sigma r\cos\vartheta\,dt,$$

where φ is the angle between the z-axis and the major axis, taken in the direction from the nucleus to the aphelium, and

where r is the length of the radius-vector from the nucleus to the electron, and ϑ the angle between this radius-vector and the major axis. By means of the well known equations for a Keplerian motion

$$r\cos\vartheta = \alpha(\cos u + \varepsilon), \quad \frac{dt}{\sigma} = (1+\varepsilon\cos u)\frac{du}{2\pi},$$

where 2α is the major axis, ε the eccentricity and u the so called eccentric anomaly, this gives

$$\begin{aligned}\psi &= eF\cos\varphi\,\frac{1}{2\pi}\int_0^{2\pi}\alpha(\cos u+\varepsilon)(1+\varepsilon\cos u)\,du \\ &= \frac{3}{2}\varepsilon\alpha eF\cos\varphi.\end{aligned} \tag{74}$$

We see thus that ψ is equal to the potential energy relative to the external field, which the system would possess, if the electron was placed at a point, situated on the major axis of the ellipse and dividing the distance $2\varepsilon a$ between the foci in the ratio 3 : 1. This point may be denoted as the "electrical centre" of the orbit. From the approximate constancy of ψ during the motion, proved in § 2, it follows therefore in the first place that, with neglect of small quantities of the same order of magnitude as the ratio between the external force and the attraction from the nucleus, *the electrical centre will during the perturbations of the orbit remain in a fixed plane perpendicular to the direction of the external force.* From the considerations in § 2 it follows further, that the total energy in the stationary states of the system

in the presence of the field, with neglect of small quantities proportional to F^2, will be equal to $E_n + \psi$, where E_n is the energy of the hydrogen atom in its undisturbed stationary state. Since both ε and $\cos\varphi$ are numerically smaller than one, we obtain therefore at once from (74) a lower and an upper limit for the possible variations of the energy in the stationary states, due to the field. Introducing from (41) the values of E_n and α_n, and neglecting, here as well as in the following calculations in this section, the small correction due to the finite mass of the nucleus—not only in the expression for the additional energy but, for the sake of brevity, also in the main term—we get for these limits

$$E = -\frac{2\pi^2 N^2 e^4 m}{h^2 n^2} \pm \frac{3h^2 n^2}{8\pi^2 N e m} F, \tag{75}$$

which formula coincides with the expression previously deduced by the writer by applying the condition $I = nh$ to the two (physically not realisable) limiting cases, corresponding to $z = 1$ and $\cos\varphi = \pm 1$, in which the orbit remains periodic in the presence of the field.[1])

In order to obtain further information as to the values of the energy in the stationary states in the presence of the

[1]) See N. Bohr, Phil. Mag. XXVII, p. 506 (1914) and XXX, p. 394 (1915). Compare also E. Warburg, Verh. d. D. Phys. Ges. XV, p. 1259 (1913), where it was pointed out, for the first time, that the effect of an electric field on the hydrogen lines to be expected on the quantum theory was of the same order of magnitude as the effect observed by Stark.

field, it is necessary to consider more closely the variation of the orbit during the perturbations. Since the external forces possess axial symmetry, the problem of the stationary states might be treated by means of the procedure indicated in § 2 on page 107. In the present special case, however, the stationary states of the atom may be very simply determined, due to the fact that the secular perturbations are simply periodic independent of the initial shape and position of the orbit, so that we are concerned with a *degenerate* case of a perturbed periodic system. This property of the perturbations follows already from some calculations given by SCHWARZSCHILD[1]) in a previous attempt to explain the STARK effect of the hydrogen lines, without the help of the quantum theory, by means of a direct consideration of the harmonic vibrations into which the motion may be resolved, according to the analytical theory of conditionally periodic systems. Starting from the above result, that the electrical centre moves in a plane perpendicular to the direction of the external field, the periodicity of the perturbations may also be proved in the following way, by means of a simple consideration of the variation of the angular momentum of the electron round the nucleus, due to the effect of the external electric force.

Using again rectangular coordinates with the nucleus at the origin and the z-axis parallel to the direction of the electric force, and calling the coordinates of the electrical centre ξ, η, ζ, we have

[1]) K. SCHWARZSCHILD, Verh. d. D. Phys. Ges. XVI, p. 20 (1914).

according to formula (74)

$$\xi^2 + \eta^2 + \zeta^2 = \left(\frac{3}{2}\varepsilon\alpha\right)^2, \qquad \zeta = \text{const.} \tag{1*}$$

Denoting the components parallel to the x y and z-axis of the angular momentum of the electron round the nucleus, considered as a vector, by P_x, P_y and P_z, we have next

$$P_x^2 + P_y^2 + P_z^2 = (1-\varepsilon^2)(2\pi m\alpha^2\omega)^2, \qquad P_z = \text{const.} \tag{2*}$$

Since the angular momentum is perpendicular to the plane of the orbit, we have further

$$\xi P_x + \eta P_y + \zeta P_z = 0. \tag{3*}$$

Now we have for the mean values of the rates of variation of P_x and P_y with the time

$$\frac{DP_x}{Dt} = eF\eta, \quad \frac{DP_y}{Dt} = -eF\xi. \tag{4*}$$

From this we get, differentiating (1*) and (2*) with respect to the time, and remembering that α and ω remain constant during the perturbations,

$$\begin{aligned} \xi\frac{D\xi}{Dt} + \eta\frac{D\eta}{Dt} &= -K^2\left(P_x\frac{DP_x}{Dt} + P_y\frac{DP_y}{Dt}\right) \\ &= -eFK^2(\eta P_x - \xi P_y), \end{aligned} \tag{5*}$$

where

$$K = \frac{3}{4\pi m\alpha\omega}. \tag{6*}$$

On the other hand we have, differentiating (3*) and introducing (4*),

$$P_x \frac{DP_x}{Dt} + P_y \frac{DP_y}{Dt} = 0,$$

which together with (5*) gives

$$\frac{D\xi}{Dt} = eFK^2 P_y, \quad \frac{D\eta}{Dt} = -eFK^2 P_x,$$

from which we get, by means of (4*),

$$\frac{D^2\xi}{Dt^2} = -e^2F^2K^2\xi, \quad \frac{D^2\eta}{Dt^2} = -e^2F^2K^2\eta,$$

the solution of which is

$$\xi = \mathfrak{A}\cos 2\pi(\mathfrak{v}t + \mathfrak{a}), \quad \eta = \mathfrak{B}\cos 2\pi(\mathfrak{v}t + \mathfrak{b}), \tag{7*}$$

where $\mathfrak{A}$, $\mathfrak{a}$, $\mathfrak{B}$ and $\mathfrak{b}$ are constants, and where, introducing (6*), we have

$$\mathfrak{v} = \frac{eFK}{2\pi} = \frac{3eF}{8\pi^2 m\alpha\omega}. \tag{8*}$$

During the perturbations the electrical centre will thus perform slow harmonic vibrations perpendicular to the direction of the electric force, with a frequency which is proportional to the intensity of the electric field, but, for a given value of F, quite independent of the initial shape of the orbit and its position relative to the direction of the field. For the value of this frequency in the multitude of states of the perturbed system, for which the mean value of the inner energy is equal to the energy E_n in a stationary state of the

undisturbed system corresponding to a given value of n, we get from the above calculation, introducing for α and ω the values of α_n and ω_n given by (41),

$$\mathfrak{v}_F = \frac{3hn}{8\pi^2 Nem} F. \tag{76}$$

Now from the periodic motion of the electrical centre we may conclude that, in the presence of the field, the system will be able to emit or absorb a radiation of frequency $\mathfrak{v}_F$, and that accordingly the possible values of the additional energy of the system in the presence of the field will be given directly by PLANCK's fundamental formula (9), holding for the possible values of the total energy of a linear harmonic vibrator, if in this formula ω is replaced by the above frequency $\mathfrak{v}_F$. Since further a circular orbit, perpendicular to the direction of the electric force, will not undergo secular perturbations during a slow establishment of the field, and therefore must be included among the stationary states of the perturbed system, we get for the total energy of the atom in the presence of the field

$$E = E_n + n\mathfrak{v}_F h = -\frac{2\pi^2 N^2 e^4 m}{n^2 h^2} + \frac{3h^2 n\mathfrak{n}}{8\pi^2 Nem} F, \tag{77}$$

where $\mathfrak{n}$ is an entire number which in the present case may be taken positive as well as negative. From a comparison between (75) and (77), we see that the presence of the external field imposes the restriction on the motion of the atom in the stationary states, that the plane in which the electrical centre of the orbit moves must have a distance from the nucleus

equal to an entire multiple of the nth part of its maximum distance $\frac{3}{2}\alpha_n$.

The result, contained in formula (77), is in agreement with the expression for the total energy in the stationary states, deduced by EPSTEIN and SCHWARZSCHILD by means of the general theory of conditionally periodic systems based on the conditions (22). The treatment of these authors rests upon the fact, that, as mentioned in Part I, the equations of motion for the electron in the present problem may be solved by means of separation of variables in parabolic coordinates (compare page 36). Taking for q_1 and q_2 the parameters of the two paraboloids of revolution, which pass through the instantaneous position of the electron and which have their foci at the nucleus and their axes parallel to the direction of the field, and for q_3 the angular distance between the plane through the electron and the axis of the system and a fixed plane through this axis, the momenta p_1, p_2, p_3 will during the motion depend on the corresponding q's only, and the stationary states will be fixed by three conditions of the type (22). With neglect of small quantities proportional to higher powers of F, the final formula for the total energy, obtained by EPSTEIN in this way, is given by

$$E = -\frac{2\pi^2 N^2 e^4 m}{h^2(n_1+n_2+n_3)^2} - \frac{3h^2(n_1+n_2+n_3)(n_1-n_2)}{8\pi^2 Nem}F, \quad ^1) \tag{78}$$

where n_1, n_2, n_3 are the positive entire numbers which occur as factors to PLANCK's constant on the right sides of the mentioned three conditions.

As regards the possible values of the total energy of the hydrogen atom in the presence of the electric field, it will be seen that (78) coincides with (77) if we put $n_1 + n_2 + n_3 = n$ and $n_1 - n_2 = \mathfrak{n}$. At the same time it will be observed, however, that the motion in the stationary states, as fixed by the procedure followed by EPSTEIN, is more restricted than was necessary in order to secure the right relation between the additional energy and the frequency of the secular perturbations. Thus, in addition to the condition which fixes the plane in which the electrical centre moves, EPSTEIN's theory involves the further condition, that the angular momentum of the electron round the axis of the perturbed system is equal to an entire multiple of $h/2\pi$; which multiple is seen to be even or uneven, according as $n + \mathfrak{n}$ is an even or an uneven number respectively. This circumstance is intimately connected with the fact that, although the perturbed system under consideration is degenerate if we look apart from small quantities proportional to the square of the intensity of the external force, the degenerate character of the system does not reveal itself from the point of view of the theory of stationary states based on the conditions (22), because the system under consideration allows of separation of variables only in one set of positional coordinates. On the other

[1]) P. EPSTEIN. Ann. d. Phys. L, p. 508 (1916).

hand, this degenerate character of the system has been emphasised by SCHWARZSCHILD[1]) on the basis of the theory of stationary states based on the introduction of angle variables, in which the periodicity properties of the motion play an essential part. In a later discussion of this point EPSTEIN[2]) calls attention to the fact that, if small quantities proportional to the square of the electric force are taken into account, the system appears no more as degenerate; and he finds therein a justification of the fixation of the stationary states by means of (22). From the point of view of perturbed systems, this would mean that the motion in the stationary states of the system in question, as fixed by (22), would certainly be stable for infinitely small disturbances, but that we should expect finite deviations from the motion in these states, already if the system was exposed to a second perturbing field, the intensity of which was only of the same order as the product of the external electric force with the ratio between this force and the attraction from the nucleus. A closer consideration, however, in which regard is taken to the influence of the relativity modifications, learns that the degree of stability of the motion in the stationary states, as determined by (22), actually is often much higher, the order of magnitude of the external force, necessary to cause finite deviations from this motion, being of the same order as the product of the attraction from the nucleus with the square

[1]) K. SCHWARZSCHILD, Ber. Akad. Berlin, 1916, p. 548.
[2]) P. EPSTEIN, Ann. d. Phys. LI, p. 168 (1916).

of the ratio of the velocity of the electron and the velocity of light. To this point we shall come back at the end of this section, when considering the simultaneous perturbing influence on the motion of the election in the hydrogen atom, due to the relativity modifications and an external electric field.

In the deduction of formula (78) there is looked apart, not only from the effect on the motion of the electron due to the small modifications in the laws of mechanics claimed by the theory of relativity, but also from the effect of possible forces which might act on the electron, corresponding to the reaction from the radiation in ordinary electrodynamics. If, however, for the moment we exclude all stationary states for which the angular momentum round the axis of the system would be equal to zero ($n_3 = 0$), the total angular momentum of the electron round the nucleus will during the perturbations always remain larger than or equal to $h/2\pi$, just as in the stationary states considered in the theory of the fine structure; and, according to the considerations on page 128, we shall therefore expect that the effect of the neglect of possible "radiation" forces will be small compared with the effect of the relativity modifications. On the other hand, if the intensity of the electric field is of the same order of magnitude as that applied in STARK's experiments, the effect of these modifications must again be expected to be very small compared with the total effect of the electric force on the hydrogen lines, since the perturbing effect of this force on the Keplerian motion of the electron will be very large compared with the corresponding effects of the relativity modifications.

If, on the contrary, we would consider a state of the atom for which n_2 was equal to zero, the orbit would be plane and would during the perturbations assume shapes, for which the total angular momentum round the nucleus was very small, and in which the electron during the revolution would pass within a very short distance from the nucleus. In such a state the effect of the relativity modifications on the motion of the electron would be considerable, but quite apart from this a rough calculation shows that the amount of energy, which, on ordinary electrodynamics, would be emitted during the intervals in which the angular momentum during the perturbations of the orbit remains small, is so large that it would hardly seem justifiable to calculate the motion and the energy in these states by neglecting all forces corresponding to the radiation forces in ordinary electrodynamics. We need not, however, enter more closely on these difficulties, because, on the general considerations in Part I about the a-priori probability of the different stationary states, we are forced to conclude that, for any value of the external electric field, *no state which would correspond to* $n_3 = 0$ *will be physically possible*; since any such state might be transformed continuously, and without passing through a degenerate system, into a state which obviously cannot represent a physically realisable stationary state (compare page 49). In fact, if we imagine that an external central field of force, varying as the inverse cube of the distance from the nucleus, is slowly established, it would be possible to compensate the secular effect of the relativity modifications and to obtain

orbits in which the electron would pass within any given, however small, distance from the nucleus. As regards the other stationary states fixed by (22), which correspond to $n_3 > 1$, we shall according to the considerations in Part I expect that their a-priori probabilities are all equal.[1])

[1]) By a simple enumeration it follows from this result, that the total number of different stationary states of the hydrogen atom, subject to a small homogeneous electric field, which corresponds to a stationary state of the undisturbed atom, characterised by a given value of n in the condition $I = nh$, is equal to $n(n+1)$. This expression is directly obtained, if we remember that $n = n_1 + n_2 + n_3$ and if we count each state, characterised by a given combination of the positive integers n_1, n_2, n_3, as double, corresponding to the two possible opposite directions of rotation of the electron round the axis of the field. With reference to the necessary stability for a small variation of the external conditions of the statistical distribution of the values of the energy among a large number of atoms in temperature equilibrium (see Note on page 82), it will be seen that the expression $n(n+1)$ may be taken as a measure for *the relative value of the a-priori probability of the different stationary states of the undisturbed hydrogen atom*, corresponding to different values of n. The problem of the determination of this a-priori probability has been discussed by K. Herzfeld (Ann. d. Phys. LI, p. 261 (1916)) who, by an examination of the volumes of the different extensions in the phase space which might be considered as belonging to the different stationary states of the hydrogen atom, has arrived at an expression for the a-priori probability of these states which differs from the above. From the point of view, as regards the principles of the quantum theory, taken in the present paper, a consideration of this kind, however, does not, as explained in Part I on page 47, afford a rational means of determining the a-priori probability of the stationary states of an atomic system.

As regards the comparison between the theory and the experiments, it will be remembered that STARK found that every hydrogen line in the presence of an electric field was split up in a number of polarised components, in a way different for the different lines. When viewed parallel to the direction of the field, there appeared a number of components polarised parallel to the field and a number of components polarised perpendicular to the field; when viewed in the direction of the field, only the latter components appeared, but without showing characteristic polarisation. Apart from the marked symmetry of the resolution of every line, the distances between successive components and their relative intensities varied in an apparently irregular way from component to component. As pointed out by EPSTEIN and SCHWARZSCHILD, however, it is possible by means of (78), in connection with relation (1), to account in a convincing way for STARK's measurements as regards the *frequencies* of the components. Especially a closer examination of these measurements showed that all the differences between the frequencies of the components were equal to entire multipla of a certain quantity, which was the same for all lines in the spectrum and, within the limits of experimental errors, equal to the theoretical value $\frac{3hF}{8\pi^2 Nem}$. On the other hand, the theories of EPSTEIN and SCHWARZSCHILD gave no direct information as regards the question of the polarisation and intensity of the different components. Comparing formula (78) with STARK's observations, EPSTEIN pointed out, however,

that the *polarisation* of the different components observed could apparently be accounted for by the rule: that a transition between two stationary states gives rise to a component polarised parallel to the field, if n_3 remains unchanged or is changed by an even number of units; while a component, corresponding to a transition in which n_2 is changed by an uneven number of units, is polarised perpendicular to the field. This result may be simply interpreted on the basis of the general formal relation between the quantum theory of line spectra and the ordinary theory of radiation. In fact, it was shown in Part I that, for a conditionally periodic system possessing an axis of symmetry, we shall expect only two types of transitions to be possible. In transitions of the first type n_3 remains unchanged, and the emitted radiation is polarised parallel to the axis of symmetry, while the transitions of the second type, in which n_3 varies by one unit, give rise to a radiation of circular polarisation in a plane perpendicular to this axis (see page 64). In order to show that this agrees with the empirical rule of EPSTEIN, it may be noted in the first place that, for any component which might be ascribed to a certain transition in which n_3 changes by a given entire number of units, there exists always another transition which will give rise to a radiation of the same frequency but in which n_3 remains unchanged or changes by one unit, according to whether the given number is even or uneven. Next it will be seen that, in case of the effect of an electric field on the hydrogen spectrum, we cannot detect by means of direct observations the circular

polarisation of the radiation corresponding to transitions of the second type; because, for each transition giving rise to a radiation of circular polarisation in one direction, there will exist another transition giving rise to a radiation which possesses the same frequency but is polarised in the opposite direction. Besides on the problem of the polarisations of the different components into which the hydrogen lines are split up in the presence of the electric field, the general considerations in Part I allow also to throw light on the question of the *relative intensities* of these components, by considering the harmonic vibrations into which the motion of the electron in the stationary states can be resolved. Compared with the problem of the relative intensities of the components of the fine structure of the hydrogen lines, the present problem is simpler in that respect, that the stationary states may be assumed to be a-priori equally probable. Since the different components, into which a given hydrogen line is split up in the electric field, correspond to transitions between pairs of states which for all components have very nearly the same values for the total energy, these states may therefore be expected to be of approximately equal occurrence in the luminous gas. According to the considerations in Part I, we shall consequently assume that for a given hydrogen line the relative intensities of the different STARK effect components, corresponding to transitions between different pairs of stationary states characterised by $n_1 = n'_1$, $n_2 = n'_2$, $n_3 = n'_3$ and $n_1 = n''_1$, $n_2 = n''_2$, $n_3 = n''_3$ respectively, will be intimately connected with the intensities of the radiations of

frequency $(n'_1 - n''_1)\omega_1 + (n'_2 - n''_2)\omega_2 + (n'_3 - n''_3)\omega_3$, which on ordinary electrodynamics would be emitted by the atom in the two states involved in the transition in question; ω_1, ω_2, and ω_3 being the fundamental frequencies entering in the expression (31) for the displacement of the electron. In order to test how far such a connection is actually brought out by the observations, it is necessary to determine the numerical values of the amplitudes of the harmonic vibrations into which the motion of the electron can be resolved. The examination of this problem has been undertaken by Mr. H. A. KRAMERS, who has deduced complete expressions for these amplitudes, by means of which it was found possible, for each of the hydrogen lines H_α, H_β, H_γ and H_δ, to account in a convincing way for the apparently capricious laws which govern the intensities of the components observed by STARK.[1]) This agreement offered at the same time a direct experimen-

[1]) *Note added during the proof.* In recent papers H. NYQUIST (Phys. Rev. X, p. 226 (1917)) and J. STARK (Ann. d. Physik, LVI, p. 569 (1918)) have published measurements on the effect of an electric field on certain lines of the helium spectrum which is given by (35), if in (40) we put $N = 2$. As will be seen from (78), the differences between the frequencies of the components into which these lines are split up will, for the same intensity of the external electric field, be smaller than for the hydrogen lines. In conformity with this it was not possible, with the experimental arrangement used by the authors mentioned, to observe separately the numerous components to be expected on the theory, but only to obtain certain rough features of the resolution of the lines in question. For the interpretation of these observations a detailed consideration of the relative intensities to be expected for the different theoretical components is therefore essential; and, as it will be

tal support for the conclusions mentioned above: that there exist no stationary states corresponding to $n_3 = 0$, while the stationary states corresponding to other values of n_2 are a-priori equally probable; and that transitions can only take place between pairs of stationary states for which n_3 is the same or differs by one unit. A general discussion of these problems will be given by KRAMERS in the paper, mentioned on page 136 in the last section, in which also the problem of the intensity of the fine structure components is treated in detail.

In the former section and in the present we have seen, how the problems of the influence of the relativity modifications on the lines of the hydrogen spectrum and of the influence of an external electric field on this spectrum can be treated, by regarding the motion of the electron as a perturbed periodic motion, and by fixing the stationary states on the basis of the relation between the energy and the frequencies of the secular perturbations. As it was done originally by SOMMERFELD and EPSTEIN, both these problems can also be treated by means of the theory of the stationary states of conditionally periodic systems which allow of separation of variables in a fixed set of positional coordinates. If, however, we consider the problem of *the simultaneous influence on the hydrogen spectrum of the relativity modifications and a homogeneous*

shown in KRAMERS' paper, it is possible, on the basis of the calculation of the amplitudes of the harmonic vibrations into which the motion of the electron in the stationary states can be resolved, to account satisfactorily for NYQUIST's and STARK's results.

electric field of any given intensity, there does not exist a set of coordinates for which a separation of variables can be obtained. On the other hand it is possible, also in this case, to apply the general considerations about perturbed periodic systems developed in the preceding. In fact, with reference to the treatment given in § 3 of the problem of the fine structure of the hydrogen lines, it will be seen that the deviations of the orbit of the electron from a Keplerian ellipse in the problem under consideration will be the same as the secular perturbations produced on a Keplerian motion by the simultaneous influence of an external homogeneous field of force and an external central force proportional to the inverse cube of the distance from the nucleus. Since these two fields together form a perturbing field possessing axial symmetry, it follows therefore that the secular perturbations, when the relativity modifications are taken into account, will be conditionally periodic and that the problem of the stationary states may be treated by means of the method mentioned in § 2 on page 107. In this way we obtain in the first place the result, that, for any value of the intensity of the external electric field, we must expect that the hydrogen lines will be split up in a number of sharp components. Next, since for any value of this intensity different from zero the system will be non-degenerate, it follows from the conditions (61), that we must assume that the angular momentum round the axis of the field is always equal to an entire multiple of $h/2\pi$; in consistence with the assumption of the validity of the analogous condition involved in the fixation of the stationary

states by means of the method of separation of variables, when applied to an explanation of the STARK effect with neglect of the relativity modifications (compare page 145). On the basis of the conditions (61) it is possible to predict in detail, how the fine structure of the hydrogen lines will be influenced by an increasing electric field until, for a sufficiently large intensity of this field, the phenomenon develops gradually into the ordinary STARK effect. The problem of this transmutation will be treated in a later paper by Mr. H. A. KRAMERS,[1]) who has kindly drawn my attention to this interesting application of the method of perturbations, and has thereby given a valuable impetus to the detailed elaboration of this method as regard the treatment of more complicate problems.

§ 5. The effect of a magnetic field on the hydrogen spectrum.

A theory of the ZEEMAN effect of the hydrogen fines based on the quantum theory of line spectra has, as mentioned in the introduction, been given independently by SOMMERFELD and by DEBYE. The calculations of these authors rest upon the fact, that it is possible, also in the presence of a magnetic field, to write the equations of mo-

[1]) Besides the discussion of this problem, the paper in question will contain a general treatment of the theory of perturbed periodic systems from the point of view of the possibility of describing the motion by means of angle variables (compare Note on page 112).

tion of the electron in the canonical Hamiltonian form given by (4), if the momenta p_1, p_2, p_3 which are conjugated to the positional coordinates of the electron q_1, q_2, q_3, are defined in a suitable way. In complete analogy to the problem of the fixation of the stationary states of an atomic system when the relativity modifications are taken into account, it follows therefore that, if these equations can be solved by the method of separation of variables, we obtain, by fixing the stationary states by means of the conditions (22), a relation between the total energy of the atom in the presence of a magnetic field and the fundamental frequencies characterising the motion of the electron, which is exactly the same as that holding between the energy and frequencies in the stationary states of an ordinary conditionally periodic system. By a procedure analogous to that applied by BURGERS in his proof of the mechanical invariance of the relations (22) for slow changes of the external conditions, mentioned in Part I on page 36, it may further be proved that also in the presence of a magnetic field these relations are invariant, when regard is taken to the effect of the induced electric forces which, according to the ordinary theory of electrodynamics, will accompany a variation of the magnetic field. In the following, however, we shall not treat the problem of the influence of an external magnetic field on the hydrogen spectrum by means of the method of separation of variables, but in analogy to the treatment of the problems of the fine structure and of the STARK effect of the hydrogen lines, given in the preceding sections, we shall treat the problem

from the point of view of the theory of perturbed periodic systems. Before entering on the detailed discussion of the necessary modifications to be introduced in the general considerations in § 2, in order that they may be applied also to the problem of the fixation of the stationary states of the atom in the presence of external magnetic forces, we shall for the sake of illustration first show how it is possible in certain cases to treat the problem of the effect of a homogeneous magnetic field on the hydrogen spectrum in a simple way, which will be seen to present a close formal analogy with the theory originally devised by LORENTZ on the basis of the classical theory of electrons.

In these considerations we shall make use of a well known theorem of LARMOR, which states that, if we look apart from small quantities proportional to the square of the intensity of the magnetic field, the motion of a system of electrons moving in a conservative field of force possessing axial symmetry round a fixed axis will, in the presence of an external homogeneous magnetic field parallel to this axis, differ from a mechanically possible motion of the system without field, only by a superposed uniform rotation of the entire system round the axis, the frequency of which is given by

$$\mathfrak{v}_H = \frac{e}{4\pi m e} H, \tag{79}$$

where H is the intensity of the magnetic field and c the velocity of light, while $-e$ and m represent the charge and

the mass of an electron.[1]) If the magnetic field is not constant, but if its intensity increases slowly and uniformly from zero, it is further simply shown that the electric induction forces, which will accompany the change in the intensity of the magnetic force, will just effect that a rotation as that described will be impressed on the original motion of the system.[2]) Moreover, as regards the effect of the magnetic field on the total energy of the system,[3]) it will be observed

[1]) J. LARMOR, Aether & Matter, Cambridge 1900, p. 341. This theorem, which was established in connection with an attempt to develop a general theory of the ZEEMAN effect based on the ordinary theory of electrodynamics, is directly proved by observing that, with the degree of approximation in question, the accellerations of the electrons due to the presence of the magnetic field are equal to the changes in the accellerations of the particles due to the superposed rotation of the system.

[2]) Compare P. LANGEVIN, Ann. de Chim. et de Phys. V, p. 70 (1905), who has deduced this result in connection with his well known theory of the magnetic properties of atomic systems based on the classical theory of electrons.

[3]) In an earlier paper (Phil. Mag. XXVII, p. 506 (1914)) the writer had assumed that the total energy in the stationary states of the hydrogen atom in the presence of a magnetic field would not be different from the energy in the corresponding states without field, as far as small quantities proportional to the intensity of the magnetic force are concerned; the effect on the kinetic energy of the electron due to the superposed rotation being assumed to be compensated by some kind of "potential" energy of the whole atom relative to the magnetic field. This assumption seemed not only suggested by the absence of *paramagnetism* in many elements, the atoms and molecules of which, according to the theory to be discussed in Part IV, must be expected to possess

a resultant angular momentum, but it was especially thought to be supported by the fact, that the spectrum, emitted by hydrogen in the presence of a magnetic field, apparently did not form a combination spectrum of the type which should be expected, it the frequency of the radiation, emitted during a transition between two stationary states of the atom in the presence of the field, could be calculated directly from the values of the energy in these states by means of relation (1). As remarked by DEBYE (Phys. Zeitschr. XVII, p. 511 (1916)), this view, however, would not be reconcilable with EINSTEIN's theory of temperature radiation (see Part I, page 8) which implies the general validity of relation (1); and, moreover, as will be shown in the following, the ZEEMAN effect of the hydrogen lines may actually be considered, not as involving a deviation from the combination principle, but rather as affording an instructive example of a systematic disappearance of certain possible combination lines, for which a simple explanation can be obtained from a consideration based on the general formal relation between the quantum theory of line spectra and the ordinary theory of radiation. Further, with reference to this relation—and remembering that on ordinary electrodynamics the magnetic field will not directly influence the exchange of energy during a process of radiation, since the forces due to this field, being always perpendicular to the direction of the velocity, will not perform work on the moving electron—it seems also natural to assume that it is possible, simply from the effect of the superposed rotation on the kinetic energy of the electron, to determine the effect of the magnetic field, as regards the *differences* between the values of the energy in the different stationary states of the atom. Now, in a discussion of the spectrum to be expected on the quantum theory, we are concerned only with these differences and not with the *absolute values* of the additional energy of the system due to the presence of the magnetic field. It would therefore be possible to escape from the difficulty, mentioned above, as regards the absence of paramagnetism, by assuming that only the energy in the so called "normal" state of an atomic system (i. e. the stationary state of the system which possesses

that the superposed rotation under consideration will not affect the mutual potential energy of the particles, while, with neglect of small quantities proportional to H^2, it will produce a change in the kinetic energy equal to $2\pi P\mathfrak{v}_H$, where P represents the total angular momentum of the system round the axis, taken in the same direction as that of the superposed rotation.

From these results it follows that the motion of the electron in any stationary state of a hydrogen atom, which is exposed to a *homogeneous magnetic field*, will—if we look apart from small quantities proportional to the square of the intensity of the magnetic force and to the product of this intensity with the ratio between the mass of the electron and that of the nucleus—differ from the motion in some stationary state of the atom in the absence of the field, only by a superposed uniform rotation round an axis through the nucleus parallel to the magnetic force with a frequency given by (79). Due to the degenerate character of the system formed by the atom

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the smallest value for the total energy; see Part IV) is not altered in the presence of a magnetic field, as far as small quantities proportional to the intensity of the magnetic force are concerned. On this view, the absence of paramagnetism would thus be a special property of the normal state, connected with the impossibility of spontaneous transitions from this state to other stationary states of the system. To this question we shall come back in the following parts of this paper; for the sake of simplicity, however, we shall not, in the considerations of this section, enter more closely on the consequences of the mentioned hypothesis, which would imply small modifications in the form of the following considerations, but would not affect the results.

in the absence of the magnetic field, it is not possible, however, from a consideration of the mechanical effect produced on the motion of the electron by a slow and uniform establishment of the magnetic field, to fix the stationary states of the perturbed atom completely, but in order to fix these states we must consider more closely the relation between the additional energy of the system due to the presence of the magnetic field and the character of the secular perturbations produced by this field on the orbit of the electron. On the basis of LARMOR's theorem the discussion of this problem is very simple. In fact, since the frequency $\mathfrak{v}_H$ is independent of the shape and position of the orbit, we may proceed in a manner which is completely analogous to that applied in the fixation of the stationary states of the hydrogen atom in the presence of a homogeneous electric field. Thus, looking apart from the effect of the relativity modifications, we may conclude at once that the total energy in the stationary states of the atom will be given by

$$E = E_n + \mathfrak{n}\mathfrak{v}_H h, \tag{80}$$

where $\mathfrak{n}$ is an entire number which can be positive as well as negative, while E_n will be equal to the energy in the corresponding stationary state of the undisturbed atom, which is given by $-W_n$ in (41). As in the case of the STARK effect, it will moreover be seen that this formula includes the values of the energy in such states of the atom, in which the electron moves in a circular orbit perpendicular to the direction of the field, and which beforehand must be expected

to be included among the stationary states of the perturbed system, since such orbits during a slow and uniform establishment of the external field will not undergo secular perturbations as regards shape and position (compare page 143). In fact, since in these cases we have $P = \pm nh/2\pi$, where n is the entire number characterising the stationary states of the undisturbed hydrogen atom, it follows from the above that the total energy in the special stationary states under consideration will just be represented by the formula (80), if we put $\mathfrak{n} = \pm n$. From this formula it will be seen at the same time, that the presence of the external magnetic field imposes the restriction on the motion in the stationary states of the hydrogen atom, that, with neglect of small quantities proportional to H, the angular momentum of the electron round the axis of the field will be equal to an entire multiple of $h/2\pi$.

As regards the expression for the total energy of the hydrogen atom in the presence of the magnetic field, formula (80) is in agreement with the formulæ obtained by SOMMERFELD and DEBYE on the basis of the conditions (22), holding for conditionally periodic systems which allow of separation of variables. As shown by these authors, a system, which consists of an electron moving under the influence of the attraction from a fixed nucleus and of a homogeneous magnetic field, allows of separation of variables in polar coordinates, if the polar axis is chosen parallel to the magnetic field. Looking apart from the effect of the relativity modifications, and choosing for q_1, q_2, and q_3 the length

of the radius vector from the nucleus to the electron, the angle between this radius vector and the axis of the system, and the angle which the plane through the electron and this axis makes with a fixed plane through the axis respectively, they obtain the following expression for the total energy:[1])

$$E = -\frac{2\pi^2 N^2 e^4 m}{h^2(n_1+n_2+n_3)^2} \pm \frac{ehn_3}{4\pi mc}H, \qquad (81)$$

where n_1, n_2, and n_3 are the integers which appear as factors to PLANCK's constant on the right side of the conditions (22). As mentioned this formula gives the same result as (80); in fact, if we put $n = n_1 + n_2 + n_3$ and if we look apart from the small correction due to the finite mass of the nucleus, the first term in (81) is seen to coincide with the expression for $-W_n$ given by (41), while the last term in (81) coincides with the last term in (80), if we put $\mathfrak{n} = n_3$. It will be observed, however, that, while in the theories of SOMMERFELD and DEBYE the stationary states are characterised by three conditions, only two conditions were necessary on the above considerations in order to secure the right

[1]) A. SOMMERFELD, Phys. Zeitschr. XVII, p. 491 (1916) and P. DEBYE, Phys. Zeitschr. XVII, p. 507 (1916). While DEBYE proceeds directly by the application of the conditions (22) in a fixed set of positional polar coordinates, SOMMERFELD determines the stationary states by applying these conditions to the motion of the system relative to a set of coordinates which rotates uniformly round the polar axis with the frequency $\mathfrak{v}_H$; a procedure which in the special case under consideration is simply shown to give the same result as the direct application of (22) to fixed polar coordinates.

relation between the energy and frequencies of the system in the stationary states. Thus, besides the conditions which prescribe the length of the major axis of the rotating orbit and the value of the angular momentum of the system round the axis of the field, the theories of the mentioned authors involve the further condition, that the value of the total angular momentum of the electron round the nucleus must be equal to an entire multiple of $h/2\pi$; and that consequently the minor axis of the orbit has the same values as in a hydrogen atom perturbed by a small external central field (compare page 110). This is due to the circumstance, that the perturbed atom forms a *degenerate system* if we look apart from the effect of the relativity modifications, because the secular perturbations are simply periodic. From the point of view of separation of variables, this degenerate character of the system is in the present case, in contrast to the analogous case of the STARK effect, also directly revealed by the fact, that a separation can be obtained, not only in polar coordinates, but in any set of axial elliptical coordinates for which one focus is placed at the nucleus and the other at some point on the axis of the field. Just as in the case of the STARK effect, however, the system is no more degenerate as soon as the relativity modifications are taken into account, in which case a separation of variables will still be possible but only in polar coordinates. To this point we shall come back below.

The observations on the ZEEMAN effect of the hydrogen lines show that, if the fine structure is neglected, each line

is in the presence of a magnetic field split up in a normal LORENTZ triplet; i. e. each line is resolved in three components of which the one is undisplaced and polarised parallel to the direction of the field, while the two other components possess frequencies, which differ from that of the original line by $\mathfrak{v}_H$, and are circularly polarised in opposite directions in a plane perpendicular to the direction of the field. As pointed out by SOMMERFELD and by DEBYE, the frequencies of a LORENTZ triplet are included among the frequencies of the components deduced from (81) by application of relation (1). In addition to the observed components, however, we might from (81) and (1) expect the appearance of a number of components, displaced from the original positions of the lines by higher multipla of $\mathfrak{v}_H$. For the non-appearance of these components the theories of SOMMERFELD and DEBYE offered no explanation, no more than for the polarisation of the components observed; except that SOMMERFELD in this connection draws attention to the fact, that the law governing the observed polarisations exhibits a certain analogy to the empirical rule of EPSTEIN concerning the observed polarisations of the components of the STARK effect of the hydrogen lines (see page 149). On the other hand, just as in case of the latter effect, an explanation of the number of the components observed and their characteristic polarisations is directly obtained on the basis of the general formal relation between the quantum theory of line spectra and the ordinary theory of radiation. In the first place we have at once from LARMOR's theorem, denoting the frequency of revolution of the

electron in a stationary state of the undisturbed hydrogen atom by ω, that the motion of the electron, in a corresponding stationary state of the atom in the presence of the field, may be resolved in a number of linear harmonic vibrations parallel to the direction of the magnetic force with frequencies $\tau\omega$, where τ is a positive integer, and in a number of circular harmonic rotations perpendicular to this direction with frequencies $\tau\omega + \mathfrak{v}_H$ or $\tau\omega - \mathfrak{v}_H$, according as the direction of rotation is the same as or the opposite of that of the superposed rotation. Next, with neglect of small quantities proportional to H^2 we have for the difference in the total energy between two neighbouring states of the perturbed system under consideration

$$\delta E = \delta E_0 + \delta \mathfrak{E} = \omega\, \delta I + \mathfrak{v}_H\, \delta \mathfrak{J}, \tag{82}$$

where E_n and ω are the values of the energy and frequency and I is the value of the quantity defined by (5), all corresponding to the state of the undisturbed system which would appear if the magnetic force vanished at a slow and uniform rate, while $\mathfrak{E}$ is the additional energy due to the presence of the magnetic field and $\mathfrak{J}$ the angular momentum of the system round the axis of the field multiplied by 2π and taken in the same direction as that of the superposed rotation. Since (82) has exactly the same form as relation (66), and since in the stationary states we have $I = nh$ and $\mathfrak{J} = \mathfrak{n}h$, we are therefore from a consideration, quite analogous to that given in § 2 on page 114, led to the conclusion, that, in the presence of the magnetic field, only two types of transitions between

stationary states are possible. For both types of transitions the integer n may change by any number of units, but in transitions of the first type the integer $\mathfrak{n}$ will remain constant and the emitted radiation will be polarised parallel to the direction of the field, while in transitions of the second type $\mathfrak{n}$ will decrease or increase by one unit and the emitted radiation will be circularly polarised in a plane perpendicular to the field, the direction of the polarisation being the same as or the opposite of that of the superposed rotation respectively. Remembering that, with neglect of small quantities proportional to the magnetic force, the angular momentum of the system round the axis of the field remains unaltered in transitions of the first type and changes by $h/2\pi$ in transitions of the second type, it will be seen that this conclusion is independently supported by a consideration of conservation of angular momentum during the transitions, like that given in Part I on page 64.

With reference to formula (80), it will be seen that the above results are in complete agreement with the experiments on the ZEEMAN effect of the hydrogen lines, as regards the *frequencies and polarisations* of the observed components. On the other hand, the observed *intensities* are directly accounted for, independent of any special theory about the origin of the lines. In fact, from a consideration of the necessary "stability" of spectral phenomena, it follows that the total radiation of the components, in which a spectral line, which originally is unpolarised, is split up in the presence of a small external field, cannot show charac-

teristic polarisation with respect to any direction. In case of the ZEEMAN effect of the hydrogen lines, it is therefore necessary beforehand to expect that the intensity of the radiation, summed over all directions, corresponding to each of the three components in which every line is split up must be the same. From the point of view of the quantum theory of line spectra, it will be seen that by means of considerations of this kind we may inversely obtain a certain amount of direct quantitative information as regards the probabilities of spontaneous transition between different sets of stationary states, holding also in the region where the integers characterising these states are not large and where consequently the estimate of the values of these probabilities, based on the formal relation between the quantum theory and the ordinary theory of radiation, gives results which are only of an approximative character. This point will be discussed more closely in KRAMERS' paper on the relative intensities of the components of the fine structure and the STARK effect of the hydrogen lines.

A procedure quite analogous to that applied above may be used to treat the problem of the effect of a homogeneous magnetic field on the hydrogen spectrum, also when the relativity modifications are taken into account, and when the atoms at the same time are exposed to a small external field of force of constant potential, which possesses axial symmetry round an axis through the nucleus parallel to the magnetic force; because also in this case we can obviously make direct use of LARMOR's theorem. We shall not, however,

proceed in this way, but shall come back to these questions when we have shown how, by a simple modification of the general considerations of perturbed periodic systems given in § 2, it is possible to represent the theory of the stationary states of the hydrogen atom in the presence of a small magnetic field on a form, which allows to discuss the effect on the hydrogen spectrum also if the atom is exposed to a magnetic field which is not homogeneous, or to discuss the effect of a homogeneous magnetic field if electric forces, which do not possess axial symmetry round an axis through the nucleus parallel to the magnetic field, are acting on the atom at the same time.

In order to examine the general problem of the secular perturbations of the orbit of the electron in the hydrogen atom which take place if the atom is exposed to small external forces which, entirely or partly, are of magnetic origin, we shall, as in the usual theory of planetary perturbations, take our starting point in the equations of motion in their canonical form. Now the equations of motion of an electron of charge $-e$, which besides by an electric field of potential V is acted upon by a magnetic field of vector potential $\mathfrak{A}$ (defined by $\operatorname{div}\mathfrak{A} = 0$ and $\operatorname{curl}\mathfrak{A} = \mathfrak{H}$, where $\mathfrak{H}$ is the magnetic force considered as a vector), can be written in the Hamiltonian form given by (4), if, just as in the absence of the magnetic field, E is taken equal to the sum of the kinetic energy T of the electron and its potential energy $-eV$ relative to the electric field, while the momenta which are conjugated to the positional coordinates q_1, q_2, q_3 of the electron in space

are defined by the equations[1])

$$p'_k = p_k - \frac{e}{c}\frac{\partial(\mathfrak{v}\mathfrak{A})}{\partial \dot{q}_k}, \qquad (k = 1, 2, 3) \tag{83}$$

where the p's are the momenta defined in the usual way (compare page 15), and where $(\mathfrak{v}\mathfrak{A})$ represents the scalar product of the velocity of the electron $\mathfrak{v}$ and the vector potential $\mathfrak{A}$, considered as a function of the q's and of the generalised velocities $\dot{q}_1$, $\dot{q}_2$, $\dot{q}_3$. If we now assume that the effect of the magnetic forces on the motion of the electron is so small compared with the effect of the electric forces, that in the calculations we may look apart from all terms proportional to $\mathfrak{H}^2$, it is simply seen that the energy function E in (4), obtained by introducing the momenta defined by (83), will differ from the corresponding function, holding in the absence of the magnetic field, only by the addition of a term which is linear in the momenta and equal to $\frac{e}{c}(\mathfrak{v}\mathfrak{A})$. In fact, denoting E expressed as a function of the q's and p's by $\varphi(p, q)$, we get from (83) together with (4), with the approximation under consideration,

$$E - \varphi(p', q) = -\sum_1^3 \frac{\partial \varphi}{\partial p'_k}(p'_k - p_k) = \sum_1^3 \frac{\partial E}{\partial p'_k}\frac{e}{c}\frac{\partial(\mathfrak{v}\mathfrak{A})}{\partial \dot{q}_k}$$

$$= \frac{e}{c}\sum_1^3 \dot{q}_k \frac{\partial(\mathfrak{v}\mathfrak{A})}{\partial \dot{q}_k} = \frac{e}{c}(\mathfrak{v}\mathfrak{A}).$$

[1]) See f. inst. G. A. Schott: Electromagnetic Radiation, App. F (Cambridge, 1912).

From this it follows that, with neglect of small quantities proportional to the square of the magnetic forces, the perturbations of the orbit of the electron in a hydrogen atom, which besides to a small external electric field of potential is exposed to a small external magnetic field of vector potential $\mathfrak{A}$, are given by a set of equations of the same form as (44) in § 2, but where the α's and β's are replaced by a set of quantities α'_1, α'_2, α'_3, β'_1, β'_2, β'_3, which are related to the q's and p''s and the time in the same way as the orbital constants α_1, α_2, α_3, β_1, β_2, β_3 for the undisturbed atom are related to the q's and p's and the time through the equations (18), and where Ω is replaced by the expression $-e\Phi + \frac{e}{c}(\mathfrak{v}\mathfrak{A})$, considered as a function of the α''s and β''s and the time. Since now, at any moment, the quantities α'_1, α'_2, α'_3, β'_1, β'_2, β'_3 differ from the corresponding orbital constants α_1, α_2, α_3, β_1, β_2, β_3 only by small terms proportional to the intensity of the magnetic field, we see therefore that, with neglect of small quantities of the same order as the variation in the orbital constants within a single period, the *secular perturbations of the shape and position of the orbit of the electron will again be given by the equations* (46), if in the present case ψ is taken equal to the sum of the mean value ψ_E of the potential energy $-e\Phi$ of the electron relative to the external electric forces and the mean value ψ_M of the quantity $\frac{e}{c}(\mathfrak{v}\mathfrak{A})$, both taken over an osculating orbit corresponding to some moment during the revolution and expressed as functions of

α_1, α_2, α_3, β_1, β_2, β_3.[1]) The latter mean value, however, is easily seen to allow of a simple interpretation. In fact, we have

$$\psi_M = \frac{e}{c}\frac{1}{\sigma}\int_0^\sigma (\mathfrak{v}\mathfrak{A})\,dt = -\frac{e\omega}{c}B, \tag{84}$$

where ω is the frequency of revolution of the electron in the osculating orbit, and where B represents the total flux of magnetic force through this orbit, taken in the same direction as that of the magnetic force which would arise from the motion of the electron according to ordinary electrodynamics.

From the considerations in § 2 it follows now in the first place that, with neglect of small quantities proportional to the square of the external forces, $\psi = \psi_E + \psi_M$ will remain *constant* during the perturbations within a time interval, sufficiently long for the perturbing forces to produce a considerable change in the shape and position of the orbit of the electron; i. e. in a time interval of the same order as σ/λ, if λ, just as in § 2, denotes a small quantity of the same order as the ratio between the external forces acting on the electron and the attraction from the nucleus. From a consideration analogous to that given in § 2, we may further

[1]) If the relativity modifications are taken into account, the orbit of the electron in the undisturbed atom is not strictly periodic, but it will be seen that the secular variations of this orbit are still obtained from the equations (46), if only, to the expression for ψ as defined in the text, a term is added which is equal to the expression for ψ given by formula (70) in § 3.

conclude that, in the stationary states of the perturbed system, the quantity $\psi = \psi_E + \psi_M$ may be taken equal to the *additional energy* of the system due to the presence of the external fields. In fact, let us imagine that these fields are slowly established at a uniform rate within a time interval from $t = 0$ to $t = \vartheta$, where ϑ is a quantity of the same order as σ/λ. For the total alteration in the inner energy of the system during this process we get then, with neglect of small quantities proportional to λ^2,

$$\Delta_\vartheta \alpha_1 = e \int_0^\vartheta \frac{t}{\vartheta} \sum_1^3 \frac{\partial \Phi}{\partial q_k} \dot{q}_k \, dt - \frac{e}{c} \int_0^\vartheta \frac{\omega B}{\vartheta} \, dt,$$

where the first term represents the work done on the system by the slowly increasing external electric forces, while the second term represents the work performed by the induced electric forces which accompany the variation in the intensity of the magnetic field. By partial integration of the first term, we get from this equation, with the approximation under consideration,

$$\begin{aligned} \Delta_\vartheta \alpha_1 - e\Phi_\vartheta &= -\frac{e}{\vartheta} \int_0^\vartheta \left(\Phi + \frac{\omega}{c} B \right) dt \\ &= \frac{1}{\vartheta} \int_0^\vartheta (\psi_E + \psi_M) \, dt = \frac{1}{\vartheta} \int_0^\vartheta \psi \, dt. \end{aligned} \tag{85}$$

Now the expression on the left side of this equation is equal to the change in the total energy of the system due to the

establishment of the external field. Since the expression on the right side is seen to be a small quantity of the same order as $\lambda\alpha_1$, it follows therefore from(85) in the first place that the secular variations of α_2, α_3, β_2, β_3 during the increase of the fields will, just as in the case considered in § 2 (see page 91), be given by a set of equations of the same form as (46), where ψ is replaced by $\frac{t}{\vartheta}\psi$, and where again α_1 may be considered as a constant. Also in the present case it follows therefore that ψ will remain constant during the establishment of the external fields, and we see consequently that the expression on the right side of (85) will be simply equal to ψ, a result which, with reference to the principle of the mechanical transformability of the stationary states, leads to the conclusion mentioned above, that the value of the additional energy in the stationary states of the perturbed system is given by the value of ψ in these states.

From the above considerations it follows that the problem of the stationary states of the hydrogen atom in the presence of external electric and magnetic forces may be treated in a manner, which is exactly analogous to that applied in § 2 in case of a periodic system exposed to a small external field of constant potential. Thus, if the secular perturbations as determined by (46) are of conditionally periodic type, we shall expect that, with neglect of small quantities proportional to λ, the cycles of shapes and positions which the orbit of the electron passes through in the stationary states of the perturbed system will be characterised by the conditions (55),

and that the possible values of the additional energy of the atom in the stationary states will be fixed by these conditions with neglect of small quantities proportional to λ^2. We shall therefore conclude that, also in the presence of external magnetic forces, the lines of the hydrogen spectrum will, if only the secular perturbations are of conditionally periodic type, be split up in a number of sharp components, the frequencies of which are determined by means of the conditions (67) together with relation (1). As regards the problem of the intensities and polarisation of these components, we may further proceed in a way quite analogous to that followed in § 2. In fact, if the secular perturbations are of conditionally periodic type, the displacement of the electron in any given direction may be represented as a sum of harmonic vibrations by an expression of the same type as (65). Moreover it can be proved that the difference in the total energy of two neighbouring states of the perturbed atom will again be given by the expression (66).[1]) The general considerations in § 2 will therefore apply without alterations to the problem of the intensity and polarisation of the components into which the hydrogen lines are split up in the presence of small external forces, also if these forces are entirely or partly of magnetic origin. Similarly, it will be seen that the effect on the spectrum of a perturbed hydrogen atom, which

[1]) Compare Note on page 112. Also in the presence of small magnetic forces, it will be possible to describe the motion of the perturbed system by means of a suitably chosen set of angle variables, if only the secular perturbations are of conditionally periodic type.

will be due to the presence of a second external field small compared with the first, also in this case may be discussed directly by means of the considerations at the end of § 2.

We meet with a direct application of the preceding considerations, if the hydrogen atom is exposed to *the simultaneous influence of an external electric and an external magnetic field, which possess axial symmetry round a common axis through the nucleus.* Introducing the same set of orbital constants as described in § 2 on page 104, we get in this case that ψ_M, as well as ψ_E, and consequently the function $\psi = \psi_E + \psi_M$ which enters in the equations (46), will, besides on α_1, depend on α_2, β_2 and α_3 but not on β_3. The general character of the secular perturbations of the orbit of the electron will therefore be the same as in the case, considered in § 2, where the atom is exposed only to an electric field of axial symmetry, and the conditions which fix the stationary states of the perturbed atom will again be expressed by the relations (61). As regards the question of the probability of spontaneous transition between the stationary states, we get moreover, just as in § 2, from a consideration of the harmonic vibrations into which the motion of the electron can be resolved, that only two types of transitions will be possible; in transitions of the first type $\mathfrak{n}_2$ remains unaltered, and the accompanying radiation is polarised parallel to the direction of the common axis of the perturbing fields; in transitions of the second type $\mathfrak{n}_2$ decreases or increases by one unit, and the accompanying radiation will be circularly polarised in a plane perpendicular to this axis. In this connection it may

be remarked, however, that the number of components, into which a given hydrogen line is split up in the presence of a magnetic field, will in general be double as large as the number of components which appear in the presence of an external electric field of axial symmetry. In fact, in the latter case the motions of the electron in two stationary states of the perturbed atom, corresponding to the same value of n, will be symmetrical with respect to a plane through the axis, and these states will possess the same values for the additional energy, if $\mathfrak{n}_1$ is the same while the values of $\mathfrak{n}_2$ are numerically equal but have opposite signs. On the other hand, if the atom is exposed also to a magnetic field, this will not hold, because the value of the function ψ_M, in contrast to the value of ψ_E, will not possess the same sign for two orbits which have the same shape and position relative to the axis, but for which the direction of revolution of the electron is reversed. Considering two states of the perturbed atom for which the values of $\mathfrak{n}_1$ are the same and the values of $\mathfrak{n}_2$ are numerically equal but have opposite signs, we get therefore, if the atom is exposed only to a magnetic field of axial symmetry, that the values of the additional energy will be equal with exception of the sign; while, if the atom is exposed to a magnetic as well as to an electric field, the additional energy in two such states will in general differ also as regards its numerical value. In contrast to what in general will take place if the atom is exposed to an electric field of axial symmetry, it will thus be seen that, if the hydrogen atom is exposed only to a magnetic field possessing axial symmetry, the ensemble

of components into which a given hydrogen line is resolved will be completely symmetrical with respect to the position of the original line, as regards the frequencies as well as the intensities and polarisations. Moreover it follows from the above, that if we consider a hydrogen atom exposed to an electric field of axial symmetry and imagine that an external magnetic field, which possesses symmetry round the same axis, is gradually established, each component which appears in the presence of the first field only will split up into two components, in such a way that each component polarised parallel to the axis will split up into two components of the same polarisation, while each component polarised perpendicular to the axis, and which originally showed no polarisation when viewed in a direction parallel to the axis, will split up into two components showing circular polarisations in opposite directions. If the magnetic field is small, the new components will be placed symmetrically with respect to the position of the original components and their intensities will be approximately equal, but when the perturbing influence of the magnetic forces on the motion of the electron becomes of the same order of magnitude as that of the external electric forces, the components in question will in general be placed unsymmetrically with respect to their original position, and their intensities may differ considerably.

An especially simple example of a magnetic field which possesses axial symmetry is afforded by the case of a *homogeneous magnetic field*, discussed in the beginning of this section. In this case we have that the total magnetic flux of

force through the orbit of the electron is equal to the product of the intensity H of the magnetic field and the area of the projection of the orbit on a plane perpendicular to this field. Since this area is equal to $\alpha_3/2m\omega$, we get consequently from (84)

$$\psi_M = \frac{e\alpha_3}{2cm} H. \tag{86}$$

From the equations (46) it follows therefore that the effect of a homogeneous magnetic field, which acts upon a hydrogen atom which at the same time is exposed to an external electric field possessing axial symmetry round an axis through the nucleus parallel to the magnetic force, will consist in a superposition of a uniform rotation of the orbit round the axis with a frequency equal to

$$\mathfrak{v}_H = \frac{1}{2\pi} \frac{\partial \psi_M}{\partial \alpha_3} = \frac{e}{4\pi mc} H$$

on the secular perturbations which would take place in the absence of the magnetic field. This result follows also directly from LARMOR's theorem, on which the simple considerations about the effect of a homogeneous magnetic field in the beginning of this section were based. Since a superposed rotation as that in question will not influence the shape of the orbit of the electron or its position relative to the axis, it follows from (61) that the value of ψ_E in the stationary states of the atom will not be affected by the presence of the magnetic field, and that consequently the effect of this field on the additional energy of the system will simply consist in

the addition of a term given by

$$\psi_M = \frac{e}{2mc}\frac{\mathfrak{n}_2 h}{2\pi}H = \mathfrak{n}_2\mathfrak{v}_H h. \tag{87}$$

This result was also to be expected from a simple consideration of the mechanical effect produced on the motion by a slow and uniform establishment of the magnetic field (compare page 159). With reference to the above considerations as regards the probability of transition between stationary states, it will be seen to follow from (87), that the presence of the homogeneous magnetic field will leave the components polarised parallel to the axis unaltered, but will cause every component, which in the absence of the field was polarised perpendicular to the axis, to split up in a symmetrical doublet the members of which will show circular polarisation in opposite directions, when viewed in the direction of the axis, and will be displaced from the position of the original component by an amount corresponding to a frequency difference equal to $\mathfrak{v}_H$.

A simple application of the last result is afforded by the problem of *the simultaneous effect on the hydrogen lines of a homogeneous electric and a homogeneous magnetic field which have the same direction.* Thus, if the intensities of the fields are so large that we may look apart from the small modifications claimed by the theory of relativity, we shall from the above expect that the effect in question will differ from the ordinary STARK effect of the hydrogen lines, only therein that every component polarised perpendicular to the

field is split up in two symmetrical components corresponding to the outer members of a LORENTZ triplet. This seems to agree with some observations of the effect of two such fields on the hydrogen line H_α, published by GARBASSO.[1]) The problem in question might also have been treated by means of the method of separation of variables, because, as may be easily shown, the perturbed system—if the relativity modifications are neglected—allows of separation of variables in parabolic coordinates, just as in the presence of the electric field only. If, on the other hand, the relativity modifications are taken into account, the method of separation of variables cannot be applied, but, with reference to the considerations at the end of the last section, it will be seen that it is possible, also in this case, to predict at once the modification in the effect of an electric field on the fine structure of the hydrogen lines, which would result from the simultaneous presence of a parallel magnetic field. Passing to the limiting case where the intensity of the electric field is equal to zero, it will thus be seen at once from the preceding, that *the effect of a homogeneous magnetic field on the fine structure of the hydrogen lines* will consist in the splitting up of every component in a normal LORENTZ triplet. As far as the frequencies of the components are concerned, this result has been predicted by SOMMERFELD and DEBYE, who have treated the problem under consideration by means of separation of variables in polar coordinates (compare page 165). In connection with

[1]) A. GARBASSO, Phys. Zeitschr. XV, p. 123 (1914).

the fixation of the stationary states in this problem, it may be remarked that we must assume that no stationary state will exist for which the angular momentum round an axis through the nucleus parallel to the magnetic field would be equal to zero. In fact, as seen in § 4, we must assume that, in case of a hydrogen atom exposed to a homogeneous electric field, no such states will be possible; and by imagining that the electric field decreases slowly to zero, while at the same time a magnetic field parallel to the electric field is slowly established, it would be possible, without passing through a degenerate system, to obtain a continuous transformation of the stationary states of the perturbed atom during which the angular momentum of the electron round the axis would remain unaltered. With reference to the invariance of the a-priori probability of the stationary states during such a transformation (see Part I, page 14 and page 49), we must therefore conclude that, also in the case of a hydrogen atom in the presence of a magnetic field, no stationary states exist for which the angular momentum round the axis would be equal to zero, although these slates in mechanical respect do not exhibit singularities from which we might anticipate that they are physically unrealisable.[1])

[1]) *Note added during the proof.* In a dissertation which has just appeared, J. M. Burgers (Het Atoommodel van Rutherford-Bohr, Haarlem 1918) has given a very interesting general survey of the applications of the quantum theory to the problem of the constitution of atoms, and has in this connection entered upon several of the questions discussed in the present paper; for instance on the question of

the relation between the spectrum of an atomic system, deduced by application of relation (1) from the values of the energy in the stationary states, and the frequencies of the harmonic vibrations into which the motion in these states can be resolved; and on the question of the determination of the relative values for the a-priori probability of the different stationary states of an atomic system by means of EHRENFEST's principle of the invariance of these values during a continuous transformation of the system. As an illustration of the latter considerations, BURGERS has deduced an expression for the relative values of the a-priori probability of the different stationary states of the undisturbed hydrogen atom, by means of an enumeration of the states, determined by the conditions (22) when applied in connection with a separation of variables in polar coordinates, which correspond to a stationary state of the undisturbed atom, characterised by a given value of n in the condition $I = nh$. Excluding only such states for which the total angular momentum of the electron round the nucleus would be equal to zero, BURGERS (loc. cit. p. 259) finds in this way for the value of the a-priori probability in question $(n+1)^2 - 1$. In connection with the analogous consideration, given in the Note on page 149 of the present paper, which leads to a different result, it may be of interest to remark that the necessary conformity between the relative values for the a-priori probability of the different stationary states of the undisturbed hydrogen atom, deduced from an enumeration of the stationary states of the atom which appear in the presence of a small external electric field or in the presence of a small magnetic field respectively, cannot be obtained if in both cases we would exclude only such states in which the angular momentum of the electron round the nucleus is always equal to zero. In fact, while in case of a magnetic field this would give $(n+1)^2 - 1$ different states corresponding to a given value of n, it would in case of an electric field give only $(n+1)^2 - 2$ such states. On the other hand, if the possible stationary states are selected in the manner explained in the text, the conformity in question will obviously be obtained.

In case we consider the general problem of the effect on a hydrogen atom of a small electric or magnetic field, which do not possess axial symmetry round an axis through the nucleus, or of the simultaneous effect of two such fields, which do not possess such symmetry round a common axis, we must expect that the secular perturbations of the orbit of the electron will in general not be of conditionally periodic type. In such a case we cannot obtain a complete fixation of the stationary states, and we may conclude that the presence of the external forces will not give rise to the splitting up of the hydrogen lines into a number of sharp components but to a diffusion of these lines. With a simple example, in which the secular perturbations of the atom seem not to be of conditionally periodic type, we meet if we consider *the simultaneous effect on the hydrogen spectrum of an external homogeneous electric field and a homogeneous magnetic field, the directions of which make an angle with each other.* If the effects of the two fields on the motion of the electron are of the same order of magnitude we may in this case expect that the hydrogen lines will not be resolved into sharp components but will become diffuse. From the considerations on page 117 of the effect on the spectrum of a perturbed periodic system due to a second external field, the perturbing effect of which is small compared with that of the first, we may conclude, however, that, if the effect of one of the fields on the motion of the electron is large compared with that of the other, the hydrogen lines will still show a resolution in a number of components, the spectral widths of which

are small compared with the displacements which they have undergone due to the presence of the weaker of the external fields. In the discussion of this problem we shall for simplicity neglect the influence of the relativity modifications, assuming that the effect on the spectrum produced by each external field separately is large compared with the inherent fine structure of the hydrogen lines. Denoting, as in § 2, by μ a small constant of the same order as the ratio between the forces on the electron due to the weaker of the external fields and those due to the stronger of these fields, and by λ a small constant of the same order as the ratio between the latter forces and the attraction from the nucleus, we have, as shown on page 119, that, with neglect of small quantities of the same order of magnitude as $\lambda\mu^2$,[1]) the change in the additional energy of the atom due to the presence of the weaker field is, in general, directly obtained by taking the mean value of the function ψ, corresponding to the weaker field, over the cycle of shapes and positions which the orbit of the electron passes through in the stationary states of the atom in the presence of the stronger field only. In the special case under consideration, however, the perturbed system, formed by the atom in the presence of the stronger field only, is degenerate, the secular perturbations of the orbit of the electron being of a simple periodic character. The mean value in question will

[1]) Rigorously this result holds with neglect of small quantities of the same order of magnitude as the largest of the quantities λ^2 and $\lambda\mu^2$, but for the sake of simplicity it is here and in the following assumed that μ is not smaller than $\sqrt{\lambda}$ (compare page 119).

therefore not be completely determined, but will be different for the different periodic cycles of shapes and positions of the orbit, which represent the continuous multitude of stationary motions which the electron may perform in each of the stationary states of the atom in the presence of the stronger field only. In order to fix the stationary states in the presence of both fields and the change in the additional energy of the atom due to the presence of the weaker field, it will thus, as mentioned on page 121, be necessary to examine the relation between the mean value in question and the frequency of the slow periodic "secular" variations which the cycles under consideration will undergo under the influence of the weaker of the external fields. Now, in the special case under consideration this problem may be treated very simply, if we imagine the weaker field as composed of two homogeneous fields of which the one is parallel and the other perpendicular to the stronger field, and if we consider separately the secular effect due to each of these fields. In fact, due to the symmetry with respect to the axis of the stronger field, exhibited by the periodic cycle of shapes and positions which the orbit of the electron would pass through if the atom were exposed to this field only, it is easily seen that the contribution, which the perpendicular component of the weaker field gives to the mean value of ψ corresponding to the latter field, will vanish. From this it follows that the secular effect of the weaker field, with neglect of small quantities proportional to μ^2 will be the same as if only the parallel component of this field was acting on the atom; and we see consequently

that, in the stationary states of the atom in the presence of both fields, the possible cycles of shapes and positions of the orbit of the electron will be characterised in the same way as if the weaker field was parallel to the stronger. The problem, however, of the fixation of the stationary states of a hydrogen atom in the presence of a homogeneous electric field and a homogeneous magnetic field, which are parallel to each other, is very simple. In fact, as it appears from the considerations on page 179, the stationary states will in this case be fixed completely by two conditions, of which the one, in the same way as in the simple theory of the STARK effect, defines the position of the plane in which the electrical centre of the orbit of the electron moves, while the other defines the value of the angular momentum of the electron round the axis of the fields in the same way as in the simple theory of the ZEEMAN effect. In connection with the problem under consideration here, it may be useful for the sake of illustration to note, that, if the perturbing effect of the electric field is large compared with that of the magnetic, the second of these conditions may be said to be imposed on the system by the slow and uniform rotation, which the magnetic field produces on the periodic cycle of shapes and positions of the orbit of the electron, which would appear if the atom was exposed to the electric field only. If, on the other hand, the effect of the magnetic field is large compared with that of the electric field, the first condition may be said to be imposed on the system by the slow periodic oscillation in the shape and position relative to the axis, which the

electric field produces on the uniformly rotating orbit which the electron would describe if the atom was exposed to the magnetic field only.

If we consider a hydrogen atom which is exposed to the simultaneous influence of a homogeneous electric field of intensity F and a homogeneous magnetic field of intensity H, the direction of which makes an angle φ with the direction of the electric field, it follows from the above that, if the perturbing influence of the electric field is large compared with that of the magnetic field, the main effect produced by the latter field on the spectrum may be described as the splitting up of each STARK effect component, polarised perpendicular to the axis of the electric field, into two circularly polarised components, corresponding to the outer members of a LORENTZ triplet which would be produced by a magnetic field of intensity $H \cos \varphi$. On the other hand, if the perturbing effect of the magnetic field is large compared with that of the electric, it follows that the main effect, produced by the latter field on the spectrum, may be described as the resolution of the middle component and of each of the outer components of the normal ZEEMAN effect into a number of components, corresponding to the parallel and perpendicular components respectively of a STARK effect produced by an electric field of intensity $F \cos \varphi$.

The effects just described, however, which are the same as would take place if only the parallel component of the weaker field was acting on the atom, will not be the only effects of the presence of the weaker field on the spectrum. In

fact, although the perpendicular component of the weaker field, apart from small quantities proportional to μ^2, will not have any secular effect on the cycle of shapes and positions which the orbit of the electron would pass through if the atom was exposed to the stronger field only, it will obviously produce alterations in the motion of the electron within this cycle which are proportional to μ. Thus, if the weaker field was parallel to the stronger, the motion of the electron in the perturbed atom would be composed of a number of linear harmonic vibrations parallel to the direction of the fields, the frequencies of which are of the type $|\tau\omega_P + \mathfrak{k}_1\mathfrak{v}_1|$, and of a number of circular harmonic rotations perpendicular to this direction, the frequencies of which are of the type $|\tau\omega_P + \mathfrak{k}_1\mathfrak{v}_1 + \mathfrak{v}_2|$ (compare page 114). In the general case, however, where the weaker field is not parallel to the stronger, there will, in the expression for the displacement of the electron in any given direction, in addition appear a number of harmonic vibrations the amplitudes of which are proportional to μ and the frequencies of which, as a closer consideration of the perturbations learns, are equal to the sum or difference of the frequency of one of the harmonic vibrations, in which the motion in this direction could be resolved if the external fields were parallel to each other, and one of the small frequencies of type $|\mathfrak{k}_1\mathfrak{v}_1 + \mathfrak{v}_2|$, which appear in the expression for the secular perturbations of the electron in this case. A part of these additional vibrations will again possess frequencies of the types $|\tau\omega_P + \mathfrak{k}_1\mathfrak{v}_1|$ and $|\tau\omega_P + \mathfrak{k}_1\mathfrak{v}_1 + \mathfrak{v}_2|$, and will cause that the motion, instead of

consisting of vibrations which are exactly linear and exactly circular as in the case where the external fields are parallel to each other, will be composed of elliptical harmonic vibrations which partly are nearly linear and parallel to the direction of the stronger field and partly nearly circular and perpendicular to this direction. On account of this we shall expect that, due to the presence of the perpendicular component of the weaker field, the different components mentioned above will not be sharply polarised. Further there will, in the motion of the perturbed atom, also appear a number of circular harmonic rotations perpendicular to the stronger field, the amplitudes of which are small quantities proportional to μ, and the frequencies of which are of the type $|\tau\omega_P + \mathfrak{k}_1\mathfrak{v}_1 + 2\mathfrak{v}_2|$. From this we shall expect the appearance in the spectrum of a number of new weak components, corresponding to a type of transition between stationary states which would not be possible if the two external fields were parallel to each other. When considering more closely the frequencies of these new components, it must be remembered, however, that, as mentioned above, the present method of treating the problem of the perturbations assures us of the conditionally periodic character of the motion of the electron within a time interval of the same order of magnitude as σ/λ, only if we look apart from small quantities of the same order as μ^2; and we must therefore be prepared to find that the frequencies of the vibrations of small amplitudes will not be defined with the same degree of approximation as the frequencies of the vibrations of large amplitudes. Thus, while the frequencies of

the latter vibrations are defined with neglect of small quantities proportional to $\lambda\mu^2$, the frequencies of the small vibrations under consideration are obviously defined only with neglect of small quantities proportional to $\lambda\mu$. In intimate connection with the general want of definition of the energy in the stationary states for perturbed systems of the type in question, we must accordingly be prepared to find that, in contrast to the strong components, for which we may expect that by far the larger part of the intensity is contained within a spectral interval of a width proportional to $\lambda\mu^2$, the new components will be diffused over spectral intervals of a width proportional to $\lambda\mu$.[1]) Thus, in case the effect of the exter-

[1]) Compare Note on page 119. With reference to the general validity of relation (1), it will be seen that the assumption, that the weak components possess this degree of diffusion, implies the assumption, that the corresponding transitions (the probability of occurrence of which is very small compared with the probability of the transitions responsible for the strong components) will generally take place between two states of the perturbed atom, which do not both belong to the well defined ensemble of stationary states in which at any moment the great majority among a large number of atoms will be present. Thus, in case the effect of the external electric field is large compared with that of the magnetic field, we may expect that, in both states involved in the transitions in question, the positions of the plane in which the electrical centre moves will coincide with positions of this plane in states belonging to the ensemble just mentioned, while the angular momentum of the electron round the axis of the electric field will generally change by an amount which will not be equal to an entire multiple of $h/2\pi$. On the other hand, if the effect of the magnetic field is the larger, the angular momentum of the electron round the axis of this field will, in

nal electric field is large compared with that of the magnetic field, we might expect at first sight that, on each side of every of the STARK effect components polarised parallel to the electric force, there would appear a weak component which would be circularly polarised and be displaced from this component by an amount twice that of the displacement of the strong components into which the perpendicularly polarised STARK effect components are split up as a consequence of the small magnetic field. We must be prepared, however, to find that these weak components will be so diffuse, that they are not separated from the weak perpendicular component which has the same frequency as the strong parallel components on each side of which the weak components under consideration would lie, and which appears as a consequence of the above mentioned want of sharpness as regards the polarisation of the strong components. On the other hand, if the effect of the magnetic field is large compared with that of the electric field, any weak component of the type under consideration, which corresponds to transitions in which the angular momentum of the electron round the axis of the magnetic field changes by two times $h/2\pi$, will lie at a distance from the original hydrogen line, which is approximately twice as large as that of the outer components of the normal ZEE-

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the transitions in question, change by two times $h/2\pi$, while we may expect that the plane in which the electrical centre moves will generally, in at least one of the states involved in these transitions, differ from the positions of this plane in the ensemble of stationary states referred to.

MAN effect, and will therefore be distinctly separated from the strong components into which each of the components of the normal ZEEMAN effect is split up in the presence of the small electric field. We must be prepared, however, to find that the weak components will not, as it might be expected at first sight, form two sets of distinctly separated lines, but that they will only appear as two diffuse lines of circular polarisation in opposite directions and of a spectral width proportional to $\lambda\mu$.[1])

§ 6. The continuous hydrogen spectrum.

We shall conclude the considerations of this Part by a brief discussion of the characteristic continuous spectrum of hydrogen in the ultra violet region, which is intimately connected with the series spectrum given by (35). This spectrum

[1]) No experiments, which allow to test the preceding results in detail, seem to have been recorded, but it would appear that the above considerations afford an explanation of the general character of the remarkable deviations from a normal ZEEMAN effect, observed by F PASCHEN and E. BACK (Ann. d. Phys. XXXIX, p. 897 (1912)) in experiments in which the hydrogen lines were excited by passing a powerful condensed discharge through a capillary tube placed at right angles with the direction of the magnetic field. Besides the characteristic want of sharpness of the polarisation of the middle component, exhibited by all the spectrograms published by PASCHEN and BACK, especially one of their photographs (Tafel VIII, Bild 4) seems to suggest the presence of a weak, perpendicularly polarised, diffuse line on each side of the original line and at a distance from it twice that of the outer components of the normal effect.

consists of a radiation, the frequencies of which are continuously distributed over a spectral interval extending from the head of the Balmer series in the direction of higher frequencies.[1]) The existence of a continuous spectrum of this type is just what should be expected from a natural generalisation of the principles underlying the quantum theory of series spectra.[2]) Thus the spectrum under consideration may be directly explained by application of relation (1), if we assume that the complete spectrum, emitted by a system consisting of a nucleus and of an electron, originates not only from radiations, emitted during transitions between two states belonging to the multitude of stationary states in which the electron describes a closed orbit, characterised by the condition $I = nh$, but also from radiations emitted during transitions between two states, one (or both) of which belong to the multitude of states in which the electron possesses sufficient energy to remove to infinite distance from the nucleus. While the electron in the states of the type first mentioned

[1]) This spectrum has been observed as an emission spectrum in spectra of solar protuberances and planetary nebulae (See J. EVERSHED, Phil. Trans. Roy. Soc. 197 A, p. 399 (1901) and W. H. WRIGHT, Lick Observatory Bulletin, No. 291 (1917)) as well as in direct laboratory experiments on spectra excited by positive rays (See J. STARK, Ann. d. Phys. LII, p. 255 (1917)). Further it has been observed as an absorption spectrum in the spectra of several stars (see W. HUGGINS, An Atlas of Representative Stellar Spectra, p. 85 (1899) and J. HARTMANN, Phys. Zeitschr. XVIII p. 429 (1917)).

[2]) Compare N. BOHR, Phil. Mag. XXVI, p. 17 (1913); and also P. DEBYE, Phys. Zeitschr. XVIII, p. 428 (1917).

may be said to be "bound" by the nucleus to form an atom, it may in the states of the last mentioned type be described as "free". In order to account for the appearance of the continuous spectrum, it is necessary to assume that the motions in the latter states are not restricted by extra-mechanical conditions of the type holding for the former states, but that all motions, which are consistent with the application of ordinary mechanics, will represent physically possible states. This assumption would also seem to present itself naturally from the point of view on the principles of the quantum theory, taken in the present paper.[1]) Thus it will in the first place be observed that any attempt to discriminate between the different states of the type in question, by means of considerations of the mechanical stability of stationary states for slow transformations of the external conditions, would fail on account of the essentially non-periodic character of the motion, which is irreconcilable with the idea of invariance of extra-mechanical conditions for such transformations. Next, with reference to the formal analogy between the quantum theory and the ordinary theory of radiation, it will be seen that the fact, that the motion of a free electron in its hyper-

[1]) A view contrary to this has been taken by EPSTEIN, who in a recent paper (Ann. d. Phys. L, p. 815 (1916)) has made an attempt to obtain an explanation of certain observations on the photoelectric effect of hydrogen occluded in metals, by applying conditions of the same type as (22) to states of the hydrogen atom in which the electron describes a hyperbolic orbit, and has tried in a similar way to develop a theory of the characteristic β-ray spectra of radioactive substances.

bolic orbit cannot be resolved in a sum of harmonic vibrations of discontinuously varying frequencies but can only be represented by a Fourier integral extended over a continuous range of frequencies, suggests beforehand that the free electron may pass, under emission or absorption of radiation, to any one among a continuous multitude of other states corresponding to a continuous multitude of values for the energy of the system. From the preceding considerations we may infer, by application of (1), that the complete spectrum emitted by the hydrogen atom will, besides the series spectrum and the continuous ultra-violet spectrum mentioned above, which corresponds to transitions from a state in which the electron is free to a stationary state characterised by $n = 2$ in (41), contain a set of continuous spectra, corresponding to transitions from free states to other stationary states, and each extending in the direction of larger frequencies from one of the values of the frequency, given by (35) if we put $n' = \infty$. Moreover, we may expect the presence of a weak continuous spectrum, extending as a continuous back ground over the whole region of frequencies, which will correspond to transitions between two different states in both of which the electron is free. The relative intensities of these different continuous spectra, and the laws according to which the intensity is distributed within each of them, may be expected to vary to a large extent according to the different conditions under which the radiation is excited. Thus, while the continuous spectrum of hydrogen, when observed as emission spectrum in stars, shows a abrupt beginning at the head

of the Balmer series, the continuous spectrum, observed by STARK in his experiments referred to above, was not sharply limited but showed a pronounced maximum in the spectral region which corresponds to transitions between two states, in the first of which the velocity of the free electron relative to the nucleus, before the "collision" with the latter, was of the same order of magnitude as the velocity of the positive rays by means of which the spectrum was excited.

Besides the series spectrum and the connected continuous spectrum just considered, there exists, as well known, another hydrogen spectrum, the so called many-line spectrum, which on account of its complex structure and its resemblance with the band spectra, emitted by other elements and combinations of elements, is generally ascribed to the hydrogen molecule and not to the atom. This assumption would also seem to present itself directly from the point of view of the quantum theory, according to which the simple structure of the series spectrum is directly connected with the simple periodic character of the motion of the particles in the atom, while a spectrum of a complexity of the order exhibited by the many-line spectrum must be assumed to originate from a system the motion of which does not show such simple periodicity properties. The problem of the constitution of the hydrogen molecule, to be expected on the quantum theory, and the possible motions of the particles of this system will be treated in Part IV. In this connection we shall also consider the problem of dispersion of light in hydrogen gas and the problem of the voltage necessary to

produce the lines of the series spectrum of hydrogen by an electric discharge in this gas.

Færdig fra Trykkeriet d. 30, December 1918.

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The Theory of Spectra and Atomic Constitution

THREE ESSAYS
BY

NIELS BOHR

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in the University of Copenhagen

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PREFACE

THE three essays which here appear in English all deal with the application of the quantum theory to problems of atomic structure, and refer to the different stages in the development of this theory.

The first essay "On the spectrum of hydrogen" is a translation of a Danish address given before the Physical Society of Copenhagen on the 20th of December 1913, and printed in *Fysisk Tidsskrift*, XII. p. 97, 1914. Although this address was delivered at a time when the formal development of the quantum theory was only at its beginning, the reader will find the general trend of thought very similar to that expressed in the later addresses, which form the other two essays. As emphasized at several points the theory does not attempt an "explanation" in the usual sense of this word, but only the establishment of a connection between facts which in the present state of science are unexplained, that is to say the usual physical conceptions do not offer sufficient basis for a detailed description.

The second essay "On the series spectra of the elements" is a translation of a German address given before the Physical Society of Berlin on the 27th of April 1920, and printed in *Zeitschrift für Physik*, VI. p. 423, 1920. This address falls into two main parts. The considerations in the first part are closely related to the contents of the first essay; especially no use is made of the new formal conceptions established through the later development of the quantum theory. The

second part contains a survey of the results reached by this development. An attempt is made to elucidate the problems by means of a general principle which postulates a formal correspondence between the fundamentally different conceptions of the classical electrodynamics and those of the quantum theory. The first germ of this correspondence principle may be found in the first essay in the deduction of the expression for the constant of the hydrogen spectrum in terms of Planck's constant and of the quantities which in Rutherford's atomic model are necessary for the description of the hydrogen atom.

The third essay "The structure of the atom and the physical and chemical properties of the elements" is based on a Danish address, given before a joint meeting of the Physical and Chemical Societies of Copenhagen on the 18th of October 1921, and printed in *Fysisk Tidsskrift*, XIX. p. 153, 1921. While the first two essays form verbal translations of the respective addresses, this essay differs from the Danish original in certain minor points. Besides the addition of a few new figures with explanatory text, certain passages dealing with problems discussed in the second essay are left out, and some remarks about recent contributions to the subject are inserted. Where such insertions have been introduced will clearly appear from the text. This essay is divided into four parts. The first two parts contain a survey of previous results concerning atomic problems and a short account of the theoretical ideas of the quantum theory. In the following parts it is shown how these ideas lead to a view of atomic

constitution which seems to offer an explanation of the observed physical and chemical properties of the elements, and especially to bring the characteristic features of the periodic table into close connection with the interpretation of the optical and high frequency spectra of the elements.

For the convenience of the reader all three essays are subdivided into smaller paragraphs, each with a headline. Conforming to the character of the essays there is, however, no question of anything like a full account or even a proportionate treatment of the subject stated in these headlines, the principal object being to emphasize certain general views in a freer form than is usual in scientific treatises or text books. For the same reason no detailed references to the literature are given, although an attempt is made to mention the main contributions to the development of the subject. As regards further information the reader in the case of the second essay is referred to a larger treatise "On the quantum theory of line spectra," two parts of which have appeared in the Transactions of the Copenhagen Academy (*D. Kgl. Danske Vidensk. Selsk. Skrifter*, 8. Række, IV. 1, I and II, 1918),* where full references to the literature may be found. The proposed continuation of this treatise, mentioned at several places in the second essay, has for various reasons been delayed, but in the near future the work will be completed by the publication of a third part. It is my intention to deal more fully with the problems discussed in the third essay by a larger systematic

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account of the application of the quantum theory to atomic problems, which is under preparation.

As mentioned both in the beginning and at the end of the third essay, the considerations which it contains are clearly still incomplete in character. This holds not only as regards the elaboration of details, but also as regards the development of the theoretical ideas. It may be useful once more to emphasize, that—although the word "explanation" has been used more liberally than for instance in the first essay—we are not concerned with a description of the phenomena, based on a well-defined physical picture. It may rather be said that hitherto every progress in the problem of atomic structure has tended to emphasize the well-known "mysteries" of the quantum theory more and more. I hope the exposition in these essays is sufficiently clear, nevertheless, to give the reader an impression of the peculiar charm which the study of atomic physics possesses just on this account.

I wish to express my best thanks to Dr A. D. Udden, University of Pennsylvania, who has undertaken the translation of the original addresses into English, and to Mr C. D. Ellis, Trinity College, Cambridge, who has looked through the manuscript and suggested many valuable improvements in the exposition of the subject.

N. BOHR.

COPENHAGEN,
May 1922.

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ESSAY I*

ON THE SPECTRUM OF HYDROGEN

Empirical spectral laws. Hydrogen possesses not only the smallest atomic weight of all the elements, but it also occupies a peculiar position both with regard to its physical and its chemical properties. One of the points where this becomes particularly apparent is the hydrogen line spectrum.

The spectrum of hydrogen observed in an ordinary Geissler tube consists of a series of lines, the strongest of which lies at the red end of the spectrum, while the others extend out into the ultra-violet, the distance between the various lines, as well as their intensities, constantly decreasing. In the ultra-violet the series converges to a limit.

Balmer, as we know, discovered (1885) that it was possible to represent the wave lengths of these lines very accurately by the simple law

$$\frac{1}{\lambda_n} = R\left(\frac{1}{4} - \frac{1}{n^2}\right), \tag{1}$$

where R is a constant and n is a whole number. The wave lengths of the five strongest hydrogen lines, corresponding to $n = 3, 4, 5, 6, 7$, measured in air at ordinary pressure and

*Address delivered before the Physical Society in Copenhagen, Dec. 20, 1913.

temperature, and the values of these wave lengths multiplied by $\left(\frac{1}{4} - \frac{1}{n^2}\right)$ are given in the following table:

n	$\lambda \cdot 10^8$	$\lambda \cdot \left(\frac{1}{4} - \frac{1}{n^2}\right) \cdot 10^{10}$
3	6563.04	91153.3
4	4861.49	91152.9
5	4340.66	91153.9
6	4101.85	91152.2
7	3970.25	91153.7

The table shows that the product is nearly constant, while the deviations are not greater than might be ascribed to experimental errors.

As you already know, Balmer's discovery of the law relating to the hydrogen spectrum led to the discovery of laws applying to the spectra of other elements. The most important work in this connection was done by Rydberg (1890) and Ritz (1908). Rydberg pointed out that the spectra of many elements contain series of lines whose wave lengths are given approximately by the formula

$$\frac{1}{\lambda_n} = A - \frac{R}{(n+\alpha)^2},$$

where A and α are constants having different values for the various series, while R is a universal constant equal to the constant in the spectrum of hydrogen. If the wave lengths

are measured in vacuo Rydberg calculated the value of R to be 109675. In the spectra of many elements, as opposed to the simple spectrum of hydrogen, there are several series of lines whose wave lengths are to a close approximation given by Rydberg's formula if different values are assigned to the constants A and α. Rydberg showed, however, in his earliest work, that certain relations existed between the constants in the various series of the spectrum of one and the same element. These relations were later very successfully generalized by Ritz through the establishment of the "combination principle." According to this principle, the wave lengths of the various lines in the spectrum of an element may be expressed by the formula

$$\frac{1}{\lambda} = F_r(n_1) - F_s(n_2). \tag{2}$$

In this formula n_1 and n_2 are whole numbers, and $F_1(n)$, $F_2(n)$, ... is a series of functions of n, which may be written approximately

$$F_r(n) = \frac{R}{(n+\alpha_r)^2},$$

where R is Rydberg's universal constant and α_r is a constant which is different for the different functions. A particular spectral line will, according to this principle, correspond to each combination of n_1 and n_2, as well as to the functions F_1, F_2, ... The establishment of this principle led therefore to the prediction of a great number of lines which were not included in the spectral formulae previously considered, and

in a large number of cases the calculations were found to be in close agreement with the experimental observations. In the case of hydrogen Ritz assumed that formula (1) was a special case of the general formula

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \tag{3}$$

and therefore predicted among other things a series of lines in the infra-red given by the formula

$$\frac{1}{\lambda} = R\left(\frac{1}{9} - \frac{1}{n^2}\right).$$

In 1909 Paschen succeeded in observing the first two lines of this series corresponding to $n = 4$ and $n = 5$.

The part played by hydrogen in the development of our knowledge of the spectral laws is not solely due to its ordinary simple spectrum, but it can also be traced in other less direct ways. At a time when Rydberg's laws were still in want of further confirmation Pickering (1897) found in the spectrum of a star a series of lines whose wave lengths showed a very simple relation to the ordinary hydrogen spectrum, since to a very close approximation they could be expressed by the formula

$$\frac{1}{\lambda} = R\left(\frac{1}{4} - \frac{1}{(n+\frac{1}{2})^2}\right).$$

Rydberg considered these lines to represent a new series of lines in the spectrum of hydrogen, and predicted according

to his theory the existence of still another series of hydrogen lines the wave lengths of which would be given by

$$\frac{1}{\lambda} = R\left(\frac{1}{(\frac{3}{2})^2} - \frac{1}{n^2}\right).$$

By examining earlier observations it was actually found that a line had been observed in the spectrum of certain stars which coincided closely with the first line in this series (corresponding to $n = 2$); from analogy with other spectra it was also to be expected that this would be the strongest line. This was regarded as a great triumph for Rydberg's theory and tended to remove all doubt that the new spectrum was actually due to hydrogen. Rydberg's view has therefore been generally accepted by physicists up to the present moment. Recently however the question has been reopened and Fowler (1912) has succeeded in observing the Pickering lines in ordinary laboratory experiments. We shall return to this question again later.

The discovery of these beautiful and simple laws concerning the line spectra of the elements has naturally resulted in many attempts at a theoretical explanation. Such attempts are very alluring because the simplicity of the spectral laws and the exceptional accuracy with which they apply appear to promise that the correct explanation will be very simple and will give valuable information about the properties of matter. I should like to consider some of these theories somewhat more closely, several of which are extremely interesting

and have been developed with the greatest keenness and ingenuity, but unfortunately space does not permit me to do so here. I shall have to limit myself to the statement that not one of the theories so far proposed appears to offer a satisfactory or even a plausible way of explaining the laws of the line spectra. Considering our deficient knowledge of the laws which determine the processes inside atoms it is scarcely possible to give an explanation of the kind attempted in these theories. The inadequacy of our ordinary theoretical conceptions has become especially apparent from the important results which have been obtained in recent years from the theoretical and experimental study of the laws of temperature radiation. You will therefore understand that I shall not attempt to propose an explanation of the spectral laws; on the contrary I shall try to indicate a way in which it appears possible to bring the spectral laws into close connection with other properties of the elements, which appear to be equally inexplicable on the basis of the present state of the science. In these considerations I shall employ the results obtained from the study of temperature radiation as well as the view of atomic structure which has been reached by the study of the radioactive elements.

Laws of temperature radiation. I shall commence by mentioning the conclusions which have been drawn from experimental and theoretical work on temperature radiation.

Let us consider an enclosure surrounded by bodies which

are in temperature equilibrium. In this space there will be a certain amount of energy contained in the rays emitted by the surrounding substances and crossing each other in every direction. By making the assumption that the temperature equilibrium will not be disturbed by the mutual radiation of the various bodies Kirchhoff (1860) showed that the amount of energy per unit volume as well as the distribution of this energy among the various wave lengths is independent of the form and size of the space and of the nature of the surrounding bodies and depends only on the temperature. Kirchhoff's result has been confirmed by experiment, and the amount of energy and its distribution among the various wave lengths and the manner in which it depends on the temperature are now fairly well known from a great amount of experimental work; or, as it is usually expressed, we have a fairly accurate experimental knowledge of the "laws of temperature radiation."

Kirchhoff's considerations were only capable of predicting the existence of a law of temperature radiation, and many physicists have subsequently attempted to find a more thorough explanation of the experimental results. You will perceive that the electromagnetic theory of light together with the electron theory suggests a method of solving this problem. According to the electron theory of matter a body consists of a system of electrons. By making certain definite assumptions concerning the forces acting on the electrons it is possible to calculate their motion and consequently the energy radiated from the body per second in the form of

electromagnetic oscillations of various wave lengths. In a similar manner the absorption of rays of a given wave length by a substance can be determined by calculating the effect of electromagnetic oscillations upon the motion of the electrons. Having investigated the emission and absorption of a body at all temperatures, and for rays of all wave lengths, it is possible, as Kirchhoff has shown, to determine immediately the laws of temperature radiation. Since the result is to be independent of the nature of the body we are justified in expecting an agreement with experiment, even though very special assumptions are made about the forces acting upon the electrons of the hypothetical substance. This naturally simplifies the problem considerably, but it is nevertheless sufficiently difficult and it is remarkable that it has been possible to make any advance at all in this direction. As is well known this has been done by Lorentz (1903). He calculated the emissive as well as the absorptive power of a metal for long wave lengths, using the same assumptions about the motions of the electrons in the metal that Drude (1900) employed in his calculation of the ratio of the electrical and thermal conductivities. Subsequently, by calculating the ratio of the emissive to the absorptive power, Lorentz really obtained an expression for the law of temperature radiation which for long wave lengths agrees remarkably well with experimental facts. In spite of this beautiful and promising result, it has nevertheless become apparent that the electromagnetic theory is incapable of explaining the law of temperature radiation. For, it is possible to show, that, if the inves-

tigation is not confined to oscillations of long wave lengths, as in Lorentz's work, but is also extended to oscillations corresponding to small wave lengths, results are obtained which are contrary to experiment. This is especially evident from Jeans' investigations (1905) in which he employed a very interesting statistical method first proposed by Lord Rayleigh.

We are therefore compelled to assume, that the classical electrodynamics does not agree with reality, or expressed more carefully, that it cannot be employed in calculating the absorption and emission of radiation by atoms. Fortunately, the law of temperature radiation has also successfully indicated the direction in which the necessary changes in the electrodynamics are to be sought. Even before the appearance of the papers by Lorentz and Jeans, Planck (1900) had derived theoretically a formula for the black body radiation which was in good agreement with the results of experiment. Planck did not limit himself exclusively to the classical electrodynamics, but introduced the further assumption that a system of oscillating electrical particles (elementary resonators) will neither radiate nor absorb energy continuously, as required by the ordinary electrodynamics, but on the contrary will radiate and absorb discontinuously. The energy contained within the system at any moment is always equal to a whole multiple of the so-called quantum of energy the magnitude of which is equal to $h\nu$, where h is Planck's constant and ν is the frequency of oscillation of the system per second. In formal respects Planck's theory leaves much to be desired; in certain calculations the ordinary electro-

dynamics is used, while in others assumptions distinctly at variance with it are introduced without any attempt being made to show that it is possible to give a consistent explanation of the procedure used. Planck's theory would hardly have acquired general recognition merely on the ground of its agreement with experiments on black body radiation, but, as you know, the theory has also contributed quite remarkably to the elucidation of many different physical phenomena, such as specific heats, photoelectric effect, X-rays and the absorption of heat rays by gases. These explanations involve more than the qualitative assumption of a discontinuous transformation of energy, for with the aid of Planck's constant h it seems to be possible, at least approximately, to account for a great number of phenomena about which nothing could be said previously. It is therefore hardly too early to express the opinion that, whatever the final explanation will be, the discovery of "energy quanta" must be considered as one of the most important results arrived at in physics, and must be taken into consideration in investigations of the properties of atoms and particularly in connection with any explanation of the spectral laws in which such phenomena as the emission and absorption of electromagnetic radiation are concerned.

The nuclear theory of the atom. We shall now consider the second part of the foundation on which we shall build, namely the conclusions arrived at from experiments

with the rays emitted by radioactive substances. I have previously here in the Physical Society had the opportunity of speaking of the scattering of α rays in passing through thin plates, and to mention how Rutherford (1911) has proposed a theory for the structure of the atom in order to explain the remarkable and unexpected results of these experiments. I shall, therefore, only remind you that the characteristic feature of Rutherford's theory is the assumption of the existence of a positively charged nucleus inside the atom. A number of electrons are supposed to revolve in closed orbits around the nucleus, the number of these electrons being sufficient to neutralize the positive charge of the nucleus. The dimensions of the nucleus are supposed to be very small in comparison with the dimensions of the orbits of the electrons, and almost the entire mass of the atom is supposed to be concentrated in the nucleus.

According to Rutherford's calculation the positive charge of the nucleus corresponds to a number of electrons equal to about half the atomic weight. This number coincides approximately with the number of the particular element in the periodic system and it is therefore natural to assume that the number of electrons in the atom is exactly equal to this number. This hypothesis, which was first stated by van den Broek (1912), opens the possibility of obtaining a simple explanation of the periodic system. This assumption is strongly confirmed by experiments on the elements of small atomic weight. In the first place, it is evident that according to Rutherford's theory the α particle is the same

as the nucleus of a helium atom. Since the α particle has a double positive charge it follows immediately that a neutral helium atom contains two electrons. Further the concordant results obtained from calculations based on experiments as different as the diffuse scattering of X-rays and the decrease in velocity of α rays in passing through matter render the conclusion extremely likely that a hydrogen atom contains only a single electron. This agrees most beautifully with the fact that J. J. Thomson in his well-known experiments on rays of positive electricity has never observed a hydrogen atom with more than a single positive charge, while all other elements investigated may have several charges.

Let us now assume that a hydrogen atom simply consists of an electron revolving around a nucleus of equal and opposite charge, and of a mass which is very large in comparison with that of the electron. It is evident that this assumption may explain the peculiar position already referred to which hydrogen occupies among the elements, but it appears at the outset completely hopeless to attempt to explain anything at all of the special properties of hydrogen, still less its line spectrum, on the basis of considerations relating to such a simple system.

Let us assume for the sake of brevity that the mass of the nucleus is infinitely large in proportion to that of the electron, and that the velocity of the electron is very small in comparison with that of light. If we now temporarily disregard the energy radiation, which, according to the ordinary electrodynamics, will accompany the accelerated motion of

the electron, the latter in accordance with Kepler's first law will describe an ellipse with the nucleus in one of the foci. Denoting the frequency of revolution by ω, and the major axis of the ellipse by $2a$ we find that

$$\omega^2 = \frac{2W^3}{\pi^2 e^4 m}, \quad 2a = \frac{e^2}{W}, \tag{4}$$

where e is the charge of the electron and m its mass, while W is the work which must be added to the system in order to remove the electron to an infinite distance from the nucleus.

These expressions are extremely simple and they show that the magnitude of the frequency of revolution as well as the length of the major axis depend only on W, and are independent of the eccentricity of the orbit. By varying W we may obtain all possible values for ω and $2a$. This condition shows, however, that it is not possible to employ the above formulae directly in calculating the orbit of the electron in a hydrogen atom. For this it will be necessary to assume that the orbit of the electron cannot take on all values, and in any event, the line spectrum clearly indicates that the oscillations of the electron cannot vary continuously between wide limits. The impossibility of making any progress with a simple system like the one considered here might have been foretold from a consideration of the dimensions involved; for with the aid of e and m alone it is impossible to obtain a quantity which can be interpreted as a diameter of an atom or as a frequency.

If we attempt to account for the radiation of energy in

the manner required by the ordinary electrodynamics it will only make matters worse. As a result of the radiation of energy W would continually increase, and the above expressions (4) show that at the same time the frequency of revolution of the system would increase, and the dimensions of the orbit decrease. This process would not stop until the particles had approached so closely to one another that they no longer attracted each other. The quantity of energy which would be radiated away before this happened would be very great. If we were to treat these particles as geometrical points this energy would be infinitely great, and with the dimensions of the electrons as calculated from their mass (about 10^{-13} cm.), and of the nucleus as calculated by Rutherford (about 10^{-12} cm.), this energy would be many times greater than the energy changes with which we are familiar in ordinary atomic processes.

It can be seen that it is impossible to employ Rutherford's atomic model so long as we confine ourselves exclusively to the ordinary electrodynamics. But this is nothing more than might have been expected. As I have mentioned we may consider it to be an established fact that it is impossible to obtain a satisfactory explanation of the experiments on temperature radiation with the aid of electrodynamics, no matter what atomic model be employed. The fact that the deficiencies of the atomic model we are considering stand out so plainly is therefore perhaps no serious drawback; even though the defects of other atomic models are much better concealed they must nevertheless be present and will be just

as serious.

Quantum theory of spectra. Let us now try to overcome these difficulties by applying Planck's theory to the problem.

It is readily seen that there can be no question of a direct application of Planck's theory. This theory is concerned with the emission and absorption of energy in a system of electrical particles, which oscillate with a given frequency per second, dependent only on the nature of the system and independent of the amount of energy contained in the system. In a system consisting of an electron and a nucleus the period of oscillation corresponds to the period of revolution of the electron. But the formula (4) for ω shows that the frequency of revolution depends upon W, i.e. on the energy of the system. Still the fact that we cannot immediately apply Planck's theory to our problem is not as serious as it might seem to be, for in assuming Planck's theory we have manifestly acknowledged the inadequacy of the ordinary electrodynamics and have definitely parted with the coherent group of ideas on which the latter theory is based. In fact in taking such a step we cannot expect that all cases of disagreement between the theoretical conceptions hitherto employed and experiment will be removed by the use of Planck's assumption regarding the quantum of the energy momentarily present in an oscillating system. We stand here almost entirely on virgin ground, and upon introducing new

assumptions we need only take care not to get into contradiction with experiment. Time will have to show to what extent this can be avoided; but the safest way is, of course, to make as few assumptions as possible.

With this in mind let us first examine the experiments on temperature radiation. The subject of direct observation is the distribution of radiant energy over oscillations of the various wave lengths. Even though we may assume that this energy comes from systems of oscillating particles, we know little or nothing about these systems. No one has ever seen a Planck's resonator, nor indeed even measured its frequency of oscillation; we can observe only the period of oscillation of the radiation which is emitted. It is therefore very convenient that it is possible to show that to obtain the laws of temperature radiation it is not necessary to make any assumptions about the systems which emit the radiation except that the amount of energy emitted each time shall be equal to $h\nu$, where h is Planck's constant and ν is the frequency of the radiation. Indeed, it is possible to derive Planck's law of radiation from this assumption alone, as shown by Debye, who employed a method which is a combination of that of Planck and of Jeans. Before considering any further the nature of the oscillating systems let us see whether it is possible to bring this assumption about the emission of radiation into agreement with the spectral laws.

If the spectrum of some element contains a spectral line corresponding to the frequency ν it will be assumed that one of the atoms of the element (or some other elementary sys-

tem) can emit an amount of energy $h\nu$. Denoting the energy of the atom before and after the emission of the radiation by E_1 and E_2 we have

$$h\nu = E_1 - E_2 \text{ or } \nu = \frac{E_1}{h} - \frac{E_2}{h}. \tag{5}$$

During the emission of the radiation the system may be regarded as passing from one state to another; in order to introduce a name for these states, we shall call them "stationary" states, simply indicating thereby that they form some kind of waiting places between which occurs the emission of the energy corresponding to the various spectral lines. As previously mentioned the spectrum of an element consists of a series of lines whose wave lengths may be expressed by the formula (2). By comparing this expression with the relation given above it is seen that—since $\nu = \frac{c}{\lambda}$, where c is the velocity of light—each of the spectral lines may be regarded as being emitted by the transition of a system between two stationary states in which the energy apart from an additive arbitrary constant is given by $-chF_r(n_1)$ and $-chF_s(n_2)$ respectively. Using this interpretation the combination principle asserts that a series of stationary states exists for the given system, and that it can pass from one to any other of these states with the emission of a monochromatic radiation. We see, therefore, that with a simple extension of our first assumption it is possible to give a formal explanation of the most general law of line spectra.

Hydrogen spectrum. This result encourages us to make an attempt to obtain a clear conception of the stationary states which have so far only been regarded as formal. With this end in view, we naturally turn to the spectrum of hydrogen. The formula applying to this spectrum is given by the expression

$$\frac{1}{\lambda} = \frac{R}{n_1^2} - \frac{R}{n_2^2}.$$

According to our assumption this spectrum is produced by transitions between a series of stationary states of a system, concerning which we can for the present only say that the energy of the system in the nth state, apart from an additive constant, is given by $-\frac{Rhc}{n^2}$. Let us now try to find a connection between this and the model of the hydrogen atom. We assume that in the calculation of the frequency of revolution of the electron in the stationary states of the atom it will be possible to employ the above formula for ω. It is quite natural to make this assumption; since, in trying to form a reasonable conception of the stationary states, there is, for the present at least, no other means available besides the ordinary mechanics.

Corresponding to the nth stationary state in formula (4) for ω, let us by way of experiment put $W = \frac{Rhc}{n^2}$. This gives us

$$\omega_n^2 = \frac{2}{\pi^2} \frac{R^3 h^3 c^3}{e^4 m n^6}. \tag{6}$$

The radiation of light corresponding to a particular spec-

tral line is according to our assumption emitted by a transition between two stationary states, corresponding to two different frequencies of revolution, and we are not justified in expecting any simple relation between these frequencies of revolution of the electron and the frequency of the emitted radiation. You understand, of course, that I am by no means trying to give what might ordinarily be described as an explanation; nothing has been said here about how or why the radiation is emitted. On one point, however, we may expect a connection with the ordinary conceptions; namely, that it will be possible to calculate the emission of slow electromagnetic oscillations on the basis of the classical electrodynamics. This assumption is very strongly supported by the result of Lorentz's calculations which have already been described. From the formula for ω it is seen that the frequency of revolution decreases as n increases, and that the expression $\dfrac{\omega_n}{\omega_{n+1}}$ approaches the value 1.

According to what has been said above, the frequency of the radiation corresponding to the transition between the $(n+1)$th and the nth stationary state is given by

$$\nu = Rc\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right).$$

If n is very large this expression is approximately equal to

$$\nu = 2Rc/n^3.$$

In order to obtain a connection with the ordinary electrodynamics let us now place this frequency equal to the fre-

quency of revolution, that is

$$\omega_n = 2Rc/n^3.$$

Introducing this value of ω_n in (6) we see that n disappears from the equation, and further that the equation will be satisfied only if

$$R = \frac{2\pi^2 e^4 m}{ch^3}. \tag{7}$$

The constant R is very accurately known, and is, as I have said before, equal to 109675. By introducing the most recent values for e, m and h the expression on the right-hand side of the equation becomes equal to $1.09 \cdot 10^5$. The agreement is as good as could be expected, considering the uncertainty in the experimental determination of the constants e, m and h. The agreement between our calculations and the classical electrodynamics is, therefore, fully as good as we are justified in expecting.

We cannot expect to obtain a corresponding explanation of the frequency values of the other stationary states. Certain simple formal relations apply, however, to all the stationary states. By introducing the expression, which has been found for R, we get for the nth state $W_n = \frac{1}{2}nh\omega_n$. This equation is entirely analogous to Planck's assumption concerning the energy of a resonator. W in our system is readily shown to be equal to the average value of the kinetic energy of the electron during a single revolution. The energy of a resonator was shown by Planck you may remember to

be always equal to $nh\nu$. Further the average value of the kinetic energy of Planck's resonator is equal to its potential energy, so that the average value of the kinetic energy of the resonator, according to Planck, is equal to $\frac{1}{2}nh\omega$. This analogy suggests another manner of presenting the theory, and it was just in this way that I was originally led into these considerations. When we consider how differently the equation is employed here and in Planck's theory it appears to me misleading to use this analogy as a foundation, and in the account I have given I have tried to free myself as much as possible from it.

Let us continue with the elucidation of the calculations, and in the expression for $2a$ introduce the value of W which corresponds to the nth stationary state. This gives us

$$2a = n^2 \cdot \frac{e^2}{chR} = n^2 \cdot \frac{h^2}{2\pi^2 me^2} = n^2 \cdot 1.1 \cdot 10^{-8}. \qquad (8)$$

It is seen that for small values of n, we obtain values for the major axis of the orbit of the electron which are of the same order of magnitude as the values of the diameters of the atoms calculated from the kinetic theory of gases. For large values of n, $2a$ becomes very large in proportion to the calculated dimensions of the atoms. This, however, does not necessarily disagree with experiment. Under ordinary circumstances a hydrogen atom will probably exist only in the state corresponding to $n = 1$. For this state W will have its greatest value and, consequently, the atom will have emitted the largest amount of energy possible; this will therefore

represent the most stable state of the atom from which the system cannot be transferred except by adding energy to it from without. The large values for $2a$ corresponding to large n need not, therefore, be contrary to experiment; indeed, we may in these large values seek an explanation of the fact, that in the laboratory it has hitherto not been possible to observe the hydrogen lines corresponding to large values of n in Balmer's formula, while they have been observed in the spectra of certain stars. In order that the large orbits of the electrons may not be disturbed by electrical forces from the neighbouring atoms the pressure will have to be very low, so low, indeed, that it is impossible to obtain sufficient light from a Geissler tube of ordinary dimensions. In the stars, however, we may assume that we have to do with hydrogen which is exceedingly attenuated and distributed throughout an enormously large region of space.

The Pickering lines. You have probably noticed that we have not mentioned at all the spectrum found in certain stars which according to the opinion then current was assigned to hydrogen, and together with the ordinary hydrogen spectrum was considered by Rydberg to form a connected system of lines completely analogous to the spectra of other elements. You have probably also perceived that difficulties would arise in interpreting this spectrum by means of the assumptions which have been employed. If such an attempt were to be made it would be necessary to give up the simple

considerations which lead to the expression (7) for the constant R. We shall see, however, that it appears possible to explain the occurrence of this spectrum in another way. Let us suppose that it is not due to hydrogen, but to some other simple system consisting of a single electron revolving about a nucleus with an electrical charge Ne. The expression for ω becomes then

$$\omega^2 = \frac{2}{\pi^2}\frac{W^3}{N^2e^4m}.$$

Repeating the same calculations as before only in the inverse order we find, that this system will emit a line spectrum given by the expression

$$\frac{1}{\lambda} = \frac{2\pi^2N^2e^4m}{ch^3}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = R\left(\frac{1}{\left(\frac{n_1}{N}\right)^2} - \frac{1}{\left(\frac{n_2}{N}\right)^2}\right). \quad (9)$$

By comparing this formula with the formula for Pickering's and Rydberg's series, we see that the observed lines can be explained on the basis of the theory, if it be assumed that the spectrum is due to an electron revolving about a nucleus with a charge $2e$, or according to Rutherford's theory around the nucleus of a helium atom. The fact that the spectrum in question is not observed in an ordinary helium tube, but only in stars, may be accounted for by the high degree of ionization which is required for the production of this spectrum; a neutral helium atom contains of course two electrons while the system under consideration contains only one.

These conclusions appear to be supported by experiment. Fowler, as I have mentioned, has recently succeeded in observing Pickering's and Rydberg's lines in a laboratory experiment. By passing a very heavy current through a mixture of hydrogen and helium Fowler observed not only these lines but also a new series of lines. This new series was of the same general type, the wave length being given approximately by

$$\frac{1}{\lambda} = R\left(\frac{1}{(\frac{3}{2})^2} - \frac{1}{(n+\frac{1}{2})^2}\right).$$

Fowler interpreted all the observed lines as the hydrogen spectrum sought for. With the observation of the latter series of lines, however, the basis of the analogy between the hypothetical hydrogen spectrum and the other spectra disappeared, and thereby also the foundation upon which Rydberg had founded his conclusions; on the contrary it is seen, that the occurrence of the lines was exactly what was to be expected on our view.

In the following table the first column contains the wave lengths measured by Fowler, while the second contains the limiting values of the experimental errors given by him; in the third column we find the products of the wave lengths by the quantity $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \cdot 10^{10}$; the values employed for n_1 and n_2 are enclosed in parentheses in the last column.

The values of the products are seen to be very nearly equal, while the deviations are of the same order of magnitude as the limits of experimental error. The value of the

$\lambda \cdot 10^8$	Limit of error	$\lambda \cdot \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \cdot 10^{10}$	
4685.98	0.01	22779.1	(3 : 4)
3203.30	0.05	22779.0	(3 : 5)
2733.34	0.05	22777.8	(3 : 6)
2511.31	0.05	22778.3	(3 : 7)
2385.47	0.05	22777.9	(3 : 8)
2306.20	0.10	22777.3	(3 : 9)
2252.88	0.10	22779.1	(3 : 10)
5410.5	1.0	22774	(4 : 7)
4541.3	0.25	22777	(4 : 9)
4200.3	0.5	22781	(4 : 11)

product

$$\lambda \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

should for this spectrum, according to the formula (9), be exactly $\frac{1}{4}$ of the corresponding product for the hydrogen spectrum. From the tables on pages 2 and 25 we find for these products 91153 and 22779, and dividing the former by the latter we get 4.0016. This value is very nearly equal to 4; the deviation is, however, much greater than can be accounted for in any way by the errors of the experiments. It has been easy, however, to find a theoretical explanation of this point. In all the foregoing calculations we have assumed that the

mass of the nucleus is infinitely great compared to that of the electron. This is of course not the case, even though it holds to a very close approximation; for a hydrogen atom the ratio of the mass of the nucleus to that of the electron will be about 1850 and for a helium atom four times as great.

If we consider a system consisting of an electron revolving about a nucleus with a charge Ne and a mass M, we find the following expression for the frequency of revolution of the system:

$$\omega^2 = \frac{2}{\pi^2} \frac{W^3(M+m)}{N^2 e^4 M m}.$$

From this formula we find in a manner quite similar to that previously employed that the system will emit a line spectrum, the wave lengths of which are given by the formula

$$\frac{1}{\lambda} = \frac{2\pi^2 N^2 e^4 m M}{c h^3 (M+m)} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right). \tag{10}$$

If with the aid of this formula we try to find the ratio of the product for the hydrogen spectrum, to that of the hypothetical helium spectrum we get the value 4.00163 which is in complete agreement with the preceding value calculated from the experimental observations.

I must further mention that Evans has made some experiments to determine whether the spectrum in question is due to hydrogen or helium. He succeeded in observing one of the lines in very pure helium; there was, at any rate, not enough hydrogen present to enable the hydrogen lines to be

observed. Since in any event Fowler does not seem to consider such evidence as conclusive it is to be hoped that these experiments will be continued. There is, however, also another possibility of deciding this question. As is evident from the formula (10), the helium spectrum under consideration should contain, besides the lines observed by Fowler, a series of lines lying close to the ordinary hydrogen lines. These lines may be obtained by putting $n_1 = 4$, $n_2 = 6$, 8, 10, etc. Even if these lines were present, it would be extremely difficult to observe them on account of their position with regard to the hydrogen lines, but should they be observed this would probably also settle the question of the origin of the spectrum, since no reason would seem to be left to assume the spectrum to be due to hydrogen.

Other spectra. For the spectra of other elements the problem becomes more complicated, since the atoms contain a larger number of electrons. It has not yet been possible on the basis of this theory to explain any other spectra besides those which I have already mentioned. On the other hand it ought to be mentioned that the general laws applying to the spectra are very simply interpreted on the basis of our assumptions. So far as the combination principle is concerned its explanation is obvious. In the method we have employed our point of departure was largely determined by this particular principle. But a simple explanation can be also given of the other general law, namely, the occurrence of Rydberg's

constant in all spectral formulae. Let us assume that the spectra under consideration, like the spectrum of hydrogen, are emitted by a neutral system, and that they are produced by the binding of an electron previously removed from the system. If such an electron revolves about the nucleus in an orbit which is large in proportion to that of the other electrons it will be subjected to forces much the same as the electron in a hydrogen atom, since the inner electrons individually will approximately neutralize the effect of a part of the positive charge of the nucleus. We may therefore assume that for this system there will exist a series of stationary states in which the motion of the outermost electron is approximately the same as in the stationary states of a hydrogen atom. I shall not discuss these matters any further, but shall only mention that they lead to the conclusion that Rydberg's constant is not exactly the same for all elements. The expression for this constant will in fact contain the factor $\frac{M}{M+m}$, where M is the mass of the nucleus. The correction is exceedingly small for elements of large atomic weight, but for hydrogen it is, from the point of view of spectrum analysis, very considerable. If the procedure employed leads to correct results, it is not therefore permissible to calculate Rydberg's constant directly from the hydrogen spectrum; the value of the universal constant should according to the theory be 109735 and not 109675.

I shall not tire you any further with more details; I hope to return to these questions here in the Physical Society,

and to show how, on the basis of the underlying ideas, it is possible to develop a theory for the structure of atoms and molecules. Before closing I only wish to say that I hope I have expressed myself sufficiently clearly so that you have appreciated the extent to which these considerations conflict with the admirably coherent group of conceptions which have been rightly termed the classical theory of electrodynamics. On the other hand, by emphasizing this conflict, I have tried to convey to you the impression that it may be also possible in the course of time to discover a certain coherence in the new ideas.

ESSAY II*

ON THE SERIES SPECTRA OF THE ELEMENTS

I. INTRODUCTION

The subject on which I have the honour to speak here, at the kind invitation of the Council of your society, is very extensive and it would be impossible in a single address to give a comprehensive survey of even the most important results obtained in the theory of spectra. In what follows I shall try merely to emphasize some points of view which seem to me important when considering the present state of the theory of spectra and the possibilities of its development in the near future. I regret in this connection not to have time to describe the history of the development of spectral theories, although this would be of interest for our purpose. No difficulty, however, in understanding this lecture need be experienced on this account, since the points of view underlying previous attempts to explain the spectra differ fundamentally from those upon which the following considerations rest. This difference exists both in the development of our ideas about the structure of the atom and in the manner in which these ideas are used in explaining the spectra.

We shall assume, according to Rutherford's theory, that an atom consists of a positively charged nucleus with a num-

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ber of electrons revolving about it. Although the nucleus is assumed to be very small in proportion to the size of the whole atom, it will contain nearly the entire mass of the atom. I shall not state the reasons which led to the establishment of this nuclear theory of the atom, nor describe the very strong support which this theory has received from very different sources. I shall mention only that result which lends such charm and simplicity to the modern development of the atomic theory. I refer to the idea that the number of electrons in a neutral atom is exactly equal to the number, giving the position of the element in the periodic table, the so-called "atomic number." This assumption, which was first proposed by van den Broek, immediately suggests the possibility ultimately of deriving the explanation of the physical and chemical properties of the elements from their atomic numbers. If, however, an explanation of this kind is attempted on the basis of the classical laws of mechanics and electrodynamics, insurmountable difficulties are encountered. These difficulties become especially apparent when we consider the spectra of the elements. In fact, the difficulties are here so obvious that it would be a waste of time to discuss them in detail. It is evident that systems like the nuclear atom, if based upon the usual mechanical and electrodynamical conceptions, would not even possess sufficient stability to give a spectrum consisting of sharp lines.

In this lecture I shall use the ideas of the quantum theory. It will not be necessary, particularly here in Berlin, to consider in detail how Planck's fundamental work on tem-

perature radiation has given rise to this theory, according to which the laws governing atomic processes exhibit a definite element of discontinuity. I shall mention only Planck's chief result about the properties of an exceedingly simple kind of atomic system, the Planck "oscillator." This consists of an electrically charged particle which can execute harmonic oscillations about its position of equilibrium with a frequency independent of the amplitude. By studying the statistical equilibrium of a number of such systems in a field of radiation Planck was led to the conclusion that the emission and absorption of radiation take place in such a manner, that, so far as a statistical equilibrium is concerned only certain distinctive states of the oscillator are to be taken into consideration. In these states the energy of the system is equal to a whole multiple of a so-called "energy quantum," which was found to be proportional to the frequency of the oscillator. The particular energy values are therefore given by the well-known formula

$$E_n = nh\omega, \tag{1}$$

where n is a whole number, ω the frequency of vibration of the oscillator, and h is Planck's constant.

If we attempt to use this result to explain the spectra of the elements, however, we encounter difficulties, because the motion of the particles in the atom, in spite of its simple structure, is in general exceedingly complicated compared with the motion of a Planck oscillator. The question then arises, how Planck's result ought to be generalized in order

to make its application possible. Different points of view immediately suggest themselves. Thus we might regard this equation as a relation expressing certain characteristic properties of the distinctive motions of an atomic system and try to obtain the general form of these properties. On the other hand, we may also regard equation (1) as a statement about a property of the process of radiation and inquire into the general laws which control this process.

In Planck's theory it is taken for granted that the frequency of the radiation emitted and absorbed by the oscillator is equal to its own frequency, an assumption which may be written

$$\nu \equiv \omega, \tag{2}$$

if in order to make a sharp distinction between the frequency of the emitted radiation and the frequency of the particles in the atoms, we here and in the following denote the former by ν and the latter by ω. We see, therefore, that Planck's result may be interpreted to mean, that the oscillator can emit and absorb radiation only in "radiation quanta" of magnitude

$$\Delta E = h\nu. \tag{3}$$

It is well known that ideas of this kind led Einstein to a theory of the photoelectric effect. This is of great importance, since it represents the first instance in which the quantum theory was applied to a phenomenon of non-statistical character. I shall not here discuss the familiar difficulties to which the "hypothesis of light quanta" leads in connection with the

phenomena of interference, for the explanation of which the classical theory of radiation has shown itself to be so remarkably suited. Above all I shall not consider the problem of the nature of radiation, I shall only attempt to show how it has been possible in a purely formal manner to develop a spectral theory, the essential elements of which may be considered as a simultaneous rational development of the two ways of interpreting Planck's result.

II. GENERAL PRINCIPLES OF THE QUANTUM THEORY OF SPECTRA

In order to explain the appearance of line spectra we are compelled to assume that the emission of radiation by an atomic system takes place in such a manner that it is not possible to follow the emission in detail by means of the usual conceptions. Indeed, these do not even offer us the means of calculating the frequency of the emitted radiation. We shall see, however, that it is possible to give a very simple explanation of the general empirical laws for the frequencies of the spectral lines, if for each emission of radiation by the atom we assume the fundamental law to hold, that during the entire period of the emission the radiation possesses one and the same frequency ν, connected with the total energy emitted by the *frequency relation*

$$h\nu = E' - E''. \tag{4}$$

Here E' and E'' represent the energy of the system before and after the emission.

If this law is assumed, the spectra do not give us information about the motion of the particles in the atom, as is supposed in the usual theory of radiation, but only a knowledge of the energy changes in the various processes which can occur in the atom. From this point of view the spectra show the existence of certain, definite energy values corresponding to certain distinctive states of the atoms. These states will be called the *stationary states* of the atoms, since we shall assume that the atom can remain a finite time in each state, and can leave this state only by a process of transition to another stationary state. Notwithstanding the fundamental departure from the ordinary mechanical and electrodynamical conceptions, we shall see, however, that it is possible to give a rational interpretation of the evidence provided by the spectra on the basis of these ideas.

Although we must assume that the ordinary mechanics cannot be used to describe the transitions between the stationary states, nevertheless, it has been found possible to develop a consistent theory on the assumption that the motion in these states can be described by the use of the ordinary mechanics. Moreover, although the process of radiation cannot be described on the basis of the ordinary theory of electrodynamics, according to which the nature of the radiation emitted by an atom is directly related to the harmonic components occurring in the motion of the system, there is found, nevertheless, to exist a far-reaching *correspondence*

between the various types of possible transitions between the stationary states on the one hand and the various harmonic components of the motion on the other hand. This correspondence is of such a nature, that the present theory of spectra is in a certain sense to be regarded as a rational generalization of the ordinary theory of radiation.

Hydrogen spectrum. In order that the principal points may stand out as clearly as possible I shall, before considering the more complicated types of series spectra, first consider the simplest spectrum, namely, the series spectrum of hydrogen. This spectrum consists of a number of lines whose frequencies are given with great exactness by Balmer's formula

$$\nu = \frac{K}{(n'')^2} - \frac{K}{(n')^2}, \tag{5}$$

where K is a constant, and n' and n'' are whole numbers. If we put $n'' = 2$ and give to n' the values 3, 4, etc., we get the well-known Balmer series of hydrogen. If we put $n'' = 1$ or $n'' = 3$ we obtain respectively the ultra-violet and infra-red series. We shall assume the hydrogen atom simply to consist of a positively charged nucleus with a single electron revolving about it. For the sake of simplicity we shall suppose the mass of the nucleus to be infinite in comparison with the mass of the electron, and further we shall disregard the small variations in the motion due to the change in mass of the electron with its velocity. With these simplifications

the electron will describe a closed elliptical orbit with the nucleus at one of the foci. The frequency of revolution ω and the major axis $2a$ of the orbit will be connected with the energy of the system by the following equations:

$$\omega = \sqrt{\frac{2W^3}{\pi^2 e^4 m}}, \quad 2a = \frac{e^2}{W}. \tag{6}$$

Here e is the charge of the electron and m its mass, while W is the work required to remove the electron to infinity.

The simplicity of these formulae suggests the possibility of using them in an attempt to explain the spectrum of hydrogen. This, however, is not possible so long as we use the classical theory of radiation. It would not even be possible to understand how hydrogen could emit a spectrum consisting of sharp lines; for since ω varies with W, the frequency of the emitted radiation would vary continuously during the emission. We can avoid these difficulties if we use the ideas of the quantum theory. If for each line we form the product $h\nu$ by multiplying both sides of (5) by h, then, since the right-hand side of the resulting relation may be written as the difference of two simple expressions, we are led by comparison with formula (4) to the assumption that the separate lines of the spectrum will be emitted by transitions between two stationary states, forming members of an infinite series of states, in which the energy in the nth state apart from an arbitrary additive constant is determined by the expression

$$E_n = -\frac{Kh}{n^2}. \tag{7}$$

The negative sign has been chosen because the energy of the atom will be most simply characterized by the work W required to remove the electron completely from the atom. If we now substitute $\frac{Kh}{n^2}$ for W in formula (6), we obtain the following expression for the frequency and the major axis in the nth stationary state:

$$\omega_n = \frac{1}{n^3}\sqrt{\frac{2h^3K^3}{\pi^2e^4m}}, \quad 2a_n = \frac{n^2e^2}{hK}. \tag{8}$$

A comparison between the motions determined by these equations and the distinctive states of a Planck resonator may be shown to offer a theoretical determination of the constant K. Instead of doing this I shall show how the value of K can be found by a simple comparison of the spectrum emitted with the motion in the stationary states, a comparison which at the same time will lead us to the principle of correspondence.

We have assumed that each hydrogen line is the result of a transition between two stationary states of the atom corresponding to different values of n. Equations (8) show that the frequency of revolution and the major axis of the orbit can be entirely different in the two states, since, as the energy decreases, the major axis of the orbit becomes smaller and the frequency of revolution increases. In general, therefore, it will be impossible to obtain a relation between the frequency of revolution of the electrons and the frequency of the radiation as in the ordinary theory of radiation. If,

however, we consider the ratio of the frequencies of revolution in two stationary states corresponding to given values of n' and n'', we see that this ratio approaches unity as n' and n'' gradually increase, if at the same time the difference $n' - n''$ remains unchanged. By considering transitions corresponding to large values of n' and n'' we may therefore hope to establish a certain connection with the ordinary theory. For the frequency of the radiation emitted by a transition, we get according to (5)

$$\nu = \frac{K}{(n'')^2} - \frac{K}{(n')^2} = (n' - n'')K\,\frac{n' + n''}{(n')^2(n'')^2}. \qquad (9)$$

If now the numbers n' and n'' are large in proportion to their difference, we see that by equations (8) this expression may be written approximately,

$$\nu \sim (n' - n'')\omega\sqrt{\frac{2\pi^2 e^4 m}{Kh^3}}, \qquad (10)$$

where ω represents the frequency of revolution in the one or the other of the two stationary states. Since $n' - n''$ is a whole number, we see that the first part of this expression, i.e. $(n' - n'')\omega$, is the same as the frequency of one of the harmonic components into which the elliptical motion may be decomposed. This involves the well-known result that for a system of particles having a periodic motion of frequency ω, the displacement ξ of the particles in a given direction in space may be represented as a function of the time by a

trigonometric series of the form

$$\xi = \sum C_\tau \cos 2\pi(\tau\omega t + c_\tau), \tag{11}$$

where the summation is to be extended over all positive integral values of τ.

We see, therefore, that the frequency of the radiation emitted by a transition between two stationary states, for which the numbers n' and n'' are large in proportion to their difference, will coincide with the frequency of one of the components of the radiation, which according to the ordinary ideas of radiation would be expected from the motion of the atom in these states, provided the last factor on the right-hand side of equation (10) is equal to 1. This condition, which is identical to the condition

$$K = \frac{2\pi^2 e^4 m}{h^3}, \tag{12}$$

is in fact fulfilled, if we give to K its value as found from measurements on the hydrogen spectrum, and if for e, m and h we use the values obtained directly from experiment. This agreement clearly gives us a *connection between the spectrum and the atomic model of hydrogen*, which is as close as could reasonably be expected considering the fundamental difference between the ideas of the quantum theory and of the ordinary theory of radiation.

The correspondence principle. Let us now consider somewhat more closely this relation between the spectra one

would expect on the basis of the quantum theory, and on the ordinary theory of radiation. The frequencies of the spectral lines calculated according to both methods agree completely in the region where the stationary states deviate only little from one another. We must not forget, however, that the mechanism of emission in both cases is different. The different frequencies corresponding to the various harmonic components of the motion are emitted simultaneously according to the ordinary theory of radiation and with a relative intensity depending directly upon the ratio of the amplitudes of these oscillations. But according to the quantum theory the various spectral lines are emitted by entirely distinct processes, consisting of transitions from one stationary state to various adjacent states, so that the radiation corresponding to the τth "harmonic" will be emitted by a transition for which $n' - n'' = \tau$. The relative intensity with which each particular line is emitted depends consequently upon the relative probability of the occurrence of the different transitions.

This correspondence between the frequencies determined by the two methods must have a deeper significance and we are led to anticipate that it will also apply to the intensities. This is equivalent to the statement that, when the quantum numbers are large, the relative probability of a particular transition is connected in a simple manner with the amplitude of the corresponding harmonic component in the motion.

This peculiar relation suggests a *general law for the oc-*

currence of transitions between stationary states. Thus we shall assume that even when the quantum numbers are small the possibility of transition between two stationary states is connected with the presence of a certain harmonic component in the motion of the system. If the numbers n' and n'' are not large in proportion to their difference, the numerical value of the amplitudes of these components in the two stationary states may be entirely different. We must be prepared to find, therefore, that the exact connection between the probability of a transition and the amplitude of the corresponding harmonic component in the motion is in general complicated like the connection between the frequency of the radiation and that of the component. From this point of view, for example, the green line H_β of the hydrogen spectrum which corresponds to a transition from the fourth to the second stationary state may be considered in a certain sense to be an "octave" of the red line H_α, corresponding to a transition from the third to the second state, even though the frequency of the first line is by no means twice as great as that of the latter. In fact, the transition giving rise to H_β may be regarded as due to the presence of a harmonic oscillation in the motion of the atom, which is an octave higher than the oscillation giving rise to the emission of H_α.

Before considering other spectra, where numerous opportunities will be found to use this point of view, I shall briefly mention an interesting application to the Planck oscillator. If from (1) and (4) we calculate the frequency, which would correspond to a transition between two particular states of

such an oscillator, we find

$$\nu = (n' - n'')\omega, \tag{13}$$

where n' and n'' are the numbers characterizing the states. It was an essential assumption in Planck's theory that the frequency of the radiation emitted and absorbed by the oscillator is always equal to ω. We see that this assumption is equivalent to the assertion that transitions occur only between two successive stationary states in sharp contrast to the hydrogen atom. According to our view, however, this was exactly what might have been expected, for we must assume that the essential difference between the oscillator and the hydrogen atom is that the motion of the oscillator is simple harmonic. We can see that it is possible to develop a formal theory of radiation, in which the spectrum of hydrogen and the simple spectrum of a Planck oscillator appear completely analogous. This theory can only be formulated by one and the same condition for a system as simple as the oscillator. In general this condition breaks up into two parts, one concerning the fixation of the stationary states, and the other relating to the frequency of the radiation emitted by a transition between these states.

General spectral laws. Although the series spectra of the elements of higher atomic number have a more complicated structure than the hydrogen spectrum, simple laws have been discovered showing a remarkable analogy to the

Balmer formula. Rydberg and Ritz showed that the frequencies in the series spectra of many elements can be expressed by a formula of the type

$$\nu = f_{k''}(n'') - f_{k'}(n'), \tag{14}$$

where n' and n'' are two whole numbers and $f_{k'}$ and $f_{k''}$ are two functions belonging to a series of functions characteristic of the element. These functions vary in a simple manner with n and in particular converge to zero for increasing values of n. The various series of lines are obtained from this formula by allowing the first term $f_{k''}(n'')$ to remain constant, while a series of consecutive whole numbers are substituted for n' in the second term $f_{k'}(n')$. According to the Ritz *combination principle* the entire spectrum may then be obtained by forming every possible combination of two values among all the quantities $f_k(n)$.

The fact that the frequency of each line of the spectrum may be written as the difference of two simple expressions depending upon whole numbers suggests at once that the terms on the right-hand side multiplied by h may be placed equal to the energy in the various stationary states of the atom. The existence in the spectra of the other elements of a number of separate functions of n compels us to assume the presence not of one but of a number of series of stationary states, the energy of the nth state of the kth series apart from an arbitrary additive constant being given by

$$E_k(n) = -hf_k(n). \tag{15}$$

This complicated character of the ensemble of stationary states of atoms of higher atomic number is exactly what was to be expected from the relation between the spectra calculated on the quantum theory, and the decomposition of the motions of the atoms into harmonic oscillations. From this point of view we may regard the simple character of the stationary states of the hydrogen atom as intimately connected with the simple periodic character of this atom. Where the neutral atom contains more than one electron, we find much more complicated motions with correspondingly complicated harmonic components. We must therefore expect a more complicated ensemble of stationary states, if we are still to have a corresponding relation between the motions in the atom and the spectrum. In the course of the lecture we shall trace this correspondence in detail, and we shall be led to a simple explanation of the apparent capriciousness in the occurrence of lines predicted by the combination principle.

The following figure gives a survey of the stationary states of the sodium atom deduced from the series terms.

The stationary states are represented by black dots whose distance from the vertical line a—a is proportional to the numerical value of the energy in the states. The arrows in the figure indicate the transitions giving those lines of the sodium spectrum which appear under the usual conditions of excitation. The arrangement of the states in horizontal rows corresponds to the ordinary arrangement of the "spectral terms" in the spectroscopic tables. Thus, the states in the first row (S) correspond to the variable term in the

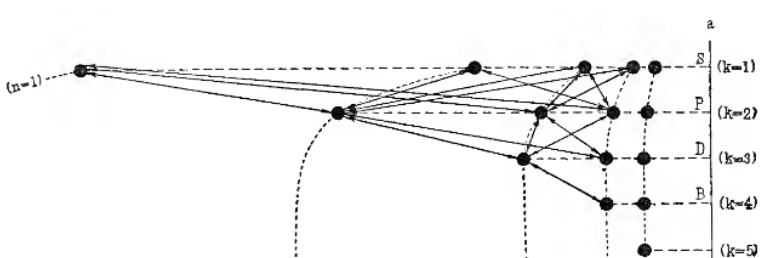


Diagram of the series spectrum of sodium.

"sharp series," the lines of which are emitted by transitions from these states to the first state in the second row. The states in the second row (P) correspond to the variable term in the "principal series" which is emitted by transitions from these states to the first state in the S row. The D states correspond to the variable term in the "diffuse series," which like the sharp series is emitted by transitions to the first state in the P row, and finally the B states correspond to the variable term in the "Bergmann" series (fundamental series), in which transitions take place to the first state in the D row. The manner in which the various rows are arranged with reference to one another will be used to illustrate the more detailed theory which will be discussed later. The apparent capriciousness of the combination principle, which I mentioned, consists in the fact that under the usual conditions of excitation not all the lines belonging to possible combinations of the terms of the sodium spectrum appear,

but only those indicated in the figure by arrows.

The general question of the fixation of the stationary states of an atom containing several electrons presents difficulties of a profound character which are perhaps still far from completely solved. It is possible, however, to obtain an immediate insight into the stationary states involved in the emission of the series spectra by considering the empirical laws which have been discovered about the spectral terms. According to the well-known law discovered by Rydberg for the spectra of elements emitted under the usual conditions of excitation the functions $f_k(n)$ appearing in formula (14) can be written in the form

$$f_k(n) = \frac{K}{n^2}\phi_k(n), \tag{16}$$

where $\phi_k(n)$ represents a function which converges to unity for large values of n. K is the same constant which appears in formula (5) for the spectrum of hydrogen. This result must evidently be explained by supposing the atom to be electrically neutral in these states and one electron to be moving round the nucleus in an orbit the dimensions of which are very large in proportion to the distance of the other electrons from the nucleus. We see, indeed, that in this case the electric force acting on the outer electron will to a first approximation be the same as that acting upon the electron in the hydrogen atom, and the approximation will be the better the larger the orbit.

On account of the limited time I shall not discuss how

this explanation of the universal appearance of Rydberg's constant in the arc spectra is convincingly supported by the investigation of the "spark spectra." These are emitted by the elements under the influence of very strong electrical discharges, and come from ionized not neutral atoms. It is important, however, that I should indicate briefly how the fundamental ideas of the theory and the assumption that in the states corresponding to the spectra one electron moves in an orbit around the others, are both supported by investigations on selective absorption and the excitation of spectral lines by bombardment by electrons.

Absorption and excitation of radiation. Just as we have assumed that each emission of radiation is due to a transition from a stationary state of higher to one of lower energy, so also we must assume absorption of radiation by the atom to be due to a transition in the opposite direction. For an element to absorb light corresponding to a given line in its series spectrum, it is therefore necessary for the atom of this element to be in that one of the two states connected with the line possessing the smaller energy value. If we now consider an element whose atoms in the gaseous state do not combine into molecules, it will be necessary to assume that under ordinary conditions nearly all the atoms exist in that stationary state in which the value of the energy is a minimum. This state I shall call the *normal state*. We must therefore expect that the absorption spectrum of

a monatomic gas will contain only those lines of the series spectrum, whose emission corresponds to transitions to the normal state. This expectation is completely confirmed by the spectra of the alkali metals. The absorption spectrum of sodium vapour, for example, exhibits lines corresponding only to the principal series, which as mentioned in the description of the figure corresponds with transitions to the state of minimum energy. Further confirmation of this view of the process of absorption is given by experiments on *resonance radiation*. Wood first showed that sodium vapour subjected to light corresponding to the first line of the principal series—the familiar yellow line—acquires the ability of again emitting a radiation consisting only of the light of this line. We can explain this by supposing the sodium atom to have been transferred from the normal state to the first state in the second row. The fact that the resonance radiation does not exhibit the same degree of polarization as the incident light is in perfect agreement with our assumption that the radiation from the excited vapour is not a resonance phenomenon in the sense of the ordinary theory of radiation, but on the contrary depends on a process which is not directly connected with the incident radiation.

The phenomenon of the resonance radiation of the yellow sodium line is, however, not quite so simple as I have indicated, since, as you know, this line is really a doublet. This means that the variable terms of the principal series are not simple but are represented by two values slightly different from one another. According to our picture of the

origin of the sodium spectrum this means that the P states in the second row in the figure—as opposed to the S states in the first row—are not simple, but that for each place in this row there are two stationary states. The energy values differ so little from one another that it is impossible to represent them in the figure as separate dots. The emission (and absorption) of the two components of the yellow line are, therefore, connected with two different processes. This was beautifully shown by some later researches of Wood and Dunoyer. They found that if sodium vapour is subjected to radiation from only one of the two components of the yellow line, the resonance radiation, at least at low pressures, consists only of this component. These experiments were later continued by Strutt, and were extended to the case where the exciting line corresponded to the second line in the principal series. Strutt found that the resonance radiation consisted apparently only to a small extent of light of the same frequency as the incident light, while the greater part consisted of the familiar yellow line. This result must appear very astonishing on the ordinary ideas of resonance, since, as Strutt pointed out, no rational connection exists between the frequencies of the first and second lines of the principal series. It is however easily explained from our point of view. From the figure it can be seen that when an atom has been transferred into the second state in the second row, in addition to the direct return to the normal state, there are still two other transitions which may give rise to radiation, namely the transitions to the second state in the first

row and to the first state in the third row. The experiments seem to indicate that the second of these three transitions is most probable, and I shall show later that there is some theoretical justification for this conclusion. By this transition, which results in the emission of an infra-red line which could not be observed with the experimental arrangement, the atom is taken to the second state of the first row, and from this state only one transition is possible, which again gives an infra-red line. This transition takes the atom to the first state in the second row, and the subsequent transition to the normal state then gives rise to the yellow line. Strutt discovered another equally surprising result, that this yellow resonance radiation seemed to consist of both components of the first line of the principal series, even when the incident light consisted of only one component of the second line of the principal series. This is in beautiful agreement with our picture of the phenomenon. We must remember that the states in the first row are simple, so when the atom has arrived in one of these it has lost every possibility of later giving any indication from which of the two states in the second row it originally came.

Sodium vapour, in addition to the absorption corresponding to the lines of the principal series, exhibits a *selective absorption in a continuous spectral region* beginning at the limit of this series and extending into the ultra-violet. This confirms in a striking manner our assumption that the absorption of the lines of the principal series of sodium results in final states of the atom in which one of the electrons re-

volves in larger and larger orbits. For we must assume that this continuous absorption corresponds to transitions from the normal state to states in which the electron is in a position to remove itself infinitely far from the nucleus. This phenomenon exhibits a complete analogy with the *photoelectric effect* from an illuminated metal plate in which, by using light of a suitable frequency, electrons of any velocity can be obtained. The frequency, however, must always lie above a certain limit connected according to Einstein's theory in a simple manner with the energy necessary to bring an electron out of the metal.

This view of the origin of the emission and absorption spectra has been confirmed in a very interesting manner by experiments on the *excitation of spectral lines and production of ionization by electron bombardment.* The chief advance in this field is due to the well-known experiments of Franck and Hertz. These investigators obtained their first important results from their experiments on mercury vapour, whose properties particularly facilitate such experiments. On account of the great importance of the results, these experiments have been extended to most gases and metals that can be obtained in a gaseous state. With the aid of the figure I shall briefly illustrate the results for the case of sodium vapour. It was found that the electrons upon colliding with the atoms were thrown back with undiminished velocity when their energy was less than that required to transfer the atom from the normal state to the next succeeding stationary state of higher energy value. In the case of sodium vapour this means

from the first state in the first row to the first state in the second row. As soon, however, as the energy of the electron reaches this critical value, a new type of collision takes place, in which the electron loses all its kinetic energy, while at the same time the vapour is excited and emits a radiation corresponding to the yellow line. This is what would be expected, if by the collision the atom was transferred from the normal state to the first one in the second row. For some time it was uncertain to what extent this explanation was correct, since in the experiments on mercury vapour it was found that, together with the occurrence of non-elastic impacts, ions were always formed in the vapour. From our figure, however, we would expect ions to be produced only when the kinetic energy of the electrons is sufficiently great to bring the atom out of the normal state to the common limit of the states. Later experiments, especially by Davis and Goucher, have settled this point. It has been shown that ions can only be directly produced by collisions when the kinetic energy of the electrons corresponds to the limit of the series, and that the ionization found at first was an indirect effect arising from the photoelectric effect produced at the metal walls of the apparatus by the radiation arising from the return of the mercury atoms to the normal state. These experiments provide a direct and independent proof of the reality of the distinctive stationary states, whose existence we were led to infer from the series spectra. At the same time we get a striking impression of the insufficiency of the ordinary electrodynamical and mechanical conceptions

for the description of atomic processes, not only as regards the emission of radiation but also in such phenomena as the collision of free electrons with atoms.

III. DEVELOPMENT OF THE QUANTUM THEORY OF SPECTRA

We see that it is possible by making use of a few simple ideas to obtain a certain insight into the origin of the series spectra. But when we attempt to penetrate more deeply, difficulties arise. In fact, for systems which are not simply periodic it is not possible to obtain sufficient information about the motions of these systems in the stationary states from the numerical values of the energy alone; more determining factors are required for the fixation of the motion. We meet the same difficulties when we try to explain in detail the characteristic effect of external forces upon the spectrum of hydrogen. A foundation for further advances in this field has been made in recent years through a development of the quantum theory, which allows a fixation of the stationary states not only in the case of simple periodic systems, but also for certain classes of non-periodic systems. These are the *conditionally periodic systems* whose equations of motion can be solved by a "separation of the variables." If generalized coordinates are used the description of the motion of these systems can be reduced to the consideration of a number of generalized "components of motion." Each of these corresponds to the change of only one of the co-

ordinates and may therefore in a certain sense be regarded as "independent." The method for the fixation of the stationary states consists in fixing the motion of each of these components by a condition, which can be considered as a direct generalization of condition (1) for a Planck oscillator, so that the stationary states are in general characterized by as many whole numbers as the number of the degrees of freedom which the system possesses. A considerable number of physicists have taken part in this development of the quantum theory, including Planck himself. I also wish to mention the important contribution made by Ehrenfest to this subject on the limitations of the applicability of the laws of mechanics to atomic processes. The decisive advance in the application of the quantum theory to spectra, however, is due to Sommerfeld and his followers. However, I shall not further discuss the systematic form in which these authors have presented their results. In a paper which appeared some time ago in the Transactions of the Copenhagen Academy, I have shown that the spectra, calculated with the aid of this method for the fixation of the stationary states, exhibit a correspondence with the spectra which should correspond to the motion of the system similar to that which we have already considered in the case of hydrogen. With the aid of this general correspondence I shall try in the remainder of this lecture to show how it is possible to present the theory of series spectra and the effects produced by external fields of force upon these spectra in a form which may be considered as the natural generalization of the foregoing considerations.

This form appears to me to be especially suited for future work in the theory of spectra, since it allows of an immediate insight into problems for which the methods mentioned above fail on account of the complexity of the motions in the atom.

Effect of external forces on the hydrogen spectrum. We shall now proceed to investigate the effect of small perturbing forces upon the spectrum of the simple system consisting of a single electron revolving about a nucleus. For the sake of simplicity we shall for the moment disregard the variation of the mass of the electron with its velocity. The consideration of the small changes in the motion due to this variation has been of great importance in the development of Sommerfeld's theory which originated in the explanation of the *fine structure of the hydrogen lines.* This fine structure is due to the fact, that taking into account the variation of mass with velocity the orbit of the electron deviates a little from a simple ellipse and is no longer exactly periodic. This deviation from a Keplerian motion is, however, very small compared with the perturbations due to the presence of external forces, such as occur in experiments on the Zeeman and Stark effects. In atoms of higher atomic number it is also negligible compared with the disturbing effect of the inner electrons on the motion of the outer electron. The neglect of the change in mass will therefore have no important influence upon the explanation of the Zeeman and Stark ef-

fects, or upon the explanation of the difference between the hydrogen spectrum and the spectra of other elements.

We shall therefore as before consider the motion of the unperturbed hydrogen atom as simply periodic and inquire in the first place about the stationary states corresponding to this motion. The energy in these states will then be determined by expression (7) which was derived from the spectrum of hydrogen. The energy of the system being given, the major axis of the elliptical orbit of the electron and its frequency of revolution are also determined. Substituting in formulae (7) and (8) the expression for K given in (12), we obtain for the energy, major axis and frequency of revolution in the nth state of the unperturbed atom the expressions

$$\begin{aligned} E_n = -W_n = -\frac{1}{n^2}\,\frac{2\pi^2 e^4 m}{h^2}, \\ 2a_n = n^2\,\frac{h^2}{2\pi^2 e^2 m}, \qquad \omega_n = \frac{1}{n^3}\,\frac{4\pi^2 e^4 m}{h^3}. \end{aligned} \tag{17}$$

We must further assume that in the stationary states of the unperturbed system the form of the orbit is so far undetermined that the eccentricity can vary continuously. This is not only immediately indicated by the principle of correspondence,—since the frequency of revolution is determined only by the energy and not by the eccentricity,—but also by the fact that the presence of any small external forces will in general, in the course of time, produce a finite change in the position as well as in the eccentricity of the periodic orbit, while in the major axis it can produce only small changes

proportional to the intensity of the perturbing forces.

In order to fix the stationary states of systems in the presence of a given conservative external field of force, we shall have to investigate, on the basis of the principle of correspondence, how these forces affect the decomposition of the motion into harmonic oscillations. Owing to the external forces the form and position of the orbit will vary continuously. In the general case these changes will be so complicated that it will not be possible to decompose the perturbed motion into discrete harmonic oscillations. In such a case we must expect that the perturbed system will not possess any sharply separated stationary states. Although each emission of radiation must be assumed to be monochromatic and to proceed according to the general frequency condition we shall therefore expect the final effect to be a broadening of the sharp spectral lines of the unperturbed system. In certain cases, however, the perturbations will be of such a regular character that the perturbed system can be decomposed into harmonic oscillations, although the ensemble of these oscillations will naturally be of a more complicated kind than in the unperturbed system. This happens, for example, when the variations of the orbit with respect to time are periodic. In this case harmonic oscillations will appear in the motion of the system the frequencies of which are equal to whole multiples of the period of the orbital perturbations, and in the spectrum to be expected on the basis of the ordinary theory of radiation we would expect components corresponding to these frequencies. According to the principle of correspondence

we are therefore immediately led to the conclusion, that to each stationary state in the unperturbed system there corresponds a number of stationary states in the perturbed system in such a manner, that for a transition between two of these states a radiation is emitted, whose frequency stands in the same relationship to the periodic course of the variations in the orbit, as the spectrum of a simple periodic system does to its motion in the stationary states.

The Stark effect. An instructive example of the appearance of periodic perturbations is obtained when hydrogen is subjected to the effect of a homogeneous electric field. The eccentricity and the position of the orbit vary continuously under the influence of the field. During these changes, however, it is found that the centre of the orbit remains in a plane perpendicular to the direction of the electric force and that its motion in this plane is simply periodic. When the centre has returned to its starting point, the orbit will resume its original eccentricity and position, and from this moment the entire cycle of orbits will be repeated. In this case the determination of the energy of the stationary states of the disturbed system is extremely simple, since it is found that the period of the disturbance does not depend upon the original configuration of the orbits nor therefore upon the position of the plane in which the centre of the orbit moves, but only upon the major axis and the frequency of revolution. From a simple calculation it is found that the period a

is given by the following formula

$$\sigma = \frac{3eF}{8\pi^2 ma\omega}, \tag{18}$$

where F is the intensity of the external electric field. From analogy with the fixation of the distinctive energy values of a Planck oscillator we must therefore expect that the energy difference between two different states, corresponding to the same stationary state of the unperturbed system, will simply be equal to a whole multiple of the product of h by the period σ of the perturbations. We are therefore immediately led to the following expression for the energy of the stationary states of the perturbed system,

$$E = E_n + kh\sigma, \tag{19}$$

where E_n depends only upon the number n characterizing the stationary state of the unperturbed system, while k is a new whole number which in this case may be either positive or negative. As we shall see below, consideration of the relation between the energy and the motion of the system shows that k must be numerically less than n, if, as before, we place the quantity E_n equal to the energy $-W_n$ of the nth stationary state of the undisturbed atom. Substituting the values of W_n, ω_n and a_n given by (17) in formula (19) we get

$$E = -\frac{1}{n^2}\,\frac{2\pi^2 e^4 m}{h^2} + nk\,\frac{3h^2 F}{8\pi^2 em}. \tag{20}$$

To find the effect of an electric field upon the lines of the hydrogen spectrum, we use the frequency condition (4) and obtain for the frequency ν of the radiation emitted by a transition between two stationary states defined by the numbers n', k' and n'', k''

$$\nu = \frac{2\pi^2 e^4 m}{h^3}\left(\frac{1}{(n'')^2} - \frac{1}{(n'')^2}\right) + \frac{3h \cdot F}{8\pi^2 em}(n'k' - n''k''). \quad (21)$$

It is well known that this formula provides a complete explanation of the Stark effect of the hydrogen lines. It corresponds exactly with the one obtained by a different method by Epstein and Schwarzschild. They used the fact that the hydrogen atom in a homogeneous electric field is a conditionally periodic system permitting a separation of variables by the use of parabolic coordinates. The stationary states were fixed by applying quantum conditions to each of these variables.

We shall now consider more closely the correspondence between the changes in the spectrum of hydrogen due to the presence of an electric field and the decomposition of the perturbed motion of the atom into its harmonic components. Instead of the simple decomposition into harmonic components corresponding to a simple Kepler motion, the displacement ξ of the electron in a given direction in space can be expressed in the present case by the formula

$$\xi = \sum C_{\tau,\kappa} \cos 2\pi\{t(\tau\omega + \kappa\sigma) + c_{\tau,\kappa}\}, \quad (22)$$

where ω is the average frequency of revolution in the perturbed orbit and σ is the period of the orbital perturbations, while $C_{\tau,\kappa}$ and $c_{\tau,\kappa}$ are constants. The summation is to be extended over all integral values for τ and κ.

If we now consider a transition between two stationary states characterized by certain numbers n', k' and n'', k'', we find that in the region where these numbers are large compared with their differences $n' - n''$ and $k' - k''$, the frequency of the spectral line which is emitted will be given approximately by the formula

$$\nu \sim (n' - n'')\omega + (k' - k'')\sigma. \tag{23}$$

We see, therefore, that we have obtained a relation between the spectrum and the motion of precisely the same character as in the simple case of the unperturbed hydrogen atom. We have here a similar correspondence between the harmonic component in the motion, corresponding to definite values for τ and κ in formula (22), and the transition between two stationary states for which $n' - n'' = \tau$ and $k' - k'' = \kappa$.

A number of interesting results can be obtained from this correspondence by considering the motion in more detail. Each harmonic component in expression (22) for which $\tau + \kappa$ is an even number corresponds to a linear oscillation parallel to the direction of the electric field, while each component for which $\tau + \kappa$ is odd corresponds to an elliptical oscillation perpendicular to this direction. The correspondence principle suggests at once that these facts are connected with the *characteristic polarization* observed in the

Stark effect. We would anticipate that a transition for which $(n' - n'') + (k' - k'')$ is even would give rise to a component with an electric vector parallel to the field, while a transition for which $(n' - n'') + (k' - k'')$ is odd would correspond to a component with an electric vector perpendicular to the field. These results have been fully confirmed by experiment and correspond to the empirical rule of polarization, which Epstein proposed in his first paper on the Stark effect.

The applications of the correspondence principle that have so far been described have been purely qualitative in character. It is possible however to obtain a quantitative estimate of the relative intensity of the various components of the Stark effect of hydrogen, by correlating the numerical values of the coefficients $C_{\tau,\kappa}$ in formula (22) with the probability of the corresponding transitions between the stationary states. This problem has been treated in detail by Kramers in a recently published dissertation. In this he gives a thorough discussion of the application of the correspondence principle to the question of the intensity of spectral lines.

The Zeeman effect. The problem of the effect of a homogeneous magnetic field upon the hydrogen lines may be treated in an entirely analogous manner. The effect on the motion of the hydrogen atom consists simply of the superposition of a uniform rotation upon the motion of the electron in the unperturbed atom. The axis of rotation is parallel

with the direction of the magnetic force, while the frequency of revolution is given by the formula

$$\sigma = \frac{eH}{4\pi mc}, \tag{24}$$

where H is the intensity of the field and c the velocity of light.

Again we have a case where the perturbations are simply periodic and where the period of the perturbations is independent of the form and position of the orbit, and in the present case, even of the major axis. Similar considerations apply therefore as in the case of the Stark effect, and we must expect that the energy in the stationary states will again be given by formula (19), if we substitute for σ the value given in expression (24). This result is also in complete agreement with that obtained by Sommerfeld and Debye. The method they used involved the solution of the equations of motion by the method of the separation of the variables. The appropriate coordinates are polar ones about an axis parallel to the field.

If we try, however, to calculate directly the effect of the field by means of the frequency condition (4), we immediately meet with an apparent disagreement which for some time was regarded as a grave difficulty for the theory. As both Sommerfeld and Debye have pointed out, lines are not observed corresponding to every transition between the stationary states included in the formula. We overcome this difficulty, however, as soon as we apply the principle of cor-

respondence. If we consider the harmonic components of the motion we obtain a simple explanation both of the non-occurrence of certain transitions and of the observed polarization. In the magnetic field each elliptic harmonic component having the frequency $\tau\omega$ splits up into three harmonic components owing to the uniform rotation of the orbit. Of these one is rectilinear with frequency $\tau\omega$ oscillating parallel to the magnetic field, and two are circular with frequencies $\tau\omega+\sigma$ and $\tau\omega-\sigma$ oscillating in opposite directions in a plane perpendicular to the direction of the field. Consequently the motion represented by formula (22) contains no components for which κ is numerically greater than 1, in contrast to the Stark effect, where components corresponding to all values of κ are present. Now formula (23) again applies for large values of n and k, and shows the asymptotic agreement between the frequency of the radiation and the frequency of a harmonic component in the motion. We arrive, therefore, at the conclusion that transitions for which k changes by more than unity cannot occur. The argument is similar to that by which transitions between two distinctive states of a Planck oscillator for which the values of n in (1) differ by more than unity are excluded. We must further conclude that the various possible transitions consist of two types. For the one type corresponding to the rectilinear component, k remains unchanged, and in the emitted radiation which possesses the same frequency ν_0 as the original hydrogen line, the electric vector will oscillate parallel with the field. For the second type, corresponding to the circular components, k will in-

crease or decrease by unity, and the radiation viewed in the direction of the field will be circularly polarized and have frequencies $\nu_0 + \sigma$ and $\nu_0 - \sigma$ respectively. These results agree with those of the familiar Lorentz theory. The similarity in the two theories is remarkable, when we recall the fundamental difference between the ideas of the quantum theory and the ordinary theories of radiation.

Central perturbations. An illustration based on similar considerations which will throw light upon the spectra of other elements consists in finding the effect of a small perturbing field of force radially symmetrical with respect to the nucleus. In this case neither the form of the orbit nor the position of its plane will change with time, and the perturbing effect of the field will simply consist of a uniform rotation of the major axis of the orbit. The perturbations are periodic, so that we may assume that to each energy value of a stationary state of the unperturbed system there belongs a series of discrete energy values of the perturbed system, characterized by different values of a whole number k. The frequency σ of the perturbations is equal to the frequency of rotation of the major axis. For a given law of force for the perturbing field we find that σ depends both on the major axis and on the eccentricity. The change in the energy of the stationary states, therefore, will not be given by an expression as simple as the second term in formula (19), but will be a function of k, which is different for different fields.

It is possible, however, to characterize by one and the same condition the motion in the stationary states of a hydrogen atom which is perturbed by any central field. In order to show this we must consider more closely the fixation of the motion of a perturbed hydrogen atom.

In the stationary states of the unperturbed hydrogen atom only the major axis of the orbit is to be regarded as fixed, while the eccentricity may assume any value. Since the change in the energy of the atom due to the external field of force depends upon the form and position of its orbit, the fixation of the energy of the atom in the presence of such a field naturally involves a closer determination of the orbit of the perturbed system.

Consider, for the sake of illustration, the change in the hydrogen spectrum due to the presence of homogeneous electric and magnetic fields which was described by equation (19). It is found that this energy condition can be given a simple geometrical interpretation. In the case of an electric field the distance from the nucleus to the plane in which the centre of the orbit moves determines the change in the energy of the system due to the presence of the field. In the stationary states this distance is simply equal to $\frac{k}{n}$ times half the major axis of the orbit. In the case of a magnetic field it is found that the quantity which determines the change of energy of the system is the area of the projection of the orbit upon a plane perpendicular to the magnetic force. In the various stationary states this area is equal to

$\frac{k}{n}$ times the area of a circle whose radius is equal to half the major axis of the orbit. In the case of a perturbing central force the correspondence between the spectrum and the motion which is required by the quantum theory leads now to the simple condition that in the stationary states of the perturbed system the minor axis of the rotating orbit is simply equal to $\frac{k}{n}$ times the major axis. This condition was first derived by Sommerfeld from his general theory for the determination of the stationary states of a central motion. It is easily shown that this fixation of the value of the minor axis is equivalent to the statement that the parameter $2p$ of the elliptical orbit is given by an expression of exactly the same form as that which gives the major axis $2a$ in the unperturbed atom. The only difference from the expression for $2a_n$ in (17) is that n is replaced by k, so that the value of the parameter in the stationary states of the perturbed atom is given by

$$2p_k = k^2 \frac{h^2}{2\pi^2 e^2 m}. \tag{25}$$

The frequency of the radiation emitted by a transition between two stationary states determined in this way for which n' and n'' are large in proportion to their difference is given by an expression which is the same as that in equation (23), if in this case ω is the frequency of revolution of the electron in the slowly rotating orbit and σ represents the frequency of rotation of the major axis.

Before proceeding further, it might be of interest to note

that this fixation of the stationary states of the hydrogen atom perturbed by external electric and magnetic forces does not coincide in certain respects with the theories of Sommerfeld, Epstein and Debye. According to the theory of conditionally periodic systems the stationary states for a system of three degrees of freedom will in general be determined by three conditions, and therefore in these theories each state is characterized by three whole numbers. This would mean that the stationary states of the perturbed hydrogen atom corresponding to a certain stationary state of the unperturbed hydrogen atom, fixed by one condition, should be subject to two further conditions and should therefore be characterized by two new whole numbers in addition to the number n. But the perturbations of the Keplerian motion are simply periodic and the energy of the perturbed atom will therefore be fixed completely by one additional condition. The introduction of a second condition will add nothing further to the explanation of the phenomenon, since with the appearance of new perturbing forces, even if these are too small noticeably to affect the observed Zeeman and Stark effects, the forms of motion characterized by such a condition may be entirely changed. This is completely analogous to the fact that the hydrogen spectrum as it is usually observed is not noticeably affected by small forces, even when they are large enough to produce a great change in the form and position of the orbit of the electron.

Relativity effect on hydrogen lines. Before leaving the hydrogen spectrum I shall consider briefly the effect of the variation of the mass of the electron with its velocity. In the preceding sections I have described how external fields of force split up the hydrogen lines into several components, but it should be noticed that these results are only accurate when the perturbations are large in comparison with the small deviations from a pure Keplerian motion due to the variation of the mass of the electron with its velocity. When the variation of the mass is taken into account the motion of the unperturbed atom will not be exactly periodic. Instead we obtain a motion of precisely the same kind as that occurring in the hydrogen atom perturbed by a small central field. According to the correspondence principle an intimate connection is to be expected between the frequency of revolution of the major axis of the orbit and the difference of the frequencies of the fine structure components, and the stationary states will be those orbits whose parameters are given by expression (25). If we now consider the effect of external forces upon the fine structure components of the hydrogen lines it is necessary to keep in mind that this fixation of the stationary states only applies to the unperturbed hydrogen atom, and that, as mentioned, the orbits in these states are in general already strongly influenced by the presence of external forces, which are small compared with those with which we are concerned in experiments on the Stark and Zeeman effects. In general the presence of such forces will lead to a great complexity of perturbations, and the atom will no longer possess a group of

sharply defined stationary states. The fine structure components of a given hydrogen line will therefore become diffuse and merged together. There are, however, several important cases where this does not happen on account of the simple character of the perturbations. The simplest example is a hydrogen atom perturbed by a central force acting from the nucleus. In this case it is evident that the motion of the system will retain its centrally symmetrical character, and that the perturbed motion will differ from the unperturbed motion only in that the frequency of rotation of the major axis will be different for different values of this axis and of the parameter. This point is of importance in the theory of the spectra of elements of higher atomic number, since, as we shall see, the effect of the forces originating from the inner electrons may to a first approximation be compared with that of a perturbing central field. We cannot therefore expect these spectra to exhibit a separate effect due to the variation of the mass of the electron of the same kind as that found in the case of the hydrogen lines. This variation will not give rise to a splitting up into separate components but only to small displacements in the position of the various lines.

We obtain still another simple example in which the hydrogen atom possesses sharp stationary states, although the change of mass of the electron is considered, if we take an atom subject to a homogeneous magnetic field. The effect of such a field will consist in the superposition of a rotation of the entire system about an axis through the nucleus and

parallel with the magnetic force. It follows immediately from this result according to the principle of correspondence that each fine structure component must be expected to split up into a normal Zeeman effect (Lorentz triplet). The problem may also be solved by means of the theory of conditionally periodic systems, since the equations of motion in the presence of a magnetic field, even when the change in the mass is considered, will allow of a separation of the variables using polar coordinates in space. This has been pointed out by Sommerfeld and Debye.

A more complicated case arises when the atom is exposed to a homogeneous electric field which is not so strong that the effect due to the change in the mass may be neglected. In this case there is no system of coordinates by which the equations of motion can be solved by separation of the variables, and the problem, therefore, cannot be treated by the theory of the stationary states of conditionally periodic systems. A closer investigation of the perturbations, however, shows them to be of such a character that the motion of the electrons may be decomposed into a number of separate harmonic components. These fall into two groups for which the direction of oscillation is either parallel with or perpendicular to the field. According to the principle of correspondence, therefore, we must expect that also in this case in the presence of the field each hydrogen line will consist of a number of sharp, polarized components. In fact by means of the principles I have described, it is possible to give a unique fixation of the stationary states. The problem of the

effect of a homogeneous electric field upon the fine structure components of the hydrogen lines has been treated in detail from this point of view by Kramers in a paper which will soon be published. In this paper it will be shown how it appears possible to predict in detail the manner in which the fine structure of the hydrogen lines gradually changes into the ordinary Stark effect as the electric intensity increases.

Theory of series spectra. Let us now turn our attention once more to the problem of the series spectra of elements of higher atomic number. The general appearance of the Rydberg constant in these spectra is to be explained by assuming that the atom is neutral and that one electron revolves in an orbit the dimensions of which are large in comparison with the distance of the inner electrons from the nucleus. In a certain sense, therefore, the motion of the outer electron may be compared with the motion of the electron of the hydrogen atom perturbed by external forces, and the appearance of the various series in the spectra of the other elements is from this point of view to be regarded as analogous to the splitting up of the hydrogen lines into components on account of such forces.

In his theory of the structure of series spectra of the type exhibited by the alkali metals, Sommerfeld has made the assumption that the orbit of the outer electron to a first approximation possesses the same character as that produced by a simple perturbing central field whose intensity dimin-

ishes rapidly with increasing distance from the nucleus. He fixed the motion of the external electron by means of his general theory for the fixation of the stationary states of a central motion. The application of this method depends on the possibility of separating the variables in the equations of motion. In this manner Sommerfeld was able to calculate a number of energy values which can be arranged in rows just like the empirical spectral terms shown in the diagram of the sodium spectrum (p. 46). The states grouped together by Sommerfeld in the separate rows are exactly those which were characterized by one and the same value of k in our investigation of the hydrogen atom perturbed by a central force. The states in the first row of the figure (row S) correspond to the value $k = 1$, those of the second row (P) correspond to $k = 2$, etc. The states corresponding to one and the same value of n are connected by dotted lines which are continued so that their vertical asymptotes correspond to the energy value of the stationary states of the hydrogen atom. The fact that for a constant n and increasing values of k the energy values approach the corresponding values for the unperturbed hydrogen atom is immediately evident from the theory since the outer electron, for large values of the parameter of its orbit, remains at a great distance from the inner system during the whole revolution. The orbit will become almost elliptical and the period of rotation of the major axis will be very large. It can be seen, therefore, that the effect of the inner system on the energy necessary to remove this electron from the atom must become less for

increasing values of k.

These beautiful results suggest the possibility of finding laws of force for the perturbing central field which would account for the spectra observed. Although Sommerfeld in this way has in fact succeeded in deriving formulae for the spectral terms which vary with n for a constant k in agreement with Rydberg's formulae, it has not been possible to explain the simultaneous variation with both k and n in any actual case. This is not surprising, since it is to be anticipated that the effect of the inner electrons on the spectrum could not be accounted for in such a simple manner. Further consideration shows that it is necessary to consider not only the forces which originate from the inner electrons but also to consider the effect of the presence of the outer electron upon the motion of the inner electrons.

Before considering the series spectra of elements of low atomic number I shall point out how the occurrence or non-occurrence of certain transitions can be shown by the correspondence principle to furnish convincing evidence in favour of Sommerfeld's assumption about the orbit of the outer electron. For this purpose we must describe the motion of the outer electron in terms of its harmonic components. This is easily performed if we assume that the presence of the inner electrons simply produces a uniform rotation of the orbit of the outer electron in its plane. On account of this rotation, the frequency of which we will denote by σ, two circular rotations with the periods $\tau\omega + \sigma$ and $\tau\omega - \sigma$ will appear in the motion of the perturbed electron, instead of

each of the harmonic elliptical components with a period $\tau\omega$ in the unperturbed motion. The decomposition of the perturbed motion into harmonic components consequently will again be represented by a formula of the type (22), in which only such terms appear for which κ is equal to $+1$ or -1. Since the frequency of the emitted radiation in the regions where n and k are large is again given by the asymptotic formula (23), we at once deduce from the correspondence principle that the only transitions which can take place are those for which the values of k differ by unity. A glance at the figure for the sodium spectrum shows that this agrees exactly with the experimental results. This fact is all the more remarkable, since in Sommerfeld's theory the arrangement of the energy values of the stationary states in rows has no special relation to the possibility of transition between these states.

Correspondence principle and conservation of angular momentum. Besides these results the correspondence principle suggests that the radiation emitted by the perturbed atom must exhibit circular polarization. On account of the indeterminateness of the plane of the orbit, however, this polarization cannot be directly observed. The assumption of such a polarization is a matter of particular interest for the theory of radiation emission. On account of the general correspondence between the spectrum of an atom and the decomposition of its motion into harmonic compo-

nents, we are led to compare the radiation emitted during the transition between two stationary states with the radiation which would be emitted by a harmonically oscillating electron on the basis of the classical electrodynamics. In particular the radiation emitted according to the classical theory by an electron revolving in a circular orbit possesses an angular momentum and the energy ΔE and the angular momentum ΔP of the radiation emitted during a certain time are connected by the relation

$$\Delta E = 2\pi\omega \cdot \Delta P. \tag{26}$$

Here ω represents the frequency of revolution of the electron, and according to the classical theory this is equal to the frequency ν of the radiation. If we now assume that the total energy emitted is equal to $h\nu$ we obtain for the total angular momentum of the radiation

$$\Delta P = \frac{h}{2\pi}. \tag{27}$$

It is extremely interesting to note that this expression is equal to the change in the angular momentum which the atom suffers in a transition where k varies by unity. For in Sommerfeld's theory the general condition for the fixation of the stationary states of a central system, which in the special case of an approximately Keplerian motion is equivalent to the relation (25), asserts that the angular momentum of the system must be equal to a whole multiple of $\frac{h}{2\pi}$, a condition

which may be written in our notation

$$P = k\,\frac{h}{2\pi}. \tag{28}$$

We see, therefore, that this condition has obtained direct support from a simple consideration of the conservation of angular momentum during the emission of the radiation. I wish to emphasize that this equation is to be regarded as a rational generalization of Planck's original statement about the distinctive states of a harmonic oscillator. It may be of interest to recall that the possible significance of the angular momentum in applications of the quantum theory to atomic processes was first pointed out by Nicholson on the basis of the fact that for a circular motion the angular momentum is simply proportional to the ratio of the kinetic energy to the frequency of revolution.

In a previous paper which I presented to the Copenhagen Academy I pointed out that these results confirm the conclusions obtained by the application of the correspondence principle to atomic systems possessing radial or axial symmetry. Rubinowicz has independently indicated the conclusions which may be obtained directly from a consideration of conservation of angular momentum during the radiation process. In this way he has obtained several of our results concerning the various types of possible transitions and the polarization of the emitted radiation. Even for systems possessing radial or axial symmetry, however, the conclusions which we can draw by means of the correspondence principle are of a more

detailed character than can be obtained solely from a consideration of the conservation of angular momentum. For example, in the case of the hydrogen atom perturbed by a central force we can only conclude that k cannot change by more than unity, while the correspondence principle requires that k shall vary by unity for every possible transition and that its value cannot remain unchanged. Further, this principle enables us not only to exclude certain transitions as being impossible—and can from this point of view be considered as a "selection principle"—but it also enables us to draw conclusions about the relative probabilities of the various possible types of transitions from the values of the amplitudes of the harmonic components. In the present case, for example, the fact that the amplitudes of those circular components which rotate in the same sense as the electron are in general greater than the amplitudes of those which rotate in the opposite sense leads us to expect that lines corresponding to transitions for which k decreases by unity will in general possess greater intensity than lines during the emission of which k increases by unity. Simple considerations like this, however, apply only to spectral lines corresponding to transitions from one and the same stationary state. In other cases when we wish to estimate the relative intensities of two spectral lines it is clearly necessary to take into consideration the relative number of atoms which are present in each of the two stationary states from which the transitions start. While the intensity naturally cannot depend upon the number of atoms in the final state, it is to be noticed,

however, that in estimating the probability of a transition between two stationary states it is necessary to consider the character of the motion in the final as well as in the initial state, since the values of the amplitudes of the components of oscillation of both states are to be regarded as decisive for the probability.

To show how this method can be applied I shall return for a moment to the problem which I mentioned in connection with Strutt's experiment on the resonance radiation of sodium vapour. This involved the discussion of the relative probability of the various possible transitions which can start from that state corresponding to the second term in the second row of the figure on p. 46. These were transitions to the first and second states in the first row and to the first state in the third row, and the results of experiment indicate, as we saw, that the probability is greatest for the second transitions. These transitions correspond to those harmonic components having frequencies $2\omega + \sigma$, $\omega + \sigma$ and σ, and it is seen that only for the second transition do the amplitudes of the corresponding harmonic component differ from zero in the initial as well as in the final state. [In the next essay the reader will find that the values of quantum number n assigned in fig. 1 to the various stationary states must be altered. While this correction in no way influences the other conclusions in this essay it involves that the reasoning in this passage cannot be maintained.]

I have shown how the correspondence between the spectrum of an element and the motion of the atom enables us

to understand the limitations in the direct application of the combination principle in the prediction of spectral lines. The same ideas give an immediate explanation of the interesting discovery made in recent years by Stark and his collaborators, that certain *new series of combination lines* appear with considerable intensity when the radiating atoms are subject to a strong external electric field. This phenomenon is entirely analogous to the appearance of the so-called combination tones in acoustics. It is due to the fact that the perturbation of the motion will not only consist in an effect upon the components originally present, but in addition will give rise to new components. The frequencies of these new components may be $\tau\omega + \kappa\sigma$, where κ is different from ± 1. According to the correspondence principle we must therefore expect that the electric field will not only influence the lines appearing under ordinary circumstances, but that it will also render possible new types of transitions which give rise to the "new" combination lines observed. From an estimate of the amplitudes of the particular components in the initial and final states it has even been found possible to account for the varying facility with which the new lines are brought up by the external field.

The general problem of the effect of an electric field on the spectra of elements of higher atomic number differs essentially from the simple Stark effect of the hydrogen lines, since we are here concerned not with the perturbation of a purely periodic system, but with the effect of the field on a periodic motion already subject to a perturbation. The

problem to a certain extent resembles the effect of a weak electric force on the fine structure components of the hydrogen atom. In much the same way the effect of an electric field upon the series spectra of the elements may be treated directly by investigating the perturbations of the external electron. A continuation of my paper in the Transactions of the Copenhagen Academy will soon appear in which I shall show how this method enables us to understand the interesting observations Stark and others have made in this field.

The spectra of helium and lithium. We see that it has been possible to obtain a certain general insight into the origin of the series spectra of a type like that of sodium. The difficulties encountered in an attempt to give a detailed explanation of the spectrum of a particular element, however, become very serious, even when we consider the spectrum of helium whose neutral atom contains only two electrons. The spectrum of this element has a simple structure in that it consists of single lines or at any rate of double lines whose components are very close together. We find, however, that the lines fall into two groups each of which can be described by a formula of the type (14). These are usually called the (ortho) helium and parhelium spectra. While the latter consists of simple lines, the former possesses narrow doublets. The discovery that helium, as opposed to the alkali metals, possesses two complete spectra of the Rydberg type which do not exhibit any mutual combinations was so surprising that

at times there has been a tendency to believe that helium consisted of two elements. This way out of the difficulty is no longer open, since there is no room for another element in this region of the periodic system, or more correctly expressed, for an element possessing a new spectrum. The existence of the two spectra can, however, be traced back to the fact that in the stationary states corresponding to the series spectra we have to do with a system possessing only one inner electron and in consequence the motion of the inner system, in the absence of the outer electron, will be simply periodic and therefore easily perturbed by external forces.

In order to illustrate this point we shall have to consider more carefully the stationary states connected with the origin of a series spectrum. We must assume that in these states one electron revolves in an orbit outside the nucleus and the other electrons. We might now suppose that in general a number of different groups of such states might exist, each group corresponding to a different stationary state of the inner system considered by itself. Further consideration shows, however, that under the usual conditions of excitation those groups have by far the greatest probability for which the motion of the inner electrons corresponds to the "normal" state of the inner system, i.e. to that stationary state having the least energy. Further the energy required to transfer the inner system from its normal state to another stationary state is in general very large compared with the energy which is necessary to transfer an electron from the normal state of the neutral atom to a stationary orbit

of greater dimensions. Lastly the inner system is in general capable of a permanent existence only in its normal state. Now, the configuration of an atomic system in its stationary states and also in the normal state will, in general, be completely determined. We may therefore expect that the inner system under the influence of the forces arising from the presence of the outer electron can in the course of time suffer only small changes. For this reason we must assume that the influence of the inner system upon the motion of the external electron will, in general, be of the same character as the perturbations produced by a constant external field upon the motion of the electron in the hydrogen atom. We must therefore expect a spectrum consisting of an ensemble of spectral terms, which in general form a connected group, even though in the absence of external perturbing forces not every combination actually occurs. The case of the helium spectrum, however, is quite different since here the inner system contains only one electron the motion of which in the absence of the external electron is simple periodic provided the small changes due to the variation in the mass of the electron with its velocity are neglected. For this reason the form of the orbit in the stationary states of the inner system considered by itself will not be determined. In other words, the stability of the orbit is so slight, even if the variation in the mass is taken into account, that small external forces are in a position to change the eccentricity in the course of time to a finite extent. In this case, therefore, it is possible to have several groups of stationary states, for

which the energy of the inner system is approximately the same while the form of the orbit of the inner electron and its position relative to the motion of the other electrons are so essentially different, that no transitions between the states of different groups can occur even in the presence of external forces. It can be seen that these conclusions summarize the experimental observations on the helium spectra.

These considerations suggest an investigation of the nature of the perturbations in the orbit of the inner electron of the helium atom, due to the presence of the external electron. A discussion of the helium spectrum from this point of view has recently been given by Landé. The results of this work are of great interest particularly in the demonstration of the large back effect on the outer electron due to the perturbations of the inner orbit which themselves arise from the presence of the outer electron. Nevertheless, it can scarcely be regarded as a satisfactory explanation of the helium spectrum. Apart from the serious objections which may be raised against his calculation of the perturbations, difficulties arise if we try to apply the correspondence principle to Landé's results in order to account for the occurrence of two distinct spectra showing no mutual combinations. To explain this fact it seems necessary to base the discussion on a more thorough investigation of the mutual perturbations of the outer and the inner orbits. As a result of these perturbations both electrons move in such an extremely complicated way that the stationary states cannot be fixed by the methods developed for conditionally periodic systems. Dr Kramers and I

have in the last few years been engaged in such an investigation, and in an address on atomic problems at the meeting of the Dutch Congress of Natural and Medical Sciences held in Leiden, April 1919, I gave a short communication of our results. For various reasons we have up to the present time been prevented from publishing, but in the very near future we hope to give an account of these results and of the light which they seem to throw upon the helium spectrum.

The problem presented by the spectra of elements of higher atomic number is simpler, since the inner system is better defined in its normal state. On the other hand the difficulty of the mechanical problem of course increases with the number of the particles in the atom. We obtain an example of this in the case of lithium with three electrons. The differences between the spectral terms of the lithium spectrum and the corresponding spectral terms of hydrogen are very small for the variable term of the principal series ($k = 2$) and for the diffuse series ($k = 3$), on the other hand it is very considerable for the variable term of the sharp series ($k = 1$). This is very different from what would be expected if it were possible to describe the effect of the inner electron by a central force varying in a simple manner with the distance. This must be because the parameter of the orbit of the outer electron in the stationary states corresponding to the terms of the sharp series is not much greater than the linear dimensions of the orbits of the inner electrons. According to the principle of correspondence the frequency of rotation of the major axis of the orbit of the outer electron is to be regarded

as a measure of the deviation of the spectral terms from the corresponding hydrogen terms. In order to calculate this frequency it appears necessary to consider in detail the mutual effect of all three electrons, at all events for that part of the orbit where the outer electron is very close to the other two electrons. Even if we assumed that we were fully acquainted with the normal state of the inner system in the absence of the outer electron—which would be expected to be similar to the normal state of the neutral helium atom—the exact calculation of this mechanical problem would evidently form an exceedingly difficult task.

Complex structure of series lines. For the spectra of elements of still higher atomic number the mechanical problem which has to be solved in order to describe the motion in the stationary states becomes still more difficult. This is indicated by the extraordinarily complicated structure of many of the observed spectra. The fact that the series spectra of the alkali metals, which possess the simplest structure, consist of double lines whose separation increases with the atomic number, indicates that here we have to do with systems in which the motion of the outer electron possesses in general a somewhat more complicated character than that of a simple central motion. This gives rise to a more complicated ensemble of stationary states. It would, however, appear that in the sodium atom the major axis and the parameter of the stationary states corresponding to each pair

of spectral terms are given approximately by formulae (17) and (25). This is indicated not only by the similar part played by the two states in the experiments on the resonance radiation of sodium vapour, but is also shown in a very instructive manner by the peculiar effect of magnetic fields on the doublets. For small fields each component splits up into a large number of sharp lines instead of into the normal Lorentz triplet. With increasing field strength Paschen and Back found that this *anomalous Zeeman effect* changed into the normal Lorentz triplet of a single line by a gradual fusion of the components.

This effect of a magnetic field upon the doublets of the alkali spectrum is of interest in showing the intimate relation of the components and confirms the reality of the simple explanation of the general structure of the spectra of the alkali metals. If we may again here rely upon the correspondence principle we have unambiguous evidence that the effect of a magnetic field on the motion of the electrons simply consists in the superposition of a uniform rotation with a frequency given by equation (24) as in the case of the hydrogen atom. For if this were the case the correspondence principle would indicate under all conditions a normal Zeeman effect for each component of the doublets. I want to emphasize that the difference between the simple effect of a magnetic field, which the theory predicts for the fine structure of components of the hydrogen lines, and the observed effect on the alkali doublets is in no way to be considered as a contradiction. The fine structure components are not analogous to the individ-

ual doublet components, but each single fine structure component corresponds to the ensemble of components (doublet, triplet) which makes up one of the series lines in Rydberg's scheme. The occurrence in strong fields of the effect observed by Paschen and Back must therefore be regarded as a strong support for the theoretical prediction of the effect of a magnetic field on the fine structure components of the hydrogen lines.

It does not appear necessary to assume the "anomalous" effect of small fields on the doublet components to be due to a failure of the ordinary electrodynamical laws for the description of the motion of the outer electron, but rather to be connected with an effect of the magnetic field on that intimate interaction between the motion of the inner and outer electrons which is responsible for the occurrence of the doublets. Such a view is probably not very different from the "coupling theory" by which Voigt was able to account formally for the details of the anomalous Zeeman effect. We might even expect it to be possible to construct a theory of these effects which would exhibit a formal analogy with the Voigt theory similar to that between the quantum theory of the normal Zeeman effect and the theory originally developed by Lorentz. Time unfortunately does not permit me to enter further into this interesting problem, so I must refer you to the continuation of my paper in the Transactions of the Copenhagen Academy, which will contain a general discussion of the origin of series spectra and of the effects of electric and magnetic fields.

IV. CONCLUSION

In this lecture I have purposely not considered the question of the structure of atoms and molecules although this is of course most intimately connected with the kind of spectral theory I have developed. We are encouraged to use results obtained from the spectra, since even the simple theory of the hydrogen spectrum gives a value for the major axis of the orbit of the electron in the normal state ($n = 1$) of the same order of magnitude as that derived from the kinetic theory of gases. In my first paper on the subject I attempted to sketch a theory of the structure of atoms and of molecules of chemical compounds. This theory was based on a simple generalization of the results for the stationary states of the hydrogen atom. In several respects the theory was supported by experiment, especially in the general way in which the properties of the elements change with increasing atomic number, shown most clearly by Moseley's results. I should like, however, to use this occasion to state, that in view of the recent development of the quantum theory, many of the special assumptions will certainly have to be changed in detail. This has become clear from various sides by the lack of agreement of the theory with experiment. It appears no longer possible to justify the assumption that in the normal states the electrons move in orbits of special geometrical simplicity, like "electronic rings." Considerations relating to the stability of atoms and molecules against external influences and concerning the possibility of the formation of an atom by

successive addition of the individual electrons compel us to claim, first that the configurations of electrons are not only in mechanical equilibrium but also possess a certain stability in the sense required by ordinary mechanics, and secondly that the configurations employed must be of such a nature that transitions to these from other stationary states of the atom are possible. These requirements are not in general fulfilled by such simple configurations as electronic rings and they force us to look about for possibilities of more complicated motions. It will not be possible here to consider further these still open questions and I must content myself by referring to the discussion in my forthcoming paper. In closing, however, I should like to emphasize once more that in this lecture I have only intended to bring out certain general points of view lying at the basis of the spectral theory. In particular it was my intention to show that, in spite of the fundamental differences between these points of view and the ordinary conceptions of the phenomena of radiation, it still appears possible on the basis of the general correspondence between the spectrum and the motion in the atom to employ these conceptions in a certain sense as guides in the investigation of the spectra.

ESSAY III*

THE STRUCTURE OF THE ATOM AND THE PHYSICAL AND CHEMICAL PROPERTIES OF THE ELEMENTS

I. PRELIMINARY

In an address which I delivered to you about a year ago I described the main features of a theory of atomic structure which I shall attempt to develop this evening. In the meantime this theory has assumed more definite form, and in two recent letters to *Nature* I have given a somewhat further sketch of the development.† The results which I am about to present to you are of no final character; but I hope to be able to show you how this view renders a correlation of the various properties of the elements in such a way, that we avoid the difficulties which previously appeared to stand in the way of a simple and consistent explanation. Before proceeding, however, I must ask your forbearance if initially I deal with matters already known to you, but in order to introduce you to the subject it will first be necessary to give a brief description of the most important results which have been obtained in recent years in connection with the work on atomic structure.

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*Address delivered before a joint meeting of the Physical and Chemical Societies in Copenhagen, October 18, 1921.

†*Nature*, March 24, and October 13, 1921.

The nuclear atom. The conception of atomic structure which will form the basis of all the following remarks is the so-called nuclear atom according to which an atom is assumed to consist of a nucleus surrounded by a number of electrons whose distances from one another and from the nucleus are very large compared to the dimensions of the particles themselves. The nucleus possesses almost the entire mass of the atom and has a positive charge of such a magnitude that the number of electrons in a neutral atom is equal to the number of the element in the periodic system, the so-called *atomic number*. This idea of the atom, which is due principally to Rutherford's fundamental researches on radioactive substances, exhibits extremely simple features, but just this simplicity appears at first sight to present difficulties in explaining the properties of the elements. When we treat this question on the basis of the ordinary mechanical and electrodynamical theories it is impossible to find a starting point for an explanation of the marked properties exhibited by the various elements, indeed not even of their permanency. On the one hand the particles of the atom apparently could not be at rest in a state of stable equilibrium, and on the other hand we should have to expect that every motion which might be present would give rise to the emission of electromagnetic radiation which would not cease until all the energy of the system had been emitted and all the electrons had fallen into the nucleus. A method of escaping from these difficulties has now been found in the application of ideas belonging to the quantum theory, the basis of

which was laid by Planck in his celebrated work on the law of temperature radiation. This represented a radical departure from previous conceptions since it was the first instance in which the assumption of a discontinuity was employed in the formulation of the general laws of nature.

The postulates of the quantum theory. The quantum theory in the form in which it has been applied to the problems of atomic structure rests upon two postulates which have a direct bearing on the difficulties mentioned above. According to the first postulate there are certain states in which the atom can exist without emitting radiation, although the particles are supposed to have an accelerated motion relative to one another. These *stationary states* are, in addition, supposed to possess a peculiar kind of stability, so that it is impossible either to add energy to or remove energy from the atom except by a process involving a transition of the atom into another of these states. According to the second postulate each emission of radiation from the atom resulting from such a transition always consists of a train of purely harmonic waves. The frequency of these waves does not depend directly upon the motion of the atom, but is determined by a *frequency relation*, according to which the frequency multiplied by the universal constant introduced by Planck is equal to the total energy emitted during the process. For a transition between two stationary states for which the values of the energy of the atom before

and after the emission of radiation are E' and E'' respectively, we have therefore

$$h\nu = E' - E'', \tag{1}$$

where h is Planck's constant and ν is the frequency of the emitted radiation. Time does not permit me to give a systematic survey of the quantum theory, the recent development of which has gone hand in hand with its applications to atomic structure. I shall therefore immediately proceed to the consideration of those applications of the theory which are of direct importance in connection with our subject.

Hydrogen atom. We shall commence by considering the simplest atom conceivable, namely, an atom consisting of a nucleus and one electron. If the charge on the nucleus corresponds to that of a single electron and the system consequently is neutral we have a hydrogen atom. Those developments of the quantum theory which have made possible its application to atomic structure started with the interpretation of the well-known simple spectrum emitted by hydrogen. This spectrum consists of a series of lines, the frequencies of which are given by the extremely simple Balmer formula

$$\nu = K\left(\frac{1}{(n'')^2} - \frac{1}{(n')^2}\right), \tag{2}$$

where n'' and n' are integers. According to the quantum theory we shall now assume that the atom possesses a series

of stationary states characterized by a series of integers, and it can be seen how the frequencies given by formula (2) may be derived from the frequency relation if it is assumed that a hydrogen line is connected with a radiation emitted during a transition between two of these states corresponding to the numbers n' and n'', and if the energy in the nth state apart from an arbitrary additive constant is supposed to be given by the formula

$$E_n = -\frac{Kh}{n^2}. \tag{3}$$

The negative sign is used because the energy of the atom is measured most simply by the work required to remove the electron to infinite distance from the nucleus, and we shall assume that the numerical value of the expression on the right-hand side of formula (3) is just equal to this work.

As regards the closer description of the stationary states we find that the electron will very nearly describe an ellipse with the nucleus in the focus. The major axis of this ellipse is connected with the energy of the atom in a simple way, and corresponding to the energy values of the stationary states given by formula (3) there are a series of values for the major axis $2a$ of the orbit of the electron given by the formula

$$2a_n = \frac{n^2 e^2}{hK}, \tag{4}$$

where e is the numerical value of the charge of the electron and the nucleus.

On the whole we may say that the spectrum of hydrogen shows us the *formation of the hydrogen atom*, since the stationary states may be regarded as different stages of a process by which the electron under the emission of radiation is bound in orbits of smaller and smaller dimensions corresponding to states with decreasing values of n. It will be seen that this view has certain characteristic features in common with the binding process of an electron to the nucleus if this were to take place according to the ordinary electrodynamics, but that our view differs from it in just such a way that it is possible to account for the observed properties of hydrogen. In particular it is seen that the final result of the binding process leads to a quite definite stationary state of the atom, namely that state for which $n = 1$. This state which corresponds to the minimum energy of the atom will be called the *normal state* of the atom. It may be stated here that the values of the energy of the atom and the major axis of the orbit of the electron which are found if we put $n = 1$ in formulae (3) and (4) are of the same order of magnitude as the values of the firmness of binding of electrons and of the dimensions of the atoms which have been obtained from experiments on the electrical and mechanical properties of gases. A more accurate check of formulae (3) and (4) can however not be obtained from such a comparison, because in such experiments hydrogen is not present in the form of simple atoms but as molecules.

The formal basis of the quantum theory consists not only of the frequency relation, but also of conditions which per-

mit the determination of the stationary states of atomic systems. The latter conditions, like that assumed for the frequency, may be regarded as natural generalizations of that assumption regarding the interaction between simple electrodynamic systems and a surrounding field of electromagnetic radiation which forms the basis of Planck's theory of temperature radiation. I shall not here go further into the nature of these conditions but only mention that by their means the stationary states are characterized by a number of integers, the so-called *quantum numbers*. For a purely periodic motion like that assumed in the case of the hydrogen atom only a single quantum number is necessary for the determination of the stationary states. This number determines the energy of the atom and also the major axis of the orbit of the electron, but not its eccentricity. The energy in the various stationary states, if the small influence of the motion of the nucleus is neglected, is given by the following formula:

$$E_n = -\frac{2\pi^2 N^2 e^4 m}{n^2 h^2}, \tag{5}$$

where e and m are respectively the charge and the mass of the electron, and where for the sake of subsequent applications the charge on the nucleus has been designated by Ne.

For the atom of hydrogen $N = 1$, and a comparison with equation (3) leads to the following theoretical expression for K in formula (2), namely

$$K = \frac{2\pi^2 e^4 m}{h^3}. \tag{6}$$

This agrees with the empirical value of the constant for the spectrum of hydrogen within the limit of accuracy with which the various quantities can be determined.

Hydrogen spectrum and X-ray spectra. If in the above formula we put $N = 2$ which corresponds to an atom consisting of an electron revolving around a nucleus with a double charge, we get values for the energies in the stationary states, which are four times larger than the energies in the corresponding states of the hydrogen atom, and we obtain the following formula for the spectrum which would be emitted by such an atom:

$$\nu = 4K\left(\frac{1}{(n'')^2} - \frac{1}{(n')^2}\right). \tag{7}$$

This formula represents certain lines which have been known for some time and which had been attributed to hydrogen on account of the great similarity between formulae (2) and (7) since it had never been anticipated that two different substances could exhibit properties so closely resembling each other. According to the theory we may, however, expect that the emission of the spectrum given by (7) corresponds to the *first stage of the formation of the helium atom*, i.e. to the binding of a first electron by the doubly charged nucleus of this atom. This interpretation has been found to agree with more recent information. For instance it has been possible to obtain this spectrum from pure helium. I have dwelt

on this point in order to show how this intimate connection between the properties of two elements, which at first sight might appear quite surprising, is to be regarded as an immediate expression of the characteristic simple structure of the nuclear atom. A short time after the elucidation of this question, new evidence of extraordinary interest was obtained of such a similarity between the properties of the elements. I refer to Moseley's fundamental researches on the X-ray spectra of the elements. Moseley found that these spectra varied in an extremely simple manner from one element to the next in the periodic system. It is well known that the lines of the X-ray spectra may be divided into groups corresponding to the different characteristic absorption regions for X-rays discovered by Barkla. As regards the K group which contains the most penetrating X-rays, Moseley found that the strongest line for all elements investigated could be represented by a formula which with a small simplification can be written

$$\nu = N^2 K \left(\frac{1}{1^2} - \frac{1}{2^2} \right). \tag{8}$$

K is the same constant as in the hydrogen spectrum, and N the atomic number. The great significance of this discovery lies in the fact that it would seem firmly to establish the view that this atomic number is equal to the number of electrons in the atom. This assumption had already been used as a basis for work on atomic structure and was first stated by van den Broek. While the significance of this aspect of Moseley's discovery was at once clear to all, it has on the

other hand been more difficult to understand the very great similarity between the spectrum of hydrogen and the X-ray spectra. This similarity is shown, not only by the lines of the K group, but also by groups of less penetrating X-rays. Thus Moseley found for all the elements he investigated that the frequencies of the strongest line in the L group may be represented by a formula which with a simplification similar to that employed in formula (8) can be written

$$\nu = N^2 K \left(\frac{1}{2^2} - \frac{1}{3^2} \right). \tag{9}$$

Here again we obtain an expression for the frequency which corresponds to a line in the spectrum which would be emitted by the *binding of an electron to a nucleus, whose charge is* Ne.

The fine structure of the hydrogen lines. This similarity between the structure of the X-ray spectra and the hydrogen spectrum was still further extended in a very interesting manner by Sommerfeld's important theory of the fine structure of the hydrogen lines. The calculation given above of the energy in the stationary states of the hydrogen system, where each state is characterized by a single quantum number, rests upon the assumption that the orbit of the electron in the atom is simply periodic. This is, however, only approximately true. It is found that if the change in the mass of the electron due to its velocity is taken into

consideration the orbit of the electron no longer remains a simple ellipse, but its motion may be described as a *central motion* obtained by superposing a slow and uniform rotation upon a simple periodic motion in a very nearly elliptical orbit. For a central motion of this kind the stationary states are characterized by *two quantum numbers*. In the case under consideration one of these may be so chosen that to a very close approximation it will determine the energy of the atom in the same manner as the quantum number previously used determined the energy in the case of a simple elliptical orbit. This quantum number which will always be denoted by n will therefore be called the "principal quantum number." Besides this condition, which to a very close approximation determines the major axis in the rotating and almost elliptical orbit, a second condition will be imposed upon the stationary states of a central orbit, namely that the angular momentum of the electron about the centre shall be equal to a whole multiple of Planck's constant divided by 2π. The whole number, which occurs as a factor in this expression, may be regarded as the second quantum number and will be denoted by k. The latter condition fixes the eccentricity of the rotating orbit which in the case of a simple periodic orbit was undetermined. It should be mentioned that the possible importance of the angular momentum in the quantum theory was pointed out by Nicholson before the application of this theory to the spectrum of hydrogen, and that a determination of the stationary states for the hydrogen atom similar to that employed by Sommerfeld was proposed almost simul-

taneously by Wilson, although the latter did not succeed in giving a physical application to his results.

The simplest description of the form of the rotating nearly elliptical electronic orbit in the hydrogen atom is obtained by considering the chord which passes through the focus and is perpendicular to the major axis, the so-called "parameter." The length $2p$ of this parameter is given to a very close approximation by an expression of exactly the same form as the expression for the major axis, except that k takes the place of n. Using the same notation as before we have therefore

$$2a = n^2 \frac{h^2}{2\pi^2 N e^2 m}, \quad 2p = k^2 \frac{h^2}{2\pi^2 N e^2 m}. \tag{10}$$

For each of the stationary states which had previously been denoted by a given value of n, we obtain therefore a set of stationary states corresponding to values of k from 1 to n. Instead of the simple formula (5) Sommerfeld found a more complicated expression for the energy in the stationary states which depends on k as well as n. Taking the variation of the mass of the electron with velocity into account and neglecting terms of higher order of magnitude he obtained

$$E_{n,k} = -\frac{2\pi^2 N^2 e^4 m}{n^2 h^2}\left[1 + \frac{4\pi^2 N^2 e^4}{h^2 c^2}\left(-\frac{3}{4n^2} + \frac{1}{nk}\right)\right], \tag{11}$$

where c is the velocity of light.

Corresponding to each of the energy values for the stationary states of the hydrogen atom given by the simple formula (5) we obtain n values differing only very little from

one another, since the second term within the bracket is very small. With the aid of the general frequency relation (1) we therefore obtain a number of components with nearly coincident frequencies instead of each hydrogen line given by the simple formula (2). Sommerfeld has now shown that this calculation actually agrees with measurements of the fine structure. This agreement applies not only to the fine structure of the hydrogen lines which is very difficult to measure on account of the extreme proximity of the components, but it is also possible to account in detail for the fine structure of the helium lines given by formula (7) which has been very carefully investigated by Paschen. Sommerfeld in connection with this theory also pointed out that formula (11) could be applied to the X-ray spectra. Thus he showed that in the K and L groups pairs of lines appeared the differences of whose frequencies could be determined by the expression (11) for the energy in the stationary states which correspond to the binding of a single electron by a nucleus of charge Ne.

Periodic table. In spite of the great formal similarity between the X-ray spectra and the hydrogen spectrum indicated by these results a far-reaching difference must be assumed to exist between the processes which give rise to the appearance of these two types of spectra. While the emission of the hydrogen spectrum, like the emission of the ordinary optical spectra of other elements, may be assumed to be connected with the binding of an electron by an atom, ob-

servations on the appearance and absorption of X-ray spectra clearly indicate that these spectra are connected with a process which may be described as a *reorganization of the electronic arrangement* after a disturbance within the atom due to the effect of external agencies. We should therefore expect that the appearance of the X-ray spectra would depend not only upon the direct interaction between a single electron and the nucleus, but also on the manner in which the electrons are arranged in the completely formed atom.

The peculiar manner in which the properties of the elements vary with the atomic number, as expressed in the periodic system, provides a guide of great value in considering this latter problem. A simple survey of this system is given in fig. 1. The number preceding each element indicates the atomic number, and the elements within the various vertical columns form the different "periods" of the system. The lines, which connect pairs of elements in successive columns, indicate homologous properties of such elements. Compared with usual representations of the periodic system, this method, proposed more than twenty years ago by Julius Thomsen, of indicating the periodic variations in the properties of the elements is more suited for comparison with theories of atomic constitution. The meaning of the frames round certain sequences of elements within the later periods of the table will be explained later. They refer to certain characteristic features of the theory of atomic constitution.

In an explanation of the periodic system it is natural to assume a division of the electrons in the atom into dis-

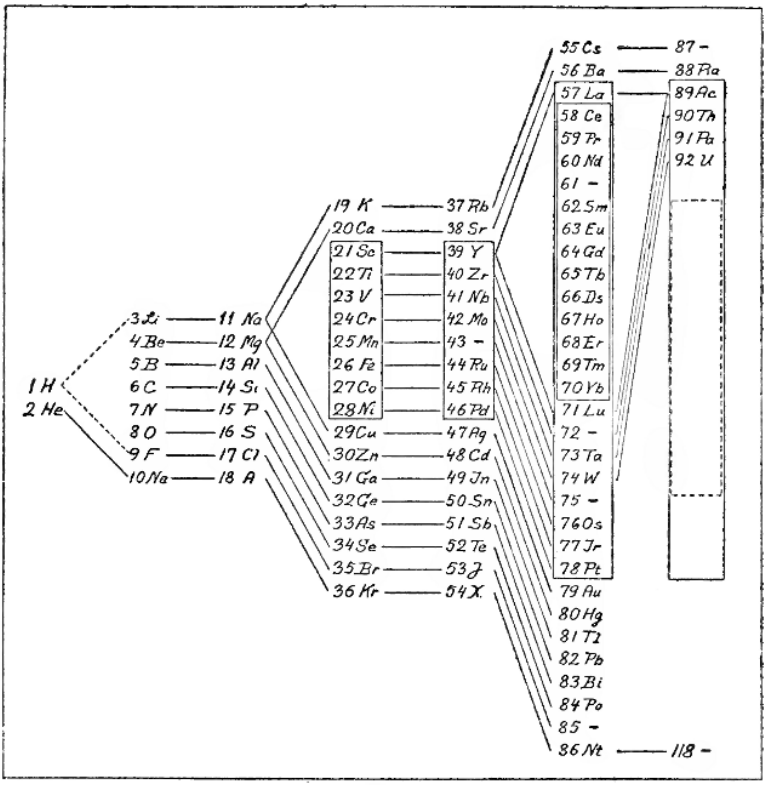


Fig. 1.

tinct groups in such a manner that the grouping of the elements in the system is attributed to the gradual formation of the groups of electrons in the atoms as the atomic number increases. Such a grouping of the electrons in the atom

has formed a prominent part of all more detailed views of atomic structure ever since J. J. Thomson's famous attempt to explain the periodic system on the basis of an investigation of the stability of various electronic configurations. Although Thomson's assumption regarding the distribution of the positive electricity in the atom is not consistent with more recent experimental evidence, nevertheless his work has exerted great influence upon the later development of the atomic theory on account of the many original ideas which it contained.

With the aid of the information concerning the binding of electrons by the nucleus obtained from the theory of the hydrogen spectrum I attempted in the same paper in which this theory was set forth to sketch in broad outlines a picture of the structure of the nucleus atom. In this it was assumed that each electron in its normal state moved in a manner analogous to the motion in the last stages of the binding of a single electron by a nucleus. As in Thomson's theory, it was assumed that the electrons moved in circular orbits and that the electrons in each separate group during this motion occupied positions with reference to one another corresponding to the vertices of plane regular polygons. Such an arrangement is frequently described as a distribution of the electrons in "rings." By means of these assumptions it was possible to account for the orders of magnitude of the dimensions of the atoms as well as the firmness with which the electrons were bound by the atom, a measure of which may be obtained by means of experiments on the excitation of the various types

of spectra. It was not possible, however, in this way to arrive at a detailed explanation of the characteristic properties of the elements even after it had become apparent from the results of Moseley and the work of Sommerfeld and others that this simple picture ought to be extended to include orbits in the fully formed atom characterized by higher quantum numbers corresponding to previous stages in the formation of the hydrogen atom. This point has been especially emphasized by Vegard.

The difficulty of arriving at a satisfactory picture of the atom is intimately connected with the difficulty of accounting for the pronounced "stability" which the properties of the elements demand. As I emphasized when considering the formation of the hydrogen atom, the postulates of the quantum theory aim directly at this point, but the results obtained in this way for an atom containing a single electron do not permit of a direct elucidation of problems like that of the distribution in groups of the electrons in an atom containing several electrons. If we imagine that the electrons in the groups of the atom are orientated relatively to one another at any moment, like the vertices of regular polygons, and rotating in either circles or ellipses, the postulates do not give sufficient information to determine the difference in the stability of electronic configurations with different numbers of electrons in the groups.

The peculiar character of stability of the atomic structure, demanded by the properties of the elements, is brought out in an interesting way by Kossel in two important papers.

In the first paper he shows that a more detailed explanation of the origin of the high frequency spectra can be obtained on the basis of the group structure of the atom. He assumes that a line in the X-ray spectrum is due to a process which may be described as follows: an electron is removed from the atom by some external action after which an electron in one of the other groups takes its place; this exchange of place may occur in as many ways as there are groups of more loosely bound electrons. This view of the origin of the characteristic X-rays afforded a simple explanation of the peculiar absorption phenomena observed. It has also led to the prediction of certain simple relations between the frequencies of the X-ray lines from one and the same element and has proved to be a suitable basis for the classification of the complete spectrum. However it has not been possible to develop a theory which reconciles in a satisfactory way Sommerfeld's work on the fine structure of the X-ray lines with Kossel's general scheme. As we shall see later the adoption of a new point of view when considering the stability of the atom renders it possible to bring the different results in a natural way in connection with one another.

In his second paper Kossel investigates the possibilities for an explanation of the periodic system on the basis of the atomic theory. Without entering further into the problem of the causes of the division of the electrons into groups, or the reasons for the different stability of the various electronic configurations, he points out in connection with ideas which had already played a part in Thomson's theory, how the pe-

riodic system affords evidence of a periodic appearance of especially stable configurations of electrons. These configurations appear in the neutral atoms of elements occupying the final position in each period in fig. 1, and the stability in question is assumed in order to explain not only the inactive chemical properties of these elements but also the characteristic active properties of the immediately preceding or succeeding elements. If we consider for instance an inactive gas like argon, the atomic number of which is 18, we must assume that the 18 electrons in the atom are arranged in an exceedingly regular configuration possessing a very marked stability. The pronounced electronegative character of the preceding element, chlorine, may then be explained by supposing the neutral atom which contains only 17 electrons to possess a tendency to capture an additional electron. This gives rise to a negative chlorine ion with a configuration of 18 electrons similar to that occurring in the neutral argon atom. On the other hand the marked electropositive character of potassium may be explained by supposing one of the 19 electrons in the neutral atom to be as it were superfluous, and that this electron therefore is easily lost; the rest of the atom forming a positive ion of potassium having a constitution similar to that of the argon atom. In a corresponding manner it is possible to account for the electronegative and electropositive character of elements like sulphur and calcium, whose atomic numbers are 16 and 20. In contrast to chlorine and potassium these elements are divalent, and the stable configuration of 18 electrons is formed by the addi-

tion of two electrons to the sulphur atom and by the loss of two electrons from the calcium atom. Developing these ideas Kossel has succeeded not only in giving interesting explanations of a large number of chemical facts, but has also been led to certain general conclusions about the grouping of the electrons in elements belonging to the first periods of the periodic system, which in certain respects are in conformity with the results to be discussed in the following paragraphs. Kossel's work was later continued in an interesting manner by Ladenburg with special reference to the grouping of the electrons in atoms of elements belonging to the later periods of the periodic table. It will be seen that Ladenburg's conclusions also exhibit points of similarity with the results which we shall discuss later.

Recent atomic models. Up to the present time it has not been possible to obtain a satisfactory account based upon a consistent application of the quantum theory to the nuclear atom of the ultimate cause of the pronounced stability of certain arrangements of electrons. Nevertheless it has been apparent for some time that the solution should be sought for by investigating the possibilities of a *spatial distribution of the electronic orbits* in the atom instead of limiting the investigation to configurations in which all electrons belonging to a particular group move in the same plane as was assumed for simplicity in my first papers on the structure of the atom. The necessity of assuming a spatial dis-

tribution of the configurations of electrons has been drawn attention to by various writers. Born and Landé, in connection with their investigations of the structure and properties of crystals, have pointed out that the assumption of spatial configurations appears necessary for an explanation of these properties. Landé has pursued this question still further, and as will be mentioned later has proposed a number of different "spatial atomic models" in which the electrons in each separate group of the atom at each moment form configurations possessing regular polyhedral symmetry. These models constitute in certain respects a distinct advance, although they have not led to decisive results on questions of the stability of atomic structure.

The importance of spatial electronic configurations has, in addition, been pointed out by Lewis and Langmuir in connection with their atomic models. Thus Lewis, who in several respects independently came to the same conclusions as Kossel, suggested that the number 8 characterizing the first groups of the periodic system might indicate a constitution of the outer atomic groups where the electrons within each group formed a configuration like the corners of a cube. He emphasized how a configuration of this kind leads to instructive models of the molecular structure of chemical combinations. It is to be remarked, however, that such a "static" model of electronic configuration will not be possible if we assume the forces within the atom to be due exclusively to the electric charges of the particles. Langmuir, who has attempted to develop Lewis' conceptions still further and to

account not only for the occurrence of the first octaves, but also for the longer periods of the periodic system, supposes therefore the structure of the atoms to be governed by forces whose nature is unknown to us. He conceives the atom to possess a "cellular structure," so that each electron is in advance assigned a place in a cell and these cells are arranged in shells in such a manner, that the various shells from the nucleus of the atom outward contain exactly the same number of places as the periods in the periodic system proceeding in the direction of increasing atomic number. Langmuir's work has attracted much attention among chemists, since it has to some extent thrown light on the conceptions with which empirical chemical science is concerned. On his theory the explanation of the properties of the various elements is based on a number of postulates about the structure of the atoms formulated for that purpose. Such a descriptive theory is sharply differentiated from one where an attempt is made to explain the specific properties of the elements with the aid of general laws applying to the interaction between the particles in each atom. The principal task of this lecture will consist in an attempt to show that an advance along these lines appears by no means hopeless, but on the contrary that with the aid of a consistent application of the postulates of the quantum theory it actually appears possible to obtain an insight into the structure and stability of the atom.

II. SERIES SPECTRA AND THE CAPTURE OF ELECTRONS BY ATOMS

We attack the problem of atomic constitution by asking the question: "How may an atom be formed by the successive capture and binding of the electrons one by one in the field of force surrounding the nucleus?"

Before attempting to answer this question it will first be necessary to consider in more detail what the quantum theory teaches us about the general character of the binding process. We have already seen how the hydrogen spectrum gives us definite information about the course of this process of binding the electron by the nucleus. In considering the formation of the atoms of other elements we have also in their spectra sources for the elucidation of the formation processes, but the direct information obtained in this way is not so complete as in the case of the hydrogen atom. For an element of atomic number N the process of formation may be regarded as occurring in N stages, corresponding with the successive binding of N electrons in the field of the nucleus. A spectrum must be assumed to correspond to each of these binding processes; but only for the first two elements, hydrogen and helium, do we possess a detailed knowledge of these spectra. For other elements of higher atomic number, where several spectra will be connected with the formation of the atom, we are at present acquainted with only two types, called the "arc" and "spark" spectra respectively, according to the experimental conditions of excitation. Although

these spectra show a much more complicated structure than the hydrogen spectrum, given by formula (2) and the helium spectrum given by formula (7), nevertheless in many cases it has been possible to find simple laws for the frequencies exhibiting a close analogy with the laws expressed by these formulae.

Arc and spark spectra. If for the sake of simplicity we disregard the complex structure shown by the lines of most spectra (occurrence of doublets, triplets etc.), the frequency of the lines of many arc spectra can be represented to a close approximation by the Rydberg formula

$$\nu = \frac{K}{(n'' + \alpha_{k''})^2} - \frac{K}{(n' + \alpha_{k'})^2}, \tag{12}$$

where n' and n'' are integral numbers, K the same constant as in the hydrogen spectrum, while $\alpha_{k'}$ and $\alpha_{k''}$ are two constants belonging to a set characteristic of the element. A spectrum with a structure of this kind is, like the hydrogen spectrum, called a series spectrum, since the lines can be arranged into series in which the frequencies converge to definite limiting values. These series are for example represented by formula (12) if, using two definite constants for $\alpha_{k''}$ and $\alpha_{k'}$, n'' remains unaltered, while n' assumes a series of successive, gradually increasing integral values.

Formula (12) applies only approximately, but it is always found that the frequencies of the spectral lines can be writ-

ten, as in formulae (2) and (12), as a difference of two functions of integral numbers. Thus the latter formula applies accurately, if the quantities α_k are not considered as constants, but as representatives of a set of series of numbers $\alpha_k(n)$ characteristic of the element, whose values for increasing n within each series quickly approach a constant limiting value. The fact that the frequencies of the spectra always appear as the difference of two terms, the so-called "spectral terms," from the combinations of which the complete spectrum is formed, has been pointed out by Ritz, who with the establishment of the combination principle has greatly advanced the study of the spectra. The quantum theory offers an immediate interpretation of this principle, since, according to the frequency relation we are led to consider the lines as due to transitions between stationary states of the atom, just as in the hydrogen spectrum, only in the spectra of the other elements we have to do not with a single series of stationary states, but with a set of such series. From formula (12) we thus obtain for an arc spectrum—if we temporarily disregard the structure of the individual lines—information about an ensemble of stationary states, for which the energy of the atom in the nth state of the kth series is given by

$$E_k(n) = -\frac{Kh}{(n+\alpha_k)^2} \tag{13}$$

very similar to the simple formula (3) for the energy in the stationary states of the hydrogen atom.

As regards the spark spectra, the structure of which has

been cleared up mainly by Fowler's investigations, it has been possible in the case of many elements to express the frequencies approximately by means of a formula of exactly the same type as (12), only with the difference that K, just as in the helium spectrum given by formula (7), is replaced by a constant, which is four times as large. For the spark spectra, therefore, the energy values in the corresponding stationary states of the atom will be given by an expression of the same type as (13), only with the difference that K is replaced by $4K$.

This remarkable similarity between the structure of these types of spectra and the simple spectra given by (2) and (7) is explained simply by assuming the arc spectra to be connected with the *last stage in the formation of the neutral atom* consisting in the capture and binding of the Nth electron. On the other hand the spark spectra are connected with the *last stage but one in the formation of the atom*, namely the binding of the $(N-1)$th electron. In these cases the field of force in which the electron moves will be much the same as that surrounding the nucleus of a hydrogen or helium atom respectively, at least in the earlier stages of the binding process, where during the greater part of its revolution it moves at a distance from the nucleus which is large in proportion to the dimensions of the orbits of the electrons previously bound. From analogy with formula (3) giving the stationary states of the hydrogen atom, we shall therefore assume that the numerical value of the expression on the right-hand side of (13) will be equal to the work required to

remove the last captured electron from the atom, the binding of which gives rise to the arc spectrum of the element.

Series diagram. While the origin of the arc and spark spectra was to this extent immediately interpreted on the basis of the original simple theory of the hydrogen spectrum, it was Sommerfeld's theory of the fine structure of the hydrogen lines which first gave us a clear insight into the characteristic difference between the hydrogen spectrum and the spark spectrum of helium on the one hand, and the arc and spark spectra of other elements on the other. When we consider the binding not of the first but of the subsequent electrons in the atom, the orbit of the electron under consideration—at any rate in the latter stages of the binding process where the electron last bound comes into intimate interaction with those previously bound—will no longer be to a near approximation a closed ellipse, but on the contrary will to a first approximation be a central orbit of the same type as in the hydrogen atom, when we take into account the change with velocity in the mass of the electron. This motion, as we have seen, may be resolved into a plane periodic motion upon which a uniform rotation is superposed in the plane of the orbit; only the superposed rotation will in this case be comparatively much more rapid and the deviation of the periodic orbit from an ellipse much greater than in the case of the hydrogen atom. For an orbit of this type the stationary states, just as in the theory of the fine structure,

will be determined by two quantum numbers which we shall denote by n and k, connected in a very simple manner with the kinematic properties of the orbit. For brevity I shall only mention that while the quantum number k is connected with the value of the constant angular momentum of the electron about the centre in the simple manner previously indicated, the determination of the principal quantum number n requires an investigation of the whole course of the orbit and for an arbitrary central orbit will not be related in a simple way to the dimensions of the rotating periodic orbit if this deviates essentially from a Keplerian ellipse.

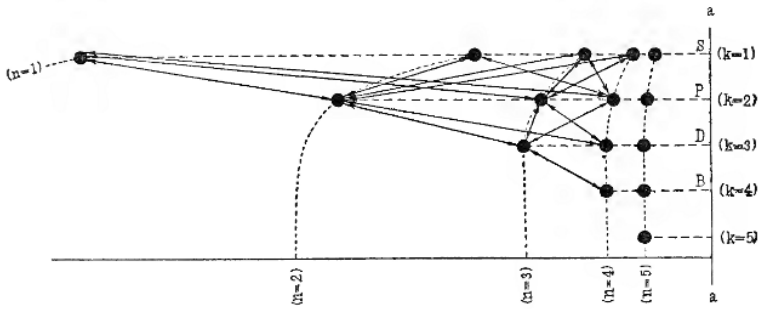


Fig. 2.

These results are represented in fig. 2 which is a reproduction of an illustration I have used on a previous occasion (see Essay II, page 46), and which gives a survey of the origin of the sodium spectrum. The black dots represent the stationary states corresponding to the various series of spectral terms, shown on the right by the letters S, P, D and B.

These letters correspond to the usual notations employed in spectroscopic literature and indicate the nature of the series (sharp series, principal series, diffuse series, etc.) obtained by combinations of the corresponding spectral terms. The distances of the separate points from the vertical line at the right of the figure are proportional to the numerical value of the energy of the atom given by equation (13). The oblique, black arrows indicate finally the transitions between the stationary states giving rise to the appearance of the lines in the commonly observed sodium spectrum. The values of n and k attached to the various states indicate the quantum numbers, which, according to Sommerfeld's theory, from a preliminary consideration might be regarded as characterizing the orbit of the outer electron. For the sake of convenience the states which were regarded as corresponding to the same value of n are connected by means of dotted lines, and these are so drawn that their vertical asymptotes correspond to the terms in the hydrogen spectrum which belong to the same value of the principal quantum number. The course of the curves illustrates how the deviation from the hydrogen terms may be expected to decrease with increasing values of k, corresponding to states, where the minimum distance between the electron in its revolution and the nucleus constantly increases.

It should be noted that even though the theory represents the principal features of the structure of the series spectra it has not yet been possible to give a detailed account of the spectrum of any element by a closer investigation of the

electronic orbits which may occur in a simple field of force possessing central symmetry. As I have mentioned already the lines of most spectra show a complex structure. In the sodium spectrum for instance the lines of the principal series are doublets indicating that to each P-term not one stationary state, but two such states correspond with slightly different values of the energy. This difference is so little that it would not be recognizable in a diagram on the same scale as fig. 2. The appearance of these doublets is undoubtedly due to the small deviations from central symmetry of the field of force originating from the inner system in consequence of which the general type of motion of the external electron will possess a more complicated character than that of a simple central motion. As a result the stationary states must be characterized by more than two quantum numbers, in the same way that the occurrence of deviations of the orbit of the electron in the hydrogen atom from a simple periodic orbit requires that the stationary states of this atom shall be characterized by more than one quantum number. Now the rules of the quantum theory lead to the introduction of a third quantum number through the condition that the resultant angular momentum of the atom, multiplied by 2π, is equal to an entire multiple of Planck's constant. This determines the orientation of the orbit of the outer electron relative to the axis of the inner system.

In this way Sommerfeld, Landé and others have shown that it is possible not only to account in a formal way for the complex structure of the lines of the series spectra, but

also to obtain a promising interpretation of the complicated effect of external magnetic fields on this structure. We shall not enter here on these problems but shall confine ourselves to the problem of the fixation of the two quantum numbers n and k, which to a first approximation describe the orbit of the outer electron in the stationary states, and whose determination is a matter of prime importance in the following discussion of the formation of the atom. In the determination of these numbers we at once encounter difficulties of a profound nature, which—as we shall see—are intimately connected with the question of the remarkable stability of atomic structure. I shall here only remark that the values of the quantum number n, given in the figure, undoubtedly cannot be retained, neither for the S nor the P series. On the other hand, so far as the values employed for the quantum number k are concerned, it may be stated with certainty, that the interpretation of the properties of the orbits, which they indicate, is correct. A starting point for the investigation of this question has been obtained from considerations of an entirely different kind from those previously mentioned, which have made it possible to establish a close connection between the motion in the atom and the appearance of spectral lines.

Correspondence principle. So far as the principles of the quantum theory are concerned, the point which has been emphasized hitherto is the radical departure of these principles from our usual conceptions of mechanical and elec-

trodynamical phenomena. As I have attempted to show in recent years, it appears possible, however, to adopt a point of view which suggests that the quantum theory may, nevertheless, be regarded as a rational generalization of our ordinary conceptions. As may be seen from the postulates of the quantum theory, and particularly the frequency relation, a direct connection between the spectra and the motion of the kind required by the classical dynamics is excluded, but at the same time the form of these postulates leads us to another relation of a remarkable nature. Let us consider an electrodynamic system and inquire into the nature of the radiation which would result from the motion of the system on the basis of the ordinary conceptions. We imagine the motion to be decomposed into purely harmonic oscillations, and the radiation is assumed to consist of the simultaneous emission of series of electromagnetic waves possessing the same frequency as these harmonic components and intensities which depend upon the amplitudes of the components. An investigation of the formal basis of the quantum theory shows us now, that it is possible to trace the question of the origin of the radiation processes which accompany the various transitions back to an investigation of the various harmonic components, which appear in the motion of the atom. The possibility, that a particular transition shall occur, may be regarded as being due to the presence of a definitely assignable "corresponding" component in the motion. This principle of correspondence at the same time throws light upon a question mentioned several times previously,

namely the relation between the number of quantum numbers, which must be used to describe the stationary states of an atom, and the types to which the orbits of the electrons belong. The classification of these types can be based very simply on a decomposition of the motion into its harmonic components. Time does not permit me to consider this question any further, and I shall confine myself to a statement of some simple conclusions, which the correspondence principle permits us to draw concerning the occurrence of transitions between various pairs of stationary states. These conclusions are of decisive importance in the subsequent argument.

The simplest example of such a conclusion is obtained by considering an atomic system, which contains a particle describing a *purely periodic orbit*, and where the stationary states are characterized by a single quantum number n. In this case the motion can according to Fourier's theorem be decomposed into a simple series of harmonic oscillations whose frequency may be written $\tau\omega$, where τ is a whole number, and ω is the frequency of revolution in the orbit. It can now be shown that a transition between two stationary states, for which the values of the quantum number are respectively equal to n' and n'', will correspond to a harmonic component, for which $\tau = n' - n''$. This throws at once light upon the remarkable difference which exists between the possibilities of transitions between the stationary states of a hydrogen atom on the one hand and of a simple system consisting of an electric particle capable of executing simple harmonic oscillations about a position of equilibrium on the

other. For the latter system, which is frequently called a Planck oscillator, the energy in the stationary states is determined by the familiar formula $E = nh\omega$, and with the aid of the frequency relation we obtain therefore for the radiation which will be emitted during a transition between two stationary states $\nu = (n' - n'')\omega$. Now, an important assumption, which is not only essential in Planck's theory of temperature radiation, but which also appears necessary to account for the molecular absorption in the infra-red region of radiation, states that a harmonic oscillator will only emit and absorb radiation, for which the frequency ν is equal to the frequency of oscillation ω of the oscillator. We are therefore compelled to assume that in the case of the oscillator transitions can occur only between stationary states which are characterized by quantum numbers differing by only one unit, while in the hydrogen spectrum represented by formula (2) all possible transitions could take place between the stationary states given by formula (5). From the point of view of the principle of correspondence it is seen, however, that this apparent difficulty is explained by the occurrence in the motion of the hydrogen atom, as opposed to the motion of the oscillator, of harmonic components corresponding to values of τ, which are different from 1; or using a terminology well known from acoustics, there appear overtones in the motion of the hydrogen atom.

Another simple example of the application of the correspondence principle is afforded by a *central motion*, to the investigation of which the explanation of the series spectra

in the first approximation may be reduced. Referring once more to the figure of the sodium spectrum, we see that the black arrows, which correspond to the spectral lines appearing under the usual conditions of excitation, only connect pairs of points in consecutive rows. Now it is found that this remarkable limitation of the occurrence of combinations between spectral terms may quite naturally be explained by an investigation of the harmonic components into which a central motion can be resolved. It can readily be shown that such a motion can be decomposed into two series of harmonic components, whose frequencies can be expressed by $\tau\omega + \sigma$ and $\tau\omega - \sigma$ respectively, where τ is a whole number, ω the frequency of revolution in the rotating periodic orbit and σ the frequency of the superposed rotation. These components correspond with transitions where the principal number n decreases by τ units, while the quantum number k decreases or increases, respectively, by one unit, corresponding exactly with the transitions indicated by the black arrows in the figure. This may be considered as a very important result, because we may say, that the quantum theory, which for the first time has offered a simple interpretation of the fundamental principle of combination of spectral lines has at the same time removed the mystery which has hitherto adhered to the application of this principle on account of the apparent capriciousness of the appearance of predicted combination lines. Especially attention may be drawn to the simple interpretation which the quantum theory offers of the appearance observed by Stark and his collaborators

of certain new series of lines, which do not appear under ordinary circumstances, but which are excited when the emitting atoms are subject to intense external electric fields. In fact, on the correspondence principle this is immediately explained from an examination of the perturbations in the motion of the outer electron which give rise to the appearance in this motion—besides the harmonic components already present in a simple central orbit—of a number of constituent harmonic vibrations of new type and of amplitudes proportional to the intensity of the external forces.

It may be of interest to note that an investigation of the limitation of the possibility of transitions between stationary states, based upon a simple consideration of conservation of angular momentum during the process of radiation, does not, contrary to what has previously been supposed (compare Essay II, page 94), suffice to throw light on the remarkably simple structure of series spectra illustrated by the figure. As mentioned above we must assume that the "complexity" of the spectral terms, corresponding to given values of n and k, which we witness in the fine structure of the spectral lines, may be ascribed to states, corresponding to different values of this angular momentum, in which the plane of the electronic orbit is orientated in a different manner, relative to the configuration of the previously bound electrons in the atom. Considerations of conservation of angular momentum can, in connection with the series spectra, therefore only contribute to an understanding of the limitation of the possibilities of combination observed in the peculiar laws applying to the

number of components in the complex structure of the lines. So far as the last question is concerned, such considerations offer a direct support for the consequences of the correspondence principle.

III. FORMATION OF ATOMS AND THE PERIODIC TABLE

A correspondence has been shown to exist between the motion of the electron last captured and the occurrence of transitions between the stationary states corresponding to the various stages of the binding process. This fact gives a point of departure for a choice between the numerous possibilities which present themselves when considering the formation of the atoms by the successive capture and binding of the electrons. Among the processes which are conceivable and which according to the quantum theory might occur in the atom we shall reject those whose occurrence cannot be regarded as consistent with a correspondence of the required nature.

First Period. Hydrogen—Helium. It will not be necessary to concern ourselves long with the question of the constitution of the hydrogen atom. From what has been said previously we may assume that the final result of the process of *binding of the first electron* in any atom will be a stationary state, where the energy of the atom is given by (5), if we put $n = 1$, or more precisely by formula (11), if we put $n = 1$ and $k = 1$. The orbit of the electron will be a circle whose

radius will be given by formulae (10), if n and k are each put equal to 1. Such an orbit will be called a 1-quantum orbit, and in general an orbit for which the principal quantum number has a given value n will be called an n-quantum orbit. Where it is necessary to differentiate between orbits corresponding to various values of the quantum number k, a central orbit, characterized by given values of the quantum numbers n and k, will be referred to as an n_k orbit.

In the question of the constitution of the helium atom we meet the much more complicated problem of the *binding of the second electron*. Information about this binding process may, however, be obtained from the arc spectrum of helium. This spectrum, as opposed to most other simple spectra, consists of two complete systems of lines with frequencies given by formulae of the type (12). On this account helium was at first assumed to be a mixture of two different gases, "orthohelium" and "parhelium," but now we know that the two spectra simply mean that the binding of the second electron can occur in two different ways. A theoretical explanation of the main features of the helium spectrum has recently been attempted in an interesting paper by Landé. He supposes the emission of the orthohelium spectrum to be due to transitions between stationary states where both electrons move in the same plane and revolve in the same sense. The parhelium spectrum, on the other hand, is ascribed by him to stationary states where the planes of the orbits form an angle with each other. Dr Kramers and I have made a closer investigation of the interaction between the two orbits

in the different stationary states. The results of our investigation which was begun several years before the appearance of Landé's work have not yet been published. Without going into details I may say, that even though our results in several respects differ materially from those of Landé (compare Essay II, page 85), we agree with his general conclusions concerning the origin of the orthohelium and parhelium spectra.

The final result of the binding of the second electron is intimately related to the origin of the two helium spectra. Important information on this point has been obtained recently by Franck and his co-workers. As is well known he has thrown light upon many features of the structure of the atom and of the origin of spectra by observing the effect of bombarding atoms by electrons of various velocities. A short time ago these experiments showed that the impact of electrons could bring helium into a "metastable" state from which the atom cannot return to its normal state by means of a simple transition accompanied by the emission of radiation, but only by means of a process analogous to a chemical reaction involving interaction with atoms of other elements. This result is closely connected with the fact that the binding of the second electron can occur in two different ways, as is shown by the occurrence of two distinct spectra. Thus it is evident from Franck's experiments that the normal state of the atom is the last stage in the binding process involving the emission of the parhelium spectrum by which the electron last captured as well as the one first captured will be bound in a 1_1 orbit. The metastable state, on the contrary,

is the final stage of the process giving the orthohelium spectrum. In this case the second electron, as opposed to the first, will move in a 2_1 orbit. This corresponds to a firmness of binding which is about six times less than for the electron in the normal state of the atom.

If we now consider somewhat more closely this apparently surprising result, it is found that a clear grasp of it may be obtained from the point of view of correspondence. It can be shown that the coherent class of motions to which the orthohelium orbits belong does not contain a 1_1 orbit. If on the whole we would claim the existence of a state where the two electrons move in 1_1 orbits in the same plane, and if in addition it is claimed that the motion should possess the periodic properties necessary for the definition of stationary states, then there seems that no possibility is afforded other than the assumption that the two electrons move around the nucleus in one and the same orbit, in such a manner that at each moment they are situated at the ends of a diameter. This extremely simple ring-configuration might be expected to correspond to the firmest possible binding of the electrons in the atom, and it was on this account proposed as a model for the helium atom in my first paper on atomic structure. If, however, we inquire about the possibility of a transition from one of the orthohelium states to a configuration of this type we meet conditions which are very different from those which apply to transitions between two of the orthohelium orbits. In fact, the occurrence of each of these transitions is due to the existence of well-defined corresponding con-

stituent harmonic vibration in the central orbits which the outer electron describes in the class of motions to which the stationary states belong. The transition we have to discuss, on the other hand, is one by which the last captured electron is transferred from a state in which it is moving "outside" the other to a state in which it moves round the nucleus on equal terms with the other electron. Now it is impossible to find a series of simple intermediate forms for the motion of those two electrons in which the orbit of the last captured electron exhibits a sufficient similarity to a central motion that for this transition there could be a correspondence of the necessary kind. It is therefore evident, that where the two electrons move in the same plane, the electron captured last cannot be bound firmer than in a 2_1 orbit. If, on the other hand, we consider the binding process which accompanies the emission of the parhelium spectrum and where the electrons in the stationary states move in orbits whose planes form angles with one another we meet essentially different conditions. A corresponding intimate change in the interaction between the electron last captured and the one previously bound is not required here for the two electrons in the atom to become equivalent. We may therefore imagine the last stage of the binding process to take place in a manner similar to those stages corresponding to transitions between orbits characterized by greater values of n and k.

In the *normal state of the helium atom* the two electrons must be assumed to move in equivalent 1_1 orbits. As a first approximation these may be described as two circular or-

bits, whose planes make an angle of 120° with one another, in agreement with the conditions which the angular momentum of an atom according to the quantum theory must satisfy. On account of the interaction between the two electrons these planes at the same time turn slowly around the fixed impulse axis of the atom. Starting from a distinctly different point of view Kemble has recently suggested a similar model for the helium atom. He has at the same time directed attention to a possible type of motion of very marked symmetry in which the electrons during their entire revolution assume symmetrical positions with reference to a fixed axis. Kemble has not, however, investigated this motion further. Previous to the appearance of this paper Kramers had commenced a closer investigation of precisely this type of motion in order to find out to what extent it was possible from such a calculation to account for the firmness with which the electrons are bound in the helium atom, that is to account for the ionization potential. Early measurements of this potential had given values corresponding approximately to that which would result from the ring-configuration already mentioned. This requires 17/8 as much work to remove a single electron as is necessary to remove an electron from the hydrogen atom in its normal state. As the theoretical value for the latter amount of work—which for the sake of simplicity will be represented by W—corresponds to an ionization potential of 13.53 volts, the ionization potential of helium would be expected to be 28.8 volts. Recent and more accurate determinations, however, have given a value for the ionization

potential of helium which is considerably lower and lies in the neighbourhood of 25 volts. This showed therefore the untenability of the ring-configuration quite independently of any other considerations. A careful investigation of the spatial atomic configuration requires elaborate calculation, and Kramers has not yet obtained final results. With the approximation to which they have been so far completed the calculations point to the possibility of an agreement with the experimental results. The final result may be awaited with great interest, since it offers in the simplest case imaginable a test of the principles by which we are attempting to determine stationary states of atoms containing more than one electron.

Hydrogen and helium, as seen in the survey of the periodic system given in fig. 1, together form the first period in the system of elements, since helium is the first of the inactive gases. The great difference in the chemical properties of hydrogen and helium is closely related to the great difference in the nature of the binding of the electron. This is directly indicated by the spectra and ionization potentials. While helium possesses the highest known ionization potential of all the elements, the binding of the electron in the hydrogen atom is sufficiently loose to account for the tendency of hydrogen to form positive ions in aqueous solutions and chemical combinations. Further consideration of this particular question requires, however, a comparison between the nature and firmness of the electronic configurations of other atoms, and it can therefore not be discussed at the moment.

Second Period. Lithium—Neon. When considering the atomic structure of elements which contain more than two electrons in the neutral atom, we shall assume first of all that what has previously been said about the formation of the helium atom will in the main features also apply to the capture and binding of the first two electrons. These electrons may, therefore, in the normal state of the atom be regarded as moving in equivalent orbits characterized by the quantum symbol 1_1. We obtain direct information about the *binding of the third electron* from the spectrum of lithium. This spectrum shows the existence of a number of series of stationary states, where the firmness with which the last captured electron is bound is very nearly the same as in the stationary states of the hydrogen atom. These states correspond to orbits where k is greater than or equal to 2, and where the last captured electron moves entirely outside the region where the first two electrons move. But in addition this spectrum gives us information about a series of states corresponding to $k = 1$ in which the energy differs essentially from the corresponding stationary states of the hydrogen atom. In these states the last captured electron, even if it remains at a considerable distance from the nucleus during the greater part of its revolution, will at certain moments during the revolution approach to a distance from the nucleus which is of the same order of magnitude as the dimensions of the orbits of the previously bound electrons. On this account the electrons will be bound with a firmness which is considerably greater than that with which the

electrons are bound in the stationary states of the hydrogen atom corresponding to the same value of n.

Now as regards the lithium spectrum as well as the other alkali spectra we are so fortunate (see page 48) as to possess definite evidence about the normal state of the atom from experiments on selective absorption. In fact these experiments tell us that the first member of the sequence of S-terms corresponds to this state. This term corresponds to a strength of binding which is only a little more than a third of that of the hydrogen atom. We must therefore conclude that the outer electron in the normal state of the lithium atom moves in a 2_1 orbit, just as the outer electron in the metastable state of the helium atom. The reason why the binding of the outer electron cannot proceed to an orbit characterized by a smaller value for the total quantum number may also be considered as analogous in the two cases. In fact, a transition by which the third electron in the lithium atom was ultimately bound in a 1_1 orbit would lead to a state in the atom in which the electron would play an equivalent part with the two electrons previously bound. Such a process would be of a type entirely different from the transitions between the stationary states connected with the emission of the lithium spectrum, and would, contrary to these, not exhibit a correspondence with a harmonic component in the motion of the atom.

We obtain, therefore, a picture of the formation and structure of the lithium atom which offers a natural explanation of the great difference of the chemical properties of

lithium from those of helium and hydrogen. This difference is at once explained by the fact that the firmness by which the last captured electron is bound in its 2_1 orbit in the lithium atom is only about a third of that with which the electron in the hydrogen atom is held, and almost five times smaller than the firmness of the binding of the electrons in the helium atom.

What has been said here applies not alone to the formation of the lithium atom, but may also be assumed to apply to the binding of the third electron in every atom, so that in contrast to the first two electrons which move in 1_1 orbits this may be assumed to move in a 2_1 orbit. As regards the *binding of the fourth, fifth and sixth electrons* in the atom, we do not possess a similar guide as no simple series spectra are known of beryllium, boron and carbon. Although conclusions of the same degree of certainty cannot be reached it seems possible, however, to arrive at results consistent with general physical and chemical evidence by proceeding by means of considerations of the same kind as those applied to the binding of the first three electrons. In fact, we shall assume that the fourth, fifth and sixth electrons will be bound in 2_1 orbits. The reason why the binding of a first electron in an orbit of this type will not prevent the capture of the others in two quanta orbits may be ascribed to the fact that 2_1 orbits are not circular but very eccentric; For example, the 3rd electron cannot keep the remaining electrons away from the inner system in the same way in which the first two electrons bound in the lithium atom prevent the

third from being bound in a 1-quantum orbit. Thus we shall expect that the 4th, 5th and 6th electrons in a similar way to the 3rd will at certain moments of their revolution enter into the region where the first two bound electrons move. We must not imagine, however, that these visits into the inner system take place at the same time, but that the four electrons visit the nucleus separately at equal intervals of time. In earlier work on atomic structure it was supposed that the electrons in the various groups in the atom moved in separate regions within the atom and that at each moment the electrons within each separate group were arranged in configurations possessing symmetry like that of a regular polygon or polyhedron. Among other things this involved that the electrons in each group were supposed to be at the point of the orbit nearest the nucleus at the same time. A structure of this kind may be described as one where the motions of the electrons within the groups are coupled together in a manner which is largely independent of the interaction between the various groups. On the contrary, the characteristic feature of a structure like that I have suggested is the *intimate coupling between the motions of the electrons in the various groups* characterized by different quantum numbers, as well as the *greater independence in the mode of binding within one and the same group of electrons* the orbits of which are characterized by the same quantum number. In emphasizing this last feature I have two points in mind. Firstly the smaller effect of the presence of previously bound electrons on the firmness of binding of succeeding electrons in the same

group. Secondly the way in which the motions of the electrons within the group reflect the independence both of the processes by which the group can be formed and by which it can be reorganized by change of position of the different electrons in the atom after a disturbance by external forces. The last point will be considered more closely when we deal with the origin and nature of the X-ray spectra; for the present we shall continue the consideration of the structure of the atom to which we are led by the investigation of the processes connected with the successive capture of the electrons.

The preceding considerations enable us to understand the fact that the two elements beryllium and boron immediately succeeding lithium can appear electropositively with 2 and 3 valencies respectively in combination with other substances. For like the third electron in the lithium atom, the last captured electrons in these elements will be much more lightly bound than the first two electrons. At the same time we understand why the electropositive character of these elements is less marked than in the case of lithium, since the electrons in the 2-quanta orbits will be much more firmly bound on account of the stronger field in which they are moving. New conditions arise, however, in the case of the next element, carbon, as this element in its typical chemical combinations cannot be supposed to occur as an ion, but rather as a neutral atom. This must be assumed to be due not only to the great firmness in the binding of the electrons but also to be an essential consequence of the symmetrical configuration of the electrons.

With the binding of the 4th, 5th and 6th electrons in 2_1 orbits, the spatial symmetry of the regular configuration of the orbits must be regarded as steadily increasing, until with the binding of the 6th electron the orbits of the four last bound electrons may be expected to form an exceptionally symmetrical configuration in which the normals to the planes of the orbits occupy positions relative to one another nearly the same as the lines from the centre to the vertices of a regular tetrahedron. Such a configuration of groups of 2-quanta orbits in the carbon atom seems capable of furnishing a suitable foundation for explaining the structure of organic compounds. I shall not discuss this question any further, for it would require a thorough study of the interaction between the motions of the electrons in the atoms forming the molecule. I might mention, however, that the types of molecular models to which we are led are very different from the molecular models which were suggested in my first papers. In these the chemical "valence bonds" were represented by "electron rings" of the same type as those which were assumed to compose the groups of electrons within the individual atoms. It is nevertheless possible to give a general explanation of the chemical properties of the elements without touching on those matters at all. This is largely due to the fact that the structures of combinations of atoms of the same element and of many organic compounds do not have the same significance for our purpose as those molecular structures in which the individual atoms occur as electrically charged ions. The latter kind of compounds, to which

the greater number of simple inorganic compounds belong, is frequently called "heteropolar" and possesses a far more typical character than the first compounds which are called "homoeopolar," and whose properties to quite a different degree exhibit the individual peculiarities of the elements. My main purpose will therefore be to consider the fitness which the configurations of the electrons in the various atoms offer for the formation of ions.

Before leaving the carbon atom I should mention, that a model of this atom in which the orbits of the four most lightly bound electrons possess a pronounced tetrahedric symmetry had already been suggested by Landé. In order to agree with the measurements of the size of the atoms he also assumed that these electrons moved in 2_1 orbits. There is, however, this difference between Landé's view and that given here, that while Landé deduced the characteristic properties of the carbon atom solely from an investigation of the simplest form of motion which four electrons can execute employing spatial symmetry, our view originates from a consideration of the stability of the whole atom. For our assumptions about the orbits of the electrons are based directly on an investigation of the interaction between these electrons and the first two bound electrons. The result is that our model of the carbon atom has dynamic properties which are essentially different from the properties of Landé's model.

In order to account for the properties of *the elements in the second half of the second period* it will first of all be necessary to show why the configuration of ten electrons oc-

curring in the neutral atom of neon possesses such a remarkable degree of stability. Previously it has been assumed that the properties of this configuration were due to the interaction between eight electrons which moved in equivalent orbits outside the nucleus and an inner group of two electrons like that in the helium atom. It will be seen, however, that the solution must be sought in an entirely different direction. It cannot be expected that *the 7th electron* will be bound in a 2_1 orbit equivalent to the orbits of the four preceding electrons. The occurrence of five such orbits would so definitely destroy the symmetry in the interaction of these electrons that it is inconceivable that a process resulting in the accession of a fifth electron to this group would be in agreement with the correspondence principle. On the contrary it will be necessary to assume that the four electrons in their exceptionally symmetrical orbital configuration will keep out later captured electrons with the result that these electrons will be bound in orbits of other types.

The orbits which come into consideration for the 7th electron in the nitrogen atom and the 7th, 8th, 9th and 10th electrons in the atoms of the immediately following elements will be circular orbits of the type 2_2. The diameters of these orbits are considerably larger than those of the l_1 orbits of the first two electrons; on the other hand the outermost part of the eccentric 2_1 orbits will extend some distance beyond these circular 2_2 orbits. I shall not here discuss the capture and binding of these electrons. This requires a further investigation of the interaction between the motions of the

electrons in the two types of 2-quanta orbits. I shall simply mention, that in the atom of neon in which we will assume that there are four electrons in 2_2 orbits the planes of these orbits must be regarded not only as occupying a position relative to one another characterized by a high degree of spatial symmetry, but also as possessing a configuration harmonizing with the four elliptical 2_1 orbits. An interaction of this kind in which the orbital planes do not coincide can be attained only if the configurations in both subgroups exhibit a systematic deviation from tetrahedral symmetry. This will have the result that the electron groups with 2-quanta orbits in the neon atom will have only a single axis of symmetry which must be supposed to coincide with the axis of symmetry of the innermost group of two electrons.

Before leaving the description of the elements within the second period it may be pointed out that the above considerations offer a basis for interpreting that tendency of the neutral atoms of oxygen and fluorine for capturing further electrons which is responsible for the marked electronegative character of these elements. In fact, this tendency may be ascribed to the fact that the orbits of the last captured electrons will find their place within the region, in which the previously captured electrons move in 2_1 orbits. This suggests an explanation of the great difference between the properties of the elements in the latter half of the second period of the periodic system and those of the elements in the first half, in whose atoms there is only a single type of 2-quanta orbits.

Third Period. Sodium—Argon. We shall now consider the structure of atoms of elements in the third period of the periodic system. This brings us immediately 'to the question of *the binding of the* 11*th electron* in the atom. Here we meet conditions which in some respects are analogous to those connected with the binding of the 7th electron. The same type of argument that applied to the carbon atom shows that the symmetry of the configuration in the neon atom would be essentially, if not entirely, destroyed by the addition of another electron in an orbit of the same type as that in which the last captured electrons were bound. Just as in the case of the 3rd and 7th electrons we may therefore expect to meet a new type of orbit for the 11th electron in the atom, and the orbits which present themselves this time are the 3_1 orbits. An electron in such an orbit will for the greater part of the time remain outside the orbits of the first ten electrons. But at certain moments during the revolution it will penetrate not only into the region of the 2-quanta orbits, but like the 2_1 orbits it will penetrate to distances from the nucleus which are smaller than the radii of the 1-quantum orbits of the two electrons first bound. This fact, which has a most important bearing on the stability of the atom, leads to a peculiar result as regards the binding of the 11th electron. In the sodium atom this electron will move in a field which so far as the outer part of the orbit is concerned deviates only very little from that surrounding the nucleus in the hydrogen atom, but the dimensions of this part of the orbit will, nevertheless, be essentially different from the dimensions of the

corresponding part of a 3_1 orbit in the hydrogen atom. This arises from the fact, that even though the electron only enters the inner configuration of the first ten electrons for short intervals during its revolution, this part of the orbit will nevertheless exert an essential influence upon the determination of the principal quantum number. This is directly related to the fact that the motion of the electron in the first part of the orbit deviates only a little from the motion which each of the previously bound electrons in 2_1 orbits executes during a complete revolution. The uncertainty which has prevailed in the determination of the quantum numbers for the stationary states corresponding to a spectrum like that of sodium is connected with this. This question has been discussed by several physicists. From a comparison of the spectral terms of the various alkali metals, Roschdestwensky has drawn the conclusion that the normal state does not, as we might be inclined to expect a priori, correspond to a 1_1 orbit as shown in fig. 2 on page 119, but that this state corresponds to a 2_1 orbit. Schrödinger has arrived at a similar result in an attempt to account for the great difference between the S terms and the terms in the P and D series of the alkali spectra. He assumes that the "outer" electron in the states corresponding to the S terms—in contrast to those corresponding to the P and D terms—penetrates partly into the region of the orbits of the inner electrons during the course of its revolution. These investigations contain without doubt important hints, but in reality the conditions must be very different for the different alkali spectra. Instead of a 2_1 orbit as in lithium we

must thus assume for the spectrum of sodium not only that the first spectral term in the S series corresponds to a 3_1 orbit, but also, as a more detailed consideration shows, that the first term in the P series corresponds not to a 2_2 orbit as indicated in fig. 2, but to a 3_2 orbit. If the numbers in this figure were correct, it would require among other things that the P terms should be smaller than the hydrogen terms corresponding to the same principal quantum number. This

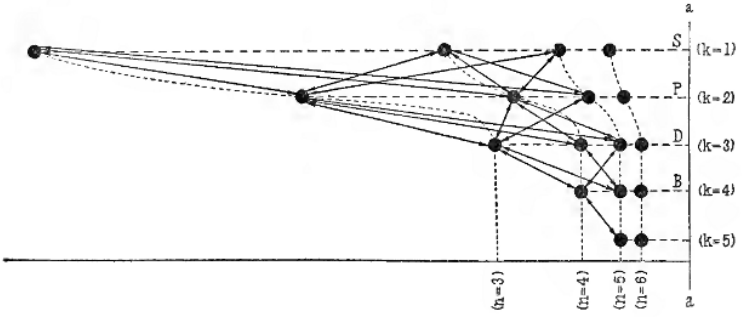


Fig. 3.

would mean that the average effect of the inner electrons could be described as a repulsion greater than would occur if their total electrical charge were united in the nucleus. This, however, cannot be expected from our view of atomic structure. The fact that the last captured electron, at any rate for low values of k, revolves partly inside the orbits of the previously bound electrons will on the contrary involve that the presence of these electrons will give rise to a virtual repulsion which is considerably smaller than that which would be

due to their combined charges. Instead of the curves drawn between points in fig. 2 which represent stationary states corresponding to the same value of the principal quantum number running from right to left, we obtain curves which run from left to right, as is indicated in fig. 3. The stationary states are labelled with quantum numbers corresponding to the structure I have described. According to the view underlying fig. 2 the sodium spectrum might be described simply as a distorted hydrogen spectrum, whereas according to fig. 3 there is not only distortion but also complete disappearance of certain terms of low quantum numbers. It may be stated, that this view not only appears to offer an explanation of the magnitude of the terms, but that the complexity of the terms in the P and D series finds a natural explanation in the deviation of the configuration of the ten electrons first bound from a purely central symmetry. This lack of symmetry has its origin in the configuration of the two innermost electrons and "transmits" itself to the outer parts of the atomic structure, since the 2_1 orbits penetrate partly into the region of these electrons.

This view of the sodium spectrum provides at the same time an immediate explanation of the pronounced electropositive properties of sodium, since the last bound electron in the sodium atom is still more loosely bound than the last captured electron in the lithium atom. In this connection it might be mentioned that the increase in atomic volume with increasing atomic number in the family of the alkali metals finds a simple explanation in the successively

looser binding of the valency electrons. In his work on the X-ray spectra Sommerfeld at an earlier period regarded this increase in the atomic volumes as supporting the assumption that the principal quantum number of the orbit of the valency electrons increases by unity as we pass from one metal to the next in the family. His later investigations on the series spectra have led him, however, definitely to abandon this assumption. At first sight it might also appear to entail a far greater increase in the atomic volume than that actually observed. A simple explanation of this fact is however afforded by realizing that the orbit of the electron will run partly inside the region of the inner orbit and that therefore the "effective" quantum number which corresponds to the outer almost elliptical loop will be much smaller than the principal quantum number, by which the whole central orbit is described. It may be mentioned that Vegard in his investigations on the X-ray spectra has also proposed the assumption of successively increasing quantum numbers for the electronic orbits in the various groups of the atom, reckoned from the nucleus outward. He has introduced assumptions about the relations between the numbers of electrons in the various groups of the atom and the lengths of the periods in the periodic system which exhibit certain formal similarities with the results presented here. But Vegard's considerations do not offer points of departure for a further consideration of the evolution and stability of the groups, and consequently no basis for a detailed interpretation of the properties of the elements.

When we consider the elements following sodium in the third period of the periodic system we meet in *the binding of the* 12*th,* 13*th and* 14*th electrons* conditions which are analogous to those we met in the binding of the 4th, 5th and 6th electrons. In the elements of the third periods, however, we possess a far more detailed knowledge of the series spectra. Too little is known about the beryllium spectrum to draw conclusions about the binding of the fourth electron, but we may infer directly from the well-known arc spectrum of magnesium that the 12th electron in the atom of this element is bound in a 3_1 orbit. As regards the binding of the 13th electron we meet in aluminium an absorption spectrum different in structure to that of the alkali metals. In fact here not the lines of the principal series but the lines of the sharp and diffuse series are absorption lines. Consequently it is the first member of the P terms and not of the S terms which corresponds to the normal state of the aluminium atom, and we must assume that the 13th electron is bound in a 3_2 orbit. This, however, would hardly seem to be a general property of the binding of the 13th electron in atoms, but rather to arise from the special conditions for the binding of the last electron in an atom, where already there are two other electrons bound as loosely as the valency electron of aluminium. At the present state of the theory it seems best to assume that in the silicon atom the four last captured electrons will move in 3_1 orbits forming a configuration possessing symmetrical properties similar to the outer configuration of the four electrons in 2_1 orbits in carbon. Like what we assumed for the

latter configuration we shall expect that the configuration of the 3_1 orbits occurring for the first time in silicon possesses such a completion, that the addition of a further electron in a 3_1 orbit to the atom of the following elements is impossible, and that *the* 15*th electron* in the elements of higher atomic number will be bound in a new type of orbit. In this case, however, the orbits with which we meet will not be circular, as in the capture of the 7th electron, but will be rotating eccentric orbits of the type 3_2. This is very closely related to the fact, mentioned above, that the non-circular orbits will correspond to a firmer binding than the circular orbits having the same value for the principal quantum number, since the electrons will at certain moments penetrate much farther into the interior of the atom. Even though a 3_2 orbit will not penetrate into the innermost configuration of 1_1 orbits, it will penetrate to distances from the nucleus which are considerably less than the radii of the circular 22 orbits. In the case of the 16th, 17th and 18th electrons the conditions are similar to those for the 15th. So for argon we may expect a configuration in which the ten innermost electrons move in orbits of the same type as in the neon atom while the last eight electrons will form a configuration of four 3_1 orbits and four 3_2 orbits, whose symmetrical properties must be regarded as closely corresponding to the configuration of 2-quanta orbits in the neon atom. At the same time, as this picture suggests a qualitative explanation of the similarity of the chemical properties of the elements in the latter part of the second and third periods, it also opens up the possibility

of a natural explanation of the conspicuous difference from a quantitative aspect.

Fourth Period. Potassium—Krypton. In the fourth period we meet at first elements which resemble chemically those at the beginning of the two previous periods. This is also what we should expect. We must thus assume that *the* 19*th electron* is bound in a new type of orbit, and a closer consideration shows that this will be a 4_1 orbit. The points which were emphasized in connection with the binding of the last electron in the sodium atom will be even more marked here on account of the larger quantum number by which the orbits of the inner electrons are characterized. In fact, in the potassium atom the 4_1 orbit of the 19th electron will, as far as inner loops are concerned, coincide closely with the shape of a 3_1 orbit. On this account, therefore, the dimensions of the outer part of the orbit will not only deviate greatly from the dimensions of a 4_1 orbit in the hydrogen atom, but will coincide closely with a hydrogen orbit of the type 2_1, the dimensions of which are about four times smaller than the 4_1 hydrogen orbit. This result allows an immediate explanation of the main features of the chemical properties and the spectrum of potassium. Corresponding results apply to calcium, in the neutral atom of which there will be two valency electrons in equivalent 4_1 orbits.

After calcium the properties of the elements in the fourth

period of the periodic system deviate, however, more and more from the corresponding elements in the previous periods, until in the family of the iron metals we meet elements whose properties are essentially different. Proceeding to still higher atomic numbers we again meet different conditions. Thus we find in the latter part of the fourth period a series of elements whose chemical properties approach more and more to the properties of the elements at the end of the preceding periods, until finally with atomic number 36 we again meet one of the inactive gases, namely krypton. This is exactly what we should expect. The formation and stability of the atoms of the elements in the first three periods require that each of the first 18 electrons in the atom shall be bound in each succeeding element in an orbit of the same principal quantum number as that possessed by the particular electron, when it first appeared. It is readily seen that this is no longer the case for the 19th electron. With increasing nuclear charge and the consequent decrease in the difference between the fields of force inside and outside the region of the orbits of the first 18 bound electrons, the dimensions of those parts of a 4_1 orbit which fall outside will approach more and more to the dimensions of a 4-quantum orbit calculated on the assumption that the interaction between the electrons in the atom may be neglected. *With increasing atomic number a point will therefore be reached where a* 3_3 *orbit will correspond to a firmer binding of the* 19*th electron than a* 4_1 *orbit*, and this occurs as early as at the beginning of the fourth period. This cannot only be anticipated from a simple calcula-

tion but is confirmed in a striking way from an examination of the series spectra. While the spectrum of potassium indicates that the 4_1 orbit corresponds to a binding which is more than twice as firm as in a 3_3 orbit corresponding to the first spectral term in the D series, the conditions are entirely different as soon as calcium is reached. We shall not consider the arc spectrum which is emitted during the capture of the 20th electron but the spark spectrum which corresponds to the capture and binding of the 19th electron. While the spark spectrum of magnesium exhibits great similarity with the sodium spectrum as regards the values of the spectral terms in the various series—apart from the fact that the constant appearing in formula (12) is four times as large as the Rydberg constant—we meet in the spark spectrum of calcium the remarkable condition that the first term of the D series is larger than the first term of the P series and is only a little smaller than the first term of the S series, which may be regarded as corresponding to the binding of the 19th electron in the normal state of the calcium atom. These facts are shown in figure 4 which gives a survey of the stationary states corresponding to the arc spectra of sodium and potassium. As in figures 2 and 3 of the sodium spectrum, we have disregarded the complexity of the spectral terms, and the numbers characterizing the stationary states are simply the quantum numbers n and k. For the sake of comparison the scale in which the energy of the different states is indicated is chosen four times as small for the spark spectra as for the arc spectra. Consequently the vertical lines indicated

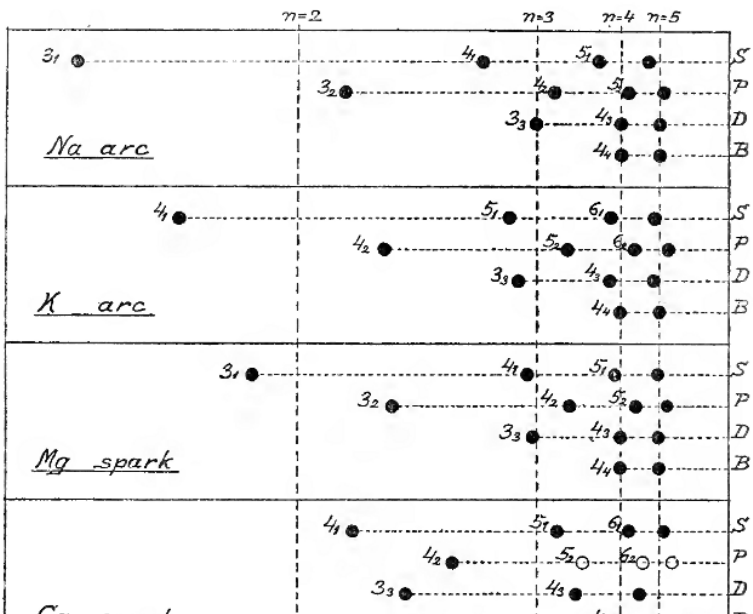


Fig. 4.

with various values of n correspond for the arc spectra to the spectral terms of hydrogen, for the spark spectra to the terms of the helium spectrum given by formula (7). Comparing the change in the relative firmness in the binding of the 19th electron in a 4_1 and 3_3 orbit for potassium and calcium we see that we must be prepared already for the next element, scandium, to find that the 3_3 orbit will correspond to a stronger binding of this electron than a 4_1 orbit. On the

other hand it follows from previous remarks that the binding will be much lighter than for the first 18 electrons which agrees that in chemical combinations scandium appears electropositively with three valencies.

If we proceed to the following elements, a still larger number of 3_3 orbits will occur in the normal state of these atoms, since the number of such electron orbits will depend upon the firmness of their binding compared to the firmness with which an electron is bound in a 4_1 orbit, in which type of orbit at least the last captured electron in the atom may be assumed to move. We therefore meet conditions which are essentially different from those which we have considered in connection with the previous periods, so that here we have to do with *the successive development of one of the inner groups of electrons in the atom*, in this case with groups of electrons in 3-quanta orbits. Only when the development of this group has been completed may we expect to find once more a corresponding change in the properties of the elements with increasing atomic number such as we find in the preceding periods. The properties of the elements in the latter part of the fourth period show immediately that the group, when completed, will possess 18 electrons. Thus in krypton, for example, we may expect besides the groups of 1, 2 and 3-quanta orbits a markedly symmetrical configuration of 8 electrons in 4-quanta orbits consisting of four 4_1 orbits and four 4_2 orbits.

The question now arises: In which way will the gradual formation of the group of electrons having 3-quanta orbits

take place? From analogy with the constitution of the groups of electrons with 2-quanta orbits we might at first sight be inclined to suppose that the complete group of 3-quanta orbits would consist of three subgroups of four electrons each in orbits of the types 3_1, 3_2 and 3_3 respectively, so that the total number of electrons would be 12 instead of 18. Further consideration shows, however, that such an expectation would not be justified. The stability of the configuration of eight electrons with 2-quanta orbits occurring in neon must be ascribed not only to the symmetrical configuration of the electronic orbits in the two subgroups of 2_1 and 2_2 orbits respectively, but fully as much to the possibility of bringing the orbits inside these subgroups into harmonic relation with one another. The situation is different, however, for the groups of electrons with 3-quanta orbits. Three subgroups of four orbits each cannot in this case be expected to come into interaction with one another in a correspondingly simple manner. On the contrary we must assume that the presence of electrons in 3_3 orbits will diminish the harmony of the orbits within the first two 3-quanta subgroups, at any rate when a point is reached where the 19th electron is no longer, as was the case with scandium, bound considerably more lightly than the previously bound electrons in 3-quanta orbits, but has been drawn so far into the atom that it revolves within essentially the same region of the atom where these electrons move. We shall now assume that this decrease in the harmony will so to say "open" the previously "closed" configuration of electrons in orbits of these types. As regards

the final result, the number 18 indicates that after the group is finally formed there will be three subgroups containing six electrons each. Even if it has not at present been possible to follow in detail the various steps in the formation of the group this result is nevertheless confirmed in an interesting manner by the fact that it is possible to arrange three configurations having six electrons each in a simple manner relative to one another. The configuration of the subgroups does not exhibit a tetrahedral symmetry like the groups of 2-quanta orbits in carbon, but a symmetry which, so far as the relative orientation of the normals to the planes of the orbits is concerned, may be described as trigonal.

In spite of the great difference in the properties of the elements of this period, compared with those of the preceding period, the completion of the group of 18 electrons in 3-quanta orbits in the fourth period may to a certain extent be said to have the same characteristic results as the completion of the group of 2-quanta orbits in the second period. As we have seen, this determined not only the properties of neon as an inactive gas, but in addition the electronegative properties of the preceding elements and the electropositive properties of the elements which follow. The fact that there is no inactive gas possessing an outer group of 18 electrons is very easily accounted for by the much larger dimensions which a 3_3 orbit has in comparison with a 2_2 orbit revolving in the same field of force. On this account a complete 3-quanta group cannot occur as the outermost group in a neutral atom, but only in positively charged ions. The character-

istic decrease in valency which we meet in copper, shown by the appearance of the singly charged cuprous ions, indicates the same tendency towards the completion of a symmetrical configuration of electrons that we found in the marked electronegative character of an element like fluorine. Direct evidence that a complete group of 3-quanta orbits is present in the cuprous ion is given by the spectrum of copper which, in contrast to the extremely complicated spectra of the preceding elements resulting from the unsymmetrical character of the inner system, possesses a simple structure very much like that of the sodium spectrum. This may no doubt be ascribed to a simple symmetrical structure present in the cuprous ion similar to that in the sodium ion, although the great difference in the constitution of the outer group of electrons in these ions is shown both by the considerable difference in the values of the spectral terms and in the separation of the doublets in the P terms of the two spectra. The occurrence of the cupric compounds shows, however, that the firmness of binding in the group of 3-quanta orbits in the copper atom is not as great as the firmness with which the electrons are bound in the group of 2-quanta orbits in the sodium atom. Zinc, which is always divalent, is the first element in which the groups of the electrons are so firmly bound that they cannot be removed by ordinary chemical processes.

The picture I have given of the formation and structure of the atoms of the elements in the fourth period gives an explanation of the chemical and spectral properties. In addition it is supported by evidence of a different nature to

that which we have hitherto used. It is a familiar fact, that the elements in the fourth period differ markedly from the elements in the preceding periods partly in their *magnetic properties* and partly in the *characteristic colours* of their compounds. Paramagnetism and colours do occur in elements belonging to the foregoing periods, but not in simple compounds where the atoms considered enter as ions. Many elements of the fourth period, on the contrary, exhibit paramagnetic properties and characteristic colours even in dissociated aqueous solutions. The importance of this has been emphasized by Ladenburg in his attempt to explain the properties of the elements in the long periods of the periodic system (see page 111). Langmuir in order to account for the difference between the fourth period and the preceding periods simply assumed that the atom, in addition to the layers of cells containing 8 electrons each, possesses an outer layer of cells with room for 18 electrons which is completely filled for the first time in the case of krypton. Ladenburg, on the other hand, assumes that for some reason or other an intermediate layer is developed between the inner electronic configuration in the atom appearing already in argon, and the external group of valency electrons. This layer commences with scandium and is completed exactly at the end of the family of iron metals. In support of this assumption Ladenburg not only mentions the chemical properties of the elements in the fourth period, but also refers to the paramagnetism and colours which occur exactly in the elements, where this intermediate layer should be in development. It is

seen that Ladenburg's ideas exhibit certain formal similarities with the interpretation I have given above of the appearance of the fourth period, and it is interesting to note that our view, based on a direct investigation of the conditions for the formation of the atoms, enables us to understand the relation emphasized by Ladenburg.

Our ordinary electrodynamic conceptions are probably insufficient to form a basis for an explanation of atomic magnetism. This is hardly to be wondered at when we remember that they have not proved adequate to account for the phenomena of radiation which are connected with the intimate interaction between the electric and magnetic forces arising from the motion of the electrons. In whatever way these difficulties may be solved it seems simplest to assume that the occurrence of magnetism, such as we meet in the elements of the fourth period, results from a lack of symmetry in the internal structure of the atom, thus preventing the magnetic forces arising from the motion of the electrons from forming a system of closed lines of force running wholly within the atom. While it has been assumed that the ions of the elements in the previous periods, whether positively or negatively charged, contain configurations of marked symmetrical character, we must, however, be prepared to encounter a definite lack of symmetry in the electronic configurations in ions of those elements within the fourth period which contain a group of electrons in 3-quanta orbits in the transition stage between symmetrical configurations of 8 and 18 electrons respectively. As pointed out by Kossel, the experimental re-

sults exhibit an extreme simplicity, the magnetic moment of the ions depending only on the number of electrons in the ion. Ferric ions, for example, exhibit the same atomic magnetism as manganous ions, while manganic ions exhibit the same atomic magnetism as chromous ions. It is in beautiful agreement with what we have assumed about the structure of the atoms of copper and zinc, that the magnetism disappears with those ions containing 28 electrons which, as I stated, must be assumed to contain a complete group of 3-quanta orbits. On the whole a consideration of the magnetic properties of the elements within the fourth period gives us a vivid impression of how a wound in the otherwise symmetrical inner structure is first developed and then healed as we pass from element to element. It is to be hoped that a further investigation of the magnetic properties will give us a clue to the way in which the group of electrons in 3-quanta orbits is developed step by step.

Also the colours of the ions directly support our view of atomic structure. According to the postulates of the quantum theory absorption as well as emission of radiation is regarded as taking place during transitions between stationary states. The occurrence of colours, that is to say the absorption of light in the visible region of the spectrum, is evidence of transitions involving energy changes of the same order of magnitude as those giving the usual optical spectra of the elements. In contrast to the ions of the elements of the preceding periods where all the electrons are assumed to be very firmly bound, the occurrence of such processes in

the fourth period is exactly what we should expect. For the development and completion of the electronic groups with 3-quanta orbits will proceed, so to say, in competition with the binding of electrons in orbits of higher quanta, since the binding of electrons in 3-quanta orbits occurs when the electrons in these orbits are bound more firmly than electrons in 4_1 orbits. The development of the group will therefore proceed to the point where we may say there is equilibrium between the two kinds of orbits. This condition may be assumed to be intimately connected not only with the colour of the ions, but also with the tendency of the elements to form ions with different valencies. This is in contrast to the elements of the first periods where the charge of the ions in aqueous solutions is always the same for one and the same element.

Fifth Period. Rubidium—Xenon. The structure of the atoms in the remaining periods may be followed up in complete analogy with what has already been said. Thus we shall assume that the 37th and 38th electrons in the elements of the fifth period are bound in 5_1 orbits. This is supported by the measurements of the arc spectrum of rubidium and the spark spectrum of strontium. The latter spectrum indicates at the same time that 4_3 orbits will soon appear, and therefore in this period, which like the 4th contains 18 elements, we must assume that we are witnessing a *further stage in the development of the electronic group of* 4*-quanta*

orbits. The first stage in the formation of this group may be said to have been attained in krypton with the appearance of a symmetrical configuration of eight electrons consisting of two subgroups each of four electrons in 4_1 and 4_2 orbits. A second preliminary completion must be regarded as having been reached with the appearance of a symmetrical configuration of 18 electrons in the case of silver, consisting of three subgroups with six electrons each in orbits of the types 4_1, 4_2 and 4_3. Everything that has been said about the successive formation of the group of electrons with 3-quanta orbits applies unchanged to this stage in the transformation of the group with 4-quanta orbits. For in no case have we made use of the absolute values of the quantum numbers nor of assumptions concerning the form of the orbits but only of the number of possible types of orbits which might come into consideration. At the same time it may be of interest to mention that the properties of these elements compared with those of the foregoing period nevertheless show a difference corresponding exactly to what would be expected from the difference in the types of orbits. For instance, the divergencies from the characteristic valency conditions of the elements in the second and third periods appear later in the fifth period than for elements in the fourth period. While an element like titanium in the fourth period already shows a marked tendency to occur with various valencies, on the other hand an element like zirconium is still quadri-valent like carbon in the second period and silicon in the third. A simple investigation of the kinematic properties of the orbits

of the electrons shows in fact that an electron in an eccentric 4_3 orbit of an element in the fifth period will be considerably more loosely bound than an electron in a circular 3_3 orbit of the corresponding element in the fourth period, while electrons which are bound in eccentric orbits of the types 5_1 and 4_1 respectively will correspond to a binding of about the same firmness.

At the end of the fifth period we may assume that xenon, the atomic number of which is 54, has a structure which in addition to the two 1-quantum, eight 2-quanta, eighteen 3-quanta and eighteen 4-quanta orbits already mentioned contains a symmetrical configuration of eight electrons in 5-quanta orbits consisting of two subgroups with four electrons each in 5_1 and 5_2 orbits respectively.

Sixth Period. Caesium—Niton. If we now consider the atoms of elements of still higher atomic number, we must first of all assume that the 55th and 56th electrons in the atoms of caesium and barium are bound in 6_1 orbits. This is confirmed by the spectra of these elements. It is clear, however, that we must be prepared shortly to meet entirely new conditions. With increasing nuclear charge we shall have to expect not only that an electron in a 5_3 orbit will be bound more firmly than in a 6_1 orbit, but we must also expect that a moment will arrive when during the formation of the atom a 4_4 orbit will represent a firmer binding of the electron than an orbit of 5 or 6-quanta, in much the same

way as in the elements of the fourth period a new stage in the development of the 3-quanta group was started when a point was reached where for the first time the 19th electron was bound in a 3_3 orbit instead of in a 4_1 orbit. We shall thus expect in the sixth period to meet with a new stage in the development of the group with 4-quanta orbits. Once this point has been reached we must be prepared to find with increasing atomic number a number of elements following one another, which as in the family of the iron metals have very nearly the same properties. The similarity will, however, be still more pronounced, since in this case we are concerned with the successive transformation of a configuration of electrons which lies deeper in the interior of the atom. You will have already guessed that what I have in view is a simple explanation of the occurrence of the *family of rare earths* at the beginning of the sixth period. As in the case of the transformation and completion of the group of 3-quanta orbits in the fourth period and the partial completion of groups of 4-quanta orbits in the fifth period, we may immediately deduce from the length of the sixth period the number of electrons, namely 32, which are finally contained in the 4-quanta group of orbits. Analogous to what applied to the group of 3-quanta orbits it is probable that, when the group is completed, it will contain eight electrons in each of the four subgroups. Even though it has not yet been possible to follow the development of the group step by step, we can even here give some theoretical evidence in favour of the occurrence of a symmetrical configuration of exactly this

number of electrons. I shall simply mention that it is not possible without coincidence of the planes of the orbits to arrive at an interaction between four subgroups of six electrons each in a configuration of simple trigonal symmetry, which is equally simple as that shown by three subgroups. The difficulties which we meet make it probable that a harmonic interaction can be attained precisely by four groups each containing eight electrons the orbital configurations of which exhibit axial symmetry.

Just as in the case of the family of the iron metals in the fourth period, the proposed explanation of the occurrence of the family of rare earths in the sixth period is supported in an interesting manner by an investigation of the magnetic properties of these elements. In spite of the great chemical similarity the members of this family exhibit very different magnetic properties, so that while some of them exhibit but very little magnetism others exhibit a greater magnetic moment per atom than any other element which has been investigated. It is also possible to give a simple interpretation of the peculiar colours exhibited by the compounds of these elements in much the same way as in the case of the family of iron metals in the fourth period. The idea that the appearance of the group of the rare earths is connected with the development of inner groups in the atom is not in itself new and has for instance been considered by Vegard in connection with his work on X-ray spectra. The new feature of the present considerations lies, however, in the emphasis laid on the peculiar way in which the relative strength of the

binding for two orbits of the same principal quantum number but of different shapes varies with the nuclear charge and with the number of electrons previously bound. Due to this fact the presence of a group like that of the rare earths in the sixth period may be considered as a direct consequence of the theory and might actually have been predicted on a quantum theory, adapted to the explanation of the properties of the elements within the preceding periods in the way I have shown.

Besides *the final development of the group of* 4*-quanta orbits* we observe in the sixth period in the family of the platinum metals *the second stage in the development of the group of* 5*-quanta orbits.* Also in the radioactive, chemically inactive gas niton, which completes this period, we observe the first preliminary step in the development of a group of electrons with 6-quanta orbits. In the atom of this element, in addition to the groups of electrons of two 1-quantum, eight 2-quanta, eighteen 3-quanta, thirty-two 4-quanta and eighteen 5-quanta orbits respectively, there is also an outer symmetrical configuration of eight electrons in 6-quanta orbits, which we shall assume to consist of two subgroups with four electrons each in 6_1 and 6_2 orbits respectively.

Seventh Period. In the seventh and last period of the periodic system we may expect the appearance of 7-quanta orbits in the normal state of the atom. Thus in the neutral atom of radium in addition to the electronic structure

of niton there will be two electrons in 7_1 orbits which will penetrate during their revolution not only into the region of the orbits of electrons possessing lower values for the principal quantum number, but even to distances from the nucleus which are less than the radii of the orbits of the innermost 1-quantum orbits. The properties of the elements in the seventh period are very similar to the properties of the elements in the fifth period. Thus, in contrast to the conditions in the sixth period, there are no elements whose properties resemble one another like those of the rare earths. In exact analogy with what has already been said about the relations between the properties of the elements in the fourth and fifth periods this may be very simply explained by the fact that an eccentric 5_4 orbit will correspond to a considerably looser binding of an electron in the atom of an element of the seventh period than the binding of an electron in a circular 4_4 orbit in the corresponding element of the sixth period, while there will be a much smaller difference in the firmness of the binding of these electrons in orbits of the types 7_1 and 6_1 respectively.

It is well known that the seventh period is not complete, for no atom has been found having an atomic number greater than 92. This is probably connected with the fact that the last elements in the system are radioactive and that nuclei of atoms with a total charge greater than 92 will not be sufficiently stable to exist under conditions where the elements can be observed. It is tempting to sketch a picture of the atoms formed by the capture and binding of electrons around nuclei having higher charges, and thus to obtain some idea

of the properties which the corresponding hypothetical elements might be expected to exhibit. I shall not develop this matter further, however, since the general results we should get will be evident to you from the views I have developed to explain the properties of the elements actually observed. A survey of these results is given in the following table, which gives a symbolical representation of the atomic structure of the inactive gases which complete the first six periods in the periodic system. In order to emphasize the progressive change the table includes the probable arrangement of electrons in the next atom which would possess properties like the inactive gases.

The view of atomic constitution underlying this table, which involves configurations of electrons moving with large velocities between each other, so that the electrons in the "outer" groups penetrate into the region of the orbits of the electrons of the "inner" groups, is of course completely different from such statical models of the atom as are proposed by Langmuir. But quite apart from this it will be seen that the arrangement of the electronic groups in the atom, to which we have been lead by tracing the way in which each single electron has been bound, is essentially different from the arrangement of the groups in Langmuir's theory. In order to explain the properties of the elements of the sixth period Langmuir assumes for instance that, in addition to the inner layers of cells containing 2, 8, 8, 18 and 18 electrons respectively, which are employed to account for the properties of the elements in the earlier periods, the atom also possesses

a layer of cells with room for 32 electrons which is just completed in the case of niton.

Element	Atomic number	Number of Electrons in n_k-orbits																								
		1_1	2_1	2_2	3_1	3_2	3_3	4_1	4_2	4_3	4_4	5_1	5_2	5_3	5_4	5_5	6_1	6_2	6_3	6_4	6_5	6_6	7_1	7_2	7_3
Helium	2	2																								
Neon	10	2	4	4																						
Argon	18	2	4	4	4	4	–																			
Krypton	36	2	4	4	6	6	6	4	4	–	–															
Xenon	54	2	4	4	6	6	6	6	6	6	–	4	4	–	–	–										
Niton	86	2	4	4	6	6	6	8	8	8	8	6	6	6	–	–	4	4	–	–	–	–				
?	118	2	4	4	6	6	6	8	8	8	8	8	8	8	8	–	6	6	6	–	–	–	4	4	–	

In this connection it may be of interest to mention a recent paper by Bury, to which my attention was first drawn after the deliverance of this address, and which contains an interesting survey of the chemical properties of the elements based on similar conceptions of atomic structure as those applied by Lewis and Langmuir. From purely chemical considerations Bury arrives at conclusions which as regards the arrangement and completion of the groups in the main coincide with those of the present theory, the outlines of which were given in my letters to Nature mentioned in the introduction.

Survey of the periodic table. The results given in this address are also illustrated by means of the representation of the periodic system given in fig. 1. In this figure the

frames are meant to indicate such elements in which one of the "inner" groups is in a stage of development. Thus there will be found in the fourth and fifth periods a single frame indicating the final completion of the electronic group with 3-quanta orbits, and the last stage but one in the development of the group with 4-quanta orbits respectively. In the sixth period it has been necessary to introduce two frames, of which the inner one indicates the last stage of the evolution of the group with 4-quanta orbits, giving rise to the rare earths. This occurs at a place in the periodic system where the third stage in the development of an electronic group with 5-quanta orbits, indicated by the outer frame, has already begun. In this connection it will be seen that the inner frame encloses a smaller number of elements than is usually attributed to the family of the rare earths. At the end of this group an uncertainty exists, due to the fact that no element of atomic number 72 is known with certainty. However, as indicated in fig. 1, we must conclude from the theory that the group with 4-quanta orbits is finally completed in lutetium (71). This element therefore ought to be the last in the sequence of consecutive elements with similar properties in the first half of the sixth period, and at the place 72 an element must be expected which in its chemical and physical properties is homologous with zirconium and thorium. This, which is already indited on Julius Thomsen's old table, has also been pointed out by Bury. [Quite recently Dauvillier has in an investigation of the X-ray spectrum excited in preparations containing rare earths, observed certain faint

lines which he ascribes to an element of atomic number 72. This element is identified by him as the element celtium, belonging to the family of rare earths, the existence of which had previously been suspected by Urbain. Quite apart from the difficulties which this result, if correct, might entail for atomic theories, it would, since the rare earths according to chemical view possess three valencies, imply a rise in positive valency of two units when passing from the element 72 to the next element 73, tantalum. This would mean an exception from the otherwise general rule, that the valency never increases by more than one unit when passing from one element to the next in the periodic table.] In the case of the incomplete seventh period the full drawn frame indicates the third stage in the development of the electronic group with 6-quanta orbits, which must begin in actinium. The dotted frame indicates the last stage but one in the development of the group with 5-quanta orbits, which hitherto has not been observed, but which ought to begin shortly after uranium, if it has not already begun in this element.

With reference to the homology of the elements the exceptional position of the elements enclosed by frames in fig. 1 is further emphasized by taking care that, in spite of the large similarity many elements exhibit, no connecting lines are drawn between two elements which occupy different positions in the system with respect to framing. In fact, the large chemical similarity between, for instance, aluminium and scandium, both of which are trivalent and pronounced electropositive elements, is directly or indirectly emphasized

in the current representations of the periodic table. While this procedure is justified by the analogous structure of the trivalent ions of these elements, our more detailed ideas of atomic structure suggest, however, marked differences in the physical properties of aluminium and scandium, originating in the essentially different character of the way in which the last three electrons in the neutral atom are bound. This fact gives probably a direct explanation of the marked difference existing between the spectra of aluminium and scandium. Even if the spectrum of scandium is not yet sufficiently cleared up, this difference seems to be of a much more fundamental character than for instance the difference between the arc spectra of sodium and copper, which apart from the large difference in the absolute values of the spectral terms possess a completely analogous structure, as previously mentioned in this essay. On the whole we must expect that the spectra of elements in the later periods lying inside a frame will show new features compared with the spectra of the elements in the first three periods. This expectation seems supported by recent work on the spectrum of manganese by Catalan, which appeared just before the printing of this essay.

Before I leave the interpretation of the chemical properties by means of this atomic model I should like to remind you once again of the fundamental principles which we have used. The whole theory has evolved from an investigation of the way in which electrons can be captured by an atom. The formation of an atom was held to consist in the successive binding of electrons, this binding resulting in radiation

according to the quantum theory. According to the fundamental postulates of the theory this binding takes place in stages by transitions between stationary states accompanied by emission of radiation. For the problem of the stability of the atom the essential problem is at what stage such a process comes to an end. As regards this point the postulates give no direct information, but here the correspondence principle is brought in. Even though it has been possible to penetrate considerably further at many points than the time has permitted me to indicate to you, still it has not yet been possible to follow in detail all stages in the formation of the atoms. We cannot say, for instance, that the above table of the atomic constitution of the inert gases may in every detail be considered as the unambiguous result of applying the correspondence principle. On the other hand it appears that our considerations already place the empirical data in a light which scarcely permits of an essentially different interpretation of the properties of the elements based upon the postulates of the quantum theory. This applies not only to the series spectra and the close relationship of these to the chemical properties of the elements, but also to the X-ray spectra, the consideration of which leads us into an investigation of interatomic processes of an entirely different character. As we have already mentioned, it is necessary to assume that the emission of the latter spectra is connected with processes which may be described as a reorganization of the completely formed atom after a disturbance produced in the interior of the atom by the action of external forces.

IV. REORGANIZATION OF ATOMS AND X-RAY SPECTRA

As in the case of the series spectra it has also been possible to represent the frequency of each line in the X-ray spectrum of an element as the difference of two of a set of spectral terms. We shall therefore assume that each X-ray line is due to a transition between two stationary states of the atom. The values of the atomic energy corresponding to these states are frequently referred to as the "energy levels" of the X-ray spectra. The great difference between the origin of the X-ray and the series spectra is clearly seen, however, in the difference of the laws applying to the absorption of radiation in the X-ray and the optical regions of the spectra. The absorption by non-excited atoms in the latter case is connected with those lines in the series spectrum which correspond to combinations of the various spectral terms with the largest of these terms. As has been shown, especially by the investigations of Wagner and de Broglie, the absorption in the X-ray region, on the other hand, is connected not with the X-ray lines but with certain spectral regions commencing at the so-called "absorption edges." The frequencies of these edges agree very closely with the spectral terms used to account for the X-ray lines. We shall now see how the conception of atomic structure developed in the preceding pages offers a simple interpretation of these facts. Let us consider the following question: What changes in the state of the atom can be produced by the absorption of radiation, and which processes of emission can be initiated by such

changes?

Absorption and emission of X-rays and correspondence principle. The possibility of producing a change at all in the motion of an electron in the interior of an atom by means of radiation must in the first place be regarded as intimately connected with the character of the interaction between the electrons within the separate groups. In contrast to the forms of motion where at every moment the position of the electrons exhibits polygonal or polyhedral symmetry, the conception of this interaction evolved from a consideration of the possible formation of atoms by successive binding of electrons has such a character that the harmonic components in the motion of an electron are in general represented in the resulting electric moment of the atom. As a result of this it will be possible to release a single electron from the interaction with the other electrons in the same group by a process which possesses the necessary analogy with an absorption process on the ordinary electrodynamic view claimed by the correspondence principle. The points of view on which we based the interpretation of the development and completion of the groups during the formation of an atom imply, on the other hand, that just as no additional electron can be taken up into a previously completed group in the atom by a change involving emission of radiation, similarly it will not be possible for a new electron to be added to such a group, when the state of the atom is changed by absorption of ra-

diation. This means that an electron which belongs to one of the inner groups of the atom, as a consequence of an absorption process—besides the case where it leaves the atom completely—can only go over either to an incompleted group, or to an orbit where the electron during the greater part of its revolution moves at a distance from the nucleus large compared to the distance of the other electrons. On account of the peculiar conditions of stability which control the occurrence of incomplete groups in the interior of the atom, the energy which is necessary to bring about a transition to such a group will in general differ very little from that required to remove the particular electron completely from the atom. We must therefore assume that the energy levels corresponding to the absorption edges indicate to a first approximation the amount of work that is required to remove an electron in one of the inner groups completely from the atom. The correspondence principle also provides a basis for understanding the experimental evidence about the appearance of the emission lines of the X-ray spectra due to transitions between the stationary states corresponding to these energy levels. Thus the nature of the interaction between the electrons in the groups of the atom implies that each electron in the atom is so to say prepared, independently of the other electrons in the same group, to seize any opportunity which is offered to become more firmly bound by being taken up into a group of electrons with orbits corresponding to smaller values of the principal quantum number. It is evident, however, that on the basis of our views of atomic structure, such an op-

portunity is always at hand as soon as an electron has been removed from one of these groups.

At the same time that our view of the atom leads to a natural conception of the phenomena of emission and absorption of X-rays, agreeing closely with that by which Kossel has attempted to give a formal explanation of the experimental observations, it also suggests a simple explanation of those quantitative relations holding for the frequencies of the lines which have been discovered by Moseley and Sommerfeld. These researches brought to light a remarkable and far-reaching similarity between the Röntgen spectrum of a given element and the spectrum which would be expected to appear upon the binding of a single electron by the nucleus. This similarity we immediately understand if we recall that in the normal state of the atom there are electrons moving in orbits which, with certain limitations, correspond to all stages of such a binding process and that, when an electron is removed from its original place in the atom, processes may be started within the atom which will correspond to all transitions between these stages permitted by the correspondence principle. This brings us at once out of those difficulties which apparently arise, when one attempts to account for the origin of the X-ray spectra by means of an atomic structure, suited to explain the periodic system. This difficulty has been felt to such an extent that it has led Sommerfeld for example in his recent work to assume that the configurations of the electrons in the various atoms of one and the same element may be different even under usual conditions. Since, in

contrast to our ideas, he supposed all electrons in the principal groups of the atom to move in equivalent orbits, he is compelled to assume that these groups are different in the different atoms, corresponding to different possible types of orbital shapes. Such an assumption, however, seems inconsistent with an interpretation of the definite character of the physical and chemical properties of the elements, and stands in marked contradiction with the points of view about the stability of the atoms which form the basis of the view of atomic structure here proposed.

X-ray spectra and atomic structure. In this connection it is of interest to emphasize that the group distribution of the electrons in the atom, on which we have based both the explanation of the periodic system and the classification of the lines in the X-ray spectra, shows itself in an entirely different manner in these two phenomena. While the characteristic change of the chemical properties with atomic number is due to the gradual development and completion of the groups of the loosest bound electrons, the characteristic absence of almost every trace of a periodic change in the X-ray spectra is due to two causes. Firstly the electronic configuration of the completed groups is repeated unchanged for increasing atomic number, and secondly the gradual way in which the incompleted groups are developed implies that a type of orbit, from the moment when it for the first time appears in the normal state of the neutral atom, always will

occur in this state and will correspond to a steadily increasing firmness of binding. The development of the groups in the atom with increasing atomic number, which governs the chemical properties of the elements shows itself in the X-ray spectra mainly in the appearance of new lines. Swinne has already referred to a connection of this kind between the periodic system and the X-ray spectra in connection with Kossel's theory. We can only expect a closer connection between the X-ray phenomena and the chemical properties of the elements, when the conditions on the surface of the atom are concerned. In agreement with what has been brought to light by investigations on absorption of X-rays in elements of lower atomic number, such as have been performed in recent years in the physical laboratory at Lund, we understand immediately that the position and eventual structure of the absorption edges will to a certain degree depend upon the physical and chemical conditions under which the element investigated exists, while such a dependence does not appear in the characteristic emission lines.

If we attempt to obtain a more detailed explanation of the experimental observations, we meet the question of the influence of the presence of the other electrons in the atom upon the firmness of the binding of an electron in a given type of orbit. This influence will, as we at once see, be least for the inner parts of the atom, where for each electron the attraction of the nucleus is large in proportion to the repulsion of the other electrons. It should also be recalled, that while the relative influence of the presence of the other

electrons upon the firmness of the binding will decrease with increasing charge of the nucleus, the effect of the variation in the mass of the electron with the velocity upon the firmness of the binding will increase strongly. This may be seen from Sommerfeld's formula (11). While we obtain a fairly good agreement for the levels corresponding to the removal of one of the innermost electrons in the atom by using the simple formula (11), it is, however, already necessary to take the influence of the other electrons into consideration in making an approximate calculation of the levels corresponding to a removal of an electron from one of the outer groups in the atom. Just this circumstance offers us, however, a possibility of obtaining information about the configurations of the electrons in the interior of the atoms from the X-ray spectra. Numerous investigations have been directed at this question both by Sommerfeld and his pupils and by Debye, Vegard and others. It may also be remarked that de Broglie and Dauvillier in a recent paper have thought it possible to find support in the experimental material for certain assumptions about the numbers of electrons in the groups of the atom to which Dauvillier had been led by considerations about the periodic system similar to those proposed by Langmuir and Ladenburg. In calculations made in connection with these investigations it is assumed that the electrons in the various groups move in separate concentric regions of the atom, so that the effect of the presence of the electrons in inner groups upon the motion of the electrons in outer groups as a first approximation may be expected to consist in a simple screen-

ing of the nucleus. On our view, however, the conditions are essentially different, since for the calculation of the firmness of the binding of the electrons it is necessary to take into consideration that the electrons in the more lightly bound groups in general during a certain fraction of their revolution will penetrate into the region of the orbits of electrons in the more firmly bound groups. On account of this fact, many examples of which we saw in the series spectra, we cannot expect to give an account of the firmness of the binding of the separate electrons, simply by means of a "screening correction" consisting in the subtraction of a constant quantity from the value for N in such formulae as (5) and (11). Furthermore in the calculation of the work corresponding to the energy levels we must take account not only of the interaction between, the electrons in the normal state of the atom, but also of the changes in the configuration and interaction of the remaining electrons, which establish themselves automatically without emission of radiation during the removal of the electron. Even though such calculations have not yet been made very accurately, a preliminary investigation has already shown that it is possible approximately to account for the experimental results.

Classification of X-ray spectra. Independently of a definite view of atomic structure it has been possible by means of a formal application of Kossel's and Sommerfeld's theories to disentangle the large amount of experimental ma-

terial on X-ray spectra. This material is drawn mainly from the accurate measurements of Siegbahn and his collaborators. From this disentanglement of the experimental observations, in which besides Sommerfeld and his students especially Smekal and Coster have taken part, we have obtained a nearly complete classification of the energy levels corresponding to the X-ray spectra. These levels are formally referred to types of orbits characterized by two quantum numbers n and k, and certain definite rules for the possibilities of combination between the various levels have also been found. In this way a number of results of great interest for the further elucidation of the origin of the X-ray spectra have been attained. First it has not only been possible to find levels, which within certain limits correspond to all possible pairs of numbers for n and k, but it has been found that in general to each such pair more than one level must be assigned. This result, which at first may appear very surprising, upon further consideration can be given a simple interpretation. We must remember that the levels depend not only upon the constitution of the atom in the normal state, but also upon the configurations which appear after the removal of one of the inner electrons and which in contrast to the normal state do not possess a uniquely completed character. If we thus consider a process in which one of the electrons in a group (subgroup) is removed we must be prepared to find that after the process the orbits of the remaining electrons in this group may be orientated in more than one way in relation to one another, and still fulfil the

conditions required of the stationary states by the quantum theory. Such a view of the "complexity" of the levels, as further consideration shows, just accounts for the manner in which the energy difference of the two levels varies with the atomic number. Without attempting to develop a more detailed picture of atomic structure, Smekal has already discussed the possibility of accounting for the multiplicity of levels. Besides referring to the possibility that the separate electrons in the principal groups do not move in equivalent orbits, Smekal suggests the introduction of three quantum numbers for the description of the various groups, but does not further indicate to what extent these quantum numbers shall be regarded as characterizing a complexity in the structure of the groups in the normal state itself or on the contrary characterizing the incompleted groups which appear when an electron is removed.

It will be seen that the complexity of the X-ray levels exhibits a close analogy with the explanation of the complexity of the terms of the series spectra. There exists, however, this difference between the complex structure of the X-ray spectra and the complex structure of the lines in the series spectra, that in the X-ray spectra there occur not only combinations between spectral terms, for which k varies by unity, but also between terms corresponding to the same value of k. This may be assumed to be due to the fact, that in the X-ray spectra in contrast to the series spectra we have to do with transitions between stationary states where, both before and after the transition, the electron concerned takes

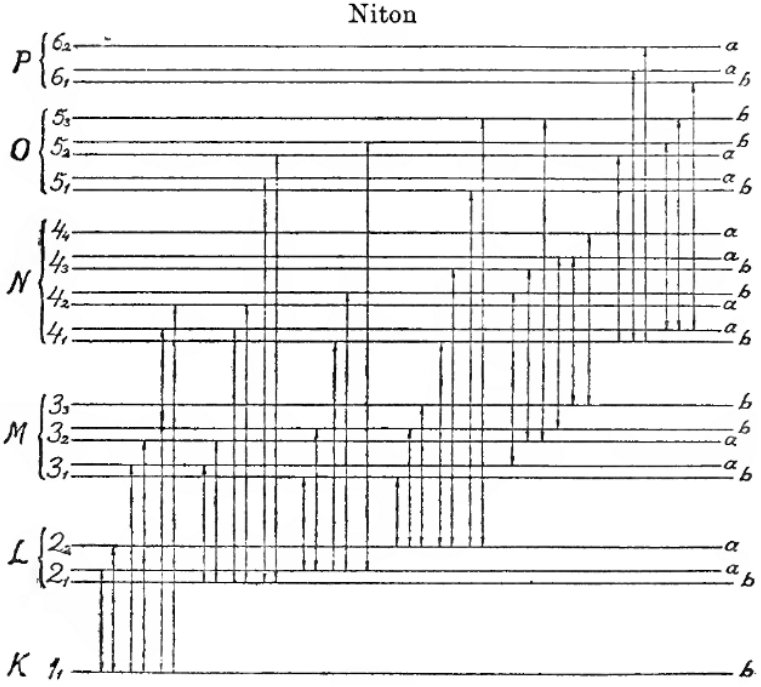


Fig. 5.

part in an intimate interaction with other electrons in orbits with the same principal quantum number. Even though this interaction may be assumed to be of such a nature that the harmonic components which would appear in the motion of an electron in the absence of the others will in general also appear in the resulting moment of the atom, we must expect that the interaction between the electrons will give rise

to the appearance in this moment of new types of harmonic components.

It may be of interest to insert here a few words about a new paper of Coster which appeared after this address was given, and in which he has succeeded in obtaining an extended and detailed connection between the X-ray spectra and the ideas of atomic structure given in this essay. The classification mentioned above was based on measurements of the spectra of the heaviest elements, and the results in their complete form, which were principally due to independent work of Coster and Wentzel, may be represented by the diagram in fig. 5, which refers to elements in the neighbourhood of niton. The vertical arrows represent the

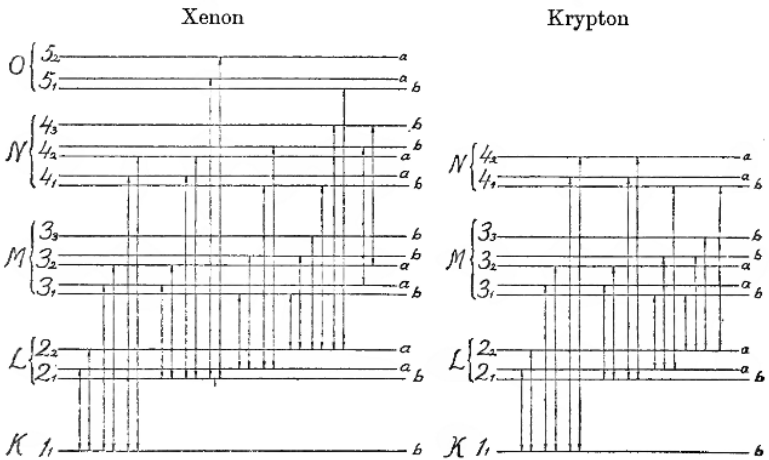


Fig. 6.

observed lines arising from combinations between the different energy levels which are represented by horizontal lines. In each group the levels are arranged in the same succession as their energy values, but their distances do not give a quantitative picture of the actual energy-differences, since this would require a much larger figure. The numbers n_k attached to the different levels indicate the type of the corresponding orbit. The letters a and b refer to the rules of combination which I mentioned. According to these rules the possibility of combination is limited (1) by the exclusion of combinations, for which k changes by more than one unit, (2) by the condition that only combinations between an a- and a b-level can take place. The latter rule was given in this form by Coster; Wentzel formulated it in a somewhat different way by the formal introduction of a third quantum number. In his new paper Coster has established a similar classification for the lighter elements. For the elements in the neighbourhood of xenon and krypton he has obtained results illustrated by the diagrams given in fig. 6. Just as in fig. 5 the levels correspond exactly to those types of orbits which, as seen from the table on page 170, according to the theory will be present in the atoms of these elements. In xenon several of the levels present in niton have disappeared, and in krypton still more levels have fallen away. Coster has also investigated in which elements these particular levels appear for the last time, when passing from higher to lower atomic number. His results concerning this point confirm in detail the predictions of the theory. Further he proves that

the change in the firmness of binding of the electrons in the outer groups in the elements of the family of the rare earths shows a dependence on the atomic number which strongly supports the assumption that in these elements a completion of an inner group of 4-quanta orbits takes place. For details the reader is referred to Coster's paper in the *Philosophical Magazine*. Another important contribution to our systematic knowledge of the X-ray spectra is contained in a recent paper by Wentzel. He shows that various lines, which find no place in the classification hitherto considered, can be ascribed in a natural manner to processes of reorganization, initiated by the removal of more than one electron from the atom; these lines are therefore in a certain sense analogous to the enhanced lines in the optical spectra.

CONCLUSION

Before bringing this address to a close I wish once more to emphasize the complete analogy in the application of the quantum theory to the stability of the atom, used in explaining two so different phenomena as the periodic system and X-ray spectra. This point is of the greatest importance in judging the reality of the theory, since the justification for employing considerations, relating to the formation of atoms by successive capture of electrons, as a guiding principle for the investigation of atomic structure might appear doubtful if such considerations could not be brought into natural agreement with views on the reorganization of the

atom after a disturbance in the normal electronic arrangement. Even though a certain inner consistency in this view of atomic structure will be recognized, it is, however, hardly necessary for me to emphasize the incomplete character of the theory, not only as regards the elaboration of details, but also so far as the foundation of the general points of view is concerned. There seems, however, to be no other way of advance in atomic problems than that which hitherto has been followed, namely to let the work in these two directions go hand in hand.

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AUTHORIZED TRANSLATION

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PREFACE.

The material contained in the following translation was given in substance by Professor Hilbert as a course of lectures on euclidean geometry at the University of Göttingen during the winter semester of 1898–1899. The results of his investigation were re-arranged and put into the form in which they appear here as a memorial address published in connection with the celebration at the unveiling of the Gauss-Weber monument at Göttingen, in June, 1899. In the French edition, which appeared soon after, Professor Hilbert made some additions, particularly in the concluding remarks, where he gave an account of the results of a recent investigation made by Dr. Dehn. These additions have been incorporated in the following translation.

As a basis for the analysis of our intuition of space, Professor Hilbert commences his discussion by considering three systems of things which he calls points, straight lines, and planes, and sets up a system of axioms connecting these elements in their mutual relations. The purpose of his investigations is to discuss systematically the relations of these axioms to one another and also the bearing of each upon the logical development of euclidean geometry. Among the important results obtained, the following are worthy of special mention:

1. The mutual independence and also the compatibility of the given system of axioms is fully discussed by the aid of various new systems of geometry which are introduced.

2. The most important propositions of euclidean geometry are demonstrated in such a manner as to show precisely what axioms underlie and make possible the demonstration.

3. The axioms of congruence are introduced and made the basis of the definition of geometric displacement.

4. The significance of several of the most important axioms and theorems in the development of the euclidean geometry is clearly shown; for example, it is shown that the whole of the euclidean geometry may be developed without the use of the axiom of continuity; the significance of Desargues's theorem, as a condition that a given plane geometry may be regarded as a part of a geometry of space, is made apparent, etc.

5. A variety of algebras of segments are introduced in accordance with the laws of arithmetic.

This development and discussion of the foundation principles of geometry is not only of mathematical but of pedagogical importance. Hoping that through an English edition these important results of Professor Hilbert's investigation may be made more accessible to English speaking students and teachers of geometry, I have undertaken, with his permission, this translation. In its preparation, I have had the assistance of many valuable suggestions from Professor Osgood of Harvard, Professor Moore of Chicago, and Professor Halsted of Texas. I am also under obligations to Mr. Henry Coar and Mr. Arthur Bell for reading the proof.

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"All human knowledge begins with intuitions, thence passes to concepts and ends with ideas."

Kant, *Kritik der reinen Vernunft, Elementariehre*, Part 2, Sec. 2.

INTRODUCTION.

Geometry, like arithmetic, requires for its logical development only a small number of simple, fundamental principles. These fundamental principles are called the axioms of geometry. The choice of the axioms and the investigation of their relations to one another is a problem which, since the time of Euclid, has been discussed in numerous excellent memoirs to be found in the mathematical literature.[1] This problem is tantamount to the logical analysis of our intuition of space.

The following investigation is a new attempt to choose for geometry a *simple* and *complete* set of *independent* axioms and to deduce from these the most important geometrical theorems in such a manner as to bring out as clearly as possible the significance of the different groups of axioms and the scope of the conclusions to be derived from the individual axioms.

[1] Compare the comprehensive and explanatory report of G. Veronese, *Grundzüge der Geometrie*, German translation by A. Schepp, Leipzig, 1894 (Appendix). See also F. Klein, "Zur ersten Verteilung des Lobatschefskiy-Preises," *Math. Ann.*, Vol. 50.

THE FIVE GROUPS OF AXIOMS.

§ 1. THE ELEMENTS OF GEOMETRY AND THE FIVE GROUPS OF AXIOMS.

Let us consider three distinct systems of things. The things composing the first system, we will call *points* and designate them by the letters A, B, C,...; those of the second, we will call *straight lines* and designate them by the letters a, b, c,...; and those of the third system, we will call *planes* and designate them by the Greek letters α, β, γ,... The points are called the *elements of linear geometry*; the points and straight lines, the *elements of plane geometry*; and the points, lines, and planes, the *elements of the geometry of space* or the *elements of space*.

We think of these points, straight lines, and planes as having certain mutual relations, which we indicate by means of such words as "are situated," "between," "parallel," "congruent," "continuous," etc. The complete and exact description of these relations follows as a consequence of the *axioms of geometry*. These axioms may be arranged in five groups. Each of these groups expresses, by itself, certain related fundamental facts of our intuition. We will name these groups as follows:

I,	1–7.	Axioms of *connection*.
II,	1–5.	Axioms of *order*.
III.		Axiom of *parallels* (Euclid's axiom).
IV,	1–6.	Axioms of *congruence*.
V.		Axiom of *continuity* (Archimedes's axiom).

§ 2. GROUP I: AXIOMS OF CONNECTION.

The axioms of this group establish a connection between the concepts indicated above; namely, points, straight lines, and planes. These axioms are as follows:

I, 1. *Two distinct points A and B always completely determine a straight line a. We write $AB = a$ or $BA = a$.*

Instead of "determine," we may also employ other forms of expression; for example, we may say A "lies upon" a, A "is a point of" a, a "goes through" A "and through" B, a "joins" A "and" or "with" B, etc. If A lies upon a and at the same time upon another straight line b, we make use also of the expression: "The straight lines" a "and" b "have the point A in common," etc.

I, 2. *Any two distinct points of a straight line completely determine that line; that is, if $AB = a$ and $AC = a$, where $B \neq C$, then is also $BC = a$.*

I, 3. *Three points A, B, C not situated in the same straight line always completely determine a plane α. We write $ABC = \alpha$.*

We employ also the expressions: A, B, C, "lie in" α; A, B, C "are points of" α, etc.

I, 4. *Any three points A, B, C of a plane α, which do not lie in the same straight line, completely determine that plane.*

I, 5. *If two points A, B of a straight line a lie in a plane α, then every point of a lies in a.*

In this case we say: "The straight line a lies in the plane α," etc.

I, 6. *If two planes α, β have a point A in common, then they have at least a second point B in common.*

I, 7. *Upon every straight line there exist at least two points, in every plane at least three points not lying in the same straight line, and in space there exist at least four points not lying in a plane.*

Axioms I, 1–2 contain statements concerning points and straight lines only; that is, concerning the elements of plane geometry. We will call them, therefore, the *plane axioms of group I*, in order to distinguish them from the axioms I, 3–7, which we will designate briefly as the *space axioms* of this group.

Of the theorems which follow from the axioms I, 3–7, we shall mention only the following:

THEOREM 1. Two straight lines of a plane have either one point or no point in common; two planes have no point in common or a straight line in common; a plane and a straight line not lying in it have no point or one point in common.

THEOREM 2. Through a straight line and a point not lying in it, or through two distinct straight lines having a common point, one and only one plane may be made to pass.

§ 3. GROUP II: AXIOMS OF ORDER.[2]

The axioms of this group define the idea expressed by the word "between," and make possible, upon the basis of this idea, an *order of sequence* of the points upon a straight line, in a plane, and in space. The points of a straight line have a certain relation to one another which the word "between" serves to describe. The axioms of this group are as follows:

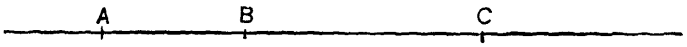


Fig. 1.

II, 1. *If A, B, C are points of a straight line and B lies between A and C, then B lies also between C and A.*

II, 2. *If A and C are two points of a straight line, then there exists at least one point B lying between A and C and at least one point D so situated that C lies between A and D.*

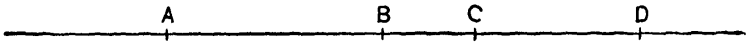


Fig. 2.

II, 3. *Of any three points situated on a straight line, there is always one and only one which lies between the other two.*

II, 4. *Any four points A, B, C, D of a straight line can always be so arranged that B shall lie between A and C and also between A and D, and, furthermore, that C shall lie between A and D and also between B and D.*

DEFINITION. We will call the system of two points A and B, lying upon a straight line, a *segment* and denote it by AB or BA. The points lying between A and B are called the *points of the segment AB* or the *points lying within the segment AB*. All other points of the straight line are referred to as the *points lying outside the segment AB*. The points A and B are called the *extremities* of the segment AB.

II, 5. *Let A, B, C be three points not lying in the same straight line and let a be a straight line lying in the plane ABC and not passing through any of the points A, B, C. Then, if the straight line a passes through a point of the segment AB, it will also pass through either a point of the segment BC or a point of the segment AC.* Axioms II, 1–4 contain statements concerning the points of a straight line only, and, hence, we will call them the *linear axioms of group II*. Axiom II, 5 relates to the elements of plane geometry and, consequently, shall be called the *plane axiom of group II*.

[2]These axioms were first studied in detail by M. Pasch in his *Vorlesungen über neuere Geometrie*, Leipsic, 1882. Axiom II, 5 is in particular due to him.

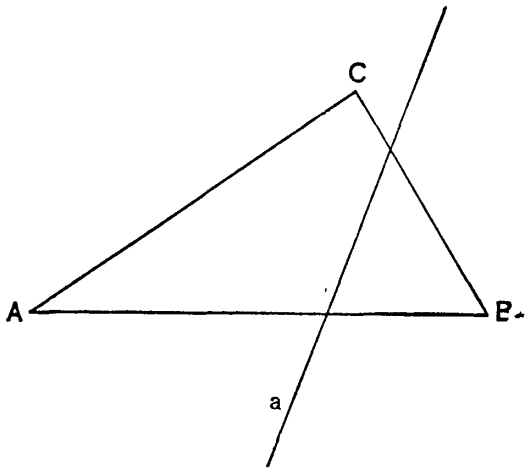


Fig. 3.

§ 4. CONSEQUENCES OF THE AXIOMS OF CONNECTION AND ORDER.

By the aid of the four linear axioms II, 1–4, we can easily deduce the following theorems:

THEOREM 3. Between any two points of a straight line, there always exists an unlimited number of points.

THEOREM 4. If we have given any finite number of points situated upon a straight line, we can always arrange them in a sequence A, B, C, D, E,..., K so that B shall lie between A and C, D, E,..., K; C between A, B and D, E,..., K; D between A, B, C and E,... K, etc. Aside from this order of sequence, there exists but one other possessing this property namely, the reverse order K,..., E, D, C, B, A.

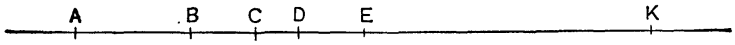


Fig. 4.

THEOREM 5. Every straight line a, which lies in a plane α, divides the remaining points of this plane into two regions having the following properties: Every point A of the one region determines with each point B of the other region a segment AB containing a point of the straight line a. On the other hand, any two points A, A' of the same region determine a segment AA' containing no point of a.

If A, A', O, B are four points of a straight line a, where O lies between A and B but not between A and A', then we may say: The points A, A' are situated *on the line a upon*

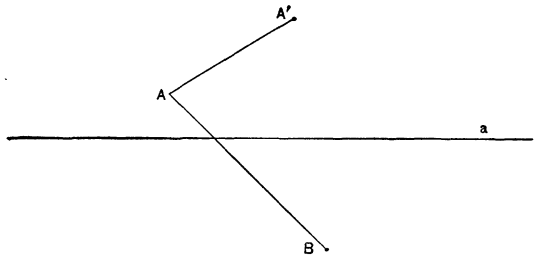


Fig. 5.

one and the same side of the point O, and the points A, B are situated *on the straight line a upon different sides of the point O.*



Fig. 6.

All of the points of a which lie upon the same side of O, when taken together, are called the *half-ray* emanating from O. Hence, each point of a straight line divides it into two half-rays.

Making use of the notation of theorem 5, we say: The points A, A' lie *in the plane* α *upon one and the same side of the straight line* a, and the points A, B lie *in the plane* α *upon different sides of the straight line* a.

Definitions. A system of segments AB, BC, CD, ..., KL is called a *broken line* joining A with L and is designated, briefly, as the broken line $ABCDE \ldots KL$. The points lying within the segments AB, BC, CD, ..., KL, as also the points A, B, C, D, ..., K, L, are called *the points of the broken line.* In particular, if the point A coincides with L, the broken line is called a *polygon* and is designated as the polygon $ABCD \ldots K$. The segments AB, BC, CD, ..., KA are called the *sides of the polygon* and the points A, B, C, D, ..., K the *vertices.* Polygons having 3, 4, 5, ..., n vertices are called, respectively, *t*riangles, *q*uadrangles, *p*entagons, ..., n-gons. If the vertices of a polygon are all distinct and none of them lie within the segments composing the sides of the polygon, and, furthermore, if no two sides have a point in common, then the polygon is called a *simple polygon.*

With the aid of theorem 5, we may now obtain, without serious difficulty, the following theorems:

> Theorem 6. Every simple polygon, whose vertices all lie in a plane α, divides the points of this plane, not belonging to the broken line constituting the sides of the polygon, into two regions, an interior and an exterior, having the following properties: If A is a point of the interior region (interior point) and B a point of the exterior region (exterior point), then any broken line joining A and B must have at least one point in common with the polygon. If, on the other hand, A, A' are two points of the interior and B, B' two points of the exterior region, then there are

always broken lines to be found joining A with A' and B with B' without having a point in common with the polygon. There exist straight lines in the plane α which lie entirely outside of the given polygon, but there are none which lie entirely within it.

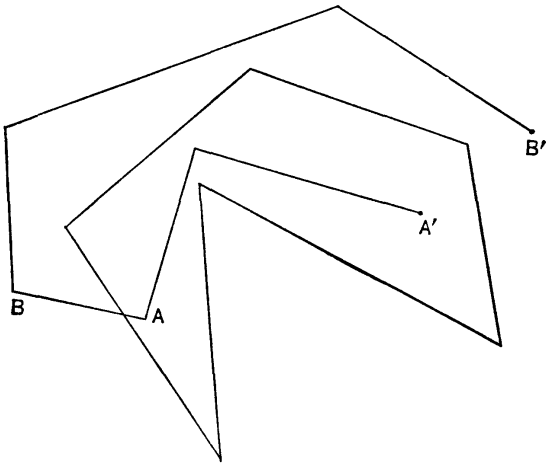


Fig. 7.

THEOREM 7. Every plane α divides the remaining points of space into two regions having the following properties: Every point A of the one region determines with each point B of the other region a segment AB, within which lies a point of α. On the other hand, any two points A, A' lying within the same region determine a segment AA' containing no point of α.

Making use of the notation of theorem 7, we may now say: The points A, A' are situated in space *upon one and the same side of the plane* α, and the points A, B are situated in space *upon different sides of the plane* α.

Theorem 7 gives us the most important facts relating to the order of sequence of the elements of space. These facts are the results, exclusively, of the axioms already considered, and, hence, no new space axioms are required in group II.

§ 5. GROUP III: AXIOM OF PARALLELS. (EUCLID'S AXIOM.)

The introduction of this axiom simplifies greatly the fundamental principles of geometry and facilitates in no small degree its development. This axiom may be expressed as follows:

III. *In a plane α there can be drawn through any point A, lying outside of a straight line a, one and only one straight line which does not intersect the line a. This straight line is called the parallel to a through the given point A.*

This statement of the axiom of parallels contains two assertions. The first of these is that, in the plane α, there is always a straight line passing through A which does not intersect the given line a. The second states that only one such line is possible. The latter of these statements is the essential one, and it may also be expressed as follows:

THEOREM 8. If two straight lines a, b of a plane do not meet a third straight line c of the same plane, then they do not meet each other.

For, if a, b had a point A in common, there would then exist in the same plane with c two straight lines a and b each passing through the point A and not meeting the straight line c. This condition of affairs is, however, contradictory to the second assertion contained in the axiom of parallels as originally stated. Conversely, the second part of the axiom of parallels, in its original form, follows as a consequence of theorem 8.

The axiom of parallels is a *plane axiom.*

§ 6. GROUP IV. AXIOMS OF CONGRUENCE.

The axioms of this group define the idea of congruence or displacement.

Segments stand in a certain relation to one another which is described by the word "*congruent.*"

IV, I. *If A, B are two points on a straight line a, and if A' is a point upon the same or another straight line a', then, upon a given side of A' on the straight line a', we can always find one and only one point B' so that the segment AB (or BA) is congruent to the segment $A'B'$. We indicate this relation by writing*

$$AB \equiv A'B'.$$

Every segment is congruent to itself; that is, we always have

$$AB \equiv AB.$$

We can state the above axiom briefly by saying that every segment can be *laid off* upon a given side of a given point of a given straight line in one and and only one way.

IV, 2. *If a segment AB is congruent to the segment $A'B'$ and also to the segment $A''B''$, then the segment $A'B'$ is congruent to the segment $A''B''$; that is, if $AB \equiv A'B$ and $AB \equiv A''B''$, then $A'B' \equiv A''B''$.*

IV, 3. *Let AB and BC be two segments of a straight line a which have no points in common aside from the point B, and, furthermore, let $A'B'$ and $B'C'$ be two segments of the same or of another straight line a' having, likewise, no point other than B' in common. Then, if $AB \equiv A'B'$ and $BC \equiv B'C'$, we have $AC \equiv A'C'$.*

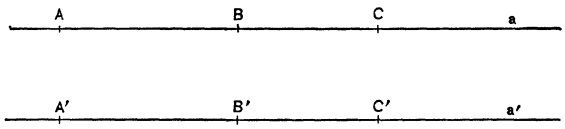


Fig. 8.

DEFINITIONS. Let α be any arbitrary plane and h, k any two distinct half-rays lying in α and emanating from the point O so as to form a part of two different straight lines. We call the system formed by these two half-rays h, k an *angle* and represent it by the symbol $\angle(h,k)$ or $\angle(k,h)$. From axioms II, 1–5, it follows readily that the half-rays h and k, taken together with the point O, divide the remaining points of the plane a into two regions having the following property: If A is a point of one region and B a point of the other, then every broken line joining A and B either passes through O or has a point in common with one of the half-rays h, k. If, however, A, A' both lie within the same region, then it is always possible to join these two points by a broken line which neither passes through O nor has a point in common with either of the half-rays h, k. One of these two regions is distinguished from the other in that the segment joining any two points of this region lies entirely within the region. The region so characterised is called the *interior of the angle* (h,k). To distinguish the other region from this, we call it the *exterior of the angle* (h,k). The half rays h and k are called the *sides of the angle*, and the point O is called the *vertex of the angle.*

IV, 4. *Let an angle (h,k) be given in the plane α and let a straight line a' be given in a plane α'. Suppose also that, in the plane α', a definite side of the straight line a' be assigned. Denote by h' a half-ray of the straight line a' emanating from a point O' of this line. Then in the plane α' there is one and only one half-ray k' such that the angle (h,k), or (k,h), is congruent to the angle (h',k') and at the same time all interior points of the angle (h',k') lie upon the given side of a'. We express this relation by means of the notation*

$$\angle(h,k) \equiv \angle(h',k')$$

Every angle is congruent to itself; that is,

$$\angle(h,k) \equiv \angle(h,k)$$

or

$$\angle(h,k) \equiv \angle(k,h)$$

We say, briefly, that every angle in a given plane can be *laid off* upon a given side of a given half-ray in one and only one way.

IV, 5. *If the angle (h,k) is congruent to the angle (h',k') and to the angle (h'',k''), then the angle (h',k') is congruent to the angle (h'',k''); that is to say, if $\angle(h,k) \equiv \angle(h',k')$ and $\angle(h,k) \equiv \angle(h'',k'')$, then $\angle(h',k') \equiv \angle(h'',k'')$.*

Suppose we have given a triangle ABC. Denote by h, k the two half-rays emanating from A and passing respectively through B and C. The angle (h, k) is then said to be the angle included by the sides AB and AC, or the one opposite to the side BC in the triangle ABC. It contains all of the interior points of the triangle ABC and is represented by the symbol $\angle BAC$, or by $\angle A$.

IV, 6. *If, in the two triangles ABC and $A'B'C'$ the congruences*

$$AB \equiv A'B',\ AC \equiv A'C',\ \angle BAC \equiv \angle B'A'C'$$

hold, then the congruences

$$\angle ABC \equiv \angle A'B'C' \text{ and } \angle ACB \equiv \angle A'C'B'$$

also hold.

Axioms IV, 1–3 contain statements concerning the congruence of segments of a straight line only. They may, therefore, be called the *linear* axioms of group IV. Axioms IV, 4, 5 contain statements relating to the congruence of angles. Axiom IV, 6 gives the connection between the congruence of segments and the congruence of angles. Axioms IV, 4–6 contain statements regarding the elements of plane geometry and may be called the *plane* axioms of group IV.

§ 7. CONSEQUENCES OF THE AXIOMS OF CONGRUENCE.

Suppose the segment AB is congruent to the segment $A'B'$. Since, according to axiom IV, 1, the segment AB is congruent to itself, it follows from axiom IV, 2 that $A'B'$ is congruent to AB; that is to say, if $AB \equiv A'B'$, then $A'B' \equiv AB$. We say, then, that the two segments are congruent to one another.

Let A, B, C, $D, \ldots$, K, L and A', B', C', $D', \ldots$, K', L' be two series of points on the straight lines a and a', respectively, so that all the corresponding segments AB and $A'B'$, AC and $A'C'$, BC and $B'C', \ldots$, KL and $K'L'$ are respectively congruent, then *the two series of points are said to be congruent to one another.* A and A', B and $B', \ldots$, L and L' are called *corresponding points* of the two congruent series of points.

From the linear axioms IV, 1–3, we can easily deduce the following theorems:

THEOREM 9. If the first of two congruent series of points A, B, C, $D, \ldots$, K, L and A', B', C', $D', \ldots$, K', L' is so arranged that B lies between A and C, $D, \ldots$, K, L, and C between A, B and $D, \ldots$, K, L, etc., then the points A', B', C', $D', \ldots$, K', L' of the second series are arranged in a similar way; that is to say, B' lies between A' and C', $D', \ldots$, K', L', and C' lies between A', B' and $D', \ldots$, K', L', etc.

Let the angle (h, k) be congruent to the angle (h', k'). Since, according to axiom IV, 4, the angle (h, k) is congruent to itself, it follows from axiom IV, 5 that the angle (h', k') is

congruent to the angle (h, k). We say, then, that the angles (h, k) and (h', k') are *congruent to one another*.

DEFINITIONS. Two angles having the same vertex and one side in common, while the sides not common form a straight line, are called *supplementary angles*. Two angles having a common vertex and whose sides form straight lines are called *vertical angles*. An angle which is congruent to its supplementary angle is called a *right angle*.

Two triangles ABC and $A'B'C'$ are said to be *congruent* to one another when all of the following congruences are fulfilled:

$$\begin{array}{lll} AB \equiv A'B', & AC \equiv A'C', & BC \equiv B'C', \\ \angle A \equiv \angle A', & \angle B \equiv \angle B', & \angle C \equiv \angle C'. \end{array}$$

THEOREM 10. (First theorem of congruence for triangles). If, for the two triangles ABC and $A'B'C'$, the congruences

$$AB \equiv A'B',\ AC \equiv A'C',\ \angle A \equiv \angle A'$$

hold, then the two triangles are congruent to each other.

PROOF. From axiom IV, 6, it follows that the two congruences

$$\angle B \equiv \angle B' \text{ and } \angle C \equiv \angle C'$$

are fulfilled, and it is, therefore, sufficient to show that the two sides BC and $B'C'$ are congruent. We will assume the contrary to be true, namely, that BC and $B'C'$ are not congruent, and show that this leads to a contradiction. We take upon $B'C'$ a point D' such that $BC \equiv B'D'$. The two triangles ABC and $A'B'D'$ have, then, two sides and the included angle of the one agreeing, respectively, to two sides and the included angle of the other. It follows from axiom IV, 6 that the two angles BAC and $B'A'D'$ are also congruent to each other. Consequently, by aid of axiom IV, 5, the two angles $B'A'C'$ and $B'A'D'$ must be congruent.

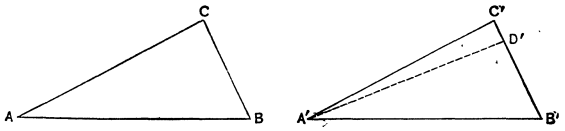


Fig. 9.

This, however, is impossible, since, by axiom IV, 4, an angle can be laid off in one and only one way on a given side of a given half-ray of a plane. From this contradiction the theorem follows.

We can also easily demonstrate the following theorem:

THEOREM 11. (Second theorem of congruence for triangles). If in any two triangles one side and the two adjacent angles are respectively congruent, the triangles are congruent.

We are now in a position to demonstrate the following important proposition.

THEOREM 12. If two angles ABC and $A'B'C'$ are congruent to each other, their supplementary angles CBD and $C'B'D'$ are also congruent.

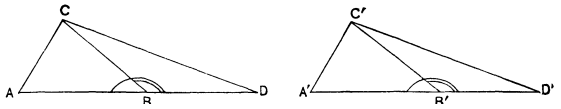


Fig. 10.

PROOF. Take the points A', C', D' upon the sides passing through B' in such a way that

$$A'B' \equiv AB,\ C'B' \equiv CB,\ D'B' \equiv DB.$$

Then, in the two triangles ABC and $A'B'C'$, the sides AB and BC are respectively congruent to $A'B'$ and $C'B'$. Moreover, since the angles included by these sides are congruent to each other by hypothesis, it follows from theorem 10 that these triangles are congruent; that is to say, we have the congruences

$$AC \equiv A'C,\ \angle BAC \equiv \angle B'A'C'.$$

On the other hand, since by axiom IV, 3 the segments AD and $A'D'$ are congruent to each other, it follows again from theorem 10 that the triangles CAD and $C'A'D'$ are congruent, and, consequently, we have the congruences:

$$CD \equiv C'D',\ \angle ADC \equiv \angle A'D'C'.$$

From these congruences and the consideration of the triangles BCD and $B'C'D'$, it follows by virtue of axiom IV, 6 that the angles CBD and $C'B'D'$ are congruent.

As an immediate consequence of theorem 12, we have a similar theorem concerning the congruence of vertical angles.

THEOREM 13. Let the angle (h, k) of the plane α be congruent to the angle (h', k') of the plane α', and, furthermore, let l be a half-ray in the plane α emanating from the vertex of the angle (h, k) and lying within this angle. Then, there always exists in the plane α' a half-ray l' emanating from the vertex of the angle (h', k') and lying within this angle so that we have

$$\angle(h, l) \equiv \angle(h', l'),\ \angle(k, l) \equiv \angle(k', l').$$

PROOF. We will represent the vertices of the angles (h, k) and (h', k') by O and O', respectively, and so select upon the sides h, k, h', k' the points A, B, A', B' that the congruences

$$OA \equiv O'A',\ OB \equiv O'B'$$

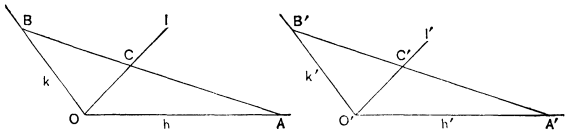


Fig. 11.

are fulfilled. Because of the congruence of the triangles OAB and $O'A'B'$, we have at once

$$AB \equiv A'B',\ \angle OAB \equiv O'A'B',\ \angle OBA \equiv \angle O'B'A'.$$

Let the straight line AB intersect l in C. Take the point C' upon the segment $A'B'$ so that $A'C' \equiv AC$. Then, $O'C'$ is the required half-ray. In fact, it follows directly from these congruences, by aid of axiom IV, 3, that $BC \equiv B'C'$. Furthermore, the triangles OAC and $O'A'C'$ are congruent to each other, and the same is true also of the triangles OCB and $O'B'C'$. With this our proposition is demonstrated.

In a similar manner, we obtain the following proposition.

THEOREM 14. Let h, k, l and h', k', l' be two sets of three half-rays, where those of each set emanate from the same point and lie in the same plane. Then, if the congruences

$$\angle(h,l) \equiv \angle(h',l'),\ \angle(k,l) \equiv \angle(k',l')$$

are fulfilled, the following congruence is also valid; viz.:

$$\angle(h,k) \equiv \angle(h',k').$$

By aid of theorems 12 and 13, it is possible to deduce the following simple theorem, which Euclid held–although it seems to me wrongly–to be an axiom.

THEOREM 15. All right angles are congruent to one another.

PROOF. Let the angle BAD be congruent to its supplementary angle CAD, and, likewise, let the angle $B'A'D'$ be congruent to its supplementary angle $C'A'D'$. Hence the angles BAD, CAD, $B'A'D'$, and $C'A'D'$ are all right angles. We will assume that the contrary of our proposition is true, namely, that the right angle $B'A'D'$ is not congruent to the right angle BAD, and will show that this assumption leads to a contradiction. We lay off the angle $B'A'D'$ upon the half-ray AB in such a manner that the side AD'' arising from this operation falls either within the angle BAD or within the angle CAD. Suppose, for example, the first of these possibilities to be true. Because of the congruence of the angles $B'A'D'$ and BAD'', it follows from theorem 12 that angle $C'A'D'$ is congruent to angle CAD'', and, as the angles $B'A'D'$ and $C'A'D'$ are congruent to each other, then, by IV, 5, the angle BAD'' must be congruent to CAD''.

Furthermore, since the angle BAD is congruent to the angle CAD, it is possible, by theorem 13, to find within the angle CAD a half-ray AD''' emanating from A, so that the angle BAD'' will be congruent to the angle CAD''', and also the angle DAD'' will be

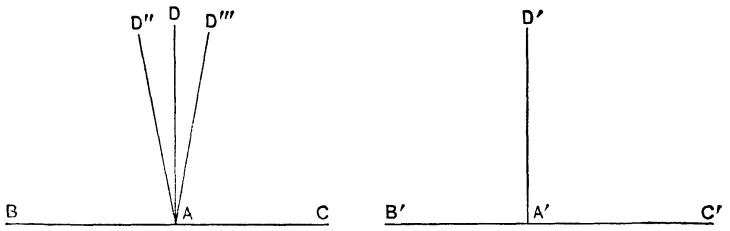


Fig. 12.

congruent to the angle DAD'''. The angle BAD'' was shown to be congruent to the angle CAD'' and, hence, by axiom IV, 5, the angle CAD'', is congruent to the angle CAD'''. This, however, is not possible; for, according to axiom IV, 4, an angle can be laid off in a plane upon a given side of a given half-ray in only one way. With this our proposition is demonstrated. We can now introduce, in accordance with common usage, the terms "*acute angle*" and "*obtuse angle.*"

The theorem relating to the congruence of the base angles A and B of an equilateral triangle ABC follows immediately by the application of axiom IV, 6 to the triangles ABC and BAC. By aid of this theorem, in addition to theorem 14, we can easily demonstrate the following proposition.

> THEOREM 16. (Third theorem of congruence for triangles.) If two triangles have the three sides of one congruent respectively to the corresponding three sides of the other, the triangles are congruent.

Any finite number of points is called a *figure.* If all of the points lie in a plane, the figure is called a *plane figure.*

Two figures are said to be *congruent* if their points can be arranged in a one-to-one correspondence so that the corresponding segments and the corresponding angles of the two figures are in every case congruent to each other.

Congruent figures have, as may be seen from theorems 9 and 12, the following properties: Three points of a figure lying in a straight line are likewise in a straight line in every figure congruent to it. In congruent figures, the arrangement of the points in corresponding planes with respect to corresponding lines is always the same. The same is true of the sequence of corresponding points situated on corresponding lines.

The most general theorems relating to congruences in a plane and in space may be expressed as follows:

> THEOREM 17. If $(A, B, C, \ldots)$ and $(A', B', C', \ldots)$ are congruent plane figures and P is a point in the plane of the first, then it is always possible to find a point P in the plane of the second figure so that $(A, B, C, \ldots, P)$ and $(A', B', C', \ldots, P')$ shall likewise be congruent figures. If the two figures have at least three points not lying in a straight line, then the selection of P' can be made in only one way.

> THEOREM 18. If $(A, B, C, \ldots)$ and $(A', B', C', \ldots =$ are congruent figures and P represents any arbitrary point, then there can always be found a point P' so that

the two figures $(A, B, C, \ldots, P)$ and $(A', B', C', \ldots, P')$ shall likewise be congruent. If the figure $(A, B, C, \ldots, P)$ contains at least four points not lying in the same plane, then the determination of P' can be made in but one way.

This theorem contains an important result; namely, that all the facts concerning space which have reference to congruence, that is to say, to displacements in space, are (by the addition of the axioms of groups I and II) exclusively the consequences of the six linear and plane axioms mentioned above. Hence, it is not necessary to assume the axiom of parallels in order to establish these facts.

If we take, in addition to the axioms of congruence, the axiom of parallels, we can then easily establish the following propositions:

THEOREM 19. If two parallel lines are cut by a third straight line, the alternate-interior angles and also the exterior-interior angles are congruent. Conversely, if the alternate-interior or the exterior-interior angles are congruent, the given lines are parallel.

THEOREM 20. The sum of the angles of a triangle is two right angles.

DEFINITIONS. If M is an arbitrary point in the plane a, the totality of all points A, for which the segments MA are congruent to one another, is called *a circle*. M is called the *centre of the circle*.

From this definition can be easily deduced, with the help of the axioms of groups III and IV, the known properties of the circle; in particular, the possibility of constructing a circle through any three points not lying in a straight line, as also the congruence of all angles inscribed in the same segment of a circle, and the theorem relating to the angles of an inscribed quadrilateral.

§ 8. GROUP V. AXIOM OF CONTINUITY. (ARCHIMEDEAN AXIOM.)

This axiom makes possible the introduction into geometry of the idea of continuity. In order to state this axiom, we must first establish a convention concerning the equality of two segments. For this purpose, we can either base our idea of equality upon the axioms relating to the congruence of segments and define as "*equal*" the correspondingly congruent segments, or upon the basis of groups I and II, we may determine how, by suitable constructions (see Chap. V, § 24), a segment is to be laid off from a point of a given straight line so that a new, definite segment is obtained "*equal*" to it. In conformity with such a convention, the axiom of Archimedes may be stated as follows:

V. Let A_1 be any point upon a straight line between the arbitrarily chosen points A and B. Take the points A_2, A_3, $A_4, \ldots$ so that A_1 lies between A and A_2, A_2 between A_1 and A_3, A_3 between A_2 and A_4 etc. Moreover, let the segments

$$AA_1,\ A_1A_2,\ A_2A_3,\ A_3A_4,\ \ldots$$

be equal to one another. Then, among this series of points, there always exists a certain point A_n such that B lies between A and A_n.

The axiom of Archimedes is a *linear* axiom.

REMARK.[3] To the preceeding five groups of axioms, we may add the following one, which, although not of a purely geometrical nature, merits particular attention from a theoretical point of view. It may be expressed in the following form:

> AXIOM OF COMPLETENESS.[4] *(Vollständigkeit): To a system of points, straight lines, and planes, it is impossible to add other elements in such a manner that the system thus generalized shall form a new geometry obeying all of the five groups of axioms. In other words, the elements of geometry form a system which is not susceptible of extension, if we regard the five groups of axioms as valid.*

This axiom gives us nothing directly concerning the existence of limiting points, or of the idea of convergence. Nevertheless, it enables us to demonstrate Bolzano's theorem by virtue of which, for all sets of points situated upon a straight line between two definite points of the same line, there exists necessarily a point of condensation, that is to say, a limiting point. From a theoretical point of view, the value of this axiom is that it leads indirectly to the introduction of limiting points, and, hence, renders it possible to establish a one-to-one correspondence between the points of a segment and the system of real numbers. However, in what is to follow, no use will be made of the "axiom of completeness."

[3] Added by Professor Hilbert in the French translation.—*Tr.*

[4] See Hilbert, "Ueber den Zahlenbegriff," *Berichte der deutschen Mathematiker-Vereinigung*, 1900.

COMPATIBILITY AND MUTUAL INDEPENDENCE OF THE AXIOMS.

§ 9. COMPATIBILITY OF THE AXIOMS.

The axioms, which we have discussed in the previous chapter and have divided into five groups, are not contradictory to one another; that is to say, it is not possible to deduce from these axioms, by any logical process of reasoning, a proposition which is contradictory to any of them. To demonstrate this, it is sufficient to construct a geometry where all of the five groups are fulfilled.

To this end, let us consider a domain Ω consisting of all of those algebraic numbers which may be obtained by beginning with the number one and applying to it a finite number of times the four arithmetical operations (addition, subtraction, multiplication, and division) and the operation $\sqrt{1+\omega^2}$, where ω represents a number arising from the five operations already given.

Let us regard a pair of numbers (x, y) of the domain Ω as defining a point and the ratio of three such numbers $(u : v : w)$ of Ω, where u, v are not both equal to zero, as defining a straight line. Furthermore, let the existence of the equation

$$ux + vy + w = 0$$

express the condition that the point (x, y) lies on the straight line $(u : v : w)$. Then, as one readily sees, axioms I, 1–2 and III are fulfilled. The numbers of the domain Ω are all real numbers. If now we take into consideration the fact that these numbers may be arranged according to magnitude, we can easily make such necessary conventions concerning our points and straight lines as will also make the axioms of order (group II) hold. In fact, if (x_1, y_1), (x_2, y_2), (x_3, y_3), ... are any points whatever of a straight line, then this may be taken as their sequence on this straight line, providing the numbers x_1, x_2, $x_3, \ldots$, or the numbers y_1, y_2, $y_3, \ldots$, either all increase or decrease in the order of sequence given here. In order that axiom II, 5 shall be fulfilled, we have merely to assume that all points corresponding to values of x and y which make $ux + vy + w$ less than zero or greater than zero shall fall respectively upon the one side or upon the other side of the straight line $(u : v : w)$. We can easily convince ourselves that this convention is in accordance with those which precede, and by which the sequence of the points on a straight line has already been determined.

The laying off of segments and of angles follows by the known methods of analytical geometry. A transformation of the form

$$\begin{aligned} x' &= x + a \\ y' &= y + b \end{aligned}$$

produces a translation of segments and of angles.

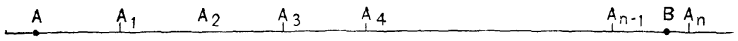


Fig. 13.

Furthermore, if, in the accompanying figure, we represent the point $(0,0)$ by O and the point $(1,0)$ by E, then, corresponding to a rotation of the angle COE about O as a center, any point (x,y) is transformed into another point (x',y') so related that

$$\begin{aligned} x' &= \frac{a}{\sqrt{a^2+b^2}}x - \frac{b}{\sqrt{a^2+b^2}}y, \\ y' &= \frac{b}{\sqrt{a^2+b^2}}x + \frac{a}{\sqrt{a^2+b^2}}y. \end{aligned}$$

Since the number

$$\sqrt{a^2+b^2} = a\sqrt{1+\left(\frac{b}{a}\right)^2}$$

belongs to the domain Ω, it follows that, under the conventions which we have made, the axioms of congruence (group IV) are all fulfilled. The same is true of the axiom of Archimedes.

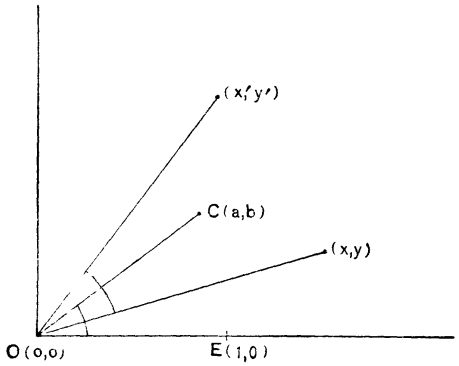


Fig. 14.

From these considerations, it follows that every contradiction resulting from our system of axioms must also appear in the arithmetic related to the domain Ω.

The corresponding considerations for the geometry of space present no difficulties.

If, in the preceding development, we had selected the domain of all real numbers instead of the domain Ω, we should have obtained likewise a geometry in which all of the axioms of groups I—V are valid. For the purposes of our demonstration, however, it was sufficient to take the domain Ω, containing on an enumerable set of elements.

§ 10. INDEPENDENCE OF THE AXIOMS OF PARALLELS. (NON-EUCLIDEAN GEOMETRY.)[5]

Having shown that the axioms of the above system are not contradictory to one another, it is of interest to investigate the question of their mutual independence. In fact, it may be shown that none of them can be deduced from the remaining ones by any logical process of reasoning.

First of all, so far as the particular axioms of groups I, II, and IV are concerned, it is easy to show that the axioms of these groups are each independent of the other of the same group.[6]

According to our presentation, the axioms of groups I and II form the basis of the remaining axioms. It is sufficient, therefore, to show that each of the groups II, IV, and V is independent of the others.

The first statement of the axiom of parallels can be demonstrated by aid of the axioms of groups I, II, and IV. In order to do this, join the given point A with any arbitrary point B of the straight line a. Let C be any other point of the given straight line. At the point A on AB, construct the angle ABC so that it shall lie in the same plane α as the point C, but upon the opposite side of AB from it. The straight line thus obtained through A does not meet the give straight line α; for, if it should cut it, say in the point D, and if we suppose B to be situated between C and D, we could then find on α a point D' so situated that B would lie between D and D', and, moreover, so that we should have

$$AD \equiv BD'$$

Because of the congruence of the two triangles ABD and BAD', we have also

$$\angle ABD \equiv \angle BAD',$$

and since the angles ABD' and ABD are supplementary, it follows from theorem 12 that the angles BAD and BAD' are also supplementary. This, however, cannot be true, as, by theorem 1, two straight lines cannot intersect in more than one point, which would be the case if BAD and BAD' were supplementary.

The second statement of the axiom of parallels is independent of all the other axioms. This may be most easily shown in the following well known manner. As the individual elements of a geometry of space, select the points, straight lines, and planes of the ordinary geometry as constructed in § 9, and regard these elements as restricted in extent to the interior of a fixed sphere. Then, define the congruences of this geometry by aid of such linear transformations of the ordinary geometry as transform the fixed sphere into itself. By suitable conventions, we can make this "*non-euclidean geometry*" obey all of the axioms of our system except the axiom of Euclid (group III). Since the possibility of the ordinary geometry has already been established, that of the non-euclidean geometry is now an immediate consequence of the above considerations.

[5]The mutual independence of Hilbert's system of axioms has also been discussed recently by Schur and Moore. Schur's paper, entitled "Ueber die Grundlagen der Geometrie" appeared in *Math. Annalem*, Vol. 55, p. 265, and that of Moore, "On the Projective Axioms of Geometry," is to be found in the Jan. (1902) number of the *Transactions of the Amer. Math. Society.—Tr.*

[6]See my lectures upon Euclidean Geometry, winter semester of 1898–1899, which were reported by Dr. Von Schaper and manifolded for the members of the class.

§ 11. INDEPENDENCE OF THE AXIOMS OF CONGRUENCE.

We shall show the independence of the axioms of congruence by demonstrating that axiom IV, 6, or what amounts to the same thing, that the first theorem of congruence for triangles (theorem 10) cannot be deduced from the remaining axioms I, II, III, IV 1–5, V by any logical process of reasoning.

Select, as the points, straight lines, and planes of our new geometry of space, the points, straight lines, and planes of ordinary geometry, and define the laying off of an angle as in ordinary geometry, for example, as explained in § 9. We will, however, define the laying off of segments in another manner. Let A_1, A_2 be two points which, in ordinary geometry, have the co-ordinates x_1, y_1, z_1 and x_2, y_2, z_2, respectively. We will now define the length of the segment A_1A_2 as the positive value of the expression

$$\sqrt{(x_1 - x_2 + y_1 - y_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

and call the two segments A_1A_2 and $A_1'A_2'$ congruent when they have equal lengths in the sense just defined.

It is at once evident that, in the geometry of space thus defined, the axioms I, II, III, IV 1–2, 4–5, V are all fulfilled.

In order to show that axiom IV, 3 also holds, we select an arbitrary straight line a and upon it three points A_1, A_2, A_3 so that A_2 shall lie between A_1 and A_3. Let the points x, y, z of the straight line a be given by means of the equations

$$\begin{aligned} x &= \lambda t + \lambda', \\ y &= \mu t + \mu', \\ z &= \nu t + \nu', \end{aligned}$$

where λ, λ', μ, μ', ν, ν' represent certain constants and T is a parameter. If t_1, t_2 $(< t_1)$, t_3 $(< t_2)$ are the values of the parameter corresponding to the points A_1, A_2, A_3 we have as the lengths of the three segments A_1A_2 A_2A_3 and A_1A_3 respectively, the following values:

$$(t_1 - t_2)\left|\sqrt{(\lambda+\mu)^2 + \mu^2 + \nu^2}\right|$$
$$(t_2 - t_3)\left|\sqrt{(\lambda+\mu)^2 + \mu^2 + \nu^2}\right|$$
$$(t_1 - t_3)\left|\sqrt{(\lambda+\mu)^2 + \mu^2 + \nu^2}\right|$$

Consequently, the length of A_1A_3 is equal to the sum of the lengths of the segments A_1A_2 and A_2A_3. But this result is equivalent to the existence of axiom IV, 3.

Axiom IV, 6, or rather the first theorem of congruence for triangles, is not always fulfilled in this geometry. Consider, for example, in the plane $z = 0$, the four points

O,	having	the	co-ordinates	$x = 0,\ y = 0$
A,	"	"	"	$x = 1,\ y = 0$
B,	"	"	"	$x = 0,\ y = 1$
C,	"	"	"	$x = \frac{1}{2},\ y = \frac{1}{2}$

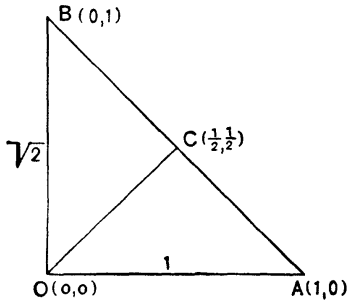


Fig. 15.

Then, in the right triangles OAC and OBC, the angles at C as also the adjacent sides AC and BC are respectively congruent; for, the side OC is common to the two triangles and the sides AC and BC have the same length, namely, $\frac{1}{2}$. However, the third sides OA and OB have the lengths 1 and $\sqrt{2}$, respectively, and are not, therefore, congruent. It is not difficult to find in this geometry two triangles for which axiom IV, 6, itself is not valid.

§ 12. INDEPENDENCE OF THE AXIOM OF CONTINUITY. (NON-ARCHIMEDEAN GEOMETRY.)

In order to demonstrate the independence of the axiom of Archimedes, we must produce a geometry in which all of the axioms are fulfilled with the exception of the one in question.[7]

For this purpose, we construct a domain $\Omega(t)$ of all those algebraic functions of t which may be obtained from t by means of the four arithmetical operations of addition, subtraction, multiplication, division, and the fifth operation $\sqrt{1+\omega^2}$, where ω represents any function arising from the application of these five operations. The elements of $\Omega(t)$—just as was previously the case for Ω—constitute an enumerable set. These five operations may all be performed without introducing imaginaries, and that in only one way. The domain $\Omega(t)$ contains, therefore, only real, single-valued functions of t.

Let c be any function of the domain $\Omega(t)$. Since this function c is an algebraic function of t, it can in no case vanish for more than a finite number of values of t, and, hence, for sufficiently large positive values of t, it must remain always positive or always negative.

Let us now regard the functions of the domain $\Omega(t)$ as a kind of complex numbers. In the system of complex numbers thus defined, all of the ordinary rules of operation evidently hold. Moreover, if a, b are any two distinct numbers of this system, then a is said to be greater than, or less than, b (written $a > b$ or $a < b$) according as the difference $c = a - b$ is always positive or always negative for sufficiently large values of t. By the adoption of this convention for the numbers of our system, it is possible to arrange them

[7] In his very scholarly book,—*Grundzüge der Geometrie*, German translation by A. Schepp, Leipzig, 1894,—G. Veronese has also attempted the construction of a geometry independent of the axiom of Archimedes.

according to their magnitude in a manner analogous to that employed for real numbers. We readily see also that, for this system of complex numbers, the validity of an inequality is not destroyed by adding the same or equal numbers to both members, or by multiplying both members by the same number, providing it is greater than zero.

If n is any arbitrary positive integral rational number, then, for the two numbers n and t of the domain $\Omega(t)$, the inequality $n < t$ certainly holds; for, the difference $n - t$, considered as a function of t, is always negative for sufficiently large values of t. We express this fact in the following manner: The two numbers l and t of the domain $\Omega(t)$, each of which is greater than zero, possess the property that any multiple whatever of the first always remains smaller than the second.

From the complex numbers of the domain $\Omega(t)$, we now proceed to construct a geometry in exactly the same manner as in § 9, where we took as the basis of our consideration the algebraic numbers of the domain Ω. We will regard a system of three numbers (x, y, z) of the domain $\Omega(t)$ as defining a point, and the ratio of any four such numbers $(u : v : w : r)$, where u, v, w are not all zero, as defining a plane. Finally, the existence of the equation

$$xu + yv + zw + r = 0$$

shall express the condition that the point (x, y, z) lies in the plane $(u : v : w : r)$. Let the straight line be defined in our geometry as the totality of all the points lying simultaneously in the same two planes. If now we adopt conventions corresponding to those of § 9 concerning the arrangement of elements and the laying off of angles and of segments, we shall obtain a *"non-archimedean" geometry* where, as the properties of the complex number system already investigated show, all of the axioms, with the exception of that of Archimedes, are fulfilled. In fact, we can lay off successively the segment 1 upon the segment t an arbitrary number of times without reaching the end point of the segment t, which is a contradiction to the axiom of Archimedes.

THE THEORY OF PROPORTION.[8]

§ 13. COMPLEX NUMBER-SYSTEMS.

At the beginning of this chapter, we shall present briefly certain preliminary ideas concerning complex number systems which will later be of service to us in our discussion.

The real numbers form, in their totality, a system of things having the following properties:

THEOREMS OF CONNECTION (1–12).

1. From the number a and the number b, there is obtained by "addition" a definite number c, which we express by writing

$$a + b = c \text{ or } c = a + b.$$

2. There exists a definite number, which we call 0, such that, for every number a, we have

$$a + 0 = a \text{ and } 0 + a = a.$$

3. If a and b are two given numbers, there exists one and only one number x, and also one and only one number y, such that we have respectively

$$a + x = b,\ y + a = b.$$

4. From the number a and the number b, there may be obtained in another way, namely, by "multiplication," a definite number c, which we express by writing

$$ab = c \text{ or } c = ab.$$

5. There exists a definite number, called 1, such that, for every number a, we have

$$a \cdot 1 = a \text{ and } 1 \cdot a = a.$$

6. If a and b are any arbitrarily given numbers, where a is different from 0, then there exists one and only one number x and also one and only one number y such that we have respectively

$$ax = b,\ ya = b.$$

If a, b, c are arbitrary numbers, the following laws of operation always hold:

7. $a + (b + c) = (a + b) + c$

8. $a + b = b + a$

9. $a(bc) = (ab)c$

10. $a(b + c) = ab + ac$

11. $(a + b)c = ac + bc$

12. $ab = ba.$

[8] See also Schur, *Math. Annalen*, Vol. 55, p. 265.—*Tr.*

THEOREMS OF ORDER (13–16).

13. If a, b are any two distinct numbers, one of these, say a, is always greater ($>$) than the other. The other number is said to be the smaller of the two. We express this relation by writing

$$a > b \text{ and } b < a.$$

14. If $a > b$ and $b > c$, then is also $a > c$.

15. If $a > b$, then is also $a + c > b + c$ and $c + a > c + b$.

16. If $a > b$ and $c > 0$, then is also $ac > bc$ and $ca > cb$.

THEOREM OF ARCHIMEDES (17).

17. If a, b are any two arbitrary numbers, such that $a > 0$ and $b > 0$, it is always possible to add a to itself a sufficient number of times so that the resulting sum shall have the property that

$$a + a + a + \cdots + a > b.$$

A system of things possessing only a portion of the above properties (1–17) is called a *complex number system*, or simply a *number system*. A number system is called *archimedean*, or *non-archimedean*, according as it does, or does not, satisfy condition (17).

Not every one of the properties (1–17) given above is independent of the others. The problem arises to investigate the logical dependence of these properties. Because of their great importance in geometry, we shall, in §§ 32, 33, pp. 65–68, answer two definite questions of this character. We will here merely call attention to the fact that, in any case, the last of these conditions (17) is not a consequence of the remaining properties, since, for example, the complex number system $\Omega(t)$, considered in § 12, possesses all of the properties (1–16), but does not fulfil the law stated in (17).

§ 14. DEMONSTRATION OF PASCAL'S THEOREM.

In this and the following chapter, we shall take as the basis of our discussion all of the plane axioms with the exception of the axiom of Archimedes; that is to say, the axioms I, 1–2 and II–IV. In the present chapter, we propose, by aid of these axioms, to establish Euclid's theory of proportion; that is, *we shall establish it for the plane and that independently of the axiom of Archimedes.*

For this purpose, we shall first demonstrate a proposition which is a special case of the well known theorem of Pascal usually considered in the theory of conic sections, and which we shall hereafter, for the sake of brevity, refer to simply as Pascal's theorem. This theorem may be stated as follows:

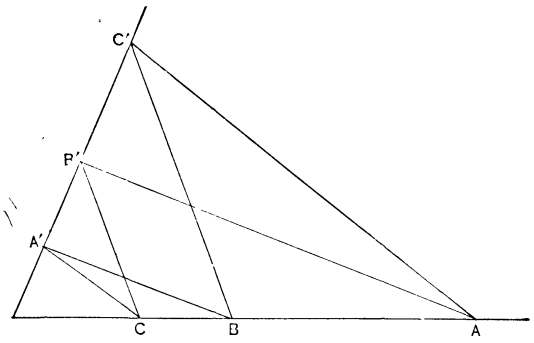


Fig. 16.

THEOREM 21. (Pascal's theorem.) Given the two sets of points A, B, C and A', B', C' so situated respectively upon two intersecting straight lines that none of them fall at the intersection of these lines. If CB' is parallel to BC' and CA' is also parallel to AC', then BA' is parallel to AB'.[9]

In order to demonstrate this theorem, we shall first introduce the following notation. In a right triangle, the base a is uniquely determined by the hypotenuse c and the base angle α included by a and c. We will express this fact briefly by writing

$$a = \alpha c.$$

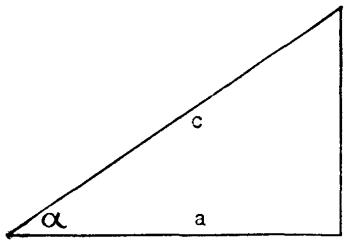


Fig. 17.

Hence, the symbol αc always represents a definite segment, providing c is any given segment whatever and α is any given acute angle.

Furthermore, if c is any arbitrary segment and α, β are any two acute angles whatever, then the two segments $\alpha\beta c$ and $\beta\alpha c$ are always congruent; that is, we have

$$\alpha\beta c = \beta\alpha c,$$

and, consequently, the symbols α and β are interchangeable.

[9]F. Schur has published in the *Math. Ann.*, Vol. 51, a very interesting proof of the theorem of Pascal, based upon the axioms I–II, IV.

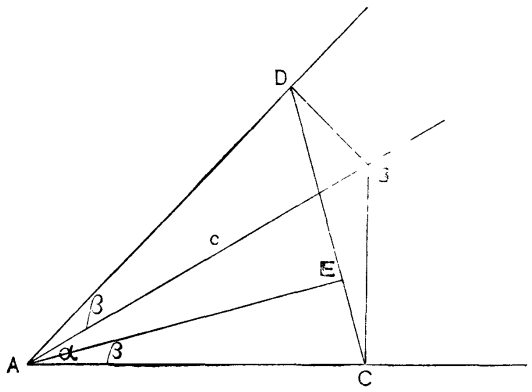


Fig. 18.

In order to prove this statement, we take the segment $c = AB$, and with A as a vertex lay off upon the one side of this segment the angle α and upon the other the angle β. Then, from the point B, let fall upon the opposite sides of the α and β the perpendiculars BC and BD, respectively. Finally, join C with D and let fall from A the perpendicular AE upon CD.

Since the two angles ACB and ADB are right angles, the four points A, B, C, D are situated upon a circle. Consequently, the angles ACD and ABD, being inscribed in the same segment of the circle, are congruent. But the angles ACD and CAE, taken together, make a right angle, and the same is true of the two angles ABD and BAD. Hence, the two angles CAE and BAD are also congruent; that is to say,

$$\angle CAE \equiv \beta$$

and, therefore,

$$\angle DAE \equiv \alpha.$$

From these considerations, we have immediately the following congruences of segments:

$$\begin{array}{rclcrcl} \beta c & \equiv & AD, & & \alpha c & \equiv & AC, \\ \alpha\beta c & \equiv & \alpha(AD) \equiv AE, & & \beta\alpha c & \equiv & \beta(AC) \equiv AE. \end{array}$$

From these, the validity of the congruence in question follows.

Returning now to the figure in connection with Pascal's theorem, denote the intersection of the two given straight lines by O and the segments OA, OB, OC, OA', OB', OC', CB', BC', CA', AC', BA', AB' by a, b, c, a', b', c', l, l^*, m, m^*, n, n^*, respectively.

Let fall from the point O a perpendicular upon each of the segments l, m, n. The perpendicular to l will form with the straight lines OA and OA' acute angles, which we shall denote by λ' and λ, respectively. Likewise, the perpendiculars to m and n form with these same lines OA and OA' acute angles, which we shall denote by μ', μ and ν', ν, respectively. If we now express, as indicated above, each of these perpendiculars in terms of the hypotenuse and base angle, we have the three following congruences of segments:

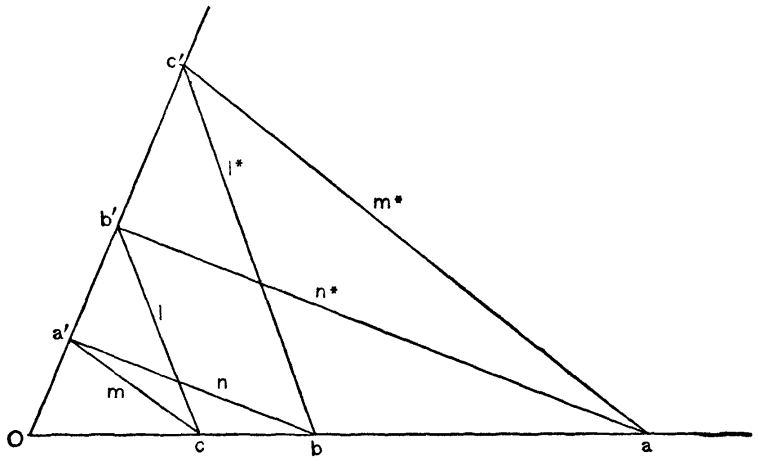


Fig. 19.

(1) $\lambda b' \equiv \lambda' c$

(2) $\mu a' \equiv \mu' c$

(3) $\nu a' \equiv \nu' b.$

But since, according to our hypothesis, l is parallel to l^* and m is parallel to m^*, the perpendiculars from O falling upon l^* and m^* must coincide with the perpendiculars from the same point falling upon l and m, and consequently, we have

(4) $\lambda c' \equiv \lambda' b,$

(5) $\mu c' \equiv \mu' a.$

Multiplying both members of congruence (3) by the symbol $\lambda'\mu$, and remembering that, as we have already seen, the symbols in question are commutative, we have

$$\nu\lambda'\mu a' \equiv \nu'\mu\lambda' b.$$

In this congruence, we may replace $\mu a'$ in the first member by its value given in (2) and $\lambda' b$ in the second member by its value given in (4), thus obtaining as a result

$$\nu\lambda'\mu' c \equiv \nu'\mu\lambda c',$$

or

$$\nu\mu'\lambda' c \equiv \nu'\lambda\mu c'.$$

Here again in this congruence we can, by aid of (1), replace $\lambda' c$ by $\lambda b'$, and, by aid of (5), we may replace in the second member $\mu c'$ by $\mu' a$. We then have

$$\nu\mu'\lambda b' \equiv \nu'\lambda\mu' a,$$

or

$$\lambda\mu'\nu b' \equiv \lambda\mu'\nu' a.$$

Because of the significance of our symbols, we can conclude at once from this congruence that

$$\mu'\nu b' \equiv \mu'\nu' a,$$

and, consequently, that

(6) $$\nu b' \equiv \nu' a.$$

If now we consider the perpendicular let fall from O upon n and draw perpendiculars to this same line from the points A and B', then congruence (6) shows that the feet of the last two perpendiculars must coincide; that is to say, the straight line $n^* = AB'$ makes a right angle with the perpendicular to n and, consequently, is parallel to n. This establishes the truth of Pascal's theorem.

Having given any straight line whatever, together with an arbitrary angle and a point lying outside of the given line, we can, by constructing the given angle and drawing a parallel line, find a straight line passing through the given point and cutting the given straight line at the given angle. By means of this construction, we can demonstrate Pascal's theorem in the following very simple manner, for which, however, I am indebted to another source.

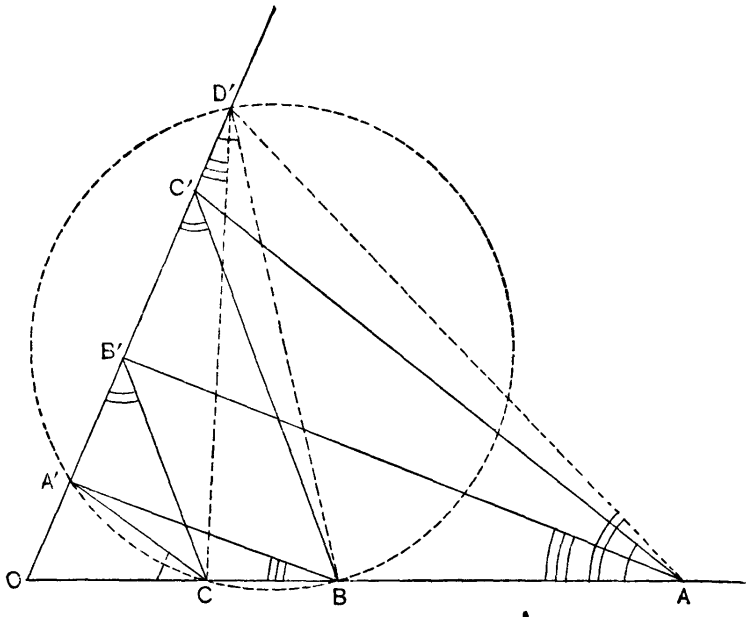


Fig. 20.

Through the point B, draw a straight line cutting OA' in the point D' and making with it the angle OCA', so that the congruence

(1∗) $$\angle OCA' \equiv \angle OD'B$$

is fulfilled. Now, according to a well known property of circles, $CBD'A'$ is an inscribed quadrilateral, and, consequently, by aid of the theorem concerning the congruence of angles inscribed in the same segment of a circle, we have the congruence

(2∗) $$\angle OBA' \equiv \angle OD'C.$$

Since, by hypothesis, CA' and AC' are parallel, we have

(3∗) $$\angle OCA' \equiv \angle OAC',$$

and from (1∗) and (3∗) we obtain the congruence

$$\angle OD'B \equiv \angle OAC'.$$

However, $BAD'C'$ is also an inscribed quadrilateral, and, consequently, by virtue of the theorem relating to the angles of such a quadrilateral, we have the congruence

(4∗) $$\angle OAD' \equiv \angle OC'B.$$

But as CB' is, by hypothesis, parallel to BC', we have also

(5∗) $$\angle OB'C \equiv \angle OC'B.$$

From (4∗) and (5∗), we obtain the congruence

$$\angle OAD' \equiv \angle OB'C,$$

which shows that $CAD'B'$ is also an inscribed quadrilateral, and, hence, the congruence

(6∗) $$\angle OAB' \equiv \angle OD'C,$$

is valid. From (2∗) and (6∗), it follows that

$$\angle OBA' \equiv \angle OAB',$$

and this congruence shows that BA' and AB' are parallel as Pascal's theorem demands. In case D' coincides with one of the points A', B', C', it is necessary to make a modification of this method, which evidently is not difficult to do.

§ 15. AN ALGEBRA OF SEGMENTS, BASED UPON PASCAL'S THEOREM.

Pascal's theorem, which was demonstrated in the last section, puts us in a position to introduce into geometry a method of calculating with segments, in which all of the rules for calculating with real numbers remain valid without any modification.

Instead of the word "congruent" and the sign ≡, we make use, in the algebra of segments, of the word "equal" and the sign =.

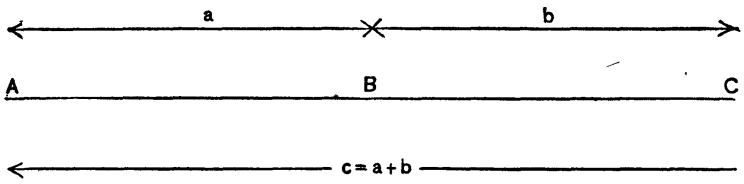


Fig. 21.

If A, B, C are three points of a straight line and if B lies between A and C, then we say that $c = AC$ is the *sum* of the two segments $a = AB$ and $b = BC$. We indicate this by writing

$$c = a + b.$$

The segments a and b are said to be smaller than c, which fact we indicate by writing

$$a < c,\ b < c.$$

On the other hand, c is said to be larger than a and b, and we indicate this by writing

$$c > a,\ c > b.$$

From the linear axioms of congruence (axioms IV, 1–3), we easily see that, for the above definition of addition of segments, the associative law

$$a + (b + c) = (a + b) + c,$$

as well as the commutative law

$$a + b = b + a$$

is valid.

In order to define geometrically the product of two segments a and b, we shall make use of the following construction. Select any convenient segment, which, having been selected, shall remain constant throughout the discussion, and denote the same by 1. Upon the one side of a right angle, lay off from the vertex O the segment 1 and also the segment b. Then, from O lay off upon the other side of the right angle the segment a. Join the extremities of the segments 1 and a by a straight line, and from the extremity of b draw a line parallel to this straight line. This parallel will cut off from the other side of the right angle a segment c. We call this segment c the *product* of the segments a and b, and indicate this relation by writing

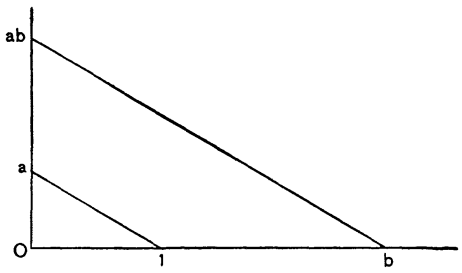


Fig. 22.

$$c = ab.$$

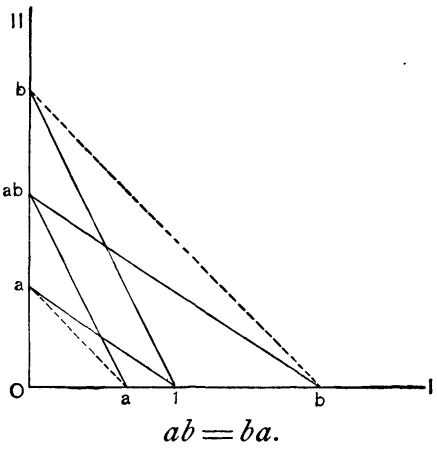


Fig. 23.

We shall now demonstrate that, for this definition of the multiplication of segments, the *commutative* law

$$ab = ba$$

holds. For this purpose, we construct in the above manner the product ab. Furthermore, lay off from 0 upon the first side (I) of the right angle the segment a and upon the other side (II) the segment b. Connect by a straight line the extremity of the segment 1 with the extremity of b, situated on II, and draw through the endpoint of a, on I, a line parallel to this straight line. This parallel will determine, by its intersection with the side II, the segment ba. But, because the two dotted lines are, by Pascal's theorem, parallel, the segment ba just found coincides with the segment ab previously constructed, and our proposition is established. In order to show that the *associative* law

$$a(bc) = (ab)c$$

holds for the multiplication of segments, we construct first of all the segment $d = be$, then da, after that the segment $e = ba$, and finally ec. By virtue of Pascal's theorem, the extremities of the segments da and ec coincide, as may be clearly seen from figure 24.

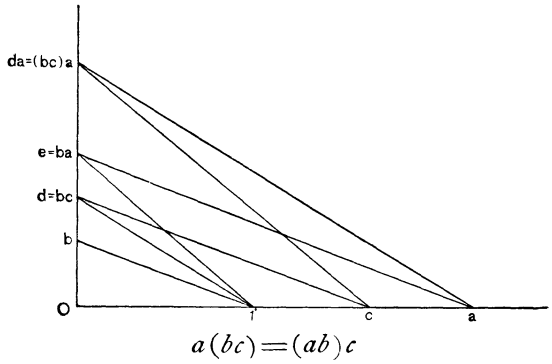


Fig. 24.

If now we apply the commutative law which we have just demonstrated, we obtain the above formula, which expresses the associative law for the multiplication of two segments.

Finally, *the distributive* law

$$a(b + c) = ab + ac$$

also holds for our algebra of segments. In order to demonstrate this, we construct the segments, ab, ac, and $a(b+c)$, and draw through the extremity of the segment c (Fig. 25) a straight line parallel to the other side of the right angle. From the congruence of the two

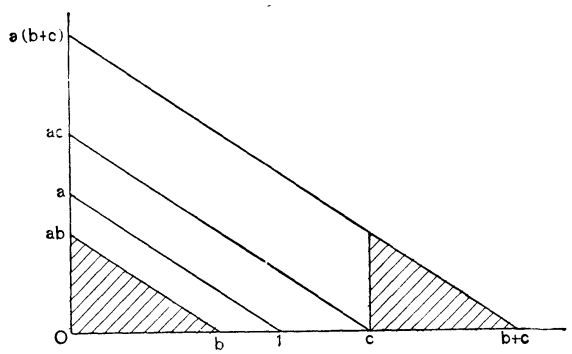


Fig. 25.

right-angled triangles which are shaded in the figure and the application of the theorem relating to the equality of the opposite sides of a parallelogram, the desired result follows. If b and c are any two arbitrary segments, there is always a segment a to be found such that $c = ab$. This segment a is denoted by $\frac{c}{b}$ and is called the *quotient* of c by b.

§ 16. PROPORTION AND THE THEOREMS OF SIMILITUDE.

By aid of the preceding algebra of segments, we can establish Euclid's theory of proportion in a manner free from objections and without making use of the axiom of Archimedes.

If a, b, a', b' are any four segments whatever, the proportion

$$a : b = a' : b'$$

expresses nothing else than the validity of equation

$$ab' = ba'.$$

DEFINITION. Two triangles are called *similar* when the corresponding angles are congruent.

THEOREM 22. If a, b and a', b' are homologous sides of two similar triangles, we have the proportion

$$a : b = a' : b'$$

PROOF. We shall first consider the special case where the angle included between a and b and the one included between a' and b' are right angles. Moreover, we shall assume that the two triangles are laid off in one and the same right angle. Upon one of the sides of this right angle, we lay off from the vertex 0 the segment 1, and through the extremity of this segment, we draw a straight line parallel to the hypotenuses of the two triangles.

This parallel determines upon the other side of the right angle a segment e. Then, according to our definition of the product of two segments, we have

$$b = ea,\ b' = ea',$$

from which we obtain

$$ab' = ba',$$

that is to say,

$$a : b = a' : b'.$$



Fig. 26.

Let us now return to the general case. In each of the two similar triangles, find the point of intersection of the bisectors of the three angles. Denote these points by S and S'. From these points let fall upon the sides of the triangles the perpendiculars r and r', respectively. Denote the segments thus determined upon the sides of the triangles by

$$a_b, a_c, b_c, b_a, c_a, c_b$$

and

$$a'_b, a'_c, b'_c, b'_a, c'_a, c'_b,$$

respectively. The special case of our proposition, demonstrated above, gives us then the following proportions:

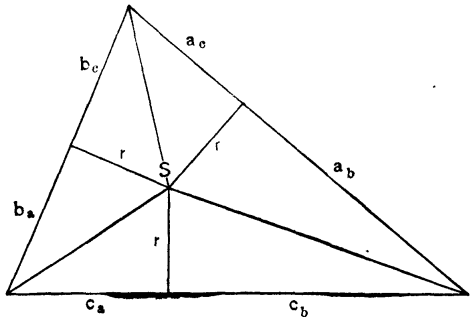


Fig. 27.

$$a_b : r = a'_b : r', \qquad b_c : r = b'_c : r',$$
$$a_c : r = a'_c : r', \qquad b_a : r = b'_a : r'.$$

By aid of the distributive law, we obtain from these proportions the following:

$$a : r = a' : r, \quad b : r = b' : r'.$$

Consequently, by virtue of the commutative law of multiplication, we have

$$a : b = a' : b'.$$

From the theorem just demonstrated, we can easily deduce the fundamental theorem in the theory of proportion. This theorem may be stated as follows:

THEOREM 23. If two parallel lines cut from the sides of an arbitrary angle the segments a, b and a', b' respectively, then we have always the proportion

$$a : b = a' : b'.$$

Conversely, if the four segments a, b, a', b fulfill this proportion and if a, a' and b, b' are laid off upon the two sides respectively of an arbitrary angle, then the straight lines joining the extremities of a and b and of a' and b' are parallel to each other.

§ 17. EQUATIONS OF STRAIGHT LINES AND OF PLANES.

To the system of segments already discussed, let us now add a second system. We will distinguish the segments of the new system from those of the former one by means of a special sign, and will call them "*negative*" segments in contradistinction to the "*positive*" segments already considered. If we introduce also the segment O, which is determined by a single point, and make other appropriate conventions, then all of the rules deduced in § 13 for calculating with real numbers will hold equally well here for calculating with segments. We call special attention to the following particular propositions:

We have always $a \cdot 1 = 1 \cdot a = a$.

If $a \cdot b = 0$, then either $a = 0$, or $b = 0$.

If $a > b$ and $c > 0$, then $ac > bc$.

In a plane α, we now take two straight lines cutting each other in O at right angles as the fixed axes of rectangular co-ordinates, and lay off from O upon these two straight lines the arbitrary segments x and y. We lay off these segments upon the one side or upon the other side of O, according as they are positive or negative. At the extremities of x and y, erect perpendiculars and determine the point P of their intersection. The segments x and y are called the co-ordinates of P. Every point of the plane α is uniquely determined by its co-ordinates x, y, which may be positive, negative, or zero.

Let l be a straight line in the plane α, such that it shall pass through O and also through a point C having the co-ordinates a, b. If x, y are the co-ordinates

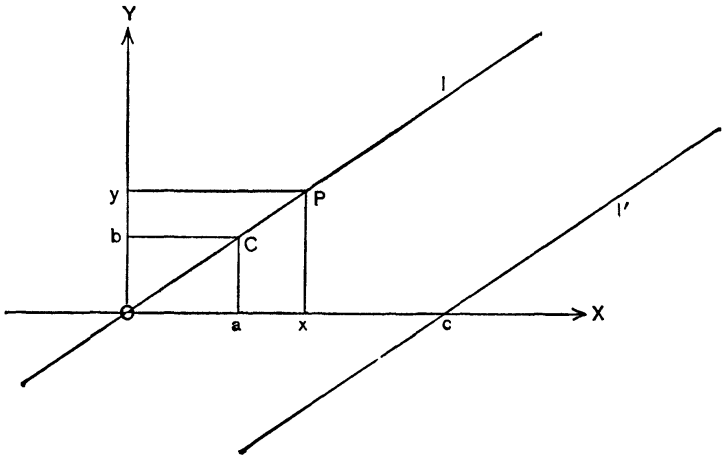


Fig. 28.

of any point on l, it follows at once from theorem 22 that

$$a : b = x : y,$$

or

$$bx - ay = 0,$$

is the equation of the straight line l. If l' is a straight line parallel to l and cutting off from the x-axis the segment c, then we may obtain the equation of the straight line l' by replacing, in the equation for l, the segment x by the segment $x - c$. The desired equation will then be of the form

$$bx - ay - bc = 0.$$

From these considerations, we may easily conclude, independently of the axiom of Archimedes, that every straight line of a plane is represented by an equation which is linear in the co-ordinates x, y, and, conversely, every such linear equation represents a straight line when the co-ordinates are segments appertaining to the geometry in question. The corresponding results for the geometry of space may be easily deduced.

The remaining parts of geometry may now be developed by the usual methods of analytic geometry.

So far in this chapter, we have made absolutely no use of the axiom of Archimedes. If now we assume the validity of this axiom, we can arrange a definite correspondence between the points on any straight line in space and the real numbers. This may be accomplished in the following manner.

We first select on a straight line any two points, and assign to these points the numbers 0 and 1. Then, bisect the segment $(0, 1)$ thus determined and denote the middle point by the number $\frac{1}{2}$. In the same way, we denote the middle of $(0, \frac{1}{2})$ by $\frac{1}{4}$, etc. After applying this process n times, we obtain a point which corresponds to $\frac{1}{2^n}$. Now, lay off m times in both directions from the point O the segment $(O, \frac{1}{2^n})$. We obtain in this manner a point corresponding to the numbers $\frac{m}{2^n}$) and $-\frac{m}{2^n}$). From the axiom of Archimedes, we now easily see that, upon the basis of this association, to each arbitrary point of a straight line there corresponds a single, definite, real number, and, indeed, such that this correspondence possesses the following property: If A, B, C are any three points on a straight line and α, β, γ are the corresponding real numbers, and, if B lies between A and C, then one of the inequalities,

$$\alpha < \beta < \gamma \text{ or } \alpha > \beta > \gamma,$$

is always fulfilled.

From the development given in § 9, p. 15, it is evident, that to every number belonging to the field of algebraic numbers Ω, there must exist a corresponding point upon the straight line. Whether to every real number there corresponds a point cannot in general be established, but depends upon the geometry to which we have reference.

However, it is always possible to generalize the original system of points, straight lines, and planes by the addition of "ideal" or "irrational" elements, so that, upon any straight line of the corresponding geometry, a point corresponds without exception to every system of three real numbers. By the adoption of suitable conventions, it may also be seen that,

in this generalized geometry, all of the axioms I–V are valid. This geometry, generalized by the addition of irrational elements, is nothing else than the ordinary analytic geometry of space.

THE THEORY OF PLANE AREAS.

§ 18. EQUAL AREA AND EQUAL CONTENT OF POLYGONS.[10]

We shall base the investigations of the present chapter upon the same axioms as were made use of in the last chapter, §§ 13–17, namely, upon the plane axioms of all the groups, with the single exception of the axiom of Archimedes. This involves then the axioms I, 1–2 and II–IV.

The theory of proportion as developed in §§ 13–17 together with the algebra of segments introduced in the same chapter, puts us now in a position to establish Euclid's theory of areas by means of the axioms already mentioned; that is to say, *for the plane geometry, and that independently of the axiom of Archimedes.*

Since, by the development given in the last chapter, pp. 23–35, the theory of proportion was made to depend essentially upon Pascal's theorem (theorem 21), the same may then be said here of the theory of areas. This manner of establishing the theory of areas seems to me a very remarkable application of Pascal's theorem to elementary geometry.

If we join two points of a polygon P by any arbitrary broken line lying entirely within the polygon, we shall obtain two new polygons P_1 and P_2 whose interior points all lie within P. We say that P is *decomposed into* P_1 and P_2, or that the polygon P is *composed of* P_1 and P_2.

DEFINITION. Two polygons are said to be of *equal area* when they can be decomposed into a finite number of triangles which are respectively congruent to one another in pairs.

DEFINITION. Two polygons are said to be of *equal content* when it is possible, by the addition of other polygons having equal area, to obtain two resulting polygons having equal area.

From these definitions, it follows at once that by combining polygons having equal area, we obtain as a result polygons having equal area. However, if from polygons having equal area we take polygons having equal area, we obtain polygons which are of equal content.

Furthermore, we have the following propositions:

> THEOREM 24. If each of two polygons P_1 and P_2 is of equal area to a third polygon P_3 then P_1 and P_2 are themselves of equal area. If each of two polygons is of equal content to a third, then they are themselves of equal content.

[10] In connection with the theory of areas, we desire to call attention to the following works of M. Gérard: *Thèse de Doctorat sur la géométrie non euclidienne* (1892) and *Géométrie plane* (Paris, 1898). M. Gérard has developed a theory concerning the measurement of polygons analogous to that presented in § 20 of the present work. The difference is that M. Gérard makes use of parallel transversals, while I use transversals emanating from the vertex. The reader should also compare the following works of F. Schur, where he will find a similar development: *Sitzungsberichte der Dorpater Naturf. Ges.*, 1892, and *Lehrbuch der analytischen Geometrie*, Leipzig, 1898 (introduction). Finally, let me refer to an article by O. Stolz in *Monatshefte für Math, und Phys.*, 1894. (Note by Professor Hilbert in French ed.)
M. Gérard has also treated the subject of areas in various ways in the following journals: *Bulletin de Math, spciales* (May, 1895), *Bulletin de la Société mathématique de France* (Dec., 1895), *Bulletin Math, élémentaires* (January, 1896, June, 1897, June, 1898). (Note in French ed.)

PROOF. By hypothesis, we can so decompose each of the polygons P_1 and P_2 into such a system of triangles that any triangle of either of these systems will be congruent to the corresponding triangle of a system into which P_3 may be decomposed. If we consider simultaneously the two decompositions of P_3 we see that, in general, each triangle of the one decomposition is broken up into polygons by the segments which belong to the other decomposition. Let ms add to these segments a sufficient number of others to reduce each of these polygons to triangles, and apply the two resulting methods of decompositions to P_1 and P_2, thus breaking them up into corresponding triangles. Then, evidently the two polygons P_1 and P_2 are each decomposed into the same number of triangles, which are respectively congruent by pairs. Consequently, the two polygons are, by definition, of equal area.

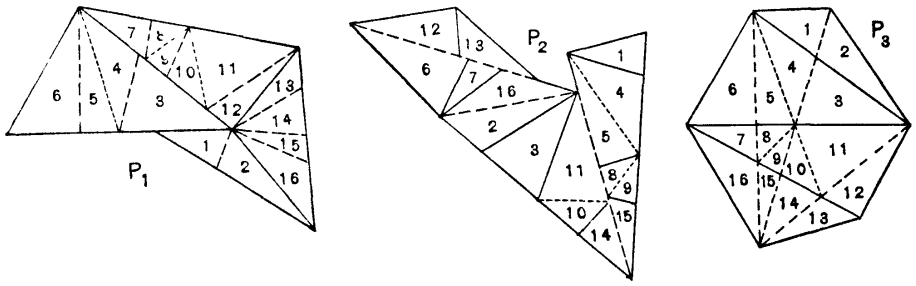


Fig. 29.

The proof of the second part of the theorem follows without difficulty.

We define, in the usual manner, the terms: *rectangle*, *base* and *height of a parallelogram*, *base* and *height of a triangle*.

§ 19. PARALLELOGRAMS AND TRIANGLES HAVING EQUAL BASES AND EQUAL ALTITUDES.

The well known reasoning of Euclid, illustrated by the accompanying figure, furnishes a proof for the following theorem:

THEOREM 25. Two parallelograms having equal bases and equal altitudes are also of equal content.

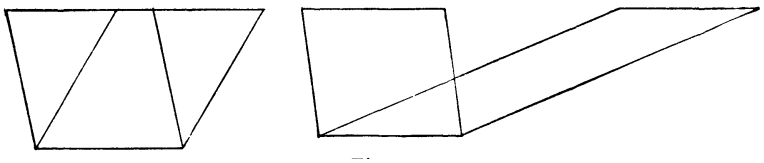


Fig. 30.

We have also the following well known proposition:

THEOREM 26. Any triangle ABC is always of equal area to a certain parallelogram having an equal base and an altitude half as great as that of the triangle.

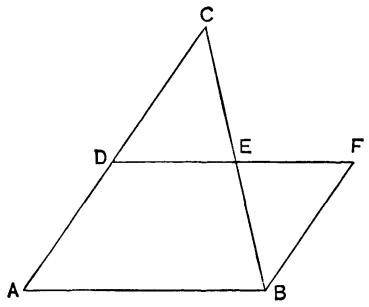


Fig. 31.

PROOF. Bisect AC in D and BC in E, and extend the line DE to F, making EF equal to DE. Then, the triangles DEC and FBE are congruent to each other, and, consequently, the triangle ABC and the parallelogram $ABFD$ are of equal area.

From theorems 25 and 26, we have at once, by aid of theorem 24, the following proposition.

THEOREM 27. Two triangles having equal bases and equal altitudes have also equal content.

It is usual to show that two triangles having equal bases and equal altitudes are always of equal area. It is to be remarked, however, that *this demonstration cannot be made without the aid of the axiom of Archimedes.* In fact, we may easily construct in our non-archimedean geometry (see § 12, p. 21) two triangles so that they shall have equal bases and equal altitudes and, consequently, by theorem 27, must be of equal content, but which are not, however, of equal area. As such an example, we may take the two triangles ABC and ABD having each the base $AB = 1$ and the altitude 1, where the vertex of the first triangle is situated perpendicularly above A, and in the second triangle the foot F of the perpendicular let fall from the vertex D upon the base is so situated that $AF = t$. The remaining propositions of elementary geometry concerning the equal content of polygons, and in particular the pythagorean theorem, are all simple consequences of the theorems which we have already given. In the further development of the theory of area, we meet, however, with an essential difficulty. In fact, the discussion so far leaves it still in doubt whether all polygons are not, perhaps, of equal content. In this case, all of the propositions which we have given would be devoid of meaning and hence of no value. Furthermore, the more general question also arises as to whether two rectangles of equal content and having one side in common, do not also have their other sides congruent; that is to say, whether a rectangle is not definitely determined by means of a side and its area. As a closer investigation shows, in order to answer this question, we need to make use of the converse of theorem 27. This may be stated as follows:

> THEOREM 28. If two triangles have equal content and equal bases, they have also equal altitudes.

This fundamental theorem is to be found in the first book of Euclid's Elements as proposition 39. In the demonstration of this theorem, however, Euclid appeals to the general proposition relating to magnitudes: "*Καὶ τὸ ὅλου τοῦ μέρους μεῖζόν ἐστιν*"—a method of procedure which amounts to the same thing as introducing a new geometrical axiom concerning areas.

It is now possible to establish the above theorem and hence the theory of areas in the manner we have proposed, that is to say, with the help of the plane axioms and without making use of the axiom of Archimedes. In order to show this, it is necessary to introduce the idea of the measure of area.

§ 20. THE MEASURE OF AREA OF TRIANGLES AND POLYGONS.

DEFINITION. If in a triangle ABC, having the sides a, b, c, we construct the two altitudes $h_a = AD$, $h_b = BE$, then, according to theorem 22, it follows from the similarity of the triangles BCE and ACD, that we have the proportion

$$a : h_b = b : ha;$$

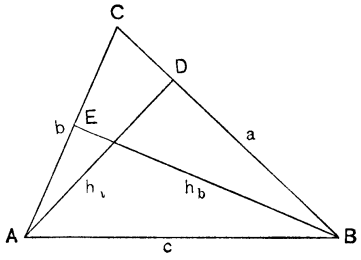


Fig. 32.

that is, we have

$$a \cdot h_a = b \cdot h_b.$$

This shows that the product of the base and the corresponding altitude of a triangle is the same whichever side is selected as the base. The half of this product of the base and the altitude of a triangle Δ is called the *measure of area of the triangle* Δ and we denote it by $F(\Delta)$. A segment joining a vertex of a triangle with a point of the opposite side is called *a transversal.* A transversal divides the given triangle into two others having the same altitude and having bases which lie in the same straight line. Such a decomposition is called a *transversal decomposition of the triangle.*

THEOREM 29. If a triangle Δ is decomposed by means of arbitrary straight lines into a finite number of triangles Δ_k, then the measure of area of Δ is equal to the sum of the measures of area of the separate triangles Δ_k.

PROOF. From the distributive law of our algebra of segments, it follows immediately that the measure of area of an arbitrary triangle is equal to the sum of the measures of area of two such triangles as arise from any transversal decomposition of the given triangle. The repeated application of this proposition shows that the measure of area of any triangle is equal to the sum of the measures of area of all the triangles arising by applying the transversal decomposition an arbitrary number of times in succession.

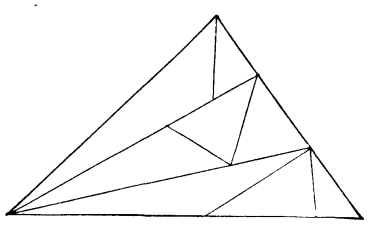


Fig. 33.

In order to establish the corresponding proof for an arbitrary decomposition of the triangle Δ into the triangles Δ_k, draw from the vertex A of the given triangle Δ a transversal through each of the points of division of the required decomposition; that is to say, draw a transversal through each vertex of the triangles Δ_k. By means of these transversals, the given triangle Δ is decomposed into certain triangles Δ_t. Each of these triangles Δ_t is broken up by the segments which determined this decomposition into certain triangles and quadrilaterals. If, now, in each of the quadrilaterals, we draw a diagonal, then each triangle Δ_t is decomposed into certain other triangles Δ_{ts}. We shall now show that the decomposition into the triangles Δ_{ts} is for the triangles Δ_t, as well as for the triangles Δ_k, nothing else than a series of

transversal decompositions. In fact, it is at once evident that any decomposition of a triangle into partial triangles may always be affected by a series of transversal decompositions, providing, in this decomposition, points of division do not exist within the triangle, and further, that at least one side of the triangle remains free from points of division.

We easily see that these conditions hold for the triangles Δ_t. In fact, the interior of each of these triangles, as also one side, namely, the side opposite the point A, contains no points of division.

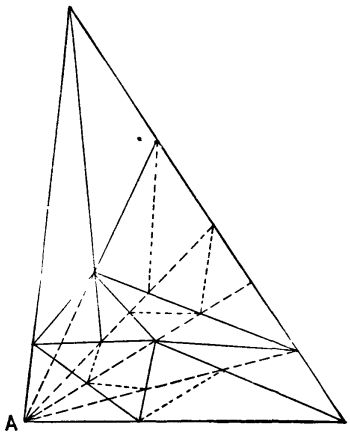


Fig. 34.

Likewise, for each of the triangles Δ_k, the decomposition into Δ_{ts} is reducible to transversal decompositions. Let us consider a triangle Δ_k. Among the transversals of the triangle Δ emanating from the point A, there is always a definite one to be found which either coincides with a side of Δ_k, or which itself divides Δ_k into two triangles. In the first case, the side in question always remains free from further points of division by the decomposition into the triangles Δ_{ts}. In the second case, the segment of the transversal contained within the triangle Δ_k is a side of the two triangles arising from the division, and this side certainly remains free from further points of division.

According to the considerations set forth at the beginning of this demonstration, the measure of area $F(\Delta)$ of the triangle Δ is equal to the sum of the measures of area $F(\Delta_t)$ of all the triangles Δ_t and this sum is in turn equal to the sum of all the measures of area $F(\Delta_{ts})$. However, the sum of the measures of area $F(\Delta_k)$ of all the triangles Δ_k is also equal to the sum of the measures of area $F(\Delta_{ts})$. Consequently, we have finally that the measure of area $F(\Delta)$ is also equal to the sum of all the measures of area $F(\Delta_k)$, and with this conclusion our demonstration is completed.

Definition. If we define the measure of area $F(P)$ of a polygon as the sum of the measures of area of all the triangles into which the polygon is, by a definite decomposition, divided, then upon the basis of theorem 29 and by a process of reasoning similar to that which we have employed in § 18 to prove theorem 24, we know that the measure of area of a polygon is independent of the manner of decomposition into triangles and, consequently, is definitely determined by the polygon itself. From this we obtain, by aid of theorem 29, the result that *polygons of equal area have also equal measures of area.*

Furthermore, if P and Q are two polygons of equal content, then there must exist, according to the above definition, two other polygons P' and Q' of equal area, such that the polygon composed of P and P' shall be of equal area with the polygon formed by combining the polygons Q and Q'. From the two equations

$$\begin{aligned} F(P+P') &= F(Q+Q') \\ F(P') &= F(Q'), \end{aligned}$$

we easily deduce the equation

$$F(P) = F(Q);$$

that is to say, *polygons of equal content have also equal measure of area.*

From this last statement, we obtain immediately the proof of theorem 28. If we denote the equal bases of the two triangles by g and the corresponding altitudes by h and h', respectively, then we may conclude from the assumed equality of content of the two triangles that they must also have equal measures of area; that is to say, it follows that

$$\frac{1}{2}gh = \frac{1}{2}gh'$$

and, consequently, dividing by $\frac{1}{2}g$, we get

$$h = h',$$

which is the statement made in theorem 28.

§ 21. EQUALITY OF CONTENT AND THE MEASURE OF AREA.

In § 20 we have found that polygons having equal content have also equal measures of area. The converse of this is also true.

In order to prove the converse, let us consider two triangles ABC and $AB'C'$ having a common right angle at A. The measures of area of these two triangles are expressed by the formulæ

$$\begin{aligned} F(ABC) &= \frac{1}{2}AB \cdot AC, \\ F(AB'C') &= \frac{1}{2}AB' \cdot AC'. \end{aligned}$$

We now assume that these measures of area are equal to each other, and consequently we have

$$AB \cdot AC = AB' \cdot AC',$$

or

$$AB : AB' = AC' : AC.$$

From this proposition, it follows, according to theorem 23, that the two straight lines BC' and $B'C$ are parallel, and hence, by theorem 27, the two triangles $BC'B'$ and $BC'C$ are of equal content. By the addition of the triangle ABC', it follows that the two triangles ABC and $AB'C'$ are of equal content. We have then shown that two right triangles having the same measure of area are always of equal content.

Take now any arbitrary triangle having the base g and the altitude h. Then, according to theorem 27, it has equal content with a right triangle having the two sides g and h. Since the original triangle had evidently the same measure of area as the right triangle, it follows that, in the above consideration, the restriction to right triangles was not necessary. Hence, two *arbitrary triangles with equal measures of area are also of equal content.*

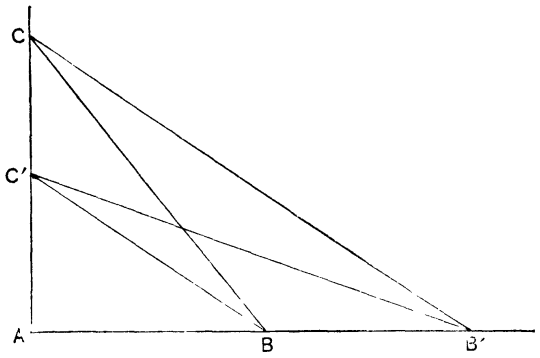


Fig. 35.

Moreover, let us suppose P to be any polygon having the measure of area g and let P be decomposed into n triangles having respectively the measures of area g_1, g_2, g_3, …, g_n. Then, we have

$$g = g_1 + g_2 + g_3 + \cdots + g_n.$$

Construct now a triangle ABC having the base $AB = g$ and the altitude $h = 1$. Take, upon the base of this triangle, the points A_1, A_2, …, A_{n-1} so that $g_1 = AA_1$, $g_1 = A_1A_2$, …, $g_{n-1} = A_{n-2}A_{n-1}$, $g_n = A_{n-1}B$.

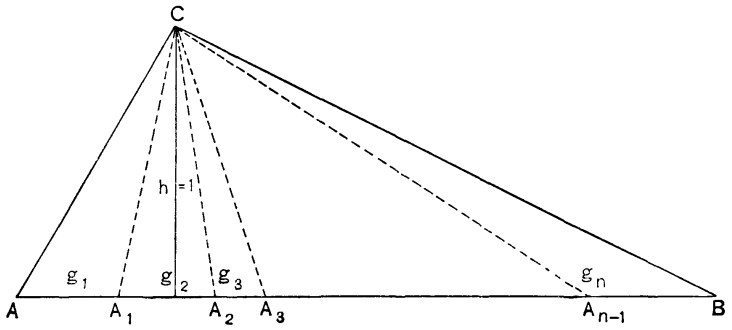


Fig. 36.

Since the triangles composing the polygon P have respectively the same measures of area as the triangles AA_1C, A_1A_2C, …, $A_{n-2}A_{n-1}C$, $A_{n-1}BC$, it follows from what has already been demonstrated that they have also the same content as these triangles. Consequently, the polygon P and a triangle, having the base g and the altitude $h = 1$ are of equal content. From this, it follows, by the application of theorem 24, that two polygons having equal measures of area are always of equal content.

We can now combine the proposition of this section with that demonstrated in the last, and thus obtain the following theorem:

THEOREM 30. Two polygons of equal content have always equal measures of area. Conversely, two polygons having equal measures of area are always of equal content.

In particular, if two rectangles are of equal content and have one side in common, then their remaining sides are respectively congruent. Hence, we have the following proposition:

> THEOREM 31. If we decompose a rectangle into several triangles by means of straight lines and then omit one of these triangles, we can no longer make up completely the rectangle from the triangles which remain.

This theorem has been demonstrated by F. Schur[11] and by W. Killing,[12] but by making use of the axiom of Archimedes. By O. Stolz,[13] it has been regarded as an axiom. In the foregoing discussion, it has been shown that it is completely independent of the axiom of Archimedes. However, when we disregard the axiom of Archimedes, this theorem (31) is not sufficient of itself to enable us to demonstrate Euclid's theorem concerning the equality of altitudes of triangles having equal content and equal bases. (Theorem 28.)

In the demonstration of theorems 28, 29, and 30, we have employed essentially the algebra of segments introduced in § 15, p. 29, and as this depends substantially upon Pascal's theorem (theorem 21), we see that this theorem is really the corner-stone in the theory of areas. We may, by the aid of theorems 27 and 28, easily establish the converse of Pascal's theorem.

Of two polygons P and Q, we call P the smaller or larger in content according as the measure of area $F(P)$ is less or greater than the measure of area $F(Q)$. From what has already been said, it is clear that the notions, equal content, smaller content, larger content, are mutually exclusive. Moreover, we easily see that a polygon, which lies wholly within another polygon, must always be of smaller content than the exterior one.

With this we have established the important theorems in the theory of areas.

[11] *Sitzungsberichte der Dorpater Naturf. Ges.* 1892.
[12] *Grundlagen der Geometrie*, Vol. 2, Chapter 5, § 5, 1898.
[13] *Monatshefte für Math, und Phys.* 1894.

DESARGUES'S THEOREM.

§ 22. DESARGUES'S THEOREM AND ITS DEMONSTRATION FOR PLANE GEOMETRY BY AID OF THE AXIOMS OF CONGRUENCE.

Of the axioms given in §§ 1–8, pp. 1–15, those of groups II–V are in part linear and in part plane axioms. Axioms 3–7 of group I are the only space axioms. In order to show clearly the significance of these axioms of space, let us assume a plane geometry and investigate, in general, the conditions for which this plane geometry may be regarded as a part of a geometry of space in which at least the axioms of groups I–III are all fulfilled.

Upon the basis of the axioms of groups I–III, it is well known that the so-called theorem of Desargues may be easily demonstrated. This theorem relates to points of intersection in a plane. Let us assume in particular that the straight line, upon which are situated the points of intersection of the homologous sides of the two triangles, is the straight line which we call the straight line at infinity. We will designate the theorem which arises in this case, together with its converse, as the theorem of Desargues. This theorem is as follows:

> THEOREM 32. (Desargues's theorem.) When two triangles are so situated in a plane that their homologous sides are respectively parallel, then the lines joining the homologous vertices pass through one and the same point, or are parallel to one another.

Conversely, if two triangles are so situated in a plane that the straight lines joining the homologous vertices intersect in a common point, or are parallel to one another, and, furthermore, if two pairs of homologous sides are parallel to each other, then the third sides of the two triangles are also parallel to each other.

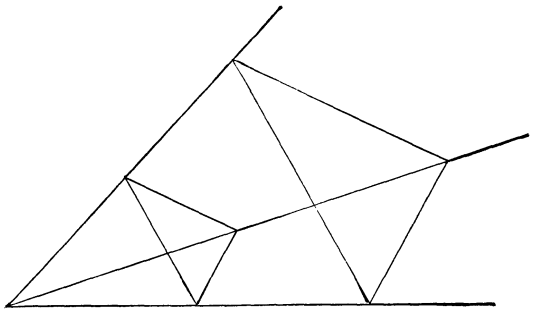


Fig. 37.

As we have already mentioned, theorem 32 is a consequence of the axioms I–III. Because of this fact, the validity of Desargues's theorem in the plane is, in any case, a necessary condition that the geometry of this plane may be regarded as a part of a geometry of space in which the axioms of groups I–III are all fulfilled.

Let us assume, as in §§ 13–21, pp. 23–45, that we have a plane geometry in which the axioms I, 1–2 and II–IV all hold and, also, that we have introduced in this geometry an algebra of segments conforming to § 15.

Now, as has already been established in § 17, there may be made to correspond to each point in the plane a pair of segments (x, y) and to each straight line a ratio of three segments $(u : v : w)$, so that the linear equation

$$ux + vy + w = 0$$

expresses the condition that the point is situated upon the straight line. The system composed of all the segments in our geometry forms, according to § 17, a domain of numbers for which the properties (1–16), enumerated in § 13, are valid. We can, therefore, by means of this domain of numbers, construct a geometry of space in a manner similar to that already employed in § 9 or in § 12, where we made use of the systems of numbers Ω and $\Omega(t)$, respectively. For this purpose, we assume that a system of three segments (x, y, z) shall represent a point, and that the ratio of four segments $(u : v : w : r)$ shall represent a plane, while a straight line is defined as the intersection of two planes. Hence, the linear equation

$$ux + vy + wz + r = 0$$

expresses the fact that the point (x, y, z) lies in the plane $(u : v : w : r)$. Finally, we determine the arrangement of the points upon a straight line, or the points of a plane with respect to a straight line situated in this plane, or the arrangement of the points in space with respect to a plane, by means of inequalities in a manner similar to the method employed for the plane in § 9.

Since we obtain again the original plane geometry by putting $z = 0$, we know that our plane geometry can be regarded as a part of geometry of space. Now, the validity of Desargues's theorem is, according to the above considerations, a necessary condition for this result. Hence, in the assumed plane geometry, it follows that Desargues's theorem must also hold.

It will be seen that the result just stated may also be deduced without difficulty from theorem 23 in the theory of proportion.

§ 23. THE IMPOSSIBILITY OF DEMONSTRATING DESARGUES'S THEOREM FOR THE PLANE WITHOUT THE HELP OF THE AXIOMS OF CONGRUENCE.[14]

We shall now investigate the question whether or no in plane geometry Desargues's theorem may be deduced without the assistance of the axioms of congruence. This leads us to the following result:

[14]See also a recent paper by F. R. Moulton on "Simple Non-desarguesian Geometry," *Transactions of the Amer. Math. Soc.*, April, 1902.—*Tr.*

THEOREM 33. A plane geometry exists in which the axioms I 1–2, II–III, IV 1–5, V, that is to say, all linear and all plane axioms with the exception of axiom IV, 6 of congruence, are fulfilled, but in which the theorem of Desargues (theorem 32) is not valid. Desargues's theorem is not, therefore, a consequence solely of the axioms mentioned; for, its demonstration necessitates either the space axioms or all of the axioms of congruence.

PROOF. Select in the ordinary plane geometry (the possibility of which has already been demonstrated in § 9, pp. 15–17) any two straight lines perpendicular to each other as the axes of x and y. Construct about the origin O of this system of co-ordinates an ellipse having the major and minor axes equal to 1 and $\frac{1}{2}$, respectively. Finally, let F denote the point situated upon the positive x-axis at the distance $\frac{3}{2}$ from O. Consider all of the circles which cut the ellipse in four real points. These points may be either distinct or in any way coincident. Of all the points situated upon these circles, we shall attempt to determine the one which lies upon the x-axis farthest from the origin. For this purpose, let us begin with an arbitrary circle cutting the ellipse in four distinct points and intersecting the positive x-axis in the point C. Suppose this circle then turned about the point C in such a manner that two or more of the four points of intersection with the ellipse finally coincide in a single point A, while the rest of them remain real. Increase now the resulting tangent circle in such a way that A always remains a point of tangency with the ellipse. In this way we obtain a circle which is either tangent to the ellipse in also a second point B, or which has with the ellipse a four-point contact in A. Moreover, this circle cuts the positive x-axis in a point more remote than C. The desired farthest point will accordingly be found among those points of intersection of the positive x-axis by circles lying exterior to the ellipse and being doubly tangent to it. All such circles must lie, as we can easily see, symmetrically with respect to the y-axis. Let a, b be the co-ordinates of any point on the ellipse. Then an easy calculation shows that the circle, which is symmetrical with respect to y-axis and tangent to the ellipse at this point, cuts off from the positive x-axis the segment

$$x = \mid \sqrt{1 + 3b^2} \mid .$$

The greatest possible value of this expression occurs for $b = \frac{1}{2}$ and, hence, is equal to $\frac{1}{2} \mid \sqrt{7} \mid$. Since the point on the x-axis which we have denoted by F has for its abscissa the value $\frac{3}{2} > \frac{1}{2} \mid \sqrt{7} \mid$, it follows that *among the circles cutting the ellipse four times there is certainly none which passes through the point F.*

We will now construct a new plane geometry in the following manner. As points in this new geometry, let us take the points of the (xy)-plane. We will define a straight line of our new geometry in the following manner. Every straight line of the (xy)-plane which is either tangent to the fixed ellipse, or does not cut it at all, is taken unchanged as a straight line of the new geometry. However, when any straight line g of the (xy)-plane cuts the ellipse, say in the points P and Q, we will then define the corresponding

straight line of the new geometry as follows. Construct a circle passing through the points P and Q and the fixed point F. From what has just been said, this circle will have no other point in common with the ellipse. We will now take the broken line, consisting of the arc PQ just mentioned and the two parts of the straight line g extending outward

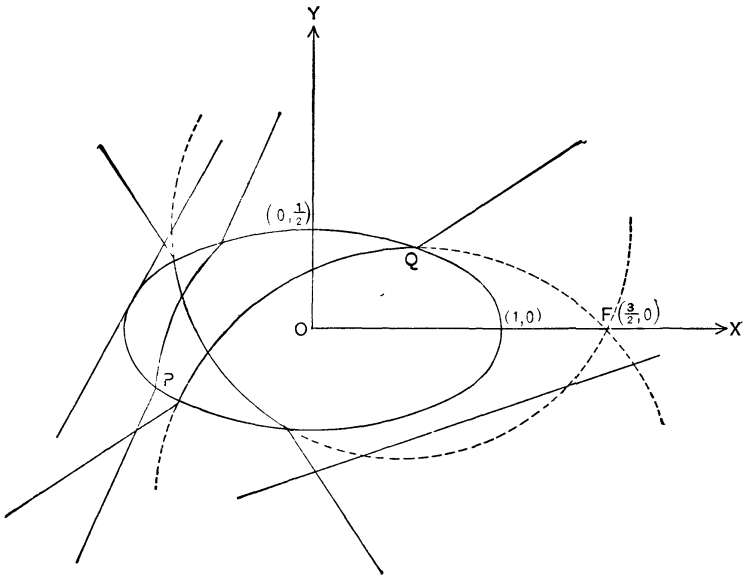


Fig. 38.

indefinitely from the points P and Q, as the required straight line in our new geometry. Let us suppose all of the broken lines constructed which correspond to straight lines of the (xy)-plane. We have then a system of broken lines which, considered as straight lines of our new geometry, evidently satisfy the axioms I, 1–2 and III. By a convention as to the actual arrangement of the points and the straight lines in our new geometry, we have also the axioms II fulfilled.

Moreover, we will call two segments AB and $A'B'$ congruent in this new geometry, if the broken line extending between A and B has equal length, in the ordinary sense of the word, with the broken line extending from A' to B'.

Finally, we need a convention concerning the congruence of angles. So long as neither of the vertices of the angles to be compared lies upon the ellipse, we call the two angles congruent to each other, if they are equal in the ordinary sense. In all other cases we make the following convention. Let A, B, C be points which follow one another in this order upon a straight line of our new geometry, and let A', B', C' be also points which lie in this order upon another straight line of our new geometry. Let D be a point lying outside of the straight line ABC and D' be a point outside of the straight $A'B'C'$. We will now say that, in our new geometry, the angles between these straight lines fulfill the congruences

$$\angle ABD \equiv \angle A'B'D' \text{ and } \angle CBD \equiv \angle C'B'D'$$

whenever the natural angles between the corresponding broken lines of the ordinary geometry fulfill the proportion

$$\angle ABD : \angle CBD = \angle A'B'D' : \angle C'B'D'.$$

These conventions render the axioms IV, 1–5 and V valid.

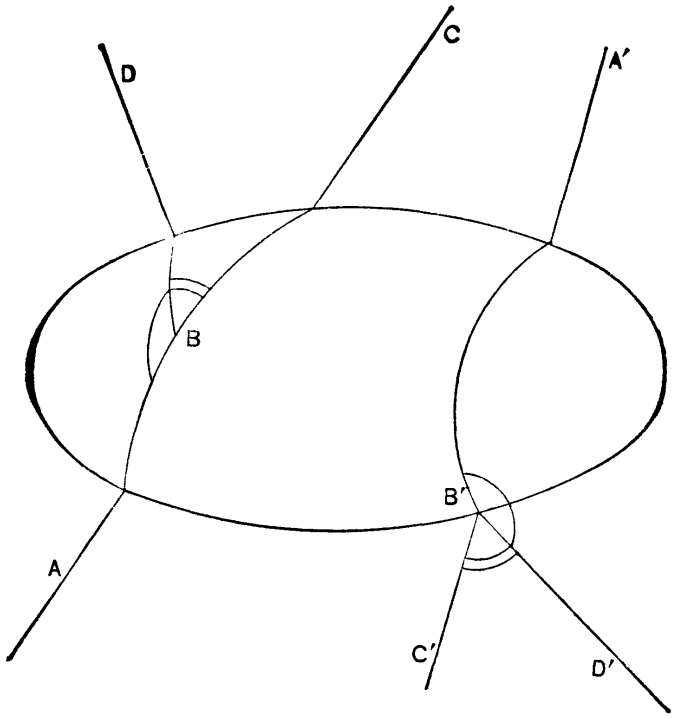


Fig. 39.

In order to see that Desargues's theorem does not hold for our new geometry, let us consider the following three ordinary straight lines of the (xy)-plane; viz., the axis of x, the axis of y, and the straight line joining the two points of the ellipse $(\frac{3}{5}, \frac{2}{5})$ and $(-\frac{3}{5}, -\frac{2}{5})$. Since these three ordinary straight lines pass through the origin, we can easily construct two triangles so that their vertices shall lie respectively upon these three straight lines and their homologous sides shall be parallel and all three sides shall lie exterior to the ellipse. As we may see from figure 40, or show by an easy calculation, the broken lines arising from the three straight lines in question do not intersect in a common point. Hence, it follows that Desargues's theorem certainly does not hold for this particular plane geometry in which we have constructed the two triangles just considered.

This new geometry serves at the same time as an example of a plane geometry in which the axioms I, 1–2, II–III, IV, 1–5, V all hold, but which cannot be considered as a part of a geometry of space.

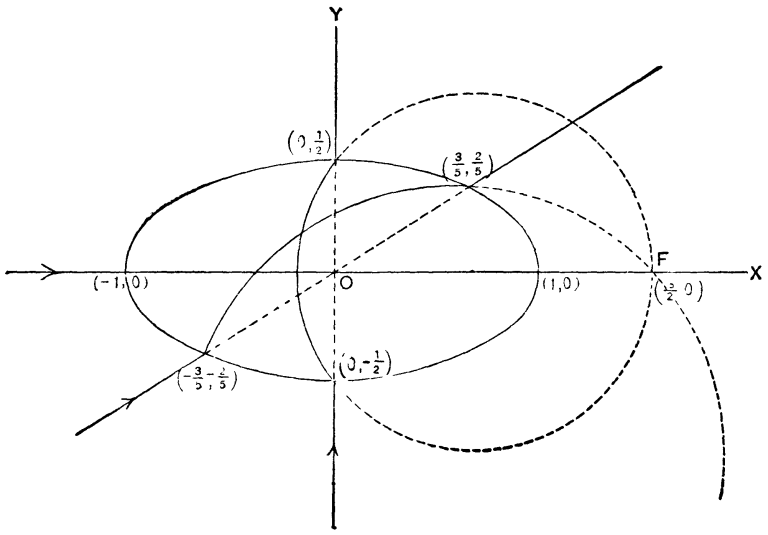


Fig. 40.

§ 24. INTRODUCTION OF AN ALGEBRA OF SEGMENTS BASED UPON DESARGUES'S THEOREM AND INDEPENDENT OF THE AXIOMS OF CONGRUENCE.[15]

In order to see fully the significance of Desargues's theorem (theorem 32), let us take as the basis of our consideration a plane geometry where all of the axioms I 1–2, II–III are valid, that is to say, where all of the plane axioms of the first three groups hold, and then introduce into this geometry, in the following manner, a new algebra of segments independent of the axioms of congruence.

Take in the plane two fixed straight lines intersecting in O, and consider only such segments as have O for their origin and their other extremity in one of the fixed lines. We will regard the point O itself as a segment and call it the segment O. We will indicate this fact by writing

$$OO = 0, \text{ or } 0 = OO.$$

Let E and E' be two definite points situated respectively upon the two fixed straight lines through O. Then, define the two segments OE and OE' as the segment 1 and write accordingly

$$OE = OE' = 1 \text{ or } 1 = OE = OE'. \tag{1}$$

We will call the straight line EE', for brevity, the unit-line. If, furthermore, A and A' are points upon

the straight lines OE and OE', respectively, and, if the straight line AA' joining them is parallel to EE', then we will say that the segments OA and OA' are equal to one

[15]Discussed also by Moore in a paper before the Am. Math. Soc., Jan, 1902. See *Trans. Am. Math. Soc.—Tr.*

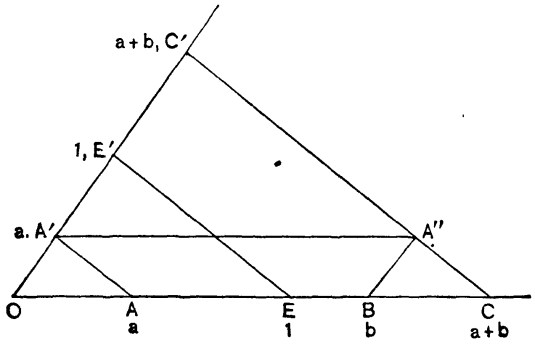


Fig. 41.

another, and write

$$OA = OA' \text{ or } OA' = OA.$$

In order now to define the sum of the segments $a = OA$ and $b = OB$, we construct AA' parallel to the unit-line EE' and draw through A' a parallel to OE and through B a parallel to OE'. Let these two parallels intersect in A''. Finally, draw through A'' a straight line parallel to the unit-line EE'. Let this parallel cut the two fixed lines OE and OE' in C and C' respectively. Then $c = OC = OC'$ is called the *sum* of the segments $a = OA$ and $b = OB$. We indicate this by writing

$$c = a + b, \text{ or } a + b = c.$$

In order to define the product of a segment $a = OA$ by a segment $b = OB$, we make use of exactly the same construction as employed in § 15, except that, in place of the sides of a right angle, we make use here of the straight lines OE and OE'. The construction

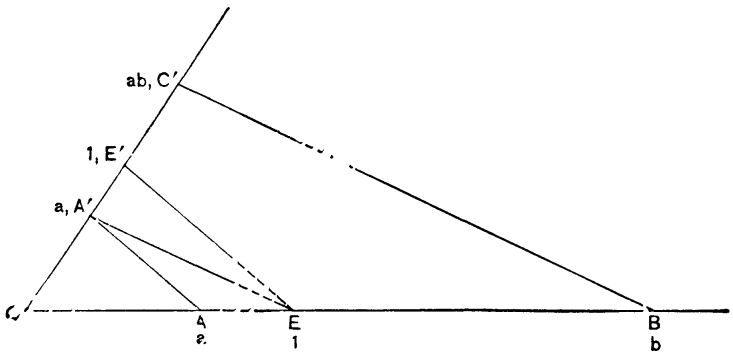


Fig. 42.

is consequently as follows. Determine upon OE' a point A' so that AA' is parallel to the unit-line EE', and join E with A'. Then draw through B a straight line parallel to EA'. This parallel will intersect the fixed straight line OE' in the point C', and we call $c = OC'$

the product of the segment $a = OA$ by the segment $b = OB$. We indicate this relation by writing

$$c = ab, \text{ or } ab = c.$$

§ 25. THE COMMUTATIVE AND THE ASSOCIATIVE LAW OF ADDITION FOR OUR NEW ALGEBRA OF SEGMENTS.

In this section, we shall investigate the laws of operation, as enumerated in § 13, in order to see which of these hold for our new algebra of segments, when we base our considerations upon a plane geometry in which axioms I 1–2, II–III are all fulfilled, and, moreover, in which Desargues's theorem also holds.

First of all, we shall show that, for the addition of segments as defined in § 24, the commutative law

$$a + b = b + a$$

holds. Let

$$a = OA = OA'$$
$$b = OB = OB'$$

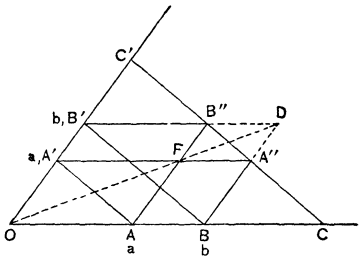


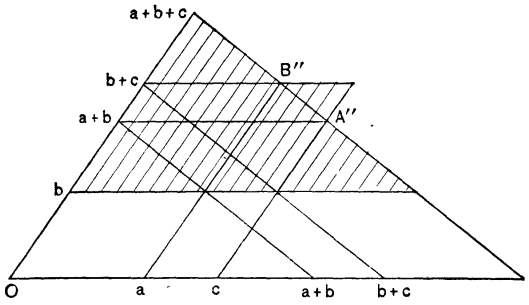
Fig. 43.

Hence, AA' and BB' are, according to our convention, parallel to the unit-line. Construct the points A'' and B'' by drawing $A'A''$ and $B'B''$ parallel to OA and also AB'' and BA' parallel to OA. We see at once that the line $A''B''$ is parallel to AA' as the commutative law requires. We shall show the validity of this statement by the aid of Desargues's theorem in the following manner. Denote the point of intersection of AB'' and $A'A''$ by F and that of BA'' and $B'B''$ by D. Then, in the triangles $AA'F$ and $BB'D$, the homologous sides are parallel to each other. By Desargues's theorem, it follows that the three points O, F, D lie in a straight line. In consequence of this condition, the two triangles OAA' and $DB''A''$ lie in such a way that the lines joining the corresponding vertices pass through the same point F, and since the homologous sides OA and DB'', as also OA' and DA'', are parallel to each other, then, according to the second part of Desargues's theorem (theorem 32), the third sides AA' and $B''A''$ are parallel to each other.

To prove the associative law of addition

$$a + (b + c) = (a + b) + c,$$

we shall make use of figure 44. In consequence of the commutative law of addition just demonstrated, the above formula states that the straight line $A''B''$ must be parallel to the unit-line. The validity of this statement is evident, since the shaded part of figure 44 corresponds exactly with figure 43.



$a + (b + c) = (a + b) + c$

Fig. 44.

§ 26. THE ASSOCIATIVE LAW OF MULTIPLICATION AND THE TWO DISTRIBUTIVE LAWS FOR THE NEW ALGEBRA OF SEGMENTS.

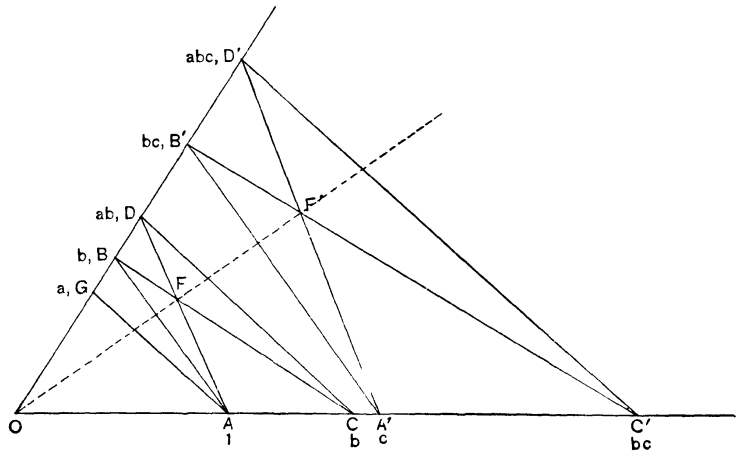
The associative law of multiplication

$$a(bc) = (ab)c$$

has also a place in our new algebra of segments.

Let there be given upon the first of the two fixed straight lines through O the segments

$$1 = OA, \quad b = OC, \quad c = OA'$$



$a(bc) = (ab)c$

Fig. 45.

and upon the second of these straight lines, the segments

$$a = OG, \quad b = OB.$$

In order to construct the segments

$$\begin{aligned} bc &= OB', \text{ and } bc = OC', \\ ab &= OD, \\ (ab)c &= OD', \end{aligned}$$

in accordance with §24, draw $A'B'$ parallel to AB, $B'C'$ parallel to BC, CD parallel to AG, and $A'D'$ parallel to AD. We see at once that the given law amounts to the same as saying that CD must also be parallel to $C'D'$. Denote the point of intersection of the straight lines $A'D'$ and $B'C'$ by F' and that of the straight lines AD and BC by F. Then the triangles ABF and $A'B'F'$ have their homologous sides parallel to each other, and, according to Desargues's theorem, the three points O, F, F' must lie in a straight line. Because of these conditions, we can apply the second part of Desargues's theorem to the two triangle CDF and $C'D'F'$, and hence show that, in fact, CD is parallel to $C'D'$.

Finally, upon the basis of Desargues's theorem, we shall show that the two distributive laws

$$a(b + c) = ab + ac$$

and

$$(a + b)c = ac + bc$$

hold for our algebra of segments.

In the proof of the first one of these laws, we shall make use of figure 46.[16] In this figure, we have

$$b = OA', c = OC',$$
$$ab = OB', ab = OA'', ac = OC'', etc.$$

In the same figure, $B''D_2$ is parallel to $C''D_1$ which is parallel to the fixed straight line OA', and $B'D_1$ is parallel to $C'D_2$, which is parallel to the fixed straight line OA''. Moreover, we have $A'A''$ parallel to $C'C''$, and $A'B''$ parallel to $B'A''$, parallel to $F'D_2$, parallel to $F''D_1$.

Our proposition amounts to asserting that we must necessarily have also

$$F'F'' \text{ parallel to } A'A'' \text{ and to } C'C''.$$

We construct the following auxiliary lines:

$$F''J \text{ parallel to the fixed straight line } OA',$$

$$F'J \quad \text{“} \quad \text{“} \quad \text{“} \quad \text{“} \quad \text{“} \quad \text{“} \quad OA''.$$

Let us denote the points of intersection of the straight lines $C''D_1$ and $C'D_2$, $C''D_1$ and $F'J$, $C'D_2$ and $F''J$ by G, H_1, H_2, respectively. Finally, we obtain the other auxiliary lines indicated in the figure by joining the points already constructed.

[16] Figures 46, 47, and 48 were designed by Dr. Von Schaper, as have also the details of the demonstrations relating to these figures.

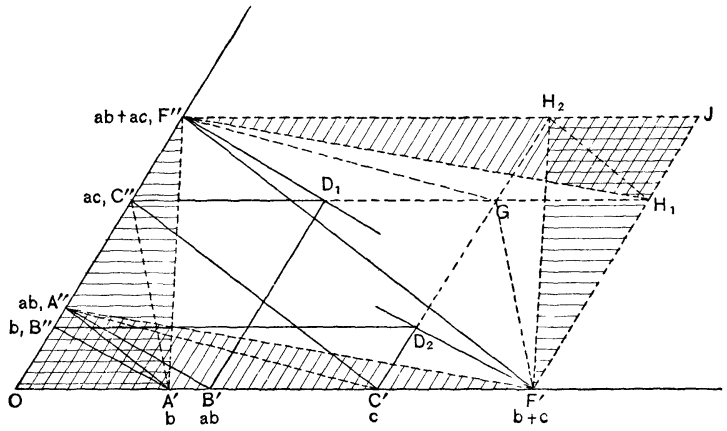


Fig. 46.

In the two triangles $A'B''C''$ and $F'D_2G$, the straight lines joining homologous vertices are parallel to each other. According to the second part of Desargues's theorem, it follows, therefore, that

$$A'C'' \text{ is parallel to } F'G.$$

In the two triangles $A'C''F''$ and $F'GH_2$, the straight lines joining the homologous vertices are also parallel to each other. From the properties already demonstrated, it follows by virtue of the second part of Desargues's theorem that we must have

$$A'F'' \text{ parallel to } F'H_2.$$

Since in the two horizontally shaded triangles $OA'F'$ and JH_2F' the homologous sides are parallel, Desargues's theorem shows that the three straight lines joining the homologous vertices, viz.:

$$OJ,\ A'H_2,\ F''F' \tag{2}$$

all intersect in one and the same point, say in P.

In the same way, we have necessarily

$$A''F' \text{ parallel to } F''H_1$$

and since, in the two obliquely shaded triangles $OA''F'$ and JH_1F'', the homologous sides are parallel, then, in consequence of Desargues's theorem, the three straight lines joining the homologous vertices, viz.:

$$OJ, A''H_1, F'F''$$

all intersect likewise in the same point, namely, in point P.

Moreover, in the triangles $OA'A''$ and JH_2H_1, the straight lines joining the homologous vertices all pass through this same point P, and, consequently, it follows that we have

$$H_1H_2 \text{ parallel to } A'A'',$$

and, therefore,

$$H_1H_2 \text{ parallel to } C'C'',$$

Finally, let us consider the figure $F''H_2C'C''H_1F'F''$. Since, in this figure, we have

$$\begin{array}{rrl} F''H_2 \text{ parallel to} & C'F', & \text{parallel to } C''H_1, \\ GH_2C' \quad \text{“ “} & F''C'', & \text{“ “ } H_1F', \\ C'C'' \quad \text{“ “} & H_1H_2, & \end{array}$$

we recognize here again figure 43, which we have already made use of in § 25 to prove the commutative law of addition. The conclusions, analogous to those which we reached there, show that we must have

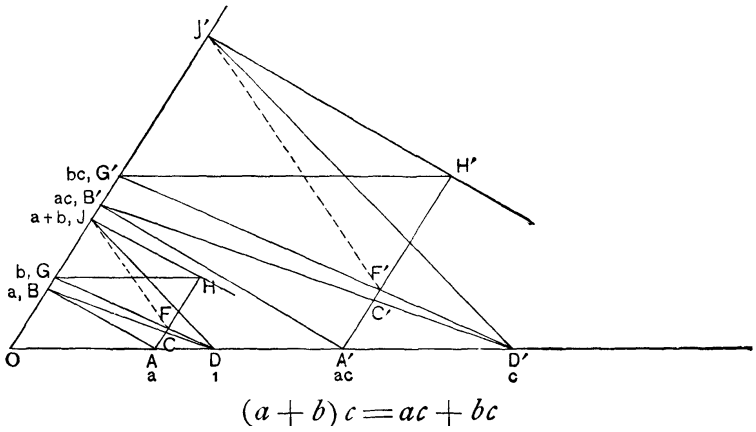
$$F'F'' \text{ parallel to } H_1H_2$$

and, consequently, we must have also

$$F'F'' \text{ parallel to} A'A'',$$

which result concludes our demonstration.

To prove the second formula of the distributive law, we make use of an entirely different figure,–figure 47. In this figure, we have



$$(a+b)\,c = ac + bc$$

Fig. 47.

$$1 = OD,\ a = OA,\ a = OB,\ b = OG,\ c = OD',$$

$$ac = OA',\ ac = OB',\ BC = OG',\ \text{etc.},$$

and, furthermore, we have

$$GH \text{ parallel to } G'H', \text{ parallel to the fixed line } OA,$$

$$GH \quad \text{“} \quad \text{“ } A'H', \quad \text{“} \quad \text{“} \quad \text{“} \quad \text{“} \quad \text{“ } OB,$$

We have also

$$\begin{array}{lcl} AB & \text{parallel to} & A'B' \\ BD & \text{“ “} & B'D' \\ DG & \text{“ “} & D'G' \\ HJ & \text{“ “} & H'J'. \end{array}$$

That which we are to prove amounts, then, to demonstrating that

$$DJ \text{ must be parallel to } D'J'.$$

Denote the points in which BD and GD intersect the straight line AH by C and F, respectively, and the points in which $B'D'$ and $G'D'$ intersect the straight line $A'H'$ by C' and F', respectively. Finally, draw the auxiliary lines FJ and $F'J'$, indicated in the figure by dotted lines.

In the triangles ABC and $A'B'C'$, the homologous sides are parallel and, consequently, by Desargues's theorem the three points O, C, C' lie on a straight line. Then, by considering in the same way the triangles CDF and $CD'F'$, it follows that the points O, F, F' lie upon the same straight line and likewise, from a consideration of the triangles FGH and $F'G'H'$, we find the points O, H, H' to be situated on a straight line. Now, in the triangles FHJ and $F'H'J'$, the straight lines joining the homologous vertices all pass through the same point O, and, hence, as a consequence of the second part of Desargues's theorem, the straight lines FJ and $F'J'$ must also be parallel to each other. Finally, a consideration of the triangles DFJ and $D'F'J'$ shows that the straight lines DJ and $D'J'$ are parallel to each other and with this our proof is completed.

§ 27. EQUATION OF THE STRAIGHT LINE, BASED UPON THE NEW ALGEBRA OF SEGMENTS.

In §§ 24–26, we have introduced into the plane geometry an algebra of segments in which the commutative law of addition and that of multiplication, as well as the two distributive laws, hold. This was done upon the assumption that the axioms cited in § 24, as also the theorem of Desargues, were valid. In this section, we shall show how an analytical representation of the point and straight line in the plane is possible upon the basis of this algebra of segments.

Definition. Take the two given fixed straight lines lying in the plane and intersecting in O as the axis of x and of y, respectively. Let us suppose any point P of the plane determined by the two segments x, y which we obtain upon the x-axis and y-axis, respectively, by drawing through P parallels to these axes. These segments are called the *co-ordinates* of the point P. Upon the basis of this new algebra of segments and by aid of Desargues's theorem, we shall deduce the following proposition.

THEOREM 34. The co-ordinates x, y of a point on an arbitrary straight line always satisfy an equation in these segments of the form

$$ax + by + c = 0.$$

In this equation, the segments a and b stand necessarily to the left of the co-ordinates x and y. The segments a and b are never both zero and c is an arbitrary segment.

Conversely, every equation in these segments and of this form represents always a straight line in the plane geometry under consideration.

PROOF. Suppose that the straight line l passes through the origin O. Furthermore, let C be a definite point upon l different from O, and P any arbitrary point of l. Let OA and OB be the co-ordinates of C and x, y be the co-ordinates of P. We will denote the straight line joining the extremities of the segments x, y by g. Finally, through the extremity of the segment 1, laid off on the x-axis, draw a straight line h parallel to AB. This parallel cuts off upon the y-axis the segment e. From the second part of

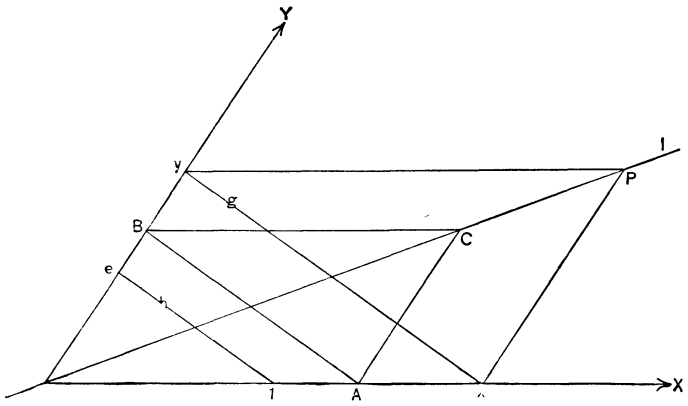


Fig. 48.

Desargues's theorem, it follows that the straight line g is also always parallel to AB. Since g is always parallel to h, it follows that the co-ordinates x, y of the point P must satisfy the equation

$$ex = g.$$

Moreover, in figure 49 let l' be any arbitrary straight line in our plane. This straight line will cut off on the x-axis the segment $c = OO'$. Now, in the same figure, draw through O the straight line l parallel to l'. Let P' be an arbitrary point on the line l'. The straight line through P', parallel to the x-axis, intersects the straight line l in P and cuts off upon the y-axis the segment $y = OB$. Finally, through P and P' let parallels to the y-axis cut off on the x-axis the segments $x = OA$ and $x' = OA'$.

We shall now undertake to show that the equation

$$x' = x + c$$

is fulfilled by the segments in question. For this purpose, draw $O'C$ parallel to the unit-line and likewise CD parallel to the x-axis and AD parallel to the y-axis.

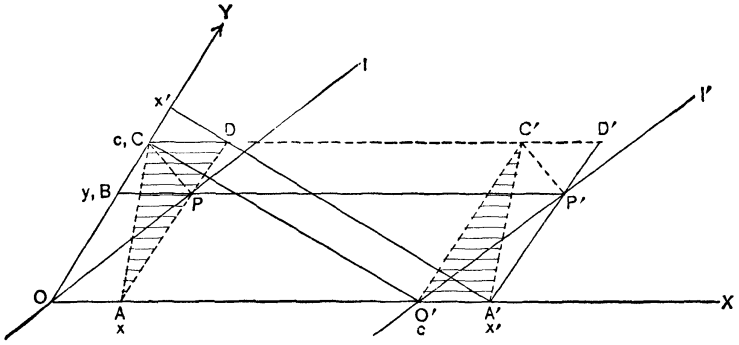


Fig. 49.

Then, to prove our proposition amounts to showing that we must have necessarily

$$A'D \text{ parallel to } O'C.$$

Let D' be the point of intersection of the straight lines CD and $A'P'$ and draw $O'C'$ parallel to the y-axis.

Since, in the triangles OCP and $O'C'P'$, the straight lines joining the homologous vertices are parallel, it follows, by virtue of the second part of Desargues's theorem, that we must have

$$CP \text{ parallel to } C'P'.$$

In a similar way, a consideration of the triangles ACP and $A'C'P'$ shows that we must have

$$AC \text{ parallel to } A'C'.$$

Since, in the triangles ACD and $C'A'O'$, the homologous sides are parallel to each other, it follows that the straight lines AC', CA' and DO' intersect in a common point. A consideration of the triangles $C'A'D$ and ACO' then shows that $A'D$ and CO' are parallel to each other. From the two equations already obtained, viz.:

$$ex = y \text{ and } x' = x + c, \tag{3}$$

follows at once the equation

$$ex' = y + ec. \tag{4}$$

If we denote, finally, by n the segment which added to the segment 1 gives the segment 0, then, from this last equation, we may easily deduce the following

$$ex' + ny + nec = 0, \tag{5}$$

and this equation is of the form required by theorem 34.

We can now show that the second part of the theorem is equally true; for, every linear equation

$$ax + by + c = 0 \tag{6}$$

may evidently be brought into the required form

$$ex + ny + nec = 0 \tag{7}$$

by a left-sided multiplication by a properly chosen segment.

It must be expressly stated, however, that, by our hypothesis, an equation of segments of the form

$$xa + yb + c = 0, \tag{8}$$

where the segments a, b stand to the right of the co-ordinates x, y does not, in general, represent a straight line.

In Section 30, we shall make an important application of theorem 34.

§ 28. THE TOTALITY OF SEGMENTS, REGARDED AS A COMPLEX NUMBER SYSTEM.

We see immediately that, for the new algebra of segments established in Section 24, theorems 1–6 of Section 13 are fulfilled. Moreover, by aid of Desargues's theorem, we have already shown in Sections 25 and 26 that the laws 7–11 of operation, as given in Section 13, are all valid in this algebra of segments. With the single exception of the commutative law of multiplication, therefore, all of the theorems of connection hold.

Finally, in order to make possible an order of magnitude of these segments, we make the following convention. Let A and B be any two distinct points of the straight line OE. Suppose then that the four points O, E, A, B stand, in conformity with axiom II, 4, in a certain sequence. If this sequence is one of the following six possible ones, viz.:

$$ABOE, AOBE, AOEB, OABE, OAEB, OEAB,$$

then we will call the segment $a = OA$ *smaller* than the segment $b = OB$ and indicate the same by writing

$$a < b.$$

On the other hand, if the sequence is one of the six following ones, viz.:

$$BAOE, BOAE, BOEA, OBAE, OBEA, OEBA,$$

then we will call the segment $a = OA$ greater than the segment $b = OB$, and we write accordingly

$$a > b.$$

This convention remains in force whenever A or B coincides with O or E, only then the coinciding points are to be regarded as a single point, and, consequently, we have only to consider the order of three points.

Upon the basis of the axioms of group II, we can easily show also that, in our algebra of segments, the laws 13–16 of operation given in Section 13 are fulfilled. Consequently, the totality of all the different segments forms a complex number system for which the laws 1–11, 13–16 of Section 13 hold; that is to say, all of the usual laws of operation except the commutative law of multiplication and the theorem of Archimedes. We will call such a system, briefly, a *desarguesian number system.*

§ 29. CONSTRUCTION OF A GEOMETRY OF SPACE BY AID OF A DESARGUESIAN NUMBER SYSTEM.

Suppose we have given a desarguesian number system D. Such a system makes possible the construction of a geometry of space in which axioms I, II, III are all fulfilled.

In order to show this, let us consider any system of three numbers (x, y, z) of the desarguesian number system D as a point, and the ratio of four such numbers $(u : v : w : r)$, of which the first three are not 0, as a plane. However, the systems $(u : v : w : r)$ and $(au : av : aw : ar)$, where a is any number of D different from 0, represent the same plane. The existence of the equation

$$ux + vy + wz + r = 0$$

expresses the condition that the point (x, y, z) shall lie in the plane $(u : v : w : r)$. Finally, we define a straight line by the aid of a system of two planes $(u' : v' : w' : r')$ and $(u'' : v'' : w'' : r'')$, where we impose the condition that it is impossible to find in D two numbers a', a'' different from zero, such that we have simultaneously the relations

$$a'u' = a''u'', a'v' = a''v'', a'w' = a''w'',$$

A point (x, y, z) is said to be situated upon this straight line $[(u' : v' : w' : r'), (u'' : v'' : w'' : r'')]$, if it is common to the two planes $(u' : v' : w' : r')$ and $(u'' : v'' : w'' : r'')$. Two straight lines which contain the same points are not regarded as being distinct.

By application of the laws 1–11 of § 13, which by hypothesis hold for the numbers of D, we obtain without difficulty the result that the geometry of space which we have just constructed satisfies all of the axioms of groups I and III.

In order that the axioms (II) of order may also be valid, we adopt the following conventions. Let

$$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$$

be any three points of a straight line

$$[(u' : v' : w' : r'), (u'' : v'' : w'' : r'')].$$

Then, the point (x_2, y_2, z_2) is said to lie between the other two, if we have fulfilled at least one of the six following double inequalities:

$$x_1 < x_2 < x_3, \quad x_1 > x_2 > x_3, \tag{9}$$
$$y_1 < y_2 < y_3, \quad y_1 > y_2 > y_3, \tag{10}$$
$$z_1 < z_2 < z_3, \quad z_1 > z_2 > z_3, \tag{11}$$

If one of the two double inequalities (1) exists, then we can easily conclude that either $y_1 = y_2 = y_3$ or one of the two double inequalities (2) exists, and, consequently, either $z_1 = z_2 = z_3$ or one of the double inequalities (3) must exist. In fact, from the equations

$$u'x_i + v'y_i + w'z_i + r' = 0,$$
$$u''x_i + v''y_i + w''z_i + r'' = 0,$$
$$(i = 1, 2, 3)$$

we may obtain, by a left-sided multiplication of these equations by numbers suitably chosen from D and then adding the resulting equations, a system of equations of the form

$$u'''x_i + v'''y_i + r''' = 0, (i = 1, 2, 3). \tag{12}$$

In this system, the coefficient v''' is certainly different from zero, since otherwise the three numbers x_1, x_2, x_3 would be mutually equal.

From

$$x_1 \lesseqgtr x_2 \lesseqgtr x_3,$$

it follows that

$$u'''x_1 \underset{>}{\overset{<}{\equiv}} x'''u_2 \underset{>}{\overset{<}{\equiv}} x'''u_3,$$

and, hence, as a consequence of (4), we have

$$v'''y_1 + r''' \underset{>}{\overset{<}{\equiv}} v'''y_2 + r''' \underset{>}{\overset{<}{\equiv}} v'''y_3 + r'''$$

and, therefore,

$$v'''y_1 \underset{>}{\overset{<}{\equiv}} v'''y_2 \underset{>}{\overset{<}{\equiv}} v'''y_3.$$

Since v''' is different from zero, we have

$$y_1 \underset{>}{\overset{<}{\equiv}} y_2 \underset{>}{\overset{<}{\equiv}} y_3.$$

In each of these double inequalities, we must take either the upper sign throughout, or the middle sign throughout, or the lower sign throughout.

The preceding considerations show, that, in our geometry, the linear axioms II, 1–4 of order are all valid. However, it remains yet to show that, in this geometry, the plane axiom II, 5 is also valid.

For this purpose let a plane $(u : v : w : r)$ and a straight line $[(u : v : w : r), (u' : v' : w' : r')]$ in this plane be given. Let us assume that all the points (x, y, z) of the plane $(u : v : w : r)$, for which we have the expression $u'x + v'y + w'z + r'$ greater than or less

than zero, lie respectively upon the one side or upon the other side of the given straight line. We have then only to show that this convention is in accordance with the preceding statements. This, however, is easily done. We have thus shown that all of the axioms of groups I, II, III are fulfilled in the geometry of space which we have obtained in the above indicated manner from the desarguesian number system D. Remembering now that the theorem of Desargues is a consequence of the axioms I, II, III, we see that the proposition just stated is exactly the converse of the result reached in Section 28.

§ 30. SIGNIFICANCE OF DESARGUES'S THEOREM.

If, in a plane geometry, axioms I, 1–2, II, III are all fulfilled and, moreover, if the theorem of Desargues holds, then, according to §§ 24–28, it is always possible to introduce into this geometry an algebra of segments to which the laws 1–11, 13–16 of § 13 are applicable. We will now consider the totality of these segments as a complex number system and construct, upon the basis of this system, a geometry of space, in accordance with § 29, in which all of the axioms I, II, III hold.

In this geometry of space, we shall consider only the points $(x, y, 0)$ and those straight lines upon which only such points lie. We have then a plane geometry which must, if we take into account the proposition established in § 27, coincide exactly with the plane geometry proposed at the beginning. Hence, we are led to the following proposition, which may be regarded as the objective point of the entire discussion of the present chapter.

> THEOREM 35. If, in a plane geometry, axioms I, 1–2, II, III are all fulfilled, then the existence of Desargues's theorem is the necessary and sufficient condition that this plane geometry may be regarded as a part of a geometry of space in which all of the axioms I, II, III are fulfilled.

The theorem of Desargues may be characterized for plane geometry as being, so to speak, the result of the elimination of the space axioms.

The results obtained so far put us now in the position to show that every geometry of space in which axioms I, II, III are all fulfilled may be always regarded as a part of a "geometry of any number of dimensions whatever." By a geometry of an arbitrary number of dimensions is to be understood the totality of all points, straight lines, planes, and other linear elements, for which the corresponding axioms of connection and of order, as well as the axiom of parallels, are all valid.

PASCAL's THEOREM.

§ 31. TWO THEOREMS CONCERNING THE POSSIBILITY OF PROVING PASCAL'S THEOREM.

As is well known, Desargues's theorem (theorem 32) may be demonstrated by the aid of axioms I, II, III; that is to say, by the use, essentially, of the axioms of space. In § 23, we have shown that the demonstration of this theorem without the aid of the space axioms of group I and without the axioms of congruence (group IV) is impossible, even if we make use of the axiom of Archimedes.

Upon the basis of axioms I, 1–2, II, III, IV and, hence, by the exclusion of the axioms of space but with the assistance, essentially, of the axioms of congruence, we have, in § 14, deduced Pascal's theorem and, consequently, according to § 22, also Desargues's theorem. The question arises as to whether Pascal's theorem can be demonstrated without the assistance of the axioms of congruence. Our investigation will show that in this respect Pascal's theorem is very different from Desargues's theorem; for, in the demonstration of Pascal's theorem, the admission or exclusion of the axiom of Archimedes is of decided influence. We may combine the essential results of our investigation in the two following theorems.

> THEOREM 36. Pascal's theorem (theorem 21) may be demonstrated by means of the axioms I, II, III, V; that is to say, without the assistance of the axioms of congruence and with the aid of the axiom of Archimedes.
>
> THEOREM 37. Pascal's theorem (theorem 21) cannot be demonstrated by means of the axioms I, II, III alone; that is to say, by exclusion of the axioms of congruence and also the axiom of Archimedes.

In the statement of these two theorems, we may, by virtue of the general theorem 35, replace the space axioms I, 3–7 by the plane condition that Desargues's theorem (theorem 32) shall be valid.

§ 32. THE COMMUTATIVE LAW OF MULTIPLICATION FOR AN ARCHIMEDEAN NUMBER SYSTEM.

The demonstration of theorems 36 and 37 rests essentially upon certain mutual relations concerning the laws of operation and the fundamental propositions of arithmetic, a knowledge of which is of itself of interest. We will state the two following theorems.

> THEOREM 38. For an archimedean number system, the commutative law of multiplication is a necessary consequence of the remaining laws of operation; that is to say, if a number system possesses the properties 1–11, 13–17 given in § 13, it follows necessarily that this system satisfies also formula 12.

PROOF. Let us observe first of all that, if a is an arbitrary number of the system, and, if

$$n = 1 + 1 + \cdots + 1$$

is a positive integral rational number, then for n and a the commutative law of multiplication always holds. In fact, we have

$$\begin{aligned} an &= a(1 + 1 + \cdots + 1) \\ &= a \cdot 1 + a \cdot 1 + \cdots + a \cdot 1 \\ &= a + a + \cdots + a \end{aligned}$$

and likewise

$$\begin{aligned} na &= (1 + 1 + \cdots + 1)a \\ &= 1 \cdot a + 1 \cdot a + \cdots + 1 \cdot a \\ &= a + a + \cdots + a. \end{aligned}$$

Suppose now, in contradiction to our hypothesis, a, b to be numbers of this system, for which the commutative law of multiplication does not hold. It is then at once evident that we may make the assumption that we have

$$a > 0,\ b > 0,\ ab - ba > 0.$$

By virtue of condition 6 of § 13, there exists a number $c(> 0)$, such that

$$(a + b + 1)c = ab - ba.$$

Finally, if we select a number d, satisfying simultaneously the inequalities

$$d > 0,\ d < 1,\ d < c,$$

and denote by m and n two such integral rational numbers ≥ 0 that we have respectively

$$md < a \leq (m + 1)d$$

and

$$nd < b \leq (n + 1)d$$

then the existence of the numbers m and n is an immediate consequence of the theorem of Archimedes (theorem 17, § 13). Recalling now the remark made at the beginning of this proof, we have by the multiplication of the last inequalities

$$\begin{aligned} ab &\leqq mnd^2 + (m + n + 1)d^2 \\ ba &> mnd^2, \end{aligned}$$

and, hence, by subtraction

$$ab - ba \leqq (m + n + 1)d^2.$$

We have, however,

$$md < a,\ nd < b,,\ d < 1$$

and, consequently,

$$(m + n + 1)d < a + b + 1;$$

i.e.,

$$ab - ba < (a + b + 1)d,$$

or, since $d < c$, we have

$$ab - ba < (a + b + 1)c.$$

This inequality stands in contradiction to the definition of the number c, and, hence, the validity of the theorem 38 follows.

§ 33. THE COMMUTATIVE LAW OF MULTIPLICATION FOR A NON-ARCHIMEDEAN NUMBER SYSTEM.

THEOREM 39. For a non-archimedean number system, the commutative law of multiplication is not a necessary consequence of the remaining laws of operation; that is to say, there exists a system of numbers possessing the properties 1–11, 13–16 mentioned in § 13, but for which the commutative law (12) of multiplication is not valid. A desarguesian number system, in the sense of § 28, is such a system.

PROOF. Let t be a parameter and T any expression containing a finite or infinite number of terms, say of the form

$$T = r_0 t^n + r_1 t^{n+1} + r_2 t^{n+2} + r_3 t^{n+3} + \ldots,$$

where $r_0(\neq 0), r_1, r_2 \ldots$ are arbitrary rational numbers and n is an arbitrary integral rational number $\gtreqless 0$. Moreover, let s be another parameter and S any expression having a finite or infinite number of terms, say of the form

$$S = s^m T_0 + s^{m+1} T_1 + s^{m+2} T_2 + \ldots,$$

where $T_0(\neq 0), T_1, T_2, \ldots$ denote arbitrary expressions of the form T and m is again an arbitrary integral rational number $\lesseqgtr 0$. We will regard the totality of all the expressions of the form S as a complex number system $\Omega(s,t)$, for which we will assume the following laws of operation; namely, we will operate with s and t according to the laws 7–11 of § 13, as with parameters, while in place of rule 12 we will apply the formula

$$ts = 2st. \tag{13}$$

If, now, S', S'' are any two expressions of the form S, say

$$S' = s^{m'} T'_0 + s^{m'+1} T'_1 + s^{m'+2} T'_2 + \ldots,$$
$$S'' = s^{m''} T''_0 + s^{m''+1} T''_1 + s^{m''+2} T''_2 + \ldots,$$

then, by combination, we can evidently form a new expression $S' + S''$ which is of the form S, and is, moreover, uniquely determined. This expression $S' + S''$ is called the sum of the numbers represented by S' and S''.

By the multiplication of the two expressions S' and S'' term by term, we obtain another expression of the form

$$S'S'' = s^{m'}T'_0 s^{m''}T''_0 + (s^{m'}T'_0 s^{m''+1}T''_1 + s^{m'+1}T'_1 s^{m''}T''_0) + (s^{m'}T'_0 s^{m''+2}T''_2 + s^{m'+1}T'_1 s^{m''+1}T''_1 + s^{m'+2}T'_2 s^{m''}T''_0) + \dots$$

This expression, by the aid of formula (1), is evidently a definite single-valued expression of the form S and we will call it the product of the numbers represented by S' and S''.

This method of calculation shows at once the validity of the laws 1–5 given in Section 13 for calculating with numbers. The validity of law 6 of that section is also not difficult to establish. To this end, let us assume that

$$S' = s^{m'}T'_0 + s^{m'+1}T'_1 + s^{m'+2}T'_2 + \cdots$$

and

$$S''' = s^{m'''}T'''_0 + s^{m'''+1}T'''_1 + s^{m'''+2}T'''_2 \cdots$$

are two expressions of the form S, and let us suppose, further, that the coefficient r'_0 of T'_0 is different from zero. By equating the like powers of s in the two members of the equation

$$S'S'' = S''',$$

we find, first of all, in a definite manner an integral number m'' as exponent, and then such a succession of expressions

$$T''_0,\ T''_1,\ T''_2 \dots$$

that, by aid of formula (1), the expression

$$S'' = s^{m''}T''_0 + s^{m''+1}T''_1 + s^{m''+2}T''_2 \dots$$

satisfies equation (2). With this our theorem is established.

In order, finally, to render possible an order of sequence of the numbers of our system $\Omega(s,t)$, we make the following conventions. Let a number of this system be called greater or less than 0 according as in the expression S, which represents it, the first coefficient r_0 of T_0 is greater or less than zero. Given any two numbers a, b of the complex number system under consideration, we say that $a < b$ or $a > b$ according as we have $a - b < 0$ or > 0. It is seen immediately that, with these conventions, the laws 13–16 of § 13 are valid; that is to say, $\Omega(s,t)$ is a desarguesian number system (see § 28).

As equation (1) shows, law 12 of § 13 is not fulfilled by our complex number system and, consequently, the validity of theorem 39 is fully established.

In conformity with theorem 38, Archimedes's theorem (theorem 17, § 13) does not hold for the number system $\Omega(s,t)$ which we have just constructed.

We wish also to call attention to the fact that the number system $\Omega(s,t)$, as well as the systems Ω and $\Omega(t)$ made use of in § 9 and § 12, respectively, contains only an enumerable set of numbers.

§ 34. PROOF OF THE TWO PROPOSITIONS CONCERNING PASCAL'S THEOREM. (NON-PASCALIAN GEOMETRY.)

If, in a geometry of space, all of the axioms I, II, III are fulfilled, then Desargues's theorem (theorem 32) is also valid, and, consequently, according to §§ 24–26, pp. 50–58, it is possible to introduce into this geometry an algebra of segments for which the rules 1–11, 13–16 of § 13 are all valid. If we assume now that the axiom (V) of Archimedes is valid for our geometry, then evidently Archimedes's theorem (theorem 17 of § 13) also holds for our algebra of segments, and, consequently, by virtue of theorem 38, the commutative law of multiplication is valid. Since, however, the definition of the product of two segments, as introduced in § 24 (figure 42) and which is the definition here also under discussion, agrees with the definition in § 15 (figure 22), it follows from the construction made in § 15 that the commutative law of multiplication is here nothing else than Pascal's theorem. Consequently, the validity of theorem 36 is established.

In order to demonstrate theorem 37, let us consider again the desarguesian number system $\Omega(s,t)$ introduced in § 33, and construct, in the manner described in § 29, a geometry of space for which all of the axioms I, II, III are fulfilled. However, Pascal's theorem will not hold for this geometry; for, the commutative law of multiplication is not valid in the desarguesian number system $\Omega(s,t)$. According to theorem 36, the non-pascalian geometry is then necessarily also a non-archimedean geometry.

By adopting the hypothesis we have, it is evident that we cannot demonstrate Pascal's theorem, providing we regard our geometry of space as a part of a geometry of an arbitrary number of dimensions in which, besides the points, straight lines, and planes, still other linear elements are present, and providing there exists for these elements a corresponding system of axioms of connection and of order, as well as the axiom of parallels.

§ 35. THE DEMONSTRATION, BY MEANS OF THE THEOREMS OF PASCAL AND DESARGUES, OF ANY THEOREM RELATING TO POINTS OF INTERSECTION.

Every proposition relating to points of intersection in a plane has necessarily The form: Select, first of all, an arbitrary system of points and straight lines satisfying respectively the condition that certain ones of these points are situated on certain ones of the straight lines. If, in some known manner, we construct the straight lines joining the given points and determine the points of intersection of the given lines, we shall obtain finally a definite system of three straight lines, of which our proposition asserts that they all pass through the same point.

Suppose we now have a plane geometry in which all of the axioms I 1–2, II ..., V are valid. According to § 17, pp. 33–35, we may now find, by making use of a rectangular pair of axes, for each point a corresponding pair of numbers (x, y) and for each straight line a ratio of three definite numbers $(u : v : w)$. Here, the numbers $x,\ y,\ u,\ v,\ w$ are all

real numbers, of which $u,\ v$ cannot both be zero. The condition showing that the given point is situated upon the given straight line, viz.:

$$ux + vy + w = 0 \tag{14}$$

is an equation in the ordinary sense of the word. Conversely, in case $x,\ y,\ u,\ v,\ w$ are numbers of the algebraic domain Ω of § 9, and $u,\ v$ are not both zero, we may certainly assume that each pair of numbers (x, y) gives a point and that each ratio of three numbers $(u : v : w)$ gives a straight line in the geometry in question.

If, for all the points and straight lines which occur in connection with any theorem relating to intersections in a plane, we introduce the corresponding pairs and triples of numbers, then such a theorem asserts that a definite expression $A(p_1, p_2, \ldots, p_r)$ with real coefficients and depending rationally upon certain parameters $p_1, p_2, \ldots, p_r$ always vanishes as soon as we put for each of these parameters a number of the main Ω considered in § 9. We conclude from this that the expression $A(p_1, p_2, \ldots, p_r)$ must also vanish identically in accordance with the laws 7–12 of § 13.

Since, according to § 32, Desargues's theorem holds for the geometry in question, it follows that we certainly can make use of the algebra of segments introduced in § 24, and because Pascal's theorem is equally valid in this case, the commutative law of multiplication is also. Hence, for this algebra of segments, all of the laws 7–12 of § 13 are valid.

If we take as our axes in this new algebra of segments the co-ordinate axes already used and consider the unit points $E,\ E'$ as suitably established, we see that the new algebra of segments is nothing else than the system of co-ordinates previously employed.

In order to show that, for the new algebra of segments, the expression $A(p_1,\ p_2,\ \ldots,\ p_r)$ vanishes identically, it is sufficient to apply the theorems of Pascal and Desargues. Consequently we see that:

Every proposition relative to points of intersection in the geometry in question must always, by the aid of suitably constructed auxiliary points and straight lines, turn out to be a combination of the theorems of Pascal and Desargues. Hence for the proof of the validity of a theorem relating to points of intersection, we need not have resource to the theorems of congruence.

GEOMETRICAL CONSTRUCTIONS BASED UPON THE AXIOMS I–V.

§ 36. GEOMETRICAL CONSTRUCTIONS BY MEANS OF A STRAIGHT-EDGE AND A TRANSFERER OF SEGMENTS.

Suppose we have given a geometry of space, in which all of the axioms I–V are valid. For the sake of simplicity, we shall consider in this chapter a a plane geometry which is contained in this geometry of space and shall investigate the question as to what elementary geometrical constructions may be carried out in such a geometry.

Upon the basis of the axioms of group I, the following constructions are always possible.

PROBLEM 1. To join two points with a straight line and to find the intersection of two straight lines, the lines not being parallel.

Axiom III renders possible the following construction:

PROBLEM 2. Through a given point to draw a parallel to a given straight line.

By the assistance of the axioms (IV) of congruence, it is possible to lay off segments and angles; that is to say, in the given geometry we may solve the following problems:

PROBLEM 3. To lay off from a given point upon a given straight line a given segment.

PROBLEM 4. To lay off on a given straight line a given angle; or what is the same thing, to construct a straight line which shall cut a given straight line at a given angle.

It is impossible to make any new constructions by the addition of the axioms of groups II and V. Consequently, when we take into consideration merely the axioms of groups I–V, all of those constructions and only those are possible, which may be reduced to the problems 1–4 given above.

We will add to the fundamental problems 1–4 also the following:

PROBLEM 5. To draw a perpendicular to a given straight line.

We see at once that this construction can be made in different ways by means of the problems 1–4.

In order to carry out the construction in problem 1, we need to make use of only a *straight edge*. An instrument which enables us to make the construction in problem 3, we will call a *transferer of segments*. We shall now show that problems 2, 4, and 5 can be reduced to the constructions in problems 1 and 3 and, consequently, all of the problems 1–5 can be completely constructed by means of a straight-edge and a transferer of segments. We arrive, then, at the following result:

> THEOREM 40. Those problems in geometrical construction, which may be solved by the assistance of only the axioms I–V, can always be carried out by the use of the straight-edge and the transferer of segments.

PROOF. In order to reduce problem 2 to the solution of problems 1 and 3, we join the given point P with any point A of the given straight line and produce PA to C, making $AC = PA$. Then, join C with any other point B of the given straight line and produce CB to Q, making, $BQ = CB$. The straight line PQ is the desired parallel.

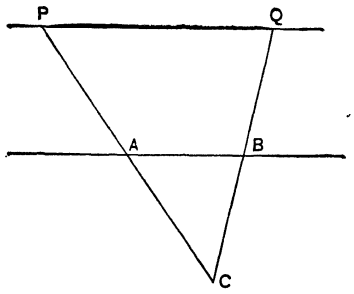


Fig. 50.

We can solve problem 5 in the following manner. Let A be an arbitrary point of the given straight line. Then upon this straight line, lay off in both directions from A the two equal segments AB and AC. Determine, upon any two straight lines passing through the point A, the points E and D so that the segments AD and AE will equal AB and AC.

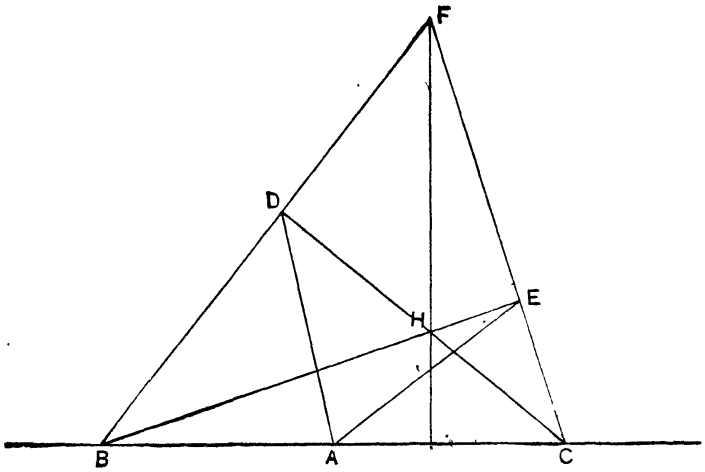


Fig. 51.

Suppose the straight lines BD and CE intersect in F and the straight lines BE and CD intersect in H. FH is then the desired perpendicular. In fact, the angles BDC and BEC, being inscribed in a semicircle having the diameter BC, are both right angles, and, hence, according to the theorem relating to the point of intersection of the altitudes of a triangle, the straight lines FH and BC are perpendicular to each other.

Moreover, we can easily solve problem 4 simply by the drawing of straight lines and the laying off of segments. We will employ the following method which requires only the drawing of parallel lines and the erection of perpendiculars. Let β be the angle to be laid off and A its vertex. Draw through A a straight line l parallel to the given straight line, upon which we are to lay off the given angle β. From an arbitrary point B of one side of the angle β, let fall a perpendicular upon the other side of this angle and also one upon l.

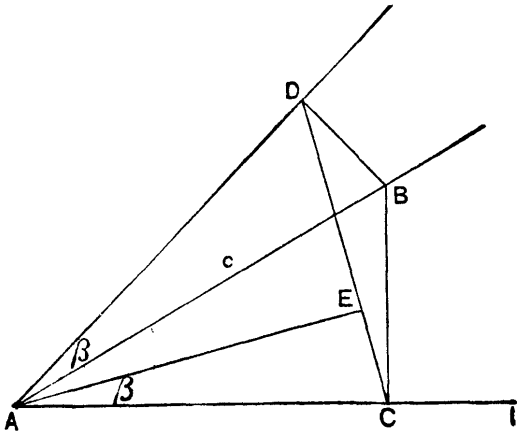


Fig. 52.

Denote the feet of these perpendiculars by D and C respectively. The construction of these perpendiculars is accomplished by means of problems 2 and 5. Then, let fall from A a perpendicular upon CD, and let its foot be denoted by E. According to the demonstration given in Section § 14, the angle CAE equals β. Consequently, the construction in 4 is made to depend upon that of 1 and 3 and with this our proposition is demonstrated.

§ 37. ANALYTICAL REPRESENTATION OF THE CO-ORDINATES OF POINTS WHICH CAN BE SO CONSTRUCTED.

Besides the elementary geometrical problems considered in § 36, there exists a long series of other problems whose solution is possible by the drawing of straight lines and the laying off of segments. In order to get a general survey of the scope of the problems which may be solved in this manner, let us take as the basis of our consideration a system of axes in rectangular co-ordinates and suppose that the co-ordinates of the points are, as usual, represented by real numbers or by functions of certain arbitrary parameters. In order to answer the question in respect to all the points capable of such a construction, we employ the following considerations.

Let a system of definite points be given. Combine the co-ordinates of these points into a domain R. This domain contains, then, certain real numbers and certain arbitrary parameters p. Consider, now, the totality of points capable of construction by the drawing of straight lines and the laying off of definite segments, making use of the system of points in question. We will call the domain formed from the co-ordinates of these points $\Omega(R)$, which will then contain real numbers and functions of the arbitrary parameters p.

The discussion in § 17 shows that the drawing of straight lines and of parallels amounts, analytically, to the addition, subtraction, multiplication, and division of segments. Furthermore, the well known formula given in § 9 for a rotation shows that the laying off of segments upon a straight line does not necessitate any other analytical operation than the

extraction of the square root of the sum of the squares of two segments whose bases have been previously constructed. Conversely, in consequence of the pythagorean theorem, we can always construct, by the aid of a right triangle, the square root of the sum of the squares of two segments by the mere laying off of segments.

From these considerations, it follows that the domain $\Omega(R)$ contains all of those and only those real numbers and functions of the parameters p, which arise from the numbers and parameters in R by means of a finite number of applications of the five operations; viz., the four elementary operations of arithmetic and, in addition, the fifth operation of extracting the square root of the sum of two squares. We may express this result as follows:

> THEOREM 41. A problem in geometrical construction is, then, possible of solution by the drawing of straight lines and the laying off of segments, that is to say, by the use of the straight-edge and a transferer of segments, when and only when, by the analytical solution of the problem, the co-ordinates of the desired points are such functions of the co-ordinates of the given points as may be determined by the rational operations and, in addition, the extraction of the square root of the sum of two squares.

From this proposition, we can at once show that not every problem which can be solved by the use of a compass can also be solved by the aid of a transferer of segments and a straight-edge. For the purpose of showing this, let us consider again that geometry which was constructed in § 9 by the help of the domain Ω of algebraic numbers. In this geometry, there exist only such segments as can be constructed by means of a straight-edge and a transferer of segments, namely, the segments determined by the numbers of the domain Ω.

Now, if ω is a number of the domain Ω, we easily see from the definition of Ω that every algebraic number conjugate to ω must also lie in Ω. Since the numbers of the domain Ω are evidently all real, it follows that it can contain only such real algebraic numbers as have their conjugates also real.

Let us now consider the following problem; viz., to construct a right triangle having the hypotenuse 1 and one side $|\sqrt{2}| - 1$. The algebraic number $\sqrt{2|\sqrt{2}| - 2}$, which expresses the numerical value of the other side, does not occur in the domain Ω, since the conjugate number $\sqrt{-2|\sqrt{2}| - 2}$ is imaginary. This problem is, therefore, not capable of solution in the geometry in question and, hence, cannot be constructed by means of a straight-edge and a transferer of segments, although the solution by means of a compass is possible.

§ 38. THE REPRESENTATION OF ALGEBRAIC NUMBERS AND OF INTEGRAL RATIONAL FUNCTIONS AS SUMS OF SQUARES.

The question of the possibility of geometrical constructions by the aid of a straight-edge and a transferer of segments necessitates, for its complete treatment, particular theorems

of an arithmetical and algebraic character, which, it appears to me, are themselves of interest. Since the time of Fermat, it has been known that every positive integral rational number can be represented as the sum of four squares. This theorem of Fermat permits the following remarkable generalization:

DEFINITION. Let k be an arbitrary number field and let m be its degree. We will denote by k', k'', ..., $k^{(m-1)}$ the $m-1$ number fields conjugate to k. If, among the m fields k, $k', k'', \ldots, k^{(m-1)}$ there is one or more formed entirely of real numbers, then we call these fields real. Suppose that the fields k, k', ..., $k^{(s-1)}$ are such. A number α of the field k is called in this case *totally positive in* k, whenever the s numbers conjugate to α, contained respectively in k, k', k'', ..., $k^{(s-1)}$ are all positive. However, if in each of the m fields k, k', k'', ..., $k^{(m-1)}$ there are also imaginary numbers present, we call every number α in k *totally positive.*

We have, then, the following proposition:

> THEOREM 42. Every totally positive number in k may be represented as the sum of four squares, whose bases are integral or fractional numbers of the field k.

The demonstration of this theorem presents serious difficulty. It depends essentially upon the theory of relatively quadratic number fields, which I have recently developed in several papers.[17] We will here call attention only to that proposition in this theory which gives the condition that a ternary diophantine equation of the form

$$\alpha\xi^2 + \beta\eta^2 + \gamma\zeta^2 = 0$$

can be solved when the coefficients α, β, γ are given numbers in k and ξ, η, ζ are the required numbers in k. The demonstration of theorem 42 is accomplished by the repeated application of the proposition just mentioned.

From theorem 42 follow a series of propositions concerning the representation of such rational functions of a variable, with rational coefficients, as never have negative values. I will mention only the following theorem, which will be of service in the following sections.

> THEOREM 43. Let, $f(x)$ be an integral rational function of x whose coefficients are rational numbers and which never becomes negative for any real value of x. Then $f(x)$ can always be represented as the quotient of two sums of squares of which the bases are all integral rational functions of x with rational coefficients.

PROOF. We will denote the degree of the function $f(x)$ by m, which, in any case, must evidently be even. When $m = 0$, that is to say, when $f(x)$ is a rational number, the validity of theorem 43 follows immediately from Fermat's theorem concerning the representation of a positive number as the sum of four squares. We will assume that the proposition is already established for functions of degree 2, 4, 6, ..., $m-2$, and show, in the following manner, its validity for the case of a function of the mth degree.

Let us, first of all, consider briefly the case where $f(x)$ breaks up into the product of two or more integral functions of x with rational coefficients. Suppose $p(x)$ to be one

[17] "Ueber die Theorie der relativquadratischen Zahlkörper," *Jahresbericht der Deutschen Math. Vereinigung*, Vol. 6, 1899, and *Math. Annalen*, Vol. 51. See, also, "Ueber die Theorie der relativ-Abelschen Zahlkörper" *Nachr. der K. Ges. der Wiss. zu Göttingen*, 1898.

of those functions contained in $f(x)$, which itself cannot be further decomposed into a product of integral functions having rational coefficients. It then follows at once from the "definite" character which we have given to the function $f(x)$, that the factor $p(x)$ must either appear in $f(x)$ to an even degree or $p(x)$ must be itself "definite"; that is to say, must be a function which never has negative values for any real values of x. In the first case, the quotient $\frac{f(x)}{\{p(x)\}^2}$ and, in the second case, both $p(x)$ and $\frac{f(x)}{p(x)}$, are "definite," and these functions have an even degree $< m$. Hence, according to our hypothesis, in the first case, $\frac{f(x)}{\{p(x)\}^2}$ and, in the last case, $p(x)$ and $\frac{f(x)}{p(x)}$ may be represented as the quotient of the sum of squares of the character mentioned in theorem 43. Consequently, in both of these cases, the function $f(x)$ admits of the required representation.

Let us now consider the case where $f(x)$ cannot be broken up into the product of two integral functions having rational coefficients. The equation $f(\theta) = 0$ defines, then, a field of algebraic numbers $k(\theta)$ of the mth degree, which, together with all their conjugate fields, are imaginary. Since, according to the definition given just before the statement of theorem 42, each number given in $k(\theta)$, and hence also -1 is totally positive in $k(\theta)$, it follows from theorem 42 that the number -1 can be represented as a sum of the squares of four definite numbers in $k(\theta)$. Let, for example

$$-1 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2, \tag{1}$$

where α, β, γ, δ are integral or fractional numbers in $k(\theta)$. Let us put

$$\begin{array}{rcccl} \alpha & = & a_1\theta^{m-1} + a_2\theta^{m-2} + \cdots + a_m & = & \phi(\theta), \\ \beta & = & b_1\theta^{m-1} + b_2\theta^{m-2} + \cdots + b_m & = & \psi(\theta), \\ \gamma & = & c_1\theta^{m-1} + c_2\theta^{m-2} + \cdots + c_m & = & \chi(\theta), \\ \theta & = & d_1\theta^{m-1} + d_2\theta^{m-2} + \cdots + d_m & = & \rho(\theta); \end{array}$$

where a_1, a_2, …, a_m, …, d_1, d_2, …, d_m are the rational numerical coefficients and $\phi(\theta)$, $\psi(\theta)$, $\chi(\theta)$, $\rho(\theta)$ the integral rational functions in question, having the degree $(m-1)$ in θ.

From (1), we have

$$1 + \{\phi(\theta)\}^2 + \{\psi(\theta)\}^2 + \{\chi(\theta)\}^2 + \{\rho(\theta)\}^2 = 0$$

Because of the irreducibility of the equation $f(x) = 0$, the expression

$$F(x) = 1 + \{\phi(\theta)\}^2 + \{\psi(\theta)\}^2 + \{\chi(\theta)\}^2 + \{\rho(\theta)\}^2$$

represents, necessarily, an integral rational function of x which is divisible by $f(x)$. $F(x)$ is, then, a "definite" function of the degree $(2m-2)$ or lower. Hence, the quotient $\frac{F(x)}{f(x)}$ is a "definite" function of the degree $(m-2)$ or lower in x, having rational coefficients. Consequently, by the hypothesis we have made, $\frac{F(x)}{f(x)}$ can be represented as the quotient of two sums of squares of the kind mentioned in theorem 43 and, since $F(x)$ is itself such a sum of squares, it follows that, $f(x)$ must also be a quotient of two sums of squares of the required kind. The validity of theorem 43 is accordingly established.

It would be perhaps difficult to formulate and to demonstrate the corresponding proposition for integral functions of two or more variables. However, I will here merely remark

that I have demonstrated in an entirely different manner the possibility of representing any "definite" integral rational function of two variables as the quotient of sums of squares of integral functions, upon the hypothesis that the functions represented may have as coefficients not only rational but *a*ny real numbers.[18]

§ 39. CRITERION FOR THE POSSIBILITY OF A GEOMETRICAL CONSTRUCTION BY MEANS OF A STRAIGHT-EDGE AND A TRANSFERER OF SEGMENTS.

Suppose we have given a problem in geometrical construction which can be affected by means of a compass. We shall attempt to find a criterion which will enable us to decide, from the analytical nature of the problem and its solutions, whether or not the construction can be carried out by means of only a straight-edge and a transferer of segments. Our investigation will lead us to the following proposition.

> THEOREM 44. Suppose we have given a problem in geometrical construction, which is of such a character that the analytical treatment of it enables us to determine uniquely the co-ordinates of the desired points from the co-ordinates of the given points by means of the rational operations and the extraction of the square root. Let n be the smallest number of square roots which suffice to calculate the co-ordinates of the points. Then, in order that the required construction shall be possible by the drawing of straight lines and the laying off of segments, it is necessary and sufficient that the given geometrical problem shall have exactly 2^n real solutions for every position of the given points; that is to say, for all values of the arbitrary parameter expressed in terms of the co-ordinates of the given points.

PROOF. We shall demonstrate this proposition merely for the case where the co-ordinates of the given points are rational functions, having rational coefficients, of a single parameter p.

It is at once evident that the proposition gives a necessary condition. In order to show that it is also sufficient, let us assume that it is fulfilled and then, among the n square roots, consider that one which, in the calculation of the co-ordinates of the desired points, is first to be extracted. The expression under this radical is a rational function $f_1(p)$, having rational coefficients, of the parameter p. This rational function cannot have a negative value for any real value of the parameter p; for, otherwise the problem must have imaginary solutions for certain values of p, which is contrary to the given hypothesis. Hence, from theorem 43, we conclude that $f_1(p)$ can be represented as a quotient of the sums of squares of integral rational functions.

Moreover, the formulæ

$$\begin{aligned} \sqrt{a^2+b^2+c^2} &= \sqrt{(\sqrt{a^2+b^2})^2+c^2} \\ \sqrt{a^2+b^2+c^2+d^2} &= \sqrt{(\sqrt{a^2+b^2+c^2})^2+d^2} \end{aligned}$$

[18] See "Ueber ternäre definite Formen," *Acta mathematica*, Vol. 17.

. . . . . .

show that, in general, the extraction of the square root of a sum of any number of squares may always be reduced to the repeated extraction of the square root of the sum of two squares.

If now we combine this conclusion with the preceding results, it follows that the expression $\sqrt{f_1(p)}$ can certainly be constructed by means of a straight-edge and a transferer of segments. Among the n square roots, consider now the second one to be extracted in the process of calculating the co-ordinates of the required points. The expression under this radical is a rational function $f_2(p, \sqrt{f_1})$ of the parameter p and the square root first considered. This function f_2 can never be negative for any real arbitrary value of the parameter p and for either sign of $\sqrt{f_1}$; for, otherwise among the 2^n solutions of our problem, there would exist for certain values of p also imaginary solutions, which is contrary to our hypothesis. It follows, therefore, that f_2 must satisfy a quadratic equation of the form

$$f_2^2 - \phi_2(p) f_2 + \psi_1(p) = 0,$$

where $\phi_1(p)$ and $\psi_1(p)$ are, necessarily, such rational functions of p as have rational coefficients and for real values of p never become negative. From this equation, we have

$$f_2 = \frac{f_2^2 + \psi_1(p)}{\phi_1(p)}.$$

Now, according to theorem 43, the functions $\phi_1(p)$ and $\psi_1(p)$ must again be the quotient of the sums of squares of rational functions, and, on the other hand, the expression f_2 may be, from the above considerations, constructed by means of a straight-edge and a transferer of segments. The expression found for f_2 shows, therefore, that f_2 is a quotient of the sum of squares of functions which may be constructed in the same way. Hence, the expression $\sqrt{f_2}$ can also be constructed by means of a straight-edge and a transferer of segments.

Just as with the expression f_2, any other rational function $\phi_2(p, \sqrt{f_1})$ of p and $\sqrt{f_1}$ may be shown to be the quotient of two sums of squares of functions which may be constructed, provided this rational function ϕ_2 possesses the property that for real values of the parameter p and for either sign of $\sqrt{f_1}$, it never becomes negative.

This remark permits us to extend the above method of reasoning in the following manner.

Let $f_3(p, \sqrt{f_1}, \sqrt{f_2})$ be such an expression as depends in a rational manner upon the three arguments $p, \sqrt{f_1}, \sqrt{f_2}$ and of which, in the analytical calculation of the co-ordinates of the desired points, the square root is the third to be extracted. As before, it follows that f_3 can never have negative values for real values of p and for either sign of $\sqrt{f_1}$ and $\sqrt{f_2}$. This condition of affairs shows again that f_3 must satisfy a quadratic equation of the form

$$f_3^2 - \phi_2(p, \sqrt{f_1}) f_3 - \psi_2(p, \sqrt{f_1}) = 0,$$

where ϕ_2 and ψ_2 are such rational functions of p and $\sqrt{f_1}$ as never become negative for any real value of p and either sign of $\sqrt{f_1}$. But, according to the preceding remark, the

functions ϕ_2 and ψ_2 are the quotients of two sums of squares of functions which may be constructed and, hence, it follows that the expression

$$f_3 = \frac{f_3^2 + \psi_2(p, \sqrt{f_1})}{\phi_2(p, \sqrt{f_1})}$$

is likewise possible of construction by aid of a straight-edge and a transferer of segments.

The continuation of this method of reasoning leads to the demonstration of theorem 44 for the case of a single parameter p.

The truth of theorem 44 for the general case depends upon whether or not theorem 43 can be generalized in a similar manner to cover the case of two or more variables.

As an example of the application of theorem 44, we may consider the regular polygons which may be constructed by means of a compass. In this case, the arbitrary parameter, p does not occur, and the expressions to be constructed all represent algebraic numbers. We easily see that the criterion of theorem 44 is fulfilled, and, consequently, it follows that the above-mentioned regular polygons can be constructed by the drawing of straight lines and the laying off of segments. We might deduce this result also directly from the theory of the division of the circle (*Kreisteilung*).

Concerning the other known problems of construction in the elementary geometry, we will here only mention that the problem of Malfatti may be constructed by means of a straight-edge and a transferer of segments. This is, however, not the case with the contact problems of Appolonius.

CONCLUSION.

The preceding work treats essentially of the problems of the euclidean geometry only; that is to say, it is a discussion of the questions which present themselves when we admit the validity of the axiom of parallels. It is none the less important to discuss the principles and the fundamental theorems when we disregard the axiom of parallels. We have thus excluded from our study the important question as to whether it is possible to construct a geometry in a logical manner, without introducing the notion of the plane and the straight line, by means of only points as elements, making use of the idea of groups of transformations, or employing the idea of distance. This last question has recently been the subject of considerable study, due to the fundamental and prolific works of Sophus Lie. However, for the complete elucidation of this question, it would be well to divide into several parts the axiom of Lie, that space is a numerical multiplicity. First of all, it would seem to me desirable to discuss thoroughly the hypothesis of Lie, that functions which produce transformations are not only continuous, but may also be differentiated. As to myself, it does not seem to me probable that the geometrical axioms included in the condition for the possibility of differentiation are all necessary.

In the treatment of all questions of this character, I believe the methods and the principles employed in the preceding work will be of value. As an example, let me call attention to an investigation undertaken at my suggestion by Mr. Dehn, and which has already appeared.[19] In this article, he has discussed the known theorems of Legendre concerning the sum of the angles of a triangle, in the demonstration of which that geometer made use of the idea of continuity.

The investigation of Mr. Dehn rests upon the axioms of connection, of order, and of congruence; that is to say, upon the axioms of groups I, II, IV. However, the axiom of parallels and the axiom of Archimedes are excluded. Moreover, the axioms of order are stated in a more general manner than in the present work, and in substance as follows: Among four points A, B, C, D of a straight line, there are always two, for example A, C, which are separated from the other two and conversely. Five points A, B, C, D, E upon a straight line may always be so arranged that A, C shall be separated from B, E and from B, D. Consequently, A, D are always separated from B, E and from C, E, etc. The (elliptic) geometry of Riemann, which we have not considered in the present work, is in this way not necessarily excluded.

Upon the basis of the axioms of connection, order, and congruence, that is to say, the axioms I, II, IV, we may introduce, in the well known manner, the elements called ideal,–ideal points, ideal straight lines, and ideal planes. Having done this, Mr. Dehn demonstrates the following theorem.

> If, with the exception of the straight line t and the points lying upon it, we regard all of the straight lines and all of the points (ideal or real) of a plane as the elements of a new geometry, we may then define a new kind of congruence so that all of the

[19] *Math. Annalen*, Vol. 53 (1900).

> axioms of connection, order, and congruence, as well as the axiom of Euclid, shall be fulfilled. In this new geometry, the straight line t takes the place of the straight line at infinity.

This euclidean geometry, superimposed upon the non-euclidean plane, may be called a *pseudo-geometry* and the new kind of congruence a *pseudo-congruence.*

By aid of the preceding theorem, we may now introduce an algebra of segments relating to the plane and depending upon the developments made in §15, pp. 29–31. This algebra of segments permits the demonstration of the following important theorem:

> If, in any triangle whatever, the sum of the angles is greater than, equal to, or less than, two right angles, then the same is true for all triangles.

The case where the sum of the angles is equal to two right angles gives the well known theorem of Legendre. However, in his demonstration, Legendre makes use of continuity.

Mr. Dehn then discusses the connection between the three different hypotheses relative to the sum of the angles and the three hypotheses relative to parallels.

He arrives in this manner at the following remarkable propositions.

> Upon the hypothesis that through a given point we may draw an infinity of lines parallel to a given straight line, it does not follow, when we exclude the axiom of Archimedes, that the sum of the angles of a triangle is less than two right angles, but on the contrary, this sum may be
>
> (a) greater than two right angles, or
>
> (b) equal to two right angles.

In order to demonstrate part (a) of this theorem, Mr. Dehn constructs a geometry where we may draw through a point an infinity of lines parallel to a given straight line and where, moreover, all of the theorems of Riemann's (elliptic) geometry are valid. This geometry may be called non-legendrian, for it is in contradiction with that theorem of Legendre by virtue of which the sum of the angles a triangle is never greater than two right angles. From the existence of this non-legendrian geometry, it follows at once that it is impossible to demonstrate the theorem of Legendre just mentioned without employing the axiom of Archimedes, and in fact, Legendre made use of continuity in his demonstration of this theorem.

For the demonstration of case (b), Mr. Dehn constructs a geometry where the axiom of parallels does not hold, but where, nevertheless, all of the theorems of the euclidean geometry are valid. Then, we have the sum of the angles of a triangle equal to two right angles. There exist also similar triangles, and the extremities of the perpendiculars having the same length and their bases upon a straight line all lie upon the same straight line, etc. The existence of this geometry shows that, if we disregard the axiom of Archimedes, the axiom of parallels cannot be replaced by any of the propositions which we usually regard as equivalent to it.

This new geometry may be called a *semi-euclidean geometry.* As in the case of the non-legendrian geometry, it is clear that the semi-euclidean geometry is at the same time a non-archimedean geometry.

Mr. Dehn finally arrives at the following surprising theorem:

> Upon the hypothesis that there exists no parallel, it follows that the sum of the angles of a triangle is greater than two right angles.

This theorem shows that, with respect to the axiom of Archimedes, the two non-euclidean hypotheses concerning parallels act very differently.

We may combine the preceding results in the following table.

THE SUM OF THE ANGLES OF A TRIANGLE IS	THOUGH A GIVEN POINT, WE MAY DRAW		
	NO PARALLELS TO A STRAIGHT LINE	ONE PARALLEL TO A STRAIGHT LINE	AN INFINITY OF PARALLELS TO A STRAIGHT LINE
$>$ 2 right angles	Riemann's (elliptic) geometry	This case is impossible	Non-legendrian geometry
$<$ 2 right angles	This case is impossible	Euclidean (parabolic) geometry	Semi-euclidean geometry
$=$ 2 right angles	This case is impossible	This case is impossible	Geometry of Lobatschewski (hyperbolic)

However, as I have already remarked, the present work is rather a critical investigation of the principles of the euclidean geometry. In this investigation, we have taken as a guide the following fundamental principle; viz., to make the discussion of each question of such a character as to examine at the same time whether or not it is possible to answer this question by following out a previously determined method and by employing certain limited means. This fundamental rule seems to me to contain a general law and to conform to the nature of things. In fact, whenever in our mathematical investigations we encounter a problem or suspect the existence of a theorem, our reason is satisfied only when we possess a complete solution of the problem or a rigorous demonstration of the theorem, or, indeed, when we see clearly the reason of the impossibility of the success and, consequently, the necessity of failure.

Thus, in the modern mathematics, the question of the impossibility of solution of certain problems plays an important role, and the attempts made to answer such questions have often been the occasion of discovering new and fruitful fields for research. We recall in this connection the demonstration by Abel of the impossibility of solving an equation of the fifth degree by means of radicals, as also the discovery of the impossibility of demonstrating the axiom of parallels, and, finally, the theorems of Hermite and Lindeman concerning the impossibility of constructing by algebraic means the numbers e and π.

This fundamental principle, which we ought to bear in mind when we come to discuss the principles underlying the impossibility of demonstrations, is intimately connected with the condition for the "purity" of methods in demonstration, which in recent times has been considered of the highest importance by many mathematicians. The foundation of this condition is nothing else than a subjective conception of the fundamental principle given above. In fact, the preceding geometrical study attempts, in general, to explain what are the axioms, hypotheses, or means, necessary to the demonstration of a truth of elementary geometry, and it only remains now for us to judge from the point of view in which we place ourselves as to what are the methods of demonstration which we should prefer.

APPENDIX.[20]

The investigations by Riemann and Helmholtz of the foundations of geometry led Lie to take up the problem of the *axiomatic* treatment of geometry as introductory to the study of groups. This profound mathematician introduced a system of axioms which he showed by means of his theory of transformation groups to be sufficient for the complete development of geometry.[21]

As the basis of his transformation groups, Lie made the assumption that the functions defining the group can be differentiated. Hence in Lie's development, the question remains uninvestigated as to whether this assumption as to the differentiability of the functions in question is really unavoidable in developing the subject according to the axioms of geometry, or whether, on the other hand, it is not a consequence of the group-conception and of the remaining axioms of geometry. In consequence of his method of development, Lie has also necessitated the express statement of the axiom that the group of displacements is produced by infinitesimal transformations. These requirements, as well as essential parts of Lie's fundamental axioms concerning the nature of the equation defining points of equal distance, can be expressed geometrically in only a very unnatural and complicated manner. Moreover, they appear only through the analytical method used by Lie and not as a necessity of the problem itself.

In what follows, I have therefore attempted to set up for plane geometry a system of axioms, depending likewise upon the conception of a group,[22] which contains only those requirements which are simple and easily seen geometrically. In particular they do not require the differentiability of the functions defining displacement. The axioms of the system which I set up are a special division of Lie's, or, as I believe, are at once deducible from his.

My method of proof is entirely different from Lie's method. I make use particularly of Cantor's theory of assemblages of points and of the theorem of C. Jordan, according to which every closed continuous plane curve free from double points divides the plane into an inner and an outer region.

To be sure, in the system set up by me, particular parts are unnecessary. However, I have turned aside from the further investigation of these conditions to the simple statement of the axioms, and above all because I wish to avoid a comparatively too complicated proof, and one which is not at once geometrically evident.

In what follows I shall consider only the axioms relating to the plane, although I suppose that an analogous system of axioms for space can be set up which will make possible the construction of the geometry of space in a similar manner.

We establish the following convention, namely: We will understand by *number-plane* the ordinary plane having a rectangular system of co-ordinates x, y.

[20]The following is a summary of a paper by Professor Hilbert which is soon to appear in full in the *Math. Annalen.—Tr.*

[21]See Lie-Engel, *Theorie der Transformationsgruppen*, Vol. 3, Chapter 5.

[22]By the following investigation is answered also, as I believe, a general question concerning the theory of groups, which I proposed in my address on "MathematischeProbleme," *Göttinger Nachrichten*, 1900, p. 17.

A continuous curve lying in this number-plane and being free from double points and including its end points is called a *Jordan curve*. If the Jordan curve is closed, the interior of the region of the number-plane bounded by it is called a *Jordan region*.

For the sake of easier representation and comprehension, I shall in the following investigation formulate the definition of the plane in a more restricted sense than my method of proof requires,[23] namely: I shall assume that it is possible to map[24] in a reversible, single-valued manner all of the points of our geometry at the same time upon the points lying in the finite region of the number-plane, or upon a definite partial system of the same. Hence, each point of our geometry is characterized by a definite pair of numbers x, y. We formulate this statement of the idea of the plane as follows:

Definitions of the Plane. The plane is a system of points which can be mapped in a reversible, single-valued manner upon the points lying in the finite region of the number-plane, or upon a certain partial system of the same. To each point A of our geometry, there exists a Jordan curve in whose interior the map of A lies and all of whose points likewise represent points of our geometry. This Jordan region is called the domain of the point A. Each Jordan region contained in a Jordan region which includes the point A is likewise called a domain of A. If B is any point in a domain of A, then this domain is at the same time called also a domain of B.

If A and B are any two points of our geometry, then there always exists a domain which contains at the same time both of the points A and B.

We will define a *displacement* as a reversible, single-valued transformation of a plane into itself. Evidently we may distinguish two kinds of reversible, single-valued, continuous transformations of the number-plane into itself. If we take any closed Jordan curve in the number-plane and think of its being traversed in a definite sense, then by such a transformation this curve goes over into another closed Jordan curve which is also traversed in a certain sense. We shall assume in the present investigation that it is traversed in the same sense as the original Jordan curve, when we apply a transformation of the number-plane into itself, which defines a displacement. This assumption[25] necessitates the following statement of the conception of a displacement.

Definition of Displacement. A displacement is a reversible, single-valued, continuous transformation of the maps of the given points upon the number-plane into themselves in such a manner that a closed Jordan curve is traversed in the same sense after the transformation as before. A displacement by which the point M remains unchanged is called a *rotation*[26] about the point M.

In accordance with the conventions setting forth the notions "plane" and "displacement," we set up the three following axioms:

[23] Concerning the broader statement of the conception of the plane see my note, "Ueber die Grundlagen der Geometrie," *Göttinger Nachrichten*, 1901.

[24] *Abbilden.*

[25] Lie makes this assumption to contain the condition that the group of displacements be generated by infinitesimal transformations. The opposite assumption would assist essentially the demonstration in so far as the "true straight line" could then be defined as the locus of those points which remain unchanged by a displacement changing the sense in which the curve is traversed (*Umklappung*).

[26] The term "rotation" is used here in the sense of a rotatory displacement; that is to say, only the initial and final stages and not the aggregate of the intermediate stages of the transition enter into consideration.—*Tr.*

Axiom I. *If two displacements are followed out one after the other, then the resulting map of the plane upon itself is again a displacement.*

We say briefly:

Axiom I. The Displacements Form a Group.

Axiom II. *If A and M are two arbitrary points distinct from each other, then by a rotation about M we can always bring A into an infinite number of different positions.*

If in our geometry we define a true circle as the totality of those points which arise by rotating about M a point different from M, then we can express the statement made in axiom II as follows:

Axiom II. Every True Circle Consists of an Infinite Number of Points.

As preliminary to axiom III, we make the following explanations:

Let A be a definite point in our geometry and A_1, A_2, A_3, ... any infinite system of points. With the same letters we will also denote the maps of these points upon the number-plane. About the point A in the number-plane take an arbitrarily small domain α. If then any of the map-points A_i fall within the domain α, we say that there are points A_i arbitrarily near the point A.

Let A, B be a definite pair of points in our geometry, and let A_1B_1, A_2B_2, A_3B_3, ... be any infinite system of pairs of points. With the same letters we will denote the maps of these pairs of points upon the number-plane. Select about each of the points A and B in the number-plane an arbitrarily small domain α and β, respectively. If then there are pairs of points A_iB_i such that A_i falls within the domain α and at the same time B_i falls within the domain β, we say that there are segments A_iB_i lying arbitrarily near the segment AB.

Let ABC a definite triad of points in our geometry, and let $A_1B_1C_1$, $A_2B_2C_2$, $A_3B_3C_3$, ... be any infinite system of triads of points. With the same letters we will also denote the maps of these triads of points upon the number-plane. About each of the points A, B, C in the number-plane take an arbitrarily small domain α, β, γ, respectively. If then there are triads of points $A_iB_iC_i$ such that A_i falls in the domain α, and likewise B_i in the domain β and C_i in the domain γ, then we say that there are triangles $A_iB_iC_i$ lying arbitrarily near to the triangle ABC.

Axiom III. *If there are displacements of such a kind that triangles arbitrarily near the triangle ABC can be brought arbitrarily near to the triangle $A'B'C'$, then there always exists a displacement by which the triangle ABC goes over exactly into the triangle $A'B'C'$.*[27]

The content of this axiom can be briefly expressed as follows:

Axiom III. The Displacements Form a Closed System.

We call special attention to the following particular cases of axiom III.

If there are rotations about a point M of the kind that segments lying arbitrarily near the segment AB can be brought arbitrarily near the segment $A'B'$, then there is always such a rotation about M possible by which the segment AB goes over exactly into the segment $A'B'$.

[27]It is sufficient to assume that axiom III holds for sufficiently small domains as Lie has done. My method of proof may be so changed as to make use of only this narrower assumption.

If there are displacements of the kind that segments arbitrarily near the segment AB can be brought arbitrarily near to the segment $A'B'$, then there is always a displacement possible by which the segment AB goes over exactly into the segment $A'B'$.

If there are rotations about the point M of the kind that points arbitrarily near the point A can be brought arbitrarily near the point A', then there is always such a rotation about M possible by which A goes over exactly into the point A'.

I now prove the following proposition:

> A geometry in which axioms I–III are fulfilled is either the euclidean or the bolyai-lobatchefskian geometry.

If we wish to obtain only the euclidean geometry, it is necessary merely to make in connection with axiom I the additional statement that the groups of displacements shall possess an invariant sub-group. This additional statement takes the place of the axioms of parallels.

In what follows, I will briefly outline the general idea of my method of proof.[28] Within the domain of a certain point M construct in a particular manner a certain point-configuration kk, and upon this configuration construct a certain point K. We then base our investigation upon the true circle k about M and passing through K. It may be easily shown that the true circle k is an assemblage of points which is closed and in itself dense. It constitutes, therefore, a perfect assemblage of points.

The next objective point in our demonstration is to show that the true circle k is a closed Jordan curve. We do this in that we first show the possibility of a cyclical arrangement of the points of the true circle k, from which it follows that we may map in a reversible, single-valued manner the points of k upon the points of an ordinary circle. Finally, we show that this map must necessarily be a continuous one. Furthermore, it follows also that the originally constructed point-configuration kk is identical with the true circle k. Moreover, the law holds that each true circle inside of k is likewise a closed Jordan curve.

We turn now to the investigation of the group of all the displacements which by the rotation of the plane about M transforms a definite true circle k into itself. This group possesses the following properties: (1) Every displacement which leaves one point of k undisturbed, leaves all points of k undisturbed. (2) There always exists a displacement which changes any given point of k into any other given point of k. (3) The group of displacements is a continuous one. These three properties determine completely the construction of the group of transformations of all the displacements of the true circle into itself. We set up the following proposition: The group of all the displacements of the true circle into itself, which are rotations about M, is holoedric, isomorphic with the group of ordinary rotations of the ordinary circle into itself.

Moreover, we investigate the group of displacements of all the points of our plane by a rotation about M. The law holds that, aside from the identity, there is no rotation of the plane about M which leaves every point of the true circle undisturbed. We now see that every true circle is a Jordan curve and deduce formulæ for the transformation of that

[28] The complete proof will appear later in the *Math. Annalen.*

group of all the rotations. Finally, the proposition easily follows that: If any two points remain fixed by a displacement of the plane, then all points remain fixed; that is to say, the displacement is the identity. Each point of the plane may be indeed made to go over into any other point of the plane by means of a displacement.

Our further important objective point is to define the idea of the true straight line in our geometry and deduce those properties of it which are necessary in the further development of geometry. First of all, the notions "semi-rotation" and "middle of a segment" are defined. A segment has at most one middle, and, when we know the middle of one segment, then every smaller segment possesses a middle.

In order to pass judgment as to the position of the middle of a segment, we need particular propositions concerning true circles which are mutually tangent, and indeed the question depends upon the construction of two congruent circles tangent to each other externally in one and only one point. We derive also a more general proposition concerning circles which are tangent to each other internally and consequently a theorem covering the special case where the circle which is tangent internally to a second passes through the centre of that circle.

Moreover, a sufficiently small definite segment is taken as a unit segment, and from this by continued bisection and semi-rotation a system of points is constructed of the kind that to each point of this system a definite number a corresponds, which is rational and has as denominator some power of 2. By setting up a law concerning this correspondence, the points of the above system are so arranged that the above laws concerning mutually tangent circles are valid. It is now shown that the points corresponding to the numbers $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ converge toward the point 0. This result is generalized step by step until it is finally shown that every series of points of our system converges, so soon as the corresponding series of numbers converges.

From what has been said, the definition of the true straight line follows as a system of points which arise from two fundamental points, if we apply repeatedly a semi-rotation, take the middle point, and add to the assemblage the points of condensation of the system of points which arises. We can then prove that the true straight line is a continuous curve, possessing no double points and having with any other true straight line at most one point in common. Furthermore, it can be shown that the true straight line cuts each circle drawn about one of its points, and from this it follows that any two arbitrary points of the plane can always be joined by a true straight line. We see also that in our geometry the laws of congruence hold, by which however two triangles are proven to be congruent if they are traversed in the same sense.

With regard to the position of the systems of all the true straight lines with respect to one another, there are two cases to distinguish, according as the axiom of parallels holds, or through each point there exists two straight lines which separate the straight lines which cut the given straight line from those which do not cut it. In the first case we have the euclidean and in the second the bolyai-lobatschefskian geometry.

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THE THEORY OF HEAT RADIATION

PLANCK AND MASIUS

THE THEORY
OF
HEAT RADIATION

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WITH 7 ILLUSTRATIONS

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TRANSLATOR'S PREFACE

The present volume is a translation of the second edition of Professor *Planck's* WAERMESTRAHLUNG (1913). The profoundly original ideas introduced by *Planck* in the endeavor to reconcile the electromagnetic theory of radiation with experimental facts have proven to be of the greatest importance in many parts of physics. Probably no single book since the appearance of *Clerk Maxwell's* ELECTRICITY AND MAGNETISM has had a deeper influence on the development of physical theories. The great majority of English-speaking physicists are, of course, able to read the work in the language in which it was written, but I believe that many will welcome the opportunity offered by a translation to study the ideas set forth by *Planck* without the difficulties that frequently arise in attempting to follow a new and somewhat difficult line of reasoning in a foreign language.

Recent developments of physical theories have placed the quantum of action in the foreground of interest. Questions regarding the bearing of the quantum theory on the law of equipartition of energy, its application to the theory of specific heats and to photoelectric effects, attempts to form some concrete idea of the physical significance of the quantum, that is, to devise a "model" for it, have created within the last few years a large and ever increasing literature. Professor *Planck* has, however, in this book confined himself exclusively to radiation phenomena and it has seemed to me probable that a brief résumé of this literature might prove useful to the reader who wishes to pursue the subject further. I have, therefore, with Professor *Planck's* permission, given in an appendix a list of the most important papers on the subjects treated of in this book and others closely related to them. I have also added a short note on one or two derivations of formulæ where the treatment in the book seemed too brief or to present some difficulties.

In preparing the translation I have been under obligation for advice and helpful suggestions to several friends and colleagues and especially to Professor A. W. Duff who has read the manuscript and the galley proof.

Morton Masius.

Worcester, Mass.,
February, 1914.

PREFACE TO SECOND EDITION

Recent advances in physical research have, on the whole, been favorable to the special theory outlined in this book, in particular to the hypothesis of an elementary quantity of action. My radiation formula especially has so far stood all tests satisfactorily, including even the refined systematic measurements which have been carried out in the Physikalisch-technische Reichsanstalt at Charlottenburg during the last year. Probably the most direct support for the fundamental idea of the hypothesis of quanta is supplied by the values of the elementary quanta of matter and electricity derived from it. When, twelve years ago, I made my first calculation of the value of the elementary electric charge and found it to be $4.69 \cdot 10^{-10}$ electrostatic units, the value of this quantity deduced by *J. J. Thomson* from his ingenious experiments on the condensation of water vapor on gas ions, namely $6.5 \cdot 10^{-10}$ was quite generally regarded as the most reliable value. This value exceeds the one given by me by 38 per cent. Meanwhile the experimental methods, improved in an admirable way by the labors of *E. Rutherford*, *E. Regener*, *J. Perrin*, *R. A. Millikan*, *The Svedberg* and others, have without exception decided in favor of the value deduced from the theory of radiation which lies between the values of *Perrin* and *Millikan*.

To the two mutually independent confirmations mentioned, there has been added, as a further strong support of the hypothesis of quanta, the heat theorem which has been in the meantime announced by *W. Nernst*, and which seems to point unmistakably to the fact that, not only the processes of radiation, but also the molecular processes take place in accordance with certain elementary quanta of a definite finite magnitude. For the hypothesis of quanta as well as the heat theorem of *Nernst* may be reduced to the simple proposition that the

thermodynamic probability (Sec. 120) of a physical state is a definite integral number, or, what amounts to the same thing, that the entropy of a state has a quite definite, positive value, which, as a minimum, becomes zero, while in contrast therewith the entropy may, according to the classical thermodynamics, decrease without limit to minus infinity. For the present, I would consider this proposition as the very quintessence of the hypothesis of quanta.

In spite of the satisfactory agreement of the results mentioned with one another as well as with experiment, the ideas from which they originated have met with wide interest but, so far as I am able to judge, with little general acceptance, the reason probably being that the hypothesis of quanta has not as yet been satisfactorily completed. While many physicists, through conservatism, reject the ideas developed by me, or, at any rate, maintain an expectant attitude, a few authors have attacked them for the opposite reason, namely, as being inadequate, and have felt compelled to supplement them by assumptions of a still more radical nature, for example, by the assumption that any radiant energy whatever, even though it travel freely in a vacuum, consists of indivisible quanta or cells. Since nothing probably is a greater drawback to the successful development of a new hypothesis than overstepping its boundaries, I have always stood for making as close a connection between the hypothesis of quanta and the classical dynamics as possible, and for not stepping outside of the boundaries of the latter until the experimental facts leave no other course open. I have attempted to keep to this standpoint in the revision of this treatise necessary for a new edition.

The main fault of the original treatment was that it began with the classical electrodynamical laws of emission and absorption, whereas later on it became evident that, in order to meet the demand of experimental measurements, the assumption of finite energy elements must

be introduced, an assumption which is in direct contradiction to the fundamental ideas of classical electrodynamics. It is true that this inconsistency is greatly reduced by the fact that, in reality, only mean values of energy are taken from classical electrodynamics, while, for the statistical calculation, the real values are used; nevertheless the treatment must, on the whole, have left the reader with the unsatisfactory feeling that it was not clearly to be seen, which of the assumptions made in the beginning could, and which could not, be finally retained.

In contrast thereto I have now attempted to treat the subject from the very outset in such a way that none of the laws stated need, later on, be restricted or modified. This presents the advantage that the theory, so far as it is treated here, shows no contradiction in itself, though certainly I do not mean that it does not seem to call for improvements in many respects, as regards both its internal structure and its external form. To treat of the numerous applications, many of them very important, which the hypothesis of quanta has already found in other parts of physics, I have not regarded as part of my task, still less to discuss all differing opinions.

Thus, while the new edition of this book may not claim to bring the theory of heat radiation to a conclusion that is satisfactory in all respects, this deficiency will not be of decisive importance in judging the theory. For any one who would make his attitude concerning the hypothesis of quanta depend on whether the significance of the quantum of action for the elementary physical processes is made clear in every respect or may be demonstrated by some simple dynamical model, misunderstands, I believe, the character and the meaning of the hypothesis of quanta. It is impossible to express a really new principle in terms of a model following old laws. And, as regards the final formulation of the hypothesis, we should not forget that, from the classical point of view, the physics of the atom really has always remained a very obscure, inac-

cessible region, into which the introduction of the elementary quantum of action promises to throw some light.

Hence it follows from the nature of the case that it will require painstaking experimental and theoretical work for many years to come to make gradual advances in the new field. Any one who, at present, devotes his efforts to the hypothesis of quanta, must, for the time being, be content with the knowledge that the fruits of the labor spent will probably be gathered by a future generation.

THE AUTHOR.

BERLIN,
November, 1912.

PREFACE TO FIRST EDITION

In this book the main contents of the lectures which I gave at the University of Berlin during the winter semester 1906–07 are presented. My original intention was merely to put together in a connected account the results of my own investigations, begun ten years ago, on the theory of heat radiation; it soon became evident, however, that it was desirable to include also the foundation of this theory in the treatment, starting with Kirchhoff's Law on emitting and absorbing power; and so I attempted to write a treatise which should also be capable of serving as an introduction to the study of the entire theory of radiant heat on a consistent thermodynamic basis. Accordingly the treatment starts from the simple known experimental laws of optics and advances, by gradual extension and by the addition of the results of electrodynamics and thermodynamics, to the problems of the spectral distribution of energy and of irreversibility. In doing this I have deviated frequently from the customary methods of treatment, wherever the matter presented or considerations regarding the form of presentation seemed to call for it, especially in deriving Kirchhoff's laws, in calculating Maxwell's radiation pressure, in deriving Wien's displacement law, and in generalizing it for radiations of any spectral distribution of energy whatever.

I have at the proper place introduced the results of my own investigations into the treatment. A list of these has been added at the end of the book to facilitate comparison and examination as regards special details.

I wish, however, to emphasize here what has been stated more fully in the last paragraph of this book, namely, that the theory thus developed does not by any means claim to be perfect or complete, although I believe that it points out a possible way of accounting for the processes of radiant energy from the same point of view as for the processes of molecular motion.

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PART I

FUNDAMENTAL FACTS AND DEFINITIONS

RADIATION OF HEAT

CHAPTER I

GENERAL INTRODUCTION

1. Heat may be propagated in a stationary medium in two entirely different ways, namely, by conduction and by radiation. Conduction of heat depends on the temperature of the medium in which it takes place, or more strictly speaking, on the non-uniform distribution of the temperature in space, as measured by the temperature gradient. In a region where the temperature of the medium is the same at all points there is no trace of heat conduction.

Radiation of heat, however, is in itself entirely independent of the temperature of the medium through which it passes. It is possible, for example, to concentrate the solar rays at a focus by passing them through a converging lens of ice, the latter remaining at a constant temperature of 0°, and so to ignite an inflammable body. Generally speaking, radiation is a far more complicated phenomenon than conduction of heat. The reason for this is that the state of the radiation at a given instant and at a given point of the medium cannot be represented, as can the flow of heat by conduction, by a single vector (that is, a single directed quantity). All heat rays which at a given instant pass through the same point of the medium are perfectly independent of one another, and in order to specify completely the state of the radiation the intensity of radiation must be known in all the directions, infinite in number, which pass through the point in question; for this

purpose two opposite directions must be considered as distinct, because the radiation in one of them is quite independent of the radiation in the other.

2. Putting aside for the present any special theory of heat radiation, we shall state for our further use a law supported by a large number of experimental facts. This law is that, so far as their physical properties are concerned, heat rays are identical with light rays of the same wave length. The term "heat radiation," then, will be applied to all physical phenomena of the same nature as light rays. Every light ray is simultaneously a heat ray. We shall also, for the sake of brevity, occasionally speak of the "color" of a heat ray in order to denote its wave length or period. As a further consequence of this law we shall apply to the radiation of heat all the well-known laws of experimental optics, especially those of reflection and refraction, as well as those relating to the propagation of light. Only the phenomena of diffraction, so far at least as they take place in space of considerable dimensions, we shall exclude on account of their rather complicated nature. We are therefore obliged to introduce right at the start a certain restriction with respect to the size of the parts of space to be considered. Throughout the following discussion it will be assumed that the linear dimensions of all parts of space considered, as well as the radii of curvature of all surfaces under consideration, are large compared with the wave lengths of the rays considered. With this assumption we may, without appreciable error, entirely neglect the influence of diffraction caused by the bounding surfaces, and everywhere apply the ordinary laws of reflection and refraction of light. To sum up: We distinguish once for all between two kinds of lengths of entirely different orders of magnitude—dimensions of bodies and wave lengths. Moreover, even the differentials of the former, *i.e.*, elements of length, area and volume, will be regarded as large compared with the corresponding powers of

wave lengths. The greater, therefore, the wave length of the rays we wish to consider, the larger must be the parts of space considered. But, inasmuch as there is no other restriction on our choice of size of the parts of space to be considered, this assumption will not give rise to any particular difficulty.

3. Even more essential for the whole theory of heat radiation than the distinction between large and small lengths, is the distinction between long and short intervals of time. For the definition of intensity of a heat ray, as being the energy transmitted by the ray per unit time, implies the assumption that the unit of time chosen is large compared with the period of vibration corresponding to the color of the ray. If this were not so, obviously the value of the intensity of the radiation would, in general, depend upon the particular phase of vibration at which the measurement of the energy of the ray was begun, and the intensity of a ray of constant period and amplitude would not be independent of the initial phase, unless by chance the unit of time were an integral multiple of the period. To avoid this difficulty, we are obliged to postulate quite generally that the unit of time, or rather that element of time used in defining the intensity, even if it appear in the form of a differential, must be large compared with the period of all colors contained in the ray in question.

The last statement leads to an important conclusion as to radiation of variable intensity. If, using an acoustic analogy, we speak of "beats" in the case of intensities undergoing periodic changes, the "unit" of time required for a definition of the instantaneous intensity of radiation must necessarily be small compared with the period of the beats. Now, since from the previous statement our unit must be large compared with a period of vibration, it follows that the period of the beats must be large compared with that of a vibration. Without this restriction it would be impossible to distinguish properly between "beats" and simple "vibra-

tions." Similarly, in the general case of an arbitrarily variable intensity of radiation, the vibrations must take place very rapidly as compared with the relatively slower changes in intensity. These statements imply, of course, a certain far-reaching restriction as to the generality of the radiation phenomena to be considered.

It might be added that a very similar and equally essential restriction is made in the kinetic theory of gases by dividing the motions of a chemically simple gas into two classes: visible, coarse, or molar, and invisible, fine, or molecular. For, since the velocity of a single molecule is a perfectly unambiguous quantity, this distinction cannot be drawn unless the assumption be made that the velocity-components of the molecules contained in sufficiently small volumes have certain mean values, independent of the size of the volumes. This in general need not by any means be the case. If such a mean value, including the value zero, does not exist, the distinction between motion of the gas as a whole and random undirected heat motion cannot be made.

Turning now to the investigation of the laws in accordance with which the phenomena of radiation take place in a medium supposed to be at rest, the problem may be approached in two ways: We must either select a certain point in space and investigate the different rays passing through this one point as time goes on, or we must select one distinct ray and inquire into its history, that is, into the way in which it was created, propagated, and finally destroyed. For the following discussion, it will be advisable to start with the second method of treatment and to consider first the three processes just mentioned.

4. Emission.—The creation of a heat ray is generally denoted by the word emission. According to the principle of the conservation of energy, emission always takes place at the expense of other forms of

energy (heat,[1] chemical or electric energy, etc.) and hence it follows that only material particles, not geometrical volumes or surfaces, can emit heat rays. It is true that for the sake of brevity we frequently speak of the surface of a body as radiating heat to the surroundings, but this form of expression does not imply that the surface actually emits heat rays. Strictly speaking, the surface of a body never emits rays, but rather it allows part of the rays coming from the interior to pass through. The other part is reflected inward and according as the fraction transmitted is larger or smaller the surface seems to emit more or less intense radiations.

We shall now consider the interior of an emitting substance assumed to be physically homogeneous, and in it we shall select any volume-element $d\tau$ of not too small size. Then the energy which is emitted by radiation in unit time by all particles in this volume-element will be proportional to $d\tau$. Should we attempt a closer analysis of the process of emission and resolve it into its elements, we should undoubtedly meet very complicated conditions, for then it would be necessary to consider elements of space of such small size that it would no longer be admissible to think of the substance as homogeneous, and we would have to allow for the atomic constitution. Hence the finite quantity obtained by dividing the radiation emitted by a volume-element $d\tau$ by this element $d\tau$ is to be considered only as a certain mean value. Nevertheless, we shall as a rule be able to treat the phenomenon of emission as if all points of the volume-element $d\tau$ took part in the emission in a uniform manner, thereby greatly simplifying our calculation. Every point of $d\tau$ will then be the vertex of a pencil of rays diverging in all directions. Such a pencil coming from one single point of course does not represent

[1]Here as in the following the German "Körperwärme" will be rendered simply as "heat." (Tr.)

a finite amount of energy, because a finite amount is emitted only by a finite though possibly small volume, not by a single point.

We shall next assume our substance to be isotropic. Hence the radiation of the volume-element $d\tau$ is emitted uniformly in all directions of space. Draw a cone in an arbitrary direction, having any point of the radiating element as vertex, and describe around the vertex as center a sphere of unit radius. This sphere intersects the cone in what is known as the solid angle of the cone, and from the isotropy of the medium it follows that the radiation in any such conical element will be proportional to its solid angle. This holds for cones of any size. If we take the solid angle as infinitely small and of size $d\Omega$ we may speak of the radiation emitted in a certain direction, but always in the sense that for the emission of a finite amount of energy an infinite number of directions are necessary and these form a finite solid angle.

5. The distribution of energy in the radiation is in general quite arbitrary; that is, the different colors of a certain radiation may have quite different intensities. The color of a ray in experimental physics is usually denoted by its wave length, because this quantity is measured directly. For the theoretical treatment, however, it is usually preferable to use the frequency ν instead, since the characteristic of color is not so much the wave length, which changes from one medium to another, as the frequency, which remains unchanged in a light or heat ray passing through stationary media. We shall, therefore, hereafter denote a certain color by the corresponding value of ν, and a certain interval of color by the limits of the interval ν and ν', where $\nu' > \nu$. The radiation lying in a certain interval of color divided by the magnitude $\nu' - \nu$ of the interval, we shall call the mean radiation in the interval ν to ν'. We shall then assume that if, keeping ν constant, we take the interval $\nu' - \nu$ sufficiently small and denote it by $d\nu$ the value of the mean radiation approaches a definite limiting value, independent of the size of $d\nu$, and

this we shall briefly call the "radiation of frequency ν." To produce a finite intensity of radiation, the frequency interval, though perhaps small, must also be finite.

We have finally to allow for the polarization of the emitted radiation. Since the medium was assumed to be isotropic the emitted rays are unpolarized. Hence every ray has just twice the intensity of one of its plane polarized components, which could, *e.g.*, be obtained by passing the ray through a *Nicol's* prism.

6. Summing up everything said so far, we may equate the total energy in a range of frequency from ν to $\nu + d\nu$ emitted in the time dt in the direction of the conical element $d\Omega$ by a volume-element $d\tau$ to

$$dt \cdot d\tau \cdot d\Omega \cdot d\nu \cdot 2\epsilon_\nu. \tag{1}$$

The finite quantity ϵ_ν is called the coefficient of emission of the medium for the frequency ν. It is a positive function of ν and refers to a plane polarized ray of definite color and direction. The total emission of the volume-element $d\tau$ may be obtained from this by integrating over all directions and all frequencies. Since ϵ_ν is independent of the direction, and since the integral over all conical elements $d\Omega$ is 4π, we get:

$$dt \cdot d\tau \cdot 8\pi \int_0^\infty \epsilon_\nu \, d\nu. \tag{2}$$

7. The coefficient of emission ϵ depends, not only on the frequency ν, but also on the condition of the emitting substance contained in the volume-element $d\tau$, and, generally speaking, in a very complicated way, according to the physical and chemical processes which take place in the elements of time and volume in question. But the empirical law that the emission of any volume-element depends entirely on what takes place inside of this element holds true in all cases (*Prevost's* principle).

A body A at 100° C. emits toward a body B at 0° C. exactly the same amount of radiation as toward an equally large and similarly situated body B' at 1000° C. The fact that the body A is cooled by B and heated by B' is due entirely to the fact that B is a weaker, B' a stronger emitter than A.

We shall now introduce the further simplifying assumption that the physical and chemical condition of the emitting substance depends on but a single variable, namely, on its absolute temperature T. A necessary consequence of this is that the coefficient of emission ϵ depends, apart from the frequency ν and the nature of the medium, only on the temperature T. The last statement excludes from our consideration a number of radiation phenomena, such as fluorescence, phosphorescence, electrical and chemical luminosity, to which *E. Wiedemann* has given the common name "phenomena of luminescence." We shall deal with pure "temperature radiation" exclusively.

A special case of temperature radiation is the case of the chemical nature of the emitting substance being invariable. In this case the emission takes place entirely at the expense of the heat of the body. Nevertheless, it is possible, according to what has been said, to have temperature radiation while chemical changes are taking place, provided the chemical condition is completely determined by the temperature.

8. Propagation.—The propagation of the radiation in a medium assumed to be homogeneous, isotropic, and at rest takes place in straight lines and with the same velocity in all directions, diffraction phenomena being entirely excluded. Yet, in general, each ray suffers during its propagation a certain weakening, because a certain fraction of its energy is continuously deviated from its original direction and scattered in all directions. This phenomenon of "scattering," which means neither a creation nor a destruction of radiant energy but sim-

ply a change in distribution, takes place, generally speaking, in all media differing from an absolute vacuum, even in substances which are perfectly pure chemically.[1] The cause of this is that no substance is homogeneous in the absolute sense of the word. The smallest elements of space always exhibit some discontinuities on account of their atomic structure. Small impurities, as, for instance, particles of dust, increase the influence of scattering without, however, appreciably affecting its general character. Hence, so-called "turbid" media, *i.e.*, such as contain foreign particles, may be quite properly regarded as optically homogeneous,[2] provided only that the linear dimensions of the foreign particles as well as the distances of neighboring particles are sufficiently small compared with the wave lengths of the rays considered. As regards optical phenomena, then, there is no fundamental distinction between chemically pure substances and the turbid media just described. No space is optically void in the absolute sense except a vacuum. Hence a chemically pure substance may be spoken of as a vacuum made turbid by the presence of molecules.

A typical example of scattering is offered by the behavior of sunlight in the atmosphere. When, with a clear sky, the sun stands in the zenith, only about two-thirds of the direct radiation of the sun reaches the surface of the earth. The remainder is intercepted by the atmosphere, being partly absorbed and changed into heat of the air, partly, however, scattered and changed into diffuse skylight. This phenomenon is produced probably not so much by the particles suspended in the atmosphere as by the air molecules themselves.

Whether the scattering depends on reflection, on diffraction, or on

[1]See, *e.g.*, *Lobry de Bruyn* and *L. K. Wolff*, Rec. des Trav. Chim. des Pays-Bas **23**, p. 155, 1904.

[2]To restrict the word homogeneous to its absolute sense would mean that it could not be applied to any material substance.

a resonance effect on the molecules or particles is a point that we may leave entirely aside. We only take account of the fact that every ray on its path through any medium loses a certain fraction of its intensity. For a very small distance, s, this fraction is proportional to s, say

$$\beta_\nu s \tag{3}$$

where the positive quantity β_ν is independent of the intensity of radiation and is called the "coefficient of scattering" of the medium. Inasmuch as the medium is assumed to be isotropic, β_ν is also independent of the direction of propagation and polarization of the ray. It depends, however, as indicated by the subscript ν, not only on the physical and chemical constitution of the body but also to a very marked degree on the frequency. For certain values of ν, β_ν may be so large that the straight-line propagation of the rays is virtually destroyed. For other values of ν, however, β_ν may become so small that the scattering can be entirely neglected. For generality we shall assume a mean value of β_ν. In the cases of most importance β_ν increases quite appreciably as ν increases, *i.e.*, the scattering is noticeably larger for rays of shorter wave length;[1] hence the blue color of diffuse skylight.

The scattered radiation energy is propagated from the place where the scattering occurs in a way similar to that in which the emitted energy is propagated from the place of emission, since it travels in all directions in space. It does not, however, have the same intensity in all directions, and moreover is polarized in some special directions, depending to a large extent on the direction of the original ray. We need not, however, enter into any further discussion of these questions.

9. While the phenomenon of scattering means a continuous modification in the interior of the medium, a discontinuous change in both the

[1] *Lord Rayleigh*, Phil. Mag., **47**, p. 379, 1899.

direction and the intensity of a ray occurs when it reaches the boundary of a medium and meets the surface of a second medium. The latter, like the former, will be assumed to be homogeneous and isotropic. In this case, the ray is in general partly reflected and partly transmitted. The reflection and refraction may be "regular," there being a single reflected ray according to the simple law of reflection and a single transmitted ray, according to *Snell's* law of refraction, or, they may be "diffuse," which means that from the point of incidence on the surface the radiation spreads out into the two media with intensities that are different in different directions. We accordingly describe the surface of the second medium as "smooth" or "rough" respectively. Diffuse reflection occurring at a rough surface should be carefully distinguished from reflection at a smooth surface of a turbid medium. In both cases part of the incident ray goes back to the first medium as diffuse radiation. But in the first case the scattering occurs on the surface, in the second in more or less thick layers entirely inside of the second medium.

10. When a smooth surface completely reflects all incident rays, as is approximately the case with many metallic surfaces, it is termed "reflecting." When a rough surface reflects all incident rays completely and uniformly in all directions, it is called "white." The other extreme, namely, complete transmission of all incident rays through the surface never occurs with smooth surfaces, at least if the two contiguous media are at all optically different. A rough surface having the property of completely transmitting the incident radiation is described as "black."

In addition to "black surfaces" the term "black body" is also used. According to *G. Kirchhoff*[1] it denotes a body which has the property

[1] *G. Kirchhoff*, Pogg. Ann., **109**, p. 275, 1860. Gesammelte Abhandlungen, J. A. Barth, Leipzig, 1882, p. 573. In defining a black body *Kirchhoff* also assumes that the absorption of incident rays takes place in a layer "infinitely thin." We do not include this in our definition.

of allowing all incident rays to enter without surface reflection and not allowing them to leave again. Hence it is seen that a black body must satisfy three independent conditions. First, the body must have a black surface in order to allow the incident rays to enter without reflection. Since, in general, the properties of a surface depend on both of the bodies which are in contact, this condition shows that the property of blackness as applied to a body depends not only on the nature of the body but also on that of the contiguous medium. A body which is black relatively to air need not be so relatively to glass, and *vice versa*. Second, the black body must have a certain minimum thickness depending on its absorbing power, in order to insure that the rays after passing into the body shall not be able to leave it again at a different point of the surface. The more absorbing a body is, the smaller the value of this minimum thickness, while in the case of bodies with vanishingly small absorbing power only a layer of infinite thickness may be regarded as black. Third, the black body must have a vanishingly small coefficient of scattering (Sec. 8). Otherwise the rays received by it would be partly scattered in the interior and might leave again through the surface.[1]

11. All the distinctions and definitions mentioned in the two preceding paragraphs refer to rays of one definite color only. It might very well happen that, *e.g.*, a surface which is rough for a certain kind of rays must be regarded as smooth for a different kind of rays. It is readily seen that, in general, a surface shows decreasing degrees of roughness for increasing wave lengths. Now, since smooth non-reflecting surfaces do not exist (Sec. 10), it follows that all approximately black surfaces

[1]For this point see especially *A. Schuster*, Astrophysical Journal, **21**, p. 1, 1905, who has pointed out that an infinite layer of gas with a black surface need by no means be a black body.

which may be realized in practice (lamp black, platinum black) show appreciable reflection for rays of sufficiently long wave lengths.

12. Absorption.—Heat rays are destroyed by "absorption." According to the principle of the conservation of energy the energy of heat radiation is thereby changed into other forms of energy (heat, chemical energy). Thus only material particles can absorb heat rays, not elements of surfaces, although sometimes for the sake of brevity the expression absorbing surfaces is used.

Whenever absorption takes place, the heat ray passing through the medium under consideration is weakened by a certain fraction of its intensity for every element of path traversed. For a sufficiently small distance s this fraction is proportional to s, and may be written

$$\alpha_\nu s. \tag{4}$$

Here α_ν is known as the "coefficient of absorption" of the medium for a ray of frequency ν. We assume this coefficient to be independent of the intensity; it will, however, depend in general in non-homogeneous and anisotropic media on the position of s and on the direction of propagation and polarization of the ray (example: tourmaline). We shall, however, consider only homogeneous isotropic substances, and shall therefore suppose that α_ν has the same value at all points and in all directions in the medium, and depends on nothing but the frequency ν, the temperature T, and the nature of the medium.

Whenever α_ν does not differ from zero except for a limited range of the spectrum, the medium shows "selective" absorption. For those colors for which $\alpha_\nu = 0$ and also the coefficient of scattering $\beta_\nu = 0$ the medium is described as perfectly "transparent" or "diathermanous." But the properties of selective absorption and of diathermancy may for a given medium vary widely with the temperature. In general we shall assume a mean value for α_ν. This implies that the absorption

in a distance equal to a single wave length is very small, because the distance s, while small, contains many wave lengths (Sec. 2).

13. The foregoing considerations regarding the emission, the propagation, and the absorption of heat rays suffice for a mathematical treatment of the radiation phenomena. The calculation requires a knowledge of the value of the constants and the initial and boundary conditions, and yields a full account of the changes the radiation undergoes in a given time in one or more contiguous media of the kind stated, including the temperature changes caused by it. The actual calculation is usually very complicated. We shall, however, before entering upon the treatment of special cases discuss the general radiation phenomena from a different point of view, namely by fixing our attention not on a definite ray, but on a definite position in space.

14. Let $d\sigma$ be an arbitrarily chosen, infinitely small element of area in the interior of a medium through which radiation passes. At a given instant rays are passing through this element in many different directions. The energy radiated through it in an element of time dt in a definite direction is proportional to the area $d\sigma$, the length of time dt, and to the cosine of the angle θ made by the normal of $d\sigma$ with the direction of the radiation. If we make $d\sigma$ sufficiently small, then, although this is only an approximation to the actual state of affairs, we can think of all points in $d\sigma$ as being affected by the radiation in the same way. Then the energy radiated through $d\sigma$ in a definite direction must be proportional to the solid angle in which $d\sigma$ intercepts that radiation and this solid angle is measured by $d\sigma \cos\theta$. It is readily seen that, when the direction of the element is varied relatively to the direction of the radiation, the energy radiated through it vanishes when

$$\theta = \frac{\pi}{2}.$$

Now in general a pencil of rays is propagated from every point of

the element $d\sigma$ in all directions, but with different intensities in different directions, and any two pencils emanating from two points of the element are identical save for differences of higher order. A single one of these pencils coming from a single point does not represent a finite quantity of energy, because a finite amount of energy is radiated only through a finite area. This holds also for the passage of rays through a so-called focus. For example, when sunlight passes through a converging lens and is concentrated in the focal plane of the lens, the solar rays do not converge to a single point, but each pencil of parallel rays forms a separate focus and all these foci together constitute a surface representing a small but finite image of the sun. A finite amount of energy does not pass through less than a finite portion of this surface.

15. Let us now consider quite generally the pencil, which is propagated from a point of the element $d\sigma$ as vertex in all directions of space and on both sides of $d\sigma$. A certain direction may be specified by the angle θ (between 0 and π), as already used, and by an azimuth ϕ (between 0 and 2π). The intensity in this direction is the energy propagated in an infinitely thin cone limited by θ and $\theta + d\theta$ and ϕ and $\phi + d\phi$. The solid angle of this cone is

$$d\Omega = \sin\theta \cdot d\theta \cdot d\phi. \tag{5}$$

Thus the energy radiated in time dt through the element of area $d\sigma$ in the direction of the cone $d\Omega$ is:

$$dt\, d\sigma\, \cos\theta\, d\Omega\, K = K \sin\theta \cos\theta\, d\theta\, d\phi\, d\sigma\, dt. \tag{6}$$

The finite quantity K we shall term the "specific intensity" or the "brightness," $d\Omega$ the "solid angle" of the pencil emanating from a point of the element $d\sigma$ in the direction (θ, ϕ). K is a positive function of position, time, and the angles θ and ϕ. In general the specific intensities of radiation in different directions are entirely independent of one

another. For example, on substituting $\pi - \theta$ for θ and $\pi + \phi$ for ϕ in the function K, we obtain the specific intensity of radiation in the diametrically opposite direction, a quantity which in general is quite different from the preceding one.

For the total radiation through the element of area $d\sigma$ toward one side, say the one on which θ is an acute angle, we get, by integrating with respect to ϕ from 0 to 2π and with respect to θ from 0 to $\frac{\pi}{2}$

$$\int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta \, K \sin\theta \cos\theta \, d\sigma \, dt.$$

Should the radiation be uniform in all directions and hence K be a constant, the total radiation on one side will be

$$\pi K \, d\sigma \, dt. \tag{7}$$

16. In speaking of the radiation in a definite direction (θ, ϕ) one should always keep in mind that the energy radiated in a cone is not finite unless the angle of the cone is finite. No finite radiation of light or heat takes place in one definite direction only, or expressing it differently, in nature there is no such thing as absolutely parallel light or an absolutely plane wave front. From a pencil of rays called "parallel" a finite amount of energy of radiation can only be obtained if the rays or wave normals of the pencil diverge so as to form a finite though perhaps exceedingly narrow cone.

17. The specific intensity K of the whole energy radiated in a certain direction may be further divided into the intensities of the separate rays belonging to the different regions of the spectrum which travel independently of one another. Hence we consider the intensity of radiation within a certain range of frequencies, say from ν to ν'. If the

interval $\nu' - \nu$ be taken sufficiently small and be denoted by $d\nu$, the intensity of radiation within the interval is proportional to $d\nu$. Such radiation is called homogeneous or monochromatic.

A last characteristic property of a ray of definite direction, intensity, and color is its state of polarization. If we break up a ray, which is in any state of polarization whatsoever and which travels in a definite direction and has a definite frequency ν, into two plane polarized components, the sum of the intensities of the components will be just equal to the intensity of the ray as a whole, independently of the direction of the two planes, provided the two planes of polarization, which otherwise may be taken at random, are at right angles to each other. If their position be denoted by the azimuth ψ of one of the planes of vibration (plane of the electric vector), then the two components of the intensity may be written in the form

$$\begin{aligned} & \mathsf{K}_\nu \cos^2\psi + \mathsf{K}'_\nu \sin^2\psi \\ \text{and} \qquad & \mathsf{K}_\nu \sin^2\psi + \mathsf{K}'_\nu \cos^2\psi. \end{aligned} \tag{8}$$

Herein K is independent of ψ. These expressions we shall call the "components of the specific intensity of radiation of frequency ν." The sum is independent of ψ and is always equal to the intensity of the whole ray $\mathsf{K}_\nu + \mathsf{K}'_\nu$. At the same time K_ν and K'_ν represent respectively the largest and smallest values which either of the components may have, namely, when $\psi = 0$ and $\psi = \dfrac{\pi}{2}$. Hence we call these values the "principal values of the intensities," or the "principal intensities," and the corresponding planes of vibration we call the "principal planes of vibration" of the ray. Of course both, in general, vary with the time. Thus we may write generally

$$\mathsf{K} = \int_0^\infty d\nu \, (\mathsf{K}_\nu + \mathsf{K}'_\nu) \tag{9}$$

where the positive quantities K_ν and K'_ν, the two principal values of the specific intensity of the radiation (brightness) of frequency ν, depend not only on ν but also on their position, the time, and on the angles θ and ϕ. By substitution in (6) the energy radiated in the time dt through the element of area $d\sigma$ in the direction of the conical element $d\Omega$ assumes the value

$$dt\, d\sigma \cos\theta\, d\Omega \int_0^\infty d\nu\, (\mathsf{K}_\nu + \mathsf{K}'_\nu) \tag{10}$$

and for monochromatic plane polarized radiation of brightness K_ν:

$$dt\, d\sigma \cos\theta\, d\Omega\, \mathsf{K}_\nu\, d\nu = dt\, d\sigma \sin\theta \cos\theta\, d\theta\, d\phi\, \mathsf{K}_\nu\, d\nu. \tag{11}$$

For unpolarized rays $\mathsf{K}_\nu = \mathsf{K}'_\nu$, and hence

$$K = 2 \int_0^\infty d\nu\, \mathsf{K}_\nu, \tag{12}$$

and the energy of a monochromatic ray of frequency ν will be:

$$2\, dt\, d\sigma \cos\theta\, d\Omega\, \mathsf{K}_\nu\, d\nu = 2\, dt\, d\sigma \sin\theta \cos\theta\, d\theta\, d\phi\, \mathsf{K}_\nu\, d\nu. \tag{13}$$

When, moreover, the radiation is uniformly distributed in all directions, the total radiation through $d\sigma$ toward one side may be found from (7) and (12); it is

$$2\pi\, d\sigma\, dt \int_0^\infty \mathsf{K}_\nu\, d\nu. \tag{14}$$

18. Since in nature K_ν can never be infinitely large, K will not have a finite value unless K_ν differs from zero over a finite range of frequencies. Hence there exists in nature no absolutely homogeneous or monochromatic radiation of light or heat. A finite amount of radiation contains

always a finite although possibly very narrow range of the spectrum. This implies a fundamental difference from the corresponding phenomena of acoustics, where a finite intensity of sound may correspond to a single definite frequency. This difference is, among other things, the cause of the fact that the second law of thermodynamics has an important bearing on light and heat rays, but not on sound waves. This will be further discussed later on.

19. From equation (9) it is seen that the quantity K_ν, the intensity of radiation of frequency ν, and the quantity K, the intensity of radiation of the whole spectrum, are of different dimensions. Further it is to be noticed that, on subdividing the spectrum according to wave lengths λ, instead of frequencies ν, the intensity of radiation E_λ of the wave lengths λ corresponding to the frequency ν is not obtained simply by replacing ν in the expression for K_ν by the corresponding value of λ deduced from

$$\nu = \frac{q}{\lambda} \tag{15}$$

where q is the velocity of propagation. For if $d\lambda$ and $d\nu$ refer to the same interval of the spectrum, we have, not $E_\lambda = \mathsf{K}_\nu$, but $E_\lambda\, d\lambda = \mathsf{K}_\nu\, d\nu$. By differentiating (15) and paying attention to the signs of corresponding values of $d\lambda$ and $d\nu$ the equation

$$d\nu = \frac{q\, d\lambda}{\lambda^2}$$

is obtained. Hence we get by substitution:

$$E_\lambda = \frac{q\mathsf{K}_\nu}{\lambda^2}. \tag{16}$$

This relation shows among other things that in a certain spectrum the maxima of E_λ and K_ν lie at different points of the spectrum.

20. When the principal intensities K_ν and K'_ν of all monochromatic rays are given at all points of the medium and for all directions, the state of radiation is known in all respects and all questions regarding it may be answered. We shall show this by one or two applications to special cases. Let us first find the amount of energy which is radiated through any element of area $d\sigma$ toward any other element $d\sigma'$. The distance r between the two elements may be thought of as large compared with the linear dimensions of the elements $d\sigma$ and $d\sigma'$ but still so small that no appreciable amount of radiation is absorbed or scattered along it. This condition is, of course, superfluous for diathermanous media.

From any definite point of $d\sigma$ rays pass to all points of $d\sigma'$. These rays form a cone whose vertex lies in $d\sigma$ and whose solid angle is

$$d\Omega = \frac{d\sigma' \cos(\mathsf{n}', r)}{r^2}$$

where n' denotes the normal of $d\sigma'$ and the angle (n', r) is to be taken as an acute angle. This value of $d\Omega$ is, neglecting small quantities of higher order, independent of the particular position of the vertex of the cone on $d\sigma$.

If we further denote the normal to $d\sigma$ by n the angle θ of (14) will be the angle (n, r) and hence from expression (6) the energy of radiation required is found to be:

$$K \cdot \frac{d\sigma\, d\sigma' \cos(\mathsf{n}, r) \cos(\mathsf{n}', r)}{r^2} \cdot dt. \tag{17}$$

For monochromatic plane polarized radiation of frequency ν the energy will be, according to equation (11),

$$\mathsf{K}_\nu\, d\nu \cdot \frac{d\sigma\, d\sigma' \cos(\mathsf{n}, r) \cos(\mathsf{n}', r)}{r^2} \cdot dt. \tag{18}$$

The relative size of the two elements $d\sigma$ and $d\sigma'$ may have any value whatever. They may be assumed to be of the same or of a different order of magnitude, provided the condition remains satisfied that r is large compared with the linear dimensions of each of them. If we choose $d\sigma$ small compared with $d\sigma'$, the rays diverge from $d\sigma$ to $d\sigma'$, whereas they converge from $d\sigma$ to $d\sigma'$ if we choose $d\sigma$ large compared with $d\sigma'$.

21. Since every point of $d\sigma$ is the vertex of a cone spreading out toward $d\sigma'$, the whole pencil of rays here considered, which is defined by $d\sigma$ and $d\sigma'$, consists of a double infinity of point pencils or of a fourfold infinity of rays which must all be considered equally for the energy radiation. Similarly the pencil of rays may be thought of as consisting of the cones which, emanating from all points of $d\sigma$, converge in one point of $d\sigma'$ respectively as a vertex. If we now imagine the whole pencil of rays to be cut by a plane at any arbitrary distance from the elements $d\sigma$ and $d\sigma'$ and lying either between them or outside, then the cross-sections of any two point pencils on this plane will not be identical, not even approximately. In general they will partly overlap and partly lie outside of each other, the amount of overlapping being different for different intersecting planes. Hence it follows that there is no definite cross-section of the pencil of rays so far as the uniformity of radiation is concerned. If, however, the intersecting plane coincides with either $d\sigma$ or $d\sigma'$, then the pencil has a definite cross-section. Thus these two planes show an exceptional property. We shall call them the two "focal planes" of the pencil.

In the special case already mentioned above, namely, when one of the two focal planes is infinitely small compared with the other, the whole pencil of rays shows the character of a point pencil inasmuch as its form is approximately that of a cone having its vertex in that focal plane which is small compared with the other. In that case the "cross-section" of the whole pencil at a definite point has a definite meaning. Such a

pencil of rays, which is similar to a cone, we shall call an elementary pencil, and the small focal plane we shall call the first focal plane of the elementary pencil. The radiation may be either converging toward the first focal plane or diverging from the first focal plane. All the pencils of rays passing through a medium may be considered as consisting of such elementary pencils, and hence we may base our future considerations on elementary pencils only, which is a great convenience, owing to their simple nature.

As quantities necessary to define an elementary pencil with a given first focal plane $d\sigma$, we may choose not the second focal plane $d\sigma'$ but the magnitude of that solid angle $d\Omega$ under which $d\sigma'$ is seen from $d\sigma$. On the other hand, in the case of an arbitrary pencil, that is, when the two focal planes are of the same order of magnitude, the second focal plane in general cannot be replaced by the solid angle $d\Omega$ without the pencil changing markedly in character. For if, instead of $d\sigma'$ being given, the magnitude and direction of $d\Omega$, to be taken as constant for all points of $d\sigma$, is given, then the rays emanating from $d\sigma$ do not any longer form the original pencil, but rather an elementary pencil whose first focal plane is $d\sigma$ and whose second focal plane lies at an infinite distance.

22. Since the energy radiation is propagated in the medium with a finite velocity q, there must be in a finite space a finite amount of energy. We shall therefore speak of the "space density of radiation," meaning thereby the ratio of the total quantity of energy of radiation contained in a volume-element to the magnitude of the latter. Let us now calculate the space density of radiation u at any arbitrary point of the medium. When we consider an infinitely small element of volume v at the point in question, having any shape whatsoever, we must allow for all rays passing through the volume-element v. For this purpose we shall construct about any point O of v as center a sphere of radius r,

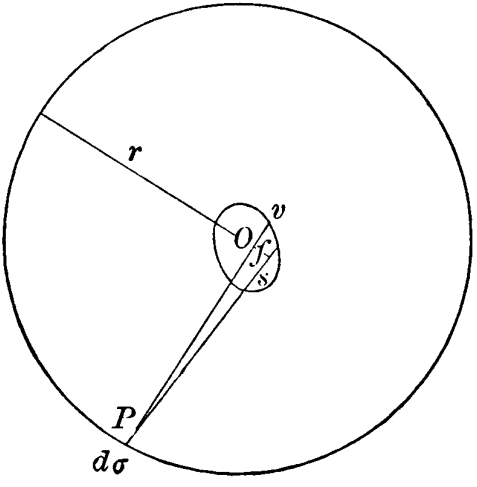


FIG. 1.

r being large compared with the linear dimensions of v but still so small that no appreciable absorption or scattering of the radiation takes place in the distance r (Fig. 1). Every ray which reaches v must then come from some point on the surface of the sphere. If, then, we at first consider only all the rays that come from the points of an infinitely small element of area $d\sigma$ on the surface of the sphere, and reach v, and then sum up for all elements of the spherical surface, we shall have accounted for all rays and not taken any one more than once.

Let us then calculate first the amount of energy which is contributed to the energy contained in v by the radiation sent from such an element $d\sigma$ to v. We choose $d\sigma$ so that its linear dimensions are small compared with those of v and consider the cone of rays which, starting at a point of $d\sigma$, meets the volume v. This cone consists of an infinite number of conical elements with the common vertex at P, a point

of $d\sigma$, each cutting out of the volume v a certain element of length, say s. The solid angle of such a conical element is $\frac{f}{r^2}$ where f denotes the area of cross-section normal to the axis of the cone at a distance r from the vertex. The time required for the radiation to pass through the distance s is:

$$\tau = \frac{s}{q}.$$

From expression (6) we may find the energy radiated through a certain element of area. In the present case $d\Omega = \frac{f}{r^2}$ and $\theta = 0$; hence the energy is:

$$\tau\, d\sigma\, \frac{f}{r^2} K = \frac{fs}{r^2 q} \cdot K\, d\sigma. \tag{19}$$

This energy enters the conical element in v and spreads out into the volume fs. Summing up over all conical elements that start from $d\sigma$ and enter v we have

$$\frac{K\, d\sigma}{r^2 q} \sum fs = \frac{K\, d\sigma}{r^2 q} v.$$

This represents the entire energy of radiation contained in the volume v, so far as it is caused by radiation through the element $d\sigma$. In order to obtain the total energy of radiation contained in v we must integrate over all elements $d\sigma$ contained in the surface of the sphere. Denoting by $d\Omega$ the solid angle $\frac{d\sigma}{r^2}$ of a cone which has its center in O and intersects in $d\sigma$ the surface of the sphere, we get for the whole energy:

$$\frac{v}{q} \int K\, d\Omega.$$

The volume density of radiation required is found from this by dividing by v. It is

$$u = \frac{1}{q} \int K \, d\Omega. \tag{20}$$

Since in this expression r has disappeared, we can think of K as the intensity of radiation at the point O itself. In integrating, it is to be noted that K in general depends on the direction (θ, ϕ). For radiation that is uniform in all directions K is a constant and on integration we get:

$$u = \frac{4\pi K}{q}. \tag{21}$$

23. A meaning similar to that of the volume density of the total radiation u is attached to the volume density of radiation of a definite frequency u_ν. Summing up for all parts of the spectrum we get:

$$u = \int_0^\infty \mathsf{u}_\nu \, d\nu. \tag{22}$$

Further by combining equations (9) and (20) we have:

$$\mathsf{u}_\nu = \frac{1}{q} \int (\mathsf{K}_\nu + \mathsf{K}'_\nu) \, d\Omega, \tag{23}$$

and finally for unpolarized radiation uniformly distributed in all directions:

$$\mathsf{u}_\nu = \frac{8\pi \mathsf{K}_\nu}{q}. \tag{24}$$

CHAPTER II

RADIATION AT THERMODYNAMIC EQUILIBRIUM. KIRCHHOFF'S LAW. BLACK RADIATION

24. We shall now apply the laws enunciated in the last chapter to the special case of thermodynamic equilibrium, and hence we begin our consideration by stating a certain consequence of the second principle of thermodynamics: A system of bodies of arbitrary nature, shape, and position which is at rest and is surrounded by a rigid cover impermeable to heat will, no matter what its initial state may be, pass in the course of time into a permanent state, in which the temperature of all bodies of the system is the same. This is the state of thermodynamic equilibrium, in which the entropy of the system has the maximum value compatible with the total energy of the system as fixed by the initial conditions. This state being reached, no further increase in entropy is possible.

In certain cases it may happen that, under the given conditions, the entropy can assume not only one but several maxima, of which one is the absolute one, the others having only a relative significance.[1] In these cases every state corresponding to a maximum value of the entropy represents a state of thermodynamic equilibrium of the system. But only one of them, the one corresponding to the absolute maximum of entropy, represents the absolutely stable equilibrium. All the others are in a certain sense unstable, inasmuch as a suitable, however small, disturbance may produce in the system a permanent change in the equilibrium in the direction of the absolutely stable equilibrium. An example of this is offered by supersaturated steam enclosed in a rigid

[1]See, *e.g.*, *M. Planck*, Vorlesungen über Thermodynamik, Leipzig, Veit and Comp., 1911 (or English Translation, Longmans Green & Co.), Secs. 165 and 189, *et seq.*

vessel or by any explosive substance. We shall also meet such unstable equilibria in the case of radiation phenomena (Sec. 52).

25. We shall now, as in the previous chapter, assume that we are dealing with homogeneous isotropic media whose condition depends only on the temperature, and we shall inquire what laws the radiation phenomena in them must obey in order to be consistent with the deduction from the second principle mentioned in the preceding section. The means of answering this inquiry is supplied by the investigation of the state of thermodynamic equilibrium of one or more of such media, this investigation to be conducted by applying the conceptions and laws established in the last chapter.

We shall begin with the simplest case, that of a single medium extending very far in all directions of space, and, like all systems we shall here consider, being surrounded by a rigid cover impermeable to heat. For the present we shall assume that the medium has finite coefficients of absorption, emission, and scattering.

Let us consider, first, points of the medium that are far away from the surface. At such points the influence of the surface is, of course, vanishingly small and from the homogeneity and the isotropy of the medium it will follow that in a state of thermodynamic equilibrium the radiation of heat has everywhere and in all directions the same properties. Then K_ν, the specific intensity of radiation of a plane polarized ray of frequency ν (Sec. 17), must be independent of the azimuth of the plane of polarization as well as of position and direction of the ray. Hence to each pencil of rays starting at an element of area $d\sigma$ and diverging within a conical element $d\Omega$ corresponds an exactly equal pencil of opposite direction converging within the same conical element toward the element of area.

Now the condition of thermodynamic equilibrium requires that the temperature shall be everywhere the same and shall not vary in time.

Therefore in any given arbitrary time just as much radiant heat must be absorbed as is emitted in each volume-element of the medium. For the heat of the body depends only on the heat radiation, since, on account of the uniformity in temperature, no conduction of heat takes place. This condition is not influenced by the phenomenon of scattering, because scattering refers only to a change in direction of the energy radiated, not to a creation or destruction of it. We shall, therefore, calculate the energy emitted and absorbed in the time dt by a volume-element v.

According to equation (2) the energy emitted has the value

$$dt\, v \cdot 8\pi \int_0^\infty \epsilon_\nu \, d\nu$$

where ϵ_ν, the coefficient of emission of the medium, depends only on the frequency ν and on the temperature in addition to the chemical nature of the medium.

26. For the calculation of the energy absorbed we shall employ the same reasoning as was illustrated by Fig. 1 (Sec. 22) and shall retain the notation there used. The radiant energy absorbed by the volume-element v in the time dt is found by considering the intensities of all the rays passing through the element v and taking that fraction of each of these rays which is absorbed in v. Now, according to (19), the conical element that starts from $d\sigma$ and cuts out of the volume v a part equal to fs has the intensity (energy radiated per unit time)

$$d\sigma \cdot \frac{f}{r^2} \cdot K$$

or, according to (12), by considering the different parts of the spectrum separately:

$$2\, d\sigma \, \frac{f}{r^2} \int_0^\infty \mathsf{K}_\nu \, d\nu.$$

Hence the intensity of a monochromatic ray is:

$$2\,d\sigma\,\frac{f}{r^2}\mathsf{K}_\nu\,d\nu.$$

The amount of energy of this ray absorbed in the distance s in the time dt is, according to (4),

$$dt\,\alpha_\nu s\,2\,d\sigma\,\frac{f}{r^2}\mathsf{K}_\nu\,d\nu.$$

Hence the absorbed part of the energy of this small cone of rays, as found by integrating over all frequencies, is:

$$dt\,2\,d\sigma\,\frac{fs}{r^2}\int_0^\infty \alpha_\nu\mathsf{K}_\nu\,d\nu.$$

When this expression is summed up over all the different cross-sections f of the conical elements starting at $d\sigma$ and passing through v, it is evident that $\sum fs = v$, and when we sum up over all elements $d\sigma$ of the spherical surface of radius r we have

$$\int\frac{d\sigma}{r^2} = 4\pi.$$

Thus for the total radiant energy absorbed in the time dt by the volume-element v the following expression is found:

$$dt\,v\,8\pi\int_0^\infty \alpha_\nu\mathsf{K}_\nu\,d\nu. \tag{25}$$

By equating the emitted and absorbed energy we obtain:

$$\int_0^\infty \epsilon_\nu\,d\nu = \int_0^\infty \alpha_\nu\mathsf{K}_\nu\,d\nu.$$

A similar relation may be obtained for the separate parts of the spectrum. For the energy emitted and the energy absorbed in the state of thermodynamic equilibrium are equal, not only for the entire radiation of the whole spectrum, but also for each monochromatic radiation. This is readily seen from the following. The magnitudes of ϵ_ν, α_ν, and K_ν are independent of position. Hence, if for any single color the absorbed were not equal to the emitted energy, there would be everywhere in the whole medium a continuous increase or decrease of the energy radiation of that particular color at the expense of the other colors. This would be contradictory to the condition that K_ν for each separate frequency does not change with the time. We have therefore for each frequency the relation:

$$\epsilon_\nu = \alpha_\nu \mathsf{K}_\nu, \text{ or} \tag{26}$$

$$\mathsf{K}_\nu = \frac{\epsilon_\nu}{\alpha_\nu}, \tag{27}$$

i.e.: *in the interior of a medium in a state of thermodynamic equilibrium the specific intensity of radiation of a certain frequency is equal to the coefficient of emission divided by the coefficient of absorption of the medium for this frequency.*

27. Since ϵ_ν and α_ν depend only on the nature of the medium, the temperature, and the frequency ν, the intensity of radiation of a definite color in the state of thermodynamic equilibrium is completely defined by the nature of the medium and the temperature. An exceptional case is when $\alpha_\nu = 0$, that is, when the medium does not at all absorb the color in question. Since K_ν cannot become infinitely large, a first consequence of this is that in that case $\epsilon_\nu = 0$ also, that is, a medium does not emit any color which it does not absorb. A second consequence is that if ϵ_ν and α_ν both vanish, equation (26) is satisfied by every value of K_ν. *In a medium which is diathermanous for a certain color thermo-*

dynamic equilibrium can exist for any intensity of radiation whatever of that color.

This supplies an immediate illustration of the cases spoken of before (Sec. 24), where, for a given value of the total energy of a system enclosed by a rigid cover impermeable to heat, several states of equilibrium can exist, corresponding to several relative maxima of the entropy. That is to say, since the intensity of radiation of the particular color in the state of thermodynamic equilibrium is quite independent of the temperature of a medium which is diathermanous for this color, the given total energy may be arbitrarily distributed between radiation of that color and the heat of the body, without making thermodynamic equilibrium impossible. Among all these distributions there is one particular one, corresponding to the absolute maximum of entropy, which represents absolutely stable equilibrium. This one, unlike all the others, which are in a certain sense unstable, has the property of not being appreciably affected by a small disturbance. Indeed we shall see later (Sec. 48) that among the infinite number of values, which the quotient $\frac{\epsilon_\nu}{\alpha_\nu}$ can have, if numerator and denominator both vanish, there exists one particular one which depends in a definite way on the nature of the medium, the frequency ν, and the temperature. This distinct value of the fraction is accordingly called the stable intensity of radiation K_ν, in the medium, which at the temperature in question is diathermanous for rays of the frequency ν.

Everything that has just been said of a medium which is diathermanous for a certain kind of rays holds true for an absolute vacuum, which is a medium diathermanous for rays of all kinds, the only difference being that one cannot speak of the heat and the temperature of an absolute vacuum in any definite sense.

For the present we again shall put the special case of diathermancy

aside and assume that all the media considered have a finite coefficient of absorption.

28. Let us now consider briefly the phenomenon of scattering at thermodynamic equilibrium. Every ray meeting the volume-element v suffers there, apart from absorption, a certain weakening of its intensity because a certain fraction of its energy is diverted in different directions. The value of the total energy of scattered radiation received and diverted, in the time dt by the volume-element v in all directions, may be calculated from expression (3) in exactly the same way as the value of the absorbed energy was calculated in Sec. 26. Hence we get an expression similar to (25), namely,

$$dt\, v\, 8\pi \int_0^\infty \beta_\nu \mathsf{K}_\nu \, d\nu. \tag{28}$$

The question as to what becomes of this energy is readily answered. On account of the isotropy of the medium, the energy scattered in v and given by (28) is radiated uniformly in all directions just as in the case of the energy entering v. Hence that part of the scattered energy received in v which is radiated out in a cone of solid angle $d\Omega$ is obtained by multiplying the last expression by $\dfrac{d\Omega}{4\pi}$. This gives

$$2\, dt\, v\, d\Omega \int_0^\infty \beta_\nu \mathsf{K}_\nu \, d\nu,$$

and, for monochromatic plane polarized radiation,

$$dt\, v\, d\Omega\, \beta_\nu \mathsf{K}_\nu \, d\nu. \tag{29}$$

Here it must be carefully kept in mind that this uniformity of radiation in all directions holds only for all rays striking the element v

taken together; a single ray, even in an isotropic medium, is scattered in different directions with different intensities and different directions of polarization. (See end of Sec. 8.)

It is thus found that, when thermodynamic equilibrium of radiation exists inside of the medium, the process of scattering produces, on the whole, no effect. The radiation falling on a volume-element from all sides and scattered from it in all directions behaves exactly as if it had passed directly through the volume-element without the least modification. Every ray loses by scattering just as much energy as it regains by the scattering of other rays.

29. We shall now consider from a different point of view the radiation phenomena in the interior of a very extended homogeneous isotropic medium which is in thermodynamic equilibrium. That is to say, we shall confine our attention, not to a definite volume-element, but to a definite pencil, and in fact to an elementary pencil (Sec. 21). Let this pencil be specified by the infinitely small focal plane $d\sigma$ at the point O (Fig. 2), perpendicular to the axis of the pencil, and by the solid angle $d\Omega$, and let the radiation take place toward the focal plane in the direction of the arrow. We shall consider exclusively rays which belong to this pencil.

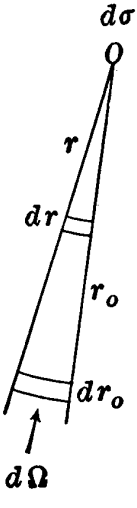


FIG. 2.

The energy of monochromatic plane polarized radiation of the pencil considered passing in unit time through $d\sigma$ is represented, according to (11), since in this case $dt = 1$, $\theta = 0$, by

$$d\sigma \, d\Omega \, \mathsf{K}_\nu \, d\nu. \tag{30}$$

The same value holds for any other cross-section of the pencil. For first, $\mathsf{K}_\nu \, d\nu$ has everywhere the same magnitude (Sec. 25), and second,

the product of any right section of the pencil and the solid angle at which the focal plane $d\sigma$ is seen from this section has the constant value $d\sigma\, d\Omega$, since the magnitude of the cross-section increases with the distance from the vertex O of the pencil in the proportion in which the solid angle decreases. Hence the radiation inside of the pencil takes place just as if the medium were perfectly diathermanous.

On the other hand, the radiation is continuously modified along its path by the effect of emission, absorption, and scattering. We shall consider the magnitude of these effects separately.

30. Let a certain volume-element of the pencil be bounded by two cross-sections at distances equal to r_0 (of arbitrary length) and $r_0 + dr_0$ respectively from the vertex O. The volume will be represented by $dr_0 \cdot r_0^2\, d\Omega$. It emits in unit time toward the focal plane $d\sigma$ at O a certain quantity E of energy of monochromatic plane polarized radiation. E may be obtained from (1) by putting

$$dt = 1, \quad d\tau = dr_0\, r_0^2\, d\Omega, \quad d\Omega = \frac{d\sigma}{r_0^2}$$

and omitting the numerical factor 2. We thus get

$$E = dr_0 \cdot d\Omega\, d\sigma\, \epsilon_\nu\, d\nu. \tag{31}$$

Of the energy E, however, only a fraction E_0 reaches O, since in every infinitesimal element of distance s which it traverses before reaching O the fraction $(\alpha_\nu + \beta_\nu)s$ is lost by absorption and scattering. Let E_r represent that part of E which reaches a cross-section at a distance r $(< r_0)$ from O. Then for a small distance $s = dr$ we have

$$E_{r+dr} - E_r = E_r(\alpha_\nu + \beta_\nu)\, dr,$$

or,

$$\frac{dE_r}{dr} = E_r(\alpha_\nu + \beta_\nu),$$

and, by integration,

$$E_r = Ee^{(\alpha_\nu+\beta_\nu)(r-r_0)}$$

since, for $r = r_0$, $E_r = E$ is given by equation (31). From this, by putting $r = 0$, the energy emitted by the volume-element at r_0 which reaches O is found to be

$$E_0 = Ee^{-(\alpha_\nu+\beta_\nu)r_0} = dr_0\, d\Omega\, d\sigma\, \epsilon_\nu e^{-(\alpha_\nu+\beta_\nu)r_0}\, d\nu. \tag{32}$$

All volume-elements of the pencils combined produce by their emission an amount of energy reaching $d\sigma$ equal to

$$d\Omega\, d\sigma\, d\nu\, \epsilon_\nu \int_0^\infty dr_0\, e^{-(\alpha_\nu+\beta_\nu)r_0} = d\Omega\, d\sigma\, \frac{\epsilon_\nu}{\alpha_\nu+\beta_\nu}\, d\nu. \tag{33}$$

31. If the scattering did not affect the radiation, the total energy reaching $d\sigma$ would necessarily consist of the quantities of energy emitted by the different volume-elements of the pencil, allowance being made, however, for the losses due to absorption on the way. For $\beta_\nu = 0$ expressions (33) and (30) are identical, as may be seen by comparison with (27). Generally, however, (30) is larger than (33) because the energy reaching $d\sigma$ contains also some rays which were not at all emitted from elements inside of the pencil, but somewhere else, and have entered later on by scattering. In fact, the volume-elements of the pencil do not merely scatter outward the radiation which is being transmitted inside the pencil, but they also collect into the pencil rays coming from without. The radiation E' thus collected by the volume-element at r_0 is found, by putting in (29),

$$dt = 1, \quad \nu = dr_0\, d\Omega\, r_0^2, \quad d\Omega = \frac{d\sigma}{r_0^2},$$

to be

$$E' = dr_0\, d\Omega\, d\sigma\, \beta_\nu \mathsf{K}_\nu\, d\nu.$$

This energy is to be added to the energy E emitted by the volume-element, which we have calculated in (31). Thus for the total energy contributed to the pencil in the volume-element at r_0 we find:

$$E + E' = dr_0\, d\Omega\, d\sigma\, (\epsilon_\nu + \beta_\nu \mathsf{K}_\nu)\, d\nu.$$

The part of this reaching O is, similar to (32):

$$dr_0\, d\Omega\, d\sigma\, (\epsilon_\nu + \beta_\nu \mathsf{K}_\nu)\, d\nu\, e^{-r_0(\alpha_\nu+\beta_\nu)}.$$

Making due allowance for emission and collection of scattered rays entering on the way, as well as for losses by absorption and scattering, all volume-elements of the pencil combined give for the energy ultimately reaching $d\sigma$

$$d\Omega\, d\sigma\, (\epsilon_\nu + \beta_\nu \mathsf{K}_\nu)\, d\nu \int_0^\infty dr_0\, e^{-r_0(\alpha_\nu+\beta_\nu)} = d\Omega\, d\sigma\, \frac{\epsilon_\nu + \beta_\nu \mathsf{K}_\nu}{\alpha_\nu + \beta_\nu}\, d\nu,$$

and this expression is really exactly equal to that given by (30), as may be seen by comparison with (26).

32. The laws just derived for the state of radiation of a homogeneous isotropic medium when it is in thermodynamic equilibrium hold, so far as we have seen, only for parts of the medium which lie very far away from the surface, because for such parts only may the radiation be considered, by symmetry, as independent of position and direction. A simple consideration, however, shows that the value of K_ν, which was already calculated and given by (27), and which depends only on the temperature and the nature of the medium, gives the correct value of the intensity of radiation of the frequency considered for all directions up to points directly below the surface of the medium. For in

the state of thermodynamic equilibrium every ray must have just the same intensity as the one travelling in an exactly opposite direction, since otherwise the radiation would cause a unidirectional transport of energy. Consider then any ray coming from the surface of the medium and directed inward; it must have the same intensity as the opposite ray, coming from the interior. A further immediate consequence of this is *that the total state of radiation of the medium is the same on the surface as in the interior.*

33. While the radiation that starts from a surface element and is directed toward the interior of the medium is in every respect equal to that emanating from an equally large parallel element of area in the interior, it nevertheless has a different history. That is to say, since the surface of the medium was assumed to be impermeable to heat, it is produced only by reflection at the surface of radiation coming from the interior. So far as special details are concerned, this can happen in very different ways, depending on whether the surface is assumed to be smooth, *i.e.*, in this case reflecting, or rough, *e.g.*, white (Sec. 10). In the first case there corresponds to each pencil which strikes the surface another perfectly definite pencil, symmetrically situated and having the same intensity, while in the second case every incident pencil is broken up into an infinite number of reflected pencils, each having a different direction, intensity, and polarization. While this is the case, nevertheless the rays that strike a surface-element from all different directions with the same intensity K_ν also produce, all taken together, a uniform radiation of the same intensity K_ν, directed toward the interior of the medium.

34. Hereafter there will not be the slightest difficulty in dispensing with the assumption made in Sec. 25 that the medium in question extends very far in all directions. For after thermodynamic equilibrium has been everywhere established in our medium, the equilibrium is,

according to the results of the last paragraph, in no way disturbed, if we assume any number of rigid surfaces impermeable to heat and rough or smooth to be inserted in the medium. By means of these the whole system is divided into an arbitrary number of perfectly closed separate systems, each of which may be chosen as small as the general restrictions stated in Sec. 2 permit. It follows from this that the value of the specific intensity of radiation K_ν given in (27) remains valid for the thermodynamic equilibrium of a substance enclosed in a space as small as we please and of any shape whatever.

35. From the consideration of a system consisting of a single homogeneous isotropic substance we now pass on to the treatment of a system consisting of two different homogeneous isotropic substances contiguous to each other, the system being, as before, enclosed by a rigid cover impermeable to heat. We consider the state of radiation when thermodynamic equilibrium exists, at first, as before, with the assumption that the media are of considerable extent. Since the equilibrium is nowise disturbed, if we think of the surface separating the two media as being replaced for an instant by an area entirely impermeable to heat radiation, the laws of the last paragraphs must hold for each of the two substances separately. Let the specific intensity of radiation of frequency ν polarized in an arbitrary plane be K_ν in the first substance (the upper one in Fig. 3), and K'_ν in the second, and, in general, let all quantities referring to the second substance be indicated by the addition of an accent. Both of the quantities K_ν and K'_ν depend, according to equation (27), only on the temperature, the frequency ν, and the nature of the two substances, and these values of the intensities of radiation hold up to very small distances from the bounding surface of the substances, and hence are entirely independent of the properties of this surface.

36. We shall now suppose, to begin with, that the bounding surface

of the media is smooth (Sec. 9). Then every ray coming from the first medium and falling on the bounding surface is divided into two rays, the reflected and the transmitted ray. The directions of these two rays vary with the angle of incidence and the color of the incident ray; the intensity also varies with its polarization. Let us denote by ρ (coefficient of reflection) the fraction of the energy reflected, then the fraction transmitted is $(1-\rho)$, ρ depending on the angle of incidence, the frequency, and the polarization of the incident ray. Similar remarks apply to ρ' the coefficient of reflection of a ray coming from the second medium and falling on the bounding surface.

Now according to (11) we have for the monochromatic plane polarized radiation of frequency ν, emitted in time dt toward the first medium (in the direction of the feathered arrow upper left hand in Fig. 3), from an element $d\sigma$ of the bounding surface and contained in the conical element $d\Omega$,

$$dt\, d\sigma\, \cos\theta\, d\Omega\, \mathsf{K}_\nu\, d\nu, \tag{34}$$

where

$$d\Omega = \sin\theta\, d\theta\, d\phi. \tag{35}$$

This energy is supplied by the two rays which come from the first and the second medium and are respectively reflected from or transmitted by the element $d\sigma$ in the corresponding direction (the unfeathered arrows). (Of the element $d\sigma$ only the one point O is indicated.) The first ray, according to the law of reflection, continues in the symmetrically situated conical element $d\Omega$, the second in the conical element

$$d\Omega' = \sin\theta'\, d\theta'\, d\phi' \tag{36}$$

where, according to the law of refraction,

$$\phi' = \phi \quad \text{and} \quad \frac{\sin\theta}{\sin\theta'} = \frac{q}{q'}. \tag{37}$$

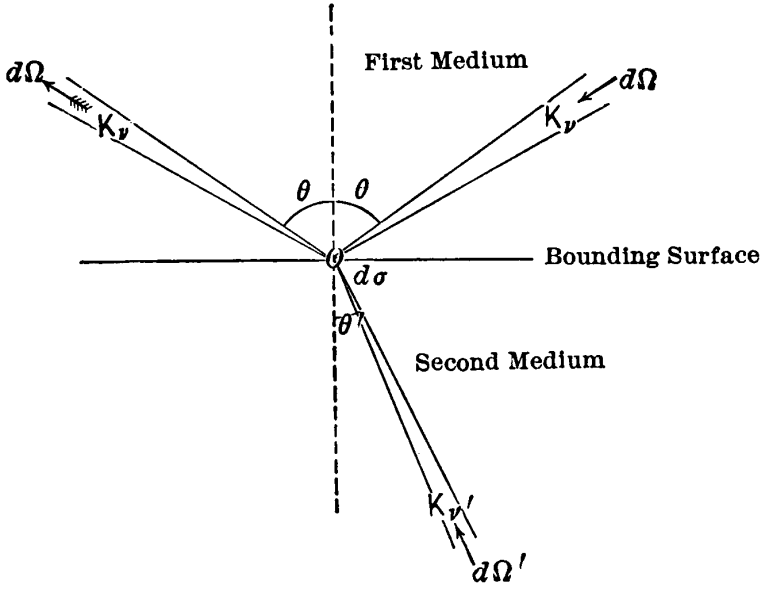


FIG. 3.

If we now assume the radiation (34) to be polarized either in the plane of incidence or at right angles thereto, the same will be true for the two radiations of which it consists, and the radiation coming from the first medium and reflected from $d\sigma$ contributes the part

$$\rho\, dt\, d\sigma\, \cos\theta\, d\Omega\, \mathsf{K}_\nu\, d\nu \tag{38}$$

while the radiation coming from the second medium and transmitted through $d\sigma$ contributes the part

$$(1-\rho')\, dt\, d\sigma\, \cos\theta'\, d\Omega'\, \mathsf{K}'_\nu\, d\nu. \tag{39}$$

The quantities dt, $d\sigma$, ν and $d\nu$ are written without the accent, because they have the same values in both media.

By adding (38) and (39) and equating their sum to the expression (34) we find

$$\rho \cos\theta \, d\Omega \, \mathsf{K}_\nu + (1-\rho') \cos\theta' \, d\Omega' \, \mathsf{K}'_\nu = \cos\theta \, d\Omega \, \mathsf{K}_\nu .$$

Now from (37) we have

$$\frac{\cos\theta \, d\theta}{q} = \frac{\cos\theta' \, d\theta'}{q'}$$

and further by (35) and (36)

$$d\Omega' \, \cos\theta' = \frac{d\Omega \, \cos\theta \, q'^2}{q^2}.$$

Therefore we find

$$\rho \mathsf{K}_\nu + (1-\rho') \frac{q'^2}{q^2} \mathsf{K}'_\nu = \mathsf{K}$$

or

$$\frac{\mathsf{K}_\nu}{\mathsf{K}'_\nu} \cdot \frac{q^2}{q'^2} = \frac{1-\rho'}{1-\rho}.$$

37. In the last equation the quantity on the left side is independent of the angle of incidence θ and of the particular kind of polarization; hence the same must be true for the right side. Hence, whenever the value of this quantity is known for a single angle of incidence and any definite kind of polarization, this value will remain valid for all angles of incidence and all kinds of polarization. Now in the special case when the rays are polarized at right angles to the plane of incidence and strike the bounding surface at the angle of polarization, $\rho = 0$, and $\rho' = 0$. The expression on the right side of the last equation then becomes 1; hence it must always be 1 and we have the general relations:

$$\rho = \rho' \tag{40}$$

and

$$q^2\mathsf{K}_\nu = q'^2\mathsf{K}'_\nu. \tag{41}$$

38. The first of these two relations, which states that the coefficient of reflection of the bounding surface is the same on both sides, is a special case of a general law of reciprocity first stated by *Helmholtz*.[1] According to this law the loss of intensity which a ray of definite color and polarization suffers on its way through any media by reflection, refraction, absorption, and scattering is exactly equal to the loss suffered by a ray of the same intensity, color, and polarization pursuing an exactly opposite path. An immediate consequence of this law is that the radiation striking the bounding surface of any two media is always transmitted as well as reflected equally on both sides, for every color, direction, and polarization.

39. The second formula (41) establishes a relation between the intensities of radiation in the two media, for it states that, when thermodynamic equilibrium exists, *the specific intensities of radiation of a certain frequency in the two media are in the inverse ratio of the squares of the velocities of propagation or in the direct ratio of the squares of the indices of refraction.*[2]

By substituting for K_ν its value from (27) we obtain the following theorem: *The quantity*

$$q^2\mathsf{K}_\nu = q^2\frac{\epsilon_\nu}{\alpha_\nu} \tag{42}$$

[1] *H. v. Helmholtz*, Handbuch d. physiologischen Optik 1. Lieferung, Leipzig, Leop. Voss, 1856, p. 169. See also *Helmholtz*, Vorlesungen über die Theorie der Wärme herausgegeben von *F. Richarz*, Leipzig, J. A. Barth, 1903, p. 161. The restrictions of the law of reciprocity made there do not bear on our problems, since we are concerned with temperature radiation only (Sec. 7).

[2] *G. Kirchhoff*, Gesammelte Abhandlungen, Leipzig, J. A. Barth, 1882, p. 594. *R. Clausius*, Pogg. Ann. **121**, p. 1, 1864.

does not depend on the nature of the substance, and is, therefore, a universal function of the temperature T and the frequency ν alone.

The great importance of this law lies evidently in the fact that it states a property of radiation which is the same for all bodies in nature, and which need be known only for a single arbitrarily chosen body, in order to be stated quite generally for all bodies. We shall later on take advantage of the opportunity offered by this statement in order actually to calculate this universal function (Sec. 165).

40. We now consider the other case, that in which the bounding surface of the two media is rough. This case is much more general than the one previously treated, inasmuch as the energy of a pencil directed from an element of the bounding surface into the first medium is no longer supplied by two definite pencils, but by an arbitrary number, which come from both media and strike the surface. Here the actual conditions may be very complicated according to the peculiarities of the bounding surface, which moreover may vary in any way from one element to another. However, according to Sec. 35, the values of the specific intensities of radiation K_ν and K'_ν remain always the same in all directions in both media, just as in the case of a smooth bounding surface. That this condition, necessary for thermodynamic equilibrium, is satisfied is readily seen from *Helmholtz's* law of reciprocity, according to which, in the case of stationary radiation, for each ray striking the bounding surface and diffusely reflected from it on both sides, there is a corresponding ray at the same point, of the same intensity and opposite direction, produced by the inverse process at the same point on the bounding surface, namely by the gathering of diffusely incident rays into a definite direction, just as is the case in the interior of each of the two media.

41. We shall now further generalize the laws obtained. First, just as in Sec. 34, the assumption made above, namely, that the two media

extend to a great distance, may be abandoned since we may introduce an arbitrary number of bounding surfaces without disturbing the thermodynamic equilibrium. Thereby we are placed in a position enabling us to pass at once to the case of any number of substances of any size and shape. For when a system consisting of an arbitrary number of contiguous substances is in the state of thermodynamic equilibrium, the equilibrium is in no way disturbed, if we assume one or more of the surfaces of contact to be wholly or partly impermeable to heat. Thereby we can always reduce the case of any number of substances to that of two substances in an enclosure impermeable to heat, and, therefore, the law may be stated quite generally, that, when any arbitrary system is in the state of thermodynamic equilibrium, the specific intensity of radiation K_ν is determined in each separate substance by the universal function (42).

42. We shall now consider a system in a state of thermodynamic equilibrium, contained within an enclosure impermeable to heat and consisting of n emitting and absorbing adjacent bodies of any size and shape whatever. As in Sec. 36, we again confine our attention to a monochromatic plane polarized pencil which proceeds from an element $d\sigma$ of the bounding surface of the two media in the direction toward the first medium (Fig. 3, feathered arrow) within the conical element $d\Omega$. Then, as in (34), the energy supplied by the pencil in unit time is

$$d\sigma \cos\theta \, d\Omega \, \mathsf{K}_\nu \, d\nu = I. \tag{43}$$

This energy of radiation I consists of a part coming from the first medium by regular or diffuse reflection at the bounding surface and of a second part coming through the bounding surface from the second medium. We shall, however, not stop at this mode of division, but shall further subdivide I according to that one of the n media from which

the separate parts of the radiation I have been emitted. This point of view is distinctly different from the preceding, since, *e.g.*, the rays transmitted from the second medium through the bounding surface into the pencil considered have not necessarily been emitted in the second medium, but may, according to circumstances, have traversed a long and very complicated path through different media and may have undergone therein the effect of refraction, reflection, scattering, and partial absorption any number of times. Similarly the rays of the pencil, which coming from the first medium are reflected at $d\sigma$, were not necessarily all emitted in the first medium. It may even happen that a ray emitted from a certain medium, after passing on its way through other media, returns to the original one and is there either absorbed or emerges from this medium a second time.

We shall now, considering all these possibilities, denote that part of I which has been emitted by volume-elements of the first medium by I_1 no matter what paths the different constituents have pursued, that which has been emitted by volume-elements of the second medium by I_2, etc. Then since every part of I must have been emitted by an element of some body, the following equation must hold,

$$I = I_1 + I_2 + I_3 + \ldots I_n. \tag{44}$$

43. The most adequate method of acquiring more detailed information as to the origin and the paths of the different rays of which the radiations I_1, I_2, I_3, $\ldots I_n$ consist, is to pursue the opposite course and to inquire into the future fate of that pencil, which travels exactly in the opposite direction to the pencil I and which therefore comes from the first medium in the cone $d\Omega$ and falls on the surface element $d\sigma$ of the second medium. For since every optical path may also be traversed in the opposite direction, we may obtain by this consideration all paths along which rays can pass into the pencil I, however complicated they

may otherwise be. Let J represent the intensity of this inverse pencil, which is directed toward the bounding surface and is in the same state of polarization. Then, according to Sec. 40,

$$J = I. \tag{45}$$

At the bounding surface $d\sigma$ the rays of the pencil J are partly reflected and partly transmitted regularly or diffusely, and thereafter, travelling in both media, are partly absorbed, partly scattered, partly again reflected or transmitted to different media, etc., according to the configuration of the system. But finally the whole pencil J after splitting into many separate rays will be completely absorbed in the n media. Let us denote that part of J which is finally absorbed in the first medium by J_1, that which is finally absorbed in the second medium by J_2, etc., then we shall have

$$J = J_1 + J_2 + J_3 + \cdots + J_n.$$

Now the volume-elements of the n media, in which the absorption of the rays of the pencil J takes place, are precisely the same as those in which takes place the emission of the rays constituting the pencil I, the first one considered above. For, according to *Helmholtz's* law of reciprocity, no appreciable radiation of the pencil J can enter a volume-element which contributes no appreciable radiation to the pencil I and *vice versa*.

Let us further keep in mind that the absorption of each volume-element is, according to (42), proportional to its emission and that, according to *Helmholtz's* law of reciprocity, the decrease which the energy of a ray suffers on any path is always equal to the decrease suffered by the energy of a ray pursuing the opposite path. It will then be clear that the volume-elements considered absorb the rays of the pencil J in

just the same ratio as they contribute by their emission to the energy of the opposite pencil I. Since, moreover, the sum I of the energies given off by emission by all volume-elements is equal to the sum J of the energies absorbed by all elements, the quantity of energy absorbed by each separate volume-element from the pencil J must be equal to the quantity of energy emitted by the same element into the pencil I. In other words: *the part of a pencil I which has been emitted from a certain volume of any medium is equal to the part of the pencil J $(= I)$ oppositely directed, which is absorbed in the same volume.*

Hence not only are the sums I and J equal, but their constituents are also separately equal or

$$J_1 = I_1, \quad J_2 = I_2, \ \ldots \quad J_n = I_n. \tag{46}$$

44. Following *G. Kirchhoff*[1] we call the quantity I_2, *i.e.*, the intensity of the pencil emitted from the second medium into the first, the *emissive power* E of the second medium, while we call the ratio of J_2 to J, *i.e.*, that fraction of a pencil incident on the second medium which is absorbed in this medium, the *absorbing power* A of the second medium. Therefore

$$E = I_2 \ (\leqq I), \quad A = \frac{J_2}{J} \ (\leqq 1). \tag{47}$$

The quantities E and A depend (a) on the nature of the two media, (b) on the temperature, the frequency ν, and the direction and the polarization of the radiation considered, (c) on the nature of the bounding surface and on the magnitude of the surface element $d\sigma$ and that of the solid angle $d\Omega$, (d) on the geometrical extent and the shape of the total surface of the two media, (e) on the nature and form of all

[1] *G. Kirchhoff*, Gesammelte Abhandlungen, 1882, p. 574.

other bodies of the system. For a ray may pass from the first into the second medium, be partly transmitted by the latter, and then, after reflection somewhere else, may return to the second medium and may be there entirely absorbed.

With these assumptions, according to equations (46), (45), and (43), *Kirchhoff's* law holds,

$$\frac{E}{A} = I = d\sigma \cos\theta \, d\Omega \, \mathsf{K}_\nu \, d\nu, \tag{48}$$

i.e., the ratio of the emissive power to the absorbing power of any body is independent of the nature of the body. For this ratio is equal to the intensity of the pencil passing through the *first* medium, which, according to equation (27), does not depend on the second medium at all. The value of this ratio does, however, depend on the nature of the first medium, inasmuch as, according to (42), it is not the quantity K_ν but the quantity $q^2\mathsf{K}_\nu$, which is a universal function of the temperature and frequency. The proof of this law given by *G. Kirchhoff* l. c. was later greatly simplified by *E. Pringsheim.*[1]

45. When in particular the second medium is a black body (Sec. 10) it absorbs all the incident radiation. Hence in that case $J_2 = J$, $A = 1$, and $E = I$, *i.e., the emissive power of a black body is independent of its nature. Its emissive power is larger than that of any other body at the same temperature and, in fact, is just equal to the intensity of radiation in the contiguous medium.*

46. We shall now add, without further proof, another general law of reciprocity, which is closely connected with that stated at the end of Sec. 43 and which may be stated thus: *When any emitting and absorbing bodies are in the state of thermodynamic equilibrium, the part*

[1] *E. Pringsheim*, Verhandlungen der Deutschen Physikalischen Gesellschaft, **3**, p. 81, 1901.

of the energy of definite color emitted by a body A, which is absorbed by another body B, is equal to the part of the energy of the same color emitted by B which is absorbed by A. Since a quantity of energy emitted causes a decrease of the heat of the body, and a quantity of energy absorbed an increase of the heat of the body, it is evident that, when thermodynamic equilibrium exists, any two bodies or elements of bodies selected at random exchange by radiation equal amounts of heat with each other. Here, of course, care must be taken to distinguish between the radiation emitted and the total radiation which reaches one body from the other.

47. The law holding for the quantity (42) can be expressed in a different form, by introducing, by means of (24), the volume density u_ν of monochromatic radiation instead of the intensity of radiation K_ν. We then obtain the law that, for radiation in a state of thermodynamic equilibrium, the quantity

$$\mathsf{u}_\nu q^3 \tag{49}$$

is a function of the temperature T and the frequency ν, and is the same for all substances.[1] This law becomes clearer if we consider that the quantity

$$\mathsf{u}_\nu \, d\nu \, \frac{q^3}{\nu^3} \tag{50}$$

also is a universal function of T, ν, and $\nu + d\nu$, and that the product $\mathsf{u}_\nu \, d\nu$ is, according to (22), the volume density of the radiation whose frequency lies between ν and $\nu + d\nu$, while the quotient $\frac{q}{\nu}$ represents the wave length of a ray of frequency ν in the medium in question. The law then takes the following simple form: *When any bodies whatever are in*

[1] In this law it is assumed that the quantity q in (24) is the same as in (37). This does not hold for strongly dispersing or absorbing substances. For the generalization applying to such cases see *M. Laue*, Annalen d. Physik **32**, p. 1085, 1910.

thermodynamic equilibrium, the energy of monochromatic radiation of a definite frequency, contained in a cubical element of side equal to the wave length, is the same for all bodies.

48. We shall finally take up the case of diathermanous (Sec. 12) media, which has so far not been considered. In Sec. 27 we saw that, in a medium which is diathermanous for a given color and is surrounded by an enclosure impermeable to heat, there can be thermodynamic equilibrium for any intensity of radiation of this color. There must, however, among all possible intensities of radiation be a definite one, corresponding to the absolute maximum of the total entropy of the system, which designates the absolutely stable equilibrium of radiation. In fact, in equation (27) the intensity of radiation K_ν for $\alpha_\nu = 0$ and $\epsilon_\nu = 0$ assumes the value $\frac{0}{0}$, and hence cannot be calculated from this equation. But we see also that this indeterminateness is removed by equation (41), which states that in the case of thermodynamic equilibrium the product $q^2\mathsf{K}_\nu$ has the same value for all substances. From this we find immediately a definite value of K_ν which is thereby distinguished from all other values. Furthermore the physical significance of this value is immediately seen by considering the way in which that equation was obtained. It is that intensity of radiation which exists in a diathermanous medium, if it is in thermodynamic equilibrium when in contact with an arbitrary absorbing and emitting medium. The volume and the form of the second medium do not matter in the least, in particular the volume may be taken as small as we please. Hence we can formulate the following law: *Although generally speaking thermodynamic equilibrium can exist in a diathermanous medium for any intensity of radiation whatever, nevertheless there exists in every diathermanous medium for a definite frequency at a definite temperature an intensity of radiation defined by the universal function (42). This may be called*

the stable intensity, inasmuch as it will always be established, when the medium is exchanging stationary radiation with an arbitrary emitting and absorbing substance.

49. According to the law stated in Sec. 45, the intensity of a pencil, when a state of stable heat radiation exists in a diathermanous medium, is equal to the emissive power E of a black body in contact with the medium. On this fact is based the possibility of measuring the emissive power of a black body, although absolutely black bodies do not exist in nature.[1] A diathermanous cavity is enclosed by strongly emitting walls[2] and the walls kept at a certain constant temperature T. Then the radiation in the cavity, when thermodynamic equilibrium is established for every frequency ν, assumes the intensity corresponding to the velocity of propagation q in the diathermanous medium, according to the universal function (42). Then any element of area of the walls radiates into the cavity just as if the wall were a black body of temperature T. The amount lacking in the intensity of the rays actually emitted by the walls as compared with the emission of a black body is supplied by rays which fall on the wall and are reflected there. Similarly every element of area of a wall receives the same radiation.

In fact, the radiation I starting from an element of area of a wall consists of the radiation E emitted by the element of area and of the radiation reflected from the element of area from the incident radiation I, *i.e.*, the radiation which is not absorbed $(1 - A)I$. We have, therefore, in agreement with *Kirchhoff's* law (48),

$$I = E + (1 - A)I.$$

[1] *W. Wien* and *O. Lummer*, Wied. Annalen, **56**, p. 451, 1895.

[2] The strength of the emission influences only the time required to establish stationary radiation, but not its character. It is essential, however, that the walls transmit no radiation to the exterior.

If we now make a hole in one of the walls of a size $d\sigma$, so small that the intensity of the radiation directed toward the hole is not changed thereby, then radiation passes through the hole to the exterior where we shall suppose there is the same diathermanous medium as within. This radiation has exactly the same properties as if $d\sigma$ were the surface of a black body, and this radiation may be measured for every color together with the temperature T.

50. Thus far all the laws derived in the preceding sections for diathermanous media hold for a definite frequency, and it is to be kept in mind that a substance may be diathermanous for one color and adiathermanous for another. Hence the radiation of a medium completely enclosed by absolutely reflecting walls is, when thermodynamic equilibrium has been established for all colors for which the medium has a finite coefficient of absorption, always the stable radiation corresponding to the temperature of the medium such as is represented by the emission of a black body. Hence this is briefly called "black" radiation.[1] On the other hand, the intensity of colors for which the medium is diathermanous is not necessarily the stable black radiation, unless the medium is in a state of stationary exchange of radiation with an absorbing substance.

There is but one medium that is diathermanous for all kinds of rays, namely, the absolute vacuum, which to be sure cannot be produced in nature except approximately. However, most gases, *e.g.*, the air of the atmosphere, have, at least if they are not too dense, to a sufficient approximation the optical properties of a vacuum with respect to waves of not too short length. So far as this is the case the velocity

[1] *M. Thiesen*, Verhandlungen d. Deutschen Physikal. Gesellschaft, **2**, p. 65, 1900.

of propagation q may be taken as the same for all frequencies, namely,

$$c = 3 \times 10^{10} \frac{\text{cm}}{\text{sec}}. \tag{51}$$

51. Hence in a vacuum bounded by totally reflecting walls any state of radiation may persist. But as soon as an arbitrarily small quantity of matter is introduced into the vacuum, a stationary state of radiation is gradually established. In this the radiation of every color which is appreciably absorbed by the substance has the intensity K_ν corresponding to the temperature of the substance and determined by the universal function (42) for $q = c$, the intensity of radiation of the other colors remaining indeterminate. If the substance introduced is not diathermanous for any color, *e.g.*, a piece of carbon however small, there exists at the stationary state of radiation in the whole vacuum for all colors the intensity K_ν of black radiation corresponding to the temperature of the substance. The magnitude of K_ν regarded as a function of ν gives the spectral distribution of black radiation in a vacuum, or the so-called *normal energy spectrum*, which depends on nothing but the temperature. In the normal spectrum, since it is the spectrum of emission of a black body, the intensity of radiation of every color is the largest which a body can emit at that temperature at all.

52. It is therefore possible to change a perfectly arbitrary radiation, which exists at the start in the evacuated cavity with perfectly reflecting walls under consideration, into black radiation by the introduction of a minute particle of carbon. The characteristic feature of this process is that the heat of the carbon particle may be just as small as we please, compared with the energy of radiation contained in the cavity of arbitrary magnitude. Hence, according to the principle of the conservation of energy, the total energy of radiation remains essentially constant during the change that takes place, because the changes in

the heat of the carbon particle may be entirely neglected, even if its changes in temperature should be finite. Herein the carbon particle exerts only a releasing (auslösend) action. Thereafter the intensities of the pencils of different frequencies originally present and having different frequencies, directions, and different states of polarization change at the expense of one another, corresponding to the passage of the system from a less to a more stable state of radiation or from a state of smaller to a state of larger entropy. From a thermodynamic point of view this process is perfectly analogous, since the time necessary for the process is not essential, to the change produced by a minute spark in a quantity of oxy-hydrogen gas or by a small drop of liquid in a quantity of supersaturated vapor. In all these cases the magnitude of the disturbance is exceedingly small and cannot be compared with the magnitude of the energies undergoing the resultant changes, so that in applying the two principles of thermodynamics the cause of the disturbance of equilibrium, *viz.* the carbon particle, the spark, or the drop, need not be considered. It is always a case of a system passing from a more or less unstable into a more stable state, wherein, according to the first principle of thermodynamics, the energy of the system remains constant, and, according to the second principle, the entropy of the system increases.

PART II

DEDUCTIONS FROM ELECTRODYNAMICS AND THERMODYNAMICS

CHAPTER I

MAXWELL'S RADIATION PRESSURE

53. While in the preceding part the phenomena of radiation have been presented with the assumption of only well known elementary laws of optics summarized in Sec. 2, which are common to all optical theories, we shall hereafter make use of the electromagnetic theory of light and shall begin by deducing a consequence characteristic of that theory. We shall, namely, calculate the magnitude of the mechanical force, which is exerted by a light or heat ray passing through a vacuum on striking a reflecting (Sec. 10) surface assumed to be at rest.

For this purpose we begin by stating Maxwell's general equations for an electromagnetic process in a vacuum. Let the vector E denote the electric field-strength (intensity of the electric field) in electric units and the vector H the magnetic field-strength in magnetic units. Then the equations are, in the abbreviated notation of the vector calculus,

$$\begin{aligned} \dot{\mathsf{E}} &= c\,\mathrm{curl}\,\mathsf{H} & \dot{\mathsf{H}} &= -c\,\mathrm{curl}\,\mathsf{E} \\ \mathrm{div.}\,\mathsf{E} &= 0 & \mathrm{div.}\,\mathsf{H} &= 0. \end{aligned} \tag{52}$$

Should the reader be unfamiliar with the symbols of this notation, he may readily deduce their meaning by working backward from the subsequent equations (53).

54. In order to pass to the case of a plane wave in any direction we assume that all the quantities that fix the state depend only on the time t and on one of the coordinates x', y', z', of an orthogonal right-handed system of coordinates, say on x'. Then the equations (52)

reduce to

$$\begin{aligned}
\frac{\partial \mathsf{E}_{x'}}{\partial t} &= 0 & \frac{\partial \mathsf{H}_{x'}}{\partial t} &= 0 \\
\frac{\partial \mathsf{E}_{y'}}{\partial t} &= -c\frac{\partial \mathsf{H}_{z'}}{\partial x'} & \frac{\partial \mathsf{H}_{y'}}{\partial t} &= \quad c\frac{\partial \mathsf{E}_{z'}}{\partial x'} \\
\frac{\partial \mathsf{E}_{x'}}{\partial t} &= \quad c\frac{\partial \mathsf{H}_{y'}}{\partial x'} & \frac{\partial \mathsf{H}_{z'}}{\partial t} &= -c\frac{\partial \mathsf{E}_{y'}}{\partial x'} \\
\frac{\partial \mathsf{E}_{x'}}{\partial x'} &= 0 & \frac{\partial \mathsf{H}_{x'}}{\partial x'} &= 0.
\end{aligned} \tag{53}$$

Hence the most general expression for a plane wave passing through a vacuum in the direction of the positive x'-axis is

$$\begin{aligned}
\mathsf{E}_{x'} &= 0 & \mathsf{H}_{x'} &= 0 \\
\mathsf{E}_{y'} &= f\left(t - \frac{x'}{c}\right) & \mathsf{H}_{y'} &= -g\left(t - \frac{x'}{c}\right) \\
\mathsf{E}_{z'} &= g\left(t - \frac{x'}{c}\right) & \mathsf{H}_{z'} &= \quad f\left(t - \frac{x'}{c}\right)
\end{aligned} \tag{54}$$

where f and g represent two arbitrary functions of the same argument.

55. Suppose now that this wave strikes a reflecting surface, *e.g.*, the surface of an absolute conductor (metal) of infinitely large conductivity. In such a conductor even an infinitely small electric field-strength produces a finite conduction current; hence the electric field-strength E in it must be always and everywhere infinitely small. For simplicity we also suppose the conductor to be non-magnetizable, *i.e.*, we assume the magnetic induction B in it to be equal to the magnetic field-strength H, just as is the case in a vacuum.

If we place the x-axis of a right-handed coordinate system (xyz) along the normal of the surface directed toward the interior of the conductor, the x-axis is the normal of incidence. We place the $(x'y')$ plane

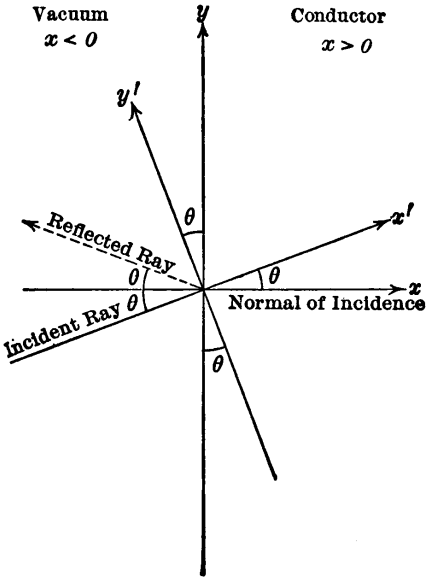


FIG. 4.

in the plane of incidence and take this as the plane of the figure (Fig. 4). Moreover, we can also, without any restriction of generality, place the y-axis in the plane of the figure, so that the z-axis coincides with the z'-axis (directed from the figure toward the observer). Let the common origin O of the two coordinate systems lie in the surface. If finally θ represents the angle of incidence, the coordinates with and without accent are related to each other by the following equations:

$$\begin{aligned} x &= x'\cos\theta - y'\sin\theta & \qquad x' &= x\cos\theta + y\sin\theta \\ y &= x'\sin\theta + y'\cos\theta & \qquad y' &= -x\sin\theta + y\cos\theta \\ z &= z' & \qquad z' &= z. \end{aligned}$$

By the same transformation we may pass from the components of the electric or magnetic field-strength in the first coordinate system to their components in the second system. Performing this transformation the following values are obtained from (54) for the components of the electric and magnetic field-strengths of the incident wave in the coordinate system without accent,

$$\begin{aligned} \mathsf{E}_x &= -\sin\theta \cdot f & \mathsf{H}_x &= \sin\theta \cdot g \\ \mathsf{E}_y &= \cos\theta \cdot f & \mathsf{H}_y &= -\cos\theta \cdot g \\ \mathsf{E}_z &= g & \mathsf{H}_z &= f. \end{aligned} \tag{55}$$

Herein the argument of the functions f and g is

$$t - \frac{x'}{c} = t - \frac{x\cos\theta + y\sin\theta}{c}. \tag{56}$$

56. In the surface of separation of the two media $x = 0$. According to the general electromagnetic boundary conditions the components of the field-strengths in the surface of separation, *i.e.*, the four quantities E_y, E_z, H_y, H_z must be equal to each other on the two sides of the surface of separation for this value of x. In the conductor the electric field-strength E is infinitely small in accordance with the assumption made above. Hence E_y and E_z must vanish also in the vacuum for $x = 0$. This condition cannot be satisfied unless we assume in the vacuum, besides the incident, also a reflected wave superposed on the former in such a way that the components of the electric field of the two waves in the y and z direction just cancel at every instant and at every point in the surface of separation. By this assumption and the condition that the reflected wave is a plane wave returning into the interior of the vacuum, the other four components of the reflected wave are also completely determined. They are all functions of the single

argument

$$t - \frac{-x\cos\theta + y\sin\theta}{c}. \tag{57}$$

The actual calculation yields as components of the total electromagnetic field produced in the vacuum by the superposition of the two waves, the following expressions valid for points of the surface of separation $x = 0$,

$$\begin{aligned}
\mathsf{E}_x &= -\sin\theta \cdot f - \sin\theta \cdot f = -2\sin\theta \cdot f \\
\mathsf{E}_y &= \quad \cos\theta \cdot f - \cos\theta \cdot f = 0 \\
\mathsf{E}_z &= g - g = 0 \\
\mathsf{H}_x &= \quad \sin\theta \cdot g - \sin\theta \cdot g = 0 \\
\mathsf{H}_y &= -\cos\theta \cdot g - \cos\theta \cdot g = -2\cos\theta \cdot g \\
\mathsf{H}_z &= f + f = 2f.
\end{aligned} \tag{58}$$

In these equations the argument of the functions f and g is, according to (56) and (57),

$$t - \frac{y\sin\theta}{c}.$$

From these values the electric and magnetic field-strength within the conductor in the immediate neighborhood of the separating surface $x = 0$ is obtained:

$$\begin{aligned}
\mathsf{E}_x &= 0 \qquad & \mathsf{H}_x &= 0 \\
\mathsf{E}_y &= 0 \qquad & \mathsf{H}_y &= -2\cos\theta \cdot g \\
\mathsf{E}_z &= 0 \qquad & \mathsf{H}_z &= 2f
\end{aligned} \tag{59}$$

where again the argument $t - \frac{y\sin\theta}{c}$ is to be substituted in the functions f and g. For the components of E all vanish in an absolute conductor and the components H_x, H_y, H_z are all continuous at the separating

surface, the two latter since they are tangential components of the field-strength, the former since it is the normal component of the magnetic induction B (Sec. 55), which likewise remains continuous on passing through any surface of separation.

On the other hand, the normal component of the electric field-strength E_x is seen to be discontinuous; the discontinuity shows the existence of an electric charge on the surface, the surface density of which is given in magnitude and sign as follows:

$$\frac{1}{4\pi} 2 \sin\theta \cdot f = \frac{1}{2\pi} \sin\theta \cdot f. \tag{60}$$

In the interior of the conductor at a finite distance from the bounding surface, *i.e.*, for $x > 0$, all six field components are infinitely small. Hence, on increasing x, the values of H_y and H_z, which are finite for $x = 0$, approach the value 0 at an infinitely rapid rate.

57. A certain mechanical force is exerted on the substance of the conductor by the electromagnetic field considered. We shall calculate the component of this force normal to the surface. It is partly of electric, partly of magnetic, origin. Let us first consider the former, F_e. Since the electric charge existing on the surface of the conductor is in an electric field, a mechanical force equal to the product of the charge and the field-strength is exerted on it. Since, however, the field-strength is discontinuous, having the value $-2\sin\theta f$ on the side of the vacuum and 0 on the side of the conductor, from a well-known law of electrostatics the magnitude of the mechanical force F_e acting on an element of surface $d\sigma$ of the conductor is obtained by multiplying the electric charge of the element of area calculated in (60) by the arithmetic mean of the electric field-strength on the two sides. Hence

$$\mathsf{F}_e = \frac{\sin\theta}{2\pi} f \, d\sigma \, (-\sin\theta f) = -\frac{\sin^2\theta}{2\pi} f^2 \, d\sigma.$$

This force acts in the direction toward the vacuum and therefore exerts a tension.

58. We shall now calculate the mechanical force of magnetic origin F_m. In the interior of the conducting substance there are certain conduction currents, whose intensity and direction are determined by the vector I of the current density

$$\mathsf{I} = \frac{c}{4\pi} \operatorname{curl} \mathsf{H}. \tag{61}$$

A mechanical force acts on every element of space $d\tau$ of the conductor through which a conduction current flows, and is given by the vector product

$$\frac{d\tau}{c}[\mathsf{I} \times \mathsf{H}]. \tag{62}$$

Hence the component of this force normal to the surface of the conductor $x = 0$ is equal to

$$\frac{d\tau}{c}(\mathsf{I}_y \mathsf{H}_z - \mathsf{I}_z \mathsf{H}_y).$$

On substituting the values of I_y and I_z from (61) we obtain

$$\frac{d\tau}{4\pi}\left[\mathsf{H}_z\left(\frac{\partial \mathsf{H}_x}{\partial z} - \frac{\partial \mathsf{H}_z}{\partial x}\right) - \mathsf{H}_y\left(\frac{\partial \mathsf{H}_y}{\partial x} - \frac{\partial \mathsf{H}_x}{\partial y}\right)\right].$$

In this expression the differential coefficients with respect to y and z are negligibly small in comparison to those with respect to x, according to the remark at the end of Sec. 56; hence the expression reduces to

$$-\frac{d\tau}{4\pi}\left(\mathsf{H}_y \frac{\partial \mathsf{H}_y}{\partial x} + \mathsf{H}_z \frac{\partial \mathsf{H}_z}{\partial x}\right).$$

Let us now consider a cylinder cut out of the conductor perpendicular to the surface with the cross-section $d\sigma$, and extending from $x = 0$ to

$x = \infty$. The entire mechanical force of magnetic origin acting on this cylinder in the direction of the x-axis, since $d\tau = d\sigma\, x$, is given by

$$\mathsf{F}_m = -\frac{d\sigma}{4\pi}\int_0^\infty dx\left(\mathsf{H}_y\frac{\partial \mathsf{H}_y}{\partial x} + \mathsf{H}_z\frac{\partial \mathsf{H}_z}{\partial x}\right).$$

On integration, since H vanishes for $x = \infty$, we obtain

$$\mathsf{F}_m = \frac{d\sigma}{8\pi}\left(\mathsf{H}_y^2 + \mathsf{H}_z^2\right)_{x=0}$$

or by equation (59)

$$\mathsf{F}_m = \frac{d\sigma}{2\pi}(\cos^2\theta \cdot g^2 + f^2).$$

By adding F_e and F_m the total mechanical force acting on the cylinder in question in the direction of the x-axis is found to be

$$\mathsf{F} = \frac{d\sigma}{2\pi}\cos^2\theta(f^2 + g^2). \tag{63}$$

This force exerts on the surface of the conductor a pressure, which acts in a direction normal to the surface toward the interior and is called "*Maxwell's* radiation pressure." The existence and the magnitude of the radiation pressure as predicted by the theory was first found by delicate measurements with the radiometer by *P. Lebedew.*[1]

59. We shall now establish a relation between the radiation pressure and the energy of radiation $I\,dt$ falling on the surface element $d\sigma$ of the conductor in a time element dt. The latter from *Poynting's* law of energy flow is

$$I\,dt = \frac{c}{4\pi}(\mathsf{E}_y\mathsf{H}_z - \mathsf{E}_z\mathsf{H}_y)\,d\sigma\,dt,$$

[1] *P. Lebedew*, Annalen d. Phys. **6**, p. 433, 1901. See also *E. F. Nichols* and *G. F. Hull*, Annalen d. Phys. **12**, p. 225, 1903.

hence from (55)

$$I\,dt = \frac{c}{4\pi}\cos\theta(f^2+g^2)\,d\sigma\,dt.$$

By comparison with (63) we obtain

$$\mathsf{F} = \frac{2\cos\theta}{c}I. \tag{64}$$

From this we finally calculate the total pressure p, *i.e.*, that mechanical force, which an arbitrary radiation proceeding from the vacuum and totally reflected upon incidence on the conductor exerts in a normal direction on a unit surface of the conductor. The energy radiated in the conical element

$$d\Omega = \sin\theta\,d\theta\,d\phi$$

in the time dt on the element of area $d\sigma$ is, according to (6),

$$I\,dt = K\cos\theta\,d\Omega\,d\sigma\,dt,$$

where K represents the specific intensity of the radiation in the direction $d\Omega$ toward the reflector. On substituting this in (64) and integrating over $d\Omega$ we obtain for the total pressure of all pencils which fall on the surface and are reflected by it

$$p = \frac{2}{c}\int K\cos^2\theta\,d\Omega, \tag{65}$$

the integration with respect to ϕ extending from 0 to 2π and with respect to θ from 0 to $\frac{\pi}{2}$.

In case K is independent of direction as in the case of black radiation, we obtain for the radiation pressure

$$p = \frac{2K}{c}\int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta\cos^2\theta\sin\theta = \frac{4\pi K}{3c}$$

or, if we introduce instead of K the volume density of radiation u from (21)

$$p = \frac{u}{3}. \tag{66}$$

This value of the radiation pressure holds only when the reflection of the radiation occurs at the surface of an absolute non-magnetizable conductor. Therefore we shall in the thermodynamic deductions of the next chapter make use of it only in such cases. Nevertheless it will be shown later on (Sec. 66) that equation (66) gives the pressure of uniform radiation against any totally reflecting surface, no matter whether it reflects uniformly or diffusely.

60. In view of the extraordinarily simple and close relation between the radiation pressure and the energy of radiation, the question might be raised whether this relation is really a special consequence of the electromagnetic theory, or whether it might not, perhaps, be founded on more general energetic or thermodynamic considerations. To decide this question we shall calculate the radiation pressure that would follow by Newtonian mechanics from *Newton's* (emission) theory of light, a theory which, in itself, is quite consistent with the energy principle. According to it the energy radiated onto a surface by a light ray passing through a vacuum is equal to the kinetic energy of the light particles striking the surface, all moving with the constant velocity c. The decrease in intensity of the energy radiation with the distance is then explained simply by the decrease of the volume density of the light particles.

Let us denote by n the number of the light particles contained in a unit volume and by m the mass of a particle. Then for a beam of parallel light the number of particles impinging in unit time on the element $d\sigma$ of a reflecting surface at the angle of incidence θ is

$$nc \cos\theta \, d\sigma. \tag{67}$$

Their kinetic energy is given according to Newtonian mechanics by

$$I = nc\cos\theta\, d\sigma\, \frac{mc^2}{2} = nm\cos\theta \frac{c^3}{2}\, d\sigma. \tag{68}$$

Now, in order to determine the normal pressure of these particles on the surface, we may note that the normal component of the velocity $c\cos\theta$ of every particle is changed on reflection into a component of opposite direction. Hence the normal component of the momentum of every particle (impulse-coordinate) is changed through reflection by $-2mc\cos\theta$. Then the change in momentum for all particles considered will be, according to (67),

$$-2nm\cos^2\theta\, c^2\, d\sigma. \tag{69}$$

Should the reflecting body be free to move in the direction of the normal of the reflecting surface and should there be no force acting on it except the impact of the light particles, it would be set into motion by the impacts. According to the law of action and reaction the ensuing motion would be such that the momentum acquired in a certain interval of time would be equal and opposite to the change in momentum of all the light particles reflected from it in the same time interval. But if we allow a separate constant force to act from outside on the reflector, there is to be added to the change in momenta of the light particles the impulse of the external force, *i.e.*, the product of the force and the time interval in question.

Therefore the reflector will remain continuously at rest, whenever the constant external force exerted on it is so chosen that its impulse for any time is just equal to the change in momentum of all the particles reflected from the reflector in the same time. Thus it follows that the force F itself which the particles exert by their impact on the surface

element $d\sigma$ is equal and opposite to the change of their momentum in unit time as expressed in (69)

$$\mathsf{F} = 2nm\cos^2\theta\, c^2\, d\sigma$$

and by making use of (68),

$$\mathsf{F} = \frac{4\cos\theta}{c} I.$$

On comparing this relation with equation (64) in which all symbols have the same physical significance, it is seen that *Newton's* radiation pressure is twice as large as *Maxwell's* for the same energy radiation. A necessary consequence of this is that the magnitude of *Maxwell's* radiation pressure cannot be deduced from general energetic considerations, but is a special feature of the electromagnetic theory and hence all deductions from *Maxwell's* radiation pressure are to be regarded as consequences of the electromagnetic theory of light and all confirmations of them are confirmations of this special theory.

CHAPTER II

STEFAN-BOLTZMANN LAW OF RADIATION

61. For the following we imagine a perfectly evacuated hollow cylinder with an absolutely tight-fitting piston free to move in a vertical direction with no friction. A part of the walls of the cylinder, say the rigid bottom, should consist of a black body, whose temperature T may be regulated arbitrarily from the outside. The rest of the walls including the inner surface of the piston may be assumed as totally reflecting. Then, if the piston remains stationary and the temperature, T, constant, the radiation in the vacuum will, after a certain time, assume the character of black radiation (Sec. 50) uniform in all directions. The specific intensity, K, and the volume density, u, depend only on the temperature, T, and are independent of the volume, V, of the vacuum and hence of the position of the piston.

If now the piston is moved downward, the radiation is compressed into a smaller space; if it is moved upward the radiation expands into a larger space. At the same time the temperature of the black body forming the bottom may be arbitrarily changed by adding or removing heat from the outside. This always causes certain disturbances of the stationary state. If, however, the arbitrary changes in V and T are made sufficiently slowly, the departure from the conditions of a stationary state may always be kept just as small as we please. Hence the state of radiation in the vacuum may, without appreciable error, be regarded as a state of thermodynamic equilibrium, just as is done in the thermodynamics of ordinary matter in the case of so-called infinitely slow processes, where, at any instant, the divergence from the state of equilibrium may be neglected, compared with the changes which the total system considered undergoes as a result of the entire process.

If, *e.g.*, we keep the temperature T of the black body forming the bottom constant, as can be done by a suitable connection between it and a heat reservoir of large capacity, then, on raising the piston, the black body will emit more than it absorbs, until the newly made space is filled with the same density of radiation as was the original one. *Vice versa*, on lowering the piston the black body will absorb the superfluous radiation until the original radiation corresponding to the temperature T is again established. Similarly, on raising the temperature T of the black body, as can be done by heat conduction from a heat reservoir which is slightly warmer, the density of radiation in the vacuum will be correspondingly increased by a larger emission, etc. To accelerate the establishment of radiation equilibrium the reflecting mantle of the hollow cylinder may be assumed white (Sec. 10), since by diffuse reflection the predominant directions of radiation that may, perhaps, be produced by the direction of the motion of the piston, are more quickly neutralized. The reflecting surface of the piston, however, should be chosen for the present as a perfect metallic reflector, to make sure that the radiation pressure (66) on the piston is *Maxwell's*. Then, in order to produce mechanical equilibrium, the piston must be loaded by a weight equal to the product of the radiation pressure p and the cross-section of the piston. An exceedingly small difference of the loading weight will then produce a correspondingly slow motion of the piston in one or the other direction.

Since the effects produced from the outside on the system in question, the cavity through which the radiation travels, during the processes we are considering, are partly of a mechanical nature (displacement of the loaded piston), partly of a thermal nature (heat conduction away from and toward the reservoir), they show a certain similarity to the processes usually considered in thermodynamics, with the difference that the system here considered is not a material system, *e.g.*, a

gas, but a purely energetic one. If, however, the principles of thermodynamics hold quite generally in nature, as indeed we shall assume, then they must also hold for the system under consideration. That is to say, in the case of any change occurring in nature the energy of all systems taking part in the change must remain constant (first principle), and, moreover, the entropy of all systems taking part in the change must increase, or in the limiting case of reversible processes must remain constant (second principle).

62. Let us first establish the equation of the first principle for an infinitesimal change of the system in question. That the cavity enclosing the radiation has a certain energy we have already (Sec. 22) deduced from the fact that the energy radiation is propagated with a finite velocity. We shall denote the energy by U. Then we have

$$U = Vu, \tag{70}$$

where u the volume density of radiation depends only on the temperature T of the black body at the bottom.

The work done by the system, when the volume V of the cavity increases by dV against the external forces of pressure (weight of the loaded piston), is $p\,dV$, where p represents *Maxwell's* radiation pressure (66). This amount of mechanical energy is therefore gained by the surroundings of the system, since the weight is raised. The error made by using the radiation pressure on a stationary surface, whereas the reflecting surface moves during the volume change, is evidently negligible, since the motion may be thought of as taking place with an arbitrarily small velocity.

If, moreover, Q denotes the infinitesimal quantity of heat in mechanical units, which, owing to increased emission, passes from the black body at the bottom to the cavity containing the radiation, the bottom or the heat reservoir connected to it loses this heat Q, and

its internal energy is decreased by that amount. Hence, according to the first principle of thermodynamics, since the sum of the energy of radiation and the energy of the material bodies remains constant, we have

$$dU + p\,dV - Q = 0. \tag{71}$$

According to the second principle of thermodynamics the cavity containing the radiation also has a definite entropy. For when the heat Q passes from the heat reservoir into the cavity, the entropy of the reservoir decreases, the change being

$$-\frac{Q}{T}.$$

Therefore, since no changes occur in the other bodies—inasmuch as the rigid absolutely reflecting piston with the weight on it does not change its internal condition with the motion—there must somewhere in nature occur a compensation of entropy having at least the value $\frac{Q}{T}$, by which the above diminution is compensated, and this can be nowhere except in the entropy of the cavity containing the radiation. Let the entropy of the latter be denoted by S.

Now, since the processes described consist entirely of states of equilibrium, they are perfectly reversible and hence there is no increase in entropy. Then we have

$$dS - \frac{Q}{T} = 0, \tag{72}$$

or from (71)

$$dS = \frac{dU + p\,dV}{T}. \tag{73}$$

In this equation the quantities U, p, V, S represent certain properties of the heat radiation, which are completely defined by the instantaneous state of the radiation. Therefore the quantity T is also a certain

property of the state of the radiation, *i.e.*, the black radiation in the cavity has a certain temperature T and this temperature is that of a body which is in heat equilibrium with the radiation.

63. We shall now deduce from the last equation a consequence which is based on the fact that the state of the system considered, and therefore also its entropy, is determined by the values of two independent variables. As the first variable we shall take V, as the second either T, u, or p may be chosen. Of these three quantities any two are determined by the third together with V. We shall take the volume V and the temperature T as independent variables. Then by substituting from (66) and (70) in (73) we have

$$dS = \frac{V}{T}\frac{du}{dT}\,dT + \frac{4u}{3T}\,dV. \tag{74}$$

From this we obtain

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{V}{T}\frac{du}{dT} \qquad \left(\frac{\partial S}{\partial V}\right)_T = \frac{4u}{3T}.$$

On partial differentiation of these equations, the first with respect to V, the second with respect to T, we find

$$\frac{\partial^2 S}{\partial T\,\partial V} = \frac{1}{T}\frac{du}{dT} = \frac{4}{3T}\frac{du}{dT} - \frac{4u}{3T^2}$$

or

$$\frac{du}{dT} = \frac{4u}{T}$$

and on integration

$$u = aT^4 \tag{75}$$

and from (21) for the specific intensity of black radiation

$$K = \frac{c}{4\pi}\cdot u = \frac{ac}{4\pi}T^4. \tag{76}$$

Moreover for the pressure of black radiation

$$p = \frac{a}{3}T^4, \tag{77}$$

and for the total radiant energy

$$U = aT^4 \cdot V. \tag{78}$$

This law, which states that the volume density and the specific intensity of black radiation are proportional to the fourth power of the absolute temperature, was first established by *J. Stefan*[1] on a basis of rather rough measurements. It was later deduced by *L. Boltzmann*[2] on a thermodynamic basis from *Maxwell's* radiation pressure and has been more recently confirmed by *O. Lummer* and *E. Pringsheim*[3] by exact measurements between 100° and 1300° C., the temperature being defined by the gas thermometer. In ranges of temperature and for requirements of precision for which the readings of the different gas thermometers no longer agree sufficiently or cannot be obtained at all, the *Stefan-Boltzmann* law of radiation can be used for an absolute definition of temperature independent of all substances.

64. The numerical value of the constant a is obtained from measurements made by *F. Kurlbaum*.[4] According to them, if we denote by S_t the total energy radiated in one second into air by a square centimeter of a black body at a temperature of $t°$ C., the following equation holds

$$S_{100} - S_0 = 0.0731 \,\frac{\text{watt}}{\text{cm}^2} = 7.31 \times 10^5 \,\frac{\text{erg}}{\text{cm}^2\,\text{sec}}. \tag{79}$$

[1] *J. Stefan*, Wien. Berichte, **79**, p. 391, 1879.

[2] *L. Boltzmann*, Wied. Annalen, **22**, p. 291, 1884.

[3] *O. Lummer* und *E. Pringsheim*, Wied. Annalen, **63**, p. 395, 1897. Annalen d. Physik **3**, p. 159, 1900.

[4] *F. Kurlbaum*, Wied. Annalen, **65**, p. 759, 1898.

Now, since the radiation in air is approximately identical with the radiation into a vacuum, we may according to (7) and (76) put

$$S_t = \pi K = \frac{ac}{4}(273 + t)^4$$

and from this

$$S_{100} - S_0 = \frac{ac}{4}(373^4 - 273^4),$$

therefore

$$a = \frac{4 \times 7.31 \times 10^5}{3 \times 10^{10} \times (373^4 - 273^4)} = 7.061 \times 10^{-15} \frac{\text{erg}}{\text{cm}^3\,\text{degree}^4}.$$

Recently *Kurlbaum* has increased the value measured by him by 2.5 per cent.,[1] on account of the bolometer used being not perfectly black, whence it follows that $a = 7.24 \cdot 10^{-15}$.

Meanwhile the radiation constant has been made the object of as accurate measurements as possible in various places. Thus it was measured by *Féry*, *Bauer* and *Moulin*, *Valentiner*, *Féry* and *Drecq*, *Shakespear*, *Gerlach*, with in some cases very divergent results, so that a mean value may hardly be formed.

For later computations we shall use the most recent determination made in the physical laboratory of the University of Berlin[2]

$$\frac{ac}{4} = \sigma = 5.46 \cdot 10^{-12} \frac{\text{watt}}{\text{cm}^2\,\text{degree}^4}.$$

[1] *F. Kurlbaum*, Verhandlungen d. Deutsch. physikal. Gesellschaft, **14**, p. 580, 1912.

[2] According to private information kindly furnished by my colleague *H. Rubens* (July, 1912). (These results have since been published. See *W. H. Westphal*, Verhandlungen d. Deutsch. physikal. Gesellschaft, **14**, p. 987, 1912, Tr.)

From this a is found to be

$$a = \frac{4 \cdot 5.46 \cdot 10^{-12} \cdot 10^{7}}{3 \cdot 10^{10}} = 7.28 \cdot 10^{-15} \frac{\text{erg}}{\text{cm}^3 \, \text{degree}^4}$$

which agrees rather closely with *Kurlbaum's* corrected value.

65. The magnitude of the entropy S of black radiation found by integration of the differential equation (73) is

$$S = \frac{4}{3} a T^3 V. \tag{80}$$

In this equation the additive constant is determined by a choice that readily suggests itself, so that at the zero of the absolute scale of temperature, that is to say, when u vanishes, S shall become zero. From this the entropy of unit volume or the volume density of the entropy of black radiation is obtained,

$$\frac{S}{V} = s = \frac{4}{3} a T^3. \tag{81}$$

66. We shall now remove a restricting assumption made in order to enable us to apply the value of *Maxwell's* radiation pressure, calculated in the preceding chapter. Up to now we have assumed the cylinder to be fixed and only the piston to be free to move. We shall now think of the whole of the vessel, consisting of the cylinder, the black bottom, and the piston, the latter attached to the walls in a definite height above the bottom, as being free to move in space. Then, according to the principle of action and reaction, the vessel as a whole must remain constantly at rest, since no external force acts on it. This is the conclusion to which we must necessarily come, even without, in this case, admitting *a priori* the validity of the principle of action and reaction. For if the vessel

should begin to move, the kinetic energy of this motion could originate only at the expense of the heat of the body forming the bottom or the energy of radiation, as there exists in the system enclosed in a rigid cover no other available energy; and together with the decrease of energy the entropy of the body or the radiation would also decrease, an event which would contradict the second principle, since no other changes of entropy occur in nature. Hence the vessel as a whole is in a state of mechanical equilibrium. An immediate consequence of this is that the pressure of the radiation on the black bottom is just as large as the oppositely directed pressure of the radiation on the reflecting piston. Hence the pressure of black radiation is the same on a black as on a reflecting body of the same temperature and the same may be readily proven for any completely reflecting surface whatsoever, which we may assume to be at the bottom of the cylinder without in the least disturbing the stationary state of radiation. Hence we may also in all the foregoing considerations replace the reflecting metal by any completely reflecting or black body whatsoever, at the same temperature as the body forming the bottom, and it may be stated as a quite general law that the radiation pressure depends only on the properties of the radiation passing to and fro, not on the properties of the enclosing substance.

67. If, on raising the piston, the temperature of the black body forming the bottom is kept constant by a corresponding addition of heat from the heat reservoir, the process takes place isothermally. Then, along with the temperature T of the black body, the energy density u, the radiation pressure p, and the density of the entropy s also remain constant; hence the total energy of radiation increases from $U = uV$ to $U' = uV'$, the entropy from $S = sV$ to $S' = sV'$ and the heat supplied from the heat reservoir is obtained by integrating (72) at constant T,

$$Q = T(S' - S) = Ts(V' - V)$$

or, according to (81) and (75),

$$Q = \frac{4}{3}aT^4(V' - V) = \frac{4}{3}(U' - U).$$

Thus it is seen that the heat furnished from the outside exceeds the increase in energy of radiation $(U' - U)$ by $\frac{1}{3}(U' - U)$. This excess in the added heat is necessary to do the external work accompanying the increase in the volume of radiation.

68. Let us also consider a reversible adiabatic process. For this it is necessary not merely that the piston and the mantle but also that the bottom of the cylinder be assumed as completely reflecting, *e.g.*, as white. Then the heat furnished on compression or expansion of the volume of radiation is $Q = 0$ and the energy of radiation changes only by the value $p\,dV$ of the external work. To insure, however, that in a finite adiabatic process the radiation shall be perfectly stable at every instant, *i.e.*, shall have the character of black radiation, we may assume that inside the evacuated cavity there is a carbon particle of minute size. This particle, which may be assumed to possess an absorbing power differing from zero for all kinds of rays, serves merely to produce stable equilibrium of the radiation in the cavity (Sec. 51 *et seq.*) and thereby to insure the reversibility of the process, while its heat contents may be taken as so small compared with the energy of radiation, U, that the addition of heat required for an appreciable temperature change of the particle is perfectly negligible. Then at every instant in the process there exists absolutely stable equilibrium of radiation and the radiation has the temperature of the particle in the cavity. The volume, energy, and entropy of the particle may be entirely neglected.

On a reversible adiabatic change, according to (72), the entropy S of the system remains constant. Hence from (80) we have as a condition

for such a process

$$T^3V = \text{const.},$$

or, according to (77),

$$pV^{\frac{4}{3}} = \text{const.},$$

i.e., on an adiabatic compression the temperature and the pressure of the radiation increase in a manner that may be definitely stated. The energy of the radiation, U, in such a case varies according to the law

$$\frac{U}{T} = \frac{3}{4}S = \text{const.},$$

i.e., it increases in proportion to the absolute temperature, although the volume becomes smaller.

69. Let us finally, as a further example, consider a simple case of an irreversible process. Let the cavity of volume V, which is everywhere enclosed by absolutely reflecting walls, be uniformly filled with black radiation. Now let us make a small hole through any part of the walls, *e.g.*, by opening a stopcock, so that the radiation may escape into another completely evacuated space, which may also be surrounded by rigid, absolutely reflecting walls. The radiation will at first be of a very irregular character; after some time, however, it will assume a stationary condition and will fill both communicating spaces uniformly, its total volume being, say, V'. The presence of a carbon particle will cause all conditions of black radiation to be satisfied in the new state. Then, since there is neither external work nor addition of heat from the outside, the energy of the new state is, according to the first principle, equal to that of the original one, or $U' = U$ and hence from (78)

$$T'^4V' = T^4V$$

$$\frac{T'}{T} = \sqrt[4]{\frac{V}{V'}}$$

which defines completely the new state of equilibrium. Since $V' > V$ the temperature of the radiation has been lowered by the process.

According to the second principle of thermodynamics the entropy of the system must have increased, since no external changes have occurred; in fact we have from (80)

$$\frac{S'}{S} = \frac{T'^3 V'}{T^3 V} = \sqrt[4]{\frac{V'}{V}} > 1. \tag{82}$$

70. If the process of irreversible adiabatic expansion of the radiation from the volume V to the volume V' takes place as just described with the single difference that there is no carbon particle present in the vacuum, after the stationary state of radiation is established, as will be the case after a certain time on account of the diffuse reflection from the walls of the cavity, the radiation in the new volume V' will not any longer have the character of black radiation, and hence no definite temperature. Nevertheless the radiation, like every system in a definite physical state, has a definite entropy, which, according to the second principle, is larger than the original S, but not as large as the S' given in (82). The calculation cannot be performed without the use of laws to be taken up later (see Sec. 103). If a carbon particle is afterward introduced into the vacuum, absolutely stable equilibrium is established by a second irreversible process, and, the total energy as well as the total volume remaining constant, the radiation assumes the normal energy distribution of black radiation and the entropy increases to the maximum value S' given by (82).

CHAPTER III

WIEN'S DISPLACEMENT LAW

71. Though the manner in which the volume density u and the specific intensity K of black radiation depend on the temperature is determined by the *Stefan-Boltzmann* law, this law is of comparatively little use in finding the volume density u_ν corresponding to a definite frequency ν, and the specific intensity of radiation K_ν of monochromatic radiation, which are related to each other by equation (24) and to u and K by equations (22) and (12). There remains as one of the principal problems of the theory of heat radiation the problem of determining the quantities u_ν and K_ν for black radiation in a vacuum and hence, according to (42), in any medium whatever, as functions of ν and T, or, in other words, to find the distribution of energy in the normal spectrum for any arbitrary temperature. An essential step in the solution of this problem is contained in the so-called "displacement law" stated by *W. Wien*,[1] the importance of which lies in the fact that it reduces the functions u_ν and K_ν of the two arguments ν and T to a function of a single argument.

The starting point of *Wien's* displacement law is the following theorem. If the black radiation contained in a perfectly evacuated cavity with absolutely reflecting walls is compressed or expanded adiabatically and infinitely slowly, as described above in Sec. 68, *the radiation always retains the character of black radiation, even without the presence of a*

[1] *W. Wien*, Sitzungsberichte d. Akad. d. Wissensch. Berlin, Febr. 9, 1893, p. 55. Wiedemann's Annal., **52**, p. 132, 1894. See also among others *M. Thiesen*, Verhandl. d. Deutsch. phys. Gesellsch., **2**, p. 65, 1900. *H. A. Lorentz*, Akad. d. Wissensch. Amsterdam, May 18, 1901, p. 607. *M. Abraham*, Annal. d. Physik **14**, p. 236, 1904.

carbon particle. Hence the process takes place in an absolute vacuum just as was calculated in Sec. 68 and the introduction, as a precaution, of a carbon particle is shown to be superfluous. But this is true only in this special case, not at all in the case described in Sec. 70.

The truth of the proposition stated may be shown as follows: Let the completely evacuated hollow cylinder, which is at the start filled with black radiation, be compressed adiabatically and infinitely slowly to a finite fraction of the original volume. If, now, the compression being completed, the radiation were no longer black, there would be no stable thermodynamic equilibrium of the radiation (Sec. 51). It would then be possible to produce a finite change at constant volume and constant total energy of radiation, namely, the change to the absolutely stable state of radiation, which would cause a finite increase of entropy. This change could be brought about by the introduction of a carbon particle, containing a negligible amount of heat as compared with the energy of radiation. This change, of course, refers only to the spectral density of radiation u_ν, whereas the total density of energy u remains constant. After this has been accomplished, we could, leaving the carbon particle in the space, allow the hollow cylinder to return adiabatically and infinitely slowly to its original volume and then remove the carbon particle. The system will then have passed through a cycle without any external changes remaining. For heat has been neither added nor removed, and the mechanical work done on compression has been regained on expansion, because the latter, like the radiation pressure, depends only on the total density u of the energy of radiation, not on its spectral distribution. Therefore, according to the first principle of thermodynamics, the total energy of radiation is at the end just the same as at the beginning, and hence also the temperature of the black radiation is again the same. The carbon particle and its changes do not enter into the calculation, for its energy and entropy are vanish-

ingly small compared with the corresponding quantities of the system. The process has therefore been reversed in all details; it may be repeated any number of times without any permanent change occurring in nature. This contradicts the assumption, made above, that a finite increase in entropy occurs; for such a finite increase, once having taken place, cannot in any way be completely reversed. Therefore no finite increase in entropy can have been produced by the introduction of the carbon particle in the space of radiation, but the radiation was, before the introduction and always, in the state of stable equilibrium.

72. In order to bring out more clearly the essential part of this important proof, let us point out an analogous and more or less obvious consideration. Let a cavity containing originally a vapor in a state of saturation be compressed adiabatically and infinitely slowly.

"Then on an arbitrary adiabatic compression the vapor remains always just in the state of saturation. For let us suppose that it becomes supersaturated on compression. After the compression to an appreciable fraction of the original volume has taken place, condensation of a finite amount of vapor and thereby a change into a more stable state, and hence a finite increase of entropy of the system, would be produced at constant volume and constant total energy by the introduction of a minute drop of liquid, which has no appreciable mass or heat capacity. After this has been done, the volume could again be increased adiabatically and infinitely slowly until again all liquid is evaporated and thereby the process completely reversed, which contradicts the assumed increase of entropy."

Such a method of proof would be erroneous, because, by the process described, the change that originally took place is not at all completely reversed. For since the mechanical work expended on the compression of the supersaturated steam is not equal to the amount gained on expanding the saturated steam, there corresponds to a definite volume of

the system when it is being compressed an amount of energy different from the one during expansion and therefore the volume at which all liquid is just vaporized cannot be equal to the original volume. The supposed analogy therefore breaks down and the statement made above in quotation marks is incorrect.

73. We shall now again suppose the reversible adiabatic process described in Sec. 68 to be carried out with the black radiation contained in the evacuated cavity with white walls and white bottom, by allowing the piston, which consists of absolutely reflecting metal, to move downward infinitely slowly, with the single difference that now there shall be no carbon particle in the cylinder. The process will, as we now know, take place exactly as there described, and, since no absorption or emission of radiation takes place, we can now give an account of the changes of color and intensity which the separate pencils of the system undergo. Such changes will of course occur only on reflection from the moving metallic reflector, not on reflection from the stationary walls and the stationary bottom of the cylinder.

If the reflecting piston moves down with the constant, infinitely small, velocity v, the monochromatic pencils striking it during the motion will suffer on reflection a change of color, intensity, and direction. Let us consider these different influences in order.[1]

74. To begin with, we consider the *change of color* which a monochromatic ray suffers by reflection from the reflector, which is moving with an infinitely small velocity. For this purpose we consider first the case of a ray which falls normally from below on the reflector

[1]The complete solution of the problem of reflection of a pencil from a moving absolutely reflecting surface including the case of an arbitrarily large velocity of the surface may be found in the paper by *M. Abraham* quoted in Sec. 71. See also the text-book by the same author. Electromagnetische Theorie der Strahlung, 1908 (Leipzig, B. G. Teubner).

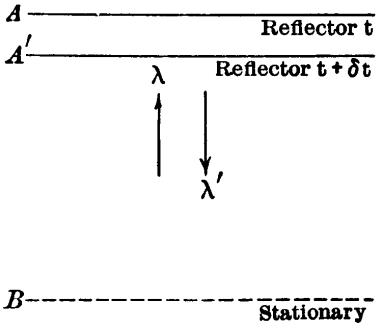


FIG. 5.

and hence is reflected normally downward. Let the plane A (Fig. 5) represent the position of the reflector at the time t, the plane A' the position at the time $t + \delta t$, where the distance AA' equals $v\,\delta t$, v denoting the velocity of the reflector. Let us now suppose a stationary plane B to be placed parallel to A at a suitable distance and let us denote by λ the wave length of the ray incident on the reflector and by λ' the wave length of the ray reflected from it. Then at a time t there are in the interval AB in the vacuum containing the radiation $\frac{AB}{\lambda}$ waves of the incident and $\frac{AB}{\lambda'}$ waves of the reflected ray, as can be seen, *e.g.*, by thinking of the electric field-strength as being drawn at the different points of each of the two rays at the time t in the form of a sine curve. Reckoning both incident and reflected ray there are at the time t

$$AB\left(\frac{1}{\lambda}+\frac{1}{\lambda'}\right)$$

waves in the interval between A and B. Since this is a large number, it is immaterial whether the number is an integer or not.

Similarly at the time $t + \delta t$, when the reflector is at A', there are

$$A'B\left(\frac{1}{\lambda} + \frac{1}{\lambda'}\right)$$

waves in the interval between A' and B all told.

The latter number will be smaller than the former, since in the shorter distance $A'B$ there is room for fewer waves of both kinds than in the longer distance AB. The remaining waves must have been expelled in the time δt from the space between the stationary plane B and the moving reflector, and this must have taken place through the plane B downward; for in no other way could a wave disappear from the space considered.

Now $\nu\,\delta t$ waves pass in the time δt through the stationary plane B in an upward direction and $\nu'\,\delta t$ waves in a downward direction; hence we have for the difference

$$(\nu' - \nu)\,\delta t = (AB - A'B)\left(\frac{1}{\lambda} + \frac{1}{\lambda'}\right)$$

or, since

$$AB - A'B = v\,\delta t,$$

and

$$\lambda = \frac{c}{\nu} \quad \lambda' = \frac{c}{\nu'}$$
$$\nu' = \frac{c + v}{c - v}\nu$$

or, since v is infinitely small compared with c,

$$\nu' = \nu\left(1 + \frac{2v}{c}\right).$$

75. When the radiation does not fall on the reflector normally but at an acute angle of incidence θ, it is possible to pursue a very similar line of reasoning, with the difference that then A, the point of intersection of a definite ray BA with the reflector at the time t, has not the same position on the reflector as the point of intersection, A', of the same ray with the reflector at the time $t+\delta t$ (Fig. 6). The number of waves which lie in the interval BA at the time t is $\frac{BA}{\lambda}$. Similarly, at the time t the number of waves in the interval AC representing the distance of the point A from a wave plane CC', belonging to the reflected ray and stationary in the vacuum, is $\frac{AC}{\lambda'}$.

Hence there are, all told, at the time t in the interval BAC

$$\frac{BA}{\lambda}+\frac{AC}{\lambda'}$$

waves of the ray under consideration. We may further note that the angle of reflection θ' is not exactly equal to the angle of incidence, but is a little smaller as can be shown by a simple geometric consideration based on *Huyghens'* principle. The difference of θ and θ', however, will be shown to be non-essential for our calculation. Moreover there are at the time $t+\delta t$, when the reflector passes through A',

$$\frac{BA'}{\lambda}+\frac{A'C'}{\lambda'}$$

waves in the distance $BA'C'$. The latter number is smaller than the former and the difference must equal the total number of waves which are expelled in the time δt from the space which is bounded by the stationary planes BB' and CC'.

Now $\nu\,\delta t$ waves enter into the space through the plane BB' in the time δt and $\nu'\,\delta t$ waves leave the space through the plane CC'. Hence

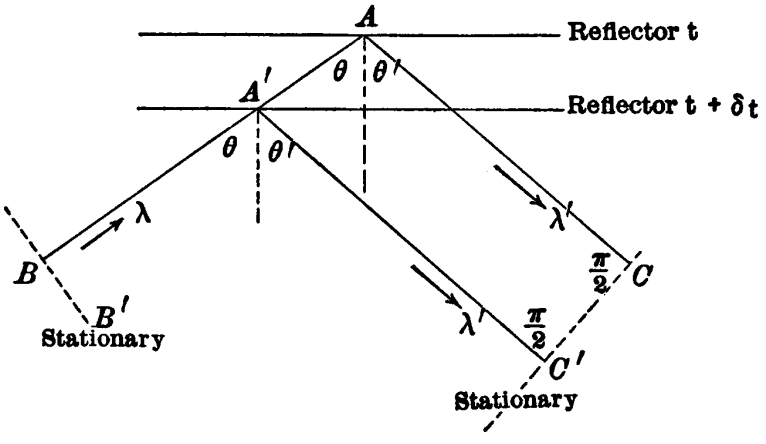


FIG. 6.

we have

$$(\nu' - \nu)\,\delta t = \left(\frac{BA}{\lambda} + \frac{AC}{\lambda'}\right) - \left(\frac{BA'}{\lambda} + \frac{A'C'}{\lambda'}\right)$$

but

$$BA - BA' = AA' = \frac{v\,\delta t}{\cos\theta}$$
$$AC - A'C' = AA'\cos(\theta + \theta')$$
$$\lambda = \frac{c}{\nu}, \quad \lambda' = \frac{c}{\nu'}.$$

Hence

$$\nu' = \frac{c\cos\theta + v}{c\cos\theta - v\cos(\theta + \theta')}\nu.$$

This relation holds for any velocity v of the moving reflector. Now, since in our case v is infinitely small compared with c, we have the

simpler expression

$$\nu' = \nu(1 + \frac{v}{c\cos\theta}[1 + \cos(\theta + \theta')]).$$

The difference between the two angles θ and θ' is in any case of the order of magnitude $\frac{v}{c}$; hence we may without appreciable error replace θ' by θ, thereby obtaining the following expression for the frequency of the reflected ray for oblique incidence

$$\nu' = \nu\left(1 + \frac{2v\cos\theta}{c}\right). \tag{83}$$

76. From the foregoing it is seen that the frequency of all rays which strike the moving reflector are increased on reflection, when the reflector moves toward the radiation, and decreased, when the reflector moves in the direction of the incident rays ($v < 0$). However, the total radiation of a definite frequency ν striking the moving reflector is by no means reflected as monochromatic radiation but the change in color on reflection depends also essentially on the angle of incidence θ. Hence we may not speak of a certain spectral "displacement" of color except in the case of a single pencil of rays of definite direction, whereas in the case of the entire monochromatic radiation we must refer to a spectral "dispersion." The change in color is the largest for normal incidence and vanishes entirely for grazing incidence.

77. Secondly, let us calculate the *change in energy*, which the moving reflector produces in the incident radiation, and let us consider from the outset the general case of oblique incidence. Let a monochromatic, infinitely thin, unpolarized pencil of rays, which falls on a surface element of the reflector at the angle of incidence θ, transmit the energy $I\,\delta t$ to the reflector in the time δt. Then, ignoring vanishingly small quantities, the mechanical pressure of the pencil of rays normally to the

reflector is, according to equation (64),

$$\mathsf{F} = \frac{2\cos\theta}{c} I,$$

and to the same degree of approximation the work done from the outside on the incident radiation in the time δt is

$$\mathsf{F} v\,\delta t = \frac{2v\cos\theta}{c} I\,\delta t. \tag{84}$$

According to the principle of the conservation of energy this amount of work must reappear in the energy of the reflected radiation. Hence the reflected pencil has a larger intensity than the incident one. It produces, namely, in the time δt the energy[1]

$$I\,\delta t + \mathsf{F} v\,\delta t = I\left(1 + \frac{2v\cos\theta}{c}\right)\delta t = I'\,\delta t. \tag{85}$$

Hence we may summarize as follows: By the reflection of a monochromatic unpolarized pencil, incident at an angle θ on a reflector moving toward the radiation with the infinitely small velocity v, the radiant energy $I\,\delta t$, whose frequencies extend from ν to $\nu + d\nu$, is in the time δt changed into the radiant energy $I'\,\delta t$ with the interval of frequency $(\nu', \nu' + d\nu')$, where I' is given by (85), ν' by (83), and accordingly $d\nu'$, the spectral breadth of the reflected pencil, by

$$d\nu' = d\nu\left(1 + \frac{2v\cos\theta}{c}\right). \tag{86}$$

[1]It is clear that the change in intensity of the reflected radiation caused by the motion of the reflector can also be derived from purely electrodynamical considerations, since electrodynamics are consistent with the energy principle. This method is somewhat lengthy, but it affords a deeper insight into the details of the phenomenon of reflection.

A comparison of these values shows that

$$\frac{I'}{I} = \frac{\nu'}{\nu} = \frac{d\nu'}{d\nu}. \tag{87}$$

The absolute value of the radiant energy which has disappeared in this change is, from equation (13),

$$I\,\delta t = 2\mathsf{K}_\nu\, d\sigma \cos\theta\, d\Omega\, d\nu\, \delta t, \tag{88}$$

and hence the absolute value of the radiant energy which has been formed is, according to (85),

$$I'\,\delta t = 2\mathsf{K}_\nu\, d\sigma \cos\theta\, d\Omega\, d\nu \left(1 + \frac{2v\cos\theta}{c}\right) \delta t. \tag{89}$$

Strictly speaking these last two expressions would require an infinitely small correction, since the quantity I from equation (88) represents the energy radiation on a stationary element of area $d\sigma$, while, in reality, the incident radiation is slightly increased by the motion of $d\sigma$ toward the incident pencil. The corresponding additional terms may, however, be omitted here without appreciable error, since the correction caused by them would consist merely of the addition to the energy change here calculated of a comparatively infinitesimal energy change of the same kind with an external work that is infinitesimal of the second order.

78. As regards finally the *changes in direction*, which are imparted to the incident ray by reflection from the moving reflector, we need not calculate them at all at this stage. For if the motion of the reflector takes place sufficiently slowly, all irregularities in the direction of the radiation are at once equalized by further reflection from the walls of

the vessel. We may, indeed, think of the whole process as being accomplished in a very large number of short intervals, in such a way that the piston, after it has moved a very small distance with very small velocity, is kept at rest for a while, namely, until all irregularities produced in the directions of the radiation have disappeared as the result of the reflection from the white walls of the hollow cylinder. If this procedure be carried on sufficiently long, the compression of the radiation may be continued to an arbitrarily small fraction of the original volume, and while this is being done, the radiation may be always regarded as uniform in all directions. This continuous process of equalization refers, of course, only to difference in the direction of the radiation; for changes in the color or intensity of the radiation of however small size, having once occurred, can evidently never be equalized by reflection from totally reflecting stationary walls but continue to exist forever.

79. With the aid of the theorems established we are now in a position to calculate the change of the density of radiation for every frequency for the case of infinitely slow adiabatic compression of the perfectly evacuated hollow cylinder, which is filled with uniform radiation. For this purpose we consider the radiation at the time t in a definite infinitely small interval of frequencies, from ν to $\nu + d\nu$, and inquire into the change which the total energy of radiation contained in this definite constant interval suffers in the time δt.

At the time t this radiant energy is, according to Sec. 23, $V\mathsf{u}\,d\nu$; at the time $t+\delta t$ it is $\big(V\mathsf{u}+\delta(V\mathsf{u})\big)\,d\nu$, hence the change to be calculated is

$$\delta(V\mathsf{u})\,d\nu. \tag{90}$$

In this the density of monochromatic radiation u is to be regarded as a function of the mutually independent variables ν and t, the differentials of which are distinguished by the symbols d and δ.

The change of the energy of monochromatic radiation is produced only by the reflection from the moving reflector, that is to say, firstly by certain rays, which at the time t belong to the interval $(\nu, d\nu)$, leaving this interval on account of the change in color suffered by reflection, and secondly by certain rays, which at the time t do not belong to the interval $(\nu, d\nu)$, coming into this interval on account of the change in color suffered on reflection. Let us calculate these influences in order. The calculation is greatly simplified by taking the width of this interval $d\nu$ so small that

$$d\nu \text{ is small compared with } \frac{v}{c}\nu, \tag{91}$$

a condition which can always be satisfied, since $d\nu$ and v are mutually independent.

80. The rays which at the time t belong to the interval $(\nu, d\nu)$ and leave this interval in the time δt on account of reflection from the moving reflector, are simply those rays which strike the moving reflector in the time δt. For the change in color which such a ray undergoes is, from (83) and (91), large compared with $d\nu$, the width of the whole interval. Hence we need only calculate the energy, which in the time δt is transmitted to the reflector by the rays in the interval $(\nu, d\nu)$.

For an elementary pencil, which falls on the element $d\sigma$ of the reflecting surface at the angle of incidence θ, this energy is, according to (88) and (5),

$$I\,\delta t = 2\mathsf{K}_\nu\, d\sigma \cos\theta\, d\Omega\, d\nu\, \delta t = 2\mathsf{K}_\nu\, d\sigma \sin\theta \cos\theta\, d\theta\, d\phi\, d\nu\, \delta t.$$

Hence we obtain for the total monochromatic radiation, which falls on the whole surface F of the reflector, by integration with respect to ϕ from 0 to 2π, with respect to θ from 0 to $\frac{\pi}{2}$, and with respect to $d\sigma$

from 0 to F,

$$2\pi F \mathsf{K}_\nu \, d\nu \, \delta t. \tag{92}$$

Thus this radiant energy leaves, in the time δt, the interval of frequencies $(\nu, d\nu)$ considered.

81. In calculating the radiant energy which enters the interval $(\nu, d\nu)$ in the time δt on account of reflection from the moving reflector, the rays falling on the reflector at different angles of incidence must be considered separately. Since in the case of a positive v, the frequency is increased by the reflection, the rays which must be considered have, at the time t, the frequency $\nu_1 < \nu$. If we now consider at the time t a monochromatic pencil of frequency $(\nu_1, d\nu_1)$, falling on the reflector at an angle of incidence θ, a necessary and sufficient condition for its entrance, by reflection, into the interval $(\nu, d\nu)$ is

$$\nu = \nu_1 \left(1 + \frac{2v\cos\theta}{c}\right) \quad \text{and} \quad d\nu = d\nu_1 \left(1 + \frac{2v\cos\theta}{c}\right).$$

These relations are obtained by substituting ν_1 and ν respectively in the equations (83) and (86) in place of the frequencies before and after reflection ν and ν'.

The energy which this pencil carries into the interval $(\nu_1, d\nu)$ in the time δt is obtained from (89), likewise by substituting ν_1 for ν. It is

$$2\mathsf{K}_{\nu_1} \, d\sigma \cos\theta \, d\Omega \, d\nu_1 \left(1 + \frac{2v\cos\theta}{c}\right) \delta t = 2\mathsf{K}_{\nu_1} \, d\sigma \cos\theta \, d\Omega \, d\nu \, \delta t.$$

Now we have

$$\mathsf{K}_{\nu_1} = \mathsf{K}_\nu + (\nu_1 - \nu)\frac{\partial \mathsf{K}}{\partial \nu} + \dots$$

where we shall assume $\dfrac{\partial \mathsf{K}}{\partial \nu}$ to be finite.

Hence, neglecting small quantities of higher order,

$$\mathsf{K}_{\nu_1} = \mathsf{K}_\nu - \frac{2\nu v \cos\theta}{c}\,\frac{\partial \mathsf{K}}{\partial \nu}.$$

Thus the energy required becomes

$$2\,d\sigma \left(\mathsf{K}_\nu - \frac{2\nu v\cos\theta}{c}\,\frac{\partial \mathsf{K}}{\partial \nu}\right) \sin\theta\cos\theta\,d\theta\,d\phi\,d\nu\,\delta t,$$

and, integrating this expression as above, with respect to $d\sigma$, ϕ, and θ, the total radiant energy which enters into the interval $(\nu, d\nu)$ in the time δt becomes

$$2\pi F \left(\mathsf{K}_\nu - \frac{4}{3}\,\frac{\nu v}{c}\,\frac{\partial \mathsf{K}}{\partial \nu}\right) d\nu\,\delta t. \tag{93}$$

82. The difference of the two expressions (93) and (92) is equal to the whole change (90), hence

$$-\frac{8\pi}{3} F \frac{\nu v}{c}\,\frac{\partial \mathsf{K}}{\partial \nu}\,\delta t = \delta(V\mathsf{u}),$$

or, according to (24),

$$-\frac{1}{3} F \nu v \frac{\partial \mathsf{u}}{\partial \nu}\,\delta t = \delta(V\mathsf{u}),$$

or, finally, since $Fv\,\delta t$ is equal to the decrease of the volume V,

$$\frac{1}{3}\nu\frac{\partial \mathsf{u}}{\partial \nu}\,\delta V = \delta(V\mathsf{u}) = \mathsf{u}\,\delta V + V\,\delta\mathsf{u}, \tag{94}$$

whence it follows that

$$\delta\mathsf{u} = \left(\frac{\nu}{3}\,\frac{\partial \mathsf{u}}{\partial \nu} - \mathsf{u}\right)\frac{\delta V}{V}. \tag{95}$$

This equation gives the change of the energy density of any definite frequency ν, which occurs on an infinitely slow adiabatic compression of the radiation. It holds, moreover, not only for black radiation, but also for *radiation originally of a perfectly arbitrary distribution of energy*, as is shown by the method of derivation.

Since the changes taking place in the state of the radiation in the time δt are proportional to the infinitely small velocity v and are reversed on changing the sign of the latter, this equation holds for any sign of δV; *hence the process is reversible.*

83. Before passing on to the general integration of equation (95) let us examine it in the manner which most easily suggests itself. According to the energy principle, the change in the radiant energy

$$U = Vu = V\int_0^\infty \mathfrak{u}\, d\nu,$$

occurring on adiabatic compression, must be equal to the external work done against the radiation pressure

$$-p\,\delta V = -\frac{u}{3}\,\delta V = -\frac{\delta V}{3}\int_0^\infty \mathfrak{u}\, d\nu. \tag{96}$$

Now from (94) the change in the total energy is found to be

$$\delta U = \int_0^\infty d\nu\,\delta(V\mathfrak{u}) = \frac{\delta V}{3}\int_0^\infty \nu\frac{\partial \mathfrak{u}}{\partial \nu}\, d\nu,$$

or, by partial integration,

$$\delta U = \frac{\delta V}{3}\Big([\nu\mathfrak{u}]_0^\infty - \int_0^\infty \mathfrak{u}\, d\nu\Big),$$

and this expression is, in fact, identical with (96), since the product $\nu\mathfrak{u}$ vanishes for $\nu = 0$ as well as for $\nu = \infty$. The latter might at first seem

doubtful; but it is easily seen that, if $\nu\mathsf{u}$ for $\nu = \infty$ had a value different from zero, the integral of u with respect to ν taken from 0 to ∞ could not have a finite value, which, however, certainly is the case.

84. We have already emphasized (Sec. 79) that u must be regarded as a function of two independent variables, of which we have taken as the first the frequency ν and as the second the time t. Since, now, in equation (95) the time t does not explicitly appear, it is more appropriate to introduce the volume V, which depends only on t, as the second variable instead of t itself. Then equation (95) may be written as a partial differential equation as follows:

$$V\frac{\partial \mathsf{u}}{\partial V} = \frac{\nu}{3}\frac{\partial \mathsf{u}}{\partial \nu} - \mathsf{u}. \tag{97}$$

From this equation, if, for a definite value of V, u is known as a function of ν, it may be calculated for all other values of V as a function of ν. The general integral of this differential equation, as may be readily seen by substitution, is

$$\mathsf{u} = \frac{1}{V}\phi(\nu^3 V), \tag{98}$$

where ϕ denotes an arbitrary function of the single argument $\nu^3 V$. Instead of this we may, on substituting $\nu^3 V\phi(\nu^3 V)$ for $\phi(\nu^3 V)$, write

$$\mathsf{u} = \nu^3\phi(\nu^3 V). \tag{99}$$

Either of the last two equations is the general expression of *Wien's* displacement law.

If for a definitely given volume V the spectral distribution of energy is known (*i.e.*, u as a function of ν), it is possible to deduce therefrom the dependence of the function ϕ on its argument, and thence the distribution of energy for any other volume V', into which the radiation

filling the hollow cylinder may be brought by a reversible adiabatic process.

84a. The characteristic feature of this new distribution of energy may be stated as follows: If we denote all quantities referring to the new state by the addition of an accent, we have the following equation in addition to (99)

$$\mathsf{u}' = \nu'^3 \phi(\nu'^3 V').$$

Therefore, if we put

$$\nu'^3 V' = \nu^3 V, \tag{99a}$$

we shall also have

$$\frac{\mathsf{u}'}{\nu'^3} = \frac{\mathsf{u}}{\nu^3} \quad \text{and} \quad \mathsf{u}'V' = \mathsf{u}V, \tag{99b}$$

i.e., if we coordinate with every frequency ν in the original state that frequency ν' which is to ν in the inverse ratio of the cube roots of the respective volumes, the corresponding energy densities u' and u will be in the inverse ratio of the volumes.

The meaning of these relations will be more clearly seen, if we write

$$\frac{V'}{\lambda'^3} = \frac{V}{\lambda^3}.$$

This is the number of the cubes of the wave lengths, which correspond to the frequency ν and are contained in the volume of the radiation. Moreover $\mathsf{u}\, d\nu\, V = \mathsf{U}\, d\nu$ denotes the radiant energy lying between the frequencies ν and $\nu + d\nu$, which is contained in the volume V. Now since, according to (99a),

$$\sqrt[3]{V'}\, d\nu' = \sqrt[3]{V}\, d\nu \quad \text{or} \quad \frac{d\nu'}{\nu'} = \frac{d\nu}{\nu} \tag{99c}$$

we have, taking account of (99b),

$$\mathsf{U}'\frac{d\nu'}{\nu'} = \mathsf{U}\frac{d\nu}{\nu}.$$

These results may be summarized thus: On an infinitely slow reversible adiabatic change in volume of radiation contained in a cavity and uniform in all directions, the frequencies change in such a way that the number of cubes of wave lengths of every frequency contained in the total volume remains unchanged, and the radiant energy of every infinitely small spectral interval changes in proportion to the frequency. These laws hold for any original distribution of energy whatever; hence, *e.g.*, an originally monochromatic radiation remains monochromatic during the process described, its color changing in the way stated.

85. Returning now to the discussion of Sec. 73 we introduce the assumption that at first the spectral distribution of energy is the normal one, corresponding to black radiation. Then, according to the law there proven, the radiation retains this property without change during a reversible adiabatic change of volume and the laws derived in Sec. 68 hold for the process. The radiation then possesses in every state a definite temperature T, which depends on the volume V according to the equation derived in that paragraph,

$$T^3V = \text{const.} = T'^3V'. \tag{100}$$

Hence we may now write equation (99) as follows:

$$\mathsf{u} = \nu^3\phi\left(\frac{\nu^3}{T^3}\right)$$

or

$$\mathsf{u} = \nu^3\phi\left(\frac{T}{\nu}\right).$$

Therefore, if for a single temperature the spectral distribution of black radiation, *i.e.*, u as a function of ν, is known, the dependence of the function ϕ on its argument, and hence the spectral distribution for any other temperature, may be deduced therefrom.

If we also take into account the law proved in Sec. 47, that, for the black radiation of a definite temperature, the product $\mathsf{u}q^3$ has for all media the same value, we may also write

$$\mathsf{u} = \frac{\nu^3}{c^3} F\left(\frac{T}{\nu}\right) \tag{101}$$

where now the function F no longer contains the velocity of propagation.

86. For the total radiation density in space of the black radiation in the vacuum we find

$$u = \int_0^\infty \mathsf{u}\, d\nu = \frac{1}{c^3} \int_0^\infty \nu^3 F\left(\frac{T}{\nu}\right) d\nu, \tag{102}$$

or, on introducing $\dfrac{T}{\nu} = x$ as the variable of integration instead of ν,

$$u = \frac{T^4}{c^3} \int_0^\infty \frac{F(x)}{x^5}\, dx. \tag{103}$$

If we let the absolute constant

$$\frac{1}{c^3} \int_0^\infty \frac{F(x)}{x^5}\, dx = a \tag{104}$$

the equation reduces to the form of the *Stefan-Boltzmann* law of radiation expressed in equation (75).

87. If we combine equation (100) with equation (99a) we obtain

$$\frac{\nu'}{T'} = \frac{\nu}{T}. \tag{105}$$

Hence the laws derived at the end of Sec. 84a assume the following form: On infinitely slow reversible adiabatic change in volume of black radiation contained in a cavity, the temperature T varies in the inverse ratio of the cube root of the volume V, the frequencies ν vary in proportion to the temperature, and the radiant energy $\mathsf{U}\,d\nu$ of an infinitely small spectral interval varies in the same ratio. Hence the total radiant energy U as the sum of the energies of all spectral intervals varies also in proportion to the temperature, a statement which agrees with the conclusion arrived at already at the end of Sec. 68, while the space density of radiation, $u = \dfrac{U}{V}$, varies in proportion to the fourth power of the temperature, in agreement with the *Stefan-Boltzmann* law.

88. *Wien's* displacement law may also in the case of black radiation be stated for the specific intensity of radiation K_ν of a plane polarized monochromatic ray. In this form it reads according to (24)

$$\mathsf{K}_\nu = \frac{\nu^3}{c^2} F\left(\frac{T}{\nu}\right). \tag{106}$$

If, as is usually done in experimental physics, the radiation intensity is referred to wave lengths λ instead of frequencies ν, according to (16), namely

$$\mathsf{E}_\lambda = \frac{c\mathsf{K}_\nu}{\lambda^2},$$

equation (106) takes the following form:

$$E_\lambda = \frac{c^2}{\lambda^5} F\left(\frac{\lambda T}{c}\right). \tag{107}$$

This form of *Wien's* displacement law has usually been the starting-point for an experimental test, the result of which has in all cases been a fairly accurate verification of the law.[1]

89. Since E_λ vanishes for $\lambda = 0$ as well as for $\lambda = \infty$, E_λ must have a maximum with respect to λ, which is found from the equation

$$\frac{dE_\lambda}{d\lambda} = 0 = -\frac{5}{\lambda^6} F\left(\frac{\lambda T}{c}\right) + \frac{1}{\lambda^5}\frac{T}{c}\dot{F}\left(\frac{\lambda T}{c}\right)$$

where $\dot{F}$ denotes the differential coefficient of F with respect to its argument. Or

$$\frac{\lambda T}{c}\dot{F}\left(\frac{\lambda T}{c}\right) - 5F\left(\frac{\lambda T}{c}\right) = 0. \tag{108}$$

This equation furnishes a definite value for the argument $\frac{\lambda T}{c}$, so that for the wave length λ_m corresponding to the maximum of the radiation intensity E_λ the relation holds

$$\lambda_m T = b. \tag{109}$$

With increasing temperature the maximum of radiation is therefore displaced in the direction of the shorter wave lengths.

The numerical value of the constant b as determined by *Lummer* and *Pringsheim*[2] is

$$b = 0.294 \,\text{cm degree}. \tag{110}$$

Paschen[3] has found a slightly smaller value, about 0.292.

[1] *E.g.*, *F. Paschen*, Sitzungsber. d. Akad. d. Wissensch. Berlin, pp. 405 and 959, 1899. *O. Lummer* und *E. Pringsheim*, Verhandlungen d. Deutschen physikalischen Gesellschaft **1**, pp. 23 and 215, 1899. Annal. d. Physik **6**, p. 192, 1901.

[2] *O. Lummer* und *E. Pringsheim*, l. c.

[3] *F. Paschen*, Annal. d. Physik **6**, p. 657, 1901.

We may emphasize again at this point that, according to Sec. 19, the maximum of E_λ does not by any means occur at the same point in the spectrum as the maximum of K_ν and that hence the significance of the constant b is essentially dependent on the fact that the intensity of monochromatic radiation is referred to wave lengths, not to frequencies.

90. The value also of the maximum of E_λ is found from (107) by putting $\lambda = \lambda_m$. Allowing for (109) we obtain

$$E_{max} = \text{const. } T^5, \tag{111}$$

i.e., the value of the maximum of radiation in the spectrum of the black radiation is proportional to the fifth power of the absolute temperature.

Should we measure the intensity of monochromatic radiation not by E_λ but by K_ν, we would obtain for the value of the radiation maximum a quite different law, namely,

$$\mathsf{K}_{max} = \text{const. } T^3. \tag{112}$$

CHAPTER IV

RADIATION OF ANY ARBITRARY SPECTRAL DISTRIBUTION OF ENERGY. ENTROPY AND TEMPERATURE OF MONOCHROMATIC RADIATION

91. We have so far applied *Wien's* displacement law only to the case of black radiation; it has, however, a much more general importance. For equation (95), as has already been stated, gives, for any original spectral distribution of the energy radiation contained in the evacuated cavity and radiated uniformly in all directions, the change of this energy distribution accompanying a reversible adiabatic change of the total volume. Every state of radiation brought about by such a process is perfectly stationary and can continue infinitely long, subject, however, to the condition that no trace of an emitting or absorbing substance exists in the radiation space. For otherwise, according to Sec. 51, the distribution of energy would, in the course of time, change through the releasing action of the substance irreversibly, *i.e.*, with an increase of the total entropy, into the stable distribution corresponding to black radiation.

The difference of this general case from the special one dealt with in the preceding chapter is that we can no longer, as in the case of black radiation, speak of a definite temperature of the radiation. Nevertheless, since the second principle of thermodynamics is supposed to hold quite generally, the radiation, like every physical system which is in a definite state, has a definite entropy, $S = Vs$. This entropy consists of the entropies of the monochromatic radiations, and, since the separate kinds of rays are independent of one another, may be obtained by

addition. Hence

$$s = \int_0^\infty \mathsf{s}\, d\nu, \qquad S = V \int_0^\infty \mathsf{s}\, d\nu, \tag{113}$$

where $\mathsf{s}\, d\nu$ denotes the entropy of the radiation of frequencies between ν and $\nu + d\nu$ contained in unit volume. s is a definite function of the two independent variables ν and u and in the following will always be treated as such.

92. If the analytical expression of the function s were known, the law of energy distribution in the normal spectrum could immediately be deduced from it; for the normal spectral distribution of energy or that of black radiation is distinguished from all others by the fact that it has the maximum of the entropy of radiation S.

Suppose then we take s to be a known function of ν and u. Then as a condition for black radiation we have

$$\delta S = 0, \tag{114}$$

for any variations of energy distribution, which are possible with a constant total volume V and constant total energy of radiation U. Let the variation of energy distribution be characterized by making an infinitely small change $\delta\mathsf{u}$ in the energy u of every separate definite frequency ν. Then we have as fixed conditions

$$\delta V = 0 \quad \text{and} \quad \int_0^\infty \delta\mathsf{u}\, d\nu = 0. \tag{115}$$

The changes d and δ are of course quite independent of each other.

Now since $\delta V = 0$, we have from (114) and (113)

$$\int_0^\infty \delta\mathsf{s}\, d\nu = 0,$$

or, since ν remains unvaried

$$\int_0^\infty \frac{\partial \mathsf{s}}{\partial \mathsf{u}}\,\delta \mathsf{u}\,d\nu = 0,$$

and, by allowing for (115), the validity of this equation for all values of $\delta \mathsf{u}$ whatever requires that

$$\frac{\partial \mathsf{s}}{\partial \mathsf{u}} = \text{const.} \tag{116}$$

for all different frequencies. This equation states the law of energy distribution in the case of black radiation.

93. The constant of equation (116) bears a simple relation to the temperature of black radiation. For if the black radiation, by conduction into it of a certain amount of heat at constant volume V, undergoes an infinitely small change in energy δU, then, according to (73), its change in entropy is

$$\delta S = \frac{\delta U}{T}.$$

However, from (113) and (116),

$$\delta S = V\int_0^\infty \frac{\partial \mathsf{s}}{\partial \mathsf{u}}\,\delta \mathsf{u}\,d\nu = \frac{\partial \mathsf{s}}{\partial \mathsf{u}} V\int_0^\infty \delta \mathsf{u}\,d\nu = \frac{\partial \mathsf{s}}{\partial \mathsf{u}}\,\delta U$$

hence

$$\frac{\partial \mathsf{s}}{\partial \mathsf{u}} = \frac{1}{T} \tag{117}$$

and the above quantity, which was found to be the same for all frequencies in the case of black radiation, is shown to be the reciprocal of the temperature of black radiation.

Through this law the concept of temperature gains significance also for radiation of a quite arbitrary distribution of energy. For since s depends only on u and ν, *monochromatic radiation, which is uniform in all directions and has a definite energy density* u, *has also a definite temperature given by* (117), *and, among all conceivable distributions of energy, the normal one is characterized by the fact that the radiations of all frequencies have the same temperature.*

Any change in the energy distribution consists of a passage of energy from one monochromatic radiation into another, and, if the temperature of the first radiation is higher, the energy transformation causes an increase of the total entropy and is hence possible in nature without compensation; on the other hand, if the temperature of the second radiation is higher, the total entropy decreases and therefore the change is impossible in nature, unless compensation occurs simultaneously, just as is the case with the transfer of heat between two bodies of different temperatures.

94. Let us now investigate *Wien's* displacement law with regard to the dependence of the quantity s on the variables u and ν. From equation (101) it follows, on solving for T and substituting the value given in (117), that

$$\frac{1}{T} = \frac{1}{\nu} F\left(\frac{c^3 \mathsf{u}}{\nu^3}\right) = \frac{\partial \mathsf{s}}{\partial \mathsf{u}} \tag{118}$$

where again F represents a function of a single argument and the constants do not contain the velocity of propagation c. On integration with respect to the argument we obtain

$$\mathsf{s} = \frac{\nu^2}{c^3} F_1\left(\frac{c^3 \mathsf{u}}{\nu^3}\right) \tag{119}$$

the notation remaining the same. In this form *Wien's* displacement

law has a significance for every separate monochromatic radiation and hence also for radiations of any arbitrary energy distribution.

95. According to the second principle of thermodynamics, the total entropy of radiation of quite arbitrary distribution of energy must remain constant on adiabatic reversible compression. We are now able to give a direct proof of this proposition on the basis of equation (119). For such a process, according to equation (113), the relation holds:

$$\begin{aligned}\delta S &= \int_0^\infty d\nu\,(V\,\delta \mathsf{s} + \mathsf{s}\,\delta V)\\ &= \int_0^\infty d\nu\,(V\frac{\partial \mathsf{s}}{\partial \mathsf{u}}\,\delta \mathsf{u} + \mathsf{s}\,\delta V).\end{aligned} \tag{120}$$

Here, as everywhere, s should be regarded as a function of u and ν, and $\delta\nu = 0$.

Now for a reversible adiabatic change of state the relation (95) holds. Let us take from the latter the value of $\delta\mathsf{u}$ and substitute. Then we have

$$\delta S = \delta V \int_0^\infty d\nu \left\{ \frac{\partial \mathsf{s}}{\partial \mathsf{u}} \left(\frac{\nu\, d\mathsf{u}}{3\, d\nu} - \mathsf{u} \right) + \mathsf{s} \right\}.$$

In this equation the differential coefficient of u with respect to ν refers to the spectral distribution of energy originally assigned arbitrarily and is therefore, in contrast to the partial differential coefficients, denoted by the letter d.

Now the complete differential is:

$$\frac{d\mathsf{s}}{d\nu} = \frac{\partial \mathsf{s}}{\partial \mathsf{u}} \frac{d\mathsf{u}}{d\nu} + \frac{\partial \mathsf{s}}{\partial \nu}.$$

Hence by substitution:

$$\delta S = \delta V \int_0^\infty d\nu \left\{ \frac{\nu}{3} \left(\frac{d\mathsf{s}}{d\nu} - \frac{\partial \mathsf{s}}{\partial \nu} \right) - \mathsf{u} \frac{\partial \mathsf{s}}{\partial \mathsf{u}} + \mathsf{s} \right\}. \tag{121}$$

But from equation (119) we obtain by differentiation

$$\frac{\partial \mathsf{s}}{\partial \mathsf{u}} = \frac{1}{\nu}\dot{F}\left(\frac{c^3\mathsf{u}}{\nu^3}\right) \quad \text{and} \quad \frac{\partial \mathsf{s}}{\partial \nu} = \frac{2\nu}{c^3}F\left(\frac{c^3\mathsf{u}}{\nu^3}\right) - \frac{3\mathsf{u}}{\nu^2}\dot{F}\left(\frac{c^3\mathsf{u}}{\nu^3}\right). \tag{122}$$

Hence

$$\nu\frac{\partial \mathsf{s}}{\partial \nu} = 2\mathsf{s} - 3\mathsf{u}\frac{\partial \mathsf{s}}{\partial \mathsf{u}}. \tag{123}$$

On substituting this in (121), we obtain

$$\delta S = \delta V \int_0^\infty d\nu \left(\frac{\nu}{3}\frac{d\mathsf{s}}{d\nu} + \frac{1}{3}\mathsf{s}\right) \tag{124}$$

or,

$$\delta S = \frac{\delta V}{3}[\nu\mathsf{s}]_0^\infty = 0,$$

as it should be. That the product $\nu\mathsf{s}$ vanishes also for $\nu = \infty$ may be shown just as was done in Sec. 83 for the product $\nu\mathsf{u}$.

96. By means of equations (118) and (119) it is possible to give to the laws of reversible adiabatic compression a form in which their meaning is more clearly seen and which is the generalization of the laws stated in Sec. 87 for black radiation and a supplement to them. It is, namely, possible to derive (105) again from (118) and (99b). Hence the laws deduced in Sec. 87 for the change of frequency and temperature of the monochromatic radiation energy remain valid for a radiation of an originally quite arbitrary distribution of energy. The only difference as compared with the black radiation consists in the fact that now every frequency has its own distinct temperature.

Moreover it follows from (119) and (99b) that

$$\frac{\mathsf{s}'}{\nu'^2} = \frac{\mathsf{s}}{\nu^2}. \tag{125}$$

Now $\mathsf{s}\,d\nu\,V = \mathsf{S}\,d\nu$ denotes the radiation entropy between the frequencies ν and $\nu + d\nu$ contained in the volume V. Hence on account of (125), (99a), and (99c)

$$\mathsf{S}'\,d\nu' = \mathsf{S}\,d\nu, \tag{126}$$

i.e., the radiation entropy of an infinitely small spectral interval remains constant. This is another statement of the fact that the total entropy of radiation, taken as the sum of the entropies of all monochromatic radiations contained therein, remains constant.

97. We may go one step further, and, from the entropy s and the temperature T of an unpolarized monochromatic radiation which is uniform in all directions, draw a certain conclusion regarding the entropy and temperature of a single, plane polarized, monochromatic pencil. That every separate pencil also has a certain entropy follows by the second principle of thermodynamics from the phenomenon of emission. For since, by the act of emission, heat is changed into radiant heat, the entropy of the emitting body decreases during emission, and, along with this decrease, there must be, according to the principle of increase of the total entropy, an increase in a different form of entropy as a compensation. This can only be due to the energy of the emitted radiation. Hence every separate, plane polarized, monochromatic pencil has its definite entropy, which can depend only on its energy and frequency and which is propagated and spreads into space with it. We thus gain the idea of entropy radiation, which is measured, as in the analogous case of energy radiation, by the amount of entropy which passes in unit time through unit area in a definite direction. Hence statements, exactly similar to those made in Sec. 14 regarding energy radiation, will hold for the radiation of entropy, inasmuch as every pencil possesses and conveys, not only its energy, but also its entropy. Referring the reader to the discussions of Sec. 14, we shall, for the present, merely

enumerate the most important laws for future use.

98. In a space filled with any radiation whatever the entropy radiated in the time dt through an element of area $d\sigma$ in the direction of the conical element $d\Omega$ is given by an expression of the form

$$dt\, d\sigma \cos\theta\, d\Omega\, L = L \sin\theta \cos\theta\, d\theta\, d\phi\, d\sigma\, dt. \tag{127}$$

The positive quantity L we shall call the "specific intensity of entropy radiation" at the position of the element of area $d\sigma$ in the direction of the solid angle $d\Omega$. L is, in general, a function of position, time, and direction.

The total radiation of entropy through the element of area $d\sigma$ toward one side, say the one where θ is an acute angle, is obtained by integration with respect to ϕ from 0 to 2π and with respect to θ from 0 to $\frac{\pi}{2}$. It is

$$d\sigma\, dt \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta\, L \sin\theta \cos\theta.$$

When the radiation is uniform in all directions, and hence L constant, the entropy radiation through $d\sigma$ toward one side is

$$\pi L\, d\sigma\, dt. \tag{128}$$

The specific intensity L of the entropy radiation in every direction consists further of the intensities of the separate rays belonging to the different regions of the spectrum, which are propagated independently of one another. Finally for a ray of definite color and intensity the nature of its polarization is characteristic. When a monochromatic ray of frequency ν consists of two mutually independent[1] components,

[1] "Independent" in the sense of "non-coherent." If, *e.g.*, a ray with the principal

polarized at right angles to each other, with the principal intensities of energy radiation (Sec. 17) K_ν and K'_ν, the specific intensity of entropy radiation is of the form

$$L = \int_0^\infty d\nu\,(\mathsf{L}_\nu + \mathsf{L}'_\nu). \tag{129}$$

The positive quantities L_ν and L'_ν in this expression, the principal intensities of entropy radiation of frequency ν, are determined by the values of K_ν and K'_ν. By substitution in (127), this gives for the entropy which is radiated in the time dt through the element of area $d\sigma$ in the direction of the conical element $d\Omega$ the expression

$$dt\,d\sigma\cos\theta\,d\Omega\int_0^\infty d\nu\,(\mathsf{L}_\nu + \mathsf{L}'_\nu),$$

and, for monochromatic plane polarized radiation,

$$dt\,d\sigma\cos\theta\,d\Omega\,\mathsf{L}_\nu\,d\nu = \mathsf{L}_\nu\,d\nu\,\sin\theta\cos\theta\,d\theta\,d\phi\,d\sigma\,dt. \tag{130}$$

For unpolarized rays $\mathsf{L}_\nu = \mathsf{L}'_\nu$ and (129) becomes

$$L = 2\int_0^\infty \mathsf{L}_\nu\,d\nu.$$

For radiation which is uniform in all directions the total entropy radiation toward one side is, according to (128),

$$2\pi\,d\sigma\,dt\int_0^\infty \mathsf{L}_\nu\,d\nu.$$

intensities K and K' is elliptically polarized, its entropy is not equal to $\mathsf{L}+\mathsf{L}'$, but equal to the entropy of a plane polarized ray of intensity $\mathsf{K}+\mathsf{K}'$. For an elliptically polarized ray may be transformed at once into a plane polarized one, *e.g.*, by total reflection. For the entropy of a ray with coherent components see below Sec. 104, *et seq.*

99. From the intensity of the propagated entropy radiation the expression for the *space density* of the radiant entropy may also be obtained, just as the space density of the radiant energy follows from the intensity of the propagated radiant energy. (Compare Sec. 22.) In fact, in analogy with equation (20), the space density, s, of the entropy of radiation at any point in a vacuum is

$$s = \frac{1}{c}\int L\, d\Omega, \tag{131}$$

where the integration is to be extended over the conical elements which spread out from the point in question in all directions. L is constant for uniform radiation and we obtain

$$s = \frac{4\pi L}{c}. \tag{132}$$

By spectral resolution of the quantity L, according to equation (129), we obtain from (131) also the space density of the monochromatic radiation entropy:

$$\mathsf{s} = \frac{1}{c}\int (\mathsf{L} + \mathsf{L}')\, d\Omega,$$

and for unpolarized radiation, which is uniform in all directions

$$\mathsf{s} = \frac{8\pi \mathsf{L}}{c}. \tag{133}$$

100. As to how the entropy radiation L depends on the energy radiation K *Wien's* displacement law in the form of (119) affords immediate information. It follows, namely, from it, considering (133) and (24), that

$$\mathsf{L} = \frac{\nu^2}{c^2} F\left(\frac{c^2 \mathsf{K}}{\nu^3}\right) \tag{134}$$

and, moreover, on taking into account (118),

$$\frac{\partial \mathsf{L}}{\partial \mathsf{K}} = \frac{\partial \mathsf{s}}{\partial \mathsf{u}} = \frac{1}{T}. \tag{135}$$

Hence also

$$T = \nu F_1\left(\frac{c^2 \mathsf{K}}{\nu^3}\right) \tag{136}$$

or

$$\mathsf{K} = \frac{\nu^3}{c^2} F_2\left(\frac{T}{\nu}\right). \tag{137}$$

It is true that these relations, like the equations (118) and (119), were originally derived for radiation which is unpolarized and uniform in all directions. They hold, however, generally in the case of any radiation whatever for each separate monochromatic plane polarized ray. For, since the separate rays behave and are propagated quite independently of one another, the intensity, L, of the entropy radiation of a ray can depend only on the intensity of the energy radiation, K, of the same ray. Hence every separate monochromatic ray has not only its energy but also its entropy defined by (134) and its temperature defined by (136).

101. The extension of the conception of temperature to a single monochromatic ray, just discussed, implies that at the same point in a medium, through which any rays whatever pass, there exist in general an infinite number of temperatures, since every ray passing through the point has its separate temperature, and, moreover, even the rays of different color traveling in the same direction show temperatures that differ according to the spectral distribution of energy. In addition to all these temperatures there is finally the temperature of the medium itself, which at the outset is entirely independent of the temperature of the radiation. This complicated method of consideration lies in the nature of the case and corresponds to the complexity of the physical

processes in a medium through which radiation travels in such a way. It is only in the case of stable thermodynamic equilibrium that there is but one temperature, which then is common to the medium itself and to all rays of whatever color crossing it in different directions.

In practical physics also the necessity of separating the conception of radiation temperature from that of body temperature has made itself felt to a continually increasing degree. Thus it has for some time past been found advantageous to speak, not only of the real temperature of the sun, but also of an "apparent" or "effective" temperature of the sun, *i.e.*, that temperature which the sun would need to have in order to send to the earth the heat radiation actually observed, if it radiated like a black body. Now the apparent temperature of the sun is obviously nothing but the actual temperature of the solar rays,[1] depending entirely on the nature of the rays, and hence a property of the rays and not a property of the sun itself. Therefore it would be, not only more convenient, but also more correct, to apply this notation directly, instead of speaking of a fictitious temperature of the sun, which can be made to have a meaning only by the introduction of an assumption that does not hold in reality.

Measurements of the brightness of monochromatic light have recently led *L. Holborn* and *F. Kurlbaum*[2] to the introduction of the concept of "black" temperature of a radiating surface. The black temperature of a radiating surface is measured by the brightness of the rays which it emits. It is in general a separate one for each ray of definite color, direction, and polarization, which the surface emits, and, in fact, merely represents the temperature of such a ray. It is, according

[1] On the average, since the solar rays of different color do not have exactly the same temperature.

[2] *L. Holborn* und *F. Kurlbaum*, Annal. d. Physik **10**, p. 229, 1903.

to equation (136), determined by its brightness (specific intensity), K, and its frequency, ν, without any reference to its origin and previous states. The definite numerical form of this equation will be given below in Sec. 166. Since a black body has the maximum emissive power, the temperature of an emitted ray can never be higher than that of the emitting body.

102. Let us make one more simple application of the laws just found to the special case of black radiation. For this, according to (81), the total space density of entropy is

$$s = \frac{4}{3}a^3T. \tag{138}$$

Hence, according to (132), the specific intensity of the total entropy radiation in any direction is

$$L = \frac{c}{3\pi}aT^3, \tag{139}$$

and the total entropy radiation through an element of area $d\sigma$ toward one side is, according to (128),

$$\frac{c}{3}aT^3\,d\sigma\,dt. \tag{140}$$

As a special example we shall now apply the two principles of thermodynamics to the case in which the surface of a black body of temperature T and of infinitely large heat capacity is struck by black radiation of temperature T' coming from all directions. Then, according to (7) and (76), the black body emits per unit area and unit time the energy

$$\pi K = \frac{ac}{4}T^4,$$

and, according to (140), the entropy

$$\frac{ac}{3}T^3.$$

On the other hand, it absorbs the energy

$$\frac{ac}{4}T'^4$$

and the entropy

$$\frac{ac}{3}T'^3.$$

Hence, according to the first principle, the total heat added to the body, positive or negative according as T' is larger or smaller than T, is

$$Q = \frac{ac}{4}T'^4 - \frac{ac}{4}T^4 = \frac{ac}{4}(T'^4 - T^4),$$

and, according to the second principle, the change of the entire entropy is positive or zero. Now the entropy of the body changes by $\frac{Q}{T}$, the entropy of the radiation in the vacuum by

$$\frac{ac}{3}(T^3 - T'^3).$$

Hence the change per unit time and unit area of the entire entropy of the system considered is

$$\frac{ac}{4}\frac{T'^4 - T^4}{T} + \frac{ac}{3}(T^3 - T'^3) \geqq 0.$$

In fact this relation is satisfied for all values of T and T'. The minimum value of the expression on the left side is zero; this value is reached when $T = T'$. In that case the process is reversible. If, however, T differs

from T', we have an appreciable increase of entropy; hence the process is irreversible. In particular we find that if $T = 0$ the increase in entropy is ∞, *i.e.*, the absorption of heat radiation by a black body of vanishingly small temperature is accompanied by an infinite increase in entropy and cannot therefore be reversed by any finite compensation. On the other hand for $T' = 0$, the increase in entropy is only equal to $\frac{ac}{12}T^3$, *i.e.*, the emission of a black body of temperature T without simultaneous absorption of heat radiation is irreversible without compensation, but can be reversed by a compensation of at least the stated finite amount. For example, if we let the rays emitted by the body fall back on it, say by suitable reflection, the body, while again absorbing these rays, will necessarily be at the same time emitting new rays, and this is the compensation required by the second principle.

Generally we may say: Emission without simultaneous absorption is irreversible, while the opposite process, absorption without emission, is impossible in nature.

103. A further example of the application of the two principles of thermodynamics is afforded by the irreversible expansion of originally black radiation of volume V and temperature T to the larger volume V' as considered above in Sec. 70, but in the absence of any absorbing or emitting substance whatever. Then not only the total energy but also the energy of every separate frequency ν remains constant; hence, when on account of diffuse reflection from the walls the radiation has again become uniform in all directions, $\mathsf{u}_\nu V = \mathsf{u}'_\nu V'$; moreover by this relation, according to (118), the temperature T_ν' of the monochromatic radiation of frequency ν in the final state is determined. The actual calculation, however, can be performed only with the help of equation (275) (see below). The total entropy of radiation, *i.e.*, the sum of

the entropies of the radiations of all frequencies,

$$V' \int_0^\infty \mathsf{s}'_\nu \, d\nu,$$

must, according to the second principle, be larger in the final state than in the original state. Since T'_ν has different values for the different frequencies ν, the final radiation is no longer black. Hence, on subsequent introduction of a carbon particle into the cavity, a finite change of the distribution of energy is obtained, and simultaneously the entropy increases further to the value S' calculated in (82).

104. In Sec. 98 we have found the intensity of entropy radiation of a definite frequency in a definite direction by adding the entropy radiations of the two independent components K and K', polarized at right angles to each other, or

$$\mathsf{L}(\mathsf{K}) + \mathsf{L}(\mathsf{K}'), \tag{141}$$

where L denotes the function of K given in equation (134). This method of procedure is based on the general law that the entropy of two mutually independent physical systems is equal to the sum of the entropies of the separate systems.

If, however, the two components of a ray, polarized at right angles to each other, are not independent of each other, this method of procedure no longer remains correct. This may be seen, *e.g.*, on resolving the radiation intensity, not with reference to the two principal planes of polarization with the principal intensities K and K', but with reference to any other two planes at right angles to each other, where, according to equation (8), the intensities of the two components assume the following values

$$\begin{aligned} \mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi &= \mathsf{K}'' \\ \mathsf{K}\sin^2\psi + \mathsf{K}'\cos^2\psi &= \mathsf{K}'''. \end{aligned} \tag{142}$$

In that case, of course, the entropy radiation is not equal to $\mathsf{L}(\mathsf{K}'') + \mathsf{L}(\mathsf{K}''')$.

Thus, while the energy radiation is always obtained by the summation of any two components which are polarized at right angles to each other, no matter according to which azimuth the resolution is performed, since always

$$\mathsf{K}'' + \mathsf{K}''' = \mathsf{K} + \mathsf{K}', \tag{143}$$

a corresponding equation does not hold in general for the entropy radiation. The cause of this is that the two components, the intensities of which we have denoted by K'' and K''', are, unlike K and K', not independent or non-coherent in the optic sense. In such a case

$$\mathsf{L}(\mathsf{K}'') + \mathsf{L}(\mathsf{K}''') > \mathsf{L}(\mathsf{K}) + \mathsf{L}(\mathsf{K}'), \tag{144}$$

as is shown by the following consideration.

Since in the state of thermodynamic equilibrium all rays of the same frequency have the same intensity of radiation, the intensities of radiation of any two plane polarized rays will tend to become equal, *i.e.*, the passage of energy between them will be accompanied by an increase of entropy, when it takes place in the direction from the ray of greater intensity toward that of smaller intensity. Now the left side of the inequality (144) represents the entropy radiation of two non-coherent plane polarized rays with the intensities K'' and K''', and the right side the entropy radiation of two non-coherent plane polarized rays with the intensities K and K'. But, according to (142), the values of K'' and K''' lie between K and K'; therefore the inequality (144) holds.

At the same time it is apparent that the error committed, when the entropy of two coherent rays is calculated as if they were non-coherent, is always in such a sense that the entropy found is too large.

The radiations K'' and K''' are called "partially coherent," since they have some terms in common. In the special case when one of the two principal intensities K and K' vanishes entirely, the radiations K'' and K''' are said to be "completely coherent," since in that case the expression for one radiation may be completely reduced to that for the other. The entropy of two completely coherent plane polarized rays is equal to the entropy of a single plane polarized ray, the energy of which is equal to the sum of the two separate energies.

105. Let us for future use solve also the more general problem of calculating the entropy radiation of a ray consisting of an arbitrary number of plane polarized non-coherent components K_1, K_2, K_3, ..., the planes of vibration (planes of the electric vector) of which are given by the azimuths ψ_1, ψ_2, ψ_3, ... This problem amounts to finding the principal intensities K_0 and K'_0 of the whole ray; for the ray behaves in every physical respect as if it consisted of the non-coherent components K_0 and K'_0. For this purpose we begin by establishing the value K_ψ of the component of the ray for an azimuth ψ taken arbitrarily. Denoting by f the electric vector of the ray in the direction ψ, we obtain this value K_ψ from the equation

$$f = f_1 \cos(\psi_1 - \psi) + f_2 \cos(\psi_2 - \psi) + f_3 \cos(\psi_3 - \psi) + \ldots,$$

where the terms on the right side denote the projections of the vectors of the separate components in the direction ψ, by squaring and averaging and taking into account the fact that f_1, f_2, f_3, ... are non-coherent

$$\begin{aligned} & \mathsf{K}_\psi = \mathsf{K}_1 \cos^2(\psi_1 - \psi) + \mathsf{K}_2 \cos^2(\psi_2 - \psi) + \ldots \\ \text{or} \quad & \mathsf{K}_\psi = A \cos^2 \psi + B \sin^2 \psi + C \sin \psi \cos \psi \\ \text{where} \quad & A = \mathsf{K}_1 \cos^2 \psi_1 + \mathsf{K}_2 \cos^2 \psi_2 + \ldots \\ & B = \mathsf{K}_1 \sin^2 \psi_1 + \mathsf{K}_2 \sin^2 \psi_2 + \ldots \\ & C = 2(\mathsf{K}_1 \sin \psi_1 \cos \psi_1 + \mathsf{K}_2 \sin \psi_2 \cos \psi_2 + \ldots). \end{aligned} \tag{145}$$

The principal intensities K_0 and K'_0 of the ray follow from this expression as the maximum and the minimum value of K_ψ according to the equation

$$\frac{d\mathsf{K}_\psi}{d\psi} = 0 \quad \text{or,} \quad \tan 2\psi = \frac{C}{A-B}.$$

Hence it follows that the principal intensities are

$$\left.\begin{matrix}\mathsf{K}_0\\ \mathsf{K}'_0\end{matrix}\right\} = \tfrac{1}{2}(A+B \pm \sqrt{(A-B)^2+C^2}), \tag{146}$$

or, by taking (145) into account,

$$\left.\begin{matrix}\mathsf{K}_0\\ \mathsf{K}'_0\end{matrix}\right\} = \frac{1}{2}\Bigg(\mathsf{K}_1+\mathsf{K}_2+\dots \pm \sqrt{\begin{matrix}(\mathsf{K}_1\cos 2\psi_1+\mathsf{K}_2\cos 2\psi_2+\dots)^2\\ +(\mathsf{K}_1\sin 2\psi_1+\mathsf{K}_2\sin 2\psi_2+\dots)^2\end{matrix}}\Bigg). \tag{147}$$

Then the entropy radiation required becomes:

$$\mathsf{L}(\mathsf{K}_0)+\mathsf{L}(\mathsf{K}'_0). \tag{148}$$

106. When two ray components K and K', polarized at right angles to each other, are non-coherent, K and K' are also the principal intensities, and the entropy radiation is given by (141). The converse proposition, however, does not hold in general, that is to say, the two components of a ray polarized at right angles to each other, which correspond to the principal intensities K and K', are not necessarily non-coherent, and hence the entropy radiation is not always given by (141).

This is true, *e.g.*, in the case of elliptically polarized light. There the radiations K and K' are completely coherent and their entropy is

equal to $\mathsf{L}(\mathsf{K} + \mathsf{K}')$. This is caused by the fact that it is possible to give the two ray components an arbitrary displacement of phase in a reversible manner, say by total reflection. Thereby it is possible to change elliptically polarized light to plane polarized light and *vice versa*.

The entropy of completely or partially coherent rays has been investigated most thoroughly by *M. Laue*.[1] For the significance of optical coherence for thermodynamic probability see the next part, Sec. 119.

[1] *M. Laue*, Annalen d. Phys. **23**, p. 1, 1907.

CHAPTER V

ELECTRODYNAMICAL PROCESSES IN A STATIONARY FIELD OF RADIATION

107. We shall now consider from the standpoint of pure electrodynamics the processes that take place in a vacuum, which is bounded on all sides by reflecting walls and through which heat radiation passes uniformly in all directions, and shall then inquire into the relations between the electrodynamical and the thermodynamic quantities.

The electrodynamical state of the field of radiation is determined at every instant by the values of the electric field-strength E and the magnetic field-strength H at every point in the field, and the changes in time of these two vectors are completely determined by *Maxwell's* field equations (52), which we have already used in Sec. 53, together with the boundary conditions, which hold at the reflecting walls. In the present case, however, we have to deal with a solution of these equations of much greater complexity than that expressed by (54), which corresponds to a plane wave. For a plane wave, even though it be periodic with a wave length lying within the optical or thermal spectrum, can never be interpreted as heat radiation. For, according to Sec. 16, a finite intensity K of heat radiation requires a finite solid angle of the rays and, according to Sec. 18, a spectral interval of finite width. But an absolutely plane, absolutely periodic wave has a zero solid angle and a zero spectral width. Hence in the case of a plane periodic wave there can be no question of either entropy or temperature of the radiation.

108. Let us proceed in a perfectly general way to consider the components of the field-strengths E and H as functions of the time at a definite point, which we may think of as the origin of the coordinate system. Of these components, which are produced by all rays passing

through the origin, there are six; we select one of them, say E_z, for closer consideration. However complicated it may be, it may under all circumstances be written as a *Fourier's* series for a limited time interval, say from $t = 0$ to $t = \mathsf{T}$; thus

$$\mathsf{E}_z = \sum_{n=1}^{n=\infty} C_n \cos\left(\frac{2\pi n t}{\mathsf{T}} - \theta_n\right) \tag{149}$$

where the summation is to extend over all positive integers n, while the constants C_n (positive) and θ_n may vary arbitrarily from term to term. The time interval T, the fundamental period of the *Fourier's* series, we shall choose so large that all times t which we shall consider hereafter are included in this time interval, so that $0 < t < \mathsf{T}$. Then we may regard E_z as identical in all respects with the *Fourier's* series, *i.e.*, we may regard E_z as consisting of "partial vibrations," which are strictly periodic and of frequencies given by

$$\nu = \frac{n}{\mathsf{T}}.$$

Since, according to Sec. 3, the time differential dt required for the definition of the intensity of a heat ray is necessarily large compared with the periods of vibration of all colors contained in the ray, a single time differential dt contains a large number of vibrations, *i.e.*, the product $\nu\, dt$ is a large number. Then it follows *a fortiori* that νt and, still more,

$$\nu\mathsf{T} = n \text{ is enormously large} \tag{150}$$

for all values of ν entering into consideration. From this we must conclude that all amplitudes C_n with a moderately large value for the ordinal number n do not appear at all in the *Fourier's* series, that is to say, they are negligibly small.

109. Though we have no detailed special information about the function E_z, nevertheless its relation to the radiation of heat affords some important information as to a few of its general properties. Firstly, for the space density of radiation in a vacuum we have, according to Maxwell's theory,

$$u = \frac{1}{8\pi}(\overline{\mathsf{E}_x^2} + \overline{\mathsf{E}_y^2} + \overline{\mathsf{E}_z^2} + \overline{\mathsf{H}_x^2} + \overline{\mathsf{H}_y^2} + \overline{\mathsf{H}_z^2}).$$

Now the radiation is uniform in all directions and in the stationary state, hence the six mean values named are all equal to one another, and it follows that

$$u = \frac{3}{4\pi}\overline{\mathsf{E}_z^2}. \tag{151}$$

Let us substitute in this equation the value of E_z as given by (149). Squaring the latter and integrating term by term through a time interval, from 0 to t, assumed large in comparison with all periods of vibration $\frac{1}{\nu}$ but otherwise arbitrary, and then dividing by t, we obtain, since the radiation is perfectly stationary,

$$u = \frac{3}{8\pi}\sum C_n^2. \tag{152}$$

From this relation we may at once draw an important conclusion as to the nature of E_z as a function of time. Namely, since the *Fourier's* series (149) consists, as we have seen, of a great many terms, the squares, C_n^2, of the separate amplitudes of vibration the sum of which gives the space density of radiation, must have exceedingly small values. Moreover in the integral of the square of the *Fourier's* series the terms which depend on the time t and contain the products of any two different amplitudes all cancel; hence the amplitudes C_n and the phase-constants θ_n must vary from one ordinal number to another in a quite

irregular manner. We may express this fact by saying that the separate partial vibrations of the series are very small and in a "chaotic"[1] state.

For the specific intensity of the radiation travelling in any direction whatever we obtain from (21)

$$K = \frac{cu}{4\pi} = \frac{3c}{32\pi^2} \sum C_n^2. \tag{153}$$

110. Let us now perform the spectral resolution of the last two equations. To begin with we have from (22):

$$u = \int_0^\infty \mathsf{u}_\nu \, d\nu = \frac{3}{8\pi} \sum_1^\infty C_n^2. \tag{154}$$

On the right side of the equation the sum $\sum$ consists of separate terms, every one of which corresponds to a separate ordinal number n and to a simple periodic partial vibration. Strictly speaking this sum does not represent a continuous sequence of frequencies ν, since n is an integral number. But n is, according to (150), so enormously large for all frequencies which need be considered that the frequencies ν corresponding to the successive values of n lie very close together. Hence the interval $d\nu$, though infinitesimal compared with ν, still contains a large number of partial vibrations, say n', where

$$d\nu = \frac{n'}{\mathsf{T}}. \tag{155}$$

If now in (154) we equate, instead of the total energy densities, the energy densities corresponding to the interval $d\nu$ only, which are independent of those of the other spectral regions, we obtain

$$\mathsf{u}_\nu \, d\nu = \frac{3}{8\pi} \sum_n^{n+n'} C_n^2,$$

[1] Compare footnote to page 137 (Tr.).

or, according to (155),

$$\mathsf{u}_\nu = \frac{3\mathsf{T}}{8\pi} \cdot \frac{1}{n'} \sum_{n}^{n+n'} C_n^2 = \frac{3\mathsf{T}}{8\pi} \cdot \overline{C_n^2}, \tag{156}$$

where we denote by $\overline{C_n^2}$ the average value of C_n^2 in the interval from n to $n + n'$. The existence of such an average value, the magnitude of which is independent of n, provided n' be taken small compared with n, is, of course, not self-evident at the outset, but is due to a special property of the function E_z which is peculiar to stationary heat radiation. On the other hand, since many terms contribute to the mean value, nothing can be said either about the magnitude of a separate term C_n^2, or about the connection of two consecutive terms, but they are to be regarded as perfectly independent of each other.

In a very similar manner, by making use of (24), we find for the specific intensity of a monochromatic plane polarized ray, travelling in any direction whatever,

$$\mathsf{K}_\nu = \frac{3c\mathsf{T}}{64\pi^2} \overline{C_n^2}. \tag{157}$$

From this it is apparent, among other things, that, according to the electromagnetic theory of radiation, a monochromatic light or heat ray is represented, not by a simple periodic wave, but by a superposition of a large number of simple periodic waves, the mean value of which constitutes the intensity of the ray. In accord with this is the fact, known from optics, that two rays of the same color and intensity but of different origin never interfere with each other, as they would, of necessity, if every ray were a simple periodic one.

Finally we shall also perform the spectral resolution of the mean

value of E_z^2, by writing

$$\mathsf{E}_z^2 = J = \int_0^\infty \mathsf{J}_\nu \, d\nu. \tag{158}$$

Then by comparison with (151), (154), and (156) we find

$$\mathsf{J}_\nu = \frac{4\pi}{3}\mathsf{u}_\nu = \frac{\mathsf{T}}{2}\overline{C_n^2}. \tag{159}$$

According to (157), J_ν is related to K_ν, the specific intensity of radiation of a plane polarized ray, as follows:

$$\mathsf{K}_\nu = \frac{3c}{32\pi^2}\mathsf{J}_\nu. \tag{160}$$

111. Black radiation is frequently said to consist of a large number of regular periodic vibrations. This method of expression is perfectly justified, inasmuch as it refers to the resolution of the total vibration in a *Fourier's* series, according to equation (149), and often is exceedingly well adapted for convenience and clearness of discussion. It should, however, not mislead us into believing that such a "regularity" is caused by a special physical property of the elementary processes of vibration. For the resolvability into a Fourier's series is mathematically self-evident and hence, in a physical sense, tells us nothing new. In fact, it is even always possible to regard a vibration which is damped to an arbitrary extent as consisting of a sum of regular periodic partial vibrations with constant amplitudes and constant phases. On the contrary, it may just as correctly be said that in all nature there is no process more complicated than the vibrations of black radiation. In particular, these vibrations do not depend in any characteristic manner on the special processes that take place in the centers of emission of

the rays, say on the period or the damping of the emitting particles; for the normal spectrum is distinguished from all other spectra by the very fact that all individual differences caused by the special nature of the emitting substances are perfectly equalized and effaced. Therefore to attempt to draw conclusions concerning the special properties of the particles emitting the rays from the elementary vibrations in the rays of the normal spectrum would be a hopeless undertaking.

In fact, black radiation may just as well be regarded as consisting, not of regular periodic vibrations, but of absolutely irregular separate impulses. The special regularities, which we observe in monochromatic light resolved spectrally, are caused merely by the special properties of the spectral apparatus used, *e.g.*, the dispersing prism (natural periods of the molecules), or the diffraction grating (width of the slits). Hence it is also incorrect to find a characteristic difference between light rays and Roentgen rays (the latter assumed as an electromagnetic process in a vacuum) in the circumstance that in the former the vibrations take place with greater regularity. Roentgen rays may, under certain conditions, possess more selective properties than light rays. The resolvability into a *Fourier's* series of partial vibrations with constant amplitudes and constant phases exists for both kinds of rays in precisely the same manner. What especially distinguishes light vibrations from Roentgen vibrations is the much smaller frequency of the partial vibrations of the former. To this is due the possibility of their spectral resolution, and probably also the far greater regularity of the changes of the radiation intensity in every region of the spectrum in the course of time, which, however, is not caused by a special property of the elementary processes of vibration, but merely by the constancy of the mean values.

112. The elementary processes of radiation exhibit regularities only when the vibrations are restricted to a narrow spectral region, that is

to say in the case of spectroscopically resolved light, and especially in the case of the natural spectral lines. If, *e.g.*, the amplitudes C_n of the *Fourier's* series (149) differ from zero only between the ordinal numbers $n = n_0$ and $n = n_1$, where $\frac{n_1 - n_0}{n_0}$ is small, we may write

$$\mathsf{E}_z = C_0 \cos\left(\frac{2\pi n_0 t}{\mathsf{T}} - \theta_0\right), \tag{161}$$

where

$$C_0 \cos\theta_0 = \sum_{n_0}^{n_1} C_n \cos\left(\frac{2\pi(n-n_0)t}{\mathsf{T}} - \theta_n\right)$$

$$C_0 \sin\theta_0 = -\sum_{n_0}^{n_1} C_n \sin\left(\frac{2\pi(n-n_0)t}{\mathsf{T}} - \theta_n\right)$$

and E_z may be regarded as a single approximately periodic vibration of frequency $\nu_0 = \frac{n_0}{\mathsf{T}}$ with an amplitude C_0 and a phase-constant θ_0 which vary slowly and irregularly.

The smaller the spectral region, and accordingly the smaller $\frac{n_1 - n_0}{n_0}$, the slower are the fluctuations ("Schwankungen") of C_0 and θ_0, and the more regular is the resulting vibration and also the larger is the difference of path for which radiation can interfere with itself. If a spectral line were absolutely sharp, the radiation would have the property of being capable of interfering with itself for differences of path of any size whatever. This case, however, according to Sec. 18, is an ideal abstraction, never occurring in reality.

PART III

ENTROPY AND PROBABILITY

CHAPTER I

FUNDAMENTAL DEFINITIONS AND LAWS. HYPOTHESIS OF QUANTA

113. Since a wholly new element, entirely unrelated to the fundamental principles of electrodynamics, enters into the range of investigation with the introduction of probability considerations into the electrodynamic theory of heat radiation, the question arises at the outset, whether such considerations are justifiable and necessary. At first sight we might, in fact, be inclined to think that in a purely electrodynamical theory there would be no room at all for probability calculations. For since, as is well known, the electrodynamic equations of the field together with the initial and boundary conditions determine uniquely the way in which an electrodynamical process takes place, in the course of time, considerations which lie outside of the equations of the field would seem, theoretically speaking, to be uncalled for and in any case dispensable. For either they lead to the same results as the fundamental equations of electrodynamics and then they are superfluous, or they lead to different results and in this case they are wrong.

In spite of this apparently unavoidable dilemma, there is a flaw in the reasoning. For on closer consideration it is seen that what is understood in electrodynamics by "initial and boundary" conditions, as well as by the "way in which a process takes place in the course of time," is entirely different from what is denoted by the same words in thermodynamics. In order to make this evident, let us consider the case of radiation *in vacuo*, uniform in all directions, which was treated in the last chapter.

From the standpoint of thermodynamics the state of radiation is completely determined, when the intensity of monochromatic radia-

tion K_ν is given for all frequencies ν. The electrodynamical observer, however, has gained very little by this single statement; because for him a knowledge of the state requires that every one of the six components of the electric and magnetic field-strength be given at all points of the space; and, while from the thermodynamic point of view the question as to the way in which the process takes place in time is settled by the constancy of the intensity of radiation K_ν, from the electrodynamical point of view it would be necessary to know the six components of the field at every point as functions of the time, and hence the amplitudes C_n and the phase-constants θ_n of all the several partial vibrations contained in the radiation would have to be calculated. This, however, is a problem whose solution is quite impossible, for the data obtainable from the measurements are by no means sufficient. The thermodynamically measurable quantities, looked at from the electrodynamical standpoint, represent only certain mean values, as we saw in the special case of stationary radiation in the last chapter.

We might now think that, since in thermodynamic measurements we are always concerned with mean values only, we need consider nothing beyond these mean values, and, therefore, need not take any account of the particular values at all. This method is, however, impracticable, because frequently and that too just in the most important cases, namely, in the cases of the processes of emission and absorption, we have to deal with mean values which cannot be calculated unambiguously by electrodynamical methods from the measured mean values. For example, the mean value of C_n cannot be calculated from the mean value of C_n^2, if no special information as to the particular values of C_n is available.

Thus we see that the electrodynamical state is not by any means determined by the thermodynamic data and that in cases where, according to the laws of thermodynamics and according to all experience,

an unambiguous result is to be expected, a purely electrodynamical theory fails entirely, since it admits not one definite result, but an infinite number of different results.

114. Before entering on a further discussion of this fact and of the difficulty to which it leads in the electrodynamical theory of heat radiation, it may be pointed out that exactly the same case and the same difficulty are met with in the mechanical theory of heat, especially in the kinetic theory of gases. For when, for example, in the case of a gas flowing out of an opening at the time $t = 0$, the velocity, the density, and the temperature are given at every point, and the boundary conditions are completely known, we should expect, according to all experience, that these data would suffice for a unique determination of the way in which the process takes place in time. This, however, from a purely mechanical point of view is not the case at all; for the positions and velocities of all the separate molecules are not at all given by the visible velocity, density, and temperature of the gas, and they would have to be known exactly, if the way in which the process takes place in time had to be completely calculated from the equations of motion. In fact, it is easy to show that, with given initial values of the visible velocity, density, and temperature, an infinite number of entirely different processes is mechanically possible, some of which are in direct contradiction to the principles of thermodynamics, especially the second principle.

115. From these considerations we see that, if we wish to calculate the way in which a thermodynamic process takes place in time, such a formulation of initial and boundary conditions as is perfectly sufficient for a unique determination of the process in thermodynamics, does not suffice for the mechanical theory of heat or for the electrodynamical theory of heat radiation. On the contrary, from the standpoint of pure mechanics or electrodynamics the solutions of the problem are infinite in number. Hence, unless we wish to renounce entirely the possibility

of representing the thermodynamic processes mechanically or electrodynamically, there remains only one way out of the difficulty, namely, to supplement the initial and boundary conditions by special hypotheses of such a nature that the mechanical or electrodynamical equations will lead to an unambiguous result in agreement with experience. As to how such an hypothesis is to be formulated, no hint can naturally be obtained from the principles of mechanics or electrodynamics, for they leave the question entirely open. Just on that account any mechanical or electrodynamical hypothesis containing some further specialization of the given initial and boundary conditions, which cannot be tested by direct measurement, is admissible *a priori*. What hypothesis is to be preferred can be decided only by testing the results to which it leads in the light of the thermodynamic principles based on experience.

116. Although, according to the statement just made, a decisive test of the different admissible hypotheses can be made only *a posteriori*, it is nevertheless worth while noticing that it is possible to obtain *a priori*, without relying in any way on thermodynamics, a definite hint as to the nature of an admissible hypothesis. Let us again consider a flowing gas as an illustration (Sec. 114). The mechanical state of all the separate gas molecules is not at all completely defined by the thermodynamic state of the gas, as has previously been pointed out. If, however, we consider all conceivable positions and velocities of the separate gas molecules, consistent with the given values of the visible velocity, density, and temperature, and calculate for every combination of them the mechanical process, assuming some simple law for the impact of two molecules, we shall arrive at processes, the vast majority of which agree completely in the mean values, though perhaps not in all details. Those cases, on the other hand, which show appreciable deviations, are vanishingly few, and only occur when certain very special and far-reaching conditions between the coordinates

and velocity-components of the molecules are satisfied. Hence, if the assumption be made that such special conditions do not exist, however different the mechanical details may be in other respects, a form of flow of gas will be found, which may be called quite definite with respect to all measurable mean values—and they are the only ones which can be tested experimentally—although it will not, of course, be quite definite in all details. And the remarkable feature of this is that it is just the motion obtained in this manner that satisfies the postulates of the second principle of thermodynamics.

117. From these considerations it is evident that the hypotheses whose introduction was proven above to be necessary completely answer their purpose, if they state nothing more than that exceptional cases, corresponding to special conditions which exist between the separate quantities determining the state and which cannot be tested directly, do not occur in nature. In mechanics this is done by the hypothesis[1] that the heat motion is a "molecular chaos";[2] in electrodynamics the same thing is accomplished by the hypothesis of "natural radiation," which states that there exist between the numerous different partial vibrations (149) of a ray no other relations than those caused by the measurable mean values (compare below, Sec. 148). If, for brevity, we denote any condition or process for which such an hypothesis holds as an "elemental chaos," the principle, *that in nature any state or any process containing numerous elements not in themselves measurable is an elemental chaos*, furnishes the necessary condition for a unique determination of the measurable processes in mechanics as well as in elec-

[1] *L. Boltzmann*, Vorlesungen über Gastheorie **1**, p. 21, 1896. Wiener Sitzungsberichte **78**, Juni, 1878, at the end. Compare also *S. H. Burbury*, Nature, **51**, p. 78, 1894.

[2] Hereafter *Boltzmann's* "Unordnung" will be rendered by chaos, "ungeordnet" by chaotic (Tr.).

trodynamics and also for the validity of the second principle of thermodynamics. This must also serve as a mechanical or electrodynamical explanation of the conception of entropy, which is characteristic of the second law and of the closely allied concept of temperature.[1] It also follows from this that the significance of entropy and temperature is, according to their nature, connected with the condition of an elemental chaos. The terms entropy and temperature do not apply to a purely periodic, perfectly plane wave, since all the quantities in such a wave are in themselves measurable, and hence cannot be an elemental chaos any more than a single rigid atom in motion can. The necessary condition for the hypothesis of an elemental chaos and with it for the existence of entropy and temperature can consist only in the irregular simultaneous effect of very many partial vibrations of different periods, which are propagated in the different directions in space independent of one another, or in the irregular flight of a multitude of atoms.

118. But what mechanical or electrodynamical quantity represents the entropy of a state? It is evident that this quantity depends in some way on the "probability" of the state. For since an elemental chaos and the absence of a record of any individual element forms an essential feature of entropy, the tendency to neutralize any existing temperature differences, which is connected with an increase of entropy, can mean nothing for the mechanical or electrodynamical observer but that uniform distribution of elements in a chaotic state is more probable than any other distribution.

Now since the concept of entropy as well as the second principle of

[1]To avoid misunderstanding I must emphasize that the question, whether the hypothesis of elemental chaos is really everywhere satisfied in nature, is not touched upon by the preceding considerations. I intended only to show at this point that, wherever this hypothesis does not hold, the natural processes, if viewed from the thermodynamic (macroscopic) point of view, do not take place unambiguously.

thermodynamics are of universal application, and since on the other hand the laws of probability have no less universal validity, it is to be expected that the connection between entropy and probability should be very close. Hence we make the following proposition the foundation of our further discussion: *The entropy of a physical system in a definite state depends solely on the probability of this state.* The fertility of this law will be seen later in several cases. We shall not, however, attempt to give a strict general proof of it at this point. In fact, such an attempt evidently would have no meaning at this point. For, so long as the "probability" of a state is not numerically defined, the correctness of the proposition cannot be quantitatively tested. One might, in fact, suspect at first sight that on this account the proposition has no definite physical meaning. It may, however, be shown by a simple deduction that it is possible by means of this fundamental proposition to determine quite generally the way in which entropy depends on probability, without any further discussion of the probability of a state.

119. For let S be the entropy, W the probability of a physical system in a definite state; then the proposition states that

$$S = f(W) \tag{162}$$

where $f(W)$ represents a universal function of the argument W. In whatever way W may be defined, it can be safely inferred from the mathematical concept of probability that the probability of a system which consists of two entirely independent[1] systems is equal to the product of the probabilities of these two systems separately. If we

[1]It is well known that the condition that the two systems be independent of each other is essential for the validity of the expression (163). That it is also a necessary condition for the additive combination of the entropy was proven first by *M. Laue* in the case of optically coherent rays. Annalen d. Physik **20**, p. 365, 1906.

think, *e.g.*, of the first system as any body whatever on the earth and of the second system as a cavity containing radiation on Sirius, then the probability that the terrestrial body be in a certain state 1 and that simultaneously the radiation in the cavity in a definite state 2 is

$$W = W_1 W_2, \tag{163}$$

where W_1 and W_2 are the probabilities that the systems involved are in the states in question.

If now S_1 and S_2 are the entropies of the separate systems in the two states, then, according to (162), we have

$$S_1 = f(W_1) \qquad S_2 = f(W_2).$$

But, according to the second principle of thermodynamics, the total entropy of the two systems, which are independent (see preceding footnote) of each other, is $S = S_1 + S_2$ and hence from (162) and (163)

$$f(W_1 W_2) = f(W_1) + f(W_2).$$

From this functional equation f can be determined. For on differentiating both sides with respect to W_1, W_2 remaining constant, we obtain

$$W_2 \dot{f}(W_1 W_2) = \dot{f}(W_1).$$

On further differentiating with respect to W_2, W_1 now remaining constant, we get

$$\dot{f}(W_1 W_2) + W_1 W_2 \ddot{f}(W_1 W_2) = 0$$

or

$$\dot{f}(W) + W \ddot{f}(W) = 0.$$

The general integral of this differential equation of the second order is

$$f(W) = k \log W + \text{const.}$$

Hence from (162) we get

$$S = k \log W + \text{const.}, \tag{164}$$

an equation which determines the general way in which the entropy depends on the probability. The universal constant of integration k is the same for a terrestrial as for a cosmic system, and its value, having been determined for the former, will remain valid for the latter. The second additive constant of integration may, without any restriction as regards generality, be included as a constant multiplier in the quantity W, which here has not yet been completely defined, so that the equation reduces to

$$S = k \log W.$$

120. The logarithmic connection between entropy and probability was first stated by *L. Boltzmann*[1] in his kinetic theory of gases. Nevertheless our equation (164) differs in its meaning from the corresponding one of Boltzmann in two essential points.

Firstly, *Boltzmann's* equation lacks the factor k, which is due to the fact that *Boltzmann* always used gram-molecules, not the molecules themselves, in his calculations. Secondly, and this is of greater consequence, *Boltzmann* leaves an additive constant undetermined in the entropy S as is done in the whole of classical thermodynamics, and accordingly there is a constant factor of proportionality, which remains undetermined in the value of the probability W.

[1] *L. Boltzmann*, Vorlesungen über Gastheorie, **1**, Sec. 6.

In contrast with this we assign a definite absolute value to the entropy S. This is a step of fundamental importance, which can be justified only by its consequences. As we shall see later, this step leads necessarily to the "hypothesis of quanta" and moreover it also leads, as regards radiant heat, to a definite law of distribution of energy of black radiation, and, as regards heat energy of bodies, to *Nernst's* heat theorem.

From (164) it follows that with the entropy S the probability W is, of course, also determined in the absolute sense. We shall designate the quantity W thus defined as the "thermodynamic probability," in contrast to the "mathematical probability," to which it is proportional but not equal. For, while the mathematical probability is a proper fraction, the thermodynamic probability is, as we shall see, always an integer.

121. The relation (164) contains a general method for calculating the entropy S by probability considerations. This, however, is of no practical value, unless the thermodynamic probability W of a system in a given state can be expressed numerically. The problem of finding the most general and most precise definition of this quantity is among the most important problems in the mechanical or electrodynamical theory of heat. It makes it necessary to discuss more fully what we mean by the "state" of a physical system.

By the state of a physical system at a certain time we mean the aggregate of all those mutually independent quantities, which determine uniquely the way in which the processes in the system take place in the course of time for given boundary conditions. Hence a knowledge of the state is precisely equivalent to a knowledge of the "initial conditions." If we now take into account the considerations stated above in Sec. 113, it is evident that we must distinguish in the theoretical treatment two entirely different kinds of states, which we may denote

as "microscopic" and "macroscopic" states. The microscopic state is the state as described by a mechanical or electrodynamical observer; it contains the separate values of all coordinates, velocities, and field-strengths. The microscopic processes, according to the laws of mechanics and electrodynamics, take place in a perfectly unambiguous way; for them entropy and the second principle of thermodynamics have no significance. The macroscopic state, however, is the state as observed by a thermodynamic observer; any macroscopic state contains a large number of microscopic ones, which it unites in a mean value. Macroscopic processes take place in an unambiguous way in the sense of the second principle, when, and only when, the hypothesis of the elemental chaos (Sec. 117) is satisfied.

122. If now the calculation of the probability W of a state is in question, it is evident that the state is to be thought of in the macroscopic sense. The first and most important question is now: How is a macroscopic state defined? An answer to it will dispose of the main features of the whole problem.

For the sake of simplicity, let us first consider a special case, that of a very large number, N, of simple similar molecules. Let the problem be solely the distribution of these molecules in space within a given volume, V, irrespective of their velocities, and further the definition of a certain macroscopic distribution in space. The latter cannot consist of a statement of the coordinates of all the separate molecules, for that would be a definite microscopic distribution. We must, on the contrary, leave the positions of the molecules undetermined to a certain extent, and that can be done only by thinking of the whole volume V as being divided into a number of small but finite *space elements*, G, each containing a specified number of molecules. By any such statement a definite macroscopic distribution in space is defined. The manner in which the molecules are distributed within every separate space element is im-

material, for here the hypothesis of elemental chaos (Sec. 117) provides a supplement, which insures the unambiguity of the macroscopic state, in spite of the microscopic indefiniteness. If we distinguish the space elements in order by the numbers 1, 2, 3, ... and, for any particular macroscopic distribution in space, denote the number of the molecules lying in the separate space elements by N_1, N_2, N_3 ..., then to every definite system of values N_1, N_2, N_3 ..., there corresponds a definite macroscopic distribution in space. We have of course always:

$$N_1 + N_2 + N_3 + \cdots = N \tag{165}$$

or if

$$\frac{N_1}{N} = w_1, \quad \frac{N_2}{N} = w_2, \ \ldots \tag{166}$$

$$w_1 + w_2 + w_3 + \cdots = 1. \tag{167}$$

The quantity w_i may be called the density of distribution of the molecules, or the mathematical probability that any molecule selected at random lies in the ith space element.

If we now had, *e.g.*, only 10 molecules and 7 space elements, a definite space distribution would be represented by the values:

$$N_1 = 1, \ N_2 = 2, \ N_3 = 0, \ N_4 = 0, \ N_5 = 1, \ N_6 = 4, \ N_7 = 2, \tag{168}$$

which state that in the seven space elements there lie respectively 1, 2, 0, 0, 1, 4, 2 molecules.

123. The definition of a macroscopic distribution in space may now be followed immediately by that of its thermodynamic probability W. The latter is founded on the consideration that a certain distribution in space may be realized in many different ways, namely, by many different individual coordinations or "complexions," according as a certain

molecule considered will happen to lie in one or the other space element. For, with a given distribution of space, it is of consequence only how many, not which, molecules lie in every space element.

The number of all complexions which are possible with a given distribution in space we equate to the thermodynamic probability W of the space distribution.

In order to form a definite conception of a certain complexion, we can give the molecules numbers, write these numbers in order from 1 to N, and place below the number of every molecule the number of that space element to which the molecule in question belongs in that particular complexion. Thus the following table represents one particular complexion, selected at random, for the distribution in the preceding illustration

1	2	3	4	5	6	7	8	9	10
6	1	7	5	6	2	2	6	6	7

(169)

By this the fact is exhibited that the

Molecule 2 lies in space element 1.

Molecules 6 and 7 lie in space element 2.

Molecule 4 lies in space element 5.

Molecules 1, 5, 8, and 9 lie in space element 6.

Molecules 3 and 10 lie in space element 7.

As becomes evident on comparison with (168), this complexion does, in fact, correspond in every respect to the space distribution given above, and in a similar manner it is easy to exhibit many other complexions, which also belong to the same space distribution. The number of all possible complexions required is now easily found by inspecting the lower of the two lines of figures in (169). For, since the number of the molecules is given, this line of figures contains a definite number of places. Since, moreover, the distribution in space is also given, the

number of times that every figure (*i.e.*, every space element) appears in the line is equal to the number of molecules which lie in that particular space element. But every change in the table gives a new particular coordination between molecules and space elements and hence a new complexion. Hence the number of the possible complexions, or the thermodynamic probability, W, of the given space distribution, is equal to the number of "permutations with repetition" possible under the given conditions. In the simple numerical example chosen, we get for W, according to a well-known formula, the expression

$$\frac{10!}{1!\,2!\,0!\,0!\,1!\,4!\,2!} = 37,800.$$

The form of this expression is so chosen that it may be applied easily to the general case. The numerator is equal to factorial N, N being the total number of molecules considered, and the denominator is equal to the product of the factorials of the numbers, N_1, N_2, N_3, ... of the molecules, which lie in every separate space element and which, in the general case, must be thought of as large numbers. Hence we obtain for the required probability of the given space distribution

$$W = \frac{N!}{N_1!\,N_2!\,N_3!\,\ldots}. \tag{170}$$

Since all the N's are large numbers, we may apply to their factorials Stirling's formula, which for a large number may be abridged[1] to[2]

$$n! = \left(\frac{n}{e}\right)^n. \tag{171}$$

[1]Abridged in the sense that factors which in the logarithmic expression (173) would give rise to small additive terms have been omitted at the outset. A brief derivation of equation (173) may be found on p. 258 (Tr.).

[2]See for example *E. Czuber*, Wahrscheinlichkeitsrechnung (Leipzig, B. G. Teubner) p. 22, 1903; *H. Poincaré*, Calcul des Probabilités (Paris, Gauthier-Villars), p. 85, 1912.

Hence, by taking account of (165), we obtain

$$W = \left(\frac{N}{N_1}\right)^{N_1} \left(\frac{N}{N_2}\right)^{N_2} \left(\frac{N}{N_3}\right)^{N_3} \dots . \tag{172}$$

124. Exactly the same method as in the case of the space distribution just considered may be used for the definition of a macroscopic state and of the thermodynamic probability in the general case, where not only the coordinates but also the velocities, the electric moments, etc., of the molecules are to be dealt with. Every thermodynamic state of a system of N molecules is, in the macroscopic sense, defined by the statement of the number of molecules, N_1, N_2, N_3, ..., which are contained in the region elements 1, 2, 3, ... of the "state space." This state space, however, is not the ordinary three-dimensional space, but an ideal space of as many dimensions as there are variables for every molecule. In other respects the definition and the calculation of the thermodynamic probability W are exactly the same as above and the entropy of the state is accordingly found from (164), taking (166) also into account, to be

$$S = -kN \sum w_1 \log w_1, \tag{173}$$

where the sum $\sum$ is to be taken over all region elements. It is obvious from this expression that the entropy is in every case a *positive quantity.*

125. By the preceding developments the calculation of the entropy of a system of N molecules in a given thermodynamic state is, in general, reduced to the single problem of finding the magnitude G of the region elements in the state space. That such a definite finite quantity really exists is a characteristic feature of the theory we are developing, as contrasted with that due to *Boltzmann*, and forms the content of the so-called *hypothesis of quanta.* As is readily seen, this is an immediate

consequence of the proposition of Sec. 120 that the entropy S has an absolute, not merely a relative, value; for this, according to (164), necessitates also an absolute value for the magnitude of the thermodynamic probability W, which, in turn, according to Sec. 123, is dependent on the number of complexions, and hence also on the number and size of the region elements which are used. Since all different complexions contribute uniformly to the value of the probability W, the region elements of the state space represent also *regions of equal probability.* If this were not so, the complexions would not be all equally probable.

However, not only the magnitude, but also the shape and position of the region elements must be perfectly definite. For since, in general, the distribution density w is apt to vary appreciably from one region element to another, a change in the shape of a region element, the magnitude remaining unchanged, would, in general, lead to a change in the value of w and hence to a change in S. We shall see that only in special cases, namely, when the distribution densities w are very small, may the absolute magnitude of the region elements become physically unimportant, inasmuch as it enters into the entropy only through an additive constant. This happens, *e.g.*, at high temperatures, large volumes, slow vibrations (state of an ideal gas, Sec. 132, *Rayleigh's* radiation law, Sec. 159). Hence it is permissible for such limiting cases to assume, without appreciable error, that G is infinitely small in the macroscopic sense, as has hitherto been the practice in statistical mechanics. As soon, however, as the distribution densities w assume appreciable values, the classical statistical mechanics fail.

126. If now the problem be to determine the magnitude G of the region elements of equal probability, the laws of the classical statistical mechanics afford a certain hint, since in certain limiting cases they lead to correct results.

Let $\phi_1, \phi_2, \phi_3, \ldots$ be the "generalized coordinates," $\psi_1, \psi_2, \psi_3, \ldots$

the corresponding "impulse coordinates" or "moments," which determine the microscopic state of a certain molecule; then the state space contains as many dimensions as there are coordinates ϕ and moments ψ for every molecule. Now the region element of probability, according to classical statistical mechanics, is identical with the infinitely small element of the state space (in the macroscopic sense)[1]

$$d\phi_1\, d\phi_2\, d\phi_3 \ldots d\psi_1\, d\psi_2\, d\psi_3 \ldots . \tag{174}$$

According to the hypothesis of quanta, on the other hand, every region element of probability has a definite finite magnitude

$$G = \int d\phi_1\, d\phi_2\, d\phi_3 \ldots d\psi_1\, d\psi_2\, d\psi_3 \ldots \tag{175}$$

whose value is the same for all different region elements and, moreover, depends on the nature of the system of molecules considered. The shape and position of the separate region elements are determined by the limits of the integral and must be determined anew in every separate case.

[1]Compare, for example, *L. Boltzmann*, Gastheorie, **2**, p. 62 *et seq.*, 1898, or *J. W. Gibbs*, Elementary principles in statistical mechanics, Chapter I, 1902.

CHAPTER II

IDEAL MONATOMIC GASES

127. In the preceding chapter it was proven that the introduction of probability considerations into the mechanical and electrodynamical theory of heat is justifiable and necessary, and from the general connection between entropy S and probability W, as expressed in equation (164), a method was derived for calculating the entropy of a physical system in a given state. Before we apply this method to the determination of the entropy of radiant heat we shall in this chapter make use of it for calculating the entropy of an ideal monatomic gas in an arbitrarily given state. The essential parts of this calculation are already contained in the investigations of *L. Boltzmann*[1] on the mechanical theory of heat; it will, however, be advisable to discuss this simple case in full, firstly to enable us to compare more readily the method of calculation and physical significance of mechanical entropy with that of radiation entropy, and secondly, what is more important, to set forth clearly the differences as compared with *Boltzmann's* treatment, that is, to discuss the meaning of the universal constant k and of the finite region elements G. For this purpose the treatment of a special case is sufficient.

128. Let us then take N similar monatomic gas molecules in an arbitrarily given thermodynamic state and try to find the corresponding entropy. The state space is six-dimensional, with the three coordinates x, y, z, and the three corresponding moments $m\xi$, $m\eta$, $m\zeta$, of a molecule, where we denote the mass by m and velocity components by ξ, η, ζ. Hence these quantities are to be substituted for the ϕ and ψ in

[1] *L. Boltzmann*, Sitzungsber. d. Akad. d. Wissensch. zu Wien (II) **76**, p. 373, 1877. Compare also Gastheorie, **1**, p. 38, 1896.

Sec. 126. We thus obtain for the size of a region element G the sextuple integral

$$G = m^3 \int d\sigma, \tag{176}$$

where, for brevity

$$dx\, dy\, dz\, d\xi\, d\eta\, d\zeta = d\sigma. \tag{177}$$

If the region elements are known, then, since the macroscopic state of the system of molecules was assumed as known, the numbers N_1, N_2, N_3, ... of the molecules which lie in the separate region elements are also known, and hence the distribution densities w_1, w_2, w_3, ... (166) are given and the entropy of the state follows at once from (173).

129. The theoretical determination of G is a problem as difficult as it is important. Hence we shall at this point restrict ourselves from the very outset to the special case in which the distribution density varies but slightly from one region element to the next—the characteristic feature of the state of an ideal gas. Then the summation over all region elements may be replaced by the integral over the whole state space. Thus we have from (176) and (167)

$$\sum w_1 = \sum w_1 \frac{m^3}{G} \int d\sigma = \frac{m^3}{G} \int w\, d\sigma = 1, \tag{178}$$

in which w is no longer thought of as a discontinuous function of the ordinal number, i, of the region element, where $i = 1, 2, 3, \ldots n$, but as a continuous function of the variables, x, y, z, ξ, η, ζ, of the state space. Since the whole state region contains very many region elements, it follows, according to (167) and from the fact that the distribution density w changes slowly, that w has everywhere a small value.

Similarly we find for the entropy of the gas from (173):

$$S = -kN \sum w_1 \log w_1 = -kN \frac{m^3}{G} \int w \log w\, d\sigma. \tag{179}$$

Of course the whole energy E of the gas is also determined by the distribution densities w. If w is sufficiently small in every region element, the molecules contained in any one region element are, on the average, so far apart that their energy depends only on the velocities. Hence:

$$E = \sum N_1 \frac{1}{2} m(\xi_1^2 + \eta_1^2 + \zeta_1^2) + E_0$$
$$= N \sum w_1 \frac{1}{2} m(\xi_1^2 + \eta_1^2 + \zeta_1^2) + E_0, \tag{180}$$

where $\xi_1\, \eta_1\, \zeta_1$ denotes any velocity lying within the region element 1 and E_0 denotes the internal energy of the stationary molecules, which is assumed constant. In place of the latter expression we may write, again according to (176),

$$E = \frac{m^4 N}{2G} \int (\xi^2 + \eta^2 + \zeta^2) w \, d\sigma + E_0. \tag{181}$$

130. Let us consider the state of thermodynamic equilibrium. According to the second principle of thermodynamics this state is distinguished from all others by the fact that, for a given volume V and a given energy E of the gas, the entropy S is a maximum. Let us then regard the volume

$$V = \iiint dx\, dy\, dz \tag{182}$$

and the energy E of the gas as given. The condition for equilibrium is $\delta S = 0$, or, according to (179),

$$\sum (\log w_1 + 1)\, \delta w_1 = 0,$$

and this holds for any variations of the distribution densities whatever, provided that, according to (167) and (180), they satisfy the conditions

$$\sum \delta w_1 = 0 \quad \text{and} \quad \sum (\xi_1^2 + \eta_1^2 + \zeta_1^2)\, \delta w_1 = 0.$$

This gives us as the necessary and sufficient condition for thermodynamic equilibrium for every separate distribution density w:

$$\log w + \beta(\xi^2 + \eta^2 + \zeta^2) + \text{const.} = 0$$

or

$$w = \alpha e^{-\beta(\xi^2+\eta^2+\zeta^2)}, \tag{183}$$

where α and β are constants. Hence in the state of equilibrium the distribution of the molecules in space is independent of x, y, z, that is, macroscopically uniform, and the distribution of velocities is the well-known one of *Maxwell*.

131. The values of the constants α and β may be found from those of V and E. For, on substituting the value of w just found in (178) and taking account of (177) and (182), we get

$$\frac{G}{m^3} = \alpha V \iiint_{-\infty}^{\infty} e^{-\beta(\xi^2+\eta^2+\zeta^2)}\, d\xi\, d\eta\, d\zeta = \alpha V \left(\frac{\pi}{\beta}\right)^{\frac{3}{2}},$$

and on substituting w in (181) we get

$$E = E_0 + \frac{\alpha m^4 NV}{2G} \iiint_{-\infty}^{\infty} (\xi^2 + \eta^2 + \zeta^2) e^{-\beta(\xi^2+\eta^2+\zeta^2)}\, d\xi\, d\eta\, d\zeta,$$

or

$$E = E_0 + \frac{3\alpha m^4 NV}{4G} \frac{1}{\beta} \left(\frac{\pi}{\beta}\right)^{\frac{3}{2}}.$$

Solving for α and β we have

$$\alpha = \frac{G}{V} \left(\frac{3N}{4\pi m(E - E_0)}\right)^{\frac{3}{2}} \tag{184}$$

$$\beta = \frac{3}{4} \frac{Nm}{E - E_0}. \tag{185}$$

From this finally we find, as an expression for the entropy S of the gas in the state of equilibrium with given values of N, V, and E,

$$S = kN \log \left\{ \frac{V}{G} \left(\frac{4\pi em(E - E_0)}{3N} \right)^{\frac{3}{2}} \right\}. \tag{186}$$

132. This determination of the entropy of an ideal monatomic gas is based solely on the general connection between entropy and probability as expressed in equation (164); in particular, we have at no stage of our calculation made use of any special law of the theory of gases. It is, therefore, of importance to see how the entire thermodynamic behavior of a monatomic gas, especially the equation of state and the values of the specific heats, may be deduced from the expression found for the entropy directly by means of the principles of thermodynamics. From the general thermodynamic equation defining the entropy, namely,

$$dS = \frac{dE + p\,dV}{T}, \tag{187}$$

the partial differential coefficients of S with respect to E and V are found to be

$$\left(\frac{\partial S}{\partial E} \right)_V = \frac{1}{T}, \quad \left(\frac{\partial S}{\partial V} \right)_E = \frac{p}{T}.$$

Hence, by using (186), we get for our gas

$$\left(\frac{\partial S}{\partial E} \right)_V = \frac{3}{2} \frac{kN}{E - E_0} = \frac{1}{T} \tag{188}$$

and

$$\left(\frac{\partial S}{\partial V} \right)_E = \frac{kN}{V} = \frac{p}{T}. \tag{189}$$

The second of these equations

$$p = \frac{kNT}{V} \tag{190}$$

contains the laws of *Boyle*, *Gay Lussac*, and *Avogadro*, the last named because the pressure depends only on the number N, not on the nature of the molecules. If we write it in the customary form:

$$p = \frac{RnT}{V}, \tag{191}$$

where n denotes the number of gram molecules or mols of the gas, referred to $O_2 = 32\,\text{gr.}$, and R represents the absolute gas constant

$$R = 831 \times 10^5 \,\frac{\text{erg}}{\text{degree}}, \tag{192}$$

we obtain by comparison

$$k = \frac{Rn}{N}. \tag{193}$$

If we now call the ratio of the number of mols to the number of molecules ω, or, what is the same thing, the ratio of the mass of a molecule to that of a mol, $\omega = \frac{n}{N}$, we shall have

$$k = \omega R. \tag{194}$$

From this the universal constant k may be calculated, when ω is given, and *vice versa*. According to (190) this constant k is nothing but the absolute gas constant, if it is referred to molecules instead of mols.

From equation (188)

$$E - E_0 = \tfrac{3}{2}kNT. \tag{195}$$

Now, since the energy of an ideal gas is also given by

$$E = Anc_vT + E_0 \tag{196}$$

where c_v is the heat capacity of a mol at constant volume in calories and A is the mechanical equivalent of heat:

$$A = 419 \times 10^5 \frac{\text{erg}}{\text{cal}} \tag{197}$$

it follows that

$$c_v = \frac{3}{2}\frac{kN}{An}$$

and further, by taking account of (193)

$$c_v = \frac{3}{2}\frac{R}{A} = \frac{3}{2} \cdot \frac{831 \times 10^5}{419 \times 10^5} = 3.0 \tag{198}$$

as an expression for the heat capacity per mol of any monatomic gas at constant volume in calories.[1]

For the heat capacity per mol at constant pressure, c_p, we have as a consequence of the first principle of thermodynamics:

$$c_p - c_v = \frac{R}{A}$$

and hence by (198)

$$c_p = \frac{5}{2}\frac{R}{A}, \qquad \frac{c_p}{c_v} = \frac{5}{3}, \tag{199}$$

as is known to be the case for monatomic gases. It follows from (195) that the kinetic energy L of the gas molecules is equal to

$$L = E - E_0 = \tfrac{3}{2}NkT. \tag{200}$$

[1]Compare *F. Richarz*, Wiedemann's Annal., **67**, p. 705, 1899.

133. The preceding relations, obtained simply by identifying the mechanical expression of the entropy (186) with its thermodynamic expression (187), show the usefulness of the theory developed. In them an additive constant in the expression for the entropy is immaterial and hence the size G of the region element of probability does not matter. The hypothesis of quanta, however, goes further, since it fixes the absolute value of the entropy and thus leads to the same conclusion as the heat theorem of *Nernst*. According to this theorem the "characteristic function" of an ideal gas[1] is in our notation

$$\Phi = S - \frac{E + pV}{T} = n\left(Ac_p \log T - R \log p + a - \frac{b}{T}\right),$$

where a denotes *Nernst's* chemical constant, and b the energy constant.

On the other hand, the preceding formulæ (186), (188), and (189) give for the same function Φ the following expression:

$$\Phi = N\left(\frac{5}{2}k \log T - k \log p + a'\right) - \frac{E_0}{T}$$

where for brevity a' is put for:

$$a' = k \log\left\{\frac{kN}{eG}(2\pi mk)^{\frac{3}{2}}\right\}.$$

From a comparison of the two expressions for Φ it is seen, by taking account of (199) and (193), that they agree completely, provided

$$a = \frac{N}{n}a' = R \log\left\{\frac{Nk^{\frac{5}{2}}}{eG}(2\pi m)^{\frac{3}{2}}\right\}, \qquad b = \frac{E_0}{n}. \tag{201}$$

[1] *E.g.*, *M. Planck*, Vorlesungen über Thermodynamik, Leipzig, Veit und Comp., 1911, Sec. 287, equation 267.

This expresses the relation between the chemical constant a of the gas and the region element G of the probability.[1]

It is seen that G is proportional to the total number, N, of the molecules. Hence, if we put $G = Ng$, we see that g, the molecular region element, depends only on the chemical nature of the gas.

Obviously the quantity g must be closely connected with the law, so far unknown, according to which the molecules act microscopically on one another. Whether the value of g varies with the nature of the molecules or whether it is the same for all kinds of molecules, may be left undecided for the present.

If g were known, *Nernst's* chemical constant, a, of the gas could be calculated from (201) and the theory could thus be tested. For the present the reverse only is feasible, namely, to calculate g from a. For it is known that a may be measured directly by the tension of the saturated vapor, which at sufficiently low temperatures satisfies the simple equation[2]

$$\log p = \frac{5}{2}\log T - \frac{Ar_0}{RT} + \frac{a}{R} \tag{202}$$

(where r_0 is the heat of vaporization of a mol at 0° in calories). When a has been found by measurement, the size g of the molecular region element is found from (201) to be

$$g = (2\pi m)^{\frac{3}{2}} k^{\frac{5}{2}} e^{-\frac{a}{R}-1}. \tag{203}$$

Let us consider the dimensions of g.

According to (176) g is of the dimensions [erg^3 sec^3]. The same follows from the present equation, when we consider that the dimension

[1] Compare also *O. Sackur*, Annal. d. Physik **36**, p. 958, 1911, Nernst-Festschrift, p. 405, 1912, and *H. Tetrode*, Annal. d. Physik **38**, p. 434, 1912.

[2] *M. Planck*, l. c., Sec. 288, equation 271.

of the chemical constant a is not, as might at first be thought, that of R, but, according to (202), that of $R \log \frac{p}{T^{\frac{5}{2}}}$.

134. To this we may at once add another quantitative relation. All the preceding calculations rest on the assumption that the distribution density w and hence also the constant α in (183) are small (Sec. 129). Hence, if we take the value of α from (184) and take account of (188), (189) and (201), it follows that

$$\frac{p}{T^{\frac{5}{2}}} e^{-\frac{a}{R}-1} \quad \text{must be small.}$$

When this relation is not satisfied, the gas cannot be in the ideal state. For the saturated vapor it follows then from (202) that $e^{-\frac{Ar_0}{RT}}$ is small. In order, then, that a saturated vapor may be assumed to be in the state of an ideal gas, the temperature T must certainly be less than $\frac{A}{R} r_0$ or $\frac{r_0}{2}$. Such a restriction is unknown to the classical thermodynamics.

CHAPTER III

IDEAL LINEAR OSCILLATORS

135. The main problem of the theory of heat radiation is to determine the energy distribution in the normal spectrum of black radiation, or, what amounts to the same thing, to find the function which has been left undetermined in the general expression of *Wien's* displacement law (119), the function which connects the entropy of a certain radiation with its energy. The purpose of this chapter is to develop some preliminary theorems leading to this solution. Now since, as we have seen in Sec. 48, the normal energy distribution in a diathermanous medium cannot be established unless the medium exchanges radiation with an emitting and absorbing substance, it will be necessary for the treatment of this problem to consider more closely the processes which cause the creation and the destruction of heat rays, that is, the processes of emission and absorption. In view of the complexity of these processes and the difficulty of acquiring knowledge of any definite details regarding them, it would indeed be quite hopeless to expect to gain any certain results in this way, if it were not possible to use as a reliable guide in this obscure region the law of *Kirchhoff* derived in Sec. 51. This law states that a vacuum completely enclosed by reflecting walls, in which any emitting and absorbing bodies are scattered in any arrangement whatever, assumes in the course of time the stationary state of black radiation, which is completely determined by one parameter only, namely, the temperature, and in particular does not depend on the number, the nature, and the arrangement of the material bodies present. Hence, for the investigation of the properties of the state of black radiation the nature of the bodies which are assumed to be in the vacuum is perfectly immaterial. In fact, it does not even matter

whether such bodies really exist somewhere in nature, provided their existence and their properties are consistent with the laws of thermodynamics and electrodynamics. If, for any special arbitrary assumption regarding the nature and arrangement of emitting and absorbing systems, we can find a state of radiation in the surrounding vacuum which is distinguished by absolute stability, this state can be no other than that of black radiation.

Since, according to this law, we are free to choose any system whatever, we now select from all possible emitting and absorbing systems the simplest conceivable one, namely, one consisting of a large number N of similar stationary oscillators, each consisting of two poles, charged with equal quantities of electricity of opposite sign, which may move relatively to each other on a fixed straight line, the axis of the oscillator.

It is true that it would be more general and in closer accord with the conditions in nature to assume the vibrations to be those of an oscillator consisting of two poles, each of which has three degrees of freedom of motion instead of one, *i.e.*, to assume the vibrations as taking place in space instead of in a straight line only. Nevertheless we may, according to the fundamental principle stated above, restrict ourselves from the beginning to the treatment of one single component, without fear of any essential loss of generality of the conclusions we have in view.

It might, however, be questioned as a matter of principle, whether it is really permissible to think of the centers of mass of the oscillators as stationary, since, according to the kinetic theory of gases, all material particles which are contained in substances of finite temperature and free to move possess a certain finite mean kinetic energy of translatory motion. This objection, however, may also be removed by the consideration that the velocity is not fixed by the kinetic energy alone. We need only think of an oscillator as being loaded, say at its positive

pole, with a comparatively large inert mass, which is perfectly neutral electrodynamically, in order to decrease its velocity for a given kinetic energy below any preassigned value whatever. Of course this consideration remains valid also, if, as is now frequently done, all inertia is reduced to electrodynamic action. For this action is at any rate of a kind quite different from the one to be considered in the following, and hence cannot influence it.

Let the state of such an oscillator be completely determined by its moment $f(t)$, that is, by the product of the electric charge of the pole situated on the positive side of the axis and the pole distance, and by the derivative of f with respect to the time or

$$\frac{df(t)}{dt} = \dot{f}(t). \tag{204}$$

Let the energy of the oscillator be of the following simple form:

$$U = \tfrac{1}{2}Kf^2 + \tfrac{1}{2}L\dot{f}^2, \tag{205}$$

where K and L denote positive constants, which depend on the nature of the oscillator in some way that need not be discussed at this point.

If during its vibration an oscillator neither absorbed nor emitted any energy, its energy of vibration, U, would remain constant, and we would have:

$$dU = Kf\,df + L\dot{f}\,d\dot{f} = 0, \tag{205a}$$

or, on account of (204),

$$Kf(t) + L\ddot{f}(t) = 0. \tag{206}$$

The general solution of this differential equation is found to be a purely periodical vibration:

$$f = C\cos(2\pi\nu t - \theta) \tag{207}$$

where C and θ denote the integration constants and ν the number of vibrations per unit time:

$$\nu = \frac{1}{2\pi}\sqrt{\frac{K}{L}}. \tag{208}$$

136. If now the assumed system of oscillators is in a space traversed by heat rays, the energy of vibration, U, of an oscillator will not in general remain constant, but will be always changing by absorption and emission of energy. Without, for the present, considering in detail the laws to which these processes are subject, let us consider any one arbitrarily given thermodynamic state of the oscillators and calculate its entropy, irrespective of the surrounding field of radiation. In doing this we proceed entirely according to the principle advanced in the two preceding chapters, allowing, however, at every stage for the conditions caused by the peculiarities of the case in question.

The first question is: What determines the thermodynamic state of the system considered? For this purpose, according to Sec. 124, the numbers N_1, N_2, N_3, ... of the oscillators, which lie in the region elements 1, 2, 3, ... of the "state space" must be given. The state space of an oscillator contains those coordinates which determine the microscopic state of an oscillator. In the case in question these are only two in number, namely, the moment f and the rate at which it varies, $\dot{f}$, or instead of the latter the quantity

$$\psi = L\dot{f}, \tag{209}$$

which is of the dimensions of an impulse. The region element of the state plane is, according to the hypothesis of quanta (Sec. 126), the double integral

$$\iint df\, d\psi = h. \tag{210}$$

The quantity h is the same for all region elements. *A priori*, it might, however, depend also on the nature of the system considered, for example, on the frequency of the oscillators. The following simple consideration, however, leads to the assumption that h is a universal constant. We know from the generalized displacement law of *Wien* (equation (119)) that in the universal function, which gives the entropy radiation as dependent on the energy radiation, there must appear a universal constant of the dimension $\frac{c^3 \mathsf{u}}{\nu^3}$ and this is of the dimension of a quantity of action[1] (erg sec.). Now, according to (210), the quantity h has precisely this dimension, on which account we may denote it as "element of action" or "quantity element of action." Hence, unless a second constant also enters, h cannot depend on any other physical quantities.

137. The principal difference, compared with the calculations for an ideal gas in the preceding chapter, lies in the fact that we do not now assume the distribution densities w_1, w_2, w_3 ... of the oscillators among the separate region elements to vary but little from region to region as was assumed in Sec. 129. Accordingly the w's are not small, but finite proper fractions, and the summation over the region elements cannot be written as an integration.

In the first place, as regards the shape of the region elements, the fact that in the case of undisturbed vibrations of an oscillator the phase is always changing, whereas the amplitude remains constant, leads to the conclusion that, for the macroscopic state of the oscillators, the amplitudes only, not the phases, must be considered, or in other words the region elements in the $f\psi$ plane are bounded by the curves $C =$

[1] The quantity from which the principle of *least action* takes its name. (Tr.)

const., that is, by ellipses, since from (207) and (209)

$$\left(\frac{f}{C}\right)^2 + \left(\frac{\psi}{2\pi\nu LC}\right)^2 = 1. \tag{211}$$

The semi-axes of such an ellipse are:

$$a = C \quad \text{and} \quad b = 2\pi\nu LC. \tag{212}$$

Accordingly the region elements 1, 2, 3, ... n ... are the concentric, similar, and similarly situated elliptic rings, which are determined by the increasing values of C:

$$0,\ C_1,\ C_2,\ C_3,\ \ldots\ C_{n-1},\ C_n\ \ldots. \tag{213}$$

The nth region element is that which is bounded by the ellipses $C = C_{n-1}$ and $C = C_n$. The first region element is the full ellipse C_1. All these rings have the same area h, which is found by subtracting the area of the full ellipse C_{n-1} from that of the full ellipse C_n; hence

$$h = (a_n b_n - a_{n-1} b_{n-1})\pi$$

or, according to (212),

$$h = (C_n^2 - C_{n-1}^2)2\pi^2\nu L,$$

where $n = 1, 2, 3, \ldots$.

From the additional fact that $C_0 = 0$, it follows that:

$$C_n^2 = \frac{nh}{2\pi^2\nu L}. \tag{214}$$

Thus the semi-axes of the bounding ellipses are in the ratio of the square roots of the integral numbers.

138. The thermodynamic state of the system of oscillators is fixed by the fact that the values of the distribution densities w_1, w_2, w_3, ... of the oscillators among the separate region elements are given. *Within* a region element the distribution of the oscillators is according to the law of elemental chaos (Sec. 122), *i.e.*, it is approximately *uniform*.

These data suffice for calculating the entropy S as well as the energy E of the system in the given state, the former quantity directly from (173), the latter by the aid of (205). It must be kept in mind in the calculation that, since the energy varies appreciably within a region element, the energy E_n of all those oscillators which lie in the nth region element is to be found by an integration. Then the whole energy E of the system is:

$$E = E_1 + E_2 + \dots E_n + \dots . \tag{215}$$

E_n may be calculated with the help of the law that within every region element the oscillators are uniformly distributed. If the nth region element contains, all told, N_n oscillators, there are per unit area $\frac{N_n}{h}$ oscillators and hence $\frac{N_n}{h}\,df \cdot d\psi$ per element of area. Hence we have:

$$E_n = \frac{N_n}{h} \iint U\, df\, d\psi.$$

In performing the integration, instead of f and ψ we take C and ϕ, as new variables, and since according to (211),

$$f = C\cos\phi \qquad \psi = 2\pi\nu LC\sin\phi \tag{216}$$

we get:

$$E_n = 2\pi\nu L\frac{N_n}{h} \iint UC\, dC\, d\phi$$

to be integrated with respect to ϕ from 0 to 2π and with respect to C from C_{n-1} to C_n. If we substitute from (205), (209) and (216)

$$U = \tfrac{1}{2}KC^2, \tag{217}$$

we obtain by integration

$$E_n = \frac{\pi^2}{2}\nu LK\frac{N_n}{h}(C_n^4 - C_{n-1}^4)$$

and from (214) and (208):

$$E_n = N_n(n - \tfrac{1}{2})h\nu = Nw_n(n - \tfrac{1}{2})h\nu, \tag{218}$$

that is, the mean energy of an oscillator in the nth region element is $(n - \frac{1}{2})h\nu$. This is exactly the arithmetic mean of the energies $(n-1)h\nu$ and $nh\nu$ which correspond to the two ellipses $C = C_{n-1}$ and $C = C_n$ bounding the region, as may be seen from (217), if the values of C_{n-1} and C_n are therein substituted from (214).

The total energy E is, according to (215),

$$E = Nh\nu \sum_{n=1}^{n=\infty}(n - \tfrac{1}{2})w_n. \tag{219}$$

139. Let us now consider the state of thermodynamic equilibrium of the oscillators. According to the second principle of thermodynamics, the entropy S is in that case a maximum for a given energy E. Hence we assume E in (219) as given. Then from (179) we have for the state of equilibrium:

$$\delta S = 0 = \sum_{1}^{\infty}(\log w_n + 1)\,\delta w_n,$$

where according to (167) and (219)

$$\sum_1^\infty \delta w_n = 0 \quad \text{and} \quad \sum_1^\infty (n - \tfrac{1}{2})\,\delta w_n = 0.$$

From these relations we find:

$$\log w_n + \beta n + \text{const.} = 0$$

or

$$w_n = \alpha \gamma^n. \tag{220}$$

The values of the constants α and γ follow from equations (167) and (219):

$$\alpha = \frac{2Nh\nu}{2E - Nh\nu} \qquad \gamma = \frac{2E - Nh\nu}{2E + Nh\nu}. \tag{221}$$

Since w_n is essentially positive it follows that equilibrium is not possible in the system of oscillators considered unless the total energy E has a greater value than $\frac{Nh\nu}{2}$, that is unless the mean energy of the oscillators is at least $\frac{h\nu}{2}$. This, according to (218), is the mean energy of the oscillators lying in the first region element. In fact, in this extreme case all N oscillators lie in the first region element, the region of smallest energy; within this element they are arranged uniformly.

The entropy S of the system, which is in thermodynamic equilibrium, is found by combining (173) with (220) and (221)

$$S = kN\left\{\left(\frac{E}{Nh\nu} + \frac{1}{2}\right)\log\left(\frac{E}{Nh\nu} + \frac{1}{2}\right) - \left(\frac{E}{Nh\nu} - \frac{1}{2}\right)\log\left(\frac{E}{Nh\nu} - \frac{1}{2}\right)\right\}. \tag{222}$$

140. The connection between energy and entropy just obtained allows furthermore a certain conclusion as regards the temperature. For from the equation of the second principle of thermodynamics, $dS = \frac{dE}{T}$ and from differentiation of (222) with respect to E it follows that

$$E = N\frac{h\nu}{2}\frac{1+e^{-\frac{h\nu}{kT}}}{1-e^{-\frac{h\nu}{kT}}} = Nh\nu\left(\frac{1}{2}+\frac{1}{e^{\frac{h\nu}{kT}}-1}\right). \tag{223}$$

Hence, for the zero point of the absolute temperature E becomes, not 0, but $N\frac{h\nu}{2}$. This is the extreme case discussed in the preceding paragraph, which just allows thermodynamic equilibrium to exist. That the oscillators are said to perform vibrations even at the temperature zero, the mean energy of which is as large as $\frac{h\nu}{2}$ and hence may become quite large for rapid vibrations, may at first sight seem strange. It seems to me, however, that certain facts point to the existence, inside the atoms, of vibrations independent of the temperature and supplied with appreciable energy, which need only a small suitable excitation to become evident externally. For example, the velocity, sometimes very large, of secondary cathode rays produced by Roentgen rays, and that of electrons liberated by photoelectric effect are independent of the temperature of the metal and of the intensity of the exciting radiation. Moreover the radioactive energies are also independent of the temperature. It is also well known that the close connection between the inertia of matter and its energy as postulated by the relativity principle leads to the assumption of very appreciable quantities of intra-atomic energy even at the zero of absolute temperature.

For the extreme case, $T = \infty$, we find from (223) that

$$E = NkT, \tag{224}$$

i.e., the energy is proportional to the temperature and independent of the size of the quantum of action, h, and of the nature of the oscillators. It is of interest to compare this value of the energy of vibration E of the system of oscillators, which holds at high temperatures, with the kinetic energy L of the molecular motion of an ideal monatomic gas at the same temperature as calculated in (200). From the comparison it follows that

$$E = \tfrac{2}{3}L. \tag{225}$$

This simple relation is caused by the fact that for high temperatures the contents of the hypothesis of quanta coincide with those of the classical statistical mechanics. Then the absolute magnitude of the region element, G or h respectively, becomes physically unimportant (compare Sec. 125) and we have the simple law of equipartition of the energy among all variables in question (see below Sec. 169). The factor $\frac{2}{3}$ in equation (225) is due to the fact that the kinetic energy of a moving molecule depends on three variables (ξ, η, ζ,) and the energy of a vibrating oscillator on only two (f, ψ).

The heat capacity of the system of oscillators in question is, from (223),

$$\frac{dE}{dT} = Nk\left(\frac{h\nu}{kT}\right)^2 \frac{e^{\frac{h\nu}{kT}}}{(e^{\frac{h\nu}{kT}} - 1)^2}. \tag{226}$$

It vanishes for $T = 0$ and becomes equal to Nk for $T = \infty$. *A. Einstein*[1] has made an important application of this equation to the heat capacity of solid bodies, but a closer discussion of this would be beyond the scope of the investigations to be made in this book.

[1] *A. Einstein*, Ann. d. Phys. **22**, p. 180, 1907. Compare also *M. Born* und *Th. von Kárman*, Phys. Zeitschr. **13**, p. 297, 1912.

For the constants α and γ in the expression (220) for the distribution density w we find from (221):

$$\alpha = e^{\frac{h\nu}{kT}} - 1 \qquad \gamma = e^{-\frac{h\nu}{kT}} \tag{227}$$

and finally for the entropy S of our system as a function of temperature:

$$S = kN \left\{ \frac{\frac{h\nu}{kT}}{e^{\frac{h\nu}{kT}} - 1} - \log\left(1 - e^{-\frac{h\nu}{kT}}\right) \right\}. \tag{228}$$

CHAPTER IV

DIRECT CALCULATION OF THE ENTROPY IN THE CASE OF THERMODYNAMIC EQUILIBRIUM

141. In the calculation of the entropy of an ideal gas and of a system of resonators, as carried out in the preceding chapters, we proceeded in both cases, by first determining the entropy for an arbitrarily given state, then introducing the special condition of thermodynamic equilibrium, *i.e.*, of the maximum of entropy, and then deducing for this special case an expression for the entropy.

If the problem is only the determination of the entropy in the case of thermodynamic equilibrium, this method is a roundabout one, inasmuch as it requires a number of calculations, namely, the determination of the separate distribution densities w_1, w_2, w_3, ... which do not enter separately into the final result. It is therefore useful to have a method which leads directly to the expression for the *entropy* of a system in the state of thermodynamic equilibrium, without requiring any consideration of the *state* of thermodynamic equilibrium. This method is based on an important general property of the thermodynamic probability of a state of equilibrium.

We know that there exists between the entropy S and the thermodynamic probability W in any state whatever the general relation (164). In the state of thermodynamic equilibrium both quantities have maximum values; hence, if we denote the maximum values by a suitable index:

$$S_m = k \log W_m. \tag{229}$$

It follows from the two equations that:

$$\frac{W_m}{W} = e^{\frac{S_m - S}{k}}.$$

Now, when the deviation from thermodynamic equilibrium is at all appreciable, $\frac{S_m - S}{k}$ is certainly a very large number. Accordingly W_m is not only large but of a very high order large, compared with W, that is to say: The thermodynamic probability of the state of equilibrium is enormously large compared with the thermodynamic probability of all states which, in the course of time, change into the state of equilibrium.

This proposition leads to the possibility of calculating W_m with an accuracy quite sufficient for the determination of S_m, without the necessity of introducing the special condition of equilibrium. According to Sec. 123, *et seq.*, W_m is equal to the number of all different complexions possible in the state of thermodynamic equilibrium. This number is so enormously large compared with the number of complexions of all states deviating from equilibrium that we commit no appreciable error if we think of the number of complexions of all states, which as time goes on change into the state of equilibrium, *i.e.*, all states which are at all possible under the given external conditions, as being included in this number. The total number of all possible complexions may be calculated much more readily and directly than the number of complexions referring to the state of equilibrium only.

142. We shall now use the method just formulated to calculate the entropy, in the state of equilibrium, of the system of ideal linear oscillators considered in the last chapter, when the total energy E is given. The notation remains the same as above.

We put then W_m equal to the number of complexions of all stages which are at all possible with the given energy E of the system. Then according to (219) we have the condition:

$$E = h\nu \sum_{n=1}^{\infty} (n - \tfrac{1}{2}) N_n. \tag{230}$$

Whereas we have so far been dealing with the number of complexions with given N_n, now the N_n are also to be varied in all ways consistent with the condition (230).

The total number of all complexions is obtained in a simple way by the following consideration. We write, according to (165), the condition (230) in the following form:

$$\frac{E}{h\nu} - \frac{N}{2} = \sum_{n=1}^{\infty}(n-1)N_n$$

or

$$0 \cdot N_1 + 1 \cdot N_2 + 2 \cdot N_3 + \cdots + (n-1)N_n + \cdots = \frac{E}{h\nu} - \frac{N}{2} = P. \quad (231)$$

P is a given large positive number, which may, without restricting the generality, be taken as an integer.

According to Sec. 123 a complexion is a definite assignment of every individual oscillator to a definite region element 1, 2, 3, ... of the state plane (f, ψ). Hence we may characterize a certain complexion by thinking of the N oscillators as being numbered from 1 to N and, when an oscillator is assigned to the nth region element, writing down the number of the oscillator $(n-1)$ times. If in any complexion an oscillator is assigned to the first region element its number is not put down at all. Thus every complexion gives a certain row of figures, and *vice versa* to every row of figures there corresponds a certain complexion. The position of the figures in the row is immaterial.

What makes this form of representation useful is the fact that according to (231) the number of figures in such a row is always equal to P. Hence we have "combinations with repetitions of N elements taken P at a time," whose total number is

$$\frac{N(N+1)(N+2)\ldots(N+P-1)}{1 \quad 2 \quad 3 \quad \ldots \quad P} = \frac{(N+P-1)!}{(N-1)!\,P!}. \quad (232)$$

If for example we had $N = 3$ and $P = 4$ all possible complexions would be represented by the rows of figures:

1111	1133	2222
1112	1222	2223
1113	1223	2233
1122	1233	2333
1123	1333	3333

The first row denotes that complexion in which the first oscillator lies in the 5th region element and the two others in the first. The number of complexions in this case is 15, in agreement with the formula.

143. For the entropy S of the system of oscillators which is in the state of thermodynamic equilibrium we thus obtain from equation (229) since N and P are large numbers:

$$S = k \log \frac{(N+P)!}{N!\,P!}$$

and by making use of Stirling's formula (171)[1]

$$S = kN \left\{ \left(\frac{P}{N} + 1 \right) \log \left(\frac{P}{N} + 1 \right) - \frac{P}{N} \log \frac{P}{N} \right\}.$$

If we now replace P by E from (231) we find for the entropy exactly the same value as given by (222) and thus we have demonstrated in a special case both the admissibility and the practical usefulness of the method employed.[2]

[1] Compare footnote to page 146. See also page 258.

[2] A complete mathematical discussion of the subject of this chapter has been given by *H. A. Lorentz.* Compare, *e.g.*, Nature, **92**, p. 305, Nov. 6, 1913. (Tr.)

PART IV

A SYSTEM OF OSCILLATORS IN A STATIONARY FIELD OF RADIATION

CHAPTER I

THE ELEMENTARY DYNAMICAL LAW FOR THE VIBRATIONS OF AN IDEAL OSCILLATOR. HYPOTHESIS OF EMISSION OF QUANTA

144. All that precedes has been by way of preparation. Before taking the final step, which will lead to the law of distribution of energy in the spectrum of black radiation, let us briefly put together the essentials of the problem still to be solved. As we have already seen in Sec. 93, the whole problem amounts to the determination of the temperature corresponding to a monochromatic radiation of given intensity. For among all conceivable distributions of energy the normal one, that is, the one peculiar to black radiation, is characterized by the fact that in it the rays of all frequencies have the same temperature. But the temperature of a radiation cannot be determined unless it be brought into thermodynamic equilibrium with a system of molecules or oscillators, the temperature of which is known from other sources. For if we did not consider any emitting and absorbing matter there would be no possibility of defining the entropy and temperature of the radiation, and the simple propagation of free radiation would be a reversible process, in which the entropy and temperature of the separate pencils would not undergo any change. (Compare below Sec. 166.)

Now we have deduced in the preceding section all the characteristic properties of the thermodynamic equilibrium of a system of ideal oscillators. Hence, if we succeed in indicating a state of radiation which is in thermodynamic equilibrium with the system of oscillators, the temperature of the radiation can be no other than that of the oscillators, and therewith the problem is solved.

145. Accordingly we now return to the considerations of Sec. 135 and assume a system of ideal linear oscillators in a stationary field of radiation. In order to make progress along the line proposed, it is necessary to know the elementary dynamical law, according to which the mutual action between an oscillator and the incident radiation takes place, and it is moreover easy to see that this law cannot be the same as the one which the classical electrodynamical theory postulates for the vibrations of a linear Hertzian oscillator. For, according to this law, all the oscillators, when placed in a stationary field of radiation, would, since their properties are exactly similar, assume the same energy of vibration, if we disregard certain irregular variations, which, however, will be smaller, the smaller we assume the damping constant of the oscillators, that is, the more pronounced their natural vibration is. This, however, is in direct contradiction to the definite discrete values of the distribution densities w_1, w_2, w_3, ... which we have found in Sec. 139 for the stationary state of the system of oscillators. The latter allows us to conclude with certainty that in the dynamical law to be established the quantity element of action h must play a characteristic part. Of what nature this will be cannot be predicted *a priori*; this much, however, is certain, that the only type of dynamical law admissible is one that will give for the stationary state of the oscillators exactly the distribution densities w calculated previously. It is in this problem that the question of the dynamical significance of the quantum of action h stands for the first time in the foreground, a question the answer to which was unnecessary for the calculations of the preceding sections, and this is the principal reason why in our treatment the preceding section was taken up first.

146. In establishing the dynamical law, it will be rational to proceed in such a way as to make the deviation from the laws of classical electrodynamics, which was recognized as necessary, as slight as possible.

Hence, as regards the influence of the field of radiation on an oscillator, we follow that theory closely. If the oscillator vibrates under the influence of any external electromagnetic field whatever, its energy U will not in general remain constant, but the energy equation (205a) must be extended to include the work which the external electromagnetic field does on the oscillator, and, if the axis of the electric doublet coincides with the z-axis, this work is expressed by the term $\mathsf{E}_z\, df = \mathsf{E}_z \dot{f}\, dt$. Here E_z denotes the z component of the external electric field-strength at the position of the oscillator, that is, that electric field-strength which would exist at the position of the oscillator, if the latter were not there at all. The other components of the external field have no influence on the vibrations of the oscillator.

Hence the complete energy equation reads:

$$K f\, df + L \dot{f}\, d\dot{f} = \mathsf{E}_z\, df$$

or:

$$K f + L \ddot{f} = \mathsf{E}_z, \tag{233}$$

and the energy absorbed by the oscillator during the time element dt is:

$$\mathsf{E}_z \dot{f}\, dt. \tag{234}$$

147. While the oscillator is absorbing it must also be emitting, for otherwise a stationary state would be impossible. Now, since in the law of absorption just assumed the hypothesis of quanta has as yet found no room, it follows that it must come into play in some way or other in the emission of the oscillator, and this is provided for by the introduction of the hypothesis of emission of quanta. That is to say, we shall assume that the emission does not take place continuously, as does the absorption, but that it occurs only at certain definite times, suddenly, in pulses, and in particular we assume that an oscillator can

emit energy only at the moment when its energy of vibration, U, is an integral multiple n of the quantum of energy, $\epsilon = h\nu$. Whether it then really emits or whether its energy of vibration increases further by absorption will be regarded as a matter of chance. This will not be regarded as implying that there is no causality for emission; but the processes which cause the emission will be assumed to be of such a concealed nature that for the present their laws cannot be obtained by any but statistical methods. Such an assumption is not at all foreign to physics; it is, *e.g.*, made in the atomistic theory of chemical reactions and the disintegration theory of radioactive substances.

It will be assumed, however, that if emission does take place, the entire energy of vibration, U, is emitted, so that the vibration of the oscillator decreases to zero and then increases again by further absorption of radiant energy.

It now remains to fix the law which gives the probability that an oscillator will or will not emit at an instant when its energy has reached an integral multiple of ϵ. For it is evident that the statistical state of equilibrium, established in the system of oscillators by the assumed alternations of absorption and emission will depend on this law; and evidently the mean energy U of the oscillators will be larger, the larger the probability that in such a critical state no emission takes place. On the other hand, since the mean energy U will be larger, the larger the intensity of the field of radiation surrounding the oscillators, we shall state the law of emission as follows: *The ratio of the probability that no emission takes place to the probability that emission does take place is proportional to the intensity* | of the vibration which excites the oscillator and which was defined in equation (158). The value of the constant of proportionality we shall determine later on by the application of the theory to the special case in which the energy of vibration is very large. For in this case, as we know, the familiar formulæ of the

classical dynamics hold for any period of the oscillator whatever, since the quantity element of action h may then, without any appreciable error, be regarded as infinitely small.

These statements define completely the way in which the radiation processes considered take place, as time goes on, and the properties of the stationary state. We shall now, in the first place, consider in the second chapter the absorption, and, then, in the third chapter the emission and the stationary distribution of energy, and, lastly, in the fourth chapter we shall compare the stationary state of the system of oscillators thus found with the thermodynamic state of equilibrium which was derived directly from the hypothesis of quanta in the preceding part. If we find them to agree, the hypothesis of emission of quanta may be regarded as admissible.

It is true that we shall not thereby prove that this hypothesis represents the only possible or even the most adequate expression of the elementary dynamical law of the vibrations of the oscillators. On the contrary I think it very probable that it may be greatly improved as regards form and contents. There is, however, no method of testing its admissibility except by the investigation of its consequences, and as long as no contradiction in itself or with experiment is discovered in it, and as long as no more adequate hypothesis can be advanced to replace it, it may justly claim a certain importance.

CHAPTER II

ABSORBED ENERGY

148. Let us consider an oscillator which has just completed an emission and which has, accordingly, lost all its energy of vibration. If we reckon the time t from this instant then for $t = 0$ we have $f = 0$ and $\frac{df}{dt} = 0$, and the vibration takes place according to equation (233). Let us write E_z as in (149) in the form of a Fourier's series:

$$\mathsf{E}_z = \sum_{n=1}^{n=\infty} \left[A_n \cos \frac{2\pi n t}{\mathsf{T}} + B_n \sin \frac{2\pi n t}{\mathsf{T}} \right], \tag{235}$$

where T may be chosen very large, so that for all times t considered $t < \mathsf{T}$. Since we assume the radiation to be stationary, the constant coefficients A_n and B_n depend on the ordinal numbers n in a wholly irregular way, according to the hypothesis of natural radiation (Sec. 117). The partial vibration with the ordinal number n has the frequency ν, where

$$\omega = 2\pi\nu = \frac{2\pi n}{\mathsf{T}}, \tag{236}$$

while for the frequency ν_0 of the natural period of the oscillator

$$\omega_0 = 2\pi\nu_0 = \sqrt{\frac{K}{L}}.$$

Taking the initial condition into account, we now obtain as the solution of the differential equation (233) the expression

$$f = \sum_{1}^{\infty} [a_n(\cos \omega t - \cos \omega_0 t) + b_n(\sin \omega t - \frac{\omega}{\omega_0} \sin \omega_0 t)], \tag{237}$$

where

$$a_n = \frac{A_n}{L(\omega_0^2 - \omega^2)}, \qquad b_n = \frac{B_n}{L(\omega_0^2 - \omega^2)}. \tag{238}$$

This represents the vibration of the oscillator up to the instant when the next emission occurs.

The coefficients a_n and b_n attain their largest values when ω is nearly equal to ω_0. (The case $\omega = \omega_0$ may be excluded by assuming at the outset that $\nu_0 \mathsf{T}$ is not an integer.)

149. Let us now calculate the total energy which is absorbed by the oscillator in the time from $t = 0$ to $t = \tau$, where

$$\omega_0 \tau \text{ is large.} \tag{239}$$

According to equation (234), it is given by the integral

$$\int_0^\tau \mathsf{E}_z \frac{df}{dt}\, dt, \tag{240}$$

the value of which may be obtained from the known expression for E_z (235) and from

$$\frac{df}{dt} = \sum_1^\infty [a_n(-\omega \sin \omega t + \omega_0 \sin \omega_0 t) + b_n(\omega \cos \omega t - \omega \cos \omega_0 t)]. \tag{241}$$

By multiplying out, substituting for a_n and b_n their values from (238), and leaving off all terms resulting from the multiplication of two constants A_n and B_n, this gives for the absorbed energy the following value:

$$\frac{1}{L} \int_0^\tau dt \sum_1^\infty \left[\frac{A_n^2}{\omega_0^2 - \omega^2} \cos \omega t(-\omega \sin \omega t + \omega_0 \sin \omega_0 t) \right.$$
$$\left. + \frac{B_n^2}{\omega_0^2 - \omega^2} \sin \omega t(\omega \cos \omega t - \omega \cos \omega_0 t) \right]. \tag{241a}$$

In this expression the integration with respect to t may be performed term by term. Substituting the limits τ and 0 it gives

$$\frac{1}{L}\sum_1^\infty \frac{A_n^2}{\omega_0^2-\omega^2}\left[-\frac{\sin^2\omega\tau}{2}+\omega_0\left(\frac{\sin^2\frac{\omega_0+\omega}{2}\tau}{\omega_0+\omega}+\frac{\sin^2\frac{\omega_0-\omega}{2}\tau}{\omega_0-\omega}\right)\right]$$
$$+\frac{1}{L}\sum_1^\infty \frac{B_n^2}{\omega_0^2-\omega^2}\left[\frac{\sin^2\omega\tau}{2}-\omega\left(\frac{\sin^2\frac{\omega_0+\omega}{2}\tau}{\omega_0+\omega}-\frac{\sin^2\frac{\omega_0-\omega}{2}\tau}{\omega_0-\omega}\right)\right].$$

In order to separate the terms of different order of magnitude, this expression is to be transformed in such a way that the difference $\omega_0-\omega$ will appear in all terms of the sum. This gives

$$\frac{1}{L}\sum_1^\infty \frac{A_n^2}{\omega_0^2-\omega^2}\left[\frac{\omega_0-\omega}{2(\omega_0+\omega)}\sin^2\omega\tau+\frac{\omega_0}{\omega_0+\omega}\sin\frac{\omega_0-\omega}{2}\tau\cdot\sin\frac{\omega_0+3\omega}{2}\tau\right.$$
$$\left.+\frac{\omega_0}{\omega_0-\omega}\sin^2\frac{\omega_0-\omega}{2}\tau\right]$$
$$+\frac{1}{L}\sum_1^\infty \frac{B_n^2}{\omega_0^2-\omega^2}\left[\frac{\omega_0-\omega}{2(\omega_0+\omega)}\sin^2\omega\tau-\frac{\omega}{\omega_0+\omega}\sin\frac{\omega_0-\omega}{2}\tau\cdot\sin\frac{\omega_0+3\omega}{2}\tau\right.$$
$$\left.+\frac{\omega}{\omega_0-\omega}\sin^2\frac{\omega_0-\omega}{2}\tau\right].$$

The summation with respect to the ordinal numbers n of the Fourier's series may now be performed. Since the fundamental period T of the series is extremely large, there corresponds to the difference of two consecutive ordinal numbers, $\Delta n = 1$ only a very small difference of the corresponding values of ω, $d\omega$, namely, according to (236),

$$\Delta n = 1 = \mathsf{T}\,d\nu = \frac{\mathsf{T}\,d\omega}{2\pi}, \tag{242}$$

and the summation with respect to n becomes an integration with respect to ω.

The last summation with respect to A_n may be rearranged as the sum of three series, whose orders of magnitude we shall first compare. So long as only the order is under discussion we may disregard the variability of the A_n^2 and need only compare the three integrals

$$\int_0^\infty d\omega \, \frac{\sin^2 \omega\tau}{2(\omega_0+\omega)^2} = J_1,$$

$$\int_0^\infty d\omega \, \frac{\omega_0}{(\omega_0+\omega)^2(\omega_0-\omega)} \sin\frac{\omega_0-\omega}{2}\tau \cdot \sin\frac{\omega_0+3\omega}{2}\tau = J_2,$$

and

$$\int_0^\infty d\omega \, \frac{\omega_0}{(\omega_0+\omega)(\omega_0-\omega)^2} \sin^2\frac{\omega_0-\omega}{2}\tau = J_3.$$

The evaluation of these integrals is greatly simplified by the fact that, according to (239), $\omega_0\tau$ and therefore also $\omega\tau$ are large numbers, at least for all values of ω which have to be considered. Hence it is possible to replace the expression $\sin^2\omega\tau$ in the integral J_1 by its mean value $\frac{1}{2}$ and thus we obtain:

$$J_1 = \frac{1}{4\omega_0}.$$

It is readily seen that, on account of the last factor, we obtain

$$J_2 = 0$$

for the second integral.

In order finally to calculate the third integral J_3 we shall lay off in the series of values of ω on both sides of ω_0 an interval extending from

ω_1 ($< \omega_0$) to ω_2 ($> \omega_0$) such that

$$\frac{\omega_0 - \omega_1}{\omega_0} \quad \text{and} \quad \frac{\omega_2 - \omega_0}{\omega_0} \quad \text{are small,} \tag{243}$$

and simultaneously

$$(\omega_0 - \omega_1)\tau \quad \text{and} \quad (\omega_2 - \omega_0)\tau \quad \text{are large.} \tag{244}$$

This can always be done, since $\omega_0\tau$ is large. If we now break up the integral J_3 into three parts, as follows:

$$J_3 = \int_0^\infty = \int_0^{\omega_1} + \int_{\omega_1}^{\omega_2} + \int_{\omega_2}^\infty,$$

it is seen that in the first and third partial integral the expression $\sin^2 \frac{\omega_0 - \omega}{2}\tau$ may, because of the condition (244), be replaced by its mean value $\frac{1}{2}$. Then the two partial integrals become:

$$\int_0^{\omega_1} \frac{\omega_0\, d\omega}{2(\omega_0 + \omega)(\omega_0 - \omega)^2} \quad \text{and} \quad \int_{\omega_2}^\infty \frac{\omega_0\, d\omega}{2(\omega_0 + \omega)(\omega_0 - \omega)^2}. \tag{245}$$

These are certainly smaller than the integrals:

$$\int_0^{\omega_1} \frac{d\omega}{2(\omega_0 - \omega)^2} \quad \text{and} \quad \int_{\omega_2}^\infty \frac{d\omega}{2(\omega_0 - \omega)^2}$$

which have the values

$$\frac{1}{2}\,\frac{\omega_1}{\omega_0(\omega_0 - \omega_1)} \quad \text{and} \quad \frac{1}{2(\omega_2 - \omega_0)} \tag{246}$$

respectively. We must now consider the middle one of the three partial integrals:

$$\int_{\omega_1}^{\omega_2} d\omega\, \frac{\omega_0}{(\omega_0 + \omega)(\omega_0 - \omega)^2} \cdot \sin^2 \frac{\omega_0 - \omega}{2}\tau.$$

Because of condition (243) we may write instead of this:

$$\int_{\omega_1}^{\omega_2} d\omega \cdot \frac{\sin^2 \frac{\omega_0 - \omega}{2}\tau}{2(\omega_0 - \omega)^2}$$

and by introducing the variable of integration x, where

$$x = \frac{\omega - \omega_0}{2}\tau$$

and taking account of condition (244) for the limits of the integral, we get:

$$\frac{\tau}{4}\int_{-\infty}^{+\infty} \frac{\sin^2 x\, dx}{x^2} = \frac{\tau}{4}\pi.$$

This expression is of a higher order of magnitude than the expressions (246) and hence of still higher order than the partial integrals (245) and the integrals J_1 and J_2 given above. Thus for our calculation only those values of ω will contribute an appreciable part which lie in the interval between ω_1 and ω_2, and hence we may, because of (243), replace the separate coefficients A_n^2 and B_n^2 in the expression for the total absorbed energy by their mean values A_0^2 and B_0^2 in the neighborhood of ω_0 and thus, by taking account of (242), we shall finally obtain for the total value of the energy absorbed by the oscillator in the time τ:

$$\frac{1}{L}\frac{\tau}{8}(A_0^2 + B_0^2)\mathsf{T}. \tag{247}$$

If we now, as in (158), define I, the "intensity of the vibration exciting the oscillator," by spectral resolution of the mean value of the square of the exciting field-strength E_z:

$$\overline{\mathsf{E}_z^2} = \int_0^\infty \mathsf{I}_\nu\, d\nu \tag{248}$$

we obtain from (235) and (242):

$$\overline{\mathsf{E}_z^2} = \tfrac{1}{2}\sum_1^\infty (A_n^2 + B_n^2) = \tfrac{1}{2}\int_0^\infty (A_n^2 + B_n^2)\mathsf{T}\, d\nu,$$

and by comparison with (248):

$$\mathsf{I} = \tfrac{1}{2}(A_0^2 + B_0^2)\mathsf{T}.$$

Accordingly from (247) the energy absorbed in the time τ becomes:

$$\frac{\mathsf{I}}{4L}\tau,$$

that is, *in the time between two successive emissions, the energy U of the oscillator increases uniformly with the time*, according to the law

$$\frac{dU}{dt} = \frac{\mathsf{I}}{4L} = a. \tag{249}$$

Hence the energy absorbed by all N oscillators in the time dt is:

$$\frac{N\mathsf{I}}{4L}\, dt = Na\, dt. \tag{250}$$

CHAPTER III

EMITTED ENERGY. STATIONARY STATE

150. Whereas the absorption of radiation by an oscillator takes place in a perfectly continuous way, so that the energy of the oscillator increases continuously and at a constant rate, for its emission we have, in accordance with Sec. 147, the following law: The oscillator emits in irregular intervals, subject to the laws of chance; it emits, however, only at a moment when its energy of vibration is just equal to an integral multiple n of the elementary quantum $\epsilon = h\nu$, and then it always emits its whole energy of vibration $n\epsilon$.

We may represent the whole process by the following figure in which the abscissæ represent the time t and the ordinates the energy

$$U = n\epsilon + \rho, \quad (\rho < \epsilon) \tag{251}$$

of a definite oscillator under consideration. The oblique parallel lines indicate the continuous increase of energy at a constant rate

$$\frac{dU}{dt} = \frac{d\rho}{dt} = a, \tag{252}$$

which is, according to (249), caused by absorption at a constant rate. Whenever this straight line intersects one of the parallels to the axis of abscissæ $U = \epsilon$, $U = 2\epsilon$, ... emission may possibly take place, in which case the curve drops down to zero at that point and immediately begins to rise again.

151. Let us now calculate the most important properties of the state of statistical equilibrium thus produced. Of the N oscillators situated in the field of radiation the number of those whose energy at the time t

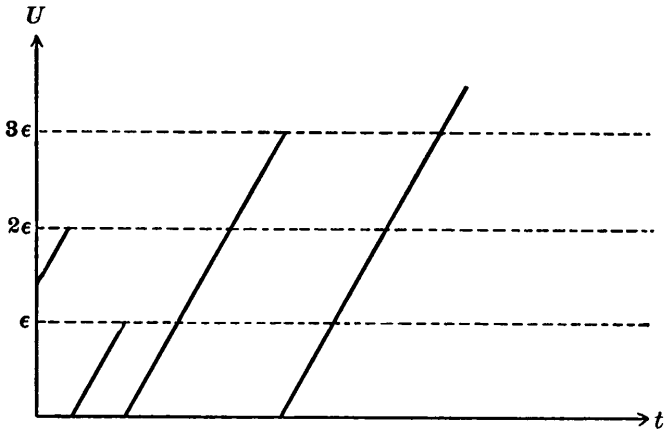


FIG. 7.

lies in the interval between $U = n\epsilon + \rho$ and $U + dU = n\epsilon + \rho + d\rho$ may be represented by

$$NR_{n,\rho}\, d\rho, \tag{253}$$

where R depends in a definite way on the integer n and the quantity ρ which varies continuously between 0 and ϵ.

After a time $dt = \dfrac{d\rho}{a}$ all the oscillators will have their energy increased by $d\rho$ and hence they will all now lie outside of the energy interval considered. On the other hand, during the same time dt, all oscillators whose energy at the time t was between $n\epsilon+\rho-d\rho$ and $n\epsilon+\rho$ will have entered that interval. The number of all these oscillators is, according to the notation used above,

$$NR_{n,\rho-d\rho}\, d\rho. \tag{254}$$

Hence this expression gives the number of oscillators which are at the time $t + dt$ in the interval considered.

Now, since we assume our system to be in a state of statistical equilibrium, the distribution of energy is independent of the time and hence the expressions (253) and (254) are equal, *i.e.*,

$$R_{n,\rho-d\rho} = R_{n,\rho} = R_n. \tag{255}$$

Thus R_n does not depend on ρ.

This consideration must, however, be modified for the special case in which $\rho = 0$. For, in that case, of the oscillators, $N = R_{n-1}\, d\rho$ in number, whose energy at the time t was between $n\epsilon$ and $n\epsilon - d\rho$, during the time $dt = \dfrac{d\rho}{a}$ some enter into the energy interval (from $U = n\epsilon$ to $U + dU = n\epsilon + d\rho$) considered; but all of them do not necessarily enter, for an oscillator may possibly emit all its energy on passing through the value $U = n\epsilon$. If the probability that emission takes place be denoted by η (< 1) the number of oscillators which pass through the critical value without emitting will be

$$N R_{n-1}(1-\eta)\, d\rho, \tag{256}$$

and by equating (256) and (253) it follows that

$$R_n = R_{n-1}(1-\eta),$$

and hence, by successive reduction,

$$R_n = R_0(1-\eta)^n. \tag{257}$$

To calculate R_0 we repeat the above process for the special case when $n = 0$ and $\rho = 0$. In this case the energy interval in question extends from $U = 0$ to $dU = d\rho$. Into this interval enter in the time $dt = \dfrac{d\rho}{a}$ all the oscillators which perform an emission during this time, namely,

those whose energy at the time t was between $\epsilon - d\rho$ and ϵ, $2\epsilon - d\rho$ and 2ϵ, $3\epsilon - d\rho$ and 3ϵ .... The numbers of these oscillators are respectively

$$NR_0\,d\rho, \quad NR_1\,d\rho, \quad NR_2\,d\rho,$$

hence their sum multiplied by η gives the desired number of emitting oscillators, namely,

$$N\eta(R_0 + R_1 + R_2 + \dots)\,d\rho, \tag{258}$$

and this number is equal to that of the oscillators in the energy interval between 0 and $d\rho$ at the time $t + dt$, which is $NR_0\,d\rho$. Hence it follows that

$$R_0 = \eta(R_0 + R_1 + R_2 + \dots). \tag{259}$$

Now, according to (253), the whole number of all the oscillators is obtained by integrating with respect to ρ from 0 to ϵ, and summing up with respect to n from 0 to ∞. Thus

$$N = N\sum_{n=0}^{n=\infty}\int_0^{\epsilon} R_{n,\rho}\,d\rho = N\sum R_n\epsilon \tag{260}$$

and

$$\sum R_n = \frac{1}{\epsilon}. \tag{261}$$

Hence we get from (257) and (259)

$$R_0 = \frac{\eta}{\epsilon}, \quad R_n = \frac{\eta}{\epsilon}(1-\eta)^n. \tag{262}$$

152. The total energy emitted in the time element $dt = \dfrac{d\rho}{a}$ is found from (258) by considering that every emitting oscillator expends all its

energy of vibration and is

$$\begin{aligned} &N\eta\, d\rho\,(R_0 + 2R_1 + 3R_2 + \dots)\epsilon \\ =&N\eta\, d\rho\,\eta(1 + 2(1-\eta) + 3(1-\eta)^2 + \dots) \\ =&N\, d\rho = Na\, dt. \end{aligned}$$

It is therefore equal to the energy absorbed in the same time by all oscillators (250), as is necessary, since the state is one of statistical equilibrium.

Let us now consider the *mean energy* $\overline{U}$ of an oscillator. It is evidently given by the following relation, which is derived in the same way as (260):

$$N\overline{U} = N\sum_0^\infty \int_0^\epsilon (n\epsilon + \rho)R_n\, d\rho. \tag{263}$$

From this it follows by means of (262), that

$$\overline{U} = \left(\frac{1}{\eta} - \frac{1}{2}\right)\epsilon = \left(\frac{1}{\eta} - \frac{1}{2}\right)h\nu. \tag{264}$$

Since $\eta < 1$, $\overline{U}$ lies between $\dfrac{h\nu}{2}$ and ∞. Indeed, it is immediately evident that $\overline{U}$ can never become less than $\dfrac{h\nu}{2}$ since the energy of *every* oscillator, however small it may be, will assume the value $\epsilon = h\nu$ within a time limit, which can be definitely stated.

153. The probability constant η contained in the formulæ for the stationary state is determined by the law of emission enunciated in Sec. 147. According to this, the ratio of the probability that no emission takes place to the probability that emission does take place is

proportional to the intensity I of the vibration exciting the oscillator, and hence

$$\frac{1-\eta}{\eta} = p\mathsf{I} \tag{265}$$

where the constant of proportionality is to be determined in such a way that for very large energies of vibration the familiar formulæ of classical dynamics shall hold.

Now, according to (264), η becomes small for large values of $\overline{U}$ and for this special case the equations (264) and (265) give

$$\overline{U} = ph\nu\mathsf{I},$$

and the energy emitted or absorbed respectively in the time dt by all N oscillators becomes, according to (250),

$$\frac{N\mathsf{I}}{4L}\,dt = \frac{N\overline{U}}{4Lph\nu}\,dt. \tag{266}$$

On the other hand, *H. Hertz* has already calculated from *Maxwell's* theory the energy emitted by a linear oscillator vibrating periodically. For the energy emitted in the time of one-half of one vibration he gives the expression[1]

$$\frac{\pi^4 E^2 l^2}{3\lambda^3}$$

where λ denotes half the wave length, and the product El (the C of our notation) denotes the amplitude of the moment f (Sec. 135) of the vibrations. This gives for the energy emitted in the time of a whole vibration

$$\frac{16\pi^4 C^2}{3\lambda^3}$$

[1] *H. Hertz*, Wied. Ann. **36**, p. 12, 1889.

where λ denotes the whole wave length, and for the energy emitted by N similar oscillators in the time dt

$$N\frac{16\pi^4 C^2\nu^4}{3c^3}\,dt$$

since $\lambda = \frac{c}{\nu}$. On introducing into this expression the energy U of an oscillator from (205), (207), and (208), namely

$$U = 2\pi^2\nu^2 LC^2,$$

we have for the energy emitted by the system of oscillators

$$N\frac{8\pi^2\nu^2 U}{3c^3 L}\,dt \tag{267}$$

and by equating the expressions (266) and (267) we find for the factor of proportionality p

$$p = \frac{3c^3}{32\pi^2 h\nu^3}. \tag{268}$$

154. By the determination of p the question regarding the properties of the state of statistical equilibrium between the system of the oscillators and the vibration exciting them receives a general answer. For from (265) we get

$$\eta = \frac{1}{1+p\mathsf{l}}$$

and further from (262)

$$R_n = \frac{1}{\epsilon}\frac{(p\mathsf{l})^n}{(1+p\mathsf{l})^{n+1}}. \tag{269}$$

Hence in the state of stationary equilibrium the number of oscillators whose energy lies between $nh\nu$ and $(n+1)h\nu$ is, from equation (253),

$$N\int_0^{\epsilon} R_n\, d\rho = NR_n\epsilon = N\frac{(p\mathsf{l})^n}{(1+p\mathsf{l})^{n+1}} \tag{270}$$

where $n = 0,\ 1,\ 2,\ 3,\ \ldots.$

CHAPTER IV

THE LAW OF THE NORMAL DISTRIBUTION OF ENERGY. ELEMENTARY QUANTA OF MATTER AND ELECTRICITY

155. In the preceding chapter we have made ourselves familiar with all the details of a system of oscillators exposed to uniform radiation. We may now develop the idea put forth at the end of Sec. 144. That is to say, we may identify the stationary state of the oscillators just found with the state of maximum entropy of the system of oscillators which was derived directly from the hypothesis of quanta in the preceding part, and we may then equate the temperature of the radiation to the temperature of the oscillators. It is, in fact, possible to obtain perfect agreement of the two states by a suitable coordination of their corresponding quantities.

According to Sec. 139, the "distribution density" w of the oscillators in the state of statistical equilibrium changes abruptly from one region element to another, while, according to Sec. 138, the distribution within a single region element is uniform. The region elements of the state plane $(f\psi)$ are bounded by concentric similar and similarly situated ellipses which correspond to those values of the energy U of an oscillator which are integral multiples of $h\nu$. We have found exactly the same thing for the stationary state of the oscillators when they are exposed to uniform radiation, and the distribution density w_n in the nth region element may be found from (270), if we remember that the nth region element contains the energies between $(n-1)h\nu$ and $nh\nu$. Hence:

$$w_n = \frac{(p\mathfrak{l})^{n-1}}{(1+p\mathfrak{l})^n} = \frac{1}{p\mathfrak{l}}\left(\frac{p\mathfrak{l}}{1+p\mathfrak{l}}\right)^n. \tag{271}$$

This is in perfect agreement with the previous value (220) of w_n if we put

$$\alpha = \frac{1}{p\mathsf{I}} \quad \text{and} \quad \gamma = \frac{p\mathsf{I}}{1 + p\mathsf{I}},$$

and each of these two equations leads, according to (221), to the following relation between the intensity of the exciting vibration I and the total energy E of the N oscillators:

$$p\mathsf{I} = \frac{E}{Nh\nu} - \frac{1}{2}. \tag{272}$$

156. If we finally introduce the temperature T from (223), we get from the last equation, by taking account of the value (268) of the factor of proportionality p,

$$\mathsf{I} = \frac{32\pi^2 h\nu^3}{3c^3} \frac{1}{e^{\frac{h\nu}{kT}} - 1}. \tag{273}$$

Moreover the specific intensity K of a monochromatic plane polarized ray of frequency ν is, according to equation (160),

$$\mathsf{K} = \frac{h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{kT}} - 1} \tag{274}$$

and the space density of energy of uniform monochromatic unpolarized radiation of frequency ν is, from (159),

$$\mathsf{u} = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{kT}} - 1}. \tag{275}$$

Since, among all the forms of radiation of differing constitutions, black radiation is distinguished by the fact that all monochromatic rays contained in it have the same temperature (Sec. 93) these equations also

give the law of distribution of energy in the normal spectrum, *i.e.*, in the emission spectrum of a body which is black with respect to the vacuum.

If we refer the specific intensity of a monochromatic ray not to the frequency ν but, as is usually done in experimental physics, to the wave length λ, by making use of (15) and (16) we obtain the expression

$$E_\lambda = \frac{c^2 h}{\lambda^5} \frac{1}{e^{\frac{ch}{k\lambda T}} - 1} = \frac{c_1}{\lambda^5} \frac{1}{e^{\frac{c_2}{\lambda T}} - 1}. \tag{276}$$

This is the specific intensity of a monochromatic plane polarized ray of the wave length λ which is emitted from a black body at the temperature T into a vacuum in a direction perpendicular to the surface. The corresponding space density of unpolarized radiation is obtained by multiplying E_λ by $\frac{8\pi}{c}$.

Experimental tests have so far confirmed equation (276).[1] According to the most recent measurements made in the Physikalisch-technische Reichsanstalt[2] the value of the second radiation constant c_2 is approximately

$$c_2 = \frac{ch}{k} = 1.436 \text{ cm degree}. \tag{277}$$

More detailed information regarding the history of the equation of radiation is to be found in the original papers and in the first edition

[1] See among others *H. Rubens* und *F. Kurlbaum*, Sitz. Ber. d. Akad. d. Wiss. zu Berlin vom 25. Okt., 1900, p. 929. Ann. d. Phys. **4**, p. 649, 1901. *F. Paschen*, Ann. d. Phys. **4**, p. 277, 1901. *O. Lummer* und *E. Pringsheim*, Ann. d. Phys. **6**, p. 210, 1901. Tätigkeitsbericht der Phys.-Techn. Reichsanstalt vom J. 1911, Zeitschr. f. Instrumentenkunde, 1912, April, p. 134 ff.

[2] According to private information kindly furnished by the president, *Mr. Warburg*.

of this book. At this point it may merely be added that equation (276) was not simply extrapolated from radiation measurements, but was originally found in a search after a connection between the entropy and the energy of an oscillator vibrating in a field, a connection which would be as simple as possible and consistent with known measurements.

157. The entropy of a ray is, of course, also determined by its temperature. In fact, by combining equations (138) and (274) we readily obtain as an expression for the entropy radiation L of a monochromatic plane polarized ray of the specific intensity of radiation K and the frequency ν,

$$\mathsf{L} = \frac{k\nu^2}{c^2}\left\{\left(1+\frac{c^2\mathsf{K}}{h\nu^3}\right)\log\left(1+\frac{c^2\mathsf{K}}{h\nu^3}\right) - \frac{c^2\mathsf{K}}{h\nu^3}\log\frac{c^2\mathsf{K}}{h\nu^3}\right\} \tag{278}$$

which is a more definite statement of equation (134) for *Wien's* displacement law.

Moreover it follows from (135), by taking account of (273), that the space density of the entropy s of uniform monochromatic unpolarized radiation as a function of the space density of energy u is

$$\mathsf{s} = \frac{8\pi k\nu^2}{c^3}\left\{\left(1+\frac{c^3\mathsf{u}}{8\pi h\nu^3}\right)\log\left(1+\frac{c^3\mathsf{u}}{8\pi h\nu^3}\right) - \frac{c^3\mathsf{u}}{8\pi h\nu^3}\log\frac{c^3\mathsf{u}}{8\pi h\nu^3}\right\}. \tag{279}$$

This is a more definite statement of equation (119).

158. For *small* values of λT (*i.e.*, small compared with the constant $\frac{ch}{k}$) equation (276) becomes

$$E_\lambda = \frac{c^2h}{\lambda^5}e^{-\frac{ch}{k\lambda T}} \tag{280}$$

an equation which expresses *Wien's*[1] law of energy distribution.

[1] *W. Wien*, Wied. Ann. **58**, p. 662, 1896.

The specific intensity of radiation K then becomes, according to (274),

$$\mathsf{K} = \frac{h\nu^3}{c^2} e^{-\frac{h\nu}{kT}} \tag{281}$$

and the space density of energy u is, from (275),

$$\mathsf{u} = \frac{8\pi h\nu^3}{c^3} e^{-\frac{h\nu}{kT}}. \tag{282}$$

159. On the other hand, for *large* values of λT (276) becomes

$$E_\lambda = \frac{ckT}{\lambda^4}, \tag{283}$$

a relation which was established first by *Lord Rayleigh*[1] and which we may, therefore, call "*Rayleigh's* law of radiation."

We then find for the specific intensity of radiation K from (274)

$$\mathsf{K} = \frac{k\nu^2 T}{c^2} \tag{284}$$

and from (275) for the space density of monochromatic radiation we get

$$\mathsf{u} = \frac{8\pi k\nu^2 T}{c^3}. \tag{285}$$

Rayleigh's law of radiation is of very great theoretical interest, since it represents that distribution of energy which is obtained for radiation in statistical equilibrium with material molecules by means of the classical dynamics, and without introducing the hypothesis of quanta.[2] This

[1] *Lord Rayleigh*, Phil. Mag. **49**, p. 539, 1900.

[2] *J. H. Jeans*, Phil. Mag. Febr., 1909, p. 229, *H. A. Lorentz*, Nuovo Cimento V, vol. **16**, 1908.

may also be seen from the fact that for a vanishingly small value of the quantity element of action, h, the general formula (276) degenerates into *Rayleigh's* formula (283). See also below, Sec. 168 *et seq.*

160. For the total space density, u, of black radiation at any temperature T we obtain, from (275),

$$u = \int_0^\infty \mathsf{u}\, d\nu = \frac{8\pi h}{c^3} \int_0^\infty \frac{\nu^3\, d\nu}{e^{\frac{h\nu}{kT}} - 1}$$

or

$$u = \frac{8\pi h}{c^3} \int_0^\infty \left(e^{-\frac{h\nu}{kT}} + e^{-\frac{2h\nu}{kT}} + e^{-\frac{3h\nu}{kT}} + \dots \right) \nu^3\, d\nu$$

and, integrating term by term,

$$u = \frac{48\pi h}{c^3} \left(\frac{kT}{h} \right)^4 \alpha \tag{286}$$

where α is an abbreviation for

$$\alpha = 1 + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots = 1.0823. \tag{287}$$

This relation expresses the *Stefan-Boltzmann* law (75) and it also tells us that the constant of this law is given by

$$a = \frac{48\pi\alpha k^4}{c^3 h^3}. \tag{288}$$

161. For that wave length λ_m the maximum of the intensity of radiation corresponds in the spectrum of black radiation, we find from (276)

$$\left(\frac{dE_\lambda}{d\lambda} \right)_{\lambda=\lambda_m} = 0.$$

On performing the differentiation and putting as an abbreviation

$$\frac{ch}{k\lambda_m T} = \beta,$$

we get

$$e^{-\beta} + \frac{\beta}{5} - 1 = 0.$$

The root of this transcendental equation is

$$\beta = 4.9651, \tag{289}$$

and accordingly $\lambda_m T = \dfrac{ch}{\beta k}$, and this is a constant, as demanded by *Wien's* displacement law. By comparison with (109) we find the meaning of the constant b, namely,

$$b = \frac{ch}{\beta k}, \tag{290}$$

and, from (277),

$$b = \frac{c_2}{\beta} = \frac{1.436}{4.9651} = 0.289 \text{ cm degree}, \tag{291}$$

while *Lummer* and *Pringsheim* found by measurements 0.294 and *Paschen* 0.292.

162. By means of the measured values[1] of a and c_2 the universal constants h and k may be readily calculated. For it follows from

[1]Here as well as later on the value given above (79) has been replaced by $a = 7.39 \cdot 10^{-15}$, obtained from $\sigma = \dfrac{ac}{4} = 5.54 \cdot 10^{-5}$. This is the final result of the newest measurements made by *W. Westphal*, according to information kindly furnished by him and *Mr. H. Rubens*. (Nov., 1912). [Compare p. 75, footnote. Tr.]

equations (277) and (288) that

$$h = \frac{ac_2^4}{48\pi\alpha c} \qquad k = \frac{ac_2^3}{48\pi\alpha}. \tag{292}$$

Substituting the values of the constants a, c_2, α, c, we get

$$h = 6.415 \cdot 10^{-27}\,\text{erg sec.}, \qquad k = 1.34 \cdot 10^{-16}\,\frac{\text{erg}}{\text{degree}}. \tag{293}$$

163. To ascertain the full physical significance of the quantity element of action, h, much further research work will be required. On the other hand, the value obtained for k enables us readily to state numerically in the C. G. S. system the general connection between the entropy S and the thermodynamic probability W as expressed by the universal equation (164). The general expression for the entropy of a physical system is

$$S = 1.34 \cdot 10^{-16} \log W\,\frac{\text{erg}}{\text{degree}}. \tag{294}$$

This equation may be regarded as the most general definition of entropy. Herein the thermodynamic probability W is an integral number, which is completely defined by the macroscopic state of the system. Applying the result expressed in (293) to the kinetic theory of gases, we obtain from equation (194) for the ratio of the mass of a molecule to that of a mol,

$$\omega = \frac{k}{R} = \frac{1.34 \times 10^{-16}}{831 \times 10^5} = 1.61 \times 10^{-24}, \tag{295}$$

that is to say, there are in one mol

$$\frac{1}{\omega} = 6.20 \times 10^{23}$$

molecules, where the mol of oxygen, O_2, is always assumed as 32 gr. Hence, for example, the absolute mass of a hydrogen atom ($\frac{1}{2}H_2 = 1.008$) equals 1.62×10^{-24} gr. With these numerical values the number of molecules contained in 1 cm^3 of an ideal gas at 0° C. and 1 atmosphere pressure becomes

$$N = \frac{76 \cdot 13.6 \cdot 981}{831 \cdot 10^5 \cdot 273\omega} = 2.77 \cdot 10^{19}. \tag{296}$$

The mean kinetic energy of translatory motion of a molecule at the absolute temperature $T = 1$ is, in the absolute C. G. S. system, according to (200),

$$\tfrac{3}{2}k = 2.01 \cdot 10^{-16}. \tag{297}$$

In general the mean kinetic energy of translatory motion of a molecule is expressed by the product of this number and the absolute temperature T.

The elementary quantity of electricity or the free charge of a monovalent ion or electron is, in electrostatic units,

$$e = \omega \cdot 9654 \cdot 3 \cdot 10^{10} = 4.67 \cdot 10^{-10}. \tag{298}$$

Since absolute accuracy is claimed for the formulæ here employed, the degree of approximation to which these numbers represent the corresponding physical constants depends only on the accuracy of the measurements of the two radiation constants a and c_2.

164. Natural Units.—All the systems of units which have hitherto been employed, including the so-called absolute C. G. S. system, owe their origin to the coincidence of accidental circumstances, inasmuch as the choice of the units lying at the base of every system has been made, not according to general points of view which would necessarily retain their importance for all places and all times, but essentially with reference to the special needs of our terrestrial civilization.

Thus the units of length and time were derived from the present dimensions and motion of our planet, and the units of mass and temperature from the density and the most important temperature points of water, as being the liquid which plays the most important part on the surface of the earth, under a pressure which corresponds to the mean properties of the atmosphere surrounding us. It would be no less arbitrary if, let us say, the invariable wave length of Na-light were taken as unit of length. For, again, the particular choice of Na from among the many chemical elements could be justified only, perhaps, by its common occurrence on the earth, or by its double line, which is in the range of our vision, but is by no means the only one of its kind. Hence it is quite conceivable that at some other time, under changed external conditions, every one of the systems of units which have so far been adopted for use might lose, in part or wholly, its original natural significance.

In contrast with this it might be of interest to note that, with the aid of the two constants h and k which appear in the universal law of radiation, we have the means of establishing units of length, mass, time, and temperature, which are independent of special bodies or substances, which necessarily retain their significance for all times and for all environments, terrestrial and human or otherwise, and which may, therefore, be described as "natural units."

The means of determining the four units of length, mass, time, and temperature, are given by the two constants h and k mentioned, together with the magnitude of the velocity of propagation of light in a vacuum, c, and that of the constant of gravitation, f. Referred to centimeter, gram, second, and degrees Centigrade, the numerical values of these four constants are as follows:

$$h = 6.415 \cdot 10^{-27} \frac{\text{gr cm}^2}{\text{sec}}$$

$$k = 1.34 \cdot 10^{-16} \frac{\text{gr cm}^2}{\text{sec}^2 \text{ degree}}$$
$$c = 3 \cdot 10^{10} \frac{\text{cm}}{\text{sec}}$$
$$f = 6.685 \cdot 10^{-8} \frac{\text{cm}^3}{\text{gr sec}^2} .^{1}$$

If we now choose the natural units so that in the new system of measurement each of the four preceding constants assumes the value 1, we obtain, as unit of length, the quantity

$$\sqrt{\frac{fh}{c^3}} = 3.99 \cdot 10^{-33} \text{ cm},$$

as unit of mass

$$\sqrt{\frac{ch}{f}} = 5.37 \cdot 10^{-5} \text{ gr},$$

as unit of time

$$\sqrt{\frac{fh}{c^5}} = 1.33 \cdot 10^{-43} \text{ sec},$$

as unit of temperature

$$\frac{1}{k}\sqrt{\frac{c^5 h}{f}} = 3.60 \cdot 10^{32} \text{ degree}.$$

These quantities retain their natural significance as long as the law of gravitation and that of the propagation of light in a vacuum and the two principles of thermodynamics remain valid; they therefore must be

[1] *F. Richarz* and *O. Krigar-Menzel*, Wied. Ann. **66**, p. 190, 1898.

found always the same, when measured by the most widely differing intelligences according to the most widely differing methods.

165. The relations between the intensity of radiation and the temperature expressed in Sec. 156 hold for radiation in a pure vacuum. If the radiation is in a medium of refractive index n, the way in which the intensity of radiation depends on the frequency and the temperature is given by the proposition of Sec. 39, namely, the product of the specific intensity of radiation K_ν and the square of the velocity of propagation of the radiation has the same value for all substances. The form of this universal function (42) follows directly from (274)

$$\mathsf{K}q^2 = \frac{\epsilon_\nu}{\alpha_\nu} q^2 = \frac{h\nu^3}{e^{\frac{h\nu}{kT}} - 1}. \tag{299}$$

Now, since the refractive index n is inversely proportional to the velocity of propagation, equation (274) is, in the case of a medium with the index of refraction n, replaced by the more general relation

$$\mathsf{K}_\nu = \frac{h\nu^3 n^2}{c^2} \frac{1}{e^{\frac{h\nu}{kT}} - 1} \tag{300}$$

and, similarly, in place of (275) we have the more general relation

$$\mathsf{u} = \frac{8\pi h\nu^3 n^3}{c_3} \frac{1}{e^{\frac{h\nu}{kT}} - 1}. \tag{301}$$

These expressions hold, of course, also for the emission of a body which is black with respect to a medium with an index of refraction n.

166. We shall now use the laws of radiation we have obtained to calculate the temperature of a monochromatic unpolarized radiation of given intensity in the following case. Let the light pass normally

through a small area (slit) and let it fall on an arbitrary system of diathermanous media separated by spherical surfaces, the centers of which lie on the same line, the axis of the system. Such radiation consists of homocentric pencils and hence forms behind every refracting surface a real or virtual image of the emitting surface, the image being likewise normal to the axis. To begin with, we assume the last as well as the first medium to be a pure vacuum. Then, for the determination of the temperature of the radiation according to equation (274), we need calculate only the specific intensity of radiation K_ν in the last medium, and this is given by the total intensity of the monochromatic radiation I_ν, the size of the area of the image F, and the solid angle Ω of the cone of rays passing through a point of the image. For the specific intensity of radiation K_ν is, according to (13), determined by the fact that an amount

$$2\mathsf{K}_\nu \, d\sigma \, d\Omega \, d\nu \, dt$$

of energy of unpolarized light corresponding to the interval of frequencies from ν to $\nu + d\nu$ is, in the time dt, radiated in a normal direction through an element of area $d\sigma$ within the conical element $d\Omega$. If now $d\sigma$ denotes an element of the area of the surface image in the last medium, then the total monochromatic radiation falling on the image has the intensity

$$I_\nu = 2\mathsf{K}_\nu \int d\sigma \int d\Omega.$$

I_ν is of the dimensions of energy, since the product $d\nu \, dt$ is a mere number. The first integral is the whole area, F, of the image, the second is the solid angle, Ω, of the cone of rays passing through a point of the surface of the image. Hence we get

$$I_\nu = 2\mathsf{K}_\nu F \Omega, \tag{302}$$

and, by making use of (274), for the temperature of the radiation

$$T = \frac{h\nu}{k} \cdot \frac{1}{\log\left(\frac{2h\nu^3 F\Omega}{c^2 I_\nu} + 1\right)}. \tag{303}$$

If the diathermanous medium considered is not a vacuum but has an index of refraction n, (274) is replaced by the more general relation (300), and, instead of the last equation, we obtain

$$T = \frac{h\nu}{k} \frac{1}{\log\left(\frac{2h\nu^3 F\Omega n^2}{c^2 I_\nu} + 1\right)} \tag{304}$$

or, on substituting the numerical values of c, h, and k,

$$T = \frac{0.479 \cdot 10^{-10}\nu}{\log\left(\frac{1.43 \cdot 10^{-47}\nu^3 F\Omega n^2}{I_\nu} + 1\right)} \text{ degree Centigrade.}$$

In this formula the natural logarithm is to be taken, and I_ν is to be expressed in ergs, ν in "reciprocal seconds," *i.e.*, $(\text{seconds})^{-1}$, F in square centimeters. In the case of visible rays the second term, 1, in the denominator may usually be omitted.

The temperature thus calculated is retained by the radiation considered, so long as it is propagated without any disturbing influence in the diathermanous medium, however great the distance to which it is propagated or the space in which it spreads. For, while at larger distances an ever decreasing amount of energy is radiated through an element of area of given size, this is contained in a cone of rays starting from the element, the angle of the cone continually decreasing in

such a way that the value of K remains entirely unchanged. Hence the free expansion of radiation is a perfectly reversible process. (Compare above, Sec. 144.) It may actually be reversed by the aid of a suitable concave mirror or a converging lens.

Let us next consider the temperature of the radiation in the other media, which lie between the separate refracting or reflecting spherical surfaces. In every one of these media the radiation has a definite temperature, which is given by the last formula when referred to the real or virtual image formed by the radiation in that medium.

The frequency ν of the monochromatic radiation is, of course, the same in all media; moreover, according to the laws of geometrical optics, the product $n^2 F\Omega$ is the same for all media. Hence, if, in addition, the total intensity of radiation I_ν remains constant on refraction (or reflection), T also remains constant, or in other words: The temperature of a homocentric pencil is not changed by regular refraction or reflection, unless a loss in energy of radiation occurs. Any weakening, however, of the total intensity I_ν by a subdivision of the radiation, whether into two or into many different directions, as in the case of diffuse reflection, leads to a lowering of the temperature of the pencil. In fact, a certain loss of energy by refraction or reflection does occur, in general, on a refraction or reflection, and hence also a lowering of the temperature takes place. In these cases a fundamental difference appears, depending on whether the radiation is weakened merely by free expansion or by subdivision or absorption. In the first case the temperature remains constant, in the second it decreases.[1]

[1]Nevertheless regular refraction and reflection are not irreversible processes; for the refracted and the reflected rays are coherent and the entropy of two coherent rays is not equal to the sum of the entropies of the separate rays. (Compare above, Sec. 104.) On the other hand, diffraction is an irreversible process. *M. Laue*, Ann. d. Phys. **31**, p. 547, 1910.

167. The laws of emission of a black body having been determined, it is possible to calculate, with the aid of *Kirchhoff's* law (48), the emissive power E of any body whatever, when its absorbing power A or its reflecting power $1 - A$ is known. In the case of metals this calculation becomes especially simple for long waves, since *E. Hagen* and *H. Rubens*[1] have shown experimentally that the reflecting power and, in fact, the entire optical behavior of the metals in the spectral region mentioned is represented by the simple equations of *Maxwell* for an electromagnetic field with homogeneous conductors and hence depends only on the specific conductivity for steady electric currents. Accordingly, it is possible to express completely the emissive power of a metal for long waves by its electric conductivity combined with the formulæ for black radiation.[2]

168. There is, however, also a method, applicable to the case of long waves, for the direct theoretical determination of the electric conductivity and, with it, of the absorbing power, A, as well as the emissive power, E, of metals. This is based on the ideas of the electron theory, as they have been developed for the thermal and electrical processes in metals by *E. Riecke*[3] and especially by *P. Drude*.[4] According to these, all such processes are based on the rapid irregular motions of the negative electrons, which fly back and forth between the positively charged molecules of matter (here of the metal) and rebound on impact with them as well as with one another, like gas molecules when they strike a rigid obstacle or one another. The velocity of the heat motions of the material molecules may be neglected compared with that of the electrons, since in the stationary state the mean kinetic energy of motion

[1] *E. Hagen* und *H. Rubens*, Ann. d. Phys. **11**, p. 873, 1903.
[2] *E. Aschkinass*, Ann. d. Phys. **17**, p. 960, 1905.
[3] *E. Riecke*, Wied. Ann. **66**, p. 353, 1898.
[4] *P. Drude*, Ann. d. Phys. **1**, p. 566, 1900.

of a material molecule is equal to that of an electron, and since the mass of a material molecule is more than a thousand times as large as that of an electron. Now, if there is an electric field in the interior of the metal, the oppositely charged particles are driven in opposite directions with average velocities depending on the mean free path, among other factors, and this explains the conductivity of the metal for the electric current. On the other hand, the emissive power of the metal for the radiant heat follows from the calculation of the impacts of the electrons. For, so long as an electron flies with constant speed in a constant direction, its kinetic energy remains constant and there is no radiation of energy; but, whenever it suffers by impact a change of its velocity components, a certain amount of energy, which may be calculated from electrodynamics and which may always be represented in the form of a *Fourier's* series, is radiated into the surrounding space, just as we think of *Roentgen rays* as being caused by the impact on the anticathode of the electrons ejected from the cathode. From the standpoint of the hypothesis of quanta this calculation cannot, for the present, be carried out without ambiguity except under the assumption that, during the time of a partial vibration of the *Fourier* series, a large number of impacts of electrons occurs, *i.e.*, for comparatively long waves, for then the fundamental law of impact does not essentially matter.

Now this method may evidently be used to derive the laws of black radiation in a new way, entirely independent of that previously employed. For if the emissive power, E, of the metal, thus calculated, is divided by the absorbing power, A, of the same metal, determined by means of its electric conductivity, then, according to *Kirchhoff's* law (48), the result must be the emissive power of a black body, irrespective of the special substance used in the determination. In this

manner *H. A. Lorentz*[1] has, in a profound investigation, derived the law of radiation of a black body and has obtained a result the contents of which agree exactly with equation (283), and where also the constant k is related to the gas constant R by equation (193). It is true that this method of establishing the laws of radiation is, as already said, restricted to the range of long waves, but it affords a deeper and very important insight into the mechanism of the motions of the electrons and the radiation phenomena in metals caused by them. At the same time the point of view described above in Sec. 111, according to which the normal spectrum may be regarded as consisting of a large number of quite irregular processes as elements, is expressly confirmed.

169. A further interesting confirmation of the law of radiation of black bodies for long waves and of the connection of the radiation constant k with the absolute mass of the material molecules was found by *J. H. Jeans*[2] by a method previously used by *Lord Rayleigh*,[3] which differs essentially from the one pursued here, in the fact that it entirely avoids making use of any special mutual action between matter (molecules, oscillators) and the ether and considers essentially only the processes in the vacuum through which the radiation passes. The starting point for this method of treatment is given by the following proposition of statistical mechanics. (Compare above, Sec. 140.) When irreversible processes take place in a system, which satisfies *Hamilton's* equations of motion, and whose state is determined by a large number of independent variables and whose total energy is found by addition of different parts depending on the squares of the variables of state, they do so, on the average, in such a sense that the partial energies

[1] *H. A. Lorentz*, Proc. Kon. Akad. v. Wet. Amsterdam, 1903, p. 666.
[2] *J. H. Jeans*, Phil. Mag. **10**, p. 91, 1905.
[3] *Lord Rayleigh*, Nature **72**, p. 54 and p. 243, 1905.

corresponding to the separate independent variables of state tend to equality, so that finally, on reaching statistical equilibrium, their mean values have become equal. From this proposition the stationary distribution of energy in such a system may be found, when the independent variables which determine the state are known.

Let us now imagine a perfect vacuum, cubical in form, of edge l, and with metallically reflecting sides. If we take the origin of coordinates at one corner of the cube and let the axes of coordinates coincide with the adjoining edges, an electromagnetic process which may occur in this cavity is represented by the following system of equations:

$$\begin{aligned}
\mathsf{E}_x &= \cos\frac{\mathsf{a}\pi x}{l}\sin\frac{\mathsf{b}\pi y}{l}\sin\frac{\mathsf{c}\pi z}{l}(e_1\cos 2\pi\nu t + e_1'\sin 2\pi\nu t),\\
\mathsf{E}_y &= \sin\frac{\mathsf{a}\pi x}{l}\cos\frac{\mathsf{b}\pi y}{l}\sin\frac{\mathsf{c}\pi z}{l}(e_2\cos 2\pi\nu t + e_2'\sin 2\pi\nu t),\\
\mathsf{E}_z &= \sin\frac{\mathsf{a}\pi x}{l}\sin\frac{\mathsf{b}\pi y}{l}\cos\frac{\mathsf{c}\pi z}{l}(e_3\cos 2\pi\nu t + e_3'\sin 2\pi\nu t),\\
\mathsf{H}_x &= \sin\frac{\mathsf{a}\pi x}{l}\cos\frac{\mathsf{b}\pi y}{l}\cos\frac{\mathsf{c}\pi z}{l}(h_1\sin 2\pi\nu t - h_1'\cos 2\pi\nu t),\\
\mathsf{H}_y &= \cos\frac{\mathsf{a}\pi x}{l}\sin\frac{\mathsf{b}\pi y}{l}\cos\frac{\mathsf{c}\pi z}{l}(h_2\sin 2\pi\nu t - h_2'\cos 2\pi\nu t),\\
\mathsf{H}_z &= \cos\frac{\mathsf{a}\pi x}{l}\cos\frac{\mathsf{b}\pi y}{l}\sin\frac{\mathsf{c}\pi z}{l}(h_3\sin 2\pi\nu t - h_3'\cos 2\pi\nu t),
\end{aligned} \tag{305}$$

where a, b, c represent any three positive integral numbers. The boundary conditions in these expressions are satisfied by the fact that for the six bounding surfaces $x = 0$, $x = l$, $y = 0$, $y = l$, $z = 0$, $z = l$ the tangential components of the electric field-strength E vanish. Maxwell's equations of the field (52) are also satisfied, as may be seen on substitution, provided there exist certain conditions between the constants which may be stated in a single proposition as follows: Let a be a

certain positive constant, then there exist between the nine quantities written in the following square:

$$\begin{array}{ccc} \dfrac{\mathsf{a}c}{2l\nu} & \dfrac{\mathsf{b}c}{2l\nu} & \dfrac{\mathsf{c}c}{2l\nu} \\ \dfrac{h_1}{a} & \dfrac{h_2}{a} & \dfrac{h_3}{a} \\ \dfrac{e_1}{a} & \dfrac{e_2}{a} & \dfrac{e_3}{a} \end{array}$$

all the relations which are satisfied by the nine so-called "direction cosines" of two orthogonal right-handed coordinate systems, *i.e.*, the cosines of the angles of any two axes of the systems.

Hence the sum of the squares of the terms of any horizontal or vertical row equals 1, for example,

$$\begin{gathered} \frac{c^2}{4l^2}\nu^2(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2)=1 \\ h_1^2+h_2^2+h_3^2=a^2=e_1^2+e_2^2+e_3^2. \end{gathered} \tag{306}$$

Moreover the sum of the products of corresponding terms in any two parallel rows is equal to zero, for example,

$$\begin{aligned} \mathsf{a}e_1+\mathsf{b}e_2+\mathsf{c}e_3 &= 0 \\ \mathsf{a}h_1+\mathsf{b}h_2+\mathsf{c}h_3 &= 0. \end{aligned} \tag{307}$$

Moreover there are relations of the following form:

$$\frac{h_1}{a}=\frac{e_2}{a}\cdot\frac{\mathsf{c}c}{2l\nu}-\frac{e_3}{a}\frac{\mathsf{b}c}{2l\nu},$$

and hence

$$h_1=\frac{c}{2l\nu}(\mathsf{c}e_2-\mathsf{b}e_3),\ \text{etc.} \tag{308}$$

If the integral numbers a, b, c are given, then the frequency ν is immediately determined by means of (306). Then among the six quantities e_1, e_2, e_3, h_1, h_2, h_3, only two may be chosen arbitrarily, the others then being uniquely determined by them by linear homogeneous relations. If, for example, we assume e_1 and e_2 arbitrarily, e_3 follows from (307) and the values of h_1, h_2, h_3 are then found by relations of the form (308). Between the quantities with accent e_1', e_2', e_3', h_1', h_2', h_3' there exist exactly the same relations as between those without accent, of which they are entirely independent. Hence two also of them, say h_1' and h_2', may be chosen arbitrarily so that in the equations given above for given values of a, b, c four constants remain undetermined. If we now form, for all values of a b c whatever, expressions of the type (305) and add the corresponding field components, we again obtain a solution for *Maxwell's* equations of the field and the boundary conditions, which, however, is now so general that it is capable of representing any electromagnetic process possible in the hollow cube considered. For it is always possible to dispose of the constants e_1, e_2, h_1', h_2' which have remained undetermined in the separate particular solutions in such a way that the process may be adapted to any initial state ($t = 0$) whatever.

If now, as we have assumed so far, the cavity is entirely void of matter, the process of radiation with a given initial state is uniquely determined in all its details. It consists of a set of stationary vibrations, every one of which is represented by one of the particular solutions considered, and which take place entirely independent of one another. Hence in this case there can be no question of irreversibility and hence also none of any tendency to equality of the partial energies corresponding to the separate partial vibrations. As soon, however, as we assume the presence in the cavity of only the slightest trace of matter which can influence the electrodynamic vibrations, *e.g.*, a few gas molecules, which emit or absorb radiation, the process becomes chaotic and a passage

from less to more probable states will take place, though perhaps slowly. Without considering any further details of the electromagnetic constitution of the molecules, we may from the law of statistical mechanics quoted above draw the conclusion that, among all possible processes, that one in which the energy is distributed uniformly among all the independent variables of the state has the stationary character.

From this let us determine these independent variables. In the first place there are the velocity components of the gas molecules. In the stationary state to every one of the three mutually independent velocity components of a molecule there corresponds on the average the energy $\frac{1}{3}\overline{L}$ where $\overline{L}$ represents the mean energy of a molecule and is given by (200). Hence the partial energy, which on the average corresponds to any one of the independent variables of the electromagnetic system, is just as large.

Now, according to the above discussion, the electromagnetic state of the whole cavity for every stationary vibration corresponding to any one system of values of the numbers a b c is determined, at any instant, by four mutually independent quantities. Hence for the radiation processes the number of independent variables of state is four times as large as the number of the possible systems of values of the positive integers a, b, c.

We shall now calculate the number of the possible systems of values a, b, c, which correspond to the vibrations within a certain small range of the spectrum, say between the frequencies ν and $\nu + d\nu$. According to (306), these systems of values satisfy the inequalities

$$\left(\frac{2l\nu}{c}\right)^2 < \mathsf{a}^2 + \mathsf{b}^2 + \mathsf{c}^2 < \left(\frac{2l(\nu + d\nu)}{c}\right)^2, \tag{309}$$

where not only $\dfrac{2l\nu}{c}$ but also $\dfrac{2l\,d\nu}{c}$ is to be thought of as a large number.

If we now represent every system of values of a, b, c graphically by a point, taking a, b, c as coordinates in an orthogonal coordinate system, the points thus obtained occupy one octant of the space of infinite extent, and condition (309) is equivalent to requiring that the distance of any one of these points from the origin of the coordinates shall lie between $\frac{2l\nu}{c}$ and $\frac{2l(\nu+d\nu)}{c}$. Hence the required number is equal to the number of points which lie between the two spherical surface-octants corresponding to the radii $\frac{2l\nu}{c}$ and $\frac{2l(\nu+d\nu)}{c}$. Now since to every point there corresponds a cube of volume 1 and *vice versa*, that number is simply equal to the space between the two spheres mentioned, and hence equal to

$$\frac{1}{8}\,4\pi\left(\frac{2l\nu}{c}\right)^2\frac{2l\,d\nu}{c},$$

and the number of the independent variables of state is four times as large or

$$\frac{16\pi l^3\nu^2\,d\nu}{c^3}.$$

Since, moreover, the partial energy $\frac{\overline{L}}{3}$ corresponds on the average to every independent variable of state in the state of equilibrium, the total energy falling in the interval from ν to $\nu+d\nu$ becomes

$$\frac{16\pi l^3\nu^2\,d\nu}{3c^3}\overline{L}.$$

Since the volume of the cavity is l^3, this gives for the space density of the energy of frequency ν

$$\mathsf{u}\,d\nu=\frac{16\pi\nu^2\,d\nu}{3c^3}\overline{L},$$

and, by substitution of the value of $\overline{L} = \frac{L}{N}$ from (200),

$$\mathsf{u} = \frac{8\pi\nu^2 kT}{c^3}, \tag{310}$$

which is in perfect agreement with *Rayleigh's* formula (285).

If the law of the equipartition of energy held true in all cases, *Rayleigh's* law of radiation would, in consequence, hold for all wave lengths and temperatures. But since this possibility is excluded by the measurements at hand, the only possible conclusion is that the law of the equipartition of energy and, with it, the system of *Hamilton's* equations of motion does not possess the general importance attributed to it in classical dynamics. Therein lies the strongest proof of the necessity of a fundamental modification of the latter.

PART V

IRREVERSIBLE RADIATION PROCESSES

CHAPTER I

FIELDS OF RADIATION IN GENERAL

170. According to the theory developed in the preceding section, the nature of heat radiation within an isotropic medium, when the state is one of stable thermodynamic equilibrium, may be regarded as known in every respect. The intensity of the radiation, uniform in all directions, depends for all wave lengths only on the temperature and the velocity of propagation, according to equation (300), which applies to black radiation in any medium whatever. But there remains another problem to be solved by the theory. It is still necessary to explain how and by what processes the radiation which is originally present in the medium and which may be assigned in any way whatever, passes gradually, when the medium is bounded by walls impermeable to heat, into the stable state of black radiation, corresponding to the maximum of entropy, just as a gas which is enclosed in a rigid vessel and in which there are originally currents and temperature differences assigned in any way whatever gradually passes into the state of rest and of uniform distribution of temperature.

To this much more difficult question only a partial answer can, at present, be given. In the first place, it is evident from the extensive discussion in the first chapter of the third part that, since irreversible processes are to be dealt with, the principles of pure electrodynamics alone will not suffice. For the second principle of thermodynamics or the principle of increase of entropy is foreign to the contents of pure electrodynamics as well as of pure mechanics. This is most immediately shown by the fact that the fundamental equations of mechanics as well as those of electrodynamics allow the direct reversal of every process as regards time, which contradicts the principle of increase of entropy. Of

course all kinds of friction and of electric conduction of currents must be assumed to be excluded; for these processes, since they are always connected with the production of heat, do not belong to mechanics or electrodynamics proper.

This assumption being made, the time t occurs in the fundamental equations of mechanics only in the components of acceleration; that is, in the form of the square of its differential. Hence, if instead of t the quantity $-t$ is introduced as time variable in the equations of motion, they retain their form without change, and hence it follows that if in any motion of a system of material points whatever the velocity components of all points are suddenly reversed at any instant, the process must take place in the reverse direction. For the electrodynamic processes in a homogeneous non-conducting medium a similar statement holds. If in *Maxwell's* equations of the electrodynamic field $-t$ is written everywhere instead of t, and if, moreover, the sign of the magnetic field-strength H is reversed, the equations remain unchanged, as can be readily seen, and hence it follows that if in any electrodynamic process whatever the magnetic field-strength is everywhere suddenly reversed at a certain instant, while the electric field-strength keeps its value, the whole process must take place in the opposite sense.

If we now consider any radiation processes whatever, taking place in a perfect vacuum enclosed by reflecting walls, it is found that, since they are completely determined by the principles of classical electrodynamics, there can be in their case no question of irreversibility of any kind. This is seen most clearly by considering the perfectly general formulæ (305), which hold for a cubical cavity and which evidently have a periodic, *i.e.*, reversible character. Accordingly we have frequently (Sec. 144 and 166) pointed out that the simple propagation of free radiation represents a reversible process. An irreversible element is introduced by the addition of emitting and absorbing substance.

171. Let us now try to define for the general case the state of radiation in the thermodynamic-macroscopic sense as we did above in Sec. 107, *et seq.*, for a stationary radiation. Every one of the three components of the electric field-strength, *e.g.*, E_z may, for the long time interval from $t = 0$ to $t = \mathsf{T}$, be represented at every point, *e.g.*, at the origin of coordinates, by a *Fourier's* integral, which in the present case is somewhat more convenient than the *Fourier's* series (149):

$$\mathsf{E}_z = \int_0^\infty d\nu\, C_\nu \cos(2\pi\nu t - \theta_\nu), \tag{311}$$

where C_ν (positive) and θ_ν denote certain functions of the positive variable of integration ν. The values of these functions are not wholly determined by the behavior of E_z in the time interval mentioned, but depend also on the manner in which E_z varies as a function of the time beyond both ends of that interval. Hence the quantities C_ν and θ_ν possess separately no definite physical significance, and it would be quite incorrect to think of the vibration E_z as, say, a continuous spectrum of periodic vibrations with the constant amplitudes C_ν. This may, by the way, be seen at once from the fact that the character of the vibration E_z may vary with the time in any way whatever. How the spectral resolution of the vibration E_z is to be performed and to what results it leads will be shown below (Sec. 174).

172. We shall, as heretofore (158), define J, the "intensity of the exciting vibration,"[1] as a function of the time to be the mean value of E_z^2 in the time interval from t to $t + \tau$, where τ is taken as large compared with the time $\frac{1}{\nu}$, which is the duration of one of the periodic partial vibrations contained in the radiation, but as small as possible

[1]Not to be confused with the "field intensity" (field-strength) E_z of the exciting vibration.

compared with the time T. In this statement there is a certain indefiniteness, from which results the fact that J will, in general, depend not only on t but also on τ. If this is the case one cannot speak of the intensity of the exciting vibration at all. For it is an essential feature of the conception of the intensity of a vibration that its value should change but unappreciably within the time required for a single vibration. (Compare above, Sec. 3.) Hence we shall consider in future only those processes for which, under the conditions mentioned, there exists a mean value of E_z^2 depending only on t. We are then obliged to assume that the quantities C_ν in (311) are negligible for all values of ν which are of the same order of magnitude as $\frac{1}{\tau}$ or smaller, *i.e.*,

$$\nu\tau \text{ is large.} \tag{312}$$

In order to calculate J we now form from (311) the value of E_z^2 and determine the mean value $\overline{\mathsf{E}_z^2}$ of this quantity by integrating with respect to t from t to $t+\tau$, then dividing by τ and passing to the limit by decreasing τ sufficiently. Thus we get

$$\mathsf{E}_z^2 = \int_0^\infty \int_0^\infty d\nu'\, d\nu\, C_{\nu'} C_\nu \cos(2\pi\nu' t - \theta_{\nu'}) \cos(2\pi\nu t - \theta_\nu).$$

If we now exchange the values of ν and ν', the function under the sign of integration does not change; hence we assume

$$\nu' > \nu$$

and write:

$$\mathsf{E}_z^2 = 2 \iint d\nu'\, d\nu\, C_{\nu'} C_\nu \cos(2\pi\nu' t - \theta_{\nu'}) \cos(2\pi\nu t - \theta_\nu),$$

or

$$\mathsf{E}_z^2 = \iint d\nu'\, d\nu\, C_{\nu'} C_\nu \{\cos[2\pi(\nu'-\nu)t - \theta_{\nu'} + \theta_\nu] + \cos[2\pi(\nu'+\nu)t - \theta_{\nu'} - \theta_\nu]\}.$$

And hence

$$J = \overline{\mathsf{E}_z^2} = \frac{1}{\tau}\int_t^{t+\tau} \mathsf{E}_z^2\, dt$$
$$= \iint d\nu'\, d\nu\, C_{\nu'} C_\nu \left\{ \frac{\sin \pi(\nu'-\nu)\tau \cdot \cos[\pi(\nu'-\nu)(2t+\tau) - \theta_{\nu'} + \theta_\nu]}{\pi(\nu'-\nu)\tau} + \frac{\sin \pi(\nu'+\nu)\tau \cdot \cos[\pi(\nu'+\nu)(2t+\tau) - \theta_{\nu'} - \theta_\nu]}{\pi(\nu'+\nu)\tau} \right\}.$$

If we now let τ become smaller and smaller, since $\nu\tau$ remains large, the denominator $(\nu'+\nu)\tau$ of the second fraction remains large under all circumstances, while that of the first fraction $(\nu'-\nu)\tau$ may decrease with decreasing value of τ to less than any finite value. Hence for sufficiently small values of $(\nu'-\nu)$ the integral reduces to

$$\iint d\nu'\, d\nu\, C_{\nu'} C_\nu \cos[2\pi(\nu'-\nu)t - \theta_{\nu'} + \theta_\nu]$$

which is in fact independent of τ. The remaining terms of the double integral, which correspond to larger values of $\nu'-\nu$, *i.e.*, to more rapid changes with the time, depend in general on τ and therefore must vanish, if the intensity J is not to depend on τ. Hence in our case on introducing as a second variable of integration instead of ν

$$\mu = \nu' - \nu \ (>0)$$

we have

$$J = \iint d\mu\, d\nu\, C_{\nu+\mu} C_\nu \cos(2\pi\mu t - \theta_{\nu+\mu} + \theta_\nu) \tag{313}$$

or

$$J = \int d\mu\, (A_\mu \cos 2\pi\mu t + B_\mu \sin 2\pi\mu t)$$

where

$$A_\mu = \int d\nu\, C_{\nu+\mu} C_\nu \cos(\theta_{\nu+\mu} - \theta_\nu) \tag{314}$$

$$B_\mu = \int d\nu\, C_{\nu+\mu} C_\nu \sin(\theta_{\nu+\mu} - \theta_\nu).$$

By this expression the intensity J of the exciting vibration, if it exists at all, is expressed by a function of the time in the form of a *Fourier's* integral.

173. The conception of the intensity of vibration J necessarily contains the assumption that this quantity varies much more slowly with the time t than the vibration E_z itself. The same follows from the calculation of J in the preceding paragraph. For there, according to (312), $\nu\tau$ and $\nu'\tau$ are large, but $(\nu' - \nu)\tau$ is small for all pairs of values C_ν and $C_{\nu'}$ that come into consideration; hence, *a fortiori*,

$$\frac{\nu' - \nu}{\nu} = \frac{\mu}{\nu} \quad \text{is small,} \tag{315}$$

and accordingly the *Fourier's* integrals E_z in (311) and J in (314) vary with the time in entirely different ways. Hence in the following we shall have to distinguish, as regards dependence on time, two kinds of quantities, which vary in different ways: Rapidly varying quantities, as E_z, and slowly varying quantities as J and I the spectral intensity of the exciting vibration, whose value we shall calculate in the next paragraph. Nevertheless this difference in the variability with respect

to time of the quantities named is only relative, since the absolute value of the differential coefficient of J with respect to time depends on the value of the unit of time and may, by a suitable choice of this unit, be made as large as we please. It is, therefore, not proper to speak of $J(t)$ simply as a slowly varying function of t. If, in the following, we nevertheless employ this mode of expression for the sake of brevity, it will always be in the relative sense, namely, with respect to the different behavior of the function $\mathsf{E}_z(t)$.

On the other hand, as regards the dependence of the phase constant θ_ν on its index ν it necessarily possesses the property of rapid variability in the *absolute* sense. For, although μ is small compared with ν, nevertheless the difference $\theta_{\nu+\mu} - \theta_\nu$ is in general not small, for if it were, the quantities A_μ and B_μ in (314) would have too special values and hence it follows that $\dfrac{\partial\theta_\nu}{\partial\nu} \cdot \nu$ must be large. This is not essentially modified by changing the unit of time or by shifting the origin of time.

Hence the rapid variability of the quantities θ_ν and also C_ν with ν is, in the absolute sense, a necessary condition for the existence of a definite intensity of vibration J, or, in other words, for the possibility of dividing the quantities depending on the time into those which vary rapidly and those which vary slowly—a distinction which is also made in other physical theories and upon which all the following investigations are based.

174. The distinction between rapidly variable and slowly variable quantities introduced in the preceding section has, at the present stage, an important physical aspect, because in the following we shall assume that only slow variability with time is capable of direct measurement. On this assumption we approach conditions as they actually exist in optics and heat radiation. Our problem will then be to establish re-

lations between slowly variable quantities exclusively; for these only can be compared with the results of experience. Hence we shall now determine the most important one of the slowly variable quantities to be considered here, namely, the "spectral intensity" I of the exciting vibration. This is effected as in (158) by means of the equation

$$J = \int_0^\infty \mathsf{I}\, d\nu.$$

By comparison with (313) we obtain:

$$\left\{\begin{array}{ll} \mathsf{I} = \int d\mu\, (\mathsf{A}_\mu \cos 2\pi\mu t + \mathsf{B}_\mu \sin 2\pi\mu t) & \\ \text{where} \qquad \mathsf{A}_\mu = \overline{C_{\nu+\mu} C_\nu \cos(\theta_{\nu+\mu} - \theta_\nu)} & \\ \qquad\qquad\quad \mathsf{B}_\mu = \overline{C_{\nu+\mu} C_\nu \sin(\theta_{\nu+\mu} - \theta_\nu)}. & \end{array}\right. \tag{316}$$

By this expression the spectral intensity, I, of the exciting vibration at a point in the spectrum is expressed as a slowly variable function of the time t in the form of a *Fourier's* integral. The dashes over the expressions on the right side denote the mean values extended over a narrow spectral range for a given value of μ. If such mean values do not exist, there is no definite spectral intensity.

CHAPTER II

ONE OSCILLATOR IN THE FIELD OF RADIATION

175. If in any field of radiation whatever we have an ideal oscillator of the kind assumed above (Sec. 135), there will take place between it and the radiation falling on it certain mutual actions, for which we shall again assume the validity of the elementary dynamical law introduced in the preceding section. The question is then, how the processes of emission and absorption will take place in the case now under consideration.

In the first place, as regards the emission of radiant energy by the oscillator, this takes place, as before, according to the hypothesis of emission of quanta (Sec. 147), where the probability quantity η again depends on the corresponding spectral intensity I through the relation (265).

On the other hand, the absorption is calculated, exactly as above, from (234), where the vibrations of the oscillator also take place according to the equation (233). In this way, by calculations analogous to those performed in the second chapter of the preceding part, with the difference only that instead of the Fourier's series (235) the Fourier's integral (311) is used, we obtain for the energy absorbed by the oscillator in the time τ the expression

$$\frac{\tau}{4L}\int d\mu\,(\mathsf{A}_\mu \cos 2\pi\mu t + \mathsf{B}_\mu \sin 2\pi\mu t),$$

where the constants A_μ and B_μ denote the mean values expressed in (316), taken for the spectral region in the neighborhood of the natural frequency ν_0 of the oscillator. Hence the law of absorption will again be given by equation (249), which now holds also for an intensity of vibration I varying with the time.

176. There now remains the problem of deriving the expression for I, the spectral intensity of the vibration exciting the oscillator, when the thermodynamic state of the field of radiation at the oscillator is given in accordance with the statements made in Sec. 17.

Let us first calculate the total intensity $J = \overline{\mathsf{E}_z^2}$ of the vibration exciting an oscillator, from the intensities of the heat rays striking the oscillator from all directions.

For this purpose we must also allow for the polarization of the monochromatic rays which strike the oscillator. Let us begin by considering a pencil which strikes the oscillator within a conical element whose vertex lies in the oscillator and whose solid angle, $d\Omega$, is given by (5), where the angles θ and ϕ, polar coordinates, designate the direction of the propagation of the rays. The whole pencil consists of a set of monochromatic pencils, one of which may have the principal values of intensity K and K' (Sec. 17). If we now denote the angle which the plane of vibration belonging to the principal intensity K makes with the plane through the direction of the ray and the z-axis (the axis of the oscillator) by ψ, no matter in which quadrant it lies, then, according to (8), the specific intensity of the monochromatic pencil may be resolved into the two plane polarized components at right angles with each other,

$$\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi$$
$$\mathsf{K}\sin^2\psi + \mathsf{K}'\cos^2\psi,$$

the first of which vibrates in a plane passing through the z-axis and the second in a plane perpendicular thereto.

The latter component does not contribute anything to the value of E_z^2, since its electric field-strength is perpendicular to the axis of the oscillator. Hence there remains only the first component whose electric

field-strength makes the angle $\frac{\pi}{2} - \theta$ with the z-axis. Now according to *Poynting's* law the intensity of a plane polarized ray in a vacuum is equal to the product of $\frac{c}{4\pi}$ and the mean square of the electric field-strength. Hence the mean square of the electric field-strength of the pencil here considered is

$$\frac{4\pi}{c}(\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\,d\nu\,d\Omega,$$

and the mean square of its component in the direction of the z-axis is

$$\frac{4\pi}{c}(\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\sin^2\theta\,d\nu\,d\Omega. \tag{317}$$

By integration over all frequencies and all solid angles we then obtain the value required

$$\overline{\mathsf{E}_z^2} = \frac{4\pi}{c}\int \sin^2\theta\,d\Omega \int d\nu\,(\mathsf{K}_\nu\cos^2\psi + \mathsf{K}'_\nu\sin^2\psi) = J. \tag{318}$$

The space density u of the electromagnetic energy at a point of the field is

$$u = \frac{1}{8\pi}(\overline{\mathsf{E}_x^2} + \overline{\mathsf{E}_y^2} + \overline{\mathsf{E}_z^2} + \overline{\mathsf{H}_x^2} + \overline{\mathsf{H}_y^2} + \overline{\mathsf{H}_z^2}),$$

where E_x^2, E_y^2, E_z^2, H_x^2, H_y^2, H_z^2 denote the squares of the field-strengths, regarded as "slowly variable" quantities, and are hence supplied with the dash to denote their mean value. Since for every separate ray the mean electric and magnetic energies are equal, we may always write

$$u = \frac{1}{4\pi} = (\overline{\mathsf{E}_x^2} + \overline{\mathsf{E}_y^2} + \overline{\mathsf{E}_z^2}). \tag{319}$$

If, in particular, all rays are unpolarized and if the intensity of radiation is constant in all directions, $\mathsf{K}_\nu = \mathsf{K}'_\nu$ and, since

$$\int \sin^2\theta\, d\Omega = \iint \sin^3\theta\, d\theta\, d\phi = \frac{8\pi}{3}$$
$$\overline{\mathsf{E}_z^2} = \frac{32\pi^2}{3c}\int \mathsf{K}_\nu\, d\nu = \overline{\mathsf{E}_x^2} = \overline{\mathsf{E}_y^2} \tag{319a}$$

and, by substitution in (319),

$$u = \frac{8\pi}{c}\int \mathsf{K}_\nu\, d\nu,$$

which agrees with (22) and (24).

177. Let us perform the spectral resolution of the intensity J according to Sec. 174; namely,

$$J = \int \mathsf{I}_\nu\, d\nu.$$

Then, by comparison with (318), we find for the intensity of a definite frequency ν contained in the exciting vibration the value

$$\mathsf{I} = \frac{4\pi}{c}\int \sin^2\theta\, d\Omega\, (\mathsf{K}_\nu \cos^2\psi + \mathsf{K}'_\nu \sin^2\psi). \tag{320}$$

For radiation which is unpolarized and uniform in all directions we obtain again, in agreement with (160),

$$\mathsf{I} = \frac{32\pi^2}{3c}\mathsf{K}.$$

178. With the value (320) obtained for I the total energy absorbed by the oscillator in an element of time dt from the radiation falling on

it is found from (249) to be

$$\frac{\pi\, dt}{cL} \int \sin^2\theta\, d\Omega\, (\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi).$$

Hence the oscillator absorbs in the time dt from the pencil striking it within the conical element $d\Omega$ an amount of energy equal to

$$\frac{\pi\, dt}{cL} \sin^2\theta(\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\, d\Omega. \tag{321}$$

CHAPTER III

A SYSTEM OF OSCILLATORS

179. Let us suppose that a large number N of similar oscillators with parallel axes, acting quite independently of one another, are distributed irregularly in a volume-element of the field of radiation, the dimensions of which are so small that within it the intensities of radiation K do not vary appreciably. We shall investigate the mutual action between the oscillators and the radiation which is propagated freely in space.

As before, the state of the field of radiation may be given by the magnitude and the azimuth of vibration ψ of the principal intensities K_ν and K'_ν of the pencils which strike the system of oscillators, where K_ν and K'_ν depend in an arbitrary way on the direction angles θ and ϕ. On the other hand, let the state of the system of oscillators be given by the densities of distribution w_1, w_2, w_3, ... (166), with which the oscillators are distributed among the different region elements, w_1, w_2, w_3, ... being any proper fractions whose sum is 1. Herein, as always, the nth region element is supposed to contain the oscillators with energies between $(n-1)h\nu$ and $nh\nu$.

The energy absorbed by the system in the time dt within the conical element $d\Omega$ is, according to (321),

$$\frac{\pi N\, dt}{cL} \sin^2\theta(\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\, d\Omega. \tag{322}$$

Let us now calculate also the energy emitted within the same conical element.

180. The total energy emitted in the time element dt by all N oscillators is found from the consideration that a single oscillator, according

to (249), takes up an energy element $h\nu$ during the time

$$\frac{4h\nu L}{\mathsf{I}} = \tau, \tag{323}$$

and hence has a chance to emit once, the probability being η. We shall assume that the intensity I of the exciting vibration does not change appreciably in the time τ. Of the Nw_n oscillators which at the time t are in the nth region element a number $Nw_n\eta$ will emit during the time τ, the energy emitted by each being $nh\nu$. From (323) we see that the energy emitted by all oscillators during the time element dt is

$$\sum Nw_n\,\eta\,nh\nu\frac{dt}{\tau} = \frac{N\eta\mathsf{I}\,dt}{4L}\sum nw_n,$$

or, according to (265),

$$\frac{N(1-\eta)\,dt}{4pL}\sum nw_n. \tag{324}$$

From this the energy emitted within the conical element $d\Omega$ may be calculated by considering that, in the state of thermodynamic equilibrium, the energy emitted in every conical element is equal to the energy absorbed and that, in the general case, the energy emitted in a certain direction is independent of the energy simultaneously absorbed. For the stationary state we have from (160) and (265)

$$\mathsf{K} = \mathsf{K}' = \frac{3c}{32\pi^2}\mathsf{I} = \frac{3c}{32\pi^2}\,\frac{1-\eta}{p\eta} \tag{325}$$

and further from (271) and (265)

$$w_n = \frac{1}{p\mathsf{I}}\left(\frac{p\mathsf{I}}{1+p\mathsf{I}}\right)^n = \eta(1-\eta)^{n-1}, \tag{326}$$

and hence

$$\sum nw_n = \eta \sum n(1-\eta)^{n-1} = \frac{1}{\eta}. \tag{327}$$

Thus the energy emitted (324) becomes

$$\frac{N(1-\eta)\,dt}{4Lp\eta}. \tag{328}$$

This is, in fact, equal to the total energy absorbed, as may be found by integrating the expression (322) over all conical elements $d\Omega$ and taking account of (325).

Within the conical element $d\Omega$ the energy emitted or absorbed will then be

$$\frac{\pi N\,dt}{c} \sin^2\theta\, \mathsf{K}\, d\Omega,$$

or, from (325), (327) and (268),

$$\frac{\pi h\nu^3(1-\eta)N}{c^3 L} \sum nw_n \sin^2\theta\, d\Omega\, dt, \tag{329}$$

and this is the general expression for the energy emitted by the system of oscillators in the time element dt within the conical element $d\Omega$, as is seen by comparison with (324).

181. Let us now, as a preparation for the following deductions, consider more closely the properties of the different pencils passing the system of oscillators. From all directions rays strike the volume-element that contains the oscillators; if we again consider those which come toward it in the direction (θ, ϕ) within the conical element $d\Omega$, the vertex of which lies in the volume-element, we may in the first place think of them as being resolved into their monochromatic constituents, and then we need consider further only that one of these constituents

which corresponds to the frequency ν of the oscillators; for all other rays simply pass the oscillators without influencing them or being influenced by them. The specific intensity of a monochromatic ray of frequency ν is

$$\mathsf{K} + \mathsf{K}'$$

where K and K' represent the principal intensities which we assume as non-coherent. This ray is now resolved into two components according to the directions of its principal planes of vibration (Sec. 176).

The first component,

$$\mathsf{K} \sin^2 \psi + \mathsf{K}' \cos^2 \psi,$$

passes by the oscillators and emerges on the other side with no change whatever. Hence it gives a plane polarized ray, which starts from the system of oscillators in the direction (θ, ϕ) within the solid angle $d\Omega$ and whose vibrations are perpendicular to the axis of the oscillators and whose intensity is

$$\mathsf{K} \sin^2 \psi + \mathsf{K}' \cos^2 \psi = K''. \tag{330}$$

The second component,

$$\mathsf{K} \cos^2 \psi + \mathsf{K}' \sin^2 \psi,$$

polarized at right angles to the first consists again, according to Sec. 176, of two parts

$$(\mathsf{K} \cos^2 \psi + \mathsf{K}' \sin^2 \psi) \cos^2 \theta \tag{331}$$

and

$$(\mathsf{K} \cos^2 \psi + \mathsf{K}' \sin^2 \psi) \sin^2 \theta, \tag{332}$$

of which the first passes by the system without any change, since its direction of vibration is at right angles to the axes of the oscillators,

while the second is weakened by absorption, say by the small fraction β. Hence on emergence this component has only the intensity

$$(1-\beta)(\mathsf{K}\cos^2\psi + K'\sin^2\psi)\sin^2\theta. \tag{333}$$

It is, however, strengthened by the radiation emitted by the system of oscillators (329), which has the value

$$\beta'(1-\eta)\sum nw_n\sin^2\theta, \tag{334}$$

where β' denotes a certain other constant, which depends only on the nature of the system and whose value is obtained at once from the condition that, in the state of thermodynamic equilibrium, the loss is just compensated by the gain.

For this purpose we make use of the relations (325) and (327) corresponding to the stationary state, and thus find that the sum of the expressions (333) and (334) becomes just equal to (332); and thus for the constant β' the following value is found:

$$\beta' = \beta\frac{3c}{32\pi^2 p} = \beta\frac{h\nu^3}{c^2}.$$

Then by addition of (331), (333) and (334) the total specific intensity of the radiation which emanates from the system of oscillators within the conical element $d\Omega$, and whose plane of vibration is parallel to the axes of the oscillators, is found to be

$$\mathsf{K}''' = \mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi + \beta\sin^2\theta\big(\mathsf{K}_e - (\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\big) \tag{335}$$

where for the sake of brevity the term referring to the emission is written

$$\frac{h\nu^3}{c^2}(1-\eta)\sum nw_n = \mathsf{K}_e. \tag{336}$$

Thus we finally have a ray starting from the system of oscillators in the direction (θ, ϕ) within the conical element $d\Omega$ and consisting of two components K'' and K''' polarized perpendicularly to each other, the first component vibrating at right angles to the axes of the oscillators.

In the state of thermodynamic equilibrium

$$\mathsf{K} = \mathsf{K}' = \mathsf{K}'' = \mathsf{K}''' = \mathsf{K}_e,$$

a result which follows in several ways from the last equations.

182. The constant β introduced above, a small positive number, is determined by the spacial and spectral limits of the radiation influenced by the system of oscillators. If q denotes the cross-section at right angles to the direction of the ray, $\Delta\nu$ the spectral width of the pencil cut out of the total incident radiation by the system, the energy which is capable of absorption and which is brought to the system of oscillators within the conical element $d\Omega$ in the time dt is, according to (332) and (11),

$$q\,\Delta\nu\,(\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\sin^2\theta\,d\Omega\,dt. \tag{337}$$

Hence the energy actually absorbed is the fraction β of this value. Comparing this with (322) we get

$$\beta = \frac{\pi N}{q \cdot \Delta\nu \cdot cL}. \tag{338}$$

CHAPTER IV

CONSERVATION OF ENERGY AND INCREASE OF ENTROPY. CONCLUSION

183. It is now easy to state the relation of the two principles of thermodynamics to the irreversible processes here considered. Let us consider first the *conservation of energy*. If there is no oscillator in the field, every one of the elementary pencils, infinite in number, retains, during its rectilinear propagation, both its specific intensity K and its energy without change, even though it be reflected at the surface, assumed as plane and reflecting, which bounds the field (Sec. 166). The system of oscillators, on the other hand, produces a change in the incident pencils and hence also a change in the energy of the radiation propagated in the field. To calculate this we need consider only those monochromatic rays which lie close to the natural frequency ν of the oscillators, since the rest are not altered at all by the system.

The system is struck in the direction (θ, ϕ) within the conical element $d\Omega$ which converges toward the system of oscillators by a pencil polarized in some arbitrary way, the intensity of which is given by the sum of the two principal intensities K and K'. This pencil, according to Sec. 182, conveys the energy

$$q\,\Delta\nu\,(\mathsf{K}+\mathsf{K}')\,d\Omega\,dt$$

to the system in the time dt; hence this energy is taken from the field of radiation on the side of the rays arriving within $d\Omega$. As a compensation there emerges from the system on the other side in the same direction (θ, ϕ) a pencil polarized in some definite way, the intensity of which is given by the sum of the two components K'' and K'''. By it an amount

of energy

$$q\,\Delta\nu\,(\mathsf{K}'' + \mathsf{K}''')\,d\Omega\,dt,$$

is added to the field of radiation. Hence, all told, the change in energy of the field of radiation in the time dt is obtained by subtracting the first expression from the second and by integrating with respect to $d\Omega$. Thus we get

$$dt\,\Delta\nu \int (\mathsf{K}'' + \mathsf{K}''' - \mathsf{K} - \mathsf{K}')q\,d\Omega,$$

or by taking account of (330), (335), and (338)

$$\frac{\pi N\,dt}{cL} \int d\Omega \sin^2\theta\big(\mathsf{K}_e - (\mathsf{K}\cos^2\psi + \mathsf{K}'\sin^2\psi)\big). \tag{339}$$

184. Let us now calculate the change in energy of the system of oscillators which has taken place in the same time dt. According to (219), this energy at the time t is

$$E = Nh\nu \sum_1^\infty (n - \tfrac{1}{2})w_n,$$

where the quantities w_n whose total sum is equal to 1 represent the densities of distribution characteristic of the state. Hence the energy change in the time dt is

$$dE = Nh\nu \sum_1^\infty (n - \tfrac{1}{2})\,dw_n = Nh\nu \sum_1^\infty n\,dw_n. \tag{340}$$

To calculate dw_n we consider the nth region element. All of the oscillators which lie in this region at the time t have, after the lapse of time τ, given by (323), left this region; they have either passed into the

$(n+1)$st region, or they have performed an emission at the boundary of the two regions. In compensation there have entered $(1-\eta)Nw_{n-1}$ oscillators during the time τ, that is, all oscillators which, at the time t, were in the $(n-1)$st region element, excepting such as have lost their energy by emission. Thus we obtain for the required change in the time dt

$$N\,dw_n = \frac{dt}{\tau}N\big((1-\eta)w_{n-1} - w_n\big). \tag{341}$$

A separate discussion is required for the first region element $n=1$. For into this region there enter in the time τ all those oscillators which have performed an emission in this time. Their number is

$$\eta(w_1 + w_2 + w_3 + \dots)N = \eta N.$$

Hence we have

$$N\,dw_1 = \frac{dt}{\tau}N(\eta - w_1).$$

We may include this equation in the general one (341) if we introduce as a new expression

$$w_0 = \frac{\eta}{1-\eta}. \tag{342}$$

Then (341) gives, substituting τ from (323),

$$dw_n = \frac{\mathsf{I}\,dt}{4h\nu L}\big((1-\eta)w_{n-1} - w_n\big), \tag{343}$$

and the energy change (340) of the system of oscillators becomes

$$dE = \frac{N\mathsf{I}\,dt}{4L}\sum_1^\infty n\big((1-\eta)w_{n-1} - w_n\big).$$

The sum $\sum$ may be simplified by recalling that

$$\sum_1^\infty n w_{n-1} = \sum_1^\infty (n-1) w_{n-1} + \sum_1^\infty w_{n-1}$$
$$= \sum_1^\infty n w_n + w_0 + 1 = \sum_1^\infty n w_n + \frac{1}{1-\eta}.$$

Then we have

$$dE = \frac{N \mathsf{l}\, dt}{4L}(1 - \eta \sum_1^\infty n w_n). \tag{344}$$

This expression may be obtained more readily by considering that dE is the difference of the total energy absorbed and the total energy emitted. The former is found from (250), the latter from (324), by taking account of (265).

The principle of the conservation of energy demands that the sum of the energy change (339) of the field of radiation and the energy change (344) of the system of oscillators shall be zero, which, in fact, is quite generally the case, as is seen from the relations (320) and (336).

185. We now turn to the discussion of the second principle, the principle of the *increase of entropy*, and follow closely the above discussion regarding the energy. When there is no oscillator in the field, every one of the elementary pencils, infinite in number, retains during rectilinear propagation both its specific intensity and its entropy without change, even when reflected at the surface, assumed as plane and reflecting, which bounds the field. The system of oscillators, however, produces a change in the incident pencils and hence also a change in the entropy of the radiation propagated in the field. For the calculation of this change we need to investigate only those monochromatic rays

which lie close to the natural frequency ν of the oscillators, since the rest are not altered at all by the system.

The system of oscillators is struck in the direction (θ, ϕ) within the conical element $d\Omega$ converging toward the system by a pencil polarized in some arbitrary way, the spectral intensity of which is given by the sum of the two principal intensities K and K' with the azimuth of vibration ψ and $\frac{\pi}{2} + \psi$ respectively, which are assumed to be non-coherent. According to (141) and Sec. 182 this pencil conveys the entropy

$$q\,\Delta\nu\,[\mathsf{L}(\mathsf{K}) + \mathsf{L}(\mathsf{K}')]\,d\Omega\,dt \tag{345}$$

to the system of oscillators in the time dt, where the function $\mathsf{L}(\mathsf{K})$ is given by (278). Hence this amount of entropy is taken from the field of radiation on the side of the rays arriving within $d\Omega$. In compensation a pencil starts from the system on the other side in the same direction (θ, ϕ) within $d\Omega$ having the components K'' and K''' with the azimuth of vibration $\frac{\pi}{2}$ and 0 respectively, but its entropy radiation is not represented by $\mathsf{L}(\mathsf{K}'') + \mathsf{L}(\mathsf{K}''')$, since K'' and K''' are not non-coherent, but by

$$\mathsf{L}(\mathsf{K}_0) + \mathsf{L}(\mathsf{K}'_0) \tag{346}$$

where K_0 and K'_0 represent the principal intensities of the pencil.

For the calculation of K_0 and K'_0 we make use of the fact that, according to (330) and (335), the radiation K'' and K''', of which the component K''' vibrates in the azimuth 0, consists of the following three components, non-coherent with one another:

$$\mathsf{K}_1 = \mathsf{K}\sin^2\psi + \mathsf{K}\cos^2\psi(1 - \beta\sin^2\theta) = \mathsf{K}(1 - \beta\sin^2\theta\cos^2\psi)$$

with the azimuth of vibration $\operatorname{tg}^2\psi_1 = \dfrac{\operatorname{tg}^2\psi}{1 - \beta\sin^2\theta}$,

$$\mathsf{K}_2 = \mathsf{K}'\cos^2\psi + \mathsf{K}'\sin^2\psi(1 - \beta\sin^2\theta) = \mathsf{K}'(1 - \beta\sin^2\theta\sin^2\psi)$$

with the azimuth of vibration $\operatorname{tg}^2 \psi_2 = \dfrac{\cot^2 \psi}{1 - \beta \sin^2 \theta}$, and,

$$\mathsf{K}_3 = \beta \sin^2 \theta \, \mathsf{K}_e$$

with the azimuth of vibration $\operatorname{tg} \psi_3 = 0$.

According to (147) these values give the principal intensities K_0 and K'_0 required and hence the entropy radiation (346). Thereby the amount of entropy

$$q \, \Delta\nu \, [\mathsf{L}(\mathsf{K}_0) + \mathsf{L}(\mathsf{K}'_0)] \, d\Omega \, dt \tag{347}$$

is added to the field of radiation in the time dt. All told, the entropy change of the field of radiation in the time dt, as given by subtraction of the expression (345) from (347) and integration with respect to $d\Omega$, is

$$dt \, \Delta\nu \int q \, d\Omega \, [\mathsf{L}(\mathsf{K}_0) + \mathsf{L}(\mathsf{K}'_0) - \mathsf{L}(\mathsf{K}) - \mathsf{L}(\mathsf{K}')]. \tag{348}$$

Let us now calculate the entropy change of the system of oscillators which has taken place in the same time dt. According to (173) the entropy at the time t is

$$S = -kN \sum_1^\infty w_n \log w_n.$$

Hence the entropy change in the time dt is

$$dS = -kN \sum_1^\infty \log w_n \, dw_n,$$

and, by taking account of (343), we have:

$$dS = \frac{Nk\mathsf{I} \, dt}{4h\nu L} \sum_1^\infty \big(w_n - (1-\eta) w_{n-1}\big) \log w_n. \tag{349}$$

186. The principle of increase of entropy requires that the sum of the entropy change (348) of the field of radiation and the entropy change (349) of the system of oscillators be always positive, or zero in the limiting case. That this condition is in fact satisfied we shall prove only for the special case when all rays falling on the oscillators are unpolarized, *i.e.*, when $\mathsf{K}' = \mathsf{K}$.

In this case we have from (147) and Sec. 185.

$$\left.\begin{matrix}\mathsf{K}_0 \\ \mathsf{K}'_0\end{matrix}\right\} = \tfrac{1}{2}\{2\mathsf{K} + \beta \sin^2\theta(\mathsf{K}_e - \mathsf{K}) \pm \beta \sin^2\theta(\mathsf{K}_e - \mathsf{K})\},$$

and hence

$$\mathsf{K}_0 = \mathsf{K} + \beta \sin^2\theta(\mathsf{K}_e - \mathsf{K}), \quad \mathsf{K}'_0 = \mathsf{K}.$$

The entropy change (348) of the field of radiation becomes

$$dt\,\Delta\nu \int q\,d\Omega\,\{\mathsf{L}(\mathsf{K}_o) - \mathsf{L}(\mathsf{K})\} = dt\,\Delta\nu \int q\,d\Omega\,\beta \sin^2\theta(\mathsf{K}_e - \mathsf{K})\,\frac{d\mathsf{L}(\mathsf{K})}{d\mathsf{K}}$$

or, by (338) and (278),

$$= \frac{\pi k N\,dt}{hc\nu L} \int d\Omega \sin^2\theta(\mathsf{K}_e - \mathsf{K}) \log\left(1 + \frac{h\nu^3}{c^2\mathsf{K}}\right).$$

On adding to this the entropy change (349) of the system of oscillators and taking account of (320), the total increase in entropy in the time dt is found to be equal to the expression

$$\frac{\pi k N\,dt}{ch\nu L} \int d\Omega \sin^2\theta \left\{\mathsf{K}\sum_1^\infty (w_n - \zeta w_{n-1}) \log w_n + (\mathsf{K}_e - \mathsf{K}) \log\left(1 + \frac{h\nu^3}{c^2\mathsf{K}}\right)\right\}$$

where

$$\zeta = 1 - \eta. \tag{350}$$

We now must prove that the expression

$$F = \int d\Omega \sin^2\theta \Big\{ \mathsf{K} \sum_1^\infty (w_n - \zeta w_{n-1}) \log w_n + (\mathsf{K}_e - \mathsf{K}) \log \left(1 + \frac{h\nu^3}{c^2 \mathsf{K}}\right) \Big\} \quad (351)$$

is always positive and for that purpose we set down once more the meaning of the quantities involved. K is an arbitrary positive function of the polar angles θ and ϕ. The positive proper fraction ζ is according to (350), (265), and (320) given by

$$\frac{\zeta}{1-\zeta} = \frac{3c^2}{8\pi h\nu^3} \int \mathsf{K} \sin^2\theta \, d\Omega. \quad (352)$$

The quantities w_1, w_2, w_3, ... are any positive proper fractions whatever which, according to (167), satisfy the condition

$$\sum_1^\infty w_n = 1 \quad (353)$$

while, according to (342),

$$w_0 = \frac{1-\zeta}{\zeta}. \quad (354)$$

Finally we have from (336)

$$\mathsf{K}_e = \frac{h\nu^3 \zeta}{c^2} \sum_1^\infty n w_n. \quad (355)$$

187. To give the proof required we shall show that the least value which the function F can assume is positive or zero. For this purpose we consider first that positive function, K, of θ and ϕ, which, with fixed values of ζ, w_1, w_2, w_3, ... and K_e, will make F a minimum. The necessary condition for this is $\delta F = 0$, where according to (352)

$$\int \delta \mathsf{K} \sin^2\theta \, d\Omega = 0.$$

This gives, by considering that the quantities w and ζ do not depend on θ and ϕ, as a necessary condition for the minimum,

$$\delta F = 0 = \int d\Omega \sin^2\theta \, \delta \mathsf{K} \left\{ -\log\left(1 + \frac{h\nu^3}{c^2 \mathsf{K}}\right) - \frac{\mathsf{K}_e - \mathsf{K}}{\frac{c^2 \mathsf{K}}{h\nu^3} + 1} \cdot \frac{1}{\mathsf{K}} \right\}$$

and it follows, therefore, that the quantity in brackets, and hence also K itself is independent of θ and ϕ. That in this case F really has a minimum value is readily seen by forming the second variation

$$\delta^2 F = \int d\Omega \sin^2\theta \, \delta \mathsf{K} \, \delta \left\{ -\log\left(1 + \frac{h\nu^3}{c^2 \mathsf{K}}\right) - \frac{\mathsf{K}_e - \mathsf{K}}{\frac{c^2 \mathsf{K}}{h\nu^3} + 1} \cdot \frac{1}{\mathsf{K}} \right\}$$

which may by direct computation be seen to be positive under all circumstances.

In order to form the minimum value of F we calculate the value of K, which, from (352), is independent of θ and ϕ. Then it follows, by taking account of (319a), that

$$\mathsf{K} = \frac{h\nu^3}{c^2} \frac{\zeta}{1 - \zeta}$$

and, by also substituting K_e from (355),

$$F = \frac{8\pi h\nu^3}{3c^2}\,\frac{\zeta}{1-\zeta}\sum_1^\infty (w_n - \zeta w_{n-1})\log w_n - [(1-\zeta)n - 1]w_n \log\zeta.$$

188. It now remains to prove that the sum

$$\Phi = \sum_1^\infty (w_n - \zeta w_{n-1})\log w_n - [(1-\zeta)n - 1]w_n\log\zeta, \tag{356}$$

where the quantities w_n are subject only to the restrictions that (353) and (354) can never become negative. For this purpose we determine that system of values of the w's which, with a fixed value of ζ, makes the sum Φ a minimum. In this case $\delta\Phi = 0$, or

$$\sum_1^\infty (\delta w_n - \zeta\,\delta w_{n-1})\log w_n + (w_n - \zeta w_{n-1})\frac{\delta w_n}{w_n} - [(1-\zeta)n - 1]\,\delta w_n \log\zeta = 0, \tag{357}$$

where, according to (353) and (354),

$$\sum_1^\infty \delta w_n = 0 \quad \text{and} \quad \delta w_0 = 0. \tag{358}$$

If we suppose all the separate terms of the sum to be written out, the equation may be put into the following form:

$$\sum_1^\infty \delta w_n\{\log w_n - \zeta\log w_{n+1} + \frac{w_n - \zeta w_{n-1}}{w_n} - [(1-\zeta)n - 1]\log\zeta\} = 0. \tag{359}$$

From this, by taking account of (358), we get as the condition for a minimum, that

$$\log w_n - \zeta \log w_{n+1} + \frac{w_n - \zeta w_{n-1}}{w_n} - [(1-\zeta)n - 1] \log \zeta \qquad (360)$$

must be independent of n.

The solution of this functional equation is

$$w_n = (1-\zeta)\zeta^{n-1} \qquad (361)$$

for it satisfies (360) as well as (353) and (354). With this value (356) becomes

$$\Phi = 0. \qquad (362)$$

189. In order to show finally that the value (362) of Φ is really the minimum value, we form from (357) the second variation

$$\delta^2\Phi = \sum_1^\infty (\delta w_n - \zeta\, \delta w_{n-1}) \frac{\delta w_n}{w_n} - \frac{\zeta\, \delta w_{n-1}}{w_n}\, \delta w_n + \frac{\zeta w_{n-1}}{w_n^2}\, \delta w_n^2,$$

where all terms containing the second variation $\delta^2 w_n$ have been omitted since their coefficients are, by (360), independent of n and since

$$\sum_1^\infty \delta^2 w_n = 0.$$

This gives, taking account of (361),

$$\delta^2\Phi = \sum_1^\infty \frac{2\delta w_n^2}{(1-\zeta)\zeta^{n-1}} - \frac{2\zeta\, \delta w_{n-1}\, \delta w_n}{(1-\zeta)\zeta^{n-1}}$$

or

$$\delta^2\Phi = \frac{2\zeta}{1-\zeta}\sum_1^\infty \frac{\delta w_n^2}{\zeta^n} - \frac{\delta w_{n-1}\,\delta w_n}{\zeta^{n-1}}.$$

That the sum which occurs here, namely,

$$\frac{\delta w_1^2}{\zeta} - \frac{\delta w_1\,\delta w_2}{\zeta} + \frac{\delta w_2^2}{\zeta^2} - \frac{\delta w_2\,\delta w_3}{\zeta^2} + \frac{\delta w_3^2}{\zeta^3} - \frac{\delta w_3\,\delta w_4}{\zeta^3} + \ldots \tag{363}$$

is essentially positive may be seen by resolving it into a sum of squares. For this purpose we write it in the form

$$\sum_1^\infty \frac{1-\alpha_n}{\zeta^n}\,\delta w_n^2 - \frac{\delta w_n\,\delta w_{n+1}}{\zeta^n} + \frac{\alpha_{n+1}}{\zeta^{n+1}}\,\delta w_{n+1}^2,$$

which is identical with (363) provided $\alpha_1 = 0$. Now the α's may be so determined that every term of the last sum is a perfect square, *i.e.*, that

$$4\cdot\frac{1-\alpha_n}{\zeta^n}\cdot\frac{\alpha_{n+1}}{\zeta^{n+1}} = \left(\frac{1}{\zeta^n}\right)^2$$

or

$$\alpha_{n+1} = \frac{\zeta}{4(1-\alpha_n)}. \tag{364}$$

By means of this formula the α's may be readily calculated. The first values are:

$$\alpha_1 = 0, \quad \alpha_2 = \frac{\zeta}{4}, \quad \alpha_3 = \frac{\zeta}{4-\zeta}, \ \ldots.$$

Continuing the procedure α_n remains always positive and less than $\alpha' = \frac{1}{2}\left(1-\sqrt{1-\zeta}\right)$. To prove the correctness of this statement we show that, if it holds for α_n, it holds also for α_{n+1}. We assume, therefore, that α_n is positive and $< \alpha'$. Then from (364) α_{n+1} is positive

and $< \frac{\zeta}{4(1-\alpha')}$. But $\frac{\zeta}{4(1-\alpha')} = \alpha'$. Hence $\alpha_{n+1} < \alpha'$. Now, since the assumption made does actually hold for $n = 1$, it holds in general. The sum (363) is thus essentially positive and hence the value (362) of Φ really is a minimum, so that the increase of entropy is proven generally.

The limiting case (361), in which the increase of entropy vanishes, corresponds, of course, to the case of thermodynamic equilibrium between radiation and oscillators, as may also be seen directly by comparison of (361) with (271), (265), and (360).

190. Conclusion.—The theory of irreversible radiation processes here developed explains how, with an arbitrarily assumed initial state, a stationary state is, in the course of time, established in a cavity through which radiation passes and which contains oscillators of all kinds of natural vibrations, by the intensities and polarizations of all rays equalizing one another as regards magnitude and direction. But the theory is still incomplete in an important respect. For it deals only with the mutual actions of rays and vibrations of oscillators of the same period. For a definite frequency the increase of entropy in every time element until the maximum value is attained, as demanded by the second principle of thermodynamics, has been proven directly. But, for all frequencies taken together, the maximum thus attained does not yet represent the absolute maximum of the entropy of the system and the corresponding state of radiation does not, in general, represent the absolutely stable equilibrium (compare Sec. 27). For the theory gives no information as to the way in which the intensities of radiation corresponding to different frequencies equalize one another, that is to say, how from any arbitrary initial spectral distribution of energy the normal energy distribution corresponding to black radiation is, in the course of time, developed. For the oscillators on which the consideration was based

influence only the intensities of rays which correspond to their natural vibration, but they are not capable of changing their frequencies, so long as they exert or suffer no other action than emitting or absorbing radiant energy.[1]

To get an insight into those processes by which the exchange of energy between rays of different frequencies takes place in nature would require also an investigation of the influence which the motion of the oscillators and of the electrons flying back and forth between them exerts on the radiation phenomena. For, if the oscillators and electrons are in motion, there will be impacts between them, and, at every impact, actions must come into play which influence the energy of vibration of the oscillators in a quite different and much more radical way than the simple emission and absorption of radiant energy. It is true that the final result of all such impact actions may be anticipated by the aid of the probability considerations discussed in the third section, but to show in detail how and in what time intervals this result is arrived at will be the problem of a future theory. It is certain that, from such a theory, further information may be expected as to the nature of the oscillators which really exist in nature, for the very reason that it must give a closer explanation of the physical significance of the universal elementary quantity of action, a significance which is certainly not second in importance to that of the elementary quantity of electricity.

[1]Compare *P. Ehrenfest*, Wien. Ber. **114** [2a], p. 1301, 1905. Ann. d. Phys. **36**, p. 91, 1911. *H. A. Lorentz*, Phys. Zeitschr. **11**, p. 1244, 1910. *H. Poincaré*, Journ. de Phys. (5) **2**, p. 5, p. 347, 1912.

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APPENDIX I

On Deductions from Stirling's Formula

The formula is

$$\lim_{n=\infty} \frac{n!}{n^n e^{-n}\sqrt{2\pi n}} = 1, \tag{a}$$

or, to an approximation quite sufficient for all practical purposes, provided that n is larger than 7

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n}. \tag{b}$$

For a proof of this relation and a discussion of its limits of accuracy a treatise on probability must be consulted.

On substitution in (170) this gives

$$W = \frac{\left(\frac{N}{e}\right)^N}{\left(\frac{N_1}{e}\right)^{N_1} \cdot \left(\frac{N_2}{e}\right)^{N_2} \dots} \cdot \frac{\sqrt{2\pi N}}{\sqrt{2\pi N_1} \cdot \sqrt{2\pi N_2} \dots}.$$

On account of (165) this reduces at once to

$$\frac{N_N}{N_1^{N_1} N_2^{N_2} \dots} \cdot \frac{\sqrt{2\pi N}}{\sqrt{2\pi N_1} \cdot \sqrt{2\pi N_2} \dots}.$$

Passing now to the logarithmic expression we get

$$S = k \log W = k[N \log N - N_1 \log N_1 - N_2 \log N_2 - \dots + \log \sqrt{2\pi N} - \log \sqrt{2\pi N_1} - \log \sqrt{2\pi N_2} - \dots],$$

or,

$$S = k \log W = k[(N \log N + \log \sqrt{2\pi N}) - (N_1 \log N_1 + \log \sqrt{2\pi N_1}) - (N_2 \log N_2 + \log \sqrt{2\pi N_2}) - \dots].$$

Now, for a large value of N_i, the term $N_i \log N_i$ is very much larger than $\log \sqrt{2\pi N_i}$, as is seen by writing the latter in the form $\frac{1}{2} \log 2\pi + \frac{1}{2} \log N_i$. Hence the last expression will, with a fair approximation, reduce to

$$S = k \log W = k[N \log N - N_1 \log N_1 - N_2 \log N_2 - \dots].$$

Introducing now the values of the densities of distribution w by means of the relation

$$N_i = w_i N$$

we obtain

$$S = k \log W = kN[\log N - w_1 \log N_1 - w_2 \log N_2 - \dots],$$

or, since

$$w_1 + w_2 + w_3 + \dots = 1,$$

and hence

$$(w_1 + w_2 + w_3 + \dots) \log N = \log N,$$

and

$$\log N - \log N_1 = \log \frac{N}{N_1} = \log \frac{1}{w_1} = -\log w_1,$$

we obtain by substitution, after one or two simple transformations

$$S = k \log W = -kN \sum w_1 \log w_1,$$

a relation which is identical with (173).

The statements of Sec. 143 may be proven in a similar manner. From (232) we get at once

$$S = k \log W_m = k \log \frac{(N+P-1)!}{(N-1)!\,P!}$$

Now

$$\log(N-1)! = \log N! - \log N,$$

and, for large values of N, $\log N$ is negligible compared with $\log N!$. Applying the same reasoning to the numerator we may without appreciable error write

$$S = k \log W_m = k \log \frac{(N+P)!}{N!\,P!}.$$

Substituting now for $(N+P)!$, $N!$, and $P!$ their values from (b) and omitting, as was previously shown to be approximately correct, the terms arising from the $\sqrt{2\pi(N+P)}$ etc., we get, since the terms containing e cancel out

$$\begin{aligned} S &= k[(N+P)\log(N+P) - N\log N - P\log P] \\ &= k[(N+P)\log\frac{N+P}{N} + P\log N - P\log P] \\ &= kN\left[\left(\frac{P}{N}+1\right)\log\left(\frac{P}{N}+1\right) - \frac{P}{N}\log\frac{P}{N}\right]. \end{aligned}$$

This is the relation of Sec. 143.

APPENDIX II

References

Among general papers treating of the application of the theory of quanta to different parts of physics are:

1. *A. Sommerfeld*, Das Planck'sche Wirkungsquantum und seine allgemeine Bedeutung für die Molekularphysik, Phys. Zeitschr., **12**, p. 1057. Report to the Versammlung Deutscher Naturforscher und Aerzte. Deals especially with applications to the theory of specific heats and to the photoelectric effect. Numerous references are quoted.

2. Meeting of the British Association, Sept., 1913. See Nature, **92**, p. 305, Nov. 6, 1913, and Phys. Zeitschr., **14**, p. 1297. Among the principal speakers were *J. H. Jeans* and *H. A. Lorentz*.

(Also American Phys. Soc., Chicago Meeting, 1913.[1])

3. *R. A. Millikan*, Atomic Theories of Radiation, Science, **37**, p. 119, Jan. 24, 1913. A non-mathematical discussion.

4. *W. Wien*, Neuere Probleme der Theoretischen Physik, 1913. (*Wien's* Columbia Lectures, in German.) This is perhaps the most complete review of the entire theory of quanta.

H. A. Lorentz, Alte und Neue Probleme der Physik, Phys. Zeitschr., **11**, p. 1234. Address to the Versammlung Deutscher Naturforscher und Aerzte, Königsberg, 1910, contains also some discussion of the theory of quanta.

Among the papers on radiation are:

E. Bauer, Sur la théorie du rayonnement, Comptes Rendus, **153**, p. 1466. Adheres to the quantum theory in the original form, namely, that emission and absorption both take place in a discontinuous manner.

[1]Not yet published (Jan. 26, 1914. Tr.)

E. Buckingham, Calculation of c_2 in Planck's equation, Bull. Bur. Stand. **7**, p. 393.

E. Buckingham, On *Wien's* Displacement Law, Bull. Bur. Stand. **8**, p. 543. Contains a very simple and clear proof of the displacement law.

P. Ehrenfest, Strahlungshypothesen, Ann. d. Phys. **36**, p. 91.

A. Joffé, Theorie der Strahlung, Ann. d. Phys. **36**, p. 534.

Discussions of the method of derivation of the radiation formula are given in many papers on the subject. In addition to those quoted elsewhere may be mentioned:

C. Benedicks, Ueber die Herleitung von *Planck's* Energieverteilungsgesetz Ann. d. Phys. **42**, p. 133. Derives *Planck's* law without the help of the quantum theory. The law of equipartition of energy is avoided by the assumption that solids are not always monatomic, but that, with decreasing temperature, the atoms form atomic complexes, thus changing the number of degrees of freedom. The equipartition principle applies only to the free atoms.

P. Debye, *Planck's* Strahlungsformel, Ann. d. Phys. **33**, p. 1427. This method is fully discussed by *Wien* (see 4, above). It somewhat resembles *Jeans'* method (Sec. 169) since it avoids all reference to resonators of any particular kind and merely establishes the most probable energy distribution. It differs, however, from *Jeans'* method by the assumption of discrete energy quanta $h\nu$. The physical nature of these units is not discussed at all and it is also left undecided whether it is a property of matter or of the ether or perhaps a property of the energy exchange between matter and the ether that causes their existence. (Compare also some remarks of *Lorentz* in 2.)

P. Frank, Zur Ableitung der Planckschen Strahlungsformel, Phys. Zeitschr., **13**, p. 506.

L. Natanson, Statistische Theorie der Strahlung, Phys. Zeitschr., **12**, p. 659.

W. Nernst, Zur Theorie der specifischen Wärme und über die Anwendung der Lehre von den Energiequanten auf physikalisch-chemische Fragen überhaupt, Zeitschr. f. Elektrochemie, **17**, p. 265.

The experimental facts on which the recent theories of specific heat (quantum theories) rely, were discovered by *W. Nernst* and his fellow workers. The results are published in a large number of papers that have appeared in different periodicals. See, *e.g.*, *W. Nernst*, Der Energieinhalt fester Substanzen, Ann. d. Phys. **36**, p. 395, where also numerous other papers are quoted. (See also references given in 1.) These experimental facts give very strong support to the heat theorem of *Nernst* (Sec. 120), according to which the entropy approaches a definite limit (perhaps the value zero, see *Planck's* Thermodynamics, 3. ed., sec. 282, *et seq.*) at the absolute zero of temperature, and which is consistent with the quantum theory. This work is in close connection with the recent attempts to develop an equation of state applicable to the solid state of matter. In addition to the papers by *Nernst* and his school there may be mentioned:

K. Eisenmann, Canonische Zustandsgleichung einatomiger fester Körper und die Quantentheorie, Verhandlungen der Deutschen Physikalischen Gesellschaft, **14**, p. 769.

W. H. Keesom, Entropy and the Equation of State, Konink. Akad. Wetensch. Amsterdam Proc., **15**, p. 240.

L. Natanson, Energy Content of Bodies, Acad. Science Cracovie Bull. Ser. A, p. 95. In *Einstein's* theory of specific heats (Sec. 140) the atoms of actual bodies in nature are apparently identified with the ideal resonators of *Planck*. In this paper it is pointed out that this is implying too special features for the atoms of real bodies, and also, that such far-reaching specializations do not seem necessary for deriving the laws of specific heat from the quantum theory.

L. S. Ornstein, Statistical Theory of the Solid State, Konink. Akad.

Wetensch. Amsterdam Proc., **14**, p. 983.

S. Ratnowsky, Die Zustandsgleichung einatomiger fester Körper und die Quantentheorie, Ann. d. Phys. **38**, p. 637.

Among papers on the law of equipartition of energy (Sec. 169) are:

J. H. Jeans, Planck's Radiation Theory and *Non-Newtonian* Mechanics, Phil. Mag., **20**, p. 943.

S. B. McLaren, Partition of Energy between Matter and Radiation, Phil. Mag., **21**, p. 15.

S. B. McLaren, Complete Radiation, Phil. Mag., **23**, p. 513. This paper and the one of Jeans deal with the fact that from Newtonian Mechanics (Hamilton's Principle) the equipartition principle necessarily follows, and that hence either Planck's law or the fundamental principles of mechanics need a modification.

For the law of equipartition compare also the discussion at the meeting of the British Association (see 2).

In many of the papers cited so far deductions from the quantum theory are compared with experimental facts. This is also done by:

F. Haber, Absorptionsspectra fester Körper und die Quantentheorie, Verhandlungen der Deutschen Physikalischen Gesellschaft, **13**, p. 1117.

J. Franck und G. Hertz, Quantumhypothese und Ionisation, Ibid., **13**, p. 967.

Attempts of giving a concrete physical idea of Planck's constant h are made by:

A. Schidlof, Zur Aufklärung der universellen electrodynamischen Bedeutung der Planckschen Strahlungsconstanten h, Ann. d. Phys. **35**, p. 96.

D. A. Goldhammer, Ueber die Lichtquantenhypothese, Phys. Zeitschr., **13**, p. 535.

J. J. Thomson, On the Structure of the Atom, Phil. Mag., **26**, p. 792.

N. Bohr, On the Constitution of the Atom, Phil. Mag., **26**, p. 1.

S. B. McLaren, The Magneton and Planck's Universal Constant, Phil. Mag., **26**, p. 800.

The line of reasoning may be briefly stated thus: Find some quantity of the same dimension as h, and then construct a model of an atom where this property plays an important part and can be made, by a simple hypothesis, to vary by finite amounts instead of continuously. The simplest of these is *Bohr's*, where h is interpreted as angular momentum.

The logical reason for the quantum theory is found in the fact that the *Rayleigh-Jeans* radiation formula does not agree with experiment. Formerly *Jeans* attempted to reconcile theory and experiment by the assumption that the equilibrium of radiation and a black body observed and agreeing with *Planck's* law rather than his own, was only apparent, and that the true state of equilibrium which really corresponds to his law and the equipartition of energy among all variables, is so slowly reached that it is never actually observed. This standpoint, which was strongly objected to by authorities on the experimental side of the question (see, *e.g.*, *E. Pringsheim* in 2), he has recently abandoned. *H. Poincaré*, in a profound mathematical investigation (*H. Poincaré*, Sur la Théorie des Quanta, Journal de Physique (5), **2**, p. 1, 1912) reached the conclusion that whatever the law of radiation may be, it must always, if the total radiation is assumed as finite, lead to a function presenting similar discontinuities as the one obtained from the hypothesis of quanta.

While most authorities have accepted the quantum theory for good (see *J. H. Jeans* and *H. A. Lorentz* in 2), a few still entertain doubts as to the general validity of Poincaré's conclusion (see above *C. Benedicks* and *R. A. Millikan* 3). Others still reject the quantum theory on account of the fact that the experimental evidence in favor of *Planck's*

law is not absolutely conclusive (see *R. A. Millikan* 3); among these is *A. E. H. Love* (2), who suggests that *Korn's* (*A. Korn*, Neue Mechanische Vorstellungen über die schwarze Strahlung und eine sich aus denselben ergebende Modification des Planckschen Verteilungsgesetzes, Phys. Zeitschr., **14**, p. 632) radiation formula fits the facts as well as that of Planck.

H. A. Callendar, Note on Radiation and Specific Heat, Phil. Mag., **26**, p. 787, has also suggested a radiation formula that fits the data well. Both Korn's and Callendar's formulæ conform to *Wien's* displacement law and degenerate for large values of λT into the Rayleigh-Jeans, and for small values of λT into Wien's radiation law. Whether Planck's law or one of these is the correct law, and whether, if either of the others should prove to be right, it would eliminate the necessity of the adoption of the quantum theory, are questions as yet undecided. Both Korn and Callendar have promised in their papers to follow them by further ones.

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Project Gutenberg's Eight Lectures on Theoretical Physics, by Max Planck

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DELIVERED AT COLUMBIA UNIVERSITY
IN 1909

BY

MAX PLANCK

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LECTURER IN MATHEMATICAL PHYSICS IN COLUMBIA UNIVERSITY FOR 1909

TRANSLATED BY

A. P. WILLS

PROFESSOR OF MATHEMATICAL PHYSICS IN COLUMBIA UNIVERSITY



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On the seventeenth day of December, nineteen hundred and four, Edward Dean Adams, of New York, established in Columbia University "The Ernest Kempton Adams Fund for Physical Research" as a memorial to his son, Ernest Kempton Adams, who received the degrees of Electrical Engineering in 1897 and Master of Arts in 1898, and who devoted his life to scientific research. The income of this fund is, by the terms of the deed of gift, to be devoted to the maintenance of a research fellowship and to the publication and distribution of the results of scientific research on the part of the fellow. A generous interpretation of the terms of the deed on the part of Mr. Adams and of the Trustees of the University has made it possible to issue these lectures as a publication of the Ernest Kempton Adams Fund.

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Number Three. **Eight Lectures on Theoretical Physics.** By Max Planck, Professor of Theoretical Physics in the University of Berlin. A course of lectures delivered at Columbia University in 1909, translated by A. P. Wills, Professor of Mathematical Physics in Columbia University.

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These publications are distributed under the Adams Fund to many libraries and to a limited number of individuals, but may also be bought at cost from the Columbia University Press.

PREFACE TO ORIGINAL EDITION.

The present book has for its object the presentation of the lectures which I delivered as foreign lecturer at Columbia University in the spring of the present year under the title: "The Present System of Theoretical Physics." The points of view which influenced me in the selection and treatment of the material are given at the beginning of the first lecture. Essentially, they represent the extension of a theoretical physical scheme, the fundamental elements of which I developed in an address at Leyden entitled: "The Unity of the Physical Concept of the Universe." Therefore I regard it as advantageous to consider again some of the topics of that lecture. The presentation will not and can not, of course, claim to cover exhaustively in all directions the principles of theoretical physics.

The Author.

Berlin, 1909

TRANSLATOR'S PREFACE.

At the request of the Adams Fund Advisory Committee, and with the consent of the author, the following translation of Professor Planck's Columbia Lectures was undertaken. It is hoped that the translation will be of service to many of those interested in the development of theoretical physics who, in spite of the inevitable loss, prefer a translated text in English to an original text in German. Since the time of the publication of the original text, some of the subjects treated, particularly that of heat radiation, have received much attention, with the result that some of the points of view taken at that time have undergone considerable modifications. The author considers it desirable, however, to have the translation conform to the original text, since the nature and extent of these modifications can best be appreciated by reference to the recent literature relating to the matters in question.

A. P. Wills.

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FIRST LECTURE.

Introduction: Reversibility and Irreversibility.

Colleagues, ladies and gentlemen: The cordial invitation, which the President of Columbia University extended to me to deliver at this prominent center of American science some lectures in the domain of theoretical physics, has inspired in me a sense of the high honor and distinction thus conferred upon me and, in no less degree, a consciousness of the special obligations which, through its acceptance, would be imposed upon me. If I am to count upon meeting in some measure your just expectations, I can succeed only through directing your attention to the branches of my science with which I myself have been specially and deeply concerned, thus exposing myself to the danger that my report in certain respects shall thereby have somewhat too subjective a coloring.

From those points of view which appear to me the most striking, it is my desire to depict for you in these lectures the present status of the system of theoretical physics. I do not say: the present status of theoretical physics; for to cover this far broader subject, even approximately, the number of lecture hours at my disposal would by no means suffice. Time limitations forbid the extensive consideration of the details of this great field of learning; but it will be quite possible to develop for you, in bold outline, a representation of the system as a whole, that is, to give a sketch of the fundamental laws which rule in the physics of today, of the most important hypotheses employed, and of the great ideas which have recently forced themselves into the subject. I will often gladly endeavor to go into details, but not in the sense of a thorough treatment of the subject, and only with the object of making the general laws more clear, through appropriate specially chosen examples. I shall select these examples from the most varied branches of physics.

If we wish to obtain a correct understanding of the achievements of theoretical physics, we must guard in equal measure against the mistake of overestimating these achievements, and on the other hand, against the corresponding mistake of underestimating them. That the second mistake is actually often made, is shown by the circumstance that quite recently voices have been loudly raised maintaining the bankruptcy and, débâcle of the whole of natural science. But I think such assertions may easily be refuted by reference to the simple fact that with each decade the number and the significance of the means increase, whereby mankind learns directly through the aid of theoretical physics to make nature useful for its own purposes. The technology of today would be impossible without the aid of theoretical physics. The development of the whole of electro-technics from galvanoplasty to wireless telegraphy is a striking proof of this, not to mention aerial navigation. On the other hand, the mistake of overestimating the achievements of theoretical physics appears to me to be much more dangerous, and this danger is particularly threatened by those who have penetrated comparatively little into the heart of the subject. They maintain that some time, through a proper improvement of our science, it will be possible, not only to represent completely through physical formulae the inner constitution of the atoms, but also the laws of mental life. I think that there is nothing in the world entitling us to the one or the other of these expectations. On the other hand, I believe that there is much which directly opposes them. Let us endeavor then to follow the middle course and not to deviate appreciably

toward the one side or the other.

When we seek for a solid immovable foundation which is able to carry the whole structure of theoretical physics, we meet with the questions: What lies at the bottom of physics? What is the material with which it operates? Fortunately, there is a complete answer to this question. The material with which theoretical physics operates is measurements, and mathematics is the chief tool with which this material is worked. All physical ideas depend upon measurements, more or less exactly carried out, and each physical definition, each physical law, possesses a more definite significance the nearer it can be brought into accord with the results of measurements. Now measurements are made with the aid of the senses; before all with that of sight, with hearing and with feeling. Thus far, one can say that the origin and the foundation of all physical research are seated in our sense perceptions. Through sense perceptions only do we experience anything of nature; they are the highest court of appeal in questions under dispute. This view is completely confirmed by a glance at the historical development of physical science. Physics grows upon the ground of sensations. The first physical ideas derived were from the individual perceptions of man, and, accordingly, physics was subdivided into: physics of the eye (optics), physics of the ear (acoustics), and physics of heat sensation (theory of heat). It may well be said that so far as there was a domain of sense, so far extended originally the domain of physics. Therefore it appears that in the beginning the division of physics was based upon the peculiarities of man. It possessed, in short, an anthropomorphic character. This appears also, in that physical research, when not occupied with special sense perceptions, is concerned with practical life, and particularly with the practical needs of men. Thus, the art of geodesy led to geometry, the study of machinery to mechanics, and the conclusion lies near that physics in the last analysis had only to do with the sense perceptions and needs of mankind.

In accordance with this view, the sense perceptions are the essential elements of the world; to construct an object as opposed to sense perceptions is more or less an arbitrary matter of will. In fact, when I speak of a tree, I really mean only a complex of sense perceptions: I can see it, I can hear the rustling of its branches, I can smell its fragrance, I experience pain if I knock my head against it, but disregarding all of these sensations, there remains nothing to be made the object of a measurement, wherewith, therefore, natural science can occupy itself. This is certainly true. In accordance with this view, the problem of physics consists only in the relating of sense perceptions, in accordance with experience, to fixed laws; or, as one may express it, in the greatest possible economic accommodation of our ideas to our sensations, an operation which we undertake solely because it is of use to us in the general battle of existence.

All this appears extraordinarily simple and clear and, in accordance with it, the fact may readily be explained that this positivist view is quite widely spread in scientific circles today. It permits, so far as it is limited to the standpoint here depicted (not always done consistently by the exponents of positivism), no hypothesis, no metaphysics; all is clear and plain. I will go still further; this conception never leads to an actual contradiction. I may even say, it can lead to no contradiction. But, ladies and gentlemen, this view has never contributed to any advance in physics. If physics is to advance, in a certain sense its problem must be stated in quite the inverse way, on account of the fact that this conception is inadequate and at bottom possesses only a formal meaning.

The proof of the correctness of this assertion is to be found directly from a consid-

eration of the process of development which theoretical physics has actually undergone, and which one certainly cannot fail to designate as essential. Let us compare the system of physics of today with the earlier and more primitive system which I have depicted above. At the first glance we encounter the most striking difference of all, that in the present system, as well in the division of the various physical domains as in all physical definitions, the historical element plays a much smaller rôle than in the earlier system. While originally, as I have shown above, the fundamental ideas of physics were taken from the specific sense perceptions of man, the latter are today in large measure excluded from physical acoustics, optics, and the theory of heat. The physical definitions of tone, color, and of temperature are today in no wise derived from perception through the corresponding senses; but tone and color are defined through a vibration number or wave length, and the temperature through the volume change of a thermometric substance, or through a temperature scale based on the second law of thermodynamics; but heat sensation is in no wise mentioned in connection with the temperature. With the idea of force it has not been otherwise. Without doubt, the word force originally meant bodily force, corresponding to the circumstance that the oldest tools, the ax, hammer, and mallet, were swung by man's hands, and that the first machines, the lever, roller, and screw, were operated by men or animals. This shows that the idea of force was originally derived from the sense of force, or muscular sense, and was, therefore, a specific sense perception. Consequently, I regard it today as quite essential in a lecture on mechanics to refer, at any rate in the introduction, to the original meaning of the force idea. But in the modern exact definition of force the specific notion of sense perception is eliminated, as in the case of color sense, and we may say, quite in general, that in modern theoretical physics the specific sense perceptions play a much smaller rôle in all physical definitions than formerly. In fact, the crowding into the background of the specific sense elements goes so far that the branches of physics which were originally completely and uniquely characterized by an arrangement in accordance with definite sense perceptions have fallen apart, in consequence of the loosening of the bonds between different and widely separated branches, on account of the general advance towards simplification and coordination. The best example of this is furnished by the theory of heat. Earlier, heat formed a separate and unified domain of physics, characterized through the perceptions of heat sensation. Today one finds in well nigh all physics textbooks dealing with heat a whole domain, that of radiant heat, separated and treated under optics. The significance of heat perception no longer suffices to bring together the heterogeneous parts.

In short, we may say that the characteristic feature of the entire previous development of theoretical physics is a definite elimination from all physical ideas of the anthropomorphic elements, particularly those of specific sense perceptions. On the other hand, as we have seen above, if one reflects that the perceptions form the point of departure in all physical research, and that it is impossible to contemplate their absolute exclusion, because we cannot close the source of all our knowledge, then this conscious departure from the original conceptions must always appear astonishing or even paradoxical. There is scarcely a fact in the history of physics which today stands out so clearly as this. Now, what are the great advantages to be gained through such a real obliteration of personality? What is the result for the sake of whose achievement are sacrificed the directness and succinctness such as only the special sense perceptions

vouchsafe to physical ideas?

The result is nothing more than the attainment of unity and compactness in our system of theoretical physics, and, in fact, the unity of the system, not only in relation to all of its details, but also in relation to physicists of all places, all times, all peoples, all cultures. Certainly, the system of theoretical physics should be adequate, not only for the inhabitants of this earth, but also for the inhabitants of other heavenly bodies. Whether the inhabitants of Mars, in case such actually exist, have eyes and ears like our own, we do not know,—it is quite improbable; but that they, in so far as they possess the necessary intelligence, recognize the law of gravitation and the principle of energy, most physicists would hold as self evident: and anyone to whom this is not evident had better not appeal to the physicists, for it will always remain for him an unsolvable riddle that the same physics is made in the United States as in Germany.

To sum up, we may say that the characteristic feature of the actual development of the system of theoretical physics is an ever extending emancipation from the anthropomorphic elements, which has for its object the most complete separation possible of the system of physics and the individual personality of the physicist. One may call this the objectiveness of the system of physics. In order to exclude the possibility of any misunderstanding, I wish to emphasize particularly that we have here to do, not with an absolute separation of physics from the physicist—for a physics without the physicist is unthinkable,—but with the elimination of the individuality of the particular physicist and therefore with the production of a common system of physics for all physicists.

Now, how does this principle agree with the positivist conceptions mentioned above? Separation of the system of physics from the individual personality of the physicist? Opposed to this principle, in accordance with those conceptions, each particular physicist must have his special system of physics, in case that complete elimination of all metaphysical elements is effected; for physics occupies itself only with the facts discovered through perceptions, and only the individual perceptions are directly involved. That other living beings have sensations is, strictly speaking, but a very probable, though arbitrary, conclusion from analogy. The system of physics is therefore primarily an individual matter and, if two physicists accept the same system, it is a very happy circumstance in connection with their personal relationship, but it is not essentially necessary. One can regard this view-point however he will; in physics it is certainly quite fruitless, and this is all that I care to maintain here. Certainly, I might add, each great physical idea means a further advance toward the emancipation from anthropomorphic ideas. This was true in the passage from the Ptolemaic to the Copernican cosmical system, just as it is true at the present time for the apparently impending passage from the so-called classical mechanics of mass points to the general dynamics originating in the principle of relativity. In accordance with this, man and the earth upon which he dwells are removed from the centre of the world. It may be predicted that in this century the idea of time will be divested of the absolute character with which men have been accustomed to endow it (cf. the final lecture). Certainly, the sacrifices demanded by every such revolution in the intuitive point of view are enormous; consequently, the resistance against such a change is very great. But the development of science is not to be permanently halted thereby; on the contrary, its strongest impetus is experienced through precisely those forces which attain success in the struggle against the old points of view, and to this extent such a struggle is constantly necessary and useful.

Now, how far have we advanced today toward the unification of our system of physics? The numerous independent domains of the earlier physics now appear reduced to two; mechanics and electrodynamics, or, as one may say: the physics of material bodies and the physics of the ether. The former comprehends acoustics, phenomena in material bodies, and chemical phenomena; the latter, magnetism, optics, and radiant heat. But is this division a fundamental one? Will it prove final? This is a question of great consequence for the future development of physics. For myself, I believe it must be answered in the negative, and upon the following grounds: mechanics and electrodynamics cannot be permanently sharply differentiated from each other. Does the process of light emission, for example, belong to mechanics or to electrodynamics? To which domain shall be assigned the laws of motion of electrons? At first glance, one may perhaps say: to electrodynamics, since with the electrons ponderable matter does not play any rôle. But let one direct his attention to the motion of free electrons in metals. There he will find, in the study of the classical researches of H. A. Lorentz, for example, that the laws obeyed by the electrons belong rather to the kinetic theory of gases than to electrodynamics. In general, it appears to me that the original differences between processes in the ether and processes in material bodies are to be considered as disappearing. Electrodynamics and mechanics are not so remarkably far apart, as is considered to be the case by many people, who already speak of a conflict between the mechanical and the electrodynamic views of the world. Mechanics requires for its foundation essentially nothing more than the ideas of space, of time, and of that which is moving, whether one considers this as a substance or a state. The same ideas are also involved in electrodynamics. A sufficiently generalized conception of mechanics can therefore also well include electrodynamics, and, in fact, there are many indications pointing toward the ultimate amalgamation of these two subjects, the domains of which already overlap in some measure.

If, therefore, the gulf between ether and matter be once bridged, what is the point of view which in the last analysis will best serve in the subdivision of the system of physics? The answer to this question will characterize the whole nature of the further development of our science. It is, therefore, the most important among all those which I propose to treat today. But for the purposes of a closer investigation it is necessary that we go somewhat more deeply into the peculiarities of physical principles.

We shall best begin at that point from which the first step was made toward the actual realization of the unified system of physics previously postulated by the philosophers only; at the principle of conservation of energy. For the idea of energy is the only one besides those of space and time which is common to all the various domains of physics. In accordance with what I have stated above, it will be apparent and quite self evident to you that the principle of energy, before its general formularization by Mayer, Joule, and Helmholz, also bore an anthropomorphic character. The roots of this principle lay already in the recognition of the fact that no one is able to obtain useful work from nothing; and this recognition had originated essentially in the experiences which were gathered in attempts at the solution of a technical problem: the discovery of perpetual motion. To this extent, perpetual motion has come to have for physics a far reaching significance, similar to that of alchemy for the chemist, although it was not the positive, but rather the negative results of these experiments, through which science was advanced. Today we speak of the principle of energy quite

without reference to the technical viewpoint or to that of man. We say that the total amount of energy of an isolated system of bodies is a quantity whose amount can be neither increased nor diminished through any kind of process within the system, and we no longer consider the accuracy with which this law holds as dependent upon the refinement of the methods, which we at present possess, of testing experimentally the question of the realization of perpetual motion. In this, strictly speaking, unprovable generalization, impressed upon us with elemental force, lies the emancipation from the anthropomorphic elements mentioned above.

While the principle of energy stands before us as a complete independent structure, freed from and independent of the accidents appertaining to its historical development, this is by no means true in equal measure in the case of that principle which R. Clausius introduced into physics; namely, the second law of thermodynamics. This law plays a very peculiar rôle in the development of physical science, to the extent that one is not able to assert today that for it a generally recognized, and therefore objective formularization, has been found. In our present consideration it is therefore a matter of particular interest to examine more closely its significance.

In contrast to the first law of thermodynamics, or the energy principle, the second law may be characterized as follows. While the first law permits in all processes of nature neither the creation nor destruction of energy, but permits of transformations only, the second law goes still further into the limitation of the possible processes of nature, in that it permits, not all kinds of transformations, but only certain types, subject to certain conditions. The second law occupies itself, therefore, with the question of the kind and, in particular, with the direction of any natural process.

At this point a mistake has frequently been made, which has hindered in a very pronounced manner the advance of science up to the present day. In the endeavor to give to the second law of thermodynamics the most general character possible, it has been proclaimed by followers of W. Ostwald as the second law of energetics, and the attempt made so to formulate it that it shall determine quite generally the direction of every process occurring in nature. Some weeks ago I read in a public academic address of an esteemed colleague the statement that the import of the second law consists in this, that a stone falls downwards, that water flows not up hill, but down, that electricity flows from a higher to a lower potential, and so on. This is a mistake which at present is altogether too prevalent not to warrant mention here.

The truth is, these statements are false. A stone can just as well rise in the air as fall downwards; water can likewise flow upwards, as, for example, in a spring; electricity can flow very well from a lower to a higher potential, as in the case of oscillating discharge of a condenser. The statements are obviously quite correct, if one applies them to a stone originally at rest, to water at rest, to electricity at rest; but then they follow immediately from the energy principle, and one does not need to add a special second law. For, in accordance with the energy principle, the kinetic energy of the stone or of the water can only originate at the cost of gravitational energy, i. e., the center of mass must descend. If, therefore, motion is to take place at all, it is necessary that the gravitational energy shall decrease. That is, the center of mass must descend. In like manner, an electric current between two condenser plates can originate only at the cost of electrical energy already present; the electricity must therefore pass to a lower potential. If, however, motion and current be already present, then one is not able to

say, a priori, anything in regard to the direction of the change; it can take place just as well in one direction as the other. Therefore, there is no new insight into nature to be obtained from this point of view.

Upon an equally inadequate basis rests another conception of the second law, which I shall now mention. In considering the circumstance that mechanical work may very easily be transformed into heat, as by friction, while on the other hand heat can only with difficulty be transformed into work, the attempt has been made so to characterize the second law, that in nature the transformation of work into heat can take place completely, while that of heat into work, on the other hand, only incompletely and in such manner that every time a quantity of heat is transformed into work another corresponding quantity of energy must necessarily undergo at the same time a compensating transformation, as, e. g., the passage of heat from a higher to a lower temperature. This assertion is in certain special cases correct, but does not strike in general at the true import of the matter, as I shall show by a simple example.

One of the most important laws of thermodynamics is, that the total energy of an ideal gas depends only upon its temperature, and not upon its volume. If an ideal gas be allowed to expand while doing work, and if the cooling of the gas be prevented through the simultaneous addition of heat from a heat reservoir at higher temperature, the gas remains unchanged in temperature and energy content, and one may say that the heat furnished by the heat reservoir is completely transformed into work without exchange of energy. Not the least objection can be urged against this assertion. The law of incomplete transformation of heat into work is retained only through the adoption of a different point of view, but which has nothing to do with the status of the physical facts and only modifies the way of looking at the matter, and therefore can neither be supported nor contradicted through facts; namely, through the introduction ad hoc of new particular kinds of energy, in that one divides the energy of the gas into numerous parts which individually can depend upon the volume. But it is a priori evident that one can never derive from so artificial a definition a new physical law, and it is with such that we have to do when we pass from the first law, the principle of conservation of energy, to the second law.

I desire now to introduce such a new physical law: "It is not possible to construct a periodically functioning motor which in principle does not involve more than the raising of a load and the cooling of a heat reservoir." It is to be understood, that in one cycle of the motor quite arbitrary complicated processes may take place, but that after the completion of one cycle there shall remain no other changes in the surroundings than that the heat reservoir is cooled and that the load is raised a corresponding distance, which may be calculated from the first law. Such a motor could of course be used at the same time as a refrigerating machine also, without any further expenditure of energy and materials. Such a motor would moreover be the most efficient in the world, since it would involve no cost to run it; for the earth, the atmosphere, or the ocean could be utilized as the heat reservoir. We shall call this, in accordance with the proposal of W. Ostwald, perpetual motion of the second kind. Whether in nature such a motion is actually possible cannot be inferred from the energy principle, and may only be determined by special experiments.

Just as the impossibility of perpetual motion of the first kind leads to the principle of the conservation of energy, the quite independent principle of the impossibility of

perpetual motion of the second kind leads to the second law of thermodynamics, and, if we assume this impossibility as proven experimentally, the general law follows immediately: *there are processes in nature which in no possible way can be made completely reversible*. For consider, e. g., a frictional process through which mechanical work is transformed into heat with the aid of suitable apparatus, if it were actually possible to make in some way such complicated apparatus completely reversible, so that everywhere in nature exactly the same conditions be reestablished as existed at the beginning of the frictional process, then the apparatus considered would be nothing more than the motor described above, furnishing a perpetual motion of the second kind. This appears evident immediately, if one clearly perceives what the apparatus would accomplish: transformation of heat into work without any further outstanding change.

We call such a process, which in no wise can be made completely reversible, an irreversible process, and all other processes reversible processes; and thus we strike the kernel of the second law of thermodynamics when we say that irreversible processes occur in nature. In accordance with this, the changes in nature have a unidirectional tendency. With each irreversible process the world takes a step forward, the traces of which under no circumstances can be completely obliterated. Besides friction, examples of irreversible processes are: heat conduction, diffusion, conduction of electricity in conductors of finite resistance, emission of light and heat radiation, disintegration of the atom in radioactive substances, and so on. On the other hand, examples of reversible processes are: motion of the planets, free fall in empty space, the undamped motion of a pendulum, the frictionless flow of liquids, the propagation of light and sound waves without absorption and refraction, undamped electrical vibrations, and so on. For all these processes are already periodic or may be made completely reversible through suitable contrivances, so that there remains no outstanding change in nature; for example, the free fall of a body whereby the acquired velocity is utilized to raise the body again to its original height; a light or sound wave which is allowed in a suitable manner to be totally reflected from a perfect mirror.

What now are the general properties and criteria of irreversible processes, and what is the general quantitative measure of irreversibility? This question has been examined and answered in the most widely different ways, and it is evident here again how difficult it is to reach a correct formularization of a problem. Just as originally we came upon the trail of the energy principle through the technical problem of perpetual motion, so again a technical problem, namely, that of the steam engine, led to the differentiation between reversible and irreversible processes. Long ago Sadi Carnot recognized, although he utilized an incorrect conception of the nature of heat, that irreversible processes are less economical than reversible, or that in an irreversible process a certain opportunity to derive mechanical work from heat is lost. What then could have been simpler than the thought of making, quite in general, the measure of the irreversibility of a process the quantity of mechanical work which is unavoidably lost in the process. For a reversible process then, the unavoidably lost work is naturally to be set equal to zero. This view, in accordance with which the import of the second law consists in a dissipation of useful energy, has in fact, in certain special cases, e. g., in isothermal processes, proved itself useful. It has persisted, therefore, in certain of its aspects up to the present day; but for the general case, however, it has shown itself as fruitless and, in fact, misleading. The reason for this lies in the fact that the question concerning the lost work in a given

irreversible process is by no means to be answered in a determinate manner, so long as nothing further is specified with regard to the source of energy from which the work considered shall be obtained.

An example will make this clear. Heat conduction is an irreversible process, or as Clausius expresses it: Heat cannot without compensation pass from a colder to a warmer body. What now is the work which in accordance with definition is lost when the quantity of heat Q passes through direct conduction from a warmer body at the temperature T_1 to a colder body, at the temperature T_2? In order to answer this question, we make use of the heat transfer involved in carrying out a reversible Carnot cyclical process between the two bodies employed as heat reservoirs. In this process a certain amount of work would be obtained, and it is just the amount sought, since it is that which would be lost in the direct passage by conduction; but this has no definite value so long as we do not know whence the work originates, whether, e. g., in the warmer body or in the colder body, or from somewhere else. Let one reflect that the heat given up by the warmer body in the reversible process is certainly not equal to the heat absorbed by the colder body, because a certain amount of heat is transformed into work, and that we can identify, with exactly the same right, the quantity of heat Q transferred by the direct process of conduction with that which in the cyclical process is given up by the warmer body, or with that absorbed by the colder body. As one does the former or the latter, he accordingly obtains for the quantity of lost work in the process of conduction:

$$Q \cdot \frac{T_1 - T_2}{T_1} \quad \text{or} \quad Q \cdot \frac{T_1 - T_2}{T_2}.$$

We see, therefore, that the proposed method of expressing mathematically the irreversibility of a process does not in general effect its object, and at the same time we recognize the peculiar reason which prevents its doing so. The statement of the question is too anthropomorphic. It is primarily too much concerned with the needs of mankind, in that it refers directly to the acquirement of useful work. If one require from nature a determinate answer, he must take a more general point of view, more disinterested, less economic. We shall now seek to do this.

Let us consider any typical process occurring in nature. This will carry all bodies concerned in it from a determinate initial state, which I designate as state A, into a determinate final state B. The process is either reversible or irreversible. A third possibility is excluded. But whether it is reversible or irreversible depends solely upon the nature of the two states A and B, and not at all upon the way in which the process has been carried out; for we are only concerned with the answer to the question as to whether or not, when the state B is once reached, a complete return to A in any conceivable manner may be accomplished. If now, the complete return from B to A is not possible, and the process therefore irreversible, it is obvious that the state B may be distinguished in nature through a certain property from state A. Several years ago I ventured to express this as follows: that nature possesses a greater "preference" for state B than for state A. In accordance with this mode of expression, all those processes of nature are impossible for whose final state nature possesses a smaller preference than for the original state. Reversible processes constitute a limiting case; for such, nature possesses an equal preference for the initial and for the final state, and the passage between them takes place as well in one direction as the other.

We have now to seek a physical quantity whose magnitude shall serve as a general measure of the preference of nature for a given state. This quantity must be one which is directly determined by the state of the system considered, without reference to the previous history of the system, as is the case with the energy, with the volume, and with other properties of the system. It should possess the peculiarity of increasing in all irreversible processes and of remaining unchanged in all reversible processes, and the amount of change which it experiences in a process would furnish a general measure for the irreversibility of the process.

R. Clausius actually found this quantity and called it "entropy." Every system of bodies possesses in each of its states a definite entropy, and this entropy expresses the preference of nature for the state in question. It can, in all the processes which take place within the system, only increase and never decrease. If it be desired to consider a process in which external actions upon the system are present, it is necessary to consider those bodies in which these actions originate as constituting part of the system; then the law as stated in the above form is valid. In accordance with it, the entropy of a system of bodies is simply equal to the sum of the entropies of the individual bodies, and the entropy of a single body is, in accordance with Clausius, found by the aid of a certain reversible process. Conduction of heat to a body increases its entropy, and, in fact, by an amount equal to the ratio of the quantity of heat given the body to its temperature. Simple compression, on the other hand, does not change the entropy.

Returning to the example mentioned above, in which the quantity of heat Q is conducted from a warmer body at the temperature T_1 to a colder body at the temperature T_2, in accordance with what precedes, the entropy of the warmer body decreases in this process, while, on the other hand, that of the colder increases, and the sum of both changes, that is, the change of the total entropy of both bodies, is:

$$-\frac{Q}{T_1}+\frac{Q}{T_2}>0.$$

This positive quantity furnishes, in a manner free from all arbitrary assumptions, the measure of the irreversibility of the process of heat conduction. Such examples may be cited indefinitely. Every chemical process furnishes an increase of entropy.

We shall here consider only the most general case treated by Clausius: an arbitrary reversible or irreversible cyclical process, carried out with any physico-chemical arrangement, utilizing an arbitrary number of heat reservoirs. Since the arrangement at the conclusion of the cyclical process is the same as that at the beginning, the final state of the process is to be distinguished from the initial state solely through the different heat content of the heat reservoirs, and in that a certain amount of mechanical work has been furnished or consumed. Let Q be the heat given up in the course of the process by a heat reservoir at the temperature T, and let A be the total work yielded (consisting, e. g., in the raising of weights); then, in accordance with the first law of thermodynamics:

$$\sum Q = A.$$

In accordance with the second law, the sum of the changes in entropy of all the heat reservoirs is positive, or zero. It follows, therefore, since the entropy of a reservoir is decreased by the amount Q/T through the loss of heat Q that:

$$\sum \frac{Q}{T} \leqq 0.$$

This is the well-known inequality of Clausius.

In an isothermal cyclical process, T is the same for all reservoirs. Therefore:

$$\sum Q \leqq 0, \quad \text{hence:} \quad A \leqq 0.$$

That is: in an isothermal cyclical process, heat is produced and work is consumed. In the limiting case, a reversible isothermal cyclical process, the sign of equality holds, and therefore the work consumed is zero, and also the heat produced. This law plays a leading rôle in the application of thermodynamics to physical chemistry.

The second law of thermodynamics including all of its consequences has thus led to the principle of increase of entropy. You will now readily understand, having regard to the questions mentioned above, why I express it as my opinion that in the theoretical physics of the future the first and most important differentiation of all physical processes will be into reversible and irreversible processes.

In fact, all reversible processes, whether they take place in material bodies, in the ether, or in both together, show a much greater similarity among themselves than to any irreversible process. In the differential equations of reversible processes the time differential enters only as an even power, corresponding to the circumstance that the sign of time can be reversed. This holds equally well for vibrations of the pendulum, electrical vibrations, acoustic and optical waves, and for motions of mass points or of electrons, if we only exclude every kind of damping. But to such processes also belong those infinitely slow processes of thermodynamics which consist of states of equilibrium in which the time in general plays no rôle, or, as one may also say, occurs with the zero power, which is to be reckoned as an even power. As Helmholtz has pointed out, all these reversible processes have the common property that they may be completely represented by the principle of least action, which gives a definite answer to all questions concerning any such measurable process, and, to this extent, the theory of reversible processes may be regarded as completely established. Reversible processes have, however, the disadvantage that singly and collectively they are only ideal: in actual nature there is no such thing as a reversible process. Every natural process involves in greater or less degree friction or conduction of heat. But in the domain of irreversible processes the principle of least action is no longer sufficient; for the principle of increase of entropy brings into the system of physics a wholly new element, foreign to the action principle, and which demands special mathematical treatment. The unidirectional course of a process in the attainment of a fixed final state is related to it.

I hope the foregoing considerations have sufficed to make clear to you that the distinction between reversible and irreversible processes is much broader than that between mechanical and electrical processes and that, therefore, this difference, with better right than any other, may be taken advantage of in classifying all physical processes, and that it may eventually play in the theoretical physics of the future the principal rôle.

However, the classification mentioned is in need of quite an essential improvement, for it cannot be denied that in the form set forth, the system of physics is still suffering from a strong dose of anthropomorphism. In the definition of irreversibility, as well as in that of entropy, reference is made to the possibility of carrying out in nature certain changes, and this means, fundamentally, nothing more than that the division of physical processes is made dependent upon the manipulative skill of man in the art of experimentation, which certainly does not always remain at a fixed stage, but

is continually being more and more perfected. If, therefore, the distinction between reversible and irreversible processes is actually to have a lasting significance for all times, it must be essentially broadened and made independent of any reference to the capacities of mankind. How this may happen, I desire to state one week from tomorrow. The lecture of tomorrow will be devoted to the problem of bringing before you some of the most important of the great number of practical consequences following from the entropy principle.

SECOND LECTURE.

Thermodynamic States of Equilibrium in Dilute Solutions.

In the lecture of yesterday I sought to make clear the fact that the essential, and therefore the final division of all processes occurring in nature, is into reversible and irreversible processes, and the characteristic difference between these two kinds of processes, as I have further separated them, is that in irreversible processes the entropy increases, while in all reversible processes it remains constant. Today I am constrained to speak of some of the consequences of this law which will illustrate its rich fruitfulness. They have to do with the question of the laws of thermodynamic equilibrium. Since in nature the entropy can only increase, it follows that the state of a physical configuration which is completely isolated, and in which the entropy of the system possesses an absolute maximum, is necessarily a state of stable equilibrium, since for it no further change is possible. How deeply this law underlies all physical and chemical relations has been shown by no one better and more completely than by John Willard Gibbs, whose name, not only in America, but in the whole world will be counted among those of the most famous theoretical physicists of all times; to whom, to my sorrow, it is no longer possible for me to tender personally my respects. It would be gratuitous for me, here in the land of his activity, to expatiate fully on the progress of his ideas, but you will perhaps permit me to speak in the lecture of today of some of the important applications in which thermodynamic research, based on Gibbs works, can be advanced beyond his results.

These applications refer to the theory of dilute solutions, and we shall occupy ourselves today with these, while I show you by a definite example what fruitfulness is inherent in thermodynamic theory. I shall first characterize the problem quite generally. It has to do with the state of equilibrium of a material system of any number of arbitrary constituents in an arbitrary number of phases, at a given temperature T and given pressure p. If the system is completely isolated, and therefore guarded against all external thermal and mechanical actions, then in any ensuing change the entropy of the system will increase:

$$dS > 0.$$

But if, as we assume, the system stands in such relation to its surroundings that in any change which the system undergoes the temperature T and the pressure p are maintained constant, as, for instance, through its introduction into a calorimeter of great heat capacity and through loading with a piston of fixed weight, the inequality would suffer a change thereby. We must then take account of the fact that the surrounding bodies also, e. g., the calorimetric liquid, will be involved in the change. If we denote the entropy of the surrounding bodies by S_0, then the following more general equation holds:

$$dS + dS_0 > 0.$$

In this equation

$$dS_0 = -\frac{Q}{T},$$

if Q denote the heat which is given up in the change by the surroundings to the system. On the other hand, if U denote the energy, V the volume of the system, then, in

accordance with the first law of thermodynamics,

$$Q = dU + pdV.$$

Consequently, through substitution:

$$dS - \frac{dU + pdV}{T} > 0$$

or, since p and T are constant:

$$d\left(S - \frac{U + pV}{T}\right) > 0.$$

If, therefore, we put:

$$S - \frac{U + pV}{T} = \Phi, \tag{1}$$

then

$$d\Phi > 0,$$

and we have the general law, that in every isothermal-isobaric ($T = \text{const.}$, $p = \text{const.}$) change of state of a physical system the quantity Φ increases. The absolutely stable state of equilibrium of the system is therefore characterized through the maximum of Φ:

$$\delta\Phi = 0. \tag{2}$$

If the system consist of numerous phases, then, because Φ, in accordance with (1), is linear and homogeneous in S, U and V, the quantity Φ referring to the whole system is the sum of the quantities Φ referring to the individual phases. If the expression for Φ is known as a function of the independent variables for each phase of the system, then, from equation (2), all questions concerning the conditions of stable equilibrium may be answered. Now, within limits, this is the case for dilute solutions. By "solution" in thermodynamics is meant each homogeneous phase, in whatever state of aggregation, which is composed of a series of different molecular complexes, each of which is represented by a definite molecular number. If the molecular number of a given complex is great with reference to all the remaining complexes, then the solution is called dilute, and the molecular complex in question is called the solvent; the remaining complexes are called the dissolved substances.

Let us now consider a dilute solution whose state is determined by the pressure p, the temperature T, and the molecular numbers n_0, n_1, n_2, n_3, $\cdots$, wherein the subscript zero refers to the solvent. Then the numbers n_1, n_2, n_3, $\cdots$ are all small with respect to n_0, and on this account the volume V and the energy U are linear functions of the molecular numbers:

$$V = n_0 v_0 + n_1 v_1 + n_2 v_2 + \cdots,$$
$$U = n_0 u_0 + n_1 u_1 + n_2 u_2 + \cdots,$$

wherein the v's and u's depend upon p and T only.

From the general equation of entropy:

$$dS = \frac{dU + pdV}{T},$$

in which the differentials depend only upon changes in p and T, and not in the molecular numbers, there results therefore:

$$dS = n_0 \frac{du_0 + pdv_0}{T} + n_1 \frac{du_1 + pdv_1}{T} + \cdots ,$$

and from this it follows that the expressions multiplied by n_0, n_1 $\cdots$, dependent upon p and T only, are complete differentials. We may therefore write:

$$\frac{du_0 + pdv_0}{T} = ds_0, \quad \frac{du_1 + pdv_1}{T} = ds_1, \ \cdots \tag{3}$$

and by integration obtain:

$$S = n_0 s_0 + n_1 s_1 + n_2 s_2 + \cdots + C.$$

The constant C of integration does not depend upon p and T, but may depend upon the molecular numbers n_0, n_1, n_2, $\cdots$. In order to express this dependence generally, it suffices to know it for a special case, for fixed values of p and T. Now every solution passes, through appropriate increase of temperature and decrease of pressure, into the state of a mixture of ideal gases, and for this case the entropy is fully known, the integration constant being, in accordance with Gibbs:

$$C = -R(n_0 \log c_0 + n_1 \log c_1 + \cdots),$$

wherein R denotes the absolute gas constant and c_0, c_1, c_2, $\cdots$ denote the "molecular concentrations":

$$c_0 = \frac{n_0}{n_0 + n_1 + n_2 + \cdots}, \quad c_1 = \frac{n_1}{n_0 + n_1 + n_2 + \cdots}, \ \cdots .$$

Consequently, quite in general, the entropy of a dilute solution is:

$$S = n_0(s_0 - R \log c_0) + n_1(s_1 - R \log c_1) + \cdots ,$$

and, finally, from this it follows by substitution in equation (1) that:

$$\Phi = n_0(\varphi_0 - R \log c_0) + n_1(\varphi_1 - R \log c_1) + \cdots , \tag{4}$$

if we put for brevity:

$$\varphi_0 = s_0 - \frac{u_0 + pv_0}{T}, \quad \varphi_1 = s_1 - \frac{u_1 + pv_1}{T}, \ \cdots \tag{5}$$

all of which quantities depend only upon p and T.

With the aid of the expression obtained for Φ we are enabled through equation (2) to answer the question with regard to thermodynamic equilibrium. We shall first find the general law of equilibrium and then apply it to a series of particularly interesting special cases.

Every material system consisting of an arbitrary number of homogeneous phases may be represented symbolically in the following way:

$$n_0 m_0,\ n_1 m_1,\ \cdots \mid n_0' m_0',\ n_1' m_1',\ \cdots \mid n_0'' m_0'',\ n_1'' m_1'',\ \cdots \mid \cdots .$$

Here the molecular numbers are denoted by n, the molecular weights by m, and the individual phases are separated from one another by vertical lines. We shall now suppose that each phase represents a dilute solution. This will be the case when each phase contains only a single molecular complex and therefore represents an absolutely pure substance; for then the concentrations of all the dissolved substances will be zero.

If now an isobaric-isothermal change in the system of such kind is possible that the molecular numbers

$$n_0,\, n_1,\, n_2,\, \cdots ,\quad n_0',\, n_1',\, n_2',\, \cdots ,\quad n_0'',\, n_1'',\, n_2'',\, \cdots$$

change simultaneously by the amounts

$$\delta n_0,\, \delta n_1,\, \delta n_2, \cdots ,\quad \delta n_0',\, \delta n_1',\, \delta n_2', \cdots ,\quad \delta n_0'',\, \delta n_1'',\, \delta n_2'', \cdots$$

then, in accordance with equation (2), equilibrium obtains with respect to the occurrence of this change if, when T and p are held constant, the function

$$\Phi + \Phi' + \Phi'' + \cdots$$

is a maximum, or, in accordance with equation (4):

$$\sum(\varphi_0 - R\log c_0)\delta n_0 + (\varphi_1 - R\log c_1)\delta n_1 + \cdots = 0$$

(the summation $\sum$ being extended over all phases of the system). Since we are only concerned in this equation with the ratios of the δn's, we put

$$\delta n_0 : \delta n_1 : \cdots : \delta n_0' : \delta n_1' : \cdots : \delta n_0'' : \delta n_1'' : \cdots \\ = \nu_0 : \nu_1 : \cdots : \nu_0' : \nu_1' : \cdots : \nu_0'' : \nu_1'' : \cdots ,$$

wherein we are to understand by the simultaneously changing ν's, in the variation considered, simple integer positive or negative numbers, according as the molecular complex under consideration is formed or disappears in the change. Then the condition for equilibrium is:

$$\sum\nu_0 \log c_0 + \nu_1 \log c_1 + \cdots = \frac{1}{R}\sum\nu_0\varphi_0 + \nu_1\varphi_1 + \cdots = \log K. \tag{6}$$

K and the quantities φ_0, φ_1, φ_2, $\cdots$ depend only upon p and T, and this dependence is to be found from the equations:

$$\frac{\partial \log K}{\partial p} = \frac{1}{R}\sum\nu_0\frac{\partial\varphi_0}{\partial p} + \nu_1\frac{\partial\varphi_1}{\partial p} + \cdots ,$$
$$\frac{\partial \log K}{\partial T} = \frac{1}{R}\sum\nu_0\frac{\partial\varphi_0}{\partial T} + \nu_1\frac{\partial\varphi_1}{\partial T} + \cdots .$$

Now, in accordance with (5), for any infinitely small change of p and T:

$$d\varphi_0 = ds_0 - \frac{du_0 + pdv_0 + v_0dp}{T} + \frac{u_0 + pv_0}{T^2}\cdot dT,$$

and consequently, from (3):

$$d\varphi_0 = \frac{u_0 + pv_0}{T^2} dT - \frac{v_0 dp}{T},$$

and hence:

$$\frac{\partial \varphi_0}{\partial p} = -\frac{v_0}{T}, \quad \frac{\partial \varphi_0}{\partial T} = \frac{u_0 + pv_0}{T^2}.$$

Similar equations hold for the other φ's, and therefore we get:

$$\frac{\partial \log K}{\partial p} = -\frac{1}{RT} \sum \nu_0 v_0 + \nu_1 v_1 + \cdots,$$

$$\frac{\partial \log K}{\partial T} = -\frac{1}{RT^2} \sum \nu_0 u_0 + \nu_2 u_2 + \cdots + p(\nu_0 v_0 + \nu_1 v_1 + \cdots)$$

or, more briefly:

$$\frac{\partial \log K}{\partial p} = -\frac{1}{RT} \cdot \Delta V, \quad \frac{\partial \log K}{\partial T} = \frac{\Delta Q}{RT^2}, \tag{7}$$

if ΔV denote the change in the total volume of the system and ΔQ the heat which is communicated to it from outside, during the isobaric isothermal change considered. We shall now investigate the import of these relations in a series of important applications.

I. *Electrolytic Dissociation of Water.*

The system consists of a single phase:

$$n_0 H_2O, \quad n_1 \overset{+}{H}, \quad n_2 \overset{-}{HO}.$$

The transformation under consideration

$$\nu_0 : \nu_1 : \nu_2 = \delta n_0 : \delta n_1 : \delta n_2$$

consists in the dissociation of a molecule H_2O into a molecule $\overset{+}{H}$ and a molecule $\overset{-}{HO}$, therefore:

$$\nu_0 = -1, \quad \nu_1 = 1, \quad \nu_2 = 1.$$

Hence, in accordance with (6), for equilibrium:

$$-\log c_0 + \log c_1 + \log c_2 = \log K,$$

or, since $c_1 = c_2$ and $c_0 = 1$, approximately:

$$2 \log c_1 = \log K.$$

The dependence of the concentration c_1 upon the temperature now follows from (7):

$$2 \frac{\partial \log c_1}{\partial T} = \frac{\Delta Q}{RT^2}.$$

ΔQ, the quantity of heat which it is necessary to supply for the dissociation of a molecule of H_2O into the ions $\overset{+}{H}$ and $\overset{-}{HO}$, is, in accordance with Arrhenius, equal to the heat of

ionization in the neutralization of a strong univalent base and acid in a dilute aqueous solution, and, therefore, in accordance with the recent measurements of Wörmann,[1]

$$\Delta Q = 27{,}857 - 48.5T \text{ gr. cal.}$$

Using the number 1.985 for the ratio of the absolute gas constant R to the mechanical equivalent of heat, it follows that:

$$\frac{\partial \log c_1}{\partial T} = \frac{1}{2 \cdot 1.985}\left(\frac{27{,}857}{T^2} - \frac{48.5}{T}\right),$$

and by integration:

$$\overset{10}{\log} c_1 = -\frac{3047.3}{T} - 12.125 \overset{10}{\log} T + \text{const.}$$

This dependence of the degree of dissociation upon the temperature agrees very well with the measurements of the electric conductivity of water at different temperatures by Kohlrausch and Heydweiller, Noyes, and Lundén.

II. *Dissociation of a Dissolved Electrolyte.*

Let the system consists of an aqueous solution of acetic acid:

$$n_0 H_2O, \quad n_1 H_4C_2O_2, \quad n_2 \overset{+}{H}, \quad n_3 H_3 \bar{C}_2 O_2.$$

The change under consideration consists in the dissociation of a molecule $H_4C_2O_2$ into its two ions, therefore

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 1, \quad \nu_3 = 1.$$

Hence, for the state of equilibrium, in accordance with (6):

$$-\log c_1 + \log c_2 + \log c_3 = \log K,$$

or, since $c_2 = c_3$:

$$\frac{{c_2}^2}{c_1} = K.$$

Now the sum $c_1 + c_2 = c$ is to be regarded as known, since the total number of the undissociated and dissociated acid molecules is independent of the degree of dissociation. Therefore c_1 and c_2 may be calculated from K and c. An experimental test of the equation of equilibrium is possible on account of the connection between the degree of dissociation and electrical conductivity of the solution. In accordance with the electrolytic dissociation theory of Arrhenius, the ratio of the molecular conductivity λ of the solution in any dilution to the molecular conductivity λ_∞ of the solution in infinite dilution is:

$$\frac{\lambda}{\lambda_\infty} = \frac{c_2}{c_1 + c_2} = \frac{c_2}{c},$$

[1] Ad Heydweiller, Ann. d. Phys., 28, 506, 1909.

since electric conduction is accounted for by the dissociated molecules only. It follows then, with the aid of the last equation, that:

$$\frac{\lambda^2 c}{\lambda_\infty - \lambda} = K \cdot \lambda_\infty = \text{const.}$$

With unlimited decreasing c, λ increases to λ_∞. This "law of dilution" for binary electrolytes, first enunciated by Ostwald, has been confirmed in numerous cases by experiment, as in the case of acetic acid.

Also, the dependence of the degree of dissociation upon the temperature is indicated here in quite an analogous manner to that in the example considered above, of the dissociation of water.

III. *Vaporization or Solidification of a Pure Liquid.*

In equilibrium the system consists of two phases, one liquid, and one gaseous or solid:

$$n_0 m_0 \mid n_0{}' m_0'.$$

Each phase contains only a single molecular complex (the solvent), but the molecules in both phases do not need to be the same. Now, if a liquid molecule evaporates or solidifies, then in our notation

$$\nu_0 = -1, \quad \nu_0{}' = \frac{m_0}{m_0{}'}, \quad c_0 = 1, \quad c_0{}' = 1,$$

and consequently the condition for equilibrium, in accordance with (6), is:

$$0 = \log K. \tag{8}$$

Since K depends only upon p and T, this equation therefore expresses a definite relation between p and T: the law of dependence of the pressure of vaporization (or melting pressure) upon the temperature, or vice versa. The import of this law is obtained through the consideration of the dependence of the quantity K upon p and T. If we form the complete differential of the last equation, there results:

$$0 = \frac{\partial \log K}{\partial p} dp + \frac{\partial \log K}{\partial T} dT,$$

or, in accordance with (7):

$$0 = -\frac{\Delta V}{T} dp + \frac{\Delta Q}{T^2} dT.$$

If v_0 and $v_0{}'$ denote the molecular volumes of the two phases, then:

$$\Delta V = \frac{m_0 v_0{}'}{m_0{}'} - v_0,$$

consequently:

$$\Delta Q = T \left(\frac{m_0 v_0{}'}{m_0{}'} - v_0 \right) \frac{dp}{dT},$$

or, referred to unit mass:

$$\frac{\Delta Q}{m_0} = T \left(\frac{v_0{}'}{m_0{}'} - \frac{v_0}{m_0} \right) \cdot \frac{dp}{dT},$$

the well-known formula of Carnot and Clapeyron.

IV. *The Vaporization or Solidification of a Solution of Non-Volatile Substances.*

Most aqueous salt solutions afford examples. The symbol of the system in this case is, since the second phase (gaseous or solid) contains only a single molecular complex:

$$n_0 m_0,\ n_1 m_1,\ n_2 m_2,\ \cdots \mid n_0' m_0'.$$

The change is represented by:

$$\nu_0 = -1, \quad \nu_1 = 0, \quad \nu_2 = 0, \quad \cdots \quad \nu_0' = \frac{m_0}{m_0'},$$

and hence the condition of equilibrium, in accordance with (6), is:

$$-\log c_0 = \log K,$$

or, since to small quantities of higher order:

$$c_0 = \frac{n_0}{n_0 + n_1 + n_2 + \cdots} = 1 - \frac{n_1 + n_2 + \cdots}{n_0},$$

$$\frac{n_1 + n_2 + \cdots}{n_0} = \log K. \tag{9}$$

A comparison with formula (8), found in example III, shows that through the solution of a foreign substance there is involved in the total concentration a small proportionate departure from the law of vaporization or solidification which holds for the pure solvent. One can express this, either by saying: at a fixed pressure p, the boiling point or the freezing point T of the solution is different than that (T_0) for the pure solvent, or: at a fixed temperature T the vapor pressure or solidification pressure p of the solution is different from that (p_0) of the pure solvent. Let us calculate the departure in both cases.

1. If T_0 be the boiling (or freezing temperature) of the pure solvent at the pressure p, then, in accordance with (8):

$$(\log K)_{T=T_0} = 0,$$

and by subtraction of (9) there results:

$$\log K - (\log K)_{T=T_0} = \frac{n_1 + n_2 + \cdots}{n_0}.$$

Now, since T is little different from T_0, we may write in place of this equation, with the aid of (7):

$$\frac{\partial \log K}{\partial T}(T - T_0) = \frac{\Delta Q}{R T_0^2}(T - T_0) = \frac{n_1 + n_2 + \cdots}{n_0},$$

and from this it follows that:

$$T - T_0 = \frac{n_1 + n_2 + \cdots}{n_0} \cdot \frac{R T_0^2}{\Delta Q}. \tag{10}$$

This is the law for the raising of the boiling point or for the lowering of the freezing point, first derived by van't Hoff: in the case of freezing ΔQ (the heat taken from the

surroundings during the freezing of a liquid molecule) is negative. Since n_0 and ΔQ occur only as a product, it is not possible to infer anything from this formula with regard to the molecular number of the liquid solvent.

2. If p_0 be the vapor pressure of the pure solvent at the temperature T, then, in accordance with (8):

$$(\log K)_{p=p_0} = 0,$$

and by subtraction of (9) there results:

$$\log K - (\log K)_{p=p_0} = \frac{n_1 + n_2 + \cdots}{n_0}.$$

Now, since p and p_0 are nearly equal, with the aid of (7) we may write:

$$\frac{\partial \log K}{\partial p}(p - p_0) = -\frac{\Delta V}{RT}(p - p_0) = \frac{n_1 + n_2 + \cdots}{n_0},$$

and from this it follows, if ΔV be placed equal to the volume of the gaseous molecule produced in the vaporization of a liquid molecule:

$$\Delta V = \frac{m_0}{m_0'}\frac{RT}{p},$$
$$\frac{p_0 - p}{p} = \frac{m_0'}{m_0} \cdot \frac{n_1 + n_2 + \cdots}{n_0}.$$

This is the law of relative depression of the vapor pressure, first derived by van't Hoff. Since n_0 and m_0 occur only as a product, it is not possible to infer from this formula anything with regard to the molecular weight of the liquid solvent. Frequently the factor m_0'/m_0 is left out in this formula; but this is not allowable when m_0 and m_0' are unequal (as, e. g., in the case of water).

V. *Vaporization of a Solution of Volatile Substances.*

(*E. g.., a Sufficiently Dilute Solution of Propyl Alcohol in Water.*)

The system, consisting of two phases, is represented by the following symbol:

$$n_0 m_0,\ n_1 m_1,\ n_2 m_2,\ \cdots \mid n_0' m_0',\ n_1' m_1',\ n_2' m_2',\ \cdots,$$

wherein, as above, the figure 0 refers to the solvent and the figures 1, 2, 3 $\cdots$ refer to the various molecular complexes of the dissolved substances. By the addition of primes in the case of the molecular weights (m_0', m_1', m_2' $\cdots$) the possibility is left open that the various molecular complexes in the vapor may possess a different molecular weight than in the liquid.

Since the system here considered may experience various sorts of changes, there are also various conditions of equilibrium to fulfill, each of which relates to a definite sort of transformation. Let us consider first that change which consists in the vaporization of the solvent. In accordance with our scheme of notation, the following conditions hold:

$$\nu_0 = -1,\ \nu_1 = 0,\ \nu_2 = 0,\ \cdots\ \nu_0' = \frac{m_0}{m_0'},\ \nu_1' = 0,\ \nu_2' = 0,\ \cdots,$$

and, therefore, the condition of equilibrium (6) becomes:

$$-\log c_0 + \frac{m_0}{m_0{}'}\log c_0{}' = \log K,$$

or, if one substitutes:

$$c_0 = 1 - \frac{n_1 + n_2 + \cdots}{n_0} \quad \text{and} \quad c_0{}' = 1 - \frac{n_1{}' + n_2{}' + \cdots}{n_0{}'},$$
$$\frac{n_1 + n_2 + \cdots}{n_0} - \frac{m_0}{m_0{}'} \cdot \frac{n_1{}' + n_2{}' + \cdots}{n_0{}'} = \log K.$$

If we treat this equation upon equation (9) as a model, there results an equation similar to (10):

$$T - T_0 = \left(\frac{n_1 + n_2 + \cdots}{n_0 m_0} - \frac{n_1{}' + n_2{}' + \cdots}{n_0{}' m_0{}'}\right)\frac{RT_0^2 m_0}{\Delta Q}.$$

Here ΔQ is the heat effect in the vaporization of one molecule of the solvent and, therefore, $\Delta Q/m_0$ is the heat effect in the vaporization of a unit mass of the solvent.

We remark, once more, that the solvent always occurs in the formula through the mass only, and not through the molecular number or the molecular weight, while, on the other hand, in the case of the dissolved substances, the molecular state is characteristic on account of their influence upon vaporization. Finally, the formula contains a generalization of the law of van't Hoff, stated above, for the raising of the boiling point, in that here in place of the number of dissolved molecules in the liquid, the difference between the number of dissolved molecules in unit mass of the liquid and in unit mass of the vapor appears. According as the unit mass of liquid or the unit mass of vapor contains more dissolved molecules, there results for the solution a raising or lowering of the boiling point; in the limiting case, when both quantities are equal, and the mixture therefore boils without changing, the change in boiling point becomes equal to zero. Of course, there are corresponding laws holding for the change in the vapor pressure.

Let us consider now a change which consists in the vaporization of a dissolved molecule. For this case we have in our notation

$$\nu_0 = 0,\ \nu_1 = -1,\ \nu_2 = 0\ \cdots,\ \nu_0{}' = 0,\ \nu_1{}' = \frac{m_1}{m_1{}'},\ \nu_2{}' = 0,\ \cdots$$

and, in accordance with (6), for the condition of equilibrium:

$$-\log c_1 + \frac{m_1}{m_1{}'}\log c_1{}' = \log K$$

or:

$$\frac{{c_1{}'}^{\frac{m_1}{m_1{}'}}}{c_1} = K.$$

This equation expresses the Nernst law of distribution. If the dissolved substance possesses in both phases the same molecular weight ($m_1 = m_1{}'$), then, in a state of equilibrium a fixed ratio of the concentrations c_1 and $c_1{}'$ in the liquid and in the vapor exists, which depends only upon the pressure and temperature. But, if the dissolved substance polymerises somewhat in the liquid, then the relation demanded in the last equation appears in place of the simple ratio.

VI. *The Dissolved Substance only Passes over into the Second Phase.*

This case is in a certain sense a special case of the one preceding. To it belongs that of the solubility of a slightly soluble salt, first investigated by van't Hoff, e. g., succinic acid in water. The symbol of this system is:

$$n_0 H_2O,\ n_1 H_6C_4O_4 \mid n_0' H_6C_4O_4,$$

in which we disregard the small dissociation of the acid solution. The concentrations of the individual molecular complexes are:

$$c_0 = \frac{n_0}{n_0 + n_1}, \quad c_1 = \frac{n_1}{n_0 + n_1}, \quad c_0' = \frac{n_0'}{n_0'} = 1.$$

For the precipitation of solid succinic acid we have:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_0' = 1,$$

and, therefore, from the condition of equilibrium (6):

$$-\log c_1 = \log K,$$

hence, from (7):

$$\Delta Q = -RT^2 \frac{\partial \log c_1}{\partial T}.$$

By means of this equation van't Hoff calculated the heat of solution ΔQ from the solubility of succinic acid at 0° and at 8.5° C. The corresponding numbers were 2.88 and 4.22 in an arbitrary unit. Approximately, then:

$$\frac{\partial \log c_1}{\partial T} = \frac{\overset{e}{\log}\ 4.22 - \overset{e}{\log}\ 2.88}{8.5} = 0.04494,$$

from which for $T = 273$:

$$\Delta Q = -1.98 \cdot 273^2 \cdot 0.04494 = -6{,}600 \text{ cal.},$$

that is, in the precipitation of a molecule of succinic acid, 6,600 cal. are given out to the surroundings. Berthelot found, however, through direct measurement, 6,700 calories for the heat of solution.

The absorption of a gas also comes under this head, e. g. carbonic acid, in a liquid of relatively unnoticeable smaller vapor pressure, e. g., water at not too high a temperature. The symbol of the system is then

$$n_0 H_2O,\ n_1 CO_2 \mid n_0' CO_2.$$

The vaporization of a molecule CO_2 corresponds to the values

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_0' = 1.$$

The condition of equilibrium is therefore again:

$$-\log c_1 = \log K,$$

i. e., at a fixed temperature and a fixed pressure the concentration c_1 of the gas in the solution is constant. The change of the concentration with p and T is obtained through substitution in equation (7). It follows from this that:

$$\frac{\partial \log c_1}{\partial p} = \frac{\Delta V}{RT}, \quad \frac{\partial \log c_1}{\partial T} = -\frac{\Delta Q}{RT^2}.$$

ΔV is the change in volume of the system which occurs in the isobaric-isothermal vaporization of a molecule of CO_2, ΔQ the quantity of heat absorbed in the process from outside. Now, since ΔV represents approximately the volume of a molecule of gaseous carbonic acid, we may put approximately:

$$\Delta V = \frac{RT}{p},$$

and the equation gives:

$$\frac{\partial \log c_1}{\partial p} = \frac{1}{p},$$

which integrated, gives:

$$\log c_1 = \log p + \text{const.}, \quad c_1 = C \cdot p,$$

i. e., the concentration of the dissolved gas is proportional to the pressure of the free gas above the solution (law of Henry and Bunsen). The factor of proportionality C, which furnishes a measure of the solubility of the gas, depends upon the heat effect in quite the same manner as in the example previously considered.

A number of no less important relations are easily derived as by-products of those found above, e. g., the Nernst laws concerning the influence of solubility, the Arrhenius theory of isohydric solutions, etc. All such may be obtained through the application of the general condition of equilibrium (6). In conclusion, there is one other case that I desire to treat here. In the historical development of the theory this has played a particularly important rôle.

VII. *Osmotic Pressure.*

We consider now a dilute solution separated by a membrane (permeable with regard to the solvent but impermeable as regards the dissolved substance) from the pure solvent (in the same state of aggregation), and inquire as to the condition of equilibrium. The symbol of the system considered we may again take as

$$n_0 m_0,\ n_1 m_1,\ n_2 m_2,\ \cdots \mid n_0' m_0.$$

The condition of equilibrium is also here again expressed by equation (6), valid for a change of state in which the temperature and the pressure in each phase is maintained constant. The only difference with respect to the cases treated earlier is this, that here, in the presence of a separating membrane between two phases, the pressure p in the first phase may be different from the pressure p' in the second phase, whereby by "pressure," as always, is to be understood the ordinary hydrostatic or manometric pressure.

The proof of the applicability of equation (6) is found in the same way as this equation was derived above, proceeding from the principle of increase of entropy. One has but to remember that, in the somewhat more general case here considered, the external work in a given change is represented by the sum $pdV + p'dV'$, where V and V' denote the volumes of the two individual phases, while before V denoted the total volume of all phases. Accordingly, we use, instead of (7), to express the dependence of the constant K in (6) upon the pressure:

$$\frac{\partial \log K}{\partial p} = -\frac{\Delta V}{RT}, \quad \frac{\partial \log K}{\partial p'} = -\frac{\Delta V'}{RT}. \tag{11}$$

We have here to do with the following change:

$$\nu_0 = -1, \quad \nu_1 = 0, \quad \nu_2 = 0, \quad \cdots, \quad \nu_0{}' = 1,$$

whereby is expressed, that a molecule of the solvent passes out of the solution through the membrane into the pure solvent. Hence, in accordance with (6):

$$-\log c_0 = \log K,$$

or, since

$$c_0 = 1 - \frac{n_1 + n_2 + \cdots}{n_0}, \quad \frac{n_1 + n_2 + \cdots}{n_0} = \log K.$$

Here K depends only upon T, p and p'. If a pure solvent were present upon both sides of the membrane, we should have $c_0 = 1$, and $p = p'$; consequently:

$$(\log K)_{p=p'} = 0,$$

and by subtraction of the last two equations:

$$\frac{n_1 + n_2 + \cdots}{n_0} = \log K - (\log K)_{p=p'} = \frac{\partial \log K}{\partial p}(p - p')$$

and in accordance with (11):

$$\frac{n_1 + n_2 + \cdots}{n_0} = -(p - p') \cdot \frac{\Delta V}{RT}.$$

Here ΔV denotes the change in volume of the solution due to the loss of a molecule of the solvent ($\nu_0 = -1$). Approximately then:

$$-\Delta V \cdot n_0 = V,$$

the volume of the whole solution, and

$$\frac{n_1 + n_2 + \cdots}{n_0} = (p - p') \cdot \frac{V}{RT}.$$

If we call the difference $p - p'$, the osmotic pressure of the solution, this equation contains the well known law of osmotic pressure, due to van't Hoff.

The equations here derived, which easily permit of multiplication and generalization, have, of course, for the most part not been derived in the ways described above, but have been derived, either directly from experiment, or theoretically from the consideration of special reversible isothermal cycles to which the thermodynamic law was applied, that in such a cyclic process not only the algebraic sum of the work produced and the heat produced, but that also each of these two quantities separately, is equal to zero (first lecture, p. 18). The employment of a cyclic process has the advantage over the procedure here proposed, that in it the connection between the directly measurable quantities and the requirements of the laws of thermodynamics succinctly appears in each case; but for each individual case a satisfactory cyclic process must be imagined, and one has not always the certain assurance that the thermodynamic realization of the cyclic process also actually supplies all the conditions of equilibrium. Furthermore, in the process of calculation certain terms of considerable weight frequently appear as empty ballast, since they disappear at the end in the summation over the individual phases of the process.

On the other hand, the significance of the process here employed consists therein, that the necessary and sufficient conditions of equilibrium for each individually considered case appear collectively in the single equation (6), and that they are derived collectively from it in a direct manner through an unambiguous procedure. The more complicated the systems considered are, the more apparent becomes the advantage of this method, and there is no doubt in my mind that in chemical circles it will be more and more employed, especially, since in general it is now the custom to deal directly with the energies, and not with cyclic processes, in the calculation of heat effects in chemical changes.

THIRD LECTURE.

The Atomic Theory of Matter.

The problem with which we shall be occupied in the present lecture is that of a closer investigation of the atomic theory of matter. It is, however, not my intention to introduce this theory with nothing further, and to set it up as something apart and disconnected with other physical theories, but I intend above all to bring out the peculiar significance of the atomic theory as related to the present general system of theoretical physics; for in this way only will it be possible to regard the whole system as one containing within itself the essential compact unity, and thereby to realize the principal object of these lectures.

Consequently it is self evident that we must rely on that sort of treatment which we have recognized in last week's lecture as fundamental. That is, the division of all physical processes into reversible and irreversible processes. Furthermore, we shall be convinced that the accomplishment of this division is only possible through the atomic theory of matter, or, in other words, that irreversibility leads of necessity to atomistics.

I have already referred at the close of the first lecture to the fact that in pure thermodynamics, which knows nothing of an atomic structure and which regards all substances as absolutely continuous, the difference between reversible and irreversible processes can only be defined in one way, which a priori carries a provisional character and does not withstand penetrating analysis. This appears immediately evident when one reflects that the purely thermodynamic definition of irreversibility which proceeds from the impossibility of the realization of certain changes in nature, as, e. g., the transformation of heat into work without compensation, has at the outset assumed a definite limit to man's mental capacity, while, however, such a limit is not indicated in reality. On the contrary: mankind is making every endeavor to press beyond the present boundaries of its capacity, and we hope that later on many things will be attained which, perhaps, many regard at present as impossible of accomplishment. Can it not happen then that a process, which up to the present has been regarded as irreversible, may be proved, through a new discovery or invention, to be reversible? In this case the whole structure of the second law would undeniably collapse, for the irreversibility of a single process conditions that of all the others.

It is evident then that the only means to assure to the second law real meaning consists in this, that the idea of irreversibility be made independent of any relationship to man and especially of all technical relations.

Now the idea of irreversibility harks back to the idea of entropy; for a process is irreversible when it is connected with an increase of entropy. The problem is hereby referred back to a proper improvement of the definition of entropy. In accordance with the original definition of Clausius, the entropy is measured by means of a certain reversible process, and the weakness of this definition rests upon the fact that many such reversible processes, strictly speaking all, are not capable of being carried out in practice. With some reason it may be objected that we have here to do, not with an actual process and an actual physicist, but only with ideal processes, so-called thought experiments, and with an ideal physicist who operates with all the experimental methods with absolute accuracy. But at this point the difficulty is encountered: How far do the physicist's ideal measurements of this sort suffice? It may be understood, by passing to the limit,

that a gas is compressed by a pressure which is equal to the pressure of the gas, and is heated by a heat reservoir which possesses the same temperature as the gas, but, for example, that a saturated vapor shall be transformed through isothermal compression in a reversible manner to a liquid without at any time a part of the vapor being condensed, as in certain thermodynamic considerations is supposed, must certainly appear doubtful. Still more striking, however, is the liberty as regards thought experiments, which in physical chemistry is granted the theorist. With his semi-permeable membranes, which in reality are only realizable under certain special conditions and then only with a certain approximation, he separates in a reversible manner, not only all possible varieties of molecules, whether or not they are in stable or unstable conditions, but he also separates the oppositely charged ions from one another and from the undissociated molecules, and he is disturbed, neither by the enormous electrostatic forces which resist such a separation, nor by the circumstance that in reality, from the beginning of the separation, the molecules become in part dissociated while the ions in part again combine. But such ideal processes are necessary throughout in order to make possible the comparison of the entropy of the undissociated molecules with the entropy of the dissociated molecules; for the law of thermodynamic equilibrium does not permit in general of derivation in any other way, in case one wishes to retain pure thermodynamics as a basis. It must be considered remarkable that all these ingenious thought processes have so well found confirmation of their results in experience, as is shown by the examples considered by us in the last lecture.

If now, on the other hand, one reflects that in all these results every reference to the possibility of actually carrying out each ideal process has disappeared—there are certainly left relations between directly measurable quantities only, such as temperature, heat effect, concentration, etc.—the presumption forces itself upon one that perhaps the introduction as above of such ideal processes is at bottom a round-about method, and that the peculiar import of the principle of increase of entropy with all its consequences can be evolved from the original idea of irreversibility or, just as well, from the impossibility of perpetual motion of the second kind, just as the principle of conservation of energy has been evolved from the law of impossibility of perpetual motion of the first kind.

This step: to have completed the emancipation of the entropy idea from the experimental art of man and the elevation of the second law thereby to a real principle, was the scientific life's work of Ludwig Boltzmann. Briefly stated, it consisted in general of referring back the idea of entropy to the idea of probability. Thereby is also explained, at the same time, the significance of the above (p. 17) auxiliary term used by me; "preference" of nature for a definite state. Nature prefers the more probable states to the less probable, because in nature processes take place in the direction of greater probability. Heat goes from a body at higher temperature to a body at lower temperature because the state of equal temperature distribution is more probable than a state of unequal temperature distribution.

Through this conception the second law of thermodynamics is removed at one stroke from its isolated position, the mystery concerning the preference of nature vanishes, and the entropy principle reduces to a well understood law of the calculus of probability.

The enormous fruitfulness of so "objective" a definition of entropy for all domains of physics I shall seek to demonstrate in the following lectures. But today we have

principally to do with the proof of its admissibility; for on closer consideration we shall immediately perceive that the new conception of entropy at once introduces a great number of questions, new requirements and difficult problems. The first requirement is the introduction of the atomic hypothesis into the system of physics. For, if one wishes to speak of the probability of a physical state, i. e., if he wishes to introduce the probability for a given state as a definite quantity into the calculation, this can only be brought about, as in cases of all probability calculations, by referring the state back to a variety of possibilities; i. e., by considering a finite number of a priori equally likely configurations (complexions) through each of which the state considered may be realized. The greater the number of complexions, the greater is the probability of the state. Thus, e. g., the probability of throwing a total of four with two ordinary six-sided dice is found through counting the complexions by which the throw with a total of four may be realized. Of these there are three complexions:

with the first die, 1, with the second die, 3,
with the first die, 2, with the second die, 2,
with the first die, 3, with the second die, 1.

On the other hand, the throw of two is only realized through a single complexion. Therefore, the probability of throwing a total of four is three times as great as the probability of throwing a total of two.

Now, in connection with the physical state under consideration, in order to be able to differentiate completely from one another the complexions realizing it, and to associate it with a definite reckonable number, there is obviously no other means than to regard it as made up of numerous discrete homogeneous elements—for in perfectly continuous systems there exist no reckonable elements—and hereby the atomistic view is made a fundamental requirement. We have, therefore, to regard all bodies in nature, in so far as they possess an entropy, as constituted of atoms, and we therefore arrive in physics at the same conception of matter as that which obtained in chemistry for so long previously.

But we can immediately go a step further yet. The conclusions reached hold, not only for thermodynamics of material bodies, but also possess complete validity for the processes of heat radiation, which are thus referred back to the second law of thermodynamics. That radiant heat also possesses an entropy follows from the fact that a body which emits radiation into a surrounding diathermanous medium experiences a loss of heat and, therefore, a decrease of entropy. Since the total entropy of a physical system can only increase, it follows that one part of the entropy of the whole system, consisting of the body and the diathermanous medium, must be contained in the radiated heat. If the entropy of the radiant heat is to be referred back to the notion of probability, we are forced, in a similar way as above, to the conclusion that for radiant heat the atomic conception possesses a definite meaning. But, since radiant heat is not directly connected with matter, it follows that this atomistic conception relates, not to matter, but only to energy, and hence, that in heat radiation certain energy elements play an essential rôle. Even though this conclusion appears so singular and even though in many circles today vigorous objection is strongly urged against it, in the long run physical research will not be able to withhold its sanction from it, and the less, since it is confirmed by experience in quite a satisfactory manner. We shall return to this

point in the lectures on heat radiation. I desire here only to mention that the novelty involved by the introduction of atomistic conceptions into the theory of heat radiation is by no means so revolutionary as, perhaps, might appear at the first glance. For there is, in my opinion at least, nothing which makes necessary the consideration of the heat processes in a complete vacuum as atomic, and it suffices to seek the atomistic features at the source of radiation, i. e., in those processes which have their play in the centres of emission and absorption of radiation. Then the Maxwellian electrodynamic differential equations can retain completely their validity for the vacuum, and, besides, the discrete elements of heat radiation are relegated exclusively to a domain which is still very mysterious and where there is still present plenty of room for all sorts of hypotheses.

Returning to more general considerations, the most important question comes up as to whether, with the introduction of atomistic conceptions and with the reference of entropy to probability, the content of the principle of increase of entropy is exhaustively comprehended, or whether still further physical hypotheses are required in order to secure the full import of that principle. If this important question had been settled at the time of the introduction of the atomic theory into thermodynamics, then the atomistic views would surely have been spared a large number of conceivable misunderstandings and justifiable attacks. For it turns out, in fact—and our further considerations will confirm this conclusion—that there has as yet nothing been done with atomistics which in itself requires much more than an essential generalization, in order to guarantee the validity of the second law.

We must first reflect that, in accordance with the central idea laid down in the first lecture (p. 11), the second law must possess validity as an objective physical law, independently of the individuality of the physicist. There is nothing to hinder us from imagining a physicist—we shall designate him a "microscopic" observer—whose senses are so sharpened that he is able to recognize each individual atom and to follow it in its motion. For this observer each atom moves exactly in accordance with the elementary laws which general dynamics lays down for it, and these laws allow, so far as we know, of an inverse performance of every process. Accordingly, here again the question is neither one of probability nor of entropy and its increase. Let us imagine, on the other hand, another observer, designated a "macroscopic" observer, who regards an ensemble of atoms as a homogeneous gas, say, and consequently applies the laws of thermodynamics to the mechanical and thermal processes within it. Then, for such an observer, in accordance with the second law, the process in general is an irreversible process. Would not now the first observer be justified in saying: "The reference of the entropy to probability has its origin in the fact that irreversible processes ought to be explained through reversible processes. At any rate, this procedure appears to me in the highest degree dubious. In any case, I declare each change of state which takes place in the ensemble of atoms designated a gas, as reversible, in opposition to the macroscopic observer." There is not the slightest thing, so far as I know, that one can urge against the validity of these statements. But do we not thereby place ourselves in the painful position of the judge who declared in a trial the correctness of the position of each separately of two contending parties and then, when a third contends that only one of the parties could emerge from the process victorious, was obliged to declare him also correct? Fortunately we find ourselves in a more favorable position. We can certainly mediate between the two parties without its being necessary for one or the other to give

up his principal point of view. For closer consideration shows that the whole controversy rests upon a misunderstanding—a new proof of how necessary it is before one begins a controversy to come to an understanding with his opponent concerning the subject of the quarrel. Certainly, a given change of state cannot be both reversible and irreversible. But the one observer connects a wholly different idea with the phrase "change of state" than the other. What is then, in general, a change of state? The state of a physical system cannot well be otherwise defined than as the aggregate of all those physical quantities, through whose instantaneous values the time changes of the quantities, with given boundary conditions, are uniquely determined. If we inquire now, in accordance with the import of this definition, of the two observers as to what they understand by the state of the collection of atoms or the gas considered, they will give quite different answers. The microscopic observer will mention those quantities which determine the position and the velocities of all the individual atoms. There are present in the simplest case, namely, that in which the atoms may be considered as material points, six times as many quantities as atoms, namely, for each atom the three coordinates and the three velocity components, and in the case of combined molecules, still more quantities. For him the state and the progress of a process is then first determined when all these various quantities are individually given. We shall designate the state defined in this way the "micro-state." The macroscopic observer, on the other hand, requires fewer data. He will say that the state of the homogeneous gas considered by him is determined by the density, the visible velocity and the temperature at each point of the gas, and he will expect that, when these quantities are given, their time variations and, therefore, the progress of the process, to be completely determined in accordance with the two laws of thermo-dynamics, and therefore accompanied by an increase in entropy. In this connection he can call upon all the experience at his disposal, which will fully confirm his expectation. If we call this state the "macro-state," it is clear that the two laws: "the micro-changes of state are reversible" and "the macro-changes of state are irreversible," lie in wholly different domains and, at any rate, are not contradictory.

But now how can we succeed in bringing the two observers to an understanding? This is a question whose answer is obviously of fundamental significance for the atomic theory. First of all, it is easy to see that the macro-observer reckons only with mean values; for what he calls density, visible velocity and temperature of the gas are, for the micro-observer, certain mean values, statistical data, which are derived from the space distribution and from the velocities of the atoms in an appropriate manner. But the micro-observer cannot operate with these mean values alone, for, if these are given at one instant of time, the progress of the process is not determined throughout; on the contrary: he can easily find with given mean values an enormously large number of individual values for the positions and the velocities of the atoms, all of which correspond with the same mean values and which, in spite of this, lead to quite different processes with regard to the mean values. It follows from this of necessity that the micro-observer must either give up the attempt to understand the unique progress, in accordance with experience, of the macroscopic changes of state—and this would be the end of the atomic theory—or that he, through the introduction of a special physical hypothesis, restrict in a suitable manner the manifold of micro-states considered by him. There is certainly nothing to prevent him from assuming that not all conceivable micro-states are realizable in nature, and that certain of them are in fact thinkable, but

never actually realized. In the formularization of such a hypothesis, there is of course no point of departure to be found from the principles of dynamics alone; for pure dynamics leaves this case undetermined. But on just this account any dynamical hypothesis, which involves nothing further than a closer specification of the micro-states realized in nature, is certainly permissible. Which hypothesis is to be given the preference can only be decided through comparison of the results to which the different possible hypotheses lead in the course of experience.

In order to limit the investigation in this way, we must obviously fix our attention only upon all imaginable configurations and velocities of the individual atoms which are compatible with determinate values of the density, the velocity and the temperature of the gas, or in other words: we must consider all the micro-states which belong to a determinate macro-state, and must investigate the various kinds of processes which follow in accordance with the fixed laws of dynamics from the different micro-states. Now, precise calculation has in every case always led to the important result that an enormously large number of these different micro-processes relate to one and the same macro-process, and that only proportionately few of the same, which are distinguished by quite special exceptional conditions concerning the positions and the velocities of neighboring atoms, furnish exceptions. Furthermore, it has also shown that one of the resulting macro-processes is that which the macroscopic observer recognizes, so that it is compatible with the second law of thermodynamics.

Here, manifestly, the bridge of understanding is supplied. The micro-observer needs only to assimilate in his theory the physical hypothesis that all those special cases in which special exceptional conditions exist among the neighboring configurations of interacting atoms do not occur in nature, or, in other words, that the micro-states are in elementary disorder. Then the uniqueness of the macroscopic process is assured and with it, also, the fulfillment of the principle of increase of entropy in all directions.

Therefore, it is not the atomic distribution, but rather the hypothesis of elementary disorder, which forms the real kernel of the principle of increase of entropy and, therefore, the preliminary condition for the existence of entropy. Without elementary disorder there is neither entropy nor irreversible process.[2] Therefore, a single atom can never possess an entropy; for we cannot speak of disorder in connection with it. But with a fairly large number of atoms, say 100 or 1,000, the matter is quite different. Here, one can certainly speak of a disorder, in case that the values of the coordinates and the velocity components are distributed among the atoms in accordance with the laws of accident. Then it is possible to calculate the probability for a given state. But how is it with regard to the increase of entropy? May we assert that the motion of 100 atoms is irreversible? Certainly not; but this is only because the state of 100 atoms

[2]To those physicists who, in spite of all this, regard the hypothesis of elementary disorder as gratuitous or as incorrect, I wish to refer the simple fact that in every calculation of a coefficient of friction, of diffusion, or of heat conduction, from molecular considerations, the notion of elementary disorder is employed, whether tacitly or otherwise, and that it is therefore essentially more correct to stipulate this condition instead of ignoring or concealing it. But he who regards the hypothesis of elementary disorder as self-evident, should be reminded that, in accordance with a law of H. Poincaré, the precise investigation concerning the foundation of which would here lead us too far, the assumption of this hypothesis for all times is unwarranted for a closed space with absolutely smooth walls,—an important conclusion, against which can only be urged the fact that absolutely smooth walls do not exist in nature.

cannot be defined in a thermodynamic sense, since the process does not proceed in a unique manner from the standpoint of a macro-observer, and this requirement forms, as we have seen above, the foundation and preliminary condition for the definition of a thermodynamic state.

If one therefore asks: How many atoms are at least necessary in order that a process may be considered irreversible?, the answer is: so many atoms that one may form from them definite mean values which define the state in a macroscopic sense. One must reflect that to secure the validity of the principle of increase of entropy there must be added to the condition of elementary disorder still another, namely, that the number of the elements under consideration be sufficiently large to render possible the formation of definite mean values. The second law has a meaning for these mean values only; but for them, it is quite exact, just as exact as the law of the calculus of probability, that the mean value, so far as it may be defined, of a sufficiently large number of throws with a six-sided die, is $3\frac{1}{2}$.

These considerations are, at the same time, capable of throwing light upon questions such as the following: Does the principle of increase of entropy possess a meaning for the so-called Brownian molecular movement of a suspended particle? Does the kinetic energy of this motion represent useful work or not? The entropy principle is just as little valid for a single suspended particle as for an atom, and therefore is not valid for a few of them, but only when there is so large a number that definite mean values can be formed. That one is able to see the particles and not the atoms makes no material difference; because the progress of a process does not depend upon the power of an observing instrument. The question with regard to useful work plays no rôle in this connection; strictly speaking, this possesses, in general, no objective physical meaning. For it does not admit of an answer without reference to the scheme of the physicist or technician who proposes to make use of the work in question. The second law, therefore, has fundamentally nothing to do with the idea of useful work (cf. first lecture, p. 16).

But, if the entropy principle is to hold, a further assumption is necessary, concerning the various disordered elements,—an assumption which tacitly is commonly made and which we have not previously definitely expressed. It is, however, not less important than those referred to above. The elements must actually be of the same kind, or they must at least form a number of groups of like kind, e. g., constitute a mixture in which each kind of element occurs in large numbers. For only through the similarity of the elements does it come about that order and law can result in the larger from the smaller. If the molecules of a gas be all different from one another, the properties of a gas can never show so simple a law-abiding behavior as that which is indicated by thermodynamics. In fact, the calculation of the probability of a state presupposes that all complexions which correspond to the state are a priori equally likely. Without this condition one is just as little able to calculate the probability of a given state as, for instance, the probability of a given throw with dice whose sides are unequal in size. In summing up we may therefore say: the second law of thermodynamics in its objective physical conception, freed from anthropomorphism, relates to certain mean values which are formed from a large number of disordered elements of the same kind.

The validity of the principle of increase of entropy and of the irreversible progress of thermodynamic processes in nature is completely assured in this formularization. After the introduction of the hypothesis of elementary disorder, the microscopic observer can

no longer confidently assert that each process considered by him in a collection of atoms is reversible; for the motion occurring in the reverse order will not always obey the requirements of that hypothesis. In fact, the motions of single atoms are always reversible, and thus far one may say, as before, that the irreversible processes appear reduced to a reversible process, but the phenomenon as a whole is nevertheless irreversible, because upon reversal the disorder of the numerous individual elementary processes would be eliminated. Irreversibility is inherent, not in the individual elementary processes themselves, but solely in their irregular constitution. It is this only which guarantees the unique change of the macroscopic mean values.

Thus, for example, the reverse progress of a frictional process is impossible, in that it would presuppose elementary arrangement of interacting neighboring molecules. For the collisions between any two molecules must thereby possess a certain distinguishing character, in that the velocities of two colliding molecules depend in a definite way upon the place at which they meet. In this way only can it happen that in collisions like directed velocities ensue and, therefore, visible motion.

Previously we have only referred to the principle of elementary disorder in its application to the atomic theory of matter. But it may also be assumed as valid, as I wish to indicate at this point, on quite the same grounds as those holding in the case of matter, for the theory of radiant heat. Let us consider, e. g., two bodies at different temperatures between which exchange of heat occurs through radiation. We can in this case also imagine a microscopic observer, as opposed to the ordinary macroscopic observer, who possesses insight into all the particulars of electromagnetic processes which are connected with emission and absorption, and the propagation of heat rays. The microscopic observer would declare the whole process reversible because all electrodynamic processes can also take place in the reverse direction, and the contradiction may here be referred back to a difference in definition of the state of a heat ray. Thus, while the macroscopic observer completely defines a monochromatic ray through direction, state of polarization, color, and intensity, the microscopic observer, in order to possess a complete knowledge of an electromagnetic state, necessarily requires the specification of all the numerous irregular variations of amplitude and phase to which the most homogeneous heat ray is actually subject. That such irregular variations actually exist follows immediately from the well known fact that two rays of the same color never interfere, except when they originate in the same source of light. But until these fluctuations are given in all particulars, the micro-observer can say nothing with regard to the progress of the process. He is also unable to specify whether the exchange of heat radiation between the two bodies leads to a decrease or to an increase of their difference in temperature. The principle of elementary disorder first furnishes the adequate criterion of the tendency of the radiation process, i. e., the warming of the colder body at the expense of the warmer, just as the same principle conditions the irreversibility of exchange of heat through conduction. However, in the two cases compared, there is indicated an essential difference in the kind of the disorder. While in heat conduction the disordered elements may be represented as associated with the various molecules, in heat radiation there are the numerous vibration periods, connected with a heat ray, among which the energy of radiation is irregularly distributed. In other words: the disorder among the molecules is a material one, while in heat radiation it is one of energy distribution. This is the most important difference between the two kinds of disorder;

a common feature exists as regards the great number of uncoordinated elements required. Just as the entropy of a body is defined as a function of the macroscopic state, only when the body contains so many atoms that from them definite mean values may be formed, so the entropy principle only possesses a meaning with regard to a heat ray when the ray comprehends so many periodic vibrations, i. e., persists for so long a time, that a definite mean value for the intensity of the ray may be obtained from the successive irregular fluctuating amplitudes.

Now, after the principle of elementary disorder has been introduced and accepted by us as valid throughout nature, the fundamental question arises as to the calculation of the probability of a given state, and the actual derivation of the entropy therefrom. From the entropy all the laws of thermodynamic states of equilibrium, for material substances, and also for energy radiation, may be uniquely derived. With regard to the connection between entropy and probability, this is inferred very simply from the law that the probability of two independent configurations is represented by the product of the individual probabilities:

$$W = W_1 \cdot W_2,$$

while the entropy S is represented by the sum of the individual entropies:

$$S = S_1 + S_2.$$

Accordingly, the entropy is proportional to the logarithm of the probability:

$$S = k \log W. \tag{12}$$

k is a universal constant. In particular, it is the same for atomic as for radiation configurations, for there is nothing to prevent us assuming that the configuration designated by 1 is atomic, while that designated by 2 is a radiation configuration. If k has been calculated, say with the aid of radiation measurements, then k must have the same value for atomic processes. Later we shall follow this procedure, in order to utilize the laws of heat radiation in the kinetic theory of gases. Now, there remains, as the last and most difficult part of the problem, the calculation of the probability W of a given physical configuration in a given macroscopic state. We shall treat today, by way of preparation for the quite general problem to follow, the simple problem: to specify the probability of a given state for a single moving material point, subject to given conservative forces. Since the state depends upon 6 variables: the 3 generalized coordinates φ_1, φ_2, φ_3, and the three corresponding velocity components $\dot{\varphi}_1$, $\dot{\varphi}_2$, $\dot{\varphi}_3$, and since all possible values of these 6 variables constitute a continuous manifold, the probability sought is, that these 6 quantities shall lie respectively within certain infinitely small intervals, or, if one thinks of these 6 quantities as the rectilinear orthogonal coordinates of a point in an ideal six-dimensional space, that this ideal "state point" shall fall within a given, infinitely small "state domain." Since the domain is infinitely small, the probability will be proportional to the magnitude of the domain and therefore proportional to

$$\int d\varphi_1 \cdot d\varphi_2 \cdot d\varphi_3 \cdot d\dot{\varphi}_1 \cdot d\dot{\varphi}_2 \cdot d\dot{\varphi}_3.$$

But this expression cannot serve as an absolute measure of the probability, because in general it changes in magnitude with the time, if each state point moves in accordance

with the laws of motion of material points, while the probability of a state which follows of necessity from another must be the same for the one as the other. Now, as is well known, another integral quite similarly formed, may be specified in place of the one above, which possesses the special property of not changing in value with the time. It is only necessary to employ, in addition to the general coordinates φ_1, φ_2, φ_3, the three so-called momenta ψ_1, ψ_2, ψ_3, in place of the three velocities $\dot{\varphi}_1$, $\dot{\varphi}_2$, $\dot{\varphi}_3$ as the determining coordinates of the state. These are defined in the following way:

$$\psi_1 = \left(\frac{\partial H}{\partial \dot{\varphi}_1}\right)_{\varphi}, \quad \psi_2 = \left(\frac{\partial H}{\partial \dot{\varphi}_2}\right)_{\varphi}, \quad \psi_3 = \left(\frac{\partial H}{\partial \dot{\varphi}_3}\right)_{\varphi},$$

wherein H denotes the kinetic potential (Helmholz). Then, in Hamiltonian form, the equations of motion are:

$$\dot{\psi}_1 = \frac{d\psi_1}{dt} = -\left(\frac{\partial E}{\partial \varphi_1}\right)_{\psi}, \ \cdots, \quad \dot{\varphi}_1 = \frac{d\varphi_1}{dt} = \left(\frac{\partial E}{\partial \psi_1}\right)_{\varphi}, \ \cdots,$$

(E is the energy), and from these equations follows the "condition of incompressibility":

$$\frac{\partial \dot{\varphi}_1}{\partial \varphi_1} + \frac{\partial \dot{\psi}_1}{\partial \psi_1} + \cdots = 0.$$

Referring to the six-dimensional space represented by the coordinates φ_1, φ_2, φ_3, ψ_1, ψ_2, ψ_3, this equation states that the magnitude of an arbitrarily chosen state domain, viz.:

$$\int d\varphi_1 \cdot d\varphi_2 \cdot d\varphi_3 \cdot d\psi_1 \cdot d\psi_2 \cdot d\psi_3$$

does not change with the time, when each point of the domain changes its position in accordance with the laws of motion of material points. Accordingly, it is made possible to take the magnitude of this domain as a direct measure for the probability that the state point falls within the domain.

From the last expression, which can be easily generalized for the case of an arbitrary number of variables, we shall calculate later the probability of a thermodynamic state, for the case of radiant energy as well as that for material substances.

FOURTH LECTURE.

The Equation of State for a Monatomic Gas.

My problem today is to utilize the general fundamental laws concerning the concept of irreversibility, which we established in the lecture of yesterday, in the solution of a definite problem: the calculation of the entropy of an ideal monatomic gas in a given state, and the derivation of all its thermodynamic properties. The way in which we have to proceed is prescribed for us by the general definition of entropy:

$$S = k \log W. \tag{13}$$

The chief part of our problem is the calculation of W for a given state of the gas, and in this connection there is first required a more precise investigation of that which is to be understood as the state of the gas. Obviously, the state is to be taken here solely in the sense of the conception which we have called macroscopic in the last lecture. Otherwise, a state would possess neither probability nor entropy. Furthermore, we are not allowed to assume a condition of equilibrium for the gas. For this is characterized through the further special condition that the entropy for it is a maximum. Thus, an unequal distribution of density may exist in the gas; also, there may be present an arbitrary number of different currents, and in general no kind of equality between the various velocities of the molecules is to be assumed. The velocities, as the coordinates of the molecules, are rather to be taken a priori as quite arbitrarily given, but in order that the state, considered in a macroscopic sense, may be assumed as known, certain mean values of the densities and the velocities must exist. Through these mean values the state from a macroscopic standpoint is completely characterized.

The conditions mentioned will all be fulfilled if we consider the state as given in such manner that the number of molecules in a sufficiently small macroscopic space, but which, however, contains a very large number of molecules, is given, and furthermore, that the (likewise great) number of these molecules is given, which are found in a certain macroscopically small velocity domain, i. e., whose velocities lie within certain small intervals. If we call the coordinates x, y, z, and the velocity components $\dot{x}$, $\dot{y}$, $\dot{z}$, then this number will be proportional to[3]

$$dx \cdot dy \cdot dz \cdot d\dot{x} \cdot d\dot{y} \cdot d\dot{z} = \sigma.$$

It will depend, besides, upon a finite factor of proportionality which may be an arbitrarily given function $f(x, y, z, \dot{x}, \dot{y}, \dot{z})$ of the coordinates and the velocities, and which has only the one condition to fulfill that

$$\sum f \cdot \sigma = N, \tag{14}$$

where N denotes the total number of molecules in the gas. We are now concerned with the calculation of the probability W of that state of the gas which corresponds to the arbitrarily given distribution function f.

[3]We can call σ a "macro-differential" in contradistinction to the micro-differentials which are infinitely small with reference to the dimensions of a molecule. I prefer this terminology for the discrimination between "physical" and "mathematical" differentials in spite of the inelegance of phrasing, because the macro-differential is also just as much mathematical as physical and the micro-differential just as much physical as mathematical.

The probability that a given molecule possesses such coordinates and such velocities that it lies within the domain σ is expressed, in accordance with the final result of the previous lecture, by the magnitude of the corresponding elementary domain:

$$d\varphi_1 \cdot d\varphi_2 \cdot d\varphi_3 \cdot d\psi_1 \cdot d\psi_2 \cdot d\psi_3,$$

therefore, since here

$$\varphi_1 = x, \quad \varphi_2 = y, \quad \varphi_3 = z, \quad \psi_1 = m\dot{x}, \quad \psi_2 = m\dot{y}, \quad \psi_3 = m\dot{z},$$

(m the mass of a molecule) by

$$m^3\sigma.$$

Now we divide the whole of the six dimensional "state domain" containing all the molecules into suitable equal elementary domains of the magnitude $m^3\sigma$. Then the probability that a given molecule fall in a given elementary domain is equally great for all such domains. Let P denote the number of these equal elementary domains. Next, let us imagine as many dice as there are molecules present, i. e., N, and each die to be provided with P equal sides. Upon these P sides we imagine numbers 1, 2, 3, $\cdots$ to P, so that each of the P sides indicates a given elementary domain. Then each throw with the N dice corresponds to a given state of the gas, while the number of dice which show a given number corresponds to the molecules which lie in the elementary domain considered. In accordance with this, each single die can indicate with the same probability each of the numbers from 1 to P, corresponding to the circumstance that each molecule may fall with equal probability in any one of the P elementary domains. The probability W sought, of the given state of the molecules, corresponds, therefore, to the number of different kinds of throws (complexions) through which is realized the given distribution f. Let us take, e. g., N equal to 10 molecules (dice) and $P = 6$ elementary domains (sides) and let us imagine the state so given that there are

3 molecules in 1st elementary domain
4 molecules in 2d elementary domain
0 molecules in 3d elementary domain
1 molecule in 4th elementary domain
0 molecules in 5th elementary domain
2 molecules in 6th elementary domain,

then this state, e. g., may be realized through a throw for which the 10 dice indicate the following numbers:

1st	2d	3d	4th	5th	6th	7th	8th	9th	10th
2	6	2	1	1	2	6	2	1	4.

(15)

Under each of the characters representing the ten dice stands the number which the die indicates in the throw. In fact,

3 dice show the figure 1
4 dice show the figure 2
0 dice show the figure 3
1 die shows the figure 4
0 dice show the figure 5
2 dice show the figure 6.

The state in question may likewise be realized through many other complexions of this kind. The number sought of all possible complexions is now found through consideration of the number series indicated in (15). For, since the number of molecules (dice) is given, the number series contains a fixed number of elements ($10 = N$). Furthermore, since the number of molecules falling in an elementary domain is given, each number, in all permissible complexions, appears equally often in the series. Finally, each change of the number configuration conditions a new complexion. The number of possible complexions or the probability W of the given state is therefore equal to the number of possible permutations with repetition under the conditions mentioned. In the simple example chosen, in accordance with a well known formula, the probability is

$$\frac{10!}{3!\ 4!\ 0!\ 1!\ 0!\ 2!} = 12{,}600.$$

Therefore, in the general case:

$$W = \frac{N!}{\prod(f \cdot \sigma)!}.$$

The sign $\prod$ denotes the product extended over all of the P elementary domains.

From this there results, in accordance with equation (13), for the entropy of the gas in the given state:

$$S = k \log N! - k \sum \log(f \cdot \sigma)!.$$

The summation is to be extended over all domains σ. Since $f \cdot \sigma$ is a large quantity, Stirling's formula may be employed for its factorial, which for a large number n is expressed by:

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n}, \tag{16}$$

therefore, neglecting unimportant terms:

$$\log n! = n(\log n - 1);$$

and hence:

$$S = k \log N! - k \sum f\sigma(\log[f \cdot \sigma] - 1),$$

or, if we note that σ and $N = \sum f\sigma$ remain constant in all changes of state:

$$S = \text{const} - k \sum f \cdot \log f \cdot \sigma. \tag{17}$$

This quantity is, to the universal factor $(-k)$, the same as that which L. Boltzmann denoted by H, and which he showed to vary in one direction only for all changes of state.

In particular, we will now determine the entropy of a gas in a state of equilibrium, and inquire first as to that form of the law of distribution which corresponds to thermodynamic equilibrium. In accordance with the second law of thermodynamics, a state of equilibrium is characterized by the condition that with given values of the total volume V and the total energy E, the entropy S assumes its maximum value. If we assume the total volume of the gas

$$V = \int dx \cdot dy \cdot dz,$$

and the total energy

$$E = \frac{m}{2}\sum(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) f\sigma \tag{18}$$

as given, then the condition:

$$\delta S = 0$$

must hold for the state of equilibrium, or, in accordance with (17):

$$\sum(\log f + 1) \cdot \delta f \cdot \sigma = 0, \tag{19}$$

wherein the variation δf refers to an arbitrary change in the law of distribution, compatible with the given values of N, V and E.

Now we have, on account of the constancy of the total number of molecules N, in accordance with (14):

$$\sum \delta f \cdot \sigma = 0$$

and, on account of the constancy of the total energy, in accordance with (18):

$$\sum(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) \cdot \delta f \cdot \sigma = 0.$$

Consequently, for the fulfillment of condition (19) for all permissible values of δf, it is sufficient and necessary that

$$\log f + \beta(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \text{const},$$

or:

$$f = \alpha e^{-\beta(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)},$$

wherein α and β are constants. In the state of equilibrium, therefore, the space distribution of molecules is uniform, i. e., independent of x, y, z, and the distribution of velocities is the well known Maxwellian distribution.

The values of the constants α and β are to be found from those of N, V and E. For the substitution of the value found for f in (14) leads to:

$$N = V\alpha \left(\frac{\pi}{\beta}\right)^{\frac{3}{2}},$$

and the substitution of f in (18) leads to:

$$E = \tfrac{3}{4} V m \frac{\alpha}{\beta} \left(\frac{\pi}{\beta}\right)^{\frac{3}{2}}.$$

From these equations it follows that:

$$\alpha = \frac{N}{V} \cdot \left(\frac{3mN}{4\pi E}\right)^{\frac{3}{2}}, \quad \beta = \frac{3mN}{4E},$$

and hence finally, in accordance with (17), the expression for the entropy S of the gas in a state of equilibrium with given values for N, V and E is:

$$S = \text{const} + kN(\tfrac{3}{2}\log E + \log V). \tag{20}$$

The additive constant contains terms in N and m, but not in E and V.

The determination of the entropy here carried out permits now the specification directly of the complete thermodynamic behavior of the gas, viz., of the equation of state, and of the values of the specific heats. From the general thermodynamic definition of entropy:

$$dS = \frac{dE + pdV}{T}$$

are obtained the partial differential quotients of S with regard to E and V respectively:

$$\left(\frac{\partial S}{\partial E}\right)_V = \frac{1}{T}, \quad \left(\frac{\partial S}{\partial V}\right)_E = \frac{p}{T}.$$

Consequently, with the aid of (20):

$$\left(\frac{\partial S}{\partial E}\right)_V = \frac{3}{2}\frac{kN}{E} = \frac{1}{T}, \tag{21}$$

and

$$\left(\frac{\partial S}{\partial V}\right)_E = \frac{kN}{V} = \frac{p}{T}. \tag{22}$$

The second of these equations:

$$p = \frac{kNT}{V}$$

contains the laws of Boyle, Gay Lussac and Avogadro, the latter because the pressure depends only upon the number N, and not upon the constitution of the molecules. Writing it in the ordinary form:

$$p = \frac{RnT}{V},$$

where n denotes the number of gram molecules or mols of the gas, referred to $O_2 = 32g$, and R the absolute gas constant:

$$R = 8.315 \cdot 10^7 \frac{\text{erg}}{\text{deg}},$$

we obtain by comparison:

$$k = \frac{Rn}{N}. \tag{23}$$

If we denote the ratio of the mol number to the molecular number by ω, or, what is the same thing, the ratio of the molecular mass to the mol mass:

$$\omega = \frac{n}{N},$$

and hence:

$$k = \omega R. \tag{24}$$

From this, if ω is given, we can calculate the universal constant k, and conversely.

The equation (21) gives:

$$E = \tfrac{3}{2}kNT. \tag{25}$$

Now since the energy of an ideal gas is given by:

$$E = Anc_vT,$$

wherein c_v denotes in calories the heat capacity at constant volume of a mol, A the mechanical equivalent of heat:

$$A = 4.19 \cdot 10^7 \frac{\text{erg}}{\text{cal}},$$

it follows that:

$$c_v = \frac{3kN}{2An},$$

and, having regard to (23), we obtain:

$$c_v = \frac{3}{2}\frac{R}{A} = 3.0, \tag{26}$$

the mol heat in calories of any monatomic gas at constant volume.

For the mol heat c_p at constant pressure we have from the first law of thermodynamics

$$c_p - c_v = \frac{R}{A},$$

and, therefore, having regard to (26):

$$c_p = 5, \quad \frac{c_p}{c_v} = \tfrac{5}{3},$$

a known result for monatomic gases.

The mean kinetic energy L of a molecule is obtained from (25):

$$L = \frac{E}{N} = \tfrac{3}{2}kT. \tag{27}$$

You notice that we have derived all these relations through the identification of the mechanical with the thermodynamic expression for the entropy, and from this you recognize the fruitfulness of the method here proposed.

But a method can first demonstrate fully its usefulness when we utilize it, not only to derive laws which are already known, but when we apply it in domains for whose investigation there at present exist no other methods. In this connection its application affords various possibilities. Take the case of a monatomic gas which is not sufficiently attenuated to have the properties of the ideal state; there are here, as pointed out by J. D. van der Waals, two things to consider: (1) the finite size of the atoms, (2) the forces which act among the atoms. Taking account of these involves a change in the value of the probability and in the energy of the gas as well, and, so far as can now be shown, the corresponding change in the conditions for thermodynamic equilibrium leads to an equation of state which agrees with that of van der Waals. Certainly there is here a rich field for further investigations, of greater promise when experimental tests of the equation of state exist in larger number.

Another important application of the theory has to do with heat radiation, with which we shall be occupied the coming week. We shall proceed then in a similar way

as here, and shall be able from the expression for the entropy of radiation to derive the thermodynamic properties of radiant heat.

Today we will refer briefly to the treatment of polyatomic gases. I have previously, upon good grounds, limited the treatment to monatomic molecules; for up to the present real difficulties appear to stand in the way of a generalization, from the principles employed by us, to include polyatomic molecules; in fact, if we wish to be quite frank, we must say that a satisfactory mechanical theory of polyatomic gases has not yet been found. Consequently, at present we do not know to what place in the system of theoretical physics to assign the processes within a molecule—the intra-molecular processes. We are obviously confronted by puzzling problems. A noteworthy and much discussed beginning was, it is true, made by Boltzmann, who introduced the most plausible assumption that for intra-molecular processes simple laws of the same kind hold as for the motion of the molecules themselves, *i. e.*, the general equations of dynamics. It is easy then, in fact, to proceed to the proof that for a monatomic gas the molecular heat c_v must be greater than 3 and that consequently, since the difference $c_p - c_v$ is always equal to 2, the ratio is

$$\frac{c_p}{c_v} = \frac{c_v + 2}{c_v} < \tfrac{5}{3}.$$

This conclusion is completely confirmed by experience. But this in itself does not confirm the assumption of Boltzmann; for, indeed, the same conclusion is reached very simply from the assumption that there exists intra-molecular energy which increases with the temperature. For then the molecular heat of a polyatomic gas must be greater by a corresponding amount than that of a monatomic gas.

Nevertheless, up to this point the Boltzmann theory never leads to contradiction with experience. But so soon as one seeks to draw special conclusions concerning the magnitude of the specific heats hazardous difficulties arise; I will refer to only one of them. If one assumes the Hamiltonian equations of mechanics as applicable to intra-molecular motions, he arrives of necessity at the law of "uniform distribution of energy," which asserts that under certain conditions, not essential to consider here, in a thermodynamic state of equilibrium the total energy of the gas is distributed uniformly among all the individual energy phases corresponding to the independent variables of state, or, as one may briefly say; the same amount of energy is associated with every independent variable of state. Accordingly, the mean energy of motion of the molecules $\frac{1}{2}kT$, corresponding to a given direction in space, is the same as for any other direction, and, moreover, the same for all the different kinds of molecules, and ions; also for all suspended particles (dust) in the gas, of whatever size, and, furthermore, the same for all kinds of motions of the constituents of a molecule relative to its centroid. If one now reflects that a molecule commonly contains, so far as we know, quite a large number of different freely moving constituents, certainly, that a normal molecule of a monatomic gas, e. g., mercury, possesses numerous freely moving electrons, then, in accordance with the law of uniform energy distribution, the intra-molecular energy must constitute a much larger fraction of the whole specific heat of the gas, and therefore c_p/c_v must turn out much smaller, than is consistent with the measured values. Thus, e. g., for an atom of mercury, in accordance with the measured value of $c_p/c_v = 5/3$, no part whatever of the heat added may be assigned to the intra-molecular energy. Boltzmann and others, in order to eliminate this contradiction, have fixed upon the

possibility that, within the time of observation of the specific heats, the vibrations of the constituents (of a molecule) do not change appreciably with respect to one another, and come later with their progressive motion so slowly into heat equilibrium that this process is no longer capable of detection through observation. Up to now no such delay in the establishment of a state of equilibrium has been observed. Perhaps it would be productive of results if in delicate measurements special attention were paid the question as to whether observations which take a longer time lead to a greater value of the molheat, or, what comes to the same thing, a smaller value of c_p/c_v, than observations lasting a shorter time.

If one has been made mistrustful through these considerations concerning the applicability of the law of uniform energy distribution to intra-molecular processes, the mistrust is accentuated upon the inclusion of the laws of heat radiation. I shall make mention of this in a later lecture.

When we pass from stable atoms to the unstable atoms of radioactive substances, the principles following from the kinetic gas theory lose their validity completely. For the striking failure of all attempts to find any influence of temperature upon radioactive phenomena shows us that an application here of the law of uniform energy distribution is certainly not warranted. It will, therefore, be safest meanwhile to offer no definite conjectures with regard to the nature and the laws of these noteworthy phenomena, and to leave this field for further development to experimental research alone, which, I may say, with every day throws new light upon the subject.

FIFTH LECTURE.

Heat Radiation. Electrodynamic Theory.

Last week I endeavored to point out that we find in the atomic theory a complete explanation for the whole content of the two laws of thermodynamics, if we, with Boltzmann, define the entropy by the probability, and I have further shown, in the example of an ideal monatomic gas, how the calculation of the probability, without any additional special hypothesis, enables us not only to find the properties of gases known from thermodynamics, but also to reach conclusions which lie essentially beyond those of pure thermodynamics. Thus, e. g., the law of Avogadro in pure thermodynamics is only a definition, while in the kinetic theory it is a necessary consequence; furthermore, the value of c_v, the mol-heat of a gas, is completely undetermined by pure thermodynamics, but from the kinetic theory it is of equal magnitude for all monatomic gases and, in fact, equal to 3, corresponding to our experimental knowledge. Today and tomorrow we shall be occupied with the application of the theory to radiant heat, and it will appear that we reach in this apparently quite isolated domain conclusions which a thorough test shows are compatible with experience. Naturally, we take as a basis the electro-magnetic theory of heat radiation, which regards the rays as electro-magnetic waves of the same kind as light rays.

We shall utilize the time today in developing in bold outline the important consequences which follow from the electro-magnetic theory for the characteristic quantities of heat radiation, and tomorrow seek to answer, through the calculation of the entropy, the question concerning the dependence of these quantities upon the temperature, as was done last week for ideal gases. Above all, we are concerned here with the determination of those quantities which at any place in a medium traversed by heat rays determine the state of the radiant heat. The state of radiation at a given place will not be represented by a vector which is determined by three components; for the energy flowing in a given direction is quite independent of that flowing in any other direction. In order to know the state of radiation, we must be able to specify, moreover, the energy which in the time dt flows through a surface element $d\sigma$ for every direction in space. This will be proportional to the magnitude of $d\sigma$, to the time dt, and to the cosine of the angle ϑ which the direction considered makes with the normal to $d\sigma$. But the quantity to be multiplied by $d\sigma \cdot dt \cdot \cos\vartheta$ will not be a finite quantity; for since the radiation through any point of $d\sigma$ passes in all directions, therefore the quantity will also depend upon the magnitude of the solid angle $d\Omega$, which we shall assume as the same for all points of $d\sigma$. In this manner we obtain for the energy which in the time dt flows through the surface element $d\sigma$ in the direction of the elementary cone $d\Omega$, the expression:

$$K d\sigma dt \cdot \cos\vartheta \cdot d\Omega. \tag{28}$$

K is a positive function of place, of time and of direction, and is for unpolarized light of the following form:

$$K = 2\int_0^\infty \mathfrak{K}_\nu d\nu \tag{29}$$

where ν denotes the frequency of a color of wave length λ and whose velocity of prop-

agation is q:

$$\nu = \frac{q}{\lambda},$$

and $\mathfrak{K}_\nu$ denotes the corresponding intensity of spectral radiation of the plane polarized light.

From the value of K is to be found the space density of radiation ϵ, i. e., the energy of radiation contained in unit volume. The point 0 in question forms the centre of a sphere whose radius r we take so small that in the distance r no appreciable absorption of radiation takes place. Then each element $d\sigma$ of the surface of the sphere furnishes, by virtue of the radiation traversing the same, the following contribution to the radiation density at 0:

$$\frac{d\sigma \cdot dt \cdot K \cdot d\Omega}{r^2 d\Omega \cdot qdt} = \frac{d\sigma \cdot K}{r^2 q}.$$

For the radiation cone of solid angle $d\Omega$ proceeding from a point of $d\sigma$ in the direction toward 0 has at the distance r from $d\sigma$ the cross-section $r^2 d\Omega$ and the energy passing in the time dt through this cross-section distributes itself along the distance qdt. By integration over all of the surface elements $d\sigma$ we obtain the total space density of radiation at 0:

$$\epsilon = \int \frac{d\sigma K}{r^2 q} = \frac{1}{q} \int K d\Omega,$$

wherein $d\Omega$ denotes the solid angle of an elementary cone whose vertex is 0. For uniform radiation we obtain:

$$\epsilon = \frac{4\pi K}{q} = \frac{8\pi}{q} \cdot \int_0^\infty \mathfrak{K}_\nu d\nu. \tag{30}$$

The production of radiant heat is a consequence of the act of emission, and its destruction is the result of absorption. Both processes, emission and absorption, have their origin only in material particles, atoms or electrons, not at the geometrical bounding surface; although one frequently says, for the sake of brevity, that a surface element emits or absorbs. In reality a surface element of a body is a place of entrance for the radiation falling upon the body from without and which is to be absorbed; or a place of exit for the radiation emitted from within the body and passing through the surface in the outward direction. The capacity for emission and the capacity for absorption of an element of a body depend only upon its own condition and not upon that of the surrounding elements. If, therefore, as we shall assume in what follows, the state of the body varies only with the temperature, then the capacity for emission and the capacity for absorption of the body will also vary only with the temperature. The dependence upon the temperature can of course be different for each wave length.

We shall now introduce that result following from the second law of thermodynamics which will serve us as a basis in all subsequent considerations: "a system of bodies at rest of arbitrary nature, form and position, which is surrounded by a fixed shell impervious to heat, passes in the course of time from an arbitrarily chosen initial state to a permanent state in which the temperature of all bodies of the system is the same." This is the thermodynamic state of equilibrium in which the entropy of the system, among all those values which it may assume compatible with the total energy specified by the initial conditions, has a maximum value. Let us now apply this law to a single homogeneous isotropic medium which is of great extent in all directions of space and which, as in

all cases subsequently considered, is surrounded by a fixed shell, perfectly reflecting as regards heat rays. The medium possesses for each frequency ν of the heat rays a finite capacity for emission and a finite capacity for absorption. Let us consider, now, such regions of the medium as are very far removed from the surface. Here the influence of the surface will be in any case vanishingly small, because no rays from the surface reach these regions, and on account of the homogeneity and isotropy of the medium we must conclude that the heat radiation is in thermodynamic equilibrium everywhere and has the same properties in all directions, so that $\mathfrak{K}_\nu$, the specific intensity of radiation of a plane polarized ray, is independent of the frequency ν, of the azimuth of polarization, of the direction of the ray, and of location. Thus, there will correspond to each diverging bundle of rays in an elementary cone $d\Omega$, proceeding from a surface element $d\sigma$, an exactly equal bundle oppositely directed, within the same elemental cone converging toward the surface element. This law retains its validity, as a simple consideration shows, right up to the surface of the medium. For in thermodynamic equilibrium each ray must possess exactly the same intensity as that of the directly opposite ray, otherwise, more energy would flow in one direction than in the opposite direction. Let us fix our attention upon a ray proceeding inwards from the surface, this must have the same intensity as that of the directly opposite ray coming from within, and from this it follows immediately that the state of radiation of the medium at all points on the surface is the same as that within. The nature of the bounding surface and the spacial extent of the medium are immaterial, and in a stationary state of radiation $\mathfrak{K}_\nu$ is completely determined by the nature of the medium for each temperature.

This law suffers a modification, however, in the special case that the medium is absolutely diathermanous for a definite frequency ν. It is then clear that the capacity for absorption and also that for emission must be zero, because otherwise no stationary state of radiation could exist, i. e., a medium emits no color which it does not absorb. But equilibrium can then obviously exist for every intensity of radiation of the frequency considered, i. e., $\mathfrak{K}_\nu$ is now undetermined and cannot be found without knowledge of the initial conditions. An important example of this is furnished by an absolute vacuum, which is diathermanous for all frequencies. In a complete vacuum thermodynamic equilibrium can therefore exist for each arbitrary intensity of radiation and for each frequency, i. e., for each arbitrary distribution of the spectral energy. From a general thermodynamic point of view this indeterminateness of the properties of thermodynamic states of equilibrium is explained through the presence of numerous different relative maxima of the entropy, as in the case of a vapor which is in a state of supersaturation. But among all the different maxima there is a special maximum, the absolute, which indicates stable equilibrium. In fact, we shall see that in a diathermanous medium for each temperature there exists a quite definite intensity of radiation, which is designated as the stable intensity of radiation of the frequency ν considered. But for the present we shall assume for all frequencies a finite capacity for absorption and for emission.

We consider now two homogeneous isotropic media in thermodynamic equilibrium separated from each other by a plane surface. Since the equilibrium will not be disturbed if one imagines for the moment the surface of separation between the two substances to be replaced by a surface quite non-transparent to heat radiation, all of the foregoing laws hold for each of the two substances individually. Let the specific intensity of radiation of frequency ν, polarized in any arbitrary plane within the first substance

(the upper in Fig. 1)[4], be $\mathfrak{K}_\nu$ and that within the second substance $\mathfrak{K}_\nu{}'$ (we shall in general designate with a dash those quantities which refer to the second substance). Both quantities $\mathfrak{K}_\nu$ and $\mathfrak{K}_\nu{}'$, besides depending upon the temperature and the frequency, depend only upon the nature of the two substances, and, in fact, these values of the intensity of radiation hold quite up to the boundary surface between the substances, and are therefore independent of the properties of this surface.

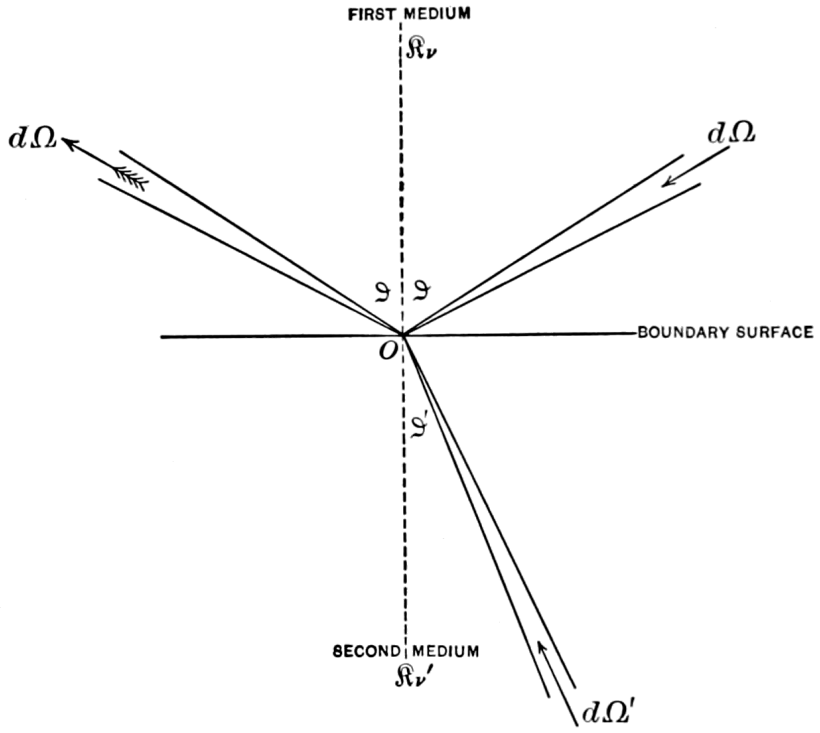


Fig. 1.

Each ray from the first medium is split into two rays at the boundary surface: the reflected and the transmitted. The directions of these two rays vary according to the angle of incidence and the color of the incident ray, and, in addition, the intensity varies according to its polarization. If we denote by ρ (the reflection coefficient) the amount of the reflected energy of radiation and consequently by $1 - \rho$ the amount of transmitted energy with respect to the incident energy, then ρ depends upon the angle of incidence, upon the frequency and upon the polarization of the incident ray. Similar remarks hold for ρ', the reflection coefficient for a ray from the second medium, upon meeting the boundary surface.

Now the energy of a monochromatic plane polarized ray of frequency ν proceeding from an element $d\sigma$ of the boundary surface within the elementary cone $d\Omega$ in a direction

[4]From my lectures upon the theory of heat radiation (Leipzig, J. A. Barth), wherein are to be found the details of the above somewhat abbreviated calculations.

toward the first medium (see the feathered arrow at the left in Fig. 1) is for the time dt, in accordance with (28) and (29):

$$dt \cdot d\sigma \cdot \cos\vartheta \cdot d\Omega \cdot \mathfrak{K}_\nu d\nu, \tag{31}$$

where

$$d\Omega = \sin\vartheta d\vartheta d\varphi. \tag{32}$$

This energy is furnished by the two rays which, approaching the surface from the first and the second medium respectively, are reflected and transmitted respectively at the surface element $d\sigma$ in the same direction. (See the unfeathered arrows. The surface element $d\sigma$ is indicated only by the point 0.) The first ray proceeds in accordance with the law of reflection within the symmetrically drawn elementary cone $d\Omega$: the second approaches the surface within the elementary cone

$$d\Omega' = \sin\vartheta' d\vartheta' d\varphi', \tag{33}$$

where, in accordance with the law of refraction,

$$\varphi' = \varphi \quad \text{and} \quad \frac{\sin\vartheta}{\sin\vartheta'} = \frac{q}{q'}. \tag{34}$$

We now assume that the ray is either polarized in the plane of incidence or perpendicular to this plane, and likewise for the two radiations out of whose energies it is composed. The radiation coming from the first medium and reflected from $d\sigma$ contributes the energy:

$$\rho \cdot dt \cdot d\sigma \cos\vartheta \cdot d\Omega \cdot \mathfrak{K}_\nu d\nu, \tag{35}$$

and the radiation coming from the second medium and transmitted through $d\sigma$ contributes the energy:

$$(1-\rho') \cdot dt \cdot d\sigma \cos\vartheta' \cdot d\Omega' \cdot \mathfrak{K}_\nu' d\nu. \tag{36}$$

The quantities dt, $d\sigma$, ν, and $d\nu$ are here written without the accent, since they have the same values in both media.

Adding the expressions (35) and (36) and placing the sum equal to the expression (31), we obtain:

$$\rho\cos\vartheta d\Omega\mathfrak{K}_\nu + (1-\rho')\cos\vartheta' d\Omega'\mathfrak{K}_\nu' = \cos\vartheta d\Omega\mathfrak{K}_\nu.$$

Now, in accordance with (34):

$$\frac{\cos\vartheta d\vartheta}{q} = \frac{\cos\vartheta' d\vartheta'}{q'},$$

and further, taking note of (32) and (33):

$$d\Omega'\cos\vartheta' = d\Omega\cos\vartheta \cdot \frac{q'^2}{q^2},$$

and it follows that:

$$\rho\mathfrak{K}_\nu + (1-\rho')\frac{q'^2}{q^2}\mathfrak{K}_\nu' = \mathfrak{K}_\nu$$

or:

$$\frac{\mathfrak{K}_\nu}{\mathfrak{K}_\nu{}'} \cdot \frac{q^2}{q'^2} = \frac{1-\rho'}{1-\rho}.$$

In the last equation the quantity on the left is independent of the angle of incidence ϑ and of the kind of polarization, consequently the quantity upon the right side must also be independent of these quantities. If one knows the value of these quantities for a single angle of incidence and for a given kind of polarization, then this value is valid for all angles of incidence and for all polarizations. Now, in the particular case that the rays are polarized at right angles to the plane of incidence and meet the bounding surface at the angle of polarization,

$$\rho = 0 \quad \text{and} \quad \rho' = 0.$$

Then the expression on the right will be equal to 1, and therefore it is in general equal to 1, and we have always:

$$\rho = \rho', \quad q^2\mathfrak{K}_\nu = q'^2\mathfrak{K}_\nu{}'. \tag{37}$$

The first of these two relations, which asserts that the coefficient of reflection is the same for both sides of the boundary surface, constitutes the special expression of a general reciprocal law, first announced by Helmholz, whereby the loss of intensity which a ray of given color and polarization suffers on its path through any medium in consequence of reflection, refraction, absorption, and dispersion is exactly equal to the loss of intensity which a ray of corresponding intensity, color and polarization suffers in passing over the directly opposite path. It follows immediately from this that the radiation meeting a boundary surface between two media is transmitted or reflected equally well from both sides, for every color, direction and polarization.

The second relation, (37), brings into connection the radiation intensities originating in both substances. It asserts that in thermodynamic equilibrium the specific intensities of radiation of a definite frequency in both media vary inversely as the square of the velocities of propagation, or directly as the squares of the refractive indices. We may therefore write

$$q^2\mathfrak{K}_\nu = F(\nu, T),$$

wherein F denotes a universal function depending only upon ν and T, the discovery of which is one of the chief problems of the theory.

Let us fix our attention again on the case of a diathermanous medium. We saw above that in a medium surrounded by a non-transparent shell which for a given color is diathermanous equilibrium can exist for any given intensity of radiation of this color. But it follows from the second law that, among all the intensities of radiation, a definite one, namely, that corresponding to the absolute maximum of the total entropy of the system, must exist, which characterizes the absolutely stable equilibrium of radiation. We now see that this indeterminateness is eliminated by the last equation, which asserts that in thermodynamic equilibrium the product $q^2\mathfrak{K}_\nu$ is a universal function. For it results immediately therefrom that there is a definite value of $\mathfrak{K}_\nu$ for every diathermanous medium which is thus differentiated from all other values. The physical meaning of this value is derived directly from a consideration of the way in which this equation was derived: it is that intensity of radiation which exists in the diathermanous medium when it is in thermodynamic equilibrium while in contact with a given absorbing and

emitting medium. The volume and the form of the second medium is immaterial; in particular, the volume may be taken arbitrarily small.

For a vacuum, the most diathermanous of all media, in which the velocity of propagation $q = c$ is the same for all rays, we can therefore express the following law: The quantity

$$\mathfrak{K}_\nu = \frac{1}{c^2} F(\nu, T) \tag{38}$$

denotes that intensity of radiation which exists in any complete vacuum when it is in a stationary state as regards exchange of radiation with any absorbing and emitting substance, whose amount may be arbitrarily small. This quantity $\mathfrak{K}_\nu$ regarded as a function of ν gives the so-called normal energy spectrum.

Let us consider, therefore, a vacuum surrounded by given emitting and absorbing bodies of uniform temperature. Then, in the course of time, there is established therein a normal energy radiation $\mathfrak{K}_\nu$ corresponding to this temperature. If now ρ_ν be the reflection coefficient of a wall for the frequency ν, then of the radiation $\mathfrak{K}_\nu$ falling upon the wall, the part $\rho_\nu \mathfrak{K}_\nu$ will be reflected. On the other hand, if we designate by E_ν the emission coefficient of the wall for the same frequency ν, the total radiation proceeding from the wall will be:

$$\rho_\nu \mathfrak{K}_\nu + E_\nu = \mathfrak{K}_\nu,$$

since each bundle of rays possesses in a stationary state the intensity $\mathfrak{K}_\nu$. From this it follows that:

$$\mathfrak{K}_\nu = \frac{E_\nu}{1 - \rho_\nu}, \tag{39}$$

i. e., the ratio of the emission coefficient E_ν to the capacity for absorption $(1 - \rho_\nu)$ of a given substance is the same for all substances and equal to the normal intensity of radiation for each frequency (Kirchoff). For the special case that ρ_ν is equal to 0, i. e., that the wall shall be perfectly black, we have:

$$\mathfrak{K}_\nu = E_\nu,$$

that is, the normal intensity of radiation is exactly equal to the emission coefficient of a black body. Therefore the normal radiation is also called "black radiation." Again, for any given body, in accordance with (39), we have:

$$E_\nu < \mathfrak{K}_\nu,$$

i. e., the emission coefficient of a body in general is smaller than that of a black body. Black radiation, thanks to W. Wien and O. Lummer, has been made possible of measurement, through a small hole bored in the wall bounding the space considered.

We proceed now to the treatment of the problem of determining the specific intensity $\mathfrak{K}_\nu$ of black radiation in a vacuum, as regards its dependence upon the frequency ν and the temperature T. In the treatment of this problem it will be necessary to go further than we have previously done into those processes which condition the production and destruction of heat rays; that is, into the question regarding the act of emission and that of absorption. On account of the complicated nature of these processes and the difficulty of bringing some of the details into connection with experience, it is certainly quite out of the question to obtain in this manner any reliable results if the following

law cannot be utilized as a dependable guide in this domain: a vacuum surrounded by reflecting walls in which arbitrary emitting and absorbing bodies are distributed in any given arrangement assumes in the course of time the stationary state of black radiation, which is completely determined by a single parameter, the temperature, and which, in particular, does not depend upon the number, the properties and the arrangement of the bodies. In the investigation of the properties of the state of black radiation the nature of the bodies which are supposed to be in the vacuum is therefore quite immaterial, and it is certainly immaterial whether such bodies actually exist anywhere in nature, so long as their existence and their properties are compatible throughout with the laws of electrodynamics and of thermodynamics. As soon as it is possible to associate with any given special kind and arrangement of emitting and absorbing bodies a state of radiation in the surrounding vacuum which is characterized by absolute stability, then this state can be no other than that of black radiation. Making use of the freedom furnished by this law, we choose among all the emitting and absorbing systems conceivable, the most simple, namely, a single oscillator at rest, consisting of two poles charged with equal quantities of electricity of opposite sign which are movable relative to each other in a fixed straight line, the axis of the oscillator. The state of the oscillator is completely determined by its moment, $f(t)$; i. e., by the product of the electric charge of the pole on the positive side of the axis into the distance between the poles, and by its differential quotient with regard to the time:

$$\frac{df(t)}{dt} = \dot{f}(t).$$

The energy of the oscillator is of the following simple form:

$$U = \tfrac{1}{2}Kf^2 + \tfrac{1}{2}L\dot{f}^2, \tag{40}$$

wherein K and L denote positive constants which depend upon the nature of the oscillator in some manner into which we need not go further at this time.

If, in the vibrations of the oscillator, the energy U remain absolutely constant, we should have: $dU = 0$ or:

$$Kf(t) + L\ddot{f}(t) = 0,$$

and from this there results, as a general solution of the differential equation, a pure periodic vibration:

$$f = C\cos(2\pi\nu_0 t - \vartheta),$$

wherein C and ϑ denote the integration constants and ν_0 the number of vibrations per unit of time:

$$\nu_0 = \frac{1}{2\pi}\sqrt{\frac{K}{L}}. \tag{41}$$

Such an oscillator vibrating periodically with constant energy would neither be influenced by the electromagnetic field surrounding it, nor would it exert any external actions due to radiation. It could therefore have no sort of influence on the heat radiation in the surrounding vacuum.

In accordance with the theory of Maxwell, the energy of vibration U of the oscillator by no means remains constant in general, but an oscillator by virtue of its vibrations sends out spherical waves in all directions into the surrounding field and, in accordance

with the principle of conservation of energy, if no actions from without are exerted upon the oscillator, there must necessarily be a loss in the energy of vibration and, therefore, a damping of the amplitude of vibration is involved. In order to find the amount of this damping we calculate the quantity of energy which flows out through a spherical surface with the oscillator at the center, in accordance with the law of Poynting. However, we may not place the energy flowing outwards in accordance with this law through the spherical surface in an infinitely small interval of time dt equal to the energy radiated in the same time interval from the oscillator. For, in general, the electromagnetic energy does not always flow in the outward direction, but flows alternately outwards and inwards, and we should obtain in this manner for the quantity of the radiation outwards, values which are alternately positive and negative, and which also depend essentially upon the radius of the supposed sphere in such manner that they increase toward infinity with decreasing radius—which is opposed to the fundamental conception of radiated energy. This energy will, moreover, be only found independent of the radius of the sphere when we calculate the total amount of energy flowing outwards through the surface of the sphere, not for the time element dt, but for a sufficiently large time. If the vibrations are purely periodic, we may choose for the time a period; if this is not the case, which for the sake of generality we must here assume, it is not possible to specify a priori any more general criterion for the least possible necessary magnitude of the time than that which makes the energy radiated essentially independent of the radius of the supposed sphere.

In this way we succeed in finding for the energy emitted from the oscillator in the time from t to $t + \mathfrak{T}$ the following expression:

$$\frac{2}{3c^3}\int_t^{t+\mathfrak{T}} \ddot{f}^2(t)dt.$$

If now, the oscillator be in an electromagnetic field which has the electric component $\mathfrak{E}_z$ at the oscillator in the direction of its axis, then the energy absorbed by the oscillator in the same time is:

$$\int_t^{t+\mathfrak{T}} \mathfrak{E}_z \dot{f} \cdot dt.$$

Hence, the principle of conservation of energy is expressed in the following form:

$$\int_t^{t+\mathfrak{T}} \left(\frac{dU}{dt} + \frac{2}{3c^3}\ddot{f}^2 - \mathfrak{E}_z \dot{f} \right) dt = 0.$$

This equation, together with the assumption that the constant

$$\frac{4\pi^2 \nu_0}{3c^3 L} = \sigma \tag{42}$$

is a small number, leads to the following linear differential equation for the vibrations of the oscillator:

$$Kf + L\ddot{f} - \frac{2}{3c^3}\dddot{f} = \mathfrak{E}_z. \tag{43}$$

In accordance with what precedes, in so far as the oscillator is excited into vibrations by an external field $\mathfrak{E}_z$, one may designate it as a resonator which possesses the natural

period ν_0 and the small logarithmic decrement σ. The same equation may be obtained from the electron theory, but I have considered it an advantage to derive it in a manner independent of any hypothesis concerning the nature of the resonator.

Now, let the resonator be in a vacuum filled with stationary black radiation of specific intensity $\mathfrak{K}_\nu$. How, then, does the mean energy U of the resonator in a state of stationary vibration depend upon the specific intensity of radiation $\mathfrak{K}_{\nu_0}$ with the natural period ν_0 of the corresponding color? It is this question which we have still to consider today. Its answer will be found by expressing on the one hand the energy of the resonator U and on the other hand the intensity of radiation $\mathfrak{K}_{\nu_0}$ by means of the component $\mathfrak{E}_z$ of the electric field exciting the resonator. Now however complicated this quantity may be, it is capable of development in any case for a very large time interval, from $t = 0$ to $t = \mathfrak{T}$, in the Fourier's series:

$$\mathfrak{E}_z = \sum_{n=1}^{n=\infty} C_n \cos\left(\frac{2\pi n t}{\mathfrak{T}} - \vartheta_n\right), \tag{44}$$

and for this same time interval $\mathfrak{T}$ the moment of the resonator in the form of a Fourier's series may be calculated as a function of t from the linear differential equation (43). The initial condition of the resonator may be neglected if we only consider such times t as are sufficiently far removed from the origin of time $t = 0$.

If it be now recalled that in a stationary state of vibration the mean energy U of the resonator is given, in accordance with (40), (41) and (42), by:

$$U = K\bar{f}^2 = \frac{16\pi^4 {\nu_0}^3}{3\sigma c^3}\bar{f}^2,$$

it appears after substitution of the value of f obtained from the differential equation (43) that:

$$U = \frac{3c^3}{64\pi^2 {\nu_0}^2}\mathfrak{T}\bar{C}_{n0}{}^2, \tag{45}$$

wherein $\bar{C}_{n0}{}^2$ denotes the mean value of C_n for all the series of numbers n which lie in the neighborhood of the value $\nu_0\mathfrak{T}$, i. e., for which $\nu_0\mathfrak{T}$ is approximately $= 1$.

Now let us consider on the other hand the intensity of black radiation, and for this purpose proceed from the space density of the total radiation. In accordance with (30), this is:

$$\epsilon = \frac{8\pi}{c}\int_0^\infty \mathfrak{K}_\nu d\nu = \frac{1}{8\pi}(\bar{\mathfrak{E}}_x{}^2 + \bar{\mathfrak{E}}_y{}^2 + \bar{\mathfrak{E}}_z{}^2 + \bar{\mathfrak{H}}_x{}^2 + \bar{\mathfrak{H}}_y{}^2 + \bar{\mathfrak{H}}_z{}^2), \tag{46}$$

and therefore, since the radiation is isotropic, in accordance with (44):

$$\frac{8\pi}{c}\int_0^\infty \mathfrak{K}_\nu d\nu = \frac{3}{4\pi}\bar{\mathfrak{E}}_z{}^2 = \frac{3}{8\pi}\sum_{n=1}^{n=\infty} C_n{}^2.$$

If we write $\Delta n/\mathfrak{T}$ on the left instead of $d\nu$, where Δn is a large number, we get:

$$\frac{8\pi}{c}\sum_{n=1}^{n=\infty}\mathfrak{K}_v\frac{\Delta n}{\mathfrak{T}} = \frac{3}{8\pi}\sum_{n=1}^{n=\infty} C_n{}^2,$$

and obtain then by "spectral" division of this equation:

$$\frac{8\pi}{c}\mathfrak{K}_{\nu_0}\frac{\Delta n}{\mathfrak{T}} = \frac{3}{8\pi}\sum_{n_0-(\Delta n/2)}^{n_0+(\Delta n/2)} {C_n}^2,$$

and, if we introduce again the mean value

$$\frac{1}{\Delta n}\cdot\sum_{n_0-(\Delta n/2)}^{n_0+(\Delta n/2)} {C_n}^2 = {\bar{C}_{n0}}^2,$$

we then get:

$$\mathfrak{K}_{\nu_0} = \frac{3c\mathfrak{T}}{64\pi^2}\cdot\bar{C}_{n0}.$$

By comparison with (45) the relation sought is now found:

$$\mathfrak{K}_{\nu_0} = \frac{{\nu_0}^2}{c^2}U, \tag{47}$$

which is striking on account of its simplicity and, in particular, because it is quite independent of the damping constant σ of the resonator.

This relation, found in a purely electrodynamic manner, between the spectral intensity of black radiation and the energy of a vibrating resonator will furnish us in the next lecture, with the aid of thermodynamic considerations, the necessary means of attack in deriving the temperature of black radiation together with the distribution of energy in the normal spectrum.

SIXTH LECTURE.

Heat Radiation. Statistical Theory.

Following the preparatory considerations of the last lecture we shall treat today the problem which we have come to recognize as one of the most important in the theory of heat radiation: the establishment of that universal function which governs the energy distribution in the normal spectrum. The means for the solution of this problem will be furnished us through the calculation of the entropy S of a resonator placed in a vacuum filled with black radiation and thereby excited into stationary vibrations. Its energy U is then connected with the corresponding specific intensity $\mathfrak{K}_\nu$ and its natural frequency ν in the radiation of the surrounding field through equation (47):

$$\mathfrak{K}_\nu = \frac{\nu^2}{c^2} U. \tag{48}$$

When S is found as a function of U, the temperature T of the resonator and that of the surrounding radiation will be given by:

$$\frac{dS}{dU} = \frac{1}{T}, \tag{49}$$

and by elimination of U from the last two equations, we then find the relationship among $\mathfrak{K}_\nu$, T and ν.

In order to find the entropy S of the resonator we will utilize the general connection between entropy and probability, which we have extensively discussed in the previous lectures, and inquire then as to the existing probability that the vibrating resonator possesses the energy U. In accordance with what we have seen in connection with the elucidation of the second law through atomistic ideas, the second law is only applicable to a physical system when we consider the quantities which determine the state of the system as mean values of numerous disordered individual values, and the probability of a state is then equal to the number of the numerous, a priori equally probable, complexions which make possible the realization of the state. Accordingly, we have to consider the energy U of a resonator placed in a stationary field of black radiation as a constant mean value of many disordered independent individual values, and this procedure agrees with the fact that every measurement of the intensity of heat radiation is extended over an enormous number of vibration periods. The entropy of a resonator is then to be calculated from the existing probability that the energy of the radiator possesses a definite mean value U within a certain time interval.

In order to find this probability, we inquire next as to the existing probability that the resonator at any fixed time possesses a given energy, or in other words, that that point (the state point) which through its coordinates indicates the state of the resonator falls in a given "state domain." At the conclusion of the third lecture (p. 43) we saw in general that this probability is simply measured through the magnitude of the corresponding state domain:

$$\int d\varphi \cdot d\psi,$$

in case one employs as coordinates of state the general coordinate φ and the corresponding momentum ψ. Now in general, the energy of the resonator, in accordance

with (40), is:

$$U = \tfrac{1}{2}Kf^2 + \tfrac{1}{2}L\dot{f}^2.$$

If we choose f as the general coordinate φ and put, therefore, $\varphi = f$, then the corresponding impulse ψ is equal

$$\frac{\partial U}{\partial \dot{f}} = L\dot{f},$$

and the energy U expressed as a function of φ and ψ is:

$$U = \tfrac{1}{2}K\varphi^2 + \frac{1}{2}\frac{\psi^2}{L}.$$

If now we desire to find the existing probability that the energy of a resonator shall lie between U and $U + \Delta U$, we have to calculate the magnitude of that state domain in the (φ, ψ)-plane which is bounded by the curves $U = \text{const.}$ and $U + \Delta U = \text{const.}$ These two curves are similar and similarly placed ellipses and the portion of surface bounded by them is equal to the difference of the areas of the two ellipses. The areas are respectively U/ν and $(U + \Delta U)/\nu$; consequently, the magnitude sought for the state domain is: $\Delta U/\nu$. Let us now consider the whole state plane so divided into elementary portions by a large number of ellipses, such that the annular areas between consecutive ellipses are equal to each other; i. e., so that:

$$\frac{\Delta U}{\nu} = \text{const} = h.$$

We thus obtain those portions ΔU of the energy which correspond to equal probabilities and which are therefore to be designated as the energy elements:

$$\epsilon = \Delta U = h\nu. \tag{50}$$

If the determination of the elementary domains is effected in a manner quite similar to that employed in the kinetic gas theory, there exist, with respect to the relationships there found, very notable differences. In the first place, the state of the physical system considered here, the resonator, does not depend as there upon the coordinates and the velocities, but upon the energy only, and this circumstance necessitates that the entropy of a state depend, not upon the distribution of the state quantities φ and ψ, but only upon the energy U. A further difference consists in this, that we have to do in the case of molecules with spacial mean values, but in the case of radiation with mean values as regards time. But this distinction may be disregarded when we reflect that the mean time value of the energy U of a given resonator is obviously identical with the mean space value at a given instant of time of a great number N of similar resonators distributed in the same stationary field of radiation. Of course these resonators must be placed sufficiently far apart in order not directly to influence one another. Then the total energy of all the resonators:

$$U_N = NU \tag{51}$$

is quite irregularly distributed among all the individual resonators, and we have referred back the disorder as regards time to a disorder as regards space.

We are now concerned with the probability W of the state determined by the energy U_N of the N resonators placed in the same stationary field of radiation; i. e., with

the number of individual arrangements or complexions which correspond to the distribution of energy U_N among the N resonators. With this in view, we subdivide the given total energy U_N into its elements ϵ so that:

$$U_N = P\epsilon. \tag{52}$$

These P energy elements are to be distributed in every possible manner among the N resonators. Let us consider, then, the N resonators to be numbered and the figures written beside one another in a series, and in such manner that the number of times each figure appears is equal to the number of energy elements which fall upon the corresponding resonator. Then we obtain through such a number series a representation of a fixed complexion, in which with each individual resonator there is associated a definite energy. For example, if there are $N = 4$ resonators and $P = 6$ energy elements present, then one of the possible complexions is represented by the number series

$$1 \quad 1 \quad 3 \quad 3 \quad 3 \quad 4$$

which asserts that the first resonator contains two, the second 0, the third 3, and the fourth 1 energy element. The totality of numbers in the series is 6, equal to the number of the energy elements present. The arrangement of figures in the series is immaterial for any complexion, since the mere interchange of figures does not change the energy of a given resonator. The number of all the possible different complexions is therefore equal to the number of possible "combinations with repetition" of 4 elements with 6 classes:

$$W = \frac{(4+6-1)!}{(4-1)!\ 6!} = \frac{9!}{3!\ 6!} = 84,$$

or, in our general case the probability sought is:

$$W = \frac{(N+P-1)!}{(N-1)!\ P!}.$$

We obtain, therefore, for the entropy S_N of the resonator system, in accordance with equation (12), since N and P are large numbers,

$$S_N = k \log \frac{(N+P)!}{N!\ P!}$$

and with the aid of Sterling's formula (16):

$$S_N = k\{(N+P)\log(N+P) - N\log N - P\log P\}.$$

If, in accordance with (52), we now write U_N/ϵ for P, NU for U_N in accordance with (51), and $h\nu$ for ϵ, in accordance with (50), we obtain, after an easy transformation, for the mean entropy of a single resonator:

$$\frac{S_N}{N} = S = k\left\{\left(1+\frac{U}{h\nu}\right)\log\left(1+\frac{U}{h\nu}\right) - \frac{U}{h\nu}\log\frac{U}{h\nu}\right\}$$

as the solution of the problem in hand.

We will now introduce the temperature T of the resonator, and will express through T the energy U of the resonator and also the intensity $\mathfrak{K}_\nu$ of the heat radiation related to it through a stationary state of energy exchange. For this purpose we utilize equation (49) and obtain then for the energy of the resonator:

$$U = \frac{h\nu}{e^{h\nu/kT} - 1}.$$

It is to be observed that we have not here to do with a uniform distribution of energy (cf. p. 50) among the various resonators.

For the specific intensity of the monochromatic plane polarized ray of frequency ν, we have, in accordance with (48):

$$\mathfrak{K}_\nu = \frac{h\nu^3}{c^2} \cdot \frac{1}{e^{h\nu/kT} - 1}. \tag{53}$$

This expression furnishes for each temperature T the energy distribution in the normal spectrum of a black body. A comparison with equation (38) of the last lecture furnishes us then with the universal function:

$$F(\nu, T) = \frac{h\nu^3}{e^{h\nu/kT} - 1}.$$

If we refer the specific intensity of a monochromatic ray, not to the frequency ν, but, as is commonly done in experimental physics, to the wave length λ, then, since between the absolute values of $d\nu$ and $d\lambda$ the relation exists:

$$|d\nu| = \frac{c \cdot |d\lambda|}{\lambda^2},$$

we obtain from

$$E_\lambda |d\lambda| = \mathfrak{K}_\nu |d\nu|,$$

the relation:

$$E_\lambda = \frac{c^2 h}{\lambda^5} \cdot \frac{1}{e^{ch/k\lambda T} - 1} \tag{54}$$

as the intensity of a monochromatic plane polarized ray of wave length λ is emitted normally to the surface of a black body in a vacuum at temperature T. For small values of λT (54) reduces to:

$$E_\lambda = \frac{c^2 h}{\lambda^5} \cdot e^{-(ch/k\lambda T)}, \tag{55}$$

which expresses Wien's Displacement Law. For large values of λT on the other hand, there results from (54):

$$E_\lambda = \frac{ckT}{\lambda^4}, \tag{56}$$

a relation first established by Lord Rayleigh and which we may here designate as the Rayleigh Law of Radiation.

From equation (30), taking account of (53), we obtain for the space density of black radiation in a vacuum:

$$\epsilon = \frac{48\pi h}{c^3} \left(\frac{kT}{h}\right)^4 \cdot \alpha = aT^4,$$

wherein

$$\alpha = 1 + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots = 1.0823.$$

The Stefan-Boltzmann law is hereby expressed. In accordance with the measurements of Kurlbaum, we have the constant

$$a = \frac{48\pi k^4}{c^3 h^3} \cdot \alpha = 7.061 \cdot 10^{-15} \frac{\text{erg}}{\text{cm}^3\text{deg}^4}.$$

For that wave length λ_m which corresponds in the spectrum of black radiation to the maximum intensity of radiation E_λ we have from equation (54):

$$\left(\frac{dE_\lambda}{d\lambda}\right)_{\lambda=\lambda_m} = 0.$$

Carrying out the differentiation, we get, after putting for brevity:

$$\frac{ch}{k\lambda_m T} = \beta, \quad e^{-\beta} + \frac{\beta}{5} - 1 = 0.$$

The root of this transcendental equation is:

$$\beta = 4.9651;$$

and $\lambda_m T = ch/k\beta = b$ is a constant (Wien's Displacement Law). In accordance with the measurements of O. Lummer and E. Pringsheim,

$$b = 0.294 \text{ cm} \cdot \text{deg}.$$

From this there follow the numerical values

$$k = 1.346 \cdot 10^{-16} \frac{\text{erg}}{\text{deg}}, \quad \text{and} \quad h = 6.548 \cdot 10^{-27} \text{erg} \cdot \text{sec}.$$

The value found for k easily permits of the specification numerically, in the C.G.S. system, of the general connection between entropy and probability, as expressed through the universal equation (12). Thus, quite in general, the entropy of a physical system is:

$$S = 1.346 \cdot 10^{-16} \overset{\text{e}}{\log} W.$$

In the application to the kinetic gas theory we obtain from equation (24) for the ratio of the molecular mass to the mol mass:

$$\omega = \frac{k}{R} = 1.62 \cdot 10^{-24},$$

i. e., to one mol there corresponds $1/\omega = 6.175 \cdot 10^{23}$ molecules, where it is supposed that the mol of oxygen

$$O_2 = 32\text{g}.$$

Accordingly, the number of molecules contained in 1 cu. cm. of an ideal gas at 0° Cels. and at atmospheric pressure is:

$$N = 2.76 \cdot 10^{19}.$$

The mean kinetic energy of the progressive motion of a molecule at the absolute temperature $T = 1$ in the absolute C.G.S. system, in accordance with (27), is:

$$L = \tfrac{3}{2}k = 2.02 \cdot 10^{-16}.$$

In general, the mean kinetic energy of progressive motion of a molecule is expressed by the product of this number and the absolute temperature T.

The elementary quantum of electricity, or the free electric charge of a monovalent ion or electron, in electrostatic measure is:

$$e = \omega \cdot 9658 \cdot 3 \cdot 10^{10} = 4.69 \cdot 10^{-10}.$$

This result stands in noteworthy agreement with the results of the latest direct measurements of the electric elementary quantum made by E. Rutherford and H. Geiger, and E. Regener.

Even if the radiation formula (54) here derived had shown itself as valid with respect to all previous tests, the theory would still require an extension as regards a certain point; for in it the physical meaning of the universal constant h remains quite unexplained. All previous attempts to derive a radiation formula upon the basis of the known laws of electron theory, among which the theory of J. H. Jeans is to be considered as the most general and exact, have led to the conclusion that h is infinitely small, so that, therefore, the radiation formula of Rayleigh possesses general validity, but, in my opinion, there can be no doubt that this formula loses its validity for short waves, and that the pains which Jeans has taken to place[5] the blame for the contradiction between theory and experiment upon the latter are unwarranted.

Consequently, there remains only the one conclusion, that previous electron theories suffer from an essential incompleteness which demands a modification, but how deeply this modification should go into the structure of the theory is a question upon which views are still widely divergent. J. J. Thompson inclines to the most radical view, as do J. Larmor, A. Einstein, and with him I. Stark, who even believe that the propagation of electromagnetic waves in a pure vacuum does not occur precisely in accordance with the Maxwellian field equations, but in definite energy quanta $h\nu$. I am of the opinion, on the other hand, that at present it is not necessary to proceed in so revolutionary a manner, and that one may come successfully through by seeking the significance of the energy quantum $h\nu$ solely in the mutual actions with which the resonators influence one another.[6] A definite decision with regard to these important questions can only be brought about as a result of further experience.

[5] In that the walls used in the measurements of hollow space radiations must be diathermanous for the shortest waves.

[6] It is my intention to give a complete presentation of these relations in Volume 31 of the Annalen der Physik.

SEVENTH LECTURE.

General Dynamics. Principle of Least Action.

Since I began three weeks ago today to depict for you the present status of the system of theoretical physics and its probable future development, I have continually sought to bring out that in the theoretical physics of the future the most important and the final division of all physical processes would likely be into reversible and irreversible processes. In succeeding lectures, with the aid of the calculus of probability and with the introduction of the hypothesis of elementary disorder, we have seen that all irreversible processes may be considered as reversible elementary processes: in other words, that irreversibility does not depend upon an elementary property of a physical process, but rather depends upon the ensemble of numerous disordered elementary processes of the same kind, each one of which individually is completely reversible, and upon the introduction of the macroscopic method of treatment. From this standpoint one can say quite correctly that in the final analysis all processes in nature are reversible. That there is herein contained no contradiction to the principle regarding the irreversibility of processes expressed in terms of the mean values of elementary processes of macroscopic changes of state, I have demonstrated fully in the third lecture. Perhaps it will be appropriate at this place to interject a more general statement. We are accustomed in physics to seek the explanation of a natural process by the method of division of the process into elements. We regard each complicated process as composed of simple elementary processes, and seek to analyse it through thinking of the whole as the sum of the parts. This method, however, presupposes that through this division the character of the whole is not changed; in somewhat similar manner each measurement of a physical process presupposes that the progress of the phenomena is not influenced by the introduction of the measuring instrument. We have here a case in which that supposition is not warranted, and where a direct conclusion with regard to the parts applied to the whole leads to quite false results. If we divide an irreversible process into its elementary constituents, the disorder and along with it the irreversibility vanishes; an irreversible process must remain beyond the understanding of anyone who relies upon the fundamental law: that all properties of the whole must also be recognizable in the parts. It appears to me as though a similar difficulty presents itself in most of the problems of intellectual life.

Now after all the irreversibility in nature thus appears in a certain sense eliminated, it is an illuminating fact that general elementary dynamics has only to do with reversible processes. Therefore we shall occupy ourselves in what follows with reversible processes exclusively. That which makes this procedure so valuable for the theory is the circumstance that all known reversible processes, be they mechanical, electrodynamical or thermal, may be brought together under a single principle which answers unambiguously all questions regarding their behavior. This principle is not that of conservation of energy; this holds, it is true, for all these processes, but does not determine unambiguously their behavior; it is the more comprehensive principle of least action.

The principle of least action has grown upon the ground of mechanics where it enjoys equal rank and regard with numerous other principles; the principle of d'Alembert, the principle of virtual displacement, Gauss's principle of least constraint, the Lagrangian Equations of the first and second kind. All these principles are equivalent to one another

and therefore at bottom are only different formularizations of the same laws; sometimes one and sometimes another is the most convenient to use. But the principle of least action has the decided advantage over all the other principles mentioned in that it connects together in a single equation the relations between quantities which possess, not only for mechanics, but also for electrodynamics and for thermodynamics, direct significance, namely, the quantities: space, time and potential. This is the reason why one may directly apply the principle of least action to processes other than mechanical, and the result has shown that such applications, as well in electrodynamics as in thermodynamics, lead to the appropriate laws holding in these subjects. Since a representation of a unified system of theoretical physics such as we have here in mind must lay the chief emphasis upon as general an interpretation as possible of physical laws, it is self evident that in our treatment the principle of least action will be called upon to play the principal rôle. I desire now to show how it is applied in simple individual cases.

The general formularization of the principle of least action in the interpretation given to it by Helmholz is as follows: among all processes which may carry a certain arbitrarily given physical system subject to given external actions from a given initial position into a given final position in a given time, the process which actually takes place in nature is that which is distinguished by the condition that the integral

$$\int_{t_0}^{t_1} (\delta H + A) dt = 0, \tag{57}$$

wherein an arbitrary displacement of the independent coordinates (and velocities) is denoted by the sign δ, and A denotes the infinitely small increase in energy (external work) which the system experiences in the displacement δ. The function H is the kinetic potential. When we speak here of the positions, the coordinates, and the velocities of the configuration, we understand thereby, not only those special ones corresponding to mechanical ideas, but also all the so-called generalized coordinates with the quantities derived therefrom; and these may represent equally well quantities of electricity, volumes, and the like.

In the applications which we shall now make of the principle of least action, we must first decide as to whether the generalized coordinates which determine the state of the system considered are present in finite number or form a continuous infinite manifold. We shall distinguish the examples here considered in accordance with this viewpoint.

1. *The Position (Configuration) is Determined by a Finite Number of Coordinates.*

In ordinary mechanics this is actually the case in every system of a finite number of material points or rigid bodies among whose coordinates there exist arbitrary fixed equations of condition. If we call the independent coordinates φ_1, φ_2, $\cdots$, then the external work is:

$$A = \Phi_1 \delta\varphi_1 + \Phi_2 \delta\varphi_2 + \cdots = \delta E, \tag{58}$$

wherein Φ_1, Φ_2, $\cdots$ are the "external force components" which correspond to the individual coordinates, and E denotes the energy of the system. Then the principle of least

action is expressed by:

$$\int_{t_0}^{t_1} dt \cdot \sum_{1,2,\cdots} \left(\frac{\partial H}{\partial \varphi_1} \delta\varphi_1 + \frac{\partial H}{\partial \dot{\varphi}_1} \delta\dot{\varphi}_1 + \Phi_1 \delta\varphi_1 \right) = 0.$$

From this follow the equations of motion:

$$\Phi_1 - \frac{d}{dt}\left(\frac{\partial H}{\partial \dot{\varphi}_1}\right) + \frac{\partial H}{\partial \varphi_1} = 0, \tag{59}$$

and so on for all the indices, $1, 2, \cdots$. Through multiplication of the individual equations by $\dot{\varphi}_1$, $\dot{\varphi}_2$, $\cdots$ addition and integration with respect to time, there results the equation of conservation of energy, whereby the energy E is given by the expression:

$$E = \sum_{1,2,\cdots} \dot{\varphi}_1 \frac{\partial H}{\partial \dot{\varphi}_1} - H. \tag{60}$$

In ordinary mechanics $H = L - U$, if L denote the kinetic and U the potential energy. Since L is a homogeneous function of the second degree with respect to the $\dot{\varphi}$'s, it follows from (60) that:

$$E = 2L - H = L + U.$$

But this expression holds by no means in general.

We pass now to the consideration of the quasi-stationary motion of a system of linear conductors carrying simple closed galvanic currents. The state of the system is given by the position and the velocities of the conductors and by the current densities in each of the same. The coordinates referring to the position of the first conductor may be represented by φ_1, φ_1', φ_1'', $\cdots$, corresponding designations holding for the remaining conductors. We inquire now as to the increase of energy or the external work, A, which corresponds to a virtual displacement of all coordinates. Energy may be conveyed to the system through mechanical actions and through electromagnetic induction as well. The former corresponds to mechanical work, the latter to electromotive work. The former will be of the familiar form:

$$\Phi_1 \delta\varphi_1 + \Phi_1' \delta\varphi_1 + \cdots + \Phi_2 \delta\varphi_2 + \cdots .$$

If we denote by E_1, E_2, $\cdots$ the electromotive forces which are induced in the individual conductors through external agencies (e. g., moving magnets which do not belong to the system), then the electromotive work done from outside upon the currents in the conductors of the system is:

$$E_1 \delta\epsilon_1 + E_2 \delta\epsilon_2 + \cdots ,$$

if $\delta\epsilon_1$, $\delta\epsilon_2$, $\cdots$ denote the quantities of electricity which pass through cross sections of the conductors due to infinitely small virtual currents. The finite current densities will then be denoted by $\dot{\epsilon}_1$, $\dot{\epsilon}_2$, $\cdots$. The electrical state of the first conductor is thus determined in general by the current density $\dot{\epsilon}_1$, the mechanical state (position and velocity) by the coordinates φ_1, φ_1', φ_1'', $\cdots$ and the corresponding velocities $\dot{\varphi}_1$, $\dot{\varphi}_1'$, $\dot{\varphi}_1''$, $\cdots$. The coordinates ϵ_1, ϵ_2, $\cdots$ are so-called "cyclical" coordinates, since the state does not depend upon their momentary values, but only upon their differential quotients

with respect to time, just as, for example, the state of a body rotatable about an axis of symmetry depends only upon the angular velocity, and not upon the angle of rotation. The scheme of notation adopted permits of the direct application of the above formularization of the principle of least action to the case here considered. Thus $H = H_\phi + H_\epsilon$, where H_ϕ, the mechanical potential, depends only upon the φ's and $\dot{\varphi}$'s, while the electrokinetic potential H_ϵ takes the following form:

$$H_\epsilon = \tfrac{1}{2} L_{11} \dot{\epsilon_1}^2 + L_{12}\dot{\epsilon_1}\dot{\epsilon_2} + L_{13}\dot{\epsilon_1}\dot{\epsilon_3} + \cdots + \tfrac{1}{2} L_{22}\dot{\epsilon_2}^2 + \cdots .$$

The quantities L_{11}, L_{12}, $L_{13} \cdots L_{22}, \cdots$ the coefficients of self induction and mutual induction depend, however, in a definite manner upon the coordinates of position φ_1, φ_1', φ_1'', $\cdots$, φ_2, φ_2', φ_2'', $\cdots$.

In accordance with (59), we have for the motion of the first conductor:

$$\Phi_1 - \frac{d}{dt}\left(\frac{\partial H_\phi}{\partial \dot{\varphi}_1}\right) + \frac{\partial H_\phi}{\partial \varphi_1} + \frac{\partial H_\epsilon}{\partial \varphi_1} = 0,$$

with corresponding equations for φ_1', φ_1'', $\cdots$, and for the electric current in it:

$$E_1 - \frac{d}{dt}\left(\frac{\partial H_\epsilon}{\partial \dot{\epsilon}_1}\right) = 0.$$

The laws for the mechanical (ponderomotive) actions may be condensed into the statement that, in addition to the ordinary force upon the first conductor expressed by Φ_1, there is a mechanical force

$$\frac{\partial H_\epsilon}{\partial \varphi_1} = \frac{1}{2}\frac{\partial L_{11}}{\partial \varphi_1}\dot{\epsilon_1}^2 + \frac{\partial L_{12}}{\partial \varphi_1}\dot{\epsilon}_1\dot{\epsilon}_2 + \frac{\partial L_{13}}{\partial \varphi_1}\dot{\epsilon}_1\dot{\epsilon}_3 + \cdots ,$$

which is composed of an action of the current upon itself (first term) and of the actions of the remaining currents upon it (following terms).

The laws of electrical action, on the other hand, are expressed by the statement, that to the external electromotive force E_1 in the first conductor there is added the electromotive force

$$-\frac{d}{dt}\left(\frac{\partial H_\epsilon}{\partial \dot{\epsilon}_1}\right) = -\frac{d}{dt}(L_{11}\dot{\epsilon}_1 + L_{12}\dot{\epsilon}_2 + L_{13}\dot{\epsilon}_3 + \cdots)$$

which likewise is composed of an action of the current upon itself (self induction) and of the inducing actions of the remaining currents, and that these two forces compensate each other.

The galvanic conductance or the galvanic resistance is not contained in these equations because the corresponding energy, Joule heat, is produced in an irreversible manner, and irreversible processes are not represented by the principle of least action. One can formally include this action, likewise any other irreversible action, in accordance with the procedure of Helmholz, by introducing it as an external force, in the present case as the electromotive force due to the resistance w, which operates to cause a diminution in the energy of the system. For an infinitely small element of time, the amount of this energy change is:

$$-(w_1\dot{\epsilon_1}^2 + w_2\dot{\epsilon_2}^2 + w_3\dot{\epsilon_3}^2 + \cdots)\cdot dt = -(w_1\dot{\epsilon}_1 d\epsilon_1 + w_2\dot{\epsilon}_2 d\epsilon_2 + \cdots).$$

Consequently, since the external work $E_1 d\epsilon_1 + E_2 d\epsilon_2 + \cdots$ now includes the Joule heat, the external force components E_1, E_2, $\cdots$ in the electromotive equations must be increased by the additional terms $-w_1\dot{\epsilon}_1$, $-w_2\dot{\epsilon}_2$, $\cdots$.

The application of the principle of least action to thermodynamic processes is of special interest, because the importance of the question relating to the fixing of the generalized coordinates, which determine the state of the system, here becomes prominent. From the standpoint of pure thermodynamics, the variables which determine the state of a body can certainly be quite arbitrarily chosen, e. g., in the case of a gas of invariable constitution any two of the following quantities may be chosen as independent variables and all others expressed through them: volume V, temperature T, pressure P, energy E, entropy S. In the present case, the matter is quite different. If we inquire, in order to apply the principle of least action, with regard to the energy change or the total work A which will be done upon the gas from without in an infinitely small virtual displacement, it may be written in the form:

$$A = -p \cdot \delta V + T \cdot \delta S.$$

$T\delta S$ is the heat added from without, $-p\delta V$ the mechanical work furnished from without. In order to bring this into agreement with the general formula for external work (58):

$$A = \Phi_1 \delta\varphi_1 + \Phi_2 \delta\varphi_2$$

it becomes necessary now to choose V and S as the generalized coordinates of state and, therefore, to identify with them the previously employed quantities φ_1 and φ_2. Then $-p$ and T are the generalized force components Φ_1 and Φ_2. Now, since in thermodynamics every reversible change of state proceeds with infinite slowness, the velocity components $\dot{V}$ and $\dot{S}$, and in general all differential coefficients with respect to time, are to be placed equal to zero, and the principle of least action (59) reduces to:

$$\Phi + \frac{\partial H}{\partial \varphi} = 0,$$

and, therefore, in our case:

$$-p + \left(\frac{\partial H}{\partial V}\right)_S = 0 \quad \text{and} \quad T + \left(\frac{\partial H}{\partial S}\right)_V = 0.$$

Further, in accordance with (60):

$$E = -H.$$

Now these equations are actually valid, since they only present other forms of the relation

$$dS = \frac{dE + pdV}{T}.$$

The view here presented is fundamentally that which is given in the energetics of Mach, Ostwald, Helm, and Wiedeburg. The generalized coordinates V and S are in this theory the "capacity factors," $-p$ and T the "intensity factors."[7] So long as one

[7]The breaking up of the energy differentials into two factors by the exponents of energetics is by no means associated with a special property of energy, but is simply an expression for the elementary law that the differential of a function $F(x)$ is equal to the product of the differential dx by the derivative $\dot{F}(x)$.

limits himself to an irreversible process, nothing stands in the way of carrying out this method completely, nor of a generalization to include chemical processes.

In opposition to it there is an essentially different method of regarding thermodynamic processes, which in its complete generality was first introduced into physics by Helmholtz. In accordance with this method, one generalized coordinate is V, and the other is not S, but a certain cyclical coordinate—we shall denote it, as in the previous example, by ϵ—which does not appear itself in the expression for the kinetic potential H and only appears through its differential coefficient, $\dot{\epsilon}$; and this differential coefficient is the temperature T. Accordingly, H is dependent only upon V and T. The equation for the total external work, in accordance with (58), is:

$$A = -p\delta V + E\delta\epsilon,$$

and agreement with thermodynamics is obviously found if we set:

$$E\delta\epsilon = T\delta S, \quad \text{and also:} \quad Ed\epsilon = TdS, \quad Edt = dS.$$

The equations (59) for the principle of least action become:

$$-p + \left(\frac{\partial H}{\partial V}\right)_T = 0 \quad \text{and} \quad E - \frac{d}{dt}\left(\frac{\partial H}{\partial T}\right)_V = 0,$$

or

$$d\left(\frac{\partial H}{\partial T}\right)_V = Edt = dS,$$

or by integration:

$$\left(\frac{\partial H}{\partial T}\right)_V = S,$$

to an additive constant, which we may set equal to 0. For the energy there results, in accordance with (60):

$$E = \dot{\epsilon}\frac{\partial H}{\partial \dot{\epsilon}} - H = T\left(\frac{\partial H}{\partial T}\right)_V - H,$$

and consequently:

$$H = -(E - TS).$$

H is therefore equal to the negative of the function which Helmholz has called the "free energy" of the system, and the above equations are known from thermodynamics.

Furthermore, the method of Helmholz permits of being carried through consistently, and so long as one limits himself to the consideration of reversible processes, it is in general quite impossible to decide in favor of the one method or the other. However, the method of Helmholz possesses a distinct advantage over the other which I desire to emphasize here. It lends itself better to the furtherance of our endeavor toward the unification of the system of physics. In accordance with the purely energetic method, the independent variables V and S have absolutely nothing to do with each other; heat is a form of energy which is distinguished in nature from mechanical energy and which in no way can be referred back to it. In accordance with Helmholz, heat energy is reduced to motion, and this certainly indicates an advance which is to be placed, perhaps, upon

exactly the same footing as the advance which is involved in the consideration of light waves as electromagnetic waves.

To be sure, the view of Helmholz is not broad enough to include irreversible processes; with regard to this, as we have earlier stated in detail, the introduction of the calculus of probability is necessary in order to throw light on the question. At the same time, this is also the real reason that the exponents of energetics will have nothing to do with the strict observance of irreversible processes, and they either declare them as doubtful or ignore them completely. In reality, the facts of the case are quite the reverse; irreversible processes are the only processes occurring in nature. Reversible processes form only an ideal abstraction, which is very valuable for the theory, but which is never completely realized in nature.

II. *The Generalized Coordinates of State Form a Continuous Manifold.*

The laws of infinitely small motions of perfectly elastic bodies furnish us with the simplest example. The coordinates of state are then the displacement components, $\mathfrak{v}_x$, $\mathfrak{v}_y$, $\mathfrak{v}_z$, of a material point from its position of equilibrium (x, y, z), considered as a function of the coordinates x, y, z. The external work is given by a surface integral:

$$A = \int d\sigma (X_\nu \delta \mathfrak{v}_x + Y_\nu \delta \mathfrak{v}_y + Z_\nu \delta \mathfrak{v}_z)$$

($d\sigma$, surface element; ν, inner normal). The kinetic potential is again given by the difference of the kinetic energy L and the potential energy U:

$$H = L - U.$$

The kinetic energy is:

$$L = \int \frac{d\tau k}{2} (\dot{\mathfrak{v}}_x^2 + \dot{\mathfrak{v}}_y^2 + \dot{\mathfrak{v}}_z^2),$$

wherein $d\tau$ denotes a volume element, k the volume density. The potential energy U is likewise a space integral of a homogeneous quadratic function f which specifies the potential energy of a volume element. This depends, as is seen from purely geometrical considerations, only upon the 6 "strain coefficients:"

$$\frac{\partial \mathfrak{v}_x}{\partial x} = x_x, \quad \frac{\partial \mathfrak{v}_y}{\partial y} = y_y, \quad \frac{\partial \mathfrak{v}_z}{\partial z} = z_z,$$

$$\frac{\partial \mathfrak{v}_y}{\partial z} + \frac{\partial \mathfrak{v}_z}{\partial y} = y_z = z_y, \quad \frac{\partial \mathfrak{v}_z}{\partial x} + \frac{\partial \mathfrak{v}_x}{\partial z} = z_x = x_z, \quad \frac{\partial \mathfrak{v}_x}{\partial y} + \frac{\partial \mathfrak{v}_y}{\partial x} = x_y = y_x.$$

In general, therefore, the function f contains 21 independent constants, which characterize the whole elastic behavior of the substance. For isotropic substances these reduce on grounds of symmetry to 2. Substituting these values in the expression for the principle of least action (57) we obtain:

$$\int dt \Big\{ \int d\tau k (\dot{\mathfrak{v}}_x \delta \dot{\mathfrak{v}}_x + \cdots) - \int d\tau \left(\frac{\partial f}{\partial x_x} \delta x_x + \frac{\partial f}{\partial x_y} \delta x_y + \cdots \right) + \int d\sigma (X_\nu \delta \mathfrak{v}_x + \cdots) \Big\} = 0.$$

If we put for brevity:

$$-\frac{\partial f}{\partial x_x} = X_x, \qquad -\frac{\partial f}{\partial y_y} = Y_y, \qquad -\frac{\partial f}{\partial z_z} = Z_z,$$
$$-\frac{\partial f}{\partial y_z} = Y_z = Z_y, \qquad -\frac{\partial f}{\partial z_x} = Z_x = X_z, \qquad -\frac{\partial f}{\partial x_y} = X_y = Y_x,$$

it turns out, as the result of purely mathematical operations in which the variations $\delta\dot{\mathfrak{v}}_x$, $\delta\dot{\mathfrak{v}}_y$, $\cdots$ and likewise the variations δx_x, δx_y, $\cdots$ are reduced through suitable partial integration with respect to the variations $\delta\mathfrak{v}_x$, $\delta\mathfrak{v}_y$, $\cdots$, that the conditions within the body are expressed by:

$$k\ddot{\mathfrak{v}}_x + \frac{\partial X_x}{\partial x} + \frac{\partial X_y}{\partial y} + \frac{\partial X_z}{\partial z} = 0, \ \cdots$$

and at the surface, by:

$$X_\nu = X_x \cos\nu x + X_y \cos\nu y + X_z \cos\nu z, \ \cdots$$

as is known from the theory of elasticity. The mechanical significance of the quantities X_x, Y_y, $\cdots$ as surface forces follows from the surface conditions.

For the last application of the principle of least action we will take a special case of electrodynamics, namely, electrodynamic processes in a homogeneous isotropic non-conductor at rest, e. g., a vacuum. The treatment is analogous to that carried out in the foregoing example. The only difference lies in the fact that in electrodynamics the dependence of the potential energy U upon the generalized coordinate $\mathfrak{v}$ is somewhat different than in elastic phenomena.

We therefore again put for the external work:

$$A = \int d\sigma(X_\nu \delta\mathfrak{v}_x + Y_\nu \delta\mathfrak{v}_y + Z_\nu \delta\mathfrak{v}_z), \tag{61}$$

and for the kinetic potential:

$$H = L - U,$$

wherein again:

$$L = \int d\tau \frac{k}{2}(\dot{\mathfrak{v}}_x{}^2 + \dot{\mathfrak{v}}_y{}^2 + \dot{\mathfrak{v}}_z{}^2) = \int d\tau \frac{k}{2}(\dot{\mathfrak{v}})^2.$$

On the other hand, we write here:

$$U = \int d\tau \frac{h}{2}(\operatorname{curl}\mathfrak{v})^2.$$

Through these assumptions the dynamical equations including the boundary conditions are now completely determined. The principle of least action (57) furnishes:

$$\int dt\{\int d\tau k(\dot{\mathfrak{v}}_x \delta\dot{\mathfrak{v}}_x + \cdots) - \int d\tau h(\operatorname{curl}_x \mathfrak{v}\delta \operatorname{curl}_x \mathfrak{v} + \cdots) + \int d\sigma(X_\nu \delta\mathfrak{v}_x + \cdots)\} = 0.$$

From this follow, in quite an analogous way to that employed above in the theory of elasticity, first, for the interior of the non-conductor:

$$k\ddot{\mathfrak{v}}_x = h\left(\frac{\partial \operatorname{curl}_y \mathfrak{v}}{\partial z} - \frac{\partial \operatorname{curl}_z \mathfrak{v}}{\partial y}\right), \ \cdots$$

or more briefly

$$k\ddot{\mathfrak{v}} = -h \operatorname{curl}\operatorname{curl} \mathfrak{v}, \tag{62}$$

and secondly, for the surface:

$$X_\nu = h(\operatorname{curl}_z \mathfrak{v} \cdot \cos \nu y - \operatorname{curl}_y \mathfrak{v} \cdot \cos \nu z), \ \cdots \tag{63}$$

These equations are identical with the known electrodynamical equations, if we identify L with the electric, and U with the magnetic energy (or conversely). If we put

$$L = \frac{1}{8\pi}\int d\tau \cdot \epsilon \mathfrak{E}^2 \quad \text{and} \quad U = \frac{1}{8\pi}\int d\tau \cdot \mu \mathfrak{H}^2,$$

($\mathfrak{E}$ and $\mathfrak{H}$, the field strengths, ϵ, the dielectric constant, μ, the permeability) and compare these values with the above expressions for L and U we may write:

$$\dot{\mathfrak{v}} = -\mathfrak{E} \cdot \sqrt{\frac{\epsilon}{4\pi k}}, \quad \operatorname{curl} \mathfrak{v} = \mathfrak{H}\sqrt{\frac{\mu}{4\pi h}}. \tag{64}$$

It follows then, by elimination of $\mathfrak{v}$, that:

$$\dot{\mathfrak{H}} = -\sqrt{\frac{\epsilon h}{\mu k}} \cdot \operatorname{curl} \mathfrak{E},$$

and further, by substitution of $\dot{\mathfrak{v}}$ and $\operatorname{curl} \mathfrak{v}$ in equation (62) found above for the interior of the non-conductor, that:

$$\dot{\mathfrak{E}} = \sqrt{\frac{\mu h}{\epsilon k}} \operatorname{curl} \mathfrak{H}.$$

Comparison with the known electrodynamical equations expressed in Gaussian units:

$$\mu\dot{\mathfrak{H}} = -c \operatorname{curl} \mathfrak{E}, \quad \epsilon\dot{\mathfrak{E}} = c \operatorname{curl} \mathfrak{H}$$

(c, velocity of light in vacuum) results in a complete agreement, if we put:

$$\frac{c}{\mu} = \sqrt{\frac{\epsilon h}{\mu k}} \quad \text{and} \quad \frac{c}{\epsilon} = \sqrt{\frac{\mu h}{\epsilon k}}.$$

From either of these two equations it follows that:

$$\frac{h}{k} = \frac{c^2}{\epsilon\mu},$$

the square of the velocity of propagation.

We obtain from (61) for the energy entering the system from without:

$$dt \cdot \int d\sigma(X_\nu \dot{\mathfrak{v}}_x + Y_\nu \dot{\mathfrak{v}}_y + Z_\nu \dot{\mathfrak{v}}_z),$$

or, taking account of the surface equation (63):

$$dt \cdot \int d\sigma h\{(\mathrm{curl}_z\, \mathfrak{v} \cos \nu y - \mathrm{curl}_y\, \mathfrak{v} \cos \nu z)\dot{\mathfrak{v}}_x + \cdots\},$$

an expression which, upon substitution of the values of $\dot{\mathfrak{v}}$ and $\mathrm{curl}\, \mathfrak{v}$ from (64), turns out to be identical with the Poynting energy current.

We have thus by an application of the principle of least action with a suitably chosen expression for the kinetic potential H arrived at the known Maxwellian field equations.

Are, then, the electromagnetic processes thus referred back to mechanical processes? By no means; for the vector $\mathfrak{v}$ employed here is certainly not a mechanical quantity. It is moreover not possible in general to interpret $\mathfrak{v}$ as a mechanical quantity, for instance, $\mathfrak{v}$ as a displacement, $\dot{\mathfrak{v}}$ as a velocity, $\mathrm{curl}\, \mathfrak{v}$ as a rotation. Thus, e. g., in an electrostatic field $\dot{\mathfrak{v}}$ is constant. Therefore, $\mathfrak{v}$ increases with the time beyond all limits, and $\mathrm{curl}\, \mathfrak{v}$ can no longer signify a rotation.[8] While from these considerations the possibility of a mechanical explanation of electrical phenomena is not proven, it does appear, on the other hand, to be undoubtedly true that the significance of the principle of least action may be essentially extended beyond ordinary mechanics and that this principle can therefore also be utilized as the foundation for general dynamics, since it governs all known reversible processes.

[8]With regard to the impossibility of interpreting electrodynamic processes in terms of the motions of a continuous medium, cf. particularly, H. Witte: "Über den gegenwärtigen Stand der Frage nach einer mechanischen Erklärung der elektrischen Erscheinungen" Berlin, 1906 (E. Ebering).

EIGHTH LECTURE.

General Dynamics. Principle of Relativity.

In the lecture of yesterday we saw, by means of examples, that all continuous reversible processes of nature may be represented as consequences of the principle of least action, and that the whole course of such a process is uniquely determined as soon as we know, besides the actions which are exerted upon the system from without, the kinetic potential H as a function of the generalized coordinates and their differential coefficients with respect to time. The determination of this function remains then as a special problem, and we recognize here a rich field for further theories and hypotheses. It is my purpose to discuss with you today an hypothesis which represents a magnificent attempt to establish quite generally the dependency of the kinetic potential H upon the velocities, and which is commonly designated as the principle of relativity. The gist of this principle is: it is in no wise possible to detect the motion of a body relative to empty space; in fact, there is absolutely no physical sense in speaking of such a motion. If, therefore, two observers move with uniform but different velocities, then each of the two with exactly the same right may assert that with respect to empty space he is at rest, and there are no physical methods of measurement enabling us to decide in favor of the one or the other. The principle of relativity in its generalized form is a very recent development. The preparatory steps were taken by H. A. Lorentz, it was first generally formulated by A. Einstein, and was developed into a finished mathematical system by H. Minkowski. However, traces of it extend quite far back into the past, and therefore it seems desirable first to say something concerning the history of its development.

The principle of relativity has been recognized in mechanics since the time of Galilee and Newton. It is contained in the form of the simple equations of motion of a material point, since these contain only the acceleration and not the velocity of the point. If, therefore, we refer the motion of the point, first to the coordinates x, y, z, and again to the coordinates x', y', z' of a second system, whose axes are directed parallel to the first and which moves with the velocity ν in the direction of the positive x-axis:

$$x' = x - \nu t, \quad y' = y, \quad z' = z, \tag{65}$$

and the form of the equations of motion is not changed in the slightest. Nothing short of the assumption of the general validity of the relativity principle in mechanics can justify the inclusion by physics of the Copernican cosmical system, since through it the independence of all processes upon the earth of the progressive motion of the earth is secured. If one were obliged to take account of this motion, I should have, e. g., to admit that the piece of chalk in my hand possesses an enormous kinetic energy, corresponding to a velocity of something like 30 kilometers per second.

It was without doubt his conviction of the absolute validity of the principle of relativity which guided Heinrich Hertz in the establishment of his fundamental equations for the electrodynamics of moving bodies. The electrodynamics of Hertz is, in fact, wholly built upon the principle of relativity. It recognizes no absolute motion with regard to empty space. It speaks only of motions of material bodies relative to one another. In accordance with the theory of Hertz, all electrodynamic processes occur in material bodies; if these move, then the electrodynamic processes occurring therein move with them. To speak of an independent state of motion of a medium outside of

material bodies, such as the ether, has just as little sense in the theory of Hertz as in the modern theory of relativity.

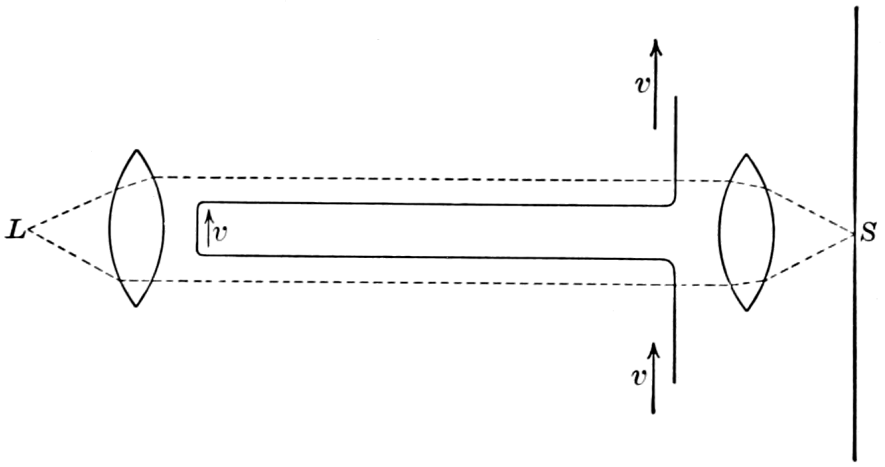


FIG. 2.

But the theory of Hertz has led to various contradictions with experience. I will refer here to the most important of these. Fizeau brought (1851) into parallelism a bundle of rays originating in a light source L by means of a lens and then brought it to a focus by means of a second lens upon a screen S (Fig. 2). In the path of the parallel light rays between the two lenses he placed a tube system of such sort that a transparent liquid could be passed through it, and in such manner that in one half (the upper) the light rays would pass in the direction of flow of the liquid while in the other half (the lower), the rays would pass in the opposite direction.

If now a liquid or a gas flow through the tube system with the velocity ν, then, in accordance with the theory of Hertz, since light must be a process in the substance, the light waves must be transported with the velocity of the liquid. The velocity of light relative to L and S is, therefore, in the upper part $q_0 + \nu$, and the lower part $q_0 - \nu$, if q_0 denote the velocity of light relative to the liquid. The difference of these two velocities, 2ν, should be observable at S through corresponding interference of the lower and the upper light rays, and quite independently of the nature of the flowing substance. Experiment did not confirm this conclusion. Moreover, it showed in gases generally no trace of the expected action; i. e., light is propagated in a flowing gas in the same manner as in a gas at rest. On the other hand, in the case of liquids an effect was certainly indicated, but notably smaller in amount than that demanded by the theory of Hertz. Instead of the expected velocity difference 2ν, the difference $2\nu(1 - 1/n^2)$ only was observed, where n is the refractive index of the liquid. The factor $(1 - 1/n^2)$ is called the Fresnel coefficient. There is contained (for $n = 1$) in this expression the result obtained in the case of gases.

It follows from the experiment of Fizeau that, as regards electrodynamic processes in a gas, the motion of the gas is practically immaterial. If, therefore, one holds that electrodynamic processes require for their propagation a substantial carrier, a special medium, then it must be concluded that this medium, the ether, remains at rest when the gas moves in an arbitrary manner. This interpretation forms the basis of the electrodynamics of Lorentz, involving an absolutely quiescent ether. In accordance with

this theory, electrodynamic phenomena have only indirectly to do with the motion of matter. Primarily all electrodynamical actions are propagated in ether at rest. Matter influences the propagation only in a secondary way, so far as it is the cause of exciting in greater or less degree resonant vibrations in its smallest parts by means of the electrodynamic waves passing through it. Now, since the refractive properties of substances are also influenced through the resonant vibrations of its smallest particles, there results from this theory a definite connection between the refractive index and the coefficient of Fresnel, and this connection is, as calculation shows, exactly that demanded by measurements. So far, therefore, the theory of Lorentz is confirmed through experience, and the principle of relativity is divested of its general significance.

The principle of relativity was immediately confronted by a new difficulty. The theory of a quiescent ether admits the idea of an absolute velocity of a body, namely the velocity relative to the ether. Therefore, in accordance with this theory, of two observers A and B who are in empty space and who move relatively to each other with the uniform velocity ν, it would be at best possible for only one rightly to assert that he is at rest relative to the ether. If we assume, e. g., that at the moment at which the two observers meet an instantaneous optical signal, a flash, is made by each, then an infinitely thin spherical wave spreads out from the place of its origin in all directions through empty space. If, therefore, the observer A remain at the center of the sphere, the observer B will not remain at the center and, as judged by the observer B, the light in his own direction of motion must travel (with the velocity $c - \nu$) more slowly than in the opposite direction (with the velocity $c + \nu$), or than in a perpendicular direction (with the velocity $\sqrt{c^2 - \nu^2}$) (cf. Fig. 3). Under suitable conditions the observer B should be able to detect and measure this sort of effect.

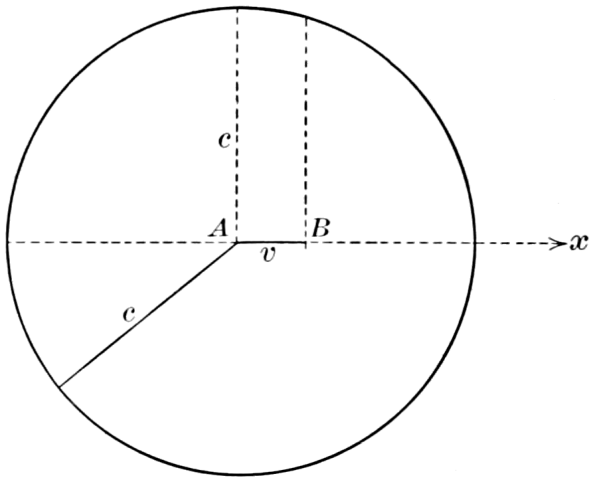


Fig. 3.

This elementary consideration led to the celebrated attempt of Michelson to measure the motion of the earth relative to the ether. A parallel beam of rays proceeding from L (Fig. 4) falls upon a transparent plane parallel plate P inclined at 45°, by which it is in part transmitted and in part reflected. The transmitted and reflected beams are brought into interference by reflection from suitable metallic mirrors S_1 and S_2, which are removed by the same distance l from P. If, now, the earth with the whole apparatus

moves in the direction PS_1 with the velocity ν, then the time which the light needs in order to go from P to S_1 and back is:

$$\frac{l}{c-\nu}+\frac{l}{c+\nu}=\frac{2l}{c}\left(1+\frac{\nu^2}{c^2}+\cdots\right).$$

On the other hand, the time which the light needs in order to pass from P to S_2 and back to P is:

$$\frac{l}{\sqrt{c^2-\nu^2}}+\frac{l}{\sqrt{c^2-\nu^2}}=\frac{2l}{c}\left(1+\frac{1}{2}\frac{\nu^2}{c^2}+\cdots\right).$$

If, now, the whole apparatus be turned through a right angle, a noticeable displacement of the interference bands should result, since the time for the passage over the path PS_2 is now longer. No trace was observed of the marked effect to be expected.

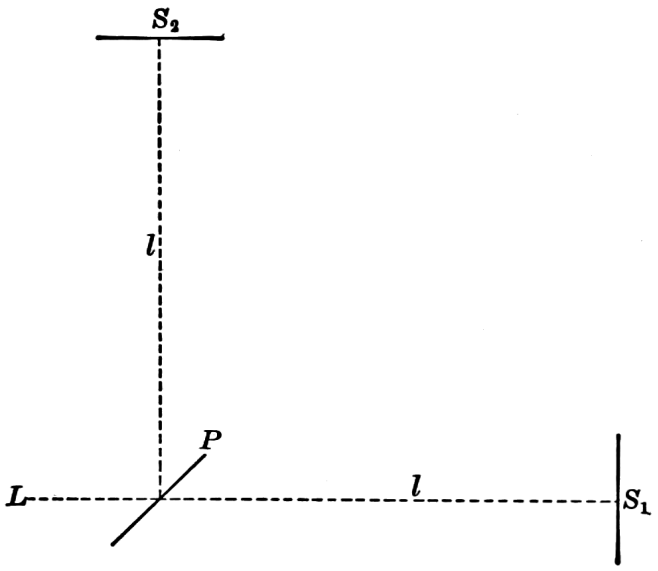


Fig. 4.

Now, how will it be possible to bring into line this result, established by repeated tests with all the facilities of modern experimental art? E. Cohn has attempted to find the necessary compensation in a certain influence of the air in which the rays are propagated. But for anyone who bears in mind the great results of the atomic theory of dispersion and who does not renounce the simple explanation which this theory gives for the dependence of the refractive index upon the color, without introducing something else in its place, the idea that a moving absolutely transparent medium, whose refractive index is absolutely = 1, shall yet have a notable influence upon the velocity of propagation of light, as the theory of Cohn demands, is not possible of assumption. For this theory distinguishes essentially a transparent medium, whose refractive index is = 1, from a perfect vacuum. For the former the velocity of propagation of light in the direction of the velocity ν of the medium with relation to an observer at rest is

$$q=c+\frac{\nu^2}{c},$$

for a vacuum, on the other hand, $q = c$. In the former medium, Cohn's theory of the Michelson experiment predicts no effect, but, on the other hand, the Michelson experiment should give a positive effect in a vacuum.

In opposition to E. Cohn, H. A. Lorentz and FitzGerald ascribe the necessary compensation to a contraction of the whole optical apparatus in the direction of the earth's motion of the order of magnitude ν^2/c^2. This assumption allows better of the introduction again of the principle of relativity, but it can first completely satisfy this principle when it appears, not as a necessary hypothesis made to fit the present special case, but as a consequence of a much more general postulate. We have to thank A. Einstein for the framing of this postulate and H. Minkowski for its further mathematical development.

Above all, the general principle of relativity demands the renunciation of the assumption which led H. A. Lorentz to the framing of his theory of a quiescent ether; the assumption of a substantial carrier of electromagnetic waves. For, when such a carrier is present, one must assume a definite velocity of a ponderable body as definable with respect to it, and this is exactly that which is excluded by the relativity principle. Thus the ether drops out of the theory and with it the possibility of mechanical explanation of electrodynamic processes, i. e., of referring them to motions. The latter difficulty, however, does not signify here so much, since it was already known before, that no mechanical theory founded upon the continuous motions of the ether permits of being completely carried through (cf. p. 78). In place of the so-called free ether there is now substituted the absolute vacuum, in which electromagnetic energy is independently propagated, like ponderable atoms. I believe it follows as a consequence that no physical properties can be consistently ascribed to the absolute vacuum. The dielectric constant and the magnetic permeability of a vacuum have no absolute meaning, only relative. If an electrodynamic process were to occur in a ponderable medium as in a vacuum, then it would have absolutely no sense to distinguish between field strength and induction. In fact, one can ascribe to the vacuum any arbitrary value of the dielectric constant, as is indicated by the various systems of units. But how is it now with regard to the velocity of propagation of light? This also is not to be regarded as a property of the vacuum, but as a property of electromagnetic energy which is present in the vacuum. Where there is no energy there can exist no velocity of propagation.

With the complete elimination of the ether, the opportunity is now present for the framing of the principle of relativity. Obviously, we must, as a simple consideration shows, introduce something radically new. In order that the moving observer B mentioned above (Fig. 3, p. 81) shall not see the light signal given by him travelling more slowly in his own direction of motion (with the velocity $c - \nu$) than in the opposite direction (with the velocity $c + \nu$), it is necessary that he shall not identify the instant of time at which the light has covered the distance $c - \nu$ in the direction of his own motion with the instant of time at which the light has covered the distance $c + \nu$ in the opposite direction, but that he regard the latter instant of time as later. In other words: the observer B measures time differently from the observer A. This is a priori quite permissible; for the relativity principle only demands that neither of the two observers shall come into contradiction with himself. However, the possibility is left open that the specifications of time of both observers may be mutually contradictory.

It need scarcely be emphasized that this new conception of the idea of time makes

the most serious demands upon the capacity of abstraction and the projective power of the physicist. It surpasses in boldness everything previously suggested in speculative natural phenomena and even in the philosophical theories of knowledge: non-euclidean geometry is child's play in comparison. And, moreover, the principle of relativity, unlike non-euclidean geometry, which only comes seriously into consideration in pure mathematics, undoubtedly possesses a real physical significance. The revolution introduced by this principle into the physical conceptions of the world is only to be compared in extent and depth with that brought about by the introduction of the Copernican system of the universe.

Since it is difficult, on account of our habitual notions concerning the idea of absolute time, to protect ourselves, without special carefully considered rules, against logical mistakes in the necessary processes of thought, we shall adopt the mathematical method of treatment. Let us consider then an electrodynamic process in a pure vacuum; first, from the standpoint of an observer A; secondly, from the standpoint of an observer B, who moves relatively to observer A with a velocity ν in the direction of the x-axis. Then, if A employ the system of reference x, y, z, t, and B the system of reference x', y', z', t', our first problem is to find the relations among the primed and the unprimed quantities. Above all, it is to be noticed that since both systems of reference, the primed and the unprimed, are to be like directed, the equations of transformation between corresponding quantities in the two systems must be so established that it is possible through a transformation of exactly the same kind to pass from the first system to the second, and conversely, from the second back to the first system. It follows immediately from this that the velocity of light c' in a vacuum for the observer B is exactly the same as for the observer A. Thus, if c' and c are different, $c' > c$, say, it would follow that: if one passes from one observer A to another observer B who moves with respect to A with uniform velocity, then he would find the velocity of propagation of light for B greater than for A. This conclusion must likewise hold quite in general independently of the direction in which B moves with respect to A, because all directions in space are equivalent for the observer A. On the same grounds, in passing from B to A, c must be greater than c', for all directions in space for the observer B are now equivalent. Since the two inequalities contradict, therefore c' must be equal to c. Of course this important result may be generalized immediately, so that the totality of the quantities independent of the motion, such as the velocity of light in a vacuum, the constant of gravitation between two bodies at rest, every isolated electric charge, and the entropy of any physical system possess the same values for both observers. On the other hand, this law does not hold for quantities such as energy, volume, temperature, etc. For these quantities depend also upon the velocity, and a body which is at rest for A is for B a moving body.

We inquire now with regard to the form of the equations of transformation between the unprimed and the primed coordinates. For this purpose let us consider, returning to the previous example, the propagation, as it appears to the two observers A and B, of an instantaneous signal creating an infinitely thin light wave which, at the instant at which the observers meet, begins to spread out from the common origin of coordinates. For the observer A the wave travels out as a spherical wave:

$$x^2 + y^2 + z^2 - c^2t^2 = 0. \tag{66}$$

For the second observer B the same wave also travels as a spherical wave with the same

velocity:

$$x'^2 + y'^2 + z'^2 - c^2t'^2 = 0; \tag{67}$$

for the first observer has no advantage over the second observer. B can exactly with the same right as A assert that he is at rest at the center of the spherical wave, and for B, after unit time, the wave appears as in Fig. 5, while its appearance for the observer A after unit time, is represented by Fig. 3 (p. 81).[9]

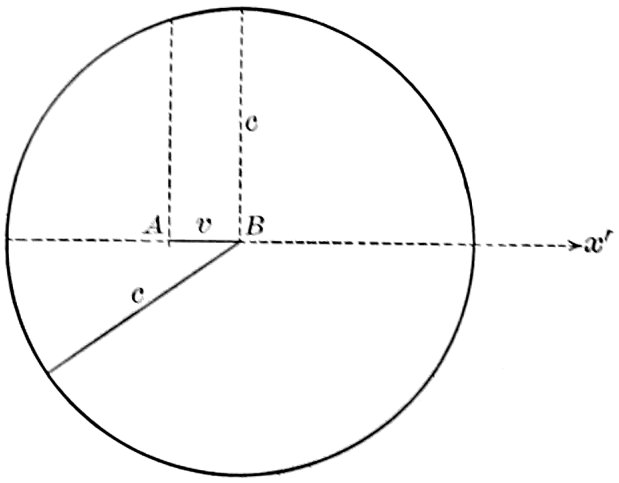


Fig. 5.

The equations of transformation must therefore fulfill the condition that the two last equations, which represent the same physical process, are compatible with each other; and furthermore: the passage from the unprimed to the primed quantities must in no wise be distinguished from the reverse passage from the primed to the unprimed quantities. In order to satisfy these conditions, we generalize the equations of transformation (65), set up at the beginning of this lecture for the old mechanical principle of relativity, in the following manner:

$$x' = \kappa(x - \nu t), \quad y' = \lambda y, \quad z' = \mu z, \quad t' = \nu t + \rho x.$$

Here ν denotes, as formerly, the velocity of the observer B relative to A and the constants κ, λ, μ, ν, ρ are yet to be determined. We must have:

$$x = \kappa'(x' - \nu' t'), \quad y = \lambda' y', \quad z = \mu' z', \quad t = \nu' t' + \rho' x'.$$

It is now easy to see that λ and λ' must both $= 1$. For, if, e. g., λ be greater than 1, then λ' must also be greater than 1; for the two transformations are equivalent with regard to the y axis. In particular, it is impossible that λ and λ' depend upon the direction of motion of the other observer. But now, since, in accordance with what precedes, $\lambda = 1/\lambda'$, each of the two inequalities contradict and therefore $\lambda = \lambda' = 1$; likewise, $\mu = \mu' = 1$. The condition for identity of the two spherical waves then demands that the expression (66):

$$x^2 + y^2 + z^2 - c^2t^2$$

[9]The circumstance that the signal is a finite one, however small the time may be, has significance only as regards the thickness of the spherical layer and not for the conclusions here under consideration.

become, through the transformation of coordinates, identical with the expression (67):

$$x'^2 + y'^2 + z'^2 - c^2t'^2,$$

and from this the equations of transformation follow without ambiguity:

$$x' = \kappa(x - \nu t), \quad y' = y, \quad z' = z, \quad t' = \kappa\left(t - \frac{\nu}{c^2}x\right), \tag{68}$$

wherein

$$\kappa = \frac{c}{\sqrt{c^2 - \nu^2}}.$$

Conversely:

$$x = \kappa(x' + \nu t'), \quad y = y', \quad z = z', \quad t = \kappa\left(t' + \frac{\nu}{c^2}x'\right). \tag{69}$$

These equations permit quite in general of the passage from the system of reference of one observer to that of the other (H. A. Lorentz), and the principle of relativity asserts that all processes in nature occur in accordance with the same laws and with the same constants for both observers (A. Einstein). Mathematically considered, the equations of transformation correspond to a rotation in the four dimensional system of reference (x, y, z, ict) through the imaginary angle $\text{arctg}(i(\nu/c))$ (H. Minkowski). Accordingly, the principle of relativity simply teaches that there is in the four dimensional system of space and time no special characteristic direction, and any doubts concerning the general validity of the principle are of exactly the same kind as those concerning the existence of the antipodians upon the other side of the earth.

We will first make some applications of the principle of relativity to processes which we have already treated above. That the result of the Michelson experiment is in agreement with the principle of relativity, is immediately evident; for, in accordance with the relativity principle, the influence of a uniform motion of the earth upon processes on the earth can under no conditions be detected.

We consider now the Fizeau experiment with the flowing liquid (see p. 80). If the velocity of propagation of light in the liquid at rest be again q_0, then, in accordance with the relativity principle, q_0 is also the velocity of the propagation of light in the flowing liquid for an observer who moves with the liquid, in case we disregard the dispersion of the liquid; for the color of the light is different for the moving observer. If we call this observer B and the velocity of the liquid as above, ν, we may employ immediately the above formulae in the calculation of the velocity of propagation of light in the flowing liquid, judged by an observer A at the screen S. We have only to put

$$\frac{dx'}{dt'} = x' = q_0,$$

to seek the corresponding value of

$$\frac{dx}{dt} = \dot{x}.$$

For this obviously gives the velocity sought.

Now it follows directly from the equations of transformation (69) that:

$$\frac{dx}{dt} = \dot{x} = \frac{\dot{x}' + \nu}{1 + \dfrac{\nu\dot{x}'}{c^2}},$$

and, therefore, through appropriate substitution, the velocity sought in the upper tube, after neglecting higher powers in ν/c and ν/q_0, is:

$$\dot{x} = \frac{q_0 + \nu}{1 + \frac{\nu q_0}{c^2}} = q_0 + \nu\left(1 - \frac{q_0^2}{c^2}\right),$$

and the corresponding velocity in the lower tube is:

$$q_0 - \nu\left(1 - \frac{q_0^2}{c^2}\right).$$

The difference of the two velocities is

$$2\nu\left(1 - \frac{q_0^2}{c^2}\right) = 2\nu\left(1 - \frac{1}{n^2}\right),$$

which is the Fresnel coefficient, in agreement with the measurements of Fizeau.

The significance of the principle of relativity extends, not only to optical and other electrodynamic phenomena, but also to all processes of ordinary mechanics; but the familiar expression $(\frac{1}{2}mq^2)$ for the kinetic energy of a mass point moving with the velocity q is incompatible with this principle.

But, on the other hand, since all mechanics as well as the rest of physics is governed by the principle of least action, the significance of the relativity principle extends at bottom only to the particular form which it prescribes for the kinetic potential H, and this form, though I will not stop to prove it, is characterized by the simple law that the expression

$$H \cdot dt$$

for every space element of a physical system is an invariant

$$= H' \cdot dt'$$

with respect to the passage from one observer A to the other observer B or, what is the same thing, the expression $H/\sqrt{c^2 - q^2}$ is in this passage an invariant $= H'/\sqrt{c^2 - q'^2}$.

Let us now make some applications of this very general law, first to the dynamics of a single mass point in a vacuum, whose state is determined by its velocity q. Let us call the kinetic potential of the mass point for $q = 0$, H_0, and consider now the point at an instant when its velocity is q. For an observer B who moves with the velocity q with respect to the observer A, $q' = 0$ at this instant, and therefore $H' = H_0$. But now since in general:

$$\frac{H}{\sqrt{c^2 - q^2}} = \frac{H'}{\sqrt{c^2 - q'^2}},$$

we have after substitution:

$$H = \sqrt{1 - \frac{q^2}{c^2}} \cdot H_0 = \sqrt{1 - \frac{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}{c^2}} \cdot H_0.$$

With this value of H, the Lagrangian equations of motion (59) of the previous lecture are applicable.

In accordance with (60), the kinetic energy of the mass point amounts to:

$$E = \dot{x}\frac{\partial H}{\partial \dot{x}} + \dot{y}\frac{\partial H}{\partial \dot{y}} + \dot{z}\frac{\partial H}{\partial \dot{z}} - H = q\frac{\partial H}{\partial q} - H = -\frac{H_0}{\sqrt{1 - \frac{q^2}{c^2}}},$$

and the momentum to:

$$G = \frac{\partial H}{\partial q} = -\frac{qH_0}{c\sqrt{c^2 - q^2}}.$$

G/q is called the transverse mass m_t, and dG/dq the longitudinal mass m_l of the point; accordingly:

$$m_t = -\frac{H_0}{c\sqrt{c^2 - q^2}}, \quad m_l = -\frac{cH_0}{(c^2 - q^2)^{3/2}}.$$

For $q = 0$, we have

$$m_t = m_l = m_0 = -\frac{H_0}{c^2}.$$

It is apparent, if one replaces in the above expressions the constant H_0 by the constant m_0, that the momentum is:

$$G = \frac{m_0 q}{\sqrt{1 - \frac{q^2}{c^2}}}$$

and the transverse mass:

$$m_t = \frac{m_0}{\sqrt{1 - \frac{q^2}{c^2}}},$$

and the longitudinal mass:

$$m_l = \frac{m_0}{\left(1 - \frac{q^2}{c^2}\right)^{3/2}},$$

and, finally, that the kinetic energy is:

$$E = \frac{m_0 c^2}{\sqrt{1 - \frac{q^2}{c^2}}} = m_0 c^2 + \tfrac{1}{2} m_0 q^2 + \cdots .$$

The familiar value of ordinary mechanics $\frac{1}{2} m_0 q^2$ appears here therefore only as an approximate value. These equations have been experimentally tested and confirmed through the measurements of A. H. Bucherer and of E. Hupka upon the magnetic deflection of electrons.

A further example of the invariance of $H \cdot dt$ will be taken from electrodynamics. Let us consider in any given medium any electromagnetic field. For any volume element V of the medium, the law holds that $V \cdot dt$ is invariant in the passage from the one to the other observer. It follows from this that H/V is invariant; i. e., the kinetic potential of a unit volume or the "*space density of kinetic potential*" is invariant.

Hence the following relation exists;

$$\mathfrak{E}\mathfrak{D} - \mathfrak{H}\mathfrak{B} = \mathfrak{E}'\mathfrak{D}' - \mathfrak{H}'\mathfrak{B}',$$

wherein $\mathfrak{E}$ and $\mathfrak{H}$ denote the field strengths and $\mathfrak{D}$ and $\mathfrak{B}$ the corresponding inductions. Obviously a corresponding law for the space energy density $\mathfrak{E}\mathfrak{D} + \mathfrak{H}\mathfrak{B}$ will not hold.

A third example is selected from thermodynamics. If we take the velocity q of a moving body, the volume V and the temperature T as independent variables, then, as I have shown in the previous lecture (p. 74), we shall have for the pressure p and the entropy S the following relations:

$$\frac{\partial H}{\partial V} = p \quad \text{and} \quad \frac{\partial H}{\partial T} = S.$$

Now since $V/\sqrt{c^2 - q^2}$ is invariant, and S likewise invariant (see p. 84), it follows from the invariance of $H/\sqrt{c^2 - q^2}$ that p is invariant and also that $T/\sqrt{c^2 - q^2}$ is invariant, and hence that:

$$p = p' \quad \text{and} \quad \frac{T}{\sqrt{c^2 - q^2}} = \frac{T'}{\sqrt{c^2 - q'^2}}.$$

The two observers A and B would estimate the pressure of a body as the same, but the temperature of the body as different.

A special case of this example is supplied when the body considered furnishes a black body radiation. The black body radiation is the only physical system whose dynamics (for quasi-stationary processes) is known with absolute accuracy. That the black body radiation possesses inertia was first pointed out by F. Hasenöhrl. For black body radiation at rest the energy $E_0 = aT^4V$ is given by the Stefan-Boltzmann law, and the entropy $S_0 = \int(dE_0/T) = \frac{4}{3}aT^3V$, and the pressure $p_0 = (a/3)T^4$, and, therefore, in accordance with the above relations, the kinetic potential is:

$$H_0 = \frac{a}{3}T^4V.$$

Let us imagine now a black body radiation moving with the velocity q with respect to the observer A and introduce an observer B who is at rest ($q = 0$) with reference to the black body radiation; then:

$$\frac{H}{\sqrt{c^2 - q^2}} = \frac{H'}{\sqrt{c^2 - q'^2}} = \frac{H_0'}{c},$$

wherein

$$H_0' = \frac{a}{3}T'^4V'.$$

Taking account of the above general relations between T' and T, V' and V, this gives for the moving black body radiation the kinetic potential:

$$H = \frac{a}{3}\frac{T^4V}{\left(1 - \frac{q^2}{c^2}\right)^2},$$

from which all the remaining thermodynamic quantities: the pressure p, the energy E, the momentum G, the longitudinal and transverse masses m_l and m_t of the moving black body radiation are uniquely determined.

Colleagues, ladies and gentlemen, I have arrived at the conclusion of my lectures. I have endeavored to bring before you in bold outline those characteristic advances in the present system of physics which in my opinion are the most important. Another in my place would perhaps have made another and better choice and, at another time, it is quite likely that I myself should have done so. The principle of relativity holds, not only for processes in physics, but also for the physicist himself, in that a fixed system of physics exists in reality only for a given physicist and for a given time. But, as in the theory of relativity, there exist invariants in the system of physics: ideas and laws which retain their meaning for all investigators and for all times, and to discover these invariants is always the real endeavor of physical research. We shall work further in this direction in order to leave behind for our successors where possible—lasting results. For if, while engaged in body and mind in patient and often modest individual endeavor, one thought strengthens and supports us, it is this, that we in physics work, not for the day only and for immediate results, but, so to speak, for eternity.

I thank you heartily for the encouragement which you have given me. I thank you no less for the patience with which you have followed my lectures to the end, and I trust that it may be possible for many among you to furnish in the direction indicated much valuable service to our beloved science.

Typographical Errors corrected in Project Gutenberg edition

p. 18. In 'the theory of reversible processes', 'the' omitted (before line-break 'the-ory').

p. 23. Eqn. (6), first term $\sum \nu_0 \log c_0$ was printed as $\sum \nu_0 \log c_1$.

p. 25. 'Let the system consist'–'consists' in text.

p. 27. 'at a fixed temperature T' was printed 'at a fixed pressure T'.

p. 28. $\nu_0' = \dfrac{m_0}{m_0'}$ was printed $v_0' = \dfrac{m_0}{m_0'}$.

p. 38. In 'give up the attempt to understand'–'undertand' in text.

p. 43. Definitions of ψ, third expression $\psi_3 = \cdots$ printed as $\dot{\varphi}_3 = \cdots$.

p. 43. Equation beginning $\dot{\psi}_1 = \dfrac{d\psi_1}{dt}$, this term was printed as $\psi_1 = \dfrac{d\psi_1}{dt}$.

p. 43. In 'we shall calculate later'–'calulate' in text.

p. 48. Eqn. before (21), the term $\left(\dfrac{\partial S}{\partial E}\right)_V$ was printed as $\left(\dfrac{dS}{\partial E}\right)_V$.

p. 48. Eqn. (21), the term $\left(\dfrac{\partial S}{\partial E}\right)_V$ was printed as $\left(\dfrac{\partial S}{dE}\right)_V$.

p. 66. 'black radiation in a vacuum'–'vaccuum' in text.

p. 67. The mass ratio symbol ω was consistently printed as ∞ on this page.

p. 71. Eqn. after (58), the term $\dfrac{\partial H}{\partial \varphi_1}$ was printed as $\dfrac{\partial H}{\delta \varphi_1}$.

p. 71. Term ${\Phi_1}'\delta{\varphi_1}'$ was printed as ${\Phi_1}'\delta\varphi_1$.

p. 72. Term $\left(\dfrac{\partial H_\phi}{\partial \dot{\varphi}_1}\right)$ was printed as $\left(\dfrac{dH_\phi}{\partial \dot{\varphi}_1}\right)$.

p. 72. Eqn. after 'The laws of electrical action', the term $\dfrac{\partial H_\epsilon}{\partial \dot{\epsilon}_1}$ was printed as $\dfrac{dH_\epsilon}{\partial \dot{\epsilon}_1}$.

p. 86. Eqn. (69), the term $\kappa(x' + \nu t')$ was printed as $\kappa(x' + vt')$.

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THE ORIGIN AND DEVELOPMENT

OF THE

QUANTUM THEORY

BY

MAX PLANCK

TRANSLATED BY

H. T. CLARKE AND **L. SILBERSTEIN**

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THE ORIGIN AND DEVELOPMENT OF THE QUANTUM THEORY

My task this day is to present an address dealing with the subjects of my publications. I feel I can best discharge this duty, the significance of which is deeply impressed upon me by my debt of gratitude to the generous founder of this Institute, by attempting to sketch in outline the history of the origin of the Quantum Theory and to give a brief account of the development of this theory and its influence on the Physics of the present day.

When I recall the days of twenty years ago, when the conception of the physical quantum of 'action' was first beginning to disentangle itself from the surrounding mass of available experimental facts, and when I look back upon the long and tortuous road which finally led to its disclosure, this development strikes me at times as a new illustration of Goethe's saying, that 'man errs, so long as he is striving.' And all the mental effort of an assiduous investigator must indeed appear vain and hopeless, if he does not occasionally run across striking facts which form incontrovertible proof of the truth he seeks, and show him that after all he has moved at least one step nearer to his objective. The pursuit of a goal, the brightness of which is undimmed by initial failure, is an indispensable condition, though by no means a guarantee, of final success.

In my own case such a goal has been for many years the solution of the question of the distribution of energy in the normal spectrum of radiant heat. The discovery by Gustav Kirchhoff that the quality of the heat radiation produced in an enclosure surrounded by any emitting or absorbing bodies whatsoever, all at the same temperature, is entirely independent of the nature of such bodies(1)[1], established the existence of a universal function, which depends only upon the temperature and the wave-length, and is entirely independent of the particular properties of the substance. And the discovery of this remarkable function promised a deeper insight into the relation between energy and temperature, which is the principal problem of thermodynamics and therefore also of the entire field of molecular physics. The only road to this function was to search among all the different bodies occurring in nature, to select one of which the emissive and absorptive powers were known, and to calculate the energy distribution in the heat radiation in equilibrium with that body. This distribution should then, according to Kirchhoff's law, be independent of the nature of the body.

A most suitable body for this purpose seemed H. Hertz's rectilinear oscillator (dipole) whose laws of emission for a given frequency he had just then fully developed(2). If a number of such oscillators be distributed in an enclosure surrounded by reflecting walls, there would take place, in analogy with sources and resonators in the case of sound, an exchange of energy by means of the emission and reception of electro-magnetic waves, and finally what is known as black body radiation corresponding to Kirchhoff's law should establish itself in the vacuum-enclosure. I expected, in a way which certainly seems at

[1]The numbers in brackets refer to the notes at the end of the article.

the present day somewhat naïve, that the laws of classical electrodynamics would suffice, if one adhered sufficiently to generalities and avoided too special hypotheses, to account in the main for the expected phenomena and thus lead to the desired goal. I thus first developed in as general terms as possible the laws of the emission and absorption of a linear resonator, as a matter of fact by a rather circuitous route which might have been avoided had I used the electron theory which had just been put forward by H. A. Lorentz. But as I had not yet complete confidence in that theory I preferred to consider the energy radiating from and into a spherical surface of a suitably large radius drawn around the resonator. In this connexion we need to consider only processes in an absolute vacuum, the knowledge of which, however, is all that is required to draw the necessary conclusions concerning the energy changes of the resonator.

The outcome of this long series of investigations, of which some could be tested and were verified by comparison with existing observations, e. g. the measurements of V. Bjerknes(3) on damping, was the establishment of a general relation between the energy of a resonator of a definite free frequency and the energy radiation of the corresponding spectral region in the surrounding field in equilibrium with it(4). The remarkable result was obtained that this relation is independent of the nature of the resonator, and in particular of its coefficient of damping—a result which was particularly welcome, since it introduced the simplification that the energy of the radiation could be replaced by the energy of the resonator, so that a simple system of one degree of freedom could be substituted for a complicated system having many degrees of freedom.

But this result constituted only a preparatory advance towards the attack on the main problem, which now towered up in all its imposing height. The first attempt to master it failed: for my original hope that the radiation emitted by the resonator would differ in some characteristic way from the absorbed radiation, and thus afford the possibility of applying a differential equation, by the integration of which a particular condition for the composition of the stationary radiation could be reached, was not realized. The resonator reacted only to those rays which were emitted by itself, and exhibited no trace of resonance to neighbouring spectral regions.

Moreover, my suggestion that the resonator might be able to exert a one-sided, i. e. irreversible, action on the energy of the surrounding radiation field called forth the emphatic protest of Ludwig Boltzmann(5), who with his more mature experience in these questions succeeded in showing that according to the laws of the classical dynamics every one of the processes I was considering could take place in exactly the opposite sense. Thus a spherical wave emitted from a resonator when reversed shrinks in concentric spherical surfaces of continually decreasing size on to the resonator, is absorbed by it, and so permits the resonator to send out again into space the energy formerly absorbed in the direction from which it came. And although I was able to exclude such singular processes as inwardly directed spherical waves by the introduction of a special restriction, to wit the hypothesis of 'natural radiation', yet in the course of these investigations it became more and more evident that in the chain of argument an essential link was missing which should lead to the comprehension of the nature of the entire question.

The only way out of the difficulty was to attack the problem from the opposite side, from the standpoint of thermodynamics, a domain in which I felt more at home. And as

a matter of fact my previous studies on the second law of thermodynamics served me here in good stead, in that my first impulse was to bring not the temperature but the entropy of the resonator into relation with its energy, more accurately not the entropy itself but its second derivative with respect to the energy, for it is this differential coefficient that has a direct physical significance for the irreversibility of the exchange of energy between the resonator and the radiation. But as I was at that time too much devoted to pure phenomenology to inquire more closely into the relation between entropy and probability, I felt compelled to limit myself to the available experimental results. Now, at that time, in 1899, interest was centred on the law of the distribution of energy, which had not long before been proposed by W. Wien(6), the experimental verification of which had been undertaken by F. Paschen in Hanover and by O. Lummer and E. Pringsheim of the Reichsanstalt, Charlottenburg. This law expresses the intensity of radiation in terms of the temperature by means of an exponential function. On calculating the relation following from this law between the entropy and energy of a resonator the remarkable result is obtained that the reciprocal value of the above differential coefficient, which I shall here denote by R, is proportional to the energy(7). This extremely simple relation can be regarded as an adequate expression of Wien's law of the distribution of energy; for with the dependence on the energy that of the wave-length is always directly given by the well-established displacement law of Wien(8).

Since this whole problem deals with a universal law of nature, and since I was then, as to-day, pervaded with a view that the more general and natural a law is the simpler it is (although the question as to which formulation is to be regarded as the simpler cannot always be definitely and unambiguously decided), I believed for the time that the basis of the law of the distribution of energy could be expressed by the theorem that the value of R is proportional to the energy(9). But in view of the results of new measurements this conception soon proved untenable. For while Wien's law was completely satisfactory for small values of energy and for short waves, on the one hand it was shown by O. Lummer and E. Pringsheim that considerable deviations were obtained with longer waves(10), and on the other hand the measurements carried out by H. Rubens and F. Kurlbaum with the infra-red residual rays (*Reststrahlen*) of fluorspar and rock salt(11) disclosed a totally different, but, under certain circumstances, a very simple relation characterized by the proportionality of the value of R not to the energy but to the square of the energy. The longer the waves and the greater the energy(12) the more accurately did this relation hold.

Thus two simple limits were established by direct observation for the function R: for small energies proportionality to the energy, for large energies proportionality to the square of the energy. Nothing therefore seemed simpler than to put in the general case R equal to the sum of a term proportional to the first power and another proportional to the square of the energy, so that the first term is relevant for small energies and the second for large energies; and thus was found a new radiation formula(13) which up to the present has withstood experimental examination fairly satisfactorily. Nevertheless it cannot be regarded as having been experimentally confirmed with final accuracy, and a renewed test would be most desirable(14).

But even if this radiation formula should prove to be absolutely accurate it would after

all be only an interpolation formula found by happy guesswork, and would thus leave one rather unsatisfied. I was, therefore, from the day of its origination, occupied with the task of giving it a real physical meaning, and this question led me, along Boltzmann's line of thought, to the consideration of the relation between entropy and probability; until after some weeks of the most intense work of my life clearness began to dawn upon me, and an unexpected view revealed itself in the distance.

Let me here make a small digression. Entropy, according to Boltzmann, is a measure of a physical probability, and the meaning of the second law of thermodynamics is that the more probable a state is, the more frequently will it occur in nature. Now what one measures are only the differences of entropy, and never entropy itself, and consequently one cannot speak, in a definite way, of the absolute entropy of a state. But nevertheless the introduction of an appropriately defined absolute magnitude of entropy is to be recommended, for the reason that by its help certain general laws can be formulated with great simplicity. As far as I can see the case is here the same as with energy. Energy, too, cannot itself be measured; only its differences can. In fact, the concept used by our predecessors was not energy but work, and even Ernst Mach, who devoted much attention to the law of conservation of energy but at the same time strictly avoided all speculations exceeding the limits of observation, always abstained from speaking of energy itself. Similarly in the early days of thermochemistry one was content to deal with heats of reaction, that is to say again with differences of energy, until Wilhelm Ostwald emphasized that many complicated calculations could be materially shortened if energies instead of calorimetric numbers were used. The additive constant which thus remained undetermined for energy was later finally fixed by the relativistic law of the proportionality between energy and inertia(15).

As in the case of energy, it is now possible to define an absolute value of entropy, and thus of physical probability, by fixing the additive constant so that together with the energy (or better still, the temperature) the entropy also should vanish. Such considerations led to a comparatively simple method of calculating the physical probability of a given distribution of energy in a system of resonators, which yielded precisely the same expression for entropy as that corresponding to the radiation law(16); and it gave me particular satisfaction, in compensation for the many disappointments I had encountered, to learn from Ludwig Boltzmann of his interest and entire acquiescence in my new line of reasoning.

To work out these probability considerations the knowledge of two universal constants is required, each of which has an independent meaning, so that the evaluation of these constants from the radiation law could serve as an a posteriori test whether the whole process is merely a mathematical artifice or has a true physical meaning. The first constant is of a somewhat formal nature; it is connected with the definition of temperature. If temperature were defined as the mean kinetic energy of a molecule in a perfect gas, which is a minute energy indeed, this constant would have the value $\frac{2}{3}$(17). But in the conventional scale of temperature the constant assumes (instead of $\frac{2}{3}$) an extremely small value, which naturally is intimately connected with the energy of a single molecule, so that its accurate determination would lead to the calculation of the mass of a molecule and of associated magnitudes. This constant is frequently termed Boltzmann's constant,

although to the best of my knowledge Boltzmann himself never introduced it (an odd circumstance, which no doubt can be explained by the fact that he, as appears from certain of his statements(18), never believed it would be possible to determine this constant accurately). Nothing can better illustrate the rapid progress of experimental physics within the last twenty years than the fact that during this period not only one, but a host of methods have been discovered by means of which the mass of a single molecule can be measured with almost the same accuracy as that of a planet.

While at the time when I carried out this calculation on the basis of the radiation law an exact test of the value thus obtained was quite impossible, and one could scarcely hope to do more than test the admissibility of its order of magnitude, it was not long before E. Rutherford and H. Geiger(19) succeeded, by means of a direct count of the α-particles, in determining the value of the electrical elementary charge as $4.65 \cdot 10^{-10}$, the agreement of which with my value $4.69 \cdot 10^{-10}$ could be regarded as a decisive confirmation of my theory. Since then further methods have been developed by E. Regener, R. A. Millikan, and others(20), which have led to a but slightly higher value.

Much less simple than that of the first was the interpretation of the second universal constant of the radiation law, which, as the product of energy and time (amounting on a first calculation to $6.55 \cdot 10^{-27}$ erg·sec.) I called the elementary quantum of action. While this constant was absolutely indispensable to the attainment of a correct expression for entropy—for only with its aid could be determined the magnitude of the 'elementary region' or 'range' of probability, necessary for the statistical treatment of the problem(21)—it obstinately withstood all attempts at fitting it, in any suitable form, into the frame of the classical theory. So long as it could be regarded as infinitely small, that is to say for large values of energy or long periods of time, all went well; but in the general case a difficulty arose at some point or other, which became the more pronounced the weaker and the more rapid the oscillations. The failure of all attempts to bridge this gap soon placed one before the dilemma: either the quantum of action was only a fictitious magnitude, and, therefore, the entire deduction from the radiation law was illusory and a mere juggling with formulae, or there is at the bottom of this method of deriving the radiation law some true physical concept. If the latter were the case, the quantum would have to play a fundamental rôle in physics, heralding the advent of a new state of things, destined, perhaps, to transform completely our physical concepts which since the introduction of the infinitesimal calculus by Leibniz and Newton have been founded upon the assumption of the continuity of all causal chains of events.

Experience has decided for the second alternative. But that the decision should come so soon and so unhesitatingly was due not to the examination of the law of distribution of the energy of heat radiation, still less to my special deduction of this law, but to the steady progress of the work of those investigators who have applied the concept of the quantum of action to their researches.

The first advance in this field was made by A. Einstein, who on the one hand pointed out that the introduction of the quanta of energy associated with the quantum of action seemed capable of explaining readily a series of remarkable properties of light action discovered experimentally, such as Stokes's rule, the emission of electrons, and the ionization of gases(22), and on the other hand, by the identification of the expression for the

energy of a system of resonators with the energy of a solid body, derived a formula for the specific heat of solid bodies which on the whole represented it correctly as a function of temperature, more especially exhibiting its decrease with falling temperature(23). A number of questions were thus thrown out in different directions, of which the accurate and many-sided investigations yielded in the course of time much valuable material. It is not my task to-day to give an even approximately complete report of the successful work achieved in this field; suffice it to give the most important and characteristic phase of the progress of the new doctrine.

First, as to thermal and chemical processes. With regard to specific heat of solid bodies, Einstein's view, which rests on the assumption of a single free period of the atoms, was extended by M. Born and Th. von Karman to the case which corresponds better to reality, viz. that of several free periods(24); while P. Debye, by a bold simplification of the assumptions as to the nature of the free periods, succeeded in developing a comparatively simple formula for the specific heat of solid bodies(25) which excellently represents its values, especially those for low temperatures obtained by W. Nernst and his pupils, and which, moreover, is compatible with the elastic and optical properties of such bodies. But the influence of the quanta asserts itself also in the case of the specific heat of gases. At the very outset it was pointed out by W. Nernst(26) that to the energy quantum of vibration must correspond an energy quantum of rotation, and it was therefore to be expected that the rotational energy of gas molecules would also vanish at low temperatures. This conclusion was confirmed by measurements, due to A. Eucken, of the specific heat of hydrogen(27); and if the calculations of A. Einstein and O. Stern, P. Ehrenfest, and others have not as yet yielded completely satisfactory agreement, this no doubt is due to our imperfect knowledge of the structure of the hydrogen atom. That 'quantized' rotations of gas molecules (i. e. satisfying the quantum condition) do actually occur in nature can no longer be doubted, thanks to the work on absorption bands in the infra-red of N. Bjerrum, E. v. Bahr, H. Rubens and G. Hettner, and others, although a completely exhaustive explanation of their remarkable rotation spectra is still outstanding.

Since all affinity properties of a substance are ultimately determined by its entropy, the quantic calculation of entropy also gives access to all problems of chemical affinity. The absolute value of the entropy of a gas is characterized by Nernst's chemical constant, which was calculated by O. Sackur by a straightforward combinatorial process similar to that applied to the case of the oscillators(28), while H. Tetrode, holding more closely to experimental data, determined, by a consideration of the process of vaporization, the difference of entropy between a substance and its vapour(29).

While the cases thus far considered have dealt with states of thermodynamical equilibrium, for which the measurements could yield only statistical averages for large numbers of particles and for comparatively long periods of time, the observation of the collisions of electrons leads directly to the dynamic details of the processes in question. Therefore the determination, carried out by J. Franck and G. Hertz, of the so-called resonance potential or the critical velocity which an electron impinging upon a neutral atom must have in order to cause it to emit a quantum of light, provides a most direct method for the measurement of the quantum of action(30). Similar methods leading to perfectly consistent results can also be developed for the excitation of the characteristic X-ray radiation

discovered by C. G. Barkla, as can be judged from the experiments of D. L. Webster, E. Wagner, and others.

The inverse of the process of producing light quanta by the impact of electrons is the emission of electrons on exposure to light-rays, or X-rays, and here, too, the energy quanta following from the action quantum and the vibration period play a characteristic rôle, as was early recognized from the striking fact that the velocity of the emitted electrons depends not upon the intensity(31) but only on the colour of the impinging light(32). But quantitatively also the relations to the light quantum, pointed out by Einstein (p. 7), have proved successful in every direction, as was shown especially by R. A. Millikan, by measurements of the velocities of emission of electrons(33), while the importance of the light quantum in inducing photochemical reactions was disclosed by E. Warburg(34).

Although the results I have hitherto quoted from the most diverse chapters of physics, taken in their totality, form an overwhelming proof of the existence of the quantum of action, the quantum hypothesis received its strongest support from the theory of the structure of atoms (Quantum Theory of Spectra) proposed and developed by Niels Bohr. For it was the lot of this theory to find the long-sought key to the gates of the wonderland of spectroscopy which since the discovery of spectrum analysis up to our days had stubbornly refused to yield. And the way once clear, a stream of new knowledge poured in a sudden flood, not only over this entire field but into the adjacent territories of physics and chemistry. Its first brilliant success was the derivation of Balmer's formula for the spectrum series of hydrogen and helium, together with the reduction of the universal constant of Rydberg to known magnitudes(35); and even the small differences of the Rydberg constant for these two gases appeared as a necessary consequence of the slight wobbling of the massive atomic nucleus (accompanying the motion of electrons around it). As a sequel came the investigation of other series in the visual and especially the X-ray spectrum aided by Ritz's resourceful combination principle, which only now was recognized in its fundamental significance.

But whoever may have still felt inclined, even in the face of this almost overwhelming agreement—all the more convincing, in view of the extreme accuracy of spectroscopic measurements—to believe it to be a coincidence, must have been compelled to give up his last doubt when A. Sommerfeld deduced, by a logical extension of the laws of the distribution of quanta in systems with several degrees of freedom, and by a consideration of the variability of inert mass required by the principle of relativity, that magic formula before which the spectra of both hydrogen and helium revealed the mystery of their 'fine structure'(36), as far as this could be disclosed by the most delicate measurements possible up to the present, those of F. Paschen(37)—a success equal to the famous discovery of the planet Neptune, the presence and orbit of which were calculated by Leverrier [and Adams] before man ever set eyes upon it. Progressing along the same road, P. Epstein achieved a complete explanation of the Stark effect of the electrical splitting of spectral lines(38), P. Debye obtained a simple interpretation of the K-series(39) of the X-ray spectrum investigated by Manne Siegbahn, and then followed a long series of further researches which illuminated with greater or less success the dark secret of atomic structure.

After all these results, for the complete exposition of which many famous names would here have to be mentioned, there must remain for an observer, who does not choose

to pass over the facts, no other conclusion than that the quantum of action, which in every one of the many and most diverse processes has always the same value, namely $6.52 \cdot 10^{-27}$ erg·sec.(40), deserves to be definitely incorporated into the system of the universal physical constants. It must certainly appear a strange coincidence that at just the same time as the idea of general relativity arose and scored its first great successes, nature revealed, precisely in a place where it was the least to be expected, an absolute and strictly unalterable unit, by means of which the amount of action contained in a space-time element can be expressed by a perfectly definite number, and thus is deprived of its former relative character.

Of course the mere introduction of the quantum of action does not yet mean that a true Quantum Theory has been established. Nay, the path which research has yet to cover to reach that goal is perhaps not less long than that from the discovery of the velocity of light by Olaf Römer to the foundation of Maxwell's theory of light. The difficulties which the introduction of the quantum of action into the well-established classical theory has encountered from the outset have already been indicated. They have gradually increased rather than diminished; and although research in its forward march has in the meantime passed over some of them, the remaining gaps in the theory are the more distressing to the conscientious theoretical physicist. In fact, what in Bohr's theory served as the basis of the laws of action consists of certain hypotheses which a generation ago would doubtless have been flatly rejected by every physicist. That with the atom certain quantized orbits [i. e. picked out on the quantum principle] should play a special rôle could well be granted; somewhat less easy to accept is the further assumption that the electrons moving on these curvilinear orbits, and therefore accelerated, radiate no energy. But that the sharply defined frequency of an emitted light quantum should be different from the frequency of the emitting electron would be regarded by a theoretician who had grown up in the classical school as monstrous and almost inconceivable.

But numbers decide, and in consequence the tables have been turned. While originally it was a question of fitting in with as little strain as possible a new and strange element into an existing system which was generally regarded as settled, the intruder, after having won an assured position, now has assumed the offensive; and it now appears certain that it is about to blow up the old system at some point. The only question now is, at what point and to what extent this will happen. If I may express at the present time a conjecture as to the probable outcome of this desperate struggle, everything appears to indicate that out of the classical theory the great principles of thermodynamics will not only maintain intact their central position in the quantum theory, but will perhaps even extend their influence. The significant part played in the origin of the classical thermodynamics by mental experiments is now taken over in the quantum theory by P. Ehrenfest's hypothesis of the adiabatic invariance(41); and just as the principle introduced by R. Clausius, that any two states of a material system are mutually interconvertible on suitable treatment by reversible processes, formed the basis for the measurement of entropy, just so do the new ideas of Bohr show a way into the midst of the wonderland he has discovered.

There is one particular question the answer to which will, in my opinion, lead to an extensive elucidation of the entire problem. What happens to the energy of a light-quantum after its emission? Does it pass outwards in all directions, according to Huygens's

wave theory, continually increasing in volume and tending towards infinite dilution? Or does it, as in Newton's emanation theory, fly like a projectile in one direction only? In the former case the quantum would never again be in a position to concentrate its energy at a spot strongly enough to detach an electron from its atom; while in the latter case it would be necessary to sacrifice the chief triumph of Maxwell's theory—the continuity between the static and the dynamic fields—and with it the classical theory of the interference phenomena which accounted for all their details, both alternatives leading to consequences very disagreeable to the modern theoretical physicist.

Whatever the answer to this question, there can be no doubt that science will some day master the dilemma, and what may now appear to us unsatisfactory will appear from a higher standpoint as endowed with a particular harmony and simplicity. But until this goal is reached the problem of the quantum of action will not cease to stimulate research, and the greater the difficulties encountered in its solution the greater will be its significance for the broadening and deepening of all our physical knowledge.

NOTES

The references to the literature are not claimed to be in any way complete, and are intended to serve only for a preliminary orientation.

(1) G. Kirchhoff, Über das Verhältnis zwischen dem Emissionsvermögen und dem Absorptionsvermögen der Körper für Wärme und Licht. *Gesammelte Abhandlungen.* Leipzig, J. A. Barth, 1882, p. 597 (§17).

(2) H. Hertz, *Ann. d. Phys.* **36**, p. 1, 1889.

(3) *Sitz.-Ber. d. Preuss. Akad. d. Wiss.* Febr. 20, 1896. *Ann. d. Phys.* **60**, p. 577, 1897.

(4) *Sitz.-Ber. d. Preuss. Akad. d. Wiss.* May 18, 1899, p. 455.

(5) L. Boltzmann, *Sitz.-Ber. d. Preuss. Akad. d. Wiss.* March 3, 1898, p. 182.

(6) W. Wien, *Ann. d. Phys.* **58**, p. 662, 1896.

(7) According to Wien's law of the distribution of energy the dependence of the energy U of the resonator upon the temperature is given by a relation of the form:

$$U = a \cdot e^{-b/t}.$$

Since

$$\frac{1}{T} = \frac{dS}{dU},$$

where S is the entropy of the resonator, we have for R as used in the text:

$$R = 1 : \frac{d^2S}{dU^2} = -bU.$$

(8) According to Wien's displacement law, the energy U of the resonator with the natural vibration period ν is expressed by:

$$U = \nu \cdot f\left(\frac{T}{\nu}\right).$$

(9) *Ann. d. Phys.* **1**, p. 719, 1900.

(10) O. Lummer und E. Pringsheim, *Verhandl. der Deutschen Physikal. Ges.*, **2**, p. 163, 1900.

(11) H. Rubens and F. Kurlbaum, *Sitz.-Ber. der Preuss. Akad. d. Wiss.* Oct. 25, 1900, p. 929.

(12) It follows from the experiments of H. Rubens and F. Kurlbaum that, for high temperatures, $U = cT$. Then, in accordance with the method quoted in (7):

$$R = 1 : \frac{d^2S}{dU^2} = -\frac{U^2}{c}.$$

(13) Put

$$R = 1 : \frac{d^2S}{dU^2} = -bU - \frac{U^2}{c},$$

then by integration,

$$\frac{1}{T} = \frac{dS}{dU} = \frac{1}{b} \log \left\{ 1 + \frac{bc}{U} \right\}$$

whence the radiation formula,

$$U = bc : (e^{-b/T} - 1).$$

Cf. *Verhandlungen der Deutschen Phys. Ges.* Oct. 19, 1900, p. 202.

(14) Cf. W. Nernst und Th. Wulf, *Verh. d. Deutsch. Phys. Ges.* **21**, p. 294, 1919.

(15) For the absolute value of the energy is equal to the product of the inert mass and the square of light velocity.

(16) *Verhandlungen der Deutschen Phys. Ges.* Dec. 14, 1900, p. 237.

(17) Generally, if k be the first radiation constant, the mean kinetic energy of a gas molecule is:

$$U = \frac{3}{2} kT$$

If we put, therefore, $T = U$, then $k = \frac{2}{3}$. In the conventional [absolute Kelvinian] temperature scale, however, T is defined by putting the temperature difference between boiling and freezing water equal to 100.

(18) Cf. for example L. Boltzmann, Zur Erinnerung an Josef Loschmidt, *Populäre Schriften*, p. 245, 1905.

(19) E. Rutherford and H. Geiger, *Proc. Roy. Soc.* A. Vol. **81**, p. 162, 1908.

(20) Cf. R. A. Millikan, *Phys. Zeitschr.* **14**, p. 796, 1913.

(21) The evaluation of the probability of a physical state is based upon counting that finite number of equally probable special cases by which the corresponding state is realized; and in order sharply to distinguish these cases from one another, a definite concept of each special case has necessarily to be introduced.

(22) A. Einstein, *Ann. d. Phys.* **17**, p. 132, 1905.

(23) A. Einstein, *Ann. d. Phys.* **22**, p. 180, 1907.

(24) M. Born und Th. v. Karman, *Phys. Zeitschr.* **14**, p. 15, 1913.

(25) P. Debye, *Ann. d. Phys.* **39**, p. 789, 1912.

(26) W. Nernst, *Phys. Zeitschr.* **13**, p. 1064, 1912.

(27) A. Eucken, *Sitz.-Ber. d. Preuss. Akad. d. Wiss.* p. 141, 1912.

(28) O. Sackur, *Ann. d. Phys.* **36**, p. 958, 1911.

(29) H. Tetrode, *Proc. Acad. Sci. Amsterdam*, Febr. 27 and March 27, 1915.

(30) J. Franck und G. Hertz, *Verh. d. Deutsch. Phys. Ges.* **16**, p. 512, 1914.

(31) Ph. Lenard, *Ann. d. Phys.* **8**, p. 149, 1902.

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ÜBER

THERMODYNAMIK

VON

Dr. MAX PLANCK
PROFESSOR DER THEORETISCHEN PHYSIK
AN DER UNIVERSITÄT BERLIN.

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Aus dem Vorwort zur ersten Auflage.

Die erste Anregung zur Abfassung des vorliegenden Buches empfing ich durch mehrfach an mich ergangene Aufforderungen, meine in das Gebiet der Thermodynamik fallenden Abhandlungen gesammelt herauszugeben bzw. zu einer zusammenfassenden Darstellung zu verarbeiten. Wenn auch das erstere Verfahren als das einfachere näher gelegen hätte, so entschied ich mich doch für eine neue Überarbeitung des ganzen Stoffes, weil sich auf diese Weise Gelegenheit bot, mittels einer entsprechenden Erweiterung des behandelten Themas das ganze Gebiet der Thermodynamik in eine einheitliche Darstellung zusammenzufassen. Hierdurch ist dem Werke allerdings der Charakter einer Forschungsarbeit genommen und ihm mehr derjenige eines Lehrbuches gegeben, bestimmt zur Einführung in das Studium der Thermodynamik für jeden, der einen Anfängerkurs in Physik und Chemie durchgemacht hat und mit den Elementen der Differential- und Integralrechnung vertraut ist.

Die in den beispielsweise durchgeführten Anwendungen der Theorie benutzten Zahlenwerte stammen fast alle aus den Originalarbeiten; nur einige durch häufige Messungen bestimmte Größen sind tabellarischen Zusammenstellungen, namentlich denen in F. Kohlrausch s Leitfaden der praktischen Physik, entnommen. Doch unterlasse ich nicht hervorzuheben, daß die benutzten Einzelzahlen, bei aller angewendeten Sorgfalt, doch bei weitem nicht denselben Grad von kritischer Sichtung erfahren haben, wie die mitgeteilten Sätze und Ableitungen allgemeineren Inhalts.

In der bisherigen Entwicklung der Thermodynamik lassen sich deutlich drei voneinander verschiedene Methoden der Forschung unterscheiden. Die erste greift am tiefsten hinein in das Wesen der betrachteten Vorgänge, sie wäre daher, wenn sie sich exakt durchführen ließe, jedenfalls als die vollkommenste zu bezeichnen. Nach ihr wird die Wärme bedingt durch bestimmte Bewegungen der als diskrete Massen gedachten chemischen Moleküle und Atome, die für gasförmige Körper verhältnismäßig einfache Eigenschaften haben, während sie sich für feste und flüssige Körper bisher nur in rohen Zügen angeben lassen. Diese kinetische Theorie hat seit ihrer Begründung durch Joule, Waterston, Krönig und Clausius besonders durch Maxwell und Boltzmann wesentliche Erweiterung und Vertiefung erfahren, scheint aber in ihrer weiteren Entwicklung auf vorläufig unüberwindliche Hindernisse zu stoßen, die nicht nur in der hochgradig komplizierten mathematischen Durchführung der angenommenen Hypothesen,

sondern vor allen Dingen in prinzipiellen, hier nicht näher zu erörternden Schwierigkeiten bei der mechanischen Deutung der thermodynamischen Hauptsätze begründet sind.

Derartige spezielle Schwierigkeiten vermeidet eine zweite, namentlich von HELMHOLTZ ausgebildete, Methode der Thermodynamik, indem sie sich auf die wichtigste Voraussetzung der mechanischen Wärmetheorie beschränkt, daß Wärme auf Bewegung beruht, dagegen auf ein Spezialisieren der Vorstellungen von der Natur dieser Bewegungen zunächst grundsätzlich verzichtet. Dieser Standpunkt ist sicherer als der vorige, er gewährt auch die volle philosophische Befriedigung, die die mechanische Naturauffassung überhaupt liefert, aber der Halt, den er bietet, ist bis jetzt nicht breit genug, um darauf eine Theorie im einzelnen aufzubauen. Alles, was man von ihm ausgehend erreichen kann, ist die Bestätigung einiger allgemeiner schon anderweitig direkt aus der Erfahrung abgeleiteter Gesetze.

Am fruchtbarsten hat sich bisher eine dritte Behandlung der Thermodynamik erwiesen. Diese Methode unterscheidet sich von den beiden zuerst besprochenen wesentlich dadurch, daß sie die mechanische Natur der Wärme nicht in den Vordergrund stellt, sondern, indem sie sich bestimmter Annahmen über das Wesen der Wärme ganz enthält, statt dessen direkt von einigen sehr allgemeinen Erfahrungstatsachen, hauptsächlich von den sogenannten beiden Hauptsätzen der Wärmelehre, ausgeht. Daraus ergeben sich dann auf rein logischem Wege eine große Reihe neuer Sätze der Physik und Chemie, die sich weitgehender Anwendungen fähig gezeigt und bis jetzt überall ausnahmslos bewährt haben.

Diese letzte Behandlungsart, welche im vorliegenden Werke ausschließlich benutzt ist, entspricht wohl am besten dem heutigen Stande der Wissenschaft, sie ist aber kaum als die abschließende zu betrachten, sondern wird wahrscheinlich künftig einmal einer mechanischen oder vielleicht auch einer elektromagnetischen Betrachtungsweise Platz machen müssen. Denn wenn es auch eine Zeitlang Vorteil gewähren mag, die einzelnen Wirkungen der Natur: Wärme, Bewegung, Elektrizität usw. zunächst als qualitativ verschieden voneinander einzuführen und die Frage nach ihrer etwaigen Wesensgemeinschaft zu unterdrücken, so wird doch unser durch die Entdeckung des Prinzips der Erhaltung der Energie so mächtig gefördertes Streben nach einer einheitlichen Naturanschauung, sei es auf mechanischer oder auf anderer Grundlage, sich niemals auf die Dauer zurückhalten lassen; würde doch schon heute ein Zurücktreten von der Annahme der Wesensgleichheit aller physikalischen Vorgänge gleichbedeutend sein mit dem Verzicht auf das Verständnis einer Reihe bereits erkannter Gesetzmäßigkeiten zwischen verschiedenen Gebieten der Natur. Dann werden selbstverständlich die hier

aus den beiden Hauptsätzen der Wärmelehre abgeleiteten Ergebnisse nicht erschüttert werden, sondern es werden nur diese beiden Sätze nicht mehr selbstständig eingeführt, sondern ihrerseits aus anderen noch allgemeineren Sätzen abgeleitet werden. Es ist aber bis jetzt die Zeit noch nicht abzusehen, in welcher der weite Weg zu diesem Ziel zurückgelegt werden kann.

Berlin, im April 1897.

Aus dem Vorwort zur zweiten Auflage.

Die zahlreichen und wertvollen Untersuchungen, durch welche seit dem Erscheinen der ersten Auflage dieses Buches die Thermodynamik bereichert worden ist, haben, besonders auf dem Gebiete der physikalischen Chemie, die Fülle der bekannten Tatsachen erheblich vermehrt, ohne jedoch die Grundlagen der Theorie irgendwie zu verändern. Da nun dies Buch hauptsächlich der Darstellung der letzteren gewidmet ist, und die speziellen hier gegebenen Anwendungen mehr den Charakter von Erläuterungsbeispielen besitzen, so glaubte ich von einer Neubearbeitung des Stoffes ganz absehen zu dürfen und habe mich darauf beschränkt, einzelne numerische Daten zu verbessern und im übrigen nur die allgemeineren Gedankengänge sorgfältig nachzuprüfen. Dabei habe ich eine Reihe von kleinen Änderungen und Zusätzen zweckmäßig gefunden, deren manche mir von wissenschaftlichen Bekannten und Fachgenossen freundlichst nahegelegt wurden.

Im Hinblick auf den Schlußpassus im Vorwort zur ersten Auflage sei mir noch die Bemerkung verstattet, daß die Theorie der Wärme auf dem dort angedeuteten Wege in der Zwischenzeit einen bemerkenswerten Schritt vorwärts getan hat. Die neueren Forschungsergebnisse auf dem Gebiete der Wärmestrahlung, bei deren Erwähnung ich hier nur auf die Namen W. Wien, F. Paschen, O. Lummer und E. Pringsheim, H. Rubens und F. Kurlbaum hinweisen möchte, haben nämlich immer deutlicher erkennen lassen, daß, ebenso wie der erste Hauptsatz der Thermodynamik nur eine Seite des universalen Prinzips der Erhaltung der Energie bildet, so auch der zweite Hauptsatz, das Prinzip der Vermehrung der Entropie, keine selbständige Bedeutung besitzt, sondern sich seinem vollen Inhalt nach verstehen lassen wird, wenn man seine Wurzel, entsprechend der von Clausius und Maxwell begründeten und dann namentlich von L. Boltzmann weiter gebildeten Auffassung, in den bekannten Sätzen der Wahrscheinlichkeitsrechnung sucht. Danach ist die Entropie irgend

eines natürlichen Zustandes ganz allgemein gleich dem Logarithmus der „Wahrscheinlichkeit“ des betreffenden Zustandes, multipliziert mit einer universellen Konstanten von der Dimension einer Energie dividiert durch eine Temperatur. Eine nähere Besprechung dieser Beziehung, welche tiefer als bisher in die Erkenntnis der Molekularvorgänge sowohl, wie auch der Strahlungsgesetze hineinführt, würde jedoch den von vornherein ausdrücklich festgelegten Rahmen des vorliegenden Werkes überschreiten und ist daher nicht in dasselbe aufgenommen worden, zumal ich dieselbe in einem besonderen Buche zu behandeln gedenke. –

Berlin, im Januar 1905.

Vorwort zur dritten Auflage.

Der Plan der Darstellung und die Anordnung des Stoffes ist auch in der neuen Auflage beibehalten. Doch finden sich in ihr, abgesehen von einer abermaligen Revision sämtlicher numerischer Daten, durch das ganze Buch verstreut eine Anzahl neuer, größerer oder kleinerer, Erläuterungen und Zusätze, deren Zweckmäßigkeit mir durch irgend einen äußeren Anlaß nahe gelegt wurde. Von derartigen Stellen erwähne ich hier beispielsweise das Gesetz der übereinstimmenden Zustände, die Definition des Molekulargewichts, den Beweis des zweiten Hauptsatzes, die charakteristische thermodynamische Funktion, die Theorie des Joule-Thomson-Effekts, die Verdampfung von Flüssigkeitsgemischen. Weitere Anregungen werde ich stets mit besonderem Dank entgegennehmen.

Eine sachliche Erweiterung von prinzipieller Bedeutung aber bildet die Einführung des im Jahre 1906 von W. Nernst aufgestellten Wärmetheorems. Wenn dieser Satz, wie es bis jetzt den Anschein hat, sich nach allen Richtungen hin bewähren sollte, so wäre damit die Thermodynamik um ein Prinzip bereichert, dessen Tragweite, nicht nur in praktischer, sondern auch in molekulartheoretischer Hinsicht, noch gar nicht abzusehen ist.

Um den wesentlichen Inhalt dieses neuen Theorems ganz rein, in einer für die experimentelle Prüfung möglichst geeigneten Form darstellen zu können, ist es aber nach meiner Meinung notwendig, seine Bedeutung für die atomistische Theorie, die heute noch keineswegs klar gestellt ist, einstweilen ganz aus dem Spiele zu lassen; und dieser Standpunkt entspricht gerade der auch sonst überall in diesem Buche befolgten Methode. Andrerseits habe ich dem Theorem, um seine Anwendungen so einfach wie umfangreich

zu gestalten, eine möglichst weitgehende Fassung geben zu sollen geglaubt, und bin dabei, nicht nur in der Form, sondern auch inhaltlich, über die von NERNST selber gegebene noch etwas hinausgegangen. Ich erwähne diesen Punkt auch an dieser Stelle, weil die Möglichkeit im Auge zu behalten ist, daß, wenn die weitergehende Fassung sich nicht bewähren sollte, die ursprüngliche NERNSTsche deswegen doch möglicherweise zu Recht bestehen bleiben könnte.

Berlin, im November 1910.

Vorwort zur fünften Auflage.

Für die fünfte Auflage habe ich den ganzen Inhalt des Buches wieder einmal überarbeitet, insbesondere den Abschnitt über das NERNSTsche Wärmetheorem, welches inzwischen, und zwar in seiner am weitesten gehenden Fassung, überall ausnahmslos Bestätigung gefunden hat und somit jetzt wohl in den sicheren Besitzstand der Theorie übergegangen ist. Seine atomistische Bedeutung, welche in den engen Beziehungen zur Quantenhypothese ihren Ausdruck findet, kann freilich im Rahmen des vorliegenden Werkes selbstverständlich nicht gewürdigt werden.

Berlin, im März 1917.

Vorwort zur sechsten und siebenten Auflage.

Unter den Ergänzungen, welche die neue Auflage erfahren hat, erwähne ich hier die von J. CH. GHOSH (Calcutta) aufgestellte Theorie für die Gefrierpunktsdepression starker Elektrolyte, welche die durch lange Zeit hindurch rätselhaft gebliebenen Abweichungen vom OSTWALDschen Verdünnungsgesetz endgültig aufzuklären scheint (§ 273), sowie die DEBYEsche Zustandsgleichung fester Körper, welche sowohl die Temperaturabhängigkeit der spezifischen Wärme als auch den GRÜNEISENschen Satz für den thermischen Ausdehnungskoeffizienten in sich enthält (§ 285).

Berlin-Grunewald, im Februar 1921 und im Juli 1922.

Der Verfasser.

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Erster Abschnitt.

Grundtatsachen und Definitionen.

Erstes Kapitel. Temperatur.

§ 1. Der Begriff „Wärme“ entspringt aus derjenigen Sinnesempfindung, die uns bei der direkten Berührung eines Körpers unmittelbaren Aufschluß über den Unterschied zwischen Warm und Kalt liefert. Ein quantitatives, wissenschaftlich brauchbares Maß für den Wärmezustand eines Körpers läßt sich aber aus der unmittelbaren Empfindung, die nur qualitative und je nach den äußeren Umständen veränderliche Resultate ergibt, nicht ableiten. Man benutzt zu diesem Zweck eine andere Erscheinung, die erfahrungsgemäß bei allen Körpern gleichzeitig mit der Erwärmung auftritt, wenn der äußere Druck konstant bleibt, und die den Vorteil einer genauen Messung darbietet: die Volumenänderung. Bei den meisten Substanzen ist mit der Erwärmung eine Volumenvergrößerung verbunden. Sonach läßt sich nicht nur durch direkte Berührung, sondern auch durch eine rein mechanische Beobachtung, und zwar durch letzteres Mittel in viel feinerem Grade, entscheiden, ob ein Körper wärmer oder kälter wird. Auch läßt sich genau angeben, wenn ein Körper einen früher einmal innegehabten Wärmezustand wiederum einnimmt.

§ 2. Wenn zwei Körper, die sich sehr verschieden warm anfühlen, z. B. eine erhitzte Metallmasse und kaltes Wasser, in Berührung gebracht werden, so findet man immer, daß der wärmere sich abkühlt, der kältere sich erwärmt, bis zu einer gewissen Grenze, wo jede Veränderung aufhört. Dann sagt man mit einem aus der Mechanik übertragenen Sprachgebrauch: die beiden Körper stehen im Wärmegleichgewicht. Ein solches Wärmegleichgewicht tritt erfahrungsgemäß schließlich immer ein, auch wenn nicht zwei, sondern beliebig viele verschieden warme Körper in beliebige wechselseitige Berührung miteinander gebracht werden. Hieraus folgt sogleich der wichtige Satz: Wenn ein Körper A mit zwei anderen Körpern B und C im Wärmegleichgewicht steht, so stehen auch B und C unter sich im Wärmegleichgewicht. Verbindet man nämlich die Körper A, B, C hintereinander zu einem

Ringe, so daß jeder der drei Körper die beiden andern berührt, so besteht nach der Voraussetzung an den Berührungsstellen (AB) und (AC) Wärmegleichgewicht, folglich auch an der Stelle (BC); denn sonst würde überhaupt kein allgemeines Wärmegleichgewicht möglich sein, was der durch den vorigen Satz angegebenen Erfahrung widerspräche.

§ 3. Hierauf beruht die Möglichkeit, den Wärmezustand irgend zweier Körper B und C zu vergleichen, ohne sie direkt miteinander in Berührung zu bringen. Man bringt nämlich jeden einzeln mit dem als Meßinstrument dienenden, zunächst beliebig ausgewählten Körper A zusammen (z. B. einem in ein enges Rohr ausmündenden Quecksilbervolumen) und kann so durch jedesmalige Beobachtung des Volumens von A entscheiden, ob B und C im Wärmegleichgewicht stehen oder nicht, bez. welcher von beiden Körpern der wärmere ist. Den Wärmezustand des Körpers A und somit auch jedes mit A im Wärmegleichgewicht befindlichen Körpers kann man einfach definieren durch das Volumen von A, oder auch, wie gewöhnlich, durch die Differenz des Volumens von A und eines nach Willkür fixierten „Normalvolumens“, nämlich desjenigen Volumens, welches der Körper A einnimmt, wenn er sich mit schmelzendem Eis unter Atmosphärendruck im Wärmegleichgewicht befindet. Ist die Einheit dieser Volumendifferenz so gewählt, daß sie gleich 100 wird, wenn sich A mit dem Dampfe siedenden Wassers unter Atmosphärendruck im Wärmegleichgewicht befindet, so heißt sie die Temperatur (in Celsiusgraden) in bezug auf den Körper A als thermometrische Substanz. Zwei Körper von gleicher Temperatur stehen also immer im Wärmegleichgewicht, und umgekehrt.

§ 4. Die Temperaturangaben zweier verschiedener thermometrischer Substanzen stimmen, außer bei 0° und bei 100°, im allgemeinen niemals überein, weshalb in der bisherigen Definition der Temperatur noch eine große Willkür herrscht. Dieselbe kann hier nur bis zu einem gewissen Grade beseitigt werden, nämlich durch die Benutzung der Erfahrung, daß die verschiedenen Gase, besonders die schwer kondensierbaren, wie Wasserstoff, Sauerstoff, Stickstoff, Kohlenoxyd und alle sogenannten Edelgase, als thermometrische Substanzen innerhalb eines beträchtlichen Temperaturbereichs eine fast vollkommene, für die meisten Messungen genügende Übereinstimmung in den Temperaturangaben liefern. Ja noch mehr: auch die absolute Größe der Ausdehnung ist bei allen diesen Gasen insofern die nämliche, als gleiche Volumina derselben sich bei gleicher Erwärmung immer um gleichviel ausdehnen, konstanten äußeren Druck vorausgesetzt. Der Betrag dieser Ausdehnung ist für eine Erwärmung von 0° auf 1° etwa der 273. Teil des Volumens. Da nun endlich auch der Einfluß des äußeren Druckes auf das Volumen eines dieser Gase durch ein sehr einfaches Gesetz dargestellt

wird, so ist der Schluß gestattet, daß diese Regelmäßigkeiten auf einer besonders einfachen Konstitution dieser Substanzen beruhen, und daß es daher rationell ist, die von ihnen angegebene gemeinschaftliche Temperatur als Temperatur schlechthin zu definieren. Es müssen also die Angaben aller anderen Thermometer auf das Gasthermometer reduziert werden.

§ 5. Bei Genauigkeitsanforderungen, für welche die Übereinstimmung in den Angaben der verschiedenen Gasthermometer nicht genügt, bleibt die Willkür in der Definition der Temperatur bestehen, da kein Grund vorliegt, ein bestimmtes Gas vor den anderen zu bevorzugen. Eine von den Eigenschaften einzelner Körper vollkommen unabhängige Definition der Temperatur, gültig für alle Wärme- und Kältegrade, wird erst möglich auf Grund des zweiten Hauptsatzes der Wärmetheorie (siehe unten § 160 ff.). Bis dahin wird daher nur von solchen Temperaturen die Rede sein, welche durch das Gasthermometer mit hinreichender Schärfe definiert sind.

§ 6. Wir beschäftigen uns im folgenden vorwiegend mit homogenen isotropen Körpern von beliebiger Form, die im Innern gleichmäßige Temperatur und Dichte besitzen und einem gleichmäßigen, überall senkrecht auf ihre Oberfläche wirkenden Druck unterworfen sind, folglich auch den nämlichen Druck nach außen hin ausüben. Von Oberflächenerscheinungen sehen wir dabei ab. Der Zustand eines solchen Körpers ist bestimmt durch seine chemische Natur, seine Masse M, sein Volumen V und seine Temperatur t. Alle anderen Eigenschaften des Zustandes sind also von den angegebenen in bestimmter Weise abhängig, vor allem der Druck, welcher gleichmäßig im ganzen Innern herrscht und ebenso nach außen hin wirkt. Der Druck p wird gemessen durch die Kraft, welche auf die Flächeneinheit der Oberfläche wirkt, also im C.G.S.-System durch Dynen pro Quadratzentimeter, wobei ein Dyn die Kraft ist, welche der Masse eines Gramms in einer Sekunde die Geschwindigkeit von einem Zentimeter in der Sekunde erteilt.

§ 7. In der Praxis mißt man gewöhnlich den Druck in Atmosphären, es soll daher hier der Wert einer Atmosphäre im absoluten C.G.S.-System berechnet werden. Der Druck einer Atmosphäre ist die Kraft, welche eine Quecksilbersäule von 0° C, 76 cm Höhe und 1 qcm Querschnitt durch ihr Gewicht auf ihre Grundfläche ausübt, wenn sie an einem Orte von der geographischen Breite 45° aufgestellt ist. Der letzte Zusatz ist notwendig, weil das durch die Erdanziehung bedingte Gewicht sich mit dem Orte ändert. Das Volumen der Quecksilbersäule beträgt 76, ihre Masse, durch Multiplikation des Volumens mit der Dichte des Quecksilbers bei 0°, $76 \cdot 13,596$; daher ihr Gewicht, durch Multiplikation der Masse mit der

Beschleunigung der Schwere an einem Orte von 45° Breite:

$$76 \cdot 13,596 \cdot 980,6 = 1\,013\,250\,\frac{\text{dyn}}{\text{cm}^2} \text{ oder } \frac{\text{g}}{\text{cm sec}^2}.$$

Dies ist also der Druck einer Atmosphäre im absoluten C.G.S.-System. Würde man als Krafteinheit nicht ein Dyn, sondern, wie es früher in der Mechanik üblich war, das Gewicht eines Gramms an einem Orte von der geographischen Breite 45° benutzen, so würde der Druck einer Atmosphäre betragen: $76 \cdot 13,596 = 1033,3$.

§ 8. Da der Druck des betrachteten Körpers offenbar nur von seiner inneren Beschaffenheit, nicht aber von seiner äußeren Form und seiner Masse abhängt, so folgt, daß p außer von der Temperatur nur von dem Verhältnis der Masse M zum Volumen V, d. h. von der Dichte, abhängt, bez. von dem umgekehrten Verhältnis, dem Volumen der Masseneinheit:

$$\frac{V}{M} = v,$$

welches wir, wie üblich, als das spezifische Volumen des Körpers bezeichnen. Es existiert also eine bestimmte, jeder Substanz eigentümliche Beziehung:

$$p = f(v, t),$$

welche die Zustandsgleichung der Substanz genannt wird. Die Funktion f besitzt für Gase stets positive, für flüssige und feste Körper unter Umständen auch negative Werte.

§ 9. Ideale Gase. Am einfachsten gestaltet sich die Form der Zustandsgleichung für diejenigen Substanzen, welche wir oben § 4 zur Definition der Temperatur benutzt haben und die, insofern sie übereinstimmende Temperaturangaben liefern, als „ideale“ oder „vollkommene“ Gase bezeichnet werden. Wird nämlich die Temperatur konstant gehalten, so ist nach dem Gesetz von Boyle (Mariotte) das Produkt aus Druck und spezifischem Volumen konstant:

$$pv = \vartheta, \tag{1}$$

wobei ϑ, außer von der Natur des Gases, allein von der Temperatur t abhängt.

Wenn aber der Druck konstant gehalten wird, so ist nach der Definition § 3 die Temperatur proportional der Differenz des jeweiligen Volumens v und des Normalvolumens: v_0, d. h.

$$t = (v - v_0)P, \tag{2}$$

worin P nur vom Druck p abhängt. Hierbei ist nach Gleichung (1)

$$pv_0 = \vartheta_0, \tag{3}$$

wenn ϑ_0 den Wert bezeichnet, den die Temperaturfunktion ϑ für $t = 0$ annimmt.

Endlich benutzen wir noch die ebenfalls schon oben, § 4, angeführte Erfahrung, daß der Betrag der Ausdehnung bei einer Erwärmung von 0° auf 1° für alle idealen Gase der nämliche Bruchteil α (etwa $= \frac{1}{273}$) des Volumens bei 0° ist. (Gesetz von Gay Lussac.) Setzt man also $t = 1$, so wird $v - v_0 = \alpha v_0$, und die Gleichung (2) geht über in:

$$1 = \alpha v_0 P. \tag{4}$$

Durch Elimination von P, v_0 und v aus den Gleichungen (1), (2), (3), (4) ergibt sich die Temperaturfunktion:

$$\vartheta = \vartheta_0(1 + \alpha t),$$

also lineär abhängig von der Temperatur, und die Zustandsgleichung (1) wird:

$$p = \frac{\vartheta_0}{v}(1 + \alpha t).$$

§ 10. Diese Gleichung nimmt eine wesentlich einfachere Form an, wenn man den im § 3 willkürlich festgesetzten Nullpunkt der Temperatur um $\frac{1}{\alpha}$ Grad verschiebt, indem man den Schmelzpunkt des Eises nicht $= 0°$, sondern $= \frac{1}{\alpha}$ (etwa $= 273°$) setzt. Schreibt man nämlich:

$$t + \frac{1}{\alpha} = T$$

(absolute Temperatur), und setzt zur Abkürzung die Konstante $\alpha\vartheta_0 = C$, so wird die Zustandsgleichung:

$$p = \frac{C}{v} \cdot T = C \cdot \frac{M}{V} \cdot T. \tag{5}$$

Die Einführung der absoluten Temperatur kommt offenbar im Grunde nur darauf hinaus, daß man die Temperatur nicht, wie in § 3, durch eine Volumendifferenz, sondern durch das Volumen selbst mißt.

Die naheliegende Frage nach der physikalischen Bedeutung des Nullpunkts der absoluten Temperatur ist offenbar dahin zu beantworten, daß die

absolute Temperatur Null diejenige Temperatur darstellt, bei welcher ein ideales Gas in einem endlichen Volumen den Druck Null, oder unter einem endlichen Druck das Volumen Null besitzt. Für wirkliche Gase hat aber dieser Satz keine Bedeutung, da dieselben bei gehöriger Abkühlung merkliche Abweichungen voneinander, also auch vom idealen Zustand zeigen. Inwieweit ein wirkliches Gas auch bei mittleren Temperaturänderungen von dem idealen Verhalten abweicht, kann natürlich erst dann geprüft werden, wenn die Definition der Temperatur von der Bezugnahme auf eine spezielle Substanz unabhängig gemacht worden ist. (Vgl. § 5.)

§ 11. Die für die Natur eines idealen Gases charakteristische Konstante C ist bestimmt, wenn man für irgendein Wertenpaar von T und p, z. B. 0° C und Atmosphärendruck, das spezifische Volumen v des Gases kennt, und zwar verhalten sich offenbar für verschiedene Gase, bei derselben Temperatur und demselben Druck genommen, die Werte der Konstanten C wie die spezifischen Volumina v, oder umgekehrt wie die Dichten $\frac{1}{v}$. Man kann also sagen: Bei derselben Temperatur und demselben Druck genommen stehen die Dichten aller idealen Gase in unveränderlichen Verhältnissen. Man charakterisiert daher oft auch ein Gas durch das konstante Verhältnis seiner Dichte zu der Dichte eines Normalgases bei demselben Druck und derselben Temperatur (spezifische Dichte in bezug auf Luft oder auf Wasserstoff). Bei 0° C ($T = 273$) und 1 Atmosphäre Druck ist die Dichte von:

Wasserstoff	0, 00008988	$\frac{\text{g}}{\text{cm}^3}$
Sauerstoff	0, 0014291	
Stickstoff	0, 0012507	
„Atmosphärischer“ Stickstoff	0, 0012567	
Luft	0, 0012928	
Argon	0, 0017809	

woraus die entsprechenden Werte von C in absolutem Maß leicht zu berechnen.

Durch die Zustandsgleichung einer Substanz lassen sich alle Fragen nach dem Verhalten der Substanz in bezug auf beliebige Änderungen der Temperatur, des Volumens und des Druckes vollständig beantworten.

§ 12. Verhalten bei konstantem Druck. (Isobare oder isopiestische Änderungen.) Ausdehnungskoeffizient heißt das Verhältnis der Zunahme des Volumens bei Erwärmung um 1° zu dem Volumen bei 0° C, d. h. die Größe: $\frac{V_{T+1} - V_T}{V_0}$, wofür man, da das Volumen sich in der Regel verhältnismäßig langsam mit der Temperatur ändert, auch $\left(\frac{\partial V}{\partial T}\right)_p \cdot \frac{1}{V_0}$ setzen kann. Für

ein ideales Gas ist nach der Zustandsgleichung (5) $V_{T+1} - V_T = \frac{CM}{p}$ und $V_0 = \frac{CM}{p} \cdot 273$, also der Ausdehnungskoeffizient des Gases: $\frac{1}{273} = \alpha$.

§ 13. Verhalten bei konstantem Volumen. (Isochore oder isopykne oder isostere Änderungen.) Spannungskoeffizient heißt das Verhältnis der Zunahme des Druckes bei Erwärmung um 1° zu dem Druck bei 0° C, d. h. die Größe: $\frac{p_{T+1} - p_T}{p_0}$ oder $\left(\frac{\partial p}{\partial T}\right)_V \cdot \frac{1}{p_0}$. Für ein ideales Gas ist nach der Zustandsgleichung (5) $p_{T+1} - p_T = \frac{CM}{V}$ und $p_0 = \frac{CM}{V} \cdot 273$, also der Spannungskoeffizient des Gases: $\frac{1}{273}$, gleich dem Ausdehnungskoeffizienten α.

§ 14. Verhalten bei konstanter Temperatur. (Isotherme Änderungen.) Elastizitätskoeffizient heißt das Verhältnis einer unendlich kleinen Zunahme des Druckes zu der dadurch bedingten Kontraktion der Volumeneinheit, d. h. die Größe: $dp : \left(-\frac{dV}{V}\right) = -\left(\frac{\partial p}{\partial v}\right)_T \cdot v$. Für ein ideales Gas ist nach der Zustandsgleichung (5)

$$\left(\frac{\partial p}{\partial v}\right)_T = -\frac{CT}{v^2},$$

und daher der Elastizitätskoeffizient des Gases:

$$\frac{CT}{v} = p, \quad \text{also gleich dem Druck.}$$

Der reziproke Wert des Elastizitätskoeffizienten, nämlich das Verhältnis einer unendlich kleinen Kontraktion der Volumeneinheit zu der entsprechenden Druckvermehrung, also $-\left(\frac{\partial v}{\partial p}\right)_T \cdot \frac{1}{v}$, heißt Kompressibilitätskoeffizient.

§ 15. Die drei Koeffizienten, welche das Verhalten einer Substanz bei isobaren, isochoren und isothermen Änderungen kennzeichnen, sind nicht unabhängig voneinander, sondern, für jede beliebige Substanz, durch eine feste Beziehung verknüpft. Durch Differentiation der Zustandsgleichung ergibt sich nämlich allgemein:

$$dp = \left(\frac{\partial p}{\partial T}\right)_v dT + \left(\frac{\partial p}{\partial v}\right)_T dv,$$

wobei, wie üblich, der angefügte Index diejenige Variable bezeichnet, welche bei der Differentiation konstant zu halten ist. Setzt man nun $dp = 0$, so erhält man die Bedingung, welche für eine isobare Änderung zwischen den

Differentialen dT und dv gilt, also entsprechend geschrieben:

$$\left(\frac{\partial v}{\partial T}\right)_p = -\frac{\left(\frac{\partial p}{\partial T}\right)_v}{\left(\frac{\partial p}{\partial v}\right)_T}. \tag{6}$$

Man kann daher für jeden Zustand einer Substanz eine der drei Größen: Ausdehnungskoeffizient, Spannungskoeffizient, Kompressibilitätskoeffizient, aus den beiden anderen berechnen. Nehmen wir z. B. Quecksilber bei 0° C und Atmosphärendruck. Der Ausdehnungskoeffizient (§ 12) ist:

$$\left(\frac{\partial v}{\partial T}\right)_p \frac{1}{v_0} = 0,00018.$$

Der Kompressibilitätskoeffizient (§ 14) ist, bezogen auf Atmosphären:

$$\left(\frac{\partial v}{\partial p}\right)_T \frac{1}{v_0} = 0,000\,0039.$$

Also nach (6) der Spannungskoeffizient (§ 13), bezogen auf Atmosphären:

$$\left(\frac{\partial p}{\partial T}\right)_v = -\left(\frac{\partial p}{\partial v}\right)_T \cdot \left(\frac{\partial v}{\partial T}\right)_p = \frac{\left(\frac{\partial v}{\partial T}\right)_p}{-\left(\frac{\partial v}{\partial p}\right)_T} = \frac{0,000\,18}{0,000\,0039} = 46,$$

d. h. um Quecksilber bei der Erwärmung von 0° auf 1° auf konstantem Volumen zu erhalten, bedarf es einer Druckzunahme von 46 Atmosphären.

§ 16. Mischungen idealer Gase. Wenn verschiedene beliebig große Quantitäten eines und desselben Gases von gleicher Temperatur und gleichem Druck, welche anfangs durch Scheidewände getrennt sind, mittels plötzlicher Beseitigung derselben in Berührung gebracht werden, so ist und bleibt selbstverständlich das Volumen des gesamten Systems gleich der Summe der Einzelvolumina. Wenn aber die in Berührung gebrachten Gase verschiedener Natur sind, so zeigt die Erfahrung, daß auch dann, bei konstant gehaltener gleichmäßiger Temperatur und gleichmäßigem Druck, das Gesamtvolumen dauernd gleich der Summe der ursprünglichen Einzelvolumina bleibt, obwohl sich gleichzeitig ein langsamer Mischungsvorgang, die Diffusion, vollzieht, der erst dann sein Ende erreicht, wenn die Zusammensetzung der Mischung in jedem Raumteil überall die nämliche, d. h. die Mischung physikalisch homogen geworden ist.

§ 17. Man kann sich das entstandene Gemisch von vornherein in zweierlei Weise konstituiert denken. Entweder könnte man annehmen, daß bei der Vermischung jedes einzelne Gas sich in unwahrnehmbar viele kleine Teile spaltet, deren jeder aber sein Volumen und seinen Druck unverändert beibehält, und daß diese kleinen Teile der verschiedenen Gase sich bei der Diffusion nebeneinandermengen, ohne sich gegenseitig zu durchdringen; dann hätte auch nach beendigter Diffusion jedes Gas im ganzen noch sein altes Volumen (Partialvolumen) und alle Gase hätten denselben gemeinsamen Druck. Oder aber — und diese Auffassung wird sich weiterhin (§ 32) als die allein berechtigte erweisen — man kann annehmen, daß die Einzelgase sich auch in ihren kleinsten Volumteilen verändern und durchdringen, daß also nach beendigter Diffusion jedes Einzelgas, soweit man überhaupt noch von einem solchen reden kann, das Volumen des ganzen Gemisches einnimmt und demzufolge unter einem geringeren Druck als früher steht. Wir wollen die Drucke, die den einzelnen Gasen zuzuschreiben wären, wenn sie, jedes für sich allein, das ganze Volumen der Mischung einnehmen würden, ihre sog. „Partialdrucke“, berechnen.

§ 18. Bezeichnet man die einzelnen Gase durch angefügte Zahlenindizes, während Temperatur T und Druck p ohne Index gelassen werden, so ist vor Beginn der Diffusion nach der Zustandsgleichung (5):

$$p = \frac{C_1 M_1 T}{V_1}, \quad p = \frac{C_2 M_2 T}{V_2} \ldots .$$

Das Gesamtvolumen:

$$V = V_1 + V_2 + \ldots$$

bleibt nach § 16 durch die Diffusion unverändert. Da nun nach beendigter Diffusion jedem einzelnen Gas das ganze Volumen V zugeschrieben wird, so sind dann die Partialdrucke nach der Gleichung (5) und nach den letzten Gleichungen:

$$p_1 = \frac{C_1 M_1 T}{V} = \frac{V_1}{V}\, p, \quad p_2 = \frac{C_2 M_2 T}{V} = \frac{V_2}{V}\, p \ldots . \tag{7}$$

Durch Addition ergibt sich:

$$p_1 + p_2 + \ldots = \frac{V_1 + V_2 + \ldots}{V}\, p = p, \tag{8}$$

das Gesetz von Dalton, welches besagt, daß in einer homogenen Gasmischung der Druck gleich ist der Summe der Partialdrucke aller einzelnen Gase. Gleichzeitig sieht man, daß

$$p_1 : p_2 : \ldots = V_1 : V_2 : \ldots = C_1 M_1 : C_2 M_2 : \ldots \tag{9}$$

d. h. die Partialdrucke der Einzelgase stehen in demselben Verhältnis wie die Volumina, welche die Gase vor der Diffusion hatten, bez. wie die Partialvolumina, welche die Gase nach der im § 17 zuerst geschilderten Auffassung in der Mischung einnehmen würden.

§ 19. Die Zustandsgleichung der Mischung lautet nach (8) und (7):

$$p = (C_1 M_1 + C_2 M_2 + \ldots)\frac{T}{V} \tag{10}$$

$$p = \frac{C_1 M_1 + C_2 M_2 + \ldots}{M} \cdot \frac{M}{V} \cdot T,$$

entspricht also ganz der Zustandsgleichung (5) eines idealen Gases, dessen charakteristische Konstante ist:

$$C = \frac{C_1 M_1 + C_2 M_2 + \ldots}{M_1 + M_2 + \ldots}. \tag{11}$$

Daher kann durch die Untersuchung der Zustandsgleichung niemals entschieden werden, ob ein ideales Gas chemisch einfach ist oder ob es eine Mischung verschiedener chemisch einfacher Gase bildet.

§ 20. Die Zusammensetzung einer Gasmischung definiert man entweder durch die Verhältnisse der Massen M_1, $M_2 \ldots$ oder durch die Verhältnisse (9) der Partialdrucke p_1, $p_2 \ldots$ bez. Partialvolumina V_1, $V_2 \ldots$ der Einzelgase. Je nachdem spricht man entweder von Gewichtsprozenten oder von Volumenprozenten. Nehmen wir z. B. atmosphärische Luft, eine Mischung von Sauerstoff (1) und von „atmosphärischem“ Stickstoff (2).

Das Verhältnis der Dichten von Sauerstoff, atmosphärischem Stickstoff und Luft ist nach § 11

$$0,001\,4291 : 0,001\,2567 : 0,001\,2928 = \frac{1}{C_1} : \frac{1}{C_2} : \frac{1}{C}.$$

Unter Berücksichtigung der Beziehung (11)

$$C = \frac{C_1 M_1 + C_2 M_2}{M_1 + M_2}$$

berechnet sich hieraus das Verhältnis

$$M_1 : M_2 = \frac{C_2 - C}{C - C_1} = 0,3009,$$

d. h. $23,1\%$ Sauerstoff und $76,9\%$ Stickstoff nach Gewichtsprozenten. Dagegen das Verhältnis

$$C_1 M_1 : C_2 M_2 = p_1 : p_2 = V_1 : V_2 = \frac{\frac{1}{C} - \frac{1}{C_2}}{\frac{1}{C_1} - \frac{1}{C}} = 0,2649,$$

d. h. $20,9\%$ Sauerstoff und $79,1\%$ Stickstoff nach Volumenprozenten.

§ 21. Zustandsgleichung anderer Substanzen. Stellt schon für die bisher beispielsweise behandelten Substanzen die Zustandsgleichung idealer Gase nur eine, wenn auch bedeutende, Annäherung an die Wirklichkeit dar, so zeigen die anderen gasförmigen Körper, besonders diejenigen, die sich leicht kondensieren lassen, und die daher früher in die besondere Klasse der Dämpfe zusammengefaßt wurden, ein von den Eigenschaften idealer Gase deutlich abweichendes Verhalten, so daß für sie eine Modifikation der Zustandsgleichung eintreten muß. Dabei ist jedoch bemerkenswert, daß die Abweichungen von dem Verhalten idealer Gase um so geringer auszufallen pflegen, je kleiner die Dichte genommen wird, weshalb man im allgemeinen sagen kann, daß sich die gasförmigen Substanzen bei genügend geringer Dichte wie ideale Gase verhalten, und zwar auch bei beliebig tiefen Temperaturen. Die Zustandsgleichung beliebiger Gase und Dämpfe wird sich also als eine Verallgemeinerung derjenigen für ideale Gase darstellen müssen, welche für große Werte von v in die spezielle oben behandelte Form (5) übergeht.

§ 22. Von dem Sinn und der Größe der Abweichungen von dem idealen Gaszustand kann man sich auf graphischem Wege eine Vorstellung verschaffen, und zwar auf verschiedene Weise. Man kann z. B. eine „isothermische“ Kurve aufzeichnen, indem man für eine beliebige konstant gehaltene Temperatur T je zwei zusammengehörige Werte von v und p als Abszisse und Ordinate eines Punktes in einer Ebene auffaßt. Die Schar aller Isothermen liefert ein vollständiges Bild der Zustandsgleichung. Je mehr nun sich das Verhalten des betrachteten Gases dem idealen nähert, um so enger schließen sich die Isothermen an die gleichseitigen Hyperbeln an, welche die Koordinatenachsen zu Asymptoten haben. Denn für ein ideales Gas ist die Gleichung einer Isotherme: $pv = \text{konst}$. Die Abweichung von der Form dieser Hyperbel gibt also zugleich ein Maß für die Abweichung von dem idealen Gaszustand.

§ 23. Augenscheinlicher noch werden diese Abweichungen, wenn man die Isotherme in der Art zeichnet, daß nicht p, sondern das Produkt pv als Ordinate, und als Abszisse etwa p erscheint. Für ein ideales Gas sind dann die Isothermen offenbar gerade, der Abszissenachse parallele Linien. Für die wirklichen Gase zeigt nun eine solche Linie ein allerdings flach verlaufendes Minimum, dessen Lage und Betrag natürlich von der Temperatur und von der Natur des Gases abhängt. Für kleinere Drucke (links vom Minimum) nimmt also das Volumen mit steigendem Druck schneller, für höhere Drucke (rechts vom Minimum) nimmt es mit steigendem Druck langsamer ab als bei idealen Gasen. Im Minimum selber ist die Kompressibilität gerade die eines idealen Gases. Beim Wasserstoff liegt das Minimum sehr weit links,

und konnte nur bei sehr tiefen Temperaturen nachgewiesen werden.

§ 24. Die erste auch für den flüssigen Zustand brauchbare analytische Formulierung der verallgemeinerten Zustandsgleichung rührt her von VAN DER WAALS, der zugleich auch eine physikalische Erklärung für die Abweichungen vom idealen Gaszustand, vom Standpunkt der kinetischen Gastheorie aus, gegeben hat. Da wir uns hier von den Voraussetzungen der kinetischen Theorie frei halten wollen, haben wir es nur mit der VAN DER WAALSschen Formel selber, als einem angenäherten Ausdruck der Tatsachen, zu tun. Sie lautet:

$$p = \frac{RT}{v-\beta} - \frac{\alpha}{v^2}, \tag{12}$$

wobei R, α und β Konstanten sind, die von der Natur der Substanz abhängen. Für große v geht die Gleichung in der Tat in die eines idealen Gases über; für kleine v und entsprechende T stellt sie die Zustandsgleichung der tropfbar flüssigen Substanz dar.

Wenn der Druck p in Atmosphären ausgedrückt und das spezifische Volumen v für $T = 273$ und $p = 1$ gleich 1 gesetzt wird, so ist nach VAN DER WAALS für Kohlensäure:

$$R = 0,00369 \qquad \alpha = 0,00874 \qquad \beta = 0,0023.$$

Da das Volumen von 1 g Kohlensäure bei 0° C und Atmosphärendruck 506 ccm beträgt, so hat man die aus der Formel sich ergebenden Werte von v noch mit 506 zu multiplizieren, um die spezifischen Volumina in absolutem Maße zu erhalten.

§ 25. Da die VAN DER WAALSsche Formel sich als nicht vollständig exakt herausgestellt hat, so ist sie von CLAUSIUS und später auch von anderen Physikern durch Einführung weiterer Konstanten einer Ergänzung unterzogen worden. Die CLAUSIUSsche Formel lautet:

$$p = \frac{RT}{v-a} - \frac{c}{T(v+b)^2}\,. \tag{12a}$$

Auch diese Formel ergibt für große v die Zustandsgleichung eines idealen Gases. In denselben Einheiten wie oben ist nach CLAUSIUS für Kohlensäure:

$$R = 0,003\,688, \quad a = 0,000\,843, \quad b = 0,000\,977, \quad c = 2,0935.$$

Die Beobachtungen über die Kompressibilität gasförmiger und flüssiger Kohlensäure bei verschiedenen Temperaturen werden durch die letzte Formel ziemlich befriedigend dargestellt.

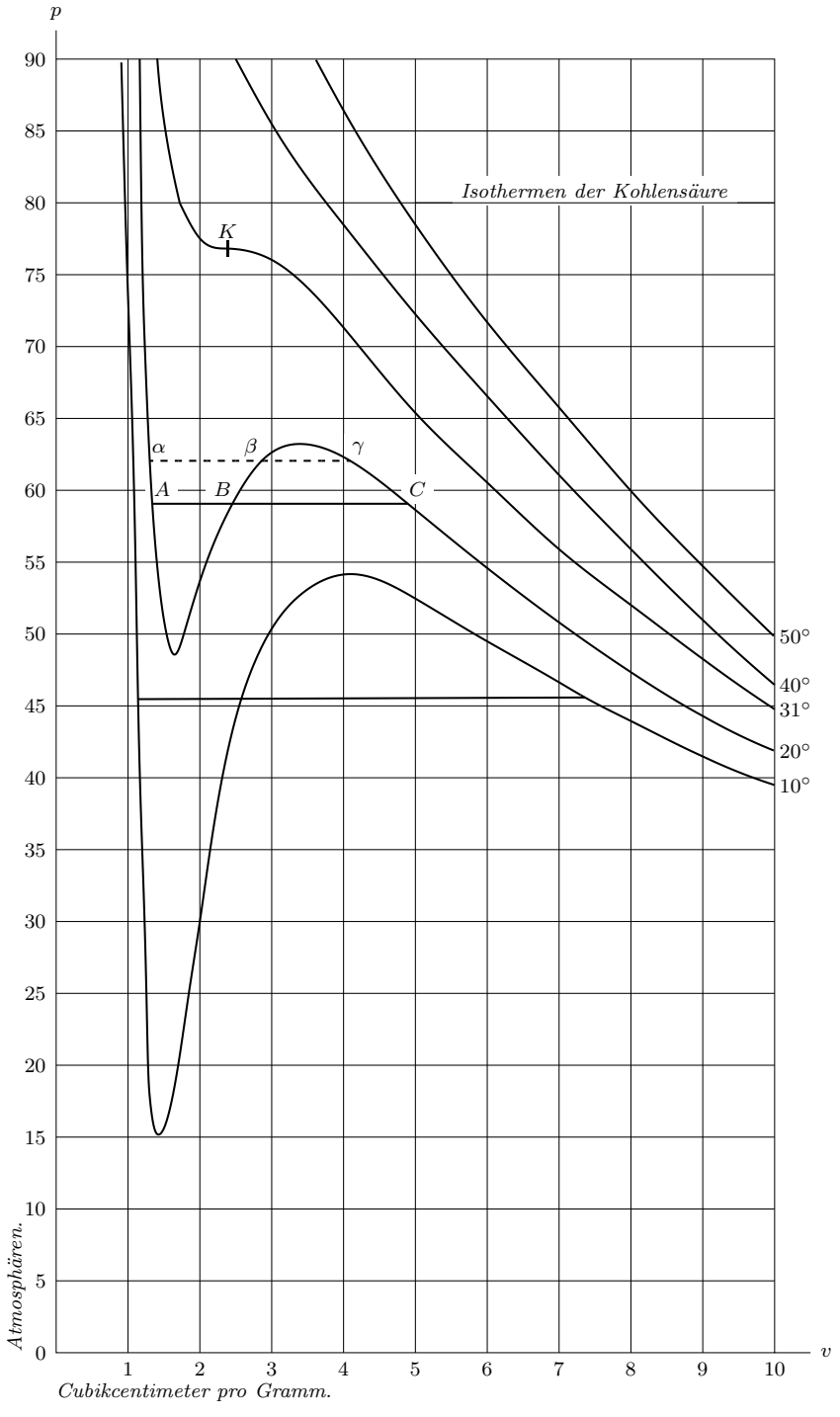


Fig. 1. Isothermen der Kohlensäure nach Clausius.

Andere Formen der Zustandsgleichung sind in großer Anzahl von verschiedenen Forschern, teils auf experimenteller, teils auf theoretischer Grundlage, abgeleitet worden. Eine besonders für Gase bei nicht zu hohen Drucken praktisch gut brauchbare Formel rührt her von D. Berthelot.

§ 26. Wenn man die Schar der Isothermen, wie sie durch die Clausiusschen Formel für Kohlensäure dargestellt werden, aufzeichnet, indem man für je einen konstant gehaltenen Wert der Temperatur die Werte von v als Abszissen, die von p als Ordinaten der Punkte einer Kurve aufträgt, so erhält man ein eigentümliches, in Figur 1 versinnlichtes Bild.[1]

Für hohe Temperaturen erscheinen gleichseitige Hyperbeln, wie auch aus der Zustandsgleichung (12a) zu erkennen; im allgemeinen aber entsprechen einem bestimmten Wert von p drei Werte von v. Mithin wird eine Isotherme im allgemeinen in 3 Punkten von einer der Abszissenachse parallelen Geraden geschnitten. Zwei derselben können aber imaginär sein, wie das für große Werte von T tatsächlich zutrifft. Für hohe Temperaturen gibt es also bei gegebenem Druck nur ein einziges reelles Volumen, während für tiefere Temperaturen einem bestimmten Wert des Druckes 3 reelle Werte des Volumens entsprechen können. Von diesen 3 Werten, die in der Figur beispielsweise durch die Punkte α, β, γ dargestellt sind, können nur der kleinste (α) und der größte (γ) einen stabilen, in der Natur herstellbaren, Zustand der Substanz bedeuten. Denn für den mittleren (β) steigt offenbar auf der Isotherme der Druck mit wachsendem Volumen an, die Kompressibilität ist also negativ. Ein derartiger Zustand hat daher nur theoretische Bedeutung.

§ 27. Der Punkt α entspricht der flüssigen, der Punkt γ der gasförmigen Kohlensäure bei der Temperatur der Isotherme und bei dem Druck der Geraden $\alpha\beta\gamma$. Doch ist im allgemeinen auch von diesen beiden Zuständen nur einer stabil (in der Figur der Zustand α). Denn wenn man gasförmige Kohlensäure, die etwa in einen Zylinder mit beweglichem Kolben eingeschlossen ist, komprimiert und dabei die Temperatur der betrachteten Isotherme (in der Figur 20°) konstant aufrecht erhält, so werden die aufeinanderfolgenden Zustände zunächst durch die ganz rechts gelegenen Punkte der Isotherme bezeichnet. Mit Verkleinerung des Volumens rückt der den Zustand bezeichnende Punkt auf der Isotherme immer weiter nach links, bis er eine bestimmte Stelle C erreicht. Bei weiterer isothermer Kompression der Substanz rückt nun der Punkt über diese Stelle nicht hinaus, sondern die Substanz kondensiert sich zum Teil, d. h. sie spaltet sich in einen flüssigen und

[1] Die Berechnung und Zeichnung der Kurven ist nach der Clausiusschen Zustandsgleichung von Herrn Dr. Richard Apt ausgeführt worden.

einen gasförmigen Teil, die selbstverständlich gemeinschaftlichen Druck und gemeinschaftliche Temperatur besitzen. Der Zustand des gasförmigen Teils wird bei fortschreitender isothermer Kompression nach wie vor immer durch den Punkt C, der des flüssigen Teils daher durch den Punkt A der nämlichen Isotherme charakterisiert. C heißt der Sättigungspunkt der gasförmigen Kohlensäure. Bei dem ganzen isothermen Kompressionsvorgang besteht die einzige Änderung darin, daß sich immer mehr Dampf niederschlägt, während die inneren Zustände der beiden Teile der Substanz (Druck, Temperatur, spezifische Volumina) während des ganzen Kondensationsprozesses immer durch die nämlichen Punkte A und C dargestellt werden. Schließlich, wenn aller Dampf kondensiert ist, befindet sich die ganze Substanz im flüssigen Zustand A, verhält sich also nun wieder homogen. Die weitere isotherme Kompression ergibt dann wieder Zunahme der Dichtigkeit und Steigerung des Druckes längs der Isotherme, wobei auch der Punkt α der Figur überschritten wird. Auf dieser Seite ist, wie aus der Figur zu erkennen, die Isotherme viel steiler als auf der andern, d. h. die Kompressibilität viel geringer.

Bisweilen gelingt es bei der Kompression eines Dampfes, die Isotherme über den Punkt C hinaus nach γ hin eine Strecke weit zu verfolgen und sogenannten übersättigten Dampf herzustellen. Man erhält aber dann nur mehr oder weniger labile Gleichgewichtszustände, wie sich daraus zu erkennen gibt, daß bei minimalen Störungen des Gleichgewichts eine plötzliche Kondensation, also ein sprungweiser Übergang in den stabilen Zustand erfolgen kann. Immerhin erhält durch das Studium der übersättigten Dämpfe auch das theoretische Stück der Isotherme zum Teil eine unmittelbare Bedeutung.

§ 28. Nach dem Gesagten besitzt jede Isotherme, die für gewisse Werte von p 3 reelle Volumina zuläßt, zwei bestimmte Stellen A und C, die den Zustand der Sättigung angeben. Ihre Lage läßt sich aus der Zeichnung der Isotherme nicht ohne weiteres ersehen. Doch führen die Sätze der Thermodynamik zu einer einfachen Konstruktion dieser Punkte, die im vierten Abschnitt (§ 172) abgeleitet werden wird. Je höher die Temperatur genommen wird, um so mehr schrumpft das Gebiet der Geraden zusammen, welche die Isothermen in 3 reellen Punkten schneiden, und um so näher rücken sich diese 3 Punkte. Den Übergang zu den hyperbelähnlichen Isothermen, welche von jeder zur Abszissenachse Parallelen nur in einem Punkt geschnitten werden, bildet eine bestimmte Isotherme, für welche jene 3 Schnittpunkte in einen einzigen zusammenfallen. Dieser Punkt stellt also einen Wendepunkt der Isotherme vor, in welchem die Tangente der Kurve parallel der Abszissenachse verläuft. Es ist der kritische Punkt K

der Substanz (s. Figur), er bezeichnet die kritische Temperatur, das kritische spezifische Volumen, und den kritischen Druck; für ihn wird der gesättigte Dampf mit seinem Niederschlag identisch. Oberhalb der kritischen Temperatur und oberhalb des kritischen Druckes gibt es überhaupt keine Kondensation, wie leicht aus der Figur zu ersehen. Daher mußten alle Versuche scheitern, Wasserstoff, Sauerstoff und Stickstoff zu kondensieren, solange die Temperatur nicht unter die kritische Temperatur, die bei diesen Substanzen sehr tief liegt, erniedrigt wurde.

§ 29. Man sieht aus der Figur 1 auch, daß es gar keine bestimmte Grenze gibt zwischen dem gasförmigen und dem flüssigen Zustand, da man leicht aus dem Bereich der entschieden gasförmigen Zustände, z. B. vom Punkte C aus, auf einer Kurve, die um den kritischen Punkt oben herumführt, in das Gebiet der entschieden flüssigen Zustände, z. B. nach A, kommen kann, ohne irgendwo einen gesättigten Zustand zu überschreiten. Man erwärme z. B. den Dampf bei konstantem Volumen über die kritische Temperatur hinaus, und kühle ihn hierauf bei konstant gehaltenem Druck bis unter das kritische Volumen ab. Dann tritt niemals Kondensation ein, und doch befindet man sich schließlich im Gebiet der unzweifelhaft flüssigen Zustände. Die frühere prinzipielle Unterscheidung zwischen Flüssigkeiten, Dämpfen und Gasen muß daher als nicht mehr durchführbar fallen gelassen werden.

Auch der in neuerer Zeit gemachte Vorschlag, diejenigen Zustände, welche einer höheren Temperatur als der kritischen angehören, als gasförmig, die übrigen dagegen als dampfförmig oder flüssig zu bezeichnen, je nachdem sie in der Figur 1 rechts oder links von den theoretischen Gebieten liegen, hat gewisse Unzuträglichkeiten im Gefolge, da hierdurch namentlich eine Grenze einerseits zwischen Flüssigkeit und Gas, andererseits zwischen Dampf und Gas festgesetzt wird, die keine unmittelbare physikalische Bedeutung hat. Denn das Überschreiten der kritischen Temperatur bei einem anderen als dem kritischen Druck unterscheidet sich in keiner wesentlichen Hinsicht von dem Überschreiten irgendeiner anderen Temperatur.

§ 30. Der kritische Punkt läßt sich leicht aus der allgemeinen Zustandsgleichung berechnen. Denn für ihn gelten nach § 28 die Gleichungen:

$$\left(\frac{\partial p}{\partial v}\right)_T = 0, \qquad \left(\frac{\partial^2 p}{\partial v^2}\right)_T = 0,$$

von denen die erste besagt, daß die Tangente der Isotherme in K parallel der Abszissenachse verläuft, die zweite, daß die Isotherme in K einen Wendepunkt besitzt.

Legt man die VAN DER WAALSsche Zustandsgleichung (12) zugrunde, so ergibt sich hiernach für den kritischen Punkt:

$$T_k = \frac{8\alpha}{27\beta R}, \qquad p_k = \frac{\alpha}{27\beta^2}, \qquad v_k = 3\beta,$$

also für Kohlensäure nach den mitgeteilten Zahlen:

$$T_k = 305 = 273^\circ + 32^\circ, \quad p_k = 61,2\text{Atm.}, \quad v_k = 0,0069,$$

und im absoluten Maß $v_k = 0,0069 \cdot 506 = 3,49 \frac{\text{ccm}}{\text{g}}$.

Statt der 3 Konstanten α, β und R kann man auch die 3 Konstanten T_k, p_k und v_k in die Zustandsgleichung einführen; dann treten in derselben außer den Variabeln T, p und v nur die Konstanten T_k, p_k und v_k auf. Da nun die Einheiten, in denen die Temperatur, der Druck und das Volumen gemessen werden, voneinander ganz unabhängig sind, so folgt, daß nur die Verhältnisse der Größen T und T_k, p und p_k, v und v_k in der Zustandsgleichung vorkommen können, oder mit anderen Worten, daß, wenn man diese Verhältnisse einführt:

$$\frac{T}{T_k} = \tau, \qquad \frac{p}{p_k} = \pi, \qquad \frac{v}{v_k} = \varphi,$$

(„reduzierte“ Temperatur, „reduzierter“ Druck, „reduziertes“ Volumen), die Zustandsgleichung außer den Variabeln τ, π und φ gar keine auf die spezielle Natur der Substanz bezügliche Konstante mehr enthält. In der Tat findet man auf die angegebene Weise für die VAN DER WAALSsche Zustandsgleichung:

$$\pi = \frac{8\tau}{3\varphi - 1} - \frac{3}{\varphi^2}.$$

Für $\tau = 1$ und $\varphi = 1$ ist natürlich $\pi = 1$. Dieses Gesetz, nach welchem die auf die reduzierten Werte von Temperatur, Druck und Volumen bezogene Zustandsgleichung für sämtliche Gase die nämliche ist, heißt das „Gesetz der übereinstimmenden Zustände“. Es ist nach dem Gesagten nicht allein der VAN DER WAALSschen, sondern überhaupt jeder Zustandsgleichung eigentümlich, in welcher nicht mehr als 3 auf die Natur der Substanz bezügliche Konstante auftreten. Doch gilt das Gesetz in Wirklichkeit nur in gewisser Annäherung, was vom molekularkinetischen Standpunkt aus deshalb einleuchten muß, weil es keine einzige Substanz gibt, die bei allen Temperatur- und Volumenänderungen im chemischen Sinne absolut einheitlich bleibt. (Vergl. unten § 33.)

§ 30a. Legt man die vierkonstantige CLAUSIUSsche Zustandsgleichung (12a) zugrunde, so ergibt sich für den kritischen Punkt:

$$T_k^2 = \frac{8c}{27(a+b)R}, \qquad p_k^2 = \frac{cR}{216(a+b)^3}, \qquad v_k = 3a + 2b,$$

also für Kohlensäure nach den mitgeteilten Zahlen:

$$T = 304 = 273^\circ + 31^\circ, \quad p = 77 \text{ Atm.}, \quad v = 2,27 \frac{\text{ccm}}{\text{g}}.$$

Diese Zahlen liegen den gemessenen Werten $T_k = 273^\circ + 31^\circ$, $p_k = 73$ Atm., $v_k = 2,22 \frac{\text{ccm}}{\text{g}}$ merklich näher als die obigen VAN DER WAALSschen.

§ 31. Auch bezüglich des Überganges aus dem flüssigen in den festen Aggregatzustand lassen sich dieselben Betrachtungen anstellen wie für den aus dem gasförmigen in den flüssigen; auch hier kann man das System der Isothermen zeichnen und „theoretische“ Gebiete der Isothermen konstatieren. In der Tat gibt es ja gewisse Erscheinungen, wie z. B. die Unterkühlung einer Flüssigkeit, die auf die Existenz mehr oder minder labiler flüssiger Zustände hinweisen.

Die vollkommenste Zustandsgleichung wäre eine solche, welche zugleich den gasförmigen, flüssigen und festen Aggregatzustand umfaßte. Die Aufstellung derselben ist aber bis jetzt noch für keine Substanz durchgeführt.

§ 32. Mischungen verschiedener Substanzen. Während sich die Zustandsgleichung einer Mischung idealer Gase, wie wir in § 19 sahen, in einfacher Weise auf die der einzelnen Komponenten zurückführen läßt, ist das bei Mischungen beliebiger Substanzen im allgemeinen nicht mehr der Fall. Nur bei Gasen und Dämpfen gilt, wenigstens mit großer Annäherung, das DALTONsche Gesetz, daß der Gesamtdruck einer Mischung gleich ist der Summe der Einzeldrucke, welche jedes Gas (oder Dampf) ausüben würde, wenn es allein bei derselben Temperatur das ganze Volumen ausfüllte. Dieser Satz gestattet es offenbar, die Zustandsgleichung einer beliebigen Gasmischung anzugeben, falls die der einzelnen Gase bekannt ist, er liefert außerdem auch die Entscheidung der oben § 17 unbeantwortet gelassenen Frage, ob man den einzelnen Gasen in einer Mischung gemeinsamen Druck und verschiedene Volumina, oder ob man ihnen gemeinsames Volumen und verschiedenen Druck zuschreiben muß. Daß die letztere Auffassung die allein zulässige ist, folgt aus der Betrachtung eines Dampfes, der sich weit von dem idealen Gaszustand entfernt: Nehmen wir z. B. eine Mischung atmosphärischer Luft und Wasserdampf bei 0° C unter Atmosphärendruck, so kann man den Wasserdampf unmöglich als unter dem Druck einer

Atmosphäre befindlich annehmen, weil Wasserdampf bei 0° C unter diesem Druck gar nicht existiert. Es bleibt also nur übrig, der Luft und dem Wasserdampf ein gemeinsames Volumen (dasjenige der Mischung) und verschiedene Drucke (Partialdrucke) zuzuschreiben.

Für Mischungen fester und flüssiger Substanzen ist kein allgemein gültiges Gesetz bekannt, welches die Zustandsgleichung auf diejenigen der einzelnen Substanzen zurückführt.

Zweites Kapitel. Molekulargewicht.

§ 33. Es ist im bisherigen immer nur von solchen Zustandsänderungen die Rede gewesen, welche allein die Temperatur, den Druck und die Dichte betreffen, während sie die chemische Natur des betreffenden Stoffes oder der Mischung ganz unberührt lassen. Es kommt aber auch häufig, — und viel häufiger, als man früher annahm — vor, daß durch eine Änderung der Temperatur oder des Druckes auch die chemische Beschaffenheit einer Substanz geändert wird. Daß auch vom thermodynamischen Standpunkt aus ein prinzipieller Unterschied zwischen physikalischen und chemischen Änderungen einer Substanz, der einen kontinuierlichen Übergang von den einen zu den andern ausschließt, konstatiert werden muß, ist im Lauf der neueren Entwicklung der Thermodynamik immer deutlicher hervorgetreten (vgl. § 42f. und § 238), wenn es sich auch bis jetzt als unmöglich gezeigt hat, ein für alle Fälle geeignetes praktisches Unterscheidungsmerkmal aufzustellen. Denn wie auffallend auch oft die chemischen Änderungen sich von den physikalischen abheben, entweder durch die Plötzlichkeit und Heftigkeit ihres Verlaufes oder durch irgend welche augenfällige Diskontinuitäten (Wärmeerzeugung, Änderungen der Farbe und anderer Eigenschaften), so gibt es doch andererseits zahlreiche Prozesse unzweifelhaft chemischer Natur, z. B. Dissoziationsvorgänge, die sich, wenigstens dem äußeren Anschein nach, vollkommen stetig und verhältnismäßig langsam abspielen. Es ist eine der Hauptaufgaben der physikalischen Chemie, diesen prinzipiellen Unterschied möglichst klar herauszuarbeiten.

§ 34. Das eigentümliche Merkmal aller chemischen Umwandlungen ist, daß sie nach konstanten Gewichtsverhältnissen erfolgen. Daher kann man als charakteristischen Ausdruck für die Natur der Reaktionen einer chemisch homogenen Substanz, sei sie ein Element oder eine Verbindung, eine Gewichts-(richtiger Massen-)größe benutzen: das Äquivalentgewicht. Für irgendein bestimmtes Element setzt man das Äquivalentgewicht willkürlich fest, z. B. für Wasserstoff $= 1$ g, und findet dann für ein anderes

Element, z. B. Sauerstoff, das zugehörige Äquivalentgewicht als diejenige Gewichtsmenge, welche sich mit 1 g Wasserstoff chemisch verbindet. Die Gewichtsmenge der entstandenen Verbindung ist dann zugleich auch das Äquivalentgewicht derselben. So fortschreitend gelangt man leicht zu Werten des Äquivalentgewichts für alle chemisch homogenen Stoffe, auch für solche Elemente, die sich gar nicht direkt mit Wasserstoff verbinden, da immer eine Anzahl von Elementen aufgefunden werden kann, welche sich sowohl mit dem fraglichen Element als auch mit Wasserstoff verbinden und so den Übergang zwischen beiden vermitteln.

Das Gesamtgewicht eines chemisch homogenen Körpers dividiert durch sein Äquivalentgewicht, heißt die im Körper enthaltene Zahl der Äquivalentgewichte oder Äquivalente. Daher kann man auch sagen: Bei jeder chemischen Umsetzung reagieren gleichviel Äquivalente der verschiedenen Stoffe aufeinander.

§ 35. Indessen leidet diese Definition an einem Mangel. Denn zwei Elemente können häufig mehr als eine einzige Verbindung miteinander eingehen, und dadurch wird die Größe des Äquivalentgewichts mehrdeutig. Doch zeigt die Erfahrung, daß in einem solchen Falle die verschiedenen möglichen Gewichtsverhältnisse immer einfache Multipla oder Submultipla eines bestimmten Verhältnisses sind. Daher reduziert sich die Vieldeutigkeit in dem Wert des Äquivalentgewichts auf einen einfachen ganzzahligen Faktor im Zähler oder Nenner dieser Größe, und man muß den Schlußsatz des vorigen Paragraphen, daß gleichviel Äquivalente aufeinander reagieren, dahin verallgemeinern, daß die Äquivalente nach einfachen ganzzahligen Verhältnissen aufeinander reagieren.[1] So z. B. verbinden sich 16 Gewichtsteile Sauerstoff mit 28 Gewichtsteilen Stickstoff zu Stickstoffoxydul, oder mit 14 Teilen zu Stickstoffoxyd, oder mit $9\frac{1}{3}$ Teilen zu Salpetrigsäureanhydrid, oder mit 7 Teilen zu Untersalpetersäure, oder mit $5\frac{3}{5}$ Teilen zu Salpetersäureanhydrid, so daß man, wenn das Äquivalentgewicht des Sauerstoffs zu 16 angenommen wird, dem Stickstoff jede beliebige der obigen Zahlen als Äquivalentgewicht zuschreiben kann. Dieselben stehen aber in einfachen rationalen Verhältnissen,

[1] Wenn diese ganzen Zahlen beliebig groß sein könnten, so wäre das Äquivalentgewicht im Grunde eine stetig veränderliche Größe; denn man kann jede Größe mit beliebiger Annäherung durch das Verhältnis zweier ganzer Zahlen ausdrücken. Dann könnte der Wert des Äquivalentgewichts aus den aufeinander reagierenden Gewichtsmengen überhaupt nicht definiert werden, und die ganze obige Betrachtung wäre illusorisch. Die Unstetigkeit in der Veränderlichkeit des Äquivalentgewichts ist also charakteristisch für die chemische Natur eines Stoffes, im Gegensatz zu seinen physikalischen Eigenschaften; ja man kann kurz zusammenfassend geradezu sagen, daß physikalische Veränderungen stetig, chemische dagegen unstetig verlaufen. Daher rechnet auch die Physik vorwiegend mit stetig veränderlichen Größen, die Chemie dagegen vorwiegend mit ganzen Zahlen.

da

$$28 : 14 : 9\tfrac{1}{3} : 7 : 5\tfrac{3}{5} = 60 : 30 : 20 : 15 : 12.$$

§ 36. Die durch die letzte Zahlenreihe illustrierte Unbestimmtheit in der Definition der für den Stickstoff charakteristischen Gewichtsgröße wird nun dadurch beseitigt, daß man aus ihr eine bestimmte Zahl herausgreift und sie als Molekulargewicht des Stickstoffs bezeichnet. In der Definition des Molekulargewichts als einer ganz bestimmten, nur von dem eigenen Zustand einer Substanz abhängigen, von etwaigen chemischen Umsetzungen mit anderen Stoffen aber unabhängigen Größe, liegt eine der wichtigsten und fruchtbarsten Errungenschaften, welche die theoretische Chemie aufzuweisen hat. Dieselbe läßt sich allerdings bis jetzt nur für spezielle Fälle exakt aussprechen, nämlich für ideale Gase und für verdünnte Lösungen. Da der letztere Fall sich, wie in der Folge gezeigt werden wird, mittels der Thermodynamik als durch den ersten mitbestimmt darstellen läßt, so haben wir es hier nur mit jenem zu tun.

Die Definition des Molekulargewichts für ein chemisch homogenes ideales Gas wird ermöglicht durch den weiteren Erfahrungssatz, daß die idealen Gase sich nicht nur, wie überhaupt alle Stoffe, nach einfachen Äquivalentzahlen, sondern auch, bei gleicher Temperatur und gleichem Druck genommen, nach einfachen Volumenverhältnissen verbinden (Gay Lussac). Daraus folgt sogleich, daß die in gleichen Volumina verschiedener Gase enthaltenen Äquivalentzahlen in einfachen Verhältnissen stehen. Die Werte dieser Verhältnisse schwanken aber, gemäß der beschriebenen Willkür in der Wahl des Äquivalentgewichts.

Diese Willkür wird nun beseitigt durch die Definition des Molekulargewichts. Setzt man nämlich diese Verhältnisse allgemein $= 1 : 1$, d. h. stellt man die Bedingung auf, daß die Zahlen der in gleichen Gasvolumina enthaltenen Äquivalente einander gleich sind, so trifft man damit eine spezielle Auswahl unter den verschiedenen Möglichkeiten, und erhält so ein bestimmtes Äquivalentgewicht für jedes Gas, das nun als Molekulargewicht des Gases bezeichnet wird, und ebenso für eine gegebene Gasmenge durch Division des Gesamtgewichts durch das Molekulargewicht eine bestimmte Äquivalentzahl, welche die Anzahl[1] der in der Gasmenge enthaltenen Moleküle genannt wird. In gleichen Volumina besitzen also alle idealen Gase gleich viel Moleküle (Avogadro). Daher werden in chemisch homogenen Gasen die Verhältnisse der Molekulargewichte direkt durch die in gleichen Volumina

[1]Dies braucht natürlich keine ganze Zahl zu sein, da es sich hier nicht um die wirklichen Moleküle im Sinne der atomistischen Theorie, sondern um die in willkürlicher Einheit definierten gr-Moleküle oder „Mole“ handelt.

enthaltenen Massen, d. h. durch die Dichten, gegeben. Das Verhältnis der Dichten ist gleich dem Verhältnis der Molekulargewichte. Nennen wir also m_1 und m_2 die Molekulargewichte zweier idealer Gase, so ist nach § 11:

$$m_1 : m_2 = \frac{1}{C_1} : \frac{1}{C_2}. \tag{12b}$$

§ 37. Setzt man das Molekulargewicht des Wasserstoffs $= m_H$, so ist mithin das Molekulargewicht irgend eines chemisch homogenen Gases gleich dem Produkte von m_H und der spezifischen Dichte des Gases, bezogen auf Wasserstoffgas (§ 11). Folgende Tabelle enthält für einige Gase und Dämpfe die spezifischen Dichten, bezogen auf Wasserstoff, in abgerundeten Zahlen, und das Molekulargewicht.

	Spezifische Dichte	Molekulargewicht
Wasserstoff	1	$1\ m_H$
Sauerstoff	16	$16\ m_H$
Stickstoff	14	$14\ m_H$
Wasserdampf	9	$9\ m_H$
Ammoniak	$8\frac{1}{2}$	$8\frac{1}{2}\, m_H$
Stickstoffoxydul	22	$22\ m_H$
Stickstoffoxyd	15	$15\ m_H$

Mit Hilfe dieser Tabelle läßt sich nun die Frage, wie sich das Molekulargewicht einer Verbindung aus den Molekulargewichten seiner Elemente zusammensetzt, vollständig eindeutig beantworten.

Da Wasserdampf sich nach der Analyse aus 1 Gewichtsteil Wasserstoff und 8 Gewichtsteilen Sauerstoff zusammensetzt, so besteht das Molekül $9\, m_H$ des Wasserdampfes notwendig aus m_H Gewichtsteilen Wasserstoff und $8\, m_H$ Gewichtsteilen Sauerstoff, d. h. nach der Tabelle aus einem Molekül Wasserstoff und einem halben Molekül Sauerstoff. Da ferner Ammoniak sich nach der Analyse aus 1 Gewichtsteil Wasserstoff und $4\frac{2}{3}$ Gewichtsteilen Stickstoff zusammensetzt, so besteht das Molekül $8\frac{1}{2}\, m_H$ des Ammoniak notwendig aus $1\frac{1}{2}\, m_H$ Gewichtsteilen Wasserstoff und aus $7\, m_H$ Gewichtsteilen Stickstoff, d. h. nach der Tabelle aus $1\frac{1}{2}$ Molekülen Wasserstoff und einem halben Molekül Stickstoff. Da ferner Stickstoffoxydul sich nach der Analyse aus 16 Gewichtsteilen Sauerstoff und 28 Gewichtsteilen Stickstoff zusammensetzt, so besteht das Molekül $22\, m_H$ des Stickstoffoxyduls notwendig aus $8\, m_H$ Gewichtsteilen Sauerstoff und $14\, m_H$ Gewichtsteilen Stickstoff, d. h. nach der Tabelle aus einem halben Molekül Sauerstoff und einem ganzen Molekül Stickstoff. In derselben Weise fortfahrend kann man auf Grund des AVOGADROschen Gesetzes für jedes chemisch homogene Gas, dessen Dichte und chemische Zusammensetzung bekannt ist, den Aufbau

des Moleküls aus den Molekülen der Elemente in ganz bestimmten Zahlen angeben.

§ 38. Die kleinste Gewichtsmenge eines chemisch einfachen Stoffes, welche in den Molekülen der Verbindungen des Stoffes vorkommt, nennt man ein Atom. Daher heißt ein halbes Molekül Wasserstoff ein Atom Wasserstoff: H, ebenso ein halbes Molekül Sauerstoff ein Atom Sauerstoff: O, und ein halbes Molekül Stickstoff ein Atom Stickstoff: N. Das Molekül jedes dieser Elemente besteht also aus zwei Atomen: H_2, O_2 und N_2. Bei Quecksilber z. B. dagegen ist das Atom gleich dem ganzen Molekül, weil in den Molekülen der Quecksilberverbindungen immer nur ganze Moleküle des Quecksilberdampfes vorkommen. Um zu bestimmten Zahlenwerten für die Atom- und Molekulargewichte zu gelangen, ist es noch nötig, das Atomgewicht eines einzelnen beliebig herausgegriffenen Elements nach Willkür zu fixieren. Früher setzte man allgemein $H = 1$, und daher $O = 16$. Nachdem sich aber gezeigt hat, daß das Verhältnis der Äquivalentgewichte von Sauerstoff und Wasserstoff nicht genau 16, sondern etwa $15,87$ beträgt, ist es mit Rücksicht auf den Umstand, daß die Sauerstoffverbindungen der meisten Elemente viel genauer analysiert sind als die Wasserstoffverbindungen, üblich geworden, von dem Atomgewicht des Sauerstoffs $O = 16$ als Definition auszugehen. Dann wird das Molekulargewicht des Sauerstoffs:

$$O_2 = 32 = 15,87\, m_H.$$

Daraus folgt das Molekulargewicht des Wasserstoffs:

$$m_H = \frac{32}{15,87} = 2,016 = H_2$$

und das Atomgewicht des Wasserstoffs: $H = 1,008$.

Die Molekulargewichte der obigen Tabelle werden dann:

Wasserstoff	$2,016$	$= H_2$
Sauerstoff	$32,00$	$= O_2$
Stickstoff	$28,02$	$= N_2$
Wasserdampf	$18,02$	$= H_2O$
Ammoniak	$17,03$	$= H_3N$
Stickstoffoxydul	$44,02$	$= N_2O$
Stickstoffoxyd	$30,01$	$= NO$

§ 39. Allgemein ist also das Molekulargewicht eines chemisch homogenen Gases gleich $2,016$ mal seiner spezifischen Dichte bezogen auf Wasserstoff, oder gleich 32 mal seiner spezifischen Dichte bezogen auf Sauerstoff. Umgekehrt läßt sich, wenn das Molekulargewicht m eines Gases bekannt

ist, seine spezifische Dichte und somit auch die Konstante C in der Zustandsgleichung (5) angeben. Bezeichnet man die auf Sauerstoff bezüglichen Größen mit dem Index 0, so ist nach der Gleichung (12b) in § 36:

$$C = \frac{m_0 C_0}{m}. \tag{13}$$

Nun ist $m_0 = 32$, während die Konstante C_0 sich aus der Dichte des Sauerstoffs bei 0° C und Atmosphärendruck berechnet. Denn hierfür ist nach der Tabelle § 11:

$$\begin{aligned} \frac{1}{v_0} &= 0,001\,4291, \\ p &= 1013250 \ \ (\S\,7), \\ T &= 273. \end{aligned}$$

Mithin

$$C_0 = \frac{p v_0}{T}$$

und nach (13)

$$C = \frac{m_0}{m}\,\frac{p v_0}{T}$$

oder:

$$C = \frac{32 \cdot 1013250}{m \cdot 273 \cdot 0,001\,4291} = \frac{83110000}{m} = \frac{m_0 C_0}{m}\,.$$

Von dem Molekulargewicht des Wasserstoffs ausgehend erhält man für mC eine nur unwesentlich größere Zahl.

Setzt man zur Abkürzung die Zahl

$$8,315 \cdot 10^7 = R, \tag{13a}$$

so ist mithin die allgemeine Zustandsgleichung eines idealen chemisch homogenen Gases mit dem Molekulargewicht m:

$$p = \frac{R}{m} \cdot \frac{T}{v}, \tag{14}$$

worin R von der Natur des Gases unabhängig ist und daher gewöhnlich als die „absolute Gaskonstante“ bezeichnet wird. Mit Hilfe von R kann man also auch das Molekulargewicht m direkt aus der Zustandsgleichung ableiten, da

$$m = \frac{R}{C}\,. \tag{15}$$

Führt man in (14) statt des spezifischen Volumens v die Masse M und das Volumen V ein, so ergibt sich:

$$V = \frac{RT}{p} \cdot \frac{M}{m}.$$

Nun ist aber $\frac{M}{m}$ die Zahl der im Gase enthaltenen Moleküle:

$$\frac{M}{m} = n,$$

folglich

$$V = \frac{RT}{p} \cdot n,$$

d. h. das Volumen eines Gases bei bestimmtem Druck und bestimmter Temperatur hängt nur von der Anzahl der darin enthaltenen Moleküle, im übrigen aber gar nicht von der Natur des Gases ab, wie es der Satz von Avogadro verlangt.

§ 40. In einer Mischung von chemisch homogenen Gasen mit den Molekulargewichten m_1, $m_2, \ldots$ ist nach (9) das Verhältnis der Partialdrucke:

$$p_1 : p_2 : \ldots = C_1 M_1 : C_2 M_2 : \ldots$$

Da aber nach (15):

$$C_1 = \frac{R}{m_1}, \quad C_2 = \frac{R}{m_2}, \ldots$$

so ist dies Verhältnis:

$$p_1 : p_2 : \ldots = \frac{M_1}{m_1} : \frac{M_2}{m_2} : \ldots = n_1 : n_2 : \ldots$$

d. h. das Verhältnis der Partialdrucke gibt zugleich das Verhältnis der in der Mischung enthaltenen Molekülzahlen n_1, $n_2, \ldots$ an. Ferner ist nach (10)

$$V = \frac{(C_1 M_1 + C_2 M_2 + \ldots)T}{p}$$

$$V = \frac{RT}{p}\left(\frac{M_1}{m_1} + \frac{M_2}{m_2} + \ldots\right)$$

$$V = \frac{RT}{p}(n_1 + n_2 + \ldots) = \frac{RT}{p} \cdot n, \tag{16}$$

d. h. das Volumen der Mischung bestimmt sich aus der Gesamtzahl n der in der Mischung enthaltenen Moleküle genau ebenso wie bei einem chemisch homogenen Gas.

§ 41. Dagegen kann man offenbar nicht von einem Molekulargewicht der Mischung sprechen, sondern höchstens von einem „scheinbaren“ oder „mittleren“ Molekulargewicht, indem man darunter dasjenige Molekulargewicht versteht, welches ein chemisch homogenes Gas haben würde, wenn es in derselben Masse dieselbe Molekülzahl wie die Mischung enthielte. Bezeichnen wir das scheinbare Molekulargewicht mit m, so ist die Molekülzahl

$$\frac{M_1 + M_2 + \ldots}{m} = \frac{M_1}{m_1} + \frac{M_2}{m_2} + \ldots$$

folglich

$$m = \frac{M_1 + M_2 + \ldots}{\frac{M_1}{m_1} + \frac{M_2}{m_2} + \ldots}.$$

Daraus berechnet sich z. B. das scheinbare Molekulargewicht der Luft folgendermaßen. Da $m_1 = \mathrm{O}_2 = 32$, $m_2 = \mathrm{N}_2 = 28$, $M_1 : M_2 = 0,30$ nach § 20, so ist

$$m = \frac{0,3 + 1}{\frac{0,3}{32} + \frac{1}{28}} = 28,9$$

etwas größer als das Molekulargewicht des Stickstoffs.

§42. Ergibt somit die Zustandsgleichung für jedes ideale Gas, sei es chemisch homogen oder nicht, nach (16) unmittelbar die Gesamtzahl der darin enthaltenen Moleküle, so liefert sie, wie schon § 19 hervorgehoben wurde, kein Mittel, um zu entscheiden, ob die Moleküle gleichartig sind oder nicht. Bei der Untersuchung dieser Frage ist man auf andere Methoden angewiesen, von denen aber keine in allen Fällen praktisch brauchbar ist. Häufig führt die Beobachtung der Diffusion, namentlich durch eine poröse oder noch besser semipermeable Wand zum Ziele, indem die einzelnen Gase einer Mischung sich durch ihre ungleiche Diffusionsgeschwindigkeit, die bei semipermeablen Wänden bis auf Null herabsinken kann, voneinander trennen und so die chemische Inhomogenität der Substanz verraten. Oft gibt auch die Entstehungsgeschichte des Gases unmittelbaren Aufschluß über seine chemische Beschaffenheit. Eine prinzipielle Definition für ein chemisch homogenes Gas liefert erst der Ausdruck der Entropie, § 237.

§ 43. Wenn ein Gas oder ein Dampf den für ideale Gase gültigen Gesetzen nicht folgt, mit anderen Worten: wenn es eine von der Temperatur oder dem Druck abhängige spezifische Dichte besitzt, so kann man dennoch die AVOGADROsche Definition des Molekulargewichts zur Anwendung bringen und nach Gleichung (16) setzen: $n = \frac{pV}{RT}$; nur ergibt sich dann offenbar keine konstante, sondern eine von dem augenblicklichen Zustand der Substanz

abhängige Molekülzahl n. Man steht also hier vor der Wahl, für diesen Fall entweder eine veränderliche Molekülzahl anzunehmen, oder aber die AVOGADROsche Definition für die Molekülzahl überhaupt nicht anzuwenden, mit anderen Worten: die Ursache der Abweichung von dem idealen Gaszustand entweder in chemischen oder in physikalischen Umständen zu suchen. Nach der letzteren Anschauung bleibt die chemische Natur des Gases erhalten, also die Moleküle auch bei veränderter Temperatur und verändertem Druck dieselben, sie unterliegen nur einer komplizierteren Zustandsgleichung als der BOYLE-GAY-LUSSACschen, z. B. der VAN DER WAALSschen oder der CLAUSIUSschen. Wesentlich davon verschieden ist aber die andere Auffassung, nach welcher ein Gas, das Abweichungen von den Gesetzen idealer Gase zeigt, nichts anderes ist als eine Mischung mehrerer verschiedener Molekülarten (bei Untersalpetersäure N_2O_4 und NO_2, bei Phosphorpentachlorid PCl_5, PCl_3 und Cl_2), deren Volumen in jedem Augenblick genau den durch die Gesamtzahl der Moleküle für eine Mischung idealer Gase nach Gleichung (16) bestimmten Wert besitzt und sich bei einer Änderung der Temperatur und des Druckes nur deshalb nicht wie bei einem idealen Gase ändert, weil durch gleichzeitige chemische Umsetzungen die verschiedenartigen Moleküle zum Teil ineinander übergehen und dadurch ihre Gesamtzahl stetig ändern. Diese Anschauung hat sich bisher am fruchtbarsten in allen den Fällen erwiesen, wo es sich um bedeutende Änderungen der Dichten handelt, um die sogenannten abnormen Dampfdichten, und dies namentlich dann, wenn die spezifische Dichte des Dampfes jenseits eines gewissen Temperatur- oder Druckintervalls wieder konstant wird. Dann ist nämlich die chemische Umsetzung vollständig geworden und die Moleküle verändern sich nicht mehr. So z. B. verhält sich Bromwasserstoffamylen sowohl unterhalb 160° als auch oberhalb 360° wie ein ideales Gas, doch im letzteren Zustand mit halber Dichte, entsprechend einer Verdoppelung der Molekülzahl:

$$C_5H_{11}Br = C_5H_{10} + HBr.$$

Sind aber die Abweichungen von den Gesetzen idealer Gase unbedeutend, so schiebt man sie gewöhnlich auf physikalische Ursachen, wie bei Wasserdampf und Kohlensäure, und faßt sie als Vorboten der Kondensation auf. Eine prinzipielle Trennung der chemischen von den physikalischen Einflüssen und damit eine Vervollständigung der Definition des Molekulargewichts für alle variablen Dampfdichten läßt sich zurzeit praktisch noch nicht durchführen; so könnte man die Zunahme der spezifischen Dichte, welche viele Dämpfe in der Nähe ihres Kondensationspunktes zeigen, ebensowohl chemischen Vorgängen zuschreiben, nämlich der Bildung von Doppelmolekülen oder

überhaupt von vielfachen Molekülen.[1] In der Tat bestehen über diesen Punkt noch öfters Meinungsverschiedenheiten, wie z. B. beim Molekulargewicht des Schwefeldampfes unterhalb 800°, das gewöhnlich zu $S_6 = 192$, von einigen aber auch gemischt mit Molekülen $S_8 = 256$ und $S_2 = 64$, von anderen noch anders angenommen wird. Im allgemeinen wird man in zweifelhaften Fällen am sichersten gehen, die Frage einstweilen noch offen zu lassen und sowohl physikalische als auch chemische Veränderungen als Ursache der Abweichungen von den Gasgesetzen anzunehmen. Nur so viel — und dies ist ein wichtiger Punkt, von dem wir später Gebrauch machen müssen — läßt sich mit Sicherheit behaupten, daß bei geringen Dichten die physikalischen Einflüsse hinter den chemischen immer mehr zurücktreten werden. Denn nach allen Erfahrungen nähern sich alle Gase mit abnehmender Dichte dem idealen Zustand (§ 21).

Drittes Kapitel. Wärmemenge.

§ 44. Taucht man zwei gleich schwere Stücke von Eisen und von Blei, beide auf 100° erhitzt, in zwei gehörig isolierte, ganz gleiche Gefäße mit gleichviel Wasser von 0° ein, und wartet für jedes Gefäß den Zustand des Wärmegleichgewichts ab, so zeigt das Gefäß mit dem Eisenstück eine bedeutend größere Temperaturerhöhung als das mit dem Bleistück. Umgekehrt wird ein Wasserbad von 100° durch ein Eisenstück von 0° bedeutend stärker abgekühlt, als durch ein gleich schweres Bleistück von 0°. Man unterscheidet daher zwischen Temperatur und W ä r m e m e n g e, und nimmt als Maß der von einem Körper bei seiner Abkühlung abgegebenen Wärmemenge diejenige Temperaturerhöhung, welche ein mit dem sich abkühlenden Körper in Berührung gebrachter Normalkörper (Wasser) erfährt, vorausgesetzt, daß andere Ursachen der Temperaturänderung, wie Kompression, ausgeschlossen sind. Zugleich setzt man dabei die von dem Körper abgegebene Wärmemenge gleich der von dem Normalkörper aufgenommenen Wärmemenge. Aus dem oben beschriebenen Experiment folgt dann, daß ein Eisenstück bei Abkühlung um ein bestimmtes Temperaturintervall eine größere (etwa die vierfache) Wärmemenge abgibt als ein Bleistück von gleichem Gewicht, und umgekehrt, daß das Eisen zu einer bestimmten Temperaturerhöhung der Zufuhr einer entsprechend größeren Wärmemenge bedarf als das Blei.

§ 45. Als Wärmeeinheit galt früher allgemein diejenige Wärmemenge,

[1] So z. B. hat W. Nernst (Verhandlungen der Deutschen Phys. Ges. 11, S. 313, 1909) die Zustandsgleichung des Wasserdampfes auf die Bildung von Doppelmolekülen $(H_2O)_2$ zurückgeführt.

welche einem Gramm Wasser zuzuführen ist, um es von 0° auf 1° zu erwärmen (Nullpunktskalorie). Dieselbe ist nahezu gleich derjenigen, welche 1 g Wasser von beliebiger Temperatur um 1° erwärmt. Seitdem aber die kalorimetrischen Messungen sich so weit verfeinert haben, daß man den Einfluß der Anfangstemperatur des Wassers berücksichtigen muß, wird häufiger die Kalorie als diejenige Wärmemenge definiert, welche 1 g Wasser von $14,5°$ auf $15,5°$ erwärmt. Dieselbe ist etwa $1,008$ mal kleiner als die Nullpunktskalorie. Endlich spricht man auch von der „mittleren Kalorie“ als dem hundertsten Teil derjenigen Wärmemenge, welche 1 g Wasser von 0° auf 100° erwärmt, und welche ungefähr ebensogroß ist wie die 15°-Kalorie. Jeder dieser sogenannten „kleinen“ Kalorien entspricht eine „große“ Kalorie, welche sich auf 1 kg Wasser bezieht, also den 1000 fachen Wert hat.

§ 46. Das Verhältnis der von einem Körper aufgenommenen Wärmemenge Q zu der durch sie bewirkten Temperaturerhöhung $T' - T = \Delta T$ heißt die mittlere Wärmekapazität des Körpers zwischen den Temperaturen T und T'

$$\frac{Q}{\Delta T} = C_m.$$

Die auf 1 g einer Substanz bezogene Wärmekapazität heißt die spezifische Wärme der Substanz:

$$c_m = \frac{C_m}{M} = \frac{Q}{M \cdot \Delta T} = \frac{q}{\Delta T}.$$

Danach ist die mittlere spezifische Wärme des Wassers zwischen 0° und 1° gleich einer Nullpunktskalorie. Geht man zu unendlich kleinen Temperaturintervallen über, so erhält man die Wärmekapazität eines Körpers bez. die spezifische Wärme einer Substanz bei der Temperatur T:

$$\frac{Q}{dT} = C \qquad \text{und} \qquad \frac{q}{dT} = c,$$

welche im allgemeinen mit der Temperatur veränderlich ist, jedoch für die meisten Substanzen sehr langsam. Daher ist es gewöhnlich gestattet, für die spezifische Wärme bei irgendeiner Temperatur die mittlere spezifische Wärme in einem benachbarten mäßig großen Temperaturintervall zu setzen.

§ 47. Die vorstehende Definition der Wärmekapazität und der spezifischen Wärme bedarf genau genommen noch einer Ergänzung. Denn da der thermodynamische Zustand eines Körpers außer von der Temperatur noch von einer zweiten Variabeln abhängt, etwa dem Druck, so ist die Zustandsänderung, welche mit einer Temperaturerhöhung verbunden ist, noch gar nicht bestimmt, solange nicht angegeben wird, wie sich die

zweite Variable dabei verhält. Nun ist allerdings bei festen Körpern und Flüssigkeiten die Wärmekapazität nahezu unabhängig davon, ob die Erwärmung bei konstantem oder bei veränderlichem äußeren Druck vollzogen wird, weshalb man dort bei der Definition der Wärmekapazität in der Regel keine besondere Bedingung hinsichtlich des Druckes hinzufügt. Bei Gasen aber wird der Wert der Wärmekapazität ganz wesentlich davon beeinflußt, unter welchen äußeren Umständen die Erwärmung erfolgt; daher muß hier die Definition der Wärmekapazität vervollständigt werden durch die Angabe dieser äußeren Umstände. Als Wärmekapazität eines Gases schlechthin gilt die Wärmekapazität bei konstantem Atmosphärendruck, welche der experimentellen Bestimmung am bequemsten zugänglich ist.

§ 48. Die Reduktion der Wärmekapazitäten verschiedener Stoffe auf die Masseneinheit ist ganz willkürlich und aus dem Umstand entsprungen, daß sich verschiedene Mengen eines Stoffes am bequemsten durch Wägen vergleichen lassen. Man könnte z. B. ebensogut die Wärmekapazitäten auf die Volumeneinheit beziehen. Am rationellsten ist aber die Vergleichung solcher Gewichtsmengen verschiedener Stoffe, welche im Verhältnis der Molekulargewichte bez. Atomgewichte stehen, weil sich hier auf den ersten Blick gewisse Regelmäßigkeiten ergeben. Die so zu vergleichenden Größen erhält man durch Multiplikation der auf 1 g bezogenen Wärmekapazität (der spezifischen Wärme) mit dem Molekulargewicht bez. Atomgewicht, und bezeichnet dann dies Produkt kurz als Molekularwärme bez. Atomwärme.

§ 49. Die Atomwärmen der chemischen Elemente im festen Zustand erweisen sich als nahezu konstant $= 6,3$ (Dulong und Petit) und zwar besonders für Elemente mit hohem Atomgewicht. Strenge Gültigkeit kann dies Gesetz schon deshalb nicht beanspruchen, weil die Wärmekapazität sowohl von der molekularen Konstitution des Elementes (z. B. für Kohle) als auch von der Temperatur abhängt, und zwar letzteres bezeichnenderweise in besonders hohem Grade bei denjenigen Stoffen (Kohle, Bor, Silicium), welche die größten Abweichungen von dem Dulong-Petitschen Gesetze zeigen. Daraus ist zu schließen, daß diesem Gesetz ein allgemeines Naturgesetz zugrunde liegt, dessen Ableitung aber über das Gebiet der allgemeinen Thermodynamik hinausführen würde.

§ 50. Wie die Atomwärmen der Elemente, so zeigen auch die Molekularwärmen der Verbindungen, besonders solche, die eine ähnliche chemische Konstitution aufweisen, gewisse Regelmäßigkeiten. Nach dem Gesetz von F. Neumann, welches später von Regnault und namentlich von Kopp bestätigt wurde, ist die Molekularwärme einer festen Verbindung einfach gleich der Summe der Atomwärmen der in ihr enthaltenen Elemente, indem jedes Element in jeder Verbindung die ihm eigentümliche Atomwärme

behält, mag sie nun dem DULONG-PETITschen Gesetz entsprechend $= 6,3$ sein oder nicht. Doch besitzt auch diese Beziehung nur angenäherte Gültigkeit.

§ 51. Da alle kalorimetrischen Messungen gemäß der in § 44 gegebenen Definition immer nur die Beträge zugeführter oder abgeleiteter Wärmemengen ergeben, so liefern sie durchaus keinen Aufschluß über die Frage nach der Größe der in einem Körper von bestimmter Temperatur im ganzen „enthaltenen“ Wärmemenge. Es würde nämlich widersinnig sein, die in einem Körper von gegebener Temperatur, Dichte usw. enthaltene Wärmemenge etwa gleich der Anzahl der Kalorien zu setzen, welche dem Körper zugeführt werden müssen, um ihn in den betrachteten Zustand zu bringen, ausgehend etwa von einem gewissen Normalzustand. Denn die Größe dieser Zahl würde ganz verschieden ausfallen je nach der Art und Weise, wie der Körper aus dem einen in den andern Zustand gebracht wird. Um z. B. ein Gas von 0° unter Atmosphärendruck auf 100° und 10fachen Atmosphärendruck zu bringen, kann man entweder so verfahren, daß man das Gas zuerst bei konstantem Atmosphärendruck auf 100° erwärmt und dann bei konstant gehaltener Temperatur bis auf den 10fachen Druck komprimiert; oder man kann das Gas zuerst bei 0° isotherm bis zu 10 Atmosphären komprimieren und dann isobar auf 100° erwärmen, oder man kann endlich Kompression und Erwärmung gleichzeitig in ganz beliebig wechselndem Verhältnis vornehmen. In jedem aller dieser unendlich vielfach verschiedenen Fälle erhält man als Gesamtzahl der zugeführten Kalorien eine andere Größe (vgl. die im § 77 ausgeführte Berechnung von Q), so daß man in diesem Sinne gar nicht von einer bestimmten Wärmemenge reden kann, die der Körper aufzunehmen hat, um aus dem alten Zustand in den neuen zu kommen. Will man also die „gesamte in einem Körper enthaltene Wärme“ als eine bestimmte Größe in die Betrachtung einführen (wie das z. B. in der kinetischen Wärmetheorie geschieht, wo die in einem Körper enthaltene Wärme als die lebendige Kraft seiner inneren Bewegungen aufgefaßt wird), so hat man dieselbe jedenfalls anders zu definieren als durch die Summation der dem Körper zugeführten Wärmemengen. Wir werden aber im folgenden dieses Begriffes gar nicht bedürfen und daher auch keine derartige Definition versuchen.

§ 52. Im Gegensatz zu der soeben geschilderten Sachlage mußte die ältere CARNOTsche Theorie der Wärme, die von der Auffassung der Wärme als eines unzerstörbaren Stoffes ausging, mit Notwendigkeit zu der Folgerung kommen, daß die in einem Körper enthaltene Wärme lediglich bedingt ist durch die Zahl der von außen aufgenommenen oder nach außen abgegebenen Kalorien. Wird daher ein Körper auf andere Weise als durch Zuleitung von Wärme, z. B. durch Kompression oder durch Reibung, erwärmt, so blieb

nach jener Theorie die im Körper enthaltene Wärme durch einen solchen Vorgang ganz ungeändert, und da doch tatsächlich eine höhere Temperatur entsteht, so war nur die Annahme übrig, daß die Wärmekapazität eines Körpers sich durch Kompression oder Reibung derartig verkleinert, daß die nämliche Wärme in ihm eine bedeutend höhere Temperatur hervorruft, ähnlich wie ein angefeuchteter Schwamm durch Kompression noch feuchter erscheint, obwohl die Menge der aufgesogenen Flüssigkeit dieselbe geblieben ist. Doch schon RUMFORD und DAVY bewiesen durch direkte Versuche, daß geriebene Körper, in denen man doch durch gehörigen Aufwand von Arbeit beliebig viel Wärme erzeugen kann, bei nachträglicher Untersuchung nicht die geringste Änderung ihrer Wärmekapazität zeigen. Auch hat zuerst REGNAULT durch genaue Messungen festgestellt, daß die Wärmekapazität von Gasen gar nicht oder nur sehr wenig vom Volumen abhängt, sich also auch durch Kompression nicht so stark verkleinern kann, wie es für die Erklärung der Kompressionswärme nach der CARNOTschen Theorie notwendig wäre. Endlich haben W. THOMSON und JOULE durch sorgfältige Versuche gezeigt, daß ein Gas, wenn es sich ohne Überwindung eines äußeren Druckes ausdehnt, keine oder eine sehr kleine Temperaturänderung erfährt (§ 70), weshalb die gewöhnlich bei der Ausdehnung eines Gases beobachtete Abkühlung nicht der Volumenvergrößerung des Gases an sich, sondern der dabei geleisteten mechanischen Arbeit zuzuschreiben ist. Jedes dieser Resultate für sich allein genommen genügt, um den Satz von der Unzerstörbarkeit der Wärme zu widerlegen und so die Haltlosigkeit jener älteren Wärmetheorie darzutun.

§ 53. Während im allgemeinen die Wärmekapazität sich stetig mit der Temperatur ändert, gibt es für jede Substanz bei bestimmtem äußeren Druck gewisse singuläre Temperaturpunkte, für welche mit anderen Eigenschaften auch die Wärmekapazität unstetig wird. In diesen Punkten kommt eine von außen zugeführte Wärmemenge nicht mehr dem ganzen Körper zugute, sondern nur einem Teil desselben, und dient außerdem nicht zur Erhöhung der Temperatur, sondern zur Veränderung des Aggregatzustandes, und zwar zum Schmelzen, Verdampfen oder Sublimieren, je nachdem die Substanz aus dem festen in den flüssigen, oder aus dem flüssigen in den gasförmigen, oder aus dem festen in den gasförmigen Zustand übergeht. Erst wenn der ganze Körper bei der nämlichen Temperatur im neuen Aggregatzustand homogen geworden ist, steigt bei weiterer Wärmezufuhr die Temperatur, und es wird wieder eine Wärmekapazität definierbar. Die Wärmemenge, welche nötig ist, um 1 g einer Substanz aus einem Aggregatzustand in einen andern zu bringen, heißt latente Wärme, speziell Schmelz-, Verdampfungs- oder Sublimationswärme. Bei der Rückkehr in den früheren Aggregatzustand

wird der nämliche Betrag von Wärme wieder frei. Auch die latente Wärme wird, ebenso wie die Wärmekapazität (§ 48), am zweckmäßigsten nicht auf die Masseneinheit, sondern auf das Molekulargewicht bez. Atomgewicht bezogen; ihr Betrag hängt übrigens wesentlich mit ab von den äußeren Bedingungen, unter denen die Umwandlung vollzogen wird (§ 47), und von denen ein konstant gehaltener Druck die wichtigste ist.

§ 54. Ähnlich wie eine Änderung des Aggregatzustandes ist auch jeder Mischungs- oder Lösungsvorgang, sowie jede chemische Umwandlung im allgemeinen von einer größeren oder geringeren, auch nach den äußeren Umständen veränderlichen, Wärmeentwicklung begleitet. Dieselbe wird nach J. Thomsen als die Wärmetönung des betr. Prozesses, speziell als Mischungs-, Lösungs-, Verbindungs-, Dissoziations- usw. Wärme bezeichnet: positiv, wenn Wärme frei oder entwickelt, d. h. nach außen abgegeben wird (exothermische Vorgänge), negativ, wenn Wärme gebunden oder absorbiert, d. h. von außen aufgenommen wird (endothermische Vorgänge).

Zweiter Abschnitt.

Der erste Hauptsatz der Wärmetheorie.

Erstes Kapitel. Allgemeine Formulierung.

§ 55. Der erste Hauptsatz der Wärmetheorie ist nichts anderes als das Prinzip der Erhaltung der Energie, angewendet auf die Erscheinungen, welche unter Wärmeproduktion oder -absorption verlaufen. Um einen allgemeinen deduktiven Beweis dieses Prinzips zu finden, kann man zwei verschiedene Wege einschlagen. Entweder: man stellt sich von vornherein auf den Boden der mechanischen Naturauffassung, d. h. man nimmt an, daß alle Veränderungen der Natur sich zurückführen lassen auf Bewegungen unveränderlicher materieller Punkte, zwischen denen Kräfte wirken, die ein Potential haben. Dann ist das Energieprinzip einfach der aus der Mechanik bekannte Satz der lebendigen Kraft, verallgemeinert auf beliebige Naturvorgänge. Oder aber: — und dieser Weg entspricht der hier eingehaltenen Darstellung — man läßt die Frage nach der Reduktion der Naturvorgänge auf Bewegungen ganz offen und geht allein aus von der durch jahrhundertelange menschliche Arbeit geprüften und in allen Fällen stets aufs neue bewährten Tatsache, daß es auf keinerlei Weise, weder mit mechanischen, noch thermischen, noch chemischen, noch anderen Apparaten möglich ist, ein perpetuum mobile zu bauen, d. h. eine periodisch wirkende Maschine zu konstruieren, durch welche fortdauernd Arbeit oder lebendige Kraft aus nichts gewonnen werden kann. Inwieweit dieser Erfahrungssatz für sich allein genommen, ganz unabhängig von der mechanischen Naturanschauung, dazu dienen kann, das Energieprinzip in seiner Allgemeinheit zu erweisen, soll jedoch an dieser Stelle nicht näher untersucht werden, und zwar namentlich aus dem Grunde, weil die Gültigkeit des Prinzips heutzutage wohl keinem ernsten Widerspruch mehr begegnet. Anders wird es mit dem zweiten Hauptsatz der Wärmetheorie sein, dessen Beweis bei dem heutigen Stande der Forschung nicht leicht sorgfältig genug geführt werden kann, da teils seine Allgemeingültigkeit noch manchmal bestritten, teils seine Bedeutung, auch von seinen Anhängern, noch recht verschieden beurteilt wird.

§ 56. Unter der Energie eines Körpers oder Körpersystems versteht man eine Größe, welche von dem augenblicklichen physikalischen Zustand abhängt, in dem sich das System befindet. Um die Energie eines Systems in einem gegebenen Zustand durch einen bestimmten Zahlenwert ausdrücken zu können, ist aber noch die Fixierung eines gewissen „Normalzustandes“ (z. B. 0° C, Atmosphärendruck) desselben Systems notwendig, welche von vornherein ganz nach Willkür erfolgen kann. Dann ist die Energie des Systems in dem gegebenen Zustand, bezogen auf den nach Willkür fixierten Normalzustand, gleich der „Summe der mechanischen Äquivalente aller Wirkungen, die außerhalb des Systems hervorgebracht werden, wenn dasselbe auf irgendeine Weise aus dem gegebenen Zustand in den Normalzustand übergeht“. Man bezeichnet daher die Energie auch kurz als die dem System innewohnende Fähigkeit, äußere Wirkungen hervorzubringen. Ob der Wert der Energie je nach der Art des Überganges in den Normalzustand verschieden ausfällt, darüber enthält diese Definition keinerlei Aussage. Dagegen ist zu ihrer Vervollständigung noch die Angabe dessen notwendig, was man unter dem mechanischen Äquivalent einer äußeren Wirkung zu verstehen hat.

§ 57. Wenn die äußere Wirkung mechanischer Natur ist, wenn sie z. B. in der Hebung eines das betrachtete Körpersystem belastenden Gewichts oder in der Überwindung des Atmosphärendrucks oder in der Erzeugung lebendiger Kraft besteht, so ist das mechanische Äquivalent der hervorgebrachten äußeren Wirkung einfach gleich der mechanischen Arbeit, welche die von dem System ausgeübte Kraft an dem äußeren Körper (Gewicht, Atmosphäre, Geschoß) leistet, positiv, wenn die Verschiebung in der Richtung der vom System ausgeübten Kraft erfolgt, also wenn das Gewicht gehoben, die Atmosphäre zurückgedrängt, das Geschoß fortgeschleudert wird; im entgegengesetzten Falle negativ.

Wenn aber die äußere Wirkung thermischer Natur ist, wenn sie also etwa in einer Erwärmung der umgebenden Körper (Atmosphäre, kalorimetrische Flüssigkeit) besteht, so setzt man das mechanische Äquivalent dieser äußeren Wirkung gleich der Anzahl Kalorien, welche in den umgebenden Körpern die nämliche Erwärmung bewirkt, multipliziert noch mit einer absoluten, nur von den Maßeinheiten der Wärmemenge und der mechanischen Arbeit abhängigen Konstanten, dem sogenannten mechanischen Wärmeäquivalent. Dieser Satz erscheint hier nur als Definition, er gewinnt aber einen tatsächlichen, an der Erfahrung zu prüfenden Inhalt durch das Prinzip der Erhaltung der Energie.

§ 58. Das Prinzip der Erhaltung der Energie besagt, und zwar allgemein und ausschließlich, daß die Energie eines Körpersystems in

einem gegebenen Zustand, bezogen auf einen bestimmten Normalzustand, einen ganz bestimmten Wert hat; oder mit anderen Worten, wenn wir den Wortlaut der Definition der Energie § 56 hier substituieren, daß die Summe der mechanischen Äquivalente aller Wirkungen, die außerhalb des Systems hervorgebracht werden, wenn dasselbe auf irgendeine Weise aus dem gegebenen Zustand in den Normalzustand übergeht, unabhängig ist von der Art des Überganges. Das System verursacht also beim Übergang in den Normalzustand eine ganz bestimmte Summe mechanisch gemessener Wirkungen, und diese Summe — auch der „Arbeitswert" der äußeren Wirkungen genannt — stellt eben die Energie des Systems dar.

§ 59. Die Gültigkeit des Energieprinzips in der Natur läßt sich also an der Erfahrung dadurch prüfen, daß man ein System aus einem bestimmten Zustand auf verschiedene Weisen in einen zweiten, hier als Normalzustand zu bezeichnenden, Zustand bringt und nun untersucht, ob die dabei jedesmal auftretenden mechanischen Äquivalente der äußeren Wirkungen, berechnet auf Grund der Festsetzungen des § 57, in allen Fällen die gleiche Summe ergeben. Dabei ist besonders darauf zu achten, daß das System in allen verglichenen Fällen auch wirklich von dem nämlichen Anfangszustand ausgeht und in den nämlichen Endzustand übergeführt wird, und daß von den äußeren Wirkungen keine übersehen und keine doppelt in Anschlag gebracht wird.

§ 60. Als erste Anwendung besprechen wir die berühmten Versuche von Joule. Derselbe verglich die äußeren Wirkungen, die entstehen, wenn gewisse Gewichte beim Herabsinken um eine gewisse Höhe einmal nur mechanische Arbeit hervorbringen (z. B. Hebung einer Last), ein anderes Mal mittels geeigneter Vorrichtungen durch Reibung Wärme erzeugen. Hierbei kann man die Anfangs- und die Endruhelage der Gewichte als ersten und zweiten Zustand des Systems, die erzeugte Arbeit und die erzeugte Wärme als äußere Wirkungen betrachten. Im ersten Falle, wo durch das Herabsinken der Gewichte nur mechanische Arbeit erzeugt wird, ist die Berechnung des mechanischen Äquivalents der äußeren Wirkungen einfach und erfordert keinen besonderen Versuch: dasselbe ist nach den Gesetzen der Mechanik immer gleich dem Produkt der Schwere der Gewichte und der durchfallenen Höhe, ganz unabhängig von der Beschaffenheit des Apparates, den man zur Erzeugung der mechanischen Arbeit benutzt (Hebel, Rolle, Flaschenzug), und von der Schwere der gehobenen Last. Diese Unabhängigkeit bildet schon eine Forderung des Prinzips der Erhaltung der Energie (§ 58). Im zweiten Falle ist eine genaue Messung der Temperaturerhöhung erforderlich, welche die geriebenen umgebenden Körper (Wasser, Quecksilber) erleiden,

sowie deren Wärmekapazität, um daraus die Anzahl Kalorien bestimmen zu können, welche in ihnen die nämliche Temperaturerhöhung bewirkt. Dabei kommt es natürlich gar nicht darauf an, welche Vorstellungen man sich über den Vorgang der Wärmeerzeugung durch Reibung im einzelnen macht, sowie über den Verbleib der in den geriebenen Körpern erzeugten Wärme, sondern einzig und allein darauf, daß der durch Reibung in der betreffenden Flüssigkeit hervorgerufene Zustand identisch ist mit einem, der durch Zuführung einer bestimmten Anzahl Kalorien herbeigeführt werden kann.

Indem nun JOULE, dem Prinzip der Erhaltung der Energie gemäß, die dem Fall der Gewichte entsprechende mechanische Arbeit gleichsetzte dem mechanischen Äquivalent der durch die Reibung erzeugten Wärme, wie sie durch die Anzahl der gewonnenen Kalorien bestimmt wird, fand er, daß das mechanische Äquivalent einer g-Kalorie unter allen Umständen gleich ist der Arbeit, welche durch die Hebung eines Gramms um $423,55^{\mathrm{m}}$ dargestellt wird. Daß sich bei allen Versuchen mit verschiedenen Gewichten, Substanzen, Temperaturen stets wieder diese nämliche Zahl ergibt, ist eine Bestätigung des Prinzips der Erhaltung der Energie.

§ 61. Bei der Berechnung des mechanischen Wärmeäquivalents im absoluten Maße ist zunächst zu berücksichtigen, daß die JOULEsche Kalorie sich auf Zimmertemperatur (§ 45) und auf die Angaben eines Quecksilberthermometers bezieht. Bei Zimmertemperatur bedeutet aber 1° des JOULEschen Quecksilberthermometers ein im Verhältnis von etwa $1 : 1,007$ kleineres Temperaturintervall als 1° des Gasthermometers; folglich hat eine auf das Gasthermometer (§ 4) bezogene Kalorie ein entsprechend größeres mechanisches Äquivalent, d. h. das Äquivalent $423,55 \cdot 1,007 = 427$.

Ferner ist noch die Größe der Beschleunigung der Schwere zu berücksichtigen, da die Hebung eines Gramms um eine bestimmte Höhe an verschiedenen Orten im allgemeinen verschiedene Arbeiten darstellt. Der absolute Betrag der geleisteten Arbeit wird erhalten durch Multiplikation der Schwerkraft, also des Produkts aus Masse und Beschleunigung der Schwere, mit der Höhe. Hieraus ergibt sich mit Rücksicht auf die oben § 45 über die Größenverhältnisse der verschiedenen Kalorien gemachten Angaben folgende Tabelle der Werte des mechanischen Wärmeäquivalents:

Wärmeeinheit bezogen auf Gasthermometer	Entsprechende Höhe in m der Hebung von 1 g an einem Orte mittlerer geogr. Breite	Absoluter Wert des Wärmeäquivalents im C.G.S.-System (Erg)
15°-Kalorie	427	$4,19 \cdot 10^7$
Nullpunktskalorie	430	$4,22 \cdot 10^7$

Die Zahlen der zweiten Kolumne entstehen aus denen der ersten durch Multiplikation mit 98 100, entsprechend der Beschleunigung der Schwere 981 und der Reduktion von Metern auf Zentimeter. Die Resultate von JOULE sind durch die neueren sorgfältigen Messungen von ROWLAND u. a. im wesentlichen bestätigt worden.

§ 62. Man kann die Kenntnis des mechanischen Wärmeäquivalents benutzen, um Wärmemengen, anstatt in Kalorien, direkt in Erg auszudrücken, und erreicht dadurch den Vorteil, daß eine Wärmemenge nicht nur proportional, sondern unmittelbar gleich ist ihrem mechanischen Äquivalent, wodurch sich der mathematische Ausdruck der Energie vereinfacht. Diese Einheit der Wärmemenge soll in den folgenden Gleichungen überall angewendet werden; bei Zahlenrechnungen kann man jeden Augenblick durch Division mit $4,19 \cdot 10^7$ zu Kalorien zurückkehren.

§ 63. Aus der oben gegebenen Formulierung des Energieprinzips ergeben sich sogleich einige weitere Sätze. Da die Energie U durch den augenblicklichen Zustand des Systems bedingt ist, so wird sich ihr Wert ändern, sobald der Zustand sich ändert. Um den Betrag der Energieänderung zu finden, die eintritt, wenn das System aus einem Zustand (1) in einen anderen Zustand (2) übergeht, und die durch die Differenz $U_1 - U_2$ bestimmt wird, hat man nach der Definition der Energie den Arbeitswert (§ 58) aller äußeren Wirkungen zu messen, welche beim Übergang des Systems, einmal aus dem Zustand 1, das andere Mal aus dem Zustand 2, in den Normalzustand eintreten, und diese Beträge, welche die Werte von U_1 und U_2 darstellen, voneinander zu subtrahieren. Denkt man sich nun den ersten dieser beiden Übergänge so eingerichtet, daß er das System aus dem Zustand 1 durch den Zustand 2 hindurch in den Normalzustand bringt, so erhellt, daß als gesuchte Differenz nur der Arbeitswert derjenigen äußeren Wirkungen übrig bleibt, welche dem Übergang des Systems aus 1 in 2 entsprechen. Daher ist $U_1 - U_2$, d. h. die Energieabnahme eines Systems bei irgendeiner Veränderung, gleich dem Arbeitswert der äußeren Wirkungen, welche bei dieser Veränderung hervorgebracht werden, oder, was dasselbe bedeutet, die Energiezunahme des Systems bei irgendeiner Veränderung ist gleich dem Arbeitswert der bei dieser Veränderung aufgewendeten

oder verbrauchten äußeren Wirkungen:

$$U_2 - U_1 = Q + A, \tag{17}$$

wo Q das mechanische Äquivalent der außerhalb des Systems verschwundenen, etwa dem System durch Leitung zugeführten Wärme, A den Betrag der von außen auf das System ausgeübten Arbeit bezeichnet, positiv, wenn die Veränderung im Sinne der von außen auf das System wirkenden Kräfte erfolgt. Man kann die Summe $Q + A$ auch den Arbeitswert aller von den umgebenden Körpern auf das System ausgeübten thermischen und mechanischen Einwirkungen nennen. In diesem Sinne werden wir die Größen Q und A stets benutzen.

Der Wert von $Q + A$ hängt nicht von der Art des Überganges aus 1 in 2 ab, und offenbar auch nicht von der Wahl des Normalzustandes des Systems; daher ist es, solange es sich nur um Energiedifferenzen eines und desselben Systems handelt, gar nicht nötig, den Normalzustand besonders zu fixieren. Dann bleibt in dem Wert der Energie selber eine additive Konstante unbestimmt.

§ 64. Die Differenz $U_2 - U_1$ läßt sich auch auffassen als die Energie des Systems im Zustand 2, bezogen auf den Zustand 1 als Normalzustand. In der Tat: nimmt man 1 als Normalzustand, so ist $U_1 = 0$, weil es dann überhaupt keiner Veränderung bedarf, um das System aus dem Zustand 1 in den Normalzustand zu bringen, und es wird $U_2 - U_1 = U_2$. Daher wird der Normalzustand manchmal auch Nullzustand genannt.

§ 65. Wenn der Zustand 2 mit dem Zustand 1 identisch gewählt wird, so macht das System beim Übergang von 1 zu 2 einen sogenannten „Kreisprozeß“ durch. Dann ist $U_2 = U_1$ und daher aus (17):

$$0 = Q + A, \tag{18}$$

d. h. bei einem Kreisprozeß ist der Arbeitswert aller äußeren Wirkungen gleich Null, oder mit anderen Worten: Die äußere Wärme ist der äußeren Arbeit gleich und entgegengesetzt. Durch diesen Satz ist die Konstruktion eines thermodynamischen perpetuum mobile, das notwendig periodisch wirkende Maschinen, also Kreisprozesse voraussetzt, ausgeschlossen.

§ 66. Wenn bei einer Zustandsänderung des Systems gar keine äußeren Wirkungen aufgewendet werden ($Q = 0,\ A = 0$), so bleibt nach (17) die Energie konstant (Erhaltung der Energie). Dabei können die einzelnen Größen, welche den Zustand des Systems bedingen, sich erheblich ändern, sie unterliegen aber stets der Bedingung $U = \text{konst.}$

Ein solches System, welches sich verändert, ohne dabei äußeren Einwirkungen zu unterliegen, heißt auch ein „vollständiges“ oder „abgeschlossenes“

System. Streng genommen gibt es in der Natur gar kein vollständiges System, weil sämtliche materielle Körper des Weltalls in steter Wechselwirkung miteinander stehen, und insofern kann man den Satz von der „Erhaltung“ der Energie auf kein wirkliches System strenge anwenden. Doch ist es wichtig zu bemerken, daß man durch passende Wahl des Systems die äußeren Wirkungen, die bei einer bestimmten ins Auge gefaßten Veränderung auftreten, im Vergleich zu den Energieänderungen der einzelnen Teile des Systems so klein machen kann, als man nur immer will. Man kann nämlich offenbar jede äußere Wirkung dadurch eliminieren, daß man nicht nur die Körper, auf welche die Wirkung ausgeübt wird, sondern auch diejenigen, von welchen dieselbe ausgeht, mit in das betrachtete System hineinbezieht. Wenn z. B. ein Gas durch ein sinkendes Gewicht komprimiert wird, so wird dabei auf das Gas, als System gedacht, durch die von dem Gewicht geleistete Arbeit eine gewisse Wirkung von außen her ausgeübt und die Energie des Systems demgemäß vergrößert. Sobald man aber das Gewicht und die Erde mit in das betrachtete System hineinbezieht, fällt jede äußere Wirkung fort, und die Energie des neuen Systems bleibt konstant. Dafür enthält aber der Ausdruck der Energie jetzt ein neues Glied: die potentielle Energie des Gewichts, deren Änderung durch die der inneren Energie des Gases gerade kompensiert wird. Ebenso kann man in allen anderen Fällen verfahren.

Zweites Kapitel. Anwendungen auf homogene Systeme.

§ 67. Wir wenden nun den ersten Hauptsatz, wie er in der Gleichung (17) ausgesprochen ist, zunächst auf eine homogene Substanz an, deren Zustand, außer durch ihre chemische Natur und durch die Masse M, durch 2 Variable, etwa die Temperatur T und das Volumen V, bestimmt ist. Dabei gebrauchen wir hier wie auch überall im folgenden das Wort „homogen“ schlechthin im Sinne von „physikalisch homogen“, d. h. wir nennen homogen ein System, welches auch in den kleinsten sichtbaren Raumteilen als vollständig gleichartig betrachtet werden kann. Es kommt hier nicht darauf an, ob die Substanz auch „chemisch homogen“ ist, d. h. ob sie aus lauter gleichartigen Molekülen besteht. So kann z. B. ein Knallgasgemenge ($O_2 + 2\,H_2$) oder ein teilweise dissoziierter Dampf, die beide chemisch inhomogen sind, sehr wohl physikalisch homogen sein. Was wir hier voraussetzen wollen, ist nur dies, daß der Zustand der betrachteten homogenen Substanz durch Temperatur und Volumen eindeutig bestimmt ist, gleichgültig ob und welche chemische Umsetzungen im Lauf der betrachteten Zustandsänderungen eintreten. Wenn

die Substanz ruht, so besteht die ganze Energie dieses Systems aus der sogenannten „inneren“ Energie U, die nur von der inneren, durch Temperatur und Dichte bedingten, Beschaffenheit der Substanz und von ihrer Masse abhängt, welch letzterer sie offenbar proportional ist. Im anderen Falle tritt in dem Ausdruck der Gesamtenergie zu der inneren Energie U noch die lebendige Kraft der Bewegung hinzu, deren Wert aus der Mechanik bekannt ist.

Um die Abhängigkeit der inneren Energie U von T und V festzustellen, muß man das System auf irgendeine Weise in einen anderen Zustand bringen und die dazu erforderlichen äußeren Wirkungen messen. Dann liefert die Gleichung (17) die eingetretene Änderung der Energie.

§ 68. Läßt man ein anfangs in Ruhe und auf gleichmäßiger Temperatur befindliches Gas (Zustand 1) aus einem Gefäß in ein anderes vorher evakuiertes Gefäß ausströmen, etwa durch Aufdrehen eines Verschlußhahnes, so werden sich bei diesem Vorgang innerhalb des Gases zunächst eine Reihe von verwickelten mechanischen und thermischen Veränderungen vollziehen. Der ausströmende Teil des Gases wird in schnelle Bewegung geraten, später beim Anprall gegen die Wände des zweiten Gefäßes und bei der Kompression durch die nachstürzenden Massen sich erwärmen, der im ersten Gefäß zurückbleibende Teil wird sich durch Ausdehnung abkühlen usw. Nimmt man nun an, daß die Wände beider Gefäße absolut fest sind und die Wärme absolut nicht leiten, und bezeichnet irgendeinen nach beliebiger Zeit eingetretenen Zustand des Gases mit 2, so ist nach Gleichung (17) die Gesamtenergie des Gases im zweiten Zustand gleich der im ersten Zustand: U_1, weil auf das Gas weder thermische noch mechanische Einwirkungen von außen stattgefunden haben. Denn auch die von den festen Wänden vermöge ihres Widerstandes ausgeübte Kraft leistet keine Arbeit. Im allgemeinen setzt sich die Energie im zweiten Zustand aus vielen Teilen zusammen, nämlich erstens aus den lebendigen Kräften der Bewegungen aller einzelnen Gasteilchen und zweitens aus ihren inneren Energien. Wartet man aber so lange, bis wieder vollständige Ruhe und thermisches Gleichgewicht eingetreten ist, und bezieht den Index 2 auf den neuen Gleichgewichtszustand, so besteht die Gesamtenergie im zweiten Zustand ebenso wie die im ersten nur aus der inneren Energie U_2, und man hat: $U_2 = U_1$. Nun sind aber die Variabeln T und V, von denen U abhängig ist, von den Werten T_1, V_1 auf die Werte T_2, V_2 übergegangen, wobei $V_2 > V_1$; man kann also hieraus durch Messung der Temperaturen und Volumina feststellen, wie sich mit verändertem Volumen die Temperatur des Gases ändert, falls die innere Energie U konstant bleibt.

§ 69. Einen derartigen Versuch hat JOULE ausgeführt[1] und dabei gefunden, daß für ideale Gase $T_2 = T_1$. Er stellte nämlich die beiden Gefäße, von denen das eine anfangs etwa mit Luft unter hohem Druck gefüllt, das andere evakuiert war, in ein gemeinsames Wasserbad von der nämlichen Temperatur und fand nach Vollendung des oben beschriebenen Ausflusses und Herstellung des Gleichgewichts die im Wasserbad eingetretene Temperaturänderung unmeßbar klein. Daraus folgt sogleich, daß auch bei thermisch isolierenden Gefäßwänden die Endtemperatur der ganzen Gasmenge gleich der Anfangstemperatur ist; denn sonst würde sich bei dem ausgeführten Versuch die Temperaturänderung dem Wasserbade mitgeteilt haben.

Wenn also die innere Energie eines nahezu idealen Gases bei stark verändertem Volumen konstant bleibt, so bleibt auch die Temperatur nahezu konstant, oder mit anderen Worten: Die innere Energie eines idealen Gases hängt nur von der Temperatur und nicht vom Volumen ab.

§ 70. Damit indessen dieser wichtige Schluß bündig erscheint, sind noch genauere Messungen notwendig. Denn bei dem beschriebenen JOULEschen Versuch ist die Wärmekapazität des Gases gegen die der Gefäßwände und des Wasserbades so klein, daß es schon einer sehr beträchtlichen Temperaturänderung des Gases bedurft hätte, um eine merkliche Temperaturänderung des Wassers hervorzurufen. Zuverlässigere Resultate liefert eine wesentliche Modifikation des Verfahrens, welche von W. THOMSON (später Lord KELVIN) ersonnen und von ihm in Gemeinschaft mit JOULE zu sorgfältigen Messungen benutzt worden ist; sie beruht darauf, daß man das Gas durch künstliche Verlangsamung des Ausflusses unmittelbar in den zweiten Gleichgewichtszustand überführt und dann die Temperatur T_2 direkt im Gase mißt. Es strömt hierbei nicht eine begrenzte Gasmasse tumultuarisch in ein Vakuum ein, sondern das Gas wird in einem unbegrenzten stationären Strom verhältnismäßig langsam aus einem Raum höheren Druckes p_1 in einen Raum niedrigeren Druckes p_2 (die Atmosphäre) übergeführt, indem es durch eine Röhre von Buchsbaumholz, welche an einer Stelle mit einem schwer durchlässigen Pfropfen von Watte oder gezupfter Seide verstopft ist, hindurchgepreßt wird. Was zunächst die Messungsresultate betrifft, so ergeben sie bei stationär gewordenem Zustand für Luft eine sehr kleine, für Wasserstoff eine noch sehr viel kleinere, kaum meßbare Temperaturänderung des Gases, weshalb man zu dem Schluß berechtigt ist, daß für ein ideales Gas die Temperaturänderung ganz verschwindet.

[1] Schon früher war derselbe Versuch von GAY LUSSAC angestellt worden.

Hieraus läßt sich nun ein Schluß auf die innere Energie idealer Gase ziehen. Wenn nach Eintritt des stationären Zustandes eine gewisse zu beiden Seiten und innerhalb des Pfropfens befindliche Masse des Gases sich so weit vorwärts bewegt hat, daß vor dem Pfropfen das Gasvolumen V_1 verschwunden, hinter dem Pfropfen das Gasvolumen V_2 hinzugekommen ist, so hat das aus der betrachteten Gasmasse und dem Pfropfen bestehende System gewisse Einwirkungen von außen erfahren, deren mechanisches Äquivalent $Q + A$ aus den in der Umgebung eingetretenen Änderungen zu berechnen ist. Nun findet in der äußeren Umgebung keine Temperaturänderung statt; denn das Holz der Röhre leitet die Wärme so gut wie gar nicht von oder nach außen; daher ist $Q = 0$. Die Arbeit ferner, welche auf das Hindurchpressen unter dem konstanten Druck p_1 verwendet worden ist, und welche, wie leicht einzusehen, durch das Produkt p_1V_1 dargestellt wird, ist für ein ideales Gas nach dem BOYLEschen Gesetz gerade gleich derjenigen Arbeit p_2V_2, welche auf der anderen Seite beim Zurückschieben des kleineren Druckes p_2 durch das größere Volumen V_2 bei der nämlichen Temperatur wieder gewonnen worden ist. Daher ist auch die Summe der von außen her auf das ganze betrachtete System ausgeübten Arbeiten: $A = 0$, und nach Gleichung (17) besitzt das oben definierte, aus der betrachteten Gasmasse und dem Pfropfen bestehende System am Schluß die nämliche Gesamtenergie wie am Anfang, d. h. die Differenz der beiden Gesamtenergien ist gleich Null. Nun verschwinden in dieser Differenz alle Glieder mit Ausnahme derjenigen, welche sich auf die Gasvolumina V_1 und V_2 beziehen. Denn auch der Pfropfen behält seinen Zustand unverändert bei, er und die verwickelten Vorgänge in ihm kommen also in der Energiegleichung gar nicht zur Geltung; von dieser bleibt somit nur übrig $U_1 - U_2 = 0$, wenn U_1 und U_2 die inneren Energien der Gasvolumina V_1 und V_2 bedeuten. Da aber nach den mitgeteilten Messungen die Temperatur wesentlich konstant geblieben ist, während das Volumen des Gases sich beträchtlich verändert hat, so kann die innere Energie eines idealen Gases nur von der Temperatur abhängen und nicht vom Volumen, d. h.

$$\left(\frac{\partial U}{\partial V}\right)_T = 0. \tag{19}$$

Für nicht vollständig ideale Gase, wie es Wasserstoff, Luft usw. tatsächlich sind, ergibt die gemessene Temperaturänderung einen Aufschluß über die Abhängigkeit der inneren Energie vom Volumen. Doch gehört hierzu auch noch die Berücksichtigung des Umstandes, daß bei solchen Gasen die äußere Arbeit

$$A = p_1V_1 - p_2V_2$$

nicht verschwindet und daher auch die innere Energie nicht konstant bleibt. Näheres darüber s. unten § 158.

§ 71. Von besonderer theoretischer Wichtigkeit sind diejenigen thermodynamischen Prozesse, welche, wie man sagt, unendlich langsam verlaufen und daher aus lauter Gleichgewichtszuständen bestehen. Wörtlich genommen ist zwar diese Ausdrucksweise undeutlich, da ein Prozeß notwendig Veränderungen, also Störungen des Gleichgewichts zur Voraussetzung hat. Aber man kann diese Störungen, wenn es nicht auf die Schnelligkeit, sondern nur auf das Resultat der Veränderungen ankommt, so klein nehmen wie man irgend will, namentlich auch beliebig klein gegen die übrigen Größen, welche im Zustand des betrachteten Systems eine Rolle spielen. So kann man ein Gas beliebig langsam um einen beliebig großen Bruchteil seines Volumens komprimieren, indem man den äußeren Druck in jedem Augenblick nur um ein äußerst Geringes größer macht als den Druck des Gases, und man begeht, wenn es sich um die Größe des äußeren Druckes handelt, z. B. bei Berechnung der zu einer bestimmten endlichen Kompression aufgewendeten Arbeit, nur einen sehr kleinen Fehler, wenn man statt des äußeren Druckes den Druck des Gases setzt. Beim Übergang zur Grenze verschwindet auch dieser kleine Fehler, d. h. bei „unendlich langsamer“ Kompression wird das so gewonnene Resultat strenge richtig.

Das Gesagte gilt sowohl für eine Kompression bei konstantem, als auch für eine solche bei veränderlichem Druck. Im letzteren Falle muß man dem äußeren Druck, etwa durch Hinzufügung oder Fortnahme kleiner Gewichtsstücke, in jedem Augenblick gerade die erforderliche Größe erteilen. Dies kann durch manuelle Eingriffe (Beiseiteschieben der Gewichtsstücke in horizontaler Richtung) oder durch eine besondere Regulierungsvorrichtung geschehen, welche nur auslösend wirkt und daher ohne Arbeitsleistung funktioniert.

§ 72. Ebenso wie bei der äußeren Arbeit ist es mit der Zuleitung oder Ableitung von Wärme. Wenn es sich nur um den Betrag der Wärmemenge handelt, welche das System aus der Umgebung empfangen oder dahin abgegeben hat, ohne daß dabei die Zeit eine Rolle spielt, so genügt es, die Temperatur der verwendeten Wärmequelle um einen beliebig kleinen Wert größer oder kleiner als die Temperatur des Systems anzunehmen, je nachdem die Wärme zu- oder abgeleitet werden soll. Dieser kleine Überschuß bestimmt lediglich die Richtung des Prozesses, seine Größe kommt aber nicht in Betracht gegen die ganze durch den Prozeß schließlich herbeigeführte Veränderung des Systems. Daher spricht man, wie von der Kompression eines Gases durch einen äußeren Druck, der dem Druck des Gases gleich ist, so auch von dem Wärmeübergang von einem Körper zu einem anderen

von der nämlichen Temperatur, und antizipiert damit nur das Resultat, das sich aus dem Grenzübergang von einer endlichen kleinen zu einer unendlich kleinen Temperaturdifferenz beider Körper ergibt.

Auch hier sind nicht nur isotherme Vorgänge, sondern auch solche von variabler Temperatur mit einbegriffen. Für letztere kommt man freilich mit einem einzigen Wärmebehälter von konstanter Temperatur nicht aus, sondern man bedarf entweder eines Körpers von willkürlich veränderlicher Temperatur, also etwa eines Gases, das man durch zweckmäßige Kompression oder Ausdehnung beliebig erwärmt oder abkühlt, oder man verwendet eine hinreichend große Zahl von Wärmebehältern verschiedener bestimmter Temperaturen und setzt in jedem Augenblick gerade denjenigen in Funktion, welcher der gleichzeitigen Temperatur des Systems möglichst nahe liegt.

§ 73. Die hohe Bedeutung dieser Betrachtungsweise besteht darin, daß man jeden „unendlich langsamen“ Prozeß auch in entgegengesetzter Richtung ausgeführt denken kann. Besteht nämlich ein Prozeß bis auf minimale Abweichungen aus lauter Gleichgewichtszuständen, so genügt offenbar immer eine ebenso minimale passend angebrachte Änderung, um ihn in entgegengesetzter Richtung ablaufen zu lassen, und diese minimale Änderung kann durch einen Grenzübergang ebenso ganz zum Verschwinden gebracht werden. Denn ein bestimmtes Resultat enthält immer auch einen ganz bestimmten Fehler, und wenn dieser Fehler kleiner ist als jede noch so klein angenommene Größe, so ist er notwendig gleich Null.

§ 74. Wir gehen nun über zur Anwendung des ersten Hauptsatzes auf einen solchen aus lauter Gleichgewichtszuständen zusammengesetzten und daher umkehrbaren Prozeß. Derselbe läßt sich in einfacher Weise graphisch versinnlichen dadurch, daß die Reihe der nacheinander durchlaufenen Gleichgewichtszustände des Systems als Kurve in eine Koordinatenebene eingetragen wird, auf deren Achsen die Werte der unabhängigen Variabeln gemessen werden. Wir wollen als unabhängige Variable zunächst das Volumen V (Abszissenachse) und den Druck p (Ordinatenachse) anwenden. Dann entspricht jedem Punkt der Ebene ein bestimmter Zustand der von bestimmter Natur und Masse angenommenen Substanz, und jeder Kurve eine bestimmte Reihe von stetig aufeinanderfolgenden Zustandsänderungen derselben. Denken wir uns also einen umkehrbaren Prozeß, der die Substanz aus einem Zustand 1 in einen Zustand 2 bringt, so wird er durch eine Kurve α bezeichnet, die vom Punkt 1 zum Punkt 2 geht (Fig. 2). Dann ist nach Gleichung (17) die Zunahme der Energie der Substanz:

$$U_2 - U_1 = A + Q,$$

wobei A die gesamte aufgewendete äußere Arbeit, Q die im ganzen von

außen zugeführte Wärme bedeutet.

§ 75. Der Wert von A läßt sich unmittelbar berechnen. Zunächst setzt sich A durch algebraische Addition aus den Elementararbeiten zusammen, welche während der aufeinanderfolgenden unendlich kleinen, den einzelnen Bogenelementen der Kurve α entsprechenden Veränderungen der Substanz von außen her auf dieselbe ausgeübt werden. Da nun der äußere Druck wegen der angenommenen Umkehrbarkeit des Prozesses in jedem Augenblick gleich dem der Substanz p zu setzen ist, so ist die von den äußeren Kräften bei einer unendlich kleinen Veränderung geleistete Arbeit, wie in der Hydrodynamik gezeigt wird, einfach gleich dem Produkte des Druckes p und der Volumenverminderung, unabhängig von der geometrischen Form der Oberfläche der Substanz, also $= -p\,dV$, und mithin die während des ganzen Prozesses geleistete äußere Arbeit:

$$A = -\int_1^2 p\,dV, \tag{20}$$

wobei die Integration über die Kurve α vom Punkt 1 bis zum Punkt 2 zu erstrecken ist. Wenn p positiv, wie bei Gasen, und $V_2 > V_1$, wie in der Fig. 2, so ist A negativ, d. h. es wird äußere Arbeit nicht aufgewendet, sondern gewonnen.

Um die Integration ausführen zu können, bedarf es der Kenntnis der Abhängigkeit des Druckes p vom Volumen V, d. h. der Kenntnis der Kurve α. Solange nur die Punkte 1 und 2, nicht aber die sie verbindende Kurve gegeben ist, hat das Integral gar keinen bestimmten Wert. Erfolgt z. B. der Übergang von 1 zu 2 auf einer anderen Kurve β, so fällt das Integral ganz anders aus. Daher ist, wie man sagt, das Differential $p\,dV$ ein „unvollständiges“ Differential. Mathematisch betrachtet rührt dieser Umstand daher, daß p außer von V im allgemeinen noch von einer anderen Variabeln, der Temperatur T, abhängt, die sich auf dem Integrationswege α in gewisser Weise mitändern wird. Solange nun α nicht gegeben ist, läßt sich auch nichts über die Abhängigkeit des T von der Integrationsvariabeln V aussagen, und die Integration liefert kein bestimmtes Resultat.

Die äußere Arbeit A hat in der Fig. 2 eine sehr anschauliche Bedeutung. Sie ist offenbar gleich dem negativ genommenen Flächeninhalt der ebenen Figur, welche durch die Kurve α, die Abszissenachse und die Ordinaten in den Punkten 1 und 2 begrenzt wird. Auch hieraus erkennt man, daß der Wert von A wesentlich durch den Verlauf der Kurve α bedingt ist. Nur für unendlich kleine Zustandsänderungen, d. h. wenn die Punkte 1 und 2 einander unendlich nahe liegen und somit α auf ein Kurvenelement

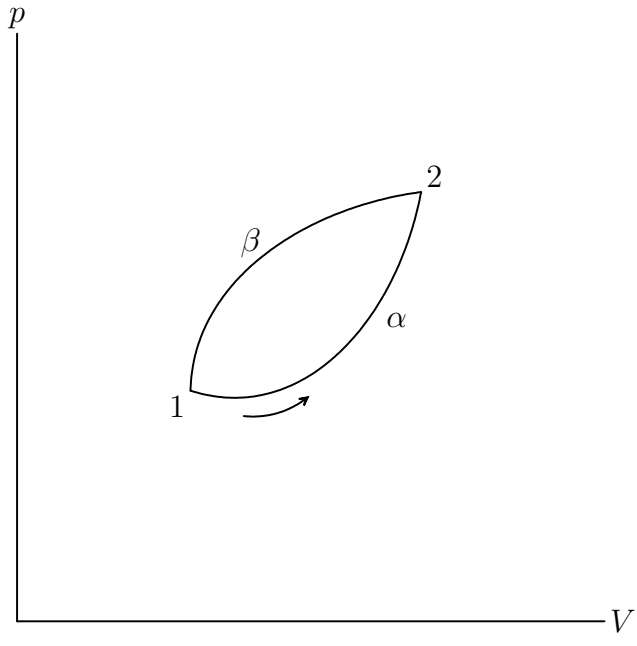


Fig. 2.

zusammenschrumpft, ist A durch den Anfangs- und Endpunkt der Kurve allein schon bestimmt.

§ 76. Die zweite der Messung zugängliche Größe ist die von außen zugeführte Wärme Q, welche durch eine kalorimetrische Bestimmung zunächst in Kalorien, und durch Multiplikation mit dem mechanischen Wärmeäquivalent auch in mechanischen Einheiten ausgedrückt werden kann. Fragen wir nun nach der theoretischen Bestimmung der zugeleiteten Wärme Q. Auch sie setzt sich, wie die äußere Arbeit A, durch algebraische Summation zusammen aus den unendlich kleinen Wärmemengen, welche während der den einzelnen Kurvenelementen entsprechenden Elementarprozesse dem Körper zugeführt werden. Doch läßt sich eine solche Elementarwärme nicht auf ähnliche Weise wie die gleichzeitige Elementararbeit durch das Produkt zweier unabhängig voneinander meßbaren Faktoren ausdrücken. Man kann zwar, um eine Analogie mit dem Ausdruck der Elementararbeit $-p\,dV$ zu schaffen, die Elementarwärme etwa gleich dem Produkt der unendlich kleinen durch sie bewirkten Temperaturerhöhung dT und einer im allgemeinen endlichen Größe C, der Wärmekapazität, setzen; aber dann hat die Größe C im allgemeinen keine bestimmte Bedeutung. Denn sie hängt nicht, wie der Faktor p in dem Ausdruck der Elementararbeit, allein

von dem augenblicklichen Zustand der Substanz, also von der Lage des betreffenden Kurvenpunktes ab, sondern zugleich auch von der Richtung des Kurvenelements. Für eine isotherme Änderung ist C offenbar $\pm\infty$, weil dann $dT = 0$, während die zugeleitete Wärme positiv oder negativ sein kann. Für eine „adiabatische“ Änderung ist $C = 0$, weil dann die zugeleitete Wärme gleich Null ist, während die Temperatur sich beliebig ändern kann. C kann also, im Gegensatz zu p, für einen und denselben Punkt alle möglichen Werte zwischen $+\infty$ und $-\infty$ haben (vgl. § 47). Aus diesem Grunde ist die durch die Zerlegung der zugeleiteten Wärme in die beiden Faktoren dT und C gesuchte Analogie mit der äußeren Arbeit in einem wesentlichen Punkte unvollständig, und führt im allgemeinen Falle nicht zu einer Vereinfachung des behandelten Problems. Übrigens beschränkt sich auch für die äußere Arbeit die Zerlegbarkeit in die beiden Faktoren p und dV natürlich auf den speziellen hier betrachteten Fall umkehrbarer Prozesse, weil sonst der äußere Druck von p verschieden sein kann.

§ 77. Wenn sich somit der Wert der zugeleiteten Wärme Q im allgemeinen nicht von vornherein bestimmen läßt, so gestattet andrerseits die Gleichung (17) des ersten Hauptsatzes einige wichtige Schlüsse auf diese Größe. Zunächst ergibt sich aus ihr, wenn man den gefundenen Wert (20) von A substituiert:

$$Q = U_2 - U_1 + \int_1^2 p\,dV. \tag{21}$$

Daraus erkennt man, daß der Wert von Q nicht allein durch die Punkte 1 und 2, sondern auch, ebenso wie A, durch den Verlauf der sie verbindenden Kurve, α oder β, bedingt wird. Mit diesem Satze ist die CARNOTsche Theorie der Wärme unvereinbar, was schon oben (§§ 51, 52) ausführlich dargelegt wurde.

§ 78. Vollständig berechnen läßt sich Q für den Fall, daß die Substanz schließlich wieder in ihren Anfangszustand 1 zurückgebracht wird, also einen Kreisprozeß durchmacht. Dies kann z. B. dadurch geschehen, daß man sie zuerst auf dem Wege α in den Zustand 2, und dann auf dem Wege β wieder in den Zustand 1 überführt. Dann ist, wie überhaupt bei Kreisprozessen, nach (18), § 65:

$$Q = -A.$$

Die gesamte äußere Arbeit ist:

$$A = -\int_1^1 p\,dV,$$

wobei das Integral über die geschlossene Kurve $1\,\alpha\,2\,\beta\,1$ zu erstrecken ist. A stellt offenbar zugleich den Inhalt des von dieser Kurve umschlossenen Flächenstücks vor, positiv, wenn der Kreisprozeß in der durch den Pfeil Fig. 2 angegebenen Richtung vor sich geht.

§ 79. Im folgenden wollen wir uns näher mit dem speziellen Fall beschäftigen, daß die für die Zustandsänderung charakteristische Kurve α in ein Element zusammenschrumpft und somit die Punkte 1 und 2 sich unendlich nahe liegen. Dann erhält A den Wert $-p\,dV$, die Energieänderung den Wert dU, und infolgedessen die von außen zugeführte Wärme nach (21) den Wert[1]

$$Q = dU + p\,dV.$$

Auf die Masseneinheit der Substanz bezogen lautet diese Gleichung

$$q = du + p\,dv, \tag{22}$$

wenn die Quotienten von Q, U und V durch die Masse M mit den entsprechenden kleinen Buchstaben bezeichnet werden. Für die folgenden Rechnungen empfiehlt es sich öfter, die Temperatur T als unabhängige

[1]Nach Clausius' Vorgang wird dieser Ausdruck gewöhnlich, um seine unendliche Kleinheit anzudeuten, mit dQ bezeichnet. Dies hat jedoch nicht selten zu dem Mißverständnis Anlaß gegeben, als ob die zugeleitete Wärme das Differential einer bestimmten endlichen Größe Q wäre. Der hierdurch nahe gelegte Trugschluß sei durch folgende kleine Rechnung illustriert. Wählt man T und V als unabhängige Variable, so ist

$$dQ = dU + p\,dV = \left(\frac{\partial U}{\partial T}\right)_V dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] dV.$$

Andrerseits ist:

$$dQ = \left(\frac{\partial Q}{\partial T}\right)_V dT + \left(\frac{\partial Q}{\partial V}\right)_T dV.$$

Folglich, da dT und dV voneinander unabhängig sind:

$$\frac{\partial Q}{\partial T} = \frac{\partial U}{\partial T} \quad \text{und} \quad \frac{\partial Q}{\partial V} = \frac{\partial U}{\partial V} + p$$

und daraus durch Differentiation der ersten Gleichung nach V, der zweiten nach T:

$$\frac{\partial^2 Q}{\partial T\,\partial V} = \frac{\partial^2 U}{\partial T\,\partial V} = \frac{\partial^2 U}{\partial T\,\partial V} + \frac{\partial p}{\partial T},$$

also $\dfrac{\partial p}{\partial T} = 0$, was sicher unrichtig ist.

Derartige Fehler sind ausgeschlossen, wenn wir bei der Bezeichnung im Text stehen bleiben, in der Erwägung, daß nicht jede unendlich kleine Größe die Differenz zweier (nahezu gleicher) endlicher Größen ist.

Variable zu benutzen, entweder neben v oder neben p. Wir werden die unabhängigen Variabeln jedesmal im Interesse der möglichsten Einfachheit wählen und überall da, wo Verwechslungen möglich sind, den Sinn der Differentiation besonders markieren.

Nun wenden wir die letzte Gleichung auf die wichtigsten umkehrbaren Vorgänge an.

§ 80. Wie schon wiederholt angeführt wurde, kann man die spezifische Wärme einer Substanz in ganz verschiedener Weise definieren, je nach der Art, in welcher man sich die Erwärmung vorgenommen denkt. In jedem Falle hat man für die spezifische Wärme nach § 46 und nach (22):

$$c = \frac{q}{dT} = \frac{du}{dT} + p\frac{dv}{dT}. \tag{23}$$

Damit nun die Differentialquotienten einen bestimmten Sinn haben, ist noch eine willkürlich festzusetzende Bedingung erforderlich, welche die Richtung der vorgenommenen Veränderungen anzeigt. Natürlich genügt eine einzige Bedingung, da der Zustand der Substanz nur von zwei Variabeln abhängt.

§ 81. Erwärmung bei konstantem Volumen. Hierfür ist:

$$dv = 0, \qquad c = c_v,$$

die spezifische Wärme bei konstantem Volumen. Also nach Gleichung (23)

$$c_v = \left(\frac{\partial u}{\partial T}\right)_v \tag{24}$$

oder auch:

$$c_v = \left(\frac{\partial u}{\partial p}\right)_v \cdot \left(\frac{\partial p}{\partial T}\right)_v. \tag{25}$$

§ 82. Erwärmung bei konstantem Druck. Hierfür ist:

$$dp = 0, \quad c = c_p,$$

die spezifische Wärme bei konstantem Druck. Also nach Gleichung (23):

$$c_p = \left(\frac{\partial u}{\partial T}\right)_p + p\left(\frac{\partial v}{\partial T}\right)_p \tag{26}$$

oder auch:

$$c_p = \left[\left(\frac{\partial u}{\partial v}\right)_p + p\right] \cdot \left(\frac{\partial v}{\partial T}\right)_p. \tag{27}$$

Durch Substitution von:

$$\left(\frac{\partial u}{\partial T}\right)_p = \left(\frac{\partial u}{\partial T}\right)_v + \left(\frac{\partial u}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_p$$

in (26) kann man c_p auch in der Form schreiben:

$$c_p = \left(\frac{\partial u}{\partial T}\right)_v + \left[\left(\frac{\partial u}{\partial v}\right)_T + p\right] \left(\frac{\partial v}{\partial T}\right)_p$$

oder mit Berücksichtigung von (24)

$$c_p = c_v + \left[\left(\frac{\partial u}{\partial v}\right)_T + p\right] \left(\frac{\partial v}{\partial T}\right)_p . \tag{28}$$

§ 83. Die Vergleichung von (25) und (27) gestattet durch Elimination von u eine direkte Prüfung der Theorie an der Erfahrung. Es ist nämlich aus (25):

$$\left(\frac{\partial u}{\partial p}\right)_v = c_v \left(\frac{\partial T}{\partial p}\right)_v$$

andrerseits aus (27)

$$\left(\frac{\partial u}{\partial v}\right)_p = c_p \left(\frac{\partial T}{\partial v}\right)_p - p.$$

Folglich durch Differentiation des ersten Ausdrucks nach v bei konstantem p und des zweiten Ausdrucks nach p bei konstantem v und Gleichsetzung beider Werte:

$$\frac{\partial}{\partial v}\left(c_v \frac{\partial T}{\partial p}\right) = \frac{\partial}{\partial p}\left(c_p \frac{\partial T}{\partial v} - p\right)$$

oder:

$$(c_p - c_v) \frac{\partial^2 T}{\partial p \, \partial v} + \frac{\partial c_p}{\partial p} \frac{\partial T}{\partial v} - \frac{\partial c_v}{\partial v} \frac{\partial T}{\partial p} = 1. \tag{29}$$

Diese Gleichung enthält nur Größen, die der Beobachtung zugänglich sind und liefert daher ein Mittel zur Prüfung des ersten Hauptsatzes der Wärmetheorie an einer beliebigen homogenen Substanz durch Messungen.

§ 84. Ideale Gase. Die obigen Gleichungen erfahren für ideale Gase beträchtliche Vereinfachungen. Zunächst ist hierfür nach (14)

$$p = \frac{R}{m} \frac{T}{v}, \tag{30}$$

wobei $R = 8,315 \cdot 10^7$ und m gleich dem (wirklichen oder mittleren) Molekulargewicht des Gases. Daher wird:

$$T = \frac{m}{R}\, pv$$

und die Gleichung (29) geht über in:

$$c_p - c_v + p\,\frac{\partial c_p}{\partial p} - v\,\frac{\partial c_v}{\partial v} = \frac{R}{m}.$$

Mehr läßt sich für ein ideales Gas, wenn nur das BOYLE-GAY LUSSAC-AVOGADROsche Gesetz als gültig vorausgesetzt wird, aus dem ersten Hauptsatz allein nicht schließen.

§ 85. Nun wollen wir die weitere, durch die im § 70 beschriebenen Versuche von THOMSON und JOULE festgestellte Eigenschaft idealer Gase benutzen, daß ihre innere Energie nur von der Temperatur, nicht vom Volumen abhängt, also nach (19), auf die Masseneinheit bezogen:

$$\left(\frac{\partial u}{\partial v}\right)_T = 0. \tag{31}$$

Dann geht die allgemeine Gleichung:

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT + \left(\frac{\partial u}{\partial v}\right)_T dv$$

für ein ideales Gas über in:

$$du = \left(\frac{\partial u}{\partial T}\right)_v dT$$

und nach (24):

$$du = c_v\, dT. \tag{32}$$

Ferner folgt dann aus (28):

$$c_p = c_v + p\left(\frac{\partial v}{\partial T}\right)_p$$

oder mit Berücksichtigung von (30):

$$c_p = c_v + \frac{R}{m},$$

d. h. die Differenz der spezifischen Wärmen eines idealen Gases bei konstantem Druck und bei konstantem Volumen ist konstant. Bezieht man die Wärmekapazität nicht auf die Masseneinheit, sondern auf das Molekulargewicht m des Gases (vgl. § 48), so ist

$$mc_p - mc_v = R, \tag{33}$$

also die Differenz sogar unabhängig von der Natur des Gases.

§ 86. Bequem messen läßt sich nur die spezifische Wärme bei konstantem Druck, weil eine in einem geschlossenen Gefäß von konstantem Volumen gehaltene Gasmenge eine zu kleine Wärmekapazität besitzt, um gegenüber den äußeren Körpern, zunächst den Gefäßwänden, hinlänglich beträchtliche thermische Wirkungen hervorzubringen. Da nun c_v nach (24) ebenso wie u nur von der Temperatur, nicht vom Volumen abhängt, so folgt aus (33) dasselbe für c_p. Dieser Schluß ist zuerst durch die Messungen von REGNAULT bestätigt worden, welcher überdies fand, daß c_p auch innerhalb eines ziemlich weiten Temperaturintervalls konstant ist. Nach (33) ist also auch c_v in demselben Bereich konstant. Wir dürfen und wollen daher ganz allgemein die Definition des idealen Gaszustandes dahin ergänzen, daß c_p und c_v vollständig unabhängig sind von Temperatur und Volumen.

Wenn man die Molekularwärmen in Kalorien ausdrückt, so ist natürlich auch die Größe R noch durch das mechanische Wärmeäquivalent a (§ 61) zu dividieren, und man hat als Differenz der Molekularwärme bei konstantem Druck und der bei konstantem Volumen:

$$\frac{R}{a} = \frac{8,315 \cdot 10^7}{4,19 \cdot 10^7} = 1,985. \tag{34}$$

§ 87. Folgende Tabelle enthält für einige Gase die direkt gemessene spezifische Wärme und die Molekularwärme bei konstantem Druck, sowie die aus Gleichung (33) durch Subtraktion von 1,985 berechnete Molekularwärme bei konstantem Volumen, endlich das Verhältnis beider Größen: $\frac{c_p}{c_v} = \gamma$.

	Spezifische Wärme bei konst. Druck	Molekulargewicht	Molekularwärme bei konst. Druck	Molekularwärme bei konst. Volumen	Verhältnis $\gamma = \frac{c_p}{c_v}$
Wasserstoff	3,410	2,016	6,87	4,88	1,41
Sauerstoff	0,220	32	7,04	5,05	1,40
Stickstoff	0,2438	28	6,83	4,85	1,41
Luft	0,2404	28,9	6,95	4,96	1,40

Bei bedeutender Temperatursteigerung nimmt die spezifische Wärme aller dieser Gase langsam zu. Innerhalb des Temperaturbereichs, in welchem die spezifische Wärme konstant ist, läßt sich die Gleichung (32) integrieren und liefert:

$$u = c_v T + \text{konst.}, \tag{35}$$

wobei die Integrationskonstante von der Wahl des Nullzustandes für die Energie abhängt. Für den idealen Gaszustand ist nach unserer in § 86 getroffenen Festsetzung c_p und c_v durchaus konstant und daher die letzte Gleichung allgemein gültig.

§ 88. Adiabatischer Vorgang. Hierfür ist charakteristisch $q = 0$, und nach Gleichung (22):

$$0 = du + p\,dv.$$

Setzen wir wieder ein ideales Gas voraus, so ergibt die Einsetzung der Werte von du aus (32) und von p aus (30):

$$0 = c_v\,dT + \frac{R}{m}\frac{T}{v}\,dv \tag{36}$$

oder integriert:

$$c_v \log T + \frac{R}{m}\log v = \text{konst.}$$

Ersetzt man hierin $\dfrac{R}{m}$ nach (33) durch $c_p - c_v$ und dividiert durch c_v, so kommt:

$$\log T + (\gamma - 1)\log v = \text{konst.} \tag{37}$$

(d. h. bei adiabatischer Ausdehnung sinkt die Temperatur)

oder, wenn man bedenkt, daß nach der Zustandsgleichung (30):

$$\log p + \log v - \log T = \text{konst.}$$

mit Elimination von v:

$$-\gamma \log T + (\gamma - 1)\log p = \text{konst.}$$

(d. h. bei adiabatischer Druckvermehrung steigt die Temperatur)

oder mit Elimination von T:

$$\log p + \gamma \log v = \text{konst.}$$

Die Werte der Integrationskonstanten ergeben sich aus dem Anfangszustand des Prozesses.

Vergleicht man die letzte Gleichung in der Form:

$$p \cdot v^{\gamma} = \text{konst.} \tag{38}$$

mit dem BOYLEschen Gesetz: $pv = \text{konst.}$, so erkennt man, daß bei adiabatischer Kompression das Volumen langsamer mit wachsendem Druck abnimmt als bei isothermer Kompression, entsprechend der gleichzeitig eintretenden Temperaturerhöhung. Die adiabatischen Kurven in der pv-Ebene (§ 22) verlaufen daher steiler als die hyperbelförmigen Isothermen.

§ 89. Die adiabatischen Vorgänge können in verschiedener Weise zur Messung des Verhältnisses γ der spezifischen Wärmen benutzt werden und liefern durch die Übereinstimmung der Resultate mit dem oben aus dem mechanischen Wärmeäquivalent berechneten Werte von γ eine wichtige Bestätigung der Theorie.

So kann z. B. die Messung der Schallgeschwindigkeit in einem Gase zur Berechnung von γ benutzt werden. Wie in der Hydrodynamik gezeigt wird, ist dieselbe in irgendeiner Flüssigkeit: $\sqrt{\frac{dp}{dk}}$, wenn $k = \frac{1}{v}$ die Dichte der Flüssigkeit bedeutet. Da nun wegen der geringen Wärmeleitungsfähigkeit der Gase die mit einer Schallbewegung verbundenen Kompressionen und Dilatationen nicht isotherm, sondern adiabatisch erfolgen, so hängt bei einem idealen Gase der Druck p von der Dichte k nicht nach dem BOYLEschen Gesetz $pv = \text{konst.}$, sondern nach der Gleichung (38) ab; also:

$$\frac{p}{k^{\gamma}} = \text{konst.},$$

woraus durch Differentiation:

$$\frac{dp}{dk} = \frac{\gamma p}{k} = \gamma p v$$

oder nach (30)

$$\frac{dp}{dk} = \gamma \frac{R}{m} T, \qquad \gamma = \frac{m}{RT} \cdot \frac{dp}{dk}.$$

Nun beträgt z. B. in atmosphärischer Luft bei 0° die Schallgeschwindigkeit: $\sqrt{\frac{dp}{dk}} = 33170 \frac{\text{cm}}{\text{sec}}$, also ist für Luft mit Hilfe der Zahlenwerte für m (§ 41), R (§ 84) und T nach der letzten Gleichung:

$$\gamma = \frac{28,9}{8,315 \cdot 10^7} \cdot \frac{33170^2}{273} = 1,40,$$

in Übereinstimmung mit dem in § 87 berechneten Wert. Natürlich kann man auch umgekehrt den aus der Schallgeschwindigkeit berechneten Wert von $\gamma = \frac{c_p}{c_v}$ dazu benutzen, um c_v in Kalorien und dann aus (33) das mechanische Wärmeäquivalent zu berechnen. Dieser Weg ist zur erstmaligen ziffernmäßigen Auswertung des Wärmeäquivalents eingeschlagen worden von Robert Mayer im Jahre 1842. Allerdings gehört wesentlich dazu die in Gleichung (31) ausgedrückte Voraussetzung, daß die innere Energie der Luft nur von der Temperatur abhängt, oder mit anderen Worten, daß die Differenz der spezifischen Wärmen bei konstantem Druck und bei konstantem Volumen lediglich durch die äußere Arbeit bedingt ist — ein Satz, der erst seit den § 70 beschriebenen Versuchen von Thomson und Joule als direkt bewiesen angesehen werden darf.

§ 90. Wir wenden uns jetzt zur Betrachtung eines zusammengesetzteren Prozesses, und zwar eines umkehrbaren Kreisprozesses von besonderer Art, der wegen seiner Bedeutung für die Technik in der Entwicklung der Thermodynamik eine wichtige Rolle gespielt hat: des sogenannten Carnotschen Kreisprozesses, um auch auf ihn den ersten Hauptsatz im einzelnen anzuwenden.

Von einem gewissen Anfangszustand, welcher durch die Werte T_1 und v_1 charakterisiert sein möge, ausgehend, werde die Substanz von der Masse 1 erstens adiabatisch komprimiert, bis die Temperatur auf $T_2 > T_1$ gestiegen und das Volumen auf $v_2 < v_1$ vermindert ist (Fig. 3). Hierauf lasse man zweitens die Substanz sich wieder ausdehnen, aber nun isotherm, indem sie in Verbindung mit einem Wärmebehälter von der konstanten Temperatur T_2 gehalten wird, welcher die Ausdehnungswärme Q_2 hergibt; dabei möge das Volumen bis v_2' wachsen. Drittens werde die Substanz noch weiter ausgedehnt, und zwar jetzt adiabatisch, so lange, bis die Temperatur wieder auf T_1 gesunken ist; dann habe das Volumen bis v_1' zugenommen. Endlich viertens werde auch das Volumen durch isotherme Kompression mit Benutzung eines Wärmebehälters von der konstanten Temperatur T_1, welcher die Kompressionswärme aufnimmt, wieder auf den Anfangswert v_1 zurückgebracht, — alles auf umkehrbarem Wege, wie es § 71 ff. beschrieben wurde.

Nach dem ersten Hauptsatz ist für diesen Kreisprozeß (§ 65) die Summe der dem System von außen zugeführten Wärme und der von außen aufgewendeten Arbeit:

$$Q + A = 0. \tag{39}$$

Die der Substanz im ganzen zugeführte Wärme beträgt hier:

$$Q = Q_1 + Q_2, \tag{40}$$

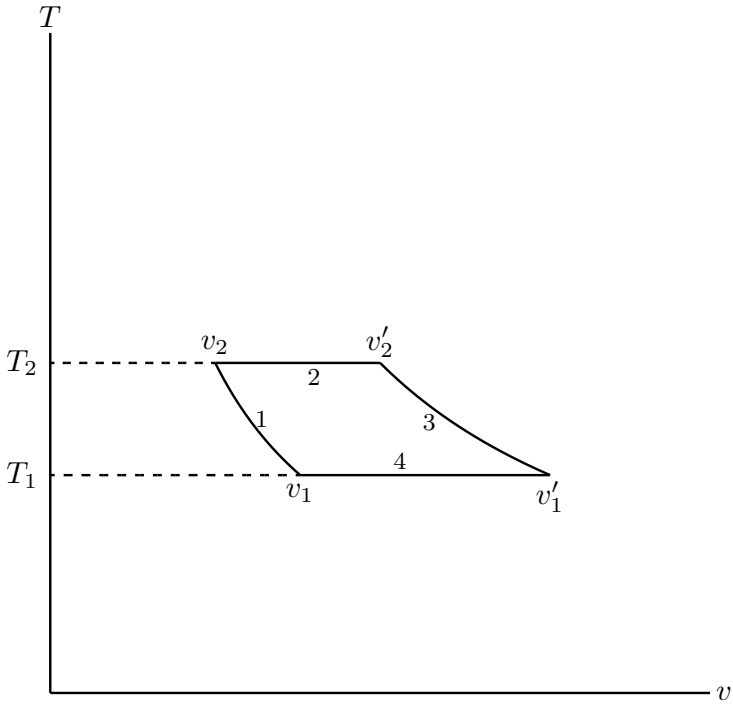


Fig. 3.

wobei Q_1 und Q_2 die von den beiden Wärmebehältern abgegebenen Wärmemengen bezeichnen (Q_1 ist hier negativ). Die äußere Arbeit A läßt sich aus der adiabatischen und aus der isothermen Kompressibilität der angewandten Substanz berechnen: dieselbe beträgt nach (20):

$$A = -\int_{v_1 T_1}^{v_2 T_2} p\,dv - \int_{v_2 T_2}^{v_2' T_2} p\,dv - \int_{v_2' T_2}^{v_1' T_1} p\,dv - \int_{v_1' T_1}^{v_1 T_1} p\,dv,$$

wobei die erste und die dritte Integration auf adiabatischem Wege, die zweite und die vierte auf isothermem Wege zu erfolgen hat.

Setzen wir von nun an ein ideales Gas voraus, so lassen sich die vier Integrale leicht berechnen. Es wird nämlich durch Berücksichtigung von (30) und (36):

$$A = \int_{T_1}^{T_2} c_v\,dT - \frac{R}{m}\int_{v_2}^{v_2'} \frac{T_2}{v}\,dv + \int_{T_2}^{T_2} c_v\,dT - \frac{R}{m}\int_{v_1'}^{v_1} \frac{T_1}{v}\,dv. \tag{41}$$

Die Arbeit bei der adiabatischen Kompression im ersten Teil des Prozesses ist also gerade gleich und entgegengesetzt der Arbeit bei der adiabatischen Ausdehnung im dritten Teil des Prozesses und hebt sich mit dieser fort. Es bleiben übrig die isothermen Arbeiten:

$$A = -\frac{R}{m}\left(T_2 \log \frac{v_2'}{v_2} + T_1 \log \frac{v_1}{v_1'}\right).$$

Nun ist aber der Zustand $v_2 T_2$ aus dem Zustand $v_1 T_1$ durch einen adiabatischen Vorgang entstanden, also ist nach (37):

$$\log T_2 + (\gamma - 1) \log v_2 = \log T_1 + (\gamma - 1) \log v_1$$

und ebenso ist für den anderen adiabatischen Vorgang, der von $v_2' T_2$ bis $v_1' T_1$ führt:

$$\log T_2 + (\gamma - 1) \log v_2' = \log T_1 + (\gamma - 1) \log v_1'.$$

Aus diesen beiden Gleichungen folgt unmittelbar:

$$\frac{v_2'}{v_2} = \frac{v_1'}{v_1}$$

und daher:

$$A = -\frac{R}{m}(T_2 - T_1) \log \frac{v_1'}{v_1}.$$

Da in unserem Falle $T_2 > T_1$ und $\frac{v_1'}{v_1} = \frac{v_2'}{v_2} > 1$, so ist hier die gesamte von außen auf das Gas ausgeübte Arbeit A negativ, d. h. es ist äußere Arbeit im ganzen nicht aufgewendet, sondern gewonnen worden. Dagegen ist infolge von (39) und (40)

$$Q = Q_1 + Q_2 = -A \tag{42}$$

positiv, d. h. der Wärmebehälter von der Temperatur T_2 hat mehr Wärme abgegeben, als der Wärmebehälter von der Temperatur T_1 aufgenommen hat.

Der Wert von A in die letzte Gleichung eingesetzt ergibt:

$$Q = Q_1 + Q_2 = \frac{R}{m}(T_2 - T_1) \log \frac{v_1'}{v_1}. \tag{43}$$

Die Richtigkeit dieser Gleichung ergibt sich auch aus der direkten Berechnung der Größen Q_1 und Q_2 einzeln. Während nämlich der Wärmebehälter T_2 in Funktion ist, dehnt sich das Gas isotherm aus, seine

Energie bleibt also ungeändert, und die von außen zugeführte Wärme ist der äußeren Arbeit gleich und entgegengesetzt. Daher hat man durch Vergleich mit dem zweiten Integral in (41):

$$Q_2 = \frac{R}{m} T_2 \log \frac{v_2'}{v_2} = \frac{R}{m} T_2 \log \frac{v_1'}{v_1}$$

und ebenso durch Vergleich mit dem vierten Integral in (41):

$$Q_1 = \frac{R}{m} T_1 \log \frac{v_1}{v_1'} = -\frac{R}{m} T_1 \log \frac{v_1'}{v_1},$$

in Übereinstimmung mit der Gleichung (43).

Es besteht also zwischen den Größen Q_1, Q_2, A außer der Relation (42) noch die andere:

$$Q_1 : Q_2 : A = (-T_1) : T_2 : (T_1 - T_2). \tag{44}$$

§ 91. Der soeben besprochene CARNOTsche Kreisprozeß, beliebig oft wiederholt, bildet den Typus einer periodisch wirkenden Maschine, durch welche fortwährend Wärme in Arbeit verwandelt wird. Um dies im einzelnen übersehen zu können, wollen wir nun genauer das Resultat aller Wirkungen ins Auge fassen, welche bei der Ausführung eines solchen Kreisprozesses in der Natur auftreten. Zu dem Zweck vergleichen wir den durch den Prozeß schließlich hervorgerufenen Zustand mit dem am Anfang des Prozesses herrschenden Zustand. Das Gas selbst ist durch den Prozeß schließlich gar nicht verändert worden, es hat gewissermaßen nur als Zwischenträger gedient, um anderweitige Änderungen zu vermitteln, wir können dasselbe also bei der Vergleichung des Endzustandes mit dem Anfangszustand ganz außer Betracht lassen. Dagegen haben die beiden Wärmebehälter ihren Zustand geändert, und außerdem ist eine gewisse positive äußere Arbeit $A' = -A$ gewonnen worden, d. h. es befinden sich am Schluß des Prozesses etwa gewisse Gewichte, die bei der Kompression und bei der Ausdehnung verwendet wurden, auf einer größeren Höhe als am Anfang, oder es ist eine elastische Feder, die zu dem gleichen Zweck diente, am Schluß stärker gespannt als sie es am Anfang war. Andrerseits hat der Wärmebehälter T_2 die Wärmemenge Q_2 abgegeben, der kältere Wärmebehälter T_1 die kleinere Wärmemenge $Q_1' = -Q_1$ empfangen, und die verschwundene Wärme ist äquivalent der gewonnenen Arbeit. Man kann das kurz so ausdrücken, daß man sagt: Die Wärmemenge Q_2 von der Temperatur T_2 ist zum einen Teil (Q_1') zur tieferen Temperatur T_1 übergegangen, zum andern Teil $(Q_2 - Q_1' = Q_2 + Q_1)$ in Arbeit verwandelt worden. Man hat also in dem

CARNOTschen Kreisprozeß, ausgeführt mit einem idealen Gase, ein Mittel, um einem Körper Wärme zu entziehen und dafür Arbeit zu gewinnen, ohne daß irgendeine andere Veränderung in der Natur zurückbleibt, als daß außerdem eine gewisse andere Wärmemenge aus einem Körper von höherer Temperatur zu einem Körper von tieferer Temperatur übergeht.

Da der beschriebene Prozeß aber umkehrbar ist, so kann man ihn auch in der Weise realisieren, daß bei unveränderten Temperaturen und Volumina die Größen Q_1, Q_2, A ihr Vorzeichen ändern. Dann ist Q_1 und A positiv, $Q_2 = -Q_2'$ negativ, d. h. der wärmere Behälter T_2 empfängt die Wärme Q_2', und zwar zum Teil (Q_1) aus dem kälteren Behälter T_1, zum Teil aus aufgewendeter Arbeit (A). Man hat also in dem umgekehrt ausgeführten CARNOTschen Prozeß ein Mittel, um Wärme aus einem kälteren in einen wärmeren Körper zu schaffen, ohne daß irgendeine andere Veränderung in der Natur zurückbleibt, als daß außerdem eine gewisse Arbeit in Wärme verwandelt wird. Wir werden später sehen, daß für den Erfolg des CARNOTschen umkehrbaren Kreisprozesses die Natur des Zwischenträgers prinzipiell unwesentlich ist, daß also ein ideales Gas darin von keiner anderen Substanz übertroffen oder unterboten wird (vgl. § 137).

Drittes Kapitel. Anwendungen auf nichthomogene Systeme.

§ 92. Ein großer Teil der im vorigen Kapitel besprochenen Sätze ist ohne weiteres auch auf den Fall anwendbar, daß die dort behandelte Substanz im Innern nicht vollständig homogen ist, und insofern kann für eine Reihe von allgemeinen Fragen auf die dortigen Ausführungen verwiesen werden. Hier werden vorwiegend nur diejenigen Erscheinungen Gegenstand der Untersuchung sein, welche für die Inhomogenität eines Systems charakteristisch sind.

Wir betrachten im folgenden ein System, welches aus einer Anzahl nebeneinander gelagerter, durch bestimmte Trennungsflächen geschiedener homogener (§ 67) Körper zusammengesetzt ist. Ein solches System kann chemisch homogen sein oder nicht. Den ersten Fall haben wir unter Umständen bei einer Flüssigkeit in Berührung mit ihrem Dampf, insofern die Flüssigkeit aus ebensolchen Molekülen bestehen kann wie der Dampf. Den zweiten Fall haben wir immer dann, wenn eine Substanz mit einer chemisch differenten in Berührung ist. Ob ein System physikalisch homogen ist oder nicht, läßt sich durch Aufsuchung etwaiger Trennungsflächen innerhalb des Systems auf optischem Wege, eventuell auch durch andere Mittel, z. B. bei Emulsionen durch Messung der Dampfspannung oder des Gefrierpunktes

(§ 223), in den meisten Fällen zu unzweifelhafter Entscheidung bringen, viel schwieriger ist häufig die Beantwortung der Frage, ob ein System chemisch homogen ist, d. h. aus lauter gleichartigen Molekülen besteht. Daher legen wir auch die erstere und nicht die letztere Einteilung unserer Untersuchung zugrunde.

§ 93. Ein Charakteristikum der Vorgänge in nichthomogenen Systemen sind die im allgemeinen beträchtlichen dabei eintretenden Temperaturänderungen, z. B. beim Verdampfen oder beim Oxydieren. Die Aufrechterhaltung der Anfangstemperatur und des Anfangsdrucks erfordert dann einen beträchtlichen Wärmeaustausch mit der Umgebung und eine entsprechende äußere Arbeit. Ersterer ist aber in der Regel viel bedeutender als letztere, die bei den meisten chemischen Vorgängen ganz vernachlässigt werden kann. Daher mißt man in der Thermochemie die äußeren Wirkungen:

$$Q + A = U_2 - U_1 \tag{45}$$

gewöhnlich in Kalorien (kalorisches Äquivalent der äußeren Wirkungen). Die äußere Arbeit A erscheint darin nur als ein untergeordnetes Glied. Da ferner die meisten chemischen Vorgänge mit Temperaturerhöhung, also, wenn die Anfangstemperatur wiederhergestellt wird, mit Wärmeabgabe nach außen verlaufen (exothermische Vorgänge), so bezeichnet man in der Thermochemie die behufs Wiederherstellung der Anfangstemperatur nach außen abzugebende Wärmemenge als „positive Wärmetönung“ des Prozesses. In unseren Rechnungen werden wir daher für einen Prozeß mit positiver Wärmetönung (z. B. Verbrennung) die von außen zugeführte Wärme Q negativ, für einen mit negativer Wärmetönung (Verdampfung, Schmelzung, Dissoziation) diese Wärme Q positiv zu nehmen haben.

§ 94. Um die Gleichung (45) thermochemisch zu verwerten, ist es zweckmäßig, zur Bezeichnung der Energie U eines Systems in einem bestimmten Zustand ein Symbol zu benutzen, welches die chemische Natur des Systems unmittelbar erkennen läßt. Ein solches Symbol hat J. Thomsen angegeben, indem er die Formel für das Atomgewicht oder Molekulargewicht einer Substanz in Klammern setzt und dadurch die Energie der entsprechenden Gewichtsmenge der Substanz, bezogen auf einen beliebigen Nullzustand, ausdrückt. Durch W. Ostwald ist dann diese Bezeichnung zur allgemeineren Benutzung gelangt. So bezeichnen [Pb], [S], [PbS] die Energien eines Atoms Blei, Schwefel und eines Moleküls Schwefelblei. Um nun auszudrücken, daß die Bildung eines Moleküls Schwefelblei aus einem Atom Blei und einem Atom Schwefel mit einer Wärmetönung von 18 400 cal verbunden ist, während dagegen die äußere Arbeit zu vernachlässigen ist (§ 98), hat man

zu setzen:

$$U_1 = [\mathrm{Pb}] + [\mathrm{S}]$$
$$U_2 = [\mathrm{PbS}]$$
$$A = 0, \qquad Q = -18\,400 \text{ cal}$$

und die Gleichung (45) wird:

$$-18\,400 \text{ cal} = [\mathrm{PbS}] - [\mathrm{Pb}] - [\mathrm{S}]$$

oder in der üblichen Schreibweise:

$$[\mathrm{Pb}] + [\mathrm{S}] - [\mathrm{PbS}] = 18\,400 \text{ cal},$$

d. h. die Energie von Blei und Schwefel in getrenntem Zustand ist um 18 400 cal größer als im chemisch verbundenen Zustand bei gleicher Temperatur. Durch die Benutzung der Formeln für das Verbindungsgewicht hat man zugleich auch eine Kontrolle dafür, daß die beiden verglichenen Energien sich auf das nämliche materielle System beziehen. Einfacher noch würde die Gleichung lauten, wenn man den getrennten Zustand der Elemente Pb und S zum Nullzustand wählt. Denn dann wird (§ 64) $[\mathrm{Pb}] = 0$ und $[\mathrm{S}] = 0$ und man hat kürzer:

$$[\mathrm{PbS}] = -18\,400 \text{ cal}.$$

§ 95. Zur genauen Definition des Zustandes einer Substanz, und damit ihrer Energie, ist aber außer der chemischen Natur und der Gewichtsmenge zunächst noch die Angabe der Temperatur und des Druckes erforderlich. Erstere wird, falls keine besondere Bemerkung darüber gemacht ist, wie auch schon in dem obigen Beispiel geschehen, als die mittlere Zimmertemperatur, also etwa 18° C., angenommen. Ebenso wird der Druck als Atmosphärendruck vorausgesetzt; derselbe hat übrigens, bei gegebener Temperatur, nur wenig Einfluß auf die Energie, bei idealen Gasen überhaupt keinen [Gleichung (35)].

Ferner bedarf es noch der Angabe des Aggregatzustandes. Man kann denselben, falls Verwechslungen zu befürchten sind, dadurch bezeichnen, daß man für den festen Zustand eckige Klammern wie oben, für den flüssigen runde, und für den gasförmigen geschweifte Klammern anwendet. So bedeutet $[\mathrm{H_2O}]$, $(\mathrm{H_2O})$, $\{\mathrm{H_2O}\}$ die Energie eines Moleküls Wasser als Eis, als Flüssigkeit, und als Dampf. Daher ist für das Schmelzen bei 0°:

$$(\mathrm{H_2O}) - [\mathrm{H_2O}] = 80 \times 18 = 1440 \text{ cal}.$$

Endlich ist es manchmal wünschenswert, z. B. bei fester Kohle, Schwefel, Arsen, oder bei isomeren Verbindungen, auch noch eine Angabe über die

spezielle Modifikation der Substanz hinzuzufügen. Das kann jedesmal in besonderer Weise geschehen.

Mit diesen Symbolen läßt sich nun wie mit bestimmten Größen rechnen, und dadurch manche Betrachtung wesentlich abkürzen, die sonst nur durch mehr oder minder verwickelte Überlegungen durchzuführen wäre. Vgl. hierzu die Beispiele weiter unten.

§ 96. Zur Bezeichnung der Energie einer Lösung oder Mischung mehrerer Verbindungen kann man die Formeln für die Molekulargewichte mit den entsprechenden Molekülzahlen direkt nebeneinander schreiben. So bedeutet:

$$(H_2SO_4) + 5(H_2O) - (H_2SO_4 \cdot 5H_2O) = 13\,100 \text{ cal},$$

daß beim Auflösen eines Moleküls Schwefelsäurehydrat in 5 Molekülen Wasser die Wärme 13 100 cal frei wird. Ähnlich gibt die Gleichung:

$$(H_2SO_4) + 10(H_2O) - (H_2SO_4 \cdot 10H_2O) = 15\,100 \text{ cal}$$

die Wärmetönung beim Auflösen in 10 Molekülen Wasser. Durch Subtraktion der beiden Gleichungen erhält man daraus:

$$(H_2SO_4 \cdot 5H_2O) + 5(H_2O) - (H_2SO_4 \cdot 10H_2O) = 2000 \text{ cal},$$

d. h. die Verdünnung einer Lösung von 1 Molekül Schwefelsäurehydrat in 5 Molekülen Wasser mit weiteren 5 Molekülen Wasser ergibt eine Wärmetönung von 2000 cal.

§ 97. Erfahrungsgemäß ruft bei sehr verdünnten Lösungen eine weitere Verdünnung keine merkliche Wärmetönung mehr hervor. Daher ist es zur Bezeichnung der Energie einer sehr verdünnten Lösung häufig gar nicht nötig, die Zahl der Moleküle des Lösungsmittels besonders anzugeben, und man schreibt kurz:

$$(H_2SO_4) + (aq) - (H_2SO_4aq) = 17\,900 \text{ cal},$$

um die Wärmetönung auszudrücken, welche bei unendlicher Verdünnung eines Moleküls Schwefelsäurehydrat mit Wasser auftritt. Hierbei bedeutet das Zeichen aq jede beliebige Wassermenge, die zur praktischen Herstellung einer unendlich verdünnten Lösung genügt.

§ 98. Das kalorische Äquivalent der äußeren Arbeit A (§ 93) ist bei chemischen Prozessen, in denen nur feste und flüssige Körper vorkommen, wegen der geringen Volumveränderungen gegen die Wärmetönung in der Regel zu vernachlässigen. Dann ergibt also die Wärmetönung allein die Energieänderung des Systems:

$$U_2 - U_1 = Q$$

und ist infolgedessen nur vom Anfangs- und Endzustand, nicht aber von dem sonstigen Verlauf des Prozesses abhängig. Anders ist es im allgemeinen, wenn gasförmige Körper an der Reaktion beteiligt sind. Nur bei den Verbrennungsprozessen in der „kalorimetrischen Bombe“, welche durch die Forschungen besonders von BERTHELOT und von STOHMANN weitgehende Anwendungen erfahren hat, bleibt das Volumen konstant und daher die äußere Arbeit $= 0$. Auch hier entspricht also die gemessene Wärmetönung direkt der eingetretenen Energiedifferenz. Aber in anderen Fällen kann bei der Mitwirkung von Gasen die äußere Arbeit A einen merklichen Betrag annehmen; derselbe ist wesentlich auch durch den Verlauf des Prozesses bedingt. So kann man z. B. ein Gas sich ausdehnen lassen mit einer äußeren Arbeitsleistung, die innerhalb gewisser Grenzen jeden beliebigen Wert, bis Null herab, haben kann. Da nun die Energiedifferenz $U_2 - U_1$ nur vom Anfangszustand und vom Endzustand des Systems abhängt, so bedingt eine größere Arbeit, die das System bei Überwindung der äußeren Kräfte leistet, immer eine geringere Wärmetönung des Prozesses und umgekehrt, und um letztere zu finden, muß man außer den Energien auch noch die äußere Arbeit kennen. Hierzu bedarf es also der Angabe der äußeren Bedingungen, unter denen der Prozeß verläuft.

§ 99. Unter allen äußeren Bedingungen, die einen chemischen Prozeß begleiten können, ist die praktisch wichtigste diejenige, daß der Druck konstant bleibt: $p = p_0$ (gewöhnlich eine Atmosphäre). Dann ist die von außen her aufgewendete Arbeit A nach Gleichung (20)

$$A = -\int_1^2 p_0\, dV = p_0(V_1 - V_2), \tag{46}$$

also gleich dem Produkt des Druckes und der Volumenabnahme des Systems. Dies gibt nach (45):

$$U_2 - U_1 = Q + p_0(V_1 - V_2). \tag{47}$$

Die Volumenabnahme $V_1 - V_2$ des Systems kann man aber in der Regel, unter Vernachlässigung der Volumenänderungen fester und flüssiger Körper, gleich setzen der Volumenabnahme der gasförmigen Teile des Systems; also nach (16):

$$V_1 - V_2 = R\frac{T}{p_0}(n_1 - n_2),$$

wobei n_1 und n_2 die Anzahl der gasförmigen Moleküle des Systems vor und nach der Reaktion bedeuten. Daraus ergibt sich das kalorische Äquivalent

der äußeren Arbeit bei konstantem Druck nach (46) und (34)

$$\frac{A}{a} = \frac{p_0(V_1 - V_2)}{a} = \frac{R}{a}T(n_1 - n_2) = 1,985 \cdot T \cdot (n_1 - n_2) \text{ cal}$$

und die Wärmetönung eines Prozesses bei konstantem Druck:

$$-Q = U_1 - U_2 + 1,985 \cdot T \cdot (n_1 - n_2) \text{ cal}. \tag{48}$$

Wenn z. B. ein Molekulargewicht Wasserstoff und ein halbes Molekulargewicht Sauerstoff, beide von 18°, sich bei konstantem Druck zu flüssigem Wasser von 18° verbinden, so ist zu setzen:

$$\begin{aligned} U_1 &= \{H_2\} + \tfrac{1}{2}\{O_2\} \\ U_2 &= (H_2O) \\ n_1 = \tfrac{3}{2}, \qquad n_2 &= 0, \qquad T = 291, \end{aligned}$$

also die Verbrennungswärme nach (48):

$$-Q = \{H_2\} + \tfrac{1}{2}\{O_2\} - (H_2O) + 860 \text{ cal},$$

um 860 cal größer, als der Abnahme der Energie, d. h. der Verbrennung ohne äußere Arbeitsleistung entspricht.

§ 100. Schreibt man die Gleichung (47) in der Form:

$$(U + p_0V)_2 - (U + p_0V)_1 = Q,$$

so erkennt man, daß bei Prozessen, die unter konstantem Druck p_0 verlaufen, die Wärmetönung nur abhängt vom Anfangszustand und vom Endzustand, ebenso wie das beim gänzlichen Fortfall der äußeren Arbeit zutrifft. Aber hier ist die Wärmetönung nicht gleich der Differenz der Energien U, sondern gleich der Differenz der Werte, welche die Größe

$$U + pV = W \tag{49}$$

am Anfang und am Ende des Prozesses besitzt. Diese Größe W hat GIBBS daher die „Wärmefunktion bei konstantem Druck“ genannt, sie spielt für die Wärmetönung bei isobaren Prozessen ganz dieselbe Rolle, wie die Energie U für die Wärmetönung bei isochoren Prozessen.

Wenn es sich also um Prozesse bei konstantem Druck handelt, so ist es zweckmäßig, die Symbole $\{H_2\}$, (H_2O) usw. ein für allemal nicht auf die Energie U, sondern auf die Wärmefunktion W zu beziehen, deren Differenz

dann immer direkt die Wärmetönung ergibt. Diese Bezeichnung werden wir daher auch im folgenden anwenden.

§ 101. Um die Wärmetönung irgendeines unter konstantem Druck verlaufenden chemischen Prozesses zu berechnen, genügt es also, die Wärmefunktion W des an dem Prozeß beteiligten materiellen Systems im Anfangszustand und im Endzustand des Prozesses zu kennen. Daher kommt die allgemeine Lösung dieser Aufgabe im wesentlichen darauf hinaus, die Wärmefunktionen aller möglichen materiellen Systeme in allen möglichen Zuständen zu finden. Sehr häufig bieten sich zur Berechnung der Wärmefunktion verschiedene Wege der Überführung aus dem einen Zustand in den andern dar, die dann entweder zur Prüfung der Theorie oder zur Kontrolle der Genauigkeit der Messungen dienen können. So fand J. Thomsen als Neutralisationswärme einer Lösung von doppeltkohlensaurem Natron mit Natronlauge:

$$(\mathrm{NaHCO_3\,aq}) + (\mathrm{NaHO\,aq}) - (\mathrm{Na_2CO_3\,aq}) = 9200 \text{ cal.}$$

Dagegen als Neutralisationswärme einer Kohlensäurelösung:

$$(\mathrm{CO_2\,aq}) + 2\,(\mathrm{NaHO\,aq}) - (\mathrm{Na_2CO_3\,aq}) = 20\,200 \text{ cal.}$$

Die Subtraktion dieser beiden Gleichungen ergibt:

$$(\mathrm{CO_2\,aq}) + (\mathrm{NaHO\,aq}) - (\mathrm{NaHCO_3\,aq}) = 11\,000 \text{ cal.}$$

Das ist die Wärmetönung, welche der Verbindung von Kohlensäure und Natronlauge zu doppeltkohlensaurem Natron entspricht, und die durch eine Messung von Berthelot direkt konstatiert wurde.

§ 102. Oft ist von zwei verschiedenen Wegen der Überführung der eine zur kalorimetrischen Verwertung besser geeignet als der andere. So läßt sich die Wärmeentwicklung bei der Zersetzung von Wasserstoffsuperoxyd in Wasser und Sauerstoff nicht gut direkt messen. Thomsen oxydierte daher eine salzsaure Lösung von Zinnchlorür einmal mit Wasserstoffsuperoxyd:

$$(\mathrm{SnCl_2 \cdot 2\,HCl\,aq}) + (\mathrm{H_2O_2\,aq}) - (\mathrm{SnCl_4\,aq}) = 88\,800 \text{ cal}$$

einmal mit Sauerstoffgas:

$$(\mathrm{SnCl_2 \cdot 2\,HCl\,aq}) + \tfrac{1}{2}\{\mathrm{O_2}\} - (\mathrm{SnCl_4\,aq}) = 65\,700 \text{ cal.}$$

Die Differenz ergibt:

$$(\mathrm{H_2O_2\,aq}) - \tfrac{1}{2}\{\mathrm{O_2}\} - (\mathrm{aq}) = 23\,100 \text{ cal}$$

als Wärmetönung bei der Zersetzung von gelöstem Wasserstoffsuperoxyd in gasförmigen Sauerstoff und Wasser.

§ 103. Die Bildungswärme des Kohlenoxydgases aus fester Kohle und Sauerstoff läßt sich deshalb nicht direkt bestimmen, weil Kohle niemals ganz zu Kohlenoxyd, sondern immer zum Teil auch zu Kohlendioxyd verbrennt. Daher bestimmten FAVRE und SILBERMANN erstens die Wärmetönung bei vollständiger Verbrennung der Holzkohle:

$$[\mathrm{C}] + \{\mathrm{O_2}\} - \{\mathrm{CO_2}\} = 97\,000 \text{ cal},$$

zweitens die Wärmetönung bei Verbrennung von Kohlenoxyd zu Kohlendioxyd:

$$\{\mathrm{CO}\} + \tfrac{1}{2}\{\mathrm{O_2}\} - \{\mathrm{CO_2}\} = 68\,000 \text{ cal}.$$

Daraus durch Subtraktion:

$$[\mathrm{C}] + \tfrac{1}{2}\{\mathrm{O_2}\} - \{\mathrm{CO}\} = 29\,000 \text{ cal},$$

die gesuchte Bildungswärme des Kohlenoxydgases.

§ 104. Hiernach führt die Anwendung der Theorie auch dazu, Wärmetönungen von Prozessen zu berechnen, die gar nicht direkt ausführbar sind. Denn sobald die Wärmefunktion eines Systems auf irgendeinem Wege gefunden worden ist, kann man sie mit beliebigen anderen Wärmefunktionen in Vergleich bringen.

Es handle sich z. B. darum, die Bildungswärme von flüssigem Schwefelkohlenstoff aus fester Kohle und festem Schwefel, die sich nicht direkt verbinden, zu bestimmen. Hierzu kann man folgende meßbare Vorgänge benutzen:

Die Verbrennung von festem Schwefel zu gasförmiger schwefliger Säure:

$$[\mathrm{S}] + \{\mathrm{O_2}\} - \{\mathrm{SO_2}\} = 71\,100 \text{ cal}.$$

Die Verbrennung von fester Kohle zu Kohlensäure:

$$[\mathrm{C}] + \{\mathrm{O_2}\} - \{\mathrm{CO_2}\} = 97\,000 \text{ cal}.$$

Die Verbrennung von gasförmigem Schwefelkohlenstoff zu Kohlensäure und schwefliger Säure:

$$\{\mathrm{CS_2}\} + 3\{\mathrm{O_2}\} - \{\mathrm{CO_2}\} - 2\{\mathrm{SO_2}\} = 265\,100 \text{ cal}.$$

Die Kondensation von Schwefelkohlenstoffdampf:

$$\{\mathrm{CS_2}\} - (\mathrm{CS_2}) = 6400 \text{ cal}.$$

Die Elimination aller übrigen Größen auf rein rechnerischem Wege ergibt die gesuchte Bildungswärme:

$$[\mathrm{C}] + 2\,[\mathrm{S}] - (\mathrm{CS_2}) = -19\,500 \text{ cal},$$

also negativ.

Es ist die wichtigste Methode der organischen Thermochemie, die Bildungswärme einer Verbindung aus ihren Bestandteilen dadurch zu bestimmen, daß man einmal die Verbindung, das andere Mal deren Bestandteile verbrennt.

Methylwasserstoff (Grubengas) liefert bei vollständiger Verbrennung zu Kohlensäure und flüssigem Wasser:

$$\{\mathrm{CH_4}\} + 2\,\{\mathrm{O_2}\} - \{\mathrm{CO_2}\} - 2\,(\mathrm{H_2O}) = 211\,900 \text{ cal}.$$

Dagegen ist

$$\{\mathrm{H_2}\} + \tfrac{1}{2}\{\mathrm{O_2}\} - (\mathrm{H_2O}) = 68\,400 \text{ cal}. \tag{50}$$

$$[\mathrm{C}] + \{\mathrm{O_2}\} - \{\mathrm{CO_2}\} = 97\,000 \text{ cal}.$$

Folglich, durch Elimination, die Bildungswärme von Methylwasserstoff aus fester Kohle und Wasserstoffgas:

$$[\mathrm{C}] + 2\,\{\mathrm{H_2}\} - \{\mathrm{CH_4}\} = 21\,900 \text{ cal}.$$

§ 105. Im allgemeinen wird die einer bestimmten, unter konstantem Druck verlaufenden Umwandlung entsprechende äußere Wärme Q von der Temperatur abhängen, bei der die Umwandlung sich vollzieht. Hierfür liefert der erste Hauptsatz der Wärmetheorie folgende Beziehung:

Aus der Gleichung (49) folgt für irgendeine Temperatur T:

$$W_2 - W_1 = Q,$$

für eine andere Temperatur T':

$$W_2' - W_1' = Q',$$

und durch Subtraktion:

$$Q' - Q = (W_2' - W_2) - (W_1' - W_1),$$

d. h. die Differenz der Wärmetönungen $(Q - Q')$ ist gleich der Differenz der Wärmemengen, die zugeführt werden müssen, um das System einmal vor der

Umwandlung (Zustand 1), das andere Mal nach vollendeter Umwandlung (Zustand 2), von T auf T' zu bringen.

Nehmen wir $T' - T$ unendlich klein, so erhalten wir hieraus für die Abhängigkeit der Wärmetönung von der Temperatur:

(50a) $$\frac{dQ}{dT} = \left(\frac{dW}{dT}\right)_2 - \left(\frac{dW}{dT}\right)_1 = C_2 - C_1,$$

wo C_1 und C_2 nach (49) und (26) die Wärmekapazitäten vor und nach der Umwandlung sind.

So findet man den Einfluß der Temperatur auf die Verbrennungswärme des Wasserstoffs zu flüssigem Wasser, wenn man die Wärmekapazität des Knallgases: $H_2 + \frac{1}{2}O_2$ vergleicht mit derjenigen des flüssigen Wassers: H_2O. Die erstere ist gleich der Molekularwärme des Wasserstoffs + der halben Molekularwärme des Sauerstoffs; also nach der Tabelle § 87:

$$C_1 = 6,87 + 3,52 = 10,39,$$

die letztere ist

$$C_2 = 1 \times 18 = 18.$$

Daraus nach (50a):

$$\frac{dQ}{dT} = 7,6.$$

Da Q hier negativ ist, so nimmt die Verbrennungswärme eines Moleküls Wasserstoff mit jedem Temperaturgrad um $7,6$ cal ab.

Dritter Abschnitt.

Der zweite Hauptsatz der Wärmetheorie.

Erstes Kapitel. Einleitung.

§ 106. Der zweite Hauptsatz der Wärmetheorie hat einen von dem ersten Hauptsatz wesentlich verschiedenen Inhalt, da er eine Frage behandelt, die von diesem gar nicht berührt wird, nämlich die Frage nach der Richtung eines in der Natur eintretenden Prozesses. Nicht jede Veränderung nämlich, welche mit dem Prinzip der Erhaltung der Energie verträglich ist, genügt auch den weitergehenden Bedingungen, die der zweite Hauptsatz den in der Natur wirklich vor sich gehenden Prozessen auferlegt; oder mit anderen Worten: das Energieprinzip reicht noch keineswegs aus zur eindeutigen Bestimmung der natürlichen Vorgänge.

Wenn z. B. zwischen zwei Körpern von verschiedener Temperatur Wärmeaustausch durch Leitung stattfindet, so verlangt der erste Hauptsatz oder das Prinzip der Erhaltung der Energie nur, daß die von dem einen Körper abgegebene Wärmemenge gleich ist der von dem anderen Körper aufgenommenen Wärmemenge. Ob aber die Wärmeleitung in der Richtung vom wärmeren zum kälteren Körper erfolgt oder umgekehrt, daraus läßt sich aus dem Energieprinzip allein nicht das mindeste schließen. Überhaupt ist die Frage nach der Größe der Temperatur dem Energieprinzip an sich fremd, wie schon daraus hervorgeht, daß man durch dieses Prinzip nicht zu einer exakten Definition der Temperatur geführt wird.

Ganz ebenso enthält die allgemeine Gleichung (17) des ersten Hauptsatzes keine Aussage über die Richtung des betreffenden Vorgangs; z. B. die Gleichung (50):

$$\{H_2\} + \tfrac{1}{2}\{O_2\} - (H_2O) = 68\,400 \text{ cal.}$$

bedeutet nur, daß, wenn sich Wasserstoff und Sauerstoff bei konstantem Druck zu flüssigem Wasser verbinden, die Herstellung der anfänglichen Temperatur eine bestimmte Wärmeabgabe an die Umgebung erfordert,

und umgekehrt, daß diese Wärme gebunden wird, wenn das Wasser sich in Wasserstoff und Sauerstoff zersetzt; sie erteilt aber keinen Aufschluß darüber, ob sich Knallgas wirklich zu Wasser verbindet, oder ob sich Wasser in Knallgas zersetzt, oder ob der Prozeß überhaupt in irgendeiner Richtung direkt vor sich gehen kann (vgl. § 104). Vom Standpunkt des ersten Hauptsatzes aus betrachtet erscheinen also Anfangszustand und Endzustand eines jeden Prozesses als vollkommen gleichwertig.

§ 107. Es gibt allerdings einen singulären Fall, wo das Energieprinzip allein einem Prozesse eine ganz bestimmte Richtung vorschreibt; dieser Fall tritt dann ein, wenn sich das betrachtete System in einem Zustand befindet, für den eine der verschiedenen Energiearten ein absolutes Maximum oder ein absolutes Minimum besitzt. Dann kann nämlich eine Veränderung offenbar nur in dem Sinne erfolgen, daß die betreffende Energie abnimmt, bez. zunimmt. Dieser singuläre Fall findet sich z. B. verwirklicht in der Mechanik, wenn ein materielles System ruht, wenn also die kinetische Energie ein absolutes Minimum ist: d. h. in einem ruhenden System ist jede Veränderung mit einer Zunahme der kinetischen, folglich, falls keine Einwirkungen von außen stattfinden, mit einer Abnahme der potentiellen Energie verknüpft. Hieraus entspringt ein wichtiger Satz der Mechanik, der die Richtung von selbst eintretender Bewegungen charakterisiert und dadurch auch zur Fixierung der allgemeinen mechanischen Gleichgewichtsbedingung führt. Denn wenn außer der kinetischen auch die potentielle Energie ein Minimum ist, so kann offenbar überhaupt keine Veränderung eintreten, da sich dann keine der beiden Energien auf Kosten der andern vergrößern kann, und das System muß in Ruhe bleiben.

Befindet sich z. B. eine schwere Flüssigkeit in zwei kommunizierenden Röhren auf verschiedenen Niveauhöhen im Ruhezustand, so muß die Bewegung in dem Sinne eintreten, daß das höhere Niveau sinkt, das tiefere steigt, weil dabei der Schwerpunkt des Systems tiefer gelegt wird und dadurch die potentielle Energie, deren Wert mit der Höhe des Schwerpunkts wächst, vermindert wird. Gleichgewicht ist dann vorhanden, wenn die Höhe des Schwerpunkts und mit ihr die potentielle Energie ein Minimum ist, d. h. wenn die Flüssigkeit in beiden Röhren gleich hoch steht. Sobald aber über den Geschwindigkeitszustand der Flüssigkeit keine besondere Voraussetzung eingeführt wird, verliert jener Satz seine Gültigkeit, die potentielle Energie braucht nicht abzunehmen, und das höhere Niveau kann ebensogut steigen wie sinken.

Würde man in der Wärme auch einen Zustand minimaler Energie kennen, so würde für diesen, aber auch nur für diesen singulären Zustand, ein ähnlicher Satz gelten. Da jedoch in Wirklichkeit nicht

einmal dieses zutrifft, so ist es aussichtslos, die allgemeinen Gesetze der Richtung thermodynamischer Veränderungen sowie des thermodynamischen Gleichgewichts auf entsprechende Sätze in der Mechanik, die nur für ruhende Systeme gelten, zurückführen zu wollen.

§ 108. Obwohl aus diesen Darlegungen ersichtlich ist, daß das Energieprinzip im allgemeinen nicht dazu dienen kann, die Richtung eines thermodynamischen Prozesses und damit auch die Bedingungen des thermodynamischen Gleichgewichts zu bestimmen, so haben doch, wesentlich im Anschluß an den im vorigen Paragraphen besprochenen Satz der Mechanik, die Versuche lange fortgedauert, das Energieprinzip in der einen oder anderen Weise zur Bestimmung der Richtung thermodynamischer Vorgänge nutzbar zu machen, und dadurch mußte der zweite Hauptsatz, der seinerseits gerade diesem Zwecke dient, in manchen Darstellungen ein ganz unklares Ansehen erhalten. Man suchte ihn gewissermaßen als einen Teil des Energieprinzips hinzustellen, indem man alle Untersuchungen, welche sich mit diesen Fragen beschäftigen, unter der hierfür zu engen Bezeichnung „Energetik“ zusammenfaßte. Der zweite Hauptsatz kommt aber mit dem Begriff der Energie nicht aus, er läßt sich keineswegs dadurch erschöpfend behandeln, daß man jeden Naturvorgang in eine Reihe Energieverwandlungen zerlegt und nun nach der Richtung jeder einzelnen Verwandlung fragt. Man kann freilich in jedem einzelnen Falle die verschiedenen Energiearten namhaft machen, die sich gegenseitig umsetzen; denn das Energieprinzip muß ja immer erfüllt sein. Aber es bleibt immer eine gewisse Willkür darin bestehen, wie man die Bedingungen der einzelnen Verwandlungen ausdrückt, und diese Willkür läßt sich durch keine allgemeine Festsetzung eindeutig beseitigen.

So findet man auch heute noch manchmal den zweiten Hauptsatz dahin charakterisiert, daß die Verwandlung von Arbeit in Wärme vollständig, die von Wärme in Arbeit dagegen nur unvollständig stattfinden könne, in der Weise, daß jedesmal, wenn ein Quantum Wärme in Arbeit verwandelt wird, zugleich notwendigerweise ein anderes Quantum Wärme eine entsprechende, als Kompensation dienende Verwandlung, z. B. Übergang von höherer in tiefere Temperatur, durchmachen müsse. Dieser Ausspruch ist in gewissen ganz speziellen Fällen richtig; ganz allgemein genommen trifft er aber durchaus nicht das Wesen der Sache, wie der Deutlichkeit halber an einem einfachen Beispiel gezeigt werden soll. Eine der allerwichtigsten mit der Entdeckung des Energieprinzips verknüpften Errungenschaften für die Wärmetheorie ist der in der Gleichung (19) (§ 70) ausgesprochene Satz, daß die gesamte innere Energie eines idealen Gases lediglich von der Temperatur abhängt und nicht vom Volumen. Läßt man nun ein ideales Gas sich unter

Arbeitsleistung ausdehnen, und verhindert man die Abkühlung des Gases durch gleichzeitige Zuleitung von Wärme aus einem Wärmebehälter von höherer Temperatur, so behält das Gas mit seiner Temperatur zugleich auch seine Energie unverändert bei, und man kann sagen, daß die vom Reservoir abgegebene Wärme vollständig in Arbeit verwandelt wird, ohne daß sonst irgendwo ein Energieumsatz stattfindet. Gegen diesen Ausspruch läßt sich nicht das mindeste Tatsächliche einwenden. Nur durch eine veränderte Betrachtungsweise, die aber nicht den physikalischen Tatbestand, sondern nur die Auffassung desselben modifiziert, also auch durch Tatsachen weder gestützt noch widerlegt werden kann, läßt sich der Satz von der „unvollständigen Verwandelbarkeit der Wärme in Arbeit“ aufrecht erhalten, nämlich mit Hilfe der Einführung neuer, nur *ad hoc* ersonnener Energiearten, indem man die Energie des Gases in mehrere Teile zerlegt, die dann einzeln auch vom Volumen abhängen können. Diese Zerlegung muß aber für verschiedene Fälle in verschiedener Weise vorgenommen werden, z. B. für isotherme Prozesse anders als für adiabatische, und erfordert auch für physikalisch einfache Fälle ziemlich verwickelte Betrachtungen. Nun ist von vornherein einleuchtend, daß man aus einer noch so künstlichen Definition, selbst wenn sie in sich keinen Widerspruch enthält, niemals eine neue Tatsache ableiten kann; und um eine solche handelt es sich, wenn man vom ersten Hauptsatz der Wärmetheorie zum zweiten Hauptsatz übergeht.

§ 109. Um die Bedeutung des zweiten Hauptsatzes klar hervortreten zu lassen, gibt es nur einen einzigen Weg: man führt ihn auf Tatsachen zurück dadurch, daß man Sätze aufstellt, die sich durch Experimente bestätigen oder widerlegen lassen. Ein solcher Satz ist nun der folgende: es ist auf keinerlei Weise möglich, einen Vorgang, in welchem Wärme durch Reibung entsteht, vollständig rückgängig zu machen. In ausführlicherer Erläuterung, etwa mit Exemplifikation auf die oben § 60 besprochenen, zur Bestimmung des mechanischen Wärmeäquivalents von Joule angestellten Reibungsversuche, soll dies besagen: wenn die herabfallenden Gewichte durch die Reibung der Schaufelräder im Wasser oder Quecksilber Wärme erzeugt haben, so läßt sich kein Verfahren ersinnen, das den Anfangszustand jenes Prozesses in der ganzen Natur genau wiederherstellt, d. h. die Gewichte wieder auf die ursprüngliche Höhe schafft, die Flüssigkeit entsprechend abkühlt, und sonst keine Veränderungen zurückläßt. Was dabei an technischen Hilfsmitteln, Apparaten mechanischer, thermischer, chemischer, elektrischer Art verwendet wird, ist ganz gleichgültig. Die in dem Worte „vollständig“ ausgesprochene Bedingung soll nur die sein, daß schließlich überall wieder genau der bekannte Anfangszustand des Reibungsprozesses hergestellt ist, wozu auch notwendig gehört, daß alle etwa benutzten Materialien und

Apparate am Schluß sich wieder genau in demselben Zustand befinden wie am Anfang, als man sie in Benutzung nahm.

Dieser Satz ist nicht *a priori* beweisbar, er stellt auch keine Definition vor, sondern er enthält eine bestimmte, in jedem Einzelfall genau zu präzisierende Behauptung, welche durch Tatsachen geprüft werden kann dadurch, daß man wirklich Versuche in der bezeichneten Richtung anstellt: er ist also entweder richtig oder falsch.

§ 110. Ein anderer derartiger, hiermit in engem Zusammenhang stehender Satz ist der folgende: Es ist auf keinerlei Weise möglich, einen Vorgang, bei welchem sich ein Gas ohne äußere Arbeitsleistung und ohne äußere Wärmezufuhr, also mit konstanter Gesamtenergie ausdehnt (wie § 68 beschrieben), vollständig rückgängig zu machen — das Wort „vollständig" immer in demselben Sinne genommen wie oben. Wollte man den Versuch dazu machen, so könnte man z. B. das Gas, nachdem es seinen neuen Gleichgewichtszustand angenommen hat, zunächst auf sein altes Volumen komprimieren, etwa durch Herabsinkenlassen eines Gewichts. Dann wird äußere Arbeit aufgewendet und zugleich das Gas entsprechend erwärmt. Damit ist an und für sich noch nichts bewiesen, es kommt vielmehr jetzt darauf an, das Gas ganz in seinen ehemaligen Zustand zu bringen und das benutzte Gewicht wieder heraufzuschaffen. Leitet man nun, um das Gas auch auf seine alte Temperatur zurückzubringen, die Kompressionswärme bei konstant gehaltenem Volumen ab, etwa in ein kühleres Wärmereservoir, so müßte, damit der Prozeß vollständig rückgängig wird, dem Reservoir die empfangene Wärme wieder entzogen und ferner das Gewicht auf seine ursprüngliche Höhe gebracht werden, ohne daß anderweitige Veränderungen zurückbleiben. Das ist aber genau dieselbe Aufgabe, deren Unausführbarkeit im vorigen Paragraphen behauptet wurde.

§ 111. Ein dritter hierhergehöriger Satz betrifft die Wärmeleitung. Gesetzt, ein Körper nehme durch Leitung eine gewisse Wärmemenge von einem anderen, höher temperierten, auf, und es handle sich darum, diesen Prozeß vollständig rückgängig zu machen, d. h. die Wärme zurückzuschaffen, ohne daß anderweitige Veränderungen in der Natur übrig bleiben. Bei der Beschreibung des CARNOTschen umkehrbaren Kreisprozesses (§ 91) ist schon darauf hingewiesen worden, daß man durch Ausführung des umgekehrt gerichteten Kreisprozesses es stets in der Hand hat, einem Wärmebehälter Wärme zu entziehen und dieselbe auf einen wärmeren Behälter zu übertragen, ohne daß irgendwelche andere Veränderungen zurückbleiben, als daß eine gewisse Arbeit verbraucht ist und die ihr äquivalente Wärme sich in einem der beiden Reservoire vorfindet. Die Aufgabe, den Prozeß der Wärmeleitung vollständig rückgängig zu machen, wäre also gelöst, wenn man auch noch

die letztgenannte Wärme wieder entfernen und dafür die entsprechende Arbeit gewinnen könnte, ohne anderweitige Veränderungen, was wiederum auf das in § 109 als unausführbar bezeichnete Problem hinauskommt.

Weitere Beispiele von Prozessen, an die sich ganz dieselben Betrachtungen knüpfen lassen, wären die Diffusion, das Gefrieren unterkühlter Flüssigkeit, die Kondensation übersättigten Dampfes, jeder explosive Vorgang, überhaupt jeder Übergang eines Systems in einen stabileren Zustand.

§ 112. Definition. Ein Prozeß, der auf keine einzige Weise vollständig rückgängig gemacht werden kann, heißt „irreversibel", alle anderen Prozesse „reversibel". Damit ein Prozeß irreversibel ist, genügt es also nicht, daß er sich nicht von selbst umkehrt — das ist auch bei vielen mechanischen Prozessen der Fall, die nicht irreversibel sind (vgl. § 113) —, sondern es wird erfordert, daß es selbst mit Anwendung sämtlicher in der Natur vorhandenen Reagenzien kein Mittel gibt, um, wenn der Prozeß abgelaufen ist, allenthalben genau den Anfangszustand wiederherzustellen, d. h. die gesamte Natur in den Zustand zurückzubringen, den sie am Anfang des Prozesses besaß. Danach besagen die in den letzten Paragraphen besprochenen Behauptungen, daß die Wärmeerzeugung durch Reibung, die Ausdehnung eines Gases ohne äußere Arbeit und äußere Wärme, die Wärmeleitung usw. irreversible Prozesse sind.[1]

§ 113. Gehen wir nun auf die Frage der tatsächlichen Existenz reversibler und irreversibler Prozesse etwas ein. Reversible Prozesse lassen sich, wenigstens in der Idee, unmittelbar in großer Anzahl angeben. So sind alle diejenigen Prozesse reversibel, welche in der § 71 auseinandergesetzten Ausdrucksweise aus lauter Gleichgewichtszuständen bestehen und daher in allen ihren Teilen direkt umgekehrt werden können, ferner alle vollkommen periodisch verlaufenden Prozesse (ideales Pendel, Planetenbewegung); denn nach Ablauf einer Periode ist der Anfangszustand überall in der ganzen Natur wiederhergestellt. Aber auch sämtliche andere rein mechanischen Prozesse,

[1] Der Satz, daß die Wärmeleitung ein irreversibler Prozeß ist, deckt sich genau mit dem von R. Clausius an die Spitze seiner Argumentation gestellten Grundsatz: „Die Wärme kann nicht von selbst aus einem kälteren in einen wärmeren Körper übergehen." Denn wie Clausius wiederholt und ausführlich hervorgehoben hat, soll mit diesem Grundsatz keineswegs nur gesagt sein, daß Wärme nicht direkt aus einem kälteren in einen wärmeren Körper übergeht, — das ist selbstverständlich und schon durch die Definition der Temperatur bedingt, — sondern es soll ausgedrückt werden, daß Wärme überhaupt auf keinerlei Weise, durch keinen irgendwie gearteten Vorgang, aus einem kälteren in einen wärmeren Körper geschafft werden kann, ohne daß anderweitige Änderungen („Kompensationen") zurückbleiben. Nur vermöge dieser weitergehenden Bedeutung des Satzes ist es möglich, aus ihm Schlüsse auf beliebige andere Naturvorgänge zu ziehen.

in denen keinerlei Reibungswärme auftritt, sind reversibel. Denn durch Einführung geeigneter, aus absolut festen Führungen, reibungslosen Gelenken und Röhren, undehnbaren Seilen usw. zusammengesetzten Maschinen kann man stets bewirken, daß die veränderten Systeme wieder vollständig in den Anfangszustand zurückgeführt werden, ohne daß an diesen Maschinen, die ja selber niemals Arbeit erzeugen, irgendeine Veränderung zurückbleibt.

Wenn z. B. eine in zwei kommunizierenden Röhren auf verschiedenen Niveauhöhen befindliche ursprünglich ruhende schwere Flüssigkeit, wie in § 107 beschrieben, durch ihr Gewicht in Bewegung gerät, so wird sie vermöge der gewonnenen lebendigen Kraft die Gleichgewichtslage überschreiten, nach der entgegengesetzten Seite pendeln und schließlich, da keine Reibung vorausgesetzt ist, genau in ihren Anfangszustand zurückkehren. Dann ist der Prozeß vollständig rückgängig geworden und gehört daher zu den reversibeln Prozessen.

Sobald aber die Reibung ins Spiel kommt, ist die Reversibilität mindestens fraglich. Ob es überhaupt irreversible Prozesse gibt, kann man von vornherein nicht wissen und auch nicht beweisen; denn rein logisch genommen ist es sehr wohl denkbar, daß eines Tages ein Mittel aufgefunden würde, durch dessen Anwendung es gelänge, einen bisher als irreversibel angenommenen Prozeß, z. B. einen Vorgang, in welchem Reibung oder Wärmeleitung vorkommt, vollständig rückgängig zu machen. Wohl aber läßt sich beweisen — und dieser Beweis wird im nächsten Kapitel geführt werden —, daß, wenn auch nur in einem einzigen Falle einer der in den §§ 109 ff. als irreversibel bezeichneten Prozesse in Wirklichkeit reversibel wäre, es notwendig auch alle übrigen in allen Fällen sein müßten. Folglich sind entweder sämtliche oben angeführte Prozesse wirklich irreversibel, oder es ist kein einziger von ihnen. Ein Drittes ist ausgeschlossen. Im letzteren Falle stürzt der ganze Bau des zweiten Hauptsatzes zusammen, keine der zahlreichen aus ihm hergeleiteten Beziehungen, so viele einzelne auch durch die Erfahrung bestätigt sind, kann mehr als allgemein bewiesen gelten und die Arbeit der Theorie muß von vorne beginnen. (Die sogenannten Beweise der „Energetik“ gewähren keinen Ersatz; denn sie stellen sich bei näherer Prüfung nur als Umschreibungen des zu Beweisenden heraus, was darzulegen hier nicht der Ort ist.)[1] Aber gerade in diesem Punkte

[1]In manchen Darstellungen findet man den Prozeß der Wärmeleitung in Parallele gestellt mit dem des Herabsinkens einer schweren Flüssigkeit von höherem auf tieferes Niveau, und den Satz, daß in dem einen Fall die Wärme von höherer zu tieferer Temperatur, im andern Fall die Flüssigkeit von höherem zu tieferem Niveau übergeht, als „zweiten Hauptsatz der Energetik“ bezeichnet. Diese Zusammenstellung zeigt besonders deutlich die Verkennung des wirklichen Sachverhalts; denn es ist hierbei

liegt auch die dem zweiten Hauptsatz innewohnende Kraft. Denn ebenso wie jede einzelne Lücke ihn völlig unhaltbar macht, so kommt auch jede einzelne Bestätigung dem Ganzen zugute und verleiht den Schlüssen auch auf scheinbar entfernten Gebieten die volle Bedeutung, die der Satz selber besitzt.

§ 114. Die Bedeutung des zweiten Hauptsatzes der Thermodynamik besteht darin, daß er ein notwendiges und hinreichendes Kriterium dafür liefert, ob ein bestimmter in der Natur stattfindender Prozeß reversibel oder irreversibel ist. Da nun die Entscheidung dieser Frage nur davon abhängt, ob der Prozeß sich auf irgendeine Weise vollständig rückgängig machen läßt oder nicht, so kommt es dabei lediglich auf die Beschaffenheit des Anfangszustandes und die des Endzustandes des Prozesses an, nicht aber auf seinen sonstigen Verlauf. Denn es handelt sich nur um die Frage, ob man, ausgehend vom Endzustand, auf irgendeine Weise den Anfangszustand, ohne anderweitig zurückbleibende Änderung, wieder erreichen kann oder nicht. Daher liefert der zweite Hauptsatz für jeden beliebigen in der Natur stattfindenden Prozeß eine Beziehung zwischen denjenigen Größen, welche sich auf den Anfangszustand beziehen, und denjenigen, welche sich auf den Endzustand beziehen. Bei irreversibeln Prozessen ist offenbar der Endzustand durch eine gewisse Eigenschaft vor dem Anfangszustand ausgezeichnet, während bei reversibeln Prozessen diese beiden Zustände in gewisser Hinsicht gleichwertig sind. Der zweite Hauptsatz lehrt diese charakteristische Eigenschaft der beiden Zustände kennen, er lehrt also auch, falls zwei Zustände eines Körpersystems beliebig gegeben sind, von vornherein entscheiden, ob in der Natur ein Übergang vom ersten zum zweiten oder vom zweiten zum ersten Zustand möglich ist, ohne daß in anderen Körpern Änderungen zurückbleiben. Dazu müssen aber die beiden Zustände vollkommen genau charakterisiert werden, insbesondere müssen außer der chemischen Beschaffenheit des in Frage kommenden Systems auch die physikalischen Bedingungen: Aggregatzustand, Temperatur, Druck, in beiden Zuständen bekannt sein, ebenso wie das bei der Anwendung des ersten Hauptsatzes erfordert wird.

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gar nicht berücksichtigt, daß der mechanische Zustand eines Körpers nicht allein von der Lage, sondern auch von der Geschwindigkeit abhängt, während dagegen der thermische Zustand allein durch die Temperatur bestimmt ist. Eine schwere Flüssigkeit kann ebensogut emporsteigen wie herabsinken, die Wärme aber kann nur „herabsinken“. Allgemein genommen ist also jener „zweite Hauptsatz der Energetik“ unrichtig. Beschränkt man den Satz aber ausdrücklich auf ruhende Körper, so ist er wie schon oben (§ 107) ausgeführt wurde, eine Folge des Energieprinzips und es ist daher durchaus unmöglich, irgend etwas Neues aus ihm abzuleiten.

Die vom zweiten Hauptsatz gelieferte Beziehung wird offenbar um so einfacher lauten, je weniger sich der Endzustand vom Anfangszustand unterscheidet. Daher rührt die große Fruchtbarkeit des zweiten Hauptsatzes für Kreisprozesse, die, so verwickelt sie in ihrem Verlauf sonst sein mögen, doch einen von dem Anfangszustand nur wenig verschiedenen Endzustand liefern. Denn da das System, welches den Kreisprozeß durchmacht, sich am Ende desselben genau in dem nämlichen Zustand befindet wie am Anfang, so kann man es bei der Vergleichung beider Zustände ganz außer Betracht lassen (vgl. § 91).

Was nun die mathematische Formulierung des zweiten Hauptsatzes betrifft, so kann die Auszeichnung des Endzustandes eines Prozesses vor dem Anfangszustand nur in einer Ungleichung bestehen, welche besagt, daß eine gewisse von dem augenblicklichen Zustand des betreffenden Körpersystems abhängige Größe im Endzustand einen größeren[1] Wert besitzt als im Anfangszustand. Der zweite Hauptsatz der Thermodynamik besagt also, daß in der Natur für jedes Körpersystem eine Größe existiert, welche die Eigenschaft besitzt, bei allen Veränderungen, die das System allein betreffen, entweder konstant zu bleiben (bei reversibeln Prozessen) oder an Wert zuzunehmen (bei irreversibeln Prozessen). Diese Größe heißt nach Clausius die „Entropie“ des Systems. Die Beweisführung im folgenden Kapitel geht darauf hinaus, den mathematischen Ausdruck der Entropie eines Systems zu bestimmen und den Nachweis ihrer Eigenschaften zu führen, und zwar zunächst für ideale Gase, weil nur für diese eine exakte Zustandsgleichung vorliegt, sodann auf Grund hiervon auch für alle übrigen Substanzen.

§ 115. Da es tatsächlich keinen Prozeß in der Natur gibt, der nicht mit Reibung oder Wärmeleitung verbunden wäre, so sind, wenn der zweite Hauptsatz der Wärmetheorie richtig ist, sämtliche Naturprozesse in Wirklichkeit irreversibel, und die reversibeln Prozesse bilden nur einen idealen Grenzfall, der aber für die theoretische Beweisführung und für die Anwendung auf Gleichgewichtszustände von erheblicher Wichtigkeit ist.

Zweites Kapitel. Beweis.

§ 116. Da der zweite Hauptsatz der Wärmetheorie, ebenso wie der erste, ein Erfahrungssatz ist, so kann man von seinem Beweise nur insofern reden, als sein gesamter Inhalt sich aus einem einzigen einfachen Erfahrungssatz von einleuchtender Gewißheit deduzieren läßt. Daher stellen wir folgenden Satz als durch die Erfahrung unmittelbar gegeben an die

[1] oder kleineren, je nach der Definition des Vorzeichens jener Größe.

Spitze: „Es ist unmöglich, eine periodisch funktionierende Maschine zu konstruieren, die weiter nichts bewirkt als Hebung einer Last und Abkühlung eines Wärmereservoirs.“[1] Eine solche Maschine könnte zu gleicher Zeit als Motor und als Kältemaschine benutzt werden, ohne jeden anderweitigen dauernden Aufwand an Energie und Materialien, sie wäre also jedenfalls die vorteilhafteste von der Welt. Zwar käme sie dem perpetuum mobile nicht gleich; denn sie erzeugt Arbeit keineswegs aus nichts, sondern aus der Wärme, die sie dem Reservoir entzieht. Deshalb steht sie auch nicht, wie das perpetuum mobile, im Widerspruch mit dem Energieprinzip. Aber sie besäße doch den für die Menschheit wesentlichsten Vorzug des perpetuum mobile: Arbeit kostenlos zu liefern. Denn die etwa in dem Erdboden, in der Atmosphäre, im Ozean enthaltene Wärme bietet sich ebenso, wie der Sauerstoff der Luft, immer in unerschöpflicher Menge einem jeden zur unmittelbaren Benutzung dar. Dieser Umstand ist der Grund, weshalb wir mit dem genannten Satz beginnen. Denn da wir aus ihm den zweiten Hauptsatz der Wärmetheorie deduzieren werden, so sichern wir uns damit zugleich die Aussicht, bei jeder etwa entdeckten Abweichung einer Naturerscheinung von dem zweiten Hauptsatz sogleich eine praktisch höchst bedeutungsvolle Nutzanwendung aus ihr ziehen zu können. Sobald nämlich irgendein Phänomen aufgefunden werden sollte, was einer einzelnen aus dem zweiten Hauptsatz gezogenen Folgerung widerspricht, so müßte der Widerspruch in einer Unrichtigkeit der gemachten allerersten Voraussetzung liegen, und man könnte, an der Hand der Beweisführung Schritt für Schritt zurückgehend, das Phänomen zur Kombination der genannten Maschine benutzen. Wir wollen dieselbe im folgenden zur Abkürzung nach einem Vorschlag von Ostwald ein perpetuum mobile zweiter Art nennen, da sie zu dem zweiten Hauptsatz in derselben Beziehung steht, wie das perpetuum mobile erster Art zum ersten Hauptsatz. Bei allen Einwänden gegen den zweiten Hauptsatz ist also daran festzuhalten, daß sie sich in letzter Linie, falls in der Beweisführung kein Fehler gefunden wird, immer gegen die Unmöglichkeit des perpetuum mobile zweiter Art richten (vgl. § 136).[2]

[1] Auf die Temperatur des Reservoirs kommt es hierbei nicht an. Wenn eine solche Maschine mit einem Wärmereservoir von 1000° C möglich ist, so ist eine solche auch mit einem Reservoir von 0° C möglich. Man braucht sich, um dies einzusehen, nur eines passend ersonnenen Carnotschen Kreisprozesses zu bedienen. (§ 91.)

[2] Der hier für den Beweis des zweiten Hauptsatzes gewählte Ausgangspunkt deckt sich vollständig mit dem, welchen Clausius, oder welchen W. Thomson, oder welchen Maxwell für denselben Zweck benutzt hat. Denn der Grundsatz, den jeder dieser drei Forscher an die Spitze seiner Deduktionen gestellt hat, spricht immer, nur in verschiedener Form, die Unmöglichkeit der Realisierung des perpetuum mobile zweiter Art aus. Der obigen Form habe ich lediglich wegen ihrer augenfälligen Bedeutung für

§ 117. Aus der Unmöglichkeit des perpetuum mobile zweiter Art folgt zunächst, daß die Erzeugung von Wärme durch Reibung „irreversibel“ ist (vgl. die Definition § 112). Gesetzt nämlich, die Wärmeerzeugung durch Reibung sei nicht irreversibel, d. h. man hätte eine Methode, um irgendeinen Vorgang, der in Erzeugung von Wärme durch Reibung besteht, auf irgendeine Weise vollständig rückgängig zu machen, so wäre diese Methode eben nichts anderes als ein perpetuum mobile zweiter Art. Denn das, was die Methode leisten würde, wäre identisch mit dem, was das perpetuum mobile zweiter Art leistet: eine Veränderung, die in nichts anderem besteht als in Erzeugung von Arbeit und Absorption der äquivalenten Wärme.

§ 118. Daraus folgt weiter, daß auch die Ausdehnung eines Gases ohne äußere Arbeitsleistung und ohne Wärmezufuhr irreversibel ist. Gesetzt nämlich, man hätte eine Methode, diesen Prozeß vollständig rückgängig zu machen, d. h. ein Gas durch irgendein Verfahren auf ein kleineres Volumen zu bringen, ohne daß irgendwelche anderweitige Veränderungen zurückbleiben, so könnte man diese Methode sogleich zur Anfertigung eines perpetuum mobile zweiter Art folgendermaßen verwerten. Man läßt das Gas sich unter Arbeitsleistung ausdehnen, ersetzt dem Gase die dabei verausgabte Energie durch Zuleitung von Wärme aus irgendeinem Reservoir von höherer Temperatur, und verkleinert dann nach der angenommenen Methode das Volumen des Gases wieder auf den Anfangswert, ohne daß anderweitige Veränderungen übrig bleiben. Dieser Prozeß, beliebig oft wiederholt, stellt eine periodisch funktionierende Maschine vor, durch die weiter nichts bewirkt wird, als daß Arbeit geleistet und außerdem dem Reservoir Wärme entzogen wird — also ein perpetuum mobile zweiter Art.

Auf Grund des so bewiesenen Satzes, daß die Ausdehnung eines Gases ohne äußere Arbeitsleistung und Wärmezufuhr irreversibel ist, wollen wir nun zunächst den Beweis des zweiten Hauptsatzes durchführen für diejenige Klasse von Körpern, deren thermodynamische Eigenschaften nach allen Richtungen bekannt sind: für ideale Gase.

§ 119. Wenn man ein ideales Gas unendlich langsam komprimiert oder dilatiert und ihm dabei gleichzeitig Wärme von außen zuführt oder entzieht, so ist nach der Gleichung (22) in jedem unendlich kleinen Teil

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die Technik den Vorzug gegeben. Doch ist bisher noch kein einziger bündiger Beweis des zweiten Hauptsatzes aufgestellt worden, der dieses oder eines genau äquivalenten Grundsatzes gar nicht bedarf, so viele Versuche in dieser Richtung auch gemacht worden sind, und ich glaube auch nicht, daß ein solcher Versuch Aussicht auf Erfolg verspricht.

des Prozesses für die Masseneinheit:

$$q = du + p\,dv$$

oder, da für ein ideales Gas nach (32)

$$du = c_v\,dT$$

und nach (14):

$$p = \frac{R}{m}\frac{T}{v},$$
$$q = c_v\,dT + \frac{R}{m}\frac{T}{v}\,dv.$$

Wenn nun die Zustandsänderung adiabatisch erfolgt, so ist $q = 0$, und durch Integration der Gleichung ergibt sich, wie in § 88, daß die Funktion

$$c_v \log T + \frac{R}{m}\log v$$

konstant bleibt. Nennen wir also den Ausdruck:

$$s = c_v \log T + \frac{R}{m}\log v + \text{konst.} \tag{51}$$

nach Clausius die Entropie der Masseneinheit des Gases, definiert bis auf eine additive Konstante, die durch Festsetzung eines Nullzustandes nach Willkür fixiert werden kann, und dementsprechend:

$$S = M \cdot s = M\left(c_v \log T + \frac{R}{m}\log v + \text{konst.}\right) \tag{52}$$

die Entropie der Masse M des Gases, so bleibt die Entropie des Gases bei der beschriebenen speziellen adiabatischen Zustandsänderung konstant.

§ 120. Bei Wärmezufuhr ändert sich die Entropie des Gases, und zwar in dem hier betrachteten Falle, da:

$$dS = M\left(c_v\frac{dT}{T} + \frac{R}{m}\frac{dv}{v}\right) = M\frac{du + p\,dv}{T} \tag{53}$$

(53a) um:
$$dS = M\frac{q}{T} = \frac{Q}{T}$$

sie nimmt also zu oder ab, je nachdem Wärme zugeführt oder abgeleitet wird.

Es ist jedoch hier besonders zu betonen, namentlich auch mit Rücksicht auf eine neuerdings geltend gemachte Ansicht, nach welcher die Zerlegung der zugeführten Wärme Q in die beiden Faktoren T und dS eine allgemeine Eigenschaft der Wärme sein soll, daß die letzte Gleichung (53a) keineswegs allgemein gilt, sondern nur dann, wenn die bei der Zustandsänderung vom Gase geleistete äußere Arbeit den Wert $p\,dV$ hat. Denn die Beziehung (53) gilt ganz allgemein für jeden beliebigen Vorgang, der das Gas auf die Temperatur $T + dT$ und das Volumen $V + dV$ bringt, da sie nur eine andere mathematische Form für die in (52) gegebene Definition der Entropie ist. Sie gilt z. B. auch, wenn das Gas, wie in dem § 68 beschriebenen Prozeß, ohne Leistung äußerer Arbeit in einen neuen Gleichgewichtszustand mit der nämlichen Temperatur T und dem größeren Volumen $V + dV$ übergeführt wird.

Dagegen gilt die Gleichung

$$Q = dU + p\,dV$$

keineswegs immer, sondern ist im allgemeinen durch die andere:

$$Q + A = dU$$

zu ersetzen, wo A, die aufgewendete äußere Arbeit, innerhalb gewisser Grenzen jeden beliebigen Wert haben kann. So ist z. B. $A = 0$, wenn das Gas sich ohne Leistung äußerer Arbeit ausdehnt. Dann ist $Q = dU$, und die Gleichung $Q = T \cdot dS$ wird ungültig.

§ 121. Nun betrachten wir zwei Gase, die sich gegenseitig durch Leitung Wärme mitteilen können, aber im allgemeinen unter verschiedenem Drucke stehen mögen. Nimmt man mit einem dieser Gase oder mit beiden irgendeine umkehrbare Volumenänderung vor, und sorgt gleichzeitig dafür, daß die Temperaturen der Gase sich in jedem Augenblick durch Wärmeleitung ausgleichen und daß mit der äußeren Umgebung keinerlei Wärmeaustausch stattfindet, so ist nach Gleichung (53) für das erste Gas in jedem Zeitelement des Prozesses:

$$dS_1 = \frac{Q_1}{T_1}.$$

Ebenso für das zweite Gas:

$$dS_2 = \frac{Q_2}{T_2};$$

aber nach der Voraussetzung ist:

$$T_1 = T_2 \qquad \text{und} \qquad Q_1 + Q_2 = 0.$$

Folglich:

$$dS_1 + dS_2 = 0$$

oder für eine endliche Zustandsänderung:

$$S_1 + S_2 = \text{konst.} \tag{54}$$

Die Summe der Entropien beider Gase bleibt also bei der beschriebenen Zustandsänderung konstant.

§ 122. Ein jeder derartiger mit den beiden Gasen ausgeführter Prozeß ist offenbar in allen Teilen reversibel, da er direkt in umgekehrter Richtung ausgeführt werden kann, ohne in anderen Körpern irgendwelche Veränderungen zu hinterlassen. Daraus folgt der Satz, daß es immer möglich ist, die beiden Gase aus einem ganz beliebig gegebenen Zustand durch einen reversibeln Prozeß in irgendeinen anderen von vornherein gegebenen Zustand zu bringen, ohne daß in anderen Körpern Änderungen zurückbleiben, wenn nur die Summe der Entropien beider Gase in den beiden Zuständen die gleiche ist.[1] Zum Beweis dieses Satzes dient folgendes: Es sei der Anfangszustand gegeben durch die Werte der Temperaturen T_1 und T_2, und der spezifischen Volumina v_1 und v_2 der beiden Gase, der zweite Zustand durch die bez. Werte $T_1'T_2'$ und $v_1'v_2'$. Voraussetzung ist, daß

$$S_1 + S_2 = S_1' + S_2'. \tag{55}$$

Man bringe nun zunächst das erste Gas durch umkehrbare adiabatische Kompression oder Dilatation auf die Temperatur T_2, stelle alsdann eine wärmeleitende Verbindung mit dem zweiten Gas her und komprimiere oder dilatiere das erste Gas unendlich langsam weiter. Dabei wird jetzt Wärme unendlich langsam aus dem ersten in das zweite Gas durch Leitung übergehen oder umgekehrt, es ändert sich daher die Entropie des ersten Gases, und man kann es dahin bringen, daß diese Entropie den Wert S_1' annimmt. Nun ist bei dem beschriebenen Vorgang nach (54) die Summe

[1] Da die Zustände der beiden Gase im übrigen beliebig gegeben sind, so werden die Gase in dem einen Zustand im allgemeinen nicht die nämliche Gesamtenergie besitzen wie im andern; dann ist die Überführung der Gase aus dem einen Zustand in den andern jedenfalls mit Aufwand oder Gewinn fremder Energie verbunden. Diese fremde Energie bringt aber keine Schwierigkeit mit sich; denn wir können sie uns immer als mechanische Arbeit denken, bewirkt etwa durch Hebung oder Senkung von Gewichten. Da die Gewichte nur ihre Lage, nicht aber ihren inneren Zustand verändern, so bleiben in ihnen niemals Änderungen zurück, wie denn überhaupt alle rein mechanischen Vorgänge wesentlich reversibel sind (§ 113).

der Entropien beider Gase konstant $= S_1 + S_2$ geblieben, folglich ist dann die Entropie des zweiten Gases gleich

$$(S_1 + S_2) - S_1'$$

geworden, d. h. nach der Voraussetzung (55) gleich S_2'.

Jetzt trennen wir die beiden Gase wieder und behandeln jedes einzelne adiabatisch umkehrbar, bis es die Temperatur T_1' bez. T_2' angenommen hat. Dann muß das spezifische Volumen v_1' bez. v_2' sein, und der verlangte Zustand ist erreicht.

Der beschriebene Prozeß ist in allen Teilen reversibel, auch sind in anderen Körpern keine Veränderungen zurückgeblieben,[1] insbesondere ist in der Umgebung kein Wärmeverlust oder -gewinn entstanden,[2] die Bedingungen der gestellten Aufgabe sind also alle erfüllt, und der ausgesprochene Satz bewiesen.

§ 123. Ein gleicher Satz läßt sich leicht beweisen für beliebig viele Gase. Es ist immer möglich, ein System von n Gasen aus einem beliebig gegebenen Zustand durch einen reversibeln Prozeß in einen anderen beliebig gegebenen Zustand zu bringen, ohne daß in anderen Körpern Änderungen zurückbleiben, wenn nur die Summe der Entropien aller Gase in beiden Zuständen die gleiche ist, d. h. wenn

$$S_1 + S_2 + \ldots + S_n = S_1' + S_2' + \ldots + S_n'. \tag{56}$$

Denn durch sukzessive Kombination je zweier Gase des Systems kann man mittels der im vorigen Paragraph beschriebenen Prozesse zunächst die Entropie des ersten Gases, dann die des zweiten, dann die des dritten, usw. auf den verlangten Wert bringen, bis auf die des $(n-1)$ten Gases einschließlich. Nun ist bei jeder der einzelnen nacheinander vorgenommenen Zustandsänderungen die Summe der Entropien sämtlicher Gase konstant geblieben. Haben also die Entropien der $n-1$ ersten Gase ihre verlangten Werte: S_1', S_2',… S_{n-1}', so nimmt die Entropie des $n$ten Gases notwendig den Wert:

$$(S_1 + S_2 + \ldots + S_n) - S_1' - S_2' \ldots - S_{n-1}',$$

d. h. nach (56) den verlangten Wert S_n' an. Alsdann kann man jedes Gas einzeln durch umkehrbare adiabatische Behandlung in den gewünschten Zustand bringen, und die Aufgabe ist vollständig gelöst.

[1] Denn die Leistung der erforderlichen mechanischen Arbeiten kann durch Heben oder Senken von unveränderlichen Gewichten erfolgen, bedingt also keine innere Veränderung.

[2] auch keine Dichtigkeitsänderung; denn man kann sich die Gasbehälter in einem Vakuum befindlich denken.

Nennen wir die Summe der Entropien aller Gase die Entropie des ganzen Systems, so können wir sagen: Wenn das Gassystem in zwei verschiedenen Zuständen den gleichen Wert der Entropie besitzt, so läßt sich das System aus dem einen Zustand in den anderen Zustand durch einen reversibeln Prozeß überführen, ohne daß in anderen Körpern Veränderungen zurückbleiben.

§ 124. Nun führen wir den im § 118 bewiesenen Satz ein, daß die Ausdehnung eines idealen Gases ohne äußere Arbeitsleistung und Wärmezufuhr, oder, was dasselbe ist, daß der Übergang eines idealen Gases in einen Zustand größeren Volumens und gleicher Temperatur, ohne äußere Wirkungen, wie in § 68 beschrieben, irreversibel ist. Einem solchen Übergang entspricht nach der Definition (52) eine Vergrößerung der Entropie des Gases. Daraus folgt sogleich, daß es überhaupt unmöglich ist, die Entropie eines idealen Gases zu verkleinern, ohne daß in anderen Körpern Änderungen zurückbleiben. Denn gäbe es hierfür irgendein Verfahren, so könnte man die irreversible Ausdehnung eines idealen Gases dadurch vollständig rückgängig machen, daß man, nachdem das Gas sich ohne äußere Wirkungen ausgedehnt und seinen neuen Gleichgewichtszustand angenommen hat, zunächst mittels des angenommenen Verfahrens die Entropie des Gases auf ihren ursprünglichen Wert verkleinert, ohne daß in anderen Körpern eine Veränderung zurückbleibt, und dann durch einen umkehrbaren adiabatischen Prozeß, bei welchem die Entropie des Gases konstant bleibt, die ursprüngliche Temperatur und damit auch das ursprüngliche Volumen wiederherstellt. Dann wäre also die erste Ausdehnung vollständig rückgängig gemacht[1] und somit nach § 118 das perpetuum mobile zweiter Art fertig.

§ 125. Ebenso verhalten sich infolgedessen auch zwei und beliebig viele ideale Gase. Es gibt in der ganzen Natur kein Mittel, um die Entropie eines Systems idealer Gase zu verkleinern, ohne daß in anderen Körpern Änderungen zurückbleiben. Denn jede Vorrichtung, welche dies leisten würde — sie sei mechanischer, thermischer, chemischer, elektrischer Art —, könnte wiederum dazu benutzt werden, um die Entropie eines einzelnen Gases zu verkleinern, ohne daß in anderen Körpern Änderungen zurückbleiben.

Gesetzt nämlich, die Entropie des Systems, oder die Summe der Entropien aller Gase, sei aus dem Zustand, in welchem die Werte der Entropien S_1, $S_2, \ldots S_n$ sind, auf irgendeine Weise in einen anderen Zustand

[1] daß auch keine mechanische Arbeit zurückbleiben kann, folgt aus dem ersten Hauptsatz der Thermodynamik, da mit dem ursprünglichen Zustand des Gases zugleich auch dessen ursprüngliche Energie wiederhergestellt ist.

mit den Entropien S'_1, S'_2, … S'_n übergeführt worden, wobei

$$S'_1 + S'_2 + \ldots + S'_n < S_1 + S_2 + \ldots + S_n, \tag{57}$$

ohne daß in anderen Körpern Änderungen zurückgeblieben sind. Dann könnte man nach dem im § 123 bewiesenen Satze stets durch einen reversibeln Prozeß, ohne in anderen Körpern Veränderungen zurückzulassen, das System in jeden beliebigen Zustand bringen, in welchem die Summe der Entropien den Wert $S'_1 + S'_2 + \ldots + S'_n$ besitzt; folglich auch in einen Zustand, in welchem das erste Gas die Entropie S_1, das zweite die Entropie S_2, …, das $(n-1)$te die Entropie S_{n-1}, und das $n$te Gas infolgedessen die Entropie:

$$(S'_1 + S'_2 + \ldots + S'_n) - S_1 - S_2 - \ldots - S_{n-1} \tag{58}$$

besitzt. Ist dies geschehen, so lassen sich alle Gase bis auf das $n$te durch umkehrbare adiabatische Prozesse einzeln in ihren ehemaligen Zustand zurückbringen. Nur das $n$te Gas besitzt die Entropie (58), und diese ist nach der Voraussetzung (57) kleiner als die ursprüngliche Entropie S_n war. So ist also im ganzen die Entropie des $n$ten Gases verkleinert worden, ohne daß in anderen Körpern irgendwelche Veränderungen zurückgeblieben sind,[1] und dies hatten wir schon im vorigen Paragraphen als unmöglich nachgewiesen.

Somit ist der am Anfang dieses Paragraphen ausgesprochene allgemeine Satz bewiesen, und wir können daran unmittelbar den folgenden knüpfen.

§ 126. Wenn ein System idealer Gase auf irgendeinem, möglicherweise gänzlich unbekannten Wege in irgendeinen anderen Zustand übergegangen ist, ohne daß in anderen Körpern Änderungen zurückgeblieben sind, so ist die Entropie des Systems im Endzustand jedenfalls nicht kleiner, also entweder größer oder, im Grenzfall, ebensogroß als im Anfangszustand, oder: die durch den Prozeß verursachte Gesamtänderung der Entropie ist $\geqq 0$. Im Falle der Ungleichung ist der Prozeß irreversibel, im Fall der Gleichung reversibel.

Die Gleichheit der Entropien in beiden Zuständen bildet also nicht nur, wie in § 123, eine hinreichende, sondern zugleich auch die notwendige Bedingung für die vollständige Reversibilität des Übergangs von dem einen Zustand in den anderen, falls in anderen Körpern keine Änderungen zurückbleiben sollen.

§ 127. Dieser Satz hat einen beträchtlichen Gültigkeitsbereich; denn da über den Weg, auf welchem das Gassystem in den Endzustand gelangt,

[1] Hebung und Senkung von Gewichtsstücken sind keine inneren Veränderungen, vgl. die Anmerkungen zu § 122.

ausdrücklich gar keine beschränkende Voraussetzung gemacht ist, so gilt er nicht etwa bloß für langsam und einfach verlaufende, sondern für beliebig komplizierte physikalische und chemische Prozesse, wenn nur am Schluß derselben in keinem Körper außerhalb des Systems Veränderungen zurückgeblieben sind. Auch darf man nicht glauben, daß die Entropie eines Gases nur für Gleichgewichtszustände Bedeutung hat. Sofern man nur in einer beliebig bewegten Gasmasse jedes hinreichend kleine Massenteilchen als homogen und von bestimmter Temperatur annehmen kann, muß man ihm auch nach (52) einen bestimmten Wert der Entropie zuschreiben, wobei dann M die Masse, v die reziproke Dichte und T die Temperatur des Teilchens sind. Die Summierung über alle Massenteilchen, wobei v und T von Teilchen zu Teilchen variieren können, ergibt dann die Entropie der ganzen Gasmasse in dem betreffenden Zustand, und der Satz bleibt bestehen, daß die Entropie des gesamten Gases bei irgend einer Zustandsänderung in jedem Augenblick zunehmen muß, falls in anderen Körpern keine Veränderungen eintreten. Die Geschwindigkeit der Gasteilchen hat, wie man sieht, gar keinen Einfluß auf den Wert der Entropie, ebensowenig wie die Höhe der als schwer gedachten Teilchen über einer bestimmten Horizontalebene.[1]

§ 128. Die bisher für ideale Gase abgeleiteten Gesetze lassen sich ganz in derselben Weise auch auf beliebige Substanzen übertragen, wobei der Hauptunterschied nur darin besteht, daß man den Ausdruck der Entropie für einen beliebigen Körper im allgemeinen nicht in endlichen Größen hinschreiben kann, weil die Zustandsgleichung nicht allgemein bekannt ist. Doch läßt sich stets beweisen — und dies allein ist der entscheidende Punkt —, daß auch für beliebige andere Körper eine Funktion mit den charakteristischen Eigenschaften der Entropie wirklich existiert.

Wir denken uns mit einem beliebigen homogenen Körper, von der Art, wie wir ihn § 67 ff. betrachtet haben, einen gewissen, reversibeln oder irreversibeln, Kreisprozeß ausgeführt, der also den Körper genau in seinen Anfangszustand zurückbringt. Die äußeren Wirkungen auf den Körper sollen in Arbeitsleistung und in Wärmezufuhr oder -abfuhr bestehen, welch letztere durch eine beliebige Anzahl geeigneter Wärmebehälter vermittelt wird. Nach Beendigung des Prozesses sind in dem Körper gar keine Änderungen zurückgeblieben, nur die Wärmebehälter haben ihren Zustand

[1] Wenn die Bewegung des Gases so tumultuarisch ist, daß man Temperatur und Dichte nicht definieren kann, so verliert natürlich auch die hier gegebene Definition der Entropie ihren Sinn. Für diesen Fall läßt sich vom Standpunkt der kinetischen Gastheorie aus, wie L. Boltzmann gezeigt hat, eine andere Definition der Entropie angeben, welche noch allgemeinere Bedeutung besitzt und für stationäre oder nahezu stationäre Zustände in die gewöhnliche übergeht.

geändert. Wir wollen nun als Träger der Wärme in den Behältern lauter ideale Gase annehmen, die etwa auf konstantem Volumen oder auch unter konstantem Druck gehalten werden mögen, jedenfalls aber nur umkehrbaren Volumenänderungen unterworfen sein sollen.[1] Nach dem zuletzt bewiesenen Satze kann die Summe der Entropien aller Gase nicht kleiner geworden sein, da nach Beendigung des Prozesses in keinem anderen Körper, auch nicht in demjenigen, welcher den Kreisprozeß durchgemacht hat, eine Veränderung zurückgeblieben ist.

Bezeichnet nun Q die von einem Wärmebehälter während eines unendlich kleinen Zeitelements an den Körper abgegebene Wärmemenge, T die Temperatur des Behälters in demselben Augenblick,[2] so ist die in demselben Zeitelement erfolgte Entropieänderung des Behälters nach Gleichung (53a):

$$-\frac{Q}{T},$$

daher die im Verlauf aller Zeitelemente erfolgte Entropieänderung sämtlicher Behälter:

$$-\sum \frac{Q}{T}$$

und es gilt nach § 126 die Bedingung:

$$-\sum \frac{Q}{T} \geqq 0$$

oder

$$\sum \frac{Q}{T} \leqq 0.$$

In dieser Form hat Clausius zuerst den zweiten Hauptsatz ausgesprochen.

Eine andere Bedingung für den betrachteten Prozeß liefert der erste Hauptsatz. Denn für jedes Zeitelement des Prozesses ist nach Gleichung (17) § 63:

$$Q + A = dU,$$

wobei U die Energie des Körpers, A die im Zeitelement gegen ihn aufgewendete äußere Arbeit bezeichnet.

§ 129. Nehmen wir nun spezieller an, daß der äußere Druck in jedem Augenblick gleich dem Druck p des ruhend gedachten Körpers ist, so wird

[1] Die Voraussetzung, daß die Gase sich im idealen Zustand befinden, ist nicht an eine bestimmte Grenze der Temperatur gebunden. Denn auch bei den tiefsten Temperaturen verhält sich jedes Gas merklich wie ein ideales, wenn nur die Dichte hinreichend klein genommen wird (§ 21).

[2] Die gleichzeitige Temperatur des Körpers ist hierbei ganz ohne Belang.

die äußere Arbeit nach (20):

$$A = -p \cdot dV,$$

woraus folgt:

$$Q = dU + p\,dV.$$

Sei ferner die Temperatur eines jeden Wärmebehälters in dem Augenblick, wo er in Funktion tritt, gerade gleich der gleichzeitigen Temperatur des Körpers, dann ist der Kreisprozeß reversibel, und die Ungleichung des zweiten Hauptsatzes verwandelt sich in die Gleichung:

$$\sum \frac{Q}{T} = 0$$

oder, mit Substitution des Wertes von Q:

$$\sum \frac{dU + p\,dV}{T} = 0.$$

In dieser Gleichung kommen nur solche Größen vor, welche sich auf den Zustand des Körpers selber beziehen, man kann dieselbe also interpretieren, ohne auf Wärmereservoire irgendwelchen Bezug zu nehmen. Es ist darin folgender Satz ausgesprochen:

§ 130. Wenn man einen homogenen Körper durch passende Behandlung eine Reihe von stetig aufeinanderfolgenden Gleichgewichtszuständen (§ 71) durchmachen läßt, und ihn so schließlich wieder in seinen Anfangszustand zurückbringt, so liefert das Differential

$$\frac{dU + p\,dV}{T}$$

über alle Zustandsänderungen summiert, den Wert Null. Daraus folgt sogleich, daß, wenn man die Zustandsänderung nicht bis zur Wiederherstellung des Anfangszustandes (1) fortsetzt, sondern bei einem beliebigen Zustand (2) stehen bleibt, der Wert der Summe:

$$\int_1^2 \frac{dU + p\,dV}{T} \tag{59}$$

lediglich abhängt von dem Endzustand (2) und dem Anfangszustand (1), nicht aber von dem Wege des Überganges von 1 zu 2. Denn faßt man zwei verschiedene von 1 zu 2 führende Reihen von Zustandsänderungen ins Auge (etwa die Kurven α und β in Fig. 2, § 75), so kann man diese beiden

Reihen zu einem unendlich langsamen Kreisprozeß kombinieren, indem man etwa die eine Reihe (α) als Hinweg von 1 zu 2, die zweite (β) als Rückweg von 2 zu 1 benutzt. Dann ist nach dem oben Bewiesenen die Summe über den gesamten Kreisprozeß:

$$\int_1^2 \frac{dU + p\,dV}{T} + \int_2^1 \frac{dU + p\,dV}{T} = 0,$$

mithin das erste Integral dem zweiten gerade entgegengesetzt, woraus sich die Richtigkeit des aufgestellten Satzes ergibt.

Der Ausdruck (59) mit den bewiesenen Eigenschaften heißt nach Clausius die Entropie des Körpers im Zustand 2, bezogen auf den Zustand 1 als Nullzustand. Die Entropie eines Körpers in einem bestimmten Zustand ist also, ebenso wie die Energie, vollständig bestimmt bis auf eine additive Konstante, welche von der Wahl des Nullzustandes abhängt.

Bezeichnen wir die Entropie wieder mit S, so ist

$$S = \int \frac{dU + p\,dV}{T}$$

und, was dasselbe bedeutet:

$$dS = \frac{dU + p\,dV}{T}, \tag{60}$$

auf die Masseneinheit bezogen:

$$ds = \frac{du + p\,dv}{T}. \tag{61}$$

Für ein ideales Gas ergibt sich hieraus wieder der bekannte Wert (51). Ebenso kann man für jeden anderen Körper, wenn seine Energie $U = Mu$ und sein Volumen $V = Mv$ als Funktionen etwa von T und p bekannt sind, unmittelbar durch Integration den Ausdruck der Entropie bestimmen (vgl. § 254). Da dies jedoch noch für keine andere Substanz vollständig der Fall ist, so muß man sich im allgemeinen mit der Differentialgleichung begnügen. Für den Beweis und für viele Anwendungen des zweiten Hauptsatzes genügt es aber, zu wissen, daß diese Differentialgleichung wirklich die eindeutige Definition der Entropie enthält.

§ 131. Hiernach kann man nun, ebenso wie bei einem idealen Gase, von der Entropie irgend einer Substanz als von einer durch die augenblicklichen Werte von Temperatur und Volumen stets bestimmten endlichen Größe reden,

also auch dann, wenn die Substanz beliebige, reversible oder irreversible, Änderungen erleidet, und die Differentialgleichung (61) für ds gilt, wie das schon oben § 120 bei einem idealen Gas hervorgehoben wurde, für jede beliebige, auch jede irreversible Änderung des Zustandes. In dieser Anwendung des Begriffes der Entropie darf man keinen Widerspruch erblicken mit der Art der Ableitung dieser Größe. Gemessen wird die Entropie in jedem Zustand eines Körpers mittels eines reversibeln Prozesses, der den Körper aus seinem jeweiligen Zustand in den Nullzustand überführt, aber dieser ideale Prozeß hat nichts zu tun mit den Zustandsänderungen, die der Körper in Wirklichkeit erlitten hat oder erleiden wird.

Dagegen ist andrerseits ausdrücklich zu betonen, daß die Differentialgleichung (60) für dS nur für Änderungen der Temperatur und des Volumens, nicht aber für solche der Masse und der chemischen Zusammensetzung des Körpers gilt. Denn von Änderungen der letzteren Art ist bei der Definition der Entropie überhaupt nicht die Rede gewesen.

Endlich bezeichnen wir die Summe der Entropien mehrerer Körper kurz als die Entropie des Systems aller Körper, woraus sich dann auch wieder, ebenso wie oben § 127 bei idealen Gasen, die Entropie eines in seinen einzelnen Teilen ungleichmäßig temperierten und bewegten Körpers durch Summation über alle einzelnen Massenelemente ergibt, solange man innerhalb jedes unendlich kleinen Massenelements die Temperatur und die Dichte als gleichmäßig annehmen kann, während dagegen die Geschwindigkeit und die Schwere gar nicht in den Ausdruck der Entropie eingeht.

§ 132. Nachdem nun die Existenz und der Wert der Entropie für jeden beliebigen Zustand eines Körpers festgestellt ist, bietet es nicht die geringste Schwierigkeit mehr, den von § 118 an beginnenden, oben nur für ideale Gase gelieferten Beweis auf jedes System von Körpern zu übertragen. Man findet wie in § 119, daß bei umkehrbarer adiabatischer Ausdehnung oder Kompression eines Körpers seine Entropie konstant bleibt, während dagegen bei Wärmezufuhr von außen die Entropieänderung beträgt:

$$dS = \frac{Q}{T}, \tag{62}$$

eine Beziehung, die jedoch, ebenso wie § 120 für ideale Gase gezeigt wurde, nur dann gilt, wenn die Volumenänderung des Körpers in umkehrbarer Weise erfolgt. Man findet ferner, wie in § 121, daß bei umkehrbarer Ausdehnung oder Kompression zweier Körper von gemeinsamer Temperatur, die untereinander, aber nicht nach außen hin Wärme durch Leitung austauschen, die Summe der Entropien konstant bleibt, und daran knüpfen

sich dann in ganz derselben Weise die entsprechenden Überlegungen,[1] so daß wir uns nun darauf beschränken können, gleich das allgemeine Resultat auszusprechen: Es ist auf keinerlei Weise möglich, die Entropie eines Systems von Körpern zu verkleinern, ohne daß in anderen Körpern Änderungen zurückbleiben. Wenn also irgend ein System von Körpern auf irgend eine Weise, durch beliebige physikalische und chemische Änderungen, in einen anderen Zustand übergegangen ist, ohne in anderen Körpern Änderungen zurückzulassen, so ist die Entropie des Systems im Endzustand entweder größer, oder, im Grenzfall, ebensogroß wie im Anfangszustand. Im ersten Fall ist der Prozeß irreversibel, im zweiten reversibel.

§ 133. Die bisher stets notwendige Beschränkung, daß in anderen Körpern keine Veränderungen zurückgeblieben sind, läßt sich einfach dadurch aufheben, daß man alle von etwaigen Veränderungen betroffenen Körper mit in das betrachtete System hineinbezieht. Dann lautet der Satz folgendermaßen: Jeder in der Natur stattfindende physikalische und chemische Prozeß verläuft in der Art, daß die Summe der Entropien sämtlicher an dem Prozeß irgendwie beteiligten Körper vergrößert wird. Im Grenzfall, für reversible Prozesse, bleibt jene Summe ungeändert. Dies ist der allgemeinste Ausdruck des zweiten Hauptsatzes der Wärmetheorie.

§134. Wie die Unmöglichkeit des perpetuum mobile erster Art zum ersten Hauptsatz, dem Prinzip der Erhaltung der Energie, führt, so hat uns die Unmöglichkeit des perpetuum mobile zweiter Art zum zweiten Hauptsatz geführt, den wir daher passend als das Prinzip der Vermehrung der Entropie bezeichnen.[2] Man kann diesem Prinzip in speziellen Fällen noch andere Formen geben, welche für die praktische Anwendung gewisse Vorzüge

[1] Bezüglich der Verallgemeinerung des in § 124 für ein ideales Gas bewiesenen Satzes auf beliebige Substanzen erhebt sich eine gewisse Schwierigkeit für den Fall einer inkompressibeln Substanz, weil eine solche ihr Volumen überhaupt nicht ändern kann und daher auch keiner irreversibeln Ausdehnung fähig ist. Für diesen Fall läßt sich aber der Beweis leicht ergänzen durch Zuhilfenahme eines idealen Gases, welches man mit der Substanz in wärmeleitende Verbindung bringt und dadurch die Entropie der Substanz in passender Weise verändert.

[2] Daß der erste Hauptsatz durch eine Gleichung, der zweite aber nur durch eine Ungleichung ausgedrückt wird, rührt natürlich daher, daß der Satz von der Unmöglichkeit des perpetuum mobile erster Art auch umkehrbar ist, d. h. Arbeit läßt sich weder absolut schaffen noch absolut vernichten, während dagegen der Satz von der Unmöglichkeit des perpetuum mobile zweiter Art keine Umkehrung zuläßt, da es sehr wohl möglich ist, eine Maschine zu konstruieren, welche weiter nichts bewirkt als Verbrauch von Arbeit und entsprechende Erwärmung eines Reservoirs.

besitzen, besonders für isotherme und isobare Prozesse. Wir werden diese Formen im nächsten Kapitel kennen lernen. Doch ist hier ausdrücklich zu betonen, daß die hier gegebene Form unter allen die einzige ist, welche sich ohne jede Beschränkung für jeden beliebigen endlichen Prozeß aussprechen läßt, und daß es daher für die Irreversibilität eines Prozesses kein anderes allgemeines Maß gibt als den Betrag der eingetretenen Vermehrung der Entropie. Jede andere Form des zweiten Hauptsatzes ist entweder nur auf unendlich kleine Zustandsänderungen anwendbar, oder sie setzt, auf endliche Zustandsänderungen ausgedehnt, eine spezielle äußere Bedingung voraus, unter welcher der Prozeß verläuft. Näheres hierüber unten § 140 ff.

Die Bedeutung des zweiten Hauptsatzes ist häufig in einer „Zerstreuung der Energie“ gesucht worden. Indes stellt diese Bezeichnung, welche an den irreversibeln Vorgang der Wärmeleitung und -strahlung anknüpft, die Sache nur von einer Seite dar. Es gibt irreversible Prozesse, deren Endzustand genau dieselben einzelnen Energieformen aufweist wie der Anfangszustand, so z. B. die Diffusion zweier idealer Gase (§ 238), oder die weitere Verdünnung einer sehr verdünnten Lösung. Ein solcher Prozeß ist von keinem merklichen Wärmeübergang, keiner äußeren Arbeit, überhaupt keinem merklichen Umsatz an Energie begleitet,[1] er geht nur deshalb vor sich, weil ihm eine merkliche Vermehrung der Entropie entspricht.[2] Ebensowenig wie von einer zerstreuten Energie kann man im allgemeinen von einer „verlorenen Arbeit“ als einem bestimmten Maß der Irreversibilität reden. Dies ist nur bei isothermen Prozessen möglich (§ 143).

§ 135. Clausius hat den ersten Hauptsatz der Wärmetheorie dahin zusammengefaßt, daß die Energie der Welt konstant bleibt, den zweiten dahin, daß die Entropie der Welt einem Maximum zustrebt. Mit Recht ist dagegen eingewendet worden, daß es keinen Sinn hat, schlechthin von der Energie oder der Entropie der Welt zu sprechen, weil eine derartige Größe gar nicht bestimmt zu definieren ist. Indessen fällt es nicht schwer, die Clausiusschen Sätze so zu formulieren, daß sie sehr wohl einen Sinn ergeben, und daß dasjenige, was an ihnen charakteristisch ist, und was Clausius offenbar mit ihnen sagen wollte, deutlicher zum Ausdruck gelangt.

Die Energie jedes Körpersystems ändert sich nach Maßgabe der Wirkungen, welche von außen her auf das System ausgeübt werden; nur bei Ausschluß aller äußeren Wirkungen bleibt sie konstant. Da nun streng genommen ein System stets äußeren Wirkungen unterliegt — denn eine

[1] Wenigstens wenn man bei der im § 56 gegebenen Definition der Energie stehen bleibt und nicht ad hoc neue Energiearten einführt.

[2] In diesem Falle würde man also viel passender von einer Zerstreuung der Materie als von einer Zerstreuung der Energie reden.

absolute Absperrung ist in der Natur unmöglich —, so tritt im strengen Sinne unter Umständen wohl eine annähernde, aber nie eine absolute Konstanz der Energie eines endlichen Systems ein. Indessen: je räumlich ausgedehnter man das System wählt, um so mehr treten im allgemeinen die äußeren Wirkungen zurück gegen die Größe der Energie des Systems und der Änderungen ihrer einzelnen Teile (vgl. § 66). Denn die äußeren Wirkungen sind von der Größenordnung der Oberfläche, die Energie des Systems aber ist von der Größenordnung des Volumens.[1] Bei sehr kleinen Systemen (Volumenelementen) ist es aus demselben Grunde gerade umgekehrt: hier überwiegen die äußeren Wirkungen derart, daß die Energie des Systems gegen jede einzelne äußere Wirkung vernachlässigt werden kann. Von diesem Satze macht man häufig Gebrauch, z. B. in der Theorie der Wärmeleitung bei der Aufstellung der Grenzbedingungen.

In dem hier besprochenen Falle wird man also sagen können: Je räumlich ausgedehnter das System angenommen wird, um so angenäherter bleibt im allgemeinen seine Energie konstant. Man wird schon einen verhältnismäßig kleinen Fehler begehen, wenn man die Energie unseres Sonnensystems konstant setzt, einen verhältnismäßig noch kleineren, wenn man dasselbe bei dem ganzen uns bekannten Fixsternsysteme tut, und in diesem Sinne hat der Satz: Die Energie eines unendlich großen Systems, oder die Energie der Welt, bleibt konstant, allerdings eine tatsächliche Bedeutung.[2]

Ganz in ähnlicher Weise läßt sich der Satz von der steten Vermehrung der Entropie der Welt verstehen. Je umfassender ein System ist, einen desto kleineren verhältnismäßigen Fehler wird man im allgemeinen begehen, wenn man den Satz ausspricht, daß die Entropie des Systems zunimmt, ganz abgesehen von allen außerhalb des Systems eingetretenen Veränderungen.

§ 136. Zum Schlusse möge noch die prinzipielle Frage nach den etwaigen Grenzen der Gültigkeit des zweiten Hauptsatzes kurz erörtert werden.[3] Wenn dem zweiten Hauptsatz der Wärmetheorie irgendwelche

[1] Dieser Satz gilt ganz allgemein für alle physikalischen Vorgänge, falls unvermittelte Fernewirkungen ausgeschlossen werden.

[2] Analytisch läßt sich dies folgendermaßen formulieren: Bezeichnet E die gesamte Energie, die in einem bestimmten sehr großen Raume R enthalten ist, so gilt die Gleichung:

$$\lim_{r=\infty} \frac{1}{E} \frac{dE}{dt} = 0.$$

D. h. die Größe $\log E$ ändert sich um so weniger mit der Zeit, je größer R genommen wird, und nähert sich unbegrenzt der Konstanz.

[3] Die folgenden Betrachtungen erörtern die Bedeutung des zweiten Hauptsatzes selbstverständlich nur so weit, als sie sich von dem in diesem Werk eingehaltenen, alle atomistischen Hypothesen vermeidenden Standpunkte aus übersehen läßt.

Schranken gesetzt sind, wie gegenwärtig noch viele Naturforscher und Philosophen wollen, so läßt sich doch jedenfalls so viel von vornherein behaupten, daß deren Existenz nur entweder in einer Unrichtigkeit unseres Ausgangspunktes: der Unmöglichkeit des perpetuum mobile zweiter Art, oder in einem Mangel unserer Beweisführung begründet sein kann. Den ersten Einwand haben wir schon am Anfang der Beweisführung (§ 116) als berechtigt anerkannt, er läßt sich durch keine Argumentation beseitigen. Der zweite Einwand aber, der wesentlich darauf hinausläuft, daß zwar die praktische Unmöglichkeit des perpetuum mobile zweiter Art zugegeben wird, nicht aber die absolute, da wir eben mit unseren beschränkten experimentellen Hilfsmitteln gar nicht immer imstande seien, eintretendenfalls die in dem Beweisgang vorausgesetzten idealen Prozesse zur wirklichen Konstruktion eines perpetuum mobile zweiter Art zu verwerten, erweist sich bei näherer Untersuchung als unstichhaltig. Denn es wäre ganz ungereimt, anzunehmen, daß die Gültigkeit des zweiten Hauptsatzes irgendwie mit der größeren oder geringeren Ausbildung der Beobachtungs- bez. Experimentierkunst des Physikers oder Chemikers zusammenhängt. Der Inhalt des zweiten Hauptsatzes hat ja mit dem Experimentieren gar nichts zu tun, er lautet in nuce: „Es existiert in der Natur eine Größe, welche bei allen in der Natur stattfindenden Veränderungen sich immer nur in demselben Sinne ändert.“ Dieser Satz, in dieser Allgemeinheit ausgesprochen, ist entweder richtig oder falsch; aber er bleibt das, was er ist, ohne Rücksicht darauf, ob auf der Erde denkende oder messende Wesen existieren, und ob diese Wesen, wenn sie existieren, die Einzelheiten physikalischer oder chemischer Prozesse um eine, zwei, oder um hundert Dezimalstellen genauer kontrollieren können, als wir das heute zu tun vermögen. Die Grenzen des Satzes, falls sie überhaupt vorhanden sind, können notwendig nur auf demselben Gebiete liegen, wo auch sein Inhalt liegt: in der beobachteten Natur, und nicht im beobachtenden Menschen. Daran ändert der Umstand nichts, daß wir uns zur Ableitung des Satzes menschlicher Erfahrungen bedienen; das ist überhaupt der einzige Weg für uns, um zur Erkenntnis von Naturgesetzen zu gelangen. Sind sie einmal erkannt, so müssen sie auch als selbständig anerkannt werden, soweit wir überhaupt davon reden können, daß ein Naturgesetz unabhängig vom denkenden Geiste Bestand hat; und wer dies leugnen wollte, müßte die Möglichkeit einer Naturwissenschaft überhaupt leugnen.

Mit dem ersten Hauptsatz verhält es sich ganz ähnlich. Der unmittelbarste unter den allgemeinen Beweisen des Energieprinzips ist wohl für die Mehrzahl der vorurteilslosen Naturforscher die Tatsache der Unmöglichkeit des perpetuum mobile erster Art, und doch wird sich heutzutage kaum jemand finden, der die

Gültigkeit des Energieprinzips mit dem Genauigkeitsgrade des experimentellen Nachweises dieses allgemeinen Erfahrungssatzes in Verbindung bringt. Ebenso wird vermutlich die Zeit kommen, wo auch das Prinzip der Vermehrung der Entropie außer Zusammenhang mit menschlicher Experimentierkunst, von einzelnen Metaphysikern wohl gar als *a priori* gültig hingestellt werden wird. Bis dahin wird es aber für den Anhänger sowohl wie für den Gegner des zweiten Hauptsatzes keine wirksamere Waffe zum Kampfe um seine Allgemeingültigkeit geben, als das unablässige Bemühen, den tatsächlichen Inhalt dieses Satzes bis in seine äußersten Konsequenzen zu verfolgen und jede derselben, korrekt formuliert, vor den Richterstuhl der höchsten Instanz, der Erfahrung, zu bringen. Wie die Entscheidung dann auch fallen möge, unter allen Umständen wird uns aus dieser Taktik sicher bleibender Gewinn erwachsen, da wir damit dem Hauptzweck der ganzen Naturforschung dienen: der Bereicherung unseres tatsächlichen Wissens.

Drittes Kapitel. Allgemeine Folgerungen.

§ 137. Die erste Anwendung, welche wir von dem im vorigen Kapitel in die allgemeinste Form gebrachten Entropieprinzip machen wollen, betrifft den früher im § 90 für ein ideales Gas eingehend beschriebenen CARNOTschen umkehrbaren Kreisprozeß zwischen zwei Wärmereservoiren, diesmal aber, statt mit einem idealen Gase, ausgeführt mit einem ganz beliebigen System, wobei auch chemische Wirkungen, wofern sie nur reversibel sind, nicht ausgeschlossen werden. Indem wir wegen der Bezeichnungen usw. auf das dort Gesagte verweisen, können wir sogleich das Resultat hier aussprechen.

Der erste Hauptsatz verlangt für den Kreisprozeß, daß die von dem wärmeren Reservoir abgegebene Wärmemenge Q_2 äquivalent ist der Summe der von dem System geleisteten Arbeiten: $A' = -A$ und der von dem kühleren Reservoir aufgenommenen Wärmemenge: $Q_1' = -Q_1$, also, ebenso wie in (42)

$$Q_2 = A' + Q_1'$$

oder:

$$Q_1 + Q_2 + A = 0. \tag{63}$$

Der zweite Hauptsatz verlangt wegen der Reversibilität des Prozesses, daß alle Körper, welche am Schluß des Prozesses irgendwelche innere Veränderungen aufweisen, — und das sind hier nur die beiden Wärmebehälter — die nämliche Entropiesumme besitzen wie am Anfang. Nun beträgt die

Entropieänderung des ersten Reservoirs nach (62):

$$\frac{Q_1'}{T_1} = -\frac{Q_1}{T_1}, \text{ die des zweiten: } -\frac{Q_2}{T_2}. \tag{64}$$

Also ist die Summe:

$$\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0, \tag{65}$$

woraus in Verbindung mit (63) folgt:

$$Q_1 : Q_2 : A = (-T_1) : T_2 : (T_1 - T_2)$$

genau die Gleichung (44), nur daß hier über die Natur des Systems, mit welchem der Kreisprozeß ausgeführt wurde, gar keine Voraussetzung gemacht ist.

Um also durch einen CARNOTschen reversibeln Kreisprozeß, ausgeführt mit einer beliebigen Substanz zwischen zwei Wärmebehältern von den Temperaturen T_1 und $T_2 > T_1$, die Arbeit A' aus dem wärmeren Behälter zu gewinnen, muß man die Wärmemenge:

$$Q_1' = \frac{T_1}{T_2 - T_1} A'$$

vom wärmeren Behälter zum kälteren übergehen lassen. Oder umgekehrt ausgedrückt: man kann den Übergang der Wärme Q_1' von T_2 auf T_1 mittels eines reversibeln Kreisprozesses dazu benutzen, um die Arbeit

$$A' = \frac{T_2 - T_1}{T_1} Q_1' \tag{66}$$

aus dem wärmeren Behälter zu gewinnen.

§ 138. Ist der Kreisprozeß nicht reversibel, kommen also in seinem Verlauf irgendwelche irreversible physikalische oder chemische Änderungen des Systems vor, so bleibt die Energiegleichung (63) bestehen, dagegen tritt für die Entropieänderung der Wärmebehälter statt (65) die Ungleichung ein:

$$-\frac{Q_1}{T_1} - \frac{Q_2}{T_2} > 0.$$

Hierbei ist wohl zu bemerken, daß die Ausdrücke (64) für die Entropieänderung der Wärmebehälter auch hier bestehen bleiben, wenn wir nur die Voraussetzung festhalten, daß T_1 und T_2 die Temperaturen der

beiden Wärmebehälter bedeuten, und daß etwaige Volumenänderungen der als Wärmebehälter dienenden Körper in umkehrbarer Weise stattfinden.

Also:

$$\frac{Q_1}{T_1} + \frac{Q_2}{T_2} < 0 \tag{67}$$

oder:

$$Q_2 < \frac{T_2}{T_1} Q_1',$$

woraus in Verbindung mit (63) hervorgeht:

$$A' = -A = Q_1 + Q_2 < Q_1 + \frac{T_2}{T_1} Q_1'$$
$$A' < \frac{T_2 - T_1}{T_1} Q_1',$$

d. h. die mittels eines Kreisprozesses durch den Übergang der Wärme Q_1' von T_2 auf T_1 aus dem wärmeren Behälter zu gewinnende Arbeit A' ist für einen irreversibeln Prozeß stets kleiner als für einen reversibeln Prozeß. Letztere, durch (66) dargestellte Arbeit gibt also zugleich das Maximum der Arbeit an, welches überhaupt durch einen Kreisprozeß mit irgend einem System durch den Übergang der Wärme Q_1' von T_2 auf T_1 aus dem wärmeren Behälter zu gewinnen ist.

Wenn speziell $A' = 0$, so folgt aus der Energiegleichung (63):

$$Q_2 = -Q_1 = Q_1'$$

und die Ungleichung (67) geht über in:

$$Q_2 \cdot \left(\frac{1}{T_2} - \frac{1}{T_1} \right) < 0.$$

In diesem Falle besteht die ganze durch den Kreisprozeß hervorgerufene Veränderung einfach in dem Übergang der Wärme Q_2 aus dem Behälter von der Temperatur T_2 in den von der Temperatur T_1, und die letzte Ungleichung besagt, daß dieser Übergang immer in der Richtung vom wärmeren zum kälteren Reservoir erfolgt.

Wiederum ein spezieller Fall eines derartigen Prozesses ist der direkte Übergang der Wärme durch Leitung von dem einen Behälter zum anderen, ohne jede tatsächliche Beteiligung des Systems, welches den Kreisprozeß

durchmacht. Die Wärmeleitung ist also, wie man hier sieht, irreversibel, da sie eine Zunahme der Summe der Entropien beider Wärmebehälter bedingt.

§ 139. Machen wir noch die Anwendung auf einen beliebigen, reversibeln oder irreversibeln, Kreisprozeß mit irgend einem System von Körpern, in dessen Verlauf nur ein einziges Wärmereservoir von der konstanten Temperatur T beliebig oft zur Benutzung kommt. Wie auch der Prozeß im einzelnen beschaffen sein mag, am Schluß desselben ist keine andere Entropieänderung in der Natur eingetreten, als diejenige, welche das benutzte Reservoir erlitten hat. Nach dem ersten Hauptsatz ist die Summe der im ganzen von außen auf das System ausgeübten Arbeit A und der im ganzen dem System aus dem Reservoir zugeführten Wärme Q:

$$A + Q = 0.$$

Nach dem zweiten Hauptsatz ist die Entropieänderung des Reservoirs, wenn wir wieder, wie immer, voraussetzen, daß etwaige Volumenveränderungen des Reservoirs in umkehrbarer Weise stattfinden:

$$-\frac{Q}{T} \geqq 0$$

$$\text{oder:} \qquad Q \leqq 0, \qquad \text{folglich:} \qquad A \geqq 0,$$

d. h. es ist Arbeit verbraucht und Wärme im Reservoir erzeugt worden. Diese Ungleichung ist die analytische Formulierung der Unmöglichkeit des perpetuum mobile zweiter Art.

Ist im Grenzfall der Prozeß reversibel, so verschwindet das Ungleichheitszeichen, und es ist sowohl die Wärme Q als auch die Arbeit A gleich Null. Auf diesem Satze beruht die große Fruchtbarkeit des zweiten Hauptsatzes in seiner Anwendung auf isotherme reversible Kreisprozesse.

§ 140. Wir wollen uns jetzt nicht mehr mit Kreisprozessen, sondern mit der allgemeinen Frage nach der Richtung irgend einer in der Natur eintretenden Veränderung eines beliebig gegebenen Systems beschäftigen. Besonders bei den chemischen Vorgängen spielt ja diese Frage eine wichtige Rolle. Der zweite Hauptsatz, in Verbindung mit dem ersten, erteilt hierauf eine allgemeine Antwort, da er eine notwendige Bedingung für jede in der Natur stattfindende Änderung enthält. Wir denken uns irgend ein homogenes oder heterogenes System von Körpern, von gemeinsamer Temperatur T, und fragen nach den Bedingungen des Eintritts irgend einer physikalischen oder chemischen Veränderung.

Nach dem ersten Hauptsatz ist für eine unendlich kleine Zustandsänderung:

$$dU = Q + A, \tag{68}$$

wenn U die Gesamtenergie des Systems, Q die während des betrachteten Vorgangs von außen in das System eintretende Wärmemenge, und A die von außen gegen das System geleistete Arbeit bezeichnet.

Nach dem zweiten Hauptsatz ist die Änderung der Summe der Entropien aller an dem Vorgang irgendwie beteiligten Körper:

$$dS + dS_0 \geqq 0,$$

wenn S die Entropie des Systems, S_0 die Entropie des umgebenden Mediums (atmosphärische Luft, kalorimetrische Flüssigkeit, Gefäßwand) bezeichnet, welches infolge von Arbeitsleistung oder Wärmeabgabe auch an dem Vorgang beteiligt sein kann.

Das Gleichheitszeichen gilt für reversible Vorgänge, die allerdings nur als idealer Grenzfall der in Wirklichkeit möglichen Vorgänge zu betrachten sind (§ 115).

Setzen wir voraus, daß etwaige Volumenänderungen des umgebenden Mediums in umkehrbarer Weise erfolgen, so ist nach (62):

$$dS_0 = -\frac{Q}{T}, \text{ also: } dS - \frac{Q}{T} \geqq 0$$

oder nach (68):

$$dS - \frac{dU - A}{T} \geqq 0. \tag{69}$$

Anders geschrieben:

$$dU - T\,dS \leqq A. \tag{70}$$

In dieser Relation gipfeln alle bisher von verschiedenen Autoren auf verschiedenen Wegen aus dem zweiten Hauptsatz für den Eintritt thermodynamisch-chemischer Veränderungen hergeleiteten Schlüsse. Da der Differentialausdruck links im allgemeinen nicht das vollständige Differential einer bestimmten Größe bildet, so läßt sich die Relation nicht allgemein integrieren, d. h. der zweite Hauptsatz gestattet keinen allgemeinen Ausspruch über eine endliche Zustandsänderung des Systems allein, falls man von den äußeren Bedingungen, denen das System unterworfen ist, nichts Näheres weiß; wie das ja auch von vornherein einleuchtend ist und ebenso auch für den ersten Hauptsatz gilt. Um zu einem Gesetz für eine endliche Zustandsänderung des Systems allein zu gelangen, muß man solche äußere Bedingungen kennen,

welche die Integration des Differentialausdrucks gestatten.[1] Unter diesen sind im folgenden die merkwürdigsten Fälle hervorgehoben.

§ 141. Erster Fall. Adiabatischer Vorgang. Bei Ausschluß des Wärmeaustausches mit der Umgebung ist $Q = 0$, also nach (68):

$$dU = A$$

und infolgedessen nach (70):

$$dS \geqq 0,$$

d. h. die Entropie des Systems nimmt zu oder bleibt konstant. Diesen Fall haben wir schon genügend erörtert.

§ 142. Zweiter Fall. Isothermer Vorgang. Bei konstant gehaltener Temperatur T geht (70) über in:

(70a) $$d(U - TS) \leqq A,$$

d. h. die Zunahme der Größe $(U - TS)$ ist kleiner, im Grenzfall ebensogroß wie die von außen her gegen das System geleistete Arbeit. Da man in der Thermochemie bei der Messung von Wärmetönungen den Endzustand stets auf die Temperatur des Anfangszustandes reduziert, so ist dieser Satz für die Anwendung auf chemische Prozesse besonders geeignet. Setzen wir:

(71) $$U - TS = F,$$

so ist für eine reversible isotherme Zustandsänderung:

$$dF = A,$$

integriert:

(72) $$F_2 - F_1 = \sum A,$$

d. h. bei einem endlichen reversibeln isothermen Vorgang ist die ganze von außen auf das System ausgeübte Arbeit gleich der Zunahme von F, oder

[1] Man spricht zwar manchmal von einem System, das „sich selbst überlassen“ ist; aber dies darf natürlich niemals so gedeutet werden, als ob in gewissen Fällen bestimmte Änderungen in dem System vor sich gehen könnten, ohne daß überhaupt äußere Bedingungen vorgeschrieben sind. Denn in der Natur gibt es kein endliches „sich selbst überlassenes“ System. Irgendwelche Bedingungen an der Oberfläche sind immer vorhanden, sei es, daß das System in ein Gefäß mit festen oder elastischen Wänden eingeschlossen ist, oder daß es an die freie Atmosphäre grenzt, oder daß es sich in einem evakuierten Raum befindet, usw.

die ganze von dem System nach außen hin geleistete Arbeit ist gleich der Abnahme von F, hängt also nur von dem Anfangs- und Endzustand des Vorgangs ab. Ist $F_1 = F_2$, wie z. B. bei einem Kreisprozeß, so ist die äußere Arbeit gleich Null. (Vgl. § 139.)

Da somit die Funktion F zu der äußeren Arbeit in ganz derselben Beziehung steht, wie die Energie U nach der Gleichung (17) zu der Summe von äußerer Arbeit und äußerer Wärme, so heißt F nach H. v. Helmholtz die „freie Energie" des Systems (vollständiger würde sie heißen: „freie Energie für isotherme Vorgänge"), und dementsprechend U die „Gesamtenergie", und der Rest:

$$U - F = TS = G$$

die „gebundene" Energie des Systems. Letztere liefert demnach für einen reversibeln isothermen Vorgang durch ihre Änderung die äußere Wärmeaufnahme. Man kann dies auch so ausdrücken: Bei reversibeln isothermen Prozessen zerfällt der Satz der Erhaltung der Energie:

$$U_2 - U_1 = \sum A + \sum Q$$

in zwei getrennte Teile: den Satz der freien Energie:

$$F_2 - F_1 = \sum A,$$

und den Satz der gebundenen Energie:

$$G_2 - G_1 = \sum Q.$$

Aber es ist wohl zu beachten, daß diese Zerlegung nur für reversible isotherme Änderungen durchführbar ist.

Bei irreversibeln isothermen Vorgängen ist nach (70a):

$$dF < A,$$

integriert:

(73) $$F_2 - F_1 < \sum A,$$

d. h. die freie Energie nimmt weniger zu als der verbrauchten Arbeit entspricht. In Verbindung mit dem obigen Resultat für reversible Prozesse kann man dies auch so formulieren: Bei irreversibeln isothermen Prozessen ist die verbrauchte Arbeit immer größer, also die gewonnene Arbeit immer kleiner als diejenige Arbeit, welche man bei der nämlichen Zustandsänderung des Systems verbrauchen bez. gewinnen würde, wenn sie auf reversibelm

Wege vor sich ginge. Denn die letztere wird eben nach (72) durch die Differenz der freien Energie am Anfang und am Ende des Prozesses gegeben.

Daher liefert ein, im übrigen beliebiger, reversibler Übergang des Systems von einem Zustand zu einem anderen immer das Maximum der Arbeit, welches überhaupt aus einem isothermen Übergang des Systems von dem einen Zustand zum anderen gewonnen werden kann, während bei jedem irreversibeln Übergang ein gewisser Arbeitsbetrag, nämlich die Differenz des Maximums der zu gewinnenden Arbeit (Abnahme der freien Energie) und der wirklich gewonnenen Arbeit, verloren geht.

Wenn hier davon gesprochen wird, daß der Übergang eines Systems aus einem Zustand in einen anderen einmal auf irreversibelm, einmal auf reversibelm Wege vorgenommen wird, so liegt darin kein Widerspruch mit dem anderen, selbstverständlichen Satze, daß zwischen zwei Zuständen eines Systems nur entweder ein reversibler oder ein irreversibler Übergang möglich ist, ohne daß in anderen Körpern Änderungen zurückbleiben. In dem hier betrachteten Falle können in der Tat Änderungen in einem anderen Körper zurückbleiben, nämlich in dem das System umgebenden Medium, welches nach § 140 im allgemeinen positive oder negative Wärme an das System abgibt, und in unserem Falle abgeben m u ß, um das System auf konstanter Temperatur zu erhalten.

§ 143. Erfolgt ein isothermer Prozeß, wie die meisten chemischen Prozesse, mit verschwindend kleiner Arbeitsleistung:

$$\sum A = 0,$$

so ist nach (73):

$$F_2 - F_1 < 0,$$

d. h. die freie Energie nimmt ab. Die Größe dieser Abnahme kann man als ein quantitatives Maß benutzen für die Arbeit der Kräfte (chemische Verwandtschaft, Affinität, Avidität), welche den Prozeß veranlassen; dieselbe geht dabei als äußere Arbeit verloren.

Es werde z. B. eine wäßrige Lösung eines nichtflüchtigen Salzes durch Zusatz von Wasser auf isothermem Wege verdünnt, indem die Verdünnungswärme von einem passenden Wärmereservoir aufgenommen oder geliefert wird, je nachdem die Energie U_2 der verdünnten Lösung (Endzustand) kleiner oder größer ist als die Summe U_1 der Energie der unverdünnten Lösung und der Energie der zugesetzten Wassermenge (Anfangszustand). Die freie Energie F_2 der verdünnten Lösung dagegen ist nach der letzten Ungleichung notwendig kleiner als die Summe F_1 der freien Energie der unverdünnten Lösung und der freien Energie des

zugesetzten Wassers. Der Betrag der Abnahme der freien Energie, oder die von der „Anziehungskraft der Lösung auf das Wasser“ beim Verdünnen geleistete Arbeit, kann gemessen werden, indem man den Verdünnungsprozeß auf irgend einem reversibeln isothermen Wege vollzieht, wobei dann nach Gleichung (72) dieser Arbeitsbetrag wirklich als äußere Arbeit gewonnen wird. Ein solcher reversibler Übergang ist z. B. folgender: Man lasse das zuzusetzende Wasser zunächst bei konstanter Temperatur unter dem Druck seines gesättigten Dampfes unendlich langsam verdampfen. Wenn alles in Dampf verwandelt ist, lasse man den Dampf sich isotherm und umkehrbar weiter ausdehnen, so lange, bis die Dichte des Dampfes derjenigen gleich ist, welche gesättigter Wasserdampf bei der betreffenden Temperatur in Berührung mit der Lösung besitzt. Nun bringe man den Dampf mit der Lösung in dauernde Berührung; das Gleichgewicht wird dadurch nicht gestört. Schließlich kondensiere man durch unendlich langsame isotherme Kompression den unmittelbar über der Lösung befindlichen Wasserdampf vollständig; er verteilt sich dann gleichmäßig durch die ganze Lösung. Dieser isotherme Prozeß besteht aus lauter Gleichgewichtszuständen, er ist also reversibel, und die durch ihn gewonnene äußere Arbeit repräsentiert daher zugleich die bei direkter Mischung eingetretene Abnahme der freien Energie: $F_1 - F_2$. Eine jede andere Methode der isothermen reversibeln Überführung von Wasser in die Lösung muß natürlich den nämlichen Wert dieser Differenz liefern. Daher kann man z. B. die elektromotorische Kraft einer galvanischen Konzentrationskette aus dem Dampfdruck der Lösung berechnen (Helmholtz).

Nehmen wir als weiteres Beispiel ein Knallgasgemenge, das durch einen elektrischen Funken zur Explosion gebracht wird. Der Funken spielt hier nur eine sekundäre Rolle, als auslösende Wirkung, da seine Energie gegen die übrigen zum Umsatz gelangenden Energiemengen nicht in Betracht kommt. Die Arbeit der chemischen Verwandtschaftskräfte, welche sich in diesem Prozeß betätigt, wird gemessen durch diejenige Arbeit, die man durch die chemische Vereinigung von Wasserstoff und Sauerstoff gewinnen könnte, wenn dieselbe auf irgend einem reversibeln Wege vorgenommen würde. Durch Division dieser Arbeit durch die Zahl der oxydierten Wasserstoffmoleküle erhält man ein Maß für die Größe der Kraft, mit welcher ein Wasserstoffmolekül sich zu oxydieren strebt. Doch hat diese Definition der chemischen Kraft zunächst nur insofern Bedeutung, als sie eben mit jener Arbeit zusammenhängt.

§ 144. In dem Ausdruck (71) der freien Energie überwiegen bei chemischen Vorgängen die Änderungen des ersten Gliedes: U oft bei weitem die des zweiten Gliedes: TS. Deshalb kann man häufig statt der Abnahme

von F auch die Abnahme von U, d. h. die Wärmetönung, als ein Maß der chemischen Arbeit ansehen, und den Satz aussprechen, daß die ohne äußere Arbeit eintretenden chemischen Umwandlungen im Sinne größter Wärmeentwickelung erfolgen (Prinzip von BERTHELOT). Indessen bei hohen Temperaturen, wo T, und bei Gasen und verdünnten Lösungen, wo S groß wird, kann man die Vernachlässigung des Gliedes TS nicht ohne merklichen Fehler begehen. Daher erfolgen bei höherer Temperatur und in Gasen und verdünnten Lösungen chemische Änderungen häufig auch in der Richtung steigender Gesamtenergie, d. h. unter Wärmeabsorption.

§ 145. Bei allen diesen Sätzen ist streng daran festzuhalten, daß sie sich nur auf isotherme Zustandsänderungen beziehen. Um die Frage zu beantworten, wie sich die freie Energie bei anderen Zustandsänderungen verhält, hat man nur aus (71) das vollständige Differential zu bilden:

$$dF = dU - TdS - SdT$$

und dies in die allgemein gültige Beziehung (70) einzusetzen. Man erhält dann für einen beliebigen physikalischen oder chemischen Vorgang:

$$dF \leqq A - SdT,$$

d. h. wenn die Temperatur sich ändert, besteht eine wesentlich verwickeltere Beziehung zwischen der geleisteten Arbeit A und der Änderung der freien Energie F, — eine Beziehung, die sich im allgemeinen wohl kaum fruchtbar verwerten läßt.

§ 146. Berechnen wir den Wert der freien Energie für ein ideales Gas. Da hierfür nach Gleichung (35)

$$U = Mu = M(c_v T + b) \qquad (b \text{ konstant})$$

und nach Gleichung (52)

$$S = M(c_v \log T + \frac{R}{m} \log v + a), \qquad (a \text{ konstant})$$

so ist nach (71)

$$F = M\left\{T(c_v - a - c_v \log T) - \frac{RT}{m} \log v + b\right\} \tag{74}$$

also behaftet mit einer additiven lineären Funktion von T, die hier ganz nach Willkür fixiert werden kann. Bezüglich der Konstanten b läßt sich

diese Willkür auf thermodynamischem Wege überhaupt nicht beseitigen. Dagegen ergibt sich ein Verfahren zur Berechnung der Konstanten a aus dem NERNSTschen Wärmetheorem. (Vgl. § 287.)

Bei einer isothermen Zustandsänderung des Gases ist nach § 142:

$$dF \leqq A$$

oder nach (74)

$$dF = -\frac{MTR}{m}\frac{dv}{v} = -pdV \leqq A.$$

Ist die Zustandsänderung reversibel, so ist also die von außen aufgewendete Arbeit $A = -pdV$. Ist aber die Änderung irreversibel, so gilt das Ungleichheitszeichen, d. h. die Kompressionsarbeit ist größer, oder die Ausdehnungsarbeit geringer als diejenige Arbeit, welche man bei reversibler Volumenänderung aufwenden bez. gewinnen würde.

§ 147. Dritter Fall. Isotherm-isobarer Vorgang. Wenn außer der Temperatur T auch der äußere Druck p, unter dem das System stehen möge, andauernd konstant gehalten wird, so läßt sich der Betrag der von außen aufgewendeten Arbeit angeben:

$$A = -pdV$$

und der Ausdruck in (69) stellt ein vollständiges Differential vor:

$$d\left(S - \frac{U + pV}{T}\right) \geqq 0.$$

Man kann also dann auch für endliche Zustandsänderungen den Satz aussprechen, daß die Funktion:

$$S - \frac{U + pV}{T} = \Phi \tag{75}$$

notwendig zunimmt, und nur im Grenzfall, für reversible Änderungen, konstant bleibt.

§ 148. Gleichgewichtsbedingungen. Die allgemeinste aus der Thermodynamik für ein Körpersystem abzuleitende Gleichgewichtsbedingung beruht auf dem Satz, daß in einem System dann keine Veränderung eintreten kann, wenn eine zu einer Veränderung notwendige Bedingung in keiner Weise erfüllbar ist.

Nun ist nach (69) für jede in Wirklichkeit eintretende Veränderung eines Systems:

$$dS = \frac{dU - A}{T} > 0.$$

Denn das Gleichheitszeichen würde nur idealen Änderungen entsprechen, und ideale Änderungen treten in der Natur nicht ein. Folglich muß Gleichgewicht bestehen, wenn für jede mit den gegebenen festen Bedingungen des Systems verträgliche Zustandsänderung:

$$\delta S - \frac{\delta U - A}{T} \leqq 0.$$

Hier bezieht sich das Zeichen δ, im Gegensatz zum Zeichen d, das der wirklichen Veränderung entspricht, auf irgend eine beliebige virtuelle unendlich kleine Zustandsänderung des Systems.

§ 149. In den meisten von uns weiter zu behandelnden Fällen ist, wenn eine gewisse virtuelle unendlich kleine Zustandsänderung mit den festen Bedingungen des Systems verträglich ist, auch die gerade entgegengesetzte, durch die entgegengesetzten Vorzeichen aller Variationen dargestellte Zustandsänderung mit ihnen verträglich. Das gilt immer dann, wenn die festen Bedingungen durch Gleichungen, nicht durch Ungleichungen ausgedrückt werden. In einem solchen Falle könnte man, falls für eine virtuelle Änderung in obiger Bedingung das Zeichen $<$ gelten würde, einfach die entgegengesetzte Variation nehmen, um eine Zustandsänderung zu erhalten, welche den Bedingungen der wirklichen Vorgänge genügt und daher in der Natur eintreten kann. Hier ist also das Gleichgewicht nur dann nach allen Richtungen hin gesichert, wenn für jede mit den festen Bedingungen verträgliche Änderung:

$$\delta S - \frac{\delta U - A}{T} = 0. \tag{76}$$

Diese Gleichung spricht eine für das Gleichgewicht hinreichende, aber, wie wir eben sahen, nicht gerade in allen Fällen notwendige Bedingung aus. Ja selbst wenn die festen Bedingungen eine Umkehrung der Vorzeichen aller Variationen gestatten, besteht erfahrungsgemäß manchmal Gleichgewicht, ohne daß die letzte Gleichung erfüllt ist, d. h. es tritt unter Umständen in der Natur eine Veränderung nicht ein, obwohl sie sowohl den festen Bedingungen als auch den Forderungen des zweiten Hauptsatzes Genüge leisten würde. Man wird dadurch zu dem Schlusse geführt, daß sich in einem solchen Falle dem Eintritt der Veränderung eine Art Widerstand entgegenstellt, der wegen der Richtung, in welcher er wirkt, in manchen Fällen auch Trägheitswiderstand oder passiver Widerstand genannt wird. Solch ein Gleichgewichtszustand ist immer in gewissem Sinne labil; denn oft genügt eine geringfügige und mit den im System vorhandenen Größen quantitativ gar nicht vergleichbare Störung, um die Veränderung, dann oft

mit großer Heftigkeit, eintreten zu lassen. Beispiele hierfür bieten eine unter ihre Gefriertemperatur abgekühlte Flüssigkeit, ein übersättigter Dampf, eine übersättigte Lösung, eine explosible Substanz usw. Wir werden uns vorwiegend mit den Bedingungen des stabilen Gleichgewichts beschäftigen, wie sie aus der Bedingung (76) folgen.

Diese Gleichung läßt sich unter gewissen Umständen als Maximum- oder Minimumbedingung aussprechen, nämlich immer, aber auch nur dann, wenn die äußeren Bedingungen, unter denen das System gehalten wird, derart sind, daß die linke Gleichungsseite als Variation einer bestimmten endlichen Funktion dargestellt werden kann. Im folgenden sind die wichtigsten derartigen Fälle hervorgehoben; sie entsprechen ganz den oben für gewisse spezielle Veränderungen abgeleiteten Sätzen, aus deren Inhalt auch unmittelbar zu erkennen ist, ob es sich hier um ein Maximum oder um ein Minimum handelt.

§ 150. Erster Fall (§ 141). Bei Ausschluß des Wärmeaustausches mit der Umgebung ist nach dem ersten Hauptsatz:

$$\delta U = A$$

und daher aus (76)

$$\delta S = 0. \tag{77}$$

D. h. unter allen Zuständen des Systems, die bei verhinderter äußerer Wärmezufuhr auseinander hervorgehen können, ist der Gleichgewichtszustand durch ein Maximum der Entropie ausgezeichnet. Wenn es mehrere Zustände gibt, in welchen die Entropie einen Maximalwert besitzt, so stellt jeder derselben einen Gleichgewichtszustand dar. Wenn aber der Wert der Entropie in einem bestimmten Zustand größer ist als in allen übrigen in Betracht kommenden, so bezeichnet dieser Zustand das absolut stabile Gleichgewicht. Denn von ihm aus ist überhaupt keine Veränderung mehr möglich.

§ 151. Zweiter Fall (§ 142). Bei konstant gehaltener Temperatur geht (76) über in:

$$\delta\left(S - \frac{U}{T}\right) + \frac{A}{T} = 0$$

oder nach (71):

$$-\delta F = -A,$$

d. h. unter allen Zuständen, die das System bei konstant gehaltener Temperatur annehmen kann, ist ein Gleichgewichtszustand dadurch ausgezeichnet, daß die freie Energie des Systems nicht abnehmen kann, ohne daß das System gleichzeitig eine äquivalente Arbeit nach außen hin leistet.

Wenn der Betrag der äußeren Arbeit zu vernachlässigen ist, wie bei konstant gehaltenem Volumen, oder bei vielen chemischen Vorgängen, so ist $A = 0$, und die Bedingung des Gleichgewichts lautet:

$$\delta F = 0,$$

d. h. unter allen Zuständen, die bei konstant gehaltener Temperatur ohne Leistung äußerer Arbeit auseinander hervorgehen können, ist der stabilste Gleichgewichtszustand durch das absolute Minimum der freien Energie ausgezeichnet.

§ 152. Dritter Fall (§ 147). Wird außer der Temperatur T der Druck p, dem das System unterworfen ist, gleichmäßig und konstant gehalten, so hat man

$$A = -p\,\delta V \tag{78}$$

und die Gleichgewichtsbedingung (76) wird:

$$\delta\left(S - \frac{U + pV}{T}\right) = 0$$

oder nach (75)

$$\delta\Phi = 0, \tag{79}$$

d. h. bei konstanter Temperatur und konstantem Druck nimmt das System im stabilsten Gleichgewicht denjenigen Zustand an, welchem das absolute Maximum der Funktion Φ entspricht.

Wir werden im nächsten Abschnitt Gleichgewichtszustände verschiedener Systeme auf Grund der hier abgeleiteten Sätze betrachten, und dabei nach der Reihe von einfacheren zu komplizierteren Fällen aufsteigen.

§ 152a. Bei der mathematischen Behandlung thermodynamischer Gleichgewichtsprobleme ist in erster Linie von Wichtigkeit die Wahl der unabhängigen Variabeln. Je nachdem diese in der einen oder der anderen Weise getroffen wird, treten andere charakteristische Funktionen in den Vordergrund, und hierdurch unterscheidet sich die Form der Darstellung bei den verschiedenen Autoren. In jedem Falle, bei jeder Wahl der unabhängigen Variabeln, existiert nämlich, wie zuerst MASSIEU gezeigt hat, eine charakteristische Funktion, aus deren Ausdruck alle thermodynamischen Eigenschaften des betrachteten im thermodynamischen Gleichgewicht befindlichen Systems eindeutig, durch einfache Differentiation, abgeleitet werden können, und zwar ist dies immer gerade diejenige Funktion, deren Maximum bzw. Minimum bei

Konstanthaltung der betreffenden unabhängigen Variabeln gemäß den Sätzen der letzten Paragraphen das thermodynamische Gleichgewicht charakterisiert.

Wenn z. B. als unabhängige Variable die Energie U und das Volumen V gewählt werden, so stellt die Entropie S die charakteristische Funktion dar. In der Tat findet man aus der Gleichung:

$$dS = \frac{dU + p\,dV}{T} = \frac{\partial S}{\partial U}\,dU + \frac{\partial S}{\partial V}\,dV$$

unmittelbar:

$$\left(\frac{\partial S}{\partial U}\right)_V = \frac{1}{V} \quad \text{und} \quad \left(\frac{\partial S}{\partial V}\right)_U = \frac{p}{T}.$$

Ist also S als Funktion von U und V bekannt, so ergeben sich für T und p die Ausdrücke:

$$T = \frac{1}{\dfrac{\partial S}{\partial U}}, \quad p = \frac{\dfrac{\partial S}{\partial V}}{\dfrac{\partial S}{\partial U}},$$

und daraus auch S und U als Funktionen von T und p.

Wenn andrerseits V und T als unabhängige Variable gewählt werden, so ist die freie Energie F die charakteristische Funktion. Denn aus (71) folgt allgemein:

$$dF = dU - T\,dS - S\,dT$$

oder:

$$dF = -p\,dV - S\,dT.$$

Also:

$$\left(\frac{\partial F}{\partial V}\right)_T = -p \quad \text{und} \quad \left(\frac{\partial F}{\partial T}\right)_V = -S.$$

Ist also F als Funktion von V und T bekannt, so ergeben sich daraus unmittelbar die Ausdrücke:

(79a) $$p = -\frac{\partial F}{\partial V}, \quad S = -\frac{\partial F}{\partial T}, \qquad U = F + TS = F - T\,\frac{\partial F}{\partial T}.$$

Wenn endlich p und T als unabhängige Variable gewählt werden, so ist die charakteristische Funktion:

$$\Phi = S - \frac{U + pV}{T}.$$

Denn es folgt allgemein:

$$d\Phi = dS - \frac{dU + p\,dV + V\,dp}{T} + \frac{U + pV}{T^2}\,dT,$$

oder:

$$d\Phi = -\frac{V}{T}\,dp + \frac{U+pV}{T^2}\,dT.$$

Also:

(79b) $$\left(\frac{\partial\Phi}{\partial p}\right)_T = -\frac{V}{T} \quad \text{und} \quad \left(\frac{\partial\Phi}{\partial T}\right)_p = \frac{U+pV}{T^2}.$$

Ist also Φ als Funktion von p und T bekannt, so ergeben sich daraus unmittelbar die Ausdrücke:

(79c) $$V = -T\,\frac{\partial\Phi}{\partial p}, \qquad U = T\left(T\,\frac{\partial\Phi}{\partial T} + p\,\frac{\partial\Phi}{\partial p}\right),$$
$$S = \Phi + T\,\frac{\partial\Phi}{\partial T},$$

wodurch alle thermodynamischen Eigenschaften des betrachteten Systems eindeutig bestimmt sind.

§ 152b. So wichtig im Prinzip genommen die Ableitbarkeit aller thermodynamischen Eigenschaften aus den charakteristischen Funktionen S, F oder Φ sich darstellt, gewährt sie doch praktisch erst dann direkten Nutzen, wenn es möglich ist, den Ausdruck der charakteristischen Funktion, wie er sich aus den unabhängigen Variabeln zusammensetzt, wirklich anzugeben. Daher ist es von besonderer Wichtigkeit zu untersuchen, inwieweit dieser Ausdruck durch direkte Wärmemessungen gewonnen werden kann.

Da Wärmetönungen in der Regel auf konstante Temperatur und konstanten Druck bezogen werden, setzen wir hier T und p als unabhängige Variable voraus. Dann läßt sich die Wärmetönung irgend eines physikalisch-chemischen Prozesses (z. B. Verdampfung, Oxydation) zurückführen auf die Gibbssche Wärmefunktion (§ 100)

$$W = U + pV,$$

deren Änderung ganz allgemein den Betrag der von außen zugeführten Wärme angibt. Der Ausdruck von W ist natürlich eindeutig bestimmt durch die charakteristische Funktion Φ nach Gleichung (79b)

(79d) $$W = T^2\,\frac{\partial\Phi}{\partial T}.$$

Will man nun umgekehrt, nachdem W durch Wärmemessungen gefunden ist, Φ durch W bestimmen, so hat man die letzte Gleichung bei konstantem Druck p zu integrieren:

(79e) $$\Phi = \int \frac{W}{T^2}\,dT.$$

In diesem Integral treten zwei zunächst unbestimmt bleibende additive Glieder auf, nämlich erstens die Integrationskonstante, zweitens ein Glied, welches davon herrührt, daß in dem Ausdruck von W, ebenso wie in dem von U, eine additive Konstante unbestimmt ist. Das erste Glied ist von der Form a, das zweite von der Form $\frac{b}{T}$. Dementsprechend bleibt in dem Ausdruck der Entropie S nach (79c) eine additive Konstante a, und in dem Ausdruck der freien Energie $F = U - TS$ ein additives Glied von der Form $b - aT$ unbestimmt. Dasselbe ergab sich schon oben im § 146, Gl. (74) für den speziellen Fall eines idealen Gases. Im allgemeinen sind a und b noch vom Druck p abhängig. Näheres über die Bestimmung von Φ aus Wärmemessungen siehe in § 210.

§ 152c. Differentiiert man die Gleichung (79d) nach p bei konstanter Temperatur, so ergibt sich mit Berücksichtigung von (79b):

$$\frac{\partial W}{\partial p} = -T^2 \frac{\partial}{\partial T}\left(\frac{V}{T}\right) = V - T\frac{\partial V}{\partial T}. \tag{79f}$$

Durch diese Gleichung wird eine ganz allgemeine Beziehung zwischen Wärmetönungen und thermischen Volumenänderungen ausgedrückt, die zur experimentellen Prüfung des zweiten Hauptsatzes benutzt werden kann. Denn bei irgend einer isotherm-isobaren Reaktion ist nach § 105 die von außen zugeführte Wärme:

$$Q = W_2 - W_1,$$

wenn die Ziffern 1 und 2 sich auf den Anfangszustand und den Endzustand des Systems beziehen. Nennen wir weiter die bei der Reaktion eintretende Volumenänderung:

$$V_2 - V_1 = \Delta V,$$

so folgt aus (79f):

$$\frac{\partial Q}{\partial p} = -T^2 \frac{\partial}{\partial T}\left(\frac{\Delta V}{T}\right), \tag{79g}$$

woraus z. B. folgt, daß, falls die Wärmetönung einer isotherm-isobaren Reaktion unabhängig ist vom äußeren Druck, die entsprechende Volumenänderung proportional ist der absoluten Temperatur $\left(\frac{\Delta V}{T} = \text{konst.}\right)$, und umgekehrt.

Vierter Abschnitt.

Anwendungen auf spezielle Gleichgewichtszustände.

Erstes Kapitel. Homogenes System.

§ 153. Den Zustand des homogenen (§ 67) Systems nehmen wir, wie früher, als bestimmt an durch seine Masse M, seine Temperatur T und entweder durch den Druck p oder durch das spezifische Volumen $v = \frac{V}{M}$. Wir wollen hier zunächst, außer M, T und v als unabhängige Variable wählen. Dann ist der Druck p, sowie die spezifische Energie $u = \frac{U}{M}$ und die spezifische Entropie $s = \frac{S}{M}$ Funktion von T und v, und zwar gilt für die spezifische Entropie die Definition (61):

$$ds = \frac{du + p\,dv}{T} = \frac{1}{T}\left(\frac{\partial u}{\partial T}\right)_v dT + \frac{\left(\frac{\partial u}{\partial v}\right)_T + p}{T} dv.$$

Andrerseits ist

$$ds = \left(\frac{\partial s}{\partial T}\right)_v dT + \left(\frac{\partial s}{\partial v}\right)_T dv.$$

Folglich, da dT und dv voneinander unabhängig sind:

(79h)
$$\left(\frac{\partial s}{\partial T}\right)_v = \frac{1}{T}\left(\frac{\partial u}{\partial T}\right)_v$$

und

$$\left(\frac{\partial s}{\partial v}\right)_T = \frac{\left(\frac{\partial u}{\partial v}\right)_T + p}{T}.$$

Diese beiden Gleichungen gestatten eine Prüfung des zweiten Hauptsatzes an der Erfahrung. Denn differentiiert man die erste nach v, die zweite

nach T, so ergibt sich:

$$\frac{\partial^2 s}{\partial T\,\partial v} = \frac{1}{T}\frac{\partial^2 u}{\partial T\,\partial v} \quad = \frac{\dfrac{\partial^2 u}{\partial T\,\partial v} + \left(\dfrac{\partial p}{\partial T}\right)_v}{T} - \frac{\left(\dfrac{\partial u}{\partial v}\right)_T + p}{T^2}$$

oder:

$$\left(\frac{\partial u}{\partial v}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_v - p \tag{80}$$

und hierdurch, sowie durch die Gleichung (24) werden die obigen Ausdrücke für die Differentialquotienten von s nach T und v:

$$\begin{cases} \left(\dfrac{\partial s}{\partial T}\right)_v = \dfrac{c_v}{T} \\ \left(\dfrac{\partial s}{\partial v}\right)_T = \left(\dfrac{\partial p}{\partial T}\right)_v \end{cases} \tag{81}$$

Differentiiert man die erste dieser Gleichungen nach v, die zweite nach T, und setzt die erhaltenen Ausdrücke einander gleich, so ergibt sich die Beziehung:

$$\left(\frac{\partial c_v}{\partial v}\right)_T = \left(\frac{\partial^2 p}{\partial T^2}\right)_v, \tag{81a}$$

welche die Abhängigkeit der spezifischen Wärme vom Volumen in Zusammenhang bringt mit der Abhängigkeit des thermischen Spannungskoeffizienten von der Temperatur. Für ideale Gase werden beide unmerklich klein.

§ 154. Die Gleichung (80) in Verbindung mit der Gleichung (28) des ersten Hauptsatzes ergibt die Beziehung:

$$c_p - c_v = T\left(\frac{\partial p}{\partial T}\right)_v \cdot \left(\frac{\partial v}{\partial T}\right)_p, \tag{82}$$

die sich entweder zur Prüfung des zweiten Hauptsatzes oder zur Berechnung von c_v aus c_p verwerten läßt.

Da man $\left(\dfrac{\partial p}{\partial T}\right)_v$ häufig nicht direkt messen kann, so empfiehlt es sich, die Relation (6) zu benutzen, aus welcher folgt:

$$c_p - c_v = -T\left(\frac{\partial p}{\partial v}\right)_T \cdot \left(\frac{\partial v}{\partial T}\right)_p^2. \tag{83}$$

Da $\left(\frac{\partial p}{\partial v}\right)_T$ notwendig negativ, so ist immer $c_p > c_v$; nur im Grenzfall, z. B. wenn der Ausdehnungskoeffizient Null ist, wie für Wasser bei 4°, ist $c_p - c_v = 0$.

Berechnen wir als Beispiel die spezifische Wärme bei konstantem Volumen für Quecksilber von 0° C. unter Atmosphärendruck. Hierfür ist zu setzen:

$$c_p = 0,0333$$
$$T = 273$$
$$\left(\frac{\partial p}{\partial v}\right)_T = -\frac{1013250}{0,000\,0039 \cdot v},$$

wobei die Zahl im Nenner den auf Atmosphären bezogenen Kompressibilitätskoeffizienten (§ 15), die im Zähler den Betrag des Druckes einer Atmosphäre im absoluten Maß (§ 7) bedeutet,

$$v = \frac{1}{13,596}, \quad \text{Volumen von 1 g Quecksilber bei } 0° \text{ C.}$$
$$\left(\frac{\partial v}{\partial T}\right)_p = 0,000\,1812 \cdot v \ (\S\,15) \ \text{(thermischer Ausdehnungskoeffizient).}$$

Um c_v in Kalorien zu erhalten, hat man noch mit dem mechanischen Wärmeäquivalent $4,19 \cdot 10^7$ (§ 61) zu dividieren und berechnet so aus (83):

$$c_p - c_v = \frac{273 \cdot 1013\,250 \cdot 0,000\,1812^2}{0,000\,0039 \cdot 13,596 \cdot 4,19 \cdot 10^7}$$
$$c_p - c_v = 0,0041$$

und daraus mit Benutzung des obigen Wertes von c_p:

$$c_v = 0,0292, \quad \frac{c_p}{c_v} = 1,1.$$

§ 155. Diese für alle Substanzen gültige Berechnung der Differenz der spezifischen Wärmen eröffnet einen Einblick in die Größenordnung der verschiedenen Einflüsse, welche für diese Differenz von Bedeutung sind. Nach der Gleichung (28) des ersten Hauptsatzes ist die Differenz der beiden spezifischen Wärmen:

$$c_p - c_v = \left\{\left(\frac{\partial u}{\partial v}\right)_T + p\right\} \cdot \left(\frac{\partial v}{\partial T}\right)_p$$

durch zwei Ursachen bedingt: erstens durch die Veränderlichkeit der Energie u mit dem Volumen, zweitens durch die bei der Ausdehnung geleistete äußere Arbeit. Die erste Ursache bedingt das Glied:

$$\left(\frac{\partial u}{\partial v}\right)_T \cdot \left(\frac{\partial v}{\partial T}\right)_p,$$

die zweite das Glied:

$$p\left(\frac{\partial v}{\partial T}\right)_p.$$

Um zu untersuchen, welchem von beiden Gliedern der größere Einfluß zukommt, bilden wir das Verhältnis des ersten zum zweiten:

$$\frac{1}{p} \cdot \left(\frac{\partial u}{\partial v}\right)_T$$

oder nach (80):

$$\frac{T}{p}\left(\frac{\partial p}{\partial T}\right)_v - 1 \tag{84}$$

oder nach (6):

$$-\frac{T}{p}\left(\frac{\partial v}{\partial T}\right)_p \cdot \left(\frac{\partial p}{\partial v}\right)_T - 1.$$

Ein Blick in die Tabellen der thermischen Ausdehnungskoeffizienten und der Kompressibilitätskoeffizienten fester und flüssiger Körper lehrt, daß unter gewöhnlichen Umständen das erste Glied dieses Ausdrucks eine große Zahl ist, wogegen das zweite Glied 1 gänzlich zu vernachlässigen ist. Für Quecksilber bei 0° C. z. B. ergeben die obigen Daten:

$$273 \cdot \frac{0,000\,1812}{0,000\,0039} = 12700.$$

Eine Ausnahme bildet z. B. Wasser bei 4° C.

Daraus folgt, daß bei festen und flüssigen Körpern die Differenz $c_p - c_v$ der beiden spezifischen Wärmen in der Regel nicht sowohl durch die bei der Ausdehnung geleistete äußere Arbeit, als vielmehr durch die Abhängigkeit der Energie vom Volumen bedingt ist. Bei idealen Gasen dagegen ist es gerade umgekehrt. Hier ist nach (19) die innere Energie unabhängig vom Volumen, d. h.

$$\left(\frac{\partial u}{\partial v}\right)_T = 0$$

und daher fällt bei der Ausdehnung der Einfluß der inneren Energie gegen den der äußeren Arbeit ganz fort. In der Tat ist aus (84) auch direkt zu entnehmen, daß für die Zustandsgleichung eines idealen Gases der ganze Ausdruck verschwindet.

Bei gewöhnlichen Gasen wird sowohl die innere Energie als auch die äußere Arbeit zu berücksichtigen sein.

§ 156. Was nun ferner die Summe der beiden besprochenen Einflüsse, also die ganze Differenz $c_p - c_v$ betrifft, so hat dieselbe für feste und flüssige Körper gewöhnlich einen verhältnismäßig kleinen Wert, oder das Verhältnis $\frac{c_p}{c_v} = \gamma$ ist nur wenig größer als 1; d. h. bei festen und flüssigen Körpern spielt die Abhängigkeit der Energie von der Temperatur eine viel größere Rolle als die vom Volumen. Bei Gasen ist γ größer und zwar im allgemeinen um so größer, aus je weniger Atomen das Gasmolekül besteht. Für Wasserstoff, Stickstoff und die meisten anderen zweiatomigen Moleküle ist $\gamma = 1,41$ (§ 87). Der größte je beobachtete Wert von γ ist der von Kundt und Warburg für den einatomigen Quecksilberdampf gefundene: 1,667, der auch für die einatomigen Edelgase gilt.

§ 157. Für manche Anwendungen des zweiten Hauptsatzes ist es bequem, statt der Variablen T und v, wie wir es bisher getan haben, die Variablen T und p als unabhängige Variable einzuführen. Dann ergibt sich aus (61):

$$\begin{aligned} ds &= \frac{du + p\,dv}{T} \\ &= \left[\left(\frac{\partial u}{\partial T}\right)_p + p\left(\frac{\partial v}{\partial T}\right)_p\right]\frac{dT}{T} + \left[\left(\frac{\partial u}{\partial p}\right)_T + p\left(\frac{\partial v}{\partial p}\right)_T\right]\frac{dp}{T}. \end{aligned}$$

Andrerseits ist

$$ds = \left(\frac{\partial s}{\partial T}\right)_p dT + \left(\frac{\partial s}{\partial p}\right)_T dp.$$

Folglich:

$$\left(\frac{\partial s}{\partial T}\right)_p = \frac{\left(\frac{\partial u}{\partial T}\right)_p + p\left(\frac{\partial v}{\partial T}\right)_p}{T}$$

$$\left(\frac{\partial s}{\partial p}\right)_T = \frac{\left(\frac{\partial u}{\partial p}\right)_T + p\left(\frac{\partial v}{\partial p}\right)_T}{T}.$$

Die erste dieser Gleichungen nach p, die zweite nach T differentiiert ergibt

$$\frac{\partial^2 s}{\partial T\,\partial p} = \frac{\dfrac{\partial^2 u}{\partial T\,\partial p} + p\dfrac{\partial^2 v}{\partial T\,\partial p} + \left(\dfrac{\partial v}{\partial T}\right)_p}{T}$$
$$= \frac{\dfrac{\partial^2 u}{\partial T\,\partial p} + p\dfrac{\partial^2 v}{\partial T\,\partial p}}{T} - \frac{\left(\dfrac{\partial u}{\partial p}\right)_T + p\left(\dfrac{\partial v}{\partial p}\right)_T}{T}$$

und daraus

$$\left(\frac{\partial u}{\partial p}\right)_T = -T\left(\frac{\partial u}{\partial T}\right)_p - p\left(\frac{\partial v}{\partial p}\right)_T.$$

Hierdurch, sowie durch Gleichung (26) werden die obigen Ausdrücke der Differentialquotienten von s nach T und p:

(84a) $$\left(\frac{\partial s}{\partial T}\right)_p = \frac{c_p}{T}$$

(84b) $$\left(\frac{\partial s}{\partial p}\right)_T = -\left(\frac{\partial v}{\partial T}\right)_p$$

und endlich durch Differentiation der ersten Gleichung nach p, der zweiten nach T, und Gleichsetzung der Werte:

(85) $$\left(\frac{\partial c_p}{\partial p}\right)_T = -T\left(\frac{\partial^2 v}{\partial T^2}\right)_p.$$

Diese Gleichung, die sich übrigens auch unmittelbar durch Differentiation der allgemeinen Gleichung (79f) in § 152c nach T gewinnen läßt, enthält nur direkt meßbare Größen; sie bringt die Abhängigkeit der spezifischen Wärme vom Druck in Beziehung zur Abhängigkeit des thermischen Ausdehnungskoeffizienten von der Temperatur, d. h. zur Abweichung vom Gay-Lussacschen Gesetz. Vgl. hierzu Gl. (81a).

§ 158. Mittels der vom zweiten Hauptsatz gelieferten Beziehungen können wir auch den früher (§ 70) beschriebenen Versuchen, welche Thomson und Joule über die Temperaturänderung eines durch einen Wattepfropf langsam hindurchgepreßten Gases anstellten, eine weitergehende Deutung geben, als dort, wo wir sie nur zur Bestimmung der Eigenschaften idealer Gase verwerteten. Damals haben wir schon ausgeführt, daß diese Versuche im wesentlichen darauf hinauskommen, einem Gase ohne Zuleitung oder Ableitung äußerer Wärme ($Q = 0$) eine Volumenvergrößerung $V_2 - V_1$, auf die

Masseneinheit bezogen: $v_2 - v_1$, zu erteilen, während die auf die Masseneinheit des Gases ausgeübte äußere Arbeit durch

$$p_1 v_1 - p_2 v_2 = A$$

ausgedrückt wird. Die Größe A verschwindet für ein ideales Gas, da dann die Temperatur konstant bleibt. Für ein wirkliches Gas aber folgt allgemein aus der Gleichung (17) des ersten Hauptsatzes:

$$u_2 - u_1 = A = p_1 v_1 - p_2 v_2$$

oder:

$$u_2 + p_2 v_2 = u_1 + p_1 v_1$$

und mit Einführung der auf die Masseneinheit bezogenen Wärmefunktion w nach (49):

$$w_2 = w_1.$$

Beim JOULE-THOMSONschen Versuch bleibt also im allgemeinen nicht die Energie u, sondern die Wärmefunktion $w = u + pv$ konstant.

Nehmen wir nun der Einfachheit halber den Druck auf beiden Seiten des Wattepfropfs als sehr wenig verschieden an und bezeichnen die Differenzen aller Zustandsgrößen auf beiden Seiten des Pfropfes durch ein hinzugefügtes Δ, so läßt sich die letzte Gleichung schreiben:

$$\Delta w = 0 = \Delta u + p\,\Delta v + v\,\Delta p$$

oder nach (61):

$$T\,\Delta s + v\,\Delta p = 0,$$

woraus:

$$T\left(\frac{\partial s}{\partial T}\right)_p \Delta T + \left\{T\left(\frac{\partial s}{\partial p}\right)_T + v\right\}\Delta p = 0.$$

Dies ergibt mit Berücksichtigung der Gleichungen (84a) und (84b):

$$c_p\,\Delta T + \left\{v - T\left(\frac{\partial v}{\partial T}\right)_p\right\}\Delta p = 0,$$

also:

$$\Delta T = \frac{T\left(\frac{\partial v}{\partial T}\right)_p - v}{c_p}\,\Delta p. \tag{86}$$

Mit Hilfe dieser einfachen Gleichung läßt sich die in dem THOMSON-JOULEschen Versuch eintretende Temperaturänderung ΔT des Gases bei bekannter Druckdifferenz Δp (< 0) in Beziehung bringen zur spezifischen Wärme c_p bei konstantem Druck und zu der Abweichung des Gases vom GAY-LUSSACschen Gesetz. Denn nach diesem Gesetz wäre v bei konstantem Druck proportional T, also nach der Gleichung (86) $\Delta T = 0$, wie es in der Tat für ideale Gase zutrifft.

§ 158a. Legt man die VAN DER WAALSsche Zustandsgleichung (12) zugrunde, so ergibt sich aus (86) ohne jede Vernachlässigung:

$$\Delta T = \frac{2\alpha(v-\beta)^2 - RT\beta v^2}{RTv^3 - 2\alpha(v-\beta)^2} \cdot \frac{v\,\Delta p}{c_p},$$

und wenn man α und β hinreichend klein annimmt:

$$\Delta T = \left(\frac{2\alpha}{RT} - \beta\right) \cdot \frac{\Delta p}{c_p},$$

eine Beziehung, die annähernd mit den Messungen übereinstimmt. Für die meisten Gase ist bei gewöhnlichen, nicht zu hohen Temperaturen der Klammerausdruck positiv, d. h. der Effekt besteht in einer Abkühlung da Δp stets negativ. Auf diesen Umstand hat C. VON LINDE die Konstruktion seiner Luftverflüssigungsmaschine gegründet.

Bei Wasserstoff ist wegen der Kleinheit von α die Klammer bei mittleren Temperaturen negativ. Das positive Glied überwiegt erst, wenn die Temperatur unter $-80°$ C. erniedrigt wird. Der sogenannte „Umkehrpunkt", in welchem der Effekt sein Vorzeichen wechselt, liegt nach der letzten Gleichung bei

$$T = \frac{2\alpha}{R\beta}.$$

Doch ist, da diese Gleichung nur angenäherte Gültigkeit besitzt, im allgemeinen die Lage des Umkehrpunktes noch vom Druck p abhängig.[1]

§ 159. Ebenso wie man die Zustandsgleichung eines Gases benutzen kann, den JOULE-THOMSON-Effekt zu berechnen, könnte man auch umgekehrt daran denken, genaue Messungen dieses Effektes zu verwerten, um die Abweichung der Zustandsgleichung von den Gesetzen idealer Gase festzustellen. THOMSON und JOULE haben seinerzeit die Resultate ihrer Messungen zusammengefaßt in die Formel:

$$\Delta T = \frac{\alpha}{T^2}\,\Delta p,$$

[1] Näheres u. a. bei F. A. SCHULZE, Ann. d. Phys. **49**, 585, 1916.

wobei α konstant. Drückt man p in Atmosphären aus, so ist z. B. für Luft:

$$\alpha = 0,276 \cdot (273)^2.$$

Diese Formel ist natürlich ebenfalls nur angenähert richtig. Innerhalb des Bereichs ihrer Gültigkeit erhält man durch Vergleichung mit (86):

$$T\left(\frac{\partial v}{\partial T}\right)_p - v = c_p \cdot \frac{\alpha}{T^2} \tag{87}$$

und durch Differentiation nach T:

$$T\left(\frac{\partial^2 v}{\partial T^2}\right)_p = \frac{\alpha}{T^2}\left(\frac{\partial c_p}{\partial T}\right)_p - \frac{2\alpha c_p}{T^3}.$$

Hieraus mit Rücksicht auf (85):

$$\left(\frac{\partial c_p}{\partial p}\right)_T + \frac{\alpha}{T^2}\left(\frac{\partial c_p}{\partial T}\right)_p - \frac{2\alpha c_p}{T^3} = 0.$$

Die allgemeine Lösung dieser Differentialgleichung ist:

$$c_p = T^2 \cdot f(T^3 - 3\alpha p),$$

wobei $f(x)$ eine ganz beliebige Funktion eines einzigen Arguments x bedeutet.

Nehmen wir nun an, daß für abnehmende Werte von p sich das Gas bei jeder Temperatur unbegrenzt dem idealen Verhalten nähert, so wird für $p = 0$ c_p konstant $= c_p^0$ (z. B. für Luft in kalorischem Maße: $0,238$) und daher allgemein:

$$c_p = c_p^0 T^2 (T^3 - 3\alpha p)^{-\frac{2}{3}}$$

$$c_p = \frac{c_p^0}{\left(1 - \frac{3\alpha p}{T^3}\right)^{\frac{2}{3}}}. \tag{88}$$

Dieser Ausdruck von c_p läßt sich nun weiter benutzen, um auch v als Funktion von T und p zu bestimmen. Es folgt nämlich aus (87)

$$T^2 \cdot \frac{\partial}{\partial T}\left(\frac{v}{T}\right)_p = \frac{\alpha c_p}{T^3} = \frac{\alpha a_p^0}{(T^3 - 3\alpha p)^{\frac{2}{3}}}$$

und daraus:

$$\frac{v}{T} = \alpha c_p^0 \int \frac{dT}{T^4 \cdot \left(1 - \frac{3\alpha p}{T^3}\right)^{\frac{2}{3}}}$$

oder:

$$v = \frac{c_p^0 T}{3p}\left\{\sqrt[3]{1 - \frac{3\alpha p}{T^3}} + \beta\right\} \tag{89}$$

als Zustandsgleichung des Gases. Die Integrationskonstante β bestimmt sich aus der Dichte bei 0° und Atmosphärendruck. Wie die THOMSON-JOULEsche Formel, so haben auch die Gleichungen (88) und (89) nur beschränkte Gültigkeit. Es ist aber prinzipiell von Interesse, zu sehen, wie diese verschiedenen Beziehungen mit Notwendigkeit auseinander hervorgehen.[1]

§ 160. Eine weitere Anwendung von prinzipiell wichtiger Bedeutung, welche der zweite Hauptsatz zu machen gestattet, ist die Bestimmung der absoluten Temperatur T eines Körpers nach einer Methode, die unabhängig ist von den Abweichungen der Gase vom idealen Zustand. Wir haben früher (§ 4) die Temperatur definiert durch ein Gasthermometer, mußten aber dort die Definition beschränken auf die Fälle, wo die verschiedenen Gasthermometer (Wasserstoff, Luft usw.) so übereinstimmende Angaben liefern, wie sie für die beabsichtigte Genauigkeit erforderlich sind. Für alle anderen Fälle aber — und bei hohen Genauigkeitsanforderungen kommen hier auch die mittleren Temperaturen in Betracht — hatten wir die Definition der absoluten Temperatur vorläufig suspendiert. Mit Hilfe der Gleichung (61), bzw. der aus ihr abgeleiteten Gleichung (80) sind wir nun imstande, eine vollständig exakte, von dem Verhalten spezieller Substanzen gänzlich unabhängige Definition der absoluten Temperatur zu liefern.

Gehen wir von irgend einem willkürlich angenommenen praktischen Thermometer aus (z. B. Quecksilberthermometer, oder auch Skalenausschlag eines Thermoelementes oder eines Bolometers), dessen Angaben wir mit t bezeichnen wollen, so handelt es sich darum, dies Thermometer auf ein absolutes zu reduzieren, d. h. die absolute Temperatur T als Funktion von t zu bestimmen. Was wir direkt messen können, ist die Abhängigkeit des Verhaltens irgend einer bequem zu behandelnden Substanz, z. B. eines Gases, von t und von v oder p. Wir führen also in (80) etwa t und v als unabhängige Variable statt T und v ein und erhalten:

$$\left(\frac{\partial u}{\partial v}\right)_t = T\left(\frac{\partial p}{\partial t}\right)_v \frac{dt}{dT} - p.$$

Hier sind $\left(\frac{\partial u}{\partial v}\right)_t$, p und $\left(\frac{\partial p}{\partial z}\right)_v$ als meßbare Funktionen von t und v

[1] Weiter ausgearbeitet und auf Versuchsdaten angewandt finden sich diese Überlegungen bei R. PLANK, Physikalische Zeitschrift **11**, 633, 1910. Ebendort **17**, 521, 1916. Vgl. auch L. SCHAMES, Phys. Zeitschr. **18**, 30, 1917.

anzusehen; daher läßt sich diese Differentialgleichung in folgender Weise integrieren:

$$\int \frac{dt}{T} = \int \frac{\left(\frac{\partial p}{\partial t}\right)_v dt}{\left(\frac{\partial u}{\partial v}\right)_t + p}.$$

Setzt man noch fest, daß für den Gefrierpunkt des Wassers, wo $t = t_0$ sein möge, $T = T_0 = 273$, so ist

$$\log \frac{T}{T_0} = \int_{t_0}^{t} \frac{\left(\frac{\partial p}{\partial t}\right)_v dt}{\left(\frac{\partial u}{\partial v}\right)_t + p}$$

und hierdurch T vollständig als Funktion von t bestimmt. Das Volumen v muß offenbar in dem Ausdruck unter dem Integralzeichen ganz herausfallen, und diese Forderung der Theorie kann daher gleichzeitig zur Prüfung der Richtigkeit des zweiten Hauptsatzes benutzt werden.

§ 161. Was nun die Messung der einzelnen Größen unter dem Integralzeichen betrifft, so ergibt sich der Zähler direkt aus der Zustandsgleichung der Substanz, der Nenner aber aus der Wärmemenge, welche die Substanz bei isothermer reversibler Ausdehnung von außen aufnimmt, bzw. bei isothermer reversibler Kompression nach außen abgibt. Denn nach der Gleichung (22) des ersten Hauptsatzes ist für isotherme reversible Ausdehnung das Verhältnis der zugeführten Wärmemenge q zur Volumenänderung dv:

$$\left(\frac{q}{dv}\right)_t = \left(\frac{\partial u}{\partial v}\right)_t + p.$$

§ 162. Statt die Wärmemenge zu messen, welche eine Substanz bei isothermer Ausdehnung von außen aufnimmt, stellt man zur Bestimmung der absoluten Temperatur bequemer Versuche an von der Art der soeben besprochenen von THOMSON und JOULE über die Temperaturänderung eines langsam ausströmenden Gases. Führen wir nämlich in der Gleichung (86), welche die Theorie dieser Versuche, bezogen auf absolute Temperaturen, darstellt, statt der absoluten Temperatur T wieder t (§ 160) ein, so ist zu

setzen:

$$\Delta T = \frac{dT}{dt}\,\Delta t$$
$$\left(\frac{\partial v}{\partial T}\right)_p = \left(\frac{\partial v}{\partial t}\right)_p \cdot \frac{dt}{dT}$$
$$c_p = \left(\frac{q}{dT}\right)_p = \left(\frac{q}{dt}\right)_p \cdot \frac{dt}{dT} = c'_p\,\frac{dt}{dT},$$

wenn c'_p die auf die Temperatur t bezogene spezifische Wärme bei konstantem Druck bezeichnet. Folglich aus (86):

$$\Delta t = \frac{T\left(\frac{\partial v}{\partial t}\right)_p \frac{dt}{dT} - v}{c'_p} \cdot \Delta p$$

und wieder durch Integration:

$$\log\frac{T}{T_0} = \int_{t_0}^{t} \frac{\left(\frac{\partial v}{\partial t}\right)_p dt}{v + c'_p\,\frac{\Delta t}{\Delta p}} = J, \tag{90}$$

wo nun wieder unter dem Integralzeichen lauter direkt und verhältnismäßig bequem meßbare Größen stehen.

§ 163. In der von uns § 160 gemachten Festsetzung, daß für t_0, den Gefrierpunkt des Wassers, $T = T_0 = 273$ sein soll, liegt die Voraussetzung, daß der Ausdehnungskoeffizient α der idealen Gase schon bekannt ist. Da aber genau genommen die wirklichen Gase sämtlich bei allen Temperaturen Abweichungen voneinander und vom idealen Verhalten zeigen, so wollen wir uns auch noch von dieser Voraussetzung befreien. Wir tun dies, indem wir zur ursprünglichen Definition der Temperatur (§ 3) zurückkehren und festsetzen, daß die Differenz der absoluten Temperatur des unter Atmosphärendruck siedenden Wassers T_1, und der des unter Atmosphärendruck gefrierenden Wassers T_0:

$$T_1 - T_0 = 100 \tag{91}$$

sein soll.

Bedeutet nun t_1 die am t-Thermometer gemessene Temperatur des Siedepunkts, so ist nach (90)

$$\log \frac{T_1}{T_0} = \int_{t_0}^{t_1} \frac{\left(\frac{\partial v}{\partial t}\right)_p dt}{v + c'_p \frac{\Delta t}{\Delta p}} = J_1 \tag{92}$$

und die Elimination von T_0 und T_1 aus (90), (91) und (92) ergibt als absolute Temperatur

$$T = \frac{100 \cdot e^J}{e^{J_1} - 1}. \tag{93}$$

Hieraus erhält man auch den Ausdehnungskoeffizienten eines idealen Gases, unabhängig von jedem Gasthermometer:

$$\alpha = \frac{1}{T_0} = \frac{e^{J_1} - 1}{100}. \tag{94}$$

Da der Ausdruck unter dem Integralzeichen in jedem der beiden Integrale J und J_1 notwendig allein von t und nicht noch von einer zweiten Variablen abhängt, so genügt es zur Berechnung des Integrals, wenn man die Messungen bei den verschiedenen Temperaturen t unter einer vereinfachenden Bedingung, z. B. immer bei dem nämlichen Druck (Atmosphärendruck) vornimmt.

§ 164. Noch einfacher wird die Formel, wenn man, unter Beschränkung auf Atmosphärendruck, für das t-Thermometer als thermometrische Substanz (§ 3) gerade dasjenige Gas nimmt, mit welchem man die Ausströmungsversuche anstellt. Dann ist nämlich der auf die Temperatur t bezogene Ausdehnungskoeffizient α' konstant, und wenn, wie gewöhnlich, $t_0 = 0$ und $t_1 = 100$ gesetzt ist:

$$v = v_0(1 + \alpha' t),$$

wobei v_0 das spezifische Volumen bei der Gefriertemperatur des Wassers und Atmosphärendruck bezeichnet.

Ferner:

$$\left(\frac{\partial v}{\partial t}\right)_p = \alpha' v_0,$$

daher aus (90):

$$J = \int_0^t \frac{\alpha' \, dt}{1 + \alpha' t + \frac{c'_p}{v_0} \frac{\Delta t}{\Delta p}}$$

und aus (92):

$$J_1 = \int_0^{100} \frac{\alpha'\,dt}{1 + \alpha' t + \frac{c_p'}{v_0}\frac{\Delta t}{\Delta p}}.$$

Für ein nahezu ideales Gas, wie z. B. Luft, ist Δt klein, und daher das Glied mit c_p' und v_0 nur ein Korrektionsglied, in welchem die Ansprüche an die Genauigkeit der Koeffizienten c_p' und v_0 entsprechend ermäßigt sind. Für ein vollkommen ideales Gas wäre $\Delta t = 0$, und aus den letzten beiden Gleichungen:

$$J = \log(1 + \alpha' t), \qquad J_1 = \log(1 + 100\alpha'),$$

mithin nach (93):

$$T = t + \frac{1}{\alpha'}$$

und nach (94):

$$\alpha = \frac{1}{T_0} = \alpha'$$

wie es sein muß.

Sobald durch eine genaue Messung, wenn auch nur mit einer einzigen Substanz, T als Funktion von t bestimmt ist, kann die Frage nach der Größe der absoluten Temperatur auch praktisch als allgemein gelöst gelten.

Wie durch Messungen an homogenen Substanzen läßt sich die absolute Temperatur auch durch Anwendungen des zweiten Hauptsatzes auf heterogene Systeme bestimmen. Vgl. unten § 177.

§ 164a. So wichtig in prinzipieller Hinsicht und so unentbehrlich bei extremen Wärmegraden der zweite Hauptsatz sich für die Festlegung der absoluten Temperaturskala erweist, so ist doch der bis jetzt wohl genaueste Zahlenwert des Ausdehnungskoeffizienten α der idealen Gase unabhängig vom zweiten Hauptsatz direkt aus der Messung des Ausdehnungskoeffizienten wirklicher Gase abgeleitet worden, unter Benutzung der von D. Berthelot (§ 25) modifizierten van der Waalsschen Zustandsgleichung (12), die um so angenäherter gilt, je weiter entfernt sich die Gase vom Kondensationspunkt befinden. Auf diesem Wege hat sich der Ausdehnungskoeffizient α idealer Gase zu $0,003\,661\,8$ und dementsprechend die absolute Temperatur des Wassergefrierpunktes

$$T_0 = \frac{1}{\alpha} = 273,09$$

ergeben.

Zweites Kapitel. System in verschiedenen Aggregatzuständen.

§ 165. Wir untersuchen im folgenden das Gleichgewicht eines Systems, dessen einzelne Teile verschiedenen Aggregatzuständen, dem festen, flüssigen oder gasförmigen, angehören können. Dabei nehmen wir überall an, daß der Zustand jedes dieser Teile durch Masse, Temperatur und Volumen vollständig bestimmt ist, oder anders ausgedrückt, daß das ganze System von einem einzigen unabhängigen Bestandteil (§ 198) gebildet wird. Es ist dazu nicht erforderlich, daß das System, oder daß auch nur ein einzelner Teil des Systems chemisch homogen ist. Die Frage nach der chemischen Homogenität läßt sich im allgemeinen gar nicht einmal strenge beantworten (vgl. § 92), z. B. ist es noch sehr dahingestellt, ob im flüssigen Wasser die Moleküle dieselben sind wie im Eis, ja es ist wegen der anomalen Eigenschaften des flüssigen Wassers in der Nähe des Gefrierpunktes sogar wahrscheinlich, daß die Moleküle schon innerhalb des flüssigen Wassers nicht alle gleichartig sind. Die Entscheidung darüber ist für die folgenden Untersuchungen ganz ohne Belang. Es kann sogar das System aus verschiedenartigen Stoffen in beliebigem Gewichtsverhältnis zusammengesetzt sein und etwa eine Lösung oder eine Legierung bilden. Was wir hier voraussetzen wollen, ist nur dies, daß der innere Zustand jedes homogenen Teiles der betrachteten Substanz bei bestimmter Temperatur T und bestimmtem spezifischen Volumen v ein ganz bestimmter ist, daß also, falls die Substanz aus verschiedenartigen Stoffen zusammengesetzt ist, das Gewichtsverhältnis derselben in allen Teilen des Systems das nämliche ist. Dann können wir die Frage, um deren Beantwortung es sich hier handelt, in folgender Form aussprechen:

Wir denken uns die Substanz, deren Gesamtmasse M gegeben ist, in eine feste Hülle vom gegebenen Volumen V eingeschlossen und ihr durch Zuleitung von Wärme eine gegebene Energie U mitgeteilt. Wird nun dies System nach außen abgeschlossen und sich selbst überlassen, so bleibt M, V und U konstant; dagegen die Entropie S nimmt zu. Nun suchen wir den, oder wenn es mehrere sind, die Gleichgewichtszustände zu bestimmen, welche das System annehmen kann, und zugleich die Bedingungen dafür anzugeben, unter denen das Gleichgewicht stabil oder labil ist. Die vollständige Durchführung dieser Untersuchung wird ermöglicht durch den in der Gleichung (77) ausgesprochenen Satz, daß unter allen Zuständen, die bei verhinderter äußerer Wärmezufuhr auseinander hervorgehen können, der stabilste Gleichgewichtszustand durch das absolute Maximum der Entropie ausgezeichnet ist. Im allgemeinen wird aber, wie wir sehen werden, die Entropie

des Systems unter den angegebenen äußeren Bedingungen mehrere relative Maxima annehmen können; dann entspricht jedem Maximum, welches nicht das absolute ist, ein mehr oder weniger labiler Gleichgewichtszustand. Wenn sich das System in einem derartigen Zustand befindet (z. B. als übersättigter Dampf), so kann unter Umständen, wenn eine gewisse beliebig kleine, aber passende Störung hinzutritt, das System sich um endliche Strecken aus dem Zustand entfernen und in einen anderen Gleichgewichtszustand übergehen, dem dann notwendig ein größerer Wert der Entropie entspricht als dem vorigen.

§ 166. Wir haben nun zunächst diejenigen Zustände aufzusuchen, in denen die Entropie S des Systems ein Maximum annimmt.

Die allgemeinste Annahme über den Zustand des Systems ist die, daß sich drei verschiedene Teile desselben in den drei verschiedenen Aggregatzuständen befinden. Bezeichnen wir demnach die Massen dieser Teile mit M_1, M_2, M_3, wobei die spezielle Bedeutung der einzelnen Indizes einstweilen offen gelassen ist, so haben wir als gegebene Masse des ganzen Systems:

$$M_1 + M_2 + M_3 = M.$$

Die Größen M sind positiv, einzelne können auch Null sein.

Ferner muß, weil der gesuchte Zustand ein Gleichgewichtszustand ist, jeder dieser drei Teile des Systems auch für sich im Gleichgewicht, d. h. von gleichmäßiger Temperatur und Dichte sein, und es gelten für ihn alle im vorigen Kapitel für ein homogenes System abgeleiteten Sätze.

Bezeichnen also v_1, v_2, v_3 die spezifischen Volumina, so ist das gegebene Volumen des Systems:

$$M_1 v_1 + M_2 v_2 + M_3 v_3 = V.$$

Analog erhält man für die gegebene Energie des Systems:

$$M_1 u_1 + M_2 u_2 + M_3 u_3 = U,$$

wobei die u die spezifischen Energien bezeichnen.

Diese drei Gleichungen entsprechen den gegebenen äußeren Bedingungen.

§ 167. Für die Entropie erhält man nun:

$$S = M_1 s_1 + M_2 s_2 + M_3 s_3,$$

wobei die s die spezifischen Entropien bezeichnen.

Aus dieser Gleichung ergibt sich für irgend eine unendlich kleine Zustandsänderung:

$$\delta S = \sum M_1 \, \delta s_1 + \sum s_1 \, \delta M_1,$$

wenn hier, wie überall im folgenden, das Zeichen $\sum$ für die Summierung über die Ziffern 1, 2, 3 gebraucht wird. Mit Rücksicht darauf, daß nach (61) allgemein:

$$\delta s = \frac{\delta u + p\,\delta v}{T},$$

erhält man:

$$\delta S = \sum \frac{M_1\,\delta u_1}{T_1} + \sum \frac{M_1 p_1\,\delta v_1}{T_1} + \sum s_1\,\delta M_1. \tag{95}$$

Die Variationen sind aber nicht alle unabhängig voneinander, vielmehr folgt aus den drei äußeren Bedingungsgleichungen des vorigen Paragraphen durch Variation:

$$\begin{cases} \sum \delta M_1 = 0 \\ \sum M_1\,\delta v_1 + \sum v_1\,\delta M_1 = 0 \\ \sum M_1\,\delta u_1 + \sum u_1\,\delta M_1 = 0. \end{cases} \tag{96}$$

Wir müssen daher mit Hilfe dieser drei Gleichungen irgend drei Variationen aus dem Ausdruck von δS eliminieren, um in demselben lauter unabhängige Variationen zu erhalten. Wenn wir z. B. aus diesen letzten Gleichungen die Werte von δM_2, δv_2 und δu_2 entnehmen und sie in (95) einsetzen, so kommt:

$$\begin{cases} \delta S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right) M_1\,\delta u_1 - \left(\frac{1}{T_2} - \frac{1}{T_3}\right) M_3\,\delta u_3 \\ \quad + \left(\frac{p_1}{T_1} - \frac{p_2}{T_2}\right) M_1\,\delta v_1 - \left(\frac{p_2}{T_2} - \frac{p_3}{T_3}\right) M_3\,\delta v_3 \\ \quad + \left(s_1 - s_2 - \frac{u_1 - u_2}{T_2} - \frac{p_2(v_1 - v_2)}{T_2}\right) \delta M_1 \\ \quad - \left(s_2 - s_3 - \frac{u_2 - u_3}{T_2} - \frac{p_2(v_2 - v_3)}{T_2}\right) \delta M_3. \end{cases} \tag{97}$$

Da nun die in diesem Ausdruck vorkommenden sechs Variationen vollständig unabhängig voneinander sind, so muß, damit nach (77) δS für alle beliebigen Zustandsänderungen $= 0$ ist, jeder der sechs Koeffizienten verschwinden. Mithin haben wir:

$$\begin{cases} T_1 = T_2 = T_3 (= T) \\ p_1 = p_2 = p_3 \\ s_1 - s_2 = \dfrac{(u_1 - u_2) + p_1(v_1 - v_2)}{T} \\ s_2 - s_3 = \dfrac{(u_2 - u_3) + p_2(v_2 - v_3)}{T}. \end{cases} \tag{98}$$

Diese sechs Gleichungen stellen notwendige Eigenschaften eines Zustandes dar, dem ein Maximum der Entropie entspricht, also eines Gleichgewichtszustandes. Die ersten vier derselben sprechen die Gleichheit von Temperatur und Druck aus, das Hauptinteresse konzentriert sich daher auf die beiden letzten Gleichungen, in welchen die thermodynamische Theorie der Schmelzung, Verdampfung und Sublimierung enthalten ist.

§ 168. Wir wollen jene beiden Gleichungen zunächst auf eine etwas einfachere Form bringen, indem wir für die spezifische Entropie s, die wir, wie auch u und p, als eindeutige Funktion der unabhängigen Variablen T und v betrachten, ihren Wert einsetzen. Da nämlich allgemein nach (61):

$$ds = \frac{du + p\,dv}{T},$$

so haben wir durch Integration dieser Gleichung:

$$s_1 - s_2 = \int_2^1 \frac{du + p\,dv}{T}.$$

Die obere Grenze des Integrals ist durch die Werte $T = T_1$, $v = v_1$, die untere durch die Werte $T = T_2$, $v = v_2$ bestimmt. Der Integrationsweg ist ganz beliebig und hat auf den Wert der Differenz $s_1 - s_2$ gar keinen Einfluß. Da nun nach (98) $T_1 = T_2 = T$, so wollen wir den isothermen Integrationsweg $T =$ konst. wählen und erhalten dadurch:

$$s_1 - s_2 = \frac{u_1 - u_2}{T} + \frac{1}{T}\int_{v_1}^{v_2} p\,dv.$$

In dem Integral ist nun die Integration bei konstantem T auszuführen, indem p als eine durch die Zustandsgleichung der homogenen Substanz bekannte Funktion von T und v anzusehen ist.

Substituiert man den Wert von $s_1 - s_2$ in die Gleichungen (98), so ergibt sich die Relation:

(99)
$$\int_{v_2}^{v_1} p\,dv = p_1(v_1 - v_2).$$

Ebenso:

$$\int_{v_1}^{v_2} p\,dv = p_2(v_2 - v_3).$$

Fügen wir noch hinzu: $p_1 = p_2 = p_3$,

so haben wir hier im ganzen 4 Gleichungen mit den 4 Unbekannten T, v_1, v_2, v_3, welchen jeder Gleichgewichtszustand genügen muß.

Die in diesen Gleichungen vorkommenden Konstanten hängen offenbar lediglich von der chemischen Beschaffenheit der Substanz, nicht aber von den gegebenen Werten der Masse M, des Volumens V und der Energie U des Systems ab. Man kann daher diese Gleichungen die „inneren" Gleichgewichtsbedingungen nennen, im Gegensatz zu den Gleichungen im § 166, welche die äußeren Umstände bezeichnen, denen das System unterworfen ist.

§ 169. Ehe wir zur Betrachtung und Vergleichung der aus den entwickelten Gleichungen sich ergebenden Werte der Unbekannten übergehen, wollen wir allgemein untersuchen, ob bzw. unter welcher Bedingung dieselben auch wirklich einen Maximalwert der Entropie, und nicht etwa z. B. einen Minimalwert liefern. Zur Beantwortung dieser Frage müssen wir den Wert der zweiten Variation $\delta^2 S$ berechnen. Ist derselbe für alle möglichen Zustandsänderungen negativ, so ist der betreffende Zustand jedenfalls ein Maximalzustand.

Wir variieren daher den Ausdruck (97) von δS und erhalten dadurch den Wert von $\delta^2 S$, welcher sich bedeutend vereinfacht, wenn wir die Gleichungen (98), die aber selber nicht variiert werden dürfen, benutzen. Berücksichtigen wir dann noch die festen Bedingungen, sowohl in unvariierter wie in der variierten Form (96), so ergibt sich schließlich:

$$\delta^2 S = -\sum \frac{M_1\, \delta s_1\, \delta T_1}{T_1} + \sum \frac{M_1\, \delta p_1\, \delta v_1}{T_1},$$

wofür man auch schreiben kann:

$$T\, \delta^2 S = -\sum M_1 (\delta s_1\, \delta T_1 - \delta p_1\, \delta v_1).$$

Um alle Variationen auf die der unabhängigen Variabeln T und v zu reduzieren, setzen wir noch nach (81):

$$\begin{aligned} \delta s &= \left(\frac{\partial s}{\partial T}\right)_v \delta T + \left(\frac{\partial s}{\partial v}\right)_T \delta v \\ &= \frac{c_v}{T}\, \delta T + \left(\frac{\partial p}{\partial T}\right)_v \delta v \end{aligned}$$

und

$$\delta p = \left(\frac{\partial p}{\partial T}\right)_v \delta T + \left(\frac{\partial p}{\partial v}\right)_T \delta v,$$

dann erhalten wir:

$$T\,\delta^2 S = -\sum M_1 \left(\frac{(c_v)_1}{T}\,\delta {T_1}^2 - \left(\frac{\partial p_1}{\partial v_1}\right)_T \delta {v_1}^2 \right). \tag{100}$$

Wenn die Größen $(c_v)_1$, $(c_v)_2$, $(c_v)_3$ alle positiv und die Größen $\left(\frac{\partial p_1}{\partial v_1}\right)_T, \ldots$ alle negativ sind, so ist $\delta^2 S$, wie man sieht, in jedem Falle negativ, also S wirklich ein Maximum, und der Zustand ein Gleichgewichtszustand. Da nun c_v als spezifische Wärme bei konstantem Volumen stets positiv ist, so hängt die Bedingung des Gleichgewichts davon ab, ob $\left(\frac{\partial p}{\partial v}\right)_T$ für alle drei Teile des Systems negativ ist oder nicht. Im letzteren Fall ist kein Gleichgewicht vorhanden. In der Tat ist aus der unmittelbaren Erfahrung ersichtlich, daß in jedem Gleichgewichtszustand $\frac{\partial p}{\partial v}$ negativ ist, da sich der Druck, sei er positiv oder negativ, bei konstanter Temperatur immer in entgegengesetzter Richtung wie das Volumen verändert. Es gibt aber, wie ein Blick auf die in Fig. 1 (§ 26) gegebene graphische Darstellung der Größe p als isotherme Funktion von v lehrt, auch Zustände, in denen $\frac{\partial p}{\partial v}$ positiv ist. Diese Zustände stellen also niemals eine Gleichgewichtslage dar, und sind deshalb auch nicht der direkten Beobachtung zugänglich. Wenn dagegen $\frac{\partial p}{\partial v}$ negativ ist, so findet Gleichgewicht statt; doch braucht dasselbe noch nicht stabil zu sein; es kommt vielmehr dann darauf an, ob nicht unter den gegebenen Bedingungen noch ein anderer Gleichgewichtszustand möglich ist, dem ein größerer Wert der Entropie entspricht.

Wir wollen nun im folgenden die Werte der Unbekannten T, v_1, v_2, v_3 untersuchen, die eine Lösung der inneren Gleichgewichtsbedingungen (98) oder (99) vorstellen; es wird dies, wie wir sehen werden, auf verschiedene Arten möglich sein. Wenn das geschehen ist, wollen wir (von § 189 an) die weitere Frage behandeln, welche der verschiedenartigen Lösungen in jedem Einzelfalle, unter den gegebenen äußeren Bedingungen, den stabilsten Gleichgewichtszustand, d. h. den größten Wert der Entropie des Systems liefert.

§ 170. Erste Lösung. Wenn wir erstens setzen:

$$v_1 = v_2 = v_3 = v,$$

so werden dadurch alle vier Gleichungen (98) befriedigt. Denn da ohnehin die Temperatur T allen drei Teilen des Systems gemeinsam ist, werden dadurch ihre Zustände vollkommen identisch, d. h. das ganze System ist

homogen. Der Zustand des Systems ist dann bestimmt, wenn man noch die Gleichungen des § 166 hinzunimmt, welche die äußeren Bedingungen aussprechen. Dieselben lauten in diesem Falle:

$$\begin{aligned} M_1 + M_2 + M_3 &= M \\ v\,(M_1 + M_2 + M_3) &= V \\ u\,(M_1 + M_2 + M_3) &= U. \end{aligned}$$

Folglich: $$v = \frac{V}{M} \quad \text{und} \quad u = \frac{U}{M}.$$

Aus v und u ergibt sich dann auch T, da u als bekannte Funktion von T und v vorausgesetzt ist.

Diese Lösung hat immer einen bestimmten Sinn, sie stellt aber, wie wir an Gleichung (100) gesehen haben, nur dann einen Gleichgewichtszustand dar, wenn $\frac{\partial p}{\partial v}$ negativ ist. Trifft dies zu, so ist das Gleichgewicht labil oder stabil, je nachdem unter den gegebenen äußeren Bedingungen (§ 166) ein Zustand existiert, dem ein noch größerer Wert der Entropie entspricht oder nicht. Wann das eine oder das andere der Fall ist, müssen wir später noch besonders untersuchen (§ 189).

§ 171. Zweite Lösung. Wenn wir zweitens setzen:

$$v_1 \neq v_2 = v_3,$$

so fallen die mit 2 und 3 bezeichneten Aggregatzustände zusammen, und die Gleichungen (98) reduzieren sich auf:

(101)
$$\begin{cases} p_1 = p_2 \\ s_1 - s_2 = \dfrac{u_1 - u_2 + p_1(v_1 - v_2)}{T} \end{cases}$$

oder statt der zweiten Gleichung:

(102)
$$\int_{v_2}^{v_1} p\,dv = p_1(v_1 - v_2).$$

In diesem Falle befindet sich das System in zwei verschiedenen Aggregatzuständen nebeneinander, z. B. als Dampf und Flüssigkeit. Die beiden Gleichungen (101) enthalten drei Unbekannte: T, v_1, v_2, sie können also dazu dienen, die Größen v_1 und v_2, und infolgedessen auch den Druck $p_1 = p_2$ und die spezifischen Energien u_1 und u_2 als bestimmte Funktionen

der Temperatur T darzustellen. Durch die Temperatur ist also der innere Zustand zweier sich im Gleichgewicht berührender heterogener Teile derselben Substanz vollständig bestimmt. Die Temperatur selber, sowie die Massen der beiden Teile des Systems ergeben sich aus den äußeren Bedingungen (§ 166), welche für diesen Fall lauten:

$$(103) \qquad \begin{cases} M_1 + (M_2 + M_3) = M \\ M_1 v_1 + (M_2 + M_3) v_2 = V \\ M_1 u_1 + (M_2 + M_3) u_2 = U. \end{cases}$$

Diese drei Gleichungen dienen zur Berechnung der drei letzten Unbekannten, nämlich T, M_1 und $(M_2 + M_3)$, wodurch dann der physikalische Zustand des Systems ganz bestimmt ist; denn bei den Massen M_2 und M_3 kommt es offenbar nur auf ihre Summe an. Natürlich hat das Resultat nur dann einen physikalischen Sinn, wenn sowohl M_1 als auch $(M_2 + M_3)$ positiv ausfällt.

§ 172. Die nähere Betrachtung der Gleichung (102) zeigt, daß sie nur dann befriedigt werden kann, wenn der Druck p, der ja für die beiden Grenzen des Integrals den nämlichen Wert $p_1 = p_2$ hat, zwischen den Grenzen Werte annimmt, die teils kleiner, teils größer als p_1 sind, und daß sich daher hier Zustände vorfinden müssen, welche nach § 169 labil sind, weil stellenweise p mit v zunimmt. Die Gleichung läßt sich sehr einfach geometrisch interpretieren, wenn man die schon dort erwähnte graphische Darstellung der Zustandsgleichung durch die Isotherme (Fig. 1, § 26) zu Hilfe nimmt. Denn da das Integral $\int_2^1 p\,dv$ den Flächenraum darstellt, der von der Isotherme, der Abszissenachse und den durch die Punkte v_1 und v_2 der Isotherme begrenzten Ordinaten umschlossen wird, während andrerseits das Produkt $p_1(v_1 - v_2)$ den Flächenraum des aus denselben Ordinaten und der Abszissenstrecke $v_1 - v_2$ gebildeten Rechtecks bezeichnet, so lehrt die Gleichung (102) folgendes: In jeder Isotherme wird der Druck, bei welchem sich zwei Aggregatzustände der Substanz dauernd berühren können, durch diejenige zur Abszissenachse parallele Gerade dargestellt, welche zu beiden Seiten der Isotherme gleiche Flächenräume abgrenzt. Eine derartige Gerade ist in der Fig. 1 durch ABC bezeichnet. Man kann also aus der für homogene, stabile und labile, Zustände aufgestellten Zustandsgleichung direkt das Gesetz der Abhängigkeit des Drucks und der Dichtigkeit des gesättigten Dampfes und der berührenden Flüssigkeit von der Temperatur ableiten.

Wenn wir z. B. die Clausiussche Zustandsgleichung (12a) als empirische Formulierung der Tatsachen zugrunde legen, so folgen aus ihr für das

spezifische Volumen v_1 des gesättigten Dampfes und v_2 der berührenden Flüssigkeit die beiden Bedingungen:

$$\frac{RT}{v_1 - a} - \frac{c}{T(v_1 + b)^2} = \frac{RT}{v_2 - a} - \frac{c}{T(v_2 + b)^2}$$

und aus (102)

$$RT \log \frac{v_1 - a}{v_2 - a} - \frac{c}{T}\left(\frac{1}{v_2 + b} - \frac{1}{v_1 + b}\right) = (v_1 - v_2)\left(\frac{RT}{v_1 - a} - \frac{c}{T(v_1 + b)^2}\right).$$

Hierdurch können v_1 und v_2, also auch $p_1 = p_2$, als Funktionen von T, oder bequemer v_1, v_2, p_1 und T als Funktionen einer einzigen passend gewählten unabhängigen Variabeln bestimmt werden. Für die kritische Temperatur ergibt sich so der kritische Zustand $v_1 = v_2$ (§ 30a), für höhere Temperaturen werden v_1 und v_2 imaginär, für unbegrenzt abnehmende Temperatur wird $v_1 = \infty,\ v_2 = a$.[1]

Mit den Clausiusschen Zahlenwerten der Konstanten für Kohlensäure (§ 25) ergeben sich aus dieser Rechnung Resultate, die mit den Andrewsschen Beobachtungen befriedigend übereinstimmen; doch besitzt nach Thiesen die Clausiussche Zustandsgleichung keine allgemeinere Bedeutung.

§ 173. Der Inhalt der Gleichungen (101) läßt sich einfacher ausdrücken, wenn man statt der Entropie S nach § 152a die freie Energie F einführt, und noch einfacher, wenn man die charakteristische Funktion Φ benutzt. Indessen wollen wir hier bei der freien Energie stehen bleiben. Dieselbe ist, auf die Masseneinheit bezogen, nach (71):

$$f = u - Ts. \tag{104}$$

Dann schreiben sich die Gleichungen (101):

$$p_1 = p_2 \tag{105}$$

$$f_2 - f_1 = p_1(v_1 - v_2). \tag{106}$$

Die Funktion f genügt folgenden einfachen Bedingungen: Nach (79a) ist:

$$\left(\frac{\partial f}{\partial T}\right)_v = -s \tag{107}$$

und:

$$\left(\frac{\partial f}{\partial v}\right)_T = -p. \tag{108}$$

[1] Näheres in Wied. Ann. **13**, 535, 1881.

Die der Berührung zweier Aggregatzustände entsprechenden Gleichgewichtsbedingungen gelten für jede der drei möglichen Kombinationen je zweier Aggregatzustände, wir wollen jedoch, um die Ideen zu fixieren, zuerst beispielsweise diejenige Lösung dieser Gleichungen im Auge behalten, welche der Berührung von Dampf und Flüssigkeit entspricht. Wenn wir hierbei den Index 1 auf den Dampf, den Index 2 auf die Flüssigkeit beziehen, so bedeutet v_1 das spezifische Volumen des bei der Temperatur T gesättigten Dampfes, $p_1 = p_2$ seinen Druck, v_2 das spezifische Volumen der berührenden Flüssigkeit. Diese Größen sind also alle Funktionen der Temperatur allein, wie es der Erfahrung entspricht.

§ 174. Wir können zunächst durch Differentiation der Gleichgewichtsbedingungen nach T zu neuen Sätzen gelangen, wobei wir, da alle Variabeln nur von T abhängen, die entsprechenden totalen Differentialquotienten kurz mit $\frac{dv_1}{dT}$, $\frac{dv_2}{dT}$, $\frac{dp_1}{dT}$ usw. bezeichnen wollen, während wir für die partiellen Differentialquotienten nach T bei konstantem v, und nach v bei konstantem T die bisherige Bezeichnung $\frac{\partial p}{\partial T}$ usw. beibehalten.

Dann ergeben die Gleichungen (105) und (106) nach T differentiiert:

$$\frac{dp_1}{dT} = \frac{dp_2}{dT}$$

und:

$$\frac{df_2}{dT} - \frac{df_1}{dT} = (v_1 - v_2)\frac{dp_1}{dT} + p_1\left(\frac{dv_1}{dT} - \frac{dv_2}{dT}\right).$$

Nun ist aber nach (107) und (108):

$$\begin{aligned}\frac{df_2}{dT} - \frac{df_1}{dT} &= \left(\frac{\partial f_2}{\partial T}\right)_v + \left(\frac{\partial f_2}{\partial v}\right)_T \frac{dv_2}{dT} - \left(\frac{\partial f_1}{\partial T}\right)_v - \left(\frac{\partial f_1}{\partial v}\right)_T \frac{dv_1}{dT} \\ &= -s_2 - p_2\frac{dv_2}{dT} + s_1 + p_1\frac{dv_1}{dT}.\end{aligned}$$

Folglich durch Substitution:

$$s_1 - s_2 = (v_1 - v_2)\frac{dp_1}{dT},$$

oder endlich nach (101):

$$(u_1 - u_2) + p_1(v_1 - v_2) = T(v_1 - v_2)\frac{dp_1}{dT}. \tag{109}$$

Der Ausdruck links bedeutet nach der Gleichung (17) des ersten Hauptsatzes der Wärmetheorie nichts anderes als die Verdampfungswärme r der Flüssigkeit,

d. h. diejenige Wärmemenge, welche der Masseneinheit Flüssigkeit von außen zuzuführen ist, damit sie bei konstant gehaltener Temperatur unter dem konstanten Druck des gesättigten Dampfes vollständig in Dampf übergeht. Denn die Veränderung der Energie ist hierbei $u_1 - u_2$, und die dabei von außen aufgewendete äußere Arbeit A, welche hier negativ ist, beträgt:

$$A = -p_1(v_1 - v_2).$$

Es ist also:

$$r = u_1 - u_2 + p_1(v_1 - v_2) \tag{110}$$

und daher

$$r = T(v_1 - v_2)\frac{dp_1}{dT}. \tag{111}$$

Diese schon von CLAPEYRON aus der CARNOTschen Theorie (§ 52) abgeleitete, zuerst von CLAUSIUS streng begründete Gleichung gestattet die Berechnung der Verdampfungswärme für eine beliebige Temperatur aus den Volumina des gesättigten Dampfes und der Flüssigkeit, sowie der Abhängigkeit der Spannung des gesättigten Dampfes von der Temperatur. Sie ist in sehr vielen Fällen durch die Erfahrung bestätigt worden.

§ 175. Als Beispiel berechnen wir die Verdampfungswärme des Wassers bei 100° C., also beim Druck einer Atmosphäre. Hierfür ist:

$$T = 273 + 100 = 373,$$

$v_1 = 1674$ nach KNOBLAUCH, LINDE und KLEBE

(Volumen eines g gesättigten Wasserdampfes bei 100° C. in ccm),

$$v_2 = 1$$

(Volumen eines g Wasser bei 100° C. in ccm),

$\frac{dp_1}{dT}$ ergibt sich aus den Messungen des Druckes gesättigten Wasserdampfes von HOLBORN und HENNING für 100° C. zu $27,12^{\text{mm}}$ Quecksilber pro Grad. Die Reduktion auf absolute Druckeinheiten liefert nach § 7:

$$\frac{dp_1}{dT} = \frac{27,12}{760} \cdot 1\,013\,250$$

und somit die gesuchte Verdampfungswärme in Kalorien, durch Division mit dem mechanischen Wärmeäquivalent:

$$r = \frac{373 \cdot 1673 \cdot 27,12 \cdot 1\,013\,250}{760 \cdot 4,19 \cdot 10^7} = 539.$$

HENNING fand durch direkte Messung für die Verdampfungswärme des Wassers bei 100° C. 538,7 cal, in ausgezeichneter Übereinstimmung mit der berechneten Zahl.

§ 176. Wie man aus (110) sieht, entspricht ein Teil der Verdampfungswärme r der Zunahme der Energie, ein anderer Teil der äußeren Arbeit. Um zu beurteilen, in welcher Beziehung diese beiden Teile stehen, bildet man am bequemsten das Verhältnis der äußeren Arbeit zur ganzen Verdampfungswärme:

$$\frac{p_1(v_1 - v_2)}{r} = \frac{p_1}{T \cdot \frac{dp_1}{dT}}.$$

Für den soeben behandelten Fall ist

$$\begin{aligned} p_2 &= 760^{\text{mm}}, \\ T &= 373, \\ \frac{dp_1}{dT} &= 27,12^{\text{mm}}, \end{aligned}$$

und man erhält daher für dies Verhältnis:

$$\frac{760}{373 \cdot 27,12} = 0,075,$$

woraus zu entnehmen ist, daß die äußere Arbeit in dem Betrag der Verdampfungswärme hier nur eine geringe Rolle spielt.

§ 177. Die Gleichung (111) gestattet auch wieder eine Berechnung der absoluten Temperatur T, sobald die Verdampfungswärme, sowie der Druck und die Dichte des gesättigten Dampfes und der berührenden Flüssigkeit als Funktion irgend einer beliebigen konventionellen Temperaturskala t (§ 160) durch Messung bestimmt sind. Es ist nämlich:

$$r = T(v_1 - v_2)\frac{dp_1}{dt} \cdot \frac{dt}{dT}$$

und daraus:

$$\log T = \int \frac{v_1 - v_2}{r} \cdot \frac{dp_1}{dt} \cdot dt,$$

woraus T in derselben Weise als Funktion von t zu berechnen ist, wie dies schon früher ausgeführt wurde. Überhaupt ist ersichtlich, daß eine jede aus dem zweiten Hauptsatz abgeleitete Gleichung zwischen meßbaren Größen dazu benutzt werden kann, eine Bestimmung der absoluten Temperatur vorzunehmen, und es handelt sich nur um die praktische Frage der Genauigkeit der Messungen in dem zu untersuchenden Temperaturintervall, um darüber zu entscheiden, welche Methode den Vorzug verdient.

§ 178. Eine einfache Annäherungsformel, die in manchen Fällen gute, in anderen dagegen nur mäßig brauchbare Resultate ergibt, erhält man, wenn in der Gleichung (111) das spezifische Volumen der Flüssigkeit v_2 gegen das des Dampfes v_1 vernachlässigt, und wenn außerdem für letzteres die Zustandsgleichung eines idealen Gases als gültig vorausgesetzt wird. Dann ist nach Gleichung (14)

$$v_1 = \frac{R}{m}\frac{T}{p_1},$$

wobei R die absolute Gaskonstante, m das Molekulargewicht des Dampfes bezeichnet, und die Formel (111) geht über in:

$$r = \frac{R}{m}\frac{T^2}{p_1}\frac{dp_1}{dT}. \tag{112}$$

Für Wasser bei 100° C. wäre z. B.

$$\begin{aligned} R &= 1,985 \qquad \text{in Kalorien nach Gleichung (34)},\\ m &= \mathrm{H_2O} = 18,\\ T &= 373,\\ p_1 &= 760^{\mathrm{mm}},\\ \frac{dp_1}{dT} &= 27,12^{\mathrm{mm}} \end{aligned}$$

und daraus die Verdampfungswärme in Kalorien:

$$r = \frac{1,985 \cdot 373^2 \cdot 27,12}{18 \cdot 760} = 547,5,$$

also etwas zu groß (§ 175). Die Ursache dieser Abweichung liegt darin, daß das Volumen des bei 100° C. gesättigten Wasserdampfes in Wirklichkeit kleiner ist als das aus der Zustandsgleichung eines idealen Gases vom Molekulargewicht 18 für diese Temperatur und Atmosphärendruck berechnete Volumen. Eben deshalb kann eine genaue Messung der Verdampfungswärme

auch dazu dienen, um aus dem zweiten Hauptsatz einen Schluß zu ziehen auf die Abweichung der Dichte eines Dampfes von dem idealen Wert.

Eine in denselben Grenzen gültige Annäherungsformel von anderer Bedeutung ergibt sich, wenn man weiter in der Gleichung (109) für die spezifische Energie des Dampfes nach (35) den für ideale Gase gültigen Wert $u_1 = c_v T + \text{konst.}$, ferner für die spezifische Energie der Flüssigkeit unter Konstantsetzung der spezifischen Wärme c_2 und Vernachlässigung der äußeren Arbeit $u_2 = c_2 T + \text{konst.}$ setzt. Dann folgt aus (109):

$$(c_v - c_2)T + \text{konst.} + \frac{RT}{m} = \frac{R}{m}\frac{T^2}{p_1}\frac{dp_1}{dT}.$$

Multipliziert man beiderseits mit $\dfrac{dT}{T^2}$, so läßt sich diese Gleichung Glied für Glied integrieren, und man erhält schließlich, unter Berücksichtigung von (33)

$$p_1 = a e^{-\frac{b}{t}} T^{\frac{m}{R}(c_p - c_2)}.$$

Hier bedeuten a und b positive Konstanten, c_p und c_2 die spezifischen Wärmen von Dampf und Flüssigkeit bei konstantem Druck. Dies gibt ein Gesetz für die Abhängigkeit der Spannkraft des gesättigten Dampfes von der Temperatur.

Eine Berechnung der Konstanten ist von H. Hertz für Quecksilber ausgeführt worden.[1] Indessen muß bemerkt werden, daß für tiefere Temperaturen die Annahme der Konstanz von c_2 nicht mehr gerechtfertigt ist. (Vgl. unten § 284 und § 288.)

§ 179. In gleicher Weise wie für den Verdampfungsprozeß läßt sich die Gleichung (111) auch auf den Schmelz- oder auch auf den Sublimationsprozeß anwenden. Im ersten Fall bedeutet r die Schmelzwärme der Substanz, falls der Index 1 dem flüssigen, der Index 2 dem festen Zustand entspricht, ferner p_1 den Schmelzdruck, d. h. den Druck, bei welchem feste und flüssige Substanz sich im Gleichgewicht berühren können. Der Schmelzdruck hängt hiernach, ebenso wie der Verdampfungsdruck, von der Temperatur ab, oder in umgekehrter Fassung: durch Veränderung des Druckes wird die Schmelztemperatur geändert:

$$\frac{dT}{dp_1} = \frac{T(v_1 - v_2)}{r}. \tag{113}$$

[1] Wied. Ann. d. Phys. **17**, 193, 1882.

Für Eis bei 0° C., also unter Atmosphärendruck, ergibt sich z. B.

$r = 80 \cdot 4,19 \cdot 10^7$ (Schmelzwärme von 1 g Eis in absoluten C.G.S.-Einheiten),

$T = 273$,

$v_1 = 1,000$ (Volumen von 1 g Wasser bei 0° C. in ccm),

$v_2 = 1,091$ (Volumen von 1 g Eis bei 0° C. in ccm).

Um $\frac{dT}{dp_1}$ in Atmosphären zu erhalten, hat man den Ausdruck noch mit 1013250 (§ 7) zu multiplizieren und erhält so aus (113):

$$\frac{dT}{dp_1} = -\frac{273 \cdot 0,091 \cdot 1013250}{80 \cdot 4,19 \cdot 10^7} = -0,0075. \tag{114}$$

Durch Erhöhung des äußeren Druckes um 1 Atmosphäre wird also die Schmelztemperatur des Eises um 0,0075° C. erniedrigt, oder: um den Schmelzpunkt des Eises um 1° C. zu erniedrigen, bedürfte es einer Druckerhöhung von ca. 130 Atmosphären, was zuerst durch Messungen von W. Thomson (Lord Kelvin) bestätigt worden ist. Für Substanzen, welche sich, entgegengesetzt dem Eis, beim Schmelzen ausdehnen, wird nach der Gleichung (113) umgekehrt die Schmelztemperatur mit wachsendem Druck erhöht. Auch dies ist durch Messungen qualitativ und quantitativ bestätigt worden.

§ 180. Die Gleichungen (101) gestatten, noch andere wichtige Eigenschaften, die eine Substanz in verschiedenen Aggregatzuständen besitzt, in Beziehung miteinander zu bringen. Wir fassen sie mit (110) in folgender Form zusammen:

$$\frac{r}{T} = s_1 - s_2$$

und differentiieren nach T. Dann ergibt sich:

$$\frac{1}{t} \cdot \frac{dr}{dT} - \frac{r}{T^2} = \left(\frac{\partial s_1}{\partial T}\right)_p + \left(\frac{\partial s_1}{\partial p}\right)_T \cdot \frac{dp_1}{dT} - \left(\frac{\partial s_2}{\partial T}\right)_p - \left(\frac{\partial s_2}{\partial p}\right)_T \frac{dp_2}{dT}$$

oder nach (84a) und (84b)

$$= \frac{(c_p)_1}{T} - \left(\frac{\partial v_1}{\partial T}\right)_p \frac{dp_1}{dT} - \frac{(c_p)_2}{T} + \left(\frac{\partial v_2}{\partial T}\right)_p \frac{dp_2}{dT}.$$

Da nun nach (111):

$$\frac{dp_1}{dT} = \frac{dp_2}{dT} = \frac{r}{T(v_1 - v_2)},$$

so erhält man schließlich:

$$(115) \qquad (c_p)_1 - (c_p)_2 = \frac{dr}{dT} - \frac{r}{T} + \frac{r}{v_1 - v_2}\left\{\left(\frac{\partial v_1}{\partial T}\right)_p - \left(\frac{\partial v_2}{\partial T}\right)_p\right\}.$$

Diese Gleichung gestattet abermals eine Prüfung des zweiten Hauptsatzes, da sie lauter Größen enthält, die unabhängig voneinander gemessen werden können.

§ 181. Nehmen wir als Beispiel wieder gesättigten Wasserdampf bei 100° C., also unter Atmosphärendruck, und berechnen hierfür die spezifische Wärme des Dampfes bei konstantem Druck: $(c_p)_1$, dann ist:

$(c_p)_2 = 1,01$ (spezifische Wärme des flüssigen Wassers bei 100°),
$r = 539$ (Verdampfungswärme des Wassers bei 100°),
$T = 373$.

Ferner[1]:

$\frac{dr}{dT} = -0,64$ (Abnahme der Verdampfungswärme mit der Temperatur),
$v_1 = 1674$ (Volumen von 1 g gesättigtem Wasserdampf bei 100°),
$\left(\frac{\partial v_1}{\partial T}\right)_p = 4,813$ (Isobarer Ausdehnungskoeffizient dieses Dampfes).

Die entsprechenden Größen für flüssiges Wasser sind:

$$v_2 = 1,0,$$
$$\left(\frac{\partial v_2}{\partial T}\right)_p = 0,001.$$

Diese Zahlen liefern nach (115) das Ergebnis:

$$(c_p)_1 - (c_p)_2 = -0,54,$$

oder:

$$(c_p)_1 = (c_p)_2 - 0,54 = 1,01 - 0,54 = 0,47,$$

[1] Vgl. O. Knoblauch, R. Linde und H. Klebe, Mitteilungen über Forschungsarbeiten, herausgegeben vom Verein Deutscher Ingenieure Heft 21, Berlin 1905. Ferner: Harvey N. Davis, Proc. of the American Academy of Arts and Sciences. Bd. 45, p. 265, 1910. F. Henning, Zeitschr. f. Physik **2**, p. 197, 1920.

in Übereinstimmung mit den direkten Messungen.

§ 182. Die Beziehung (115) vereinfacht sich bedeutend, wird aber ungenau, wenn man wieder das Volumen v_2 des flüssigen Wassers gegen das v_1 des Dampfes vernachlässigt und für letzteres die Zustandsgleichung eines idealen Gases benutzt. Denn dann wird:

$$v_1 = \frac{R}{m}\frac{T}{p_1}$$
$$\left(\frac{\partial r_1}{\partial T}\right)_p = \frac{R}{mp_1}$$

und die Gleichung (115) lautet einfach:

$$(c_p)_1 - (c_p)_2 = \frac{dr}{dT},$$

in unserem Beispiel:

$$(c_p)_1 - (c_p)_2 = -0,64$$
$$(c_p)_1 = 1,01 - 0,64 = 0,37,$$

also erheblich zu klein.

§ 183. Wenden wir nun die Beziehung (115) auch auf schmelzendes Eis bei 0° und Atmosphärendruck an, indem wir den Index 1 auf den flüssigen, den Index 2 auf den festen Aggregatzustand beziehen. Die Abhängigkeit der Schmelzwärme r des Eises von der Schmelztemperatur T ist wohl noch nicht direkt gemessen worden, sie läßt sich aber aus (115) berechnen, da diese Gleichung ergibt:

$$\frac{dr}{dt} = (c_p)_1 - (c_p)_2 + \frac{r}{T} - \frac{r}{v_1 - v_2}\left\{\left(\frac{\partial v_1}{\partial T}\right)_p - \left(\frac{\partial v_2}{\partial T}\right)_p\right\}.$$

Dabei ist:

$$(c_p)_1 = 1,01 \text{ (Spezifische Wärme des Wassers bei } 0^\circ),$$
$$(c_p)_2 = 0,50 \text{ (Spezifische Wärme von Eis bei } 0^\circ),$$
$$r = 80,$$
$$T = 273,$$
$$v_1 = 1,00,$$
$$v_2 = 1,09,$$
$$\left(\frac{\partial v_1}{\partial T}\right)_p = -0,00006 \text{ (Ausdehnungskoeffizient des Wassers bei } 0^\circ),$$
$$\left(\frac{\partial v_2}{\partial T}\right)_p = 0,0001 \text{ (Ausdehnungskoeffizient von Eis bei } 0^\circ).$$

Folglich nach der obigen Gleichung:

$$\frac{dr}{dT} = 0,66,$$

d. h. wenn der Schmelzpunkt des Eises durch entsprechende Vermehrung des äußeren Druckes um 1° erniedrigt wird, nimmt auch die Schmelzwärme um 0,66 cal ab.

§ 184. Es ist schon früher wiederholt darauf hingewiesen worden, daß man außer der spezifischen Wärme bei konstantem Druck und der bei konstantem Volumen noch beliebige andere spezifische Wärmen definieren kann, je nachdem man die äußeren Umstände, unter denen die Erwärmung stattfindet, verschieden reguliert. In jedem Falle gilt die Gleichung (23) des ersten Hauptsatzes:

$$c = \frac{du}{dT} + p\frac{dv}{dT}.$$

Bei den gesättigten Dämpfen ist nun auch diejenige Art der Erwärmung von Interesse, bei welcher der Dampf immer gerade im Zustand der Sättigung erhalten wird. Bezeichnen wir die diesem Vorgang entsprechende spezifische Wärme des Dampfes mit h_1 — Clausius nannte sie die spezifische Wärme „des gesättigten Dampfes“ — so ergibt sich in unserer Bezeichnung:

$$h_1 = \frac{du_1}{dT} + p_1\frac{dv_1}{dT}. \tag{116}$$

Über den Wert von h_1 läßt sich von vornherein nichts aussagen, ja selbst das Vorzeichen dieser Größe muß vorläufig dahingestellt bleiben. Denn wenn der Dampf während der Erwärmung um 1° gerade gesättigt bleiben soll, muß

er offenbar gleichzeitig komprimiert werden, weil das spezifische Volumen des gesättigten Dampfes mit steigender Temperatur abnimmt. Nun wird aber durch die Kompression Wärme erzeugt, und es fragt sich, ob diese Wärme nicht so beträchtlich ist, daß sogar eine Ableitung von Wärme nach außen erforderlich wird, um den Dampf nicht zu überhitzen. Daher sind hier von vornherein zwei Fälle denkbar: 1. Die Kompressionswärme ist so beträchtlich, daß der ursprünglich gesättigte Dampf bei adiabatischer Kompression überhitzt wird. Dann ist bei der Kompression des gesättigten Dampfes Ableitung von Wärme nach außen erforderlich, um bei der erhöhten Temperatur den Sättigungszustand aufrecht zu erhalten, d. h. h_1 ist negativ. 2. Die Kompressionswärme ist zu gering, um ohne Zuleitung äußerer Wärme den komprimierten Dampf vor Übersättigung zu bewahren; dann muß h_1 positiv ausfallen. Dazwischen liegt der Grenzfall $h_1 = 0$, wo die Kompressionswärme gerade hinreicht, um den komprimierten Dampf im Zustand der Sättigung zu erhalten, wo also die Sättigungskurve zusammenfällt mit der Kurve der adiabatischen Kompression. Dieser Grenzfall wurde noch von Watt als für Wasserdampf gültig angenommen.

Es ist nun leicht, h_1 aus den obigen Formeln zu berechnen. Bilden wir zunächst die entsprechende spezifische Wärme für die berührende Flüssigkeit:

$$h_2 = \frac{du_2}{dT} + p_2 \frac{dv_2}{dT}. \tag{117}$$

Diese spezifische Wärme entspricht einer Erwärmung der Flüssigkeit, die immer gerade unter dem Drucke ihres gesättigten Dampfes gehalten wird. Da nun der äußere Druck, wenn er nicht nach vielen Atmosphären mißt, auf den Zustand einer Flüssigkeit keinen wesentlichen Einfluß hat, so fällt der Wert von h_2 so gut wie ganz mit dem Wert der spezifischen Wärme der Flüssigkeit bei konstantem Druck zusammen, d. h.

$$h_2 = (c_p)_2. \tag{118}$$

Nun ergeben die Gleichungen (116) und (117) voneinander subtrahiert:

$$h_1 - h_2 = \frac{d(u_1 - u_2)}{dT} + p_1 \frac{d(v_1 - v_2)}{dT}.$$

Aber nach (110) ist durch Differentiation nach T:

$$\frac{dr}{dT} = \frac{d(u_1 - u_2)}{dT} + p_1 \frac{d(v_1 - v_2)}{dT} + (v_1 - v_2) \frac{dp_1}{dT}.$$

Folglich:

$$h_1 - h_2 = \frac{dr}{dT} - (v_1 - v_2) \frac{dp_1}{dT}$$

oder nach (118) und (111):

$$h_1 = (c_p)_2 + \frac{dr}{dT} - \frac{r}{T}.$$

Für gesättigten Wasserdampf bei 100° haben wir nun, wie oben:

$$\begin{aligned} (c_p)_2 &= 1,01, \\ \frac{dr}{dT} &= -0,64, \\ r &= 539, \\ T &= 373. \end{aligned}$$

Folglich:

$$h_1 = 1,01 - 0,64 - 1,44 = -1,07.$$

Wasserdampf bei 100° C. repräsentiert also den oben unter 1. beschriebenen Fall, d. h. gesättigter Wasserdampf bei 100°, adiabatisch komprimiert, wird überhitzt; oder umgekehrt: gesättigter Wasserdampf bei 100°, adiabatisch ausgedehnt, wird übersättigt, indem der Einfluß der Kompressions-, bez. Dilatationswärme den Einfluß der Dichtigkeitszunahme, bez. Abnahme weit überwiegt. Andere Dämpfe zeigen das entgegengesetzte Verhalten.

§ 185. Es kann der Fall eintreten, daß für einen bestimmten Wert von T die Werte der Größen v_1 und v_2, wie sie sich aus den Gleichungen (101) in ganz bestimmter Weise ergeben, einander gleich werden; dann sind die beiden Aggregatzustände, die miteinander in Berührung sind, überhaupt identisch. Ein solcher Wert von T heißt eine kritische Temperatur (§ 28) der betreffenden Substanz. Vom rein mathematischen Standpunkt aus muß man von vornherein annehmen, daß jede Substanz für jede der drei Kombinationen zweier Aggregatzustände eine solche kritische Temperatur besitzt, die allerdings nicht immer reell sein wird. Durch die kritische Temperatur T und das kritische Volumen $v_1 = v_2$ ist dann auch der ganze kritische Zustand bestimmt. Seine Berechnung erfolgt aus den Gleichungen (101), wenn man darin noch die Bedingung einführt, daß die Differenz $v_1 - v_2$ verschwindet. Nehmen wir also $v_1 - v_2$ sehr klein an, so wird für ein beliebiges Volumen v, welches zwischen den Werten v_1 und v_2 liegt, nach dem TAYLORschen Satze:

$$p = p_2 + \left(\frac{\partial p}{\partial v}\right)_2 (v - v_2) + \frac{1}{2}\left(\frac{\partial^2 p}{\partial v^2}\right)_2 (v - v_2)^2. \tag{119}$$

Dann geht die erste Gleichung (101) über in:

$$p_2 + \left(\frac{\partial p}{\partial v}\right)_2 (v_1 - v_2) + \frac{1}{2}\left(\frac{\partial^2 p}{\partial v}\right)_2 (v_1 - v_2)^2 = p_2$$

und die Gleichung (102) liefert, durch Ausführung der Integration von (119) nach v:

$$p_2(v_1 - v_2) + \frac{1}{2}\left(\frac{\partial p}{\partial v}\right)_2 (v_1 - v_2)^2 + \frac{1}{2 \cdot 3}\left(\frac{\partial^2 p}{\partial v^2}\right)(v_1 - v_2)^3 = p_2(v_1 - v_2).$$

Die letzten beiden Gleichungen ergeben:

$$\left(\frac{\partial p}{\partial v}\right)_2 = 0 \qquad \left(\frac{\partial^2 p}{\partial v^2}\right)_2 = 0$$

als Bedingung des kritischen Zustandes. Diese Bedingung stimmt überein mit der schon im § 30 für den kritischen Zustand eines Dampfes abgeleiteten Beziehung, und wird durch die dort gegebene Zeichnung der Isotherme geometrisch illustriert. Im kritischen Zustand ist die Kompressibilität unendlich groß, der Ausdehnungskoeffizient bei konstantem Druck unendlich groß, die spezifische Wärme bei konstantem Druck unendlich groß, die Verdampfungswärme Null.

Bei anderen Temperaturen als der kritischen sind die Werte von v_1 und v_2 verschieden, und zwar auf der einen Seite reell, auf der anderen komplex; im letzteren Fall verliert die hier betrachtete Lösung des Gleichgewichtsproblems ihren Sinn.

§ 186. Dritte Lösung. Setzen wir drittens in den für das innere Gleichgewicht gültigen Bedingungen (98):

$$v_1 \neq v_2 \neq v_3,$$

so haben wir ohne Vereinfachung:

$$\left\{\begin{array}{c} p_1 = p_2 = p_3 \\ s_1 - s_2 = \dfrac{u_1 - u_2 + p_1(v_1 - v_2)}{T} \\ s_2 - s_3 = \dfrac{u_2 - u_3 + p_1(v_2 - v_3)}{T}. \end{array}\right. \tag{120}$$

Dieser Fall bezeichnet einen Zustand, bei welchem sich im System alle drei Aggregatzustände nebeneinander vorfinden. Die vier Gleichungen (120) enthalten vier Unbekannte, nämlich T, v_1, v_2, v_3, so daß ihnen ganz bestimmte Werte dieser vier Größen entsprechen. Alle drei Aggregatzustände können also nur bei einer ganz bestimmten Temperatur und ganz bestimmten Dichtigkeiten, und daher auch nur bei einem ganz bestimmten Druck nebeneinander im Gleichgewicht existieren. Wir wollen diese Temperatur die „Fundamentaltemperatur“ und den entsprechenden Druck

den Fundamentaldruck der Substanz nennen. Die Fundamentaltemperatur ist nach den Gleichungen (120) durch die Bedingung charakterisiert, daß für sie der Druck des über der Flüssigkeit gesättigten Dampfes gleich ist dem Schmelzdruck. Dann folgt mit Notwendigkeit durch Addition der beiden letzten Gleichungen, daß jener Druck auch gleich dem Sublimationsdruck ist, bei welchem die feste Substanz mit der gasförmigen in Berührung ist.

Sind die Fundamentalwerte gefunden, so berechnen sich aus den äußeren Bedingungen im § 166

$$\left\{\begin{aligned} M_1 + M_2 + M_3 &= M \\ M_1 v_1 + M_2 v_2 + M_3 v_3 &= V \\ M_1 u_1 + M_2 u_2 + M_3 u_3 &= U \end{aligned}\right. \tag{121}$$

die Massen M_1, M_2, M_3 der in den verschiedenen Aggregatzuständen befindlichen Teile des Systems in eindeutiger Weise. Doch hat diese Lösung nur dann einen physikalischen Sinn, wenn die Werte von M_1, M_2, M_3 sämtlich positiv ausfallen.

§ 187. Bestimmen wir z. B. den Fundamentalzustand des Wassers. Da für 0° C. der Druck des über flüssigem Wasser gesättigten Dampfes $4,58^{\text{mm}}$, der Schmelzdruck des Eises aber 760^{mm} beträgt, so ist 0° nicht die Fundamentaltemperatur des Wassers. Nun nimmt aber der Schmelzdruck des Eises mit steigender Temperatur ab, während der Druck des über flüssigem Wasser gesättigten Dampfes wächst; folglich wird für eine etwas höhere Temperatur als 0° ein Zusammenfallen jener beiden Drucke eintreten. Nach der Gleichung (114) steigt die Schmelztemperatur des Eises bei Erniedrigung des Druckes von 760^{mm} bis $4,58^{\text{mm}}$ um nahezu $0,0075°$. Die Temperatur $0,0075°$ C. ist also sehr angenähert die Fundamentaltemperatur des Wassers, da für sie der Druck des über flüssigem Wasser gesättigten Dampfes nahe zusammenfällt mit dem Schmelzdruck des Eises, und infolgedessen auch mit dem Druck des über Eis gesättigten Dampfes. Daraus ergeben sich dann auch die Werte für das spezifische Volumen von Wasser im gasförmigen, flüssigen und festen Fundamentalzustand:

$$v_1 = 206000, \qquad v_2 = 1,00, \qquad v_3 = 1,09.$$

Für andere Temperaturen als die Fundamentaltemperatur fallen Verdampfungs-, Schmelz- und Sublimationsdruck alle verschieden aus.

§ 188. Überblicken wir nun noch einmal die inneren Gleichgewichtsbedingungen (101) für die drei Kombinationen je zweier sich berührender Aggregatzustände einer gegebenen Substanz im Zusammenhang. Für jede dieser Kombinationen ist sowohl der Druck p als auch die spezifischen

Volumina der beiden sich berührenden Teile allein von der Temperatur abhängig und durch (101) bestimmt. Hierbei ist aber wohl zu unterscheiden, ob sich z. B. gesättigter Dampf in Berührung mit flüssiger oder mit fester Substanz befindet, da für diese beiden Fälle die Funktionen, welche Druck und spezifisches Volumen des gesättigten Dampfes in ihrer Abhängigkeit von der Temperatur darstellen, ganz verschieden ausfallen. Der Zustand des gesättigten Dampfes ist erst dann bestimmt, wenn außer der Temperatur auch noch angegeben ist, mit welchem Aggregatzustand der Dampf sich in Berührung befindet, und das nämliche gilt für die beiden anderen Aggregatzustände. Wenn wir daher von jetzt an die Ziffern 1, 2, 3 der Reihe nach auf den gasförmigen, flüssigen, festen Zustand beziehen, so müssen wir zur Bezeichnung eines im Zustand der Sättigung befindlichen Körperteils zwei Indizes verwenden, von denen der erste den Aggregatzustand des betrachteten Körperteils selbst, der zweite denjenigen Aggregatzustand angibt, mit welchem der Körperteil in Berührung gedacht ist. So erhalten wir zur Bezeichnung des spezifischen Volumens des gesättigten Dampfes die beiden Ausdrücke v_{12} und v_{13}, von denen der erste den Dampf in Berührung mit flüssiger, der zweite den Dampf in Berührung mit fester Substanz darstellt. Analog ergeben sich die Bezeichnungen v_{23} und v_{21}, v_{31} und v_{32} für die spezifischen Volumina flüssiger und fester Substanz im Zustand der Sättigung; jede von diesen sechs Größen ist eine bestimmte Funktion der Temperatur allein. Die entsprechenden Drucke sind:

Verdampfungsdruck:	Schmelzdruck:	Sublimationsdruck:
$p_{12} = p_{21}$	$p_{23} = p_{32}$	$p_{31} = p_{13}$

ebenfalls Funktionen der Temperatur allein. Nur für die Fundamentaltemperatur werden zwei dieser Drucke einander gleich, und dann auch gleich dem dritten.

Stellt man also die Abhängigkeit der drei Drucke von der Temperatur durch drei Kurven dar, indem man etwa die Temperatur als Abszisse und die Drucke als Ordinaten aufträgt, so schneiden sich die drei Kurven in einem einzigen Punkt, dem Fundamentalpunkt, auch dreifacher Punkt genannt. Es ist auch leicht zu berechnen, unter welchem Winkel sich die Kurven in dem Fundamentalpunkt schneiden. Denn die Neigung der Kurven gegen die Abszissenachse wird gegeben durch die Differentialquotienten:

$$\frac{dp_{12}}{dT}, \qquad \frac{dp_{23}}{dT}, \qquad \frac{dp_{31}}{dT}.$$

Nun ist nach Gleichung (111) in entsprechender Bezeichnung

$$\frac{dp_{12}}{dT} = \frac{r_{12}}{T(v_1 - v_2)}.$$

Ebenso

$$\frac{dp_{23}}{dT} = \frac{r_{23}}{T(v_2 - v_3)}$$

und

$$\frac{dp_{31}}{dT} = \frac{r_{31}}{T(v_3 - v_1)}.$$

Dabei beziehen sich die v auf den Fundamentalzustand und sind daher nur mit einem einzigen Index versehen. Hieraus ergibt sich nun die Richtung des Verlaufs jeder Kurve im Fundamentalpunkt, sobald man die Verdampfungs-, Schmelz- und Sublimationswärme kennt.

Vergleichen wir z. B. die Kurve des Verdampfungsdrucks p_{12} mit der Kurve des Sublimationsdrucks p_{13} für Wasser in der Nähe des Fundamentalpunkts $0,0075°$ C. Hierfür ist im absoluten Maßsystem, durch Multiplikation des in Kalorien ausgedrückten Wertes mit dem mechanischen Wärmeäquivalent:

$$r_{12} = 600 \cdot 4,19 \cdot 10^7$$

(Verdampfungswärme des Wassers bei $0,0075°$ C.),

$$r_{13} = -r_{31} = (80 + 600) \cdot 4,19 \cdot 10^7$$

(Sublimationswärme des Eises bei $0,0075°$ C.),

$$v_1 = 206000, \qquad v_2 = 1,00, \qquad v_3 = 1,09, \qquad (\S\,187)$$
$$T = 273.$$

Also in Millimetern Quecksilber, durch Multiplikation des absoluten Wertes mit $\frac{760}{1013250}$:

$$\frac{dp_{12}}{dT} = \frac{600 \cdot 4,19 \cdot 10^7 \cdot 760}{273 \cdot 206000 \cdot 1013250} = 0,335,$$
$$\frac{dp_{31}}{dT} = \frac{dp_{13}}{dT} = \frac{680 \cdot 4,19 \cdot 10^7 \cdot 760}{273 \cdot 206000 \cdot 1013250} = 0,380.$$

Die Kurve des Sublimationsdrucks p_{13} verläuft also im Fundamentalpunkt steiler als die Kurve des Verdampfungsdrucks p_{12}, oder: für Temperaturen oberhalb der Fundamentaltemperatur ist $p_{13} > p_{12}$, für Temperaturen unterhalb derselben ist $p_{13} < p_{12}$. Die Differenz beträgt:

$$\frac{dp_{13}}{dT} - \frac{dp_{12}}{dT} = \frac{d(p_{13} - p_{12})}{dT} = 0,045.$$

Mißt man also die Spannung des gesättigten Wasserdampfes oberhalb des Fundamentalpunktes über Wasser, unterhalb desselben über Eis, so erleidet die Spannungskurve im Fundamentalpunkt einen Knick, dessen Größe durch den Sprung des Differentialquotienten, d. h. durch die obige Differenz angegeben wird. Bei $-1°$ $(dT = -1)$ ist demnach angenähert:

$$p_{13} - p_{12} = -0,04,$$

d. h. bei $-1°$ C. ist der Druck des gesättigten Wasserdampfes über Eis um $0,04^{\text{mm}}$ kleiner als der über Wasser, was auch experimentell bestätigt worden ist.[1] Dagegen läßt sich die Existenz eines scharfen Knicks in dem angegebenen Betrage nur aus der Theorie erschließen.

§ 189. Wir haben unsere bisherigen Untersuchungen nur auf die Betrachtung der einzelnen verschiedenartigen Lösungen derjenigen Gleichungen erstreckt, welche die inneren Gleichgewichtsbedingungen des Systems aussprechen, und daraus die wichtigsten Eigenschaften des betreffenden Gleichgewichtszustandes abgeleitet. Nunmehr kommen wir zu der weiteren Frage, welche unter den verschiedenen möglichen Lösungen der Aufgabe in jedem gegebenen Falle den Vorzug besitzt, d. h. den stabilsten Gleichgewichtszustand darstellt. Zur Beantwortung dieser Frage nehmen wir die ursprünglich in § 165 gegebene Fassung des Problems wieder auf, welche kurz folgendermaßen lautet. Gegeben ist die Gesamtmasse M, das Gesamtvolumen V, die Gesamtenergie U des Systems. Statt V und U wird es öfter bequemer sein, die Werte $\frac{V}{M} = v$ (mittleres spezifisches Volumen des Systems) und $\frac{U}{M} = u$ (mittlere spezifische Energie des Systems) zu benutzen. Gesucht ist der stabilste Gleichgewichtszustand, d. h. der Zustand des absoluten Maximums der Gesamtentropie S.

Wir fanden oben, daß im allgemeinen die Gleichgewichtsbedingungen drei verschiedene Arten von Lösungen zulassen, je nachdem das System sich in 1, 2 oder 3 Aggregatzustände spaltet. Bei der Frage, welche von

[1] Wärmetabellen von L. Holborn, K. Scheel und F. Henning. Braunschweig (Vieweg u. Sohn) 1919, p. 61.

diesen drei Lösungen in jedem gegebenen Falle den Vorzug hat, ist zunächst zu berücksichtigen, daß die zweite und die dritte Lösung nur dann einen physikalischen Sinn haben, wenn die aus den Gleichungen (103) und (121) sich ergebenden Werte der Massen positiv ausfallen. Dies führt zu einer Einschränkung des Gültigkeitsbereichs dieser beiden Lösungen. Zuerst wollen wir diesen Gültigkeitsbereich feststellen, und werden dann den Nachweis führen, daß innerhalb ihres Gültigkeitsbereichs die dritte Lösung stets den Vorzug hat vor den beiden ersten, und die zweite den Vorzug hat vor der ersten.

Zur Erleichterung der Übersicht möge die geometrische Anschauung zu Hilfe genommen werden. Zu diesem Zweck denken wir uns die von vornherein gegebenen Werte $v = \frac{V}{M}$ und $u = \frac{U}{M}$ (der Wert von M ist hier nebensächlich) dadurch graphisch dargestellt, daß wir diese Größen als die rechtwinkligen Koordinaten eines Punktes in einer Ebene (der Zeichnungsebene in Fig. 4) ansehen, so daß jedem Punkt der Ebene ein bestimmtes Wertenpaar dieser beiden Größen entspricht. Unsere Aufgabe ist dann die, für jeden beliebig gegebenen Punkt dieser Ebene die Entscheidung zu treffen, welcher Art das stabile Gleichgewicht ist, welches bei den entsprechenden Werten von v und u zustande kommt.

§ 190. Betrachten wir nun den Gültigkeitsbereich der dritten Lösung. Die sich aus den Gleichungen (121) ergebenden Werte der Massen M_1, M_2, M_3 sind:

$$M_1 : M_2 : M_3 : M = \begin{vmatrix} 1 & 1 & 1 \\ v & v_2 & v_3 \\ u & u_2 & u_3 \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ v & v_3 & v_1 \\ u & u_3 & u_1 \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ v & v_1 & v_2 \\ u & u_1 & u_2 \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \end{vmatrix} \tag{121a}$$

wobei wir hier, wie überall im folgenden, die Bezeichnung v_1, v_2, v_3, u_1, u_2, u_3 speziell auf die Fundamentalwerte der v und u anwenden.

Hieraus ersieht man, daß die Werte von M_1, M_2, M_3 nur dann alle zugleich positiv ausfallen, wenn der dem Wertenpaar (v, u) entsprechende Punkt innerhalb des Dreiecks liegt, das von den Punkten mit den respektiven Koordinaten (v_1, u_1), (v_2, u_2) und (v_3, u_3) gebildet wird. Der Gültigkeitsbereich der dritten Lösung wird daher durch dieses Dreieck dargestellt, welches wir das Fundamentaldreieck der Substanz nennen können, und ist in der Fig. 4 mit (123) bezeichnet. Die Zeichnung ist für eine Substanz ausgeführt, für die, wie bei Wasser, $v_1 > v_3 > v_2$ und $u_1 > u_2 > u_3$.

§ 191. Wir kommen nun zur Betrachtung des Gültigkeitsbereichs der zweiten Lösung, welcher die Gleichungen (101) und (103) entsprechen. Diese Gleichungen ergeben drei Arten von Wertensystemen, je nach den drei

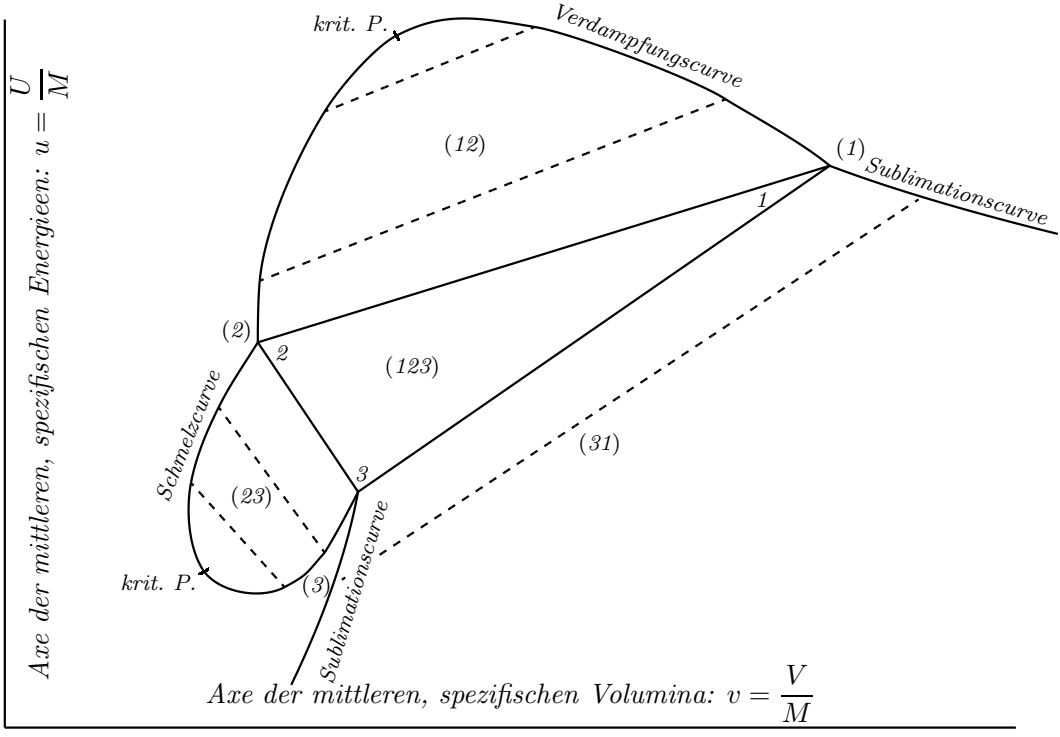


Fig. 4.

paarweisen Kombinationen der drei Aggregatzustände, deren jede den beiden andern von vornherein gleichberechtigt gegenübersteht. Wir betrachten zunächst die Kombination des gasförmigen und des flüssigen Zustandes. Dann lauten jene Gleichungen gemäß der jetzt eingeführten Bezeichnung:

$$(122) \qquad \begin{cases} T_{12} = T_{21}, \qquad p_{12} = p_{21} \\ s_{12} - s_{21} = \dfrac{u_{12} - u_{21} + p_{12}(v_{12} - v_{21})}{T_{12}} \end{cases}$$

und:

$$(123) \qquad \begin{cases} M_{12} + M_{21} = M \\ M_{12}v_{12} + M_{21}v_{21} = V = Mv \\ M_{12}u_{12} + M_{21}u_{21} = U = Mu. \end{cases}$$

Um das Gebiet zu finden, innerhalb dessen der Punkt mit den Koordinaten v, u liegen muß, damit M_{12} und M_{21} beide positiv ausfallen, suchen wir die

Grenzen dieses Gebietes auf, d. h. die Kurven, welche durch die Bedingungen $M_{12} = 0$ und $M_{21} = 0$ dargestellt werden; zunächst die Kurve: $M_{21} = 0$ (flüssige Masse $= 0$). Diese Bedingung in (123) eingeführt ergibt: $M_{12} = M$ und

$$v = v_{12} \qquad u = u_{12}. \tag{124}$$

Da v_{12} und u_{12} Funktionen einer einzigen Variabeln sind, so ist durch diese beiden Gleichungen den Größen v und u eine bestimmte Bedingung vorgeschrieben, und diese Bedingung ergibt die gesuchte Kurve, eine Grenze des gesuchten Gültigkeitsbereichs. Diese Kurve geht durch die Ecke *1* des Fundamentaldreiecks, weil für die Fundamentaltemperatur $v_{12} = v_1$ und $u_{12} = u_1$ wird. Zur Feststellung ihres weiteren Verlaufs bilden wir den Ausdruck des Differentialquotienten $\frac{du_{12}}{dv_{12}}$. Hierfür hat man:

$$\frac{du_{12}}{dv_{12}} = \left(\frac{\partial u}{\partial v}\right)_{12} + \left(\frac{\partial u}{\partial T}\right)_{12} \frac{dT_{12}}{dv_{12}}.$$

Die mit ∂ bezeichneten partiellen Differentialquotienten beziehen sich hier überall auf die unabhängigen Variabeln T und v. Daraus folgt nach (80) und (24):

$$\frac{du_{12}}{dv_{12}} = T_{12}\left(\frac{\partial p}{\partial T}\right)_{12} - p_{12} + (c_v)_{12}\frac{dT_{12}}{dv_{12}}.$$

Mittels dieser Gleichung kann man den Verlauf der Kurve (124) experimentell verfolgen, indem man T_{12} oder v_{12} oder irgend eine andere geeignete Größe als unabhängigen Parameter nimmt.

In gleicher Weise liefert die Bedingung $M_{12} = 0$ (dampfförmige Masse $= 0$) eine andere Grenze des gesuchten Gültigkeitsbereichs durch die Kurve:

$$v = v_{21} \qquad u = u_{21},$$

welche durch die Ecke *2* des Fundamentaldreiecks geht und der Gleichung genügt:

$$\frac{du_{21}}{dv_{21}} = T_{12}\left(\frac{\partial p}{\partial T}\right)_{21} - p_{12} + (c_v)_{21}\frac{dT_{12}}{dv_{21}}.$$

Hierbei ist davon Gebrauch gemacht, daß $T_21 = T_12$ und $p_21 = p_12$.

Diese beiden Kurven sind aber nichts anderes als Zweige einer und derselben Kurve, da sie für den kritischen Punkt: $v_{12} = v_{21}$, ineinander übergehen, und zwar, wie eine nähere Untersuchung des Wertes von $\frac{du_{12}}{dv_{12}}$

und $\frac{du_{21}}{dv_{21}}$ nach § 185 lehrt, ohne in diesem Punkte eine Ecke oder Spitze zu bilden.

Wir können daher beide Kurvenäste unter dem gemeinsamen Namen „Verdampfungskurve“ zusammenfassen. Dann entspricht jedem Punkt (v_{12}, u_{12}) auf dem einen Ast ein bestimmter Punkt (v_{21}, u_{21}) auf dem andern Ast, insofern beiden Punkten die nämliche Temperatur $T_{12} = T_{21}$ und der nämliche Druck $p_{12} = p_{21}$ zukommt. Diese Zuordnung je zweier Punkte auf den beiden Ästen wird bestimmt durch die Gleichungen (122) und ist in der Fig. 4 durch die Verbindungslinien einiger solcher Punktpaare angedeutet. So entsprechen sich auch die beiden Ecken des Fundamentaldreiecks (v_1, u_1) und (v_2, u_2). Der kritische Punkt entspricht sich selbst.

Die gefundene Verdampfungskurve bildet somit die Grenze des Gültigkeitsbereiches desjenigen Teils der zweiten Lösung, welchem die Berührung von Dampf und Flüssigkeit entspricht, und man überzeugt sich leicht aus (123), daß der Gültigkeitsbereich in den von der Kurve e i n g e s c h l o s s e n e n Raum der Zeichnungsebene fällt. Gleichwohl ist die Kurve nur bis zu den Ecken *1* und *2* des Fundamentaldreiecks gezeichnet, weil, wie sich später zeigen wird, die Lösung nur bis dahin das stabile Gleichgewicht angibt. Dieser Raum ist mit (*12*) bezeichnet.

Ganz analog der Verdampfungskurve ergibt sich nun auch der Verlauf der „Schmelzkurve“, deren beide Äste durch die Gleichungen:

$$v = v_{23} \qquad u = u_{23}$$

und:

$$v = v_{32} \qquad u = u_{32}$$

dargestellt werden, und der „Sublimationskurve“, für deren Äste die Gleichungen:

$$v = v_{31} \qquad u = u_{31}$$

und:

$$v = v_{13} \qquad u = u_{13}$$

gelten. Die erstere Kurve geht durch die Ecken *2* und *3*, die letztere durch die Ecken *3* und *1* des Fundamentaldreiecks. Die hierdurch abgegrenzten Gültigkeitsbereiche des 2. und 3. Teils der zweiten Lösung sind in der Fig. 4 mit (*23*) und (*31*) bezeichnet. Im übrigen gelten alle für die Verdampfungskurve abgeleiteten Beziehungen auch hier, nur mit entsprechender Vertauschung der Indizes. Einige entsprechende Punktpaare sind wieder durch Verbindungslinien angedeutet. Für die Schmelzkurve ist auch ein kritischer Punkt gezeichnet.

§ 192. Nachdem so auch für die zweite Lösung der Gültigkeitsbereich festgestellt ist, ersieht man unmittelbar, daß für alle Punkte (v, u), welche außerhalb der nun abgegrenzten Flächenräume liegen, nur die erste Lösung einen physikalischen Sinn ergibt, woraus folgt, daß für diese Punkte das stabile Gleichgewicht jedenfalls durch die erste Lösung (§ 170) dargestellt wird. Die entsprechenden Räume sind in der Fig. 4 mit *(1)*, *(2)* und *(3)* bezeichnet, je nachdem der betreffende Zustand als gasförmig, flüssig oder fest aufgefaßt wird. Wenn für zwei Aggregatzustände ein kritischer Punkt existiert, gibt es zwischen ihnen keine scharfe Grenze.

§ 193. Es handelt sich nun um die Frage: Welcher unter mehreren Gleichgewichtszuständen, die einem gegebenen Wertsystem M, v, u, also einem gegebenen Punkte der Zeichnungsebene entsprechen, besitzt den größten Wert der Entropie? Da jede der drei besprochenen Lösungen einen ganz bestimmten Zustand angibt, so erhalten wir für jedes gegebene Wertensystem (M, v, u) ebensoviel Werte der Entropie, als Lösungen für dies Wertensystem vorhanden sind. Bezeichnen wir also die den verschiedenen Lösungen entsprechenden Werte der Entropie der Reihe nach mit S, S' und S'', so haben wir:
Für die erste Lösung:

$$S = M \cdot s. \tag{125}$$

Für die zweite Lösung:

$$S' = M \cdot s' = M_{12}s_{12} + M_{21}s_{21}, \tag{126}$$

oder eine andere Kombination zweier Aggregatzustände.
Für die dritte Lösung:

$$S'' = M \cdot s'' = M_1 s_1 + M_2 s_2 + M_3 s_3. \tag{127}$$

Diese Größen sind alle vollständig bestimmt durch die gegebenen Werte von M, v und u. Es wird sich nun nachweisen lassen, daß für jedes beliebige Wertensystem (M, v, u) stets $S'' > S' > S$, oder $s'' > s' > s$, vorausgesetzt, daß sämtliche Massenteile positiv sind. Statt der Entropien selber ist es bequemer, die entsprechenden mittleren spezifischen Entropien s'', s', s zu betrachten, weil diese Größen gar nicht von M, sondern nur von v und u abhängen.

Zur geometrischen Veranschaulichung kann man sich in jedem Punkte (v, u) die entsprechenden Werte von s, s' und s'' in senkrechter Richtung zur Zeichnungsebene nach oben als Strecken aufgetragen denken, wodurch die drei Entropieflächen s, s' und s'' entstehen.

§ 194. Zunächst soll gezeigt werden, daß $s' - s$ stets positiv ist, d. h. daß die Fläche s' stets oberhalb der Fläche s liegt.

Während sich s direkt aus v und u nach der Definition (61) der Entropie für eine homogene Substanz ergibt, hat man zur Bestimmung des Wertes von s' die Gleichungen (126), (122) und (123). Durch dieselben wird s' als von v und u allein abhängig dargestellt und so die Fläche s' bestimmt, die im ganzen 3 Blätter bildet, entsprechend den 3 paarweisen Kombinationen der drei Aggregatzustände. Wir beziehen uns im folgenden zunächst wieder auf die Kombination von Dampf und Flüssigkeit.

Was nun die gegenseitige Lage der beiden Flächen s und s' anbelangt, so läßt sich leicht erkennen, daß dieselben eine Kurve gemeinsam haben, deren Projektion auf die Zeichnungsebene die Verdampfungskurve ist. Denn für irgend einen Punkt der Verdampfungskurve: $v = v_{12}$, $u = u_{12}$ hat man für die erste Fläche: $s = s_{12}$, wie selbstverständlich, und für die zweite Fläche zunächst aus (123):

$$M_{21} = 0, \qquad M_{12} = M \tag{128}$$

und aus (126): $s' = s_{12}$. In der Tat fallen ja für die Punkte der Verdampfungskurve die erste und die zweite Lösung zusammen. Die Schnittkurve der Flächen s und s' wird dargestellt durch die Gleichungen:

$$v = v_{12}, \qquad u = u_{12}, \qquad s = s_{12},$$

in denen v, u, s die drei variabeln orthogonalen Koordinaten eines Punktes im Raum vorstellen. v_{12}, u_{12}, s_{12} hängen ab von einem einzigen variabeln Parameter, z. B. der Temperatur $T_{12} = T_{21}$. Die Kurve geht auch durch den Punkt (v_1, u_1, s_1), welcher die Ecke *1* des Fundamentaldreiecks zur Projektion hat.

Ein anderer Ast derselben Schnittkurve ist gegeben durch die Gleichungen:

$$v = v_{12}, \qquad u = u_{12}, \qquad s = s_{12}.$$

Beide Äste treffen sich in einem Punkte, der den kritischen Punkt zur Projektion hat. Jedem Punkte des einen Astes ist ein bestimmter Punkt des andern zugeordnet, insofern beiden die nämliche Temperatur $T_{12} = T_{21}$ und der nämliche Druck $p_{12} = p_{21}$ entspricht. So ist dem Punkte (v_1, u_1, s_1) des ersten Astes der Punkt (v_2, u_2, s_2) des zweiten zugeordnet.

Ferner ist ersichtlich, daß die Fläche s' aus lauter Geraden besteht, und daß sie auf eine Ebene abwickelbar ist. Das erstere geht hervor aus der Betrachtung eines Punktes, dessen Koordinaten die Werte haben:

$$v = \frac{\lambda v_{12} + \mu v_{21}}{\lambda + \mu}, \quad u = \frac{\lambda u_{12} + \mu u_{21}}{\lambda + \mu}, \quad s = \frac{\lambda s_{12} + \mu s_{21}}{\lambda + \mu},$$

wobei λ und μ beliebige positive Größen sind. Für alle positiven Werte von λ und μ erhält man nämlich hieraus alle Punkte der geradlinigen Strecke, welche die beiden zugeordneten Punkte (v_{12}, u_{12}, s_{12}) und (v_{21}, u_{21}, s_{21}) verbindet. Diese Gerade liegt offenbar ganz auf der Fläche s', weil für jedes beliebige λ und μ die entsprechenden Werte von v, u, s die Gleichungen (123) und (126) befriedigen, wenn $M_{12} = \lambda$ und $M_{21} = \mu$ gesetzt wird. Also wird die Fläche s' gebildet von den geradlinigen Strecken, welche je zwei zugeordnete Punkte der Schnittkurve der Flächen s und s' verbinden. Eine solche Gerade der Fläche ist auch die Verbindungslinie der Punkte (v_1, u_1, s_1) und (v_2, u_2, s_2), deren Projektion auf die Zeichnungsebene die Seite (12) des Fundamentaldreiecks ist. Für den kritischen Punkt zieht sich diese Strecke auf einen Punkt zusammen, und hier erreicht die Fläche s' ihr Ende. Analog verhält es sich mit den beiden anderen Blättern der Fläche: das eine Blatt beginnt mit der Verbindungslinie der Punkte (v_2, u_2, s_2) und (v_3, u_3, s_3), das andere mit der der Punkte (v_3, u_3, s_3) und (v_1, u_1, s_1).

Die Abwickelbarkeit der Fläche s' ergibt sich am einfachsten aus der Betrachtung der Gleichung folgender Ebene:

$$p_{12}(v - v_{12}) + (u - u_{12}) - T_{12}(s - s_{12}) = 0,$$

worin v, u, s die drei variabeln Raumkoordinaten bedeuten, während p_{12}, v_{12}, u_{12}, T_{12}, s_{12} nach (122) von einem einzigen Parameter, etwa T_{12}, abhängen. Diese Ebene enthält erstens die zugeordneten Punkte (v_{12}, u_{12}, s_{12}) und (v_{21}, u_{21}, s_{21}), den letzteren vermöge der Gleichungen (122), also auch ihre Verbindungsstrecke, und zweitens die unendlich benachbarten zugeordneten Punkte mit den Koordinaten:

$$v_{12} + dv_{12}, \quad u_{12} + du_{12}, \quad s_{12} + ds_{12}$$

und

$$v_{21} + dv_{21}, \quad u_{21} + du_{21}, \quad s_{21} + ds_{21},$$

wie aus (61) folgt, also auch ihre Verbindungsstrecke. Mithin liegen zwei unendlich benachbarte Erzeugende der Fläche in einer Ebene, und die Fläche ist developpabel.

Zur Feststellung des Wertes von $s' - s$ möge die Änderung untersucht werden, welche diese Differenz dadurch erleidet, daß man von einem beliebigen Punkt (v, u) der Zeichnungsebene zu einem beliebigen unendlich benachbarten $(v + \delta v,\ u + \delta u)$ übergeht. Dabei lassen wir $M = M_{12} + M_{21}$ konstant, was der Allgemeinheit keinen Eintrag tut, weil s und s' nur von v und u abhängen. Nun haben wir durch Variation von (126):

$$M\,\delta s' = M_{12}\,\delta s_{12} + M_{21}\,\delta s_{21} + s_{12}\,\delta M_{12} + s_{21}\,\delta M_{21}.$$

Andrerseits nach (61):

$$\delta s = \frac{\delta u + p\,\delta v}{T}.$$

Aber nach (123) ist:

$$(129)\qquad \begin{cases} \delta M_{12} + \delta M_{21} = 0 \\ M_{12}\,\delta v_{12} + M_{21}\,\delta v_{21} + v_{12}\,\delta M_{12} + v_{21}\,\delta M_{21} = M\,\delta v \\ M_{12}\,\delta u_{12} + M_{21}\,\delta u_{21} + u_{12}\,\delta M_{12} + u_{21}\,\delta M_{21} = M\,\delta u. \end{cases}$$

Daraus ergibt sich unter Berücksichtigung von (122):

$$(130)\qquad \delta s' = \frac{\delta u + p_{12}\,\delta v}{T_{12}}$$

und daher:

$$(131)\qquad \delta(s' - s) = \left(\frac{1}{T_{12}} - \frac{1}{T}\right)\delta u + \left(\frac{p_{12}}{T_{12}} - \frac{p}{T}\right)\delta v.$$

Betrachten wir nun den Verlauf der Flächen s und s' in der Umgebung ihrer Schnittkurve, so ist aus der letzten Gleichung unmittelbar ersichtlich, daß sie sich längs dieser ganzen Kurve berühren. Denn z. B. für irgend einen Punkt der Verdampfungskurve: $v = v_{12},\ u = u_{12}$, dem nach (128) ein gemeinsamer Punkt beider Flächen entspricht, erhalten wir natürlich auch:

$$(132)\qquad T = T_{12}, \qquad p = p_{12},$$

und somit $\delta(s' - s) = 0$.

Um nun das Verhalten der beiden Flächen an diesen Berührungsstellen des Näheren zu prüfen, variieren wir die Gleichung (131) noch einmal allgemein, und wenden sie dann abermals auf dieselben Stellen an.

Zunächst erhalten wir allgemein:

$$\delta^2(s' - s) = \delta u\left(\frac{\delta T}{T^2} - \frac{\delta T_{12}}{T_{12}^2}\right) + \delta v\left(\frac{\delta p_{12}}{T_{12}} - \frac{\delta p}{T} - \frac{p_{12}\,\delta T_{12}}{T_{12}^2} + \frac{p\,\delta T}{T^2}\right)$$
$$+\delta^2 u\left(\frac{1}{T_{12}} - \frac{1}{T}\right) + \delta^2 v\left(\frac{p_{12}}{T_{12}} - \frac{p}{T}\right).$$

Dies ergibt für die Berührungspunkte der beiden Flächen nach (132):

$$T^2\delta^2(s - s') = \delta u\,(\delta T - \delta T_{12}) + \delta v\,(T\,\delta p_{12} - T\,\delta p - p\,\delta T_{12} + p\,\delta T)$$

oder kürzer, nach (61):

$$(133)\qquad T\,\delta^2(s' - s) = (\delta T - \delta T_{12})\,\delta s + (\delta p_{12} - \delta p)\,\delta v.$$

Nun reduzieren wir alle in diesem Ausdruck vorkommenden Variationen auf die beiden einzigen δT und δv, indem wir setzen:

Nach (81):
$$\delta s = \frac{c_v}{T}\,\delta T + \frac{\partial p}{\partial T}\,\delta v.$$

Ferner:
$$\delta p = \frac{\partial p}{\partial T}\,\delta T + \frac{\partial p}{\partial v}\,\delta v,$$
$$\delta p_{12} = \frac{dp_{12}}{dT_{12}}\,\delta T_{12}.$$

Es bleibt nun noch δT_{12} durch δT und δv auszudrücken. Dies geschieht durch die Gleichungen (129), welche mit Berücksichtigung der hier eintretenden Vereinfachung (128) die Bedingung liefern:

$$\frac{\delta u_{12} - \delta u}{u_{12} - u_{21}} = \frac{\delta v_{12} - \delta v}{v_{12} - v_{21}}.$$

Setzt man hierin:

(134)
$$\delta u_{12} = \frac{du_{12}}{dT_{12}}\,\delta T_{12}, \qquad \delta v_{12} = \frac{dv_{12}}{dT_{12}}\,\delta T_{12},$$
$$\delta u = c_v\,\delta T + \frac{\partial u}{\partial v}\,\delta v,$$

so ergibt sich:

$$\delta T_{12} = \frac{c_v\,\delta T + \left(\dfrac{\partial u}{\partial v} - \dfrac{u_{12} - u_{21}}{v_{12} - v_{21}}\right)\delta v}{\dfrac{du_{12}}{dT_{12}} - \dfrac{u_{12} - u_{21}}{v_{12} - v_{21}}\,\dfrac{dv_{12}}{dT_{12}}}.$$

Dieser Ausdruck läßt sich noch vereinfachen. Berücksichtigt man nämlich, daß nach (109):

(135)
$$\frac{u_{12} - u_{21}}{v_{12} - v_{21}} = T_{12}\,\frac{dp_{12}}{dT_{12}} - p_{12}$$
$$= T\,\frac{dp_{12}}{dT_{12}} - p,$$

und daß nach (80):

$$\frac{\partial u}{\partial v} = T\,\frac{\partial p}{\partial T} - p,$$

ferner daß

$$\frac{du_{12}}{dT_{12}} = \left(\frac{\partial u}{\partial T}\right)_{12} + \left(\frac{\partial u}{\partial v}\right)_{12} \frac{dv_{12}}{dT_{12}} \tag{136}$$

$$= c_v + \left(T\,\frac{\partial p}{\partial T} - p\right) \cdot \frac{dv_{12}}{dT_{12}},$$

sowie:

$$\frac{dp_{12}}{dT_{12}} = \frac{\partial p}{\partial T} + \frac{\partial p}{\partial v}\,\frac{dv_{12}}{dT_{12}},$$

so folgt:

$$\delta T_{12} = \frac{c_v\,\delta T - T\,\dfrac{\partial p}{\partial v} \cdot \dfrac{dv_{12}}{dT_{12}}\,\delta v}{c_v - T\,\dfrac{\partial p}{\partial v}\left(\dfrac{dv_{12}}{dT_{12}}\right)^2}.$$

Die so gefundenen Werte der Variationen in (133) eingesetzt ergeben schließlich für die gesuchte Variation:

$$\delta^2(s' - s) = -\frac{\partial p}{\partial v} \cdot \frac{c_v}{T} \cdot \frac{\left(\dfrac{dv_{12}}{dT_{12}}\,\delta T - \delta v\right)^2}{c_v - T\dfrac{\partial p}{\partial v}\left(\dfrac{dv_{12}}{dT_{12}}\right)^2}.$$

Dieser Ausdruck ist wesentlich positiv, da c_v seiner physikalischen Bedeutung nach stets positiv, und $\dfrac{\partial p}{\partial v}$ nach § 169 für jeden Gleichgewichtszustand wesentlich negativ ist. Ein Grenzfall tritt ein, wenn

$$\frac{dv_{12}}{dT_{12}}\,\delta T - \delta v = 0$$

genommen wird; dann wird $\delta^2(s' - s) = 0$. In diesem Falle findet die Verrückung $(\delta T, \delta v)$ in der Richtung der Berührungskurve (T_{12}, v_{12}) der beiden Flächen statt, und es ist selbstverständlich, daß dann $s' = s$ bleibt.

Hieraus folgt, daß die Fläche s' sich in der Umgebung aller Berührungsstellen mit der Fläche s über dieselbe erhebt, oder daß $s' - s$ stets > 0, und dadurch ist bewiesen, daß die zweite Lösung der Gleichgewichtsbedingungen innerhalb ihres Gültigkeitsbereichs, also in den Gebieten (*12*), (*23*), (*31*) der Fig. 4 stets das stabile Gleichgewicht darstellt.

§ 195. Auf ähnliche Weise läßt sich zeigen, daß die dritte Lösung der Gleichgewichtsbedingungen innerhalb ihres Gültigkeitsbereiches den Vorzug vor der zweiten hat. Sind v und u gegeben, so berechnet sich der dieser Lösung entsprechende Wert der mittleren spezifischen Entropie s'' eindeutig

aus den Gleichungen (127) und (121). Die Größen v_1, v_2, v_3, u_1, u_2, u_3, also auch s_1, s_2, s_3 haben ganz bestimmte Zahlenwerte, die sich aus den Gleichungen (120) ergeben.

Zunächst ist ersichtlich, daß die Fläche s'' nichts anderes ist als das ebene Dreieck, welches gebildet wird von den Punkten (v_1, u_1, s_1), (v_2, u_2, s_2) und (v_3, u_3, s_3), deren Projektionen auf die Zeichnungsebene die Ecken des Fundamentaldreiecks sind. Denn jeder Punkt mit den Koordinaten:

$$v = \frac{\lambda v_1 + \mu v_2 + \nu v_3}{\lambda + \mu + \nu}, \quad u = \frac{\lambda u_1 + \mu u_2 + \nu u_3}{\lambda + \mu + \nu}, \quad s = \frac{\lambda s_1 + \mu s_2 + \nu s_3}{\lambda + \mu + \nu},$$

wobei λ, μ, ν beliebige positive Werte haben, befriedigt die Gleichungen (121) und (127), da man nur $M_1 = \lambda$, $M_2 = \mu$, $M_3 = \nu$ zu setzen braucht. Diese Ebene s'' hat mit den drei Blättern der abwickelbaren Fläche s' die drei geradlinigen Strecken gemeinsam, welche die Punkte (v_1, u_1, s_1), (v_2, u_2, s_2) und (v_3, u_3, s_3) verbinden. In der Tat: Wird in den letzten Ausdrücken etwa $\nu = 0$ angenommen, so liefern die Gleichungen (121) $M_3 = 0$, und die dritte Lösung fällt mit der zweiten zusammen, da dann:

$$\text{(137)} \qquad \begin{cases} M_1 = M_{12} & M_2 = M_{21} \quad v_1 = v_{12} \quad u_1 = u_{12} \\ v_2 = v_{21} & T_1 = T_{12} \text{ etc.} \end{cases}$$

wird. Setzt man außerdem noch $\mu = 0$, so ergibt sich auch $M_2 = 0$, $v_1 = v$, $u_1 = u$, was ein Zusammenfallen aller drei Flächen s'', s' und s bedeutet.

Zur Untersuchung des Wertes von $s'' - s'$ bilden wir nun wieder die Variation $\delta(s'' - s')$, die durch δv und δu bestimmt wird. Hierfür ergibt sich zunächst aus (127):

$$\text{(138)} \qquad M\,\delta s'' = s_1\,\delta M_1 + s_2\,\delta M_2 + s_3\,\delta M_3,$$

wobei nach (121) die Bedingungen gelten:

$$\begin{aligned} &\delta M_1 + \delta M_2 + \delta M_3 = 0, \\ &v_1\,\delta M_1 + v_2\,\delta M_2 + v_3\,\delta M_3 = M\,\delta v, \\ &u_1\,\delta M_1 + u_2\,\delta M_2 + u_3\,\delta M_3 = M\,\delta u. \end{aligned}$$

Die Zurückführung des Ausdrucks (138) auf die unabhängigen Variationen δv und δu geschieht am bequemsten dadurch, daß man die letzte Gleichung mit $\frac{1}{T_1}$, die vorletzte mit $\frac{p_1}{T_1}$ multipliziert, und sie dann zu (138) addiert. Dann ergibt die Berücksichtigung von (120):

$$\delta s'' = \frac{\delta u + p_1\,\delta v}{T_1}.$$

Dies mit (130) verbunden ergibt für die gesuchte Variation:

$$\delta(s'' - s') = \left(\frac{1}{T_1} - \frac{1}{T_{12}}\right)\delta u + \left(\frac{p_1}{T_1} - \frac{p_{12}}{T_{12}}\right)\delta v, \tag{139}$$

wenn die Fläche s' durch das Blatt (12) vertreten ist. Aus dieser Gleichung geht hervor, daß die Ebene s'' das betreffende Blatt der Fläche s' in der den beiden Flächen gemeinsamen Geraden berührt. Denn für irgend einen Punkt dieser Geraden ist nach (137) $T_1 = T_{12}$, $p_1 = p_{12}$, so daß $\delta(s'' - s')$ verschwindet. Die Ebene s'' ist also gemeinsame Tangentialebene zu allen drei Blättern der Fläche s', und die Berührungskurven sind die drei Geraden, welche das ebene Dreieck s'' begrenzen. Für einen der Berührungspunkte haben wir nun aus (139) durch abermalige Variation, da T_1 und p_1 absolute Konstanten sind:

$$\delta^2(s'' - s') = \frac{\delta T_{12}}{T_1^2}\,\delta u + \left(\frac{p_1\,\delta T_{12}}{T_1^2} - \frac{\delta p_{12}}{T_1}\right)\delta v,$$

oder:

$$T_1^2\,\delta^2(s'' - s') = \left[\delta u - \left(T_1\frac{dp_{12}}{dT_{12}} - p_1\right)\delta v\right]\delta T_{12}. \tag{140}$$

Nun folgt aus (129) durch Elimination von δM_{12} und δM_{21}:

$$\frac{M_{12}\,\delta v_{12} + M_{21}\,\delta v_{21} - M\,\delta v}{v_{12} - v_{21}} = \frac{M_{12}\,\delta u_{12} + M_{21}\,\delta u_{21} - M\,\delta u}{u_{12} - u_{21}},$$

oder mit Rücksicht auf (135) und (134):

$$\begin{aligned} &M\left[\delta u - \left(T_1\,\frac{dp_{12}}{dT_{12}} - p_1\right)\delta v\right] \\ &= \delta T_{12}\left[M_{12}\,\frac{du_{12}}{dT_{12}} + M_{21}\,\frac{du_{21}}{dT_{12}} - \left(T_1\,\frac{dp_{12}}{dT_{12}} - p_1\right)\left(M_{12}\,\frac{dv_{12}}{dT_{12}} + M_{21}\,\frac{dv_{21}}{dT_{12}}\right)\right]. \end{aligned}$$

Dieser Ausdruck in (140) substituiert und zugleich $\dfrac{du_{12}}{dT_{12}}$, analog $\dfrac{du_{21}}{dT_{12}}$, nach (136) durch seinen Wert ersetzt, ergibt schließlich:

$$\begin{aligned} \delta^2(s'' - s') = \frac{\delta T_{12}^2}{MT_1^2}\Bigg[&M_{12}\left((c_v)_{12} - T_1\left(\frac{\partial p}{\partial v}\right)_{12}\left(\frac{dv_{12}}{dT_{12}}\right)^2\right) \\ &+M_{21}\left((c_v)_{21} - T_1\left(\frac{\partial p}{\partial v}\right)_{21}\left(\frac{dv_{21}}{dT_{12}}\right)^2\right)\Bigg]. \end{aligned}$$

Diese Größe ist wesentlich positiv, da sowohl M_{12} und M_{21} als auch c_v stets positiv, dagegen $\frac{\partial p}{\partial v}$ stets negativ ist. Ein Grenzfall tritt dann ein, wenn man $\partial T_{12} = 0$ nimmt, d. h. wenn man in der Richtung der Berührungslinie der Flächen s'' und s' fortgeht, wie es von vornherein klar ist. Daraus folgt also, daß das Ebenenstück s'' sich in allen seinen Punkten über die Fläche s' erhebt, oder daß $s'' - s'$ niemals negativ wird, und damit ist bewiesen, daß die dritte Lösung der Gleichgewichtsbedingungen innerhalb ihres Gültigkeitsbereiches, also innerhalb des Fundamentaldreiecks der Substanz: (*123*) das stabile Gleichgewicht darstellt.

§ 196. Wir sind nun imstande, die oben in § 165 betreffs des stabilen Gleichgewichts gestellte Frage allgemein zu beantworten. Ist die Gesamtmasse M, das Gesamtvolumen V und die Gesamtenergie U des Systems gegeben, so wird der entsprechende stabile Gleichgewichtszustand bestimmt durch die Lage des Punktes, dem die Koordinaten $v = \frac{V}{M}$ und $u = \frac{U}{M}$ angehören, in der Zeichnungsebene der Fig. 4.

Fällt nämlich erstens dieser Punkt in eins der Gebiete (*1*), (*2*), (*3*), so verhält sich das System ganz homogen, im gasförmigen, flüssigen oder festen Aggregatzustand. Fällt der Punkt zweitens in eins der Gebiete (*12*), (*23*), (*31*), so zerfällt das System in zwei verschiedene Aggregatformen, wie sie durch die Indizes des betreffenden Gebietes angegeben werden. Hierdurch ist aber auch sowohl die gemeinsame Temperatur, als auch die Werte der beiden heterogenen Massenteile vollständig bestimmt. Denn nach (123) liegt der Punkt (v, u) auf der geradlinigen Verbindungsstrecke zweier zugeordneter (§ 191) Punkte der Kurve, welche das betreffende Gebiet begrenzt; man ziehe also durch den gegebenen Punkt (v, u) diejenige Gerade, welche aus den beiden Ästen jener Kurve zwei zugeordnete Punkte ausschneidet. Diese beiden Punkte geben dann die Beschaffenheit der beiden Aggregatformen an, in die sich das System spaltet; sie haben natürlich gleiche Temperatur und gleichen Druck. Die Größen der Massenteile selber ergeben sich ebenfalls aus (123): ihr Quotient ist gleich dem Verhältnis, in welchem der Punkt (v, u) die Verbindungsstrecke der beiden zugeordneten Punkte teilt.

Wenn der gegebene Punkt drittens in das Gebiet des Fundamentaldreiecks (*123*) hineinfällt, wird das stabile Gleichgewicht durch eine Spaltung des Systems in alle drei Aggregatzustände bezeichnet, bei der Fundamentaltemperatur und unter dem Fundamentaldruck. Es bleibt dann nur noch übrig, die Massen der einzelnen heterogenen Teile des Systems zu bestimmen, und dies geschieht durch die Gleichungen (121a), aus denen hervorgeht, daß die Massenteile sich verhalten wie die Flächen der drei

Dreiecke, welche der Punkt (v, u) mit je einer Seite des Fundamentaldreiecks bildet.

So kann man in jedem einzelnen Falle die Bestimmungsstücke des stabilen Gleichgewichts finden, vorausgesetzt, daß das Fundamentaldreieck und die Verdampfungs-, Schmelz- und Sublimationskurve für die betreffende Substanz ein für allemal gezeichnet sind. Zur besseren Übersicht der Verhältnisse könnte man der Figur noch diejenigen Kurven beifügen, welche Stellen gleicher Temperatur oder gleichen Drucks miteinander verbinden. In den Gebieten (*12*), (*23*), (*31*) fallen die isothermen mit den isobaren Linien zusammen in die geradlinigen Verbindungsstrecken je zweier zugeordneter Punkte der Begrenzungskurven, das Gebiet (*123*) stellt selber eine singuläre Isotherme und Isobare vor. Dann erhält man z. B. für Wasser das Resultat, daß Eis im stabilen Gleichgewicht unter keinerlei Umständen eine höhere Temperatur als die Fundamentaltemperatur (0,0075° C.) annehmen kann, also auch nicht, wenn der Druck noch so sehr erniedrigt werden sollte, während flüssiges Wasser bei geeignetem Druck auf beliebig hohe und tiefe Temperatur gebracht werden kann, ohne zu verdampfen oder zu gefrieren.

Sodann läßt sich auch die Frage direkt beantworten, welche Zustände ein Körper durchmacht, wenn man ihn einer Anzahl von bestimmten äußeren Veränderungen der Reihe nach unterwirft. So z. B. erfährt man das Verhalten eines Körpers mit der Masse M, der bei konstantem Volumen V abgekühlt oder erwärmt wird, durch die Betrachtung der Geraden $v = \frac{V}{M}$, welche der Ordinatenachse parallel läuft. Die Gebiete nämlich, welche diese Gerade durchschneidet, geben an, welche Zustände der Körper in diesem Falle durchmacht, also z. B. ob er im Laufe des Prozesses schmilzt, oder ob er verdampft, oder ob er direkt sublimiert, usw.[1]

Drittes Kapitel. System von beliebig vielen unabhängigen Bestandteilen (Komponenten).

§ 197. Im folgenden untersuchen wir ganz allgemein das thermodynamische Gleichgewicht eines aus verschiedenen räumlich aneinandergrenzenden Teilen bestehenden Systems, welches, im Gegensatz zu dem im vorigen Kapitel behandelten, aus beliebig vielen unabhängigen Bestandteilen (§ 198) zusammengesetzt sein kann. Jeden der verschiedenartigen, räumlich aneinandergrenzenden Teile des Systems setzen wir als physikalisch homogen (§ 67)

[1] Wegen der tatsächlichen Verhältnisse für Wasser vgl. G. Tammann, Göttinger Nachrichten 1913, S. 99.

voraus, und nennen ihn nach GIBBS eine „Phase“ des Systems.[1] So bildet eine Wassermenge, welche sich teils im gasförmigen, teils im flüssigen, teils im festen Aggregatzustand befindet, ein System von drei Phasen. Von vornherein ist sowohl die Anzahl der Phasen als auch ihr Aggregatzustand ganz beliebig. Doch läßt sich sogleich erkennen, daß ein im Gleichgewicht befindliches System zwar beliebig viele feste und flüssige, aber nur eine einzige gasförmige Phase besitzen kann; denn zwei verschiedene frei aneinandergrenzende Gase befinden sich niemals miteinander im thermodynamischen Gleichgewicht.

§ 198. Außer der Zahl der Phasen ist für das System charakteristisch die Zahl seiner „unabhängigen Bestandteile“, oft auch „Komponenten“ genannt; von ihr hängen die wesentlichsten Eigenschaften des Gleichgewichts ab. Die Zahl der unabhängigen Bestandteile eines Systems definieren wir in folgender Weise: Man bilde zunächst die Zahl sämtlicher im System vorhandener chemisch einfachen Stoffe (Elemente) und scheide dann aus dieser Reihe diejenigen Stoffe als abhängige Bestandteile aus, deren Menge durch die der übrigen Stoffe in jeder Phase von vornherein bereits mitbestimmt ist; die Zahl der übrig bleibenden Stoffe ist die Zahl der unabhängigen Bestandteile des Systems.[2] Welche von den Bestandteilen des Systems man als unabhängige, und welche man als abhängige Bestandteile ansehen will, ist im übrigen gleichgültig, da es hier nur auf die Anzahl, nicht auf die Art der unabhängigen Bestandteile ankommt.

Mit der chemischen Konstitution der einzelnen Stoffe in den verschiedenen Phasen des Systems, insbesondere mit der Zahl der verschiedenen Molekülarten, hat die Frage nach der Anzahl der unabhängigen Bestandteile gar nichts zu tun. Eine Wassermenge in beliebigen Aggregatzuständen bildet z. B. immer einen einzigen unabhängigen Bestandteil, mögen noch so viele und verschiedenartige Assoziationen und Dissoziationen der H_2O-Moleküle, sei es in Knallgas oder in Ionen, vorkommen, insofern nämlich die Masse des Sauerstoffs durch die des Wasserstoffs, oder umgekehrt, in jeder Phase von vornherein bereits mitbestimmt ist. Ob allerdings letzteres wirklich zutrifft, muß, genau genommen, erst durch eine besondere Untersuchung festgestellt werden. Sobald man z. B. den Umstand berücksichtigt, daß Wasserdampf bei jeder Temperatur zum Teil in Knallgas dissoziiert ist

[1] Über die Anwendung des Begriffs der „Phase“ auf die beiden enantiomorphen Formen optisch-aktiver Substanzen vgl. die ausführliche Untersuchung von A. BYK, Zeitschr. f. phys. Chemie **45**, 465, 1903.

[2] Diese Definition der „unabhängigen Bestandteile“ eines Systems reicht für alle Fälle aus, in denen es sich um einen wahren Gleichgewichtszustand handelt, d. h. um einen solchen, welcher für a l l e denkbaren Veränderungen des Systems der Gleichung (76) genügt.

und daß der Sauerstoff des Knallgases von flüssigem Wasser stärker absorbiert wird als der Wasserstoff, erhält man, trotzdem nur vollständige H_2O-Moleküle zum Aufbau des Systems verwendet worden sind, verschiedene Gewichtsverhältnisse der beiden Elemente H und O in beiden Phasen des Systems Wasser — Wasserdampf, und damit nicht einen, sondern zwei unabhängige Bestandteile im System. Dasselbe gilt natürlich, wenn von vornherein H oder O im Überschuß vorhanden ist.

Eine wäßrige Lösung von Schwefelsäure bildet ein System von drei chemisch einfachen Stoffen: S, H, O, aber nur von zwei unabhängigen Bestandteilen, da die Masse des O durch die von S und von H in jeder Phase (z. B. flüssige Lösung, Dampf, Eis) von vornherein mitbestimmt ist, während S und H sich nicht in jeder Phase von vornherein gegenseitig bestimmen. Ob nun in der Lösung sich das Molekül H_2SO_4 irgendwie dissoziiert oder ob sich Molekülkomplexe oder Hydrate bilden oder nicht, ändert an der Zahl der unabhängigen Bestandteile des Systems nichts.

§ 199. Bezeichnen wir die Zahl der unabhängigen Bestandteile eines Systems mit α, so ergibt sich aus der für diese Zahl aufgestellten Definition, daß der Zustand einer bestimmten Phase des Systems im thermodynamischen Gleichgewicht bestimmt ist durch die Massen der α in ihr enthaltenen unabhängigen Bestandteile, und außerdem durch die Temperatur T und den Druck p. Dabei nehmen wir der Gleichförmigkeit halber an, daß ein jeder der α unabhängigen Bestandteile in jede Phase des Systems mit einer gewissen Menge eingeht, welche in speziellen Fällen auch verschwindend klein sein kann. Die Wahl der Temperatur und des Druckes als unabhängige Variable bedingt in der Form der folgenden Gleichungen eine gewisse Abweichung von denen des vorigen Kapitels, wo neben der Temperatur das spezifische Volumen als unabhängige Variable diente. Doch ist hier die Einführung des Drucks bequemer, weil derselbe im Gleichgewichtszustande allen frei aneinandergrenzenden Phasen des Systems gemeinsam ist und deshalb auch in der Regel leichter gemessen werden kann.

§ 200. Wir denken uns nun die Gesamtmassen der α unabhängigen Bestandteile des ganzen Systems: M_1, M_2,... M_α als gegeben und fragen nach dem thermodynamischen Gleichgewicht. Von den verschiedenen früher für ein beliebiges System aufgestellten Formen der Gleichgewichtsbedingung benutzen wir hier am besten diejenige, welche in der Gleichung (79) ausgesprochen ist:

$$\delta\Phi = 0, \tag{141}$$

gültig für jede beliebige mit den gegebenen Bedingungen verträgliche Zustandsänderung, bei der die Temperatur T und der Druck p ungeändert

bleiben. Dabei ist die Funktion Φ durch die Entropie S, die Energie U und das Volumen V des Systems nach (75) in folgender Weise bestimmt:

$$\Phi = S - \frac{U + pV}{T}.$$

§ 201. Nun sei β die Anzahl der Phasen des Systems; dann besteht die Funktion S, ebenso U, V und infolgedessen auch Φ, aus einer Summe von β Gliedern, deren jedes sich auf eine einzelne Phase, also auf einen physikalisch homogenen Körper bezieht:

$$\Phi = \Phi' + \Phi'' + \ldots + \Phi^{\beta}, \tag{142}$$

wenn wir, wie immer im folgenden, die verschiedenen Phasen durch beigesetzte Striche voneinander unterscheiden. Dabei ist für die erste Phase:

$$\Phi' = S' - \frac{U' + pV'}{T}. \tag{143}$$

S', U', V' und Φ' sind vollständig bestimmt durch T, p und die Massen M_1', $M_2', \ldots$ M_α' der in der Phase enthaltenen unabhängigen Bestandteile. Über die Art der Abhängigkeit von den einzelnen Massen läßt sich von vornherein nur so viel sagen, daß, wenn alle Massen in einem bestimmten Verhältnis verändert, z. B. verdoppelt werden, auch jede der obigen Funktionen sich in demselben Verhältnis verändert. Denn bei der genannten Veränderung bleibt die innere Beschaffenheit der Phase konstant, nur ihre Gesamtmasse ändert sich, und zwar gerade in dem angenommenen bestimmten Verhältnis, und eben dieser Gesamtmasse proportional wächst die Entropie, die Energie und das Volumen, und daher auch die Funktion Φ'. Mit anderen Worten: Φ' ist eine homogene Funktion ersten Grades der Massen M_1', $M_2', \ldots$ M_α', die natürlich nicht lineär zu sein braucht. (Ein Beispiel einer nichtlineären homogenen Funktion ersten Grades ist: $\Phi' = \sqrt{M_1'^2 + M_2'^2 + \ldots + M_\alpha'^2}$.)

Um dies analytisch auszudrücken, lassen wir alle Massen sich in dem Verhältnis $1 + \varepsilon$ vergrößern, wobei ε eine sehr kleine Zahl ist. Dann sind alle Änderungen sehr klein, und man erhält für die entsprechende Änderung von Φ':

$$\begin{aligned}\Delta\Phi' &= \frac{\partial\Phi'}{\partial M_1'}\,\Delta M_1' + \frac{\partial\Phi'}{\partial M_2'}\,\Delta M_2' + \ldots \\ &= \frac{\partial\Phi'}{\partial M_1'}\,\varepsilon M_1' + \frac{\partial\Phi'}{\partial M_2'}\,\varepsilon M_2' + \ldots.\end{aligned}$$

Aber nach der Voraussetzung ist:

$$\Delta\Phi' = \varepsilon\Phi'.$$

Folglich:

$$\Phi' = \frac{\partial\Phi'}{\partial M_1'} M_1' + \frac{\partial\Phi'}{\partial M_2'} M_2' + \ldots + \frac{\partial\Phi'}{\partial M_\alpha'} M_\alpha'. \tag{144}$$

Diese EULERsche Gleichung läßt sich durch Differentiation noch in verschiedene andere Formen bringen. Die in ihr vorkommenden Differentialkoeffizienten $\frac{\partial\Phi'}{\partial M_1'}$, $\frac{\partial\Phi'}{\partial M_2'}$,... hängen offenbar nur von der inneren Beschaffenheit der Phase, nicht von ihrer Gesamtmasse ab, da sich bei einer gleichmäßigen Vergrößerung aller Massen in ihnen Zähler und Nenner in gleichem Verhältnis ändern.

Was für die erste Phase gilt, läßt sich ohne weiteres auf jede andere Phase übertragen.

§ 202. Mit Benutzung von (142) lautet nun die Gleichgewichtsbedingung:

$$\delta\Phi' + \delta\Phi'' + \ldots + \delta\Phi^\beta = 0 \tag{145}$$

oder, da Temperatur und Druck nicht variiert werden:

$$\begin{aligned} &\frac{\partial\phi'}{\partial M_1'}\,\delta M_1' + \frac{\partial\phi'}{\partial M_2'}\,\delta M_2' + \ldots + \frac{\partial\phi'}{\partial M_\alpha'}\,\delta M_\alpha' \\ &+ \frac{\partial\phi''}{\partial M_1''}\,\delta M_1'' + \frac{\partial\phi''}{\partial M_2''}\,\delta M_2'' + \ldots + \frac{\partial\phi''}{\partial M_\alpha''}\,\partial M_\alpha'' \\ &+ \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \\ &+ \frac{\partial\phi^\beta}{\partial M_1^\beta}\,\delta M_1^\beta + \frac{\partial\phi^\beta}{\partial M_2^\beta}\,\delta M_2^\beta + \ldots + \frac{\partial\phi^\beta}{\partial M_\alpha^\beta}\,\delta M_\alpha^\beta = 0. \end{aligned} \tag{146}$$

Wären die Variationen der Massen ganz beliebig, so würde diese Gleichung nur dann erfüllt, wenn sämtliche Koeffizienten der Variationen einzeln gleich Null wären. Nun aber besteht zwischen diesen nach § 200 die Bedingung, daß:

$$\left\{\begin{aligned} M_1 &= M_1' + M_1'' + \ldots + M_1^\beta \\ M_2 &= M_2' + M_2'' + \ldots + M_2^\beta \\ &\quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \\ M_\alpha &= M_\alpha' + M_\alpha'' + \ldots + M_\alpha^\beta \end{aligned}\right. \tag{147}$$

also bei irgend einer möglichen Veränderung des Systems:

$$(148)\qquad \left\{\begin{array}{l} 0 = \delta M_1' + \delta M_1'' + \ldots + \delta M_1^{\beta} \\ 0 = \delta M_2' + \delta M_2'' + \ldots + \delta M_2^{\beta} \\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot \\ 0 = \delta M_\alpha' + \delta M_\alpha'' + \ldots + \delta M_\alpha^{\beta} \end{array}\right.$$

Daraus folgt als notwendige und hinreichende Bedingung für das Verschwinden des Ausdrucks (146):

$$(149)\qquad \left\{\begin{array}{l} \dfrac{\partial \Phi'}{\partial M_1'} = \dfrac{\partial \Phi''}{\partial M_1''} = \ldots = \dfrac{\partial \Phi^{\beta}}{\partial M_1^{\beta}} \\ \dfrac{\partial \Phi'}{\partial M_2'} = \dfrac{\partial \Phi''}{\partial M_2''} = \ldots = \dfrac{\partial \Phi^{\beta}}{\partial M_2^{\beta}} \\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot \\ \dfrac{\partial \Phi'}{\partial M_\alpha'} = \dfrac{\partial \Phi''}{\partial M_\alpha''} = \ldots = \dfrac{\partial \Phi^{\beta}}{\partial M_\alpha^{\beta}} \end{array}\right.$$

Dies sind für jeden unabhängigen Bestandteil $\beta - 1$ Gleichungen, die im Gleichgewichtszustand erfüllt sein müssen; also für alle α unabhängigen Bestandteile $\alpha(\beta - 1)$ Bedingungen. Jede dieser Gleichungen bezieht sich auf den Übertritt eines unabhängigen Bestandteils aus einer Phase in eine andere, sie spricht aus, daß das Gleichgewicht in bezug auf diesen Übertritt gesichert ist, daß also in der Natur gerade dieser Übertritt nicht stattfindet. Wie es sein muß, hängt diese Bedingung nur von der inneren Beschaffenheit der Phasen, nicht von ihrer Gesamtmasse ab.

Da man die auf einen bestimmten Bestandteil bezüglichen, in einer einzelnen Reihe befindlichen Gleichungen beliebig umstellen kann, so folgt daraus der Satz: Befindet sich eine Phase in bezug auf einen bestimmten Bestandteil mit zwei anderen Phasen im Gleichgewicht, so befinden sich in bezug auf diesen Bestandteil auch die beiden anderen Phasen unter sich im Gleichgewicht (sie „koexistieren“). Nehmen wir dazu den schon oben (§ 197) hervorgehobenen Satz, daß jedes System im Gleichgewicht höchstens eine einzige gasförmige Phase besitzt, so ergibt sich daraus ganz allgemein, daß zwei koexistierende Phasen, z. B. zwei Flüssigkeiten, die zwei getrennte Schichten bilden, den nämlichen Dampf aussenden. Denn da die eine Phase nach der Voraussetzung mit der anderen Phase und selbstverständlich auch mit ihrem eigenen Dampf koexistiert, und zwar in bezug auf jeden ihrer Bestandteile, so koexistiert auch die andere Phase mit dem nämlichen Dampf. Man kann daher die Koexistenz von festen und flüssigen Phasen dadurch prüfen, daß man ihre Dämpfe vergleicht.

§ 203. Es ist nun leicht zu sehen, wie sich im allgemeinen der Gleichgewichtszustand des Systems aus den von vornherein gegebenen äußeren Bedingungen (147) und den Gleichgewichtsbedingungen (149) bestimmt. Die Zahl der ersten beträgt α, die der zweiten $\alpha(\beta-1)$, also zusammen $\alpha\beta$ Gleichungen. Andrerseits hängt der Zustand der β Phasen von $\alpha\beta+2$ Variabeln ab, nämlich außer von den $\alpha\beta$ Massen $M'_1, \ldots M^{\beta}_{\alpha}$ noch von der Temperatur T und dem Druck p. Es bleiben daher, wenn alle Bedingungen erfüllt sind, noch zwei Variable unbestimmt. Im allgemeinen wird man also sowohl die Temperatur als auch den Druck ganz beliebig wählen können; es gibt aber, wie sogleich gezeigt werden wird, spezielle Fälle, in denen Temperatur und Druck nicht mehr willkürlich sind, so daß dann zwei andere Variable, etwa Gesamtvolumen und Gesamtenergie des Systems, unbestimmt bleiben. Verfügt man willkürlich über ihre Werte, so ist dann der Gleichgewichtszustand des Systems in allen Stücken bestimmt.

§ 204. Scheiden wir nun die $\alpha\beta+2$ Variabeln, von denen der Zustand des Systems abhängt, in solche, welche nur die innere Beschaffenheit der Phasen betreffen (innere Variable) und in solche, welche nur die Gesamtmassen der Phasen bestimmen (äußere Variable). Die Anzahl der ersteren ist $(\alpha-1)\beta+2$, nämlich in jeder der β Phasen die $\alpha-1$ Verhältnisse der α in ihr enthaltenen unabhängigen Bestandteile zueinander, und außerdem Temperatur und Druck; die Anzahl der letzteren ist β, nämlich die Gesamtmassen aller Phasen. Zusammen:

$$(\alpha-1)\beta+2+\beta=\alpha\beta+2.$$

Nun enthalten die $\alpha(\beta-1)$ Gleichungen (149) nach dem dort Gesagten nur innere Variable, also bleiben nach Befriedigung dieser Gleichungen von der Gesamtzahl der inneren Variabeln noch

$$\big[(\alpha-1)\beta+2\big]-\big[\alpha(\beta-1)\big]=\alpha-\beta+2$$

als unbestimmt zurück. Diese Zahl kann nicht negativ sein; denn sonst würden die inneren Variabeln des Systems nicht ausreichen, um die Gleichungen (149) alle zu befriedigen; es muß also sein:

$$\beta\leqq\alpha+2,$$

d. h. die Zahl der Phasen kann die Zahl der unabhängigen Bestandteile höchstens um zwei übertreffen, oder: ein System von α unabhängigen Bestandteilen kann höchstens $\alpha+2$ Phasen bilden. Im Grenzfall $\beta=\alpha+2$ reicht die Anzahl der inneren Variabeln gerade aus, um die inneren

Gleichgewichtsbedingungen (149) zu erfüllen, ihre Werte sind dann im Gleichgewichtszustand vollkommen bestimmt, ganz unabhängig von den gegebenen äußeren Bedingungen. Mit jeder Phase weniger wächst dann die Zahl der noch unbestimmten inneren Variabeln um Eins.

Dieser, zuerst von GIBBS ausgesprochene, gewöhnlich als „Phasenregel" bezeichnete Satz hat besonders durch die Untersuchungen von BAKHUIS ROOZEBOOM[1] eine weitgehende experimentelle Bestätigung erhalten.

§ 205. Betrachten wir zunächst den Grenzfall:

$$\beta = \alpha + 2.$$

Dann sind sämtliche innere Variable vollständig bestimmt, sie bilden einen „$(\alpha+2)$fachen Punkt". Durch Abänderung der äußeren Bedingungen, wie z. B. durch Wärmezufuhr, Kompression, weiteren Zusatz von Substanzmengen, werden nur die Gesamtmassen der Phasen, nicht aber ihre innere Beschaffenheit, einschließlich Temperatur und Druck, alteriert. Dies gilt so lange, bis etwa eine der Phasen die Masse Null annimmt und somit ganz aus dem System verschwindet.

Für $\alpha = 1$ hat man hier $\beta = 3$, d. h. ein einziger Bestandteil kann sich höchstens in drei Phasen spalten und bildet dann einen dreifachen Punkt. Ein Beispiel für diesen Fall liefert eine Substanz, die sich in drei Aggregatzuständen nebeneinander befindet, wie im vorigen Kapitel ausführlich untersucht wurde. Für Wasser ist z. B., wie § 187 gezeigt wurde, im dreifachen Punkt die Temperatur $0,0075°$ C., der Druck $4,58^{\text{mm}}$ Quecksilber. Die drei Phasen brauchen aber nicht verschiedenen Aggregatzuständen anzugehören, so z. B. gibt es Substanzen, wie Schwefel, die im festen Zustand verschiedene Modifikationen bilden können. Dann stellt jede Modifikation eine besondere Phase dar, und es gilt der Satz, daß, wenn zwei Modifikationen einer Substanz mit einer dritten Phase derselben Substanz, etwa mit ihrem Dampfe, koexistieren sollen, dies nur bei einer ganz bestimmten Temperatur und bei einem ganz bestimmten Druck geschehen kann.

Für $\alpha = 2$ erhält man einen vierfachen Punkt. So liefern die beiden unabhängigen Bestandteile SO_2 (schweflige Säure) und H_2O die vier koexistierenden Phasen: $SO_2 \cdot 7\,H_2O$ fest, SO_2 in H_2O gelöst flüssig, SO_2 flüssig, SO_2 gasförmig, Temperatur $12,1°$ C., Druck 1773^{mm} Quecksilber. Die Frage, ob SO_2 in wäßriger Lösung ein Hydrat bildet, berührt nach der Auseinandersetzung im § 198 die Bedeutung der Phasenregel in keiner Weise. Ein anderes Beispiel für zwei unabhängige Bestandteile in einem vierfachen

[1] Vgl. das Werk „Die heterogenen Gleichgewichte vom Standpunkte der Phasenlehre" von BAKHUIS ROOZEBOOM, Braunschweig, VIEWEG & SOHN, 1904.

Punkt wird gebildet von einer wäßrigen Kochsalzlösung in Berührung mit festem Salz, Eis und Wasserdampf.

Ferner liefern $\alpha = 3$ unabhängige Bestandteile, wie Na_2O_4, $MgSO_4$ und H_2O, einen fünffachen Punkt, nämlich das Doppelsalz $Na_2Mg(SO_4)_2 \cdot 4\,H_2O$ (Astrakanit), die beiden einfachen Salze kristallisiert, wäßrige Lösung und Wasserdampf, Temperatur $21,5°$ C., Druck etwa 10^{mm} Quecksilber.

§ 206. Nehmen wir weiter den Fall:

$$\beta = \alpha + 1,$$

d. h. die α unabhängigen Bestandteile bilden $\alpha + 1$ Phasen. Dann ist die innere Beschaffenheit aller Phasen durch eine einzige Variable vollkommen bestimmt, etwa durch die Temperatur oder durch den Druck. Dieser Fall wird gewöhnlich als der des „vollständig heterogenen" Gleichgewichts bezeichnet.

Für $\alpha = 1$ ergibt sich hier $\beta = 2$: ein einziger unabhängiger Bestandteil in zwei Phasen, z. B. als Flüssigkeit und Dampf. Sowohl der Druck als auch die Dichtigkeit der Flüssigkeit und des Dampfes hängen allein von der Temperatur ab, wie schon im vorigen Kapitel gezeigt wurde. Hierher gehört auch die mit chemischer Zersetzung verbundene Verdampfung, insofern das System nur einen einzigen unabhängigen Bestandteil enthält, wie z. B. die Verdampfung von festem NH_4Cl (Salmiak). Solange nicht Salzsäure- oder Ammoniakdampf im Überschuß zugegen sind, entspricht jeder Temperatur T eine ganz bestimmte Dissoziationsspannung p.

Für $\alpha = 2$ wird $\beta = 3$, wie z. B., wenn eine Salzlösung zugleich mit ihrem Dampf und mit dem festen Salz in Berührung ist, oder wenn zwei nicht in allen Verhältnissen mischbare Flüssigkeiten (Äther und Wasser) sich zusammen unter ihrem gemeinsamen Dampf befinden. Dampfdruck, Dichte und Konzentration in jeder Phase sind dann Funktionen der Temperatur allein.

§ 207. Außer der Temperatur wird auch der Druck häufig als diejenige Variable angenommen, durch welche beim vollständig heterogenen Gleichgewicht die innere Beschaffenheit aller Phasen bestimmt ist, namentlich bei solchen Systemen, welche keine gasförmige Phase besitzen, den sogenannten „kondensierten" Systemen. Denn bei diesen Systemen ist der Einfluß des Druckes auf den Zustand aller Phasen unter gewöhnlichen Umständen so gering, daß man dann ohne merklichen Fehler den Druck als gegeben und etwa gleich dem einer Atmosphäre annehmen kann. Daher liefert hierfür die Phasenregel den Satz: Ein kondensiertes System von α unabhängigen Bestandteilen kann höchstens $\alpha + 1$ Phasen bilden und ist dann in seiner inneren Beschaffenheit, einschließlich der Temperatur, vollkommen bestimmt.

Ein Beispiel für $\alpha = 1,\ \beta = 2$ ist der Schmelzpunkt einer Substanz, ein anderes Beispiel ist der sogenannte Umwandlungspunkt, falls eine Substanz in zwei allotropen Modifikationen existieren kann. Für $\alpha = 2,\ \beta = 3$ dient als Beispiel der Punkt, wo sich aus einer Salzlösung das Kryohydrat (Eis neben festem Salz) ausscheidet, oder auch, wie für Arsenbromür und Wasser, der Punkt, wo aus zwei sich berührenden flüssigen Schichten ein fester Bestandteil sich niederzuschlagen beginnt, für $\alpha = 3,\ \beta = 4$ der Punkt, wo eine Lösung zweier Salze, die fähig sind ein Doppelsalz zu bilden, sowohl mit den einfachen Salzen als auch mit dem Doppelsalz in Berührung ist, usw.

§ 208. Wenn ferner:

$$\beta = \alpha,$$

so bilden α unabhängige Bestandteile α Phasen. Dann ist die innere Beschaffenheit aller Phasen noch von zwei Variabeln abhängig, z. B. von Temperatur und Druck. Ein Beispiel für $\alpha = 1$ liefert jede Substanz in homogenem Zustand, eins für $\alpha = 2$ eine flüssige Lösung eines Salzes in Berührung mit ihrem Dampf. Durch Temperatur und Druck ist die Konzentration sowohl in der Lösung als auch im Dampf bestimmt. Häufig werden nicht Temperatur und Druck, sondern statt dessen die Konzentration der flüssigen Lösung, entweder zusammen mit der Temperatur, oder zusammen mit dem Druck, als die beiden unabhängigen Variabeln genommen. Im ersteren Falle sagt man, daß eine Lösung von beliebig gewählter Konzentration bei beliebig gewählter Temperatur einen Dampf von bestimmter Zusammensetzung und von bestimmter Spannung aussendet; im zweiten Falle sagt man, daß eine Lösung von beliebig gewählter Konzentration bei beliebig gewähltem Druck einen bestimmten Siedepunkt besitzt, bei dem ein Dampf von bestimmter Zusammensetzung abdestilliert.

Ganz entsprechende Gesetzmäßigkeiten ergeben sich, wenn die zweite Phase nicht dampfförmig, sondern fest (Salz, Eis) oder flüssig ist, wie bei zwei Flüssigkeiten, die sich nicht in allen Verhältnissen mischen. Immer hängt die innere Beschaffenheit der beiden Phasen, also im letzten Beispiel auch die Konzentrationen in den beiden übereinander geschichteten Lösungen, von zwei Variabeln ab, etwa Druck und Temperatur. Wenn in einem speziellen Fall diese Konzentrationen einander gleich werden, erhält man eine Erscheinung, die ganz analog ist dem Phänomen des kritischen Punktes bei einer homogenen Substanz (kritische Lösungstemperatur zweier Flüssigkeiten).

§ 209. Betrachten wir endlich noch kurz den Fall:

$$\beta = \alpha - 1.$$

Hier ist die Zahl der Phasen um Eins geringer als die der unabhängigen Bestandteile, und die innere Beschaffenheit aller Phasen hängt also außer von Temperatur und Druck auch von einer dritten willkürlich zu wählenden Variabeln ab. So ist z. B. $\alpha = 3$, $\beta = 2$ für eine wäßrige Lösung zweier isomorpher Substanzen (Kaliumchlorat und Thalliumchlorat) in Berührung mit einem Mischkristall, welcher als physikalisch homogener Körper eine einzige Phase darstellt. Bei Atmosphärendruck und einer bestimmten Temperatur wird je nach der Zusammensetzung des Mischkristalls die Konzentration der Lösung eine verschiedene sein, so daß man von einer „gesättigten" Lösung der beiden Substanzen von bestimmter Zusammensetzung gar nicht sprechen kann. Erst wenn außer dem Mischkristall sich noch eine zweite feste Phase, etwa ein zweiter anders beschaffener Mischkristall, aus der Lösung niederschlägt, wird die innere Beschaffenheit des Systems durch Temperatur und Druck allein bestimmt. Die experimentelle Untersuchung der Gleichgewichtszustände solcher Systeme kann auch umgekehrt dazu dienen, um an der Hand der Phasenregel zu entscheiden, ob ein aus einer wäßrigen Lösung zweier Stoffe erhaltener fester Niederschlag der beiden Stoffe eine einzige Phase, also z. B. einen Mischkristall von veränderlicher Konzentration bildet, oder ob die beiden Stoffe in zwei räumlich aneinandergrenzenden Phasen anzunehmen sind. Denn offenbar trifft die zweite oder die erste Annahme zu, je nachdem die Konzentration der angrenzenden wäßrigen Lösung bei bestimmter Temperatur und bestimmtem Druck eine ganz bestimmte ist oder nicht.

§ 210. Wenn die Ausdrücke der Funktionen Φ', Φ'', ... für jede einzelne Phase bekannt sind, so kann man aus den Gleichungen (149) unmittelbar alle Einzelheiten eines Gleichgewichtszustandes des Systems entnehmen. Jenes ist aber im allgemeinen durchaus nicht der Fall; wenigstens läßt sich über die Abhängigkeit jener Funktionen von den Massen der in den einzelnen Phasen enthaltenen Bestandteile im allgemeinen nur das aussagen, was schon oben (§ 201) hervorgehoben wurde, daß sie nämlich homogen und vom ersten Grade sind. Was dagegen ihre Abhängigkeit von Temperatur und Druck betrifft, so ist diese insofern bekannt, als sich, wie wir in § 152a, Gleichung (79b), gesehen haben, die Differentialquotienten der charakteristischen Funktion Φ nach p und T auf meßbare Größen zurückführen lassen, und dieser Umstand gestattet weitgehende Schlüsse zu ziehen in bezug auf die Abhängigkeit des Gleichgewichts von Druck und Temperatur (§ 211).

Wenn es sich darum handelt, die charakteristische Funktion Φ mit Hilfe physikalischer Messungen als Funktion aller unabhängigen Variabeln direkt darzustellen, so geht man am zweckmäßigsten aus von dem in

Gleichung (75) gegebenen Zusammenhang von Φ mit der Entropie S und der GIBBSschen Wärmefunktion $W = U + pV$ (§ 100):

$$\Phi = S - \frac{W}{T}. \tag{150}$$

In dieser Gleichung kann man sowohl W als auch S durch Wärmemessungen bestimmen. Denn wenn C_p die Wärmekapazität bei konstantem Druck bezeichnet, so ist nach (26):

$$C_p = \left(\frac{\partial W}{\partial T}\right)_p. \tag{150a}$$

Andrerseits ist nach (84a):

$$C_p = T\left(\frac{\partial S}{\partial T}\right)_p. \tag{150b}$$

Folglich kann man statt (150) schreiben:

$$\Phi = \int \frac{C_p}{T}\,dT - \frac{1}{T}\int C_p\,dT. \tag{150c}$$

Die beiden Integrationen sind bei konstantem Druck p auszuführen. Die Auffindung der charakteristischen Funktion Φ und somit aller thermodynamischen Eigenschaften des betrachteten Systems ist hierdurch zurückgeführt auf die Messung der Wärmekapazität C_p des Systems für alle Werte von T und p. Doch müssen noch die bei der Integration auftretenden additiven Konstanten beachtet werden. Dieselben können noch von p und außerdem von der chemischen Zusammensetzung des Systems abhängen. Die Abhängigkeit von p ergibt sich nach der ersten Gleichung (79b) durch eine Messung des Volumens V, die von der chemischen Zusammensetzung kann im allgemeinen nur aus der Messung solcher Vorgänge erschlossen werden, die mit chemischen Zustandsänderungen verbunden sind.

Schließlich bleibt in dem Ausdruck von Φ noch ein additives Glied von der Form $a + \frac{b}{T}$ ganz willkürlich wählbar. (Vgl. oben § 152b.)

§ 211. Die durch die Gleichungen (79b) ausgedrückte Abhängigkeit der charakteristischen Funktion Φ von T und p wollen wir nun benutzen, um ein allgemeines Gesetz aufzustellen für den Einfluß der Temperatur und des Druckes auf das thermodynamische Gleichgewicht.

Zu diesem Zweck wollen wir im folgenden unterscheiden zwischen zwei verschiedenartigen unendlich kleinen Zustandsänderungen. Das Zeichen δ

soll sich, ganz wie bisher, auf irgend eine virtuelle, mit den gegebenen äußeren Bedingungen verträgliche, und daher den Gleichungen (148) entsprechende Änderung der Massen M_1', M_2', M_1'',… M_α^β beziehen, wobei außerdem Temperatur und Druck konstant bleiben, d. h. $\delta T = 0$ und $\delta p = 0$. Den Gleichgewichtsbedingungen (149) braucht der so variierte Zustand nicht zu genügen. Dagegen soll sich das Zeichen d auf eine wirklich vorgenommene Veränderung, nämlich auf einen dem ursprünglichen unendlich nahe gelegenen zweiten Gleichgewichtszustand beziehen, bei dem auch alle äußeren Bedingungen, einschließlich Temperatur und Druck, in von vornherein willkürlicher Weise geändert sein können.

Es handelt sich nun darum, auch für den zweiten Zustand die Bedingungen des Gleichgewichts festzustellen und sie mit den für den ursprünglichen Zustand gültigen zu vergleichen. Zunächst haben wir für den letzteren als Bedingung des Gleichgewichts, wie oben:

$$\delta\Phi = 0.$$

Sodann für den zweiten Zustand:

$$\delta(\Phi + d\Phi) = 0,$$

daraus:

$$\delta d\Phi = 0. \tag{151}$$

Nun ist:

$$d\Phi = \frac{\partial\Phi}{\partial T}\,dT + \frac{\partial\Phi}{\partial p}\,dp + \sum^{\beta} \frac{\partial\Phi'}{\partial M_1'}\,dM_1' + \frac{\partial\Phi'}{\partial M_2'}\,dM_2' + \ldots,$$

wobei das Zeichen $\sum$ die Summierung über alle β Phasen des Systems bedeuten soll, während die Summierung über die α unabhängigen Bestandteile einer einzelnen Phase besonders ausgeschrieben ist. Oder nach (79b):

$$d\Phi = \frac{U + pV}{T^2}\,dT - \frac{V}{T}\,dp + \sum^{\beta} \frac{\partial\Phi'}{\partial M_1'}\,dM_1' + \frac{\partial\Phi'}{\partial M_2'}\,dM_2' + \ldots$$

Folglich geht die Gleichgewichtsbedingung (151) über in:

$$\frac{\delta U + p\,\delta V}{T^2}\,dT - \frac{\delta V}{T}\,dp + \sum^{\beta} dM_1'\,\delta\frac{\partial\Phi'}{\partial M_1'} + dM_2'\,\delta\frac{\partial\Phi'}{\partial M_2'} + \ldots = 0. \tag{152}$$

Es fallen nämlich alle von der Variation der Differentiale dT, dp, dM_1', $dM_2', \ldots$ herrührenden Glieder weg: die ersten beiden, weil $\delta T = 0$ und $\delta p = 0$, alle übrigen, weil in der Gesamtsumme:

$$\begin{aligned}
&\frac{\partial \Phi'}{\partial M_1'}\,\delta dM_1' + \frac{\partial \Phi'}{\partial M_2'}\,\delta dM_2' + \ldots \\
&+ \frac{\partial \Phi''}{\partial M_1''}\,\delta dM_1'' + \frac{\partial \Phi''}{\partial M_2''}\,\delta dM_2'' + \ldots \\
&+ \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \\
&+ \frac{\partial \Phi^{\beta}}{\partial M_1^{\beta}}\,\delta dM_1^{\beta} + \frac{\partial \Phi^{\beta}}{\partial M_2^{\beta}}\,\delta dM_2^{\beta} + \ldots
\end{aligned}$$

jede Vertikalkolumne für sich verschwindet. Denn z. B. für die erste Kolumne ist nach (149):

$$\frac{\partial \Phi'}{\partial M_1'} = \frac{\partial \Phi''}{\partial M_1''} = \ldots = \frac{\partial \Phi^{\beta}}{\partial M_1^{\beta}}$$

und außerdem ist nach (148):

$$\begin{aligned}
&\delta dM_1' + \delta dM_1'' + \ldots + \delta dM_1^{\beta} \\
&\qquad = d(dM_1' + dM_1'' + \ldots + dM_1^{\beta}) = 0.
\end{aligned}$$

Nun bedeutet weiter $\delta U + p\,\delta V$ nach dem ersten Hauptsatz der Wärmetheorie die bei der virtuellen Zustandsänderung δ dem System von außen zugeführte Wärme Q, also läßt sich die Gleichung (152) auch schreiben:

$$\frac{Q}{T^2}\,dT - \frac{\delta V}{T}\,dp + \sum^{\beta} dM_1'\,\delta\frac{\partial \Phi'}{\partial M_1'} + dM_2'\,\delta\frac{\partial \Phi'}{\partial M_2'} + \ldots = 0. \tag{153}$$

Hierin ist das Gesetz der Abhängigkeit des Gleichgewichts von Temperatur, Druck und von den Massen der unabhängigen Bestandteile des Systems ausgesprochen. Man ersieht daraus zunächst, daß der Einfluß der Temperatur auf das Gleichgewicht wesentlich abhängt von der Wärmetönung, welche bei einer virtuellen isotherm-isobaren Zustandsänderung eintritt. Ist diese gleich Null, so fällt das Glied mit dT fort, und eine Änderung der Temperatur führt gar keine Störung des Gleichgewichts herbei; wechselt Q das Vorzeichen, so wechselt auch der Einfluß der Temperatur auf das Gleichgewicht sein Vorzeichen. Ganz Ähnliches gilt in bezug auf den Einfluß des Druckes, welcher seinerseits wesentlich durch die Volumenänderung δV bedingt ist, die durch eine virtuelle isotherm-isobare Zustandsänderung hervorgerufen wird.

§ 212. Die Gleichung (153) soll nun auf einige spezielle Fälle angewendet werden, zunächst auf solche des vollständig heterogenen Gleichgewichts, welche nach § 206 charakterisiert werden durch die Beziehung:

$$\beta = \alpha + 1.$$

Die innere Beschaffenheit aller Phasen, einschließlich des Druckes, ist hier durch die Temperatur allein bestimmt; daher bringt z. B. eine isotherme unendlich langsame Kompression des Systems nur in den Gesamtmassen der Phasen, nicht aber in deren Zusammensetzung, und auch nicht im Druck, eine Änderung hervor. Eine derartige Änderung nun, die also in diesem speziellen Falle zu einem neuen Gleichgewichtszustand führt, wollen wir als virtuelle Zustandsänderung δ wählen. Dann bleibt außer der Temperatur und dem Druck auch die innere Beschaffenheit aller Phasen unvariiert, und die Variationen der Funktionen $\frac{\partial \Phi'}{\partial M_1'}, \frac{\partial \Phi'}{\partial M_2'}, \ldots$ werden alle gleich Null, da diese Größen nur von der inneren Beschaffenheit der Phasen abhängen. Dadurch geht die Gleichung (153) über in:

$$\frac{dp}{dT} = \frac{Q}{T \cdot \delta V}, \tag{154}$$

d. h. die durch eine virtuelle Zustandsänderung, bei der die innere Beschaffenheit aller Phasen ungeändert bleibt, bedingte Wärmetönung, dividiert durch die entsprechende Volumenänderung des Systems und durch die absolute Temperatur, ergibt die Änderung des Gleichgewichtsdrucks mit der Temperatur. Ist Wärmezufuhr von außen mit Volumenvergrößerung des Systems verbunden, wie bei der Verdampfung, so steigt der Gleichgewichtsdruck mit der Temperatur, im entgegengesetzten Fall, wie beim Schmelzen des Eises, sinkt er mit steigender Temperatur.

§ 213. Für einen einzigen unabhängigen Bestandteil: $\alpha = 1$, also $\beta = 2$, führt die letzte Gleichung unmittelbar zu den im vorigen Kapitel ausführlich behandelten Gesetzen der Verdampfungs-, Schmelz- und Sublimationswärme. Bildet z. B. die Flüssigkeit die erste Phase, der Dampf die zweite, und bezeichnet r die Verdampfungswärme der Masseneinheit, so ist:

$$Q = r\delta M''$$
$$\delta V = (v'' - v')\delta M'',$$

worin v' und v'' die spezifischen Volumina von Flüssigkeit und Dampf, $\delta M''$ die bei einer virtuellen isotherm-isobaren Zustandsänderung gebildete Menge Dampf bedeutet. Mithin aus (154):

$$r = T \frac{dp}{dT}(v'' - v')$$

identisch mit der Gleichung (111).

Selbstverständlich sind hier auch Fälle chemischer Umsetzungen mit einbegriffen, insofern das System einen einzigen unabhängigen Bestandteil in zwei Phasen enthält, wie z. B. die zuerst von HORSTMANN daraufhin untersuchte Verdampfung von Salmiak, wobei Zersetzung in Salzsäure- und Ammoniakdampf stattfindet (§ 206), oder die Verdampfung von karbaminsaurem Ammoniak, wobei Zersetzung in Ammoniak und Kohlensäure eintritt. Dann bedeutet in der letzten Gleichung r die Dissoziationswärme, p die allein von der Temperatur abhängige Dissoziationsspannung.

§ 214. Wir wollen das vollständig heterogene Gleichgewicht auch noch für zwei unabhängige Bestandteile betrachten:

$$\alpha = 2 \qquad \beta = 3,$$

z. B. Wasser (Index 1) und ein Salz (Index 2) in drei Phasen, von denen die erste eine flüssige Lösung (Masse des Wassers M_1', des Salzes M_2'), die zweite Wasserdampf (Masse M_1''), die dritte festes Salz (Masse M_2''') sein möge. Dann ist für eine virtuelle Zustandsänderung:

$$\delta M_1' + \delta M_1'' = 0 \qquad \text{und} \qquad \delta M_2' + \delta M_2''' = 0.$$

Nach der Phasenregel ist sowohl die Konzentration der Lösung:

$$\frac{M_2'}{M_1'} = c,$$

als auch der Dampfdruck p Funktion der Temperatur T allein, und nach Gleichung (154) ist die während einer unendlich kleinen virtuellen Zustandsänderung, bei der T, p und c ungeändert bleiben, von außen zugeführte Wärme:

$$Q = T \cdot \frac{dp}{dT} \cdot \delta V. \tag{155}$$

Wir lassen nun die virtuelle Zustandsänderung darin bestehen, daß eine unendlich kleine Wassermenge:

$$\delta M_1'' = -\delta M_1'$$

aus der Lösung verdampft. Dann fällt gleichzeitig, da außer T und p auch die Konzentration c unvariiert bleibt, die Salzmenge:

$$\delta M_2''' = -\delta M_2' = -c\delta M_1' = c\delta M_1''$$

aus der Lösung aus, und die Variationen aller Massen sind auf $\delta M_1''$ zurückgeführt.

Das Gesamtvolumen des Systems:

$$V = v'(M_1' + M_2') + v''M_1'' + v'''M_2''',$$

worin v', v'' und v''' die spezifischen Volumina der drei Phasen bedeuten, wird bei der Variation vergrößert um:

$$\delta V = v'(\delta M_1' + \delta M_2') + v''\delta M_1'' + v'''\delta M_2'''$$

$$\delta V = [(v'' + cv''') - (1 + c)v']\delta M_1''. \tag{156}$$

Nennen wir nun r die Wärmemenge, welche dem System von außen zuzuführen ist, damit bei konstanter Temperatur, Druck und Konzentration die Masseneinheit Wasser aus der Lösung verdampft und gleichzeitig die entsprechende Menge Salz ausscheidet, so ist in (155) zu setzen:

$$Q = r\delta M_1''$$

und wir erhalten:

$$r = T\frac{dp}{dT}(v'' + cv''' - (1 + c)v').$$

Eine in manchen Fällen gut brauchbare Annäherungsformel erhält man, wenn man die spezifischen Volumina der Lösung v' und des festen Salzes v''' gegen das des Dampfes v'' vernachlässigt und außerdem für letzteren den idealen Gaszustand voraussetzt. Dann ist nach (14):

$$v'' = \frac{R}{m}\frac{T}{p}$$

(R Gaskonstante, m Molekulargewicht des Dampfes)

und es wird:

$$r = \frac{R}{m}T^2\frac{d\log p}{dT}. \tag{157}$$

§ 215. r ist zugleich auch umgekehrt die Wärmemenge, welche nach außen abgegeben wird, wenn sich die Masseneinheit Wasserdampf mit der dazu erforderlichen Menge festem Salz bei konstanter Temperatur und konstantem Druck zu gesättigter Lösung vereinigt.

Statt die Vereinigung direkt vorzunehmen, kann man auch die Masseneinheit Wasserdampf zunächst isoliert zu reinem Wasser kondensieren, und dann das feste Salz im flüssigen Wasser auflösen. Nach dem ersten

Hauptsatz der Wärmetheorie ist, wenn in beiden Fällen der Anfangszustand und der Endzustand des Systems derselbe ist, auch die Summe der vom System im ganzen abgegebenen Wärme und Arbeit in beiden Fällen die gleiche, und aus dieser Beziehung läßt sich die Lösungswärme des festen Salzes im flüssigen Wasser berechnen wie folgt.

Im ersten Falle (direkte Vereinigung von Wasserdampf und festem Salz) haben wir für die bei der Kondensation nach außen abgegebene Wärme: r, für die nach außen abgegebene Arbeit: $-p\,\frac{\delta V}{\delta M_1''}$, also für die Summe beider, nach (157) und (156), mit den schon benutzten Annäherungen:

$$\frac{R}{m}\,T^2\,\frac{d\log p}{dT} - pv''. \tag{158}$$

Berechnen wir jene Summe nun auch für den zweiten Fall. Zunächst ist zu bemerken, daß der Wasserdampf über der Lösung im allgemeinen nicht denselben Druck besitzen wird wie über reinem Wasser bei derselben Temperatur, sondern einen anderen, und zwar keinesfalls einen größeren, sondern einen kleineren, weil sonst der Dampf über der Lösung übersättigt wäre. Bezeichnen wir also den Druck des bei der Temperatur T gesättigten Wasserdampfes über reinem Wasser mit p_0, so ist $p < p_0$.

Wir bringen nun zunächst die Masseneinheit Wasserdampf vom Drucke p und dem Volumen v'' durch isotherme Kompression auf den Druck p_0 mit dem spezifischen Volumen v_0'', und somit in den Zustand der Sättigung. Dabei wird positive Wärme und negative Arbeit nach außen abgegeben. Die Summe beider Beträge, welche zugleich die Abnahme der Energie des Dampfes angibt, ist gleich Null, wenn wir wieder annehmen, daß der Dampf sich wie ein ideales Gas verhält, daß also seine Energie bei konstant gehaltener Temperatur ungeändert bleibt. Hierauf kondensieren wir den Wasserdampf vom Volumen v_0'' bei konstanter Temperatur T und konstantem Druck p_0 zu reinem Wasser. Die Summe der hierbei nach außen abgegebenen Wärme und Arbeit ergibt sich mit Hilfe der Gleichung (112) zu:

$$\frac{R}{m}\,T^2\,\frac{d\log p_0}{dT} - p_0 v_0''. \tag{159}$$

Um weiter das flüssige Wasser vom Drucke p_0 wieder auf den Druck p zu bringen, sind keine merklichen äußeren Wirkungen erforderlich.

Endlich lösen wir in der so erhaltenen Masseneinheit von flüssigem Wasser bei konstanter Temperatur T und konstantem Druck p so viel Salz auf, als zur Sättigung der Lösung dient, und erhalten dabei als Summe der

nach außen abgegebenen Wärme und Arbeit einfach die Lösungswärme:

$$\lambda, \tag{160}$$

da der Betrag der äußeren Arbeit ganz zu vernachlässigen ist. Die Summe der Ausdrücke (159) und (160) ist nach dem ersten Hauptsatz der Wärmetheorie gleich dem Ausdruck (158). Wir haben daher:

$$\frac{R}{m} T^2 \frac{d \log p_0}{dT} - p_0 v_0'' + \lambda = \frac{R}{m} T^2 \frac{d \log p}{dT} - p v''$$

oder, wenn das BOYLEsche Gesetz: $p_0 v_0'' = p v''$ benutzt wird:

$$\lambda = \frac{R}{m} T^2 \frac{d}{dT} \log \frac{p}{p_0}. \tag{161}$$

Diese zuerst von KIRCHHOFF aufgestellte Formel ergibt also die Wärmetönung (abgegebene Wärme) bei der Auflösung von so viel Salz in 1 g reinem Wasser, als zur Sättigung der entstehenden Lösung erforderlich ist.

Um λ in Kalorien auszudrücken, hat man die Gaskonstante R noch mit dem mechanischen Wärmeäquivalent zu dividieren, also nach (34) an Stelle von R $1,985$ zu setzen. Dies ergibt mit $m = 18$:

$$\lambda = 0,11 \; T^2 \frac{d}{dT} \log \frac{p}{p_0} \text{ cal.}$$

Hierbei ist noch besonders zu beachten, daß p, der Dampfdruck über der mit Salz gesättigten Lösung, insofern Funktion der Temperatur allein ist, als auch die Konzentration c der gesättigten Lösung sich mit der Temperatur in bestimmter Weise ändert.

Die im Laufe der Rechnung vorgenommenen Vernachlässigungen lassen sich im Bedarfsfalle ohne Schwierigkeit an den einzelnen Stellen verbessern.

§ 216. Wir gehen nun zur Behandlung des wichtigen Falles über, daß zwei unabhängige Bestandteile (1) und (2) sich in zwei Phasen befinden:

$$\alpha = 2 \qquad \beta = 2$$

und entwickeln zunächst die Gleichgewichtsbedingungen unter der allgemeinsten Voraussetzung, daß beide Bestandteile in beiden Phasen mit merklichen Mengen enthalten sind, nämlich in der ersten Phase mit den Massen M_1' und M_2', in der zweiten Phase mit den Massen M_1'' und M_2''.

Die inneren Variabeln des Systems sind außer der Temperatur und dem Druck die Konzentrationen des zweiten Bestandteils in beiden Phasen:

$$c' = \frac{M_2'}{M_1'} \quad \text{und} \quad c'' = \frac{M_2''}{M_1''}, \tag{162}$$

und nach der Phasenregel sind von den vier Variabeln T, p, c', c'' zwei willkürlich, die andern dadurch bestimmt.

Bei irgend einer mit den äußeren Bedingungen vorgenommenen Änderung ergibt sich nach (153) folgendes Gesetz für die Verschiebung des Gleichgewichts:

$$\begin{aligned}\frac{Q}{T^2}\,dT - \frac{\delta V}{T}\,dp + dM_1'\,\delta\frac{\partial\Phi'}{\partial M_1'} + dM_2'\,\delta\frac{\partial\Phi'}{\partial M_2'} \\ + dM_1''\,\delta\frac{\partial\Phi''}{\partial M_1''} + dM_2''\,\delta\frac{\partial\Phi''}{\partial M_2''} = 0.\end{aligned} \tag{163}$$

Hierin ist für die erste Phase:

$$\begin{cases} \delta\dfrac{\partial\Phi'}{\partial M_1'} = \dfrac{\partial^2\Phi'}{\partial M_1'^2}\,\delta M_1' + \dfrac{\partial^2\Phi'}{\partial M_1'\,\partial M_2'}\,\delta M_2' \\ \delta\dfrac{\partial\Phi'}{\partial M_2'} = \dfrac{\partial^2\Phi'}{\partial M_1'\,\partial M_2'}\,\delta M_1' + \dfrac{\partial^2\Phi'}{\partial M_2'^2}\,\delta M_2'. \end{cases} \tag{164}$$

Zwischen den Abgeleiteten von Φ' nach M_1' und M_2' bestehen gewisse einfache Beziehungen. Da nämlich nach (144):

$$\Phi' = M_1'\,\frac{\partial\Phi'}{\partial M_1'} + M_2'\,\frac{\partial\Phi'}{\partial M_2'},$$

so folgt durch partielle Differentiation nach M_1' und nach M_2':

$$\begin{aligned} 0 &= M_1'\,\frac{\partial^2\Phi'}{\partial M_1'^2} + M_2'\,\frac{\partial^2\Phi'}{\partial M_2'\,\partial M_2'} \\ 0 &= M_1'\,\frac{\partial^2\Phi'}{\partial M_1'\,\partial M_2'} + M_2'\,\frac{\partial^2\Phi'}{\partial M_2'^2}. \end{aligned}$$

Setzen wir also zur Abkürzung:

$$M_1'\,\frac{\partial^2\Phi'}{\partial M_1'\,\partial M_2'} = \varphi', \tag{165}$$

eine Größe, die nur von der inneren Beschaffenheit der ersten Phase, also von T, p und c', nicht aber von den Massen M_1' und M_2' einzeln abhängt, so ergibt sich:

$$\begin{cases} \dfrac{\partial^2\Phi'}{\partial M_1'\,\partial M_2'} = \dfrac{\varphi'}{M_1'} \\ \dfrac{\partial^2\Phi'}{\partial M_1'^2} = -\dfrac{M_2'}{M_1'^2}\varphi' \\ \dfrac{\partial^2\Phi'}{\partial M_2'^2} = -\dfrac{\varphi'}{M_2'}. \end{cases} \tag{166}$$

Ganz dieselben Gleichungen gelten, bei Einführung der entsprechenden Größe:

$$\varphi'' = M_1'' \frac{\partial^2 \Phi''}{\partial M_1'' \, \partial M_2''},$$

für die zweite Phase.

§ 217. Von den beiden Größen φ' und φ'' läßt sich von vornherein nur über das Vorzeichen etwas aussagen. Denn nach § 147 ist im stabilen Gleichgewicht Φ ein Maximum, sofern man nur Vorgänge bei konstanter Temperatur und konstantem Druck in Betracht zieht, d. h.

$$\delta^2 \Phi < 0. \tag{167}$$

Nun ist

$$\Phi = \Phi' + \Phi'',$$

also:

$$\delta\Phi = \frac{\partial \Phi'}{\partial M_1'} \delta M_1' + \frac{\partial \Phi'}{\partial M_2'} \delta M_2' + \frac{\partial \Phi''}{\partial M_1''} \delta M_1'' + \frac{\partial \Phi''}{\partial M_2''} \delta M_2''$$

und durch abermalige Variation, wenn $\delta\Phi = 0$

$$\begin{aligned}\delta^2 \Phi = {} & \frac{\partial^2 \Phi'}{\partial M_1'^2} \delta M_1'^2 + 2 \frac{\partial^2 \Phi'}{\partial M_1' \, \partial M_2'} \delta M_1' \, \delta M_2' + \frac{\partial^2 \Phi'}{\partial M_2'^2} \delta M_2'^2 \\ & + \frac{\partial^2 \Phi''}{\partial M_1''^2} \delta M_1''^2 + 2 \frac{\partial^2 \Phi''}{\partial M_1'' \, \partial M_2''} \delta M_1'' \, \delta M_2'' + \frac{\partial^2 \Phi''}{\partial M_2''^2} \delta M_2''^2.\end{aligned}$$

Führt man hierin nach (165) die Größen φ' und φ'' ein, so ergibt sich:

$$\delta^2 \Phi = -M_2' \varphi' \left(\frac{\delta M_1'}{M_1'} - \frac{\delta M_2'}{M_2'} \right)^2 - M_2'' \varphi'' \left(\frac{\delta M_1''}{M_1''} - \frac{\delta M_2''}{M_2''} \right)^2,$$

und diese Beziehung zeigt, daß die Ungleichung (167) stets und nur dann erfüllt wird, wenn φ' und φ'' beide p o s i t i v sind.

§ 218. Im ganzen sind in dem betrachteten System zwei Arten von virtuellen isotherm-isobaren Zustandsänderungen möglich, indem entweder der erste oder der zweite Bestandteil aus der ersten in die zweite Phase übergeht. Für die erste Änderung haben wir:

$$\delta M_1' = -\delta M_1'' \qquad \delta M_2' = \delta M_2'' = 0. \tag{168}$$

für die zweite:

$$\delta M_1' = \delta M_1'' = 0 \qquad \delta M_2' = -\delta M_2''.$$

Wir wollen die dem System zugeführte Wärme Q und die Volumenänderung δV des Systems in dem ersten und in dem zweiten Falle durch die Indizes 1 und 2 unterscheiden; dann reduziert sich für den ersten Fall das für eine Verschiebung des Gleichgewichts gültige Gesetz durch Kombination von (163), (164), (168), (166) und (162) auf folgende Form:

$$\frac{Q_1}{T^2}\,dT - \frac{\delta_1 V}{T}\,dp - \delta M_1''\,(\varphi'\,dc' - \varphi''\,dc'') = 0.$$

Führen wir noch zur Abkürzung die bestimmten endlichen Größen ein:

$$r_1 = \frac{Q_1}{\delta M_1''}, \qquad v_1 = \frac{\delta_1 V}{\delta M_1''}, \tag{169}$$

d. h. das Verhältnis der von außen zugeführten Wärme bzw. der Volumenänderung des Systems zu der aus der ersten in die zweite Phase übergehenden Masse des ersten Bestandteils, so ergibt sich:

$$\frac{r_1}{T^2}\,dT - \frac{v_1}{T}\,dp - \varphi'\,dc' + \varphi''\,dc'' = 0. \tag{170}$$

Auf ganz demselben Wege findet man für den Übergang des zweiten Bestandteils aus der ersten in die zweite Phase, in entsprechender Bezeichnung:

$$\frac{r_2}{T^2}\,dT - \frac{v_2}{T}\,dp + \phi'\,\frac{dc'}{c'} - \phi''\,\frac{dc''}{c''} = 0. \tag{171}$$

Durch diese beiden Gleichungen sind bei irgend einer Verschiebung des Gleichgewichts die vier Differentiale dT, dp, dc', dc'' entsprechend der Phasenregel aneinander geknüpft.

§ 219. Um eine Anwendung dieser Gesetze zu geben, betrachten wir ein System, welches besteht aus dem Gemisch zweier Komponenten (z. B. Wasser und Alkohol) in einer flüssigen, der ersten, und einer dampfförmigen, der zweiten Phase. Dann sind nach der Phasenregel von den vier Variabeln T, p, c', c'' zwei bestimmt durch die beiden übrigen.

Läßt man also bei konstantem Druck ($dp = 0$) das flüssige Gemisch durch gehörige Wärmezuführung unendlich langsam verdampfen, so sind bei jeder Temperatur die Konzentrationen in beiden Phasen vollständig bestimmt, und zwar ändern sich bei fortschreitender Wärmezufuhr und der damit verbundenen Temperaturerhöhung ($dT > 0$) die Konzentrationen c und c' in folgender Weise:

$$\frac{dc'}{c'} = \frac{r_1 + c''r_2}{c' - c''} \cdot \frac{dT}{T^2\phi'} \tag{172}$$

$$\frac{dc''}{c''} = \frac{r_1 + c'r_2}{c' - c''} \cdot \frac{dT}{T^2\phi''}. \tag{173}$$

Hieraus ergibt sich zunächst, daß die Konzentrationen c' und c'' sich stets in demselben Sinne ändern. Denn nicht nur ϕ' und ϕ'' (vgl. § 217), sondern auch r_1 und r_2, die Verdampfungswärmen der beiden Komponenten, sind wesentlich positiv.

Wir wollen nun weiter annehmen, daß $c'' > c'$, was keine Beschränkung der Allgemeinheit bedeutet, da man stets die Ziffer 1 auf diejenige Komponente (z. B. Wasser) beziehen kann, die in der eingestrichenen Phase (Flüssigkeit) mit höherem Prozentgehalt vertreten ist als in der zweigestrichenen Phase (Dampf), und die Ziffer 2 auf diejenige Komponente (z. B. Alkohol), die in der zweigestrichenen Phase mit höherem Prozentgehalt vertreten ist als in der eingestrichenen, und die wir die „flüchtigere“ Komponente nennen wollen. Dann lehren die letzten Gleichungen, daß bei eintretender Temperaturerhöhung die Konzentration der flüchtigeren Komponente in beiden Phasen gleichzeitig abnimmt, also daß z. B. bei fortschreitender Destillation einer Mischung von Wasser und Alkohol sowohl das Destillat als auch der Rückstand mit steigender Temperatur immer alkoholärmer wird. Dieses Resultat kommt natürlich dadurch zustande, daß das Verhältnis der in irgend einem Augenblicke verdampfenden Alkoholmenge zu der gleichzeitig verdampfenden Wassermenge größer ist als c', aber kleiner ist als c''. Durch ersteren Umstand wird bedingt, daß die Flüssigkeit alkoholärmer wird ($dc' < 0$), durch den zweiten Umstand, daß der Dampf alkoholärmer wird ($dc'' < 0$).

Bei gehöriger Fortsetzung dieses isobaren Verdampfungsprozesses können zwei Fälle eintreten, je nachdem die Konzentration c', welche ja zwischen 0 und c'' liegt, schließlich mit dem Werte Null oder mit c'' zusammenfällt. Im ersten Fall, für $c' = 0$, ist schließlich die flüchtigere Komponente ganz in die zweite Phase übergegangen, und man erhält in der ersten Phase die weniger flüchtige Komponente mit jedem beliebigen Grade von Reinheit. Das trifft z. B. zu bei Wasser und Äthylalkohol. Im zweiten Fall aber wird bei einer bestimmten Temperatur $c' = c''$. Dann ändert sich, wie man aus den letzten beiden Gleichungen erkennt, der Siedepunkt T nicht mehr mit der Konzentration, d. h. die weitere Verdampfung erfolgt ohne Temperaturerhöhung, und das Gemisch siedet konstant, Destillat und Rückstand weisen dauernd dieselbe prozentische Zusammensetzung auf, z. B. für Wasser und Ameisensäure bei etwa 80% Säuregehalt (etwas verschieden je nach der Größe des angewandten Druckes).

Für dieses spezielle Mischungsverhältnis ist also die Siedetemperatur T ein Maximum: $\dfrac{dT}{dc'} = 0$. Eine jede Lösung von Ameisensäure in Wasser nähert sich daher bei steigender Siedetemperatur, d. h. bei fortgesetzter

Destillation diesem bestimmten Mischungsverhältnis immer mehr an, sei es, daß man ursprünglich von größerem oder von kleinerem Säuregehalt c' der Flüssigkeit ausgeht. Da die Konzentrationen in beiden Phasen sich in demselben Sinne ändern, so zeigen dieselben stets beide übereinstimmend entweder größeren oder kleineren Säuregehalt als 80%. Im ersten Fall nimmt bei fortschreitender Verdampfung der Säuregehalt in beiden Phasen ab, im zweiten zu. Folglich ist nach (172) und (173) im ersten Fall die Säure der flüchtigere Bestandteil ($c'' > c'$), d. h. der Dampf säurereicher als die Flüssigkeit. Im zweiten Fall (Säuregehalt kleiner als 80%) ist das Wasser der flüchtigere Bestandteil ($c' > c''$), d. h. die Flüssigkeit säurereicher als der Dampf; bei Steigerung der Temperatur werden dann beide Phasen säurereicher, bis wieder der Säuregehalt 80% erreicht ist.

Die Gleichung $\frac{dT}{dc'} = 0$ wird auch erfüllt, wenn die Siedetemperatur in bezug auf die Konzentration ein Minimum aufweist, wie z. B. bei einer Mischung von Wasser und Propylalkohol. Dann ist nach Gleichung (172) ebenfalls $c' = c''$, d. h. das Gemisch siedet ebenfalls konstant. Aber dieses Mischungsverhältnis ist insofern labil, als bei der geringsten Änderung der Konzentrationen, nach der einen oder nach der anderen Seite, die fortgesetzte Destillation eine Steigerung des Konzentrationsunterschiedes der beiden Phasen herbeiführt. Denn da mit der Destillation eine Temperaturerhöhung verbunden ist, so entfernt sich die Siedetemperatur von ihrem Minimalwert, und dementsprechend die Konzentrationen von der Zusammensetzung des konstant siedenden Gemisches.

§ 219a. Nimmt man dagegen die Verdampfung des betrachteten Flüssigkeitsgemisches bei konstanter Temperatur ($dT = 0$) vor, durch allmähliche Vergrößerung des Volumens, so ändern sich mit abnehmendem Druck ($dp < 0$) die Konzentrationen nach (170) und (171) in folgender Weise:

$$\frac{dc'}{c'} = \frac{v_1 + c''v_2}{c'' - c'} \cdot \frac{dp}{T\varphi'},$$
$$\frac{dc''}{c''} = \frac{v_1 + c'v_2}{c'' - c'} \cdot \frac{dp}{T\varphi''}.$$

Auch hier ändern sich also die beiden Konzentrationen c' und c'' gleichzeitig in demselben Sinne, da außer φ' und φ'' auch v_1 und v_2, die Volumenänderungen bei der Verdampfung der beiden Komponenten, wesentlich positiv sind. Wenn wieder wie nach unserer im vorigen Paragraphen eingeführten Festsetzung $c'' > c'$, so wird bei fortschreitender Verdampfung und der damit verbundenen Druckabnahme sowohl die Flüssigkeit als auch

der Dampf stets ärmer an dem flüchtigeren Bestandteil, bis schließlich entweder dieser Bestandteil ganz aus der flüssigen Phase verschwindet ($c' = 0$), oder diejenige Zusammensetzung der Mischung erreicht wird ($c' = c''$), bei der die fortgesetzte Verdampfung keine Druckänderung und keine Konzentrationsänderung mehr hervorbringt. Diese Verhältnisse entsprechen also ganz denen bei der Verdampfung unter konstantem Druck.

§ 219b. Die Gleichungen (172) und (173) gelten natürlich auch, wenn die zweite Phase flüssig oder fest ist, z. B. für den Gefrierpunkt einer Metalllegierung (Wismut und Blei). Daraus ergibt sich u. a., daß die Erstarrungstemperatur einer flüssigen Legierung steigt, wenn die flüssige Legierung (eingestrichene Phase) mit demjenigen Bestandteil (2) angereichert wird, der vorwiegend ausfriert. Denn da nach der Annahme $c'' > c'$, und da r_1 und r_2 beide negativ, so ist $\frac{dT}{dc'} > 0$. Nach demselben Satze steigt auch die Gefriertemperatur einer wäßrigen Salzlösung, aus der reines Eis ausfriert, wenn die Lösung mit Wasser verdünnt wird. Im Grenzfall, wenn die Erstarrungstemperatur sich mit der Konzentration nicht ändert $\left(\frac{dT}{dc'} = 0\right)$, ist $c'' = c'$, d. h. die Legierung friert ohne Konzentrationsänderung aus (Eutektische Legierung).

§ 220. Bei den weiteren Anwendungen wollen wir uns auf den speziellen Fall beschränken, daß der zweite Bestandteil nur in der ersten Phase vorhanden ist:

$$c'' = 0, \quad \text{und daher auch} \quad dc'' = 0. \tag{174}$$

Den ersten Bestandteil, welcher in der ersten Phase mit dem zweiten gemischt, in der zweiten Phase rein enthalten ist, wollen wir hier als das „Lösungsmittel“, den zweiten als den „gelösten Stoff“ bezeichnen. (Vgl. dagegen unten § 249.) Dann fällt die Gleichung (171) ganz fort, und von (170) bleibt übrig:

$$\frac{r}{T^2}\,dT - \frac{v}{T}\,dp - \varphi\,dc = 0, \tag{175}$$

wenn wir der Einfachheit halber bei r_1 und v_1 den Index 1, und bei φ' und c' den Strich fortlassen.

Nehmen wir zunächst eine Lösung eines nichtflüchtigen Salzes in Berührung mit dem Dampf des Lösungsmittels. Wir wollen die Gleichung (175) nach drei Richtungen betrachten, je nachdem wir die Konzentration c der Lösung, die Temperatur T oder den Druck p konstant halten, und die Abhängigkeit der beiden übrigen Größen voneinander betrachten.

§ 221. Konstante Konzentration. $dc = 0$.

Die Abhängigkeit des Dampfdrucks von der Temperatur ist nach (175):

$$\left(\frac{\partial p}{\partial T}\right)_c = \frac{r}{T \cdot v}. \tag{176}$$

Hierbei kann man r kurz bezeichnen als die Verdampfungswärme der Lösung. Faßt man sie nicht, wie in (169) gefordert wird, als Verhältnis zweier unendlich kleiner Größen auf, sondern bezieht sie auf die Masseneinheit des Lösungsmittels, so muß man sich die Menge der Lösung so groß denken, daß ihre Konzentration durch den Austritt der Masseneinheit des Lösungsmittels nicht merklich geändert wird. v kann man gewöhnlich einfach gleich dem Volumen der Masseneinheit des Dampfes setzen. Nimmt man außerdem für diesen das Boyle-Gay-Lussacsche Gesetz als gültig an, so ergibt sich:

$$v = \frac{R}{m}\frac{T}{p} \tag{177}$$

und aus der letzten Gleichung:

$$r = \frac{R}{m} T^2 \left(\frac{\partial \log p}{\partial T}\right)_c.$$

r ist zugleich auch umgekehrt die Wärmemenge, welche nach außen abgegeben wird, wenn sich die Masseneinheit des dampfförmigen Lösungsmittels bei konstantem T und p mit einer großen Quantität der Lösung von der Konzentration c vereinigt.

Statt die Vereinigung direkt vorzunehmen, kann man auch die Masseneinheit Dampf zunächst isoliert zum flüssigen reinen Lösungsmittel kondensieren und dann damit die Lösung verdünnen. Nach dem ersten Hauptsatz der Wärmetheorie ist, wenn in beiden Fällen der Anfangszustand und der Endzustand des Systems derselbe ist, auch die Summe der vom System im ganzen abgegebenen Wärme und Arbeit in beiden Fällen die gleiche, und aus dieser Überlegung läßt sich die „Verdünnungswärme“ der Lösung in folgender Weise ableiten.

Im ersten Fall haben wir für die Summe der nach außen abgegebenen Wärme und Arbeit:

$$r - pv = \frac{R}{m} T^2 \left(\frac{\partial \log p}{\partial T}\right)_c - pv.$$

Im zweiten Falle haben wir, wenn wir genau ebenso verfahren, wie in § 215 beschrieben wurde, als Summe der bei der Kondensation und der darauf folgenden Verdünnung nach außen abgegebenen Wärme und Arbeit:

$$\frac{R}{m} T^2 \frac{d \log p_0}{dT} - p_0 v_0 + \Delta,$$

wobei p_0 der Druck, v_0 das spezifische Volumen des bei der Temperatur T über reinem Lösungsmittel befindlichen Dampfes, Δ die Verdünnungswärme der Lösung bezeichnet, d. h. die Wärmetönung (frei werdende Wärme) beim Zusatz der Masseneinheit des flüssigen Lösungsmittels zu einer großen Quantität der Lösung von der Konzentration c. Da nun nach dem ersten Hauptsatz der Wärmetheorie die letzten beiden Ausdrücke gleich sind, so erhalten wir, mit Rücksicht auf das BOYLEsche Gesetz:

$$\Delta = \frac{R}{m} T^2 \left(\frac{\partial}{\partial T} \log \frac{p}{p_0} \right)_c , \tag{178}$$

die KIRCHHOFFsche Formel für die Verdünnungswärme.

Die im Laufe der Rechnung eingeführten Vernachlässigungen, die darauf beruhen, daß der Dampf als ideales Gas und sein spezifisches Volumen groß gegen das der Flüssigkeit angenommen ist, lassen sich nötigenfalls leicht ergänzen.

Die Ähnlichkeit des Ausdrucks für die Verdünnungswärme Δ mit dem oben aufgestellten Ausdruck (161) für die Sättigungswärme λ der Masseneinheit des Lösungsmittels mit dem festen Salz ist nur eine äußerliche, weil es sich hier um eine Lösung von ganz beliebiger Konzentration handelt und demgemäß auch die Differentiation nach der Temperatur bei konstantem c auszuführen ist, während dort die Konzentration der mit Salz gesättigten Lösung zu nehmen ist, welche sich bei der Differentiation mit der Temperatur in bestimmter Weise mitändert.

§ 222. Da bei kleinen Werten von c (verdünnte Lösung) die Verdünnungswärme Δ klein ist (§ 97), so wird nach (178) für eine verdünnte Lösung von bestimmter Konzentration das Verhältnis des Dampfdrucks p zu dem Dampfdruck über reinem Lösungsmittel p_0 merklich unabhängig von der Temperatur (Gesetz von BABO).

§ 223. Konstante Temperatur. $dT = 0$.

Die Abhängigkeit des Dampfdrucks p von der Konzentration c der Lösung ist nach (175):

$$\left(\frac{\partial p}{\partial c} \right)_T = -\frac{T\varphi}{v}, \tag{179}$$

und, wenn man das spezifische Volumen der Flüssigkeit gegen das des Dampfes vernachlässigt und letzteren als ideales Gas mit dem Molekulargewicht m betrachtet, nach (177):

$$\left(\frac{\partial p}{\partial c} \right)_T = -\frac{mp\varphi}{R},$$

oder:

$$\left(\frac{\partial \log p}{\partial c}\right)_T = -\frac{m}{R}\varphi.$$

Da φ nach § 217 notwendig positiv, so nimmt mit steigender Konzentration der Lösung der Dampfdruck immer ab. Dieser Satz liefert u. a. ein Mittel zur Entscheidung der Frage, ob eine bestimmte Flüssigkeit wirklich eine Lösung oder etwa nur eine Emulsion des einen Bestandteils in dem andern bildet. Im letzteren Falle hat die Anzahl der in der Flüssigkeit suspendierten Teile gar keinen Einfluß auf den Dampfdruck.

Näheres läßt sich im allgemeinen, solange die Größe φ nicht genauer bekannt ist, über die Abhängigkeit der Dampfspannung von der Konzentration nicht sagen.

§ 224. Für $c = 0$ (reines Lösungsmittel) sei wieder $p = p_0$. Dann ist für einen kleinen Wert von c p nur wenig von p_0 verschieden, und man kann setzen:

$$\frac{\partial p}{\partial c} = \frac{p - p_0}{c - 0} = \frac{p - p_0}{c}.$$

Folglich nach (179):

$$p_0 - p = \frac{cT\varphi}{v}. \tag{180}$$

Setzt man wieder für v nach (177) das spezifische Volumen des als ideales Gas angenommenen Dampfes, so ergibt sich:

$$\frac{p_0 - p}{p} = \frac{cm\varphi}{R}, \tag{181}$$

d. h. die relative Dampfspannungserniedrigung ist proportional der Konzentration der Lösung (Gesetz von Wüllner). Weiteres siehe § 270.

§ 225. Konstanter Druck. $dp = 0$.

Die Abhängigkeit der Temperatur (Siedetemperatur) von der Konzentration ist nach (175):

$$\left(\frac{\partial T}{\partial c}\right)_p = \frac{T^2\varphi}{r}. \tag{182}$$

Da φ positiv, so wird die Siedetemperatur bei steigender Konzentration erhöht. Durch Vergleich mit der Formel (179) für die Abnahme des Dampfdrucks ergibt sich für eine beliebige Lösung:

$$\left(\frac{\partial T}{\partial c}\right)_p : \left(\frac{\partial p}{\partial c}\right)_T = -\frac{Tv}{r},$$

d. h. bei einer unendlich kleinen Vergrößerung der Konzentration verhält sich die Zunahme der Siedetemperatur (bei konstantem Druck) zu der Abnahme des Dampfdrucks (bei konstanter Temperatur) wie das Produkt der absoluten Temperatur und des spezifischen Dampfvolumens zu der Verdampfungswärme der Lösung.

Bedenkt man, daß dieses Verhältnis der der Gleichung (6) ganz analogen Identität genügt:

$$\left(\frac{\partial T}{\partial c}\right)_p : \left(\frac{\partial p}{\partial c}\right)_T = -\left(\frac{\partial T}{\partial p}\right)_c,$$

so kommt man unmittelbar zur Gleichung (176) zurück.

§ 226. Für $c = 0$ sei $T = T_0$ (Siedepunkt des reinen Lösungsmittels), dann wird für k l e i n e Werte von c T nahezu $= T_0$, und man kann setzen:

$$\frac{\partial T}{\partial c} = \frac{T - T_0}{c - 0} = \frac{T - T_0}{c},$$

wodurch die Gleichung (182) übergeht in:

$$T - T_0 = \frac{cT^2\varphi}{r}, \tag{183}$$

d. h. die Siedepunktserhöhung ist proportional der Konzentration der Lösung. Weiteres siehe § 269.

§ 227. Betrachten wir weiter den Fall, daß die zweite Phase das reine Lösungsmittel nicht im dampfförmigen, sondern im festen Aggregatzustand enthält, wie das z. B. beim Gefrieren einer wäßrigen Salzlösung oder beim Ausfällen von Salz aus gesättigter Lösung eintritt. (Im letzteren Falle müssen wir nach den Festsetzungen in § 220 als ersten Bestandteil das Salz, als zweiten Bestandteil das Wasser rechnen, und das Salz als Lösungsmittel, das Wasser als gelösten Stoff bezeichnen.) Dann ist wieder die Gleichung (175) anwendbar, und läßt sich auch hier wieder nach drei Richtungen behandeln, je nachdem man untersucht, in welcher Weise sich der Gefrierpunkt bzw. Sättigungspunkt einer Lösung von bestimmter Konzentration mit dem Druck ändert ($dc = 0$), oder wie der Druck zu ändern ist, damit eine Lösung von geänderter Konzentration bei der nämlichen Temperatur gefriert bzw. gesättigt ist ($dT = 0$) oder endlich, wie sich der Gefrierpunkt bzw. Sättigungspunkt einer Lösung unter bestimmtem äußeren Druck mit der Konzentration ändert ($dp = 0$). Für den letzten als den wichtigsten Fall erhalten wir aus (175), wenn wir zugleich zum Unterschied gegen die Siedetemperatur die Gefriertemperatur bzw. Sättigungstemperatur als Funktion der Konzentration mit T' bezeichnen:

$$\left(\frac{\partial T'}{\partial c}\right)_p = \frac{T^2\varphi}{r'}, \tag{184}$$

wobei r' hier die Wärme bedeutet, welche von außen aufgenommen wird, wenn aus einer großen Quantität Lösung von der Konzentration c die Masseneinheit des Lösungsmittels (Eis, Salz) ausfällt. Diese Wärmemenge ist in der Regel negativ, ihr Betrag bezeichnet die „Gefrierwärme" der Lösung bzw. die „Fällungswärme" des Salzes. Die Gefrierwärme r' einer Salzlösung ist immer negativ; daher sinkt der Gefrierpunkt T' der Lösung stets mit wachsendem Salzgehalt c. Andrerseits: wenn die Fällungswärme r' eines Salzes aus einer Lösung negativ ist, so sinkt der Sättigungspunkt T' der Lösung mit wachsendem Wassergehalt c der Lösung, d. h. er steigt mit wachsendem Salzgehalt. Im entgegengesetzten Fall sinkt der Sättigungspunkt mit wachsendem Salzgehalt. Will man bei einer mit Salz gesättigten Lösung nicht den Wassergehalt, sondern den Salzgehalt der Lösung mit c bezeichnen, so hat man nach der von c in (162) und nach der von φ in (165) aufgestellten Definition in der letzten Formel statt c zu schreiben: $\frac{1}{c}$, und statt φ zu schreiben: $c\varphi$, wodurch dieselbe übergeht in:

$$\left(\frac{\partial T'}{\partial c}\right)_p = -\frac{T^2\varphi}{cr'}. \tag{185}$$

Hier haben nun c und φ dieselbe Bedeutung wie in der auf den Gefrierpunkt einer Salzlösung bezüglichen Formel (184).

§ 228. Für $c = 0$ sei $T = T_0'$ (Gefriertemperatur des reinen Lösungsmittels). Dann wird für kleine Werte von c T' nahezu $= T_0'$, und man kann setzen:

$$\frac{\partial T'}{\partial c} = \frac{T' - T_0}{c - 0} = \frac{T' - T_0'}{c},$$

so daß die Gleichung (184) übergeht in:

$$T' - T_0' = \frac{cT^2\varphi}{r'}, \tag{186}$$

d. h. die Gefrierpunktsänderung ist proportional der Konzentration. Weiteres siehe § 269.

§ 229. Da die hier überall vorkommende positive Größe φ für eine Lösung von bestimmtem c, T und p einen ganz bestimmten Wert hat, insbesondere unabhängig ist von der Beschaffenheit der zweiten Phase, so sind durch die letzten Formeln die Gesetze der Dampfdruckerniedrigung, Siedepunktserhöhung, Gefrierpunktserniedrigung und Sättigungspunktsveränderung auf allgemeine Weise miteinander verknüpft, und man braucht nur eine einzige dieser Erscheinungen für eine Lösung messend zu verfolgen, um mit Hilfe des

daraus berechneten Wertes von φ die übrigen für die nämliche Lösung ableiten zu können.

Wir wollen nun noch einen weiteren Fall betrachten, in welchem die nämliche Größe φ wiederum eine charakteristische Bedeutung besitzt, nämlich den Gleichgewichtszustand, der eintritt, wenn neben der flüssigen Lösung sich das reine Lösungsmittel weder im dampfförmigen, noch im festen, sondern ebenfalls im flüssigen Zustand befindet, aber nicht frei angrenzend, weil sonst kein Gleichgewicht möglich sein würde, sondern von der Lösung getrennt durch eine Wand, welche nur für das Lösungsmittel, nicht aber für den gelösten Stoff durchlässig ist. Derartige „semipermeable“ Wände sind zwar für keine einzige Lösung mit absoluter Vollkommenheit herzustellen, ja sie sind sogar durch die später zu entwickelnde Theorie (§ 259) prinzipiell ausgeschlossen, da der gelöste Stoff unter allen Umständen mit endlicher, wenn auch in gewissen Fällen äußerst geringer Geschwindigkeit durch die Substanz der Wand hindurchdiffundieren wird. Aber es kommt hier auch nur allein darauf an, daß man, ohne ein Gesetz der Thermodynamik zu verletzen, die Diffusionsgeschwindigkeit des gelösten Stoffes durch die Wand gegen diejenige des Lösungsmittels beliebig klein annehmen darf, und diese Annahme wird dadurch gerechtfertigt, daß sich in der Natur die Eigenschaft der Semipermeabilität für manche Substanzen in praktisch überaus großer Annäherung verwirklicht findet. Der Fehler, den man dadurch begeht, daß man die Diffusionsgeschwindigkeit des gelösten Stoffes durch die Wand direkt gleich Null setzt, sinkt daher hier ebenso unter alle meßbaren Grenzen herab, wie etwa der ganz ähnliche Fehler, der in der von uns oben gemachten Voraussetzung liegt, daß ein Salz absolut nicht aus der Lösung verdampft oder ausfriert; denn auch diese Annahme ist streng genommen unzulässig (§ 259).

Die Bedingung, daß zwei Phasen, die voneinander durch eine semipermeable Wand getrennt sind, sich im Gleichgewicht befinden, ergibt sich leicht aus der allgemeinen thermodynamischen Gleichgewichtsbedingung. Sie lautet, ebenso wie in Gleichung (145):

$$\delta\Phi' - \delta\Phi'' = 0, \tag{187}$$

gültig für jede virtuelle Zustandsänderung, bei der die Temperatur und der Druck in jeder Phase ungeändert bleibt. Der einzige Unterschied gegen den Fall freier Berührungsflächen ist der, daß hier, bei der Anwesenheit einer trennenden Wand zwischen beiden Phasen, der Druck in der zweiten Phase: p'' ein anderer sein kann als der in der ersten: p', wobei unter „Druck“ schlechthin, wie immer, der gewöhnliche hydrostatische, manometrisch wirksame Druck zu verstehen ist.

Der Nachweis für die Gültigkeit obiger Gleichgewichtsbedingung findet sich unmittelbar, wenn man von der allgemeinen Gleichung (76) ausgeht und dort anstatt der Gleichung (78) für die äußere Arbeit den Wert:

$$A = -p' \, \delta V' - p'' \, \delta V''$$

einsetzt. Die weiteren Folgerungen aus (187) schließen sich ganz den oben für eine freie Berührungsfläche abgeleiteten an. Zunächst haben wir, entsprechend der Gleichung (163), für irgend eine Verschiebung des Gleichgewichts:

$$\frac{Q}{T^2} \, dT - \frac{\delta V'}{T} \, dp' - \frac{\delta V''}{T} \, dp'' + dM_1' \, \delta \frac{\partial \Phi'}{\partial M_1'} + dM_2' \, \delta \frac{\partial \Phi'}{\partial M_2'} + \ldots = 0$$

und weiter, unter Berücksichtigung des Umstandes, daß der Bestandteil 2 nur in der ersten Phase vorkommt, anstatt der Gleichung (175) die folgende:

$$\frac{r}{T^2} \, dT - \frac{v'}{T} \, dp' - \frac{v''}{T} \, dp'' - \varphi \, dc = 0. \tag{188}$$

Hier ist, wie in § 221, r die „Austrittswärme“ des Lösungsmittels aus der Lösung, d. h. die Wärmemenge, welche von außen zuzuführen ist, wenn bei konstanter Temperatur T und bei konstanten Drucken p' und p'' die Masseneinheit des Lösungsmittels aus einer großen Quantität der Lösung durch die semipermeable Wand in das reine Lösungsmittel übergeht. Ferner ist v' die bei demselben Vorgang eintretende Volumenänderung der Lösung (negativ), v'' diejenige des angrenzenden Lösungsmittels (positiv). In der Gleichgewichtsbedingung (188) sind also von den vier Variabeln T, p', p'', c drei willkürlich und erst die vierte dadurch bestimmt.

Nehmen wir zunächst den Druck p'' im reinen Lösungsmittel als gegeben und unveränderlich an, etwa als den Druck einer Atmosphäre, so haben wir $dp'' = 0$. Setzen wir ferner $dT = 0$ und dc von Null verschieden, d. h. betrachten wir Lösungen verschiedener Konzentration bei der nämlichen Temperatur und bei dem nämlichen Druck im angrenzenden reinen Lösungsmittel, so ergibt sich aus (188):

$$\left(\frac{\partial p'}{\partial c}\right)_T = -\frac{T\varphi}{v'}.$$

Da nun $\varphi > 0$ und $v' < 0$, so wächst mit steigender Konzentration c der Druck p' im Innern der Lösung.

Man bezeichnet die Differenz der Drucke in beiden Phasen:

$$p' - p'' = P$$

als den „osmotischen Druck“ der Lösung. Da nun p'' oben als konstant angenommen ist, läßt sich schreiben:

$$\left(\frac{\partial P}{\partial c}\right)_T = -\frac{T\varphi}{v'}. \tag{189}$$

Daher wächst der osmotische Druck mit steigender Konzentration. Da aber die Differenz $p' - p''$ für $c = 0$ notwendig verschwindet, so ist der osmotische Druck immer positiv.

Für kleine Werte von c ist:

$$\frac{\partial P}{\partial c} = \frac{P - 0}{c - 0} = \frac{P}{c}$$

und $-v'$ nahezu gleich dem spezifischen Volumen der Lösung. Daraus folgt nach (189):

$$P = \frac{cT\varphi}{v}, \tag{190}$$

wenn man mit v das spezifische Volumen der Lösung bezeichnet. Weiteres siehe § 272.

Somit sind auch die Gesetze des osmotischen Druckes auf die nämliche Größe φ zurückgeführt, welche die Gesetze der Dampfspannungserniedrigung, Siedepunktserhöhung usw. bedingt, oder, wie man auch sagen kann: die letztgenannten Gesetze sind alle zurückgeführt auf die Gesetze des osmotischen Druckes, sie ergeben sich durch Elimination von φ aus der Gleichung (189) und einer der vorhergehenden Gleichungen. Es ist besonders bemerkenswert, daß der so hergestellte Zusammenhang vollständig unabhängig ist von allen molekulartheoretischen Annahmen und Vorstellungen, wiewohl gerade die letzteren in der historischen Entwicklung der Theorie eine sehr wichtige Rolle gespielt haben.

§ 230. Wir haben für verschiedene den Bedingungen des § 220 entsprechende Systeme die Gesetze des Gleichgewichts auf eine einzige für das thermodynamische Verhalten einer Lösung charakteristische Größe φ zurückgeführt. Es bietet keine Schwierigkeit, die entsprechenden Sätze auch für den Fall abzuleiten, daß die gelöste Substanz auch in der zweiten Phase enthalten ist. Man hat dann von den beiden Gleichungen (170) und (171) auszugehen, und findet die einzelnen Beziehungen durch die beiden Größen φ' und φ'' bedingt. Zu einer direkteren Kenntnis dieser Größen φ kann man aber erst dadurch gelangen, daß der Begriff des Moleküls, den wir bisher nur auf den gasförmigen Aggregatzustand angewendet haben, auch auf den flüssigen Zustand erstreckt wird. In den folgenden beiden Kapiteln wird

dieser Schritt ausgeführt, wobei sich zugleich zeigt, daß die Richtung, in der er zu erfolgen hat, durch die vorliegenden Sätze der Thermodynamik in jeder Hinsicht eindeutig vorgeschrieben ist.

§ 231. Wie für zwei unabhängige Bestandteile in zwei Phasen aus der allgemeinen Beziehung (153) die Gleichgewichtsbedingungen (170) und (171) abgeleitet wurden, so läßt sich ganz auf dem entsprechenden Wege auch für den allgemeinen Fall die nämliche Ableitung ausführen.

Hier soll zum Schluß nur kurz das Resultat angeführt werden, welches sich auf diese Weise für ein System von α unabhängigen Bestandteilen in β Phasen ergibt.

Bezeichnet man die Konzentrationen der einzelnen unabhängigen Bestandteile in den einzelnen Phasen, bezogen auf einen bestimmten, mit 1 bezeichneten Bestandteil, entsprechend den Gleichungen (162), mit:

$$\begin{array}{llll} \dfrac{M_2'}{M_1'} = c_2', & \dfrac{M_3'}{M_1'} = c_3', & \dfrac{M_4'}{M_1'} = c_4', & \dots \\ \dfrac{M_2''}{M_1''} = c_2'', & \dfrac{M_3''}{M_1''} = c_3'', & \dfrac{M_4''}{M_1''} = c_4'', & \dots \\ \dots\dots\dots\dots\dots\dots\dots\dots & & & \end{array}$$

so lautet die Bedingung dafür, daß bei irgend einer mit dem Zustand des Systems vorgenommenen unendlich kleinen Veränderung: dT, dp, dc_2', dc_3', $dc_4', \dots dc_2''$, dc_3'', $dc_4'' \dots$ das Gleichgewicht gesichert bleibt gegen den Übertritt des Bestandteils 1 aus der eingestrichenen Phase in die zweigestrichene Phase:

$$\frac{r_1}{T^2}\,dT - \frac{v_1}{T}\,dp + (\varphi_2''\,dc_2'' - \varphi_2'\,dc_2') + (\varphi_3''\,dc_3'' - \varphi_3'\,dc_3') + \dots = 0.$$

Dabei ist analog (165):

$$\begin{array}{ll} \varphi_2' = M_1' \dfrac{\partial^2 \Phi'}{\partial M_1' \partial M_2'}, & \varphi_3' = M_1' \dfrac{\partial^2 \Phi'}{\partial M_1' \partial M_3'}, \quad \dots \\ \varphi_2'' = M_1'' \dfrac{\partial^2 \Phi''}{\partial M_1'' \partial M_2''}, & \varphi_3'' = M_1'' \dfrac{\partial^2 \Phi''}{\partial M_1'' \partial M_3''}, \quad \dots \end{array}$$

und r_1 und v_1 bedeuten die Wärmezufuhr bez. die Volumenänderung des Systems bei dem isotherm-isobaren Übertritt der Masseneinheit des Bestandteils 1 aus einer großen Quantität der eingestrichenen Phase in eine große Quantität der zweigestrichenen Phase (vgl. § 221).

So läßt sich für jeden möglichen Übertritt irgend eines Bestandteils aus irgend einer Phase in irgend eine andere Phase die entsprechende Gleichgewichtsbedingung aufstellen.

Viertes Kapitel. Gasförmiges System.

§ 232. Die Beziehungen, welche wir bisher für die verschiedenen Eigenschaften thermodynamischer Gleichgewichtszustände aus der allgemeinen Gleichgewichtsbedingung (79) hergeleitet haben, beruhen im Grunde auf der Abhängigkeit der für das Gleichgewicht bei gegebener Temperatur und gegebenem Druck charakteristischen Funktion Φ von Temperatur und Druck, wie sie in den Gleichungen (79b) ausgedrückt ist. Eine vollständige Beantwortung aller auf das Gleichgewicht bezüglichen Fragen ist aber erst dann möglich, wenn Φ auch in seiner Abhängigkeit von den Massen der in den einzelnen Phasen des Systems vorhandenen Bestandteile angegeben werden kann, und hierzu dient die Einführung des Molekulargewichts. Wir haben schon früher, bei der Besprechung der Eigenschaften idealer Gase, sowohl das Molekulargewicht eines chemisch homogenen Gases, als auch die Molekülzahl einer Gasmischung aus dem AVOGADROschen Satze definiert, und wenden uns daher hier zunächst der Untersuchung eines Systems zu, welches eine einzige gasförmige Phase vorstellt.

Die Aufgabe ist vollständig gelöst, wenn es gelingt, die Funktion Φ in ihrer Abhängigkeit von den unabhängigen Variabeln, nämlich der Temperatur T, dem Druck p und den Zahlen n_1, n_2, $n_3, \ldots$ aller in der Mischung vorhandenen verschiedenartigen Moleküle anzugeben.

Da nach (75) allgemein:

$$\Phi = S - \frac{U + pV}{T},$$

so läuft die Aufgabe darauf hinaus, die Entropie S, die Energie U und das Volumen V einer Gasmischung als Funktion der obigen unabhängigen Variabeln auszudrücken. Dies läßt sich nun ganz allgemein bewerkstelligen, wenn wir die Voraussetzung einführen, daß für die Mischung die Gesetze idealer Gase gelten — eine Beschränkung, die in vielen Fällen keinen erheblichen Fehler bedingen wird. Will man sich von ihr frei machen, so muß man durch besondere Messungen die Werte der Größen S, U und V ermitteln. Hier wollen wir aber die Annahme idealer Gase festhalten.

§ 233. Was zunächst das Volumen V der Mischung betrifft, so ist dieses durch das BOYLE-GAY-LUSSAC-DALTONsche Gesetz bestimmt. Denn nach Gleichung (16) ist

$$V = \frac{RT}{p}(n_1 + n_2 + \ldots) = \frac{RT}{p}\sum n_1. \tag{191}$$

Die Energie U einer Gasmischung ferner ergibt sich aus den Energien der einzelnen getrennten Gase mit Hilfe des ersten Hauptsatzes der Wärmetheorie.

Denn nach diesem bleibt die Energie eines Systems unverändert, wenn keinerlei äußere Wirkungen auf dasselbe ausgeübt werden, einerlei welche inneren Veränderungen dabei eintreten. Läßt man nun eine beliebige Anzahl von Gasen, die auf eine gemeinsame Temperatur T und auf einen gemeinsamen Druck p gebracht sind, bei konstant gehaltener Temperatur und konstantem Druck ineinander diffundieren, so lehrt die Erfahrung, daß dann weder das Volumen des Systems sich ändert, noch Wärme von außen aufgenommen wird. Folglich ist dabei das mechanische Äquivalent der äußeren Wirkungen gleich Null, und die Energie des Systems behält ihren Anfangswert bis zur vollständigen Beendigung des Diffusionsprozesses unverändert bei. Daher ist die Energie einer Mischung idealer Gase gleich der Summe der Energien der einzelnen Gase, bei der nämlichen Temperatur und dem nämlichen Druck genommen. Die Energie U_1 eines einzelnen idealen Gases mit der Molekülzahl n_1 ist aber nur abhängig von der Temperatur, nämlich nach (35):

$$U_1 = n_1(C_{v_1}T + b_1), \tag{192}$$

wobei C_{v_1} die Molekularwärme des Gases bei konstantem Volumen, b_1 eine Konstante bedeutet. Folglich ist die Gesamtenergie der Mischung:

$$U = \sum n_1(C_{v_1}T + b_1). \tag{193}$$

§ 234. Es handelt sich nun noch um die Bestimmung der Entropie S einer Gasmischung als Funktion von T, p und den Molekülzahlen n_1, $n_2, \ldots$ Soweit S von T und p abhängt, läßt es sich aus U und V berechnen mittels der Gleichung (60):

$$dS = \frac{dU + p\,dV}{T},$$

wobei die Differentiale sich nur auf Änderungen von T und p, nicht aber auf solche der Molekülzahlen beziehen.

Nun ist nach (193):

$$dU = \sum n_1 c_{v_1}\,dT$$

und nach (191):

$$dV = R\sum n_1\,d\left(\frac{T}{p}\right).$$

Folglich durch Substitution:

$$dS = \sum n_1\left(C_{v_1}\frac{dT}{T} + \frac{R\,dT}{T} - \frac{R\,dp}{p}\right)$$

oder, da nach (33):

$$C_{v_1} + R = C_{p_1}$$

$$dS = \sum n_1 \left(C_{p_1} \frac{dT}{T} - R \frac{dp}{p} \right)$$

und durch Integration nach T und p:

$$S = \sum n_1 (C_{p_1} \log T - R \log p + k_1) = C. \tag{194}$$

Hier ist außer den Integrationskonstanten k_1, k_2, $k_3, \ldots$, die von der Natur der einzelnen Gase und von den gewählten Maßeinheiten für T und p abhängen, noch eine besondere Integrationskonstante C hinzuzufügen, weil die Konstanten k_1, $k_2, \ldots$ nur eine lineäre Abhängigkeit der Entropie S von den Molekülzahlen n_1, $n_2, \ldots$ ergeben, während die Integrationskonstante noch in komplizierterer Weise von der Zusammensetzung der Mischung, d. h. von den Verhältnissen der Molekülzahlen n abhängen kann und wird. Gerade die Untersuchung dieser Abhängigkeit bildet den wichtigsten Teil unserer jetzigen Aufgabe.

Die Bestimmung von C kann nicht einfach auf dem Wege einer Definition erfolgen, sondern nur durch die Anwendung des zweiten Hauptsatzes der Wärmetheorie auf irgend einen bekannten reversibeln Prozeß, der eine Änderung in der Zusammensetzung der Mischung herbeiführt. Denn bei einem reversibeln Prozeß ändert sich nach dem zweiten Hauptsatz die Entropie des Systems in ganz bestimmter Weise, und durch die Berücksichtigung der gleichzeitig eintretenden Änderungen der Molekülzahlen läßt sich die Abhängigkeit der Entropie von der Zusammensetzung der Mischung ermitteln. Wir werden den Prozeß derart wählen, daß während desselben keinerlei Einwirkungen von außen, weder Arbeitsleistung noch Wärmezufuhr, stattfinden; dann bleibt die Entropie des Systems während des ganzen Prozesses konstant. Den oben zur Bestimmung der Energie U der Gasmischung benutzten Diffusionsvorgang können wir aber hier nicht verwerten; denn derselbe ist, wie sich schon vermuten läßt und im § 238 zeigen wird, irreversibel, und gestattet daher von vornherein nur die eine Folgerung, daß die Entropie des Systems durch ihn vergrößert wird. Dagegen bietet sich dar als ein reversibler Prozeß, durch welchen die Zusammensetzung der Mischung geändert wird, die Behandlung der Gasmischung mittels einer semipermeabeln Wand, wie sie schon oben § 229 eingeführt und begründet wurde.

§ 235. Damit ein mit einer semipermeabeln Wand ausgeführter Prozeß für den genannten Zweck nutzbar wird, muß man zuerst wissen, welcher

Art das thermodynamische Gleichgewicht ist, das auf beiden Seiten einer Wand besteht für eine Gasart, welche die Wand durchdringen kann; für jede andere Gasart besteht natürlich keine besondere Gleichgewichtsbedingung, da sich hierfür die Wand wie eine gewöhnliche verhält.

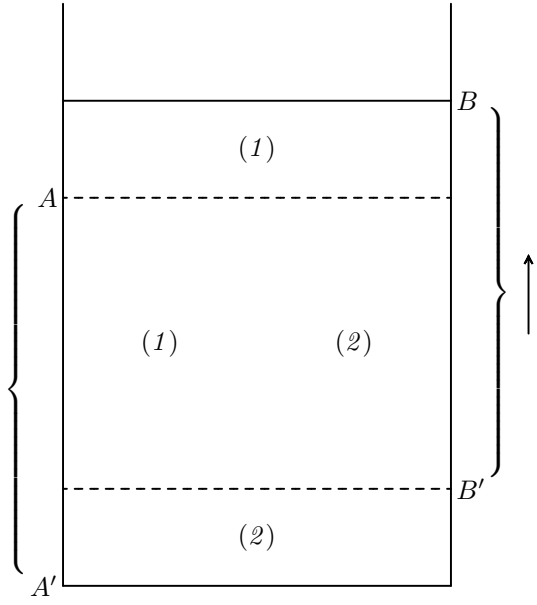
Hier liefert nun die Erfahrung den einfachen Satz, daß jede Gasart, für welche eine Wand permeabel ist, sich dann auf beiden Seiten im Gleichgewicht befindet, wenn ihr Partialdruck (§ 18) auf beiden Seiten gleich ist, ganz unabhängig von den übrigen auf beiden Seiten anwesenden Gasarten. Dieser Satz ist weder selbstverständlich noch notwendig bedingt durch das Vorhergehende, er leuchtet aber durch seine Einfachheit unmittelbar ein und hat sich auch in den allerdings wenig zahlreichen Fällen, die eine direkte Prüfung gestatten, überall bestätigt.

Eine solche Prüfung, die zu einer augenfälligen Folgerung führt, läßt sich z. B. folgendermaßen anstellen. Glühendes Platinblech ist permeabel für Wasserstoff, dagegen impermeabel für atmosphärische Luft. Füllt man also ein Gefäß, dessen Wandung an einer Stelle aus Platinblech besteht, mit reinem Wasserstoff, etwa unter Atmosphärendruck, und schließt es dann vollkommen ab, so muß, wenn das Platinblech ins Glühen gebracht wird, der innen befindliche Wasserstoff in die äußere Luft, also entgegen dem Atmosphärendruck, hinausdiffundieren, und zwar offenbar so lange, bis er vollständig aus dem Gefäß entwichen ist. Da nun andrerseits die Luft nicht hineindringen kann, so wird schließlich das Gefäß gänzlich evakuiert sein.[1]

§ 236. Wir wollen nun die besprochene Eigenschaft der semipermeablen Wände benutzen, um auf reversibelm Wege, in möglichst einfacher Weise, die Bestandteile eines Gasgemisches voneinander zu trennen. Betrachten

[1] Diese Folgerung habe ich im Winter 1882/83 im physikalischen Institut der Universität München experimentell geprüft und, soweit es die unvermeidlichen Abweichungen von den idealen Voraussetzungen erwarten ließen, bestätigt gefunden. Da über diesen Versuch bisher nichts veröffentlicht wurde, so mag eine kurze Beschreibung hier Platz finden. Ein gerades Glasrohr von etwa 5^{mm} lichtem Durchmesser, in der Mitte zu einem kleinen Ballon ausgebaucht, war am einen Ende mit einem Glashahn versehen; an das andere Ende war als Verlängerung mit Siegellack angekittet ein 10^{cm} langes Platinröhrchen, einerseits offen, andrerseits geschlossen. Mit der Quecksilberluftpumpe wurde die ganze Röhre durch den Hahn evakuiert und mit Wasserstoff unter gewöhnlichem Druck gefüllt, hierauf der Hahn geschlossen und nun unter das geschlossene Ende des horizontal gelegten Platinröhrchens ein Bunsenbrenner gestellt, wodurch die das Rohr abschließende Platinkuppe ins Glühen kam. Um das Siegellack nicht durch Erwärmung zum Erweichen zu bringen, wurde die Lackstelle beständig von dem Strahl der Wasserleitung umspült. Nach etwa 4 Stunden wurde der Apparat abgenommen, auf Zimmertemperatur gebracht, und der Hahn unter Quecksilber geöffnet. Das Quecksilber stieg rapid in die Höhe und füllte die Röhre fast gänzlich aus, — ein Beweis, daß sie bis zu einem gewissen Grade evakuiert war.

wir daher folgenden Fall.



In einem Hohlzylinder seien im ganzen 4 Stempel vorhanden, zwei davon: A und A', fest, die beiden andern: B und B', beweglich, doch so, daß der Abstand BB' konstant gleich dem Abstand AA' gehalten wird, wie in der Fig. 5 durch die beiden Klammern angedeutet ist. A', der Boden, und B, der Deckel des ganzen Gefäßes, seien beide für alle Stoffe undurchdringlich, dagegen A und B' semipermeabel, und zwar A permeabel nur für ein gewisses Gas (*1*), B' permeabel nur für ein anderes Gas (*2*). Oberhalb B sei und bleibe der Raum evakuiert.

Anfänglich befinde sich der Stempel B bei A, also B' bei A', und in dem Zwischenraum eine Mischung der Gase (*1*) und (*2*). Nun werde der Stempel B und mit ihm auch B' unendlich langsam gehoben. Das Gas (*1*) strömt in den zwischen B und A sich öffnenden Raum, das Gas (*2*) in den zwischen B' und A' sich öffnenden Raum. Wenn B' bei A angekommen ist, sind die beiden Gase gänzlich voneinander getrennt.

Berechnen wir zunächst die während des Prozesses geleistete äußere Arbeit. Auf den einen beweglichen Stempel B wirkt, da der obere Raum evakuiert ist, nur der Druck des Gases (*1*), und zwar nach oben, auf den anderen beweglichen Stempel B' wirkt nur der Partialdruck des ersten Gases in der Mischung, nach unten. Nach dem vorigen Paragraphen ist aber der erstere Druck dem letzteren im Gleichgewicht gerade gleich, und da andrerseits die beiden Stempel B und B' gleiche Wege zurücklegen, so

ist die gesamte auf die Stempel ausgeübte Arbeit gleich Null. Wenn nun, wie wir weiter annehmen wollen, auch keine Wärme von außen zugeleitet wird, so bleibt nach dem Energieprinzip die Energie des Systems konstant, und da die Energie sowohl der einzelnen Gase als auch der Mischung nach (193) nur von der Temperatur abhängt, so bleibt auch die Temperatur des Systems allenthalben konstant.

Der unendlich langsam ausgeführte Prozeß ist reversibel; also ist, beim Fehlen jeglicher äußerer Einwirkung, die Entropie im Anfangszustand gleich derjenigen im Endzustand, d. h. die Entropie der Mischung ist gleich der Summe der Entropien der beiden Einzelgase, wenn ein jedes bei der nämlichen Temperatur das ganze Volumen der Mischung allein einnimmt. Dieser Satz läßt sich leicht verallgemeinern auf eine Mischung beliebig vieler Gasarten: „Die Entropie einer Gasmischung ist gleich der Summe der Entropien der Einzelgase, wenn ein jedes bei der nämlichen Temperatur das ganze Volumen der Mischung allein einnimmt.“ Er wurde zuerst von Gibbs aufgestellt.

§ 237. Für die Entropie eines einzelnen chemisch homogenen idealen Gases von der Masse M und dem Molekulargewicht m hatten wir früher in (52) gefunden:

$$M\left(\frac{C_v}{m}\log T+\frac{R}{m}\log v+\text{konst.}\right),$$

wenn wir wieder mit C_v die Molekularwärme bezeichnen. Nach den Gasgesetzen (14) ist v das Volumen der Masseneinheit:

$$v=\frac{R}{m}\frac{T}{p},$$

und daher die Entropie, für die Molekülzahl $n=\dfrac{M}{m}$

$$\begin{aligned}&n\left(C_v\log T+R\log\frac{T}{p}+k\right)\\&=n\left(C_p\log T-R\log p+k\right),\end{aligned}\tag{195}$$

wobei das Glied mit $\log\dfrac{R}{m}$ in die Konstante k einbegriffen ist. Somit liefert der Gibbssche Satz für die Entropie der ganzen Mischung:

$$S=\sum n_1(C_{p_1}\log T-R\log p_1+k_1);$$

p_1 ist dabei der Druck der ersten Gasart, wenn sie allein das ganze Volumen der Mischung einnimmt, d. h. der Partialdruck der ersten Gasart in der Mischung.

Da nun nach (8) die Summe aller Partialdrucke: $p_1 + p_2 + \ldots$ den Gesamtdruck p der Mischung darstellt, und da ferner nach § 40 die Verhältnisse der Partialdrucke mit den Verhältnissen der Molekülzahlen übereinstimmen:

$$p_1 : p_2 : \ldots = n_1 : n_2 : \ldots,$$

so ist

$$p_1 = \frac{n_1}{n_1 + n_2 + \ldots} \cdot p$$
$$p_2 = \frac{n_2}{n_1 + n_2 + \ldots} \cdot p$$
$$\ldots\ldots\ldots\ldots$$

oder, wenn wir der Einfachheit halber von jetzt ab die Konzentrationen der einzelnen Molekülarten in der Mischung einführen:

(196)
$$c_1 = \frac{n_1}{n_1 + n_2 + \ldots}, \qquad c_2 = \frac{n_2}{n_1 + n_2 + \ldots}, \quad \ldots$$
$$p_1 = c_1 p, \qquad p_2 = c_2 p, \quad \ldots$$

Daher ergibt sich schließlich die Entropie der Mischung in der gesuchten Form als Funktion von T, p und den Molekülzahlen n in folgender Weise:

(197)
$$S = \sum n_1 (C_{p_1} \log T - R \log(c_1 p) + k_1).$$

Durch Vergleichung dieses Ausdrucks mit dem in (194) für die Entropie der Mischung gefundenen Wert ergibt sich die Größe der damals unbestimmt gebliebenen Integrationskonstanten:

(198)
$$C = -R \sum n_1 \log c_1.$$

§ 238. Nachdem einmal der Wert der Entropie einer Gasmischung festgestellt ist, läßt sich auch die oben § 234 berührte Frage beantworten, ob und in welchem Betrage die Entropie eines Systems von Gasen durch Diffusion vergrößert wird. Nehmen wir den einfachsten Fall, daß zwei Gase, mit den Molekülzahlen n_1 und n_2, auf gleiche Temperatur T und gleichen Druck p gebracht, ineinander diffundieren, indem die Temperatur und der Druck konstant gehalten wird. Vor Beginn des Prozesses ist dann die Entropie des Systems gleich der Summe der Entropien der getrennten Gase, also nach (195)

$$n_1 (C_{p_1} \log T - R \log p + k_1) + n_2 (C_{p_2} \log T - R \log p + k_2).$$

Nach Beendigung des Diffusionsvorganges ist die Entropie der Mischung nach (197)

$$n_1(C_{p_1}\log T - R\log(c_1 p) + k_1) + n_2(C_{p_2}\log T - R\log(c_2 p) + k_2).$$

Also die Änderung der Entropie des Systems:

$$-n_1 R\log c_1 - n_2 R\log c_2.$$

Das ist mit Rücksicht auf (196) eine wesentlich positive Größe, woraus folgt, daß die Diffusion immer irreversibel ist.

Zugleich ersehen wir, daß die durch Diffusion bedingte Vermehrung der Entropie nur von den Molekülzahlen n_1 und n_2 der diffundierenden Gase, nicht aber von ihrer Natur, z. B. ihrem Molekulargewicht, abhängt. Es macht also in bezug auf die Entropievermehrung durch Diffusion gar keinen Unterschied, ob die Gase sich chemisch mehr oder weniger „ähnlich“ sind. Nimmt man nun beide Gase identisch, so wird offenbar die Entropievermehrung Null, weil man dann überhaupt keine Zustandsänderung erhält. Daraus folgt, daß der chemische Unterschied zweier Gase, und überhaupt zweier Substanzen, nicht durch eine stetig veränderliche Größe dargestellt werden kann, sondern daß man hier nur von sprungweisen Beziehungen: entweder von Gleichheit oder von Ungleichheit, reden kann. In diesem Umstand liegt ein prinzipieller Gegensatz zwischen chemischen und physikalischen Eigenschaften begründet, da die letzteren immer als stetig veränderlich anzusehen sind. (Vgl. die Anmerkung zu § 35.)

§ 239. Mittels der gefundenen Werte der Entropie S (197), der Energie U (193) und des Volumens V (191) der Gasmischung ergibt sich die gesuchte Funktion Φ aus (75) zu:

$$\Phi = \sum n_1 \left(C_{p_1}\log T - R\log(c_1 p) + k_1 - C_{v_1} - \frac{b_1}{T} - R \right)$$

oder, wenn man zur Abkürzung die Konstante:

$$k_1 - C_{v_1} - R = k_1 - C_{p_1} = a_1, \tag{198a}$$

und die nur von T und p, nicht aber von den Molekülzahlen abhängige Größe:

$$C_{p_1}\log T - \frac{b_1}{T} - R\log p + a_1 = \varphi_1 \tag{199}$$

setzt:

$$\Phi = \sum n_1(\varphi_1 - R\log c_1). \tag{199a}$$

§ 240. Nun können wir zur Aufstellung der Gleichgewichtsbedingung schreiten. Wenn in der Gasmischung eine chemische Änderung möglich ist, derart daß die Molekülzahlen $n_1,\ n_2, \ldots$ sich gleichzeitig um $\delta n_1,\ \delta n_2, \ldots$ ändern, so besteht nach (79) gegen diese Änderung Gleichgewicht, wenn für $\delta T = 0$ und $\delta p = 0$

$$\delta\Phi = 0$$

oder:

$$\sum(\varphi_1 - R\log c_1)\,\delta n_1 + \sum n_1\delta(\varphi_1 - R\log c_1) = 0. \tag{200}$$

Da die Größen $\varphi_1,\ \varphi_2, \ldots$ nur von T und p abhängen, so ist $\delta\varphi_1 = \delta\varphi_2 \ldots = 0$. Ferner haben wir:

$$n_1\delta\log c_1 + n_2\delta\log c_2 + \ldots = \frac{n_1}{c_1}\,\delta c_1 + \frac{n_2}{c_2}\,\delta c_2 + \ldots$$

und nach (196):

$$= (n_1 + n_2 + \ldots)(\delta c_1 + \delta c_2 + \ldots) = 0,$$

da

$$c_1 + c_2 + \ldots = 1.$$

Daher bleibt von der Gleichgewichtsbedingung übrig:

$$\sum(\varphi_1 - R\log c_1)\,\delta n_1 = 0.$$

Da es in dieser Gleichung nicht auf die absoluten Werte der unendlich kleinen Variationen δn_1, sondern nur auf deren Verhältnisse ankommt, so setzen wir:

$$\delta n_1 : \delta n_2 : \ldots = \nu_1 : \nu_2 : \ldots \tag{201}$$

und verstehen unter $\nu_1,\ \nu_2, \ldots$ die bei der gedachten chemischen Veränderung sich gleichzeitig umsetzenden Molekülzahlen: einfache ganze, positive oder negative Zahlen, je nachdem die betreffende Molekülart bei der Veränderung sich bildet oder verbraucht wird. Dann erhalten wir als Gleichgewichtsbedingung:

$$\sum(\varphi_1 - R\log c_1)\nu_1 = 0,$$

oder:

$$\nu_1\log c_1 + \nu_2\log c_2 + \ldots = \frac{\nu_1\varphi_1 + \nu_2\varphi_2 + \ldots}{R} = \log K.$$

Die rechte Gleichungsseite $\log K$ hängt nach (199) nur von Temperatur und Druck ab; also ergibt die Gleichung eine bestimmte Beziehung zwischen

den Konzentrationen der verschiedenen Molekülarten, falls p und T gegeben sind.

§ 241. Wir wollen nun noch die Werte der Größen φ_1, $\varphi_2 \ldots$ hier einführen. Setzt man zur Abkürzung die Konstanten:

(201a) $$\nu_1 + \nu_2 + \nu_3 + \ldots = \nu$$

(201b) $$\frac{\nu_1 a_1 + \nu_2 a_2 + \ldots}{R} = \log A,$$

(202) $$\frac{\nu_1 b_1 + \nu_2 b_2 + \ldots}{R} = B,$$

(203) $$\frac{\nu_1 C_{p_1} + \nu_2 C_{p_2} + \ldots}{R} = C,$$

so ergibt der Wert von φ_1 usw. aus (199) als Gleichgewichtsbedingung:

$$\nu_1 \log c_1 + \nu_2 \log c_2 + \ldots = \log A + C \log T - \frac{B}{T} - \nu \log p = \log K,$$

oder:

(203a) $$c_1^{\nu_1} c_2^{\nu_2} \ldots = A e^{-\frac{B}{T}} T^C p^{-\nu} = K.$$

§ 242. Diese Gleichung würde sich noch vereinfachen, wenn man den Erfahrungssatz (§ 50) einführt, daß die Atomwärme eines chemischen Elements in seinen verschiedenen Verbindungen den nämlichen Wert hat. Denn nach Gleichung (203) bedeutet das Produkt RC die Änderung, welche die Summe der Molekularwärmen aller Moleküle oder die Wärmekapazität des ganzen Systems bei konstantem Druck: $n_1 C_{p_1} + n_2 C_{p_2} + \ldots$ durch die angenommene chemische Reaktion erfährt. Wenn nun für jedes Molekül die Molekularwärme bei konstantem Volumen gleich ist der Summe der Atomwärmen bei konstantem Volumen, so wäre nach dem obigen Satz die Wärmekapazität des ganzen Systems bei konstantem Volumen, als Summe aller Atomwärmen, unveränderlich, mithin $\nu_1 C_{\nu_1} + \nu_2 C_{\nu_2} + \ldots = 0$, und folglich $C = \nu_1 + \nu_2 + \nu_3 + \ldots = \nu$. Indessen scheint im allgemeinen diese Annahme doch nicht mit solcher Annäherung erfüllt zu sein, daß ihre Einführung hier gerechtfertigt wäre.

§ 243. Der Einfluß des Druckes p auf den Gleichgewichtszustand hängt nach (203a) lediglich ab von der Zahl $\nu = \nu_1 + \nu_2 + \ldots$, welche angibt, in welchem Grade die Gesamtzahl der Moleküle, also auch das Volumen der Mischung, durch die betrachtete chemische Änderung vergrößert wird. Bleibt das Volumen ungeändert, wie in dem unten behandelten Beispiel der Dissoziation von Jodwasserstoffgas, so ist der Gleichgewichtszustand unabhängig vom Druck.

Der Einfluß der Temperatur wird von den Konstanten B und C bedingt, welche beide in engem Zusammenhang stehen mit der durch den chemischen Vorgang bedingten Wärmetönung. Denn nach dem ersten Hauptsatz der Wärmetheorie ist die Wärmemenge, welche bei einer unendlich kleinen Zustandsänderung dem System von außen zuzuführen ist:

$$Q = \delta U + p\,\delta V$$

und nach (193) und (191), da T und p ungeändert bleiben:

$$Q = \sum (C_{v_1} T + b_1 + RT)\,\delta n_1 = \sum (C_{p_1} T + b_1)\,\delta n_1.$$

Beziehen wir die Wärmetönung, anstatt auf die unendlich kleinen Zahlen δn, nach (201) auf die einfachen ganzen Zahlen ν, so ergibt sich für die von außen zuzuführende endliche Wärmemenge:

$$r = \sum (C_{p_1} T + b_1)\nu_1$$

und nach (202) und (203):

$$r = R(B + CT),$$

in Kalorien nach (34):

$$r = 1,985 \cdot (B + CT) \text{ cal.}$$

Die Wärmetönung eines chemischen Umwandlungsprozesses in einem System, das vor und nach der Umwandlung vollständig gasförmig ist, hängt also gar nicht vom Druck ab und ändert sich lineär mit der Temperatur.

§ 244. Bevor wir zu einigen Anwendungen übergehen, stellen wir zur besseren Übersicht die Hauptgleichungen noch einmal zusammen.

Sei in einem gasförmigen System:

$$n_1\ m_1, \quad n_2\ m_2, \quad n_3\ m_3, \ \ldots$$

(n die Molekülzahlen, m die Molekulargewichte) irgend eine chemische Änderung möglich, bei welcher die gleichzeitigen Molekülzahländerungen betragen:

$$\delta n_1 : \delta n_2 : \delta n_3 : \ldots = \nu_1 : \nu_2 : \nu_3 : \ldots$$

(ν einfache ganze, positive oder negative Zahlen),

wobei

$$\nu_1 + \nu_2 + \nu_3 + \ldots = \nu,$$

so besteht Gleichgewicht gerade bezüglich dieser Änderung, wenn die Konzentrationen:

$$c_1 = \frac{n_1}{n_1 + n_2 + n_3 \ldots}, \qquad c_2 = \frac{n_2}{n_1 + n_2 + n_3 \ldots}, \qquad \ldots$$

der Bedingung genügen:

$$c_1^{\nu_1} c_2^{\nu_2} c_3^{\nu_3} \ldots = A e^{-\frac{B}{T}} T^C p^{-\nu}. \tag{204}$$

Die beim Eintritt der durch die Werte der ν bezeichneten Änderung bei konstanter Temperatur und konstantem Druck von außen aufzunehmende Wärme ist:

$$r = 1,985(B + CT) \text{ cal}, \tag{205}$$

während die gleichzeitig eintretende Volumenänderung beträgt:

$$v = R\nu \frac{T}{p}. \tag{206}$$

§ 245. Dissoziation von Jodwasserstoff. Da Jodwasserstoffgas sich bis zu gewissem Grade in Wasserstoff und Joddampf spaltet, so wird das System dargestellt durch drei Arten von Molekülen:

$$n_1 \text{ HJ}, \quad n_2 \text{ } H_2, \quad n_3 \text{ } J_2.$$

Die Konzentrationen sind:

$$c_1 = \frac{n_1}{n_1 + n_2 + n_3} \qquad c_2 = \frac{n_2}{n_1 + n_2 + n_3} \qquad c_3 = \frac{n_3}{n_1 + n_2 + n_3}.$$

Die chemische Änderung besteht darin, daß zwei Moleküle HJ in ein Molekül H_2 und ein Molekül J_2 übergehen; also:

$$\nu_1 = -2, \qquad \nu_2 = 1, \qquad \nu_3 = 1, \qquad \nu = \nu_1 + \nu_2 + \nu_3 = 0.$$

Dann ist nach (204) im Gleichgewichtszustand:

$$c_1^{-2} c_2^1 c_3^1 = A e^{-\frac{B}{T}} T^C$$

oder:

$$\frac{c_2 c_3}{c_1^2} = \frac{n_2 n_3}{n_1^2} = A e^{-\frac{B}{T}} T^C. \tag{207}$$

Da die im ganzen System vorhandene Anzahl der Wasserstoffatome (n_1+2n_2) und ebenso die der Jodatome (n_1+2n_3) als bekannt vorausgesetzt werden muß, so genügt diese eine Gleichung, um bei gegebener Temperatur alle drei Größen n_1, n_2 und n_3 zu bestimmen. Der Druck hat hier gar keinen Einfluß auf das Gleichgewicht, was namentlich durch die neueren Messungen von M. Bodenstein bestätigt worden ist.

Zur Berechnung der Konstanten A, B und C können die Messungen des Dissoziationsgrades bei drei verschiedenen Temperaturen dienen. Dann ist der Gleichgewichtszustand in irgend einer Mischung von Jodwasserstoff, Wasserstoff und Joddampf, auch wenn Wasserstoff und Jod nicht in äquivalenten Mengen zugegen sind, für jede Temperatur nach (207) numerisch bestimmt. Nach (205) ist dann auch die Dissoziationswärme bei der Zersetzung von zwei Molekülen Jodwasserstoff in je ein Molekül Wasserstoff und Joddampf für jede Temperatur unmittelbar anzugeben.

§ 246. Dissoziation von Joddampf. Bei höheren Temperaturen zersetzt sich Joddampf merklich, und man erhält hierfür folgendes aus zwei Molekülarten bestehendes System:

$$n_1 \ \mathrm{J}_2, \qquad n_2 \ \mathrm{J}.$$

Die Konzentrationen sind:

$$c_1 = \frac{n_1}{n_1+n_2}, \qquad c_2 = \frac{n_2}{n_1+n_2}.$$

Die chemische Umwandlung besteht in der Spaltung eines Moleküls J_2 in zwei Moleküle J, also:

$$\nu_1 = -1, \qquad \nu_2 = 2, \qquad \nu = \nu_1+\nu_2 = 1,$$

und im Gleichgewichtszustand ist nach (204)

$$c_1^{-1}c_2^2 = \frac{n_2^2}{n_1(n_1+n_2)} = A' e^{-\frac{B'}{T}} \frac{T^{C'}}{p}. \tag{208}$$

§ 247. Stufenweise Dissoziation. Da nach der Gleichung (208) auch für tiefere Temperaturen die Konzentration der einatomigen Jodmoleküle niemals Null wird, sondern stets einen endlichen, wenn auch kleinen Wert behält, so muß man, genauer genommen, die Zersetzbarkeit des Joddampfes auch schon in dem § 245 behandelten Falle, bei der Dissoziation des Jodwasserstoffgases, berücksichtigen. Praktisch wird dies zwar keinen Einfluß haben, doch sei hier wegen des prinzipiellen Interesses die theoretisch strengere Lösung der Aufgabe auch noch durchgeführt.

Das System besteht dann aus vier Molekülarten:

$$n_1 \text{ HJ}, \quad n_2 \text{ H}_2, \quad n_3 \text{ J}_2, \quad n_4 \text{ J}.$$

Die Konzentrationen sind:

$$c_1 = \frac{n_1}{n_1+n_2+n_3+n_4}, \qquad c_2 = \frac{n_2}{n_1+n_2+n_3+n_4},$$
$$c_3 = \frac{n_3}{n_1+n_2+n_3+n_4}, \qquad c_4 = \frac{n_4}{n_1+n_2+n_3+n_4}.$$

Hier sind nun zwei Arten von chemischen Umwandlungen möglich, nämlich:

1. $\nu_1 = -2 \qquad \nu_2 = 1 \qquad \nu_3 = 1 \qquad \nu_4 = 0, \qquad \nu = 0$
2. $\nu_1' = 0 \qquad \nu_2' = 0 \qquad \nu_3' = -1 \qquad \nu_4' = 2, \qquad \nu' = 1.$

Gleichgewicht gegen jede der beiden Umwandlungen ist vorhanden, wenn nach (204):

1. $$c_1^{\nu_1} c_2^{\nu_2} c_3^{\nu_3} c_4^{\nu_4} = \frac{c_2 c_3}{c_1^2} = \frac{n_2 n_3}{n_1^2} = A e^{-\frac{B}{T}} T^C$$
2. $$c_1^{\nu_1'} c_2^{\nu_2'} c_3^{\nu_3'} c_4^{\nu_4'} = \frac{c_4^2}{c_3} = \frac{n_4^2}{n_3(n_1+n_2+n_3+n_4)} = A' e^{-\frac{B'}{T}} \frac{T^{C'}}{p}.$$

Da die Gesamtzahl der im System vorhandenen Wasserstoffatome (n_1+2n_2) und ebenso die der Jodatome $(n_1+2n_3+n_4)$ als bekannt vorausgesetzt wird, so hat man im ganzen vier Gleichungen zur eindeutigen Bestimmung der vier Größen $n_1,\ n_2,\ n_3,\ n_4$.

§ 248. Aus der allgemeinen Gleichgewichtsformel (204) ersieht man, daß bei endlicher Temperatur und endlichem Druck keine der Konzentrationen c jemals gleich Null sein kann, oder mit anderen Worten, daß die Dissoziation niemals eine vollständige ist, aber auch niemals ganz verschwinden kann; es finden sich in dem System stets Moleküle von allen möglichen Arten in endlicher, wenn auch vielleicht sehr geringer Anzahl vor. So muß z. B. im Wasserdampf bei jeder Temperatur auch etwas Knallgas, wenn auch nur spurweise, vorhanden sein (vgl. unten § 259). Bei vielen Erscheinungen spielt natürlich dieser Umstand keine Rolle.

Fünftes Kapitel. Verdünnte Lösungen.

§ 249. Zur Bestimmung der für das thermodynamische Gleichgewicht charakteristischen Funktion Φ in ihrer Abhängigkeit von der Temperatur T,

dem Druck p und den Zahlen n aller verschiedenen Molekülarten in einem System, welches beliebig viele unabhängige Bestandteile in beliebig vielen Phasen enthält, kann man genau denselben Weg einschlagen, der uns bei der Untersuchung einer einzigen gasförmigen Phase im vorigen Kapitel zum Ziele geführt hat. Zunächst wird durch geeignete Messungen das Volumen V und die Energie U einer einzelnen Phase bestimmt, daraus dann gemäß der Definition (60) die Entropie S dieser Phase berechnet, und somit alle Größen gewonnen, aus denen nach (75) Φ zusammengesetzt ist. Durch einfache Addition über alle Phasen erhält man dann schließlich die charakteristische Funktion Φ des ganzen Systems.

Angesichts der mangelnden Vollständigkeit der bisherigen Messungen läßt sich aber gegenwärtig diese Rechnung, außer für eine gasförmige Phase, nur durchführen für eine verdünnte Lösung, d. h. für eine Phase, in welcher die Anzahl einer bestimmten Art von Molekülen weitaus überwiegt über die Anzahl aller übrigen in der Phase vorhandenen Molekülarten. Die so ausgezeichnete Molekülart nennen wir von jetzt an das Lösungsmittel (vgl. § 220), die übrigen Molekülarten die gelösten Stoffe. Bezeichnet also n_0 die Molekülzahl des Lösungsmittels, $n_1, n_2, n_3 \ldots$ die Molekülzahlen der gelösten Stoffe, so ist die Lösung dann als verdünnt anzusehen, wenn n_0 groß ist gegen die Summe der Zahlen $n_1, n_2, n_3 \ldots$ Der Aggregatzustand der Lösung ist vollkommen gleichgültig, sie kann fest, flüssig oder gasförmig sein.

§ 250. Berechnen wir nun, gemäß dem geschilderten Plane, zunächst die Energie U und das Volumen V einer verdünnten Lösung. Die wichtige Vereinfachung, welche die soeben angeführte Definition einer verdünnten Lösung zur Folge hat, beruht auf dem mathematischen Satze, daß eine mit ihren Differentialquotienten endliche und stetige Funktion mehrerer Variabeln, welche sehr kleine Werte haben, notwendig eine lineäre Funktion dieser Variabeln ist. Dadurch wird die Art der Abhängigkeit der Größen U und V von $n_0, n_1, n_2, \ldots$ von vornherein angebbar. Physikalisch gesprochen heißt dies, daß die Eigenschaften einer verdünnten Lösung, außer von den Wirkungen der Moleküle des Lösungsmittels aufeinander, notwendig nur von den Wechselwirkungen zwischen den Molekülen des Lösungsmittels und den Molekülen der gelösten Stoffe, nicht aber von den Wirkungen der gelösten Stoffe aufeinander abhängen können; denn diese letzteren sind klein von höherer Ordnung.

§ 251. In der Tat: Betrachten wir zunächst die Energie U der Lösung und bilden den Quotienten von U und n_0, der Molekülzahl des Lösungsmittels. Da U nach dem allgemeinen in § 201 aufgestellten Satze eine homogene Funktion ersten Grades der Molekülzahlen darstellt, so bleibt

der Quotient $\frac{U}{n_0}$ ungeändert, wenn sämtliche Molekülzahlen $n_0, n_1, n_2 \ldots$ in gleichem Verhältnis verändert werden, d. h. dieser Quotient ist eine Funktion der Verhältnisse $\frac{n_1}{n_0}, \frac{n_2}{n_0}, \ldots$ Nun sind aber alle diese Verhältnisse kleine Zahlen, folglich ist die Funktion, die wir als differentiierbar voraussetzen, eine lineäre, und daher von der Form:

$$\frac{U}{n_0} = u_0 + u_1\frac{n_1}{n_0} + u_2\frac{n_2}{n_0} + \ldots, \tag{208a}$$

wobei die Größen $u_0, u_1, u_2, \ldots$ nicht von den Molekülzahlen, sondern nur von der Temperatur T, dem Druck p und der Beschaffenheit der in der Lösung vorhandenen Molekülarten abhängen, und zwar u_0 nur von der Beschaffenheit des Lösungsmittels (denn für $n_1 = 0 = n_2 = \ldots$ reduziert sich die Energie auf $n_0 u_0$), ferner u_1 nur von der Beschaffenheit der ersten gelösten Molekülart und der des Lösungsmittels, u_2 nur von der Beschaffenheit der zweiten gelösten Molekülart und der des Lösungsmittels usw. u_0 entspricht also den Wechselwirkungen der Moleküle des Lösungsmittels aufeinander, u_1 denjenigen zwischen dem Lösungsmittel und den gelösten Molekülen erster Art, u_2 denjenigen zwischen dem Lösungsmittel und den gelösten Molekülen zweiter Art, usw. Hiermit ist zugleich ein Einwurf widerlegt, welcher der neueren Theorie verdünnter Lösungen zu wiederholten Malen gemacht worden ist, daß sie nämlich die verdünnten Lösungen einfach wie Gase behandle und keine Rücksicht nehme auf den Einfluß des Lösungsmittels.

§ 252. Wenn die Verdünnung nicht hinreichend ist, um diese einfachste Form der Funktion U zu rechtfertigen, so kann man genauere Beziehungen erhalten, wenn man die Reihenkoeffizienten $u_1, u_2, \ldots$ nicht als unabhängig von den Molekülzahlen, sondern selber als Funktionen der Verhältnisse $\frac{n_1}{n_0}, \frac{n_2}{n_0}, \ldots$ betrachtet. Dann erhält man gewisse neue Konstante, welche den Wechselwirkungen der gelösten Molekülarten untereinander entsprechen. Dies dürfte in der Tat ein gangbarer Weg sein, um zu einer rationellen thermodynamischen Theorie von Lösungen beliebiger Konzentration zu gelangen.[1] Auf ihm findet man auch eine Erklärung für das abnorme Verhalten der Lösungen starker Elektrolyte. (Vgl. unten § 273.)

§ 253. Einstweilen wollen wir jedoch hier bei der einfachsten Form stehen bleiben und schreiben:

$$\left\{\begin{array}{l} U = n_0 u_0 + n_1 u_1 + n_2 u_2 + \ldots \\ \text{Ganz ebenso:} \\ V = n_0 v_0 + n_1 v_1 + n_2 v_2 + \ldots \end{array}\right. \tag{209}$$

[1] Vgl. H. Jahn, Zeitschr. f. phys. Chemie **41**, p. 257, 1902.

Inwieweit diese Gleichungen den Tatsachen entsprechen, läßt sich aus den Folgerungen entscheiden, zu denen sie führen. Eine derselben soll hier ausführlicher besprochen werden. Verdünnt man die Lösung noch weiter, indem man ihr ein Molekül des Lösungsmittels von demselben Aggregatzustand wie die Lösung zusetzt, und hält dabei den Druck p und die Temperatur T konstant, so läßt sich mittels der letzten Gleichungen die eintretende Volumenänderung und Wärmetönung berechnen.

Ein Molekül des reinen Lösungsmittels, immer bei der nämlichen Temperatur und dem nämlichen Druck genommen, besitzt das Volumen v_0 und die Energie u_0. Nach vollzogener Verdünnung ist nun das Volumen der Lösung geworden:

$$V' = (n_0 + 1)v_0 + n_1 v_1 + n_2 v_2 + \dots$$

und die Energie ist geworden:

$$U' = (n_0 + 1)u_0 + n_1 u_1 + n_2 u_2 + \dots$$

Die durch die Verdünnung bewirkte Volumendilatation erhält man, wenn man die Summe des ursprünglichen Volumens V der Lösung und des Volumens v_0 eines Moleküls reinen Lösungsmittels subtrahiert von dem schließlichen Volumen V'. Also:

$$V' - (V + v_0),$$

d. h. die Volumendilatation ist gleich Null. Die von außen zugeführte Wärme ergibt sich nach dem ersten Hauptsatze (47) gleich:

$$U' - (U + u_0) + p(V' - (V + v_0))$$

und verschwindet ebenfalls.

Bei diesen Schlüssen ist vorausgesetzt, daß bei der Verdünnung die Molekülzahlen der gelösten Stoffe $n_1, n_2, \dots$ ungeändert bleiben, d. h. daß durch den Verdünnungsprozeß keine chemischen Änderungen der gelösten Stoffe (z. B. Änderungen des Dissoziationsgrades) bewirkt werden. In einem solchen Falle würden in den Gleichungen für U' und V' die Molekülzahlen der gelösten Stoffe andere Werte haben als in denen für U und V, und daher bei der Subtraktion nicht fortfallen. Daher läßt sich folgender Satz aussprechen: Eine verdünnte Lösung besitzt die Eigenschaft, daß eine weitere Verdünnung, die ohne chemische Änderung der gelösten Stoffe verläuft, weder merkliche Volumenänderung noch merkliche Wärmetönung hervorruft, oder mit anderen Worten: Jede Volumenänderung oder Wärmetönung, die

eine verdünnte Lösung bei weiterer Verdünnung zeigt, muß einer chemischen Umwandlung unter den Molekülen der gelösten Stoffe zugeschrieben werden. (Vgl. § 97.)

§ 254. Gehen wir nun weiter zur Berechnung der Entropie S einer verdünnten Lösung. Nach (60) ist für konstante Molekülzahlen $n_0, n_1, n_2, \ldots$

$$dS = \frac{dU + p\,dV}{T}$$

und nach (209):

$$dS = n_0 \frac{du_0 + p\,dv_0}{T} + n_1 \frac{du_1 + p\,dv_1}{T} + n_2 \frac{du_2 + p dv_2}{T} + \ldots.$$

Da nun die u und v nur von T und p, nicht aber von den n abhängen, so müssen die Koeffizienten von $n_0, n_1, n_2, \ldots$ auch einzeln vollständige Differentiale sein, d. h. es muß gewisse nur von T und p abhängige Größen s geben, derart daß

(210)
$$\left\{\begin{array}{l} ds_0 = \dfrac{du_0 + p\,dv_0}{T} \\ ds_1 = \dfrac{du_1 + p\,dv_1}{T} \\ ds_2 = \dfrac{du_2 + p\,dv_2}{T} \\ \quad . \quad . \quad . \quad . \quad . \quad . \quad . \end{array}\right.$$

Dann ist:

(211)
$$S = n_0 s_0 + n_1 s_1 + n_2 s_2 + \ldots + C,$$

wobei die Integrationskonstante C nicht von T und p, wohl aber von den Molekülzahlen abhängen kann.

Wenn man daher den Wert von C für irgend eine spezielle Temperatur und einen speziellen Druck in seiner Abhängigkeit von den Molekülzahlen $n_0, n_1, n_2 \ldots$ kennt, so ist dieser Wert zugleich auch der allgemeine Ausdruck von C für beliebige Temperaturen und Drucke.

Nun wollen wir für den speziellen Fall, daß die Temperatur groß und der Druck klein ist, C als Funktion der n berechnen. Bei gehöriger Steigerung der Temperatur und gehöriger Erniedrigung des Druckes wird die Lösung, welchem Aggregatzustand sie ursprünglich auch angehören mag, jedenfalls vollständig in den gasförmigen Zustand übergehen. Dabei werden in Wirklichkeit zugleich chemische und Aggregatzustandsänderungen eintreten, d. h. die Molekülzahlen n werden sich verändern, es wird

teilweise Verdampfung eintreten usw.; denn in der Natur sind nur solche Zustände realisierbar, welche stabilen Gleichgewichtszuständen hinreichend nahe liegen. Wir wollen aber den Vorgang derartig voraussetzen, daß alle Molekülzahlen ungeändert bleiben, und daß das ganze System stets nur eine einzige Phase bildet, weil nur dann auch die Größe C ihren Wert behält. Diese Voraussetzung ist deshalb gestattet, weil die Molekülzahlen n zusammen mit T und p die unabhängigen Variabeln des Systems bilden. Ein solcher Prozeß ist nur in idealem Sinne ausführbar, da er durch labile Zustände hindurchführt; allein es steht seiner Benutzung hier nichts im Wege, weil der obige Ausdruck von S nicht allein für stabile Gleichgewichtszustände, sondern für alle Zustände Gültigkeit besitzt, welche durch ganz beliebige Werte der unabhängigen Variabeln T, p, n_0, n_1, n_2 ... charakterisiert sind. Der stabile Gleichgewichtszustand geht ja aus diesen Zuständen erst durch eine weitere, unten aufzustellende Bedingung als spezieller Fall hervor.

Da bei genügend erhöhter Temperatur und erniedrigtem Druck jedes gasförmige System eine so geringe Dichte annimmt, daß man es als Mischung idealer Gase betrachten kann (§ 21 und § 43), so haben wir hierfür nach (194), unter Berücksichtigung des Umstandes, daß hier die erste Molekülart mit dem Index 0 bezeichnet ist:

$$\left\{\begin{aligned} S = {} & n_0(C_{p_0}\log T - R\log p + k_0) + \\ & n_1(C_{p_1}\log T - R\log p + k_1) + \ldots + C, \end{aligned}\right. \tag{212}$$

wobei C, unabhängig von T und p, den in (198) angegebenen Wert hat. Durch Vergleichung mit (211) erkennt man, daß der Ausdruck von S durch bloße Temperatur- und Druckänderungen nur dann aus (211) in (212) übergehen kann, wenn die Größe C in beiden Ausdrücken dieselbe ist, d. h. wenn nach (198)

$$C = -R(n_0\log c_0 + n_1\log c_1 + \ldots).$$

Dabei sind die Konzentrationen:

$$c_0 = \frac{n_0}{n_0 + n_1 + n_2 + \ldots}, \qquad c_1 = \frac{n_1}{n_0 + n_1 + n_2 + \ldots}, \ldots$$

Somit wird aus (211) die Entropie einer verdünnten Lösung bei beliebiger Temperatur und beliebigem Druck:

$$S = n_0(s_0 - R\log c_0) + n_1(s_1 - R\log c_1) + \ldots \tag{213}$$

Setzen wir noch zur Abkürzung die nur von T und p, nicht aber von den

Molekülzahlen n abhängigen Größen:

$$(214)\qquad \begin{cases} s_0 - \dfrac{u_0 + pv_0}{T} = \varphi_0 \\ s_1 - \dfrac{u_1 + pv_1}{T} = \varphi_1 \\ s_2 - \dfrac{u_2 + pv_2}{T} = \varphi_2 \\ \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \end{cases}$$

so wird schließlich aus (75), (213) und (209)

$$(215)\qquad \Phi = n_0(\varphi_0 - R\log c_0) + n_1(\varphi_1 - R\log c_1) + n_2(\varphi_2 - R\log c_2) + \ldots$$

und damit sind die thermodynamischen Eigenschaften einer verdünnten Lösung bestimmt. Die charakteristische Funktion Φ hat eine ganz ähnliche Form wie die einer Gasmischung in (199a). Der einzige Unterschied ist der, daß hier die Größen φ sämtlich von der Natur des Lösungsmittels abhängig sind.

§ 255. Wir können nun sogleich übergehen zur Aufstellung der Gleichgewichtsbedingung für ein aus verschiedenen Phasen bestehendes System. Was zunächst die Bezeichnung betrifft, so wollen wir, wie bisher, die verschiedenen Molekülarten innerhalb einer Phase durch Zahlenindizes, die verschiedenen Phasen aber, wie im dritten Kapitel, durch beigefügte Striche unterscheiden, wobei der Einfachheit halber die erste Phase ganz ohne Striche bleiben soll. Dann wird das ganze System dargestellt durch das Symbol:

$$(216)\qquad \left\{ \begin{array}{r} n_0m_0,\ n_1m_1,\ n_2m_2,\ldots \mid n_0'm_0',\ n_1'm_1',\ n_2'm_2'\ldots \\ \mid n_0''m_0'',\ n_1''m_1'',\ldots \mid \ldots \end{array} \right.$$

Die Molekülzahlen sind mit n, die Molekulargewichte mit m bezeichnet, und die einzelnen Phasen sind durch vertikale Striche voneinander getrennt. In den allgemeinen Formeln deuten wir die Summierung über die verschiedenen Molekülarten in einer und derselben Phase durch Anschreiben der einzelnen Summenglieder an, die Summierung über verschiedene Phasen dagegen durch das Zeichen $\sum$.

Um nun die abgeleiteten Formeln anwenden zu können, wollen wir voraussetzen, daß jede Phase entweder eine Mischung idealer Gase, oder eine verdünnte Lösung darstellt. Letzteres trifft auch dann zu, wenn die Phase überhaupt nur eine einzige Molekülart enthält, wie z. B. ein chemisch

homogener fester Niederschlag aus einer flüssigen Lösung. Denn eine einzige Molekülart stellt den speziellen Fall einer verdünnten Lösung dar, in welcher die Konzentrationen aller gelösten Stoffe gleich Null sind.

§ 256. Gesetzt nun, es sei in dem System (216) eine isotherm-isobare Änderung möglich, derart, daß die Molekülzahlen $n_0, n_1, n_2, \ldots, n_0', n_1', n_2', \ldots$ sich gleichzeitig um $\delta n_0, \delta n_1, \delta n_2, \ldots, \delta n_0', \delta n_1', \delta n_2', \ldots$ ändern; dann besteht nach (79) gegen das Eintreten dieser Änderung Gleichgewicht, wenn für konstant gehaltenes T und p

$$\delta\Phi = 0$$

oder nach (215):

$$\sum(\varphi_0 - R\log c_0)\,\delta n_0 + (\varphi_1 - R\log c_1)\,\delta n_1 + (\varphi_2 - R\log c_2)\,\delta n_2 + \ldots$$
$$+\sum n_0\,\delta(\varphi_0 - R\log c_0) + n_1\,\delta(\varphi_1 - R\log c_1) + n_2\,\delta(\varphi_2 - R\log c_2) + \ldots = 0$$

(Die Summationen $\sum$ über alle Phasen des Systems erstreckt.)

Die zweite Reihe verschwindet identisch aus denselben Gründen, die oben, im Anschluß an die Gleichung (200), entwickelt wurden. Führen wir ferner wieder die einfachen ganzzahligen Verhältnisse ein:

$$\text{(217)} \qquad \begin{cases} \delta n_0 : \delta n_1 : \delta n_2 : \ldots : \delta n_0' : \delta n_1' : \delta n_2' : \ldots \\ = \nu_0 : \nu_1 : \nu_2 : \ldots : \nu_0' : \nu_1' : \nu_2' : \ldots, \end{cases}$$

so lautet die Gleichgewichtsbedingung:

$$\sum(\varphi_0 - R\log c_0)\nu_0 + (\varphi_1 - R\log c_1)\nu_1 + (\varphi_2 - R\log c_2)\nu_2 + \ldots = 0$$

oder:

$$\text{(218)} \qquad \begin{aligned} \sum \nu_0 \log c_0 + \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots &= \frac{1}{R}\sum \nu_0\varphi_0 + \nu_1\varphi_1 + \ldots \\ &= \log K. \end{aligned}$$

K hängt, ebenso wie die Größen φ_0, φ_1, φ_2, …, nicht von den Molekülzahlen n ab.

§ 257. Die Abhängigkeit der Größe K von T und p ergibt sich aus ihrer Definition:

$$\frac{\partial \log K}{\partial T} = \frac{1}{R}\sum \nu_0 \frac{\partial \varphi_0}{\partial T} + \nu_1 \frac{\partial \varphi_1}{\partial T} + \nu_2 \frac{\partial \varphi_2}{\partial T} + \ldots$$
$$\frac{\partial \log K}{\partial p} = \frac{1}{R}\sum \nu_0 \frac{\partial \varphi_0}{\partial p} + \nu_1 \frac{\partial \varphi_1}{\partial p} + \nu_2 \frac{\partial \varphi_2}{\partial p} + \ldots$$

Nun ist nach (214) für irgend eine unendlich kleine Änderung von T und p:

$$d\varphi_0 = ds_0 - \frac{du_0 + p\,dv_0 + v_0\,dp}{T} + \frac{u_0 + pv_0}{T^2}\,dT,$$

folglich nach (210):

$$d\varphi_0 = \frac{u_0 + pv_0}{T^2}\,dT - \frac{v_0\,dp}{T},$$

und daraus:

$$\frac{\partial\varphi_0}{\partial T} = \frac{u_0 + pv_0}{T^2}, \qquad \frac{\partial\varphi_0}{\partial p} = \frac{v_0}{T}.$$

Ebenso:

$$\frac{\partial\varphi_1}{\partial T} = \frac{u_1 + pv_1}{T^2}, \qquad \frac{\partial\varphi_1}{\partial p} = \frac{v_1}{T} \quad \text{usw.}$$

Daher ergibt sich:

$$\frac{\partial \log K}{\partial T} = \frac{1}{RT^2}\sum(\nu_0 u_0 + \nu_1 u_1 + \ldots) + p(\nu_0 v_0 + \nu_1 v_1 + \ldots)$$
$$\frac{\partial \log K}{\partial p} = -\frac{1}{RT}\sum \nu_0 v_0 + \nu_1 v_1 + \ldots$$

Bezeichnen wir nun mit v die Volumenvergrößerung des Systems, mit r die von außen zugeführte Wärme, wenn bei konstanter Temperatur und konstantem Druck die Änderung (217) vor sich geht, so ist nach dem Werte von V in (209):

$$v = \sum \nu_0 v_0 + \nu_1 v_1 + \nu_2 v_2 + \ldots$$

und nach dem ersten Hauptsatz der Wärmetheorie:

$$r = \sum(\nu_0 u_0 + \nu_1 u_1 + \ldots) + p(\nu_0 v_0 + \nu_1 v_1 + \ldots).$$

Folglich:

(219) $$\frac{\partial \log K}{\partial T} = \frac{r}{RT^2},$$

(220) $$\frac{\partial \log K}{\partial p} = -\frac{v}{RT}.$$

Der Einfluß der Temperatur auf die Größe K und mithin auf die Bedingung des Gleichgewichts gegen eine bestimmte chemische Reaktion wird also durch die bei dieser Reaktion eintretende Wärmetönung, der Einfluß des Druckes durch die entsprechende Volumenänderung des Systems geregelt. Geht die

Reaktion ganz ohne Wärmetönung vor sich, so hat die Temperatur gar keinen Einfluß auf das Gleichgewicht; verursacht sie keine Volumenänderung des Systems, so hat der Druck keinen Einfluß auf das Gleichgewicht (vgl. § 211 am Schluß).

Die Elimination von K aus den letzten beiden Gleichungen ergibt eine allgemeine Beziehung zwischen der Wärmetönung r und der Volumenänderung v:

$$\frac{\partial r}{\partial p} = v - T\frac{\partial v}{\partial T},$$

übereinstimmend mit der allgemeinen Gleichung (79g).

Die früheren Gleichungen (205) und (206) sind spezielle Fälle der beiden Gleichungen (219) und (220), wie man sogleich erkennt, wenn für $\log K$ der in (203a) gegebene spezielle Wert:

$$\log K = \log A - \frac{B}{T} + C\log T - \nu \log p$$

gesetzt wird.

§ 258. Mittels der Gleichung (218) lassen sich für ein chemisch veränderliches System so viel Gleichgewichtsbedingungen aufstellen, als Arten von Veränderungen möglich sind, wobei natürlich jedesmal die Größe K einen anderen Wert hat. Dies entspricht ganz den Forderungen der allgemein gültigen GIBBSschen Phasenregel (§ 204). Dabei muß man die Zahl der im System vorhandenen Molekülarten wohl unterscheiden von der Zahl der unabhängigen Bestandteile des Systems (§ 198). Nur die letztere ist für die Bestimmung der Anzahl und Art der möglichen Phasen entscheidend, während die Zahl der Molekülarten bei der Anwendung der Phasenregel gar keine Rolle spielt. Denn durch Berücksichtigung einer neuen Molekülart wird zwar die Zahl der Variabeln vermehrt, dafür wächst aber auch die Zahl der im System möglichen chemischen Umwandlungen und damit auch die der Gleichgewichtsbedingungen in demselben Betrage, so daß die Anzahl der unabhängigen Variabeln davon ganz unberührt bleibt.

§ 259. Die Gleichung (218) lehrt ferner, daß beim Gleichgewicht, vom allgemeinen Standpunkte aus betrachtet, alle im ganzen System überhaupt möglichen Molekülarten in jeder einzelnen Phase in endlicher Zahl vertreten sind, daß z. B. in einem aus einer wäßrigen Lösung ausgefallenen festen Niederschlag immer auch Wassermoleküle vorkommen, ja daß sogar bei der Berührung fester Körper, sobald man nur hinreichend lange wartet, eine teilweise Auflösung des einen in dem andern eintritt. Denn die für das Gleichgewicht maßgebende Größe K besitzt nach ihrer Definition (218) für jede überhaupt mögliche chemische Veränderung einen bestimmten

endlichen Wert, und es kann daher nach der Gleichung (218) keine der Konzentrationen c genau gleich Null werden, solange Temperatur und Druck endlich bleiben. Diese durch die Thermodynamik bedingte prinzipielle Auffassung hat sich schon nach verschiedenen Seiten hin fruchtbar gezeigt, wie z. B. in der Erklärung der Tatsache, daß weder ein Gas, noch eine Flüssigkeit, noch auch ein fester Körper jemals vollständig von den letzten Spuren fremder gelöster Stoffe befreit werden kann. Aus ihr folgt auch, daß es keine im absoluten Sinne semipermeable Wand geben kann. Denn unter allen Umständen wird sich mit der Zeit die Substanz der Wand mit jedem der in einer angrenzenden Phase befindlichen Stoffe sättigen, und daher auch jeden Stoff nach der anderen Seite wieder abgeben (vgl. § 229).

Andrerseits wird durch die genannte Auffassung die Berechnung der thermodynamischen Eigenschaften einer Lösung beträchtlich kompliziert, da man, um sicher zu gehen, von vornherein immer alle bei den gegebenen Bestandteilen überhaupt möglichen Arten von Molekülen als in der Lösung wirklich vorhanden annehmen muß, und erst dann Vernachlässigungen eintreten lassen darf, wenn man sich durch eine besondere Untersuchung überzeugt hat, daß einzelne Molekülarten in ihr nicht in merklichem Maße vorkommen. Auf diesen Punkt ist wahrscheinlich in manchen Fällen eine scheinbar auftretende Nichtübereinstimmung der Theorie mit der Erfahrung zurückzuführen.

§ 259a. Alle vorhergehenden Sätze beziehen sich natürlich auf endliche Werte von Temperatur und Druck. Wenn sich aber die Temperatur T dem absoluten Nullpunkt nähert, so lehrt ein Blick auf die Gleichung (219), daß, wofern nur die Wärmetönung r der in Betracht kommenden Reaktion endlich bleibt, die Größe $\log K$ für unbegrenzt abnehmende Temperatur positiv oder negativ unendlich wird, je nach der Richtung der durch die Vorzeichen der Zahlen ν bezeichneten Reaktion, und daraus folgt, daß beim absoluten Nullpunkt der Temperatur die Reaktion im einen oder anderen Sinne bis zur vollständigen Beendigung verläuft, so daß schließlich die Konzentrationen der bei der Reaktion sich umbildenden Molekülarten direkt gleich Null werden.

Dies Resultat stimmt überein mit der allgemeinen Folgerung des § 144, daß bei tiefen Temperaturen die Reaktionen in der Richtung positiver Wärmetönung vor sich gehen, es präzisiert aber jene Folgerung noch weiter, da wir es hier mit einer verdünnten Lösung zu tun haben, bei der man aus dem Verhalten der Gesamtenergie im allgemeinen nicht unmittelbar auf das der freien Energie schließen kann. Wir dürfen somit ganz allgemein den Satz aussprechen: Bei unbegrenzt abnehmender Temperatur verschwinden aus einer im thermodynamischen

Gleichgewicht befindlichen Lösung alle Molekülarten, die einer Umwandlung unter Wärmeentwicklung fähig sind.

Es sollen nun einige der wichtigsten speziellen Fälle näher besprochen werden. Die Anordnung ist in erster Linie nach der Zahl der unabhängigen Bestandteile des Systems (§ 198), in zweiter nach der Zahl der Phasen eingerichtet.

§ 260. Ein unabhängiger Bestandteil in einer Phase. Nach der Phasenregel hängt der innere Zustand der Phase von zwei Variabeln ab, also z. B. von der Temperatur T und dem Druck p. Dabei kann die Phase beliebig viele Molekülarten enthalten. So wird eine Quantität flüssiges Wasser außer den einfachen H_2O-Molekülen auch Doppel- und mehrfache Moleküle, ferner Moleküle H_2 und O_2 auch H_2O_2, ferner geladene Ionen $\overset{+}{H}$, $\overset{-}{HO}$ und $\overset{--}{O}$ usw. in endlichem Betrage enthalten. Die elektrischen Ladungen der Ionen spielen in der Thermodynamik keine besondere Rolle, solange nicht die elektrischen Kräfte mit den thermodynamischen in Kollision geraten, was nur und immer dann eintritt, wenn die thermodynamische Gleichgewichtsbedingung eine Verteilung der Ionen in den verschiedenen Phasen des Systems verlangt, bei welcher vermöge der unveränderlichen Ladungen der Ionen freie Elektrizität im Innern einer Phase auftreten müßte. Einem solchen Zustande widersetzen sich die elektrischen Kräfte mit großer Stärke, und es tritt eine Abweichung von dem rein thermodynamischen Gleichgewicht ein, welche andrerseits durch entstehende Potentialdifferenzen zwischen den betreffenden Phasen kompensiert wird. Eine allgemeine Übersicht über diese elektromolekularen Erscheinungen läßt sich gewinnen, wenn man den Wert der Energie des Systems durch Hinzufügung elektrischer Glieder verallgemeinert. Doch beschränken wir uns hier auf die Betrachtung unelektrischer Zustände, und brauchen daher gar keine Rücksicht zu nehmen auf die elektrischen Ladungen der Ionen, die wir einfach wie andere Moleküle behandeln.

In dem vorliegenden Falle sind also die Konzentrationen sämtlicher Molekülarten durch T und p bestimmt. Eine Berechnung der Konzentrationen ist bisher nur für die Ionen $\overset{+}{H}$ und $\overset{-}{HO}$ gelungen (die Zahl der $\overset{--}{O}$-Ionen ist dagegen zu vernachlässigen) und zwar u. a. durch die Messung der elektrischen Leitfähigkeit der Lösung, die allein von den Ionen herrührt. Nach Kohlrausch und Heydweiller ist der Dissoziationsgrad des Wassers, d. h. das Verhältnis der Masse des in Ionen $\overset{+}{H}$ und $\overset{-}{HO}$ gespaltenen Wassers zu der Gesamtmasse des Wassers bei 18° C.

$$14,3 \cdot 10^{-10}.$$

Diese Zahl stellt zugleich das Verhältnis der Zahl der dissoziierten Moleküle

zu der Gesamtzahl der Moleküle vor. Die Thermodynamik gestattet die Abhängigkeit der Dissoziation von der Temperatur zu berechnen.

Stellen wir nämlich die Bedingung des Gleichgewichtszustandes auf. Das Symbol des Systems ist nach (216):

$$n_0 \ \mathrm{H_2O}, \quad n_1 \ \overset{+}{\mathrm{H}}, \quad n_2 \ \overset{-}{\mathrm{HO}}.$$

Die Gesamtzahl der Moleküle sei $n = n_0 + n_1 + n_2$, die Konzentrationen der einzelnen Molekülarten demnach:

$$c_0 = \frac{n_0}{n}, \qquad c_1 = \frac{n_1}{n}, \qquad c_2 = \frac{n_2}{n}.$$

Die in Betracht kommende chemische Umwandlung:

$$\nu_0 : \nu_1 : \nu_2 = \delta n_0 : \delta n_1 : \delta n_2$$

besteht in der Dissoziation eines Moleküls $\mathrm{H_2O}$ in je ein Molekül $\overset{+}{\mathrm{H}}$ und $\overset{-}{\mathrm{HO}}$, also:

$$\nu_0 = -1 \qquad \nu_1 = 1 \qquad \nu_2 = 1.$$

Also ist nach (218) im Gleichgewichtszustand:

$$-\log c_0 + \log c_1 + \log c_2 = \log K$$

oder, da $c_1 = c_2$, und c_0 nahezu $= 1$

$$2 \log c_1 = \log K.$$

Dies ergibt für die Abhängigkeit der Konzentration c_1 von der Temperatur nach (219):

$$2 \frac{\partial \log c_1}{\partial T} = \frac{1}{R} \frac{r}{T^2}. \tag{221}$$

r, die für die Dissoziation eines Moleküls $\mathrm{H_2O}$ in die Ionen $\overset{+}{\mathrm{H}}$ und $\overset{-}{\mathrm{HO}}$ nötige Wärmezufuhr, ist nach ARRHENIUS gleich der Wärmetönung bei der Neutralisation einer einwertigen starken Base und Säure in verdünnter wäßriger Lösung, also in der § 97 eingeführten Bezeichnungsweise:

$$r = (\overset{+}{\mathrm{H}}, \ \overset{-}{\mathrm{Cl}}, \ \mathrm{aq}) + (\overset{+}{\mathrm{Na}}, \ \overset{-}{\mathrm{HO}}, \ \mathrm{aq}) - (\overset{+}{\mathrm{Na}}, \ \overset{-}{\mathrm{Cl}}, \ \mathrm{aq}).$$

Dies ergibt nach neueren Messungen von WÖRMANN:

$$r = 27\,857 - 48,5T \quad \text{cal}.$$

Daraus folgt nach Gleichung (221):

$$\frac{\partial c_1}{\partial T} = \frac{1}{2 \cdot 1,985}\left(\frac{27\,857}{T^2} - \frac{48,5}{T}\right);$$

und durch Integration:

$$\log_{10} c_1 = -\frac{3047,3}{T} - 12,125 \log_{10} T + \text{konst.}$$

Diese Abhängigkeit des Dissoziationsgrades von der Temperatur stimmt gut überein mit den Messungen der galvanischen Leitfähigkeit des reinen Wassers bei verschiedenen Temperaturen von KOHLRAUSCH und HEYDWEILLER, NOYES, LUNDÉN.

Beim absoluten Nullpunkt der Temperatur verschwindet die Dissoziation völlig, in Übereinstimmung mit dem in § 259a abgeleiteten allgemeinen Satz.

§ 261. Ein unabhängiger Bestandteil in zwei Phasen. Das System besteht aus zwei Phasen, sagen wir einer flüssigen und einer gasförmigen oder festen. Das Symbol des Systems ist nach (216):

$$n_0 m_0 \mid n_0' m_0'.$$

Jede Phase enthält nur eine einzige Molekülgattung; doch brauchen die Moleküle in beiden Phasen nicht die nämlichen zu sein, namentlich kann ein flüssiges Molekül ein Vielfaches des gasförmigen Moleküls sein.

Wenn nun ein flüssiges Molekül verdampft oder erstarrt, so ist in unserer Bezeichnung:

$$\nu_0 = -1, \qquad \nu_0' = \frac{m_0}{m_0'},$$

$$c_0 = \frac{n_0}{n_0} = 1, \qquad c_0' = \frac{n_0'}{n_0'} = 1$$

folglich die Gleichgewichtsbedingung (218):

$$0 = \log K = -\varphi_0 + \frac{m_0}{m_0'}\varphi_0', \tag{221a}$$

und da K nur von T und p abhängt, so ist durch diese Gleichung eine bestimmte Beziehung zwischen p und T ausgedrückt: das Gesetz der Abhängigkeit des Verdampfungs- bzw. Schmelzdruckes von der Temperatur, und umgekehrt. Den Inhalt dieses Gesetzes erfährt man durch die Berücksichtigung der Abhängigkeit der Größe K von p und T. Differentiiert man nämlich die letzte Gleichung vollständig, so ergibt sich:

$$\frac{\partial \log K}{\partial T}\, dT + \frac{\partial \log K}{\partial p}\, dp = 0$$

oder nach (219) und (220)

$$\frac{r}{T^2}\,dT - \frac{v}{T}\,dp = 0.$$

Nun ist, wenn v_0 und v_0' die Molekularvolumina der beiden Phasen bezeichnen, die Volumenänderung des Systems bei der betrachteten Umwandlung:

$$v = \frac{m_0}{m_0'}v_0' - v_0,$$

folglich

$$r = T\left(\frac{m_0}{m_0'}v_0' - v_0\right)\frac{dp}{dT}$$

oder, auf die Masseneinheit bezogen:

$$\frac{r}{m_0} = T\left(\frac{v_0'}{m_0'} - \frac{v_0}{m_0}\right)\cdot\frac{dp}{dT},$$

die bekannte Carnot-Clapeyronsche Formel (111).

Wegen der weiteren Anwendungen vgl. oben das zweite Kapitel.

§ 262. Zwei unabhängige Bestandteile in einer Phase (Lösung eines Stoffes in einem homogenen Lösungsmittel). Nach der Phasenregel ist außer dem Druck und der Temperatur noch eine Variable beliebig, z. B. die in 1 Liter Lösung enthaltene Zahl der Moleküle des gelösten Stoffes, wie sie durch die Analyse gemessen wird. Dann ist die Konzentration jeder einzelnen Molekülart bestimmt, mag sie durch Dissoziation, durch Assoziation, durch Hydratbildung oder durch Hydrolyse der gelösten Moleküle entstehen. Betrachten wir zunächst den einfachen Fall eines binären Elektrolyten, z. B. Essigsäure in Wasser. Das Symbol des Systems ist nach (216):

$$n_0\ \mathrm{H_2O},\quad n_1\ \mathrm{H_4C_2O_2},\quad n_2\ \overset{+}{\mathrm{H}},\quad n_3\ \overset{-}{\mathrm{H_3C_2O_2}}.$$

Die Gesamtzahl der Moleküle sei:

$$n = n_0 + n_1 + n_2 + n_3 \qquad (\text{nahe gleich } n_0).$$

Die Konzentrationen sind:

$$c_0 = \frac{n_0}{n},\qquad c_1 = \frac{n_1}{n},\qquad c_2 = \frac{n_2}{n},\qquad c_3 = \frac{n_3}{n}.$$

Die einzige praktisch in Betracht kommende Umwandlung

$$\nu_0 : \nu_1 : \nu_2 : \nu_3 = \delta n_0 : \delta n_1 : \delta n_2 : \delta n_3$$

besteht in der Dissoziation eines Moleküls $H_4C_2O_2$ in seine beiden Ionen, also

$$\nu_0 = 0, \qquad \nu_1 = -1, \qquad \nu_2 = 1, \qquad \nu_3 = 1.$$

Daher ist nach (218) im Gleichgewichtszustand:

$$-\log c_1 + \log c_2 + \log c_3 = \log K$$

oder, da $c_2 = c_3$,

$$\frac{c_2^2}{c_1} = K. \tag{222}$$

Nun ist als bekannt anzusehen die Summe:

$$c_1 + c_2 = c,$$

da die Gesamtzahl (n_1+n_2) der undissoziierten und dissoziierten Säuremoleküle und auch die Gesamtzahl n_0 der Wassermoleküle, welche $= n$ gesetzt werden kann, direkt gemessen wird. Daher lassen sich c_1 und c_2 aus den letzten beiden Gleichungen berechnen. Es folgt daraus für die Konzentrationen c_1 und c_2 der undissoziierten und der dissoziierten Moleküle, im Verhältnis zu der Gesamtkonzentration c:

$$\frac{c_1}{c} = \frac{n_1}{n_1+n_2} = 1 - \frac{K}{2c}\left(\sqrt{1+\frac{4c}{K}} - 1\right),$$

$$\frac{c_2}{c} = \frac{n_2}{n_1+n_2} = \frac{K}{2c}\left(\sqrt{1+\frac{4c}{K}} - 1\right).$$

Mit wachsender Verdünnung, also abnehmendem c, wächst das Verhältnis $\frac{c_2}{c}$ in bestimmter Weise bis gegen 1, d. h. bis zur vollständigen Dissoziation, und daraus ergibt sich für die elektrische Leitfähigkeit einer Lösung von gegebener Konzentration das zuerst von OSTWALD aufgestellte sogenannte Verdünnungsgesetz der binären Elektrolyte, welches in zahlreichen Fällen durch die Erfahrung bestätigt worden ist.

Die Abhängigkeit der Dissoziationskonstante K von der Temperatur ergibt sich hier in ganz ähnlicher Weise wie in § 260 nach Gleichung (219) durch Berücksichtigung der bei der Dissoziation auftretenden Wärmetönung. Umgekehrt läßt sich aus der Veränderlichkeit der Dissoziation mit der Temperatur die Dissoziationswärme berechnen, wie zuerst von ARRHENIUS gezeigt wurde.[1]

[1] Auch die Abhängigkeit der Konstante K vom Drucke p, wie sie durch die Gleichung (220) gegeben ist, wurde experimentell von FANJUNG geprüft und bestätigt gefunden.

Die namentlich bei starken Elektrolyten beobachteten Abweichungen vom OSTWALDschen Verdünnungsgesetz lassen sich durch den Umstand erklären, daß die elektrische Leitfähigkeit einer Lösung nicht immer ein Maß für den Dissoziationszustand liefert. Nach J. C. GHOSH[1] sind nämlich in den verdünnten Lösungen starker Elektrolyte, wie KCl und NaCl, die Salzmoleküle so gut wie vollständig in ihre Ionen dissoziiert, aber von den Ionen trägt dennoch nur ein gewisser Bruchteil zur Elektrizitätsleitung bei, weil die langsamer bewegten Ionen durch die Anziehungen der ihnen benachbarten entgegengesetzt geladenen Ionen zurückgehalten werden. Mit steigender Verdünnung der Lösung wächst dieser Bruchteil, wegen der Abnahme des Einflusses jener Anziehungskräfte, und daraus ergibt sich eine Zunahme der Leitfähigkeit mit der Verdünnung, nach einem Gesetz, dessen Begründung hier nicht gegeben werden kann, weil es sich bei der elektrischen Leitung nicht um Gleichgewichtszustände handelt. Dasselbe weicht sowohl vom OSTWALDschen Verdünnungsgesetz als auch von dem unten in § 273 abgeleiteten Gesetz der Gefrierpunktserniedrigung bzw. des osmotischen Druckes starker Elektrolyte in wesentlichen Punkten ab.

§ 263. Gewöhnlich wird in der Lösung eines Stoffes nicht eine einzige, sondern eine große Anzahl von chemischen Reaktionen möglich sein, und dementsprechend enthält das vollständige System eine lange Reihe von Molekülarten. Wir wollen hier beispielsweise noch den Fall eines Elektrolyten behandeln, der sich auf verschiedene Weise in Ionen spalten kann, nämlich eine wäßrige Lösung von Schwefelsäure.

Das System ist nach (216):

$$n_0\,H_2O, \quad n_1\,H_2SO_4, \quad n_2\,\overset{+}{H}, \quad n_3\,\overset{-}{H}SO_4, \quad n_4\,\overset{--}{SO}_4.$$

Die Gesamtzahl der Moleküle ist:

$$n = n_0 + n_1 + n_2 + n_3 + n_4 \qquad \text{(nahe gleich } n_0\text{)}.$$

Die Konzentrationen sind:

$$c_0 = \frac{n_0}{n}, \quad c_1 = \frac{n_1}{n}, \quad c_2 = \frac{n_2}{n}, \quad c_3 = \frac{n_3}{n}, \quad c_4 = \frac{n_4}{n}.$$

Hier kommen zwei verschiedenartige Umwandlungen:

$$\nu_0 : \nu_1 : \nu_2 : \nu_3 : \nu_4 = \delta n_0 : \delta n_1 : \delta n_2 : \delta n_3 : \delta n_4$$

[1] Inanendra Chandra GHOSH, Trans. Chem. Soc. 1918, vol. 113, p. 449.

in Betracht, nämlich erstens die Spaltung eines Moleküls H_2SO_4 in die Ionen $\overset{+}{H}$ und $\overset{-}{HSO}_4$:

$$\nu_0 = 0 \qquad \nu_1 = -1 \qquad \nu_2 = 1 \qquad \nu_3 = 1 \qquad \nu_4 = 0,$$

zweitens die Spaltung eines Ions $\overset{-}{HSO}_4$ in die Ionen $\overset{+}{H}$ und $\overset{--}{SO}_4$:

$$\nu_0 = 0 \qquad \nu_1 = 0 \qquad \nu_2 = 1 \qquad \nu_3 = -1 \qquad \nu_4 = 1.$$

Daher gelten nach (218) im Gleichgewichtszustand die beiden Bedingungen:

$$\begin{aligned} -\log c_1 + \log c_2 + \log c_3 &= \log K \\ \log c_2 - \log c_3 + \log c_4 &= \log K' \end{aligned}$$

oder:

$$\begin{aligned} \frac{c_2 c_3}{c_1} &= K, \\ \frac{c_2 c_4}{c_3} &= K'. \end{aligned}$$

Hierzu kommt noch die Bedingung, welche ausspricht, daß in der Gesamtmenge des gelösten Stoffes die Zahl der SO_4-Radikale $(n_1 + n_3 + n_4)$ halb so groß ist als die der H-Atome $(2n_1 + n_2 + n_3)$; denn sonst enthielte das System mehr als zwei unabhängige Bestandteile. Diese Bedingung lautet:

$$2c_4 + c_3 = c_2.$$

Endlich ist als gegeben anzusehen die Gesamtmenge der gelösten Schwefelsäure, also

$$c_1 + c_3 + c_4 = c.$$

Die letzten vier Gleichungen ergeben für die vier Konzentrationen c_1, c_2, c_3, c_4 bestimmte Werte, wodurch der Gleichgewichtszustand gefunden ist.[1]

Für eine genauere Rechnung müßte man in der Lösung jedenfalls noch andere Molekülarten berücksichtigen. Jede neue Molekülart bedingt eine neue Unbekannte, aber auch eine neue Art der Umwandlung und daher eine neue Bedingung für das Gleichgewicht, so daß der Gleichgewichtszustand eindeutig bestimmt bleibt.

§ 264. Zwei unabhängige Bestandteile in zwei Phasen. Nach der Phasenregel ist der Gleichgewichtszustand durch zwei Variable, etwa

[1] Vgl. hierzu die Berechnungen von J. B. Goebel, Zeitschr. f. physikal. Chemie, **71**, S. 652, 1910.

Temperatur und Druck, bestimmt. Zur besseren Übersicht über dies weite Gebiet von Erscheinungen empfiehlt es sich, hier zwei Fälle zu unterscheiden, je nachdem nur eine der beiden Phasen beide Bestandteile in merklichen Mengen enthält, oder beide Phasen beide Bestandteile enthalten.

Nehmen wir zunächst den einfacheren Fall, daß die eine (erste) Phase beide Bestandteile, die andere (zweite) Phase dagegen nur einen einzigen Bestandteil enthält. Genau genommen ist nach § 259 diese Voraussetzung niemals zutreffend, aber sie genügt doch in sehr vielen Fällen bis auf unmeßbar kleine Fehler den beobachtbaren Tatsachen. Die Anwendung der allgemeinen Gleichgewichtsbedingung (218) auf diesen Fall führt auf ganz verschiedene Gesetze, je nachdem der in der zweiten Phase isoliert vorkommende Bestandteil in der ersten Phase als gelöster Stoff oder als Lösungsmittel (§ 249) auftritt. Wir scheiden daher den Fall noch in zwei Unterabteilungen.

§ 265. Der in der zweiten Phase isoliert vorkommende Bestandteil bildet in der ersten Phase den gelösten Stoff. Ein Beispiel dafür ist die Absorption eines Gases, z. B. Kohlensäure, in einer Flüssigkeit von verhältnismäßig unmerklich kleiner Dampfspannung, z. B. Wasser bei einer nicht zu hohen Temperatur.

Das Symbol des aus zwei Phasen bestehenden Systems ist nach (216):

$$n_0 H_2O, \quad n_1 CO_2 \,|\, n_0' CO_2.$$

Wir setzen hierbei voraus, daß das gasförmige Molekül CO_2 identisch ist mit dem gelösten Molekül CO_2. Der allgemeinere Fall wird weiter unten im § 274 behandelt werden. Die Konzentrationen der einzelnen Molekülarten des Systems in den beiden Phasen sind:

$$c_0 = \frac{n_0}{n_0 + n_1}, \qquad c_1 = \frac{n_1}{n_0 + n_1}, \qquad c_0' = \frac{n_0'}{n_0'} = 1.$$

Die in Betracht kommende Umwandlung:

$$\nu_0 : \nu_1 : \nu_0' = \delta n_0 : \delta n_1 : \delta n_0'$$

besteht hier in der Verdampfung eines Moleküls Kohlensäure aus der Lösung, also:

$$\nu_0 = 0 \qquad \nu_1 = -1 \qquad \nu_0' = 1.$$

Die Gleichgewichtsbedingung (218):

$$\nu_0 \log c_0 + \nu_1 \log c_1 + \nu_0' \log c_0' = \log K$$

wird daher hier:

$$-\log c_1 = \log K, \tag{223}$$

d. h. bei bestimmter Temperatur und bestimmtem Druck (wodurch K bestimmt ist) ist auch die Konzentration c_1 des Gases in der Lösung bestimmt. Die Änderung der Konzentration mit Druck und Temperatur ergibt sich durch Substitution der letzten Gleichung in die Gleichungen (219) und (220). Es folgt daraus:

$$\frac{\partial \log c_1}{\partial p} = \frac{1}{R}\frac{v}{T} \tag{224}$$

$$\frac{\partial \log c_1}{\partial T} = -\frac{1}{R}\frac{r}{T^2}. \tag{225}$$

v ist die bei der isotherm-isobaren Verdampfung eines Moleküls CO_2 eintretende Volumenzunahme des Systems, r die dabei von außen aufzunehmende Wärmemenge. Da nun v nahezu das Volumen eines Moleküls gasförmiger Kohlensäure darstellt, so kann man nach (16) angenähert setzen:

$$v = \frac{RT}{p}$$

und die Gleichung (224) ergibt:

$$\frac{\partial \log c_1}{\partial p} = \frac{1}{p}.$$

Integriert:

$$\log c_1 = \log p + \text{konst.}$$

oder:

$$c_1 = C \cdot p, \tag{226}$$

d. h. die Konzentration des gelösten Gases ist proportional dem Druck des freien Gases über der Lösung (Gesetz von HENRY). Der Proportionalitätsfaktor C, der ein Maß für die Löslichkeit des Gases abgibt, hängt noch von der Temperatur ab; in welcher Weise, lehrt die Gleichung (225), die mit (226) kombiniert ergibt:

$$\frac{\partial \log C}{\partial T} = -\frac{1}{R}\frac{r}{T^2}.$$

Erfolgt also die Verdampfung des Gases aus der Lösung unter Wärmezufuhr von außen, so ist r positiv, und die Löslichkeit nimmt mit steigender

Temperatur ab. Umgekehrt läßt sich aus der Veränderlichkeit von C mit der Temperatur die Wärmetönung r bei der Absorption berechnen. Es ergibt sich:

$$r = -\frac{RT^2}{C} \cdot \frac{\partial C}{\partial T}.$$

Nach den Versuchen von NACCARI und PAGLIANI ist bei 20° C. ($T = 293$) die Löslichkeit von Kohlensäure in Wasser, ausgedrückt in einer Einheit, auf die es hier nicht ankommt: $0,8928$, und ihr Temperaturkoeffizient: $-0,02483$, und daher mit Berücksichtigung von (34):

$$r = \frac{1,985 \cdot 293^2 \cdot 0,02483}{0,8928} = 4700 \text{ cal.}$$

THOMSON fand für die Wärmetönung bei der Absorption eines Moleküls Kohlensäure in Wasser 5880 cal. Der Fehler liegt (nach NERNST) wohl auf Seite der Messung des Löslichkeitskoeffizienten.

Von dem ganzen Betrage der Wärmetönung entfällt nach (48) der Teil:

$$RT \text{ oder } 1,985 \cdot 293 = 590 \text{ cal.}$$

auf die äußere Arbeit.

§ 266. Ein weiteres hierher gehöriges Beispiel ist die Sättigung eines flüssigen Lösungsmittels mit einem schwerlöslichen Salze, z. B. Bernsteinsäure in Wasser. Das Symbol dieses Systems ist nach (216):

$$n_0\,H_2O, \quad n_1\,H_6C_4O_4 \,|\, n_0'\,H_6C_4O_4$$

wenn man von der geringen Dissoziation der Säure in der Lösung absieht. Die Berechnung des Gleichgewichtszustandes ergibt genau in der nämlichen Bezeichnung wie in (223):

$$-\log c_1 = \log K,$$

also bestimmt durch Temperatur und Druck; ferner nach (219):

$$r = -RT^2 \frac{\partial \log c_1}{\partial T}. \tag{227}$$

Mittels dieser Gleichung berechnete zuerst VAN'T HOFF r aus der Löslichkeit der Bernsteinsäure bei 0° (2,88) und bei 8,5° (4,22). Es ist dann nahezu:

$$\frac{\partial \log c_1}{\partial T} = \frac{\overset{e}{\log}\, 4,22 - \overset{e}{\log}\, 2,88}{8,5} = 0,04494.$$

Daraus folgt für $T = 273$ in Kalorien:

$$r = -1,985 \cdot 273^2 \cdot 0,04494 = -6600 \text{ cal},$$

d. h. beim Ausfällen eines Moleküls fester Substanz aus der Lösung werden 6600 cal nach außen abgegeben. BERTHELOT fand dagegen als Lösungswärme 6700 cal.

Betrachtet man r als von der Temperatur unabhängig, was in manchen Fällen in erster Annäherung gestattet sein wird, so läßt sich die Gleichung (227) nach T integrieren und liefert:

$$\log c_1 = \frac{r}{RT} + \text{konst.}$$

§ 267. Die Beziehung (227) zwischen der Lösungswärme und dem Temperaturkoeffizienten der Löslichkeit wird ungültig, wenn das Salz in der Lösung eine merkliche chemische Umbildung, z. B. Dissoziation, erleidet. Dann sind in der Lösung neben den normalen Molekülen auch dissoziierte vorhanden, wie z. B. in folgendem System von Wasser und Silberacetat:

$$n_0\,H_2O, \quad n_1\,AgH_3C_2O_2, \quad n_2\,\overset{+}{A}g, \quad n_3\,H_3\overset{-}{C}_2O_2 \,|\, n_0'\,AgH_3C_2O_2.$$

Hier ist die Gesamtzahl aller Moleküle in der Lösung:

$$n = n_0 + n_1 + n_2 + n_3 \qquad (\text{nahe gleich } n_0),$$

dann sind die Konzentrationen der einzelnen Molekülarten in beiden Phasen:

$$c_0 = \frac{n_0}{n}, \quad c_1 = \frac{n_1}{n}, \quad c_2 = \frac{n_2}{n}, \quad c_3 = \frac{n_3}{n}, \quad c_0' = \frac{n_0'}{n_0'} = 1.$$

Die möglichen Umwandlungen:

$$\nu_0 : \nu_1 : \nu_2 : \nu_3 : \nu_0' = \delta n_0 : \delta n_1 : \delta n_2 : \delta n_3 : \delta n_0',$$

bestehen erstens in der Ausfällung eines Moleküls Silberacetat aus der Lösung:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 0, \quad \nu_3 = 0, \quad \nu_0' = 1,$$

zweitens in der Dissoziation eines Moleküls Silberacetat:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 1, \quad \nu_3 = 1, \quad \nu_0' = 0.$$

Es gelten also nach (218) die beiden Gleichgewichtsbedingungen:

erstens: $$-\log c_1 = \log K,$$

zweitens: $$-\log c_1 + \log c_2 + \log c_3 = \log K'$$

oder, da $c_2 = c_3$

$$\frac{c_2^2}{c_1} = K',$$

d. h. bei bestimmter Temperatur und bestimmtem Druck ist erstens die Konzentration c_1 der undissoziierten Moleküle in der mit dem Salz gesättigten Lösung eine ganz bestimmte, und zweitens bestimmt sich die Konzentration c_2 der dissoziierten Moleküle aus der der undissoziierten c_1 nach dem schon oben unter (222) abgeleiteten Dissoziationsgesetz eines Elektrolyten. Da nun durch die Messung der Löslichkeit der Wert von $c_1 + c_2$, durch die Messung der elektrischen Leitfähigkeit der Lösung aber der Wert von c_2 gefunden wird, so lassen sich hieraus die Größen K und K' für irgend eine beliebige Temperatur berechnen. Ihre Abhängigkeit von der Temperatur liefert dann nach (219) ein Maß einerseits für die bei der Ausfällung eines undissoziierten Moleküls aus der Lösung, andrerseits für die bei der Dissoziation eines gelösten Moleküls auftretende Wärmetönung, und daraus ergibt sich nach VAN'T HOFF eine Methode, um aus der gemessenen Löslichkeit des festen Salzes und der gemessenen Leitfähigkeit der gesättigten Lösung bei verschiedenen Temperaturen die wirkliche Lösungswärme des Salzes zu berechnen, d. h. die Wärmetönung, die eintritt, wenn ein Molekulargewicht des festen Salzes aufgelöst und außerdem der Bruchteil $\frac{dc_2}{(dc_1 + dc_2)}$ in seine Ionen dissoziiert wird, so wie es dem tatsächlichen Lösungsvorgang entspricht.

§ 268. Der in der zweiten Phase isoliert vorkommende Bestandteil bildet in der ersten Phase das Lösungsmittel. Dieser Fall findet sich immer dann verwirklicht, wenn sich aus einer Lösung beliebigen Aggregatzustandes das reine Lösungsmittel in einem anderen Aggregatzustand, z. B. durch Gefrieren, Verdampfen, Schmelzen, Sublimieren, ausscheidet. Der allgemeine Typus eines solchen aus zwei Phasen bestehenden Systems ist nach (216):

$$n_0 m_0,\ n_1 m_1,\ n_2 m_2,\ n_3 m_3,\ \ldots \mid n_0' m_0',$$

wobei noch offen gelassen ist, ob das Lösungsmittel in den beiden Aggregatzuständen gleiches oder verschiedenes Molekulargewicht besitzt. Die Summe der Molekülzahlen in der Lösung ist:

$$n = n_0 + n_1 + n_2 + n_3 + \ldots \qquad \text{(nahe gleich } n_0\text{)}.$$

Die Konzentrationen der einzelnen Molekülarten sind:

$$c_0 = \frac{n_0}{n}, \quad c_1 = \frac{n_1}{n}, \quad c_2 = \frac{n_2}{n}, \quad \ldots \quad c_0' = \frac{n_0'}{n_0'} = 1.$$

Eine mögliche Umwandlung:

$$\nu_0 : \nu_1 : \nu_2 : \ldots : \nu_0' = \delta n_0 : \delta n_1 : \delta n_2 : \ldots : \delta n_0'$$

ist der Austritt eines Moleküls des Lösungsmittels aus der ersten Phase in die zweite, d. h.

$$\nu_0 = -1, \quad \nu_1 = 0, \quad \nu_2 = 0, \ \ldots \ \nu_0' = \frac{m_0}{m_0'}. \tag{228}$$

Das Gleichgewicht erfordert also nach (218) die Bedingung:

$$-\log c_0 + \frac{m_0}{m_0'} \log c_0' = \log K$$

und mit Berücksichtigung der obigen Werte von c_0 und c_0'

$$\log \frac{n}{n_0} = \log K.$$

Nun ist

$$\frac{n}{n_0} = 1 + \frac{n_1 + n_2 + n_3 + \ldots}{n_0},$$

also, da der Bruch rechts sehr klein ist:

$$\frac{n_1 + n_2 + n_3 + \ldots}{n_0} = \log K. \tag{229}$$

Nach der allgemeinen Definition in (218) ist hier:

$$\log K = \frac{1}{R}(\nu_0\varphi_0 + \nu_1\varphi_1 + \nu_2\varphi_2 + \ldots + \nu_0'\varphi_0'),$$

folglich mit Berücksichtigung der Werte der ν in (228):

$$\frac{n_1 + n_2 + n_3 + \ldots}{n_0} = \frac{1}{R}\left(\frac{m_0}{m_0'}\varphi_0' - \varphi_0\right) = \log K. \tag{230}$$

Nach dieser Gleichung ist der Ausdruck rechts, oder $\log K$, ebenfalls sehr klein.

Nehmen wir einmal den speziellen Fall, daß die Zahl der gelösten Moleküle $n_1 + n_2 + \ldots$ ganz verschwindet und mithin statt der Lösung das reine Lösungsmittel vorhanden ist. Dann erhalten wir nach (230):

$$0 = \log K = \frac{m_0\varphi_0'}{m_0'} - \varphi_0,$$

also die Gleichung (221a), welche die Bedingung ausspricht, die T und p erfüllen müssen, wenn sich das reine Lösungsmittel in zwei Aggregatzuständen nebeneinander befindet. Dann kann man entweder den Druck (Dampfspannung) als abhängig von der Temperatur, oder die Temperatur (Siedepunkt, Schmelzpunkt) als abhängig vom Druck betrachten.

Kehren wir nun zur Gleichung (230) zurück. Nach ihr bewirkt jede Auflösung fremder Moleküle $n_1, n_2, n_3, \ldots$ eine entsprechende Abweichung von der für das reine Lösungsmittel gültigen Beziehung zwischen Druck und Temperatur, und zwar hängt, wie man sieht, diese Abweichung lediglich von der Gesamtzahl der gelösten Moleküle, nicht aber von ihrer Natur ab. Man kann dies ausdrücken, indem man entweder p oder T als unabhängige Variable benutzt. Im ersten Falle sagt man: Bei einem bestimmten Druck p ist die Siede- oder Gefriertemperatur T der Lösung eine andere als die des reinen Lösungsmittels (T_0); im zweiten Falle: Bei einer bestimmten Temperatur T ist der Dampfdruck oder Erstarrungsdruck p der Lösung ein anderer als der des reinen Lösungsmittels (p_0). Berechnen wir die Abweichungen in beiden Fällen.

§ 269. Ist T_0 die Siede- oder Gefriertemperatur des reinen Lösungsmittels beim Druck p, so ist nach (221a):

$$(\log K)_{T=T_0} = 0,$$

und durch Subtraktion von (230) ergibt sich:

$$\log K - (\log K)_{T=T_0} = \frac{n_1 + n_2 + \ldots}{n_0}.$$

Da nun T wenig von T_0 abweicht, so kann man nach dem TAYLORschen Satze mit Benutzung von (219) schreiben:

$$\frac{\partial \log K}{\partial T} \cdot (T - T_0) = \frac{r}{RT_0^2}(T - T_0) = \frac{n_1 + n_2 + \ldots}{n_0},$$

und daraus folgt:

$$T - T_0 = \frac{n_1 + n_2 + \ldots}{n_0} \cdot \frac{RT_0^2}{r}. \tag{231}$$

Dies ist das zuerst von VAN'T HOFF abgeleitete Gesetz der Siedepunktserhöhung bzw. Gefrierpunktserniedrigung; für das Gefrieren ist r (die beim Gefrieren eines flüssigen Moleküls von außen zugeführte Wärme) negativ. Da n_0 und r nur miteinander multipliziert vorkommen, so erfährt man aus dieser Formel nichts über die Molekülzahl n_0 und das Molekulargewicht m_0 des flüssigen

Lösungsmittels. Wenn r in Kalorien ausgedrückt wird, so hat man nach (34) für R 1,985 zu setzen.

So ist für die Verdampfung von 1 Liter Wasser unter Atmosphärendruck nahezu $n_0 r = 1000 \cdot 539$ cal, $T_0 = 373$, und daher die Siedepunktserhöhung einer verdünnten wäßrigen Lösung:

$$\begin{aligned} T - T_0 &= \frac{1,985 \cdot 373^2}{1000 \cdot 539}(n_1 + n_2 + n_3 + \ldots) \\ &= 0.51^\circ \cdot (n_1 + n_2 + n_3 + \ldots) \end{aligned} \tag{232}$$

Ferner ist für das Gefrieren von 1 Liter Wasser unter Atmosphärendruck nahezu $n_0 r' = -1000 \cdot 80$ cal, $T_0' = 273$, und daher die Gefrierpunktserniedrigung einer verdünnten wäßrigen Lösung:

$$\begin{aligned} T_0' - T &= \frac{1,985 \cdot 273^2}{1000 \cdot 80}(n_1 + n_2 + n_3 + \ldots) \\ &= 1,85^\circ (n_1 + n_2 + n_3 + \ldots) \end{aligned} \tag{233}$$

§ 270. Ist p_0 der Dampfdruck des reinen Lösungsmittels bei der Temperatur T, so ist nach (221a):

$$(\log K)_{p=p_0} = 0,$$

und durch Subtraktion von (230) ergibt sich:

$$\log K - (\log K)_{p=p_0} = \frac{n_1 + n_2 + \ldots}{n_0}.$$

Da nun p wenig von p_0 abweicht, so kann man statt dessen mit Benutzung von (220) schreiben:

$$\frac{\partial \log K}{\partial p}(p - p_0) = -\frac{v}{RT}(p - p_0) = \frac{n_1 + n_2 + \ldots}{n_0},$$

und daraus folgt, wenn v gleichgesetzt wird dem Volumen der bei der Verdampfung eines flüssigen Moleküls entstehenden gasförmigen Moleküle: $v = \dfrac{m_0}{m_0'}\dfrac{RT}{p}$

$$\frac{p_0 - p}{p} = \frac{m_0'}{m_0} \cdot \frac{n_1 + n_2 + \ldots}{n_0}. \tag{234}$$

Dies ist das zuerst von VAN'T HOFF abgeleitete Gesetz der relativen Dampfdruckerniedrigung. Da n_0 und m_0 nur miteinander multipliziert

vorkommen, so erfährt man auch aus dieser Formel nichts über das Molekulargewicht des flüssigen Lösungsmittels. Häufig findet sich diese Beziehung in der Form ausgesprochen, daß die relative Dampfdruckerniedrigung das Verhältnis der Zahl der gelösten Moleküle $(n_1+n_2+n_3+\ldots)$ zu der Zahl der Moleküle des Lösungsmittels n_0, oder, was bei verdünnten Lösungen auf dasselbe hinauskommt, aller Moleküle der Lösung n angibt. Dieser Satz ist jedoch, wie hier ersichtlich, nur dann richtig, wenn $m_0' = m$, d. h. wenn man den Molekülen des Lösungsmittels in der Lösung dasselbe Molekulargewicht zuschreiben darf, wie im Dampfe. Dies wird aber im allgemeinen, z. B. für Wasser, nicht zutreffen, und es ist daher nicht überflüssig, zu betonen, daß man durch die relative Dampfspannungserniedrigung einer verdünnten Lösung ebensowenig wie durch ihren Siedepunkt oder Gefrierpunkt irgend etwas über das Molekulargewicht des Lösungsmittels in der Lösung erfahren kann. Unter allen Umständen ergibt sich aus diesen Messungen immer nur die Gesamtzahl $(n_1 + n_2 + \ldots)$ der in der Lösung vorhandenen fremden Moleküle.

§ 271. Es ist von Interesse, die in den letzten Paragraphen abgeleiteten speziellen Beziehungen mit den früher für die nämlichen physikalischen Systeme auf Grund allgemeinerer Voraussetzungen, unabhängig von jeder Molekulartheorie, in § 224, § 226, § 228 aufgestellten entsprechenden Gleichungen zu kombinieren.

Zu diesem Zweck müssen wir nach unserer jetzigen Bezeichnung für das dortige c nach (162) setzen:

$$c = \frac{M_2}{M_1} = \frac{n_1 m_1 + n_2 m_2 + \ldots}{n_0 m_0}. \tag{235}$$

Ferner ist zu bedenken, daß das dortige m, das Molekulargewicht des dampfförmigen Lösungsmittels, hier mit m_0', und das dortige auf die Masseneinheit des Lösungsmittels bezogene r hier mit $\frac{r}{m_0}$ zu bezeichnen ist.

Durch Ausführung dieser Substitutionen werden in der Tat die damaligen Gleichungen (181) und (183) identisch mit den jetzigen (234) und (231), wenn man jedesmal der dort auftretenden für alle diese Gesetzmäßigkeiten charakteristischen positiven Größe φ den Wert beilegt:

$$\varphi = \frac{R(n_1 + n_2 + \ldots)}{n_1 m_1 + n_2 m_2 + \ldots}. \tag{236}$$

§ 272. Auch der Betrag des osmotischen Druckes P ergibt sich natürlich aus der früheren allgemeinen Formel (190), wenn man darin den soeben in (236) abgeleiteten Wert der charakteristischen Größe φ und für c den

Ausdruck (235) setzt:

$$P = \frac{RT}{n_0 m_0 v}(n_1 + n_2 + n_3 + \ldots). \tag{237}$$

Hier bedeutet v, wie früher, das spezifische Volumen der Lösung, also das Produkt $n_0 m_0 v$ nahezu das ganze Volumen der Lösung: V, und daraus folgt:

$$P = \frac{RT}{V}(n_1 + n_2 + n_3 + \ldots), \tag{238}$$

die bekannte VAN'T HOFFsche Beziehung, die nach (16) identisch ist mit der Zustandsgleichung einer Mischung idealer Gase mit den Molekülzahlen $n_1,\ n_2,\ n_3, \ldots$.

Jeder der in den letzten Paragraphen abgeleiteten Sätze enthält eine Methode zur Bestimmung der Gesamtzahl $(n_1 + n_2 + \ldots)$ der in einer verdünnten Lösung vorhandenen fremden Moleküle. Wenn diese durch eine derartige Messung gefundene Zahl eine Abweichung zeigt von der aus dem Prozentgehalt der Lösung unter der Annahme normaler Moleküle berechneten Zahl, so muß also nach der entwickelten Theorie notwendig eine chemische Veränderung der gelösten Moleküle durch Dissoziation, Assoziation oder dgl. eingetreten sein, — eine Folgerung, die große Bedeutung erlangt hat für die Beurteilung der chemischen Natur verdünnter Lösungen.

Der Zusammenhang zwischen der Gefrierpunktserniedrigung und der Molekülzahl des gelösten Stoffes ist zuerst von RAOULT experimentell mit voller Schärfe nachgewiesen worden, thermodynamisch begründet und erweitert wurde er von VAN'T HOFF mittels seiner Theorie des osmotischen Druckes. Die volle Durchführung auch für Elektrolyte hat ARRHENIUS ermöglicht durch seine Theorie der elektrolytischen Dissoziation, während unabhängig davon auch die Thermodynamik gerade auf dem hier beschriebenen Wege zu der Forderung chemischer Umwandlungen gelöster Stoffe in verdünnten Lösungen geführt hat.

§ 273. Alle in den letzten Paragraphen besprochenen Schlußfolgerungen gründen sich auf die in den Gleichungen (209) ausgedrückte Voraussetzung, daß die Energie und das Volumen einer verdünnten Lösung lineäre Funktionen der Molekülzahlen sind. Daher liegt es nahe, für die auffallenden Abweichungen, welche sich hinsichtlich des Einflusses der Verdünnung auf die Gefriertemperatur bzw. den osmotischen Druck der Lösungen starker Elektrolyte gegenüber dem OSTWALDschen Verdünnungsgesetz (§ 262) aus vielen Messungen ergeben haben, jene Gleichungen verantwortlich zu machen, und in der Tat gelangt man durch eine schon oben im § 252 angedeutete Verallgemeinerung derselben auf Grund der von J. CH. GHOSH für die

elektrische Leitfähigkeit starker Elektrolyte eingeführten Vorstellungen zu einer mit den vorliegenden Messungen gut übereinstimmenden Abhängigkeit der Gefrierpunktserniedrigung von der Verdünnung. Das soll im folgenden für das spezielle Beispiel eines binären Elektrolyten, etwa KCl oder NaCl, gezeigt werden.

Nach J. Ch. Ghosh sind alle gelösten Moleküle vollständig in ihre Ionen dissoziiert. Daher ist das Symbol des Systems nach (216):

$$n_0 m_0,\ \ n_1 m_1,\ \ n_2 m_2 \mid n_0' m_0',$$

wo n_0, n_0', m_0 und m_0' die Molekülzahlen und die Molekulargewichte des Lösungsmittels (Wasser) in den beiden aneinandergrenzenden Phasen, $n_1 = n_2$, m_1 und m_2 die Ionenzahlen und -gewichte in der verdünnten Lösung bezeichnen.

Die Konzentrationen aller Molekülzahlen sind bis auf verschwindend kleine Größen:

(238a)
$$\begin{cases} c_0 = \dfrac{n_0}{n_0 + n_1 + n_2} = 1 - \dfrac{2n_1}{n_0} = 1 - 2c \\ c_1 = c_2 = \dfrac{n_1}{n_0 + n_1 + n_2} = \dfrac{n_1}{n_0} = c \\ c_0' = \dfrac{n_0'}{n_0'} = 1. \end{cases}$$

Zur Bestimmung des Ausdrucks der charakteristischen Funktion Φ der Lösung bedienen wir uns der nämlichen Methode wie in den §§ 250 bis 254, nur mit dem Unterschied, daß wir jetzt, gemäß den Ausführungen des § 252, in dem Ausdruck der Energie der Lösung:

$$U = n_0 u_0 + n_1 (u_1 + u_2)$$

den Koeffizienten von n_1 noch als abhängig von der Konzentration $\dfrac{n_1}{n_0} = c$ annehmen. Da mit verschwindendem c, oder mit unbegrenzt wachsendem n_0 bei konstantem n_1, das Korrektionsglied jedenfalls verschwindet, so bedeutet dasselbe die zur vollständigen Trennung der Ionen erforderliche Energie, oder das elektrische Potential der Ionen aufeinander, wird also annähernd umgekehrt proportional sein der mittleren Entfernung zweier benachbarter entgegengesetzt geladener Ionen voneinander, oder, was im wesentlichen auf dasselbe hinauskommt, direkt proportional der dritten Wurzel aus der Konzentration c.

Wir setzen daher statt (209):

(238b)
$$U = n_0 u_0 + n_1 \left(\overline{u} - \beta \cdot \sqrt[3]{c}\right),$$

wo $\overline{u}$ noch von Temperatur und Druck abhängt, während wir die positive Größe β als konstant betrachten.

Damit sind alle weiteren Schritte zur Bestimmung von Φ vorgezeichnet. Die Betrachtung der mit dem Druck p multiplizierten Glieder können wir uns ersparen, indem wir p als verschwindend klein voraussetzen. Für die Entropie der Lösung ergibt sich dann wie in (211):

$$S = n_0 s_0 + n_1 \overline{s} + C,$$

wo

$$d\overline{s} = \frac{d\overline{u}}{T}$$

und

$$C = -R(n_0 \log c_0 + n_1 \log c_1 + n_2 \log c_2),$$

endlich für die charakteristische Funktion der Lösung statt (215):

$$\Phi = S - \frac{U}{T} = n_0 \varphi_0 + n_1 \left(\overline{\varphi} + \frac{\beta}{T} \sqrt[3]{c} \right) + C, \tag{238c}$$

wo

$$\overline{\varphi} = \overline{s} - \frac{\overline{u}}{T}.$$

Das thermodynamische Gleichgewicht zwischen der flüssigen und der festen Phase erfordert die Bedingung:

$$\delta\Phi + \delta\Phi' = 0,$$

also, wenn man wie in (228) setzt:

$$\delta n_0 : \delta n_1 : \delta n_2 : \delta n_0' = -1 : 0 : 0 : \frac{m_0}{m_0'}$$

und ferner bedenkt, daß dann:

$$\delta c = -\frac{n_1}{n_0^2}\,\delta n_0, \qquad \delta C = -R \log c_0 \cdot \delta n_0$$

$$-\varphi_0 + n_1 \cdot \frac{1}{3} \frac{\beta}{T} c^{-2/3} \frac{n_1}{n_0^2} + R \log c_0 + \frac{m_0}{m_0'} \varphi_0' = 0,$$

oder nach (238a):

$$\frac{1}{R} \left(\frac{m_0}{m_0'} \varphi_0' - \varphi_0 \right) = \log K = 2c - \frac{1}{3} \frac{\beta}{RT} c^{4/3}. \tag{238d}$$

Diese Beziehung tritt an die Stelle der früheren Gleichung (230) und liefert für die Gefrierpunktsänderung an Stelle des einfachen VAN'T HOFFschen Gesetzes (231) den Ausdruck:

$$T - T_0 = 2c\left(1 - \frac{1}{6}\frac{\beta}{RT}\sqrt[3]{c}\right) \cdot \frac{RT^2}{r} \tag{238e}$$

in Übereinstimmung mit den Ergebnissen der bisherigen Messungen.[1]

Die Verallgemeinerung der entwickelten Theorie auf teilweise dissoziierte Elektrolyte liegt auf der Hand und zeigt uns die vollkommene Analogie mit der Theorie der gewöhnlichen Zustandsgleichung eines Gases. Ebenso nämlich wie die Abweichungen in dem Verhalten eines Gases von den Gesetzen idealer Gase auf zwei gänzlich verschiedene Ursachen: auf chemische Wirkungen (Dissoziation) oder auf physikalische Wirkungen (Anziehungskräfte zwischen den unveränderlichen Molekülen) zurückgeführt werden können (§ 43), genau so kann auch die Veränderlichkeit, welche der Quotient der Gefrierpunktserniedrigung durch die Konzentration der gelösten Moleküle bei fortschreitender Verdünnung zeigt, bedingt sein entweder durch chemische Veränderungen oder durch physikalische Wechselwirkungen zwischen den gelösten Molekülen. Je nachdem die eine oder die andere Ursache oder beide zugleich wirksam sind, erhält man für den Einfluß der fortschreitenden Verdünnung das OSTWALDsche, das GHOSHsche, oder ein allgemeineres, die beiden genannten als spezielle Fälle enthaltendes Verdünnungsgesetz.

§ 274. Jede der beiden Phasen enthält beide Bestandteile in merklicher Menge. Der wichtigste hierhergehörige Fall ist der der Verdampfung einer flüssigen Lösung, in welcher nicht nur das Lösungsmittel, sondern auch der gelöste Stoff flüchtig ist. Da die Anwendbarkeit der allgemeinen Gleichgewichtsbedingung (218) auf eine Mischung idealer Gase nicht davon abhängt, ob die Mischung als eine verdünnte Lösung angesehen werden kann, so gilt jene Gleichung hier in entsprechender Annäherung ohne Rücksicht auf die Zusammensetzung des Dampfes, während dagegen die Flüssigkeit als verdünnte Lösung angenommen werden muß.

Im allgemeinen erhält man nach (216) als Symbol des Systems:

$$n_0\, m_0,\ \ n_1\, m_1,\ \ n_2\, m_2, \ldots \mid n_0'\, m_0',\ \ n_1'\, m_1',\ \ n_2'\, m_2', \ldots,$$

wobei wie bisher die Ziffer 0 sich auf das Lösungsmittel, die Ziffern 1, 2, 3,... sich auf die verschiedenen Molekülarten der gelösten Stoffe

[1] Vgl. NOYES und FALK, J. Amer. Chem. Soc. **32**, p. 101, 1910; J. CH. GHOSH, Trans. Chem. Soc. **113**, p. 707, 790, 1918.

beziehen. Durch die Hinzufügung der Striche bei den Molekulargewichten (m_0', m_1', $m_2', \ldots$) ist die Möglichkeit offen gelassen, daß die verschiedenen Molekülarten im Dampfe ein anderes Molekulargewicht besitzen als in der Flüssigkeit.

In der Flüssigkeit ist die Gesamtzahl der Moleküle:

$$n = n_0 + n_1 + n_2 + \ldots \qquad \text{(nahe gleich } n_0\text{)}.$$

Im Dampf sei dieselbe:

$$n' = n_0' + n_1' + n_2' + \ldots$$

Dann sind die Konzentrationen der einzelnen Molekülarten in der Flüssigkeit:

$$c_0 = \frac{n_0}{n}, \qquad c_1 = \frac{n_1}{n}, \qquad c_2 = \frac{n_2}{n}, \ \ldots,$$

im Dampf:

$$c_0' = \frac{n_0'}{n'}, \qquad c_1' = \frac{n_1'}{n'}, \qquad c_2' = \frac{n_2'}{n'}. \ \ldots$$

Da das hier betrachtete System verschiedene Arten von Veränderungen erleiden kann, so sind auch verschiedene Gleichgewichtsbedingungen zu erfüllen, deren jede sich auf eine bestimmte Umwandlungsart bezieht. Betrachten wir zuerst diejenige Veränderung:

$$\begin{aligned} &\nu_0 : \nu_1 : \nu_2 : \ldots : \nu_0' : \nu_1' : \nu_2' : \ldots \\ &= \delta n_0 : \delta n_1 : \delta n_2 : \ldots : \delta n_0' : \delta n_1' : \delta n_2' : \ldots, \end{aligned}$$

welche in der Verdampfung eines gelösten Moleküls der ersten Art besteht, so ist:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 0, \ \ldots \quad \nu_0' = 0, \quad \nu_1' = \frac{m_1}{m_1'}, \quad \nu_2' = 0, \ \ldots$$

und die Gleichgewichtsbedingung (218) wird:

$$-\log c_1 + \frac{m_1}{m_1'} \log c_1' = \log K$$

oder:

$$\frac{c_1'^{\frac{m_1}{m_1'}}}{c_1} = K.$$

Diese Gleichung spricht den NERNSTschen Verteilungssatz aus. Wenn der gelöste Stoff in beiden Phasen das nämliche Molekulargewicht besitzt ($m_1' = m_1$), so besteht im Gleichgewicht ein festes, nur von Druck und Temperatur abhängiges, von der Anwesenheit der übrigen Molekülarten aber unabhängiges Verhältnis der Konzentrationen c_1 und c_1' in der Flüssigkeit und im Dampf. Wenn aber der gelöste Stoff etwa in der Flüssigkeit sich polymerisiert, so tritt die in der letzten Gleichung geforderte Beziehung an die Stelle des einfachen Verhältnisses.

Betrachten wir andrerseits die Veränderung, welche besteht in der Verdampfung eines Moleküls des Lösungsmittels. Dann ist:

$$\nu_0 = -1, \quad \nu_1 = 0, \quad \nu_2 = 0, \ \ldots \quad \nu_0' = \frac{m_0}{m_0'}, \nu_1' = 0, \quad \nu_2' = 0, \ \ldots$$

und die Gleichgewichtsbedingung wird:

$$-\log c_0 + \frac{m_0}{m_0'} \log c_0' = \log K.$$

Hierin ist:
$$-\log c_0 = \log \frac{n}{n_0} = \log \left(1 + \frac{n_1 + n_2 + \ldots}{n_0}\right)$$

(239)
$$= \frac{n_1 + n_2 + \ldots}{n_0},$$

folglich:

(240)
$$\frac{n_1 + n_2 + \ldots}{n_0} + \frac{m_0}{m_0'} \log c_0' = \log K,$$

wobei die Quotienten $\frac{n_1}{n_0}$, $\frac{n_2}{n_0}$,..., die Konzentrationen der gelösten Moleküle in der Flüssigkeit, kleine Werte haben. Nun sind zwei Fälle zu unterscheiden:

E n t w e d e r bilden die Moleküle m_0 im Dampf nur einen kleinen oder wenigstens nur einen mäßigen Teil der Gesamtzahl der Dampfmoleküle. Dann kann man die kleinen Zahlen $\frac{n_1}{n_0}$, $\frac{n_2}{n_0} \ldots$ gegen den log vernachlässigen und schreiben:

$$\frac{m_0}{m_0'} \log c_0' = \log K.$$

Ein Beispiel hierfür bietet die Verdampfung einer verdünnten Lösung, wenn das Lösungsmittel nicht sehr stark flüchtig ist, z. B. Alkohol in Wasser.

O d e r: Die Moleküle m_0 sind im Dampf den übrigen Molekülen an Zahl stark überlegen, wie z. B. wenn in der flüssigen Phase Alkohol das

Lösungsmittel, Wasser der gelöste Stoff ist. Dann ist c_0' nahe $= 1$, und man kann jene Vereinfachung der Gleichgewichtsbedingung nicht ohne merklichen Fehler vornehmen; vielmehr hat man wie in (239)

$$\log c_0' = -\frac{n_1' + n_2' + \dots}{n_0'},$$

wodurch die Gleichung (240) wird:

$$\frac{n_1 + n_2 + \dots}{n_0} - \frac{m_0}{m_0'} \cdot \frac{n_1' + n_2' + \dots}{n_0'} = \log K.$$

Behandeln wir diese Gleichung genau nach dem Muster der Gleichung (230), so ergibt sich schließlich wie in (231):

(240a) $$T - T_0 = \left(\frac{n_1 + n_2 + \dots}{n_0 m_0} - \frac{n_1' + n_2' + \dots}{n_0' m_0'} \right) \cdot \frac{R T_0^2 m_0}{r}.$$

Hier ist r die Wärmetönung bei der Verdampfung eines Moleküls des Lösungsmittels, also $\frac{r}{m_0}$ die Wärmetönung bei der Verdampfung der Masseneinheit des Lösungsmittels.

Wir bemerken zunächst wiederum, daß das Lösungsmittel immer nur mit der Masse, nicht aber mit der Molekülzahl bzw. dem Molekulargewicht in die Formel eingeht, während dagegen bei den gelösten Stoffen der Molekularzustand charakteristisch ist für ihren Einfluß auf die Verdampfung. Im übrigen enthält die Formel eine Verallgemeinerung des obigen (§ 269) VAN'T HOFFschen Gesetzes der Siedepunktserhöhung, indem hier statt der Zahl der in der Flüssigkeit gelösten Moleküle $n_1 + n_2 + \dots$ die Differenz der in der Masseneinheit der Flüssigkeit gelösten und der in der Masseneinheit des Dampfes gelösten Moleküle auftritt. Je nachdem also die Masseneinheit der Flüssigkeit oder die Masseneinheit des Dampfes im ganzen mehr gelöste Moleküle enthält, resultiert für die Lösung eine Siedepunktserhöhung oder eine Siedepunktserniedrigung; im Grenzfall, wenn beide Größen gleich sind, also das Gemisch konstant siedet, wird die Siedepunktsänderung gleich Null, wie das schon früher (§ 219) von einem allgemeineren Standpunkte aus gefolgert wurde. Entsprechende Sätze gelten natürlich für die Dampfdruckänderung.

Ganz analoge Beziehungen ergeben sich offenbar in der nämlichen Weise auch für andere Aggregatzustände; so lautet z. B. das Gefrierpunktsgesetz in der allgemeineren Fassung: Wenn aus einer verdünnten Lösung nicht nur das Lösungsmittel, sondern auch der gelöste Stoff ausfriert, in der Weise, daß die festen Stoffe zusammen ebenfalls eine verdünnte Lösung bilden, wie z. B.

beim Erstarren mancher Legierungen, so ist die Gefrierpunktserniedrigung nicht proportional der Konzentration des gelösten Stoffes in der Flüssigkeit, sondern proportional der Differenz der Konzentrationen des gelösten Stoffes in der flüssigen und in der festen Phase, sie wechselt also auch zugleich mit dieser Differenz ihr Vorzeichen. Den Grenzfall bilden die eutektischen Mischungen, die beim Gefrieren ihre Zusammensetzung nicht ändern. Bei diesen ändert sich der Gefrierpunkt gar nicht mit der Konzentration (vgl. oben § 219b).

Während so die Verteilung jeder einzelnen Molekülart auf beide Phasen geregelt ist, stellt sich das Gleichgewicht der verschiedenen Molekülarten innerhalb einer jeden einzelnen Phase ganz nach den oben § 262f. entwickelten Gesetzmäßigkeiten her. Wir treffen also hier wieder auf die nämlichen Gesetze der Dissoziation, Assoziation usw.

§ 275. Drei unabhängige Bestandteile in einer Phase. Wenn eine verdünnte Lösung außer dem Lösungsmittel zwei verschiedene gelöste Stoffe enthält, so werden sich die letzteren, falls sie keine neuen Molekülarten miteinander bilden, durchaus nicht gegenseitig beeinflussen; denn dann ist keine Umwandlung zwischen ihnen möglich und daher auch keine besondere Gleichgewichtsbedingung zu erfüllen. Mischt man also etwa eine verdünnte wäßrige Lösung eines Elektrolyten mit der verdünnten Lösung eines andern Elektrolyten, der mit dem ersten keinerlei chemische Reaktionen eingeht, so wird sich jede Lösung so verhalten, als wenn sie mit dem entsprechenden Quantum reinen Wassers verdünnt würde; so wird auch ihr Dissoziationsgrad in einer der Verdünnung entsprechenden Weise zunehmen.

Anders ist es, wenn die beiden Elektrolyte ein Ion gemeinschaftlich haben, wie z. B. Essigsäure und essigsaures Natron. In diesem Falle hat man vor der Vermischung die beiden Systeme:

$$n_0\ H_2O, \quad n_1\ H_4C_2O_2, \quad n_2\ \overset{+}{H}, \quad n_3\ H_3\overset{-}{C}_2O_2$$

und

$$n_0'\ H_2O, \quad n_1'\ NaH_3C_2O_2, \quad n_2'\ \overset{+}{Na}, \quad n_3'\ H_3\overset{-}{C}_2O_2,$$

wobei, ganz wie in (222), für die erste Lösung die Gleichung gilt

$$\frac{c_2^2}{c_1} = K \quad \text{oder} \quad \frac{n_2^2}{n_1 n_0} = K \tag{241}$$

und ebenso für die zweite Lösung

$$\frac{c_2'^2}{c_1'} = K' \quad \text{oder} \quad \frac{n_2'^2}{n_1' n_0'} = K'. \tag{242}$$

Nach der Vermischung aber hat man das System:

$$\overline{n}_0\ \mathrm{H_2O},\quad \overline{n}_1\ \mathrm{H_4C_2O_2},\quad \overline{n}_2\ \mathrm{NaH_3C_2O_2},\quad \overline{n}_3\ \overset{+}{\mathrm{H}},\quad \overline{n}_4\ \overset{+}{\mathrm{Na}},\quad \overline{n}_5\ \mathrm{H_3\overset{-}{C}_2O_2},$$

wobei notwendig:

$$(243)\qquad \begin{cases} \overline{n}_0 = n_0 + n_0' & (\text{Anzahl der } \mathrm{H_2O}\text{-Moleküle}) \\ \overline{n}_2 + \overline{n}_4 = n_1' + n_2' & (\text{Anzahl der Na-Atome}) \\ \overline{n}_1 + \overline{n}_3 = n_1 + n_2 & (\text{Anzahl der H-Atome}) \\ \overline{n}_3 + \overline{n}_4 = \overline{n}_5 & (\text{Anzahl der freien Ionen}). \end{cases}$$

In dem letzten System ist die Gesamtzahl aller Moleküle:

$$\overline{n} = \overline{n}_0 + \overline{n}_1 + \overline{n}_2 + \overline{n}_3 + \overline{n}_4 + \overline{n}_5 \qquad (\text{nahe gleich } \overline{n}_0).$$

Die Konzentrationen der einzelnen Molekülarten sind:

$$\overline{c}_0 = \frac{\overline{n}_0}{\overline{n}},\quad \overline{c}_1 = \frac{\overline{n}_n}{\overline{n}},\quad \overline{c}_2 = \frac{\overline{n}_2}{\overline{n}},\quad \overline{c}_3 = \frac{\overline{n}_3}{\overline{n}},\quad \overline{c}_4 = \frac{\overline{n}_4}{\overline{n}},\quad \overline{c}_5 = \frac{\overline{n}_5}{\overline{n}}.$$

In dem System sind zwei verschiedene Umwandlungen:

$$\nu_0 : \nu_1 : \nu_2 : \nu_3 : \nu_4 : \nu_5 = \delta\overline{n}_0 : \delta\overline{n}_1 : \delta\overline{n}_2 : \delta\overline{n}_3 : \delta\overline{n}_4 : \delta\overline{n}_5$$

möglich, nämlich erstens die Dissoziation eines Moleküls Essigsäure:

$$\nu_0 = 0,\qquad \nu_1 = -1,\qquad \nu_2 = 0,\qquad \nu_3 = 1,\qquad \nu_4 = 0,\qquad \nu_5 = 1,$$

woraus nach (218) als Gleichgewichtsbedingung folgt:

$$-\log\overline{c}_1 + \log\overline{c}_3 + \log\overline{c}_5 = \log K,$$

oder:

$$\frac{\overline{c}_3\overline{c}_5}{\overline{c}_1} = K,$$

oder:

$$(244)\qquad \frac{\overline{n}_3 \cdot \overline{n}_5}{\overline{n}_1 \cdot \overline{n}_0} = \frac{\overline{n}_3\overline{n}_5}{\overline{n}_1(n_0 + n_0')} = K,$$

zweitens die Dissoziation eines Moleküls Natriumacetat:

$$\nu_0 = 0,\qquad \nu_1 = 0,\qquad \nu_2 = -1,\qquad \nu_3 = 0,\qquad \nu_4 = 1,\qquad \nu_5 = 1$$

woraus als Gleichgewichtsbedingung folgt:

$$-\log \bar{c}_2 + \log \bar{c}_4 + \log \bar{c}_5 = \log K',$$

oder:

$$\frac{\bar{c}_4 \bar{c}_5}{\bar{c}_2} = K',$$

oder:

$$\frac{\overline{n}_4 \cdot \overline{n}_5}{\overline{n}_2 \cdot \overline{n}_0} = \frac{\overline{n}_4 \overline{n}_5}{\overline{n}_2 (n_0 + n_0')} = K'. \tag{245}$$

Die Größen K und K' sind hier die nämlichen wie oben in (241) und (242), da sie außer von T und p nur von der Art der betreffenden Umwandlung, nicht aber von den Konzentrationen und von anderen daneben möglichen Umwandlungen abhängen. Aus den beiden Gleichgewichtsbedingungen (244) und (245) zusammen mit den vier Gleichungen (243) folgen eindeutig die Werte der sechs Molekülzahlen $\overline{n}_0$, $\overline{n}_1, \ldots \overline{n}_5$, wenn die beiden ursprünglich vorhandenen Lösungen, also auch die Molekülzahlen n_0, $n_1, \ldots$ und n_0', $n_1', \ldots$ gegeben sind.

§ 276. Die Bedingung, daß die beiden ursprünglichen Lösungen von Essigsäure und von essigsaurem Natron „isohydrisch“ sind, d. h. bei ihrer Vermischung keinerlei Änderung ihres Dissoziationsgrades erleiden, wird offenbar ausgedrückt durch die beiden Gleichungen:

$$\overline{n}_1 = n_1, \qquad \overline{n}_2 = n_1',$$

welche aussprechen, daß die Anzahl der undissoziierten Moleküle Essigsäure und Natriumacetat in den ursprünglichen Lösungen gleich der in der Mischung ist. Daraus folgt nach (243) sogleich weiter:

$$\overline{n}_3 = n_2, \qquad \overline{n}_4 = n_2', \qquad \overline{n}_5 = n_2 + n_2'.$$

Diese Werte in die Gleichungen (244) und (245) eingesetzt und mit (241) und (242) verbunden ergeben:

$$\frac{n_2(n_2 + n_2')}{n_1(n_0 + n_0')} = K = \frac{n_2^2}{n_1 n_0}$$
$$\frac{n_2'(n_2 + n_2'))}{n_1'(n_0 + n_0')} = K' = \frac{n_2'^2}{n_1' n_0'},$$

woraus als einzige Bedingung der Isohydrie folgt:

$$\frac{n_2}{n_0} = \frac{n_2'}{n_0'} \quad \text{oder} \quad c_2 = c_2'(= c_3 = c_3'),$$

d. h. die beiden Lösungen sind isohydrisch, wenn in ihnen die Konzentration des gemeinsamen Ions $H_3\bar{C}_2O_2$ die nämliche ist. Dieser Satz wurde zuerst von ARRHENIUS ausgesprochen und an zahlreichen Messungen verifiziert.

In allen Fällen, wo die genannte Bedingung der Isohydrie nicht realisiert ist, müssen bei der Vermischung der Lösungen chemische Umwandlungen: Dissoziation und Assoziation, vor sich gehen. Von der Richtung und der Größe dieser Umwandlungen gewinnt man eine Vorstellung, wenn man sich die beiden gelösten Stoffe (Essigsäure und Natriumacetat) getrennt, und die gesamte Menge des Lösungsmittels (Wasser) so auf dieselben verteilt denkt, daß die Lösungen isohydrisch werden. Befinden sich z. B. beide Lösungen ursprünglich in normaler Verdünnung (1 g-Molekül in 1 Liter Lösung), so werden sie nicht isohydrisch sein, weil Natriumacetat in normaler Verdünnung bedeutend stärker dissoziiert ist, also eine größere Konzentration der $H_3\bar{C}_2O_2$-Ionen besitzt als Essigsäure. Um nun das gesamte Lösungswasser so auf die beiden Elektrolyte zu verteilen, daß die Konzentration des gemeinsamen Ions $H_3\bar{C}_2O_2$ in beiden Lösungen dieselbe wird, muß man dem schwächer dissoziierten Elektrolyt (Essigsäure) Lösungswasser entziehen und dies dem stärker dissoziierten Natriumacetat hinzufügen. Denn infolge der abnehmenden Verdünnung geht zwar die Dissoziation der Essigsäure zurück, die Konzentration der freien Ionen in der Säure wächst aber dennoch, wie man leicht aus § 262 findet, weil die Ionen auf eine kleinere Wassermenge zusammengedrängt werden. Umgekehrt nimmt die Dissoziation des Natriumacetats mit dem Wasserzusatz zu, die Konzentration der freien Ionen des Salzes nimmt aber ab, weil die Ionen sich durch eine größere Wassermenge verbreiten. So kann man es erreichen, daß die Konzentration des gemeinsamen Ions $H_3\bar{C}_2O_2$ in beiden Lösungen gleich wird, und dann sind die Lösungen isohydrisch, d. h. sie befinden sich in demjenigen Dissoziationszustand, der bei einer Vermischung der Lösungen nicht mehr geändert wird. Dies ist also zugleich auch der Zustand, den die beiden gemischten Normallösungen schließlich im Gleichgewicht annehmen; es folgt daraus der Satz, daß bei der Vermischung zweier gleich verdünnter Lösungen binärer Elektrolyte die Dissoziation des schwächer dissoziierten Elektrolyten (Essigsäure) noch weiter zurückgeht, während die des stärker dissoziierten (Natriumacetat) noch weiter zunimmt.

§ 277. Drei unabhängige Bestandteile in zwei Phasen. Wir behandeln zunächst ein Beispiel des einfachen Falls, daß die zweite Phase nur einen einzigen Bestandteil in merklicher Menge enthält. Ein solches Beispiel bietet die teilweise Auflösung eines schwer löslichen Salzes (Silberbromat) in einer Flüssigkeit (Wasser) bei Zusatz einer geringen Menge eines dritten Bestandteils (Silbernitrat) zur Lösung.

Das aus zwei Phasen bestehende System wird nach (216) dargestellt durch:

$$n_0 H_2O,\ \ n_1 AgBrO_3,\ \ n_2 AgNO_3,\ \ n_3 \overset{+}{A}g,\ \ n_4 \overset{-}{B}rO_3,\ \ n_5 \overset{-}{N}O_3, \mid n_0' AgBrO_3.$$

Die Konzentrationen sind:

$$c_0 = \frac{n_0}{n}, \quad c_1 = \frac{n_1}{n}, \quad c_2 = \frac{n_2}{n}, \quad \ldots\ldots, \quad c_0' = \frac{n_0'}{n_0'} = 1,$$

wobei $\qquad n = n_0 + n_1 + n_2 + n_3 + n_4 + n_5 \qquad$ (nahezu gleich n_0).

Von den möglichen Umwandlungen:

$$\nu_0 : \nu_1 : \nu_2 : \nu_3 : \nu_4 : \nu_5 : \nu_0' = \delta n_0 : \delta n_1 : \delta n_2 : \delta n_3 : \delta n_4 : \delta n_5 : \delta n_0'$$

ist in Betracht zu ziehen zunächst der Austritt eines Moleküls $AgBrO_3$ aus der Lösung, also:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 0, \quad \ldots\ldots \quad \nu_0' = 1,$$

woraus nach (218) die Gleichgewichtsbedingung folgt:

$$-\log c_1 + \log c_0' = \log K$$

oder:

$$c_1 = \frac{1}{K}, \tag{246}$$

d. h. die Konzentration der undissoziierten Moleküle Silberbromat in der gesättigten Lösung hängt ausschließlich von Temperatur und Druck ab, wird also durch das Nitrat überhaupt nicht beeinflußt.

Ferner ist zu berücksichtigen die Dissoziation eines Moleküls $AgBrO_3$ in die beiden Ionen:

$$\nu_0 = 0, \quad \nu_1 = -1, \quad \nu_2 = 0, \quad \nu_3 = 1, \quad \nu_4 = 1, \quad \nu_5 = 0, \quad \nu_0' = 0,$$

was nach (218) ergibt:

$$-\log c_1 + \log c_3 + \log c_4 = \log K',$$
$$\frac{c_3 c_4}{c_1} = K',$$

oder nach (246)

$$c_3 c_4 = \frac{K'}{K}, \tag{247}$$

d. h. das Produkt der Konzentrationen der Ionen $\overset{+}{\mathrm{Ag}}$ und $\overset{-}{\mathrm{BrO}}_3$ hängt nur von Temperatur und Druck ab. Jeder Umstand also, der eine Änderung in der Konzentration c_3 der $\overset{+}{\mathrm{Ag}}$-Ionen herbeiführt, beeinflußt in umgekehrtem Verhältnis die Konzentration c_4 der $\overset{-}{\mathrm{BrO}}_3$-Ionen. Da nun durch den Zusatz von Silbernitrat die Zahl der $\overset{+}{\mathrm{Ag}}$-Ionen in der Lösung jedenfalls vergrößert wird, so wirkt eben dadurch dieser Zusatz verkleinernd auf die Zahl der $\overset{-}{\mathrm{BrO}}_3$-Ionen und somit auch verkleinernd auf die Löslichkeit des bromsauren Salzes, welche offenbar durch die Summe $c_1 + c_4$ gemessen wird.

Endlich ist noch zu betrachten die Dissoziation eines Moleküls $\mathrm{AgNO_3}$ in die beiden Ionen:

$$\nu_0 = 0, \quad \nu_1 = 0, \quad \nu_2 = -1, \quad \nu_3 = 1, \quad \nu_4 = 0, \quad \nu_5 = 1, \quad \nu_0' = 0,$$

welche für das Gleichgewicht nach (218) die Beziehung erfordert:

$$\frac{c_3 c_5}{c_2} = K''. \tag{248}$$

Zu den drei Gleichungen (246), (247) und (248) kommt als vierte noch die Bedingung:

$$c_3 = c_4 + c_5$$

und als fünfte der Wert von $c_2 + c_5$, welcher durch die Menge des zugesetzten Nitrats gegeben ist, so daß hieraus die fünf Unbekannten c_1, c_2, c_3, c_4, c_5 im Gleichgewichtszustand eindeutig bestimmt werden.

Die Theorie derartiger Löslichkeitsbeeinflussungen ist zuerst von Nernst entwickelt und von ihm, und später namentlich auch von Noyes durch Messungen bestätigt worden.

§ 278. Der allgemeinere Fall, daß jede der beiden Phasen alle drei Bestandteile enthält, wird u. a. verwirklicht bei der Verteilung eines löslichen

Stoffes zwischen zwei Lösungsmitteln, die sich selber in geringem Grade gegenseitig lösen (z. B. Wasser und Äther). Der Gleichgewichtszustand des Systems wird vollständig bestimmt durch eine Kombination derjenigen Bedingungen, welche für den Übertritt eines Moleküls aus der einen Phase in die andere, und derjenigen, welche für die chemische Umwandlung der Moleküle innerhalb einer und derselben Phase gelten. Die ersteren lassen sich zusammenfassen in den Verteilungssatz von NERNST (§ 274), wonach für jede in beiden Phasen mit dem nämlichen Molekulargewicht vorkommende Molekülgattung ein konstantes, von der Anwesenheit anderer gelöster Moleküle unabhängiges Teilungsverhältnis existiert, die letzteren in die Sätze, welche für drei unabhängige Bestandteile in einer einzigen Phase gelten (§ 275), und zu denen auch die ARRHENIUSsche Theorie isohydrischer Lösungen gehört.

§ 279. In ganz derselben Weise ist der Fall zu behandeln, daß vier oder mehr unabhängige Bestandteile zu einer oder mehreren Phasen zusammentreten. Immer läßt sich der Zustand des Systems durch das Symbol (216) ausdrücken, und immer läßt sich jede mögliche Umwandlung des Systems auf die Form (217) bringen, der dann die Gleichgewichtsbedingung (218) entspricht. Alle Gleichgewichtsbedingungen zusammen mit den festen Bedingungen ergeben dann die nach der Phasenregel vorauszusehende Anzahl Gleichungen, um den Gleichgewichtszustand des Systems zu bestimmen.

Wenn es sich um eine Lösung von mehreren gegenseitig umwandelbaren Stoffen, etwa elektrolytisch dissoziierbaren Salzen oder Säuren mit gemeinsamen Ionen handelt, so hat es im allgemeinen keinen Sinn mehr, von einem bestimmten „Dissoziationsgrad“ dieser Substanzen zu reden, da die Ionen ganz willkürlich zu dissoziierten Molekülen kombiniert werden können. Z. B. in der Lösung:

$$n_0 H_2O,\ n_1 NaCl,\ n_2 KCl,\ n_3 NaNO_3,\ n_4 KNO_3,\ n_5 \overset{+}{N}a,\ n_6 \overset{+}{K},\ n_7 \overset{-}{Cl},\ n_8 \overset{-}{N}O_3$$

läßt sich gar nicht entscheiden, welche der $\overset{+}{N}a$-Ionen dem NaCl und welche dem $NaNO_3$ zuzurechnen sind. Hier bleibt zur Charakterisierung des Zustandes nichts übrig, als zu den wirklich in der Lösung enthaltenen Molekülzahlen bez. den betreffenden Konzentrationen zurückzugehen und sich lediglich auf die Angabe dieser zu beschränken.

Das genannte System wird von Wasser und von vier Salzen gebildet, es enthält aber trotzdem außer dem Lösungsmittel nur drei unabhängige Bestandteile, weil durch die Menge des Na, des K und des Cl die des NO_3 von vornherein bereits mitbestimmt ist (§ 198). Demgemäß sind auch die Konzentrationen aller einzelnen Molekülarten nach § 204 ($\alpha = 4,\ \beta = 1$) bei

gegebenem Druck und Temperatur durch drei derselben vollkommen bestimmt. Dies gilt natürlich unabhängig davon, ob, wie es höchstwahrscheinlich zutrifft, bei der Aufstellung der Gleichgewichtsbedingungen für die betrachtete Lösung noch andere Molekülarten und andere chemische Umwandlungen als die hier vorgesehenen berücksichtigt werden müssen.

§ 280. Wenn in dem System (216) von beliebig vielen unabhängigen Bestandteilen in beliebig vielen Phasen die Gleichgewichtsbedingung (218) nicht erfüllt ist, wenn also für irgend eine virtuelle isotherm-isobare Änderung:

$$\sum \nu_0 \log c_0 + \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots \gtrless \log K,$$

so ist die Richtung der in der Natur tatsächlich eintretenden Veränderung durch die Bedingung $d\Phi > 0$ (§ 147) bestimmt. Bezeichnen wir also jetzt mit ν_0, ν_1, ν_2,... einfache ganze Zahlen, welche nicht nur proportional, sondern auch von gleichem Vorzeichen sind wie die bei der wirklichen Umwandlung eintretenden Änderungen der Molekülzahlen, so ergibt sich aus (215) für die Richtung der in der Natur eintretenden isotherm-isobaren Veränderung ganz allgemein, sei es daß es sich um eine chemische Umwandlung innerhalb einer einzelnen Phase oder um den Übergang von Molekülen zwischen verschiedenen Phasen handelt:

$$\sum \nu_0 \log c_0 + \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots < \log K, \tag{249}$$

wobei K wieder durch (218) definiert ist.

Es liegt nahe, die Differenz der Ausdrücke rechts und links in Zusammenhang zu bringen mit dem zeitlichen Verlauf der Veränderung, und in der Tat läßt sich hieraus ein allgemeiner Satz für die Geschwindigkeit irreversibler isotherm-isobarer Prozesse ableiten, indem nämlich die Geschwindigkeit der Reaktion proportional jener Differenz gesetzt wird. Indessen soll dieser Gedanke hier nicht weiter verfolgt werden, da der in der betreffenden Gleichung auftretende Proportionalitätsfaktor ohne Hinzunahme spezieller atomistischer Vorstellungen gar nicht näher zu bestimmen ist.

Sechstes Kapitel. Absoluter Wert der Entropie. Theorem von Nernst.

§ 281. Wir sind bei früheren Gelegenheiten zu wiederholten Malen auf die Bemerkung geführt worden, daß das gesamte thermodynamische Verhalten einer Substanz bestimmt ist durch eine einzige charakteristische Funktion, deren Kenntnis ein für allemal genügt, um alle Bedingungen

chemisch-physikalischer Gleichgewichtszustände, an denen die Substanz beteiligt ist, eindeutig abzuleiten. Je nach der Wahl der unabhängigen Variablen ist aber die Form der charakteristischen Funktion verschieden: Für die Energie U und das Volumen V als unabhängige Variable ist die Entropie S die charakteristische Funktion, für die Temperatur T und das Volumen V als unabhängige Variable ist es die freie Energie F, für die Temperatur T und den Druck p als unabhängige Variable ist es die Funktion Φ (vgl. § 152a).

In jedem Falle läßt sich der Ausdruck der charakteristischen Funktion finden durch Integration über gewisse Größen, deren Werte durch geeignete Messungen bestimmt werden können. Dabei bleibt aber in dem Ausdruck der Entropie S eine additive Konstante a, in dem der freien Energie F eine Funktion von der Form $aT + b$ (§ 146), in dem der Größe Φ eine Funktion von der Form $a + \frac{b}{T}$ willkürlich wählbar (vgl. § 152b, § 239). Daß diese additiven Zusatzglieder absolut genommen gar keine physikalische Bedeutung haben, hängt damit zusammen, daß bei den Vorgängen in der Natur immer nur Differenzen der Energie, der Entropie, der freien Energie usw. in verschiedenen Zuständen einer Substanz eine Rolle spielen, bei deren Bildung jene Zusatzglieder sich gegenseitig kompensieren.

Ist aber einmal das Zusatzglied für irgend einen bestimmten Zustand einer Substanz fixiert worden, so behält es selbstverständlich seinen Wert für alle andern Zustände der Substanz unverändert bei, und zwar nicht nur für denselben Aggregatzustand und dieselbe chemische Modifikation, sondern auch für andere Aggregatzustände und andere Modifikationen, in welche die Substanz übergeführt werden kann.

Will man nun, nachdem die Größe der charakteristischen Funktion einer Substanz für eine bestimmte Modifikation nach Willkür festgesetzt worden ist, dieselbe für eine andere Modifikation finden, so muß man nach einem reversiblen physikalischen Übergang zwischen den beiden Modifikationen suchen, der einer Messung zugänglich ist, damit man die zur Berechnung der charakteristischen Funktion notwendige Integration auf irgend einem Wege durch das Übergangsgebiet hindurch ausführen kann. Derartige Übergänge sind aber in vielen Fällen gar nicht realisierbar, da sie häufig durch Gebiete labiler Zustände hindurchführen, und man ist daher in der Regel gar nicht imstande, von den thermodynamischen Eigenschaften der einen Modifikation direkt auf die der andern Schlüsse zu ziehen.

§ 282. In diese Lücke der Theorie greift nun ein Theorem ein, welches W. Nernst im Jahre 1906 aufgestellt hat[1] und welches durch die

[1] Nachr. d. Ges. d. Wissensch. zu Göttingen, Math. phys. Kl. 1906. Heft I. Sitz.-Ber.

seitherigen Erfahrungen gut bestätigt worden ist. Sein Inhalt läßt sich dahin formulieren, daß bei unbegrenzt abnehmender Temperatur die Entropie eines jeden chemisch homogenen (§ 67) Körpers von endlicher Dichte sich unbegrenzt einem bestimmten, vom Druck, vom Aggregatzustand und von der speziellen chemischen Modifikation unabhängigen Wert nähert.[1]

Die interessante Frage, welche Bedeutung dieser Satz für die molekularkinetische Auffassung der Entropie besitzt, kann hier nicht erörtert werden, da wir es hier nur mit der allgemeinen Thermodynamik zu tun haben.[2] Aber die wichtigen Folgerungen, welche er für die Gesetze physikalisch-chemischer Gleichgewichtszustände enthält, müssen hier Besprechung finden.

Zunächst ist leicht einzusehen, daß man, da der Wert der Entropie eine willkürliche additive Konstante enthält, jenen für $\lim T = 0$ eintretenden Wert unbeschadet der Allgemeinheit gleich Null setzen kann, so daß das Nernstsche Wärmetheorem nun lautet: Bei unbegrenzt abnehmender Temperatur nähert sich die Entropie eines jeden chemisch homogenen Körpers von endlicher Dichte unbegrenzt dem Werte Null. Damit ist über die additive Konstante a der Entropie aller chemisch homogener Substanzen in allen Zuständen eindeutig verfügt, und man kann von nun an in diesem Sinne von einem absoluten Wert der Entropie sprechen.

Dagegen bleibt nach § 281 in dem Werte der Energie U und in dem der freien Energie F eines Körpers noch eine additive Konstante b, und in dem Werte der Funktion Φ noch ein additives Glied von der Form $\frac{b}{T}$ unbestimmt und willkürlich wählbar.

§ 283. Für die späteren Anwendungen des Nernstschen Wärmetheorems werden wir uns über die Wahl der unabhängigen Variabeln für den Zustand einer Substanz entscheiden müssen, und wollen hierfür wiederum, wie in unserer früheren Darstellung, die Temperatur T und den Druck p nehmen, weil diese Größen erstens am bequemsten meßbar sind und zweitens den

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d. preuß. Akad. d. Wiss. v. 20. Dezbr. 1906.

[1] Diese Fassung des Theorems ist inhaltlich etwas weitergehend als die von Nernst a.a.O. selber gegebene, nach welcher für $\lim T = 0$ die Differenz der Entropien eines solchen Körpers in zwei verschiedenen Modifikationen gleich Null ist. Letzterer Satz läßt nämlich noch die Möglichkeit offen, daß die Entropie selber für $\lim T = 0$ negativ unendlich wird. Praktisch wichtig wird dieser Unterschied u. a. bei der Frage nach dem Wert der spezifischen Wärme beim Nullpunkt der absoluten Temperatur. (§ 284.) Allgemeineres bei P. Gruner, Verh. d. Deutschen Physik. Gesellschaft **14**, S. 655, S. 727, 1912.

[2] Über die Beziehungen des Nernstschen Wärmetheorems zur Quantentheorie vgl. M. Planck, Vorlesungen über die Theorie der Wärmestrahlung, 4. Aufl., § 185 (Leipzig, J. A. Barth).

Vorzug besitzen, daß sie in einem aus mehreren Phasen gebildeten System in allen Phasen die nämlichen Werte haben.

Mit den unabhängigen Variabeln T und p ist notwendig verbunden die Wahl der Funktion Φ als charakteristische Funktion, und wir haben zur Bestimmung von Φ die allgemeine Gleichung (150) des § 210:

$$\Phi = S - \frac{W}{T} \tag{250}$$

wobei nach (150b)

$$S = \int^{T} \frac{C_p\, dT}{T} \tag{251}$$

und

$$W = U + pV. \tag{252}$$

Ist Φ als Funktion von T und p bekannt, so folgen daraus nach § 152a eindeutig die Werte der Wärmefunktion W, des Volumens V und der Entropie S als Funktionen von T und p:

$$W = T^2 \frac{\partial \Phi}{\partial T} \tag{253}$$

$$V = -T \frac{\partial \Phi}{\partial p} \tag{254}$$

$$S = \Phi + T \frac{\partial \Phi}{\partial T}. \tag{255}$$

Um nun in diese ganz allgemein gültigen Gleichungen das NERNSTsche Wärmetheorem einzuführen, denken wir uns speziell das betrachtete System bestehend aus einem chemisch homogenen festen oder flüssigen Körper. Auf ideale Gase ist nämlich das Theorem nicht unmittelbar anwendbar, weil ideale Gase beim absoluten Nullpunkt der Temperatur keine endliche Dichte besitzen. Dann ist nach dem genannten Theorem (§ 282) für $T = 0$ $S = 0$. Folglich nach Gleichung (251):

$$\int^{0} \frac{C_p\, dT}{T} = 0.$$

Da nun sowohl C_p als auch T wesentlich positiv sind, so ist diese Gleichung nur dann, aber auch dann immer erfüllt, wenn die untere Grenze des

Integrals Null ist. Da aber die untere Grenze nicht von T abhängt, so gilt für jede beliebige Temperatur T:

$$S = \int_0^T \frac{C_p}{T}\, dT. \tag{256}$$

Diese Gleichung ist der mathematische Ausdruck für das NERNSTsche Wärmetheorem in seiner weitergehenden Fassung (vgl. § 284). Aus ihr folgt nach (250):

$$\Phi = \int_0^T \frac{C_p}{T}\, dT - \frac{W}{T} \tag{257}$$

oder auch, mit Substitution des Wertes von W, nach (150a):

$$\Phi = \int_0^T \frac{C_p}{T}\, dT - \frac{1}{T}\int^T C_p\, dT. \tag{258}$$

Hierdurch ist die charakteristische Funktion Φ für jeden chemisch homogenen festen oder flüssigen Körper bestimmt, bzw. auf Messungen der Wärmekapazität C_p zurückgeführt, bis auf ein additives Glied von der Form $\frac{b}{T}$, welches von der unteren Grenze des zweiten Integrals herrührt. Die von T unabhängige Größe b kann noch vom Druck p und von der chemischen Zusammensetzung des Körpers abhängen. In ihr bleibt eine additive Konstante ganz willkürlich (vgl. § 282 am Schluß).

§ 284. Aus den letzten Gleichungen lassen sich wichtige Schlüsse bezüglich des thermodynamischen Verhaltens fester und flüssiger Substanzen bei tieferen Temperaturen ziehen. Zunächst ist aus der Gleichung (256) die merkwürdige Folgerung zu entnehmen, daß die Wärmekapazität C_p eines jeden chemisch homogenen festen oder flüssigen Körpers bei unbegrenzt abnehmender Temperatur sich unbegrenzt dem Werte Null nähert; denn sonst könnte die Entropie S für endliche Temperaturen keinen endlichen Wert besitzen.[1] So auffallend dieses Resultat auf den ersten Augenblick erscheint, so wird es doch durch die bisherigen Messungen durchaus bestätigt.

Auch in bezug auf das Volumen V eines Körpers ergibt das NERNSTsche Wärmetheorem einen interessanten Schluß. Denn nach der Gleichung (84b)

[1] Nach der von NERNST selber gegebenen Fassung (§ 282) tritt an die Stelle der

ist mit Benutzung von (256):

$$\frac{\partial V}{\partial T} = -\frac{\partial S}{\partial p} = -\int_0^T \frac{1}{T}\frac{\partial C_p}{\partial p}\,dT.$$

Benutzt man hier die Gleichung (85), so folgt:

$$\frac{\partial V}{\partial T} = \int_0^T \frac{\partial^2 V}{\partial T^2}\,dT = \frac{\partial V}{\partial T} - \left(\frac{\partial V}{\partial T}\right)_0,$$

und daraus: $\left(\frac{\partial V}{\partial T}\right)_0 = 0$, d. h. der Ausdehnungskoeffizient jedes chemisch homogenen festen oder flüssigen Körpers nähert sich bei abnehmender Temperatur unbegrenzt dem Werte Null. Auch diese Schlußfolgerung ist mit den vorliegenden Erfahrungen gut im Einklang, da die Ausdehnungskoeffizienten der meisten Körper mit abnehmender Temperatur eine deutliche Abnahme zeigen.

§ 285. Nach einer von P. Debye[1] aus der Quantenhypothese abgeleiteten Formel ist bei tiefen Temperaturen die freie Energie eines festen Körpers:

(258a) $$F = U_0 - AT^4,$$

wo U_0, die Nullpunktsenergie, und der positive Proportionalitätsfaktor A nur vom Volumen V abhängen. Daraus ergibt sich für den Druck p und

Gleichung (256) die folgende:

$$S' - S = \int_0^T \frac{C_p' - C_p}{T}\,dT,$$

wobei $S' - S$ die Differenz der Entropien eines festen oder flüssigen Körpers in zwei verschiedenen Modifikationen bezeichnet. Dann kann man für $T = 0$ nur schließen, daß $C_p' = C_p$, aber nicht, daß $C_p' = C_p = 0$. Würden C_p und C_p' für $T = 0$ endlich bleiben, so wäre die Entropie S für $T = 0$ nicht gleich Null, sondern negativ unendlich. Es ist daher von Wichtigkeit zu bemerken, daß, wenn die Folgerung $C_p = 0$ durch die Erfahrung nicht bestätigt werden sollte, die Nernstsche Fassung des Theorems deshalb doch aufrecht erhalten werden könnte.

[1] P. Debye, Wolfskehl-Vorträge über die kinetische Theorie der Materie und der Elektrizität, Göttingen 1914, p. 19 (Leipzig, Teubner).

für die Wärmekapazität C_v bei konstantem Volumen nach (79a):

$$p = -\frac{\partial F}{\partial V} = -\frac{\partial U_0}{\partial V} + T^4 \frac{\partial A}{\partial V} \tag{258b}$$

$$C_v = -T\frac{\partial^2 F}{\partial T^2} = 12AT^3, \tag{258c}$$

in vorzüglicher Übereinstimmung mit den Messungen.

Die Kompressibilität κ (§ 14) nimmt bei der absoluten Temperatur Null den Wert an:

$$\kappa_0 = -\frac{1}{V_0 \cdot \left(\frac{\partial p}{\partial V}\right)_0}.$$

Daher berechnet sich der thermische Ausdehnungskoeffizient bei tiefen Temperaturen nach der Beziehung (6) zu:

$$\frac{1}{V_0}\left(\frac{\partial V}{\partial T}\right)_p = -\frac{1}{V_0}\frac{\left(\frac{\partial p}{\partial T}\right)_V}{\left(\frac{\partial p}{\partial V}\right)_T} = 4T^3\,\frac{\partial A}{\partial V}\cdot\kappa_0 = \frac{C_v}{3A}\,\frac{\partial A}{\partial V}\cdot\kappa_0.$$

Der Ausdehnungskoeffizient steht also zur spezifischen Wärme in einem von der Temperatur unabhängigen Verhältnis, — ein Satz, der von E. Grüneisen empirisch abgeleitet wurde.

§ 286. Betrachten wir jetzt zwei sich berührende Phasen der nämlichen chemisch homogenen festen oder flüssigen Substanz (z. B. Schmelzpunkt, Umwandlungspunkt allotroper Modifikationen). Dann ist, wenn n und n' die Molekülzahlen der beiden Körper bedeuten:

$$\Phi = n\varphi, \qquad \Phi' = n'\varphi',$$

wobei φ und φ' sich auf je ein Molekül m und m' beziehen und nur von T, p und der Natur der betreffenden Phase abhängen. Ganz ebenso wollen wir die auf ein Molekül m bezogenen Werte der Entropie S, der Energie U, der Wärmefunktion W, des Volumens V durch entsprechende kleine Buchstaben s, u, w, v bezeichnen, ungestrichen oder gestrichen, je nachdem sie der ersten oder der zweiten Phase angehören. Dagegen soll die Molekularwärme, um Verwechslungen mit der spezifischen Wärme c_p zu vermeiden, stets mit C_p bezeichnet werden.

Dann ist die Bedingung des Gleichgewichts der beiden Phasen nach § 261, Gleichung (221a):

$$\frac{\varphi'}{m'} - \frac{\varphi}{m} = 0,$$

oder, mit Substitution von (257):

$$\int_0^T \frac{1}{T}\left(\frac{C_p'}{m'} - \frac{C_p}{m}\right) dT - \frac{1}{T}\left(\frac{w'}{m'} - \frac{w}{m}\right) = 0.$$

Hierin bedeutet:

$$\frac{C_p'}{m'} - \frac{C_p}{m} = \Delta c_p \tag{259}$$

die Differenz der spezifischen Wärmen, und

$$\frac{w'}{m'} - \frac{w}{m} = r \tag{260}$$

die beim isotherm-isobaren Übergang der Masseneinheit aus der ungestrichenen in die gestrichene Phase von außen zuzuführende Wärme; also kürzer geschrieben:

$$\int_0^T \frac{\Delta c_p}{T}\, dT - \frac{r}{T} = 0 \tag{261}$$

oder auch, da nach (150a) und (260)

$$\Delta c_p = \frac{\partial r}{\partial T} \tag{261a}$$

$$\int_0^T \frac{1}{T}\,\frac{\partial r}{\partial T}\, dT - \frac{r}{T} = 0. \tag{262}$$

Diese Gleichung kann zur Berechnung des Schmelzpunkts oder der Umwandlungstemperatur der Substanz dienen, wenn die Übergangswärme r als Funktion der Temperatur durch Wärmemessungen gefunden ist.

So ist z. B. für die Umwandlung von rhombischem in monoklinen Schwefel nach Messungen von BROENSTED angenähert:

$$r = 1,57 + 1,15 \cdot 10^{-5} \cdot T^2 \text{ cal.}$$

Diese Formel, die allerdings bei tiefen Temperaturen ungenau werden muß, da sie nach (261a) zu einem Widerspruch mit dem DEBYEschen Gesetz (258c)

der spezifischen Wärme führt, ergibt für die Umwandlungstemperatur T nach der letzten Gleichung:

$$2,3 \cdot 10^{-5} \cdot T - \frac{1,57}{T} - 1,15 \cdot 10^{-5} \cdot T = 0$$

oder:
$$T = 369,5,$$

während die Messung $368,4$ ergab.

Für den Schmelzpunkt des Eises ist ferner:

$$r = 80, \qquad T = 273.$$

Folglich nach Gleichung (261):

$$\int_0^{273} \frac{\Delta c_p}{T}\, dT = \frac{80}{273} = 0,293.$$

Nun ist für $T = 273$ die Differenz der spezifischen Wärmen von Wasser und Eis: $\Delta c_p = 1,00 - 0,51 = 0,49$.

Also nähert sich diese Differenz bei abnehmender Temperatur dem Werte Null keinesfalls geradlinig, sondern mit sinkender Temperatur immer schneller, ganz im Sinne des DEBYEschen Gesetzes.

§ 287. Während für feste und flüssige chemisch homogene Körper die Entropie nach dem NERNSTschen Wärmetheorem sich mit abnehmender Temperatur bei jedem beliebigen Druck unbegrenzt dem Werte Null nähert, hat für ideale Gase die Frage nach der Entropie für tiefe Temperaturen T nur dann einen Sinn, wenn der Druck p kleiner ist als der Sättigungsdruck p_s des Gases bei der Temperatur T; denn für höhere Drucke ist der Gaszustand gar nicht mehr stabil, also auch nicht ideal. Daher nehmen wir $p < p_s$. Nun ist nach Gleichung (195) für ein ideales Gas:

(263) $$S = n(C_p \log T - R \log p + k).$$

Folglich haben wir:

(263a) $$S > n(C_p \log T - R \log p_s + k)$$

Wegen des Wertes von p_s vgl. § 288 am Schluß.

In Verbindung mit der durch (192) ausgedrückten Energie:

(264) $$U = n(C_v T + b)$$

und dem Volumen:

$$V = n\frac{RT}{p} \tag{265}$$

eines idealen Gases ergeben sich die Ausdrücke für die Wärmefunktion:

$$W = U + pV = n(C_pT + b) \tag{266}$$

und für die charakteristische Funktion:

$$\Phi = S - \frac{W}{T} = n\left(C_p \log T - R\log p + a - \frac{b}{T}\right) = n\varphi \tag{267}$$

eines chemisch homogenen idealen Gases.

Diese schon früher abgeleiteten Formeln gewinnen hier neues Interesse durch die Bedeutung der Konstanten a. Denn nach den Erörterungen des § 282 ist diese Konstante, welche NERNST als die „chemische Konstante" des betreffenden Gases bezeichnet, und welche für alle physikalischen und chemischen Reaktionen des Gases Bedeutung besitzt, ihrem absoluten Betrage nach vollkommen bestimmt und meßbar. Um sie zu finden, muß man das Gas auf irgend einem reversiblen der Messung zugänglichen Wege in den flüssigen oder festen Zustand überführen. Das geschieht am bequemsten durch direkte Kondensation. Daher bietet die Aufsuchung der Gesetze des gesättigten Dampfes das direkteste Mittel zur Bestimmung der chemischen Konstanten a.

§ 288. Für das Gleichgewicht einer chemisch homogenen Flüssigkeit in Berührung mit ihrem chemisch homogenen Dampf gilt nach § 261, Gleichung (221a) die Beziehung:

$$\frac{\varphi'}{m'} = \frac{\varphi}{m}.$$

Ersetzen wir φ, welches sich auf ein flüssiges Molekül m bezieht, durch Gleichung (257), dagegen φ', welches sich auf ein dampfförmiges Molekül m' bezieht, durch Gleichung (267), indem wir den Dampf als ideales Gas voraussetzen, so ergibt sich:

$$C_p' \log T - R\log p + a - \frac{b}{T} = \frac{m'}{m}\left(\int_0^T \frac{C_p}{T}\,dT - \frac{w}{T}\right).$$

Diese Formel enthält die Abhängigkeit des Druckes des gesättigten Dampfes von der Temperatur. Die Konstante b, zusammen mit der in w enthaltenen

additiven Konstanten, läßt sich auf die Verdampfungswärme zurückführen. Es ist nämlich für ein dampfförmiges Molekül nach (266) die Wärmefunktion:

$$w' = C'_p T + b,$$

also die Verdampfungswärme, bezogen auf die Masse eines Dampfmoleküls m':

$$r = w' - \frac{m'}{m} w = C'_p T + b - \frac{m'}{m} w. \tag{268}$$

Dies in die letzte Gleichung substituiert ergibt:

$$C'_p \log T + C'_p - R \log p + a = \frac{r}{T} + \frac{m'}{m} \int_0^T \frac{C_p}{T}\, dT. \tag{269}$$

Beim Nullpunkt der absoluten Temperatur ist die Verdampfungswärme r_0 nach (268):

$$r_0 = b - \frac{m'}{m} w_0$$

also
$$r - r_0 = C'_p T - \frac{m'}{m}(w - w_0), \quad \text{und nach (150a):}$$

$$r - r_0 = C'_p T - \frac{m'}{m} \int_0^T C_p\, dT.$$

Dies ergibt für den Druck p_s des gesättigten Dampfes, durch Elimination von r aus (269):

$$\log p_s = \frac{C'_p}{R} \log T - \frac{r_0}{RT} + \frac{a}{R} + \frac{m'}{mR} \left(\frac{1}{T} \int_0^T C_p\, dT - \int_0^T \frac{C_p}{T}\, dT \right). \tag{270}$$

Für hinreichend tiefe Temperaturen kann man in dieser Gleichung die Integrale vernachlässigen. Dann erhält man für den Druck des gesättigten Dampfes[1]:

$$\log p_s = \frac{C'_p}{R} \log T - \frac{r_0}{RT} + \frac{a}{R}. \tag{271}$$

[1] Vgl. W. Nernst, Verb. d. Deutschen Phys. Ges. **11**, p. 313, 1909; **12**, p. 565, 1910. F. Pollitzer: Die Berechnung chemischer Affinitäten nach dem Nernstschen Wärmetheorem, Stuttgart, F. Enke, 1912.

Somit läßt sich durch Dampfdruckmessungen, besonders bei tieferen Temperaturen, die chemische Konstante a jedes chemisch homogenen Dampfes finden. Der Faktor $\frac{C'_p}{R}$ im ersten Glied ist für einatomige Gase nahezu $= \frac{5}{2}$ da hierfür $\frac{C'_p}{C'_v} = \frac{5}{3}$ und stets $C'_p - C'_v = R$.

Eine direkte Berechnung der chemischen Konstanten aus dem Molekulargewicht des Dampfes, ohne jede besondere Messung, gestatten die Methoden der Quantentheorie, auf die hier jedoch nicht eingegangen werden kann.

Eine Vergleichung von (271) mit (263a) zeigt, daß die Entropie eines idealen Gases beim absoluten Nullpunkt der Temperatur den Wert $+\infty$ annimmt.

§ 289. Die Gesetze gesättigter Dämpfe lassen sich auch noch auf andere Weise ableiten. Wir sahen nämlich früher bei den Anwendungen des zweiten Hauptsatzes (§ 172), daß der Druck des gesättigten Dampfes vollkommen bestimmt ist durch die allgemeine, für den flüssigen und gasförmigen Zustand gültige Zustandsgleichung $p = f(T, v)$ der homogenen Substanz. Bedeuten nämlich v_1 und v_2 die spezifischen Volumina im dampfförmigen und flüssigen Zustand, p_1 und p_2 die entsprechenden Drucke, so ist der Druck des gesättigten Dampfes als Funktion der Temperatur allein nach (102) bestimmt durch die beiden Gleichungen:

$$p_1 = p_2 \quad \text{und} \quad \int_{v_2}^{v_1} p\,dv = p_1(v_1 - v_2).$$

Die Integration ist bei konstanter Temperatur auszuführen.

Daraus folgt sogleich, daß die chemische Konstante a auch berechnet werden kann, wenn die allgemeine Zustandsgleichung der Substanz bekannt ist, und daß daher die chemische Konstante in der allgemeinen Zustandsgleichung vorkommen muß. — Bisher sind schon eine Anzahl von Formen für die Zustandsgleichung vorgeschlagen worden: außer der VAN DER WAALSschen (12) und der CLAUSIUSschen (12a) noch eine Reihe anderer, welche in gewissen beschränkten Gebieten ihren Zweck mit sehr guter Annäherung erfüllen; aber keine derselben ist, soweit ich sehe, umfassend genug, um auch für den flüssigen Zustand bis herab zu den tiefsten Temperaturen brauchbar zu sein, und daher liefert auch keine für den Druck des gesättigten Dampfes eine Formel, die für tiefe Temperaturen in die Gleichung (271) übergeht.

Die VAN DER WAALSsche Gleichung (12) z. B. ergibt, wie sich durch einfache Rechnungen zeigen läßt, als Druck des gesättigten Dampfes für

tiefe Temperaturen die Beziehung:

$$\log p_s = -\log\frac{\beta^2}{\alpha} - \frac{\alpha}{\beta RT}. \tag{272}$$

Hier fehlt also das Glied mit $\log T$. Dies rührt im Grunde daher, daß nach der VAN DER WAALSschen Zustandsgleichung bei unbegrenzt abnehmender Temperatur die spezifische Wärme im dampfförmigen Zustand demselben Wert zustrebt wie im flüssigen Zustand, was mit dem NERNSTschen Theorem unvereinbar ist. Eine andere Forderung des Theorems, welche keine der bisher vorgeschlagenen Zustandsgleichungen erfüllt, ist die, daß der Ausdehnungskoeffizient der flüssigen Substanz für $T = 0$ verschwindet (§ 285). Ehe man daran denken kann, die chemische Konstante aus der Zustandsgleichung zu berechnen, muß für letztere zuerst eine Form gefunden sein, welche für $T = 0$ und endliche positive Drucke mit dem NERNSTschen Theorem vereinbar ist.

§ 290. Für eine Mischung idealer Gase mit den Molekülzahlen n_1, n_2, $n_3, \ldots$ gelten die in den §§ 233–239 abgeleiteten Werte der Entropie nach (197):

$$S = \sum n_1(C_{p1}\log T - R\log(c_1 p) + k_1), \tag{273}$$

der Wärmefunktion nach (193) und (191):

$$W = U + pV = \sum n_1(C_{p1}T + b_1), \tag{274}$$

und der charakteristischen Funktion nach (199) und (199a):

$$\Phi = \sum n_1\left(C_{p1}\log T - R\log(c_1 p) + a_1 - \frac{b_1}{T}\right), \tag{275}$$

wobei c_1, c_2, $c_3, \ldots$ die molekularen Konzentrationen:

$$c_1 = \frac{n_1}{n_1 + n_2 + n_3 + \ldots}, \quad c_2 = \frac{n_2}{n_1 + n_2 + n_3 + \ldots}, \ldots$$

bedeuten.

Die chemischen Konstanten a_1, a_2, $a_3, \ldots$ haben für jede Molekülgattung besondere Werte und lassen sich daher aus dem Verhalten der einzelnen chemisch homogenen Komponenten der Mischung, z. B. durch Dampfdruckmessungen, ableiten. Wenn dies geschehen ist, kann man für jede beliebige zwischen den verschiedenen Molekülarten der Mischung stattfindende chemische Reaktion:

$$\delta n_1 : \delta n_2 : \ldots = \nu_1 : \nu_2 : \ldots$$

die für das thermodynamische Gleichgewicht charakteristische Konstante der Gleichung (201b):

$$\frac{\nu_1 a_1 + \nu_2 a_2 + \dots}{R} = \log A$$

unmittelbar angeben, und dadurch sind die Konstanten der Gleichgewichtsbedingung (203a):

$$c_1^{\nu_1} c_2^{\nu_2} c_3^{\nu_3} \dots = A e^{-\frac{B}{T}} T^C p^{-\nu} = K \tag{276}$$

sämtlich auf unabhängige Messungen zurückgeführt. Denn B berechnet sich nach (205) aus der Wärmetönung der Reaktion, und C nach (203) aus den spezifischen Wärmen der einzelnen Molekülgattungen, während ν nach (201a) die durch die Reaktion bedingte Zunahme der Molekülzahl bedeutet.

Aber nicht allein für Reaktionen innerhalb der gasförmigen Phase, sondern für jede beliebige Reaktion, an der die Gasmischung beteiligt ist, die also auch mit der Ausfällung fester oder flüssiger Stoffe verbunden sein kann, behalten die Konstanten a_1, a_2,... ihre Bedeutung, da der Ausdruck (275) der charakteristischen Funktion Φ der Gasmischung ganz allgemein die Gesetze des thermodynamischen Gleichgewichts regelt. Man hat nur den Wert von Φ in die allgemeine Gleichgewichtsbedingung (145) einzusetzen, wobei man sich für die gasförmige Phase der Gleichung (275), für chemisch homogene feste und flüssige Phasen der Gleichung (257) oder (258) bedienen kann.

§ 291. Betrachten wir nun das Verhalten fester oder flüssiger Lösungen. Das Symbol einer solchen Lösung sei nach (216) ausgedrückt durch:

$$n_1 m_1, \;\; n_2 m_2, \;\; n_3 m_3, \dots$$

Die Lösung braucht nicht verdünnt zu sein, deshalb ist hier auch, im Gegensatz zu unserer früheren Bezeichnung, der Index 0 fortgelassen.

Um einen Ausdruck für die charakteristische Funktion Φ der Lösung zu finden, fragen wir zunächst wieder nach der Entropie S derselben. Während für einen chemisch homogenen Körper nach dem NERNSTschen Wärmetheorem (§ 282) für $T = 0$ $S = 0$ wird, trifft dies für eine Lösung sicher nicht zu; denn schon in dem speziellen Fall einer verdünnten Lösung, für den sich die Entropie direkt angeben läßt, nimmt S, wie ein Blick auf den Ausdruck (213) lehrt, mit abnehmender Temperatur keineswegs unbegrenzt ab, da dieser Ausdruck additive Glieder enthält, die gar nicht von der Temperatur abhängen. Es liegt aber nahe, anzunehmen, daß die von der

Temperatur T abhängigen Glieder von S für $T = 0$ ebenso verschwinden, wie es die Entropie einer chemisch homogenen Substanz tut, und daß also für $T = 0$:

$$S = -R \sum n_1 \log c_1.$$

Für Lösungen von beliebiger Verdünnung wird dann aber dasselbe gelten, da der Ausdruck $-R \sum n_1 \log c_1$ nicht etwa nur für kleine, sondern für beliebige Werte der Konzentrationen c_1, $c_2, \ldots$ unabhängig vom Aggregatzustand, für feste, flüssige und gasförmige Körper, als additives Glied im Werte der Entropie auftritt. Denn für ideale gasförmige Körper ist dies direkt erwiesen, durch Gleichung (197), und feste und flüssige Körper lassen sich nach § 254 immer durch geeignete Temperatur- und Druckänderung, bei konstant gehaltenen Molekülzahlen n, stetig in den gasförmigen Zustand übergeführt denken.

Wir wollen daher das oben nur für chemisch homogene Körper in der Gleichung (256) ausgesprochene NERNSTsche Theorem jetzt dahin erweitern, daß für jede beliebige feste und flüssige Lösung mit den Molekülzahlen $n_1, n_2, n_3, \ldots$ die Entropie beträgt:

$$S = \int_0^T \frac{C_p}{T}\, dT - R \sum n_1 \log c_1, \tag{277}$$

wobei die Integration nach T bei konstantem Druck p und konstanten Molekülzahlen n zu erfolgen hat. Damit ist auch die charakteristische Funktion Φ der Lösung:

$$\Phi = \int_0^T \frac{C_p}{T}\, dT - R \sum n_1 \log c_1 - \frac{W}{T} \tag{278}$$

und die übrigen thermodynamischen Eigenschaften der Lösung nach § 283 eindeutig bestimmt. Für einen chemisch homogenen Körper gehen natürlich diese Gleichungen in die früher erhaltenen über, da dann $c_1 = 1$, $c_2 = 0$, $c_3 = 0, \ldots$

In dem speziellen Fall einer verdünnten Lösung gelten selbstverständlich alle im 5. Kapitel abgeleiteten Beziehungen, zu welchen jetzt noch die aus der Gleichung (277) fließende Folgerung kommt, daß die in dem Ausdruck (213) der Entropie einer verdünnten Lösung auftretenden Glieder $s_1, s_2, s_3, \ldots$ alle für $T = 0$ verschwinden.

§ 292. Aus der Gleichung (277) folgt unmittelbar, genau so wie in § 284 und § 285 für einen chemisch homogenen Körper aus der Gleichung (256),

daß sowohl die Wärmekapazität als auch der Ausdehnungskoeffizient eines jeden festen und flüssigen Körpers bei unbegrenzt abnehmender Temperatur sich unbegrenzt dem Werte Null nähert.

§ 293. Für ein aus beliebig vielen festen oder flüssigen Phasen bestehendes System, das nach folgendem Symbol dargestellt wird:

$$n_1\, m_1,\ \ n_2\, m_2, \ldots \mid n_1'\, m_1',\ \ n_2'\, m_2', \ldots \mid n_1''\, m_1'',\ \ n_2''\, m_2'', \ldots \mid \ldots,$$

in welchem eine isotherm-isobare Umwandlung von der Art:

$$\delta n_1 : \delta n_2 : \ldots : \delta n_1' : \delta n_2' : \ldots : \delta n_1'' : \delta n_2'' : \ldots = \nu_1 : \nu_2 : \ldots : \nu_1' : \nu_2' : \ldots : \nu_1'' : \nu_2'' : \ldots$$

möglich ist, gilt ganz allgemein nach (79) die Gleichgewichtsbedingung $\sum \delta\Phi = 0$. Also:

$$\sum \frac{\partial \Phi}{\partial n_1}\nu_1 + \frac{\partial \Phi}{\partial n_2}\nu_2 + \ldots = 0, \tag{279}$$

wobei die Summierung $\sum$ über alle Phasen zu erstrecken ist. Setzt man hierin den Wert von Φ aus (278), so ersieht man, daß das Gleichgewicht wesentlich bedingt ist durch die Werte der Größen $\frac{\partial C_p}{\partial n_1}, \frac{\partial C_p}{\partial n_2}, \ldots$ und $\frac{\partial W}{\partial n_1}, \frac{\partial W}{\partial n_2}, \ldots$ d. h. durch die Abhängigkeit der Wärmefunktion W und der Wärmekapazität C_p einer jeden Phase von den Molekülzahlen. C_p ist natürlich durch W mitbestimmt, da allgemein nach (150a)

$$C_p = \frac{\partial W}{\partial T}.$$

Daher erscheint die Bestimmung der Gesetze des chemischen Gleichgewichts vollständig zurückgeführt auf Messungen von Wärmekapazitäten und Wärmetönungen. Allerdings ist das vorliegende Zahlenmaterial zurzeit noch nicht umfangreich genug, um diese weitgehenden Schlüsse der Theorie jetzt schon allseitig zu prüfen.

Wir wollen im folgenden zur Abkürzung die Bezeichnungen:

$$\frac{\partial W}{\partial n_1} = w_1, \qquad \frac{\partial W}{\partial n_2} = w_2, \ldots \tag{280}$$

und daher nach (150a) auch:

$$\frac{\partial C_p}{\partial n_1} = \frac{\partial w_1}{\partial T}, \qquad \frac{\partial C_p}{\partial n_2} = \frac{\partial w_2}{\partial T}, \ldots \tag{281}$$

benutzen. Dann schreibt sich die Gleichgewichtsbedingung (279) für das ganze System, mit Rücksicht auf (278):

$$\sum \nu_1 \left(\int_0^T \frac{\partial w_1}{\partial T} \frac{dT}{T} - \frac{w_1}{T} - R \log c_1 \right)$$
$$+\nu_2 \left(\int_0^T \frac{\partial w_2}{\partial T} \frac{dT}{T} - \frac{w_2}{T} - R \log c_2 \right) + \ldots = 0.$$

Nun ist:
$$\sum \nu_1 w_1 + \nu_2 w_2 + \ldots = r$$

die Wärmemenge, welche bei der durch die ν ausgedrückten isotherm-isobaren Umwandlung von außen zuzuführen ist; folglich lautet die Gleichgewichtsbedingung einfacher:

$$\int_0^T \frac{\partial r}{\partial T} \frac{dT}{T} - \frac{r}{T} - R \left(\sum \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots \right) = 0$$

oder:

(282) $$\sum \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots = \frac{1}{R} \left(\int_0^T \frac{\partial r}{\partial T} \frac{dT}{T} - \frac{r}{T} \right) = \log K.$$

Ein Vergleich mit (218) zeigt in der Form vollständige Übereinstimmung mit der Gleichgewichtsbedingung eines Systems verdünnter Lösungen. Ein wesentlicher Unterschied ist aber erstens der, daß hier die Gleichgewichtsgröße $\log K$ durch die Wärmetönung r vollkommen bestimmt erscheint, ohne eine unbekannt bleibende additive Konstante[1], und zweitens, daß hier die Größe K, ebenso wie r, außer von T und p auch von den (endlichen) Konzentrationen $c_1, c_2, \ldots$ abhängt, was dort nicht der Fall war. Dieser Umstand bedingt es, daß man hier die Gleichgewichtskonzentrationen nicht unmittelbar als Funktionen von T und p ausdrücken kann, wie das bei verdünnten Lösungen gelungen war.

Damit die Größe K endlich ist, muß offenbar $\frac{\partial r}{\partial T}$ für $T = 0$ verschwinden, woraus folgt, daß jede Wärmetönung einer Reaktion zwischen festen und flüssigen Körpern beim Nullpunkt der absoluten Temperatur den Temperaturkoeffizienten Null besitzt.

[1]Die Integration nach T ist natürlich bei konstantem Druck p und konstanten Konzentrationen zu vollziehen.

Dieser Satz bildet eine wichtige Vorbedingung, die jede empirische Formel für die Wärmetönung erfüllen muß, wenn man aus ihr den Wert der Gleichgewichtsgröße K berechnen will.

Statt der Gleichung (282) wird man häufig mit Vorteil die folgende, ihr genau äquivalente Beziehung benutzen können:

$$\sum \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots = \frac{1}{R}\left(\int_0^T \frac{r - r_0}{T^2} dT - \frac{r_0}{T}\right) = \log K. \tag{283}$$

wo r_0 die Reaktionswärme bei $T = 0$ bedeutet.

Für $T = 0$ wird offenbar $\log K$ unendlich, was mit dem schon in § 259a für verdünnte Lösungen gezogenen Schluß übereinstimmt, daß beim Nullpunkt der absoluten Temperatur alle mit endlicher Wärmetönung verbundenen Reaktionen bis zur vollständigen Beendigung verlaufen.

§ 294. Für die Abhängigkeit der Größe K von T und p gelten auch hier wieder die Beziehungen (219) und (220):

$$\frac{\partial \log K}{\partial T} = \frac{r}{RT^2} \quad \text{und} \quad \frac{\partial \log K}{\partial p} = -\frac{v}{RT}. \tag{284}$$

Man findet dieselben direkt durch Differentiation der Gleichung (282) nach T und nach p, wenn man noch bedenkt, daß für die bei der Reaktion eintretende Volumenänderung v des Systems nach der allgemeinen Formel (79g):

$$\frac{\partial r}{\partial p} = -T^2 \frac{\partial}{\partial T}\left(\frac{v}{T}\right) = v - T\frac{\partial v}{\partial T}. \tag{285}$$

Doch geht die Gleichung (282) insofern über den Inhalt der Gleichungen (284) hinaus, als sie auch den absoluten Wert der Gleichgewichtskonstante K liefert, während diese noch eine Integrationskonstante unbestimmt lassen. Wenn z. B. $r = 0$ (thermoneutrale Reaktion), so folgt aus (284) nur, daß K unabhängig ist von der Temperatur, ihre Größe bleibt unbestimmt. Dagegen ist nach (282) $\log K = 0$. Daher befindet sich eine gegenseitige Lösung der beiden enantiomorphen Formen einer optisch aktiven Verbindung nur dann im stabilen thermodynamischen Gleichgewicht, wenn sie ein racemisches, optisch inaktives Gemisch bildet, — eine Folgerung, die dem NERNSTschen Theorem eigentümlich ist, während man ohne dasselbe nur schließen kann, daß die Zusammensetzung der Mischung sich mit der Temperatur nicht ändert.

§ 295. Berechnung des Dissoziationsgrades eines Elektrolyten aus der Dissoziationswärme. Nehmen wir einen gelösten

Elektrolyten, z. B. Essigsäure in Wasser, so ist das Symbol des Systems, ähnlich wie in § 262:

$$n_1 H_2O, \quad n_2 H_4 C_2 O_2, \quad n_3 \overset{+}{H}, \quad n_4 H_3 \overset{-}{C}_2 O_2.$$

Die Gesamtzahl der Moleküle sei:

$$n = n_1 + n_2 + n_3 + n_4.$$

Dann sind die Konzentrationen:

$$c_1 = \frac{n_1}{n}, \qquad c_2 = \frac{n_2}{n}, \qquad c_3 = \frac{n_3}{n}, \qquad c_4 = \frac{n_4}{n}.$$

Für die Dissoziation eines Moleküls $H_4C_2O_2$ ist:

$$\nu_1 = 0, \quad \nu_2 = -1, \quad \nu_3 = 1, \quad \nu_4 = 1.$$

Also ist nach (283) im Gleichgewichtszustand, da $c_3 = c_4$,

$$R \log \frac{c_3^2}{c_2} = R \log K = \int_0^T \frac{r - r_0}{T^2}\, dT - \frac{r_0}{T},$$

woraus man c_2 und c_3 einzeln berechnen kann, da die Summe $c_2 + c_3$ durch die Gesamtzahl der in der Lösung vorhandenen, undissoziierten und dissoziierten, Säuremoleküle gegeben ist. Allerdings ist die Gleichgewichtsformel nur dann allgemein brauchbar, wenn die Abhängigkeit der Dissoziationswärme r sowohl von der Temperatur als auch von den Konzentrationen bekannt ist. Bei sehr verdünnten Lösungen genügt die Abhängigkeit von der Temperatur, weil man dann r als nahezu unabhängig von der Konzentration betrachten kann.

§ 296. Berechnung der Löslichkeit aus der Lösungswärme. Nehmen wir das Gleichgewicht irgend einer Salzlösung mit einem festen Bodenkörper (Eis, Salz), so ist das Symbol des Systems:

$$n_1\, m_1, \;\; n_2\, m_2 \mid n_1'\, m_1'.$$

Hierbei bezieht sich die Ziffer 1 auf den in beiden Phasen vorhandenen Stoff, die Ziffer 2 auf den nur in der Lösung vorhandenen Stoff. Die Konzentrationen in den beiden Phasen sind:

$$c_1 = \frac{n_1}{n_1 + n_2}, \quad c_2 = \frac{n_2}{n_1 + n_2}, \quad c_1' = \frac{n_1'}{n_1'} = 1.$$

Von einem bestimmten „Lösungsmittel“ (§ 249) kann man natürlich hier nicht reden, da die beiden Stoffe in der Lösung in beliebigem Mengenverhältnis vertreten sind.

Für die Ausfällung eines flüssigen Moleküls m_1 gelten die Werte:

$$\nu_1 = -1, \quad \nu_2 = 0, \quad \nu_1' = \frac{m_1}{m_1'}.$$

Folglich ist nach (283) die Gleichgewichtsbedingung:

$$-R \log c_1 = \int_0^T \frac{r - r_0}{T^2}\, dT - \frac{r_0}{T}.$$

Hierbei ist $-r$ die bei der Ausfällung des flüssigen Moleküls frei werdende Wärmemenge.

Diese Gleichung ergibt, wenn r als Funktion der Temperatur und der Konzentration bekannt ist, den Wert der Konzentration c_1 und damit das Verhältnis $n_1 : n_2$, sie gestattet also einen Schluß auf den Molekularzustand der beiden Stoffe in der Lösung.

§ 297. Wenn in dem betrachteten System außer beliebig vielen festen und flüssigen auch noch eine gasförmige Phase vorhanden ist und wenn wir, wie früher, diese Phase als im idealen Gaszustand befindlich annehmen, so verallgemeinert sich, wie man leicht aus der Gleichung (79) mit Benutzung von (275) ableiten kann, die Gleichgewichtsbedingung (283) zu:

$$\left\{ \begin{array}{l} \sum \nu_1 \log c_1 + \nu_2 \log c_2 + \ldots = \\ \qquad \dfrac{1}{R} \displaystyle\int_0^T \frac{r'}{T^2}\, dT - \frac{r_0}{RT} + C \log T - \nu \log p + \log A = \log K. \end{array} \right. \tag{286}$$

Hierbei ist zur Abkürzung gesetzt:

$$r' = r - r_0 - RCT, \tag{287}$$

und die Konstanten A, C und ν beziehen sich, wie in (276), auf die gasförmige Phase allein.

Diese eine Gleichung (286) enthält alle in den letzten drei Kapiteln aufgestellten Gleichgewichtsbedingungen (Aggregatzustandsänderungen, Löslichkeit, Dissoziation, Gefrierpunktserniedrigung, Siedepunktserhöhung, Isohydrie usw.) als spezielle Fälle. Daher sei die Bedeutung der darin gebrauchten Bezeichnungen hier zum Schluß noch einmal zusammengestellt.

Links ist die Summation $\sum$ über alle Phasen des Systems zu erstrecken. Dabei bedeuten $c_1, c_2, c_3, \ldots$ die molekularen Konzentrationen der einzelnen Molekülgattungen mit den Molekülzahlen $n_1, n_2, n_3, \ldots$ in jeder Phase:

$$c_1 = \frac{n_1}{n_1 + n_2 + n_3 + \ldots}, \qquad c_2 = \frac{n_2}{n_1 + n_2 + n_3 + \ldots}, \ldots$$

$\nu_1, \nu_2, \nu_3, \ldots$ sind die gleichzeitigen Änderungen der Molekülzahlen $n_1, n_2, n_3, \ldots$ bei irgend einer ins Auge gefaßten isotherm-isobaren Umwandlung des Systems, sei es, daß dieselbe innerhalb einer einzigen Phase stattfindet, oder daß sie auch mit dem Übergang von Molekülen aus einer in eine andere Phase verbunden ist.

Rechts bedeutet R die absolute Gaskonstante, r' den Ausdruck (287), r die mit der isotherm-isobaren Umwandlung verbundene gesamte Reaktionswärme, positiv, wenn die Wärme von außen zugeführt wird, r_0 den Wert von r für $T = 0$. Die Konstanten A, C und ν beziehen sich allein auf die gasförmige Phase, falls eine solche vorhanden ist. Es ist nämlich ν nach (201a) die durch die Umwandlung bewirkte Änderung der Gesamtzahl der gasförmigen Moleküle, während A sich aus den einzelnen gasförmigen Molekülgattungen nach (201b), und C aus den spezifischen Wärmen nach (203) zusammensetzt. Die Integration nach T ist bei konstantem p und konstanten Konzentrationen zu vollziehen.

Damit das Temperaturintegral trotz der unteren Grenze 0 einen endlichen Wert hat, muß offenbar sowohl r' als auch $\dfrac{\partial r'}{\partial T}$ für $T = 0$ verschwinden. Das erstere sieht man unmittelbar aus (287), das letztere erkennt man, wenn man bedenkt, daß $\dfrac{\partial r}{\partial T}$ nach § 105 allgemein gleich ist der Differenz der Wärmekapazitäten des Systems nach und vor der Umwandlung, und daß diese Differenz für die festen und flüssigen Phasen bei $T = 0$ verschwindet, für die gasförmige Phase aber nach (203) gleich RC ist.

Manchmal wird es vorteilhaft sein, das Temperaturintegral in folgender Form zu schreiben:

$$\int_0^T \frac{r'}{T^2}\, dT = -\frac{r'}{T} + \int_0^T \frac{\partial r'}{\partial T}\, \frac{dT}{T}$$

$$= -\frac{r - r_0 - RCT}{T} + \int_0^T \left(\frac{\partial r}{\partial T} - RC \right) \frac{dT}{T};$$

denn $\dfrac{\partial r}{\partial T}$ läßt sich nach dem soeben Ausgeführten durch Messungen von Wärmekapazitäten bestimmen.

Für ein kondensiertes System sind ν, C und $\log A$ gleich Null, für eine einzige gasförmige Phase ist nach (205) $r' = 0$, für chemisch homogene Körper verschwinden die Glieder mit den Konzentrationen $c_1, c_2, \ldots$, so daß für jeden dieser speziellen Fälle jedesmal die früher erhaltenen einfachen Gleichungen resultieren.

Lehrbuch der Physik. Nach Vorlesungen an der Technischen Hochschule zu München. Von Dr. H. Ebert, weil. Professor der Physik an der Technischen Hochschule zu München. Groß-Oktav.

Erster Band: **Mechanik, Wärmelehre.** Mit 168 Abbildungen im Text. 2. Aufl. (Anastat. Nachdruck). XX, 661 Seiten. 1920. Grundzahl* 20.

Zweiter Band, I. Teil: **Die elektrischen Energieformen.** Fertiggestellt und herausgegeben von Dr. C. Heinke, ord. Professor der Elektrotechnik an der Technischen Hochschule zu München. Mit 341 Abbildungen im Text. XX, 687 Seiten. 1920. Grundzahl* 22.

...Für die Ausbildung der Studierenden, denen das physikalische Wissen ebensowenig Endzweck ist, wie das mathematische, sondern denen vielmehr die Anwendung zur Lösung technischer Probleme obliegt, entspricht Eberts Art der Behandlung des Stoffes geradezu einem Bedürfnis.

Riecke, Lehrbuch der Physik. Zu eigenem Studium und zum Lehrgebrauche bei Vorlesungen herausgegeben von Prof. Dr. Ernst Lecher, Vorstand des I. physikalischen Institutes der Universität Wien. Sechste, verbesserte und vermehrte Auflage. Lexikon-Oktav.

Erster Band: **Mechanik und Akustik. Wärme. Optik.** Vergriffen. Neue Auflage Ende 1922.

Zweiter Band: **Magnetismus und Elektrizität.** Mit 303 Figuren im Text, XIV, 363 Seiten. 1919. Grundzahl* 24.

Das Rieckesche Lehrbuch gehört seit dem Erscheinen der ersten Auflage zu den verbreitetsten und beliebtesten Lehrbüchern der Physik. Es wird kaum ein Institut und kaum einen selbständig denkenden Physiker geben, in dessen Bibliothek das Rieckesche Buch fehlen dürfte. Monatshefte für Mathematik und Physik.

Einführung in die theoretische Physik in zwei Bänden von Dr. Clemens Schaefer, Professor an der Universität Breslau.

Erster Band: **Mechanik materieller Punkte, Mechanik starrer Körper, Mechanik der Kontinua (Elastizität und Hydromechanik).** Mit 249 Figuren im Text. Neudruck November 1922.

Zweiter Band, I. Teil: **Theorie der Wärme, Molekular-kinetische Theorie der Materie.** Mit 71 Figuren im Text. X, 562 Seiten. 1921. Grundzahl* 15.

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Die Prinzipe der Dynamik. Von Professor Dr. Clemens Schaefer. Mit 6 Figuren im Text. Groß-Oktav. IV, 76 Seiten. 1919. Grundzahl* 2.5.

Das Werk behandelt ein außerordentlich wichtiges Kapitel der Dynamik, das in den meisten Darstellungen zu kurz kommt, nämlich das gegenseitige Verhältnis der Prinzipe der Dynamik zueinander.

Die Darstellung ist eine sehr anregende, nicht zum wenigsten infolge der Erörterung einfacher, glücklich gewählter Beispiele, mit deren Hilfe die gegenseitigen Beziehungen der einzelnen Prinzipe besonders klar zutage treten, so daß wir dem Werkchen eine recht ausgedehnte Verbreitung wünschen. Zeitschrift für technische Physik.

Physik und Hypothese. Versuch einer induktiven Wissenschaftslehre nebst einer kritischen Analyse der Fundamente der Relativitätstheorie. Groß-Oktav. XI, 200 Seiten. 1921. Grundzahl* 6.

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Einführung in die Theorie der Wärme von Dr. Heinrich Mache, o. ö. Professor an der Technischen Hochschule in Wien. Mit 96 Textfiguren. Groß-Oktav. VIII, 319 Seiten. 1921. Grundzahl* 10.

Das Buch erstrebt vor allem, die physikalische Begriffsbildung möglichst einfach und klar zu gestalten, es setzt nur physikalische Mittelschulbildung und Kenntnis der Elemente der Differential- und Integralrechnung voraus.

Die Physik der Verbrennungserscheinungen. Von Dr. Heinrich Mache, o. ö. Professor an der Technischen Hochschule in Wien. Mit 43 Abbildungen im Text und auf 2 Tafeln. Groß-Oktav. IV, 133 Seiten. 1918. Grundzahl* 4.

In diesem Werk werden zum erstenmal die Verbrennungserscheinungen von physikalischen Gesichtspunkten aus behandelt und der Verfasser hat alles zusammengestellt, was bis jetzt darüber bekannt ist. — Man wird es mit Dank begrüßen, daß der sachkundige Verfasser eine zusammenfassende Darstellung des Gebietes liefert, die über die einschlägigen Fragen gründliche Auskunft erteilt. Chemiker-Zeitung.

Einführung in die theoretische Physik mit besonderer Berücksichtigung ihrer modernen Probleme. Von Arthur Haas, Dr. phil., a. o. Professor an der Universität Leipzig. Groß-Oktav.

Erster Band. Mit 50 Abbildungen im Text. 2. Auflage. VIII, 384 Seiten. 1921. Grundzahl* 10.

Zweiter Band. Mit 30 Abbildungen im Text. 1. und 2. Auflage. IV, 286 Seiten. 1921. Grundzahl* 7.2.

Der erste, nun in zweiter Auflage vorliegende Band umfaßt die Theorie der Bewegung, der Elektrizität und des Lichtes. Der zweite, 1921 neu erschienene Band behandelt die Atomtheorie, die statistische Wärmelehre und die Relativitätstheorie.

Das Naturbild der neuen Physik. Von Arthur Haas, Dr. phil., a. o. Professor an der Universität Leipzig. Mit 6 Figuren im Text. Groß-Oktav. VI, 114 Seiten. 1920. Grundzahl* 5.2.

Der Verfasser entwickelt das Bild der Natur, wie es sich nach den großartigen Fortschritten der theoretischen Physik heute darstellt. Infolge der vielerörterten Forschungsergebnisse Einsteins ist das Interesse für die Zusammenhänge im Weltall heute ein so allgemeines, daß mit einer weitverbreiteten Vorliebe für Bücher gerechnet werden kann, die, wie das vorliegende, in leichtverständlicher und ansprechender Form das Fazit der Forscherarbeit bieten.

Die Grundlagen der Physik. Synthetische Prinzipien der mathematischen Naturphilosophie. Von Privatdozent Dr. Hugo Dingler. Groß-Oktav. XVI, 158 Seiten. 1919. Grundzahl* 6.6.

Dem Verfasser ist es gelungen, bei der Beschäftigung mit den Grundlagen der Physik umwälzende Resultate zu erzielen. Das vorliegende Werk gibt einen vollständigen Einblick in die mathematische Naturwissenschaft. Alle brennenden Fragen werden in vollständiger Weise behandelt, viele heißumstrittene Probleme gelöst. Besonderes Interesse wird das Buch Dinglers wohl deshalb finden, da er die vor allen an Einstein geknüpfte Relativitätstheorie schon in ihren ersten Grundlagen für unhaltbar erklärt, ohne damit die geniale Leistung Einsteins abstreiten zu wollen. Dingler hat sich eben zum Leitsatz gemacht, nur die Wahrheit, die er durch seine streng wissenschaftlichen Untersuchungen gefunden zu haben glaubt, zu lehren. Berichte der Deutschen Pharmazeutischen Gesellschaft.

Wörterbuch der Physik von Professor Dr. Felix Auerbach, Jena. Mit 267 Figuren. Oktav. X, 466 Seiten. 1920. Grundzahl* (9).

Im vorliegenden Buche wird das Gesamtgebiet der Physik alphabetisch nach Stichworten geordnet behandelt. Die Darstellung der einzelnen Abschnitte ist bei strenger Wissenschaftlichkeit ganz einfach gehalten und setzt besonders keine größeren mathematischen Kenntnisse voraus. Für die Gediegenheit des gebotenen Stoffes bürgt der wohlbekannte Name des Verfassers.

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AUS DER REINEN MATHEMATIK UND DER MATHEMATISCHEN PHYSIK

auf Einladung der Wolfskehl-Kommission
der Königlichen Gesellschaft der Wissenschaften
gehalten zu Göttingen vom 22.–28. April 1909

von

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Mitglied der Französischen Akademie
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Préface

L'Université de Göttingen a bien voulu m'inviter à traiter devant un savant auditoire diverses questions d'Analyse pure, de Physique mathématique, d'Astronomie théorique et de Philosophie mathématique ; les conférences que j'ai faites à cette occasion ont été recueillies par quelques étudiants qui ont eu la bonté de les rédiger en corrigeant les nombreuses offenses que j'avais faites à la grammaire allemande. Je leur en exprime ici toute ma reconnaissance.

Il convient également que je m'excuse auprès du public de la brièveté avec laquelle ces sujets sont traités. Je ne disposais pour exposer chacun d'eux que d'un temps très court, et je n'ai pu la plupart du temps que donner une idée générale des resultats, ainsi que des principes qui m'ont guidé dans les démonstrations, sans entrer dans les détails mêmes de ces démonstrations.

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Erster Vortrag

ÜBER DIE FREDHOLMSCHEN GLEICHUNGEN

Die Integralgleichung

$$\varphi(x) = \lambda \int_a^b f(x,y)\varphi(y)dy + \psi(x) \tag{1}$$

wird bekanntlich aufgelöst durch die Integralgleichung derselben Art

$$\varphi(x) = \psi(x) + \lambda \int_a^b \psi(y)G(x,y)dy, \tag{1a}$$

wobei

$$G(x,y) = \frac{N(x,y;\lambda \,|\, f)}{D(\lambda \,|\, f)}$$

gesetzt ist. N und D sind, wie aus der FREDHOLMschen Theorie bekannt ist, zwei ganze transzendente Funktionen in bezug auf λ. Um ihre Entwicklung explizite hinschreiben zu können, bezeichne man, wie FREDHOLM, mit $f\left(\begin{smallmatrix} x_1, & x_2, & \cdots & x_n \\ y_1, & y_2, & \cdots & y_n \end{smallmatrix}\right)$ diejenige n-reihige Determinante, deren allgemeines Element $f(x_i, y_k)$ ist. Setzt man dann

$$a_n = \int_a^b \int_a^b \cdots \int_a^b f\left(\begin{smallmatrix} x_1, & x_2, & \cdots & x_n \\ x_1, & x_2, & \cdots & x_n \end{smallmatrix}\right) dx_1 \ldots dx_n,$$

so hat man

$$D(\lambda) = \sum_0^\infty \frac{(-\lambda)^n}{n!} a_n.$$

Diese Gleichung formen wir um, indem wir die durch „Iteration“ aus $f(x,y)$ entstehenden Kerne heranziehen. Setzen wir zunächst

$$f(x_\alpha, x_\beta)f(x_\beta, x_\gamma)\cdots f(x_\lambda, x_\mu)f(x_\mu, x_\alpha) = f(x_\alpha, x_\beta, \cdots x_\lambda, x_\mu),$$

so ist klar, daß $f\left(\begin{smallmatrix} x_1, & x_2, & \cdots & x_n \\ x_1, & x_2, & \cdots & x_n \end{smallmatrix}\right)$ die Form hat

$$\sum \pm \prod f(x_\alpha, \ldots x_\mu),$$

wie sofort aus der Entwicklung der Determinante hervorgeht. Sei nun

$$b_k = \int_a^b \cdots \int_a^b f(x_\alpha, \cdots x_\mu)dx_\alpha \cdots dx_\mu,$$

wobei k die Anzahl der Integrationsvariabeln $x_\alpha, \ldots x_\mu$ bedeutet, so können wir offenbar auch setzen

$$b_k = \int_a^b f_k(x,x)dx,$$

wenn unter

$$f_k(x,y) = \int_a^b \cdots \int_a^b f(x,x_\alpha) f(x_\alpha,x_\beta) \cdots f(x_\lambda,y) dx_\alpha \cdots dx_\lambda$$

der "k-fach iterierte Kern" verstanden wird.

Wir haben den obigen Relationen zufolge jetzt

$$a_n = \sum \pm \prod b_k.$$

Beachten wir nun, daß gewisse unter den in einem Produkt $\prod b_k$ enthaltenen b_k einander gleich werden können, daß ferner gewisse der Produkte $\prod b_k$ selbst einander gleich sein werden, nämlich solche, die durch eine Permutation der x_i auseinander entstehen, so ergibt eine kombinatorische Betrachtung für a_n einen Ausdruck von der Form

$$a_n = \sum_{a\alpha+b\beta+c\gamma+\ldots=n} \frac{n!}{a^\alpha b^\beta c^\gamma \cdots a!b!c! \cdots} [(-1)^{\alpha+1} b_\alpha]^a [(-1)^{\beta+1} b_\beta]^b [(-1)^{\gamma+1} b_\gamma]^c \cdots$$

und also

$$D(\lambda) = \sum_{a,b,c,\ldots} \frac{1}{a!b!c! \cdots} \left(-\frac{\lambda^\alpha b_\alpha}{\alpha}\right)^a \left(-\frac{\lambda^\beta b_\beta}{\beta}\right)^b \left(-\frac{\lambda^\gamma b_\gamma}{\gamma}\right)^c \cdots$$

d. h.

$$D(\lambda) = \prod_1^\infty e^{-\frac{\lambda^\alpha b_\alpha}{\alpha}}, \tag{2}$$

also

$$\log D(\lambda) = -\sum \frac{\lambda^\alpha b_\alpha}{\alpha}, \tag{2a}$$

$$\frac{D'(\lambda)}{D(\lambda)} = -\sum \lambda^{\alpha-1} b_\alpha. \tag{2b}$$

Den Zähler $N(x,y;\lambda)$ der Funktion $G(x,y;\lambda)$ kann man auf analoge Weise durch die Gleichung

$$N(x,y;\lambda) = D(\lambda) \cdot \sum \lambda^h f_{h+1}(x,y) \tag{3}$$

definieren. Diese Gleichungen, welche sich übrigens schon bei FREDHOLM finden, sind nützlich als Ausgangspunkt für viele Betrachtungen, wie sich nun an einigen Beispielen zeigen wird.

Die FREDHOLMsche Methode ist unmittelbar gültig nur für solche Kerne $f(x,y)$, die endlich bleiben. Wird der Kern an gewissen Stellen unendlich, so

kann dennoch der Fall eintreten, daß ein iterierter Kern, etwa $f_n(x,y)$, endlich bleibt. Dann läßt sich die Integralgleichung mit dem iterierten Kerne nach FREDHOLM behandeln, und FREDHOLM zeigt, daß die ursprüngliche Integralgleichung (1) sich auf diese zurückführen läßt. Die Auflösung wird wieder durch eine Formel der Gestalt (1a) gegeben, nur ist jetzt

$$G = \frac{N_1(x,y;\lambda)}{D_n(\lambda)}$$

zu setzen, wobei

$$D_n(\lambda) = D(\lambda^n \mid f_n)$$

und

$$N_1(x,y;\lambda) = D_n(\lambda) \cdot \sum \lambda^h f_{h+1}(x,y)$$

ist. Dabei sind N_1 und D_n wieder ganze transzendente Funktionen von λ; jedoch zeigt es sich, daß sie einen gemeinsamen Teiler besitzen; wir wollen zusehen, wie sich dies aus unseren Formeln (2) bis (3) ergibt und wie wir eine Bruchdarstellung der meromorphen Funktion G erhalten, bei der Nenner und Zähler ganze Funktionen ohne gemeinsamen Teiler sind.

Aus unserer Annahme über die iterierten Kerne folgt, daß die Koeffizienten b_n, $b_{n+1}, \ldots$ endlich sind. Bilden wir nun in Anlehnung an Gleichung (2a) die Reihe

$$K(\lambda) = -\lambda^n \frac{b_n}{n} - \lambda^{n+1} \frac{b_{n+1}}{n+1} - \cdots,$$

so wird dieselbe konvergieren. Jetzt setzen wir

$$G(x,y;\lambda) = \frac{e^K \sum \lambda^h f_{h+1}}{e^K}$$

und behaupten, in dieser Formel die gewünschte Darstellung zu haben.

Um dies zu beweisen, haben wir zu zeigen, daß e^K und $e^K \cdot \sum \lambda^{h+1} f_{h+1}$ ganze Funktionen sind.

Zu diesem Zwecke bilden wir $\frac{dK}{d\lambda}$. Man berechnet leicht

$$-\frac{dK(\lambda)}{d\lambda} = \lambda^{n-1} \int_a^b \frac{N_1(x,x)}{D_n(\lambda)} dx + \sum_{k=1}^{k=n-1} \lambda^{n+k-1} \iint_a^b \frac{N_1(x,y)}{D_n} f_k(x,y)\, dx\, dy.$$

Hieraus schließt man zunächst, daß $\frac{dK}{d\lambda}$ eine meromorphe Funktion von λ ist; denn sie besitzt höchstens Pole in den Nullstellen von $D_n(\lambda)$, d. h. in den Stellen $\lambda = \alpha \cdot \lambda_i$ wo α eine n-te Einheitswurzel und λ_i ein Eigenwert des Kernes f_n ist. Man kann nun zeigen, daß in diesen möglichen Unendlichkeitsstellen das CAUCHYsche Residuum von $\frac{dK}{d\lambda}$ gleich 1 oder 0 ist, je nachdem $\alpha = 1$ oder $\alpha \neq 1$ genommen wird. Die hierzu gehörige Rechnung wollen wir jetzt nicht durchführen; man benutzt dabei den Umstand, daß das für $\lambda = \lambda_k$ genommene

Residuum von $\frac{N_1(x,y)}{D_n}$ gleich $\varphi_k(x)\psi_k(y)$ ist, wo φ_k, ψ_k, die zu $\lambda = \lambda_k$ gehörigen Eigenfunktionen, den Gleichungen

$$\int_a^b \varphi_k(x) f_p(y,x)dx = \lambda_k^{-p}\varphi_k(y)$$

$$\int_a^b \psi_k(z) f_p(z,y)dz = \lambda_k^{-p}\psi_k(y)$$

genügen. Hieraus folgt, daß $e^{K(\lambda)}$ eine ganze transzendente Funktion ist, die nur an den Stellen $\lambda = \lambda_i$ verschwindet.

Betrachtet man ebenso den Zähler von G, so sieht man zunächst, daß er eine meromorphe Funktion von λ wird, die höchstens an den Stellen $\lambda = \alpha\lambda_i$ unendlich werden kann. Die Betrachtung der Residuen zeigt jedoch, daß dies nicht geschieht, und somit, daß der Zähler $e^K \sum \lambda^h f_{k+1}$ ebenfalls eine ganze transzendente Funktion ist. Damit ist die Reduktion des FREDHOLMschen Bruches geleistet.

Die Reihenentwicklung für Zähler und Nenner des FREDHOLMschen Bruches in dieser reduzierten Gestalt erhalten wir, indem wir auf die Bildungsweise von $K(\lambda)$ zurückgehen; setzen wir den Nenner

$$e^{K(\lambda)} = \sum (-\lambda)^n \frac{a'_n}{n!},$$

so haben wir

$$a'_n = \sum_{a\alpha+b\beta+c\gamma+\cdots=n} \pm b_\alpha^a\, b_\beta^b\, b_\gamma^c\, \cdots,$$

wobei zu setzen ist

$$b_\alpha = 0 \qquad \text{für } \alpha < n \text{ und}$$
$$b_\alpha = \int_a^b f_\alpha(x,x)dx \qquad \text{für } \alpha \geq n.$$

In analoger Weise wird der Zähler gebildet. Man muß also die Determinanten in der gewöhnlichen Weise entwickeln, aber diejenigen Glieder dieser Entwicklung wegwerfen, welche einen Faktor von der Form $f(x_1, x_2, \ldots x_k)$ mit weniger als n Veränderlichen enthalten.

Unsere Formeln (2), (2a), (3) sind auch in dem Falle von Nutzen, daß außer dem Kern $f(x,y)$ auch alle iterierten Kerne unendlich werden und die FREDHOLMsche Methode also nun sicher versagt.
Seien etwa die Zahlen $b_1, b_2, \ldots b_{n-1}$ unendlich, $b_n, b_{n+1}, \ldots$ endlich. Man kann dann jedenfalls die Reihe $K(\lambda)$ bilden, fragen, ob sie konvergiert, und untersuchen, ob $e^{K(\lambda)}$ wieder eine ganze Funktion darstellt. Unter der Voraussetzung, daß $f(x,y)$ ein symmetrischer Kern ist, d. h.

$$f(x,y) = f(x,y),$$

ist mir dieser Nachweis gelungen. Ich benutze dabei die Relationen

$$b_n = \sum \lambda_i^{-n},$$

die für $n > 2$ gelten müssen, da das Geschlecht der Funktion $D(\lambda)$ einem HADAMARDschen Satze zufolge kleiner als 2 ist.

Den Beweis mitzuteilen fehlt jetzt die Zeit.

Für den Zähler des FREDHOLMschen Bruches habe ich die Betrachtung nicht durchgeführt.

Noch einige Worte über die Integralgleichung 1. Art! Auf gewisse derartige Integralgleichungen kann man, wenn man sie zuvor auf Integralgleichungen der 2. Art zurückführt, die FREDHOLMsche Methode direkt anwenden. Es liege z. B. die Gleichung

$$\int_{-\infty}^{+\infty} \varphi(y)[e^{ixy} + \lambda f(x,y)]dy = \psi(x) \qquad (-\infty < x < +\infty) \tag{1}$$

vor, in der $\psi(x)$ die gegebene, $\varphi(x)$ aber die gesuchte Funktion ist, während der Bestandteil $f(x,y)$ des Kerns eine gegebene Funktion ist, die gewissen, weiter unten angegebenen beschränkenden Voraussetzungen unterworfen ist. Für die gesuchte Funktion $\varphi(y)$ machen wir den Ansatz

$$\varphi(y) = \int_{-\infty}^{+\infty} \Phi(z)e^{-izy}dz,$$

aus dem nach dem FOURIERschen Integraltheorem, falls $\Phi(x)$ die Bedingungen für dessen Gültigkeit erfüllt, umgekehrt

$$2\pi\Phi(x) = \int_{-\infty}^{+\infty} \varphi(y)e^{ixy}dy$$

folgt. Danach verwandelt sich (1) in

$$2\pi\Phi(x) + \lambda \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} \Phi(z)f(x,y)e^{-izy}dz\,dy = \psi(x)$$

oder

$$2\pi\Phi(x) + \lambda \int_{-\infty}^{+\infty} \Phi(z)K(x,z)dz = \psi(x),$$

wenn

$$K(x,z) = \int_{-\infty}^{+\infty} f(x,y)e^{-izy}dy \tag{2}$$

gesetzt wird, und damit sind wir bereits bei einer Integralgleichung 2. Art angelangt. Der Kern (2) gestattet die Anwendung der FREDHOLMschen Methode z. B. dann, wenn $f(x,y)$ und $\frac{\partial f(x,y)}{\partial y}$ gleichmäßig in x für $y = \pm\infty$ gegen 0 konvergieren und die Ungleichung

$$\frac{\partial^2 f}{\partial y^2} < \frac{M}{1+y^2}$$

statthat, in der M eine von x und y unabhängige Konstante bedeutet. Von $\psi(x)$ genügt es etwa, anzunehmen, daß es nur endlichviele Maxima und Minima besitzt und im Intervall $-\infty \cdots +\infty$ absolut integrierbar ist.

Wir können dieselbe Methode auf eine Reihe

$$\psi(x) = \sum_{(m)} A_m \left[e^{imx} + \lambda\theta_m(x)\right]$$

anwenden; das Problem ist hier also, wenn $\psi(x)$ und die Funktionen $\theta_m(x)$ gegeben sind, die Koeffizienten A_m so zu berechnen, daß die hingeschriebene Entwicklung gültig ist. Handelte es sich soeben um eine Erweiterung des *Fourierschen Integraltheorems*, so haben wir es jetzt mit einer Verallgemeinerung der *Fourierschen Reihe* zu tun.

Setzen wir $\varphi(z)$ in der Form

$$\varphi(z) = \sum_{(m)} A_m e^{imz}; \quad 2\pi A_m = \int_0^{2\pi} \varphi(z) e^{-imz} dz$$

an, so bekommen wir

$$\psi(x) = \varphi(x) + \frac{\lambda}{2\pi} \int_0^{2\pi} \varphi(z) \cdot \sum_{(m)} e^{-imz}\theta_m(x) \cdot dz.$$

Von der Reihe, welche hier als Kern fungiert, müssen wir voraussetzen, daß sie absolut und gleichmäßig konvergiert, d. h. wir müssen annehmen, daß

$$\sum_{(m)} |\theta_m(x)| \tag{3}$$

gleichmäßig konvergiert.

Setzen wir beispielsweise

$$\lambda = 1, \quad \theta_m(x) = e^{i\mu_m x} - e^{imx},$$

so erhalten wir eine Entwicklung der Form

$$\psi(x) = \sum_{(m)} A_m e^{i\mu_m x}.$$

Die Bedingung (3) ist erfüllt, wenn wir die absolute Konvergenz von

$$\sum_{(m)}(\mu_m - m)$$

voraussetzen.

Endlich betrachten wir noch die Gleichung

$$\int_0^{2\pi} \varphi(y)[e^{ixy} + \lambda f(x,y)]dy = \psi(x), \quad (-\infty < x < +\infty) \tag{4}$$

welche sich von (1) dadurch unterscheidet, daß das Integral nicht in unendlichen, sondern in endlichen Grenzen zu nehmen ist. In diesem Fall darf $\psi(x)$ nicht willkürlich gewählt werden: es muß, falls $f(x,y)$ holomorph ist, sicher eine ganze transzendente Funktion sein, wenn die Gleichung (4) eine Auflösung besitzen soll. Dagegen dürfen die Werte $\psi(m)$ dieser Funktion ψ für alle ganzen Zahlen m im wesentlichen willkürlich angenommen werden. Setze ich nämlich

$$\varphi(z) = \sum_{(m)} A_m e^{-imz}, \quad \text{wo} \quad 2\pi A_m = \int_0^{2\pi} \varphi(y)e^{imy}dy \quad \text{ist,}$$

so verwandelt sich (4), für $x = m$ genommen, in

$$2\pi A_m + \lambda \sum_{(p)} A_p \int_0^{2\pi} e^{-ipy} f(m,y)dy = \psi(m).$$

Wir gelangen so zu einem System unendlich vieler linearer Gleichungen mit unendlich vielen Unbekannten, wie sie von Hill, H. v. Koch, Hilbert u. a. untersucht worden sind. Die Lösung dieses Systems ist, falls wir für die Reihe

$$\sum_{(p,m)} \int_0^{2\pi} e^{-ipy} f(m,y)dy \tag{5}$$

die Voraussetzung absoluter und gleichmäßiger Konvergenz machen, der Fredholmschen Lösung der Integralgleichungen durchaus analog und stellt sich wie diese als meromorphe Funktion des Parameters λ dar. Die gleichmäßige und absolute Konvergenz von (5) ist aber, wie sich durch partielle Integration ergibt, sichergestellt, falls die Summe

$$\sum_{(m)} f''(m,z)$$

oder das Integral

$$\int_{-\infty}^{+\infty} f''(x,z)dx$$

absolut und gleichmäßig konvergiert.

Man sieht die Ähnlichkeit und den Unterschied der beiden Fälle (1) und (4) deutlich: je nachdem die Integrationsgrenzen unendlich oder endlich sind — oder auch, je nachdem der Kern in den Integrationsgrenzen keine oder eine genügend hohe Singularität aufweist —, darf man die "gegebene" Funktion im wesentlichen willkürlich wählen oder ihr nur eine zwar unendliche, jedoch *diskrete* Reihe von Funktionswerten vorschreiben. Es wäre wohl nicht ohne Interesse, den hier zur Geltung kommenden Unterschied mit Hilfe der Iteration der Kerne näher zu betrachten.

Zweiter Vortrag

ANWENDUNG
DER THEORIE DER INTEGRALGLEICHUNGEN
AUF DIE FLUTBEWEGUNG DES MEERES

Ich will heute über einige Anwendungen der Integralgleichungstheorie auf die Flutbewegung berichten, die ich im letzten Semester gelegentlich einer Vorlesung über diese Erscheinung gemacht habe.

Die Differentialgleichungen des Problems sind die folgenden:

$$(1) \qquad \begin{cases} \text{a)} & k^2 \sum \frac{\partial}{\partial x}\left(h_1 \frac{\partial \varphi}{\partial x}\right) + k^2 \left(\frac{\partial \varphi}{\partial x}\frac{\partial h_2}{\partial y} - \frac{\partial \varphi}{\partial y}\frac{\partial h_2}{\partial x}\right) = \zeta, \\ \text{b)} & g \cdot \zeta = -\lambda^2 \varphi + \Pi + W. \end{cases}$$

Wir stellen uns dabei vor, daß die Kugeloberfläche der Erde etwa durch stereographische Projektion konform auf die (x, y)-Ebene bezogen sei; dann bedeute $k(x, y)$ das Ähnlichkeitsverhältnis der Abbildung zwischen Ebene und Kugel. Die Lösung des Flutproblems denken wir uns durch periodische Funktionen der Zeit t gegeben, und wir nehmen speziell an, daß unsere Gleichungen (1) einem einzigen periodischen Summanden von der Form $A\cos(\lambda t + \alpha)$ entsprechen, sodaß also λ in unseren Gleichungen die Schwingungsperiode bestimmt; es ist bequem, statt des Kosinus komplexe Exponentialgrößen einzuführen und also etwa anzunehmen, daß alle unsere Funktionen die Form

$$e^{i\lambda t} \cdot f(x, y)$$

haben; der reelle und imaginäre Teil dieser komplexen Lösungen stellt uns dann die physikalisch brauchbaren Lösungen dar.

$\varphi(x, y)$ ist definiert durch

$$-\lambda^2 \varphi = V - p,$$

wo V das hydrostatische Potential, p der Druck ist.

Ist h die Tiefe des Meeres, so definieren wir

$$h_1 = -\frac{h\lambda^2}{-\lambda^2 + 4\omega^2 \cos^2 \vartheta},$$
$$h_2 = -\frac{2\omega i \cos \vartheta}{\lambda} h_1, \qquad (i = \sqrt{-1})$$

wo ϑ die Colatitude des zu (x, y) gehörigen Punktes der Erde, ω die Winkelgeschwindigkeit der Erde bedeutet. $\zeta(x, y)$ ist die Differenz zwischen der Dicke der mittleren und der gestörten Wasserschicht, d. h. $\zeta > 0$ entspricht der Ebbe, $\zeta < 0$ der Flut. g ist die Beschleunigung der Schwerkraft, W das Potential der Störungskräfte, Π ist das Potential, welches von der Anziehung der Wassermassen von der Dicke ζ herrührt. Ist z. B.

$$\zeta = \sum A_n X_n,$$

so wird

$$\Pi = \sum \frac{4\pi A_n}{2n+1} X_n,$$

wo die X_n die Kugelfunktionen sind.

Die Einheiten sind so gewählt, daß die Dichte des Wassers gleich 1, der Radius der Erdkugel gleich 1 ist.

Die Größe Π kann man meistens vernachlässigen; tut man dies, so erhält man sofort für φ eine partielle Differentialgleichung 2. Ordnung. Um aus derselben φ zu bestimmen, muß man gewisse Grenzbedingungen vorschreiben. Wir unterscheiden da zwei Fälle:

1. Der Rand des Meeres ist eine vertikale Mauer; dann wird

$$\frac{\partial \varphi}{\partial n} + \frac{2\omega i}{\lambda} \cos \vartheta \cdot \frac{\partial \varphi}{\partial s} = 0,$$

wobei $\frac{\partial \varphi}{\partial n}$, $\frac{\partial \varphi}{\partial s}$ die normale bzw. tangentiale Ableitung von φ ist.

2. Der Rand des Meeres ist nicht vertikal; dann ist dort

$$h = 0, \quad \text{also} \quad h_1 = h_2 = 0.$$

Die Grenzbedingung lautet hier, daß φ am Rande regulär und endlich bleiben soll.

Um auf diese Probleme die Methoden der Integralgleichungen anwenden zu können, erinnern wir uns zunächst der allgemeinen Überlegungen, wie sie Hilbert und Picard für Differentialgleichungen anstellen. Sei

$$D(u) = f(x, y)$$

eine partielle Differentialgleichung 2. Ordnung für u, die elliptischen Typus hat, so ist eine, gewisse Grenzbedingungen erfüllende, Lösung u darstellbar in der Form

$$u = \int f' G \, d\sigma',$$

wobei $G(x, y; x', y')$ die zu diesen Randbedingungen gehörige Greensche Funktion des Differentialausdruckes $D(u)$ ist; f' ist $f(x', y')$, $d\sigma' = dx' \cdot dy'$, und das Integral ist über dasjenige Gebiet der (x', y')-Ebene zu erstrecken, für welches die Randwertaufgabe gestellt ist. Um die Greensche Funktion zu berechnen und so die Randwertaufgabe zu lösen, setze man

$$D(u) = D_0(u) + D_1(u),$$

wo

$$D_1(u) = a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} + cu$$

ein linearer Differentialausdruck ist. Nehmen wir nun an, wir kennen die Greensche Funktion G_0 von $D_0(u)$, so haben wir die Lösung von

$$D(\varphi) = f$$

in der Form

$$\varphi = \int G_0 \left(f' - a' \frac{\partial \varphi'}{\partial x'} - b' \frac{\partial \varphi'}{\partial y'} - c' \varphi' \right) d\sigma'.$$

Schaffen wir hieraus durch partielle Integrationen die Ableitungen $\frac{\partial\varphi'}{\partial x'}$, $\frac{\partial\varphi'}{\partial y'}$ heraus, so werden wir direkt auf eine Integralgleichung zweiter Art für φ geführt, die wir nach der FREDHOLMschen Methode behandeln können, wenn ihr Kern nicht zu stark singulär wird.

Bei unserem Probleme der Flutbewegung tritt nun gerade dieser Fall ein; der Kern wird so hoch unendlich, daß die FREDHOLMschen Methoden versagen; ich will Ihnen jedoch zeigen, in welcher Weise man diese Schwierigkeiten überwinden kann.

Betrachten wir erst den Fall der ersten Grenzbedingung

$$\frac{\partial\varphi}{\partial n} + C\frac{\partial\varphi}{\partial s} = 0,$$

wo C eine gegebene Funktion von x, y ist. Die Differentialgleichung, die sich bei Vernachlässigung von Π ergibt, hat die Form

$$A\Delta\varphi + D_1 = f,$$

und wir stehen daher vor der Aufgabe, die Gleichung

$$\Delta\varphi = F$$

mit unserer Randbedingung zu integrieren.

Diese Aufgabe ist äquivalent mit der, eine im Innern der Randkurve reguläre Potentialfunktion V, die am Rande die Bedingung $\frac{\partial V}{\partial n} + C\frac{\partial V}{\partial s} = 0$ erfüllt, als Potential einer einfachen Randbelegung zu finden. Bezeichnet s die Bogenlänge auf der Randkurve von einem festen Anfangspunkte bis zu einem Punkte P, s' die bis zum Punkte P', so erhält man für V eine Integralgleichung; jedoch wird der Kern $K(s, s')$ derselben für $s = s'$ von der ersten Ordnung unendlich, und es ist daher in dem Integrale

$$\int_A^B K(x, y)f(y)dy$$

der sogenannte CAUCHYsche Hauptwert zu nehmen, der definiert ist als das arithmetische Mittel aus den beiden Werten, die das Integral erhält, wenn ich es in der komplexen y-Ebene unter Umgehung des Punktes $y = x$ das eine mal auf einem Wege AMB oberhalb, das andere mal auf einem Wege $AM'B$ unterhalb der reellen Achse führe.

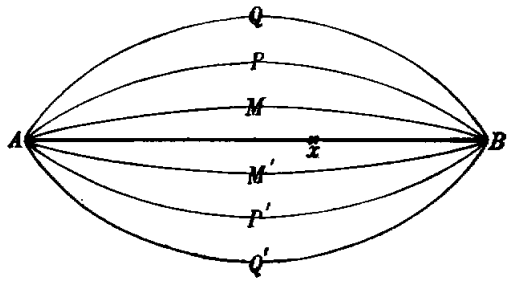
Anstatt die Methoden zu benutzen, die KELLOGG zur Behandlung solcher unstetiger Kerne angibt, will ich einen andern Weg einschlagen. Wir betrachten neben der Operation

$$S\big(f(x)\big) = \int K(x, y)f(y)dy$$

die iterierte

$$S^2\big(f(x)\big) = \iint K(x, z)K(z, y)f(y)dz\,dy,$$

bei der ebenfalls das Doppelintegral als CAUCHYscher Hauptwert zu nehmen ist; dies soll folgendermaßen verstanden werden: wir betrachten für die Variable z die Wege AMB, $AM'B$, für y die Wege APB, $AP'B$, die zueinander liegen mögen, wie in der Figur angedeutet ist. Dann bilden wir die 4 Integrale, die sich ergeben, wenn ich einen Weg für z mit einem für y kombiniere;



z : AMB, $AM'B$, AMB, $AM'B$
y : APB, APB, $AP'B$, $AP'B$,

und nehmen aus diesen 4 Integralen das arithmetische Mittel. Ziehen wir noch 2 Wege AQB, $AQ'B$ wie in der Figur, so sehen wir, daß sich in der ersten Wegkombination der Weg AMB für z ersetzen läßt durch $AQB + AMBQA$, in der zweiten $AM'B$ durch $AQ'B$, in der dritten AMB durch AQB und in der vierten $AM'B$ durch $AQ'B + AM'BQ'A$, sodaß wir jetzt die folgenden Wegkombinationen haben:

z	y
$AQB + AMBQA$	APB
$AQ'B$	APB
AQB	$AP'B$
$AQ'B + AM'BQ'A$	$AP'B$.

Führen wir jetzt die Integrale aus und wenden den Residuenkalkül auf die geschlossenen Wege an, so zeigt sich, daß unsere Operation $S^2\big(f(x)\big)$, die einer Integralgleichung 1. Art zugehört, übergeht in eine Operation, welche durch die linke Seite einer Integralgleichung 2. Art gegeben ist, deren Kern überall endlich bleibt; wenn wir zuerst die vier Kombinationen von den Wegen AQB und $AQ'B$ mit den Wegen APB und $AP'B$ nehmen, so bekommen wir ein doppeltes Integral, welches nicht unendlich werden kann, da auf diesen Wegen $x \neq y$ und $y \neq z$. Betrachten wir jetzt die beiden Wegkombinationen $AMBQA$, APB und $AM'BQ'A$, $AP'B$, oder $AMBQA$, APB und $AQ'BM'A, BP'A$, so ist leicht zu sehen, daß z eine geschlossene Kurve $AMBQA$ oder $AQ'BM'A$ um y beschreibt, und daß gleichzeitig y eine geschlossene Kurve $APBP'A$ um x beschreibt. Wir dürfen also die Residuenmethode anwenden, und wir bekommen ein Glied, wo die unbekannte Funktion ohne Integralzeichen auftritt, wie in der linken Seite einer Integralgleichung zweiter Art. Indem wir so auf eine durchaus reguläre Integralgleichung 2. Art geführt werden, die der FREDHOLMschen Methode zugänglich ist, haben wir die Schwierigkeit bei unserem

Problem überwunden.

Nur ein Punkt bedarf noch der Erläuterung: wenn x und y gleichzeitig in einen der Endpunkte A, B des Intervalles hineinfallen, so versagen zunächst die obigen Betrachtungen, und es scheint, als wären wir für diese Stellen der Endlichkeit unseres durch Iteration gewonnenen Kernes nicht sicher. Dieses Bedenken wird jedoch bei unserm Problem dadurch beseitigt, daß der Rand des Meeres, der das Integrationsintervall darstellt, geschlossen ist, woraus sich ergibt, daß die Punkte A, B keine Ausnahmestellung einnehmen können.

Durch diese Überlegungen ist also der Fall der vertikalen Meeresufer erledigt.

Wir betrachten den zweiten und schwierigeren Fall, daß das Ufer des Meeres keine vertikale Mauer ist. Dann ist am Rande

$$h = h_1 = h_2 = 0.$$

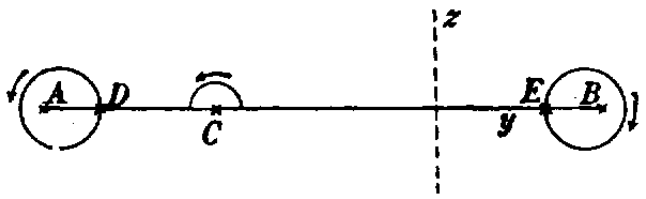
Da die Glieder 2. Ordnung unserer Differentialgleichung für φ durch den Ausdruck

$$h_1 \Delta\varphi$$

gegeben sind, so ist die Randkurve jetzt eine singuläre Linie für die Differentialgleichung. Außerdem werden h_1, h_2 gemäß ihrer Definition für die durch die Gleichung

$$4\omega^2 \cos^2 \vartheta = \lambda^2$$

gegebene *kritische geographische Breite* ϑ unendlich. Um trotz dieser Singularitäten, welche das Unendlichwerden des Kerns K zur Folge haben, das Problem durchzuführen, bin ich gezwungen gewesen, das reelle Integrationsgebiet durch ein komplexes zu ersetzen, indem ich y in eine komplexe Veränderliche $y + iz$ verwandle; x hingegen bleibt reell.



Wir deuten xyz als gewöhnliche rechtwinklige Koordinaten in einem dreidimensionalen Raum und zeichnen den Durchschnitt AB einer Ebene $x = \text{konst.}$ mit dem in der (x, y)-Ebene gelegenen Meeresbecken. Entspricht C der kritischen geographischen Breite, so ist es nicht schwer, diese Singularität durch Ausweichen in das komplexe Gebiet zu umgehen. Wählen wir ferner irgend zwei Punkte D, E zwischen A und B und umgeben A, von D ausgehend und dorthin zurückkehrend, mit einer kleinen Kurve und verfahren entsprechend bei B — räumlich gesprochen: umgeben wir die Randkurve mit einem ringförmigen Futteral —, so stellen wir uns jetzt das Problem, unsere Differentialgleichung so zu integrieren, daß φ, wenn wir seine Wertänderung längs der den Punkt

A umgebenden Kurve verfolgen, mit demselben Wert nach D zurückkehrt, mit dem es von dort ausging. Diese "veränderte" Grenzbedingung ist mit der ursprünglichen, welche verlangte, daß φ am Rande (im Punkte A) endlich bleibt und sich regulär verhält, äquivalent. Zwar sind die zu der neuen und der alten Grenzbedingung gehörigen GREENschen Funktionen G, G_1 nicht identisch, wohl aber die den betreffenden Randbedingungen unterworfenen Lösungen von

$$D(u) = f. \tag{1}$$

Hiervon überzeugen wir uns leichter im Falle nur *einer* Variablen y; dann ergeben die Gleichungen

$$u = \int G(y, y')f(y')dy', \quad u_1 = \int G_1(y, y')f(y')dy'$$

durch Anwendung des CAUCHYschen Integralsatzes, daß $u - u_1 = 0$ ist.

Um jetzt das Problem (1) zu behandeln, ziehe ich die vorige Methode heran, die hier aber in zwei Stufen zur Anwendung kommt, da unsere veränderte Randbedingung für die Gleichung $\Delta u = f$ unzulässig ist.[1] Wir können setzen

$$D(u) = \Delta(h_1 u) + D_1(u) + D_2(u);$$

dabei soll $D_1(u)$ nur die Glieder 1. Ordnung $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$, $D_2(u)$ aber nur u selbst enthalten. Indem wir

$$\Delta(v) = f$$

unter der Randbedingung $v = 0$ integrieren, erhalten wir für $u = \frac{v}{h_1}$ eine am Rande endliche und reguläre Funktion, für welche

$$\Delta(h_1 u) \equiv D_0(u) = f$$

ist. Darauf integrieren wir

$$D_0(u) + D_2(u) = f$$

unter Zugrundelegung der ursprünglichen Grenzbedingung nach der gewöhnlichen Methode. Der in der hierbei zu benutzenden Integralgleichung auftretende Kern ist zwar unendlich, aber von solcher Ordnung, daß sich die Singularität durch Iteration des Kerns beseitigen läßt: die partielle Integration, welche Glieder von einer zu hohen Ordnung des Unendlichwerdens einführen würde, bleibt uns an dieser Stelle erspart.

Das damit bewältigte Integrationsproblem ist aber der Integration von

$$D_0(u) + D_2(u) = f$$

unter der veränderten Grenzbedingung äquivalent, und infolgedessen können wir jetzt die zweite Stufe ersteigen und auch die Lösung von

$$D(u) \equiv \big(D_0(u) + D_2(u)\big) + D_1(u) = f$$

[1] Diese Randbedingung ist nicht von solcher Art, daß sie eine bestimmte Lösung von $\Delta(u) = f$ auszeichnet.

unter der veränderten Grenzbedingung bestimmen.

Wir haben bis jetzt das Glied Π als so klein vorausgesetzt, daß wir es ganz vernachlässigen durften. Heben wir diese Voraussetzung auf, so entstehen keine wesentlichen neuen Schwierigkeiten. Π ist ein von ζ erzeugtes Anziehungspotential; wir haben also

$$\Pi = \int \frac{\zeta' d\sigma'}{r},$$

wenn $d\sigma'$ ein Flächenelement der Kugel, ζ' den Wert der Funktion ζ im Schwerpunkt (x', y') dieses Flächenelementes, r aber die räumlich gemessene Entfernung der beiden Kugelpunkte (x, y); (x', y') bedeutet, und die Integration über die ganze Kugeloberfläche erstreckt wird. Wir können auch schreiben

$$\Pi = \int \frac{\zeta' dx' dy'}{k^2 r}.$$

Setzen wir dies in unsere Ausgangsgleichungen ein, von denen wir noch die erste mittels Aufstellung der zugehörigen GREENschen Funktion und unter Berücksichtigung der Randbedingung aus einer Differential- in eine Integralgleichung verwandeln, so erhalten wir zwei simultane Integralgleichungen für ζ und φ, die mit Hilfe der soeben erörterten Methoden aufgelöst werden können.

Dritter Vortrag

ANWENDUNG DER INTEGRALGLEICHUNGEN AUF HERTZSCHE WELLEN

Ich will heute über eine Anwendung der Integralgleichungen auf HERTZsche Wellen vortragen und insbesondere die äußerst merkwürdigen Beugungserscheinungen behandeln, welche bei der drahtlosen Telegraphie eine so wichtige Rolle spielen; ist es doch eine wunderbare Tatsache, daß die Krümmung der Erdoberfläche, welche eine Fortpflanzung des Lichtes verhindert, für die Ausbreitung der HERTZschen Wellen kein Hindernis darstellt, daß dieselben vielmehr auf der Erdoberfläche von Europa bis Amerika zu laufen vermögen. Der Umstand, daß die HERTZschen Wellen eine viel größere Länge haben als die Lichtwellen, kann allein diese Erscheinung noch nicht erklären. Eine solche Erklärung ergibt sich vielmehr erst durch Betrachtung der Differentialgleichungen des Problems.

Setzen wir die Lichtgeschwindigkeit gleich 1, und verstehen wir mit MAXWELL

unter	α,	β,	γ	die Komponenten der magnetischen Kraft,
unter	F,	G,	H	die Komponenten des Vektorpotentiales,
unter	f,	g,	h	die Komponenten der elektrischen Verschiebung,
unter		ψ		das skalare Potential,
unter	u,	v,	w	die Komponenten des Konduktionsstromes,
unter		ϱ		die Dichte der Elektrizität,

so gelten die Gleichungen

$$\alpha = \frac{\partial H}{\partial y} - \frac{\partial G}{\partial x}$$

$$4\pi f = -\frac{\partial F}{\partial t} - \frac{\partial \psi}{\partial x},$$

$$4\pi \left(\mu + \frac{\partial f}{\partial t}\right) = \frac{\partial \gamma}{\partial y} - \frac{\partial \beta}{\partial z},$$

$$\sum \frac{\partial f}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} = \varrho,$$

$$\sum \frac{\partial F}{\partial x} + \frac{\partial \psi}{\partial t} = 0,$$

und es folgt

$$4\pi \cdot \mu = \frac{\partial^2 F}{\partial t^2} - \Delta F,$$
$$4\pi \cdot \varrho = \frac{\partial^2 \psi}{\partial t^2} - \Delta \psi.$$

Wir betrachten nun eine gedämpfte synchrone Schwingung, indem wir annehmen, daß alle unsere Funktionen proportional sind mit der Exponentialgröße

$$e^{i\omega t}.$$

Aus den so zustande kommenden komplexen Lösungen erhalten wir die physikalischen durch Trennung in reellen und imaginären Bestandteil. Der reelle Teil von ω gibt die Schwingungsperiode, der imaginäre die Dämpfung.

Aus unserem Ansatz folgt

$$\frac{\partial F}{\partial t} = i\omega \cdot F,$$
$$\frac{\partial \psi}{\partial t} = i\omega \cdot \psi,$$

und man kann daher F und ψ als retardierte Potentiale darstellen wie folgt:

$$F = \int \mu' \frac{e^{-i\omega r}}{r} d\tau',$$
$$\psi = \int \varrho' \frac{e^{-i\omega r}}{r} d\tau';$$

$d\tau'$ ist das Raumelement im (x', y', z')-Raume, μ', ϱ' die Werte von μ, ϱ im Punkte (x', y', z'), r die Entfernung der Punkte (x', y', z') und (x, y, z).

In den meisten Problemen treten zwei verschiedene Medien auf, der freie Äther und die leitenden Körper; von den letzteren wollen wir annehmen, daß sie sich wie vollkommene Leiter verhalten, daß also in ihrem Innern das Feld verschwindet, die elektrischen Kraftlinien auf ihrer Oberfläche normal stehen, während die magnetischen in dieselbe hineinfallen; dem Umstande, daß Ladung und Strömung nur an der Oberfläche des Leiters vorhanden ist, wollen wir dadurch entsprechen, daß wir die obigen Ausdrücke für F und ψ modifizieren, indem wir an Stelle der Raumintegrale Oberflächenintegrale einführen. Wir schreiben

$$\psi = \int \varrho'' \frac{e^{-i\omega r}}{r} d\sigma',$$
$$F = \int \mu'' \frac{e^{-i\omega r}}{r} d\sigma',$$

wo ϱ'', μ'' jetzt die Flächendichte der Ladung bzw. Strömung bedeuten und $d\sigma'$ das Flächenelement ist.

Wir unterscheiden gewöhnlich zwei leitende Körper, der eine soll der äußere, der andere der innere Leiter heißen; sie erzeugen das "äußere" resp. das "innere" Feld; das äußere Feld ist gegeben, das innere gesucht. So ist z. B., wenn wir das Problem des Empfanges elektrischer Wellen betrachten, der Sender der äußere, der Empfangsapparat der innere Leiter; beim Probleme der Beugung elektrischer Wellen ist der Erreger der äußere, die Erdkugel der innere Leiter; bei dem Probleme der Schwingungserzeugung haben wir kein äußeres Feld, der Erreger wird dann als innerer Leiter anzusehen sein.

Um nun zum Ansatz einer Integralgleichung zu gelangen, wollen wir unter den oben erklärten Funktionen nur die zum unbekannten inneren Felde gehörigen verstehen, sodaß z.B. die obigen Integrale nur über die Oberfläche

des inneren Leiters zu erstrecken sind; beachten wir nun, daß die innere Normalkomponente des elektrischen Vektors am inneren Leiter unserer obigen Annahme zufolge verschwinden muß, so folgt, wenn l, m, n die Richtungskosinus der Normale bedeuten, aus unseren Ausgangs-Gleichungen:

$$4\pi f = \frac{\partial \psi}{\partial n} + i\omega\,(lF + mG + nH) = N,$$

wo N die Normalkomponente des äußeren Feldes, also eine bekannte Funktion ist.

Bezeichnen wir jetzt die Flächendichte statt mit ϱ'' mit μ', so wird zufolge unseres Ausdruckes für ψ

$$\frac{\partial \psi}{\partial n} = 2\pi\mu + \int \mu' \frac{\partial}{\partial n}\left(\frac{e^{-i\omega r}}{r}\right) d\sigma'.$$

Benutzen wir ferner unseren Ausdruck für F und die entsprechenden für G und H, so hat man

$$i\omega \sum lF = \int \frac{e^{-i\omega r}}{r} i\omega \sum l\mu''\, d\sigma'.$$

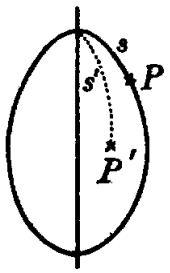
Diesen Ausdruck kann man nun in gewissen Fällen durch partielle Integrationen auf die Form

$$-i\omega \int L\mu'\, d\sigma'$$

bringen, wobei L eine bekannte Funktion ist. So haben wir schließlich

$$2\pi\mu + \int \mu' \left\{ \frac{\partial}{\partial n}\left(\frac{e^{-i\omega r}}{r}\right) - i\omega L \right\} d\sigma' = N,$$

und dies ist die Integralgleichung 2. Art für μ, auf die wir hinstrebten. Im allgemeinsten Falle bekommt man zwei Integralgleichungen mit zwei Unbekannten, welche z. B. μ und ν sein mögen, wo μ das oben definierte ist; wir setzen $\nu = \frac{dN}{dn}$, wo $\frac{d}{dn}$ die Ableitung in der Normalrichtung bezeichnet und N die Normalkomponente der magnetischen Kraft ist.



Die Funktion L läßt sich dann besonders einfach bilden, wenn der innere Leiter ein Rotationskörper ist und das äußere Feld Rotationssymmetrie besitzt.

Ist s, s' die Bogenlänge, gemessen vom Endpunkte der Rotationsachse auf einem Meridian bis zu den Punkten P, P', ist ferner ϑ der Winkel zwischen der Normale in P und der Meridiantangente in P', so wird L als Funktion von ϑ, s, s' definiert durch die Differentialgleichung

$$\frac{\partial L}{\partial s'} = \frac{e^{-i\omega r}}{r} \cos\vartheta.$$

Das Problem des Empfanges elektrischer Wellen läßt sich auf Grund der obigen Integralgleichung 2. Art behandeln.

Wollen wir nur das Problem der Erzeugung elektrischer Wellen betrachten, so haben wir das äußere Feld gleich Null zu setzen, es wird also $N = 0$, und wir haben eine homogene Integralgleichung vor uns; in ihr darf jedoch ω nicht mehr einen willkürlichen Parameterwert bedeuten, sondern ist eine zu bestimmende Zahl, die die Rolle der Eigenwerte spielt.

Ich schreibe unsere Integralgleichung in der Form

$$2\pi\mu + \int K\mu'\, d\sigma' = N$$

mit dem Kerne K; ich führe einen unbestimmten Parameter λ ein und betrachte die allgemeine Gleichung

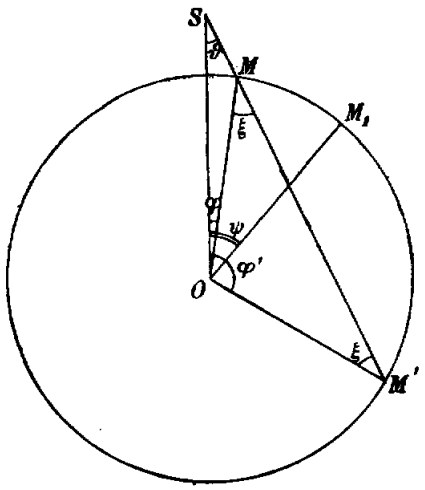
$$2\pi\mu + \lambda \int K\mu'\, d\sigma' = N.$$

Das erste Glied hängt von zwei Unbestimmten λ und ω ab. Wenn man die gewöhnliche FREDHOLMsche Methode anwendet, so erhält man die Lösung unserer obigen Integralgleichung in Gestalt einer meromorphen Funktion von λ, deren Nenner eine ganze Funktion von λ ist. Man kann nun zeigen, daß dieser Nenner auch eine ganze Funktion von ω wird, sodaß also auch hier unsere ausgezeichneten Werte ω sich als Nullstellen einer ganzen transzendenten Funktion ergeben.

Wir wollen aber jetzt das größere Problem der Beugung ausführlicher behandeln.

Nehmen wir zu diesem Ende an, daß der innere Leiter eine Kugel, die Erdkugel, vom Radius ϱ ist und das äußere Feld (dessen normale Komponente N bedeutet) von einem punktförmigen Erreger S herrührt, dessen Entfernung D vom Mittelpunkt O der Erde nur sehr wenig größer ist als der Radius ϱ. Wir wählen die Richtung OS zur z-Achse und bezeichnen die Abweichung der Richtung OM, in der M einen variablen Punkt der Kugeloberfläche bedeutet, von OS mit φ. Die Bedeutung von ϑ, ξ, φ'; r, r' ist aus der Figur ersichtlich:

$$\begin{aligned} OM = OM' = OM_1 &= \varrho, \\ OS &= D, \\ SM &= r, \\ SM' &= r'. \end{aligned}$$



Der Wert der normalen Ableitung N des äußeren Feldes berechnet sich im Punkte M, wie leicht zu sehen, nach der Formel

$$4\pi N = e^{i\omega(t-r)}\left[\frac{i\omega}{r}\sin\vartheta\sin\xi + \left(\frac{1}{r^2}+\frac{1}{i\omega r^3}\right)\cdot(\sin\vartheta\sin\xi + 2\cos\vartheta\cos\xi)\right].$$

Da ω eine sehr große Zahl ist — denn die Länge der HERTZschen Wellen ist klein gegenüber dem Radius der Erde — genügt es meistens, in dieser Formel nur das erste Glied, das in der eckigen Klammer auftritt, beizubehalten.

Im vorhergehenden haben wir die Gleichung der HERTZsehen Wellen auf die Form

$$2\pi\mu = \int \mu' K d\sigma' + N$$

gebracht und haben gezeigt, wie der Kern K berechnet werden kann. Entwickeln wir jetzt N und K nach Kugelfunktionen oder vielmehr, da unser Problem die Symmetrie eines Rotationskörpers mit der Achse OS besitzt, nach LEGENDREschen Polynomen P_n, so gewinnen wir aus dieser Integralgleichung die elektrische Flächendichte μ gleichfalls unter der Form einer nach den Funktionen P_n fortschreitenden Reihe. Es gilt zunächst

$$N = \sum K_n P_n; \qquad \int_0^\pi P_n N \sin\varphi\, d\varphi = \frac{2K_n}{2n+1}.$$

K_n ist von der Form

$$\frac{A_n J_n(\omega\varrho)}{\varrho^2},$$

wo A_n eine nur von n, nicht aber von ϱ abhängige Zahl ist, und J_n eine mit der BESSELschen verwandte Funktion bedeutet.

Wir verstehen nämlich unter $J_n(x)$ die in der Umgebung von $x = 0$ holomorphe Lösung der Gleichung

$$\frac{d^2y}{dx^2} + y\left(1 - \frac{n\,(n+1)}{x^2}\right) = 0,$$

und $I_n(x)$ sei dasjenige Integral derselben Gleichung, welches sich für große positive Werte von x angenähert wie e^{-ix} verhält. Da J_n, I_n von einander unabhängig sind, können wir außerdem dafür sorgen, daß

$$I_n{}'J_n - J_n{}'I_n = 1$$

ist, wenn unter $J_n{}'$, $I_n{}'$ die Ableitungen von J_n, I_n verstanden werden.

Die Lösung unserer Integralgleichung lautet jetzt

$$\mu = A\sum \frac{K_n P_n\,(\cos\varphi)}{I_n{}'\,(\omega\varrho)\,J_n\,(\omega\varrho)}.$$

Da aber auch der Ausdruck von K_n im Zähler $J_n\,(\omega\varrho)$ als Faktor enthält, und sich infolgedessen dieser Term $J_n\,(\omega\varrho)$ heraushebt, ist

$$I_n{}'\,(\omega\varrho) = 0$$

die für die *Eigenschwingungen* charakteristische Gleichung.

Um zu übersichtlichen Resultaten zu gelangen, benutzen wir angenäherte Formeln. Diese beruhen darauf, daß ω sehr groß, andererseits aber $\frac{D}{\varrho} - 1$ sehr klein ist. Wir stützen uns auf die folgende Näherungsformel

$$\int \eta e^{i\omega\theta}dx = \eta e^{i\theta}\sqrt{\frac{2\pi}{\omega\theta''}}e^{\pm\frac{i\pi}{4}},$$

θ, η sind gegebene Funktionen von x, ω eine sehr große Zahl, θ'' bedeutet die zweite Ableitung von θ, und auf der rechten Seite ist als Argument ein solcher Wert einzusetzen, für den θ ein Maximum oder Minimum besitzt; je nachdem der eine oder der andere Fall vorliegt, ist in dem Faktor $e^{\pm\frac{i\pi}{4}}$ das Zeichen $+$ oder das Zeichen $-$ zu nehmen. Hat θ in dem Intervall, über welches zu integrieren ist, mehrere Maxima oder Minima, so ist der Ausdruck rechts durch eine Summe analog gebildeter Terme zu ersetzen.

Durch Anwendung dieser Formel bekommen wir für die LEGENDREschen Polynome $P_n\,(\cos\varphi)$ die folgenden, für große n gültigen angenäherten Ausdrücke:

$$P_n = 2\sqrt{\frac{2\pi}{n\sin\varphi}}\cdot\cos\left(n\varphi + \frac{\varphi}{2} - \frac{\pi}{4}\right).$$

Aus ihnen folgt für die K_n, falls $n < \omega\varrho$,

$$K_n = \frac{2n+1}{8r\sqrt{n}}\left[e^{i\alpha} + e^{i\alpha'}\right]\frac{i\omega\sin\vartheta\sin\xi}{\sqrt{D\varrho\cos\vartheta\cos\xi}}\sqrt{\frac{\sin\vartheta}{\omega\varrho}}.$$

Dabei ist

$$\alpha = n\varphi - \omega r + \frac{\varphi}{2} - \frac{\pi}{2},$$
$$\alpha' = n\varphi' - \omega r' + \frac{\varphi'}{2},$$

gesetzt, und für ξ, ϑ, φ, φ', r, r' sind die aus der Figur zu entnehmenden Werte einzusetzen, für welche

$$\sin\xi = \frac{n}{\omega\varrho} \qquad \left(\xi < \frac{\pi}{2}\right)$$

wird. Die gleiche Näherungsformel gilt auch für $n > \omega\varrho$, falls in der eckigen Klammer $e^{i\alpha} + e^{i\alpha'}$ durch $e^{i\alpha}$ oder $e^{i\alpha'}$ ersetzt wird; die Diskussion darüber, welches der beiden Glieder beizubehalten ist, will ich hier nicht geben.

Auch um $I_n' J_n$ angenähert zu berechnen, müssen wir die beiden Fälle $n < \omega\varrho$ und $n > \omega\varrho$ unterscheiden. Im ersten Falle ist

$$I_n' J_n = e^{i\frac{\alpha-\alpha'}{2}} \cdot \cos\tfrac{\alpha-\alpha'}{2},$$

im zweiten

$$I_n' J_n = \tfrac{1}{2}$$

zu setzen. Daraus ergibt sich, daß sowohl für $n < \omega\varrho$ als auch für $n > \omega\varrho$ und große n

$$\frac{K_n}{I_n' J_n} = \frac{\sqrt{n}}{2r}\, e^{i\alpha}\, \frac{i\sqrt{\omega}\sin\xi(\sin\vartheta)^{\frac{3}{2}}}{\varrho\sqrt{D\cos\vartheta\cos\xi}}$$ [1]

gilt. In der Summe, durch welche wir μ dargestellt haben, geben demnach diejenigen Glieder, für welche nahezu $n = \mu$ ist, den Ausschlag. Für diese gilt näherungsweise

$$\xi = \frac{\pi}{2} \qquad \text{und} \qquad r = \sqrt{2\varrho D}.$$

Da ferner wegen der Kleinheit von $\frac{D}{\varrho} - 1$ der Winkel φ immer nahezu $= 0$ bleibt, variiert α als Funktion von n nur sehr wenig, wenn n auf die dem Werte $n = \omega$ benachbarten ganzen Zahlen beschränkt wird. Wir dürfen also, wenn wir noch die Längeneinheit so gewählt denken, daß $\varrho = 1$ ist, schreiben

$$\mu = C\sum \frac{\sqrt{\omega}\sin\xi(\sin\vartheta)^{\frac{3}{2}}}{\sqrt{\cos\vartheta\cos\xi}} \cdot \frac{1}{\sqrt{\sin\psi}}\left(\cos n\psi + \frac{\psi}{2} - \frac{\pi}{4}\right).$$

Dabei ist μ der Wert der elektrischen Oberflächendichte im Punkte M_1 (s. die Figur).

[1] Der Ausdruck von μ, kann auch auf eine einfachere Form zurückgeführt werden, nämlich

$$\mu = \frac{-i}{4\pi\omega^2\varrho^2 D^2}\sum n\,(n+1)\,(2n+1)\,\frac{I_n(\omega D)}{I_n'(\omega\varrho)} P_n(\cos\varphi)$$

und diese Formel ist nicht eine angenäherte, sondern eine strenge.

Aus

$$\sin\xi = \frac{n}{\omega}, \qquad \sin\vartheta = \frac{n}{\omega D}; \qquad \cos\xi = \sqrt{1-\frac{n^2}{\omega^2}}, \qquad \cos\vartheta = \sqrt{1-\frac{n^2}{D^2\omega^2}}$$

bekommen wir

$$\frac{\sin\xi(\sin\vartheta)^{\frac{3}{2}}}{\sqrt{\cos\vartheta\cos\xi}} = \frac{\frac{n}{\omega}\cdot\left(\frac{n}{\omega D}\right)^{\frac{3}{2}}\sqrt{D}}{\sqrt[4]{\left(1+\frac{n}{\omega}\right)\left(1+\frac{n}{D\omega}\right)}}\cdot\frac{\sqrt{\omega}}{\sqrt[4]{\omega-n}\cdot\sqrt[4]{\omega\,(D-1)}}\cdot\frac{1}{\sqrt[4]{1+\frac{\omega-n}{\omega(D-1)}}},$$

sodaß in der Nähe von $n=\omega$ der linke Ausdruck von derselben Größenordnung ist wie

$$\frac{\sqrt[4]{\omega}}{\sqrt[4]{D-1}}\cdot\frac{1}{\sqrt[4]{n-\omega}}.$$

Führen wir diese Annäherung in unsere Formel für μ ein und ersetzen $\cos\left(n\psi+\frac{\psi}{2}-\frac{\pi}{4}\right)$ zunächst durch $e^{i\left(n\psi+\frac{\psi}{2}-\frac{\pi}{4}\right)}$, so kommen wir auf die Reihe

$$\frac{\omega^{\frac{3}{4}}e^{i\left(\frac{\psi}{2}-\frac{\pi}{4}\right)}}{\sqrt{\sin\psi}\cdot\sqrt[4]{D-1}}\cdot\sum_{(n)}\frac{e^{in\psi}}{\sqrt[4]{n-\omega}}.$$

Schreiben wir

$$S=\sum_{(n)}\frac{e^{in\psi}}{\sqrt{n-\omega}},$$

so können wir

$$\int_{\nu}^{\nu+1} Se^{-i\omega\psi}d\omega \qquad\qquad (\nu \text{ ganzzahlig})$$

als einen Mittelwert der Reihe S betrachten, und ich will S durch diesen Mittelwert ersetzen. Ein solches Verfahren ist gewiß berechtigt, wenn es uns nur daran liegt, die Größenordnung von S festzustellen, umsomehr als in Wirklicheit von einer Antenne nicht bloß Schwingungen einer einzigen Wellenlänge, sondern ein ganzes kontinuierliches Spektrum von Schwingungen ausgeht. Wir erhalten

$$\begin{aligned}\int_{\nu}^{\nu+1} Se^{-i\omega\psi}d\omega &= \sum_{(n)}\int_{\nu}^{\nu+1}\frac{e^{i(n-\omega)\psi}}{\sqrt[4]{n-\omega}}d\omega\\ &= -\int_{-\omega}^{\infty}\frac{e^{iq\psi}}{\sqrt[4]{q}}dq,\end{aligned}$$

und da ω sehr groß ist, wird dieses Integral mit $\int_{-\infty}^{+\infty}\frac{e^{iq\psi}}{\sqrt[4]{q}}dq$ im wesentlichen übereinstimmen.

Auf ähnliche Weise zeigt man, daß der Mittelwert von

$$\sum \frac{e^{-in\psi}}{\sqrt[4]{n-\omega}}$$

gegen den von S zu vernachlässigen ist. Damit gewinnen wir das Resultat, daß

$$\mu \text{ von der Größenordnung } \frac{\sqrt[4]{\omega^3}}{\sqrt[4]{D-1}}$$

und also

$$\frac{\mu}{N} \text{ von der Größenordnung } \frac{1}{\sqrt[4]{\omega(D-1)}}$$

ist. Die Beugung ist daher um so größer, je näher die Quelle S der Erdoberfläche gelegen ist und je länger die entsendeten Wellen sind. Auf diese Weise wird die zunächst staunenerregende Tatsache verständlich, daß es mit Hilfe der in der drahtlosen Telegraphie verwendeten HERTZschen Wellen gelingt, vom europäischen Kontinent z. B. bis nach Amerika zu telegraphieren.

Wenn man nicht den mittleren Wert der Reihe betrachten will, welcher von einem Integral dargestellt wird, sondern den wirklichen Wert dieser Reihe, so hat man eine Diskussion durchzuführen, welche auf einem wohlbekannten ABELschen Satz beruht, und deren Resultate etwas komplizierter, aber sonst ganz ähnlich den vorliegenden sind.

Note. Je me suis aperçu que les dernières conclusions doivent être modifiées. Les formules approchées dont j'ai fait usage ne sont plus vraies lorsque n est très voisin de $\omega\varrho$. Elles doivent être alors remplacées par d'autres, où figure une transcendante entière satisfaisant à l'équation différentielle

$$y'' = xy$$

Mais les termes qui doivent être ainsi modifiés sont en petit nombre et j'avais cru d'abord que le résultat final n'en serait pas modifié. Un examen plus approfondi m'a montré qu'il n'en est rien. La somme des termes modifiés est comparable à celle des autres termes dont j'avais tenu compte et qui est donnée par la formule précédente ; il en résulte une compensation presque complète de sorte que la valeur de μ donnée par les formules définitives est notablement plus petite que celle qui résulterait des formules précédentes.

Vierter Vortrag

ÜBER DIE REDUKTION DER ABELSCHEN INTEGRALE UND DIE THEORIE DER FUCHSSCHEN FUNKTIONEN

Meine Herren! Ich habe die Absicht, Ihnen heute über die Reduktion der ABELschen Integrale im Zusammenhang mit der Theorie der automorphen und insbesondere der FUCHSschen Funktionen vorzutragen.

Ein System von ABELschen Funktionen von p Variabeln und $2p$ Perioden heißt *reduzibel*, wenn es sich auf ein System von q Variabeln und $2q$ Perioden $(q < p)$ zurückführen läßt. Hierbei ist es von vornherein von Wichtigkeit, zwei Fälle zu unterscheiden:

Im *ersten* Falle soll es möglich sein, das System S ABELscher Funktionen von p Variabeln durch eine *algebraische Kurve* C vom Geschlechte p zu erzeugen. Ebenso soll das System S' von q Variabeln aus der Theorie eines algebraischen Gebildes vom Geschlechte q entspringen.

Dieser unser erste Fall ist aber bekanntlich nicht der allgemeine; denn die Kurve C hängt nur von $3p-3$ Konstanten ab, während die allgemeinen ABELschen Funktionen von p Variabeln $\frac{p(p+1)}{2}$ Parameter enthalten. Dadurch ist der *zweite* der beiden Fälle gegeben, die wir unterscheiden. In diesem Falle nämlich soll mindestens eines der beiden Systeme S, S' nicht aus der Theorie der algebraischen Gebilde entspringen.

In meinem heutigen Vortrag will ich mich durchaus auf den erstgenannten Fall beschränken. Aber auch dann muß ich noch zwei Fälle unterscheiden. Wir knüpfen nämlich unsere Betrachtungen an die beiden algebraischen Kurven C und C' an. Im Falle der Reduzibilität besteht zwischen beiden eine algebraische *Korrespondenz*. Die Beschaffenheit derselben liegt der in Aussicht gestellten Fallunterscheidung zugrunde.

Der erste Fall ist der folgende. Vermöge der Korrespondenz ist jedem Punkte M von C ein und nur ein Punkt M' von C' zugeordnet, während umgekehrt jedem Punkte von C' n Punkte von C entsprechen. Ich nenne dann n die *charakteristische Zahl* der Korrespondenz und sage, C ist eine *vielfache Kurve* von C'.

Der eben genannte erste Fall ist aber nicht der allgemeine. Das ist vielmehr der nun folgende zweite. Hier nämlich besteht die Korrespondenz nicht zwischen einzelnen Punkten M und M', sondern zwischen Systemen von Punkten $M_1, \ldots, M_\nu$ von C mit den Koordinaten $x_1, y_1; \ldots; x_\nu, y_\nu$ und $M_1', \ldots, M_\nu'$ von C' mit den Koordinaten $x_1', y_1'; \ldots; x_\nu' y_\nu'$. Jedem System auf C soll dabei ein und nur ein System auf C' entsprechen, während umgekehrt einem System auf C' im allgemeinen mehrere Systeme auf C zugeordnet sind. Ich sage dann, C ist eine *pseudovielfache Kurve* von C'.

Im erstgenannten Falle sind x' und y' rationale Funktionen von x und y, während im zweiten nur geschlossen werden kann, daß jede rationale und symmetrische Funktion der $(x_1' y_1', \ldots, x_\nu' y_\nu')$ eine rationale Funktion der $(x_1 y_1, \ldots, x_\nu y_\nu)$ ist. Es ist leicht zu sehen, daß jede Kurve C, die eine vielfache von C' ist, auch eine pseudovielfache der Kurve C' ist. Umgekehrt aber habe ich mehrere Beispiele bilden können dafür, daß nicht jede pseudovielfache Kurve von C' auch eine vielfache von C' ist. Ich will jedoch hier nicht näher darauf eingehen, zumal da sich meine folgenden Darlegungen durchaus an den ERSTEN FALL anschließen werden.

Im Falle der Reduzibilität unserer Integrale ist es möglich, ihre Periodentabelle auf eine besondere *Normalform* zu bringen. Die folgenden beiden Beispiele mögen eine Anschauung von der Beschaffenheit derselben geben.

1) $q = 1$; $p = 3$. Die Periodentabelle kann auf die folgende Form gebracht werden:

$$\begin{array}{cccccc} 2i\pi & 0 & 0 & h & \frac{2i\pi}{\alpha} & 0 \\ 0 & 2i\pi & 0 & \frac{2i\pi}{\alpha} & a & b \\ 0 & 0 & 2i\pi & 0 & b & c. \end{array}$$

2) $q = 2$; $p = 4$. Die normierten Perioden sind hier:

$$\begin{array}{cccccccc} 2i\pi & 0 & 0 & 0 & a & b & 0 & \frac{2i\pi}{\alpha} \\ 0 & 2i\pi & 0 & 0 & b & c & \frac{2i\pi}{\alpha\beta} & 0 \\ 0 & 0 & 2i\pi & 0 & 0 & \frac{2i\pi}{\alpha\beta} & a' & b' \\ 0 & 0 & 0 & 2i\pi & \frac{2i\pi}{\alpha} & 0 & b' & c'. \end{array}$$

Die Zahlen α, β bedeuten in beiden Tabellen ganze rationale Zahlen.

Ich definiere nun noch eine *zweite charakteristische Zahl* κ. Sie gibt die Ordnung der Thetafunktion von q Variabeln an, in die eine Thetafunktion erster Ordnung von p Variabeln im Falle der Reduzibilität transformiert werden kann. Im ersten Beispiel ist $\kappa = \alpha$, im zweiten $\kappa = \alpha\beta$. *Die beiden charakteristischen Zahlen n und κ sind nun immer einander gleich.* Ich habe zwei Beweise für diesen Satz gefunden, die ich jetzt in ihren Grundzügen auseinandersetzen will.

Erster Beweis. Seien M und M' zwei ABELsche Integrale erster, zweiter oder dritter Gattung der Kurve C. Ich denke mir die zugehörige RIEMANNsche Fläche irgendwie längs $2p$ von einem Punkte ausgehenden nichtzerstückenden Rückkehrschnitten kanonisch aufgeschnitten. Dann mögen M und M' die folgenden Perioden besitzen:

$$M : x_1, x_2, \ldots, x_{2p},$$
$$M' : y_1, y_2, \ldots, y_{2p}.$$

Ich muß nun eine charakteristische fundamentale *Bilinearform* definieren. Ich setze nämlich:

$$F(x, y) = \int M dM'$$

wo das Integral längs der ganzen Kontur der Zerschneidung erstreckt werden soll. Wenn x, y Normalperioden sind, so nimmt $F(x, y)$ die folgende Form an:

$$F(x, y) = \sum_{\kappa=1}^{p} \left(x_{2\kappa-1} y_{2\kappa} - x_{2\kappa} y_{2\kappa-1} \right).$$

Nehme ich an, es sei M eines der reduziblen Integrale, dann drücken sich seine $2p$ Perioden ganzzahlig und linear durch nur $2q$ Perioden $\omega_1, \ldots, \omega_{2q}$ aus. Ich habe also dann:

$$x_\kappa = \sum_{j=1}^{2q} m_{\kappa j} \omega_j \qquad (\kappa = 1, 2, \ldots, 2p),$$

wo die m_κ ganze rationale Zahlen bedeuten. Wenn nun M und M' Integrale erster Gattung sind, dann ist bekanntlich

$$F(x, y) = 0$$

Wenn man in diese Gleichung die Ausdrücke der x durch die ω einsetzt, so bekommt man eine bilineare Gleichung zwischen den y und ω, die in der folgenden Form geschrieben werden kann:

$$\sum_{j=1}^{2q} H_j \omega_j = 0$$

Seien nun $u_1, \ldots, u_p$ p linear unabhängige Integrale erster Gattung von C. Dann können wir setzen:

$$\begin{array}{lllllll} U & = \mu_1 u_1 & + & \mu_2 u_2 & + & \ldots & + \mu_p u_p \\ U' & = \mu_1' u_1 & + & \mu_2' u_2 & + & \ldots & + \mu_p' u_p \end{array}$$

Die vorläufig noch unbestimmten Koeffizienten μ' sollen nun so bestimmt werden, daß sie den $2q$ linearen Gleichungen:

$$H_j = 0 \qquad (j = 1, 2, \ldots, 2q)$$

genügen. Wenn man dann noch beachtet, daß diese $2q$ Gleichungen nicht linear unabhängig sind, sondern daß zwischen ihnen q Relationen

$$\sum H_j \omega_j = 0$$

bestehen, so ist leicht zu erkennen, daß auch M_1 reduzierbar ist, und daß, so wie M einer Schar von q reduziblen Integralen angehört, auch M' ein Element einer $(p-q)$fach unendlichen linearen Schar von reduziblen Integralen ist. Doch dies nur nebenbei.

Ich bemerke nun, daß H_j eine lineare Funktion der y_κ ist, sodaß ich schreiben kann:

$$H_j = \sum_{i=1}^{2p} h_{ij} y_i \qquad (j = 1, 2, \ldots, 2q),$$

wo die h_{ij} ganze rationale Zahlen sind. Aus den $m_{i\kappa}$ und den $h_{i\kappa}$ kann ich nun zwei Tabellen von je $2q$ Kolonnen und $2p$ Zeilen bilden. Aus beiden kann ich gewisse q-reihige Determinanten bilden. Ich bezeichne die der m mit D und die aus denselben Zeilen der h gebildete mit D'. Dann setze ich

$$J = \sum DD'.$$

J ist nun in dem folgenden Sinne eine *invariante* Zahl: Sie bleibt ungeändert, wenn man irgendeines der Periodensysteme x oder ω durch ein äquivalentes ersetzt. Dabei heißen zwei Periodensysteme äquivalent, wenn sie sich ganzzahlig

und linear durcheinander ausdrücken lassen. Man kann nun einerseits beweisen, daß

$$J = \kappa^2,$$

andererseits aber, daß

$$J = n^2.$$

Daraus kann man folgern, daß

$$\kappa = n.$$

Das ist der erste Beweis. Der nun folgende

Zweite Beweis ist wesentlich kürzer. Er beruht auf dem Vergleich der zu S und S' gehörigen Bilinearformen $F(x, y)$ und $\Phi(\omega, \omega')$. Man hat nämlich einerseits

$$F(x, y) = n\Phi(\omega, \omega'),$$

andererseits

$$F(x, y) = \kappa\Phi(\omega, \omega').$$

Daraus schließt man

$$\kappa = n.$$

Ich komme nun zum *Zusammenhang der Reduktionstheorie mit der Theorie der* FUCHS*schen Funktionen.*

Bekanntlich definiert jede algebraische Kurve C ein System von FUCHSschen Funktionen. Nun kann man die Tatsache, daß C ein Vielfaches von C' ist, auch folgendermaßen ausdrücken. Es ist immer auf mannigfache Weise möglich, der Kurve C' eine Grenzkreisgruppe G' und C eine ebensolche Gruppe G zuzuordnen, sodaß G eine *Untergruppe* von G' ist. Ist im besonderen C ein n-faches von C', dann ist G eine Untergruppe vom Index n von G'. Man erhält daher einen Fundamentalbereich von G dadurch, daß man n geeignet gewählte Fundamentalbereiche von G', die durch die Operationen von G' auseinander hervorgehen, aneinander lagert. Das Polygon P von G erscheint dann in n Polygone $P'(\beta)$ eingeteilt, die einem Polygon P' von G' im Sinne der nichteuklidischen Geometrie kongruent sind.

Ich bezeichne die *Seiten* des Polygons P' mit $\gamma(\alpha)$ und die homologen Seiten von $P'(\beta)$ mit $\gamma(\alpha, \beta)$. Die Seiten $\gamma(\alpha, \beta)$ liegen entweder im Innern oder auf dem Rande von P. Ich nehme nun an, die Seite $\gamma(\alpha')$ gehe aus $\gamma(\alpha)$ vermöge einer Operation von G' hervor. Wenn nun $\gamma(\alpha, \beta)$ auf dem Rande von P liegt, dann gibt es eine weitere Seite $\gamma(\alpha', \beta')$ auf diesem Rande, die mit $\gamma(\alpha, \beta)$ vermöge einer Operation von G konjugiert ist. Wenn jedoch $\gamma(\alpha, \beta)$ im Innern von P liegt, so existiert eine derartige von $\gamma(\alpha, \beta)$ verschiedene Seite nicht, sondern es fallen $\gamma(\alpha, \beta)$ und $\gamma(\alpha', \beta')$ zusammen und bilden die gemeinsame Seite von $P'(\beta)$ und $P'(\beta')$. Aber wie dem auch sei, jedenfalls entspricht jeder Seite $\gamma(\alpha)$ von P' eine Permutation der n Ziffern $1, 2, \ldots, n$.

Eine der eben durchgeführten ganz ähnliche Betrachtung können wir auch für die *Ecken* von P' anstellen. So wie wir nämlich die Seiten in Paare zusammenfaßten, so können wir die Ecken in Zyklen einteilen, so daß die Ecken eines Zyklus auseinander durch die Operationen von G' hervorgehen. Jedem solchen

Zyklus kann wieder eine bestimmte Vertauschung der n Ziffern $1, 2, \ldots, n$ zugeordnet werden, die sich aus den den Seiten zugeordneten gewinnen läßt. Ich nehme nun an, es habe P $2N$ Seiten und Q Eckenzyklen. $2N'$ und Q' sollen die gleiche Bedeutung für P' haben. Die einem Eckenzyklus von P' entsprechende Permutation läßt sich in zyklische Permutationen zerlegen. Bei allen Eckenzyklen mögen dabei im ganzen λ_i zyklische Permutationen von gerade i Ziffern vorkommen. Dann bestehen die folgenden Relationen:

$$\begin{aligned} 2p &= N - Q + 1, \\ 2q &= N' - Q' + 1, \\ Q + 2p - 2 &= n(Q' + 2q - 2), \\ n(Q' - Q) &= 2(p-1) - 2n(q-1), \\ \sum \lambda_i &= Q, \\ \sum i\lambda_i &= nQ'. \end{aligned}$$

Die bisher gegebenen allgemeinen Betrachtungen setzen uns nun instand, eine Reihe schöner und wichtiger *Sätze über die nichteuklidische Geometrie der Kreisbogenpolygone, sowie über die Geometrie der algebraischen Kurven* abzuleiten. Ich will im folgenden einige Beispiele solcher Sätze anführen, ohne mich des näheren auf Beweise einzulassen, deren Grundzüge übrigens im vorstehenden enthalten sind.

1) $p = 3, q = 2, n = 2, m = m' = 4$.

Mit m und m' sind dabei die Ordnungen der Kurven C und C' bezeichnet. C hat keinen Doppelpunkt, C' hat einen Doppelpunkt. *Von den 28 Doppeltangenten von C gehen sechs durch einen Punkt außerhalb der Kurve.*

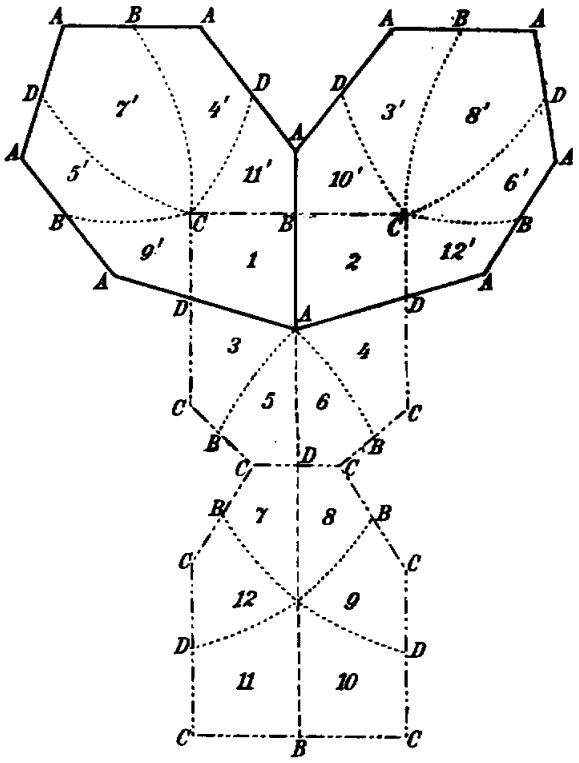
2) $p = 4, q = 2, n = 2, m = 4, m' = 5$.

C hat zwei Doppelpunkte, C' nur einen. Setzt man die Differentiale der reduziblen Integrale erster Gattung gleich Null, so erhält man ein *Kegelschnittbüschel*, dessen vier Basispunkte von den beiden Doppelpunkten von C und zwei weiteren Punkten derselben Kurve gebildet werden. Sechs dieser Kegelschnitte berühren C doppelt. Derjenige derselben, der C in einem Basispunkte berührt, oskuliert daselbst.

3) $p = 2, q = 1, n = 2$.

Die Kurve C ist ein *Vielfaches von zwei verschiedenen* Kurven C' und C''. Es existiert eine FUCHSsche Gruppe G, zu der man sowohl ein erstes Polygon P_1 konstruieren kann, das aus zwei Polygonen einer zu C' gehörigen Gruppe G' besteht, als auch ein zweites Polygon P_2, das aus zwei Polygonen einer zu C'' gehörigen Gruppe G'' besteht. G ist also sowohl in G' als in G'' als Untergruppe vom Index 2 enthalten. Die nebenstehende schematische *Figur* möge zur Veranschaulichung der Verhältnisse dienen. Die beiden eben erwähnten Fundamentalbereiche P_1 und P_2 von G sind durch die Polygone mit den Ecken A bzw. C dargestellt. Jedes derselben zerfällt in zwei Sechsecke, die bzw. Fundamentalbereiche von G' oder G'' sind. Um die Äquivalenz von P_1 und P_2 besser hervortreten zu lassen, sind die Symmetriezentren der erwähnten Sechsecke mit

den Seitenmitten verbunden, sodaß alle Polygone sich in leicht ersichtlicher Weise aus den so entstehenden Vierecken aufbauen.



Ich gehe nun zu den Sätzen aus der Geometrie der algebraischen Kurven über, die uns dieses Beispiel lehrt. Wenn ich auf C' einen Punkt M' markiere, so entsprechen diesem zwei Punkte M_a und M_b auf C. Jedem von diesen entspricht ein Punkt von C'': $M_a{}''$, $M_b{}''$. Es entsprechen also im allgemeinen jedem Punkte von C' zwei Punkte von C''. Ebenso kann man schließen, daß im allgemeinen jedem Punkte von C'' zwei Punkte von C' entsprechen. Die Korrespondenz (C', C) hat aber zwei Verzweigungspunkte $M_1{}'$, $M_2{}'$. Jedem von ihnen entspricht also nur ein Punkt von C und also auch nur ein Punkt von C'': $M_1{}''$, $M_2{}''$. Ebenso hat die Korrespondenz (C'', C) zwei Verzweigungspunkte $N_1{}''$, $N_2{}''$. Jedem von ihnen ist nur ein Punkt von C' zugeordnet: $N_1{}'$, $N_2{}'$. Wir können dann den ersten Satz, den wir anführen wollen, so aussprechen:

$N_1{}'$ und $N_2{}'$ einerseits und $M_1{}''$ und $M_2{}''$ andererseits fallen zusammen.

Ich gehe zum zweiten Satz über, der sich ergibt, wenn man C' und C'' als Kurven dritter Ordnung voraussetzt.

Ich kann in $N_1{}' = N_2{}'$ die Tangente an C' ziehen. Ich verbinde ferner $M_1{}'$ und $M_2{}'$ durch eine Sekante. Diese beiden Geraden schneiden sich auf C'. Ebenso kann ich in $M_1{}'' = M_2{}''$ die Tangente an C'' ziehen und mit der Sekante

$N_1''N_2''$ zum Schnitt bringen. Der Schnittpunkt liegt auf C''.

Diese wenigen Beispiele lassen zur Genüge erkennen, wie zahlreich die besonderen Fälle sind.

Fünfter Vortrag

ÜBER TRANSFINITE ZAHLEN

Meine Herren! Ich will heute über den Begriff der transfiniten Kardinalzahl vor Ihnen sprechen; und zwar will ich zunächst von einem *scheinbaren* Widerspruch reden, den dieser Begriff enthält. Dazu schicke ich folgendes voraus: meiner Ansicht nach ist ein Gegenstand nur dann denkbar, wenn er sich mit einer endlichen Anzahl von Worten definieren läßt. Einen Gegenstand, der in diesem Sinne endlich definierbar ist, will ich zur Abkürzung einfach "definierbar" nennen. Demnach ist also ein nicht definierbarer Gegenstand auch undenkbar. Desgleichen will ich ein Gesetz "aussagbar" nennen, wenn es in einer endlichen Anzahl von Worten ausgesagt werden kann.

Herr RICHARD hat nun bewiesen, daß die Gesamtheit der definierbaren Gegenstände abzählbar ist, d. h. daß die Kardinalzahl dieser Gesamtheit $\aleph_0$ ist. Der Beweis ist ganz einfach: sei α die Anzahl der Wörter des Wörterbuches, dann kann man mit n Wörtern höchstens α^n Gegenstände definieren. Läßt man nun n über alle Grenzen wachsen, so sieht man, daß man nie über eine abzählbare Gesamtheit hinauskommt. Die Mächtigkeit der Menge der denkbaren Gegenstände wäre also $\aleph_0$. Herr SCHOENFLIES hat gegen diesen Beweis eingewandt, daß man mit einer einzigen Definition mehrere, ja sogar unendlich viele Gegenstände definieren könne. Als Beispiel führt er die Definition der konstanten Funktionen an, deren es offenbar unendlich viele gibt. Dieser Einwand ist deshalb unzulässig, weil durch solche Definitionen gar nicht die einzelnen Gegenstände, sondern ihre Gesamtheit, in unserem Beispiel also die *Menge* der konstanten Funktionen definiert wird, und diese ist ein einziger Gegenstand. Der Einwand von Herrn SCHOENFLIES ist also nicht stichhaltig.

Nun hat bekanntlich CANTOR bewiesen, daß das Kontinuum nicht abzählbar ist; dies widerspricht dem Beweise von RICHARD. Es fragt sich also, welcher von beiden Beweisen richtig ist. Ich behaupte, sie sind beide richtig, und der Widerspruch ist nur ein scheinbarer. Zur Begründung dieser Behauptung will ich einen neuen Beweis für den CANTORschen Satz geben: Wir nehmen also an, es sei eine Strecke AB gegeben und ein Gesetz, durch welches jedem Punkte der Strecke eine ganze Zahl zugeordnet wird. Wir wollen der Einfachheit halber die Punkte durch die ihnen zugeordneten Zahlen bezeichnen. Wir teilen nun unsere Strecke durch zwei beliebige Punkte A_1 und A_2 in drei Teile, die wir als Unterstrecken 1. Stufe bezeichnen; diese teilen wir wieder in je drei Teile und erhalten Unterstrecken 2. Stufe; dieses Verfahren denken wir uns ins Unendliche fortgesetzt, wobei die Länge der Unterstrecken unter jede Grenze sinken soll. Der Punkt 1 gehört nun einer oder höchstens, wenn er mit A_1 oder A_2 zusammenfällt, zweien der Unterstrecken erster Stufe an, es gibt also sicher eine, der er nicht angehört. Auf dieser suchen wir den Punkt mit der niedrigsten Nummer, die nun mindestens 2 sein muß, auf. Unter den 3 Unterstrecken 2. Stufe, die zu derjenigen Strecke 1. Stufe gehören, auf der wir uns befinden, ist nun wieder mindestens eine, der der zuletzt betrachtete Punkt nicht angehört. Auf dieser setzen wir das Verfahren fort und erhalten so eine Folge von Strecken, die folgende Eigenschaften hat: jede von ihnen ist in allen vorhergehenden enthalten, und eine Strecke n^{ter} Stufe enthält keinen der Punkte 1 bis $n-1$. Aus der ersten Eigenschaft folgt, daß es mindestens einen Punkt geben muß, der ihnen allen gemeinsam ist; aus der zweiten Eigenschaft folgt aber, daß die Nummer dieses

Punktes größer sein muß als jede endliche Zahl, d. h. es kann ihm keine Zahl zugeordnet werden.

Was haben wir nun zu diesem Beweise vorausgesetzt? Wir haben ein Gesetz vorausgesetzt, das jedem Punkte der Strecke eine ganze Zahl zuordnet. Dann konnten wir einen Punkt definieren, dem keine ganze Zahl zugeordnet ist. In dieser Hinsicht unterscheiden sich die verschiedenen Beweise dieses Satzes nicht. Dazu mußte aber das Gesetz zuerst feststehen. Nach RICHARD müßte anscheinend ein solches Gesetz existieren, aber CANTOR hat das Gegenteil bewiesen. Wie kommen wir aus diesem Dilemma heraus? Fragen wir einmal nach der Bedeutung des Wortes "definierbar". Wir nehmen die Tafel aller endlichen Sätze und streichen daraus alle diejenigen, die keinen Punkt definieren. Die Übrigbleibenden ordnen wir den ganzen Zahlen zu. Wenn wir jetzt die Durchmusterung der Tafel von neuem vornehmen, so wird es sich im allgemeinen zeigen, daß wir jetzt einige Sätze stehen lassen müssen, die wir vorher gestrichen haben. Denn die Sätze, in welchen man von dem Zuordnungsgesetz selbst sprach, hatten früher keine Bedeutung, da die Punkte den ganzen Zahlen noch nicht zugeordnet waren. Diese Sätze haben jetzt eine Bedeutung, und müssen in unserer Tafel bleiben. Würden wir jetzt ein neues Zuordnungsgesetz aufstellen, so würde sich dieselbe Schwierigkeit wiederholen und so ad infinitum. Hierin liegt aber die Lösung des scheinbaren Widerspruchs zwischen CANTOR und RICHARD. Sei M_0 die Menge der ganzen Zahlen, M_1 die Menge der nach der ersten Durchmusterung der Tafel aller endlichen Sätze definierbaren Punkte unserer Strecke, G_1 das Gesetz der Zuordnung zwischen beiden Mengen. Durch dieses Gesetz kommt eine neue Menge M_2 von Punkten als definierbar hinzu. Zu $M_1 + M_2$ gehört aber ein neues Gesetz G_2, dadurch entsteht eine neue Menge M_3 usw. RICHARDs Beweis lehrt nun, daß, wo ich auch das Verfahren abbreche, immer ein Gesetz existiert, während CANTOR beweist, daß das Verfahren beliebig weit fortgesetzt werden kann. Es besteht also kein Widerspruch zwischen beiden.

Der Schein eines solchen rührt daher, daß dem Zuordnungsgesetz von RICHARD eine Eigenschaft fehlt, die ich mit einem von den englischen Philosophen entlehnten Ausdruck als "prädikativ" bezeichne. (Bei RUSSELL, dem ich das Wort entlehne, ist eine Definition zweier Begriffe A und A' nicht prädikativ, wenn A in der Definition von A' und umgekehrt vorkommt.) Ich verstehe darunter folgendes: Jedes Zuordnungsgesetz setzt eine bestimmte Klassifikation voraus. Ich nenne nun eine Zuordnung prädikativ, wenn die zugehörige Klassifikation prädikativ ist. Eine Klassifikation aber nenne ich prädikativ, wenn sie durch Einführung neuer Elemente nicht verändert wird. Dies ist aber bei der RICHARDschen nicht der Fall, vielmehr ändert die Einführung des Zuordnungsgesetzes die Einteilung der Sätze in solche, die eine Bedeutung haben, und solche, die keine haben. Was hier mit dem Wort "prädikativ" gemeint ist, läßt sich am besten an einem Beispiel illustrieren: wenn ich eine Menge von Gegenständen in eine Anzahl von Schachteln einordnen soll, so kann zweierlei eintreten: entweder sind die bereits eingeordneten Gegenstände endgültig an ihrem Platze, oder ich muß jedesmal, wenn ich einen neuen Gegenstand einordne, die anderen oder wenigstens einen Teil von ihnen wieder herausnehmen. Im ersten Falle nenne

ich die Klassifikation prädikativ, im zweiten nicht. Ein gutes Beispiel für eine nicht prädikative Definition hat RUSSELL gegeben: A sei die kleinste ganze Zahl, deren Definition mehr als hundert deutsche Worte erfordert. A muß existieren, da man mit hundert Worten jedenfalls nur eine endliche Anzahl von Zahlen definieren kann. Die Definition, die wir eben von dieser Zahl gegeben haben, enthält aber weniger als hundert Worte. Und die Zahl A ist also *definiert* als *undefinierbar*.

ZERMELO hat nun gegen die Verwerfung der nicht prädikativen Definitionen den Einwand erhoben, daß damit auch ein großer Teil der Mathematik hinfällig würde, z. B. der Beweis für die Existenz einer Wurzel einer algebraischen Gleichung.

Dieser Beweis lautet bekanntlich folgendermaßen:

Gegeben ist eine Gleichung $F(x) = 0$. Man beweist nun, daß $|F(x)|$ ein Minimum haben muß; sei x_0 einer der Argumentwerte, für den das Minimum eintritt, also

$$|F(x)| \geq |F(x_0)|$$

Daraus folgt dann weiter, daß $F(x_0) = 0$ ist. Hier ist nun die Definition von $F(x_0)$ nicht prädikativ, denn dieser Wert hängt ab von der Gesamtheit der Werte von $F(x)$, zu denen er selbst gehört.

Die Berechtigung dieses Einwandes kann ich nicht zugeben. Man kann den Beweis so umformen, daß die nicht prädikative Definition daraus verschwindet. Ich betrachte zu diesem Zwecke die Gesamtheit der Argumente von der Form $\frac{m+ni}{p}$, wo m, n, p ganze Zahlen sind. Dann kann ich dieselben Schlüsse wie vorher ziehen, aber der Argumentwert, für den das Minimum von $|F(x)|$ eintritt, gehört im allgemeinen nicht zu den betrachteten. Dadurch ist der Zirkel im Beweise vermieden. Man kann von jedem mathematischen Beweise verlangen, daß die darin vorkommenden Definitionen usw. prädikativ sind, sonst wäre der Beweis nicht streng.

Wie steht es nun mit dem klassischen Beweise des BERNSTEINschen Theorems? Ist er einwandfrei? Das Theorem sagt bekanntlich aus, daß, wenn drei Mengen A, B, C gegeben sind, wo A in B und B in C enthalten ist, und wenn A äquivalent C ist, auch A äquivalent B sein muß. Es handelt sich also auch hier um ein Zuordnungsgesetz. Wenn das erste Zuordnungsgesetz (zwischen A und C) prädikativ ist, so zeigt der Beweis, daß es auch ein prädikatives Zuordnungsgesetz zwischen A und B geben muß.

Was nun die zweite transfinite Kardinalzahl $\aleph_1$ betrifft, so bin ich nicht ganz überzeugt, daß sie existiert. Man gelangt zu ihr durch Betrachtung der Gesamtheit der Ordnungszahlen von der Mächtigkeit $\aleph_0$; es ist klar, daß diese Gesamtheit von höherer Mächtigkeit sein muß. Es fragt sich aber, ob sie abgeschlossen ist, ob wir also von ihrer Mächtigkeit ohne Widerspruch sprechen dürfen. Ein aktual Unendliches gibt es jedenfalls nicht.

Was haben wir von dem berühmten *Kontinuumproblem* zu halten? Kann man die Punkte des Raumes wohlordnen? Was meinen wir damit? Es sind hier zwei Fälle möglich: entweder behauptet man, daß das Gesetz der Wohlordnung endlich aussagbar ist, dann ist diese Behauptung nicht bewiesen; auch Herr

Zermelo erhebt wohl nicht den Anspruch, eine solche Behauptung bewiesen zu haben. Oder aber wir lassen auch die Möglichkeit zu, daß das Gesetz nicht endlich aussagbar ist. Dann kann ich mit dieser Aussage keinen Sinn mehr verbinden, das sind für mich nur leere Worte. Hier liegt die Schwierigkeit. Und das ist wohl auch die Ursache für den Streit über den fast genialen Satz Zermelos. Dieser Streit ist sehr merkwürdig: die einen verwerfen das Auswahlpostulat, halten aber den Beweis für richtig, die anderen nehmen das Auswahlpostulat an, erkennen aber den Beweis nicht an.

Doch ich könnte noch manche Stunde darüber sprechen, ohne die Frage zu lösen.

Sechster Vortrag

LA MÉCANIQUE NOUVELLE

Mesdames, messieurs!

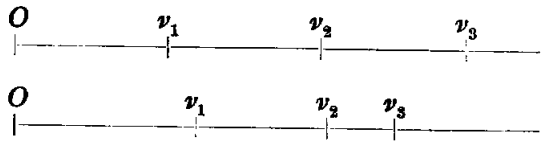
Aujourd'hui, je suis obligé de parler français, et il faut que je m'en excuse. Il est vrai que dans mes précédentes conférences je me suis exprimé en allemand, en un très mauvais allemand: parler les langues étrangères, voyez-vous, c'est vouloir marcher lorsqu'on est boiteux; il est nécessaire d'avoir des béquilles; mes béquilles, c'étaient jusqu'ici les formules mathématiques et vous ne sauriez vous imaginer quel appui elles sont pour un orateur qui ne se sent pas très solide. — Dans la conférence de ce soir, je ne veux pas user de formules, je suis sans béquilles, et c'est pourquoi je dois parler français.

En ce monde, vous le savez, il n'est rien de définitif, rien d'immuable; les empires les plus puissants, les plus solides, ne sont pas éternels: c'est là un thème que les orateurs sacrés se sont plu bien souvent à développer. — Les théories scientifiques sont comme les empires, elles ne sont pas assurées du lendemain. Si l'une d'elles semblait à l'abri des injures du temps, c'était, certes, la mécanique newtonienne: elle paraissait incontestée, c'était un monument impérissable; et voilà qu'à son tour, je ne dirai pas que le monument est par terre, ce serait prématuré, mais en tout cas il est fortement ébranlé. Il est soumis aux attaques de grands démolisseurs: vous en avez un parmi vous, M. Max Abraham, un autre est le physicien hollandais M. Lorentz. Je voudrais, en quelques mots, vous parler des ruines de l'ancien édifice et du nouveau bâtiment que l'on veut élever à leur place. —

Tout d'abord qu'est-ce qui caractérisait l'ancienne mécanique? C'était ce fait très simple: je considère un corps en repos, je lui communique une impulsion, c'est à dire je fais agir sur lui, pendant un temps donné une force donnée; le corps se met en mouvement, acquiert une certaine vitesse; le corps étant animé de cette vitesse, faisons agir encore la même force pendant le même temps, la vitesse sera doublée; si nous continuons encore, la vitesse sera triplée après que nous aurons une troisième fois donné une impulsion identique. Recommençons ainsi un nombre suffisant de fois, le corps finira par acquérir une vitesse très grande, qui pourra dépasser toute limite, une vitesse infinie.

Dans la nouvelle mécanique, au contraire, on suppose qu'il est impossible de communiquer à un corps partant du repos une vitesse supérieure à celle de la lumière. Que se passe-t-il? Je considère le même corps au repos; je lui donne une première impulsion, la même que précédemment, il prendra la même vitesse; renouvelons une seconde fois cette impulsion, la vitesse va encore augmenter, mais elle ne sera plus doublée; une troisième impulsion produira un effet analogue, la vitesse augmente mais de moins en moins, le corps oppose une résistance qui devient de plus en plus grande. Cette résistance, c'est l'inertie, c'est ce qu'on appelle communément la masse; tout ce passe alors dans cette nouvelle mécanique comme si la masse n'était pas constante, mais croissait avec la vitesse. Nous pouvons représenter graphiquement les phénomènes: dans l'ancienne mécanique, le corps prend après la première impulsion une vitesse representée par le segment $\overline{O\nu_1}$; après la deuxième impulsion $\overline{O\nu_1}$ s'accroît d'un segment $\overline{\nu_1\nu_2}$ qui lui est égal, à chaque nouvelle impulsion, la vitesse s'accroît de la même quantité, le segment qui la représente s'accroît d'une longueur constante; dans la nouvelle mécanique, le segment vitesse s'accroît de segments $\overline{\nu'_1\nu_2}$, $\overline{\nu'_2\nu'_3}$, ... qui sont de

plus en plus petits et tels que nous ne pouvons pas dépasser une certaine limite, la vitesse de la lumière.



Comment a-t-on été conduit à de telles conclusions? A-t-on fait des expériences directes? Les divergences ne se produiront que pour les corps animés de grandes vitesses; c'est alors seulement que les différences signalées deviennent sensibles. Mais, qu'est-ce qu'une très grande vitesse? Est-ce celle d'une automobile qui fait 100 kilomètres à l'heure; on s'extasie dans la rue sur une telle rapidité; à notre point de vue, c'est pourtant bien peu, une vitesse d'escargot. L'astronomie nous donne mieux: Mercure, le plus rapide des corps célestes parcourt lui aussi 100 kilomètres environ, non plus à l'heure mais à la seconde: pourtant, cela ne suffit pas encore, de telles vitesses sont trop faibles pour révéler les différences que nous voudrions observer. Je ne parle pas de nos boulets de canon, ils sont plus rapides que les automobiles, mais beaucoup plus lents que Mercure; vous savez cependant qu'on a découvert une artillerie dont les projectiles sont beaucoup plus vite: je veux parler du radium qui envoie dans tous les sens de l'énergie, des projectiles; la rapidité du tir est bien plus grande, la vitesse initiale est de 100 000 kilomètres par seconde, le tiers de la vitesse de la lumière; le calibre des projectiles, leur poids, sont, il est vrai, bien plus faibles et nous ne devons pas compter sur cette artillerie pour augmenter la puissance militaire de nos armées. Peut-on expérimenter sur ces projectiles? De telles expériences ont été effectivement tentées; sous l'influence d'un champ électrique, d'un champ magnétique il se produit une déviation qui permet de se rendre compte de l'inertie et de la mesurer. On a constaté ainsi que la masse dépend de la vitesse et énoncer cette loi: L'inertie d'un corps croît avec sa vitesse qui reste inférieure à celle de la lumière, 300 000 kilomètres par seconde.

Je passe maintenant au deuxième principe, le principe de relativité. Je suppose un observateur qui se déplace vers la droite; tout se passe pour lui comme s'il était au repos, les objets qui l'entourent se déplaçant vers la gauche: aucun moyen ne permet de savoir si les objets se déplacent réellement, si l'observateur est immobile ou en mouvement. On l'enseigne dans tous les cours de mécanique, le passager sur le bateau croit voir le rivage du fleuve se déplacer, tandis qu'il est doucement entraîné par le mouvement du navire. Examinée de plus près, cette simple notion acquiert une importance capitale; on n'a aucun moyen de trancher la question, aucune experience ne peut mettre en defaut le principe: il n'y a pas d'espace absolu, tous les déplacements que nous pouvons observer sont des deplacements relatifs. Ces considerations bien familières aux philosophes, j'ai eu quelquefois l'occasion de les exprimer: j'en ai même recueilli une publicité dont je me serais volontiers passé, tous les journaux réactionnaires français m'ont fait

démontrer que le soleil tournait autour de la terre; dans le fameux procès entre l'Inquisition et Galilée, Galilée aurait eu tous les torts.

Revenons à l'ancienne mécanique: elle admettait le principe de relativité; au lieu d'être fondées sur des expériences, ses lois étaient déduites de ce principe fondamental. Ces considerations suffisaient pour les phénomènes purement mécaniques, mais cela n'allait plus pour d'importantes parties de la physique, l'optique par exemple. On considérait comme absolue la vitesse de la lumière relativement à l'éther: cette vitesse pouvait être mesurée, on avait théoriquement le moyen de comparer le déplacement d'un mobile à un deplacement absolu, le moyen de décider si oui ou non un corps était en mouvement absolu.

Des expériences délicates, des appareils extrêmement précis, que je ne décrirai pas devant vous, ont permis d'essayer la réalisation pratique d'une pareille comparaison: le résultat a été nul. Le principe de rélativite n'admet aucune restriction dans la nouvelle mécanique; il a, si j'ose ainsi dire, une valeur absolue.

Pour comprendre le rôle que joue le principe de relativité dans la Nouvelle Mécanique, nous sommes d'abord amenés à parler du temps apparent, une invention fort ingénieuse du physicien Lorentz. Nous supposons deux observateurs l'un A à Paris, l'autre B à Berlin. A et B ont des chronomètres identiques et veulent les régler: mais ce sont des observateurs méticuleux comme il n'y en a guère; ils exigent dans leur réglage une extraordinaire exactitude: ce sera, par exemple, non une seconde, mais un milliardième de seconde. Comment pourront-ils faire? De Paris à Berlin, A envoie un signal télégraphique, avec un sans-fil, si vous voulez, pour être tout à fait moderne. B note le moment de la réception et ce sera pour les deux chronomètres l'origine des temps. Mais le signal emploie un certain temps pour aller de Paris à Berlin, il ne va qu'avec la vitesse de la lumière; la montre de B serait donc en retard; B est trop intelligent pour ne point s'en rendre compte; il va remédier a cet inconvenient. La chose semble bien simple: on croîse les signaux, A reçoit et B envoie, on prend la moyenne des corrections ainsi faites, on a l'heure exacte. Mais cela est-il bien certain? Nous supposons que de A à B le signal emploie le même temps que pour aller de B à A. Or A et B sont emportés dans le mouvement de la terre par rapport à l'éther, véhicule des ondes électriques. Quand A a envoyé son signal il fuit devant lui, B s'éloigne de même, le temps employé sera plus long que si les deux observateurs étaient au repos; si au contraire c'est B qui envoie, A qui reçoit, le temps est plus court parce que A va au devant des signaux; il leur est absolument impossible de savoir si leurs chronomètres marquent ou non la même heure. Quelle que soit la méthode employée les inconvénients restent les mêmes l'observation d'un phénomène astronomique, une méthode optique quelconque se heurtent aux mêmes difficultés, B ne pourra jamais connaître qu'une différence apparente de temps, qu'une espèce d'heure locale. Le principe de relativité s'applique intégralement.

Dans l'ancienne mécanique pourtant, on démontrait avec ce principe toutes les lois fondamentales. On pourrait être tenté de reprendre les raisonnements classiques et de raisonner comme il suit? Soit encore deux observateurs, A et B pour les nommer comme on nomme toujours deux observateurs en mathématiques; supposons les en mouvement, s'éloignant l'un de l'autre; aucun d'eux ne

peut dépasser la vitesse de la lumière; par exemple B sera animé de 200 000 kilométres vers la droite, A de 200 000 vers la gauche. A peut se croire au repos et la vitesse apparente de B sera, pour lui, 400 000 kilomètres. Si A connait la mécanique nouvelle il se dira: B a une vitesse qu'il ne peut atteindre, c'est donc que moi aussi je suis en mouvement. Il semble qu'il pourrait décider de sa situation absolue. Mais il faudrait qu'il puisse observer le mouvement de B lui meme; pour faire cette observation A et B commencent par règler leurs montres, puis B envoie à A des télégrammes pour lui indiquer ses positions successives; en les réunissant A peut se rendre compte du mouvement de B et tracer la courbe de ce mouvement. Or les signaux se propagent avec la vitesse de la lumière; les montres qui marquent le temps apparent varient à chaque instant et tout se passera comme si la montre de B avançait. B croira aller beaucoup moins vite et la vitesse apparente qu'il aura relativement à A ne dépassera pas la limite qu'elle ne doit pas atteindre. Rien ne pourra révéler à A s'il est en mouvement ou en repos absolu.

Il faut encore faire une troisième hypothèse beaucoup plus surprenante, beaucoup plus difficile à admettre, qui gêne beaucoup nos habitudes actuelles. Un corps en mouvement de translation subit une déformation dans le sens même où il se déplace; une sphère, par exemple, devient comme une espèce d'ellipsoïde aplati dont le petit axe serait parallèle à la translation. Si l'on ne s'aperçoit pas tous les jours d'une transformation pareille c'est qu'elle est d'une petitesse qui la rend presque imperceptible. La terre, emportée dans sa révolution sur son orbite se déforme environ de $\frac{1}{200\,000\,000}$: pour observer un pareil phénomène il faudrait des instruments de mesure d'une précision extrême, mais leur précision serait infinie qu'on n'en serait pas plus avancé car emportés eux aussi dans le mouvement ils subiront la même transformation. On ne s'apercevra de rien; le mètre que l'on pourrait employer deviendra plus court comme la longueur qu'on mesure. On ne peut savoir quelque chose qu'en comparant à la vitesse de la lumière la longueur de l'un de ces corps. Ce sont là de delicates experiences, réalisées par Michelson et dont je ne vous exposerai pas le détail; elles ont donné des resultats tout à fait remarquables; quelqu'étranges qu'il nous paraissent, il faut admettre que la troisième hypothèse est parfaitement vérifiée.

Telles sont les bases de la nouvelle mécanique, avec l'appui de ces hypothèses on trouve qu'elle est compatible avec le principe de relativité.

Mais il faut la rattacher alors à une conception nouvelle de la matière.

Pour le physicien moderne, l'atome n'est plus l'élément simple; il est devenu un véritable univers dans lequel des milliers de planètes gravitent autour de soleils minuscules. Soleils et planètes sont ici des particules *électrisées* soit négativement soit positivement; le physicien les appelle *électrons* et bâtit le monde avec elles. D'aucuns se représentent l'atome neutre comme une masse centrale positive autour de laquelle circulent un grand nombre d'électrons chargés négativement, dont la masse électrique totale est égale en grandeur à celle du noyau central.

Cette conception de la matière permet de rendre compte aisément de l'augmentation de la masse d'un corps avec sa vitesse, dont nous avons fait un des caractères de la mécanique nouvelle. Un corps quelconque n'étant qu'un as-

semblage d'électrons, il nous suffira de le montrer sur ces derniers. Remarquons, à cet effet, qu'un électron isolé se déplaçant à travers l'éther engendre un courant électrique, c'est-à-dire un champ électromagnétique. Ce champ correspond à une certaine quantité d'énergie localisée non dans l'électron, mais dans l'éther. Une variation en grandeur ou en direction de la vitesse de l'électron modifie le champ et se traduit par une variation de l'énergie électromagnétique de l'éther. Alors que dans la mécanique newtonienne la dépense d'énergie n'est due qu'à l'inertie du corps en mouvement, ici une partie de cette dépense est due à ce que l'on peut appeler l'inertie de l'éther relativement aux forces électromagnétiques. L'inertie de l'éther augmente avec la vitesse et sa limite devient infinie lorsque la vitesse tend vers la vitesse de la lumière. La masse apparente de l'électron augmente donc avec la vitesse; les expériences de KAUFMANN montrent que la masse réelle constante de l'électron est négligeable par rapport à la masse apparente et peut être considérée comme nulle.

Dans cette nouvelle conception, la masse constante de la matière a disparu. L'éther seul, et non plus la matière, est inerte. Seul l'éther oppose une résistance au mouvement, si bien que l'on pourrait dire: il n'y a pas de matière, il n'y a que des trous dans l'éther. Pour les mouvements stationnaires ou quasi-stationnaires, la mécanique nouvelle ne diffère pas — au degré d'approximation de nos mesures près — de la mécanique newtonienne, avec cette seule différence que la masse n'est plus indépendante ni de la vitesse, ni de l'angle que fait cette vitesse avec la direction de la force accélératrice. Si par contre la vitesse a une accélération considérable, dans le cas, par ex., d'oscillations très rapides, il y a production d'ondes hertziennes représentant une perte d'énergie de l'électron entraînant l'amortissement de son mouvement. Ainsi, dans la télégraphie sans fil, les ondes émises sont dues aux oscillations des électrons dans la décharge oscillante.

Des vibrations analogues ont lieu dans une flamme et de même encore dans un solide incandescent. Pour LORENTZ, il circule à l'intérieur d'un corps incandescent un nombre considérable d'électrons qui, ne pouvant pas en sortir, volent dans tous les sens et se réfléchissent sur sa surface. On pourrait les comparer à une nuée de moucherons enfermés dans un bocal et venant frapper de leurs ailes les parois de leur prison. Plus la temperature est élevée, plus le mouvement de ces électrons est rapide et plus les chocs mutuels et les réflexions sur la paroi sont nombreuses. A chaque choc et à chaque réflexion une onde électromagnétique est émise et c'est la perception de ces ondes qui nous fait paraître le corps incandescent.

Le mouvement des électrons est presque tangible, dans un tube de CROOKES. Il s'y produit un véritable bombardement d'électrons partant de la cathode. Ces rayons cathodiques frappent violemment l'anticathode et s'y réfléchissent en partie donnant ainsi naissance à un ébranlement électromagnétique que plusieurs physiciens identifient avec les rayons RÖNTGEN.

Il nous reste en terminant à examiner les relations de la mécanique nouvelle avec l'astronomie. La notion de masse constante d'un corps s'évanouissant, que deviendra la loi de NEWTON? Elle ne pourra subsister que pour des corps en repos. De plus il faudra tenir compte du fait que l'attraction n'est pas instantanée. On peut donc se demander avec raison si la mécanique nouvel-

le ne va réussir qu'à compliquer l'astronomie sans obtenir une approximation supérieure a celle que nous donne la mécanique céleste classique. Mr. LORENTZ a abordé la question. Partant de la loi de NEWTON supposée vraie pour deux corps électrisés au repos, il calcule l'action électrodynamique des courants engendrés par ces corps en mouvement; il obtient ainsi une nouvelle loi d'attraction contenant les vitesses des deux corps comme paramètres. Avant d'examiner comment cette loi rend compte des phénomènes astronomiques, remarquons encore que l'accélération des corps célestes a comme conséquence un rayonnement électromagnétique, donc une dissipation de l'énergie se faisant ressentir en retour par un amortissement de leur vitesse. A la longue, les planètes finiront donc par tomber sur le soleil. Mais cette perspective ne peut guère nous effrayer, la catastrophe ne pouvant arriver que dans quelques millions de milliards de siècles. Revenant maintenant à la loi d'attraction, nous voyons aisément que la différence entre les deux mécaniques sera d'autant plus grande que la vitesse des planètes sera plus grande. S'il y a une différence appréciable, ce sera donc pour Mercure qu'elle sera la plus grande, Mercure ayant de toutes les planètes la plus grande vitesse. Or il arrive justement que Mercure présente une anomalie non encore expliquée: le mouvement de son périhélie est plus rapide que le mouvement calculé par la theorie classique. L'accélération est de $38''$ trop grande. LEVERRIER attribua cette anomalie à une planète non encore découverte et un astronome amateur crut observer son passage au soleil. Depuis lors plus personne ne l'a vue et il est malheureusement certain que cette planète aperçue n'était qu'un oiseau. Or la mécanique nouvelle rend bien compte du sens de l'erreur relative à Mercure, mais elle laisse cependant encore une marge de $32''$ entre elle et l'observation. Elle ne suffit donc pas à ramener la concordance dans la théorie de Mercure. Si ce résultat n'est guère décisif en faveur de la mécanique nouvelle, il est encore moins défavorable à son acceptation puisque le sens dans lequel elle corrige l'écart de la théorie classique est le bon. La théorie des autres planètes n'est pas sensiblement modifié dans la nouvelle théorie et les résultats coïncident à l'approximation des mesures près à ceux de la théorie classique.

Pour conclure, il serait prématuré, je crois, malgré la grande valeur des arguments et des faits érigés contre elle, de regarder la mécanique classique comme définitivement condamnée. Quoiqu'il en soit d'ailleurs, elle restera la mécanique des vitesses très petites par rapport à celle de la lumière, la mécanique donc de notre vie pratique et de notre technique terrestre. Si cependant, dans quelques années sa rivale triomphe, je me permettrai de vous signaler un écueil pédagogique que n'éviteront pas nombre de maîtres, en France, tout au moins. Ces maîtres n'auront rien de plus pressé, en enseignant la mécanique élémentaire à leurs élèves, que de leur apprendre que cette mécanique là a fait son temps, qu'une mécanique nouvelle où les notions de masse et de temps ont une toute autre valeur la remplace; ils regarderont de haut cette mécanique périmée que les programmes les forcent à enseigner et feront sentir à leurs élèves le mépris qu'ils lui portent. Je crois bien cependant que cette mécanique classique dédaignée sera aussi nécessaire que maintenant et que celui qui ne la connaîtra pas à fond ne pourra comprendre la mécanique nouvelle.

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AND

RECREATIONS

BY

HERMANN SCHUBERT

PROFESSOR OF MATHEMATICS IN THE JOHANNEUM, HAMBURG, GERMANY

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THE mathematical essays and recreations in this volume are by one of the most successful teachers and text-book writers of Germany. The monistic construction of arithmetic, the systematic and organic development of all its consequences from a few thoroughly established principles, is quite foreign to the general run of American and English elementary text-books, and the first three essays of Professor Schubert will, therefore, from a logical and esthetic side, be full of suggestions for elementary mathematical teachers and students, as well as for non-mathematical readers. For the actual detailed development of the system of arithmetic here sketched, we may refer the reader to Professor Schubert's volume *Arithmetik und Algebra*, recently published in the Göschen-Sammlung (Göschen, Leipsic),—an extraordinarily cheap series containing many other unique and valuable text-books in mathematics and the sciences.

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In all these essays, which are of a simple and popular character, and designed for the general public, Professor Schubert has incorporated much of his original research.

THOMAS J. MCCORMACK.

LA SALLE, Ill., December, 1898.

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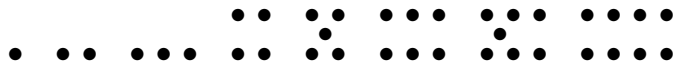
NOTION AND DEFINITION OF NUMBER.

MANY essays have been written on the definition of number. But most of them contain too many technical expressions, both philosophical and mathematical, to suit the non-mathematician. The clearest idea of what counting and numbers mean may be gained from the observation of children and of nations in the childhood of civilisation. When children count or add, they use either their fingers, or small sticks of wood, or pebbles, or similar things, which they adjoin singly to the things to be counted or otherwise ordinally associate with them. As we know from history, the Romans and Greeks employed their fingers when they counted or added. And even to-day we frequently meet with people to whom the use of the fingers is absolutely indispensable for computation.

Still better proof that the accurate association of such "other" things with the things to be counted is the essential element of numeration are the tales of travellers in Africa, telling us how African tribes sometimes inform friendly nations of the number of the enemies who have invaded their domain. The conveyance of the information is effected not by messengers, but simply by placing at spots selected for the purpose a number of stones exactly equal to the number of the invaders. No one will deny that the number of the tribe's foes is thus communicated, even though no name exists for this number in the languages of the tribes. The reason why the fingers are so universally employed as a means of numeration is, that every one possesses a definite number of fingers, sufficiently large for purposes of computation and that they are always at hand.

Besides this first and chief element of numeration which, as we have seen, is the exact, individual conjunction or association of other things with the things to be counted, is to be mentioned a second important

element, which in some respects perhaps is not so absolutely essential; namely, that the things to be counted shall be regarded as of the same kind. Thus, any one who subjects apples and nuts collectively to a process of numeration will regard them for the time being as objects of the same kind, perhaps by subsuming them under the common notion of fruit. We may therefore lay down provisionally the following as a definition of counting: to count a group of things is to regard the things as the same in kind and to associate ordinally, accurately, and singly with them other things. In writing, we associate with the things to be counted simple signs, like points, strokes, or circles. The form of the symbols we use is indifferent. Neither need they be uniform. It is also indifferent what the spatial relations or dispositions of these symbols are. Although, of course, it is much more convenient and simpler to fashion symbols growing out of operations of counting on principles of uniformity and to place them spatially near each other. In this manner are produced what I have called* natural number-pictures; for example,

 etc.

Now-a-days such natural number-pictures are rarely employed, and are to be seen only on dominoes, dice, and sometimes, also, on playing-cards.

It can be shown by archæological evidence that originally numeral writing was made up wholly of natural number-pictures. For example, the Romans in early times represented all numbers, which were written at all, by assemblages of strokes. We have remnants of this writing in the first three numerals of the modern Roman system. If we needed additional evidence that the Romans originally employed natural number-signs, we might cite the passage in Livy, VII. 3, where we are told, that, in accordance with a very ancient law, a nail was annually driven into a certain spot in the sanctuary of Minerva, the "inventrix" of counting, for the purpose of showing the number of years which had elapsed since the building of the edifice. We learn from the same source that also in the temple at Volsinii nails were shown which the Etruscans had placed there as marks for the number of years.

Also recent researches in the civilisation of ancient Mexico show that natural number-pictures were the first stage of numeral notation.

* *System der Arithmetik*. (Potsdam: Aug. Stein. 1885.)

Whosoever has carefully studied in any large ethnographical collection the monuments of ancient Mexico, will surely have remarked that the nations which inhabited Mexico before its conquest by the Spaniards, possessed natural number-signs for all numbers from one to nineteen, which they formed by combinations of circles. If in our studies of the past of modern civilised peoples, we meet with natural number-pictures only among the Greeks or Romans, and some Oriental nations, the reason is that the other nations, as the Germans, before they came into contact with the Romans and adopted the more highly developed notation of the latter, were not yet sufficiently advanced in civilisation to feel any need of expressing numbers symbolically. But since the most perfect of all systems of numeration, the Hindu system of "local value," was introduced and adopted in Europe in the twelfth century, the Roman numeral system gradually disappeared, at least from practical computation, and at present we are only reminded by the Roman characters of inscriptions of the first and primitive stage of all numeral notation. To-day we see natural number-pictures, except in the above-mentioned games, only very rarely, as where the tally-men of wharves or warehouses make single strokes with a pencil or a piece of chalk, one for each bale or sack which is counted.

As in writing it is of consequence to associate with each of the things to be counted some simple sign, so in speaking it is of consequence to utter for each single thing counted some short sound. It is quite indifferent here what this sound is called, also whether the sounds which are associated with the things to be counted are the same in kind or not, and finally, whether they are uttered at equal or unequal intervals of time. Yet it is more convenient and simpler to employ the same sound and to observe equal intervals in their utterance. We arrive thus at natural number-words. For example, utterances like,

oh, oh-oh, oh-oh-oh, oh-oh-oh-oh, oh-oh-oh-oh-oh,

are natural number-words for the numbers from one to five. Number-words of this description are not now to be found in any known language. And yet we hear such natural number-words constantly, every day and night of our lives; the only difference being that the speakers are not human beings but machines—namely, the striking-apparatus of our clocks.

Word-forms of the kind described are too inconvenient, however, for use in language, not only for the speaker, on account of their ultimate length, but also for the hearer, who must be constantly on the *qui vive* lest he misunderstand a numeral word so formed. It has thus come about that the languages of men from time immemorial have possessed numeral words which exhibit no trace of the original idea of single association. But if we should always select for every new numeral word some new and special verbal root, we should find ourselves in possession of an inordinately large number of roots, and too severely tax our powers of memory. Accordingly, the languages of both civilised and uncivilised peoples always construct their words for larger numbers from words for smaller numbers. What number we shall begin with in the formation of compound numeral words is quite indifferent, so far as the idea of number itself is concerned. Yet we find, nevertheless, in nearly all languages one and the same number taken as the first station in the formation of compound numeral words, and this number is ten. Chinese and Latins, Fins and Malays, that is, peoples who have no linguistic relationship, all display in the formation of numeral words the similarity of beginning with the number ten the formation of compound numerals. No other reason can be found for this striking agreement than the fact that all the forefathers of these nations possessed ten fingers.

Granting it were impossible to prove in any other way that people originally used their fingers in reckoning, the conclusion could be inferred with sufficient certainty solely from this agreement with regard to the first resting-point in the formation of compound numerals among the most various races. In the Indo-Germanic tongues the numeral words from ten to ninety-nine are formed by composition from smaller numeral words. Two methods remain for continuing the formation of the numerals: either to take a new root as our basis of composition (hundred) or to go on counting from ninety-nine, saying tenty, eleventy, etc. If we were logically to follow out this second method we should get tenty-ty for a thousand, tenty-ty-ty for ten thousand, etc. But in the utterance of such words, the syllable *ty* would be so frequently repeated that the same inconvenience would be produced as above in our individual number-pictures. For this reason the genius which controls the formation of speech took the first course.

But this course is only logically carried out in the old Indian numeral words. In Sanskrit we not only have for ten, hundred, and thousand a new root, but new bases of composition also exist for ten thousand, one hundred thousand, ten millions, etc., which are in no wise related with the words for smaller numbers. Such roots exist among the Hindus for all numerals up to the number expressed by a one and fifty-four appended naughts. In no other language do we find this principle carried so far. In most languages the numeral words for the numbers consisting of a one with four and five appended naughts are compounded, and in further formations use is made of the words million, billion, trillion, etc., which really exhibit only one root, before which numeral words of the Latin tongue are placed.

Besides numeral word-systems based on the number *ten*, only logical systems are found based on the number five and on the number twenty. Systems of numeral words which have the basis five occur in equatorial Africa. (See the language-tables of Stanley's books on Africa.) The Aztecs and Mayas of ancient Mexico had the base twenty. In Europe it was mainly the Celts who reckoned with twenty as base. The French language still shows some few traces of the Celtic vicenary system, as in its word for eighty, *quatre-vingt*. The choice of five and of twenty as bases is explained simply enough by the fact that each hand has five fingers, and that hands and feet together have twenty fingers and toes.

As we see, the languages of humanity now no longer possess natural number-signs and number-words, but employ names and systems of notation adopted subsequently to this first stage. Accordingly, we must add to the definition of counting above given a third factor or element which, though not absolutely necessary, is yet important, namely, that we must be able to express the results of the above-defined associating of certain other things with the things to be counted, by some conventional sign or numeral word.

Having thus established what counting or *numbering* means, we are in a position to define also the notion of *number*, which we do by simply saying that by number we understand *the results* of counting or numeration. These are naturally composed of two elements. First, of the ordinary number-word or number-sign; and secondly, of the word standing for the specific things counted. For example, eight men, seven trees, five cities. When, now, we have counted one group of things, and

subsequently also counted another group of things of the same kind, and thereupon we conceive the two groups of things combined into a single group, we can save ourselves the labor of counting the things a third time by blending the number-pictures belonging to the two groups into a single number-picture belonging to the whole. In this way we arrive on the one hand at the idea of addition, and on the other, at the notion of "unnamed" number. Since we have no means of telling from the two original number-pictures and the third one which is produced from these, the kind or character of the things counted, we are ultimately led in our conception of number to abstract wholly from the nature of the things counted, and to form the definition of unnamed number.

We thus see that to ascend from the notion of named number to the notion of unnamed number, the notion of addition, joined to a high power of abstraction, is necessary. Here again our theory is best verified by observations of children learning to count and add. A child, in beginning arithmetic, can well understand what five pens or five chairs are, but he cannot be made to understand from this alone what five abstractly is. But if we put beside the first five pens three other pens, or beside the five chairs three other chairs, we can usually bring the child to see that five things plus three things are always eight things, no matter of what nature the things are, and that accordingly we need not always specify in counting what kind of things we mean. At first we always make the answer to our question of what five plus three is, easy for the child, by relieving him of the process of abstraction, which is necessary to ascend from the named to the unnamed number, an end which we accomplish by not asking first what five plus three is, but by associating with the numbers words designating things within the sphere of the child's experience, for example, by asking how many five pens plus three pens are.

The preceding reflexions have led us to the notion of unnamed or abstract numbers. The arithmetician calls these numbers positive whole numbers, or positive integers, as he knows of other kinds of numbers, for example, negative numbers, irrational numbers, etc. Still, observation of the world of actual facts, as revealed to us by our senses, can naturally lead us only to positive whole numbers, such only, and no others, being results of actual counting. All other kinds of numbers are nothing but artificial inventions of mathematicians created for the

purpose of giving to the chief tool of the mathematician, namely, arithmetical notation, a more convenient and more practical form, so that the solution of the problems which arise in mathematics may be simplified. All numbers, excepting the results of counting above defined, are and remain mere symbols, which, although they are of incalculable value in mathematics, and, therefore, can scarcely be dispensed with, yet could, if it were a question of principle, be avoided. Kronecker has shown that any problem in which positive whole numbers are given, and only such are sought, always admits of solution without the help of other kinds of numbers, although the employment of the latter wonderfully simplifies the solution.

How these derived species of numbers, by the logical application of a single principle, flow naturally from the notion of number and of addition above deduced, I shall show in the next article entitled "Monism in Arithmetic."

MONISM IN ARITHMETIC.

In his *Primer of Philosophy*, Dr. Paul Carus defines monism as a "unitary conception of the world." Similarly, we shall understand by monism in a science the unitary conception of that science. The more a science advances the more does monism dominate it. An example of this is furnished by physics. Whereas formerly physics was made up of wholly isolated branches, like Mechanics, Heat, Optics, Electricity, and so forth, each of which received independent explanations, physics has now donned an almost absolute monistic form, by the reduction of all phenomena to the *motions* of molecules. For example, optical and electrical phenomena, we now know, are caused by the undulatory movements of the ether, and the length of the ether-waves constitutes the sole difference between light and electricity.

Still more distinctly than in physics is the monistic element displayed in pure arithmetic, by which we understand the theory of the combination of two numbers into a third by addition and the direct and indirect operations springing out of addition. Pure arithmetic is a science which has completely attained its goal, and which can prove that it has, exclusively by internal evidence. For it may be shown on the one hand that besides the seven familiar operations of addition, subtraction, multiplication, division, involution, evolution, and the finding of logarithms, no other operations are definable which present anything essentially new; and on the other hand that fresh extensions of the domain of numbers beyond irrational, imaginary, and complex numbers are arithmetically impossible. Arithmetic may be compared to a tree that has completed its growth, the boughs and branches of which may still increase in size or even give forth fresh sprouts, but whose main trunk has attained its fullest development.

Since arithmetic has arrived at its maturity, the more profound investigation of the nature of numbers and their combinations shows that a unitary conception of arithmetic is not only possible but also necessary. If we logically abide by this unitary conception, we arrive, starting from the notion of counting and the allied notion of addition, at all conceivable operations and at all possible extensions of the notion of number. Although previously expressed by Grassmann, Hankel, E. Schröder, and Kronecker, the author of the present article, in his "System of Arithmetic," Potsdam, 1885, was the first to work out the idea referred to, fully and logically and in a form comprehensible for beginners. This book, which Kronecker in his "Notion of Number," an essay published in Zeller's jubilee work, makes special mention of, is intended for persons proposing to learn arithmetic. As that cannot be the object of the readers of these essays, whose purpose will rather be the study of the logical construction of the science from some single fundamental principle, the following pages will simply give of the notions and laws of arithmetic what is absolutely necessary for an understanding of its development.

The starting-point of arithmetic is the idea of counting and of number as the result of counting. On this subject, the reader is requested to read the first essay of this collection. It is there shown that the idea of addition springs immediately from the idea of counting. As in counting it is indifferent in what order we count, so in addition it is indifferent, for the sum, or the result of the addition, whether we add the first number to the second or the second to the first. This law, which in the symbolic language of arithmetic, is expressed by the formula

$$a + b = b + a,$$

is called the *commutative law* of addition. Notwithstanding this law, however, it is evidently desirable to distinguish the two quantities which are to be summed, and out of which the sum is produced, by special names. As a fact, the two summands usually are distinguished in some way, for example, by saying a is to be increased by b, or b is to be added to a, and so forth. Here, it is plain, a is always something that is to be increased, b the increase. Accordingly it has been proposed to call the number which is regarded in addition as the passive number or the one to be changed, the *augend*, and the other which plays the active part, which accomplishes the change, so to speak, the *increment*.

Both words are derived from the Latin and are appropriately chosen. Augend is derived from *augere*, to increase, and signifies that which is to be increased; increment comes from *increscere*, to grow, and signifies as in its ordinary meaning what is added.

Besides the commutative law one other follows from the idea of counting—the *associative law* of addition. This law, which has reference not to two but to three numbers, states that having a certain sum, $a+b$, it is indifferent for the result whether we increase the increment b of that sum by a number, or whether we increase the sum itself by the same number. Expressed in the symbolic language of arithmetic this law reads,

$$a + (b + c) = (a + b) + c.$$

To obtain now all the rules of addition we have only to apply the two laws of commutation and association above stated, though frequently, in the deduction of the same rule, each must be applied many times. I may pass over here both the rules and their establishment.

In addition, two numbers, the augend a and the increment b are combined into a third number c, the sum. From this operation spring necessarily two inverse operations, the common feature of which is, that the sum sought in addition is regarded in both as known, and the difference that in the one the augend also is regarded as known, and in the other the increment. If we ask what number added to a gives c, we seek the increment. If we ask what number increased by b gives c, we seek the augend. As a matter of reckoning, the solution of the two questions is the same, since by the commutative law of addition $a + b = b + a$. Consequently, only one common name is in use for the two inverses of addition, namely, *subtraction*. But with respect to the notions involved, the two operations do differ, and it is accordingly desirable in a logical investigation of the structure of arithmetic, to distinguish the two by different names. As in all probability no terms have yet been suggested for these two kinds of subtraction, I propose here for the first time the following words for the two operations, namely, *detraction* to denote the finding of the increment, and *subtertraction* to denote the finding of the augend. We obtain these terms simply enough by thinking of the augmentation of some object already existing. For example, the cathedral at Cologne had in its tower an augend that waited centuries for its increment, which was only supplied a few decades ago. As the cathedral had originally a height of one hundred and thirty metres, but

after completion was increased in height twenty-six metres, of the total height of one hundred and fifty-six metres one hundred and thirty metres is clearly the augend and twenty-six metres the increment. If, now, we wished to recover the augend we should have to pull down (Latin, *detrahere*) the upper part along the whole height. Accordingly, the finding of the augend is called *detraction*. If we sought the increment, we should have to pull out the original part from beneath (Latin, *subtertrahere*). For this reason, the finding of the increment is called *subtertraction*. Owing to the commutative law, the two inverse operations, as matters of computation, become one, which bears the name of *subtraction*. The sign of this operation is the minus sign, a horizontal stroke. The number which originally was sum, is called in subtraction minuend; the number which in addition was increment is now called detractor; the number which in addition was augend is now called subtertractor. Comprising the two conceptually different operations in one single operation, subtraction, we employ for the number which before was increment or augend, the term subtrahend, a word which on account of its passive ending is not very good, and for which, accordingly, E. Schröder proposes to substitute the word *subtrahent*, having an active ending. The result of subtraction, or what is the same thing, the number sought, is called the *difference*. The definition-formula of subtraction reads

$$a - b + b = a,$$

that is, a minus b is the number which increased by b gives a, or the number which added to b gives a, according as the one or the other of the two operations inverse to addition is meant. From the formula for subtraction, and from the rules which hold for addition, follow now at once the rules which refer to both addition and subtraction. These rules we here omit.

From the foregoing it is plain that the minuend is necessarily larger than the subtrahent. For in the process of addition the minuend was the sum, and the sum grew out of the union of two natural number-pictures.* Thus 5 minus 9, or 11 minus 12, or 8 minus 8, are combinations of numbers *wholly destitute of meaning*; for no number, that is, no result of counting, exists that added to 9 gives the sum 5, or added to 12 gives the sum 11, or added to 8 gives 8. What, then, is to be

* See page 2, *supra*.

done? Shall we banish entirely from arithmetic such meaningless combinations of numbers; or, since they have no meaning, shall we rather invest them with one? If we do the first, arithmetic will still be confined in the strait-jacket into which it was forced by the original definition of number as the result of counting. If we adopt the latter alternative we are forced to extend our notion of number. But in doing this, we sow the first seeds of the science of pure arithmetic, an organic body of knowledge which fructifies all other provinces of science.

What significance, then, shall we impart to the symbol

$$5 - 9?$$

Since 5 minus 9 possesses no significance whatever, we may, of course, impart to it any significance we wish. But as a matter of practical convenience it should be invested with no meaning that is likely to render it subject to exceptions. As the form of the symbol $5 - 9$ is the form of a difference, it will be obviously convenient to give it a meaning which will allow us to reckon with it as we reckon with every other real difference, that is, with a difference in which the minuend is larger than the subtrahent. This being agreed upon, it follows at once that all such symbols in which the number before the minus sign is less than the number behind it by the same amount may be put equal to one another. It is practical, therefore, to comprise all these symbols under some one single symbol, and to construct this latter symbol so that it will appear unequivocally from it by how much the number before the minus sign is less than the number behind it. This difference, accordingly, is written down and the minus sign placed before it.

If the two numbers of such a differential *form* are equal, a totally new sign must be invented for the expression of the fact, having no relation to the signs which state results of counting. This invention was not made by the ancient Greeks, as one might naturally suppose from the high mathematical attainments of that people, but by Hindu Brahman priests at the end of the fourth century after Christ. The symbol which they invented they called *tsiphra*, empty, whence is derived the English *cipher*. The form of this sign has been different in different times and with different peoples. But for the last two or three centuries, since the symbolic language of arithmetic has become thoroughly established as an international character, the form of the sign has been 0 (French *zéro*, German *null*).

In calling this symbol and the symbols formed of a minus sign followed by a result of counting, *numbers*, we widen the province of numbers, which before was wholly limited to results of counting. In no other way can zero and the negative numbers be introduced into arithmetic. No man can *prove* that 7 minus 11 is equal to 1 minus 5. Originally, both are meaningless symbols. And not until we agree to impart to them a significance which allows us to reckon with them as we reckon with real differences are we led to a statement of identity between 7 minus 11 and 1 minus 5. It was a long time before the negative numbers mentioned acquired the full rights of citizenship in arithmetic. Cardan called them, in his *Ars Magna*, 1545, *numeri ficti* (imaginary numbers), as distinguished from *numeri veri* (real numbers). Not until Descartes, in the first half of the seventeenth century, was any one bold enough to substitute *numeri ficti* and *numeri veri* indiscriminately for the same letter of algebraic expressions.

We have invested, thus, combinations of signs originally meaningless, in which a smaller number stood before than after a minus sign, with a meaning which enables us to reckon with such *apparent* differences exactly as we do with ordinary differences. Now it is just this practical shift of imparting meanings to combinations, which logically applied deduces naturally the whole system of arithmetic from the idea of counting and of addition, and which we may characterise, therefore, as the *foundation-principle* of its whole construction. This principle, which Hankel once called the *principle of permanence*, but which I prefer to call the PRINCIPLE OF NO EXCEPTION, may be stated in general terms as follows:

In the construction of arithmetic every combination of two previously defined numbers by a sign for a previously defined operation (plus, minus, times, etc.) shall be invested with meaning, even where the original definition of the operation used excludes such a combination; and the meaning imparted is to be such that the combination considered shall obey the same formula of definition as a combination having from the outset a signification, so that the old laws of reckoning shall still hold good and may still be applied to it.

A person who is competent to apply this principle rigorously and logically will arrive at combinations of numbers whose results are termed irrational or imaginary with the same necessity and facility as at the combinations above discussed, whose results are termed negative

numbers and zero. To think of such combinations as *results* and to call the products reached also "numbers" is a misuse of language. It were better if we used the phrase *forms of numbers* for all numbers that are not the results of counting. But *usus tyrannus!*

It will now be my task to show how all numbers at which arithmetic ever has arrived or ever can arrive naturally flow from the simple application of the principle of no exception.

Owing to the commutative and associative laws for addition it is wholly indifferent for the result of a series of additive processes in what order the numbers to be summed are added. For example,

$$a + (b + c + d) + (e + f) = (a + b + c) + (d + e) + f.$$

The necessary consequence of this is that we may neglect the consideration of the order of the numbers and give heed only to what the quantities are that are to be summed, and, when they are equal, take note of only two things, namely, of what the quantity which is to be repeatedly summed is called and how often it occurs. We thus reach the notion of multiplication. To multiply a by b means to form the sum of b numbers each of which is called a. The number conceived summed is called the multiplicand, the number which indicates or counts how often the first is conceived summed is called the multiplier.

It appears hence, that the multiplier must be a result of counting, or a number in the original sense of the word, but that the multiplicand may be any number hitherto defined, that is, may also be zero or negative. It also follows from this definition that though the multiplicand may be a concrete number the multiplier cannot. Therefore, the commutative law of multiplication does not hold when the multiplicand is concrete. For, to take an example, though there is sense in requiring four trees to be summed three times, there is no sense in conceiving the number three summed "four trees times." When, however, multiplicand and multiplier are unnamed results of counting, (abstract numbers,) two fundamental laws hold in multiplication, exactly analogous to the fundamental laws of addition, namely, the law of commutation and the law of association. Thus,

$$a \textit{ times } b = b \textit{ times } a,$$

$$\text{and, } a \textit{ times } (b \textit{ times } c) = (a \textit{ times } b) \textit{ times } c.$$

The truth and correctness of these laws will be evident, if keeping to the definition of multiplication as an abbreviated addition of equal summands, we go back to the laws of addition. Owing to the commutative law it is unnecessary, for purposes of practical reckoning, to distinguish multiplicand and multiplier. Both have, therefore, a common name: *factor*. The result of the multiplication is called the product; the symbol of multiplication is a dot (.) or a cross ($\times$), which is read "times." Joined with the fundamental formula above written are a group of subsidiary formulæ which give directions how a sum or difference is multiplied and how multiplication is performed with a sum or difference. I need not enter, however, into any discussion of these rules here.

As the combination of two numbers by a sign of multiplication has no significance according to our definition of multiplication, when the multiplier is zero or a negative number, it will be seen that we are again in a position where it is necessary to apply the above explained principle of no exception. We revert, therefore, to what we above established, that zero and negative numbers are symbols which have the form of differences, and lay down the rule that multiplications with zero and negative numbers shall be performed exactly as with real differences. Why, then, is minus one times minus one, for example, equal to plus one? For no other reason than that minus one can be multiplied with an ordinary difference, as, for example, 8 minus 5, by first multiplying by 8, then multiplying by 5, and subtracting the differences obtained, and because agreeably to the principle of no exception we must say that the multiplication must be performed according to exactly the same rule with a symbol which has the *form* of a difference whose minuend is less by one than its subtrahent.

As from addition two inverse operations, detraction and subtertraction, spring, so also from multiplication two inverse operations must proceed which differ from each other simply in the respect that in the one the multiplicand is sought and in the other the multiplier. As matters of computation, these two inverse operations coalesce in a single operation, namely, division, owing to the validity of the commutative law in multiplication. But in so far as they are different ideas, they must be distinguished. As most civilised languages distinguish the two inverse processes of multiplication in the case in which the multiplicand is a line, we will adopt for arithmetic a name which is used in this

exception. Let us take this example,

$$4\ \textit{yards} \times 3 = 12\ \textit{yards}.$$

If twelve yards and four yards are given, and the multiplier 3 is sought, I ask, how many summands, each equal to four yards, give twelve yards, or, what is the same thing, how often I can lay a length of four yards on a length of twelve yards? But this is *measuring*. Secondly, if twelve yards and the number 3 are given, and the multiplicand four yards is sought, I ask what summand it is which taken three times gives twelve yards, or, what is the same thing, what part I shall obtain if I cut up twelve yards into three equal parts? But this is partition, or *parting*. If, therefore, the multiplier is sought we call the division *measuring*, and if the multiplicand is sought, we call it *parting*. In both cases the number which was originally the product is called the dividend, and the result the quotient. The number which originally was multiplicand is called the measure; the number which originally was multiplier is called the parter. The common name for measure and parter is divisor. The common symbol for both kinds of division is a colon, a horizontal stroke, or a combination of both. Its definitional formula reads,

$$(a \div b) \centerdot b = a, \text{ or, } \frac{a}{b} \centerdot b = a.$$

Accordingly, dividing a by b means, to find the number which multiplied by b gives a, or to find the number *with* which b must be multiplied to produce a. From this formula, together with the formulæ relative to multiplication, the well-known rules of division are derived, which I here pass over.

In the dividend of a quotient only such numbers can have a place which are the product of the divisor with some previously defined number. For example, if the divisor is 5 the dividend can only be 5, 10, 15, and so forth, and 0, -5, -10 and so forth. Accordingly, a stroke of division having underneath it 5 and above it a number different from the numbers just named is a combination of symbols having no meaning. For example, $\frac{3}{5}$ or $\frac{12}{5}$ are meaningless symbols. Now, conformably to the principle of no exception we must invest such symbols which have the form of a quotient without their dividend being the product of the divisor with any number yet defined, with a meaning such that we shall be able to reckon with such apparent quotients as with ordinary quotients. This is done by our agreeing always to put the product

of such a quotient form with its divisor equal to its dividend. In this way we reach the definition of broken numbers or *fractions*, which by the application of the principle of no exception spring from division exactly as zero and negative numbers sprang from subtraction. The latter had their origin in the impossibility of the subtraction; the former have their origin in the impossibility of the division. Putting together now both these extensions of the domain of numbers, we arrive at *negative fractional numbers*.

We pass over the easily deduced rules of computation for fractions and shall only direct the reader's attention to the connexion which exists between fractional and non-fractional or, as we usually say, whole numbers. Since the number 12 lies between the numbers 10 and 15, or, what is the same thing, $10 < 12 < 15$, and since $10 : 5 = 2$, $15 : 5 = 3$, we say also that $12 : 5$ lies between 2 and 3, or that

$$2 < \tfrac{12}{5} < 3.$$

In itself, the notion of "less than" has significance only for results of counting. Consequently, it must first be stated what is meant when it is said that 2 is less than $\frac{12}{5}$. Plainly, nothing is meant by this except that 2 times 5 is less than 12. We thus see that every broken number can be so interpolated between two whole numbers differing from each other only by 1 that the one shall be smaller and the other greater, where smaller and greater have the meaning above given.

From the above definitions and the laws of commutation and association all possible rules of computation follow, which in virtue of the principle of no exception now hold indiscriminately for all numbers hitherto defined. It is a consequence of these rules, again, that the combination of two such numbers by means of any of the operations defined must in every case lead to a number which has been already defined, that is, to a positive or negative whole or fractional number, or to zero. The sole exception is the case where such a number is to be divided by zero. If the dividend also is zero, that is, if we have the combination $\frac{0}{0}$, the expression is one which stands for any number whatsoever, because any number whatsoever, no matter what it is, if multiplied by zero gives zero. But if the dividend is not zero but some other number a, be it what it will, we get a quotient form to which *no* number hitherto defined can be equated. But we discover that if we apply the ordinary arithmetical rules to $a \div 0$ all such forms may

be equated to one another both when a is positive and also when a is negative. We may therefore invent two new signs for such quotient forms, namely $+\infty$ and $-\infty$. We find, further, that in transferring the notions greater and less to these symbols, $+\infty$ is greater than any positive number, however great, and $-\infty$ is smaller than any negative number, however small. We read these new signs, accordingly, "plus infinitely great" and "minus infinitely great."

But even here arithmetic has not reached its completion, although the combination of as many previously defined numbers as we please by as many previously defined operations as we please will still lead necessarily to some previously defined number. Every science must make every possible advance, and still one step in advance is possible in arithmetic. For in virtue of the laws of commutation and association, which also fortunately obtain in multiplication, just as we advance from addition to multiplication, so here again we may ascend from multiplication to *an operation of the third degree*. For, just as for $a+a+a+a$ we read $4\,.\,a$, so with the same reason we may introduce some more abbreviated designation for $a\,.\,a\,.\,a\,.\,a$. The introduction of this new operation is in itself simply a matter of convenience and not an extension of the ideas of arithmetic. But if after having introduced this operation we repeatedly apply the monistic principle of arithmetic, the principle of no exception, we reach new means of computation which have led to undreamt of advances not only in the hands of mathematicians but also in the hands of natural scientists. The abbreviated designation mentioned, which, fructified by the principle of no exception, can render science such incalculable services, is simply that of writing for a product of b factors of which each is called a, a^b, which we read a to the bth power. Here a new direct operation, that of *involution*, is defined, and from now on we are justified in distinguishing operations which are not inverses of others, as addition, multiplication, and involution, by *numbers of degree*. Addition is the direct operation of the first degree, multiplication that of the second degree, and involution that of the third degree. In the expression a^b the passive number a is called the *base*, the active number b the *exponent*, the result, the *power*.

But whilst in the direct operations of the first and second degree, the laws of commutation and association hold, here in involution, the operation of third degree, the two laws are inapplicable, and the result of their inapplicability is that operations of a still higher degree than the

third form no possible advancement of pure arithmetic. The product of b factors a is not equal to the product of a factors b; that is, the law of commutation does not hold. The only two different integers for which a to the bth power is equal to b to the ath power are 2 and 4, for 2 to the 4th power is 16, and 4 to the second power also is 16. So, too, the law of association as a general rule does not hold. For it is hardly the same thing whether we take the $(b^c)^{th}$ power of a or the cth power of a^b.

From the definition of involution follow the usual rules for reckoning with powers, of which we shall only mention one, namely, that the $(b-c)^{th}$ power of a is equal to the result of the division of a to the bth power by a to the cth power. If we put here c equal to b, we are obliged, by the principle of no exception, to put a to the 0th power equal to 1; a new result not contained in the original notion of involution, for that implied necessarily that the exponent should be a result of counting. Again, if we make b smaller than c we obtain a *negative exponent*, which we should not know how to dispose of if we did not follow our monistic law of arithmetic. According to the latter, a to the $(b-c)^{th}$ power must still remain equal to a^b divided by a^c even when b is smaller than c. Whence follows that a to the minus d^{th} power is equal to 1 divided by a to the dth power, or to take specific numbers, that 3 to the minus 2nd power is equal to $\frac{1}{9}$.

At this point, perhaps, the reader will inquire what a raised to a fractional power is. But this can be explained only when we have discussed the inverse processes of involution, to which we now pass.

If $a^b = c$, we may ask two questions: first, what the base is which raised to the bth power gives c; the second, what the exponent of the power is to which a must be raised to produce c. In the first case we seek the base, and term the operation which yields this result *evolution*; in the second case we seek the exponent and call the operation which yields this exponent, the *finding of the logarithm*. In the first case, we write $\sqrt[b]{c} = a$ (which we read, the bth root of c is equal to a), and call c the *radicand*, b the *exponent of the root*, and a the *root*. In the second case, we write $\log_a c = b$ (which we read, the logarithm of c to the base a is equal to b), and call c the *logarithmand* or *number*, a the *base of the logarithm*, and b the *logarithm*.

While, owing to the validity of the law of commutation in addition and multiplication, the two inverse processes of those operations are

identical so far as computation is concerned, here in the case of involution the two inverse operations are in this regard essentially different, for in this case the law of commutation does not hold.

From the definitional formulæ for evolution and the finding of logarithms, namely,

$$(\sqrt[b]{c})^b = c, \text{ and } (a)^{\log_a c} = c,$$

follow, by the application of the laws of involution, the rules for computation with roots and logarithms. These rules we pass over here, only remarking, first, that for the present $\sqrt[b]{c}$ has meaning only when c is the bth power of some number already defined; and, secondly, that for the present also $\log_a c$ has meaning only when c can be produced by raising the number a to some power which is a number already defined. In the phrase "has only meaning for the present" is contained a possibility of new extensions of the domain of number. But before we pass to those extensions we shall first make use of the idea of evolution just defined to extend the notion of power also to cases in which the exponent is a fractional number.

According to the original definition of involution, a^b was meaningless except where b was a result of counting. But afterwards, even powers which had for their exponents zero or a negative integer could be invested with meaning. Now we have to consider the arithmetical combination "a raised to the fractional power $\frac{p}{q}$." The principle of no exception compels us to give to the arithmetical combination "a to the $\frac{p}{q}^{th}$ power" a significance such that all the rules of computation will hold with respect to it. Now, one rule that holds is, that the mth power of the nth power of a is equal to the $(m \times n)^{th}$ power of a. Consequently, the qth power of a raised to the $\frac{p}{q}^{th}$ power must be equal to a raised to a power whose exponent is equal to $\frac{p}{q}$ times q. But the last-mentioned product gives, according to the definition of division, the number p. Consequently the symbol a to the $\frac{p}{q}^{th}$ power is so constituted that its qth power is equal to a to the pth power; i. e., it is equal to the qth root of a^p. Similarly, we find that the symbol "a to the minus $\frac{p}{q}^{th}$ power" must be put equal to 1 divided by the qth root of a to the pth power, if we are to reckon with this symbol as we do with real powers. Again, just as a to the bth power is invested with meaning when b is a fractional number, so some meaning harmonious with the principle of no exception must be imparted to the bth root of c where b is a positive or negative fractional number. For example, the three-fourthsth root of 8

is equal to 8 to the $\frac{4}{3}$ power, that is, to the cube root of 8 to the 4th power, or 16.

The principle underlying arithmetic now also compels us to give to the symbol the "b^{th} root of c" a meaning when c is not the bth power of any number yet defined. First, let c be any *positive* integer or fraction. Then always to be able to reckon with the bth root of c in the same way that we do with extractible roots, we must agree always to put the bth power of the bth root of c equal to c—for example, $(\sqrt[2]{3})^2$ always exactly equal to 3. A careful inspection of the new symbols, which we will also call numbers, shows, that though no one of them is exactly equal to a number hitherto defined, yet by a certain extension of the notions greater and less, two numbers of the character of numbers already defined may be found for each such new number, such that the new number is greater than the one and less than the other of the two, and that further, these two numbers may be made to differ from each other by as small a quantity as we please. For example,

$$(\tfrac{7}{5})^3 = \tfrac{343}{125} = 2\tfrac{93}{125} < 3 < 3\tfrac{3}{8} = \tfrac{27}{8} = (\tfrac{3}{2})^3.$$

The number 3, as we see, is here included between two limits which are the third powers of two numbers $\frac{7}{5}$ and $\frac{3}{2}$ whose difference is $\frac{1}{10}$. We could also have arranged it so that the difference should be equal to $\frac{1}{100}$, or to any specified number, however small. Now, instead of putting the symbol "less than" between $(\frac{7}{5})^3$ and 3, and between 3 and $(\frac{3}{2})^3$, let us put it between their third roots; for example, let us say:

$$\tfrac{7}{5} < \sqrt[3]{3} < \tfrac{3}{2}, \text{ meaning by this that } (\tfrac{7}{5})^3 < 3 < (\tfrac{3}{2})^3.$$

In this sense we may say that the new numbers always lie *between* two old numbers whose difference may be made as small as we please. Numbers possessing this property are called *irrational* numbers, in contradistinction to the numbers hitherto defined, which are termed *rational*. The considerations which before led us to negative rational numbers, now also lead us to negative irrational numbers. The repeated application of addition and multiplication as of their inverse processes to irrational numbers, (numbers which though not exactly equal to previously defined rational numbers may yet be brought as near to them as we please,) again simply leads to numbers of the same class.

A totally new domain of numbers is reached, however, when we attempt to impart meaning to *the square roots of negative numbers.*

The square root of minus 9 is neither equal to plus 3 nor to minus 3, since each multiplied by itself gives plus 9, nor is it equal to any other number hitherto defined. Accordingly, the square root of minus 9 is a new number-form, to which, harmoniously with the principle of no exception, we may give the definition that $(\sqrt[2]{-9})^2$ shall always be put equal to minus 9.* Keeping to this definition we see at once that $\sqrt{-a}$, where a is any positive rational or irrational number, is a symbol which can be put equal to the product of $\sqrt{+a}$ by $\sqrt{-1}$. In extending to these new numbers the rights of arithmetical citizenship, in calling them also "numbers," and so shaping their definition that we can reckon with them by the same rules as with already defined numbers, we obtain a fourth extension of the domain of numbers which has become of the greatest importance for the progress of all branches of mathematics. The newly defined numbers are called *imaginary*, in contradistinction to all heretofore defined, which are called *real*. Since all imaginary numbers can be represented as products of real numbers with the square root of minus one, it is convenient to introduce for this one imaginary number some concise symbol. This symbol is the first letter of the word imaginary, namely, i; so that we can always put for such an expression as $\sqrt{-9}$, $3 \cdot i$.

If we combine real and imaginary numbers by operations of the first and second degree, always supposing that we follow in our reckoning with imaginary numbers the same rules that we do in reckoning with real numbers, we always arrive again at real or imaginary numbers, excepting when we join together a real and an imaginary number by addition or its inverse operations. In this case *we reach the symbol* $a+i \cdot b$, where a and b stand for real numbers. Agreeably to the principle of no exception we are permitted to reckon with $a+ib$ according to the same rules of computation as with symbols previously defined, if for the second power of i we always substitute minus 1.

In the numerical combination $a+ib$, which we also call number, we have found the most general numerical form to which the laws of arithmetic can lead, even though we wished to extend the limits of arithmetic still further. Of course, we must represent to ourselves here by a and b either zero or positive or negative rational or irrational numbers. If b is zero, $a+ib$ represents all real numbers; if a is zero, it stands for all purely imaginary numbers. This general number $a+ib$

* Henceforward we shall use the simpler sign $\sqrt{\ }$ for $\sqrt[2]{\ }$.

is called a *complex number*, so that the complex number includes in itself as special cases all numbers heretofore defined. By the introduction of irrational, purely imaginary, and the still more general complex numbers, all combinations become invested with meaning which the operations of the third degree can produce. For example, the fifth root of 5 is an irrational number, the logarithm of 2 to the base 10 is an irrational number. The logarithm of minus 1 to the base 2 is a purely imaginary number; the fourth root of minus 1 is a complex number. Indeed, we may recognise, proceeding still further, *that every combination of two complex numbers, by means of any of the operations of the first, second, or third degree will lead in turn to a complex number*, that is to say, never furnishes occasion, by application of the principle of no exception, for inventing new forms of numbers.

A certain limit is thus reached in the construction of arithmetic. But such a limit was also twice previously reached. After the investigation of addition and its inverse operations, we reached no other numbers except zero and positive and negative whole numbers, and every combination of such numbers by operations of the first degree led to no new numbers. After the investigation of multiplication and its inverse operations, the positive or negative fractional numbers and "infinitely great" were added, and again we could say that the combination of two already defined numbers by operations of the first and second degree in turn also always led to numbers already defined. Now we have reached a point at which we can say that the combination of two complex numbers by all operations of the first, second, and third degree must again always lead to complex numbers; only that now such a combination does not necessarily always lead to a single number, but may lead to many regularly arranged numbers. For example, the combination "logarithm of minus one to a positive base" furnishes a countless number of results which form an arithmetical series of purely imaginary numbers. *Still, in no case now do we arrive at new classes of numbers.* But just as before the ascent from multiplication to involution brought in its train the definition of new numbers, so it is also possible that *some new operation springing out of involution as involution sprang from multiplication might furnish the germ of other new numbers which are not reducible to* $a+ib$. As a matter of fact, mathematicians have asked themselves this question and investigated the direct operation of the fourth degree, together with its inverse processes. The result of their

investigations was, that an operation which springs from involution as involution sprang from multiplication is incapable of performing any real mathematical service; the reason of which is, that in involution the laws of commutation and association do not hold. It also further appeared that the operations of the fourth degree could not give rise to new numbers. No more so can operations of still higher degrees. With respect to quaternions, which many might be disposed to regard as new numbers, it will be evident that though quaternions are valuable means of investigation in geometry and mechanics they are not numbers of arithmetic, because the rules of arithmetic are not unconditionally applicable to them.

The building up of arithmetic is thus completed. The extensions of the domain of number are ended. It only remains to be asked why the science of arithmetic appears in its structure so logical, natural, and unarbitrary; why zero, negative, and fractional numbers appear as much derived and as little original as irrational, imaginary, and complex numbers? We answer, wholly and alone in virtue of the logical application of the monistic principle of arithmetic, the principle of no exception.

ON THE NATURE OF MATHEMATICAL KNOWLEDGE.

"MATHEMATICALLY certain and unequivocal" is a phrase which is often heard in the sciences and in common life, to express the idea that the seal of truth is more deeply imprinted upon a proposition than is the case with ordinary acts of knowledge. We propose to investigate in this article the extent to which mathematical knowledge really is more certain and unequivocal than other knowledge.

The intrinsic character of mathematical research and knowledge is based essentially on three properties: first, on its conservative attitude towards the old truths and discoveries of mathematics; secondly, on its progressive mode of development, due to the incessant acquisition of new knowledge on the basis of the old; and thirdly, on its self-sufficiency and its consequent absolute independence.

That mathematics is the most conservative of all the sciences is apparent from the incontestability of its propositions. This last character bestows on mathematics the enviable superiority that no new development can undo the work of previous developments or substitute new in the place of old results. The discoveries that Pythagoras, Archimedes, and Apollonius made are as valid to-day as they were two thousand years ago. This is a trait which no other science possesses. The notions of previous centuries regarding the nature of heat have been disproved. Goethe's theory of colors is now antiquated. The theory of the binary combination of salts was supplanted by the theory of substitution, and this, in its turn, has also given way to newer conceptions. Think of the profound changes which the conceptions of theoretical medicine, zoölogy, botany, mineralogy, and geology have undergone. It is the same, too, in the other sciences. In philology, comparative linguistics, and history our ideas are quite different from what they formerly were.

In no other science is it so indispensable a condition that whatever is asserted must be true, as it is in mathematics. Whenever, therefore, a controversy arises in mathematics, the issue is not whether a thing is true or not, but whether the proof might not be conducted more simply in some other way, or whether the proposition demonstrated is sufficiently important for the advancement of the science as to deserve especial enunciation and emphasis, or finally, whether the proposition is not a special case of some other and more general truth which is just as easily discovered.

Let me recall the controversy which has been waged in this century regarding the eleventh axiom of Euclid, that only one line can be drawn through a point parallel to another straight line. This discussion impugned in no wise the truth of the proposition; for that things are true in mathematics is so much a matter of course that on this point it is impossible for a controversy to arise. The discussion merely touched the question whether the axiom was capable of demonstration solely by means of the other propositions, or whether it was not a special property, apprehensible only by sense-experience, of that space of three dimensions in which the organic world has been produced and which therefore is of all others alone within the reach of our powers of representation. The truth of the last supposition affects in no respect the correctness of the axiom but simply assigns to it, in an epistemological regard, a different status from what it would have if it were demonstrable, as was at one time thought, without the aid of the senses, and solely by the other propositions of mathematics.

I may recall also a second controversy which arose a few decades ago as to whether all continuous functions were differentiable. In the outcome, continuous functions were defined that possessed no differential coefficient, and it was thus learned that certain truths which were enunciated unconditionally by Newton, Leibnitz, and their mathematical successors, required qualification. But this did not invalidate at all the correctness of the method of differentiation, nor its application in all practical cases; the theoretical speculations pursued on this subject simply clarified ideas and sifted out the conditions upon which differentiability depended. Happily the gifted minds who invent the new methods and open up the new paths of research in mathematics, are not deterred by the fear that a subsequent generation gifted with unusual acumen will spy out isolated cases in which their methods fail.

Happily the creators of the differential calculus pushed onward without a thought that a critical posterity would discover exceptions to their results. In every great advance that mathematics makes, the clarification and scrutinisation of the results reached are reserved necessarily for a subsequent period, but with it the demonstration of those results is more rigorously established. Despite all this, however, in no science does cognition bear so unmistakably the imprint of truth as in pure mathematics. And this fact bestows on mathematics its conservative character.

This conservative character again is displayed in the *objects* of mathematical research. The physician, the historian, the geographer, and the philologist have to-day quite different fields of investigation from what they had centuries ago. In mathematics, too, every new age gives birth to new problems, arising partly from the advance of the science itself, and partly also from the advance of civilisation, where improvements in the other sciences bring in their train new problems that are constantly taxing afresh the resources of mathematics. But despite all this, in mathematics more than in any other science problems exist that have played a rôle for hundreds, nay, for thousands of years.

In the oldest mathematical manuscript which we possess, the Rhind Papyrus of the British Museum, which dates back to the eighteenth century before Christ, and whose decipherment we owe to the industry of Eisenlohr, we find an attempt to solve the problem of converting a circle into a square of equal area, a problem whose history covers a period of three and a half thousand years. For it was not until 1882 that a rigorous proof was given of the impossibility of solving this problem exactly by the use of straight edge and compasses alone. (See pp. 111, 133–136.)

It is, of course, the insoluble problems that have the longest history; partly because it is harder to show that a thing is impossible than that it is possible, and, on the other hand, because problems that have long defied solution are ever evoking anew the spirit of inquiry and the ambition of mathematicians, and because the uncertainty of insolubility lends to such problems a peculiar charm. Of the geometrical problems that have occupied competent and incompetent minds from the time of the ancient Greeks to the present may be mentioned in addition to the squaring of the circle two others that are also perhaps well-known

to educated readers, at least by name: the trisection of the angle and the Delic problem of the duplication of the cube. All three problems involve the condition, which is often overlooked by lay readers, that only straight edge and compasses shall be employed in the constructions. In the trisection of the angle any angle is assigned, and it is required to find the two straight lines which divide the angle into three equal parts. In the Delic problem the edge of a cube is given and the edge of a second cube is sought, containing twice the volume of the first cube. In Greece, in the golden age of the sciences, when all scholars had to understand mathematics, it was a fashionable requisite almost to have employed oneself on these famous problems.

Fortunately for us, these problems were insoluble. For in their ambition to conquer them it came to pass that men busied themselves more and more with geometry, and in this way kept constantly discovering new truths and developing new theories, all of which perhaps might never have been done if the problems had been soluble and had early received their solutions. Thus is the struggle after truth often more fruitful than the actual discovery of truth. So, too, although in a slightly different sense, the apophthegm of Lessing is confirmed here, that the search for truth is to be preferred to its possession.

Whilst the three above-named problems are now acknowledged to be insoluble, and have ceased, therefore, to stimulate mathematical inquiry, there are of course other problems in mathematics whose solution has been sought for a long time, but not yet reached, and in the case of which there is no reason for supposing that they are insoluble. Of such problems the two following perhaps have found their way out of the isolated circles of mathematicians and have become more or less known to other scholars. I refer to the astronomical Problem of Three Bodies and to the problem of the frequency of prime numbers. The first of these two problems assumes three or more heavenly bodies whose movements are mutually influenced by one another according to Newton's law of gravitation, and requires the exact determination of the path which each body describes. The second problem requires the construction of a formula which shall tell how many prime numbers there are below a certain given number. So far these two problems have been solved only approximately, and not with absolute mathematical exactness.

If the eternal and inviolable correctness of its truths lends to mathematical research, and therefore also to mathematical knowledge, a *con-*

servative character, on the other hand, by the continuous outgrowth of new truths and methods from the old, *progressiveness* is also one of its characteristics. In marvellous profusion old knowledge is augmented by new, which has the old as its necessary condition, and, therefore, could not have arisen had not the old preceded it. The indestructibility of the edifice of mathematics renders it possible that the work can be carried to ever loftier and loftier heights without fear that the highest stories shall be less solid and safe than the foundations, which are the axioms, or the lower stories, which are the elementary propositions. But it is necessary for this that all the stones should be *properly fitted together*; and it would be idle labor to attempt to lay a stone that belonged above in a place below. A good example of a stone of this character belonging in what is now the uppermost layer of the edifice, is Lindemann's demonstration of the insolubility of the quadrature of the circle, a demonstration of which interesting simplifications have been given by several mathematicians, including Weierstrass and Felix Klein. Lindemann's demonstration could not have been produced in the preceding century, because it rests necessarily on theories whose development falls in the present century. It is true, Lambert succeeded in 1761 in demonstrating the irrationality of the ratio of the circumference of a circle to its diameter, or, which is the same thing, the irrationality of the ratio of the area of a circle to the area of the square on its radius. Afterwards, Lambert also supplied a proof that it was impossible for this ratio to be the square root of a rational number. But this was the first step only in a long journey. The attempt to prove that the old problem is insoluble was still destined to fail. An astounding mass of mathematical investigations were necessary before the demonstration could be successfully accomplished.

As we see, the majority of the mathematical truths now possessed by us presuppose the intellectual toil of many centuries. A mathematician, therefore, who wishes to-day to acquire a thorough understanding of modern research in this department, must think over again in quickened tempo the mathematical labors of several centuries. This constant dependence of new results on old ones stamps mathematics as a science of uncommon exclusiveness and renders it generally impossible to lay open to uninitiated readers a speedy path to the apprehension of the higher mathematical truths. For this reason, too, the theories and results of mathematics are rarely adapted for popular presentation.

There is no royal road to the knowledge of mathematics, as Euclid once said to the first Egyptian Ptolemy. This same inaccessibility of mathematics, although it secures for it a lofty and aristocratic place among the sciences, also renders it odious to those who have never learned it, and who dread the great labor involved in acquiring an understanding of the questions of modern mathematics. Neither in the languages nor in the natural sciences are the investigations and results so closely interdependent as to make it impossible to acquaint the uninitiated student with single branches or with particular results of these sciences, without causing him to go through a long course of preliminary study.

The third trait which distinguishes mathematical research is its self-sufficiency. In philology the field of inquiry is the organic one of languages, and philology, therefore, is dependent in its investigations on the mode of development of languages, which is more or less accidental. Its task is connected with something which is given to it from without and which it cannot alter. It is much the same with the science of history, which must contemplate the history of mankind as it has actually occurred. Also zoölogy, botany, mineralogy, geology, and chemistry work with given data. In order not to become involved in futile speculations the last-mentioned sciences are constantly and inevitably obliged to revert to observations by the senses. It is then their task to link together these individual observations by bonds of causality and in this way to erect from the single stones an edifice, the view of which will render it easier for limited human intelligence to comprehend nature. Physics of all sciences stands nearest to mathematics in this respect, because unlike the other sciences she is generally in need of only a few observations in order to proceed deductively. But physics, too, must resort to observations of nature, and could not do without them for any length of time.

Mathematics alone, after certain premises have been permanently established, is able to pursue its course of development independently and unmindful of things outside of it. It can leave entirely unnoticed, questions and influences emanating from the outer world, and continue nevertheless its development. As regards geometry, the first beginnings of this science did indeed take their origin in the requirements of practical life. But it was not long before it freed itself from the restrictions of the practical art to which it owed its birth. Herodotus recounts that geometry had its origin in Egypt where the inundations of the Nile

obliterated the boundaries of the riparian estates, and by giving rise to frequent disputes constantly compelled the inhabitants to compare the areas of fields of different shapes. But with the early Greek mathematicians, who were the heirs of the Egyptian art of measurement, geometry appeared as a science which men pursued for its own sake without a thought of how their intellectual discoveries could be turned to practical account.

Nevertheless, although the workers in the domain of pure mathematics are not stimulated by the thought that their researches are likely to be of practical value, yet that result is still frequently realised, often after the lapse of centuries. The history of mathematics shows numerous instances of mathematical results which were originally the outcome of a mere desire to extend the science, suddenly receiving in astronomy, mechanics, or in physics practical applications which their originators could scarce have dreamt of. Thus Apollonius erected in ancient times the stately edifice of the properties of conic sections, without having any idea that the planets moved about the sun in conic sections, and that a Kepler and a Newton were one day to come who should apply these properties to explaining and calculating the motions of the planets about the sun. The question of the practical availability of its results in other fields has at no period exercised more than a subordinate influence on mathematical inquiry. Particularly is this true of *modern* mathematical research, whether the same consist in the extended development of isolated theories or in uniting under a higher point of view theories heretofore regarded as different.*

This independence of its character has rendered the results of pure mathematics independent also of the accidental direction which the development of civilisation has taken on our planet; so that the remark is not altogether without justification, that if beings endowed with intelligence existed on other planets, the truths of mathematics would afford the only basis of an understanding with them. Uninterruptedly and wholly from its own resources mathematics has built itself up. It is scarcely credible to a person not versed in the science, that mathematicians can derive satisfaction from the comfortless and wearisome operation of heaping up demonstration on demonstration, of rivetting

* Cf. Felix Klein, "Remarks Given at the Opening of the Mathematical and Astronomical Congress at Chicago." *The Monist* (Vol. IV, No. 1, October, 1893).

truth on truth, and of tormenting themselves with self-imposed problems, whose solution stands no one in stead, and affords satisfaction to no one but the solver himself. Yet this self-sufficiency of mathematicians becomes a little more intelligible when we reflect that the progress which has been made, particularly in the last few decades, and which is uninfluenced from without, does not consist solely in the accumulation of new truths and in the enunciation of new problems, nor solely in deductions and solutions, but culminates rather in the discovery of new methods and points of view in which the old disconnected and isolated results appear suddenly in a new connexion or as different interpretations of a common fundamental truth, or finally, as a single organic whole.

Thus, for example, the idea of representing imaginary and complex numbers in a plane, two apparently different branches, the theory of dividing the circumference of a circle into any given number of equal parts, and the theory of the solutions of the equation $x^n = 1$, have been made to exhibit an extremely simple connexion with one another which enables us to express many a truth of algebra in a corresponding truth of geometry and *vice versa*. Another example is afforded by the discovery which we chiefly owe to Alfred Clebsch, of the relation which subsists between the higher theory of functions and the theory of algebraic curves, a relation which led to the discovery of the condition under which two curves can be co-ordinated to each other, point for point, and hence also adequately represented on each other. Of course such combinations and extensions of view possess a much greater charm for the mathematician than the mere accumulation of truths and solutions, whose fascination consists entirely in their truth or correctness.

From these three cardinal characteristics, now, which distinguish mathematical *research* from research in other fields, we may gather at once the three positive characteristics that distinguish mathematical *knowledge* from other knowledge. They may be briefly expressed as follows; first, mathematical knowledge bears more distinctly the imprint of truth on all its results than any other kind of knowledge; secondly, it is always a sure preliminary step to the attainment of other correct knowledge; thirdly, it has no need of other knowledge. Naturally, however, there are associated with these characteristics which place mathematical knowledge high above all other knowledge, other characteristics which somewhat counterbalance the great superiority

which mathematics thus appears to have over the other sciences. In order to show more distinctly the nature of these characteristics, which we prefer to call negative, we shall select and confine our remarks to a branch which is commonly taken to be synonymous with mathematics, namely, to arithmetic in the broadest sense of the word.

The subject of inquiry in arithmetic is numbers and their combinations. On this account arithmetic is, of all sciences, most free from what lies outside its boundaries. Perception by the senses is necessary only in an extremely insignificant measure for the understanding of its definitions and premises. It is possible to acquaint a person who lacks both sight and hearing with the fundamental principles of arithmetic solely by the medium of "time." Such a person needs only the sense of feeling. By slight excitations of his skin, induced at equal or unequal intervals of time, he can be led to the notion of differences of time and hence also to the notion of differences of number. Uninfluenced by matter and force, independently, too, of the properties of geometrical magnitudes, arithmetic could be conducted solely by its own intrinsic potencies to its highest goals, drawing deductively truth from truth, without a break.

But what sort of a science should we arrive at by this method of procedure? Nothing but a gigantic web of self-evident truths. For, once we admit the first notions and premises to which a man thus bereft of his senses can be led, we are compelled of necessity also to admit the derivative results of arithmetic. If the beginnings of arithmetic appear self-evident, the rest of it, too, bears this character. Owing to this deductive character of arithmetic, and to its exemption from influence from without, this science appears to one person extremely attractive, while to another it appears extremely repulsive, according as each is constituted. Be that as it may, however, a finished and complete science of this character subserves no purpose in the comprehension of the world, or in the advancement of civilisation. Hence, an arithmetic which heaps up theorem on theorem with never a thought of how its results are to be turned to practical account in the acquisition of knowledge in other fields, resembles an inquisitive physician, who, taking up his abode in a desert, should arrive there at momentous results in bacteriology, but should bear them with him to his grave, without their ever redounding to the benefit of humanity. The value of arithmetical

knowledge lies entirely in its applications. But this constitutes no reason why many mathematicians, pursuing their purely deductive bent of mind, should not devote themselves exclusively to pure arithmetical developments and leave it to others at the proper time to turn to the material profit of the world the capital which they have garnered.

Geometry, on the other hand, must have recourse in a much higher degree than arithmetic to the outside world for its first notions and premises. The axioms of geometry are nothing but facts of experience perceived by our senses. The geometry which Bolyai, Lobachévski, Gauss, Riemann, and Helmholtz created and which is both independent of the eleventh axiom of Euclid and perfectly free from self-contradictions, has supplied an epistemological demonstration that geometry is a science that rests on the observation of nature, and therefore in the correct sense of the word, is a natural science.

Yet what a difference there is, for instance, between geometry and chemistry! Both derive their constructive materials from sense-perception. But whilst geometry is compelled to draw only its first results from observation and is then in a position to move forward deductively to other results without being under the necessity of making fresh observations, chemistry on the other hand is still compelled to make observations and to have recourse to nature.

It follows, therefore, that a given act of geometrical knowledge and a given act of chemical knowledge are with respect to the certainty of the truth they contain not qualitatively but only quantitatively different. In chemistry the probability of error is greater than in geometry, because more numerous and more difficult observations have to be made there than in geometry, where only the very first premises, which no man with sound senses could ever impugn, rest on observation.

The preceding reflexions deprive mathematical knowledge of that degree of certainty and incontestability which is commonly attributed to it when we say a thing is "mathematically certain." This certainty is lessened still more as we pass to the semi-mathematical sciences, where mechanics has the first claim to our attention. All the notions of mechanics, and consequently of all the other departments of physics, are composed, by multiplication or division, of three fundamental notions—length, time, and mass. That is to say, to the notions of geometry resting on length and its powers, two other fundamental notions, time and mass, are added, which, joined to that of length, lead to the notions

of force, work, horse-power, atmospheric pressure, etc. The knowledge of mechanics, thus, highly certain though it be, is rendered less certain than that of geometry and *a fortiori* than that of arithmetic. The uncertainty of knowledge continues to increase in branches which are still more remote from mathematics, owing to the increasing complexity of the observational material which must here be put to the test.

Still, although mathematical knowledge does not lead to absolutely certain results, it yet invests known results with incomparably greater trustworthiness than does the knowledge of the other sciences. But after all, it remains a useless accumulation of capital so long as it is not turned to practical account in other sciences, such as metaphysics, physics, chemistry, biology, political economy, etc. Hence also arises an obligation on the part of the other sciences, so to shape their problems and investigations that they can be made susceptible of mathematical treatment. Then will mathematics gladly perform her duty. The moment a science has advanced far enough to permit of the mathematical formulation of its problems, mathematics will not be slow to treat and to solve these problems. Mathematical knowledge, aristocratic as it may appear by the greater certainty of its results, will, so far as the advancement of human kind is concerned, never be more than a useless mass of self-evident truths, unless it constantly places itself in the service of the other sciences.

THE MAGIC SQUARE.

I.

INTRODUCTORY.

AMONG the philosophies of modern times there is none which has emphasised so much the importance of form and formal thought as the monism of *The Monist*. An expression of this philosophy is found in the following passages:

> "The order that prevails among the facts of reality is due to the laws of form. Upon the order of the world depends its cognisability.
>
> "... The laws of form are no less eternal than are matter and energy and 'Verily I say unto you, till heaven and earth pass, one jot or one tittle shall in no wise pass from the law!'
>
> "The laws of form and their origin have been a puzzle to all philosophers. 'Ay, there's the rub!' The difficulties of Hume's problem of causation, of Kant's *a priori*, of Plato's ideas, of Mill's method of deduction, etc., etc., all arise from a one-sided view of form and the laws of form and formal thought."

Considering the great results which engineering and other applied sciences accomplish through the assistance of mathematics, we must confess that the forms of thought are wonderful indeed, and it is not at all astonishing that the primitive thinkers of mankind when the importance of the laws of formal thought in some way or another first dawned on their minds, attributed magic powers to numbers and geometrical figures.

We shall devote the following pages to a brief review of magic squares, the consideration of which has made many a man believe in mysticism. And yet there is no mysticism about them unless we either consider everything mystical, even that twice two is four, or join the sceptic in his exclamation that we can truly not know whether twice two might not be five in other spheres of the universe.

The author of the short article on "Magic Squares" in the English Cyclopædia (Vol. III, p. 415), presumably Prof. De Morgan, says:

> "Though the question of magic squares be in itself of no use, yet it belongs to a class of problems which call into action a beneficial species of investigation. Without laying down any rules for their construction, we shall content ourselves with destroying their magic quality, and showing that the non-existence of such squares would be much more surprising than their existence."

This is the point. There obtains a symphonic harmony in mathematics which is the more startling the more obvious and self-evident it appears to him who understands the laws that produce this symphonic harmony.

* * *

On the wood-cut named "Melancholia"* of the famous Nuremberg painter, Albrecht Dürer, is found among a number of other emblems,

* The term melancholy meant in Dürer's time, as it did also in Shakespeare's and Milton's, "thought or thoughtfulness." Says Milton in *Il Penseroso*:

> "Hail, thou Goddess, sage and holy,
> Hail divinest melancholy
> Whose saintly visage is too bright
> To hit the sense of human sight,
> And therefore to our weaker view
> O'erlaid with black, staid Wisdom's hue.—I, 12.

Thought that does not lead to action produces a gloomy state of mind. Thoughtfulness which cannot find a way out of itself is that melancholy which engenders weakness,—a truth which is illustrated in Hamlet. Shakespeare still uses the words thought and melancholy as synonyms, saying:

> "The native hue of resolution
> Is sicklied o'er with the pale cast of thought."

Dürer's melancholy does not represent the gloominess of thought, but the power of invention. Soberness and even a certain sadness are considered only as an element of this melancholy, but on the whole the genius of thought appears



ALBERT DÜRER'S ENGRAVING

MELANCHOLY

OR THE

GENIUS OF THE INDUSTRIAL SCIENCE OF MECHANICS.

which the reader will notice in our reproduction of the cut, the subjoined square. This arrangement of the sixteen natural numbers from

1	14	15	4
12	7	6	9
8	11	10	5
13	2	3	16

Fig. 1.

1 to 16 possesses the remarkable property that the same sum 34 will always be obtained whether we add together the four figures of any of the horizontal rows or the four of any vertical row or the four which lie in either of the two diagonals. Such an arrangement of numbers is termed a magic square, and the square which we have reproduced above is the *first magic square which is met with in the Christian Occident.*

Like chess and many of the problems founded on the figure of the chess-board, the problem of constructing a magic square also probably traces its origin to Indian soil. From there the problem found its way among the Arabs, and by them it was brought to the Roman Orient. Finally, since Albrecht Dürer's time, the scholars of Western Europe also have occupied themselves with methods for the construction of squares of this character.

The oldest and the simplest magic square consists of the quadratic arrangement of the nine numbers from 1 to 9 in such a manner that the sum of each horizontal, vertical, or diagonal row, always remains

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bright, self-possessed, and strong.

Dürer represents the Science of Mechanical Invention as a winged female figure musing over some problem. Scattered on the floor around her lie some of the simple tools used in the sixteenth century. A ladder leans against the house, that assists in climbing otherwise inaccessible heights. A scale, an hour-glass, a bell, and the magic square are hanging on the wall behind her.

At a distance a bat-like creature, being the gloom of melancholy, hovers in the air like a dark cloud, but the sun rises above the horizon, and at the happy middle between these two extremes stands the rainbow of serene hope and cheerful confidence.

the same, namely 15. This square is the adjoined.

2	7	6
9	5	1
4	3	8

Fig. 2.

Here, we will find, 15 always comes out whether we add 2 and 7 and 6, or 9 and 5 and 1, or 4 and 3 and 8, or 2 and 9 and 4, or 7 and 5 and 3, or 6 and 1 and 8, or 2 and 5 and 8, or 6 and 5 and 4.

The question naturally presents itself, whether this condition of the constant equality of the added sum also remains fulfilled when the numbers are assigned different places. It may be easily shown however that 5 necessarily must occupy the middle place, and that the even numbers must stand in the corners. This being so, there are but 7 additional arrangements possible, which differ from the arrangement above given and from one another only in the respect that the rows at the top, at the left, at the bottom, and at the right, exchange places with one another and that in addition a mirror be imagined present with each arrangement. So too from Dürer's square of 4 times 4 places, by transpositions, a whole set of new correct squares may be formed. A magic square of the 4 times 4 numbers from 1 to 16 is formed in the simplest manner as follows. We inscribe the numbers from 1 to 16 in their natural order in the squares, thus:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Fig. 3.

We then leave the numbers in the four corner-squares, viz. 1, 4, 13, 16, as well also as the numbers in the four middle-squares, viz. 6, 7, 10, 11, in their original places; and in the place of the remaining eight numbers, we write the complements of the same with respect to 17: thus 15 instead of 2, 14 instead of 3, 12 instead of 5, 9 instead of 8, 8 instead of 9, 5 instead of 12, 3 instead of 14, and 2 instead of 15. We

obtain thus the magic square

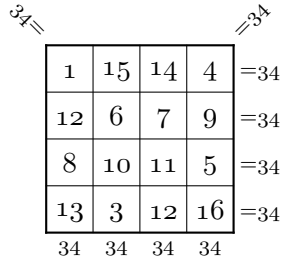


Fig. 4.

from which the same sum 34 always results. It is an interesting property of this square that any four numbers which form a rectangle or square about the centre also always give the same sum 34; for example, 1, 4, 13, 16, or 6, 7, 10, 11, or 15, 14, 3, 2, or 12, 9, 5, 8, or 15, 8, 2, 9, or 14, 12, 3, 5. We may easily convince ourselves that this square is obtainable from the square of Dürer by interchanging with one another the two middle vertical rows.

II.

EARLY METHODS FOR THE CONSTRUCTION OF ODD-NUMBERED SQUARES.

Since early times rules have also been known for the construction of magic squares of more than 3 times 3, or 4 times 4 spaces. In the first place, it is easy to calculate the sum which in the case of any given number of cells must result from the addition of each row. We take the determinate number of cells in each side of the square which we have to fill, multiply that number by itself, add 1, again multiply the number thus obtained by the number of the cells in each side, and, finally, divide the product by 2. Thus, with 4 times 4 cells or squares, we get: 4 times 4 are 16, 16 and 1 are 17, and one half of 17 times 4 is 34. Similarly, with 5 times 5 squares, we get: 5 times 5 are 25, and 1 makes 26, and the half of 26 times 5 is 65. Analogously, for 6 times 6 squares the summation 111 is obtained, for 7 times 7 squares 175, for 8 times 8 squares 260, for 9 times 9 squares 369, for 10 times 10 squares 505, and so on. The Hindu rule for the construction of magic squares whose roots are odd, may be enunciated as follows: To start with, write 1 in the centre of the topmost row, then write 2 in the lowest space of the vertical column next adjacent to the right, and then so inscribe

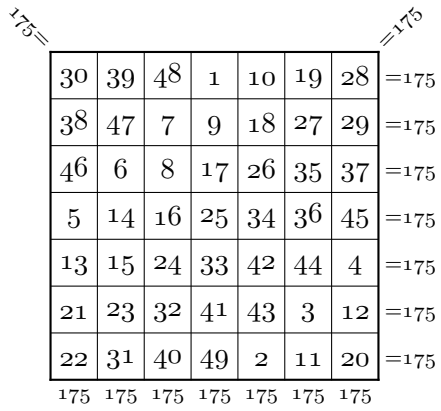


Fig. 5.

the remaining numbers in their natural order in the squares diagonally upwards towards the right, that on reaching the right-hand margin the inscription shall be continued from the left-hand margin in the row just above, and on reaching the upper margin shall be continued from the lower margin in the column next adjacent to the right, noting that whenever we are arrested in our progress by a square already occupied we are to fill out the square next beneath the one we have last filled. In this manner, for example, the last preceding square of 7 times 7 cells is formed, in which the reader is requested to follow the numbers in their natural sequence (Fig. 5).

For the next further advancements of the theory of magic squares and of the methods for their construction we are indebted to the Byzantian Greek, Moschopulus, who lived in the fourteenth century; also, after Albrecht Dürer who lived about the year 1500, to the celebrated arithmetician Adam Riese, and to the mathematician Michael Stifel, which two last lived about 1550. In the seventeenth century Bachet de Méziriac, and Athanasius Kircher employed themselves on magic squares. About 1700, finally, the French mathematicians De la Hire and Sauveur made considerable contributions to the theory. In recent times mathematicians have concerned themselves much less about magic squares, as they have indeed about mathematical recreations generally. But quite recently the Brunswick mathematician Scheffler has put forth his own and other's studies on this subject in an elegant form.

The best known of the various methods of constructing magic squares of an odd number of cells is the following. First write the numbers in diagonal succession as in the following diagram (Fig. 6).

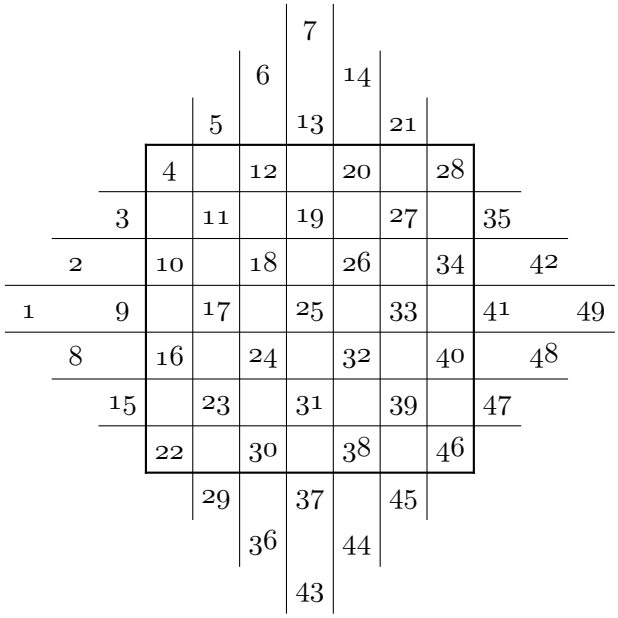


Fig. 6.

After 25 cells of the square of 49 cells which we have to fill out, have thus been occupied, transfer the six figures found outside each side of the square, without changing their configuration, into the empty cells of the side directly opposite. By this method, which we owe to Bachet de Méziriac, we obtain the following magic square of the numbers from 1 to 49:

4	29	12	37	20	45	28
35	11	36	19	44	27	3
10	42	18	43	26	2	34
41	17	49	25	1	33	9
16	48	24	7	32	8	40
47	23	6	31	14	39	15
22	5	30	13	38	21	46

Fig. 7.

III.

MODERN MODES OF CONSTRUCTION OF ODD-NUMBERED SQUARES.

The reader will justly ask whether there do not exist other correct magic squares which are constructed after a different method from that just given, and whether there do not exist modes of construction which will lead to all the imaginable and possible magic squares of a definite number of cells. A general mode of construction of this character was first given for odd-numbered squares by De la Hire, and recently perfected by Professor Scheffler.

To acquaint ourselves with this general method, let us select as our example a square of 5. First we form two auxiliary squares. In the first we write the numbers from 1 to 5 five times; and in the second, five times, the following multiples of five, viz.: 0, 5, 10, 15, 20. It is clear now that by adding each of the numbers of the series from 1 to 5 to each of the numbers 0, 5, 10, 15, 20, we shall get all the 25 numerals from 1 to 25. All that additionally remains to be done therefore, is, so to inscribe the numbers that by the addition of the two numbers in any two corresponding cells each combination shall come out once and only once; and further that in each horizontal, vertical, and diagonal row in each auxiliary square each number shall once appear. Then the required sum of 65 must necessarily result in every case, because the numbers from 1 to 5 added together make 15, and the numbers 0, 5, 10, 15, 20 make 50.

We effect the required method of inscription by imagining the numbers 1, 2, 3, 4, 5 (or 0, 5, 10, 15, 20) arranged in cyclical succession, that is 1 immediately following upon 5, and, starting from any number whatsoever, by skipping each time either none or one or two or three etc. figures. Cycles are thus obtained of the first, the second, the third etc. orders; for example 3 4 5 1 2 is a cycle of the first order, 2 4 1 3 5 is a cycle of the second order, 1 5 4 3 2 is a cycle of the fourth order, etc. The only thing then to be looked out for in the two auxiliary squares is, that the same "cycle" order be horizontally preserved in all the rows, that the same also happens for the vertical rows, but that the cycle order in the horizontal and vertical rows is different. Finally we have only additionally to take care that to the same numbers of the one auxiliary square not like numbers but *different* numbers correspond in

the other auxiliary square, that is lie in similarly situated cells. The following auxiliary squares are, for example, thus possible:

3	4	5	1	2
5	1	2	3	4
2	3	4	5	1
4	5	1	2	3
1	2	3	4	5

Fig. 8.

and

0	10	20	5	15
5	15	0	10	20
10	20	5	15	0
15	0	10	20	5
20	5	15	0	10

Fig. 9.

Adding in pairs the numbers which occupy similarly situated cells, we obtain the following correct magic square:

3	14	25	6	17
10	16	2	13	24
12	23	9	20	1
19	5	11	22	8
21	7	18	4	15

Fig. 10.

It will be seen that we are able thus to construct a very large number of magic squares of 5 times 5 spaces by varying in every possible manner the numbers in the two auxiliary squares. Furthermore, the squares thus formed possess the additional peculiarity, that every 5 numbers which fill out two rows that are parallel to a diagonal and lie on different sides of the diagonal also give the constant sum of 65. For example: 3 and 7, 11, 20, 24; or 10, 14 and 18, 22, 1. Altogether then the sum 65 is produced out of 20 rows or pairs of rows. On this peculiarity is dependent the fact that if we imagine an unlimited number of such squares placed by the side of, above, or beneath an initial one, we shall be able to obtain as many quadratic cells as we choose, so arranged that the square composed of any 25 of these cells

will form a correct magic square, as the following figure will show:

2	13	24	10	16	2	13	24	10	16	2
9	20	1	12	23	9	20	1	12	23	9
11	22	8	19	5	11	22	8	19	5	11
18	4	15	21	7	18	4	15	21	7	18
25	6	17	3	14	25	6	17	3	14	25
2	13	24	10	16	2	13	24	10	16	2
9	20	1	12	23	9	20	1	12	23	9
11	22	8	19	5	11	22	8	19	5	11
18	4	15	21	7	18	4	15	21	7	18
25	6	17	3	14	25	6	17	3	14	25
2	13	24	10	16	2	13	24	10	16	2
9	20	1	12	23	9	20	1	12	23	9
11	22	8	19	5	11	22	8	19	5	11

Fig. 11.

Every square of every 25 of these numbers, as for example the two dark-bordered ones, possesses the property that the addition of the horizontal, vertical, and diagonal rows gives each the same sum, 65.

As an example of a higher number of cells we will append here a magic square of 11 times 11 spaces formed by the general method of De la Hire from the two auxiliary squares of Figs. 12 and 13. From these two auxiliary squares we obtain by the addition of the two numbers of every two similarly situated cells, the magic square, exhibited in Diagram 14, in which each row gives the same sum 671.

IV.

EVEN-NUMBERED SQUARES.

Of magic squares having an even number of places we have hitherto had to deal only with the square of 4. To construct squares of this description having a higher even number of places, different and more complicated methods must be employed than for squares of odd numbers of places. However, in this case also, as in dealing with the square

1	2	3	4	5	6	7	8	9	10	11
3	4	5	6	7	8	9	10	11	1	2
5	6	7	8	9	10	11	1	2	3	4
7	8	9	10	11	1	2	3	4	5	6
9	10	11	1	2	3	4	5	6	7	8
11	1	2	3	4	5	6	7	8	9	10
2	3	4	5	6	7	8	9	10	11	1
4	5	6	7	8	9	10	11	1	2	3
6	7	8	9	10	11	1	2	3	4	5
8	9	10	11	1	2	3	4	5	6	7
10	11	1	2	3	4	5	6	7	8	9

Fig. 12.

0	11	22	33	44	55	66	77	88	99	110
33	44	55	66	77	88	99	110	0	11	22
66	77	88	99	110	0	11	22	33	44	55
99	110	0	11	22	33	44	55	66	77	88
11	22	33	44	55	66	77	88	99	110	0
44	55	66	77	88	99	110	0	11	22	33
77	88	99	110	0	11	22	33	44	55	66
110	0	11	22	33	44	55	66	77	88	99
22	33	44	55	66	77	88	99	110	0	11
55	66	77	88	99	110	0	11	22	33	44
88	99	110	0	11	22	33	44	55	66	77

Fig. 13.

1	13	25	37	49	61	73	85	97	109	121
36	48	60	72	84	96	108	120	11	12	24
71	83	95	107	119	10	22	23	35	47	59
106	118	9	21	33	34	46	58	70	82	94
20	32	44	45	57	69	81	93	105	117	8
55	56	68	80	92	104	116	7	19	31	43
79	91	103	115	6	18	30	42	54	66	67
114	5	17	29	41	53	65	77	78	90	102
28	40	52	64	76	88	89	101	113	4	16
63	75	87	99	100	112	3	15	27	39	51
98	110	111	2	14	26	38	50	62	74	86

Fig. 14.

of 4, we start with the natural sequence of the numbers and must then find the complements of the numbers with respect to some other certain number (as 17 in the square of 4) and also effect certain exchanges of the numbers with one another. To form, for example, a magic square of 6 times 6 places, we inscribe in the 12 diagonal cells the numbers that in the natural sequence of inscription fall into these places, then in the remaining cells the complements of the numbers that belong therein

with respect to 37, and finally effect the following six exchanges, viz. of the numbers 33 and 3, 25 and 7, 20 and 14, 18 and 13, 10 and 9, and 5 and 2. In this way the following magic square is obtained.

1	35	34	3	32	6
30	8	28	27	11	7
24	23	15	16	14	19
13	17	21	22	20	18
12	26	9	10	29	25
31	2	4	33	5	36

Fig. 15.

This square may also be constructed by the method of De la Hire, from two auxiliary squares with the numbers 1, 2, 3, 4, 5, 6 and 0, 6, 12, 18, 24, 30 respectively. In this case, however, the vertical rows of the one square and the horizontal rows of the other must each so contain two numbers three times repeated that the summation shall always remain 21 and 90 respectively. In this manner we get the magic square last given above from the two following auxiliary squares:

1	5	4	3	2	6
6	2	4	3	5	1
6	5	3	4	2	1
1	5	3	4	2	6
6	2	3	4	5	1
1	2	4	3	5	6

Fig. 16.

and

0	30	30	0	30	0
24	6	24	24	6	6
18	18	12	12	12	18
12	12	18	18	18	12
6	24	6	6	24	24
30	0	0	30	0	30

Fig. 17.

It is to be noted in connection with this example that here also as in the case of odd-numbered squares, it is possible so to inscribe six times the numbers from 1 to 6 that each number shall appear once and only once in each horizontal, vertical, and diagonal row; for example,

in the following manner:

1	2	3	4	5	6
2	4	6	1	3	5
3	6	5	2	1	4
5	3	1	6	4	2
6	5	4	3	2	1
4	1	2	5	6	3

Fig. 18.

But if we attempt so to insert, in a like manner, the other set of numbers 0, 6, 12, 18, 24, 30 in a second auxiliary square, that each number of the first auxiliary square shall stand once and once only in a corresponding cell with each number of the second square, all the attempts we may make to fulfil coincidently the last named condition will result in failure. It is therefore necessary to select auxiliary squares like the two given above. It is noteworthy, that the fulfilment of the second condition is impossible only in the case of the square of 6, but that in the case of the square of 4 or of the square of 8, for example, two auxiliary squares, such as the method of De la Hire requires, are possible. Thus, taking the square of 4 we get

1	2	3	4
4	3	2	1
2	1	4	3
3	4	1	2

Fig. 19.

and

0	4	8	12
8	12	0	4
12	8	4	0
4	0	12	8

Fig. 20.

The reader may form for himself the magic square which these give.

The existence of these two auxiliary squares furnishes a key to the solution of a pretty problem at cards. If we replace, namely, the numbers 1, 2, 3, 4 by the Ace, the King, the Queen, and the Knave, and the numbers 0, 4, 8, 12 by the four suits, clubs, spades, hearts, and diamonds, we shall at once perceive that it is possible, and must be so necessarily, quadratically to arrange in such a manner the four Aces, the four Kings, the Four Queens, and the four Knaves, that in each

horizontal, vertical, and diagonal row, each one of the four suits and each one of the four denominations shall appear once and once only. The auxiliary squares above given furnish the appended solution of this problem:

CLUBS ACE	SPADES KING	HEARTS QUEEN	DIAMONDS KNAVE
HEARTS KNAVE	DIAMONDS QUEEN	CLUBS KING	SPADES ACE
DIAMONDS KING	HEARTS ACE	SPADES KNAVE	CLUBS QUEEN
SPADES QUEEN	CLUBS KNAVE	DIAMONDS ACE	HEARTS KING

Fig. 21.

To fix the solution of the problem in the memory, observe that, starting from the several corners, each suit and each denomination must be placed in the spots of the move of a Knight. If we fix the positions of the four cards of any one row, there will be only two possibilities left of so placing the other cards that the required condition of having each suit and each denomination once and only once in each row shall be fulfilled.

Of magic squares of an even number of places we have up to this point examined only the squares of 4 and of 6. For the sake of completeness we append here one of 8 and one of 10 places. The mode of construction of these squares is similar to the method above discussed for the lower even numbers.

The magic squares of even numbers thus constructed are not the only possible ones. On the contrary, there are very many others possible, which obey different laws of formation. It has been calculated, for example, that with the square of 4 it is possible to construct 880, and with the square of 6, *several million*, different magic squares. The number of odd-numbered magic squares constructible by the method

1	63	62	4	5	59	58	8
56	10	11	53	52	14	15	49
48	18	19	45	44	22	23	41
25	39	38	28	29	35	34	32
33	31	30	36	37	27	26	40
24	42	43	21	20	46	47	17
16	50	51	13	12	54	55	9
57	7	6	60	61	3	2	64

Fig. 22.

1	99	3	97	96	5	94	8	92	10
90	12	88	14	86	85	17	83	19	11
80	79	23	77	25	26	74	28	22	71
31	69	68	34	66	65	37	33	62	40
60	42	58	57	45	46	44	53	49	51
50	52	43	47	55	56	54	48	59	41
61	32	38	64	36	35	67	63	39	70
21	29	73	27	75	76	24	78	72	30
20	82	18	84	15	16	87	13	89	81
91	9	93	4	6	95	7	98	2	100

Fig. 23.

of De la Hire is also very great. With the square of 7, the possible constructions amount to 363,916,800. With the squares of higher numbers the multitude of the possibilities increases in the same enormous ratio.

V.

MAGIC SQUARES WHOSE SUMMATION GIVES THE NUMBER OF A YEAR.

The magic squares which we have so far considered contain only the natural numbers from 1 upwards. It is possible, however, easily to deduce from a correct magic square other squares in which a different law controls the sequence of the numbers to be inscribed. Of the squares obtained in this manner, we shall devote our attention here only to such in which, although formed by the inscription of successive numbers, the sum obtained from the addition of the rows is a determinate number which we have fixed upon beforehand, as *the number of a year*. In such a case we have simply to add to the numbers of the original square a determinate number so to be calculated, that the required sum shall each time appear. If this sum is divisible by 3, magic squares will always be obtainable with 3 times 3 spaces which shall give this sum. In such a case we divide the sum required by 3 and subtract 5 from the result in order to obtain the number which we have to add to each number of the original square. If the sum desired is even but not divisible by 4, we must then subtract from it 34 and take one fourth of the result, to obtain the number which in this case is to be added in each place. If, for example, we wish to obtain the number of the year 1890 as the resulting sum of each row, we shall have to add to each of the numbers of an ordinary magic square of 4 times 4 spaces the number 464; in other words, instead of the numbers from 1 to 16 we have to insert in the squares the numbers from 465 to 480. As the number of the year 1892 is divisible by eleven, it must be possible to deduce from the magic square constructed by us at the conclusion of Section III a second magic square in which each row of 11 cells will give the number of the year 1892. To do this, we subtract from 1892 the sum of the original square, namely 671, and divide the remainder by 11, whereby we get 111 and thus perceive that the numbers from 112 to 232 are to be inscribed in the cells of the square required. We get in this way the following square, from which *one and the same sum, namely* 1892, *can*

112	124	136	148	160	172	184	196	208	220	232	=1892
147	159	171	183	195	207	219	231	122	123	135	=1892
182	194	206	218	230	121	133	134	146	158	170	=1892
217	229	120	132	144	145	157	169	181	193	205	=1892
131	143	155	156	168	180	192	204	216	228	119	=1892
166	167	179	191	203	215	227	118	130	142	154	=1892
190	202	214	226	117	129	141	153	165	177	178	=1892
225	116	128	140	152	164	176	188	189	201	213	=1892
139	151	163	175	187	199	200	212	224	115	127	=1892
174	186	198	210	211	223	114	126	138	150	162	=1892
209	221	222	113	125	137	149	161	173	185	197	=1892
1892	1892	1892	1892	1892	1892	1892	1892	1892	1892	1892	

Fig. 24.

be obtained 44 *times*, first from each of the 11 horizontal rows, secondly from each of the 11 vertical rows, thirdly from each of the two diagonal rows, and fourthly twenty additional times from each and every pair of any two rows that lie parallel to a diagonal, have together 11 cells, and lie on different sides of the diagonal, as for example, 196, 122, 158, 205, 131, 167, 214, 140, 187, 223, 149.

VI.

CONCENTRIC MAGIC SQUARES.

The acuteness of mathematicians has also discovered magic squares which possess the peculiar property that if one row after another be taken away from each side, the smaller inner squares remaining will still be magical squares, that is to say, all their rows when added will give the same sum. It will be sufficient to give two examples here of such squares, (the laws for their construction being somewhat more complicated,) of which the first has 7 times 7 and the second 8 times 8 places. The numbers within each of the dark-bordered frames form with respect to the centre smaller squares which in their own turn are magical. In the first of these two squares the internal square of 3 times

4	5	6	43	39	38	40
49	15	16	33	30	31	1
48	37	22	27	26	13	2
47	36	29	25	21	14	3
8	18	24	23	28	32	42
9	19	34	17	20	35	41
10	45	44	7	11	12	46

Fig. 25.

1	56	55	11	53	13	14	57
63	15	47	22	42	24	45	2
62	49	25	40	34	31	16	3
4	48	28	37	35	30	17	61
5	44	39	26	32	33	21	60
59	19	38	27	29	36	46	6
58	20	18	43	23	41	50	7
8	9	10	54	12	52	51	64

Fig. 26.

3 places contains the numbers from 21 to 29 in such a manner that each row gives when added the sum of 75. This square lies within a larger one of 5 times 5 spaces, which contains the numbers from 13 to 37 in such a manner that each row gives the sum of 125. Finally, this last square forms part of a square of 7 times 7 places which contains the numbers from 1 to 49 so that each row gives the sum of 175.

In the second square the inner central square of 4 times 4 places contains the numbers from 25 to 40 in such a manner that each row gives the sum of 130. This square is the middle of a square of 6 times 6 places which so contains the numbers from 15 to 50 that each row gives the sum 195. Finally, this last square is again the middle of an ordinary magic square composed of the numbers from 1 to 64.

VII.

MAGICAL SQUARES WITH MAGICAL PARTS.

If we divide a square of 8 times 8 places by means of the two middle lines parallel to its sides into 4 parts containing each 4 times 4 spaces, we may propound the problem of so inserting the numbers from 1 to 64 in these spaces that not only the whole shall form a magic square, but also that each of the 4 parts individually shall be magical, that is to say, give the same sum for each row. This problem also has been

successfully solved, as the following diagram will show.

1	4	63	62	5	8	59	58
64	61	2	3	60	57	6	7
42	43	24	21	34	35	32	29
23	22	41	44	31	30	33	36
13	16	55	50	9	12	55	54
52	49	14	15	56	53	10	11
38	39	28	25	46	47	20	17
27	26	37	40	19	18	45	48

Fig. 27.

The 4 numbers in each row of any one of the sub-squares here, gives 130; so that the sum of each one of the rows of the large square will be 260.

Finally, in further illustration of this idea, we will submit to the consideration of our readers a very remarkable square of the numbers from 1 to 81. This square, which will be found below (Fig. 28), is divided by parallel lines into 9 parts, of which each contains 9 consecutive numbers that severally make up a magic square by themselves.

31	36	29	76	81	74	13	18	11
30	32	34	75	77	79	12	14	16
35	28	33	80	73	78	17	10	15
22	27	20	40	45	38	58	63	56
21	23	25	39	41	43	57	59	61
26	19	24	44	37	42	62	55	60
67	72	65	4	9	2	49	54	47
66	68	70	3	5	7	48	50	52
71	64	69	8	1	6	53	46	51

Fig. 28.

Wonderful as the properties of this square may appear, the law by which the author constructed it is equally simple. We have simply to regard the 9 parts as the 9 cells of a magic square of the numbers from I to IX, and then to inscribe by the magic prescript in the square

designated as I the numbers from 1 to 9, in the square designated as II the numbers from 10 to 18, and so on. In this way the square above given is obtained from the following base-square:

IV	IX	II
III	V	VII
VIII	I	VI

Fig. 29.

VIII.

MAGIC SQUARES THAT INVOLVE THE MOVE OF THE CHESS-KNIGHT.

What one of our readers does not know the problems contained in the recreation columns of our magazines, the requirements of which are to compose into a verse 8 times 8 quadratically arranged syllables, of which every two successive syllables stand on spots so situated with respect to each other that a chess-knight can move from the one to the other? If we replace in such an arrangement the 64 successive syllables by the 64 numbers from 1 to 64, we shall obtain a knight-problem made up of numbers. Methods also exist indeed for the construction of such dispositions of numbers, which then form the foundation of the construction of the problems in the newspapers. But the majority of knight-problems of this class are the outcome of experiment rather than the product of methodical creation. If however it is a severe test of patience to form a knight-problem by experiment, it stands to reason that it is a still severer trial to effect at the same time the additional result that the 64 numbers which form the knight-problem shall also form a magic square.

This trial of endurance was undertaken several decades ago, by a pensioned Moravian officer named Wenzelides, who was spending the last days of his life in the country. After a series of trials which lasted years he finally succeeded in so inscribing in the 64 squares of the chess-board the numbers from 1 to 64 that successive numbers, as well also as the numbers 64 and 1, were always removed from one another in distance and direction by the move of a knight, and that in addition thereto the summation of the horizontal and the vertical rows always

gave the same sum 260. Ultimately he discovered several squares of this description, which were published in the *Berlin Chess Journal*. One of these is here appended:

47	10	23	64	49	2	59	6
22	63	48	9	60	5	50	3
11	46	61	24	1	52	7	58
62	21	12	45	8	57	4	51
19	36	25	40	13	44	53	30
26	39	20	33	56	29	14	43
35	18	37	28	41	16	31	54
38	27	34	17	32	55	42	15

Fig. 30.

The move of the knight and the equality of the summation of the horizontal and vertical rows, therefore, are the facts to be noted here. The diagonal rows do *not* give the sum 260. Perhaps some one among our readers who possesses the time and patience will be tempted to outdo Wenzelides, and to devise a numeral knight-problem of this kind which will give 260 not only in the horizontal and vertical but also in the two diagonal rows.

IX.

MAGICAL POLYGONS.

So far we have considered only such extensions of the idea underlying the construction of the magic square in which the figure of the square was retained. We may however contrive extensions of the idea in which instead of a square, a rectangle, a triangle, or a pentagon, and the like, appear. Without entering into the consideration of the methods for the construction of such figures, we will give here of magical polygons simply a few examples, all supplied by Professor Scheffler:

1) The numbers from 1 to 32 admit of being written in a rectangle of 4×8 in such a manner that the long horizontal rows give the sum

of 132 and the short vertical rows the sum of 66; thus:

1	10	11	29	28	19	18	16
9	2	30	12	20	27	7	25
24	31	3	21	13	6	26	8
32	23	22	4	5	14	15	17

Fig. 31.

2) The numbers from 1 to 27 admit of being so arranged in three regular triangles about a point which forms a common centre, that each side of the outermost triangle will present 6 numbers of the total summation 96 and each side of the middle triangle 4 numbers whose sum is 61; as the following figure shows:

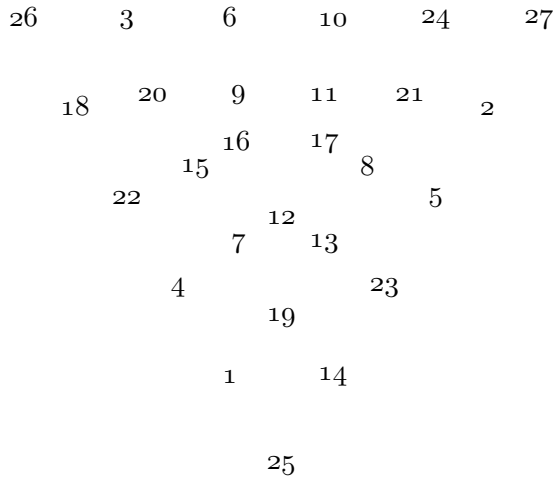


Fig. 32.

3) The numbers from 1 to 80 admit of being formed about a point as common centre into 4 pentagons, such that each side of the first pentagon from within contains two numbers, each side of the second pentagon four numbers, each of the third six numbers, and each side of the fourth, outermost pentagon eight numbers. The sum of the numbers of each side of the second pentagon is 122, the sum of those of each side of the third pentagon is 248, and that of those of each side of the fourth pentagon 254. Furthermore, the sum of any four corner numbers lying in the same straight line with the centre, is also the same; namely, 92.

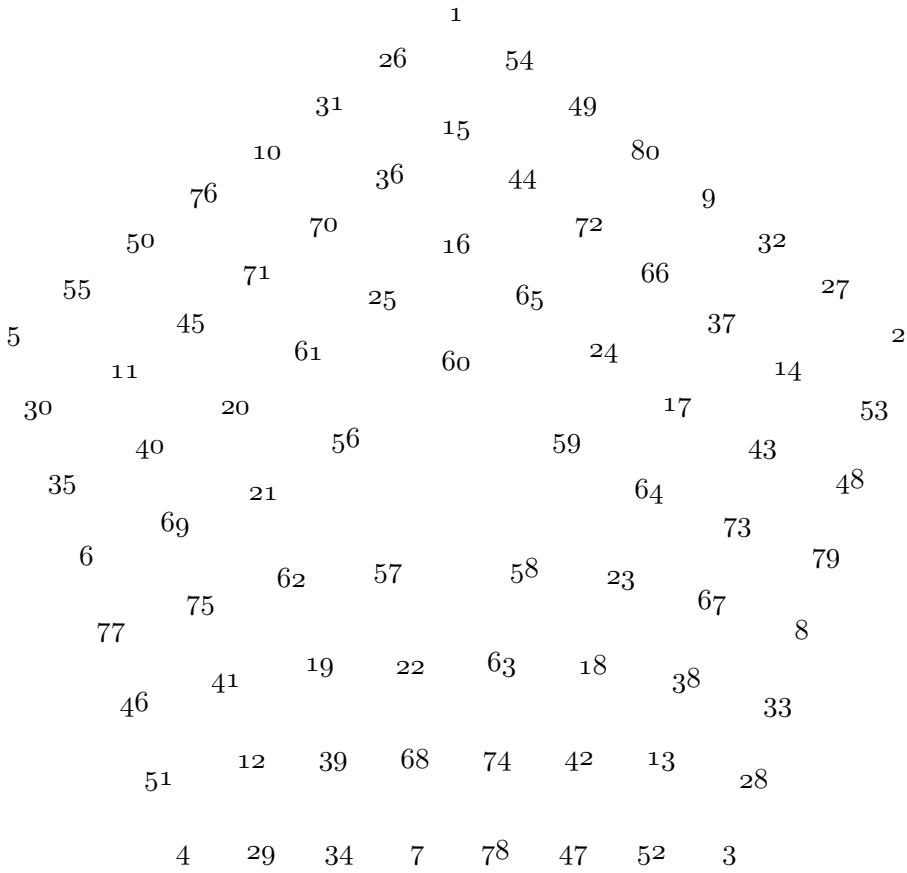


Fig. 33.

4) The numbers from 1 to 73 admit of being arranged about a centre, in which the number 37 is written, into three hexagons which contain respectively 3, 5, and 7 numbers in each side and possess the following pretty properties. Each hexagon always gives the same sum, not only when the summation is made along its six sides, but also when it is made along the six diameters that join its corners and along the six that are constructed at right angles to its sides; this sum, for the first hexagon from within, is 111, for the second 185, and for the third 259.

		1	5	6	70	60	59	58	
		63						8	
		62		19	53	46	22	45	9
	61		20				24		64
	2		48		31	42	38	49	57
	3		47		39		40	44	56
67		51		41		37		33	23	7
	66		50		34		35	54	11	
	65		25		36	32	43	26	12	
	10		30				27		13	
	17		29	21	28	52	55		72	
	18							71		
		16	69	68	4	14	15	73	

Fig. 34.

X.

MAGIC CUBES.

Several inquirers, particularly Kochansky (1686), Sauveur (1710), Hugel (1859), and Scheffler (1882), have extended the principle of the magic squares of the plane to three-dimensioned space. Imagine a cube divided by planes parallel to its sides and equidistant from one another, into cubical compartments. The problem is then, so to insert in these compartments the successive natural numbers that every row from the right to the left, every row from the front to the back, every row from the top to the bottom, every diagonal of a square, and every principal diagonal passing through the centre of the cube shall contain numbers whose sum is always the same. For 3 times 3 times 3 compartments, a magic cube of this description is not constructible. For 4 times 4 times 4 compartments a cube is constructible such that any row parallel to an edge of the cube and every principal diagonal give the sum of 130. To obtain a magic cube of 64 compartments, imagine the numbers which belong in the compartments written on the upper surface of the same and the numbers then taken off in layers of 16 from the top downwards. We obtain thus 4 squares of 16 cells each, which together

make up the magic cube; as the following diagrams will show:

First Layer from the Top.

1	48	32	49
60	21	37	12
56	25	41	8
13	36	20	61

Second Layer from the Top.

63	18	34	15
6	43	27	54
10	39	23	58
51	30	46	3

Third Layer from the Top.

62	19	35	14
7	42	26	55
1	38	22	59
50	31	47	2

Fourth Layer from the Top.

4	45	29	52
57	24	40	9
53	28	44	5
16	33	17	64

The same sum 130 here comes out not less than 52 times; viz. in the first place from the 16 rows from left to right, secondly from the 16 rows from the front to the back, thirdly from the 16 rows counting from the top to the bottom, and lastly from the 4 rows which join each two opposite corners of the cube, namely from the rows: 1, 43, 22, 64; 49, 27, 38, 16; 13, 39, 26, 52; 61, 23, 42, 4.

For a cube with 5 compartments in each edge the arrangement of the figures can so be made that all the 75 rows parallel to any and every edge, all the 30 rows lying in any diagonal of a square, and all the 4 rows forming any principal diagonal shall have one and the same summation, 315.

Just as the magic squares of an odd number of cells could be formed with the aid of *two* auxiliary squares, so also odd-numbered magic cubes can be constructed with the help of *three* auxiliary cubes.

First Layer from Top.

121	27	83	14	70
11	6	117	48	79
44	100	1	57	113
53	109	40	91	22
87	18	74	105	31

Second Layer from Top.

2	58	114	45	96
36	92	23	54	110
75	101	32	88	19
84	15	66	122	28
118	49	80	6	62

Third Layer from Top.

33	89	20	71	102
67	123	29	85	11
76	7	63	119	50
115	41	97	3	59
24	55	106	37	93

Fourth Layer from Top.

64	120	46	77	8
98	4	60	111	42
107	38	94	25	51
16	72	103	34	90
30	81	12	68	124

Lowest Layer.

95	21	52	108	39
104	35	86	17	73
13	69	125	26	82
47	78	9	65	116
56	112	43	99	5

In this manner the preceding magic cube of 5 times 5 times 5 compartments is formed, in which, it may be additionally noticed, the middle number between 1 and 125, namely 63, is placed in the central compartment; by which arrangement the attainment of the sum of 315 is assured in the four principal diagonals and the 30 sub-diagonals. The condition attained in the magic squares, that the diagonal-pairs parallel to the sub-diagonals also shall give the sum 315 is not attainable in this case but is so in the case of higher numbers of compartments.

CONCLUSION.

Musing on such problems as are the magic squares is fascinating to thinkers of a mathematical turn of mind. We take delight in discovering a harmony that abides as an intrinsic quality in the forms of our thought. The problems of the magic squares are playful puzzles, invented as it seems for mere pastime and sport. But there is a deeper problem underlying all these little riddles, and this deeper problem is of a sweeping significance. It is the philosophical problem of the world-order.

The formal sciences are creations of the mind. We build the sciences of mathematics, geometry, and algebra with our conception of pure forms which are abstract ideas. And the same order that prevails in these mental constructions permeates the universe, so that an old philosopher, overwhelmed with the grandeur of law, imagined he heard its rhythm in a cosmic harmony of the spheres.

THE FOURTH DIMENSION.

MATHEMATICAL AND SPIRITUALISTIC.

I.

INTRODUCTORY.

THE tendency to generalise long ago led mathematicians to extend the notion of three-dimensional space, which is the space of sensible representation, and to define aggregates of points, or spaces, of more than three dimensions, with the view of employing these definitions as useful means of investigation. They had no idea of requiring people to imagine four-dimensional things and worlds, and they were even still less remote from requiring them to believe in the real existence of a four-dimensioned space. In the hands of mathematicians this extension of the notion of space was a mere means devised for the discovery and expression, by shorter and more convenient ways, of truths applicable to common geometry and to algebra operating with more than three unknown quantities. At this stage, however, the spiritualists came in, and coolly took possession of this private property of the mathematicians. They were in great perplexity as to where they should put the spirits of the dead. To give them a place in the world accessible to our senses was not exactly practicable. They were compelled, therefore, to look around after some *terra incognita*, which should oppose to the spirit of research inborn in humanity an insuperable barrier. The abiding-place of the spirits had perforce to be inaccessible to the senses and full of mystery to the mind. This property the four-dimensioned space of the mathematicians possessed. With an intellectual perversity

which science has no idea of, these spiritualists boldly asserted, first, that the whole world was situated in a four-dimensioned space as a plane might be situated in the space familiar to us, secondly, that the spirits of the dead lived in such a four-dimensioned space, thirdly, that these spirits could accordingly act upon the world and, consequently, upon the human beings resident in it, exactly as we three-dimensioned creatures can produce effects upon things that are two-dimensioned; for example, such effects as that produced when we shatter a lamina of ice, and so influence some possibly existing two-dimensioned *ice*-world. Since spiritualism, under the leadership of a Leipsic Professor, Zöllner, thus proclaimed the existence of a four-dimensioned space, this notion, which the mathematicians are thoroughly master of,—for in all their operations with it, though they have forsaken the path of actual representability, they have never left that of the truth,—this notion has also passed into the heads of lay persons who have used it as a catchword, ordinarily without having any clear idea of what they or any one else mean by it. To clear up such ideas and to correct the wrong impressions of cultured people who have not a technical mathematical training, is the purpose of the following pages. A similar elucidation was aimed at in the tracts which Schlegel (Riemann, Berlin, 1888) and Cranz (Virchow-Holtzendorff's Sammlung, Nos. 112 and 113) have published on the so-called fourth dimension. Both treatises possess indubitable merits, but their methods of presentation are in many respects too concise to give to lay minds a profound comprehension of the subject. The author, accordingly, has been able to add to the reflections which these excellent treatises offer, a great deal that appears to him necessary for a thorough explanation in the minds of non-mathematicians of the notion of the fourth dimension.

II.

THE CONCEPT OF DIMENSION.

Many text-books of stereometry begin with the words: "Every body has three dimensions, length, breadth, and thickness." If we should ask the author of a book of this description to tell us the length, breadth, and thickness of an apple, of a sponge, or of a cloud of tobacco smoke, he would be somewhat perplexed and would probably say, that the definition in question referred to something different. A cubical box,

or some similar structure, whose angles are all right angles and whose bounding surfaces are consequently all rectangles is the only body of which it can at all be unmistakably asserted that there are three principal directions distinguishable in it, of which any one can be called the length, any other the breadth, and any third the thickness. We thus see that the notions of length, breadth, and thickness are not sufficiently clear and universal to enable us to derive from them any idea of what is meant when it is said that every body possesses three dimensions, or that the space of the world is three-dimensional.

This distinction may be made sharper and more evident by the following considerations: We have, let us suppose, a straight line on which a point is situated, and the problem is proposed to determine the position of the point on the line in an unequivocal manner. The simplest way to solve this is, to state how far the point is removed in the one or the other direction from some given fixed point; just as in a thermometer the position of the surface of the mercury is given by a statement of its distance in the direction of cold or heat from a predetermined fixed point—the point of freezing water. To state, therefore, the position of a point on a straight line, the sole datum necessary is a single number, if beforehand we have fixed upon some standard line, like the centimetre, and some definite point to which we give the value zero, and have also previously decided in what direction from the zero-point, points must be situated whose position is expressed by positive numbers, and also in what direction those must lie whose position is expressed by negative numbers. This last-mentioned fact, that a *single* number is sufficient to determine the place of a point in a straight line, is the real reason why we attribute to the straight line or to any part of it a single dimension.

More generally, we call every totality or system, of infinitely numerous things, *one*-dimensional, in which *one* number is all that is requisite to determine and mark out any particular one of these things from among the entire totality. Thus, time is one-dimensional. We, as inhabitants of the earth, have naturally chosen as our unit of time, the period of the rotation of the earth about its axis, namely, the day, or a definite portion of a day. The zero-point of time is regarded in Christian countries as the year of the birth of Christ, and the positive direction of time is the time *subsequent* to the birth of Christ. These data fixed, all that is necessary to establish and distinguish any definite point of time

amid the infinite totality of all the points of time, *is a single number*. Of course this number need not be a whole number, but may be made up of the sum of a whole number and a fraction in whose numerator and denominator we may have numbers as great as we please. We may, therefore, also say that the totality of all conceivable numerical magnitudes, or of only such as are greater than one definite number and smaller than some other definite number, is one-dimensional.

We shall add here a few additional examples of one-dimensional magnitudes presented by geometry. First, the circumference of a circle is a one-dimensional magnitude, as is every curved line, whether it returns into itself or not. Further, the totality of all equilateral triangles which stand on the same base is one-dimensional, or the totality of all circles that can be described through two fixed points. Also, the totality of all conceivable cubes will be seen to be one-dimensional, provided they are distinguished, not with respect to position, but with respect to magnitude.

In conformity with the fundamental ideas by which we define the notion of a one-dimensional manifoldness, it will be seen that the attribute *two*-dimensional must be applied to all totalities of things in which *two* numbers are necessary (and sufficient) to distinguish any determinate individual thing amid the totality. The simplest two-dimensioned complex which we know of is the plane. To determine accurately the position of a point in a plane, the simplest way is to take two axes at right angles to each other, that is, fixed straight lines, and then to specify the distances by which the point in question is removed from each of these axes.

This method of determining the position of a point in a plane suggested to the celebrated philosopher and mathematician Descartes the fundamental idea of analytical geometry, a branch of mathematics in which by the simple artifice of ascribing to every point in a plane two numerical values, determined by its distances from the two axes above referred to, planimetrical considerations are transformed into algebraical. So, too, all kinds of curves that graphically represent the dependence of things on time, make use of the fact that the totality of the points in a plane is two-dimensional. For example, to represent in a graphical form the increase in the population of a city, we take a horizontal axis to represent the time, and a perpendicular one to represent the numbers which are the measures of the population. Any two lines, then,

whose lengths practical considerations determine, are taken as the unit of time, which we may say is a year, and as the unit of population, which we will say is one thousand. Some definite year, say 1850, is fixed upon as the zero point. Then, from all the equally distant points on the horizontal axis, which points stand for the years, we proceed in directions parallel to the other axis, that is, in the perpendicular direction, just so much upwards as the numbers which stand for the population of that year require. The terminal points so reached, or the curve which runs through these terminal points, will then present a graphic picture of the rates of increase of the population of the town in the different years. The rectangular axes of Descartes are employed in a similar way for the construction of barometer curves, which specify for the different localities of a country the amount of variation of the atmospheric pressure during any period of time. Immediately next to the plane the surface of the earth will be recognised as a two-dimensional aggregate of points. In this case geographical latitude and longitude supply the two numbers that are requisite accurately to determine the position of a point. Also, the totality of all the possible straight lines that can be drawn through any point in space is two-dimensional, as we shall best understand if we picture to ourselves a plane which is cut in a point by each of these straight lines and then remember that by such a construction every point on the plane will belong to some one line and, *vice versa*, a line to every point, whence it follows that the totality of all the straight lines, or, as we may call them, rays, which pass through the point assigned are of the same dimensions as the totality of the points of the imagined plane.

The question might be asked, In what way and to what extent in this case is the specification of *two* numbers requisite and sufficient to determine amid all the rays which pass through the specified point a definite individual ray? To get a clear idea of the problem here involved, let us imagine the ray produced far into the heavens where some quite definite point will correspond to it. Now, the position of a point in the heavens depends, as does the position of a point on all spherical surfaces, on two numbers. In the heavens these two numbers are ordinarily supplied by the two angles called altitude, or the distance above the plane of the horizon, and azimuth, or the angular distance between the circle on which the altitude is measured and the meridian of the observer. It will be seen thus that the totality of all the

luminous rays that an eye, conceived as a point, can receive from the outer world is two-dimensional, and also that a luminous point emits a two-dimensional group of luminous rays. It will also be observed, in connexion with this example, that the two-dimensional totality of all the rays that can be drawn through a point in space is something different from the totality of the rays that pass through a point but are required to lie in a given plane. Such a group of objects as the last-named one is a one-dimensional totality.

Now that we have sufficiently discussed the attributes that are characteristic of one and two-dimensional aggregates, we may, without any further investigation of the subject, propose the following definition, that, generally, *an n-dimensional totality of infinitely numerous things is such that the specification of n numbers is necessary and sufficient to indicate definitely any individual amid all the infinitely numerous individuals of that totality.*

Accordingly, the point-aggregate made up of the world-space which we inhabit, is a three-dimensional totality. To get our bearings in this space and to define any determinate point in it, we have simply to lay through any point which we take as our zero-point three axes at right angles to each other, one running from right to left, one backwards and forwards, and one upwards and downwards. We then join each two of these axes by a plane and are enabled thus to specify the position of every point in space by the three perpendicular distances by which the point in question is removed in a positive or negative sense from these three planes. It is customary to denote the numbers which are the measures of these three distances by x, y, and z, the positive x, positive y, and positive z ordinarily being reckoned in the right hand, the hitherward, and the upward directions from the origin. If now, with direct reference to this fundamental axial system, any particular specification of x, y, and z be made, there will, by such an operation, be cut out and isolated from the three-dimensional manifoldness of all the points of space a totality of less dimensions. If, for example, z is equal to seven units or measures, this is equivalent to a statement that only the two-dimensional totality of the points is meant, which constitute the plane that can be laid at right angles to the upward-passing z-axis at a distance of seven measures from the zero-point. Consequently, every imaginable equation between x, y, and z isolates and defines a two-dimensional aggregate of points. If two different equations obtain

between x, y, and z, two such two-dimensional totalities will be isolated from among all the points of space. But as these last must have some one-dimensional totality in common, we may say that the co-existence of two equations between x, y, and z defines a one-dimensional totality of points, that is to say a straight line, a line curved in a plane, or even, perhaps, one curved in space. It is evident from this that the introduction of the three axes of reference forms a bridge between the theory of space and the theory of equations involving three variable quantities, x, y, z. The reason that the theory of space cannot thus be brought into connection with algebra in general, that is, with the theory of indefinitely numerous equations, but only with the algebra of three quantities, x, y, z, is simply to be sought in the fact that space, as we picture it, can have only three dimensions.

We have now only to supply a few additional examples of n-dimensional totalities. All particles of air are four-dimensional in magnitude when in addition to their position in space we also consider the variable densities which they assume, as they are expressed by the different heights of the barometer in the different parts of the atmosphere. Similarly, all conceivable spheres in space are four-dimensional magnitudes, for their centres form a three-dimensional point-aggregate, and around each centre there may be additionally conceived a one-dimensional totality of spheres, the radii of which can be expressed by every numerical magnitude from zero to infinity. Further, if we imagine a measuring-stick of invariable length to assume every conceivable position in space, the positions so obtained will constitute a five-dimensional aggregate. For, in the first place, one of the extremities of the measuring stick may be conceived to assume a position at every point of space, and this determines for one extremity alone of the stick a three-dimensional totality of positions; and secondly, as we have seen above, there proceeds from every such position of this extremity a two-dimensional totality of directions, and by conceiving the measuring-stick to be placed lengthwise in every one of these directions we shall obtain all the conceivable positions which the second extremity can assume, and consequently, the dimensions must be 3 plus 2 or 5. Finally, to find out how many dimensions the totality of all the possible positions of a square, invariable in magnitude, possesses, we first give one of its corners all conceivable positions in space, and we thus obtain three dimensions. One definite point in

space now being fixed for the position of one corner of the square, we imagine drawn through this point all possible lines, and on each we lay off the length of the side of the square and thus obtain two additional dimensions. Through the point obtained for the position of the second corner of the square we must now conceive all the possible directions drawn that are perpendicular to the line thus fixed, and we must lay off once more on each of these directions the side of the square. By this last determination the dimensions are only increased by one, for only one one-dimensional totality of perpendicular directions is possible to one straight line in one of its points. Three corners of the square are now fixed and therewith the position of the fourth also is uniquely determined. Accordingly, the totality of all equal squares which differ from one another only by their position in space, constitutes a manifoldness of six dimensions.

III.

THE INTRODUCTION OF THE NOTION OF FOUR-DIMENSIONAL POINT-AGGREGATES, PERMISSIBLE.

In the preceding section it was shown that we can conceive not only of manifoldnesses of one, two, and three dimensions, but also of manifoldnesses of *any* number of dimensions. But it was at the same time indicated that our world-space, that is, the totality of all conceivable *points* that differ only in respect of position, cannot in agreement with our notions of things possess more than three dimensions. But the question now arises, whether, if the progress of science tends in such a direction, it is permissible to extend the notion of space by the introduction of point-aggregates of more than three dimensions, and to engage in the study of the properties of such creations, although we know that notwithstanding the fact that we may conceptually establish and explore such aggregates of points, yet we cannot picture to ourselves these creations as we do the spatial magnitudes which surround us, that is, the regular three-dimensional aggregates of points.

To show the reader clearly that this question must be answered in the affirmative, that the extension of our notion of space is permissible, although it leads to things which we cannot perceive by our senses, I may call the reader's attention to the fact that in arithmetic we are accustomed from our youth upwards to extensions of ideas,

which, accurately viewed, as little admit of graphic conception as a four-dimensional space, that is, a point-aggregate of four dimensions. By his senses man first reaches only the idea of whole numbers—the results of counting. The observation of primitive peoples* and of children clearly proves that the essential decisive factors of counting are these three: First, we abstract, in the counting of things, completely from the individual and characteristic attributes of these things, that is, we consider them as homogeneous. Second, we associate individually with the things which we count other homogeneous things. These other things are even now, among uncivilised peoples, the ten fingers of the two hands. They may, however, be simple strokes, or, as in the case of dice and dominoes, black points on a white background. Third, we substitute for the result of this association some concise symbol or word; for example, the Romans substituted for three things counted, three strokes placed side by side, namely: III; but for greater numbers of things they employed abbreviated signs. The Aztecs, the original inhabitants of Mexico, had time enough, it seems, to express all the numbers up to nineteen by equal circles placed side by side. They had abbreviated signs only for the numbers 20, 400, 8000, and so forth. In speaking, some one same sound might be associated with the things counted; but this method of counting is nowadays employed only by clocks: the languages of men since prehistoric times have fashioned concise words for the results of the association in question. From the notion of number, thus fixed as the result of counting, man reached the notion of the addition of two numbers, and thence the notion that is the inverse of the last process, the notion of subtraction. But at this point it clearly appears that not every problem which may be propounded is soluble; for there is no number which can express the result of the subtraction of a number from one which is equally large or from one which is smaller than itself. The primary school pupil who says that 8 from 5 "won't go" is perfectly right from his point of view. For there really does not exist any result of counting which added to eight will give five.

If humanity had abided by this point of view and had rested content with the opinion that the problem "5 minus 8" is not solvable, the science of arithmetic would never have received its full development, and humanity would not have advanced as far in civilisation as it has.

* See the essay *Notion and Definition of Number* in this collection.

Fortunately, men said to themselves at this crisis: "If 5 minus 8 won't go, we'll *make it go*; if 5 minus 8 does not possess an intelligible meaning, we will simply give it one." As a fact, things which have not a meaning always afford men a pleasing opportunity of investing them with one. The question is, then, what significance is the problem "5 minus 8" to be invested with?

The most natural and therefore the most advantageous solution undoubtedly is to abide by the original notion of subtraction as the inverse of addition, and to make the significance of 5 minus 8 such, that for 5 minus 8 plus 8 we shall get our original minuend 5. By such a method all the rules of computation which apply to real differences will also hold good for unreal differences, such as 5 minus 8. But it then clearly appears that all forms expressive of differences in which the numbers that stand before the minus symbol are less by an equal amount than those which follow it may be regarded as equal; so that the simplest course seems to be to introduce as the common characteristic of all equal differential forms of this description a common sign, which will indicate at the same time the difference of the two numbers thus associated. Thus it came about that for 5 minus 8, as well as for every differential form which can be regarded as equal thereto, the sign "-3" was introduced. But in calling differential forms of this description numbers, the notion of number was extended and a new domain was opened up, namely the domain of negative numbers.

In the further development of the science of arithmetic, through the operation of division viewed as the inverse of multiplication, a second extension of the idea of number was reached, namely, the notion of fractional numbers as the outcome of divisions that had led to numbers hitherto undefined. We find, thus, that the science of arithmetic throughout its whole development has strictly adhered to the principle of conformity and consistency and has invested every association of two numbers, which before had no significance, by the introduction of new numbers, with a real significance, such that similar operations in conformity with exactly the same rules could be performed with the new numbers, viewed as the results of this association, as with the numbers which were before known and perfectly defined. Thus the science proceeded further on its way and reached the notions of irrational, imaginary, and complex numbers.

The point in all this, which the reader must carefully note, is, that all the numbers of arithmetic, with the exception of the positive whole numbers, are artificial products of human thought, invented to make the language of arithmetic more flexible, and to accelerate the progress of science. All these numbers lack the attributes of representability.

No man in the world can picture to himself "minus three trees." It is possible, of course, to know that when three trees of a garden have been cut down and carried away, three are missing, and by substituting for "missing" the inverse notion of "added," we may say, perhaps, that "minus three trees" are added. But this is quite different from the feat of imagining a negative number of trees. We can only picture to ourselves a number of trees that results from actual counting, that is, a positive whole number. Yet, notwithstanding all this, people had not the slightest hesitation in extending the notion of number. Exactly so must it be permitted us in geometry to extend the notion of space, even though such an extension can only be mentally defined and can never be brought within the range of human powers of representation.

In mathematics, in fact, the extension of any notion is admissible, provided such extension does not lead to contradictions with itself or with results which are well established. Whether such extensions are necessary, justifiable, or important for the advancement of science is a different question. It must be admitted, therefore, that the mathematician is justified in the extension of the notion of space as a point-aggregate of three dimensions, and in the introduction of space or point-aggregates of more than three dimensions, and in the employment of them as means of research. Other sciences also operate with things which they do not know exist, and which, though they are sufficiently defined, cannot be perceived by our senses. For example, the physicist employs the ether as a means of investigation, though he can have no sensory knowledge of it. The ether is nothing more than a means which enables us to comprehend mechanically the effects known as action at a distance and to bring them within the range of a common point of view. Without the assumption of a material which penetrates everything, and by means of whose undulations impulses are transmitted to the remotest parts of space, the phenomena of light, of heat, of gravitation, and of electricity would be a jumble of isolated and unconnected mysteries. The assumption of an ether, however, comprises in a systematic scheme all these isolated events, facilitates our

mental control of the phenomena of nature, and enables us to produce these phenomena at will. But it must not be forgotten in such reflexions that the ether itself is even a greater problem for man, and that the ether-hypothesis does not solve the difficulties of phenomena, but only puts them in a unitary conceptual shape. Notwithstanding all this, physicists have never had the least hesitation in employing the ether as a means of investigation. And as little do reasons exist why the mathematicians should hesitate to investigate the properties of a four-dimensioned point-aggregate, with the view of acquiring thus a convenient means of research.

IV.

THE INTRODUCTION OF THE IDEA OF FOUR-DIMENSIONED POINT-AGGREGATES OF SERVICE TO RESEARCH.

From the concession that the mathematician has the right to define and investigate the properties of point-aggregates of more than three dimensions, it does not necessarily follow that the introduction of an idea of this description is of value to science. Thus, for example, in arithmetic, the introduction of operations which spring from involution, as involution and its two inverse operations proceed from multiplication, is undoubtedly permitted. Just as for "a times a times a" we write the abbreviated symbol "a^3," (which we read, a to the third power,) and investigate in detail the operation of involution thus defined, so we might also introduce some shorthand symbol for "a to the ath power to the ath power" and thus reach an operation of the fourth degree, which would regard a as a passive number and the number 3, or any higher number, as the active number, that is, as the number which indicates how often a is taken as the base of a power whose exponent may be a, or "a to the ath, or "a to the ath to the ath power."

But the introduction of such an operation of the fourth degree has proved itself to be of no especial value to mathematics. And the reason is that in the operation of involution the law of commutation does not hold good. In addition, the numbers to be added may be interchanged and the introduction of multiplication is therefore of great value. So, also, in multiplication the numbers which are combined, that is, the factors, may be changed about in any way, and thus the introduction of involution is of value. But in involution the base and the

exponent cannot be interchanged, and consequently the introduction of any higher operation is almost valueless.

But with the introduction of the idea of point-aggregates of multiple dimensions the case is wholly different. The innovation in question has proved itself to be not only of great importance to research, but the progress of science has irresistibly forced investigators to the introduction of this idea, as we shall now set forth in detail.

In the first place, algebra, especially the algebraical theory of systems of equations, derives much advantage from the notion of multi-dimensioned spaces. If we have only three unknown quantities, x, y, z, the algebraical questions which arise from the possible problems of this class admit, as we have above seen, of geometrical representation to the eye. Owing to this possibility of geometrical representation, some certain simple geometrical ideas like "moving," "lying in," "intersecting," and so forth, may be translated into algebraical events. Now, no reason exists why algebra should stop at three variable quantities; it must in fact take into consideration any number of variable quantities.

For purposes of brevity and greater evidentness, therefore, it is quite natural to employ geometrical forms of speech in the consideration of more than three variables. But when we do this, we assume, perhaps without really intending to do so, the idea of a space of more than three dimensions. If we have four variable quantities, x, y, z, u, we arrive, by conceiving attributed to each of these four quantities every possible numerical magnitude, at a four-dimensioned manifoldness of numerical quantities, which we may just as well regard as a four-dimensioned aggregate of points. Two equations which exist on this supposition between x, y, z, and u, define two three-dimensioned aggregates of points, which intersect, as we may briefly say, in a two-dimensioned aggregate of points, that is, in a surface; and so on. In a somewhat different manner the determination of the contents of a square or a cube by the involution of a number which stands for the length of its sides, leads to the notion of four-dimensioned structures, and, consequently, to the notion of a four-dimensioned point-space. When we note that a^2 stands for the contents of a square, and a^3 for the content of a cube, we naturally inquire after the contents of a structure which is produced from the cube as the cube is produced from the square and which also will have the contents a^4. We cannot, it is true, clearly picture to ourselves a structure of this description, but we can, nevertheless,

establish its properties with mathematical exactness.* It is bounded by 8 cubes just as the cube is bounded by 6 squares; it has 16 corners, 24 squares, and 32 edges, so that from every corner 4 edges, 6 squares, and 4 cubes proceed, and from every edge 3 squares and 3 cubes.

Yet despite the great service to algebra of this idea of multi-dimensioned space, it must be conceded that the conception although convenient is yet not indispensable. It is true, algebra is in need of the idea of multiple dimensions, but it is not so absolutely in need of the idea of *point*-aggregates of multiple dimensions.

This notion is, however, necessary and serviceable for a profound comprehension of geometry. The system of geometrical knowledge which Euclid of Alexandria created about three hundred years before Christ, supplied during a period of more than two thousand years a brilliant example of a body of conclusions and truths which were mutually consistent and logical. Up to the present century the idea of elementary geometry was indissolubly bound up with the name of Euclid, so that in England where people adhered longest to the rigid deductive system of the Grecian mathematician, the task of "learning geometry" and "reading Euclid" were until a few years ago identical. Every proposition of this Euclidean system rests on other propositions, as one building-stone in a house rests upon another. Only the very lowest stones, the foundations, were without supports. These are the axioms or fundamental propositions, truths on which all other truths are, directly or indirectly, founded, but which themselves are assumed without demonstration as self-evident.

But the spirit of mathematical research grew in time more and more critical, and finally asked, whether these axioms might not possibly admit of demonstration. Especially was a rigid proof sought for the eleventh† axiom of Euclid, which treats of parallels.

After centuries of fruitless attempts to prove Euclid's eleventh axiom, Gauss, and with him Bolyai and Lobachévski, Riemann, and Helmholtz, finally stated the decisive reasons why any attempt to prove the axiom of the parallels must necessarily be futile. These reasons

* Victor Schlegel, indeed, has made models of the three-dimensional nets of all the six structures which correspond in four-dimensioned space to the five regular bodies of our space, in an analogous manner to that by which we draw in a plane the nets of five regular bodies. Schlegel's models are made by Brill of Darmstadt.

† Also called the twelfth axiom, also the fifth postulate.—*Tr.*

consist of the fact that though this axiom holds good enough in the world-space such as we do and can conceive it, yet three-dimensioned spaces are ideally conceivable though not capable of mental representation, where the axiom does not hold good. The axiom was thus shown to be a mere fact of *observation*, and from that time on there could no longer be any thought of a deductive demonstration of it. In view of the intimate connection, which both in an historical and epistemological point of view exists between the extension of the concept of space and the critical examination of the axioms of Euclid, we must enter at somewhat greater length into the discussion of the last mentioned propositions.

Of the axioms which Euclid lays at the foundation of his geometry, only the following three are really geometrical axioms:

Eighth axiom: Magnitudes which coincide with one another are equal to one another.

Eleventh axiom: If a straight line meet two straight lines so as to make the two interior angles on the same side of it taken together less than two right angles, these straight lines, being continually produced, shall at length meet on that side on which are the angles which are less than two right angles.

Twelfth axiom: Two straight lines cannot inclose a [finite] space.

The numerous proofs which in the course of time were adduced in demonstration of these axioms, especially of the eleventh, all turn out on close examination to be pseudo-proofs. Legendre drew attention to the fact that either of the following axioms might be substituted for the eleventh:

a) Given a straight line, there can be drawn through a point in the same plane with that line, one and one line only which shall not intersect the first (parallels) however far the two lines may be produced;

b) If two parallel lines are cut by a third straight line, the interior alternate angles will be equal.

c) The sum of the angles of a triangle is equal to two right angles, that is, to the angle of a straight line or to 180°.

By the aid of any one of these three assertions, the eleventh axiom of Euclid may be proved, and, *vice versa*, by the aid of the latter each of the three assertions may be proved, of course with the help of the other two axioms, eight and twelve. The perception that the eleventh axiom does not admit of demonstration without the employment of one of

the foregoing substitutes may best be gained from the consideration of congruent figures. Every reader will remember from his first instruction in geometry that the congruence of two triangles is demonstrated by the superposition of one triangle on the other and by then ascertaining whether the two completely coincide, no assumptions being made in the determination except those above mentioned.

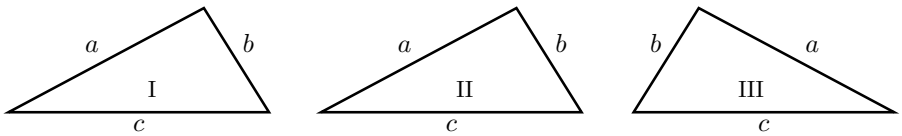


Fig. 35.

In the case of triangles which are congruent, as are I and II in the preceding cut, this coincidence may be effected by the simple *displacement* of one of the triangles; so that even a two-dimensional being, supposed to be endowed with powers of reasoning, but only capable of picturing to itself motions within a plane, also might convince itself that the two triangles I and II could be made to coincide. But a being of this description could not convince itself in like manner of the congruence of triangles I and III. It would discover the equality of the three sides and the three angles, but it could never succeed in so superposing the two triangles on each other as to make them coincide. A three-dimensional being, however, can do this very easily. It has simply to turn triangle I about one of its sides and to shove the triangle, thus brought into the position of its reflexion in a mirror, into the position of triangle III. Similarly, triangles II and III may be made to coincide by moving either out of the plane of the paper around one of its sides as axis and turning it until it again falls in the plane of the paper. The triangle thus turned over can then be brought into the position of the other.

Later on we shall revert to these two kinds of congruence: "congruence by displacement" and "congruence by circumversion." For the present we will start from the fact that it is always possible within the limits of a plane to take a triangle out of one position and bring it into another without altering its sides and angles. The question is, whether this is only possible in the plane, or whether it can also be done on other surfaces.

We find that there are certain surfaces in which this is possible, and certain others in which it is not. For instance, it is impossible to move the triangle drawn on the surface of an egg into some other position on the egg's surface without a distension or contraction of some of the triangle's parts. On the other hand, it is quite possible to move the triangle drawn on the surface of a sphere into any other position on the sphere's surface without a distension or contraction of its parts. The mathematical reason of this fact is, that the surface of a sphere, like the plane, has everywhere the same curvature, but that the surface of an egg at different places has different curvatures. Of a plane we say that it has everywhere the curvature zero; of the surface of a sphere we say it has everywhere a positive curvature, which is greater in proportion as the radius is smaller. There are surfaces also which have a constant negative curvature; these surfaces exhibit at every point in directions proceeding from the same side a partly concave and a partly convex structure, somewhat like the centre of a saddle. There is no necessity of our entering in any detail into the character and structure of the last-mentioned surfaces.

Intimately related with the plane, however, are all those surfaces, which, like the plane, have the curvature zero; in this category belong especially cylindrical surfaces and conical surfaces. A sheet of paper of the form of the sector of a circle may, for example, be readily bent into the shape of a conical surface. If two congruent triangles, now, be drawn on the sheet of paper, which may by displacement be translated the one into the other, these triangles will, it is plain, also remain congruent on the conical surface; that is, on the conical surface also we may displace the one into the other; for though a bending of the figures will take place, there will be no distension or contraction. Similarly, there are surfaces which, like the sphere, have everywhere a constant positive curvature. On such surfaces also every figure can be transferred into some other position without distension or contraction of its parts. Accordingly, on all surfaces thus related to the plane or sphere, the assumption which underlies the eighth axiom of Euclid, that it is possible to transfer into any new position any figure drawn on such surfaces without distortion, holds good.

The eleventh axiom in its turn also holds good on all surfaces of constant curvature, whether the curvature be zero or positive; only in such instances instead of "straight" line we must say "shortest" line.

On the surface of a sphere, namely, two shortest lines, that is, arcs of two great circles, always intersect, no matter whether they are produced in the direction of the side at which the third arc of a great circle makes with them angles less than two right angles, or, in the direction of the other side, where this arc makes with them angles of more than two right angles. On the plane, however, two straight lines intersect only on the side where a third straight line that meets them makes with them interior angles less than two right angles.

The twelfth axiom of Euclid, finally, only holds good on the plane and on the surfaces related to it, but not on the sphere or other surfaces which, like the sphere, have a constant positive curvature. This also accounts for the fact that one of the three postulates which we regarded as substitutes for the eleventh axiom, though valid for the plane, is not true for the surface of a sphere; namely, the postulate that defines the sum of the angles of a triangle. This sum in a plane triangle is two right angles; in a spherical triangle it is more than two right angles, the spherical triangle being greater, the greater the excess the sum of its angles is above two right angles. It will be seen, from these considerations, that in geometries in which curved surfaces and not fixed planes are studied, the axioms of Euclid are either all or partially false.

The axioms of geometry thus having been revealed as facts of experience, the question suggested itself whether in the same way in which it was shown that different two-dimensional geometries were possible, also different three-dimensional systems of geometry might not be developed; and consequently what the relations were in which these might stand to the geometry of the space given by our senses and representable to our mind. As a fact, a three-dimensional geometry can be developed, which like the geometry of the surface of an egg will exclude the axiom that a figure or body can be transferred from any one part of space to any other and yet remain congruent to itself. Of a three-dimensional space in which such a geometry can be developed we say, that it has no constant measure of curvature.

The space which is representable to us, and which we shall henceforth call the *space of experience*, possesses, as our experiences without exception confirm, the especial property that every bodily thing can be transferred from any one part of it to any other without suffering in

the transference any distension or any contraction. The space of experience, therefore, has a constant measure of curvature. The question, however, whether this measure of curvature is zero or positive, that is, whether the space of experience possesses the properties which in two-dimensional structures a plane possesses, or whether it is the three dimensional analogue of the surface of a sphere is one which future experience alone can answer. If the space of experience has a constant positive measure of curvature which is different from zero, be the difference ever so slight, a point which should move forever onward in a straight line, or, more accurately expressed, in a shortest line, would sometime, though perhaps after having traversed a distance which to us is inconceivable, ultimately have to arrive from the opposite direction at the place from which it set out, just as a point which moves forever onward in the same direction on the surface of a sphere must ultimately arrive at its starting point, the distance it traverses being longer the greater the radius of the sphere or the smaller its curvature.

It will seem, at first blush, almost incredible, that the space of experience possibly could have this property. But an example, which is the historical analogue of this modern transformation of our conceptions, will render the idea less marvellous. Let us transport ourselves to the age of Homer. At that time people believed that the earth was a great disc surrounded on all sides by oceans which were conceived to be in all directions infinitely great. Indeed, for the primitive man, who has never journeyed far from the place of his birth, this is the most natural conception. But imagine now that some scholar had come, and had informed the Homeric hero Ulysses that if he would travel forever on the earth in the same direction he would ultimately come back to the point from which he started; surely Ulysses would have gazed with as much astonishment upon this scholar as we now look upon the mathematician who tells us that it is possible that a point which moves forever onward in space in the same direction may ultimately arrive at the place from which it started. But despite the fact that Ulysses would have regarded the assertion of the scholar as false because contradictory to his familiar conceptions, that scholar, nevertheless, would have been right; for the earth is not a plane but a spherical surface. So also the mathematician may be right who bases this more recent strange view on the possible fact that the space of experience may have a measure of curvature which is not exactly zero but slightly greater than

zero. If this were really the case, the *volume* of the space of experience, though very large, would, nevertheless, be finite; just as the real spherical surface of the earth as contrasted with the Homeric plane surface is finite, having so and so many square miles. When the objection is here made that a finiteness of space is totally at variance with our modes of thought and conceptions, two ideas, "infinitely great" and "unlimited," are confounded. All that is at variance with our practical conceptions is that space can anywhere have a boundary; not that it may possibly be of tremendous but finite magnitude.

It will now be asked if we cannot determine by actual observation whether the measure of curvature of experiential space is exactly zero or slightly different therefrom. The theorem of the sum of the angles of a triangle and the conclusions which follow from this theorem do indeed supply us with a means of ascertaining this fact. And the results of observation have been, that *the measure of curvature of space is in all probability exactly equal to zero or if it is slightly different from zero it is so little so that the technical means of observation at our command and especially our telescopes are not competent to determine the amount of the deviation.* More, we cannot with any certainty say.

All these reflections, to which the criticism of the hypotheses that underlie geometry long ago led investigators, compel us to institute a comparison between the space of experience and other three-dimensional aggregates of points (spaces), which we cannot mentally represent but can in thought and word accurately define and investigate. As soon, however, as we are fully implicated in the task of accurately investigating the properties of three-dimensional aggregates of points, we find ourselves similarly forced to regard such aggregates as the component elements of a manifoldness of more than three dimensions. In this way the exact criticism of even ordinary geometry leads us to the abstract assumption of a space of more than three dimensions. And as the extension of every idea gives a clearer and more translucent form to the idea as it originally stood, here too the idea of multi-dimensioned aggregates of points and the investigation of their properties has thrown a new light on the truths of ordinary geometry and placed its properties in clearer relief. Among the numerous examples which show how the notion of a space of multiple dimensions has been of great service to science in the investigation of

three-dimensional space, we shall give one a place here which is within the comprehension of non-mathematicians.

Imagine in a plane two triangles whose angles are denoted by pairs of numbers—namely, by 1-2, 1-3, 1-4, and 2-5, 3-5, 4-5. (See Fig. 36.) Let the two triangles so lie that the three lines which join the angles 1-2 and 2-5, 1-3 and 3-5, and 1-4 and 4-5 intersect at a point, which we will call 1-5. If now we cause the sides of the triangles which are opposite to these angles to intersect, it will be found that the points of intersection so obtained possess the peculiar property of lying all in one and the same straight line. The point of intersection of the connection 1-3 and 1-4 with the connection 4-5 and 3-5 may appropriately be called 3-4. Similarly, the point of intersection 2-4 is produced by the meeting of 4-5, 2-5 and 1-2, 1-4; and the point of intersection 2-3, by the meeting of 1-3, 1-2 and 3-5, 2-5. The statement, that the three points of intersection 3-4, 2-4, 2-3, thus obtained, lie in one straight line, can be proved by the principles of plane geometry only with difficulty and great circumstantiality. But by resorting to the three-dimensional space of experience, in which the plane of the drawing lies, the proposition can be rendered almost self-evident.

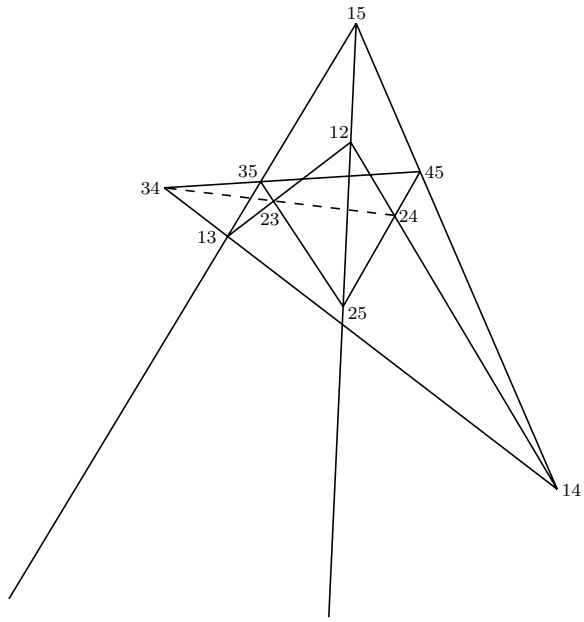


Fig. 36.

To begin with, imagine any five points in space which may be denoted by the numbers 1, 2, 3, 4, 5; then imagine all the possible ten straight lines of junction drawn between each two of these points, namely, 1-2, 1-3 . . . 4-5; and finally, also, all the ten planes of junction of every three points described, namely, the plane 1-2-3, 1-2-4, . . . 3-4-5. A spatial figure will thus be obtained, whose ten straight lines will

meet some interposed plane in ten points whose relative positions are exactly those of the ten points above described. Thus, for example, on this plane the points 1-2, 1-3, and 2-3 will lie in a straight line, for through the three spatial points 1, 2, 3, a plane can be drawn which will cut the plane of a drawing in a straight line. The reason, therefore, that the three points 3-4, 2-4, 2-3, also must ultimately lie in a straight line, consists in the simple fact that the plane of the three points 2, 3, 4, must cut the plane of the drawing in a straight line. The figure here considered consists of ten points of which sets of three so lie ten times in a straight line that conversely from every point also three straight lines proceed.

Now, just as this figure is a section of a complete three-dimensional pentagon, so another remarkable figure, of similar properties, may be

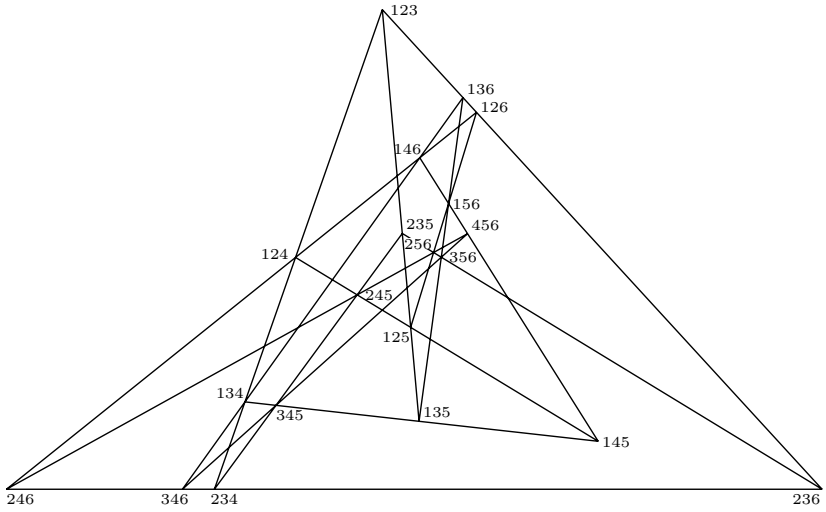


Fig. 37.

obtained from the section of a figure of four-dimensioned space. Imagine six points, 1, 2, 3, 4, 5, 6, situated in this four-dimensioned space, and every three of them connected by a plane, and every four of them by a three-dimensioned space. We shall obtain thus twenty planes and fifteen three-dimensioned spaces which will cut the plane in which the figure is to be produced in twenty points and fifteen rays which so lie that each point sends out three rays and every ray contains four

points. (See Fig. 37.) Figures of this description, which are so composed of points and rays that an equal number of rays proceed from every point and an equal number of points lie in every ray, are called *configurations*. Other configurations may, of course, be produced, by taking a different number of points and by assuming that the points taken lie in a space of different or even higher dimensions. The author of this article was the first to draw attention to configurations derived from spaces of higher dimensions. As we see, then, the notion of a space of more than three dimensions has performed an important service in the investigations of common plane geometry.

In conclusion, I should like to add a remark which Cranz makes regarding the application of the idea of multi-dimensioned space to theoretical chemistry. (See the treatise before cited.) In chemistry, the molecules of a compound body are said to consist of the atoms of the elements which are contained in the body, and these are supposed to be situated at certain distances from one another, and to be held in their relative positions by certain forces. At first, the centres of the atoms were conceived to lie in one and the same plane. But Wislicenus was led by researches in paralactic acid to explain the differences of isomeric molecules of the same structural formulæ by the different positions of the atoms in *space*. (Compare *La chimie dans l'espace* by van't Hoff, 1875, preface by J. Wislicenus). In fact four points can always be so arranged in space that every two of them may have any distance from each other; and the change of one of the six distances does not necessarily involve the alteration of any other.

But suppose our molecule consists of five atoms? Four of these may be so placed that the distance between any two of them can be made what we please. But it is no longer possible to give the fifth atom a position such that each of the four distances by which it is separated from the other atoms may be what we please. On the contrary, the fourth distance is dependent on the three remaining distances; for the space of experience has only three dimensions. If, therefore, I have a molecule which consists of five atoms I cannot alter the distance between two of them without at least altering some second distance. But if we imagine the centres of the atoms placed in a four-dimensioned space, this can be done; all the ten distances which may be conceived to exist between the five points will then be independent of one another.

To reach the same result in the case of six atoms we must assume a five-dimensional space; and so on.

Now, if the independence of all the possible distances between the atoms of a molecule is absolutely required by theoretical chemical research, the science is really compelled, if it deals with molecules of more than four atoms, to make use of the idea of a space of more than three dimensions. This idea is, in this case, simply an instrument of research, just as are, also, the ideas of molecules and atoms—means designed to embrace in a perspicuous and systematic form the phenomena of chemistry and to discover the conditions under which new phenomena can be evoked. Whether a four-dimensioned space really exists is a question whose insolubility cannot prevent research from making use of the idea, exactly as chemistry has not been prevented from making use of the notion of atom, although no one really knows whether the things we call atoms exist or not.

V.

REFUTATION OF THE ARGUMENTS ADDUCED TO PROVE THE EXISTENCE OF A FOUR-DIMENSIONED SPACE INCLUSIVE OF THE VISIBLE WORLD.

The considerations of the preceding section will have convinced the cultured non-mathematician of the service which the theory of multi-dimensioned spaces has done, and bids fair to do, for geometrical research. In addition thereto is the consideration that every extension of one branch of mathematical science is a constant source of beneficial and helpful influence to the other branches. The knowledge, however, that mathematicians can employ the notion of four-dimensioned space with good results in their researches, would never have been sufficient to procure it its present popularity; for every man of intelligence has now heard of it, and, in jest or in earnest, often speaks of it. The knowledge of a four-dimensioned space did not reach the ears of cultured non-mathematicians until the consequences which the spiritualists fancied it was permissible to draw from this mathematical notion were publicly known. But it is a tremendous step from the four-dimensioned space of the mathematicians to the space from which the spirit-friends of the spiritualistic mediums entertain us with rappings, knockings, and bad English. Before taking this step we will first discuss the question of the

real existence of a four-dimensional space, not deciding the question whether this space, if it really does exist, is inhabited by reasonable beings who consciously act upon the world in which we exist.

Among the reasons which are put forward to prove the existence of a four-dimensional space containing the world, the least reprehensible are those which are based on the existence of symmetry. We spoke above of two triangles in the same plane which have all their sides and

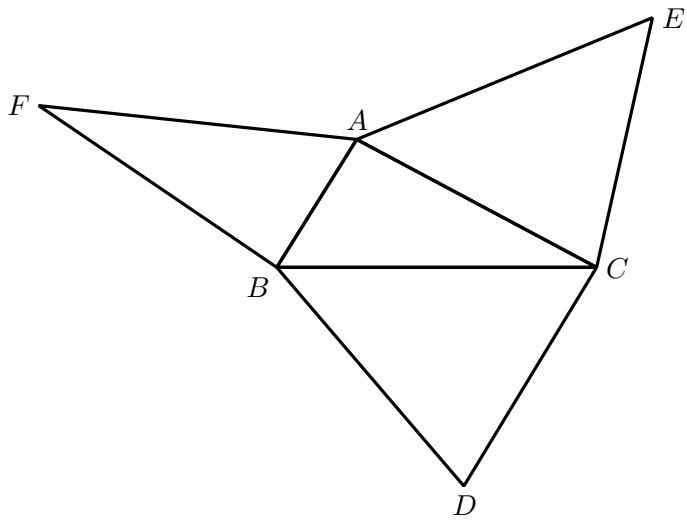


Fig. 38.

angles congruent, but which cannot be made to coincide by simple displacement within the plane; but we saw that this coincidence could be effected by holding fast one side of one triangle and moving it out of its plane until it had been so far turned round that it fell back into its plane. Now something similar to this exists in space. Cut two figures, exactly like that of Fig. 38, out of a piece of paper, and turn the triangle ABF about the side AB, ACE about the side AC, BCD about the side BC, and in one figure above and in the other below; then in both cases the points D, E, F will meet at a point, because AE is equal to AF, BF is equal to BD, CD is equal to CE. In this manner we obtain two pyramids which in all lengths and all angles are congruent, yet which cannot, no matter how we try, be made to coincide, that is, be so fitted the one into the other that they shall both stand as one pyramid. But the *reflected* image of the one could be brought into coincidence with the other. Two spatial structures whose sides and angles are thus

equal to each other, and of which each may be viewed as the reflected image of the other, are called *symmetrical*. For instance, the right and the left hand are symmetrical; or, a right and a left glove. Now just as in two dimensions it is impossible by simple displacement to bring into congruence triangles which like those above mentioned can only be made to coincide by circumversion, so also in three dimensions it is impossible to bring into congruence two symmetrical pyramids. Careful mathematical reflection, however, declares that this could be effected, if it were possible, while holding one of the surfaces, to move the pyramid out of the space of experience, and to turn it round through a four-dimensioned space until it reached a point at which it would return again into our experiential space. This process would simply be the four-dimensional analogue of the three-dimensional circumversion in the above-mentioned case of the two triangles. Further, the interior surfaces in this process would be converted into exterior surfaces, and *vice versa*, exactly as in the circumversion of a triangle the anterior and posterior sides are interchanged. If the structure which is to be converted into its symmetrical counterpart is made of a flexible material, the interchange mentioned of the interior and exterior surfaces may be effected by simply turning the structure inside out; for example, a right glove may thus be converted into a left glove.

Now from this truth, that every structure can be converted, by means of a four-dimensional space inclusive of the world, into a structure symmetrical with it, it has been sought to establish the probability of the real existence of a four-dimensioned space. Yet it will be evident, from the discussion of the preceding section, that the only inference which we can here make is, that the idea of a four-dimensioned space is competent, from a mathematical point of view, to throw some light upon the phenomena of symmetry. To conclude from these facts that a space of this kind really exists, would be as daring as to conclude from the fact that the uniform angular velocity of the apparent motions of the fixed stars is explicable from the assumption of an axial motion of the firmament, that the fixed stars are really rigidly placed in a celestial sphere rotating about its axis. It must not be forgotten that our comprehension of the phenomena of the real world consists of two elements: first, of that which the things really are; and, second, of that by which we rationally apprehend the things. This latter element is partly dependent on the sum of the experiences which we have before

acquired, and partly on the necessity, due to the imperfection of reason, of our classifying the multitudinous isolated phenomena of the world into categories which we ourselves have formed, and which, therefore, are not wholly derived from the phenomena themselves, but to a great extent are dependent on us.

Besides geometrical reasons, Zöllner has also adduced cosmological reasons to prove the existence of a four-dimensional space. To these reasons belong especially the questions, whether the number of the fixed stars is infinitely great, whether the world is finite or infinite in extension, whether the world had a beginning or will have an end, whether the world is not hastening towards a condition of equilibrium or dead level by the universal distribution of its matter and energy; the problems, also, of gravitation and action at a distance; and finally, the questions concerning the relations between the phenomena in the world of sense-perception to the unknown things-in-themselves. All these questions which can be decided in no definite sense, led Zöllner and his followers to the assumption that a four-dimensioned space inclusive of the space of experience must really exist. But more careful reflection will show that this assumption does not dispose of the difficulties but simply displaces them into another realm. Furthermore, even if four-dimensioned space did unravel and make clear all the cosmological problems which have bothered the human mind, still, its existence would not be proved thereby; it would yet remain a mere hypothesis, designed to render more intelligible to a being who can only make experiences in a three-dimensional space, the phenomena therein which are full of mystery to it. A four-dimensioned space would in such case possess for the metaphysician a value similar to that which the ether possesses for the physicist. Still more convincing than these cosmological reasons to the majority of men is the physio-psychological reason drawn from the phenomena of vision which Zöllner adduces. Into this main argument we will enter in more detail.

When we "see" an object, as we all know, the light which proceeds or is reflected therefrom produces an image on the retina of our eye; this image is conducted to our consciousness by means of the optic nerve, and our reason draws therefrom an inference respecting the object. When, now, we look at a square whose sides are a decimetre in length, and whose centre is situated at the distance of a metre from the pupil of our eye, an image is produced on the retina. But exactly the

same image will be produced there if we look at a square whose sides are parallel to the sides of the first square but two decimetres in length, and whose centre is situated at a distance of two metres from the pupil of the eye. Proceeding thus further, we readily discover that an eye can perceive in any length or line only the ratio of its magnitude to the distance at which it is situated from it, and that generally a three-dimensional world must appear to the eye two-dimensional, because all points which lie behind each other in the direction outwards from the eye produce on the retina only one image. This is due to the fact that the retinal images are themselves two-dimensional; for which reason, Zöllner says, the world must appear to a child as two-dimensional, if it be supposed to live in a primitive condition of unconscious mental activity. To such a child two objects which are moving the one behind the other, must appear as suffering displacement on a surface, which we conceive behind the objects, and on which the latter are projected. In all these apparent displacements, coincidences and changes of form also are effected. All these things must appear puzzling to a human being in the first stages of its development, and the mind thus finds itself, as Zöllner further argues, in the first years of childhood forced to adopt a hypothesis concerning the constitution of space and to assume that the world is three-dimensional, although the eye can really perceive it as only two-dimensional. Zöllner then further says, that in the explanation of the effects of the external world, man constantly finds this hypothesis of his childish years confirmed, and that in this way it has become in his mind so profound a conviction that it is no longer possible for him to think it away. Consonant with this argumentation, also, is Zöllner's remark, that the same phenomenon has presented itself in astronomical methods of knowledge. To explain the movements of the planets, which appear to describe regular paths on the surface of a celestial sphere, we were compelled in the solution of the riddles which these motions presented, to assume in the structure of the heavens a dimension of "depth," and the complicated motions in the two-dimensioned firmament were converted into very simple motions in three-dimensioned space. Zöllner also contends that our conception of the entire visible world as possessed of three dimensions is a product of our reason, which the mind was driven to form by the contradictions which would be presented to it on the assumption of only two dimensions by the perspective distortions, coincidences, and changes of magnitude of objects.

When a child moves its hand before its eyes, turns it, brings it nearer, or pushes it farther away, this child successively receives the most various impressions on the surface of its retina of one and the same object of whose identity and constancy its feelings offer it a perfect assurance. If the child regarded the changeable projection of the hand on the surface of the retina as the real object, and not the hand which lies beyond it, the child would constantly be met with contradictions in its experience, and to avoid this it makes the hypothesis that the space of experience is three-dimensional. Zöllner's contention is, therefore, that man originally had only a two-dimensional intuition of space, but was forced by experience to represent to himself the objects which on the retinal surface appeared two-dimensional, as three-dimensional, and thus to transform his two-dimensional space-intuition into a three-dimensional one. Now, in exactly the same way, according to Zöllner's notion, will man, by the advancement and increasing exactness of his knowledge of the phenomena of the outer world, also be compelled to conceive of the material world as a "shadow cast by a more real four-dimensional world," so that these conceptions will be just as trivial for the people of the twentieth century as since Copernicus's time the explanation of the motions of the heavenly bodies by means of a three-dimensional motion has been.

Zöllner's arguments from the phenomena of vision may be refuted as follows: In the first place it is incorrect to say that we see the things of the external world by means of two-dimensional retinal images. The light which penetrates the eye causes an irritation of the optic nerves, and any such effect which, though it be not powerful, is, nevertheless, a mechanical one, can only take place on things which are material. But material things are always three-dimensional. The effect of light on the sensitive plates of photography can with just as little justice be regarded as two-dimensional. Our senses can have perception of nothing but three-dimensional things, and this perception is effected by forces which in their turn act on three-dimensional things, namely our sensory nerves. It is wrong to call an image two-dimensional, for it is only by abstraction that we can conceive of a thickness so growing constantly smaller and smaller as to admit of our regarding a three-dimensional picture as two-dimensional, by giving it in mind a vanishingly small thickness. It is also wrong to say, as Zöllner says, that when we see

the shadow of a hand which is cast upon a wall we see something two-dimensional. What we really perceive is that no light falls upon our eye from the region included by the shadow, while from the entire surrounding region light does fall on our eye. But this light is reflected from the material particles which form the surface of the wall, that is, from three-dimensional particles of matter. We must always remember that our eye communicates to us only three-dimensional knowledge, and that for the comprehension of anything which has two, one, or no dimensions, *a purely intellectual act of abstraction must be added to the act of perception.* When we imagine we have made a lead-pencil mark on paper, we have, exactly viewed, simply heaped alongside of each other little particles of graphite in such a manner that there are by far fewer graphite particles in the lateral and upward directions than there are in the longitudinal direction, and thus our reason arrives by abstraction at the notion of a straight line. When we look at an object, say a cube of wood, we recognise the object as three-dimensional, and it is only by abstraction that we can conceive of its two-dimensional surfaces, of its twelve one-dimensional edges, and of its eight no-dimensional corners. For we reach the perception of its surface, for example, solely by reason of the fact that the material particles which form the cube prevent the transmission of light, and reflect it, whereby a part of the light reflected from every material particle strikes our eye. Now, by thinking exclusively of those material particles which are reflected, in contrariety to the empty space without and the hidden and therefore non-reflected particles within, we form the notion of a surface.

It is evident from this, that all that we perceive is three-dimensional, that we cannot reach anything two-dimensional without an intellectual abstraction, and that, therefore, we cannot conceive of anything two-dimensional exerting effects upon material things. But this fact is a refutation of the retinal argument of Zöllner. If vision consisted wholly and exclusively in the creation of a two-dimensional image, the things which take place in the world could never come into our consciousness. The child, therefore, does not originally apprehend the world, as Zöllner says, as two-dimensional; on the contrary, it apprehends it either not at all, or it apprehends it as three-dimensional. Of course the child must first "learn how" to see. It is found from the observation of children during the first months of their lives, and of the congenitally

blind who have suddenly acquired the power of vision by some successful operation, that seeing does not consist alone in the irritations which arise in the optic nerves, but also in the correct interpretation of these irritations by reason. This correct interpretation, however, can be accomplished only by the accumulation of a considerable stock of experience. Especially must the recognition of the distance of the object seen be gradually learned. In this, two things are especially helpful; first, the fact that we have two eyes and, consequently, that we must feel two irritations of the optic nerves which are not wholly alike; and, secondly, the fact that we are enabled by our power of motion and our sense of touch to convince ourselves of the distance and form of the bodies seen. The question now arises, what sort of an intuition of space would a creature have that had only one eye, that could neither move itself nor its eye, and also possessed no peripheral nerves. According to Zöllner's view, this creature could, owing to its two-dimensional retinal images, have only a two-dimensional intuition of space. The author's opinion, however, is, that such a creature could not see at all, as it has no possibility of collecting experiences which are adapted in any way to interpreting the effects of things on its retina. The light which proceeded from the objects roundabout and fell on the retina could produce no other effect on the being than that of a wholly unintelligible irritation, or perhaps even pain.

The reflections presented sufficiently show that neither the phenomena of symmetry nor the retinal images of the objects of vision necessarily force upon us the assumption of a four-dimensioned space. If the material world should ever present problems which could not in the progress of knowledge be solved in a natural way, the assumption that a four-dimensional space containing the world exists would also be incompetent to resolve the difficulties presented; it would simply convert these difficulties into others, and not dispose of the problems but simply displace them to another world. Yet the question might be asked, is the existence of a four-dimensional space really *impossible?* To answer this question we must first clearly know what we mean by "exist." If existence means that the intellectual *idea* of a thing can be formed and that this idea shall not lead to contradictions with other well-established ideas and with experience, we have only to say that four-dimensioned space does exist, as the arguments adduced in sections III and IV have rendered plain. If, namely, the space of four

dimensions did not exist as a clear idea in the minds of mathematicians, mathematicians could certainly not have been led by this idea to results which are recognised by the senses as true, and which really take place in our own representable space. But if existence means "material actuality," we must say that we neither now nor in the future can know anything about it. For we know material actuality only as three-dimensional, our senses can only make three-dimensional experiences, and the inferences of our reason, although they can well abstract from material things, can never ascend to the point of explaining a four-dimensional materiality. Just as little, therefore, as we can locally fix the idea of a two-dimensional material world, as little can we ever verify the notion of a four-dimensional material existence.

VI.

EXAMINATION OF THE HYPOTHESIS CONCERNING THE EXISTENCE OF FOUR-DIMENSIONAL SPIRITS.

In connection with the belief that the visible world is contained in a four-dimensioned space, Zöllner and his adherents further hold that this higher space is inhabited by intelligent beings who can act consciously and at will on the human beings who live in experiential space. To invest this opinion with greater strength, Zöllner appealed to the fact that the greatest thinkers of antiquity and of modern times were either wholly of this opinion or at least held views from which his contentions might be immediately derived. Plato's dialogue between Socrates and Glaukon in the seventh book of the Republic, is evidence, says Zöllner, that this greatest philosopher of antiquity possessed some presentiment of this extension of the notion of space. Yet any one who has connectedly studied and understood Plato's system of philosophy must concede that the so-called "ideas" of the Platonic system denote something wholly different from what Zöllner sees in them or pretends to see. Zöllner says that these Platonic ideas are spatial objects of more than three dimensions and represent "real existence" in the same sense that the material world, as contrasted with the images on the retina, represents it. Zöllner similarly deals with the Kantian "thing-in-itself," which is also regarded as an object of higher dimensions.

To show Kant in the light of a predecessor, Zöllner quotes the following passage from the former's "Träume eines Geistersehers, erläutert

durch Träume der Metaphysik" (1766, *Collected Works*, Vol. VII. page 32 et seq.): "I confess that I am very much inclined to assert the existence of immaterial beings in the world, and to rank my own soul as one of such a class. It appears, there is a spiritual essence existent which is intimately bound up with matter but which does not act on those forces of the elements by which the latter are connected, but upon some internal principle of its own condition. It will, in the time to come—I know not when or where—be proved, that the human soul, even in this life, exists in a state of uninterrupted connection with all the immaterial natures of the spiritual world; that it alternately acts on them and receives impressions from them, of which, as a human soul, it is not, in the normal state of things, conscious. It would be a great thing, if some such systematic constitution of the spiritual world, as we conceive it, could be deduced, not exclusively from our general notion of spiritual nature, which is altogether too hypothetical, but from some real and universally admitted observations,—or, for that matter, if it could even be shown to be probable."

What Kant really asserts here is, first, the partly independent and partly dependent existence of the soul, and of spiritual beings generally, on matter, and, second, that spiritual beings have some common connection with and mutually influence one another. This contention, which is that of very many thinkers, does not, however, entail the consequence that the "transcendental subject of Kant" must be four-dimensional, as Zöllner asserts it does. Kant never even hinted at the theory that the psychical features of the world owe their connection with the material features to the fact that they are four-dimensional and, therefore, include the three-dimensional. Is it a necessary conclusion that if a thing exists and is not three-dimensional, as is the case with the soul, it is therefore four-dimensional? Can it not in fact be so constituted that it is wholly meaningless to speak of dimensions at all in connection with it?

Yet still more strangely than the words of Plato and Kant do certain utterances of the mathematicians Gauss and Riemann speak in favor of Zöllner's hypothesis. S. v. Waltershausen relates of Gauss in his *Gruss zum Gedächtniss*, (Leipsic, 1856), that Gauss had often remarked that the three dimensions of space were only a specific peculiarity of the human mind. We can think ourselves, he said, into beings who are conscious of only two dimensions; similarly, perhaps, beings who

are above and outside our world may look down upon us; and there were, he continued, in a jesting tone, a number of problems which he had here indefinitely laid aside, but hoped to treat in a superior state by superior geometrical methods. Leaving aside this jest, which quite naturally suggested itself, the remarks of Gauss are quite correct. We possess the power to abstract and can think, therefore, what kind of geometry a being that is only acquainted with a two-dimensional world would have; for instance, we can imagine that such a being could not conceive of the possibility of making two triangles coincide which were congruent in the sense above explained, and so on. So, also, we can understand that a being who has control of four dimensions can only conceive of a geometry of four-dimensional space, yet may have the capacity of thinking itself into spaces of other dimensions. But it does not follow from this that a four-dimensional space exists, let alone that it is inhabited by reasonable beings.

Riemann, on the other hand, speaks directly of a world of spirits. In his *Neue mathematische Principien der Naturphilosophie* he puts forth the hypothesis that the space of the world is filled with a material that is constantly pouring into the ponderable atoms, there to disappear from the phenomenal world. In every ponderable atom, he says, at every moment of time, there enters and appears a determinate amount of matter, proportional to the force of gravitation. The ponderable bodies, according to this theory, are the place at which the spiritual world enters and acts on the material world. Riemann's world of spirits, the sole office of which is to explain the phenomenon of gravitation as a force governing matter, is, however, essentially different from the spiritual world of Zöllner, the function of which is to explain supposed supersensuous phenomena which stand in the most glaring contradiction with the established known laws of the material world.

Besides this appeal to the testimony of eminent men like Plato, Kant, Gauss, and Riemann, the scientific prophet of modern spiritualism also bases his theory on the belief, which has obtained at all times and appeared in various forms among all peoples, that there exist in the world forces which at times are competent to evoke phenomena that are exempt from the ordinary laws of nature. We have but to think of the phenomena of table-turning which once excited the Chinese as much as it has aroused, during the last few decades, the European and

American worlds; or of the divining-rod, by whose help our forefathers sought for water, in fact, as we do now in parts of Europe and America.

Cranz, in his essay on the subject, divides spiritualistic phenomena into physical and intellectual. Of the first class he enumerates the following: the moving of chairs and tables; the animation of walking-sticks, slippers, and broomsticks; the miraculous throwing of objects; spirit-rappings (Luther heard a sound in the Wartburg, "as if three score casks were hurled down the stairs"); the ecstatic suspension of persons above the floor; the diminution of the forces of gravity; the ordeals of witches; the fetching of wished-for objects; the declination of the magnetic needle by persons at a distance; the untying of knots in a closed string; insensibility to injury and exemption therefrom when tortured, as in handling red-hot coals, carrying hot irons, etc.; the music of invisible spirits; the materialisations of spirits or of individual parts of spirits (the footprints in the experiments of Slade, photographed by Zöllner); the double appearance of the same person; the penetration of matter (of closed doors, windows, and so forth). As numerous also is the selection presented by Cranz of intellectual phenomena, namely: spirit-writing (Have's instrument for the facilitation of intercourse with spirits), the clairvoyance and divination of somnambulists, of visionary, ecstatic, and hypnotised persons, prompted or controlled by narcotic medicines, by sleeping in temples, by music and dancing, by ascetic modes of life and residence in barren localities, by the exudations of the soil and of water, by the contemplation of jewels, mirrors, and crystal-pure water, and by anointing the finger-nails with consecrated oil. Also the following additional intellectual phenomena are cited: increased eloquence or suddenly acquired power of speaking in foreign languages; spirit-effects at a distance; inability to move, transferences of the will, and so forth.

All these phenomena, presented with the aspect of truth, and associated more or less with trickery, self-deception, and humbug, are adduced by the spiritualists to substantiate the belief in a world of spirits which intentionally and consciously take part in the events of the material world, and to prove that these phenomena may be sufficiently and consistently explained by the effects of the activity of such a world. It is impossible for us to discuss and put to the test here the explanations of all these supersensuous phenomena. Anything and everything can be explained by spirits who act at will upon the world.

There are only a few of these phenomena, namely, clairvoyance and Slade's experiments, whose explanations are so intimately connected with our main theme, the so-called fourth dimension, that they cannot be passed over.

First, with respect to clairvoyance, the American visionary Davis describes the experiences which he claims to have made in this condition, induced by "magnetic sleep," as follows:* "The sphere of my vision now began to expand. At first, I could only clearly discern the walls of the house. At the start they seemed to me dark and gloomy; but they soon became brighter and finally transparent. I could now see the objects, the utensils, and the persons in the adjoining house as easily as those in the room in which I sat. But my perceptions extended further still; before my wandering glance, which seemed to control a great semi-circle, the broad surface of the earth, for hundreds of miles about me, grew as transparent as water, and I saw the brains, the entrails, and the entire anatomy of the beasts that wandered about in the forests of the Eastern Hemisphere, hundreds and thousands of miles from the room in which I sat." The belief in the possibility of such states of clairvoyance is by no means new. Alexander Dumas made use of it, for example, in his *Mémoires d'un médicin*, in which Count Balsamo, afterwards called Cagliostro, is said to possess the power to throw suitable persons into this wonderful condition and thus to find out what other persons at distant places are doing. Zöllner explains clairvoyance by means of the fourth dimension thus:

A man who is accustomed to viewing things on a plain is supposed to ascend to a considerable height in a balloon. He will there enjoy a much more extended prospect than if he had remained on the plain below, and will also be able to signal to greater distances. The plain, that is, the two-dimensioned space, is accordingly viewed by him from points outside of the plain as "open" in all directions. Exactly so, in Zöllner's theory, must three-dimensioned spaces appear, when viewed from points in four-dimensioned space, namely, as "open"; and the more so in proportion as the point in question is situated at a greater distance from the place of our body, or in proportion as the soul ascends to a greater height in this fourth dimension. Zöllner thus explains clairvoyance as a condition in which the soul has displaced itself out of its three-dimensioned space and reached a point which with respect

* Quoted by Cranz.

to this space is four-dimensionally situated and whence it is able to contemplate the three-dimensional world without the interference of intervening obstacles.

The following remark is to be made to this explanation. The reason why we have a better and more extended view from a balloon than from places on the earth is simply this, that between the suspended balloon and the objects seen at a distance nothing intervenes but the air, and air allows the transmission of light, whereas, at the places below on the earth there are all kinds of material things about the observer which prevent the transmission of light and either render difficult or absolutely impossible the sight of things which lie far away. In the same way, also, from a point in four-dimensioned space, a three-dimensional object will be visible only provided there are no obstacles intervening. If, therefore, this awareness of a distant object is a real, actual vision by means of a luminous ray which strikes the eye, there is contained in the explanation of Zöllner the tacit assumption that the medium with which the four-dimensional world is filled is also pervious to light exactly as the atmosphere is.

The theory that there are four-dimensional spirits who produce the phenomena cited by the spiritualists received special support from the experiments which the prestidigitateur Slade, who claimed he was a spiritualistic medium, performed before Zöllner. Of these experiments we will speak of the two most important, the experiment with the glass sphere and the experiment with the knots. To explain the connection of the glass sphere experiment with the fourth dimension, we must first conceive of two-dimensional reasoning beings, or, let us say, two-dimensional worms, living and moving in a plane. For a creature of this kind it will be self-evident that there are no other paths between two points of its plane than such as lie within the plane. It must, accordingly, be beyond the range of conception of this worm, how any two-dimensional object which lies within a circle in its space can be brought to any other position in its space outside the circle without the object passing through the barriers formed by the circle's circumference. But if this worm could procure the services of a three-dimensional being, the transportation of the object from a position within the circle to any position outside it could be effected by the three-dimensional being simply taking the object *out* of the plane and placing it at the desired point. This object, therefore, would, in an inexplicable manner,

suddenly disappear before the eyes of the worms who were assembled as spectators, and after the lapse of an interval of time would again appear outside the circle without having passed at any point through the circle's circumference. If now we add another dimension, we shall derive from this trick, which is wholly removed from the sense-perception of the flattened worms, the following experiment, which is wholly beyond the perception of us human beings. Inside a glass sphere, which is closed all around, a grain of corn is placed; the problem is to transport the corn to some place outside the sphere without passing through the glass surface. Now we should be able to perform this trick if some four-dimensional being would render us the same aid that we before rendered the two-dimensional worm. For the four-dimensional being could take the grain of corn into his four-dimensional space and then replace it in our space in the desired spot outside of the glass sphere. Slade performed this trick before Zöllner. Its mere performance sufficed to convince this latter investigator that Slade had here made use of a four-dimensional agent, who, in respect of power of motion, controlled his four-dimensional space as we do our three-dimensional space. It never occurred to Zöllner that this experiment was the cleverly executed trick of a prestidigitateur, or, as it would at once occur to us, that the whole thing was a sensory illusion. The fact that we cannot explain a trick easily and naturally does not irrevocably prove that it is accomplished by other means than those which the world of matter presents.

Still better known than this last performance is Slade's experiments with knots. To explain this in connection with the fourth dimension, we must resort again to the plane and the flat worm inhabiting it. To two parallel lines in a plane let the two ends of a third line, which has a double point, that is, intersects itself once, be attached. Our flat worm would not be able to untie the loop formed by the doubled third line, which we will call a string, because it cannot execute motions in three dimensions. If, therefore, a two-dimensional prestidigitateur should appear and accomplish the trick of untying this loop without removing the two ends of the string from the parallel lines, he will have accomplished for our flat worm a supersensuous experiment. A human being engaged in the service of the prestidigitateur could execute for him the experiment by simply lifting the string a little out of the plane, pulling it taut, and placing it back again. This suggests the following

analogous experiment for three-dimensional beings. The two ends of a string in which a common knot has been made are sealed to the opposite walls of a room. The problem is to untie this knot without breaking the seals at the two ends of the string. Everybody knows that this problem is not soluble, but it may be calculated mathematically that the knot in the string can be untied as easily by motions in a fourth dimension of space as in the experiment above described the knot in the two-dimensional string was untied by a three-dimensional motion. Now as Slade untied the knot before Zöllner's eyes without apparently making any use of the ends fastened in the walls, Zöllner was still more firmly confirmed in the view that Slade had power over spirits who performed the experiments for him.

Still more far-reaching is the theory of Carl du Prel concerning the relations of the material and the four-dimensional world. (Compare his numerous essays in the spiritualistic magazine *Sphinx*.) Just as the shadows of three-dimensional objects cast on a wall are controlled in their movements by the things whose projections they are, in the same way it is claimed does there exist back of everything of this sense-perceptible world a real transcendental and four-dimensional "thing-in-itself" whose projection in the space of experience is what we falsely regard as the independent thing. Thus every man besides existing in his terrestrial self also exists in a spiritual or astral self which constantly accompanies him in his walks through life and whose existence is especially proclaimed in states of profound sleep, of somnambulism, and in the conditions of mediums. In this way Du Prel explains the wonderful feats of somnambulists, and the aerial journeys of sorcerers and witches. Whereas, ordinarily the separation of the material body from the astral body is only effected at the moment of death; in the case of somnambulists this separation may take place at any time, or, as Du Prel says, "the threshold of feeling may be permanently displaced."

In view of the natural relations of such theories to the dogmas of Christianity it is explainable that theologians also have raised their voices for or against spiritualism. While the *Protestant Church Times* beheld in the "repulsive thaumaturgic performances which these coryphæi of modern science offer, a lack of comprehension of real philosophy," the magazine *The Proof of Faith* expresses its delight at the discovery of spiritualism in the following manner: "Every Christian will surely rejoice at the deep and perhaps mortal wound which

these new discoveries have in all probability administered to modern materialism."

We shall pass by the childish opinion that the Bible also speaks of four dimensions, as both in Job (xi. 8–9) and in the Epistle to the Ephesians (iii. 18) only breadth, length, depth, and height, that is, four directions of extension, are mentioned. Yet we will still add, as Cranz has done, the reflections which Zöllner, as the most prominent representative of modern spiritualism, has put forward respecting its relations to the doctrines of Christianity (*Wissensch. Abhandl.*, Vol. III). By the foundation of transcendental physics on the basis of spiritualistic phenomena, the "new light" has arisen which is spoken of in the New Testament. The rending of the veil of the Temple on the crucifixion of Christ, the resurrection, the ascension, the transfiguration, the speaking with many tongues on the giving out of the Holy Ghost, all these are in Zöllner's view spiritualistic phenomena. Similarly, he sees a reference to the four-dimensional world of spirits in all those sayings of Christ in which the latter speaks to his Apostles of the impossibility of their having any image or notion of the place to which when he disappeared he would go and whence he would return. (Gospel of St. John, xii. 33, 36; xiv. 2, 3, 28; xvi. 5, 13.)

Ulrici, however, goes farthest in the mingling of spiritualistic and Christian beliefs; for he sees in the doctrine of spiritualism a means of strengthening belief in a supreme moral world-order and in the immortality of the soul. In answer to Ulrici's tract "Spiritualism So-called, a Question of Science" (1889) Wundt wrote an annihilating reply bearing the title "Spiritualism, a Question of Science So-called." Wundt criticises the future condition of our soul according to spiritualistic hypotheses in the following sarcastic yet pertinent words, which Cranz also quotes: "(1) Physically, the souls of the dead come into the thraldom of certain living beings who are called mediums. These mediums are, for the present at least, a not widely diffused class, and they appear to be almost exclusively Americans. At the command of these mediums, departed souls perform mechanical feats which possess throughout the character of absolute aimlessness. They rap, they lift tables and chairs, they move beds, they play on the harmonica, and do other similar things. (2) Intellectually, the souls of the dead enter a condition which, if we are to judge from the productions which they deposit on the slates of the mediums, must be termed a very lamentable one. These

slate-writings belong throughout in the category of imbecility; they are totally bereft of any contents. (3) The most favored, apparently, is the moral condition of the soul. According to the testimony which we have, its character cannot be said to be anything else than that of harmlessness. From brutal performances, such, for instance, as the destruction of bed-canopies, the spirits most politely refrain." Wundt then laments the demoralising effect which spiritualism exercises on people who have hitherto devoted their powers to some serious pursuit or even to the service of science. In fact it is a presumptuous and flagrant procedure to forsake the path of exact research, which slow as it is, yet always leads to a sure extension of knowledge, in the hope that some aimless, foolhardy venture into the realm of uncertainty will carry us farther than the path of slow toil, and that we can ever thus easily lift the veil which hides from man the problems of the world that are yet unsolved.

* * *

Now that we have presented the opinions of others respecting the existence of a four-dimensional world of spirits, the author would like to develop one or two ideas of his own on the subject. In the preceding section it was stated that everything that we perceive by our senses is three-dimensional and that everything that possesses four or more dimensions can only be regarded as abstractions or fictions which our reason forms in its constant efforts after an extension and generalisation of knowledge. To speak of two-dimensional matter is as self-contradictory as the notion of four-dimensional matter. But a two or a four-dimensional world might exist in some other manner than a material manner, and for all we know in one which to us does not admit of representation. But in such a case, if it were without the power of affecting the material world, we should never be able to acquire any knowledge concerning its existence, and it would be totally indifferent to the people of the three-dimensional world, whether such a world existed or not. Just as an artist during his lifetime produces a number of different works of art, so also the Creator might have created a number of different-dimensioned worlds which in no wise interfere with one another. In such a case, any one world would not be able to know anything of any other, and we must consequently regard the question whether a four-dimensional world exists which is incapable of affecting ours, as insoluble. We have only to examine, therefore, the question whether the phenomenal world perhaps is a single individual

in a great layer of worlds of which every successive one has one more dimension than the preceding and which are so connected with one another that each successive world contains and includes the preceding world, and, therefore, can produce effects in it. For our reason, which draws its inferences from the phenomena of this world, tells us, that if outside the three-dimensional world there exists a second four-dimensional world, containing ours, there is no reason why worlds of more or less dimensions should not, with the same right, also exist. But if now, as Zöllner and his adherents maintain, four-dimensional spirits exist which can act by the mere efforts of their own wills on our world, there is surely no reason why we three-dimensional beings should not also be able to produce effects on some two-dimensional animated world. Whether we have, generally, any such influence we do not know, but we certainly do know that we do not act purposely and consciously on a two-dimensional world. If, therefore, Zöllner were right, the plan of creation would possess the wonderful feature that four-dimensional beings are capable of arbitrarily affecting the three-dimensional world, but that three-dimensional beings have no right in their turn consciously to affect a two-dimensional world.

The supporters of Zöllner's hypothesis will perhaps reply to the objection just made, that the plan of creation might, after all, possibly possess this wonderful peculiarity, that we human beings perhaps, in some higher condition of culture, will be able to act consciously on two-dimensional worlds, and that at any rate it is simply an inference by analogy to conclude from the non-existence of a relation between three and two dimensions that the same relation is also wanting in the case of four and three dimensions. As a matter of fact, the objection above made is not intended to refute Zöllner's hypothesis, but only to stamp it as very improbable. But despite this improbability Zöllner would still be right if the phenomena of the material world actually made his hypothesis necessary. That, however, is by no means the case. Although most of the phenomena to which the spiritualists appeal are probably founded on sense-illusions, humbug, and self-deception, it cannot be denied that there possibly do exist phenomena which cannot be brought into harmony with the natural laws now known. There always have been mysterious phenomena, and there always will be. Yet, as we have often seen that the progress of science has again and again revealed as natural what former generations held to be supernatural, it

is certainly wholly wrong to bring in for the explanation of phenomena which now seem mysterious an hypothesis like that of Zöllner, by which everything in the world can be explained. If we adopt a point of view which regards it as natural for spirits arbitrarily to interfere in the workings of the world, all scientific investigation will cease, for we could never more trust or rely upon any chemical or physical experiment, or any botanical or zoölogical culture. If the spirits are the authors of the phenomena that are mysterious to us, why should they also not have control of the phenomena which are not mysterious? The existence of mysterious phenomena justifies in no manner or form the assumption that spirits exist which produce them. Would it not be much simpler, if we *must* have supernatural influences, to adopt the naïve religious point of view, according to which everything that happens is traceable to the direct, actual, and personal interference of a single being which we call God? Things which formerly stood beyond the sphere of our knowledge and were regarded as marvellous events, as a storm, for example, now stand in the most intimate connection with known natural laws. Things that formerly were mysterious are so no longer. If one hundred and fifty years ago some scientists were in the possession of our present knowledge of inductional electricity and had connected Paris and Berlin with a wire by whose aid one could clearly interpret in Berlin what another person had at that very moment said in Paris, people would have regarded this phenomenon as supernatural and assumed that the originator of this long-distance speaking was in league with spirits.

We recognise, thus, that the things which are termed supernatural depend to a great extent on the stage of culture which humanity has reached. Things which now appear to us mysterious, may, in a very few decades, be recognised as quite natural. This knowledge, however, is not to be obtained by the lazy assumption of bands of spirits as the authors of mysterious phenomena, but by performing what in physics and chemistry is called experiment. But the first and essential condition of all scientific experimenting is that the experimenter shall be absolutely master of the conditions under which the experiment is or is not to succeed. Now, this criterion of scientific experimenting is totally lacking in all spiritualistic experiments. We can never assign in their case the conditions under which they will or will not succeed. When all the preparations in a spiritualistic *séance* have been properly made, but nothing takes place, the beautiful excuse is always forthcoming that

the "spirits were not willing," that there were "too many incredulous persons present," and so forth. Fortunately, in physical experiments these pretexts are not necessary. By the path of experiment, and not by that of transcendental speculation, physics has thus made incredible progress and has piled new knowledge strata on strata upon the old. Accordingly, the prospect is left that the mysteries which the conditions and properties of the human soul still present can be solved more and more by the methods of scientific experiment. To this end, however, it is especially necessary that the physio-psychological experiments in question should only be performed by men who possess the critical eye of inquiry, who are free from the dangers of self-illusion, and who are competent to keep apart from their experiments all superstition and deception. The history of natural science clearly teaches that it is only by this road that man can arrive at certain and well-established knowledge. If, therefore, there really is behind such phenomena as mind-reading, telepathy, and similar psychical phenomena, something besides humbug and self-illusion, what we have to do is to study privately and carefully by serious experiments the success or non-success of such phenomena, and not allow ourselves to be influenced by the public and dramatic performances of psychical artists, like Cumberland and his ilk.

The high eminence on which the knowledge and civilisation of humanity now stands was not reached by the thoughtless employment of fanciful ideas, nor by recourse to four-dimensional worlds, but by hard, serious labor, and slow, unceasing research. Let all men of science, therefore, band themselves together and oppose a solid front to methods that explain everything that is now mysterious to us by the interference of independent spirits. For these methods, owing to the fact that they can explain everything, explain nothing, and thus oppose dangerous obstacles to the progress of real research, to which we owe the beautiful temple of modern knowledge.

THE SQUARING OF THE CIRCLE.

AN HISTORICAL SKETCH OF THE PROBLEM FROM THE REMOTEST TIMES.

I.

UNIVERSAL INTEREST IN THE PROBLEM.

For two and a half thousand years, both competent and incompetent minds have striven in vain to solve the problem known as the squaring of the circle. Now that geometers have at last succeeded in giving a rigorous demonstration of the impossibility of solving the problem with straight edge and compasses, it seems fitting and opportune to cast a glance into the nature and history of this very ancient problem. And this will be found the more justifiable in view of the fact that the squaring of the circle, at least in name, is very widely known outside of the narrow circle of professional mathematicians.

The *Proceedings of the French Academy* for the year 1775 contain at page 61 a resolution of the Academy not to examine from that time on, any so-called solutions of the quadrature of the circle. The Academy was driven to this determination by the overwhelming multitude of professed solutions of the famous problem which were sent to it every month in the year,—solutions which of course were an invariable attestation of the ignorance and self-conceit of their authors, but which suffered collectively from the very important drawback in mathematics of being *wrong*. Since that time all professed solutions of the problem received by the Academy find a sure and permanent resting-place in the waste-basket, and remain unanswered for all time. The circle-squarer,

however, sees in this high-handed manner of rejection only the envy of the great and powerful at his grand intellectual discovery. He is determined to secure recognition, and appeals therefore to the public. The newspapers must obtain for him the appreciation that scientific societies have denied. And every year the old mathematical sea-serpent more than once disports itself in the columns of our newspapers in the shape of an announcement that Mr. N. N., of P. P., has at last solved the problem of the quadrature of the circle.

But what manner of people are these circle-squarers, when examined by the light? Almost always they will be found to be imperfectly educated persons, whose mathematical knowledge does not exceed that of a modern high-school student. It is seldom that they know accurately what the requirements of the problem are and what its nature; they are totally ignorant of the two and a half thousand years' history of the problem; and they have no idea whatever of the important investigations which have been made with regard to it by great and real mathematicians in every century down to our own time.*

Yet great as is the quantum of ignorance that circle-squarers intermix with their intellectual products, the lavish supply of conceit and egotism with which they season their performances is still greater. I have not far to go to furnish a verification of this. A book printed in Hamburg in the year 1840 lies before me, in which the author thanks Almighty God at every second page that He has selected him and no one else to solve the "problem phenomenal" of mathematics, "so long sought for, so fervently desired, and attempted by millions." After this modest author has proclaimed himself the unmasker of Archimedes's deceit, he says: "And thus it hath pleased our mother Nature to withhold this precious mathematical jewel from the eye of human investigation, until she thought it fitting to reveal truth to simplicity."

This will suffice to show the great fatuity of the author. But it does not suffice to prove his ignorance. He has no conception of mathematical demonstration; he takes it for granted that things are so because they seem so to him. Errors of logic, also, abound in his book. But, minor fallacies apart, wherein does the real error of this "unmasker" of Archimedes consist? It requires considerable labor to extricate the kernel of the demonstration from the turgid language and bombastic

* For the full psychogeny and psychiatry of the circle-squarer see A. De Morgan, *A Budget of Paradoxes* (London, 1872).—*Tr.*

style in which the author has buried his conclusions. But it is this. The author inscribes a square in a circle, circumscribes another about it, then points out that the inside square is made up of four congruent triangles, whereas the circumscribed square is made up of eight such triangles; from which fact, seeing that the circle is larger than the one square and smaller than the other, he draws the bold conclusion that the circle is equal in area to six such triangles. It is hardly conceivable that a rational being could infer that something which is greater than 4 and less than 8 must necessarily be 6. But with a man that attempts the squaring of the circle this kind of ratiocination *is* possible.

It is the same with all the other attempted solutions of the problem; in all of them either logical fallacies or violations of elementary arithmetical or geometrical truths can be pointed out. Only they are not always of such a trivial nature as in the book just mentioned.

Let us now inquire into the origin of this propensity which leads people to occupy themselves with the quadrature of the circle.

Attention must first be called to the antiquity of the problem. A quadrature was attempted in Egypt 500 years before the exodus of the Israelites. Among the Greeks the problem never ceased to play a part that greatly influenced the progress of mathematics. And in the middle ages also the squaring of the circle sporadically appears as the philosopher's stone of mathematics. The problem has thus never ceased to be dealt with and considered. But it is not by the antiquity of the problem that circle-squarers are enticed, but by the allurement which everything exerts that is calculated to raise the individual above the mass of ordinary humanity, and to bind about his temples the laurel crown of celebrity. Ambition spurred men on in ancient Greece and still spurs them on in modern times to crack this primeval mathematical nut. Whether they are competent thereto is a secondary consideration. They look upon the squaring of the circle as the grand prize of a lottery that can just as well fall to their lot as that of any other man. They do not remember that—

"Toil before honor is placed by sagacious decrees of Immortals."

and that it requires years of consecutive study to gain possession of the mathematical weapons that are indispensably necessary to attack the problem, but which even in the hands of the most distinguished mathematical strategists did not suffice to take the stronghold.

But why is it, we must further ask, that it happens to be the squaring of the circle and not some other unsolved mathematical problem upon which the efforts of people are bestowed who have no knowledge of mathematics yet busy themselves with mathematical questions? The question is answered by the fact that the squaring of the circle is about the only mathematical problem that is known to the unprofessional world,—at least by name. Even among the Greeks the problem was very widely known outside of mathematical circles. In the eyes of the Grecian layman, as at present among many of his modern brethren, occupation with this problem was regarded as the most important and essential business of mathematicians. In fact, they had a special word to designate this species of activity, namely, τετραγωνίζειν, which means to busy one's self with the quadrature. In modern times, also, every educated person, though he be not a mathematician, knows the problem by name, and knows that it is insoluble, or at least, that despite the efforts of the most famous mathematicians it has not yet been solved. For this reason the phrase "to square the circle," is now generally used in the sense of attempting the impossible.

But in addition to the antiquity of the problem, and the fact also that it is known to the lay world, there is an important third factor that induces people to occupy themselves with it. This is the report that has been current for more than a century now, that the Academies, the Queen of England, or some other influential person has offered a large prize to be given to the one that first solves the problem. As a matter of fact, the hope of obtaining this large prize of money is with many circle-squarers the principal incitement to action. And the author of the book above referred to begs his readers to lend him their assistance in obtaining the prizes offered.

Although the opinion is widely current in the unprofessional world, that professional mathematicians are still busied with the solution of the problem, this is by no means the case. On the contrary, for some two hundred years, the endeavors of many great mathematicians have been directed solely towards demonstrating with exactness that the problem is insoluble. It is, as a rule,—and naturally,—more difficult to prove that a thing is impossible than to prove that it is possible. And thus it has happened, that up to within a few years ago, despite the employment of the most varied and the most comprehensive methods

of modern mathematics, no one succeeded in supplying the wished-for demonstration of the problem's impossibility. At last, Professor Lindemann, of Königsberg, in June, 1882, succeeded in furnishing a demonstration,—and the first demonstration,—that it is impossible by employing only straight edge and compasses to construct a square that is mathematically exactly equal in area to a given circle. The demonstration, naturally, was not effected with the help of the old elementary methods; for if it were, it would have been accomplished centuries ago; but methods were requisite that were first furnished by the theory of definite integrals and departments of higher algebra developed in the last few decades; in other words, it required the direct and indirect preparatory labor of many centuries to make finally possible a demonstration of the insolubility of this historic problem.

Of course, this demonstration will have no more effect than the resolution of the Paris Academy of 1775, in causing the fecund race of circle-squarers to vanish from the face of the earth. In the future as in the past, there will be people who know nothing of this demonstration and will not care to know anything, and who believe that they cannot help succeeding in a matter in which others have failed, and that just they have been appointed by Providence to solve the famous puzzle. But unfortunately the ineradicable mania for solving the quadrature of the circle has also its serious side. Circle-squarers are not always so self-satisfied as the author of the book above mentioned. They often see, or at least divine, the insuperable difficulties that tower up before them, and the conflict between their aspirations and their performances, the consciousness that the problem they long to solve they are unable to solve, darkens their soul and, lost to the world, they become interesting subjects for the science of psychiatry.

II.

NATURE OF THE PROBLEM.

It is easy to determine the length of the radius of a circle, or the length of its diameter, which must be double that of the radius; and the question next arises, what is the number that tells how many times larger the circumference of the circle, that is the length of the circular line, is than its radius or its diameter. From the fact that all circles have the same shape it follows that this proportion will be the same for all

circles both large and small. Now, since the time of Archimedes, all civilised nations that have cultivated mathematics have denoted the number that tells how many times larger the circumference of a circle is than the diameter by the symbol π,—the Greek initial letter of the word periphery.* To compute π, therefore, means to calculate how many times larger the circumference of a circle is than its diameter. This calculation is called "the numerical rectification of the circle."

Next to the calculation of the circumference, the calculation of the superficial contents of a circle by means of its radius or diameter is perhaps most important; that is, the computation of how great an area that part of a plane which lies within a circle measures. This calculation is called the "numerical quadrature." It depends, however, upon the problem of numerical rectification; that is, upon the calculation of the magnitude of π. For it is demonstrated in elementary geometry, that the area of a circle is equal to the area of a triangle produced by drawing in the circle a radius, erecting at the extremity of the same a tangent,—that is, in this case, a perpendicular,—cutting off upon the latter the length of the circumference, measuring from the extremity, and joining the point thus obtained with the centre of the circle. It follows from this that the area of a circle is as many times larger than the square upon its radius as the number π amounts to.

The numerical rectification and numerical quadrature of the circle based upon the computation of the number π are to be clearly distinguished from problems that require a straight line equal in length to the circumference of a circle, or a square equal in area to a circle, to be *constructively* produced from its radius or its diameter; problems which might properly be called "constructive rectification" or "constructive quadrature." Approximately, of course, by employing an approximate value for π, these problems are easily solvable. But to solve a problem of construction in geometry, means to solve it with mathematical exactitude. If the value π were exactly equal to the ratio of two whole numbers to each other, the constructive rectification would present no difficulties. For example, suppose the circumference of a circle were exactly $3\frac{1}{7}$ times greater than its diameter; then the diameter could be divided into seven equal parts, which could easily be done by the

* The Greek symbol π was first employed by W. Jones in 1706 and did not come into general use until about the middle of the eighteenth century through the works of Euler.—*Trans.*

principles of planimetry with straight edge and compasses; then by prolonging to the amount of such a part a straight line exactly three times as long as the diameter, we should obtain a straight line exactly equal to the circumference of the circle. But as a matter of fact,—and this has actually been demonstrated,—there do not exist two whole numbers, be they ever so great, that exactly represent by their proportion to each other the number π. Consequently, a rectification of the kind just described does not attain the object desired.

It might be asked here, whether from the demonstrated fact that the number π is not equal to the ratio of two whole numbers however great, it does not immediately follow that it is impossible to construct a straight line exactly equal in length to the circumference of a circle; thus demonstrating at once the impossibility of solving the problem. This question is to be answered in the negative. For in geometry there can easily exist pairs of lines of which the one can be readily constructed from the other, notwithstanding the fact that no two whole numbers can be found to represent the ratio of the two lines. The side and the diagonal of a square, for instance, are so constituted. It is true the ratio of the latter two magnitudes is nearly that of 5 to 7. But this proportion is not exact, and there are in fact no two numbers that represent the ratio exactly. Nevertheless, either of these two lines can be readily constructed from the other by employing only straight edge and compasses. This might be the case, too, with the rectification of the circle; and consequently from the impossibility of representing π by the ratio between two whole numbers the impossibility of the problem of rectification is not inferable.

The quadrature of the circle stands and falls with the problem of rectification. This rests upon the truth above mentioned, that a circle is equal in area to a right-angled triangle, in which one side is equal to the radius of the circle and the other to the circumference. Supposing, accordingly, that the circumference of the circle had been rectified, then we could construct this triangle. But every triangle, as we know from plane geometry, can, with the help of straight edge and compasses be converted into a square exactly equal to it in area. So that, supposing the rectification of the circumference of a circle to have been successfully effected, a square could be constructed that would be exactly equal in area to the circle.

The dependence upon one another of the three problems of the computation of the number π, the quadrature of the circle, and its rectification, thus obliges us, in dealing with the history of the quadrature, to regard investigations with respect to the value of π and attempts to rectify the circle as of equal importance, and to consider them accordingly.

We have used repeatedly in the course of this discussion the expression "to construct with straight edge and compasses." It will be necessary to explain what is meant by the specification of these two instruments. When to a requirement in geometry to construct a figure there are so large a number of conditions annexed that the construction of only *one* figure or a limited number of figures is possible in accordance with those conditions; such a full and stated requirement is called a problem of construction, or briefly a problem. When a problem of this kind is presented for solution it is necessary to reduce it to simpler problems, already recognised as solvable; and since these latter depend in their turn upon other, still simpler problems, we are finally brought back to certain fundamental problems upon which the rest are based but which are not themselves reducible to problems less simple. These fundamental problems are, so to speak, the lowermost stones of the edifice of geometrical construction. The question next arises as to what problems may be properly regarded as fundamental; and it has been found, that the solution of a great part of the problems that arise in elementary plane geometry rests upon the solution of only five original problems. They are:

1. The construction of a straight line that shall pass through two given points.

2. The construction of a circle the centre of which is a given point and the radius of which has a given length.

3. The determination of the point lying coincidently on two given straight lines prolonged as far as necessary,—in case such a point (point of intersection) exists.

4. The determination of the two points that lie coincidently on a given straight line and a given circle,—in case such common points (points of intersection) exist.

5. The determination of the two points that lie coincidently on two given circles,—in case such common points (points of intersection) exist.

For the solution of the three last of these five problems the eye alone is needed, while for the solution of the first two problems, besides pencil, ink, chalk, or the like, additional special instruments are required: for the solution of the first problem a straight edge or ruler is most generally used, and for the solution of the second a pair of compasses. But it must be remembered that it is no concern of geometry what mechanical instruments are employed in the solution of the five problems mentioned. Geometry simply limits itself to the presupposition that these problems are solvable, and regards a complicated problem as solved if, upon a specification of the constructions of which the solution consists, no other requirements are demanded than the five above mentioned. Since, accordingly, geometry does not itself furnish the solution of these five problems, but rather exacts them, they are termed *postulates*.* All problems of plane geometry are not reducible to these five problems alone. There are problems that can be solved only by assuming other problems as solvable which are not included in the five given; for example, the construction of an ellipse, having given its centre and its major and minor axes. Many problems, however, possess the property of being solvable with the assistance of the above-formulated five postulates alone, and where this is the case they are said to be "constructible with straight edge and compasses," or "elementarily" constructible.

After these general remarks upon the solvability of problems of geometrical construction, which an understanding of the history of the squaring of the circle makes indispensable, the significance of the question whether the quadrature of the circle is or is not solvable, that is elementarily solvable, will become intelligible. But the conception of elementary solvability only gradually took clear form, and we therefore find among the Greeks as well as among the Arabs endeavors, successful in some respects, that aimed at solving the quadrature of the circle with other expedients than the five postulates. We have also to take these endeavors into consideration, and especially so as they, no less than the unsuccessful efforts at elementary solution, have upon the

* Usually geometers mention only two postulates (Nos. 1 and 2). But since to geometry proper it is indifferent whether only the eye, or additional special mechanical instruments are necessary, the author has regarded it more correct in point of method to assume five postulates.

whole advanced the science of geometry, and contributed much to the clarification of geometrical ideas.

III.

THE EGYPTIANS, BABYLONIANS, AND GREEKS.

In the oldest mathematical work that we possess we find a rule telling us how to construct a square which is equal in area to a given circle. This celebrated book, the Rhind Papyrus of the British Museum, translated and explained by Eisenlohr (Leipsic, 1877), was written, as stated in the work itself, in the thirty-third year of the reign of King Ra-a-us, by a scribe of that monarch, named Ahmes. The composition of the work falls accordingly in the period of the two Hyksos dynasties, that is, in the period between 2000 and 1700 B. C. But there is another important circumstance to be noted. Ahmes mentions in his introduction that he composed his work after the model of old treatises, written in the time of King Raenmat; whence it appears that the originals of the mathematical expositions of Ahmes are half a thousand years older still than the Rhind Papyrus.

The rule given in this papyrus for obtaining a square equal to a circle specifies that the diameter of the circle shall be shortened one-ninth of its length and upon the shortened line thus obtained a square erected. Of course, the area of a square of this construction is only approximately equal to the area of the circle. An idea may be obtained of the degree of exactness of this original, primitive quadrature by remarking, that if the diameter of the circle in question is one metre in length, the square that is supposed to be equal to the circle is a little less than half a square decimetre too large; an approximation not so accurate as that made by Archimedes, yet much more correct than many a one later employed. It is not known how Ahmes or his predecessors arrived at this approximate quadrature; but it is certain that it was handed down in Egypt from century to century, and in late Egyptian times it appears repeatedly.

In addition to the effort of the Egyptians, we also find in pre-Grecian antiquity an attempt at circle-computation among the Babylonians. This is not a quadrature, but is intended as a rectification of the circumference. The Babylonian mathematicians had discovered, that if the radius of a circle be successively inscribed as a chord within

its circumference, after the sixth inscription we arrive at the point from which we set out, and they concluded from this that the circumference of a circle must be a little larger than a line which is six times as long as the radius, that is three times as long as the diameter. A trace of this Babylonian method of computation may even be found in the Bible; for in 1 Kings vii. 23, and 2 Chron. iv. 2, the great laver is described, which under the name of the "molten sea" constituted an ornament of the temple of Solomon; and it is said of this vessel that it measured ten cubits from brim to brim, and thirty cubits round about. The number 3 as the ratio of the circumference to the diameter is still more plainly given in the Talmud, where we read that "that which measures three lengths in circumference is one length across."

With regard to the earlier Greek mathematicians—as Thales and Pythagoras—we know that they acquired their elementary mathematical knowledge in Egypt. But nothing has been handed down to us which shows that they knew of the old Egyptian quadrature, or that they dealt with the problem at all. But tradition says, that, subsequently, the teacher of Euripides and Pericles, the great philosopher and mathematician Anaxagoras, whom Plato so highly praised, "drew the quadrature of the circle" in prison, in the year 434 B. C. This is the account of Plutarch in the seventeenth chapter of his work *De Exilio*. The method is not told us in which Anaxagoras is supposed to have solved the problem, and it is not said whether, knowingly or unknowingly, he gave an approximate solution after the manner of Ahmes. But at any rate, to Anaxagoras belongs the merit of having called attention to a problem that was to bear rich fruit by inciting Grecian scholars to busy themselves with geometry, and thus more and more to advance that science.

Again, it is reported that the mathematician Hippias of Elis invented a curved line that could be made to serve a double purpose: first, to trisect an angle, and second to square the circle. This curved line is the τετραγωνίζουσα so often mentioned by the later Greek mathematicians, and by the Romans called "quadratrix." Regarding the nature of this curve we have exact knowledge from Pappus. But it will be sufficient here to state that the quadratrix is not a circle nor a portion of a circle, so that its construction is not possible by means of the postulates enumerated in the preceding section. And therefore the solution of the quadrature of the circle founded on the construction

of the quadratrix is not an elementary solution in the sense discussed in the last section. We can, it is true, conceive a mechanism that will draw this curve as well as compasses draw a circle; and with the assistance of a mechanism of this description the squaring of the circle is solvable with exactitude. But if it be allowed to employ in a solution an apparatus especially adapted thereto, every problem may be said to be solvable. Strictly taken, the invention of the curve of Hippias substitutes for one insuperable difficulty another equally insuperable. Some time afterwards, about the year 350 B. C., the mathematician Dinostratus showed that the quadratrix could also be used to solve the problem of rectification, and from that time on this problem plays almost the same rôle in Grecian mathematics as the related problem of quadrature.

As these problems gradually became known to the non-mathematicians of Greece, attempts at solution at once sprang up that are worthy of a place by the side of the solutions of modern amateur circle-squarers. The Sophists especially believed themselves competent by seductive dialectic to take the stronghold that had defied the intellectual onslaughts of the greatest mathematicians. With verbal nicety, amounting to puerility, it was said that the squaring of the circle depended upon the finding of a number which represented in itself both a square and a circle; a square by being a square number, a circle in that it ended with the same number as the root number from which, by multiplication with itself, it was produced. The number 36, accordingly, was, as they thought, the one that embodied the solution of the famous problem.

Contrasted with this twisting of words the speculations of Bryson and Antiphon, both contemporaries of Socrates, though inexact, appear in a high degree intelligent. Antiphon divided the circle into four equal arcs, and by joining the points of division obtained a square; he then divided each arc again into two equal parts and thus obtained an inscribed octagon; thence he constructed an inscribed 16-gon, and perceived that the figure so inscribed more and more approached the shape of a circle. In this way, he said, one should proceed, until there was inscribed in the circle a polygon whose sides by reason of their smallness should coincide with the circle. Now this polygon could, by methods already taught by the Pythagoreans, be converted into a square of equal

area; and upon the basis of this fact Antiphon regarded the squaring of the circle as solved.

Nothing can be said against this method except that, however far the bisection of the arcs is carried, the result still remains an approximate one.

The attempt of Bryson of Heraclea was better still; for this scholar did not rest content with finding a square that was very little smaller than the circle, but obtained by means of circumscribed polygons another square that was very little larger than the circle. Only Bryson committed the error of believing that the area of the circle was the arithmetical mean between an inscribed and a circumscribed polygon of an equal number of sides. Notwithstanding this error, however, to Bryson belongs the merit—first, of having introduced into mathematics by his emphasis of the necessity of a square which was too large and one which was too small, the conception of upper and lower "limits" in approximations; and secondly, by his comparison of the regular inscribed and circumscribed polygons with a circle, of having indicated to Archimedes the way by which an approximate value of π was to be reached.

Not long after Antiphon and Bryson, Hippocrates of Chios treated the problem, which had now become more and more famous, from a new point of view. Hippocrates was not satisfied with approximate equalities, and searched for curvilinearly bounded plane figures which should be mathematically equal to a rectilinearly bounded figure, and which therefore could be converted by straight edge and compasses into a square equal in area. First, Hippocrates found that the crescent-shaped plane figure produced by drawing two perpendicular radii in a circle and describing upon the line joining their extremities a semicircle, is exactly equal in area to the triangle that is formed by this line of junction and the two radii; and upon the basis of this fact the endeavors of this untiring scholar were directed towards converting a circle into a crescent. Naturally he was unable to attain this object, but by his efforts he discovered many new geometrical truths; among others being the generalised form of the theorem mentioned, which bears to the present day the name of *lunulae Hippocratis*, the lunes of Hippocrates. Thus, in the case of Hippocrates, it appears in the plainest light, how precisely the insolvable problems of science are qualified to

advance science; in that they incite investigators to devote themselves with persistence to its study and thus to fathom its utmost depths.

Following Hippocrates in the historical line of the great Grecian geometricians comes the systematist Euclid, whose rigid formulation of geometrical principles has remained the standard presentation down to the present century. The Elements of Euclid, however, contain nothing relating to the quadrature of the circle or to circle-computation. Comparisons of surfaces which relate to the circle are indeed found in the work, but nowhere a computation of the circumference of a circle or of the area of a circle. This palpable gap in Euclid's system was filled by Archimedes, the greatest mathematician of antiquity.

Archimedes was born in Syracuse in the year 287 B. C., and devoted his life, which was spent in that city, to the mathematical and the physical sciences, which he enriched with invaluable contributions. He lived in Syracuse till the taking of the town by Marcellus, in the year 212 B. C., when he fell by the hand of a Roman soldier whom he had forbidden to destroy the figures he had drawn in the sand. To the greatest performances of Archimedes the successful computation of the number π unquestionably belongs. Like Bryson he started with regular inscribed and circumscribed polygons. He showed how it was possible, beginning with the perimeter of an inscribed hexagon, which is equal to six radii, to obtain by calculation the perimeter of a regular dodecagon, and then the perimeter of a figure having double the number of sides of that, and so on. Treating, then, the circumscribed polygons in a similar manner, and proceeding with both series of polygons up to a regular 96-sided polygon, he discovered on the one hand that the ratio of the perimeter of the inscribed 96-sided polygon to the diameter was greater than $6336 : 2017\,{}^1\!/_4$, and on the other hand, that the corresponding ratio with respect to the circumscribed 96-sided polygon was smaller than $14688 : 4673\,{}^1\!/_2$. He inferred from this, that the number π, the ratio of the circumference to the diameter, was greater than the fraction $\frac{6336}{2017\frac{1}{4}}$ and smaller than $\frac{14688}{4673\frac{1}{2}}$. Reducing the two limits thus found for the value of π, Archimedes then showed that the first fraction was greater than $3\frac{10}{71}$, and that the second fraction was smaller than $3\frac{1}{7}$, whence it followed with certainty that the value sought for π lay between $3\frac{1}{7}$ and $3\frac{10}{71}$. The larger of these two approximate values is the only one usually learned and employed. That which fills us with most astonishment in the case of Archimedes's computation of π, is, first, the great acumen

and accuracy displayed by him in all the details of the computation, and secondly the unwearied perseverance which he exercised in calculating the limits of π without the help of the Arabian system of numerals and the decimal notation. For it must be considered that at many stages of the computation what we call the extraction of roots was necessary, and that Archimedes could only by extremely tedious calculations obtain ratios that expressed approximately the roots of given numbers and fractions.*

With regard to the mathematicians of Greece that follow Archimedes, all refer to and employ the approximate value of $3\frac{1}{7}$ for π, without, however, contributing anything essentially new to the problems of quadrature and of cyclometry. Thus Hero of Alexandria, the father of surveying, who flourished about the year 100 B. C., employs for purposes of practical measurement sometimes the value $3\frac{1}{7}$ for π and sometimes even the rougher approximation $\pi = 3$. The astronomer Ptolemy, who lived in Alexandria about the year 150 A. D., and who was famous as being the author of the planetary system universally recognised as correct down to the time of Copernicus, was the only one who furnished a more exact value; this he designated, in the sexagesimal system of fractional notation which he employed, by 3, 8, 30,—that is 3 and $\frac{8}{60}$ and $\frac{30}{3600}$, or as we now say 3 degrees, 8 minutes (*partes minutae primae*), and 30 seconds (*partes minutae secundae*). As a matter of fact, the expression $3 + \frac{8}{60} + \frac{30}{3600} = 3\frac{17}{120}$ represents the number π more exactly than $3\frac{1}{7}$; but on the other hand, is, by reason of the magnitude of the numbers 17 and 120 as compared with the numbers 1 and 7, more cumbersome.

IV.

THE ROMANS, HINDUS, CHINESE, ARABS, AND THE CHRISTIAN NATIONS TO THE TIME OF NEWTON.

In the mathematical sciences, more than in any other, the Romans stood upon the shoulders of the Greeks. Indeed, with respect to cyclometry, they not only did not add anything new to the Grecian discoveries, but frequently even evinced that they either did not know

* For Archimedes's actual researches, see Rudio, *Archimedes, Huygens, Lambert, Legendre, vier Abhand. über die Kreismessung* (Leipsic, 1892), where translations of the works of these four authors on cyclometry will be found.—*Tr.*

of the beautiful result obtained by Archimedes, or at least could not appreciate it. For instance, Vitruvius, who lived during the time of Augustus, computed that a wheel 4 feet in diameter must measure $12\frac{1}{2}$ feet in circumference; in other words, he made π equal to $3\frac{1}{8}$. And, similarly, a treatise on surveying, preserved to us in the Gudian manuscript of the library of Wolfenbüttel, contains the following instructions for squaring the circle: Divide the circumference of a circle into four parts and make one part the side of a square; this square will be equal in area to the circle. Apart from the fact that the rectification of the arc of a circle is requisite to the construction of a square of this kind, the Roman quadrature, viewed as a calculation, is more inexact even than any other known computation; for its result is that $\pi = 4$.

The mathematical performances of the Hindus were not only greater than those of the Romans, but in certain directions surpassed even those of the Greeks. In the most ancient source of the mathematics of India that we know of, the Culvasûtras, which date back to a little before our chronological era, we do not find, it is true, the squaring of the circle treated of, but the opposite problem is dealt with, which might fittingly be termed the circling of the square. The half of the side of a given square is prolonged in length one third of the excess of half the diagonal over half the side, and the line thus obtained is taken as the radius of the circle equal in area to the square. The simplest way to obtain an idea of the exactness of this construction is to compute how great π would have to be if the construction were exactly correct. We find out in this way that the value of π upon which the Indian circling of the square is based, is about from five to six hundredths smaller than the true value, whereas the approximate π of Archimedes, $3\frac{1}{7}$, is only from one to two thousandths too large, and that the old Egyptian value exceeds the true value by from one to two hundredths.

Cyclometry very probably made great advances among the Hindus in the first four or five centuries of our era; for Aryabhatta, who lived about the year 500 after Christ, states, that the ratio of the circumference to the diameter is 62832 : 20000, an approximation that in exactness surpasses even that of Ptolemy. The Hindu result gives 3.1416 for π, while π really lies between 3.141592 and 3.141593. How the Hindus obtained this excellent value is told by Ganeça, the commentator of Bhâskara, an author of the twelfth century. Ganeça says

that the method of Archimedes was carried still farther by the Hindu mathematicians; that by continually doubling the number of sides they proceeded from the hexagon to a polygon of 384 sides, and that by the comparison of the circumferences of the inscribed and circumscribed 384-sided polygons they found that π was equal to $3927 : 1250$. It will be seen that the value given by Bhâskara is identical with the value of Aryabhatta. It is further worthy of remark that the earlier of these two Hindu mathematicians does not mention either the value $3\frac{1}{7}$ of Archimedes or the value $3\frac{17}{120}$ of Ptolemy, but that the later one knows of both values and especially recommends that of Archimedes as the most useful for practical applications. Strange to say, the good approximate value of Aryabhatta does not occur in Brahmagupta, the great Hindu mathematician who flourished in the beginning of the seventh century; but we find the curious information in this author that the area of a circle is exactly equal to the square root of 10 when the radius is unity. The value of π as derivable from this formula,—a value from two to three hundredths too large,—has unquestionably arisen on Hindu soil. For it occurs in no Grecian mathematician; and Arabian authors, who were in a better position than we to know Greek and Hindu mathematical literature, declare that the approximation which makes π equal to the square root of 10, is of Hindu origin. It is possible that the Hindu people, who were addicted more than any other to numeral mysticism, sought to find in this approximation some connection with the fact that man has ten fingers, and that accordingly ten is the basis of their numeral system.

Reviewing the achievements of the Hindus generally with respect to the problem of quadrature, we are brought to recognise that this people, whose talents lay more in the line of arithmetical computation than in the perception of spatial relations, accomplished as good as nothing on the purely geometrical side of the problem, but that the merit belongs to them of having carried the Archimedean method of computing π several stages farther, and of having obtained in this way a much more exact value for it—a circumstance that is explainable when we consider that the Hindus are the inventors of our present system of numeral notation, possessing which they easily outdid Archimedes, who employed the awkward Greek system.

With regard to the Chinese, this people operated in ancient times with the Babylonian value for π, or 3; but they possessed knowledge of

the approximate value of Archimedes at least since the end of the sixth century. Besides this, there appears in a number of Chinese mathematical treatises an approximate value peculiarly their own, in which $\pi = 3\frac{7}{50}$; a value, however, which notwithstanding it is written in larger figures, is no better than that of Archimedes. Attempts at the *constructive* quadrature of the circle are not found among the Chinese.

Greater were the merits of the Arabians in the advancement of mathematics; and especially in virtue of the fact that they preserved from oblivion the results of both Greek and Hindu research and handed them down to the Christian countries of the West. The Arabians expressly distinguished between the Archimedean approximate value and the two Hindu values, the square root of 10 and the ratio 62832 : 20000. This distinction occurs also in Muhammed Ibn Musa Alchwarizmî, the same scholar who in the beginning of the ninth century brought the principles of our present system of numerical notation from India and introduced it into the Mohammedan world. The Arabians, however, did not study the numerical quadrature of the circle only, but also the constructive; for instance, an attempt of this kind was made by Ibn Alhaitam, who lived in Egypt about the year 1000 and whose treatise upon the squaring of the circle is preserved in a Vatican codex, which unfortunately has not yet been edited.

Christian civilisation, to which we are now about to pass, produced up to the second half of the fifteenth century extremely insignificant results in mathematics. Even with regard to our present problem we have but a single important work to mention; the work, namely, of Frankos von Lüttich on the squaring of the circle, published in six books, but preserved only in fragments. The author, who lived in the first half of the eleventh century, was probably a pupil of Pope Sylvester II., who was himself a not inconsiderable mathematician for his time and the author of the most celebrated geometrical treatise of the period.

Greater interest came to be bestowed upon mathematics, and especially on the problem of the quadrature of the circle, in the second half of the fifteenth century, when the sciences again began to revive. This interest was principally aroused by Cardinal Nicolas de Cusa, a man highly esteemed for his astronomical and calendarial studies. He claimed to have discovered the quadrature of the circle by employing only straight edge and compasses and thus attracted the attention of

scholars to the historic problem. People believed the famous Cardinal, and marvelled at his wisdom, until Regiomontanus, in letters written in 1464 and 1465 and published in 1533, rigorously demonstrated that the Cardinal's quadrature was incorrect. The construction of Cusa was as follows. The radius of a circle is prolonged a distance equal to the side of the inscribed square; the line so obtained is taken as the diameter of a second circle, and in the latter an equilateral triangle is described; then the perimeter of the latter is equal to the circumference of the original circle. If this construction, which its inventor regarded as exact, be considered as a construction of approximation, it will be found to be more inexact even than the construction resulting from the value $\pi = 3\frac{1}{7}$. For by Cusa's method π would be from five to six thousandths smaller than it really is.

In the beginning of the sixteenth century a certain Bovillius appears, who also gave the construction of Cusa,—this time without notice. But about the middle of the sixteenth century a book was published which the scholars of the time at first received with interest. It bore the proud title *De Rebus Mathematicis Hactenus Desideratis*. Its author, Orontius Finaeus, represented that he had overcome all the difficulties that had ever stood in the way of geometrical investigators; and incidentally he also communicated to the world the "true quadrature" of the circle. His fame was short-lived. For soon afterwards, in a book entitled *De Erratis Orontii*, the Portuguese Petrus Nonius demonstrated that Orontius's quadrature, like most of his other professed discoveries, was incorrect.

In the succeeding period the number of circle-squarers so increased that we shall have to limit ourselves to those whom mathematicians recognise. And particularly is Simon Van Eyck to be mentioned, who towards the close of the sixteenth century published a quadrature which was so approximate that the value of π derived from it was more exact even than that of Archimedes; and to disprove it the mathematician Peter Metius was obliged to seek a still more accurate value than $3\frac{1}{7}$. The erroneous quadrature of Van Eyck was thus the occasion of Metius's discovery that the ratio $355 : 113$, or $3\frac{16}{113}$, varied from the true value of π by less than one one-millionth, eclipsing accordingly all values hitherto obtained. Moreover, it is demonstrable by the theory of continued fractions, that, admitting figures to four places only, no two numbers more exactly represent the value of π than 355 and 113.

In the same way the quadrature of the great philologist Joseph Scaliger led to refutations. Like most circle-squarers who believe in their discovery, Scaliger also was little versed in the elements of geometry. He solved the famous problem, however,—at least in his own opinion,—and published in 1592 a book upon it, which bore the pretentious title *Nova Cyclometria*, and in which the name of Archimedes was derided. The baselessness of his supposed discovery was demonstrated to him by the greatest mathematicians of his time; namely, Vieta, Adrianus Romanus, and Clavius.

Of the erring circle-squarers that flourished before the middle of the seventeenth century three others deserve particular mention,—Longomontanus of Copenhagen, who rendered such great services to astronomy, the Neapolitan John Porta, and Gregory of St. Vincent. Longomontanus made $\pi = 3\frac{14185}{100000}$ and was so convinced of the correctness of his result as to thank God fervently, in the preface to his work *Inventio Quadraturae Circuli*, that He had granted him in his old age the strength to conquer the celebrated difficulty. John Porta followed the example of Hippocrates and endeavored to solve the problem by a comparison of lunes. Gregory of St. Vincent published a quadrature, the error of which was very hard to detect but was finally discovered by Descartes.

Of the famous mathematicians who dealt with our problem in the period between the close of the fifteenth century and the time of Newton, we first meet with Peter Metius, before mentioned, who succeeded in finding in the fraction $355 : 113$ the best approximate value for π involving small numbers only. The problem received a different advancement at the hands of the famous mathematician Vieta. Vieta was the first to whom the idea occurred of representing π with mathematical exactness by an infinite series of definitely prescribed operations. By comparing inscribed and circumscribed polygons, Vieta found that we approach nearer and nearer to π if we cause the operations of extracting the square root of $\frac{1}{2}$, and certain related additions and multiplications, to succeed each other in a certain manner, and that π must come out exactly, if this series of operations could be continued indefinitely. Vieta thus found that to a diameter of 10000 million units a circumference belongs of from 31415 million 926535 units to 31415 million 926536 units of the same length.

But Vieta was outdone by the Netherlander Adrianus Romanus, who added five additional decimal places to the ten of Vieta. To accomplish this he computed with unspeakable labor the circumference of a regular circumscribed polygon of 1073741824 sides. This number is the thirtieth power of 2. Yet great as the labor of Adrianus Romanus was, that of Ludolf Van Ceulen was still greater; for the latter calculator succeeded in carrying the Archimedean process of approximation for the value of π to 35 decimal places; that is, the deviation from the true value was smaller than one one-thousand quintillionth, a degree of exactness that we can have scarcely any conception of. Ludolf published the figures of the tremendous computation that led to his result. His calculation was carefully examined by the mathematician Griemberger and declared to be correct. Ludolf was justly proud of his work, and following the example of Archimedes, requested in his will that the result of his most important mathematical performance, the computation of π to 35 decimal places, be engraved upon his tombstone; a request which is said to have been carried out. In honor of Ludolf, π is called to-day in Germany the Ludolfian number.

Although through the labor of Ludolf a degree of exactness for cyclometrical operations was now obtained that was more than sufficient for any practical purpose that could ever arise, neither the problem of constructive rectification nor that of constructive quadrature had been in any respect theoretically advanced thereby. The investigations conducted by the famous mathematicians and physicists Huygens and Snell about the middle of the seventeenth century, were more important from a mathematical point of view than the work of Ludolf. In his book *Cyclometricus* Snell took the position that the method of comparison of polygons, which originated with Archimedes and was employed by Ludolf, was not necessarily the best method of attaining the end sought; and he succeeded by employing propositions which state that certain arcs of a circle are greater or smaller than certain straight lines connected with the circle, in obtaining methods that make it possible to reach results like the Ludolfian with much less labor of calculation. The beautiful theorems of Snell were proved a second time, and better proved, by the celebrated Dutch promoter of the science of optics, Huygens (*Opera Varia*, p. 365 et seq.; *Theoremata De Circuli et Hyperbolae Quadratura*, 1651), as well as perfected in many ways by him. Snell and

Huygens were fully aware that they had advanced the problem of numerical quadrature only, and not that of the constructive quadrature. This plainly appeared in Huygens's case from the vehement dispute which he conducted with the English mathematician James Gregory. This controversy is significant for the history of our problem, from the fact that Gregory made the first attempt to prove that the squaring of the circle with straight edge and compasses was impossible. The result of the controversy, to which we owe many valuable tracts, was, that Huygens finally demonstrated in an incontrovertible manner the incorrectness of Gregory's proof of impossibility, adding that he also was of opinion that the solution of the problem with straight edge and compasses was impossible, but nevertheless was not himself able to demonstrate this fact. And Newton later expressed himself to the same effect. As a matter of fact a period of over 200 years elapsed before higher mathematics was far enough advanced to furnish a rigorous demonstration of impossibility.

V.

FROM NEWTON TO THE PRESENT.

Before we proceed to consider the promotive influence which the invention of the differential and the integral calculus exercised upon our problem, we shall enumerate a few at least of that never-ending succession of erring quadrators who delighted the world with the products of their ingenuity from the time of Newton to the present; and out of a pious and sincere regard for the contemporary world, we shall omit entirely to speak of the circle-squarers of our own time.

First to be mentioned is the celebrated English philosopher Hobbes. In his book *De Problematis Physicis*, in which he proposes to explain the phenomena of gravity and of ocean tides, he also takes up the quadrature of the circle and gives a very trivial construction, which in his opinion definitively solved the problem. It made $\pi = 3\frac{1}{5}$. In view of Hobbes's importance as a philosopher, two mathematicians, Huygens and Wallis, thought it proper to refute him at length. But Hobbes defended his position in a special treatise, where to sustain at least the appearance of being right, he disputed the fundamental principles of geometry and the theorem of Pythagoras.

In the last century France especially was rich in circle-squarers. We will mention: Oliver de Serres, who by means of a pair of scales determined that a circle weighed as much as the square upon the side of the equilateral triangle inscribed in it, that therefore they must have the same area, an experiment in which $\pi = 3$; Mathulon, who offered in legal form a reward of a thousand dollars to the person who would point out an error in his solution of the problem, and who was actually compelled by the courts to pay the money; Basselin, who believed that his quadrature must be right because it agreed with the approximate value of Archimedes, and who anathematised his ungrateful contemporaries, in the confidence that he would be recognised by posterity; Liger, who proved that a part is greater than the whole and to whom therefore the quadrature of the circle was child's play; Clerget, who based his solution upon the principle that a circle is a polygon of a definite number of sides, and who calculated, also, among other things, how large the point is at which two circles touch.

Germany and Poland also furnish their contingent to the army of circle-squarers. Lieutenant-Colonel Corsonich produced a quadrature in which π equalled $3\frac{1}{8}$, and promised fifty ducats to the person who could prove that it was incorrect. Hesse of Berlin wrote an arithmetic in 1776, in which a true quadrature was also "made known," π being exactly equal to $3\frac{14}{99}$. About the same time Professor Bischoff of Stettin defended a quadrature previously published by Captain Leistner, Preacher Merkel, and Schoolmaster Böhm, which virtually made π equal to the square of $\frac{62}{35}$, not even attaining the approximation of Archimedes.

From attempts of this character are to be clearly distinguished constructions of approximation in which the inventor is aware that he has not found a mathematically exact construction, but only an approximate one. The value of such a construction will depend upon two things—first, upon the degree of exactness with which it is numerically expressed, and secondly on whether the construction can be easily made with straight edge and compasses. Constructions of this kind, simple in form and yet sufficiently exact for practical purposes, have been produced for centuries in great numbers. The great mathematician Euler, who died in 1783, did not think it out of place to attempt an approximate construction of this kind. A very simple construction for the

rectification of the circle and one which has passed into many geometrical text-books is that published by Kochansky in 1685 in the *Leipziger Berichte*. It is as follows: "Erect upon the diameter of a circle at its extremities perpendiculars; with the centre as vertex and the diameter as side construct an angle of 30°; find the point of intersection of the line last drawn with the perpendicular, and join this point of intersection with that point on the other perpendicular which is distant three radii from the base of the perpendicular. The line of junction so obtained is very approximately equal to one-half of the circumference of the given circle." Calculation shows that the difference between the true length of the circumference and the line thus constructed is less than $\frac{3}{100000}$ of the diameter.

Although such constructions of approximation are very interesting in themselves, they nevertheless play but a subordinate rôle in the history of the squaring of the circle; for on the one hand they can never furnish greater exactness for circle-computation than the thirty-five decimal places which Ludolf found, and on the other hand they are not adapted to advance in any way the question whether the exact quadrature of the circle with straight edge and compasses is possible.

The numerical side of the problem, however, was considerably advanced by the new mathematical methods perfected by Newton and Leibnitz, and known as the differential and the integral calculus.

About the middle of the seventeenth century, before Newton and Leibnitz represented π by series of powers, the English mathematicians Wallis and Lord Brouncker, Newton's predecessors in certain lines, succeeded in representing π by an infinite series of figures combined according to the first four rules of arithmetic. A new method of computation was thus opened. Wallis found that the fourth part of π is represented by the regularly formed product

$$\tfrac{2}{3} \times \tfrac{4}{3} \times \tfrac{4}{5} \times \tfrac{6}{5} \times \tfrac{6}{7}\tfrac{8}{7}\tfrac{8}{9} \times \text{etc.}$$

more and more exactly the farther the multiplication is continued, and that the result always comes out too small if we stop at a proper fraction but too large if we stop at an improper fraction. Lord Brouncker, on the other hand, represents the value in question by a continued fraction in which the denominators are all 2 and the numerators are the squares of the odd numbers. Wallis, to whom Brouncker had communicated his

elegant result without proof, demonstrated the same in his *Arithmetic of Infinites*.

The computation of π could scarcely have been pushed to a greater degree of exactness by these results than that to which Ludolf and others had carried it by the older and more laborious methods. But the series of powers derived from the differential calculus of Newton and Leibnitz furnished a means of computing π to hundreds of decimal places.

Gregory, Newton, and Leibnitz found that the fourth part of π was equal exactly to

$$1 - \tfrac{1}{3} + \tfrac{1}{5} - \tfrac{1}{7} + \tfrac{1}{9} - \tfrac{1}{11} + \tfrac{1}{13} - \dots$$

if we conceive this series, which is called the *Leibnitz series*, continued indefinitely. This series is wonderfully simple but is not adapted to the computation of π, for the reason that entirely too many members have to be taken into account to obtain π accurately to a few decimal places only. The original formula, however, from which this series is derived, gives other formulæ which are excellently adapted to the actual computation. The original formula is the general series:

$$\alpha = a - \tfrac{1}{3}a^3 + \tfrac{1}{5}a^5 - \tfrac{1}{7}a^7 + \dots,$$

where α is the length of the arc belonging to any central angle in a circle of radius 1, and a the tangent to this angle. From this we derive the following:

$$\frac{\pi}{4} = (a + b + c + \dots) - \tfrac{1}{3}(a^3 + b^3 + c^3 + \dots) + \tfrac{1}{5}(a^5 + b^5 + c^5 + \dots) - \dots,$$

where $a, b, c \dots$ are the tangents of angles whose sum is $45°$. Determining, therefore, the values of $a, b, c \dots$, which are equal to small and convenient fractions and fulfil the conditions just mentioned, we obtain series of powers which are adapted to the computation of π.

The first to add by the aid of series of this description additional decimal places to the old 35 in the number π was the English arithmetician Abraham Sharp, who, following Halley's instructions, in 1700 worked out π to 72 decimal places. A little later Machin, professor of astronomy in London, computed π to 100 decimal places, by putting,

in the series given above, $a = b = c = d = \frac{1}{5}$ and $e = -\frac{1}{239}$; that is, by employing the following series:

$$\frac{\pi}{4} = 4 \cdot \left[\frac{1}{5} - \frac{1}{3 \cdot 5^2} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} + \dots\right] - \left[\frac{1}{239} - \frac{1}{3 \cdot 239^3} + \frac{1}{5 \cdot 239^5} - \dots\right]$$

In the year 1819, Lagny of Paris outdid the computation of Machin, determining in two different ways the first 127 decimal places of π. Vega then obtained as many as 140 places, and the Hamburg arithmetician Zacharias Dase went as far as 200 places. The latter did not use Machin's series in his calculation, but the series produced by putting in the general series above given $a = \frac{1}{2}$, $b = \frac{1}{5}$, $c = \frac{1}{8}$. Finally, at a recent date, π has been computed to 500 places.*

The computation to so many decimal places may serve as an illustration of the excellence of the modern methods as contrasted with those anciently employed, but it has otherwise neither a theoretical nor a practical value. That the computation of π to say 15 decimal places more than sufficiently satisfies the subtlest requirements of practice may be gathered from a concrete example of the degree of exactness thus obtainable. Imagine a circle to be described with Berlin as centre, and the circumference to pass through Hamburg; then let the circumference of the circle be computed by multiplying its diameter by the value of π to 15 decimal places, and then conceive it to be actually measured. The deviation from the true length in so large a circle as this even could not be as great as the 18 millionth part of a millimetre.

An idea can hardly be obtained of the degree of exactness produced by 100 decimal places. But the following example may possibly give us some conception of it. Conceive a sphere constructed with the earth as centre, and imagine its surface to pass through Sirius, which is $134\frac{1}{2}$ millions of millions of kilometres distant from the earth. Then imagine this enormous sphere to be so packed with microbes that in every cubic millimetre millions of millions of these diminutive animalcula are present. Now conceive these microbes to be all unpacked and so distributed singly along a straight line, that every two microbes are as far distant from each other as Sirius from us, that is $134\frac{1}{2}$ million million

* In 1873 the approximation was carried by Shanks to 707 places of decimals.—*Trans.*

kilometres. Conceive the long line thus fixed by all the microbes, as the diameter of a circle, and imagine the circumference of it to be calculated by multiplying its diameter by π to 100 decimal places. Then, in the case of a circle of this enormous magnitude even, the circumference so calculated would not vary from the real circumference by a millionth part of a millimetre.

This example will suffice to show that the calculation of π to 100 or 500 decimal places is wholly useless.

Before we close this chapter upon the evaluation of π, we must mention the method, less fruitful than curious, which Professor Wolff of Zurich employed some decades ago to compute the value of π to 3 places.* The floor of a room is divided up into equal squares, so as to resemble a huge chess-board, and a needle exactly equal in length to the side of each of these squares, is cast haphazard upon the floor. If we calculate, now, the probabilities of the needle so falling as to lie wholly within one of the squares, that is so that it does not cross any of the parallel lines forming the squares, the result of the calculation for this probability will be found to be exactly equal to $\pi - 3$. Consequently, a sufficient number of casts of the needle according to the law of large numbers must give the value of π approximately. As a matter of fact, Professor Wolff, after 10000 trials, obtained the value of π correctly to 3 decimal places.

Fruitful as the calculus of Newton and Leibnitz was for the evaluation of π, the problem of converting a circle into a square having exactly the same area was in no wise advanced thereby. Wallis, Newton, Leibnitz, and their immediate followers distinctly recognised this. The quadrature of the circle could not be solved; but it also could not be proved that the problem was insolvable with straight edge and compasses, although everybody was convinced of its insolvability. In mathematics, however, a conviction is only justified when supported by incontrovertible proof; and in the place of endeavors to solve the quadrature there accordingly now come endeavors to prove the impossibility of solving the celebrated problem.

The first step in this direction, small as it was, was made by the French mathematician Lambert, who proved in the year 1761 that π was neither a rational number nor even the square root of a rational number; that is, that neither π nor the square of π could be exactly

* See also A. De Morgan, *A Budget of Paradoxes*, pp. 169–171.—*Tr.*

represented by a fraction the denominator and numerator of which are whole numbers, however great the numbers be taken. Lambert's proof* showed, indeed, that the rectification and the quadrature of the circle could not be accomplished in one particular simple way, but it still did not exclude the possibility of the problem being solvable in some other more complicated way, and without requiring further aids than straight edge and compasses.

Proceeding slowly but surely it was next sought to discover the essential properties which distinguish problems solvable with straight edge and compasses from problems the construction of which is elementarily impossible, that is by employing the postulates only. Slight reflection showed, that a problem, to be elementarily solvable, must always be such that the unknown lines of its figure are connected with the known lines by an equation for the solution of which equations of the first and second degree only are requisite, and which can be so arranged that the measures of the known lines will appear as integers only. The conclusion to be drawn from this was that if the quadrature of the circle and consequently its rectification were solvable elementarily, the number π, which represents the ratio of the unknown circumference to the known diameter, must be the root of a certain equation, of a very high degree perhaps, but in which all the numbers are whole numbers; that is, there would have to exist an equation, made up entirely of whole numbers, which would be correct if its unknown quantity were made equal to π.

Since the beginning of this century, consequently, the efforts of a number of mathematicians have been bent upon proving that π generally is not algebraical, that is, that it cannot be the root of an equation having whole numbers for coefficients. But mathematics had to make tremendous strides forward before the means were at hand to accomplish this demonstration. After the French Academician, Professor Hermite, had furnished important preparatory assistance in his treatise *Sur la Fonction Exponentielle*, published in the seventy-seventh volume of the *Comptes Rendus*, Professor Lindemann, at that time of Freiburg, now of Munich, finally succeeded, in June 1882, in rigorously

* Given in Legendre's *Geometry*, in the Appendix to De Morgan, *op. cit.*, p. 495, and in Rudio, *op. cit.*—*Tr.*

demonstrating that the number π is not algebraical,* and so supplied the first proof that the problems of the rectification and squaring of the circle with the help only of algebraical instruments like straight edge and compasses are insolvable. Lindemann's proof appeared successively in the *Reports of the Berlin Academy* (June, 1882), in the *Comptes Rendus* of the French Academy (Vol. 115, pp. 72 to 74), and in the *Mathematische Annalen* (Vol. 20, pp. 213 to 225).

"It is impossible with straight edge and compasses to construct a square equal in area to a given circle." These are the words of the final determination of a controversy which is as old as the history of the

* For the benefit of my mathematical readers I shall present here the most important steps of Lindemann's demonstration. M. Hermite in order to prove the transcendental character of

$$e = 1 + \frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.2.3} + \frac{1}{1.2.3.4} + \dots$$

developed relations between certain definite integrals (*Comptes Rendus* of the Paris Academy, Vol. 77, 1873). Proceeding from the relations thus established, Professor Lindemann first demonstrates the following proposition: If the coefficients of an equation of the nth degree are all real or complex whole numbers and the n roots of this equation $z_1, z_2, \dots, z_n$ are different from zero and from each other it is impossible for

$$e^{z_1} + e^{z_2} + e^{z_3} \cdots + e^{z_n}$$

to be equal to $\frac{a}{b}$, where a and b are real or complex whole numbers. It is then shown that also between the functions

$$e^{rz_1} + e^{rz_2} + e^{rz_3} + \dots e^{rz_n},$$

where r denotes an integer, no linear equation can exist with rational coefficients different from zero. Finally the beautiful theorem results: If z is the root of an irreducible algebraic equation the coefficients of which are real or complex whole numbers, then e^z cannot be equal to a rational number. Now in reality $e^{\pi\sqrt{-1}}$ is equal to a rational number, namely, -1. Consequently, $\pi\sqrt{-1}$, and therefore π itself, cannot be the root of an equation of the nth degree having whole numbers for coefficients, and therefore also not of such an equation having rational coefficients. The property last mentioned, however, π would have if the squaring of the circle with straight edge and compasses were possible. [The questions involved in the discussions of the last three pages have been excellently treated by Klein in *Famous Problems of Elementary Geometry* recently translated by Beman and Smith (Ginn & Co., Boston). Lindemann's proof is here presented in a simplified form, and so brought within the comprehension of students conversant only with algebra.—*Tr.*]

human mind. But the race of circle-squarers, unmindful of the verdict of mathematics, the most infallible of arbiters, will never die out as long as ignorance and the thirst for glory remain united.

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